

# NOTES

## SECTION G2

### TIRES AND RIMS

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# NOTES

## TIRES AND RIMS

The truck tires should be inspected and tire pressure checked with an accurate pressure gauge before each working shift. Tire pressure will vary according to manufacturer and local working conditions. Consult the tire manufacturer for recommended tire pressure.

Insure valve caps are securely applied to valve stems. The caps protect valves from dirt build up and damage. DO NOT bleed air from tires which are hot due to operation; under such circumstances, it is normal for pressure to increase in the tire due to expansion.

A bent or damaged rim which does not support the bead properly may cause abnormal strain on the tire resulting in tire damage. If a tire becomes deeply cut, it should be removed and repaired. Neglected cuts cause many tire problems; water, sand, dirt and other foreign materials work into the tire through a cut, eventually causing tread or ply separation.

Tires should be stored indoors if possible. If stored outdoors, cover tires with tarpaulin to keep out dirt, water and other foreign materials. Long exposure to the sun will cause ozone cracks. Storage should be in a cool, dry, dark, draft free location. Tires should be stored vertically. If they must be laid on their sides for a short period, avoid distortion by stacking no more than three tires on top of one another. Avoid contact with oil, grease and other petroleum products.

Before storing used tires, clean thoroughly and inspect for damage. Repair as necessary. When a truck is placed in storage, it should be blocked to remove the weight from the tires. If a stored truck cannot be blocked, check air pressure and inspect tires twice a month for proper inflation pressure.

### **⚠ WARNING**

**DO NOT weld or apply heat on the rim assembly with the tire mounted on the rim. Remaining gases inside the tire may ignite causing explosion of tire and rim.**

**DO NOT go near a tire if a brake or wheel motor has experienced a fire until the tire has cooled.**

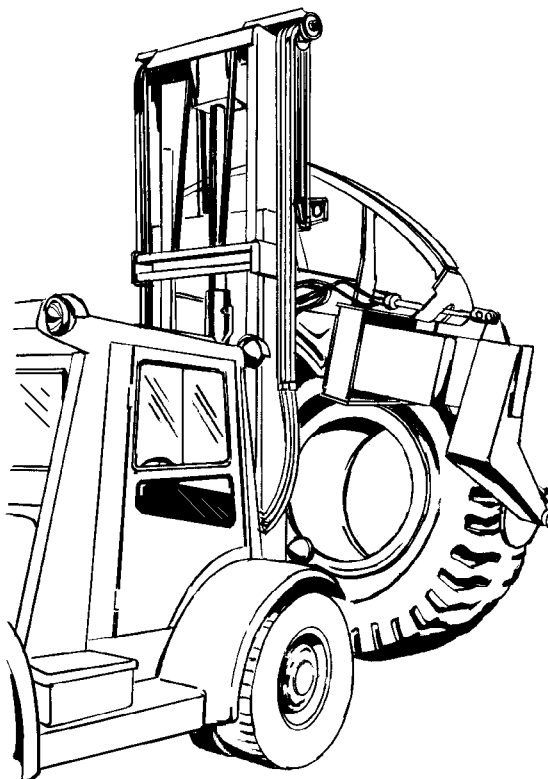
### **⚠ DANGER**

***When inflating tires always use a safety cage. Never inflate a tire until the locking is securely in place. Do not stand in front of or over the locking during inflation procedures. Never overinflate a tire. Refer to tire manufacturer's recommendations.***

***Always keep personnel away from a wheel and tire assembly when it is being removed or installed.***

***The tire and rim weigh approximately 7 165 kg (15,800 lb). Make sure that tire handling equipment is capable of lifting and maneuvering the load.***

Due to the size and weight of the tire and rim assemblies, special handling equipment, such as a modified fork lift called a "tire handler" as shown in Figure 2-1, is desirable. Consult local tire vendors for sources of equipment designed especially to remove, repair, and install large off-highway truck tires.



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FIGURE 2-1. TYPICAL TIRE HANDLER



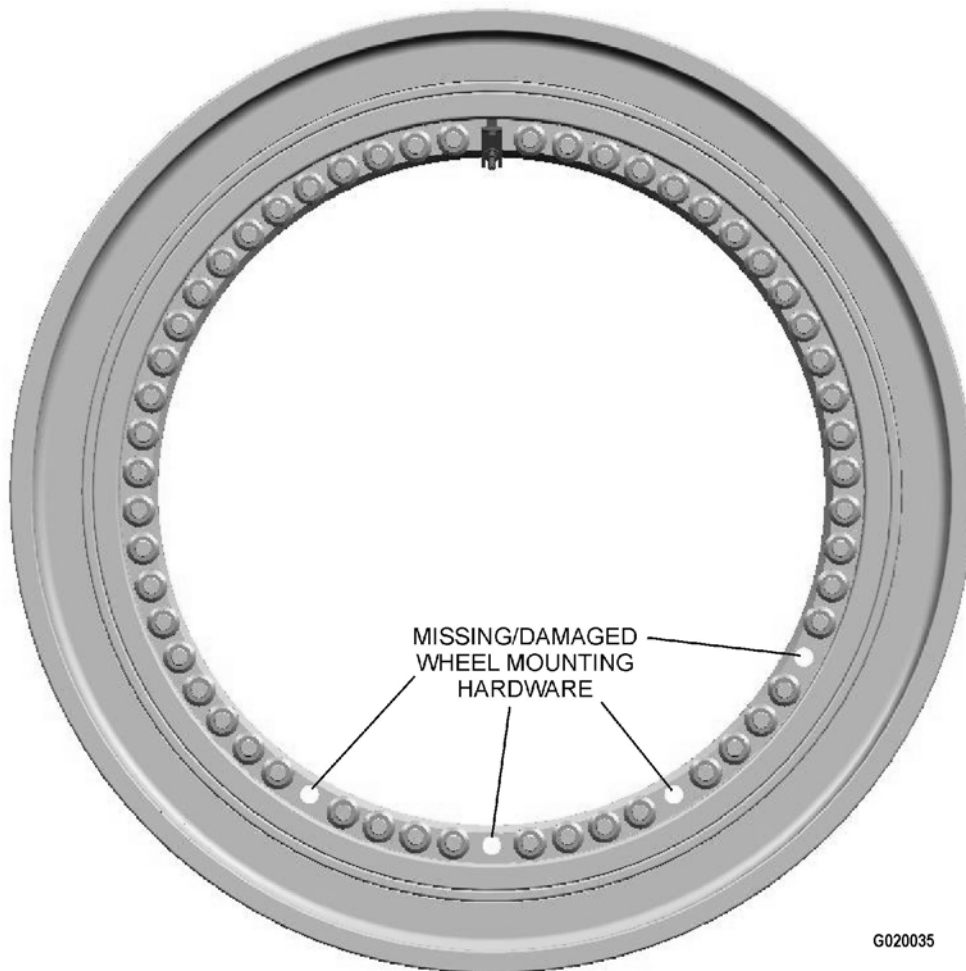
## WHEEL STUD MAINTENANCE

The wheel mounting hardware used on some trucks may no longer be able to maintain the specified clamping force (tightening torque). Inadequate clamping force may lead to broken studs, stripped threads, and/or failure of the stud knurl which would allow the stud to rotate when the nut is loosened or tightened. Stud replacement is necessary based on the following guidelines:

- The maximum allowable number of missing or damaged studs in any one bolt circle is four.
- The position of missing or damaged studs must not be in consecutive locations. In addition, there must be a minimum of four properly functioning studs between each missing stud. Refer to Figure 2-2.

### **CAUTION**

*While replacing missing or damaged studs, the condition of all other stud threads should be inspected. If minor corrosion or dirt is found in the threads, wire brush thoroughly. If damaged (knicked or highly corroded) threads are found, the stud(s) must be replaced.*



G020035

FIGURE 2-2. MISSING/DAMAGED WHEEL STUD ALLOWANCE

## FRONT TIRES AND RIMS

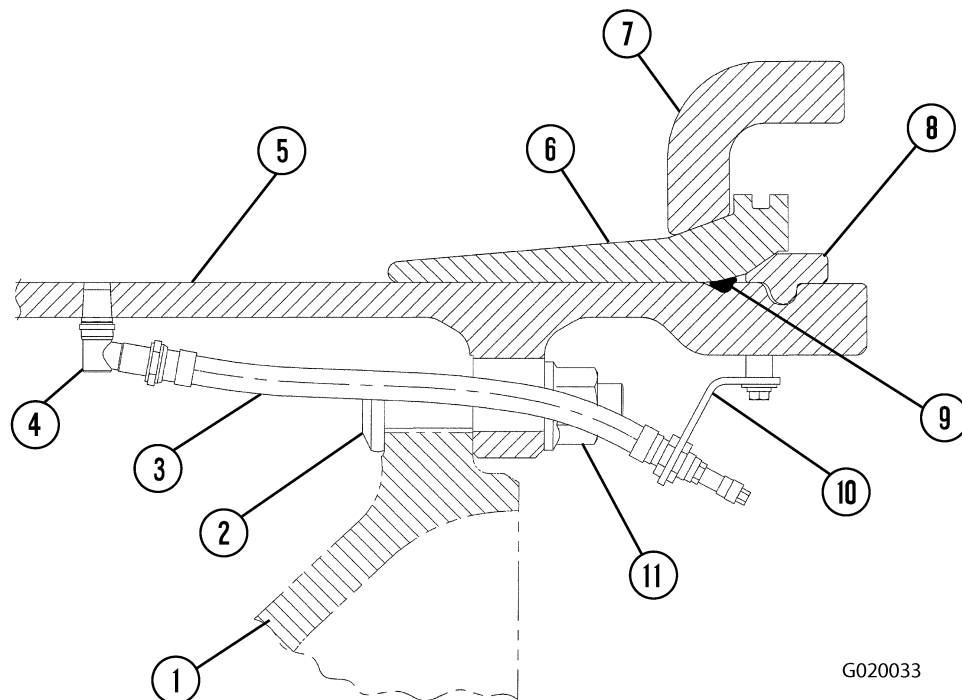
### Removal

1. Apply parking brake and block rear wheels to prevent movement of truck.
2. Following normal shutdown procedures, place the rest switch in the ON position, shut down the engine and verify that the link voltage lights are OFF. Allow at least 90 seconds for the accumulators to bleed down. Turn the steering wheel to ensure that no pressure remains. As a safety precaution, bleed down the brake accumulators.
3. Place a jack under the spindle or frame at the front cross tube.
4. Raise the front end of the truck until the tire clears the ground. Block up the truck securely under the frame.
5. Inspect the hydraulic brake lines for damage or leaking fittings.
6. Grip the tire and wheel assembly with tire a handler. Remove nuts (11, Figure 2-3) that secure the wheel assembly.
7. Be careful not to damage the inflation hose during tire removal. Move the wheel assembly away from the wheel hub and into a clean work area.

### **WARNING**

***Do not attempt to disassemble wheel assembly until all air pressure is bled off.***

***Always keep personnel away from a wheel assembly when it is being removed or installed.***



G020033

FIGURE 2-3. FRONT WHEEL HUB AND RIM ASSEMBLY

- |                        |                   |                   |
|------------------------|-------------------|-------------------|
| 1. Wheel Hub           | 5. Rim            | 9. O-Ring         |
| 2. Stud                | 6. Bead Seat Band | 10. Clamp Bracket |
| 3. Tire Inflation Hose | 7. Side Flange    | 11. Flanged Nut   |
| 4. Swivel Connector    | 8. Lock Ring      |                   |

## Installation

*NOTE: Remove all dirt and rust from mating parts before installing wheel assembly.*

1. Grip the wheel assembly with the tire handler and align the tire inflation hose and wheel hub (1, Figure 2-3). Position the rim onto the wheel hub studs.



***Grease containing molybdenum disulphide must never be used on wheel mounting hardware. Use of this type of grease on wheel mounting hardware may result in wheel mounting studs stretching beyond their elastic limit, making them susceptible to breakage.***

2. Lubricate all stud threads and nut seating flanges with a lithium based grease that does not contain molybdenum disulphide. Install and tighten the nuts in the following sequence:
  - a. Install six nuts at the 12 o'clock and six nuts at the 6 o'clock positions. Tighten each nut to **2 325 ± 136 N·m (1,715 ± 100 ft lb)**.
  - b. Install three nuts directly below the 3 o'clock and three nuts directly above the 9 o'clock positions. Tighten the nuts to **2 325 ± 136 N·m (1,715 ± 100 ft lb)**.
  - c. Install three nuts directly above the 3 o'clock and three nuts directly below the 9 o'clock positions. Tighten these nuts to **2 325 ± 136 N·m (1,715 ± 100 ft lb)**.
  - d. Install the remaining nuts and torque in a clockwise direction to **2 325 ± 136 N·m (1,715 ± 100 ft lb)**.
  - e. Retighten all nuts in a clockwise direction to **2 325 ± 136 N·m (1,715 ± 100 ft lb)**.
3. Remove the blocking and lower the jack.
4. Operate the truck for one load and retighten the wheel nuts to the specified torque.

## REAR TIRES AND RIMS

### Removal

1. Park the truck on level ground and block the front wheels. Position a jack under the rear suspension mounting plates as shown in Figure 2-4.

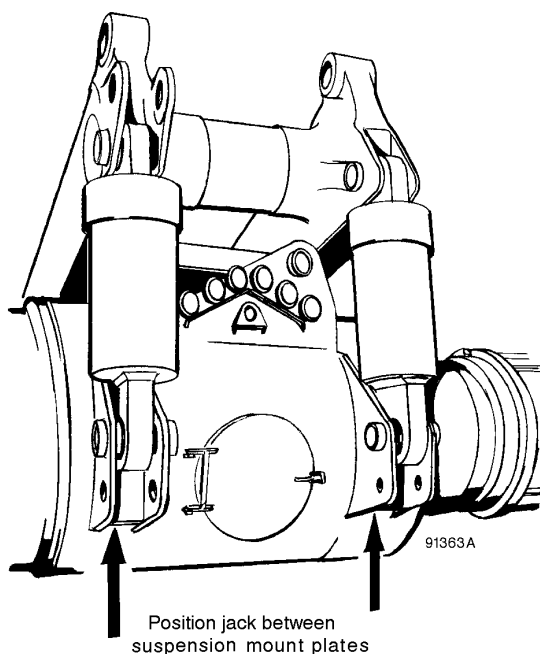


FIGURE 2-4. REAR AXLE JACK LOCATION

2. Raise the rear axle housing until the tires clear the ground. Securely block up the rear axle housing near the wheel motor mounting flange.
3. If the rear inner tire is to be removed, remove the inner tire inflation hose at the clamp on the outer wheel and disconnect it from inner extension (7, Figure 2-6).
4. Grip outer wheel and tire with tire handler arms as shown in Figure 2-5. Remove flanged nuts (10, Figure 2-6) from adaptor ring (9) securing outer rim to wheel motor hub.
5. Pull straight out on the outer wheel assembly and remove it.
6. If inner wheel removal is necessary, disconnect inner tire inflation extension (7) and remove it.

7. Position a tire handler to grip the inner wheel. Remove flanged nuts (6).
8. Pull straight out to remove the tire from the wheel hub.

### ⚠ IMPORTANT ⚠

*If the wheel motor is to be removed from the truck for service, install approximately eight flanged nuts with appropriate spacers in place of the outer wheel adaptor ring and nuts removed in Step 4. This will provide additional support for the wheel motor transmission during removal and transportation.*

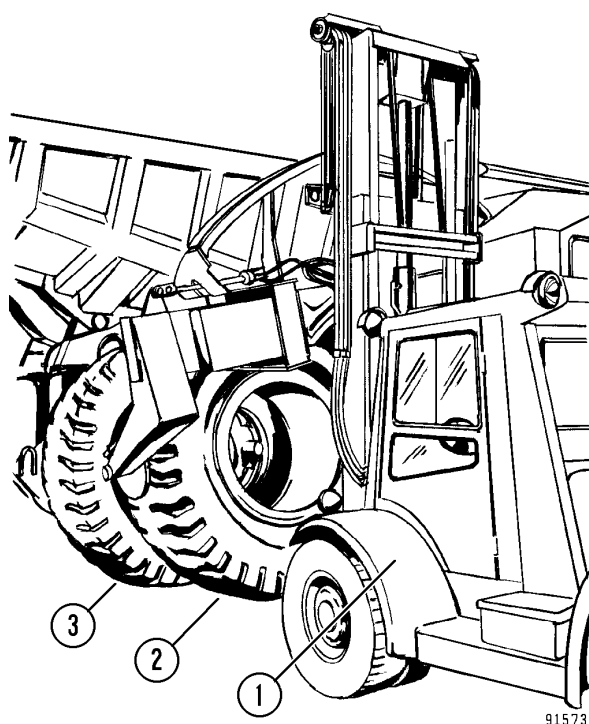
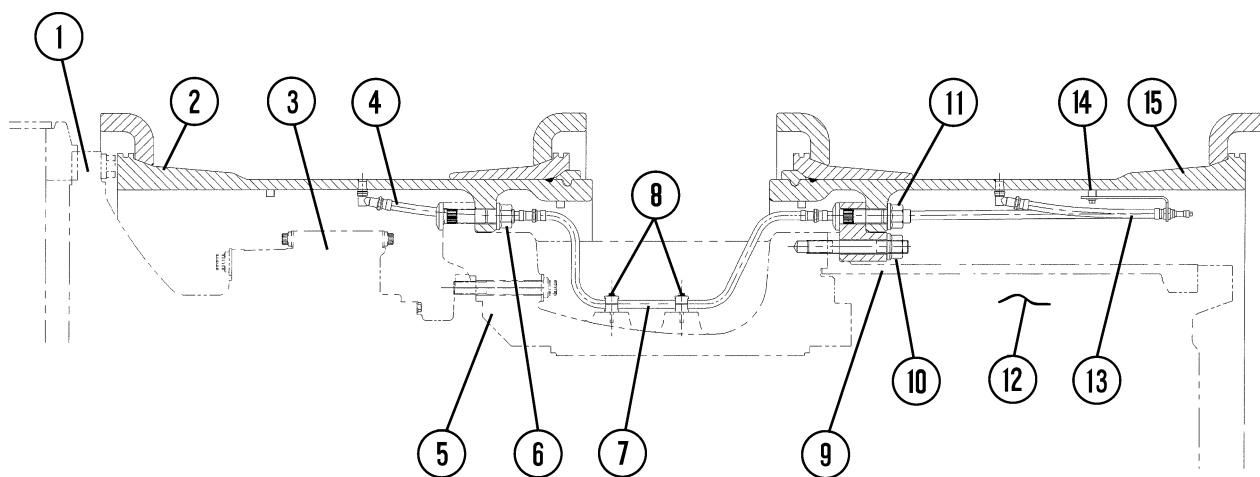


FIGURE 2-5. REAR TIRE REMOVAL

- |                    |                    |
|--------------------|--------------------|
| 1. Tire Handler    | 3. Inner Rear Tire |
| 2. Outer Rear Tire |                    |



G020034

FIGURE 2-6. REAR WHEEL ASSEMBLY

- |                                |                    |                              |
|--------------------------------|--------------------|------------------------------|
| 1. Wheel Motor Mounting Flange | 6. Flanged Nut     | 11. Flanged Nut              |
| 2. Inner Wheel Rim             | 7. Inner Extension | 12. Wheel Motor Transmission |
| 3. Disc Brake Assembly         | 8. Clamps          | 13. Extension                |
| 4. Extension                   | 9. Adapter Ring    | 14. Bracket                  |
| 5. Wheel Hub                   | 10. Flanged Nut    | 15. Outer Wheel Rim          |

## Installation



**Always keep personnel away from a wheel assembly when it is being removed and installed.**

**NOTE:** Clean all mating surfaces and check the stud threads before installing the wheel assemblies.

1. If either wheel motor has been removed, bleed the disc brakes before installing the rear tires. Refer to Section J, Wet Disc Brake Assembly - Wet Disc Brake Bleeding Procedure.
2. Grip the inner wheel assembly with a tire handler and install it onto wheel hub (5, Figure 2-6). Carefully align tire inflation extension line (4) for mating with inner extension (7).
3. Lubricate all stud threads and nut seating flanges with lithium based grease. Install and tighten the nuts in the following sequence:
  - a. Install six nuts at the 12 o'clock and six nuts at the 6 o'clock positions. Tighten each nut to **2 325 ± 136 N·m (1,715 ± 100 ft lb)**.
  - b. Install three nuts directly below the 3 o'clock and three nuts directly above the 9 o'clock positions. Tighten these nuts to **2 325 ± 136 N·m (1,715 ± 100 ft lb)**.
  - c. Install three nuts directly above the 3 o'clock and three nuts directly below the 9 o'clock positions. Tighten these nuts to **2 325 ± 136 N·m (1,715 ± 100 ft lb)**.
  - d. Install the remaining nuts and tighten in a clockwise direction to **2 325 ± 136 N·m (1,715 ± 100 ft lb)**.
  - e. Retighten all nuts in a clockwise direction to the required **2 325 ± 136 N·m (1,715 ± 100 ft lb)**.
4. If adapter ring (9) requires installation, install the adapter onto the wheel hub/wheel motor. Lubricate the studs and nut flanges with lithium based grease.
5. Using the procedures in Step 3, install flanged nuts (10) in the sequence described and tighten to the same torque value as the inner wheel.

6. Grip the outer wheel assembly with a tire handler and position it onto the wheel motor hub.

**NOTE:** Position the outer dual wheel to align the tire valve bracket with the inner wheel inflation line.

7. Using the procedures in Step 3, install flanged nuts (11) in the sequence described and tighten to the same torque value as the inner wheel.
8. Secure the inner and outer dual tire inflation lines to bracket (14) on the outer rim.
9. Remove the blocks from under the truck and lower the truck to the ground. Operate the truck for one load and retighten outer wheel nuts (11) and adapter flange nuts (10) to **2 325 ± 136 N·m (1,715 ± 100 ft lb)**.

**NOTE:** Inner flanged nuts (5 & 7) and studs should be inspected for breakage or missing nuts during scheduled maintenance checks by inserting a mirror between the rear tires.

## RIM

### Tire Removal



**DO NOT weld or apply heat on the rim assembly with the tire mounted on the rim. Resulting gases inside the tire may ignite, causing an explosion.**

**When inflating tires always use a safety cage. Never inflate a tire until the lockring is securely in place. Do not stand in front of or over the lockring during inflation procedures. Never overinflate a tire. Refer to tire manufacturers recommendations.**

1. Place the tire and wheel assembly in a safety cage and discharge all air pressure from the tire.
2. Attach a hydraulic bead breaker to the rim by slipping the jaws of the frame assembly over the outer edge of flange (7, Figure 2-7). Make sure that the jaws of the frame are as near to bead seat band (6) as possible.
3. Following the tool manufacturer's instructions, move the tire bead in far enough to permit placing a wedge between the tire and the flange at the side of the tool.

4. Repeat this procedure at locations approximately 90° from the first application. Continue this procedure until the tire bead is free from the rim.
5. After the bead is broken loose, insert the flat of a tire tool in the beading notch on lockring (8). Pry the lockring up and out of the groove on the rim.
6. Pry in on bead seat band (6) until O-ring (9) is exposed. Remove the O-ring.
7. Remove bead seat band (6) from rim (5) and remove flange (7).
8. Reposition wheel assembly and repeat removal procedure on opposite side of tire. Remove the tire from the rim.

### Tire Installation

1. Before mounting the tire to the rim, remove all dirt and rust from the rim parts, particularly the O-ring groove and bead seats. Also touch up all metal parts with anti-rust paint to prevent bare metal from being exposed to the weather.

**NOTE:** Do not allow paint, rust or other contamination to cover the mating faces of lock ring (8, Figure 2-7) and rim (5).



**Make sure that proper rim parts are used for reassembly. Use of incompatible parts may not properly secure the assembly, resulting in violently flying parts upon inflation.**

2. If the tire inflation hose and hardware were removed, reinstall them in the rim. Position the hose assembly for proper routing.
3. Install the inner flange on the rim. Coat the beads of the tire with tire mounting soap solution.



**Prying against tire bead may cause damage to tire bead and will cause air leaks.**

4. Position the tire over the rim and work the tire on as far as possible without prying against the beads.

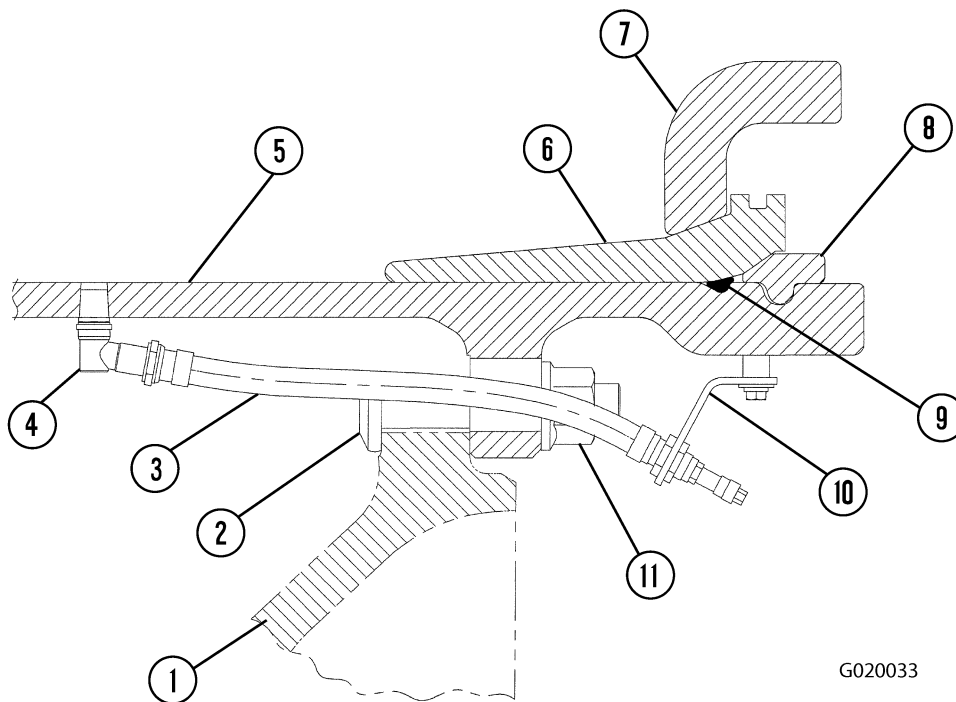


FIGURE 2-7. FRONT WHEEL HUB AND RIM ASSEMBLY

- |                        |                   |                   |
|------------------------|-------------------|-------------------|
| 1. Wheel Hub           | 5. Rim            | 9. O-Ring         |
| 2. Stud                | 6. Bead Seat Band | 10. Clamp Bracket |
| 3. Tire Inflation Hose | 7. Side Flange    | 11. Flanged Nut   |
| 4. Swivel Connector    | 8. Lock Ring      |                   |

5. Install outer flange (7, Figure 2-7) in position and install bead seat band (6). Push in on the bead seat band to expose the O-ring groove in the rim.

6. Lubricate new O-ring (9) with soap solution and install it in the groove of the rim.

7. Install lockring (8) and tap it into place with a lead hammer. The lockring lug must fit into the slot of the rim.

9. If the beads of the tire and O-ring do not seat within one minute, raise the tire slightly and tap the bead seat band. This will help the air pressure to push the tire bead out into position.

10. As soon as the seating has been accomplished, install the valve core and inflate the tire to the recommended tire pressure.

## **⚠ DANGER**

***Use a safety cage whenever possible. Stand to one side as tire is being inflated. Never start inflating unless the lockring is securely in place. DO NOT stand in front of or over the lockring when inflating.***

8. With the tire inside a safety cage, remove the valve core from the valve stem and inflate the tire to seat the beads of the tire and the O-ring as specified by the tire manufacturer.

# SECTION G3

## FRONT WHEEL HUB AND SPINDLE

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# NOTES

# FRONT WHEEL HUB AND SPINDLE

## WHEEL HUB AND SPINDLE ASSEMBLY

The following instructions will cover the complete removal, installation, disassembly, assembly and bearing adjustment of the front wheel hub and spindle. If only brake service is to be performed, refer to Section J, Brake Circuit.

### **⚠ WARNING**

***Do not loosen or disconnect any hydraulic line or component until the engine is stopped, the key switch is OFF for 90 seconds, and the brake accumulators are depressurized.***

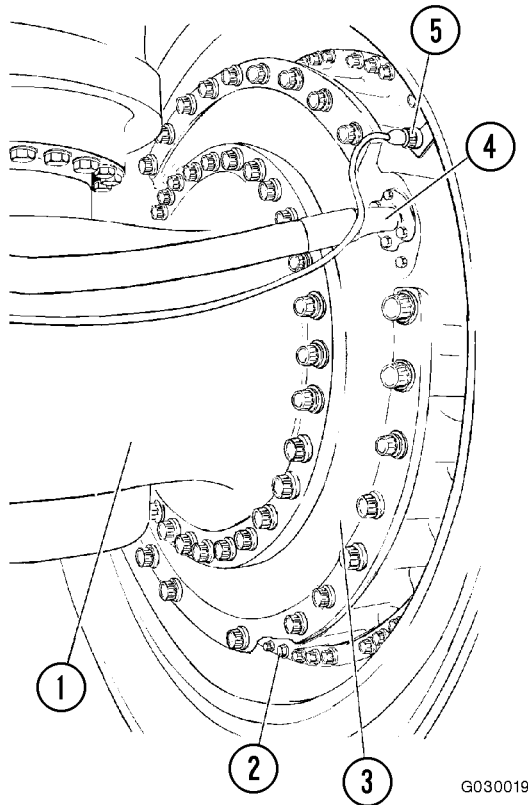


FIGURE 3-1. FRONT WHEEL & SPINDLE INSTALLATION

- |                       |                       |
|-----------------------|-----------------------|
| 1. Spindle            | 4. Brake Cooling Line |
| 2. Drain Plug         | 5. Speed Sensor       |
| 3. Disc Brake Housing | Connector             |

## Removal

1. Shut down the truck. Refer to "Normal Engine Shutdown Procedure" in Section A3, General Safety & Operating Instructions.
2. Depressurize the brake accumulators by using the bleed valves on the brake manifold.
3. Activate the battery disconnect switches.
4. Remove the front tire and rim assembly. Refer to "Front Tire and Rim - Removal" in Section G2.
5. Close the hydraulic pump shutoff valves.
6. Disconnect speed sensor connector(s) (5, Figure 3-1). Tie the cables back away from the spindle to prevent damage during spindle removal.

**NOTE:** The left wheel has two speed sensors installed.

7. Disconnect brake cooling oil lines (4) at the inlet and outlet ports on the brake housing. Plug the hoses and ports to help prevent contamination.
8. Disconnect the brake apply line. Cap the hoses and ports to prevent contamination.
9. Remove drain plug (2) and drain the oil from the brake housing and hub bearings into a suitable container.
10. Remove the lubrication lines from the tie rod and steering cylinder.
11. Disconnect the steering cylinder and the tie rod from the spindle being removed. Refer to "Steering Cylinders and Tie Rod - Removal" in this section.

12. Position a fork lift under the wheel hub and spindle assembly as shown in Figure 3-2. The weight of the spindle assembly is approximately 6 652 kg (14,665 lb). Ensure the lifting device can handle the load safely. Ensure the spindle and hub are securely fastened to the lifting device to prevent tipping when it is free from the suspension.
13. Remove capscrews and washers (1, Figure 3-2) that secure retainer plate (2) to the spindle structure and suspension. To prevent thread damage, loosen the capscrews in a circular pattern in torque increments of **678 N·m (500 ft lb)**. Remove the retainer plate.
14. Remove 16 steering arm capscrews (5, Figure 3-2) as follows:
  - a. Refer to Figure 3-4. Remove only the capscrews that are designated by an "X".
  - b. To prevent thread damage, remove these capscrews using a circular pattern in torque increments of **678 N·m (500 ft lb)**. DO NOT attempt to remove each capscrew in one sequence.

## CAUTION

**DO NOT remove the cap screws in one step with an air wrench. Revolve around the circular pattern at least 3 times, gradually loosening the cap screws during each revolution.**

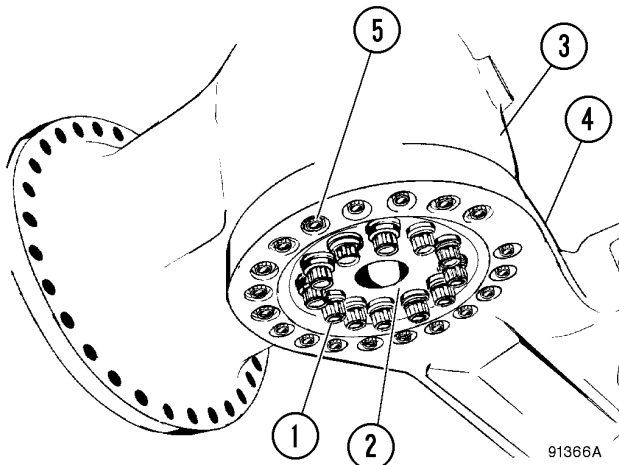


FIGURE 3-2. STEERING ARM ASSEMBLY

- |                        |                 |
|------------------------|-----------------|
| 1. Capscrews & Washers | 4. Steering Arm |
| 2. Retainer Plate      | 5. Capscrews    |
| 3. Spindle             |                 |

15. Run the proper size tap into the threaded holes to ensure clean, good quality threads.

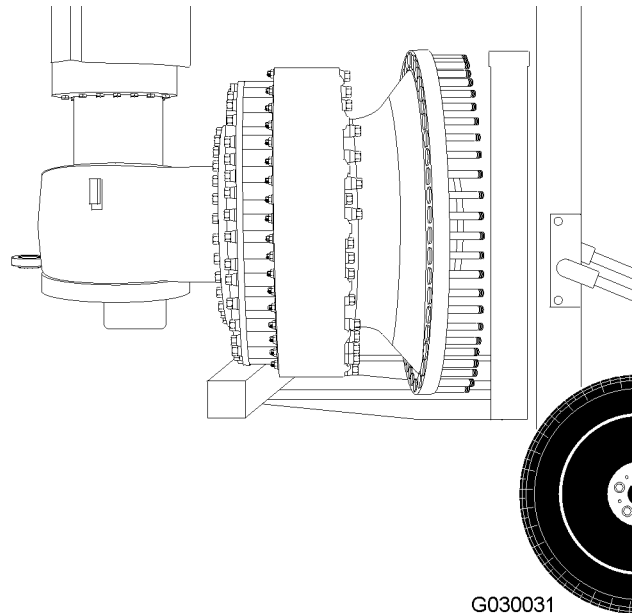


FIGURE 3-3. TYPICAL LIFTING DEVICE (FORKLIFT)

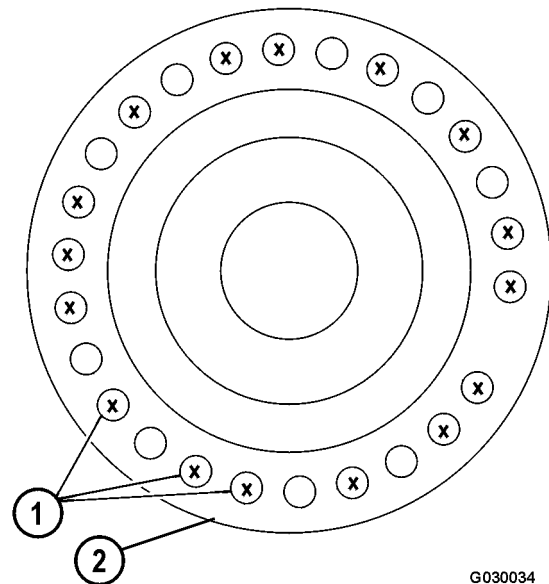


FIGURE 3-4. SPINDLE REMOVAL TOOL

- |                       |                   |
|-----------------------|-------------------|
| 1. Capscrew Insertion | 2. Tool Structure |
| Holes ("X")           |                   |

*NOTE: Older pusher tools (5, Figure 3-5) may have been made with a round tube (shown as a dotted line) welded on one side. The tool can be installed upside down and used with a hydraulic ram as shown.*

Refer to Section M8, Special Tools, for dimensions for fabricating the spindle pusher tool and spacers (6, Figure 3-5). Special cap screws and washers are also required.

The hydraulic ram must be a low profile hydraulic ram (pancake jack) to fit in this area. Most hydraulic rams that will fit in this area are rated at 150 tons.

16. Locate 16 cap screws (1, Figure 3-5) and washers (2) as specified below:

Cap Screw (P/N KC7095). . . . . 1 1/4 x 8 in.

Hardened Washer (P/N WA0366) . . . . . 1 1/4 in.

17. Lubricate the cap screw threads and flat washers with chassis grease.

18. Install pusher tool (5) as shown in Figure 3-5 using the only three cap screws (1), three washers (2) and three spacers (6). The three cap screws must be evenly spaced. Tighten the three cap screws to **68 N·m (50 ft lb)**.

*NOTE: Verify minimum thread engagement is 41.1 mm (1.62 in.) on pusher cap screws when inserted.*

19. Ensure hydraulic ram (4) is completely retracted. Install hydraulic ram with reaction plate (3) on top of the hydraulic ram.

*NOTE: If there is a large gap between the reaction plate and the suspension, it may be necessary to add steel spacers below the hydraulic ram.*



**Do not apply pressure to the hydraulic ram until all 16 cap screws are installed.**

20. Install the remaining 13 cap screws and tighten them to **68 N·m (50 ft lb)**. This is to ensure all cap screws will pull evenly when pressure is applied by the hydraulic ram.



**The maximum force that can be applied during the spindle removal process is not to exceed 701 000 kgf (1,545,600 lb).**

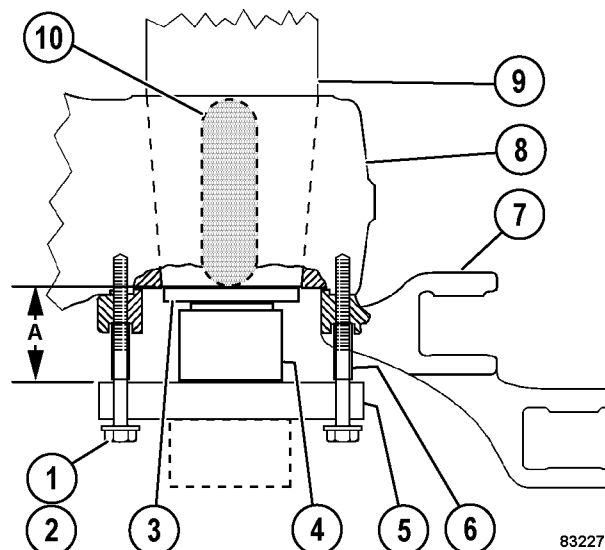


FIGURE 3-5. PULLER TOOL WITH RAM

- |                             |                      |
|-----------------------------|----------------------|
| 1. Cap Screw (KC7095)       | 5. Pusher Tool       |
| 2. Hardened Washer (WA0366) | 6. Spacer            |
| 3. Reaction Plate           | 7. Steering Arm      |
| 4. Hydraulic Ram            | 8. Spindle           |
|                             | 9. Suspension Piston |
|                             | 10. Area to heat     |

## **WARNING**

***Heavy components and high forces are involved in this procedure. Use caution at all times when applying force to these parts. Sudden release of the spindle could cause components to move forcefully and unexpectedly.***

21. Start applying pressure to the hydraulic ram to separate the parts. If the specified maximum force of the hydraulic ram is reached and the tapered parts have not separated, slowly and uniformly apply heat to the spindle as shown (10, Figure 3-5). Heat must be applied in two locations 180 degrees apart. Allow heat to penetrate into the spindle. Reapply heat as required. **Do not exceed 454 °C (850 °F) anywhere on the spindle.**

## **CAUTION**

***Heating the spindle in excess of 454° C (850° F) may cause serious damage to the spindle.***

22. Use heat as specified in the previous step and a large hammer to carefully tap the top surface of the spindle until the spindle breaks free.

***NOTE: If the spindle does not separate from the suspension, the spindle and suspension must be removed from the truck as an assembly. Then use the "Spindle Removal Procedure (off of the truck)" procedure to separate the spindle from the suspension.***

23. After separation, lower the wheel hub and spindle assembly away from suspension piston rod. Be careful during removal to prevent damage to the suspension piston rod taper and the tapered spindle bore.
24. Move the spindle and hub assembly to a clean work area for repair.

## **Spindle Removal Procedure (off of the truck)**

If the hydraulic ram method (with heat) did not separate the spindle from the suspension, then the suspension and spindle must be removed from the truck as an assembly. With the suspension on the ground, the pusher tool can be used to separate the spindle from the suspension.

## **WARNING**

***Heavy structures and high forces are involved in this procedure. Use caution at all times when applying force to these parts. Sudden release of the spindle could cause components to move forcefully and unexpectedly.***

Refer to Section M8, Special Tools for dimensions for fabricating the spindle pusher tool and the cap screws and washers required. Multiple flatwashers may be required under the cap screws to be effective.

***NOTE: Hardened flat washers must be used under the pusher cap screws to prevent galling. Lubricate cap screw threads and washers with a lubricant such as chassis lube.***

1. Position the suspension and spindle assembly on the work floor as shown in Figure 3-6. The combined weight of the spindle, hub and brake assembly and front suspension is approximately 9 375 kg (20,665 lb). Ensure the lifting device can handle the load safely.
2. The wheel hub must be supported with an overhead hoist. The spindle, hub and brake assembly weighs approximately 6 652 kg (14,665 lb). Ensure the lifting device can handle the load safely.

***NOTE: If the spindle is turned so the wheel hub is resting on the ground, the spindle must still be attached to an overhead hoist.***

3. Install pusher tool (2, Figure 3-6) using cap screws and washers (1) specified below:

Cap Screw (P/N KC7095) . . . . . 1 1/4 x 8 in.

Hardened Washer (P/N WA0366) . . . . . 1 1/4 in.

***NOTE: Verify minimum thread engagement is 41.1 mm (1.62 in.) on pusher cap screws when inserted.***

4. After the pusher tool has been installed, progressively increase the torque on the cap screws in a circular pattern until the tapered piston breaks loose, or until the maximum specified torque on the cap screws of **2 142 N·m (1,580 ft lb)** is reached.
5. If the specified torque is reached and the parts have not separated, slightly loosen the cap screws and slowly and uniformly apply heat to the spindle as shown (3, Figure 3-6). Heat must be applied in two locations 180 degrees apart. Allow heat to penetrate into the spindle. Reapply heat as required. **Do not exceed 454 °C (850 °F) anywhere on the spindle.**

6. Tighten the cap screws again to the maximum specified torque as described in Step 4.
7. Using a large hammer and heat at the specified locations, carefully tap on the top surface of the spindle until the piston breaks free.

*NOTE: In extreme cases, it may be necessary to remove additional steering arm retaining cap screws and use additional pusher cap screws to apply more force.*

8. After separation, use the lifting device to move the spindle assembly to clean work area for repair.



**Heating the spindle in excess of 454° C (850° F) may cause serious damage to the spindle.**

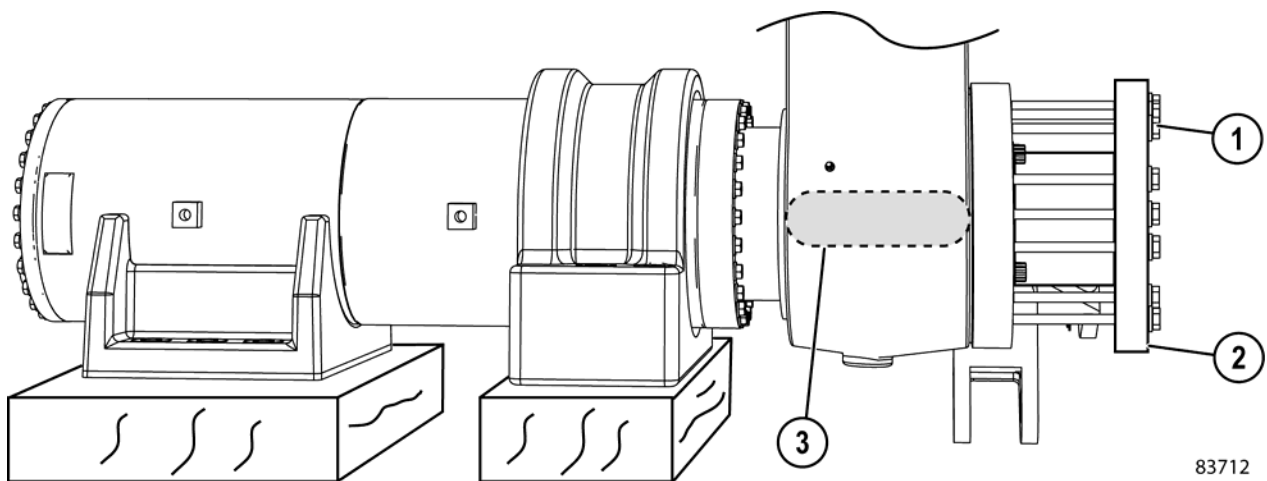


FIGURE 3-6. PUSHER TOOL INSTALLATION

- |                       |                 |
|-----------------------|-----------------|
| 1. Cap Screw & Washer | 3. Area to heat |
| 2. Pusher Tool        |                 |

## Installation

1. Clean the spindle bore and suspension rod taper so that they are free of rust, dirt, etc.
2. Lubricate the spindle bore and suspension rod taper with multi-purpose grease Number 2 with a minimum of 5% molybdenum disulphide.



***Use of anti-seize compounds that contain copper are prohibited from use on spindle bores and rod tapers. These lubricants can cause severe damage. Only use multi-purpose grease Number 2 with a minimum of 5% molybdenum disulphide.***

3. Position the spindle and wheel hub assembly on a fork lift or similar lifting device as shown in Figure 3-3.
4. Raise the spindle and wheel hub assembly into position.
5. Lubricate capscrews (1, Figure 3-2) and (5) on the threads and seats with lithium based chassis grease (multi-purpose, EP NLGI).
6. Secure the spindle to the suspension using retainer plate (2) and capscrews (1). Tighten the capscrews using the following procedure:
  - a. Tighten the capscrews uniformly to **678 N·m (500 ft lb)**.
  - b. Continue to tighten the capscrews in increments of **339 N·m (250 ft lb)** to obtain a final torque of **2 705 ± 135 N·m (1,995 ± 100 ft lb)**.
7. If removed, install the steering arm. Tighten the capscrews to **2 705 ± 135 N·m (1,995 ± 100 ft lb)**.
8. Install the steering cylinder into the steering arm. Use EH4638 sleeve alignment tool (included in tool group) to position the spacers and bearing during pin insertion. Tighten the pin retaining nut to **1 017 ± 102 N·m (750 ± 75 ft lb)**.
9. Use EH4638 alignment tool to install the tie rod on the steering arm. Tighten the pin retaining nut to **1 017 ± 102 N·m (750 ± 75 ft lb)**.
10. Connect the lubrication lines.
11. Install the brake system cooling hoses using new O-rings in the flange fittings. Install the brake apply line. Ensure that drain plug (2, Figure 3-1) is installed.
12. Reconnect the speed sensor cables.
13. Install the wheel and tire as described in "Front Tires and Rims - Installation" in Section G2.
14. Ensure that the hydraulic pump shut-off valves are open.
15. Start the engine. Move the hoist control lever to the FLOAT position to allow hydraulic oil to circulate through the brake cooling system and fill the front wheel hub and disc brake assembly with oil.
16. Bleed the air from the brake apply lines as described in "Wet Disc Brake Bleeding Procedure" in Section J5.
17. Shut off the engine. Check the hydraulic tank oil level and refill if necessary.
18. Inspect the brake assembly and all hose connections for leaks.

## Disassembly

1. Remove the wheel hub and spindle as covered in the Removal procedure before proceeding to Step 2. Remove any dirt and mud from the assembly.
2. To aid in the complete disassembly of the wheel hub and spindle assembly, support the assembly in a vertical (hub cover up) position using a fabricated spindle stand.
3. Install six 0.50 in. -13 x 0.75 in. socket head capscrews (34, Figure 3-7) through the disc brake back plate into the mating holes in the seal carrier. Tighten securely.

*NOTE: The capscrews installed in Step 3 will secure the seal carrier and face seal assembly to the brake housing during brake removal. DO NOT rotate the wheel hub.*

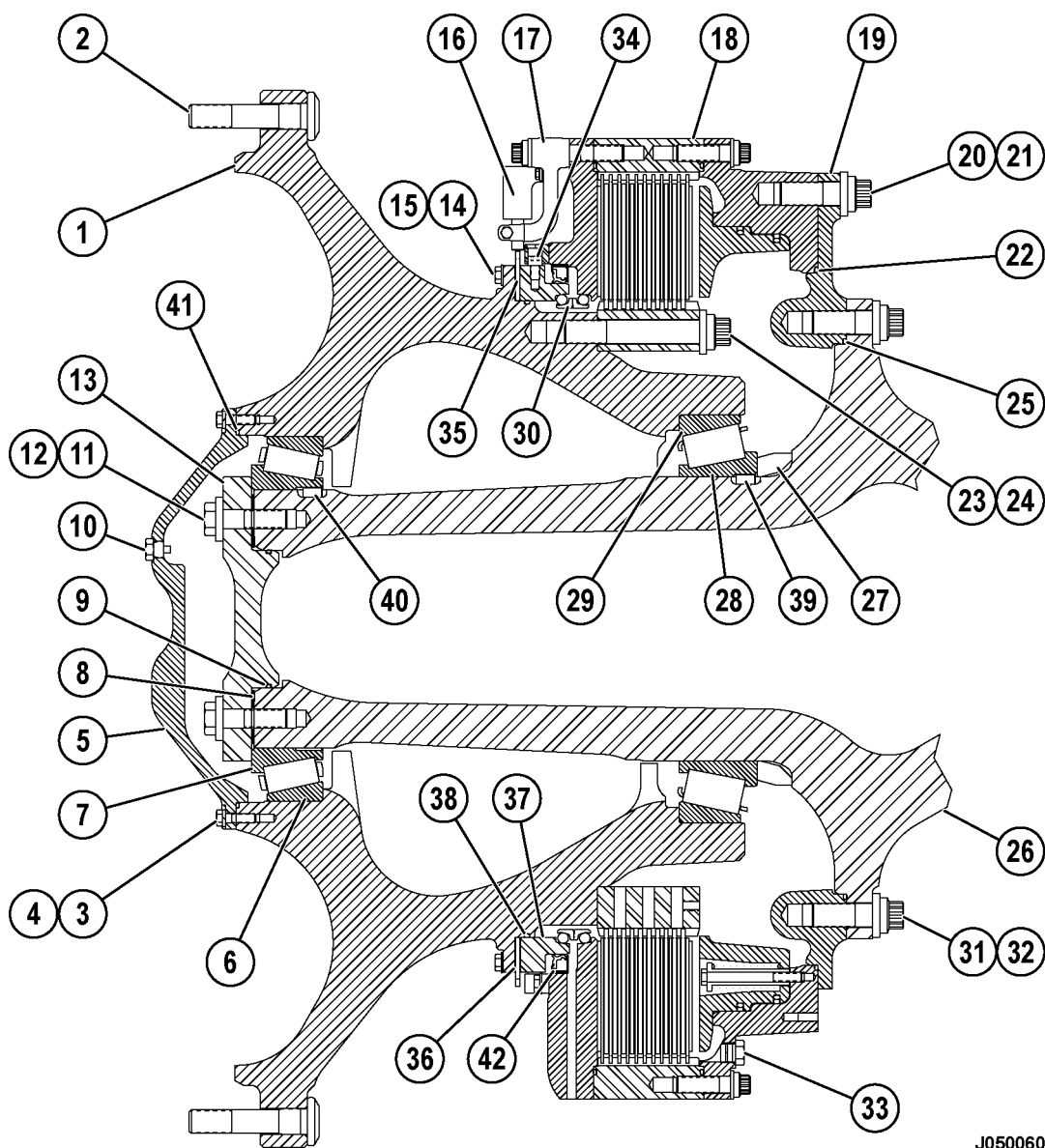
4. Disconnect the speed sensor cables. Loosen the sensor clamping capscrews and remove speed sensors (16).
5. Remove capscrews (20) and hardened flat washers (21) that secure brake adapter (19) to the brake housing.
6. Remove the capscrews and flat washers that secure cover (5). Remove the cover and discard O-ring seal (41).
7. Remove capscrews (11) and hardened flat washers (12).
8. Remove bearing retainer (13), O-ring (9), and shims (8). Discard the O-ring.

9. Attach a lifting device to the wheel hub/brake assembly and carefully lift it straight up and off the spindle.
10. Remove outer bearing cone (7) and retainer pin (40).
11. Rotate the hub vertically 180°, and place it on blocking to prevent damage to wheel studs and machined surfaces.
12. Remove capscrews (14) and flat washers (15) that secure seal carrier (37), sensor gear (35), and shims (36) to the wheel hub.
13. Remove capscrews (23) and hardened flat washers (24) that secure the brake assembly inner gear to the wheel hub.
14. Attach lifting eyes and an overhead hoist to the brake assembly. Carefully lift the assembly off the hub.

Refer to Section J for brake assembly rebuild instructions.

15. Remove and discard seal carrier O-ring (38).
16. Remove speed sensor gear (35) and shims (36).
17. If bearings require replacement, press cups (6) and (29) from the wheel hub.
18. Remove capscrews (31) and hardened flat washers (32). Remove brake adapter (19) and discard O-rings (22) and (25).
19. Remove inner bearing cone (28) and retainer pin (39).
20. Remove bearing spacer (27).





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FIGURE 3-7. FRONT WHEEL SPINDLE, HUB & BRAKE ASSEMBLY

- |                          |                          |                          |
|--------------------------|--------------------------|--------------------------|
| 1. Wheel Hub             | 15. Flat Washer          | 29. Inner Bearing Cup    |
| 2. Stud                  | 16. Speed Sensor         | 30. Seal Assembly        |
| 3. Capscrew              | 17. Sensor Bracket       | 31. Capscrew             |
| 4. Washer                | 18. Disc Brake Assembly  | 32. Hardened Flat Washer |
| 5. Cover                 | 19. Brake Adapter        | 33. Drain Plug           |
| 6. Outer Bearing Cup     | 20. Capscrew             | 34. Socket Head Capscrew |
| 7. Outer Bearing Cone    | 21. Hardened Washer      | 35. Speed Sensor Gear    |
| 8. Shims                 | 22. O-Ring               | 36. Shims                |
| 9. O-Ring                | 23. Capscrew             | 37. Seal Carrier         |
| 10. Magnetic Plug        | 24. Hardened Flat Washer | 38. O-Ring               |
| 11. Capscrew             | 25. O-Ring               | 39. Bearing Retainer Pin |
| 12. Hardened Flat Washer | 26. Spindle              | 40. Bearing Retainer Pin |
| 13. Bearing Retainer     | 27. Bearing Spacer       | 41. O-Ring               |
| 14. Capscrew             | 28. Inner Bearing Cone   | 42. Oil Seal             |

## Cleaning and Inspection

1. Clean all metal parts in fresh cleaning solvent.
2. Remove and clean magnetic plug (10, Figure 3-7).
3. Inspect wheel hub studs (2). Replace if damaged or broken.

*NOTE: If new studs are installed, coat the hole in the hub and serrated portion of the stud with an anti-seize compound prior to installation. DO NOT coat threads.*

4. Inspect the tapped hole threads. Re-tap if necessary.
5. Inspect the bearing seating surfaces in the hub and on the spindle. Inspect the bearing spacer. Use a stone to carefully dress high spots that may interfere with re-assembly.
6. Inspect all other machined surfaces for damage.
7. Always use new O-ring seals during assembly.

## Assembly

### **IMPORTANT**

***All mating surfaces of the wheel and brake components must be clean and dry during assembly. No thread lubricant is permitted on these surfaces.***

1. Position the spindle vertically (hub end up).
2. Check bearing cones (7) and (28, Figure 3-7) for a slip fit on spindle (26).
3. Check the threads in the bearing retainer holes in the end of the spindle to ensure that the capscrews will thread freely. If not, re-tap the threads.
4. Install bearing cups (6) and (29) in wheel hub (1) as follows:
  - a. Preshrink the cups by packing them in dry ice or by placing them in a deep-freeze unit.

*NOTE: Do not cool below -54°C (-65°F).*

- b. Install the cups in the wheel hub bores.
- c. After the cups have warmed to ambient temperature, press the cups tight against the hub shoulder as follows:

Inner cup (29) - Apply **20 860 kg (23 tons)** force.

Outer cup (6) - Apply **19 050 kg (21 tons)** force.

5. Install spacer (27). If necessary, tap lightly to seat the spacer against the spindle. The spacer must fit tightly against the spindle shoulder.
6. Install pin (39) in the pin groove and install inner bearing cone (28) over the pin and against the spacer. The cone is a loose fit on the spindle.
7. Install wheel hub (1) onto spindle (26). Install outer pin (40) and outer bearing cone (7).

*NOTE: To ensure bearing lubrication during initial operation, lightly lubricate the bearings with clean hydraulic oil.*

### **IMPORTANT**

***To facilitate the bearing adjustment procedure, DO NOT install the brake assembly at this time.***

## Wheel Bearing Adjustment

8. Install bearing retainer (13, Figure 3-7) onto the spindle without O-ring (9) and shims (8). The thickness of the retainer is etched on the surface of the retainer. This side should be facing outward. Use four equally-spaced capscrews in order to secure the retainer. Two capscrews must be positioned adjacent to the 13 mm (0.50 in.) diameter access holes in the retainer.

9. Seat the wheel bearings using the following procedure:

a. Tighten the four capscrews to the proper torque, then rotate the wheel hub at least three full revolutions.

Standard bearings . . . . . **108 N·m (80 ft lb)**

Heavy duty bearings . . . **135 N·m (100 ft lb)**

b. Tighten the four capscrews to the proper torque, then rotate the wheel hub at least three full revolutions.

Standard bearings . . . . . **217 N·m (160 ft lb)**

Heavy duty bearings . . . **271 N·m (200 ft lb)**

c. Repeat Step b until the torque is maintained.

d. Loosen the four capscrews and rotate the hub at least three revolutions.

e. Tighten the four capscrews to the proper torque, then rotate the wheel hub at least three full revolutions.

Standard bearings . . . . . **81 N·m (60 ft lb)**

Heavy duty bearings . . . . . **54 N·m (40 ft lb)**

f. Tighten the four capscrews to the proper torque, then rotate the wheel hub at least three full revolutions.

Standard bearings . . . . . **136 N·m (100 ft lb)**

Heavy duty bearings . . . . . **95 N·m (70 ft lb)**

g. Repeat Step f until the torque is maintained.

10. Use a depth micrometer to measure and record the distance between the face of bearing retainer (13) and spindle (26) through each of the two 13 mm (0.50 in.) diameter access holes.

11. Add the two dimensions measured in the Step 10. Divide the sum by 2 to obtain the average depth.

12. Subtract the retainer plate thickness, which is etched on the surface of bearing retainer (13), from the result calculated in Step 11.

13. Assemble a shim pack that equals the dimension calculated in Step 12 within 0.0254 mm (0.001 in.).

*NOTE: Measure the shims individually for accuracy. The above procedure results in a shim pack which will provide a nominal 0.508 mm (0.020 in.) preload for standard bearings, or 0.305 mm (0.012 in.) preload for heavy duty bearings, after assembly.*

### Example:

Step 12 (average depth) 1.416 in.

Step 13 subtract etched dim. -1.375 in.

Step 14 Required shim pack = 0.041 in.

*NOTE: After a shim pack has been determined, the shim pack, spindle, wheel hub, spacer, and bearings are now an interdependent group. If any of the parts are replaced, the shim pack is no longer valid and a new pack must be calculated.*

14. Remove bearing retainer (13), outer cone (7), retainer pin (40), and wheel hub (1) from spindle (26) for installation of the brake assembly.

## Brake Installation

15. Check the integrity of the seals before installing the brakes on the truck.
  - a. Cap the cooling oil ports and pressurize the brake to a maximum of 103 kPa (15 psi).
  - b. Close off the air supply and monitor the interior brake pressure for 15 minutes.
  - c. If the air pressure decreases rapidly, a displaced or damaged brake seal is indicated. Disassemble the brake and reinstall or replace the seal.
16. Position the wheel hub vertically on blocks with the inner bearing bore at the top.
17. Assemble the brake to the hub as follows:
  - a. Install nine 0.51 mm (0.020 in.) thick shims (36, Figure 3-7) on top of the wheel hub flange at each of the six gear/seal carrier mounting locations. Shims may be installed on either side of speed sensor gear (35) for speed sensor alignment.
  - b. Install speed sensor gear (35).
  - c. Install new O-ring (38) in the groove in the hub. Ensure that the O-ring is not twisted. Lubricate the O-ring.
  - d. Install lifting eyes on the brake assembly and attach it to an overhead hoist.
  - e. Lower the brake assembly onto the hub while aligning the hub and seal carrier mounting holes.

**NOTE:** Do not remove the shipping bars until the inner gear ring of the brake assembly is attached to the hub.

- f. Install capscrews (14) and flat washers (15) from under hub flange. Tighten the capscrews to the standard torque.
  - g. Align the brake hub holes with the wheel hub and install hardened flat washers (24) and capscrews (23). Tighten the capscrews to **2 705 ± 135 N·m (1,995 ± 100 ft lb)**.
  - h. Remove the shipping bars from the brake assembly.

18. Install new O-ring (25) to brake adapter (19). Place the assembly into position on the spindle.
  19. Install hardened flat washers (32) and capscrews (31). Tighten capscrews to **2 705 ± 135 N·m (1,995 ± 100 ft lb)**.

## **IMPORTANT**

**Socket head capscrews (34) are installed to position the seal carrier on the brake back plate after assembly and prior to installation on the hub/spindle. In the next step, it may be necessary to loosen these capscrews slightly to align the brake assembly holes with the brake adapter.**

**NOTE:** The following instructions must be performed carefully to prevent damage to brake adaptor O-ring seal (22) during assembly.

20. Prior to installing the hub and brake assembly on the spindle, install four alignment studs on the brake adapter in place of capscrews (20). Space the studs 90° apart to ensure that the brake assembly is properly aligned during the following steps.
    - a. Install new O-ring (22) onto the brake adapter shoulder. Ensure that the O-ring is not twisted and is properly seated.
    - b. Lubricate the O-ring with petroleum jelly or chassis grease.

21. Attach an overhead hoist to the wheel hub and brake assembly. Rotate the hub 180° in preparation for lowering it onto the spindle.

**NOTE:** The hoist must be rigid enough to prevent "springing" or "jerking" as the hub and brake assembly is lowered into position.

22. Lift the hub over the spindle and slowly lower while aligning the brake mounting holes with the alignment studs.
  - a. When the brake piston housing is approximately 13 mm (0.50 in.) from the adaptor, install four capscrews (20) and flat washers (21) spaced evenly in between the alignment studs.
  - b. Carefully pull the brake and wheel hub assembly into position by tightening the capscrews evenly and in small increments.
  - c. DO NOT allow the assembly to "cock" (angle). If the assembly is cocked, lift the assembly and inspect O-ring (22) for damage. Replace the O-ring if necessary, and repeat the procedure.
23. After the brake assembly is properly positioned on the adapter and the four capscrews are snug, remove the alignment studs.
24. Remove socket head capscrews (34) that secure seal carrier (37) to brake assembly (18).
25. Install remaining brake adapter/brake assembly mounting capscrews (20) and hardened flat washers (21). Tighten the capscrews to **2 705 ± 135 N·m (1,995 ± 100 ft lb)**.
26. Install bearing retainer pin (40) in the groove in the spindle. Align and slide outer bearing cone (7) over the pin. Lubricate the bearing with clean hydraulic oil.
27. Install shim pack (8), retainer (13), capscrews (11), and flat washers (12). Tighten the capscrews alternately in several successive increments while rotating the hub to a final torque of **1 017 ± 102 N·m (750 ± 75 ft lb)**.
28. Install new O-ring (41) on cover (5). Install the cover, capscrews, and flat washers. Tighten the capscrews to the standard torque.

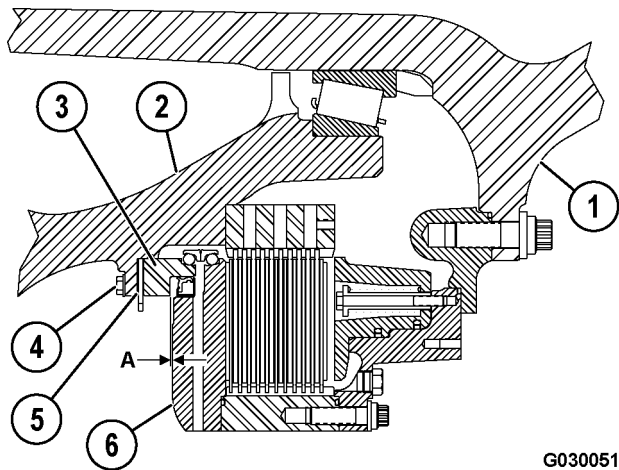
### Seal Assembly Gap Check

After the assembly of the wheel and brake is complete, the gap between seal carrier (3, Figure 3-8) and back plate (6) must be measured and adjusted if necessary.

The ideal gap is 4.81 mm (0.189 in). If necessary, use shims to maintain a gap of 4.56 - 5.06 mm (0.179 - 0.199 in.).

29. Measure the seal gap as follows:

- a. Measure gap (dimension "A", Figure 3-8) at three equally-spaced places and record the results.
  - b. Add the three dimensions and divide the result by 3 to obtain the average gap width.
30. If the average gap width is not within the allowable range, loosen seal carrier capscrews (4) and add shims (5) as required to reduce the gap, or remove shims to increase the gap. **The quantity and thickness of shims at each of the six locations must be equal.**
  31. Re-tighten the seal carrier capscrews to the standard torque and measure the seal gap as described in Step 29. If necessary, repeat Step 30 until the proper gap is maintained.



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FIGURE 3-8. MEASURING SEAL GAP

- |                 |               |
|-----------------|---------------|
| 1. Spindle      | 4. Capscrews  |
| 2. Wheel Hub    | 5. Shims      |
| 3. Seal Carrier | 6. Back Plate |

32. Install speed sensor(s) (16, Figure 3-7) in support bracket(s) (17). Adjust the sensor as follows:

- a. Rotate the hub to position the center line of a gear tooth directly under the sensor tip.
- b. Turn in the sensor until the tip contacts the gear tooth. Then turn out the sensor until the flats of the sensor housing are perpendicular to the gear tooth motion (see Figure 3-9). Clearance between the sensor tip and gear tooth must be  $2.0 \pm 0.1$  mm ( $0.079 \pm 0.004$  in.).
- c. Lock the sensor in place. Rotate the hub  $180^\circ$  to check whether there is enough sensor clearance.

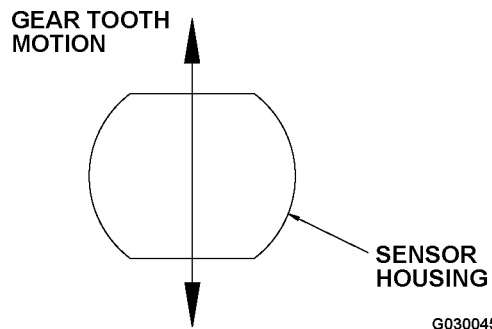


FIGURE 3-9. SPEED SENSOR ADJUSTMENT

33. Install the speed sensor cables.
34. Install the hub and spindle assembly according to the Installation instructions.

## WHEEL SPEED SENSOR TESTING

**NOTE:** The wheel speed sensor can be tested while on the truck or bench tested. Testing the speed sensor requires power in either instance.

1. Connect a ground wire to pin C (black wire) on the speed sensor connector. This is TB21, circuit 0LF/0RF in the auxiliary control cabinet.
2. Connect  $15 \pm 1$  VDC to pin A (red wire) on the speed sensor connector. This is TB21, circuit 15LWS/15RWS in the auxiliary control cabinet.
3. Connect the positive lead of a digital voltmeter to pin B (white wire) on the speed sensor connector. This is TB21, circuit 33LF/33RF in the auxiliary control cabinet.
4. Connect the common lead of the digital voltmeter to pin C (black wire) on the speed sensor connector. This is TB21, circuit 0LF/0RF in the auxiliary control cabinet.
5. Pass a ferrous metal target near the tip of the sensing tip on the speed sensor. A 6.4 mm (0.25 in.) diameter chisel or punch will work.

The sensor is a non-directional sensor. This means the ferrous metal target can pass by the sensor in any direction and the sensor will generate a voltage. If the target is moved toward the sensing tip and stopped in front of the tip, the voltmeter will display 15V. If the target passes by the sensing tip quickly, because of response time, the meter will likely only display a positive voltage, and not necessarily 15V.

6. If voltage is detected, the sensor is good. If voltage is not detected as the target passes by the sensing tip, the speed sensor is faulty.
7. If the speed sensor is faulty, replace the speed sensor.

STEERING CYLINDERS

The steering cylinders and tie rod mounting arrangements are similar. The removal, installation, and bearing wear limits are applicable to both.

Spherical Bearing Wear Limits

For optimum steering performance, it is necessary to determine the condition of the spherical bearings on the steering linkage components. Ball diameter, new dimensions, and maximum allowable wear specifications are listed in Table 1. Bearings that exceed the maximum wear limits must be replaced.

If premature wear of the bearings is evident, check the automatic lubrication system to ensure that the proper amount of lubrication is being received at the joint(s) in question. If lubrication is done manually, ensure that a sufficient amount of grease is being applied on a regular basis. Refer to Section P2, Lubrication and Service, for information on proper lubrication intervals.

It is also important to ensure that the steering linkage components are tightened to the proper torque. Use the torque specifications for steering linkage components listed in this section.



*Use extreme caution when performing maintenance on any vehicle with an active steering system. Serious injury or death can result from contact with moving parts. Always keep a safe distance from crush points.*

| Table 1: STEERING SPHERICAL BEARING WEAR SPECIFICATIONS |                      |
|---------------------------------------------------------|----------------------|
| Spherical Bearing Ball Diameter (New)                   | 109.47 mm (4.31 in.) |
| Maximum Allowable Wear                                  | 1.09 mm (0.043 in.)  |

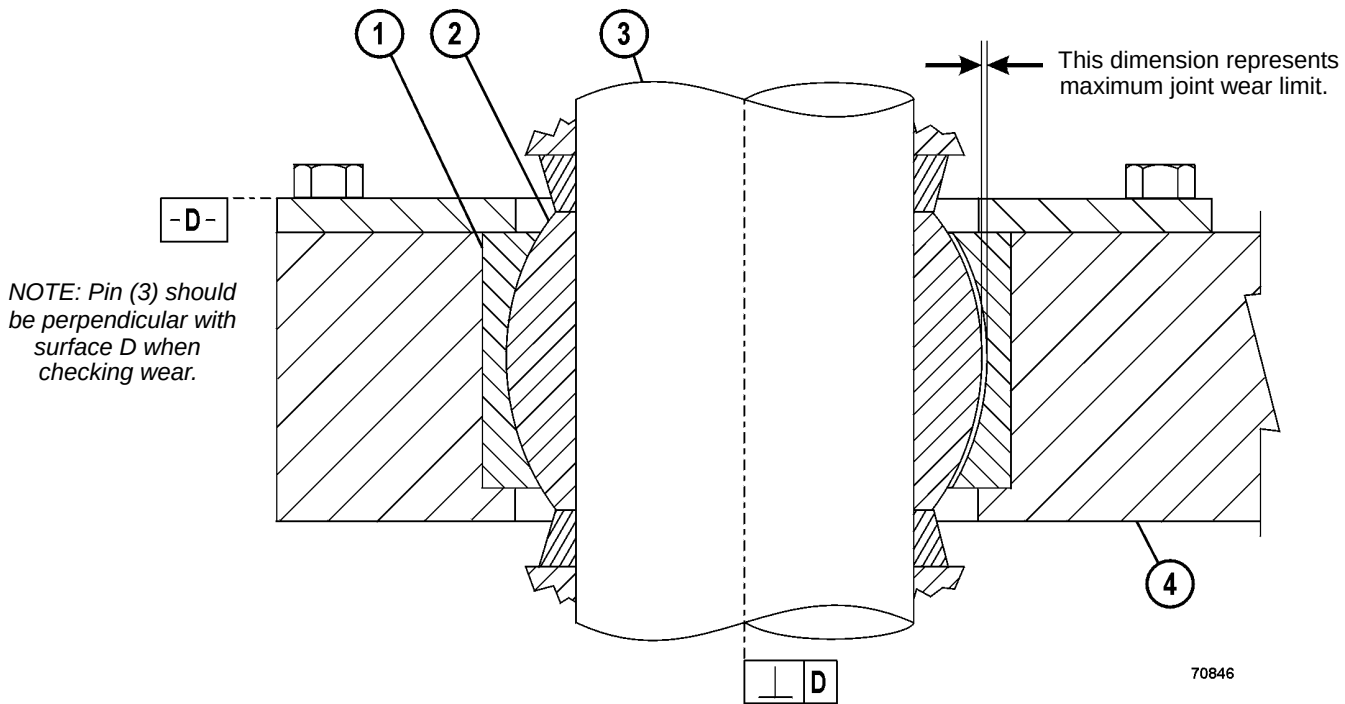


FIGURE 3-10. SPHERICAL BEARING WEAR LIMITS

1. Outer Race
2. Ball
3. Pin
4. Bearing Housing

## Removal

1. With engine shut down and key switch OFF, allow at least 90 seconds for the accumulator to bleed down. Turn the steering wheel to ensure no hydraulic pressure is present.
2. Block front and back of rear wheels to prevent truck movement.
3. Disconnect hydraulic lines at the steering cylinders. Plug all line connections and cylinder ports to prevent contamination of hydraulic system.
4. Remove locknuts (12, Figure 3-11), retainers (11), and cap screws (1) from both ends of steering cylinder.
5. The steering cylinder weighs approximately 126 kg (278 lb). Attach a suitable lifting device that can safely handle the load safely to the steering cylinder and take up the slack.
6. Remove pins (2) from each end of the cylinder and move them to a clean work area. Bearing spacers (4) will be free when the pin is removed. **Ensure the bearing spacers do not drop out and become damaged when removing the pin.**
7. Use the lifting device to lift the cylinder from the truck.
8. Remove seals (3), spacers (4) and washer (10).

## Installation

*NOTE: Use sleeve alignment tool EH4638 (included in the tool group) to hold the bearing spacers and spherical bearings in position when the pins are inserted during assembly.*

1. The steering cylinder weighs approximately 126 kg (278 lb). Use a suitable lifting device to lift the cylinder into position on the truck.

*NOTE: Ensure anti-rotation retainer (7, Figure 3-11) is installed facing downwards.*

2. Align bearing spacers (4), seals (3) and the barrel end of the steering cylinder with pin bores on truck frame.



***Always install pin retaining capscrews from the top with the locknut on the bottom side at the steering arm.***

3. Install pin (2), cap screw (1) and retainer (11) and secure with locknut (12).
4. Align bearing spacers (4), seals (3), washer (10) and rod end with pin bores on steering arm.
5. Install pin (2), cap screw (1) and retainer (11) and secure with locknut (12).
6. Tighten both locknuts (12) to **1 017 ± 102 N·m (750 ± 75 ft lb)**.
7. Connect the grease lines to their respective ports. Operate the steering and check for leaks and proper operation.

## Bearing Replacement (Steering Cylinder and Tie Rod)

1. Remove cap screws (8, Figure 3-11) and lockwashers (9). Remove bearing retainer (7).
2. Press bearing (5) out of the bore in the steering cylinder or tie rod end.
3. Press a new bearing into the bore.
4. Install anti-rotation retainers (7) with cap screws and lockwashers. Tighten the cap screws to standard torque.



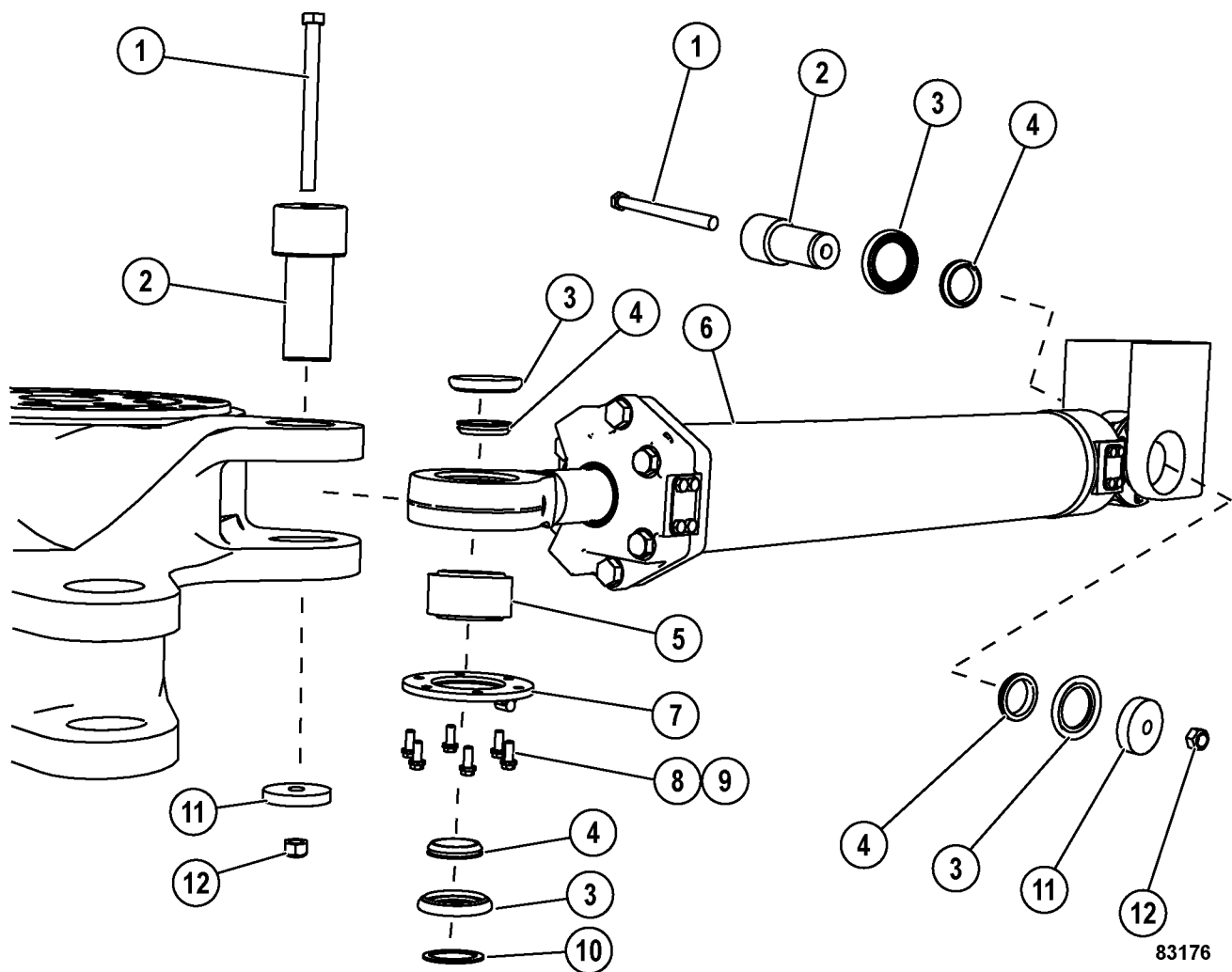


FIGURE 3-11. STEERING CYLINDER AND TIE ROD INSTALLATION

- |              |                            |                |
|--------------|----------------------------|----------------|
| 1. Cap Screw | 5. Bearing                 | 9. Lock Washer |
| 2. Pin       | 6. Steering Cylinder       | 10. Washer     |
| 3. Seal      | 7. Retainer, Anti-Rotation | 11. Retainer   |
| 4. Spacer    | 8. Cap Screw               | 12. Locknut    |

## TIE ROD

### Removal

1. With engine shut down and key switch OFF, allow at least 90 seconds for the accumulator to bleed down. Turn the steering wheel to ensure no hydraulic pressure is present.
2. Block front and back of rear wheels to prevent truck movement.
3. The tie rod weighs approximately 165 kg (364 lb). Attach a suitable lifting device that can handle the load safely to the tie rod and take up the slack.
4. Remove locknuts (8, Figure 3-12), retainers (7), and cap screws (1) from both ends of the tie rod.
5. Remove pins (2) from each end of the tie rod and move them to a clean work area. Bearing spacers (4) will be free when the pin is removed. **Ensure the bearing spacers do not drop out and become damaged when removing the pin.**
6. Use the lifting device to lift the tie rod from the truck.
7. Remove seals (3), spacers (4) and washer (6).

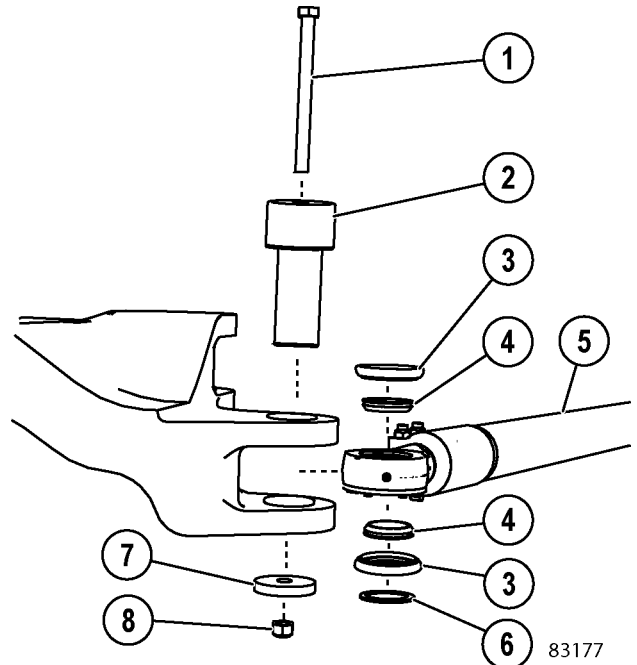


FIGURE 3-12. TIE ROD INSTALLATION (TYPICAL)

- |              |                     |
|--------------|---------------------|
| 1. Cap Screw | 5. Tie Rod Assembly |
| 2. Pin       | 6. Washer           |
| 3. Seal      | 7. Retainer         |
| 4. Spacer    | 8. Locknut          |

### Installation

1. The tie rod (5, Figure 3-12) weighs approximately 165 kg (364 lb). Use a suitable lifting device to lift the tie into position on the truck.

*NOTE: Ensure both bearing retainers are installed facing downwards, and the tie rod clamping bolts are facing to the rear of the truck.*

2. Align bearing spacers (4), seals (3) and washers (6) with the pin bores on both steering arms.
3. Install pins (2), cap screws (1) and retainers (7) and secure with locknuts (8).
4. Tighten both locknuts (8) to **1 017 ± 102 N·m (750 ± 75 ft lb)**.
5. Connect the grease lines to their respective ports. Operate the steering and check for proper operation.
6. Perform the toe-in adjustment procedure.

## Disassembly

1. Remove capscrews (4, Figure 3-13 or Figure 3-14) and retainer (5).
2. Push or press out bearing (6) from rod end (7).
3. Remove locknuts (1) and capscrews (2). New style tie rod clamps also have two flat washers (9, Figure 3-14) on each capscrew.
4. Rotate rod end (7) to remove it from the tie rod tube.

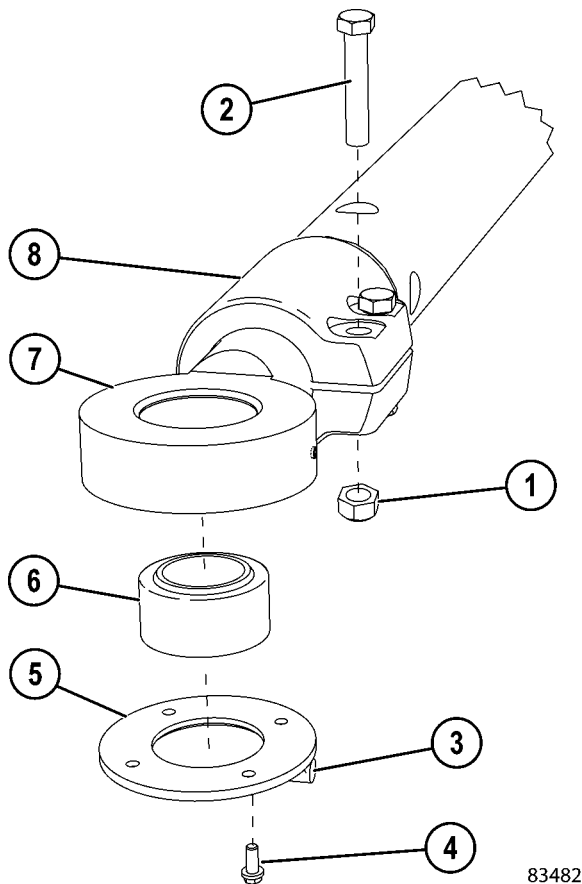


FIGURE 3-13.TIE ROD ASSEMBLY (OLD STYLE)

- |                      |                       |
|----------------------|-----------------------|
| 1. Locknut           | 5. Retainer           |
| 2. Capscrew          | 6. Bearing            |
| 3. Anti-Rotation Pad | 7. Rod End (threaded) |
| 4. Cap Screw         | 8. Tie Rod            |

## Assembly

1. Install bearing (6, Figure 3-13 or Figure 3-14) into rod end (7).
2. Install retainer (5) with capscrews (4) and tighten to standard torque. Ensure that anti-rotation pads (3) on the retainer are perpendicular to the shaft of the rod end. This will prevent the rod end from twisting from side to side once it is installed on the truck.
3. Coat the threads on rod end (7) with a lithium based chassis grease (multi-purpose, EP, NLGI #2).
4. Install rod ends (7) into the tie rod tube.
5. Lubricate the threads and seats of capscrews (2) with a lithium based chassis grease (multi-purpose, EP, NLGI #2). Install capscrews (2) and locknuts (1). New style tie rod clamps also have two flat washers (9, Figure 3-14) on each capscrew. A final torque is not needed until the tie rod assembly is installed on the truck and the toe-in adjustment is performed.

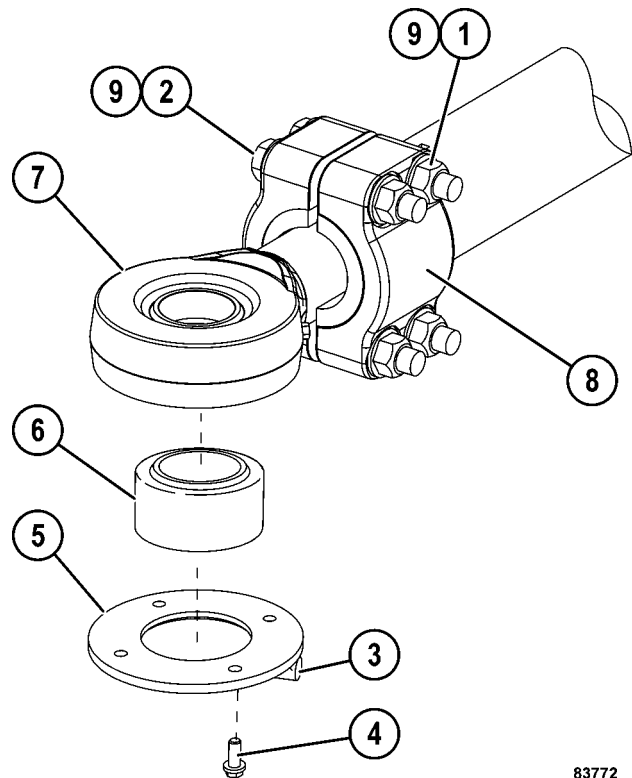


FIGURE 3-14.TIE ROD ASSEMBLY (NEW STYLE)

- |                      |                       |
|----------------------|-----------------------|
| 1. Locknut           | 6. Bearing            |
| 2. Capscrew          | 7. Rod End (threaded) |
| 3. Anti-Rotation Pad | 8. Clamp              |
| 4. Cap Screw         | 9. Flat Washer        |
| 5. Retainer          |                       |

## Toe-In Adjustment

1. Center the steering system in the straight ahead position. Turn the key switch OFF. Allow at least 90 seconds for the accumulators to bleed down. DO NOT turn the steering wheel.
2. Block the front and back of the rear wheels.
3. Check the toe-in by measuring the distance between the centers of the front tires. Obtain the measurements on the horizontal centerline (Dimension "A", Figure 3-15) at the front and rear of the tires.

Radial tires should have equal measurements (zero toe-in).

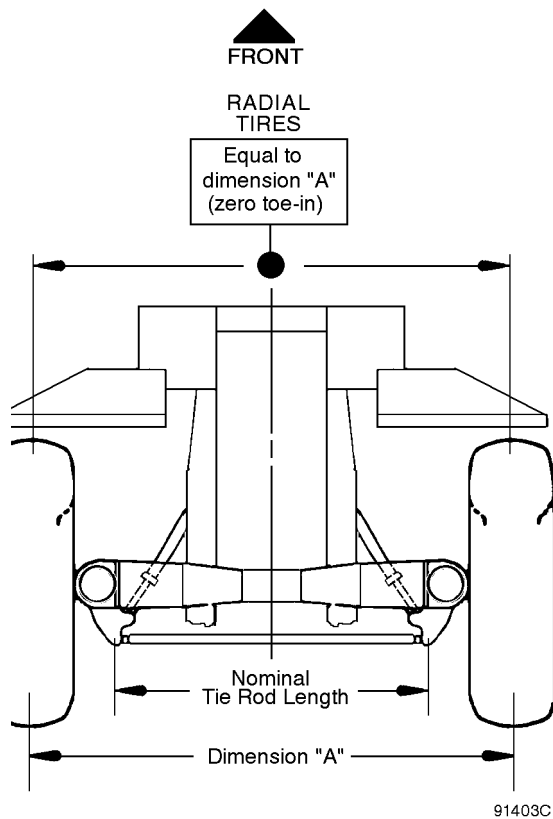


FIGURE 3-15. MEASURING TOE-IN

4. To adjust an old style tie rod:
  - a. Loosen clamp locknuts (5, Figure 3-16) on the tie rod. Rotate the tie rod as necessary to obtain the correct toe-in setting of 3 667 mm (144.37 in.).
  - b. When the adjustment is complete, tighten the clamp locknuts on the tie rod to **420 ± 42 N·m (310 ± 31 ft lb)**.
5. To adjust a new style tie rod:
  - a. Loosen clamp locknuts (4, Figure 3-17) on the tie rod. Rotate the tie rod structure as necessary to obtain the correct toe-in setting of 3 667 mm (144.37 in.).
  - b. Make sure that capscrews (5) are parallel to the ground as shown, and the gap between clamps (3) is aligned with the slot in the tie rod structure.
  - c. When the adjustment is complete, tighten the clamp locknuts on the tie rod to **1017 ± 102 N·m (750 ± 75 ft lb)**.
6. Reapply torque to each capscrew until the specified torque is maintained.
7. Remove the blocks from the wheels.

## TIE ROD INSPECTION AND TORQUE PROCEDURE

### Old Style Tie Rod

Perform this inspection at every 500 hour maintenance interval.

1. When inspecting this area or making adjustments, turn off the engine and make sure that the key switch is in the OFF position.
2. After the engine has been turned off, allow at least 90 seconds for the accumulators to bleed down.
3. Turn the steering wheel to the left and to the right to ensure that the steering accumulator pressure has been bled down.
4. Block the front and back of the rear wheels.
5. Tighten locknuts (5, Figure 3-16) to **420 ± 42 N·m (310 ± 31 ft lb)**.

*NOTE: Some slight movement of the capscrew heads may be observed during the first couple of checks due to torque tolerance.*

6. Reapply torque to each capscrew until the specified torque is maintained.
7. Apply an upwards force to each end of the tie rod assembly near the rod end threads. Look for lateral movement between the tie rod ends and tie rod structure.

If lateral movement is detected, visually inspect the internal and external threads on both ends. Replace parts that have damaged threads. Use new capscrews and locknuts where thread damage to the tie rod end and tie rod structure is observed.

8. Ensure that sufficient grease is being supplied by the auto lube system to the spherical bearings of tie rod (2) and steering cylinders (1).

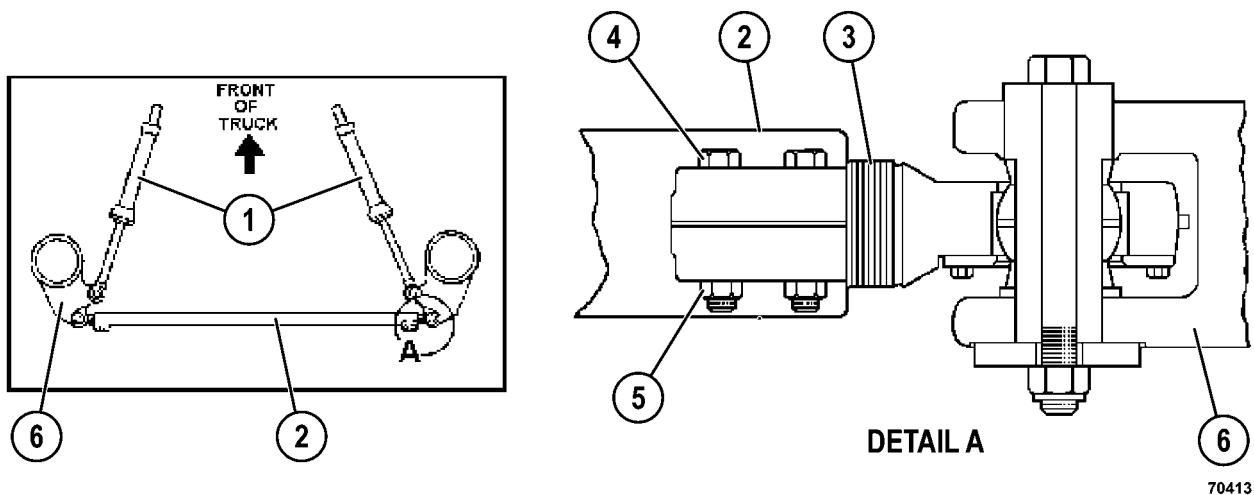


FIGURE 3-16. TYPICAL TIE ROD ASSEMBLY

- |                      |                     |                |
|----------------------|---------------------|----------------|
| 1. Steering Cylinder | 3. Threaded Rod End | 5. Locknut     |
| 2. Tie Rod           | 4. Capscrew         | 6. Spindle Arm |

*NOTE: The threads of both threaded rod ends (3) and all four capscrews (4) are lubricated with a lithium based multipurpose grease (EP NLGI No. 2).*

*NOTE: Locknuts (5) must be installed on the same side of the tie rod as the bearing retainer (downward side) as shown in Figure 3-16.*

## New Style Tie Rod

Perform this inspection at every 500 hour maintenance interval.

1. When inspecting this area or making adjustments, turn off the engine and make sure that the key switch is in the OFF position.
2. After the engine has been turned off, allow at least 90 seconds for the accumulators to bleed down.
3. Turn the steering wheel to the left and to the right to ensure that the steering accumulator pressure has been bled down.
4. Block the front and back of the rear wheels.
5. Tighten capscrews (5, Figure 3-17) to **1017 ± 102 N·m (750 ± 75 ft lb)**.

*NOTE: Some slight movement of the capscrew heads may be observed during the first couple of checks due to torque tolerance.*

6. Reapply torque to each capscrew until the specified torque is maintained.
7. Apply an upwards force to each end of the tie rod assembly near the rod end threads. Look for lateral movement between rod ends (2) and tie rod structure (1).

If lateral movement is detected, visually inspect the internal and external threads on both ends. Replace parts that have damaged threads. Use new capscrews and locknuts where thread damage to the rod end and tie rod structure is observed.

8. Ensure that sufficient grease is being supplied by the auto lube system to the spherical bearings of the tie rod and steering cylinders.

*NOTE: The threads of both rod ends and all four capscrews (5) are lubricated with a lithium based multipurpose grease (EP NLGI No. 2).*

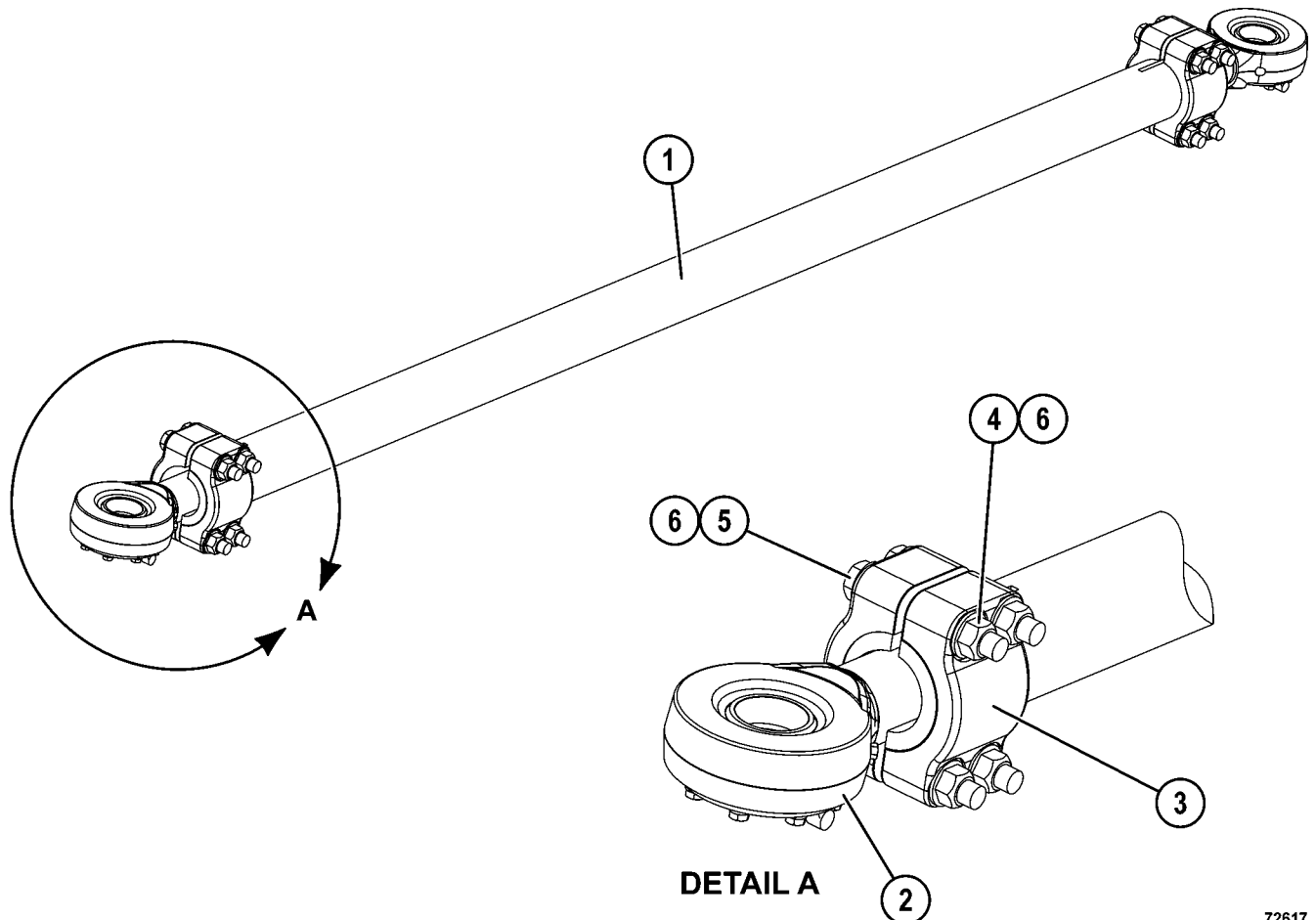


FIGURE 3-17. NEW TIE ROD ASSEMBLY

- |                      |            |             |
|----------------------|------------|-------------|
| 1. Tie Rod Structure | 3. Clamp   | 5. Capscrew |
| 2. Threaded Rod End  | 4. Locknut |             |

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# NOTES

# SECTION G4

## REAR AXLE MOUNTING

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# NOTES

## REAR AXLE MOUNTING

### PIVOT PIN

#### Removal

1. Park the truck on firm, level surface and block the front and rear of all tires.

### **WARNING**

**The truck body must be empty and resting on the frame before attempting this procedure.**

2. Release all brakes.
3. Charge the rear suspensions with nitrogen until the pistons are fully extended.
4. Place blocks or stands under each frame member beneath the hoist cylinders.

### **WARNING**

**Blocks must be securely in place before lowering the frame. Check the blocks on the wheels to make sure they are in place.**

5. Release the nitrogen from the front suspensions.
6. Release the nitrogen from the rear suspensions.
7. Attach a hoist to the lift eye at the top of the pivot eye to control downward movement of the front of the axle housing.
8. Disconnect the pivot eye bearing lube line. Remove the ground wire between the pivot eye and frame.
9. Remove capscrew (2, Figure 4-1) and lockwasher (3).

10. Remove capscrews (1). Remove locking plate (16).

11. Install puller using tapped holes (11) in head of pin. Remove pin (10).

*NOTE: Placing a pry bar or jack between the mounting structure and pivot eye may be necessary to push the pivot eye downward and away from the mount structure. Spacers (13) will fall free.*

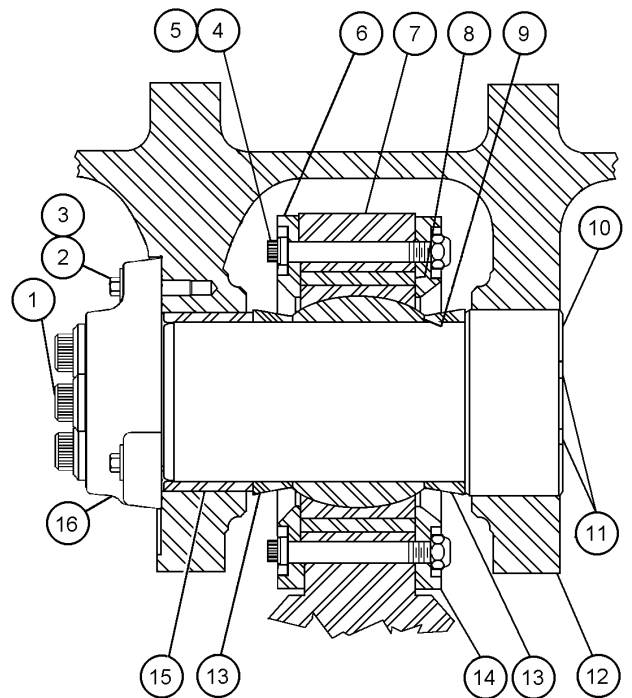


FIGURE 4-1. PIVOT PIN ASSEMBLY

- |                     |                      |
|---------------------|----------------------|
| 1. Capscrew         | 9. Bearing           |
| 2. Capscrew         | 10. Pivot Pin        |
| 3. Lockwasher       | 11. Puller Holes     |
| 4. Capscrew         | 12. Truck Frame      |
| 5. Locknut          | 13. Spacer           |
| 6. Bearing Retainer | 14. Bearing Retainer |
| 7. Pivot Eye        | 15. Sleeve           |
| 8. Bearing Carrier  | 16. Locking Plate    |

*NOTE: Some trucks may not be equipped with bearing carrier (8).*

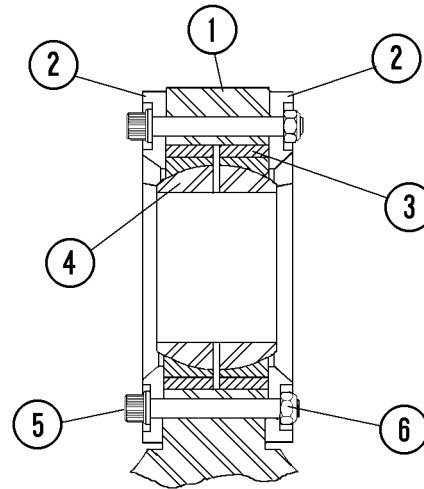
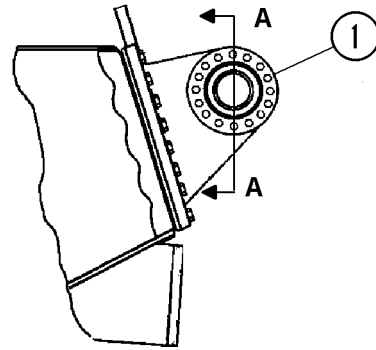
## Installation

1. Raise pivot eye (7) into position.
2. Make sure that the inner race of the spherical bearing is aligned.
3. Install spacers (13). Install pin (10).
4. Position locking plate (16) by lining up the capscrews with the capscrew holes in the pin. Install the capscrews.
5. Rotate the pin and lock plate to align capscrews (2) with holes in frame mounting structure.
  - a. Install capscrews (2) and lockwashers (3).
  - b. Tighten capscrews (2) to **237 N•m (175 ft. lbs.)**.
  - c. Tighten capscrews (1) to **2325 N•m (1715 ft. lbs.)**.
6. Install the ground wire and the lubrication line. Pressurize the lube line to ensure that bearing (9) receives grease.
7. Charge the front suspension as described in Section H, Oiling and Charging Procedures.
8. Charge the rear suspensions with nitrogen to fully extend the pistons.
9. Remove the blocks or stands from beneath the frame.
10. Release the nitrogen from the rear suspension and charge according to the procedure in Section H, Oiling and Charging Procedures.



***Before removing the blocks from the wheels, make sure the parking brake is applied.***

11. Remove the blocks from the wheels.



**SECTION A-A**

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**FIGURE 4-2. PIVOT EYE BEARING INSTALLATION**

- |                        |                      |
|------------------------|----------------------|
| 1. Pivot Eye Structure | 4. Spherical Bearing |
| 2. Bearing Retainer    | 5. 12 Pt. Capscrew   |
| 3. Bearing Carrier     | 6. Locknut           |

**NOTE:** Some trucks may not be equipped with bearing carrier (3).

## PIVOT EYE BEARING

### Removal

1. Remove capscrews and locknuts (5 and 6, Figure 4-2).
2. Remove bearing retainers (2).
3. Setup an appropriate tool to press spherical bearing (4) from the pivot eye.
4. Inspect all parts for wear or damage. Replace any parts showing excessive wear or damage.

### **Spherical bearing outer race O.D.:**

222.25 - 222.22 mm (8.7500 - 8.7488 in.)

### **Bearing bore I.D.:**

152.37 - 152.40 mm (5.9990 - 6.0000 in.)

5. If bearing carrier (3) is damaged or worn, refer to "Pivot Eye Repair" in this section for repair procedure.

*NOTE: Some trucks may not be equipped with bearing carrier (3).*

### Installation

1. Set up an appropriate tool to press spherical bearing (4, Figure 4-2) into the pivot eye. Make sure that the bearing is centered and properly installed in the pivot eye to allow proper lubrication.
2. Install bearing retainers (2) using capscrews (5) and locknuts (6). Tighten the capscrews to the standard torque.

## PIVOT EYE REPAIR

If damage occurs to pivot eye (4, Figure 4-3), it may be necessary to remove it from rear axle structure (1) to facilitate repair and bearing replacement.

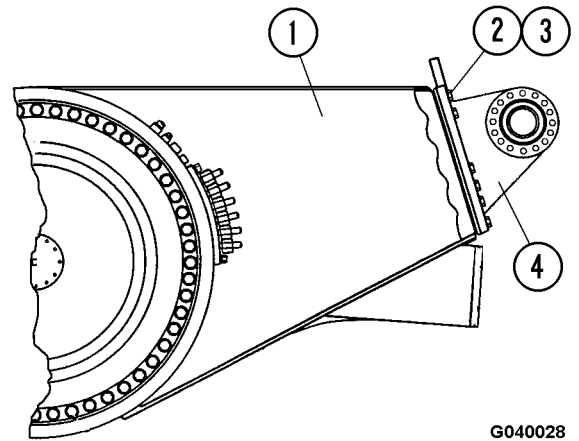


FIGURE 4-3. PIVOT EYE ATTACHMENT

- |                        |                |
|------------------------|----------------|
| 1. Rear Axle Structure | 3. Flat Washer |
| 2. Capscrew            | 4. Pivot Eye   |

### Removal

1. Follow all preceding instructions in Pivot Pin - Removal. Make sure that axle housing (1) and the wheels are blocked securely.
2. Attach a lifting device to pivot eye (4). The weight of the pivot eye is 346 kg (762 lbs).
3. Remove capscrews (2) and flat washers (3). Remove the pivot eye.

### Disassembly

1. Remove spherical bearing (4, Figure 4-2) as described in Pivot Eye Bearing - Disassembly.
2. If bearing carrier (3) (if equipped) is damaged or worn, setup an appropriate tool to press the bearing carrier out of the pivot eye bore.

### **Bearing carrier (new) (if equipped):**

I.D. = 222.209 ± 0.013 mm (8.7484 ± 0.0005 in.)

O.D.= 247.701 ± 0.013 mm (9.7520 ± 0.0005 in.)

3. Inspect the pivot eye structure bore for excessive wear or damage.

### **Pivot eye bore (new):**

247.650 ± 0.013 mm (9.7500 ± 0.0005 in.)  
(pivot eyes equipped with bearing carrier)

222.164 ± 0.013 mm (8.7466 ± 0.0005 in.)  
(pivot eyes **not** equipped with bearing carrier)

## Assembly

1. Setup an appropriate tool to press bearing carrier (3, Figure 4-2) into the bore of the pivot eye structure. (Some trucks may not be equipped with the bearing carrier.)

*NOTE: With parts to correct size, the fit of the bearing carrier into the bore of the pivot eye structure may be 0.025 - 0.08 mm (0.001 - 0.003 in.) interference fit. Freezing the bearing carrier will ease installation.*

## ! IMPORTANT !

**The lubrication groove in the bearing carrier outer diameter must be aligned with the lubrication fitting hole in the pivot eye structure.**

2. Press the bearing carrier into the bore. Make sure that the carrier is pressed fully into the pivot eye bore (flush with the sides) to allow proper lubrication.
3. Install spherical bearing (4) as described in Pivot Eye Bearing - Assembly.

## Installation

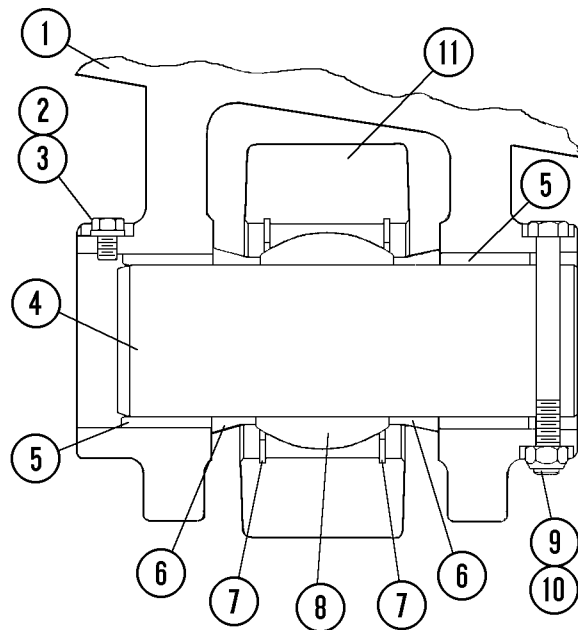
1. Make sure that the mating surfaces of axle housing (1, Figure 4-3) and pivot eye (4) are clean and not damaged.
2. Install an appropriate lifting device to the pivot eye. The weight of the pivot eye is 346 kg (762 lbs).
3. Lift the pivot eye into position on the front of the axle housing. Insert several capscrews (2) and flat washers (3) to align the parts. Remove the lifting device.
4. Install the remaining capscrews and flat washers. Tighten the capscrews alternately until the pivot eye is properly seated. Tighten the capscrews to **2325 ± 231 N•m (1715 ± 170 ft. lbs.)**.

## ANTI-SWAY BAR

*NOTE: The anti-sway bar mounting arrangement is identical at each end.*

## Removal

1. Position the frame and the rear axle housing to allow the use of a puller arrangement to remove pins (4, Figure 4-4) from the rear axle housing and frame.
2. Securely install blocking between the frame and the axle housing.
3. Disconnect the lubrication lines.
4. Position a fork lift or attach a lifting device to anti-sway bar (11). The weight of the anti-sway bar is 147 kg (325 lbs).
5. Remove capscrew or shoulder bolt (9) and locknut (10) at each mount.
6. Attach a puller and remove pin (4) from each end of the anti-sway bar.
7. Remove the anti-sway bar from the mounting brackets.
8. Remove bearing spacers (6).



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FIGURE 4-4. ANTI-SWAY BAR ASSEMBLY

- |                       |                           |
|-----------------------|---------------------------|
| 1. Mounting Structure | 7. Retainer Ring          |
| 2. Capscrew           | 8. Bearing                |
| 3. Lockwasher         | 9. Capscrew/Shoulder Bolt |
| 4. Pin                | 10. Locknut/Nut           |
| 5. Sleeve             | 11. Anti-Sway Bar         |
| 6. Bearing Spacer     |                           |

## Disassembly

1. Remove retainer rings (7, Figure 4-4) from the bores of both ends of anti-sway bar (11).
2. Press out spherical bearings (8).

## Cleaning and Inspection

1. Inspect the bearing bores of the anti-sway bar. If the bores are damaged, repair or replace the anti-sway bar.
2. Inspect bearing spacers (6) for damage or wear. Replace as needed.

## Assembly

1. Press new bearings into the anti-sway bar.
2. Install retainer rings (7). Make sure that the rings are properly seated in the grooves.

## Installation

1. Place pin (4, Figure 4-4) into position at the front of the frame mount. Push the pin through spacer (6), and rotate the pin to align retaining capscrew (9) hole with the hole in mounting structure (1).
2. Raise the anti-sway bar into position. The weight of the anti-sway bar is 147 kg (325 lbs).
3. Install capscrew or shoulder bolt (2) and locknut (3).
  - For a capscrew, tighten the locknut to **135 Nm (100 ft lb)**.
  - For a shoulder bolt, tighten the locknut to **68 Nm (50 ft lb)**.
4. Repeat the previous steps to install the remaining pin and spacers at the opposite end of the anti-sway bar. Start the pin into the bore of the axle housing mount from the rear of the truck.
5. Attach the lubrication lines. Pump grease into the bearing to verify that the line and the system are operational.
6. Remove the blocking from between the frame and the axle housing.
7. If necessary, recharge the suspensions. Refer to Section H, Oiling and Charging Procedures.

# NOTES

# SECTION G5

## REAR AXLE AND WHEEL MOTOR

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# NOTES

## REAR AXLE AND WHEEL MOTOR

### REAR AXLE HOUSING

Read and observe the following instructions before attempting any repairs on propulsion system components!



- **Do not step on or use any power cable as a hand hold when the engine is running.**
- **All removal, repair and installation of propulsion system electrical components, and cables must be performed by an electrical maintenance technician who is properly trained to service the system.**
- **In the event of a propulsion system malfunction, a qualified technician should inspect the truck and verify that the propulsion system does not have dangerous voltage levels present before repairs are started.**

After the truck is parked for the repairs, the truck must be shut down properly to ensure the safety of personnel working in the area of the wheel motors, control cabinet and retarding grids. The following procedure will ensure that the electrical system is properly discharged before repairs are started.

#### Preparation

1. Reduce the engine speed to idle. Move the directional selector lever to PARK. Make sure that the parking brake applied indicator light in the overhead display panel is illuminated.
2. Place the drive system in the “rest” mode by turning on the rest switch on the instrument panel. Make sure that the rest mode indicator light in the overhead display panel is illuminated.
3. Shut off the engine by turning the key switch OFF. If the engine does not stop, use the emergency shutdown switch on the center console.

4. Wait 90 seconds for the steering accumulators to bleed down. Ensure that the steering accumulators have bled down completely by trying to turn the steering wheel.
5. Verify that the link voltage lights on the DID panel in the cab are off. **If they remain on for longer than five minutes after engine shutdown, the propulsion system must be inspected to find the cause.**
6. To ensure that the link will not be energized during test and repair procedures, turn GF cutout switch (2, Figure 5-1) in the control cabinet to the CUTOUT position by pulling the switch out and moving the switch downward as shown.

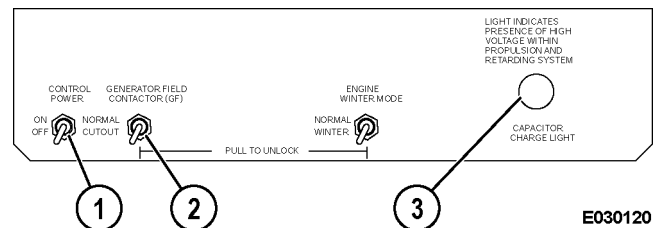


FIGURE 5-1. INFORMATION DISPLAY PANEL

1. Control Power Switch
2. GF Cutout Switch
3. Capacitor Charge Light

7. Bleed down the brake accumulators using the manual bleed valves on the brake manifold.
8. Activate the battery disconnect switches.
9. Block the front wheels to prevent truck movement.
10. Remove the dump body as outlined in Section B, Dump Body.
11. Block up the truck and remove the rear tires as described in Section G, Tires and Rims.
12. Mark all electrical cables and hydraulic hoses before removal.

## Removal

1. Loosen the hose clamps and disconnect axle air blower hose (1, Figure 5-2) from wheel motor cooling air duct (2).
2. Close the pump shutoff valves at the hydraulic tank.
3. Remove the drain plugs at the bottom of the rear brake assemblies and drain the oil from the disc brakes into a suitable container.
4. Disconnect and unclamp the hoses at hydraulic hose manifold (6) on the left side of the rear axle housing. Cap the fittings and plug the hoses to prevent contamination.
5. Remove the cable grips at power cable mounts (3) on the right side of the rear axle housing. Slide the cable grips forward on the cables.
6. Disconnect the brake system cooling lines at each brake assembly back plate. Disconnect the brake apply lines. Cap and plug all openings.
7. Remove the clamps that secure the brake hoses to the spindles.

**NOTE:** For access to components inside the rear axle housing, the rear access tube and the top cover can be removed.

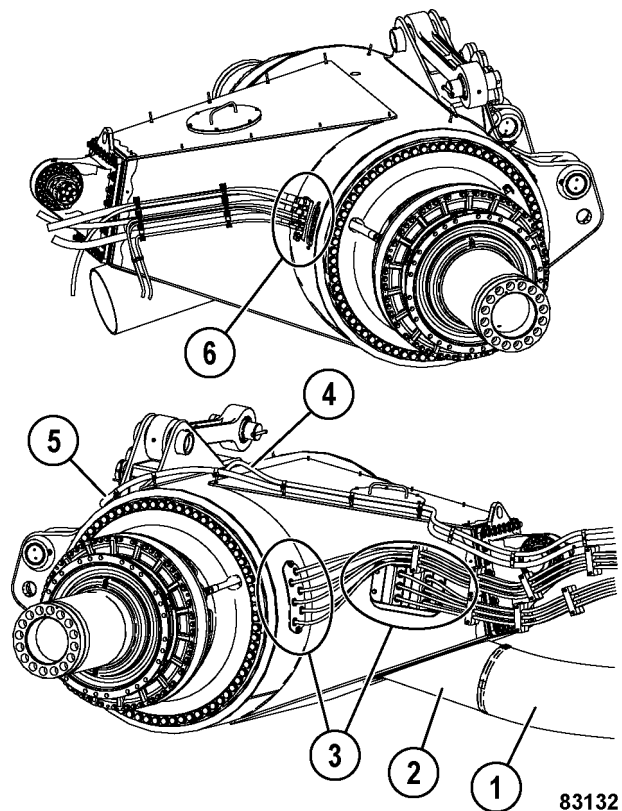
8. Disconnect and cap the parking brake supply lines to the parking brake on each wheel motor.
9. Remove the wheel motor cooling air exhaust duct from between the wheel motors.
10. Disconnect all wheel motor power cables and electrical harnesses at their terminals. Remove the cable clamps and pull the cables and harnesses from the rear axle housing.
11. Disconnect wheel speed harness (4) and rear frame harness (5) from the axle housing and remove the clamps.
12. Remove all automatic lubrication system hoses and clamps that are attached to the rear axle housing. Secure the hoses to the frame to prevent interference during rear axle housing removal.
13. Remove the wheel motors as described later in this section.
14. Remove the rear suspensions. Refer to Section H, Rear Suspensions.

15. Remove the anti-sway bar. Refer to Section G, Rear Axle Mounting.
16. Remove the pivot pin. Refer to Section G, Rear Axle Mounting.

## WARNING

**The axle housing weighs approximately 6 200 kg (13,670 lb). Ensure that the lifting devices are capable of handling the load safely.**

17. Attach an overhead hoist or crane to the lifting eyes on the rear axle housing and move it from the rear of the frame.



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FIGURE 5-2. REAR AXLE HOUSING REMOVAL

- |                                 |                            |
|---------------------------------|----------------------------|
| 1. Axle Air Blower Hose         | 4. Wheel Speed Harness     |
| 2. Wheel Motor Cooling Air Duct | 5. Rear Frame Harness      |
| 3. Power Cable Mount            | 6. Hydraulic Hose Manifold |

## Cleaning and Inspection

1. Thoroughly clean the capscrew holes and wheel motor mounting faces. Re-tap any holes that have damaged threads.
2. Check the wheel motor mounting faces for nicks, scratches and other damage. Inspect all welds and repair as necessary.
3. Inspect the brake system relay valve and the hoses for leaks.
4. Inspect all hose connections and repair any leaks and damaged hoses.

## Installation

1. Position the axle housing under the frame.
2. Align the pivot pin bores of the rear axle housing and frame. Install the pivot pin. Refer to Section G, Rear Axle Mounting.
3. Install the anti-sway bar. Refer to Section G, Rear Axle Mounting.
4. Install the rear suspensions. Refer to Section H, Rear Suspensions.
5. Connect the automatic lubrication system hoses and clamps.
6. Route the power cables into the power cable mounts (3, Figure 5-2) and clamp them in place. Install the cable grips.
7. Install wheel speed harness (4) and rear frame harness (5). Install the clamps.
8. Install the wheel motors, cables, brake lines and tires as described later in this section.
10. Connect the hoses to manifold (6) on the left side of the axle housing.
11. Install wheel motor cooling air duct (2) between the wheel motors.
12. Install axle air blower hose (1) on wheel motor cooling air duct (2). Install the hose clamps
13. Open the pump shutoff valves and service the hydraulic system.

## BLOWER PRESSURE SWITCH ADJUSTMENT

*NOTE: If a new blower pressure switch is being installed, set the switch to operate or change state at a pressure of approximately 2.0 in H<sub>2</sub>O (0.5 kPa).*

1. Connect a (+) lead of a digital voltmeter to circuit 75A2 in the auxiliary control cabinet. Connect a (-) lead to ground.
2. Turn the key switch to ON, but do not start the engine. The voltmeter should read 0V.
3. Start the engine and slowly depress the accelerator pedal to increase the engine speed to 1000 rpm. The voltmeter should read approximately +24 to 28V before the engine speed reaches 1000 rpm. If not, adjust the switch as follows:
  - a. Remove the rear axle access cover.
  - b. Remove the cover from the blower pressure switch box.
  - c. Use a 3/8" nut driver to turn adjustment nut (1, Figure 5-3) to increase (clockwise) or decrease (counterclockwise) the rpm at which the blower engages.
  - d. Install the switch box cover and rear axle access cover. Repeat Step 3.

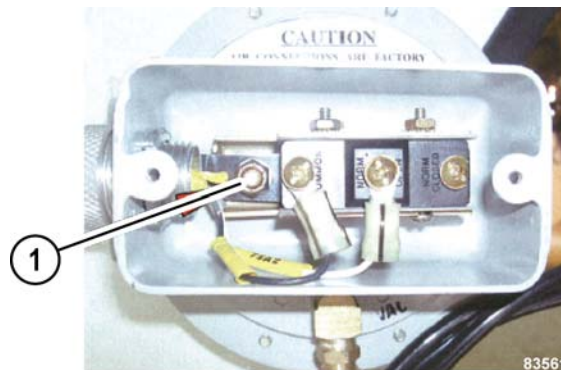


FIGURE 5-3. BLOWER PRESSURE SWITCH

1. Adjustment Nut

4. Disconnect the voltmeter.
5. To verify that the amber propulsion system warning light operates correctly, disconnect circuit 75A2. Operate the truck. The light should illuminate and the DID panel should display fault #630-1 after approximately 30 seconds. A truck speed limit restriction is imposed while this event is active.
6. Reconnect circuit 75A2, then operate the truck to verify that the amber light does not illuminate and the fault code is not displayed.

## WHEEL MOTOR

Read and observe the following instructions before attempting removal of the wheel motors or any repairs on the propulsion system components!

### **⚠ DANGER**

- ***Do not step on or use any power cable as a hand hold when the engine is running.***
- ***All removal, repair and installation of propulsion system electrical components, and cables must be performed by an electrical maintenance technician who is properly trained to service the system.***
- ***In the event of a propulsion system malfunction, a qualified technician should inspect the truck and verify that the propulsion system does not have dangerous voltage levels present before repairs are started.***

After the truck is parked for the repairs, the truck must be shut down properly to ensure the safety of personnel working in the area of the wheel motors, control cabinet and retarding grids. The following procedure will ensure that the electrical system is properly discharged before repairs are started.

### Preparation

1. Reduce the engine speed to idle. Place the directional control lever in PARK. Make sure that the parking brake applied indicator light is illuminated.
2. Place the drive system in the rest mode by turning the rest switch on the instrument panel ON. Make sure that the rest indicator light is illuminated.
3. Turn the key switch OFF. If the engine does not shut down, use the engine shutdown switch on the center console.

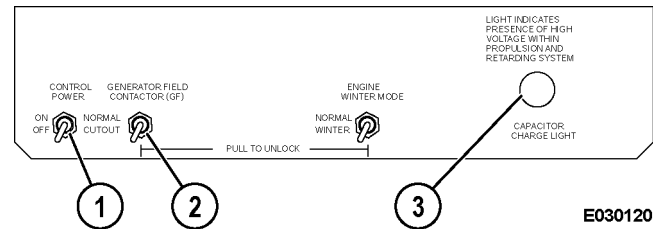


FIGURE 5-4. INFORMATION DISPLAY PANEL

1. Control Power Switch
  2. GF Cutout Switch
  3. Capacitor Charge Light
4. Place GF cutout switch (2, Figure 5-3) on the control cabinet in the CUTOUT position.
  5. Verify that the link voltage lights on the DID panel are off. If they remain on longer than five minutes after shutdown, notify the electrical department.
  6. Verify that the steering accumulators have bled down by attempting to steer. The wheels should not turn.
  7. Bleed down the brake accumulators using the manual bleed valves on the brake manifold.
  8. Activate the battery disconnect switches.
  9. Block the front wheels to prevent movement as brakes will be released.
  10. Mark all electrical cables and hydraulic hoses before removal.
  11. Close the shutoff valves on the hydraulic pumps to reduce the amount of oil loss from the brake system.

## Removal

*NOTE: If suitable equipment is available to lift the wheel motor assembly from the axle housing, it is not necessary to remove the dump body. The lifting equipment must be capable of lifting and supporting the weight of the complete wheel motor assembly. It must also be mobile and capable of aligning the wheel motor to the axle housing mounting flange during installation. If the wheel motor must be removed by use of a crane or overhead hoist, refer to Section B, Dump Body, for removal instructions.*

1. Raise the rear of truck as described in this section until tires clear the ground. Use support stands or cribbing to block up the rear axle housing.
2. Remove the inner and outer wheels from the wheel motor. Refer to Section G, Tires and Rims, for instructions. Make sure that additional capscrews have been installed to provide support for the wheel motor transmission housing while the rear tires are removed.
3. With the pump shutoff valves in the closed position, remove the drain plug at the bottom of the brake assembly and drain the oil from the disc brake housing into a suitable container.
4. Remove access covers (2, Figure 5-4) and access tube (1).
5. Disconnect brake system cooling lines (4) and brake apply lines (3) at the brake back plate. Cap and plug all openings.
6. Remove the clamps that secure the brake hoses to the spindles.
7. Remove air duct (9) from between the wheel motors.
8. Disconnect parking brake hoses (8) for the motor that is being removed. Disconnect the nylon tube to the blower pressure switch, if necessary.
9. Disconnect wheel motor power cables (6) at the wheel motor terminals. Disconnect speed sensor harness (5) at the connector in the center of the housing.
10. Tie up cables and hoses as necessary to prevent damage during wheel motor removal.



***Each complete wheel motor assembly weighs approximately 18 132 kg (39,975 lb). Make sure that the lifting device is capable of handling the load safely.***

11. Attach the lifting device to the wheel motor assembly. **Do not allow the lifting device to contact the brake housing.**
12. Remove 70 capscrews and hardened flat washers that secure the wheel motor to the rear axle housing.
13. Move the wheel motor assembly out of the axle housing. Be careful to prevent damage to the brake hoses if not removed previously.
14. Refer to the appropriate GE service manual for wheel motor repair instructions.

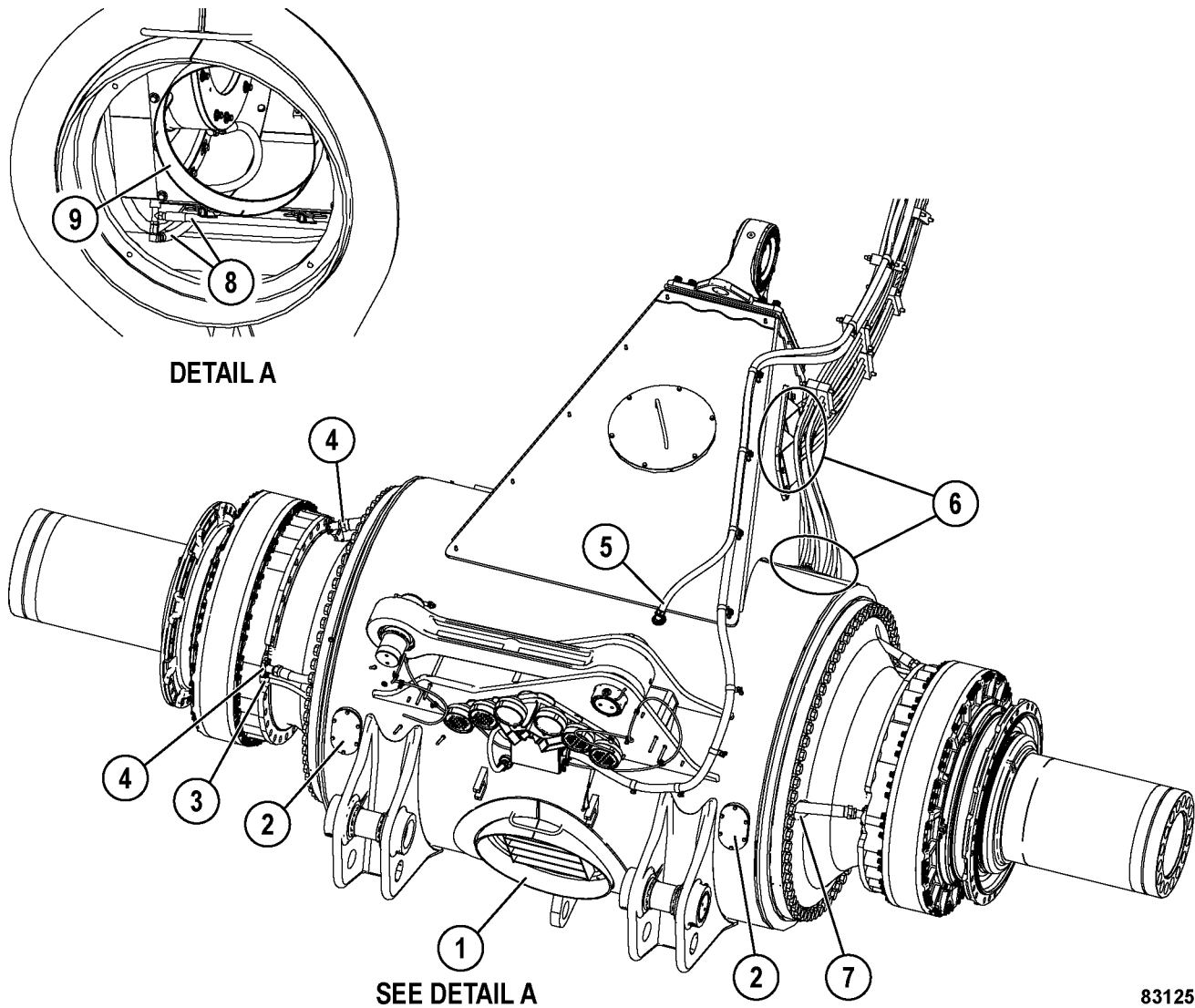


FIGURE 5-5. WHEEL MOTOR INSTALLATION

- |                     |                        |                        |
|---------------------|------------------------|------------------------|
| 1. Access Tube      | 4. Brake Cooling Hose  | 7. Spindle Hole        |
| 2. Access Cover     | 5. Wheel Speed Harness | 8. Parking Brake Hoses |
| 3. Brake Apply Hose | 6. Power Cables        | 9. Air Duct            |

## Cleaning and Inspection

1. Thoroughly clean the capscrew holes and wheel motor mounting faces. Re-tap any holes that have damaged threads.
2. Check the wheel motor mounting faces for nicks, scratches and other damage. Inspect all welds and repair as necessary.
3. Inspect the brake system relay valve and the hoses for leaks.
4. Inspect all hose connections and repair any leaks and damaged hoses.
5. Inspect the wheel motor power cables, terminals, cable grips, and clamps. Replace any cables or hardware that is worn or damaged.

## IMPORTANT

***All propulsion system power cables must be properly secured in non-ferrous cable cleats. If any clamps are cracked or broken, replace them with new parts. Inspect the cable insulation and replace the entire cable if the insulation is damaged.***

## Installation

1. Thoroughly clean the cap screw holes and the mounting faces of the rear axle housing and wheel motor.
2. Inspect the cap screw holes in the rear axle housing for damage. Re-tap any holes that have damaged threads. Thoroughly clean any cap screw holes that have been re-tapped.
3. Check the mounting faces of the rear axle housing and the wheel motor for surface defects (nicks, scratches, etc). Repair any defects before installing the wheel motor.
4. Check the flatness of both mounting faces of the rear axle housing. Check the flatness at four equally spaced intervals (for example, 0°, 45°, 90° and 135°).

The maximum allowable variation in flatness is 2.29 mm (0.09 in.). Do not use a rear axle housing that does not meet this specification.

5. Apply a multi-purpose grease with a minimum of 5% molybdenum disulphide to the threaded cap screw holes in the rear axle housing.
6. Install two guide pins 180° apart (3 o'clock and 9 o'clock positions) in the rear axle housing.

## WARNING

***Each complete wheel motor assembly weighs approximately 18 132 kg (39,975 lb). Ensure that the lifting devices are capable of handling the load safely.***

7. Lift the wheel motor into position on the rear axle housing. Ensure that all cables and hoses are clear before installation.

If the brake system hoses are installed, guide them through the spindle holes during installation.

## IMPORTANT

***The wheel motors must be properly aligned before installing them onto the rear axle housing. The wheel motor has markings which help determine installation orientation. Two sets of dimples are located at the 3 o'clock and 9 o'clock positions. A centerline symbol marks the 12 o'clock position of the wheel motor. Refer to Figure 5-5.***

***The axle housing also contains dimples at the 3 o'clock and 9 o'clock positions next to the wheel motor mounting rings. Refer to Figure 5-6.***

***When installing the wheel motors, make sure that the markings on both components line up. The top cap screw hole on the rear axle housing may also be determined by counting the holes in between the two sets of punch marks. The top hole on the rear axle housing should line up with the "CL" stamping on the wheel motor.***



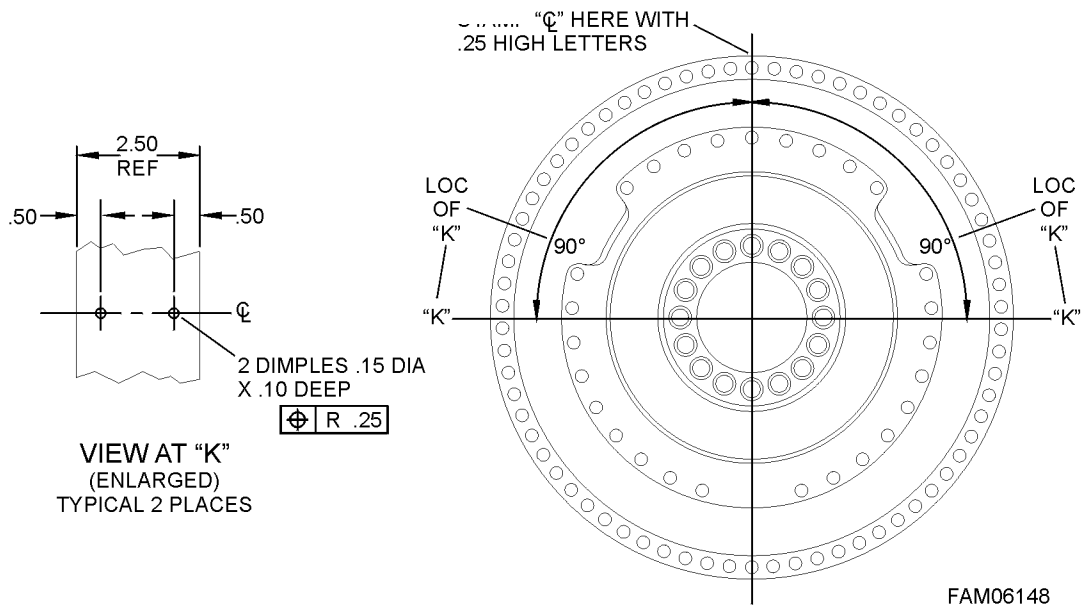


FIGURE 5-6. WHEEL MOTOR DIMPLE MARKINGS

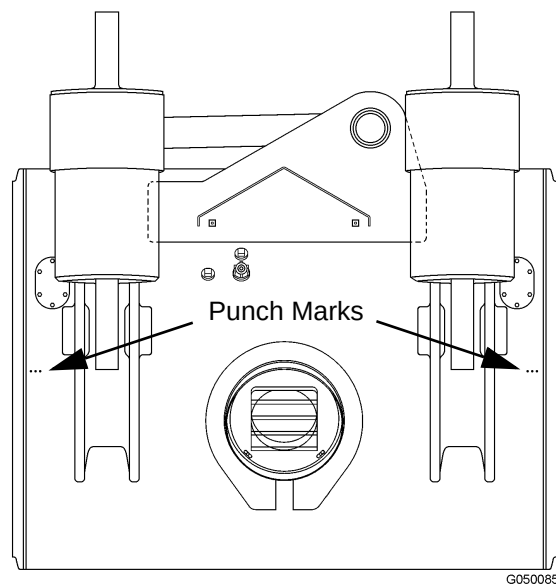


FIGURE 5-7. REAR AXLE HOUSING DIMPLE MARKINGS

## ! CAUTION

**High tightening force is required to install the cap screws that secure the wheel motors to the rear axle housing. Repeated tightening will result in cap screw fatigue and damage.**

**Do not reuse any wheel motor mounting hardware (cap screws and hardened washers). Replace the hardware after one use.**

**Do not retighten any wheel motor mounting cap screw that has loosened after the truck has been placed into operation. If any wheel motor mounting cap screw has loosened during truck operation, all of the cap screws and hardened washers must be replaced with new hardware.**

**The cap screws that are used to secure the wheel motors to the rear axle housing are specially hardened to meet or exceed Grade 8 specifications. Replace these cap screws with only new cap screws of the correct hardness. Refer to the appropriate parts book for the correct part number.**

**The use of dry threads in this application is not recommended. Due to the high tightening force that is required to install the wheel motor mounting cap screws, dry threads may cause damage to tools, cap screws or the rear axle housing.**

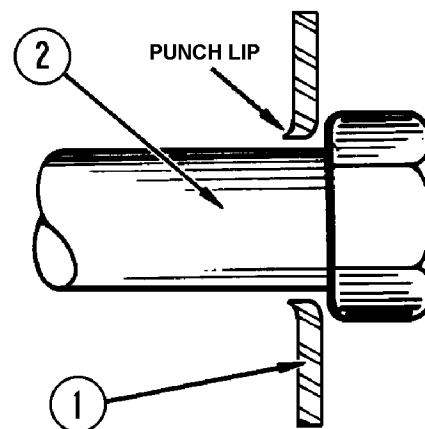
**Komatsu does not recommend the use of special friction-reducing lubricants, such as Copper Coat, Never-Seez® or other similar products, on the threads of standard fasteners where standard torque values are applied. The use of special friction-reducing lubricants will significantly alter the clamping force during the tightening process. If a special friction-reducing lubricant is used, excessive stress and possible breakage of the fasteners may result.**

8. Select the hardware to be installed in Group 1 of the tightening sequence. Refer to Figure 5-8.

Inspect each cap screw for rust, corrosion and surface defects on any seat or thread. Do not use any cap screw if a defect is suspected.

9. Lubricate the cap screw threads, cap screw head seats and washer faces with a multi-purpose grease with a minimum of 5% molybdenum disulphide.

**NOTE:** The special hardened washers that are used in this application may have a punch lip on one side due to the manufacturing process. When placing this washer under the cap screw head, the washers must be installed with the punch lip facing away from the cap screw head to prevent damage to the fillet between the cap screw head and the shank. Refer to Figure 5-7.



90011

FIGURE 5-8. INSTALLATION OF HARDENED WASHER

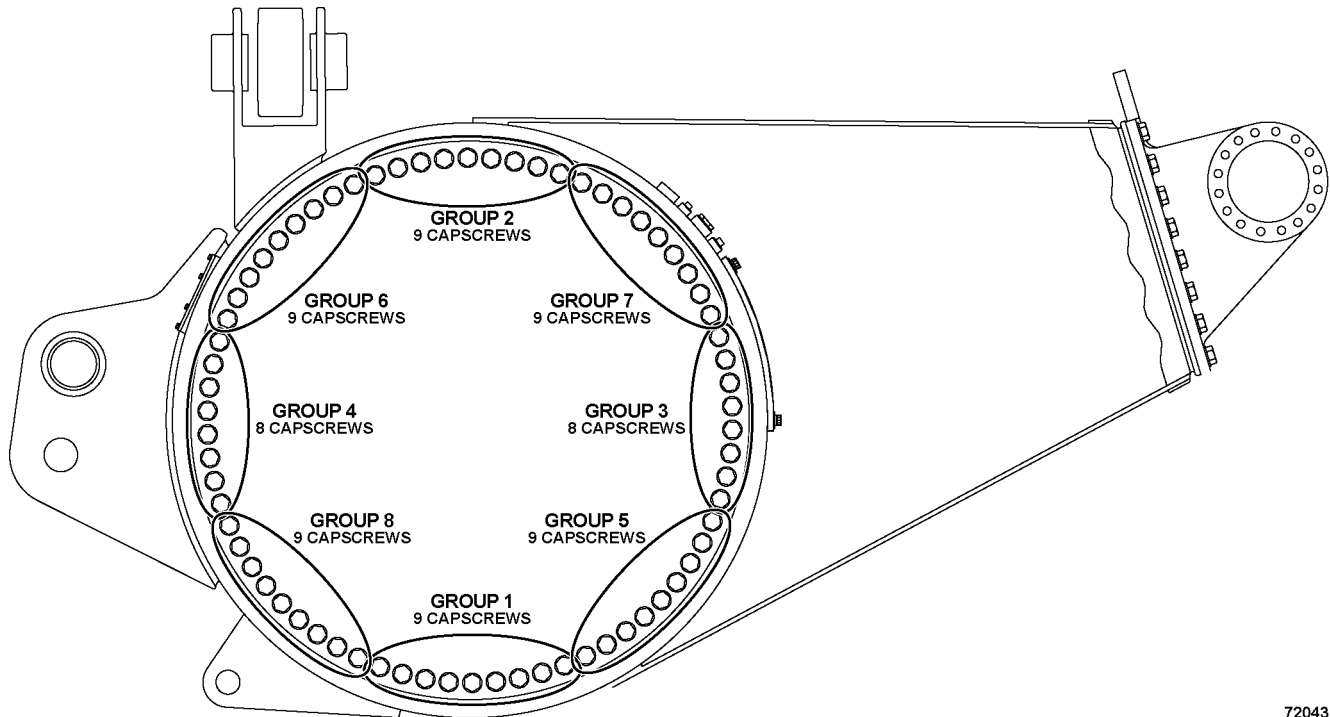
1. Washer                      2. Cap Screw

10. Install the hardware for Group 1. Tighten each cap screw to **542 N·m (400 ft lb)**.
11. Repeat steps 8 - 10 for each remaining group in the tightening sequence. Refer to Figure 5-8.

## ⚠ IMPORTANT ⚠

*Install and tighten all of the hardware in a group before proceeding to the next group in the tightening sequence. Remove the guide pins that were installed in step 6 when installing the hardware for Groups 3 and 4.*

12. After all of the hardware in all of the groups has been installed and tightened to **542 N·m (400 ft lb)**, move back to Group 1. Increase the torque on each cap screw in Group 1 to **2 006 N·m (1,480 ft lb)**. Repeat this step for each remaining group in the tightening sequence. Refer to Figure 5-8.



72043

FIGURE 5-9. WHEEL MOTOR MOUNTING HARDWARE INSTALLATION SEQUENCE

## ⚠ IMPORTANT ⚠

*All propulsion system power cables must be properly secured in wood or other non-ferrous cable clamps. If any clamps are cracked or broken, replace them with new parts. Inspect the cable insulation and replace an entire cable if the insulation is damaged.*

13. Connect power cables (6, Figure 5-4) and wheel speed harness (5) to their appropriate locations on the wheel motor.

14. Connect brake apply hose (3) and brake cooling hoses (4) to the brake assembly. Use new O-rings.
15. To prevent cooling air loss, use Uniseal 310S sealant to seal the gap around the brake cooling hoses where the hoses pass through spindle holes (7).

Uniseal 310S sealant is available from:

Uniseal  
1800 W. Maryland Street  
Evansville, IN 47712

Uniseal 310S sealant may be available from other sources throughout the world.

16. After both wheel motors are installed, remove rear axle housing access covers (2, Figure 5-4) and access tube (1) at the rear of the axle housing.
17. Install the remaining hose clamps. Connect parking brake hoses (8). If removed, connect the nylon tube to the blower pressure switch.
18. Install air duct (9) between the parking brakes.
19. Install rear axle housing access covers (2) and access tube (1) at the rear of the axle housing.
20. Open all of the hydraulic pump shutoff valves.
21. Check the oil level in each wheel motor. Rotate one of the eight plugs to the 6 o'clock position and remove the plug. The oil level should be even with the bottom of the plug opening. Refer to Figure 5-9. Fill the wheel motor with the recommended oil as necessary.

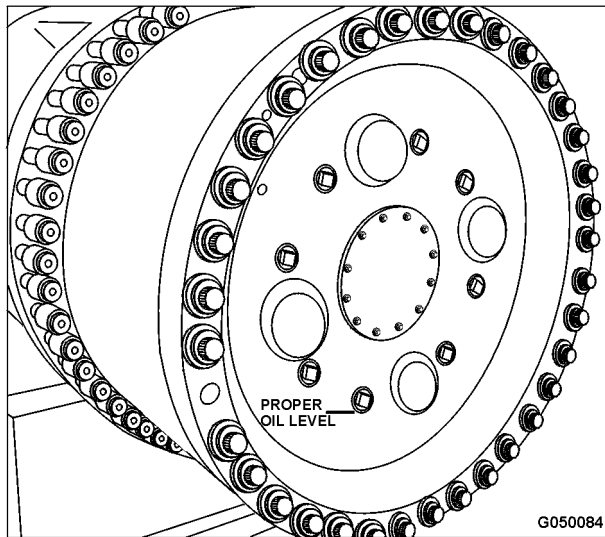


FIGURE 5-10. WHEEL MOTOR OIL LEVEL

22. Check the hydraulic tank oil level before and after engine start-up. Service the hydraulic tank oil as necessary.
23. Bleed the brake apply lines according to the brake bleeding procedure in the shop manual.
24. Check the hydraulic tank oil level after the brake bleeding procedure. Service the hydraulic tank oil as necessary.
25. Install the tires and rims according to the procedure in the shop manual.
26. Raise the truck and remove the support stands. Lower the truck and remove the jack.

## **CAUTION**

***High tightening force is required to install the cap screws that secure the wheel motors to the rear axle housing. Repeated tightening will result in cap screw fatigue and damage.***

***Do not retighten any wheel motor mounting cap screw that has loosened after the truck has been placed into operation. If any wheel motor mounting cap screw has loosened during truck operation, all of the cap screws and hardened washers must be replaced with new hardware.***

## WHEEL MOTOR GEAR OIL

### Filtering Requirements

Wheel motor gear oil must be changed (or filtered) every 500 operating hours or sooner if determined from oil sample analysis. Filtering synthetic oil for reuse is recommended, and can significantly extend the life of the oil. Due to their additive formulation, GE approved oils may last up to a year, or possibly longer. After a year, the synthetic oil must be analyzed by the oil manufacturer if continued use is desired. A 10 micron filter must be used for filtering the oil.

Viscosity of filtered oil must be monitored by oil analysis to ensure proper lubrication. Viscosities below the condemning limit indicates entry of contamination that will affect the oil film and lubricating quality. The condemning limit for 680 oil, is 53cSt when tested at 100° C (212° F), and 588cSt at 40° C (104° F). This value can be used along with element tracking of phosphorus, calcium, and zinc to verify correct oil viscosity. Ensure the appropriate viscosity value is being maintained for the approved oil you are using. If the viscosity of the oil is below the acceptable limit, the oil should be rechecked. If the second test fails, dispose of the oil. Fill the wheel motor with new, approved oil. The contaminated oil should be properly disposed of according to local regulations.



***If the viscosity of the oil is below the condemning limits, the oil may be contaminated with hydraulic oil. Make sure that all joints are properly sealed before filling the wheel motor with fresh oil. If hydraulic oil continues to contaminate the wheel motor, serious damage to internal components may result.***

### Particle Size Analysis

When filtering synthetic oil for reuse, the oil must be monitored to determine when filtering is necessary. The process of monitoring the oil for this purpose is particle size analysis. The test consists of passing a beam of light through the oil as it passes through a clear tube. A computer analyzes the quantity and size of particles contained in the oil.

Particles that are smaller than 5 microns in size are not large enough to cause any gear or bearing failures. Particles over 25 microns can damage bearings and must be filtered from the oil.

Test the oil according to the International Standards Organization's ISO Particle Analysis 4406. This standard documents the size and number of particles permitted in the lubricant. This is an extremely important measurement of the condition of the oil.

An "ISO Cleanliness Code" is used to document the number and size of the contaminants that can be tolerated in a specific application of a lubricant. An example of an "ISO Cleanliness Code" would be 18/13. The first number (18 in this example) applies to the allowable number of particles between 5 - 15 microns in size. The second number (13 in this example) applies to the allowable number of particles between 15 - 25 microns. If the particle count identified by the "ISO Cleanliness Code" is exceeded or there are particles present larger than 25 microns, the oil should be filtered.

Use Cleanliness Code of 18/13 for motorized wheels. Oil qualified for reuse must be filtered to 10 microns absolute.

If particle analysis yields a result of greater than 18 for the top number or greater than 13 for the lower number the motorized wheel oil should be filtered to 10 microns absolute. On occasion, one should check the results of the filtering process to assure the filtering process is yielding the desired results. Should the result be unacceptable, check the filter type, particle size, and condition prior to filtering again.

### Flushing

The transmission of the wheel motor must be flushed using the same oil that is used in normal operation. The use of solvents or low viscosity oils are not acceptable.

After flushing to remove contamination from the wheel motor, the transmission should be filled with new or filtered oil. A sample should be taken for analysis after 100 hours of operation. If results of the oil analysis are within the acceptance range, the wheel motor may return to the 250 hour interval for oil analysis. If the results of the analysis are not within the acceptance range, immediately contact your area Komatsu service representative. Large quantities of loose material in the oil may be an indication of imminent wheel motor failure.

## Oil Sample Analysis

Spectrographic oil analysis results should be examined prior to sun pinion gear inspection. The trends of element parts per million (ppm) or sudden ppm element changes can estimate the present condition of the wheel motor and also predict the immediate future of the mechanical components within the wheel motor.

While examining oil sample history, the most important characteristic to look for is a rapid increase in the presence of an element in the oil such as iron (Fe). Rapid changes in element content indicate that a component(s) may be deteriorating. In many cases oil analysis can help detect which component may be deteriorating and the cause of the failure.

Oil sampling must be done every 250 hours of operation. The sample must be taken within a half an hour of truck shut down and within an hour of actual truck operation. Oil samples that are not taken under this stipulation will not convey an accurate measurement of wheel motor condition.

When taking an oil sample, position one of the magnetic plugs in the six o'clock position on the wheel. Oil samples should be taken from this plugged hole using a flexible tube that is 18 in. (457 mm) in length. The tube must be inserted downward exactly 12 in. (305 mm) toward the bottom of the torque tube.

When a sample is extracted from the wheel motor and placed in a clear bottle, the appearance, as well as odor of the oil should be inspected. Note any burning smells or other odors typical of fatigued or contaminated oil. Check the oil for cloudiness, large particles, and filming. Any of these conditions indicate a problem.

A gradual rise of elements between oil samples is normal. Only when a sharp increase occurs is there evidence that a mechanical failure is imminent. Levels of elements that indicate mechanical stress differ from element to element. Additionally, oil sampling criteria established for each element may differ at each mine site due to factors such as uphill or downhill hauling, grades, and speeds.

Elements for the wheel motor that can be detected with oil analysis are iron (Fe), chromium (Cr), nickel (Ni), copper (Cu), silicon (Si), phosphorus (P), zinc (Zn), and calcium (Ca). Monitoring and plotting levels of these elements should aid in identifying operational status of the wheel motor. Rapid changes or high levels of these elements are indicators that problems exist in the mechanical components.

As stated above, some guidelines for analyzing oil samples can be given, but specific levels should be established at the operating mine site. The element type, maximum ppm, maximum spike allowable between 250 hour interval oil samples, oil sample indicators, and possible causes and actions are listed in the following charts. These charts provide basic guidelines for analysis of oil sampling information.

## ELEMENT SPECIFICATION CHARTS

| Element   | Max ppm | Max spike | Indications of oil samples                                           | Possible cause of elevated ppm's                                         |
|-----------|---------|-----------|----------------------------------------------------------------------|--------------------------------------------------------------------------|
| Iron (Fe) | 100     | 200       | If iron levels are high and other elements are slightly elevated:    | "Soft" iron contamination. Sources are likely to be casting material.    |
|           |         | 100       | If iron levels are high and other elements ppm's are rising rapidly: | "Hard" iron contamination. Sources are likely to be bearing, gears, etc. |

**Action:** A sun pinion and magnetic plug inspection should be done immediately. The magnetic plugs should be inspected for metal chips and shavings. The sun pinion and low speed planet gears should be inspected for wear or spalling per GE GDY-85/106 AC Motorized Wheel Component Inspection and Reuse Criteria (GEK-91680). If the low speed planet gears fail to meet the inspection criteria the wheel should be removed, disassembled, and gears replaced. If the low speed planet gears meet the re-use criteria, but the sun pinion fails to meet the reuse criteria, then replace the sun pinion. Once the sun pinion has been replaced drain the oil and replace with fresh oil. This wheel should then be monitored closely and replaced if the iron level spikes again.

| Element       | Max ppm | Max spike | Indications of oil samples            | Possible cause of elevated ppm's         |
|---------------|---------|-----------|---------------------------------------|------------------------------------------|
| Chromium (Cr) | 10      | 7         | If chromium is high and iron is high: | A bearing failure is likely in progress. |

**Action:** The oil should be resampled immediately and a gear train inspection completed. If resampling shows high chrome, the wheel should be removed from service, disassembled, and the bearings and gears inspected.

| Element     | Max ppm | Max spike | Indications of oil samples                    | Possible cause of elevated ppm's                |
|-------------|---------|-----------|-----------------------------------------------|-------------------------------------------------|
| Nickel (Ni) | 10      | 5         | If nickel is high and iron is high or rising: | Nickel is likely from one or more of the gears. |

**Action:** This could indicate a condition such as gear tooth pitting or spalling. The oil should be resampled immediately and a gear train inspection completed. If the resampling of the oil shows high nickel and chromium, the wheel should be removed from service, disassembled, and the bearings and gears inspected.

| Element     | Max ppm | Max spike | Indications of oil samples                 | Possible cause of elevated ppm's                                                                                |
|-------------|---------|-----------|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Copper (Cu) | 50      | 15        | If copper is high with iron high:          | High iron will produce high copper from the thrust washers and bearings.                                        |
|             |         | 20        | If copper is high with other elements low: | A thrust washer is rapidly wearing with the likely cause of the sun pinion gear thrusting inboard and outboard. |

**Action:** This often indicates a normal wear condition such as minor thrust washer wear. It can also be an indicator of bearing wear. If bearing wear is a concern review the chromium readings in conjunction with the copper readings. If both the copper and chromium readings spike; remove the wheel from service, disassemble the wheel, and inspect the bearings. If the only elevated reading is that of copper or lead; remove the sun pinion and check the outer thrust washer for wear. If the outer thrust washer has worn less than the depth of the oil lubricating grooves then reassemble the sun pinion, filter the oil and continue to monitor. If the outer thrust washers have worn more than the depth of the oil lubricating grooves then replace the outer thrust washers, reassemble the sun pinion, filter the oil and continue to monitor.

| Element      | Max ppm | Max spike | Indications of oil samples                | Possible cause of elevated ppm's           |
|--------------|---------|-----------|-------------------------------------------|--------------------------------------------|
| Calcium (Ca) | 40      | 20        | If calcium, zinc and phosphorus are high: | Hydraulic oil has entered the drive train. |

**Action:** If calcium and zinc spike then look at the oil viscosity. If the oil viscosity is below the 53 cSt at 100° C (212° F) or 588 cSt at 40° C (104° F) condemning limit, the oil should be resampled immediately. If the resampling still shows a spike in calcium and low viscosity remove the wheel, disassemble, and replace the internal brake seals. The oil removed from a low viscosity wheel should be disposed of and not reused under any circumstances. If the oil viscosity is above 53 cSt at 100° C (212° F) or 588 cSt at 40° C (104° F) then monitor the wheel for any further changes in viscosity

| Element   | Max ppm | Max spike | Indications of oil samples                           | Possible cause of elevated ppm's                 |
|-----------|---------|-----------|------------------------------------------------------|--------------------------------------------------|
| Zinc (Zn) | 75      | 40        | If zinc, phosphorus and calcium are high:            | Hydraulic oil has entered the drive train.       |
|           |         | 25        | If zinc is high without high calcium and phosphorus: | High mechanical wear of components is occurring. |

**Action:** If zinc and calcium increase proportionally then look at the oil viscosity. If the oil viscosity is below the 53 cSt at 100° C (212° F) or 588 cSt at 40° C (104° F) condemning limit, the oil should be resampled immediately. If the resampling still shows a spike in calcium and low viscosity remove the wheel, disassemble, and replace the internal brake seals. The oil removed from a low viscosity wheel should be disposed of and not reused under any circumstances. If the oil viscosity is above 53 cSt at 100° C (212° F) or 588 cSt at 40° C (104° F) then monitor the wheel for any further changes in viscosity.

| Element        | Max ppm | Max spike | Indications of oil samples                | Possible cause of elevated ppm's           |
|----------------|---------|-----------|-------------------------------------------|--------------------------------------------|
| Phosphorus (P) | 400     | 150       | If phosphorus, zinc and calcium are high: | Hydraulic oil has entered the drive train. |

**Action:** If the oil viscosity is below the 53 cSt at 100° C (212° F) or 588 cSt at 40° C (104° F) condemning limit, the oil should be resampled immediately. If the resampling still shows a spike in phosphorus and low viscosity, remove the wheel, disassemble, and replace the internal brake seals. The oil removed from a low viscosity wheel should be disposed of and not reused under any circumstances. If the oil viscosity is above 53 cSt at 100° C (212° F) or 588 cSt at 40° C (104° F) then monitor the wheel for any further changes in viscosity.

| Element      | Max ppm | Max spike | Indications of oil samples                   | Possible cause of elevated ppm's                         |
|--------------|---------|-----------|----------------------------------------------|----------------------------------------------------------|
| Silicon (Si) | 50      | 20        | If silicon is high with other elements high: | High wear of components.                                 |
|              |         | 25        | If silicon is high with other elements low:  | Dirt has entered the drive train. Flush the wheel motor. |

**Action:** Flush the wheel motor and monitor element levels.

To fully utilize oil sampling analysis, complete information should be kept with the truck. Necessary information includes: the truck identification number at the mine site, truck serial number, side of truck the wheel motor is on, dates when wheel motors are changed (include serial numbers of wheels), the type of oil, viscosity of the oil, sample date, hours on wheel, hours on oil, and element count. Refer to the chart below. Information in this format can be easily sent to equipment manufactures and gives required information to analyze the results of oil samples. Separate records for each wheel motor should be kept in this format.

## ⚠ IMPORTANT ⚠

**Contamination can reduce sun pinion gear life significantly, as well as the rest of the drive train. Additional component (sun pinion gear, etc.) inspections may be required, in addition to normal maintenance interval inspections, if oil samples from a wheel motor show elements are above normal or rapidly elevating.**



[illegible]

**SECTION H**  
**HYDRAIR® II SUSPENSIONS**  
**INDEX**

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SUSPENSION OILING AND CHARGING PROCEDURES ..... H4-1

# NOTES

## SECTION H2

### FRONT SUSPENSIONS

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# NOTES

## FRONT SUSPENSION

Hydrair® II suspensions are hydro-pneumatic components containing oil and nitrogen gas. The oil and gas in the four suspensions carry the gross truck weight less wheels, spindles and rear axle assembly.

The front suspension cylinders consist of two basic components: a suspension housing attached to the truck frame and a suspension rod attached to the front spindle.

Check valves and orifice dampening holes control suspension travel to provide good ride qualities on haul roads under loaded and empty conditions.

The front suspension rods also act as kingpins for steering the truck.

The Hydrair® II suspension cylinder requires only normal care when handling as a unit. However, after being disassembled, these parts must be handled carefully to prevent damage to the machined surfaces. Surfaces are machined to extremely close tolerances and are precisely fitted. All parts must be completely clean during assembly.

### Removal

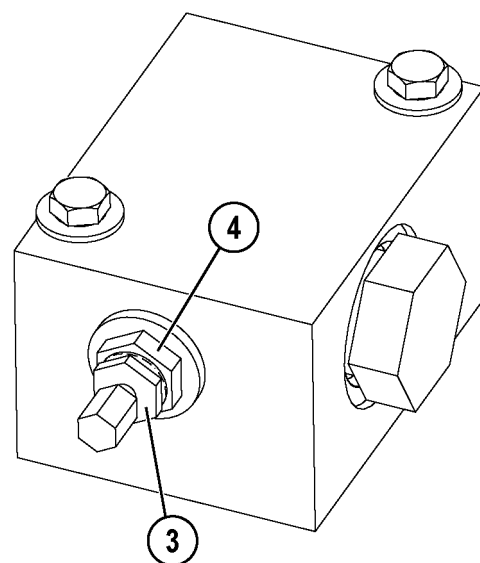
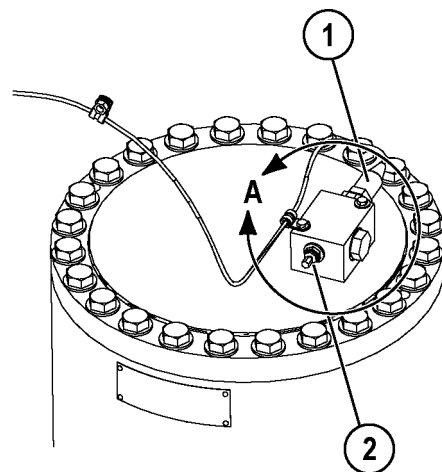
1. Remove the front wheel and tire. Refer to Section G2, Tires and Rims.
2. Remove the front wheel hub and spindle. Refer to Section G3, Front Wheel Hub and Spindle.
3. Remove the boot clamp and boot from the front suspension.



***Wear a face mask or goggles. Make sure that only the swivel nut (small hex) turns. Turning the entire charging valve body (large hex) may result in the valve assembly being forced out of the suspension by the gas pressure inside.***

4. Discharge the nitrogen pressure from the suspension as follows:
  - a. Remove the cap from charging valve (2, Figure 2-1).
  - b. Turn swivel nut (small hex) (3) counterclockwise three full turns to unseat the valve seal. DO NOT turn more than three turns. DO NOT turn charging valve body (large hex) (4).
  - c. Depress the valve stem until all nitrogen pressure has been relieved.

5. After all nitrogen pressure has been relieved, loosen charging valve body (4) and remove the charging valve assembly. Discard the O-ring seal.
6. Remove pressure sensor (1) from the charging valve block.



DETAIL A

63222

FIGURE 2-1. SUSPENSION CHARGING VALVE

- |                           |                                    |
|---------------------------|------------------------------------|
| 1. Pressure Sensor        | 4. Charging Valve Body (Large Hex) |
| 2. Charging Valve         |                                    |
| 3. Swivel Nut (Small Hex) |                                    |

7. Place a suitable container under the suspension cylinder. Remove the drain plug at the bottom of the suspension piston and allow the suspension oil to drain completely from the cylinder. A properly charged front suspension cylinder contains 98 L (25.8 gal) of oil.
8. Install the drain plug. Tighten the plug to **17 N·m (13 ft lb)**.

*NOTE: Front Hydrair® II suspensions are equipped with lower bearing retainer puller holes. If rod wiper, rod seals, bearing, O-ring and backup ring replacement is required, it is not necessary to remove the suspension from the truck. Refer to Lower Bearing & Seals for bearing retainer removal and installation.*

*If major suspension rebuild is required, continue with the removal procedure.*

9. Attach a fork truck or suitable lifting device to the suspension. Secure the suspension to the lifting device.

## ⚠ WARNING

***The front Hydrair® II suspension weighs approximately 2 790 kg (6,150 lb). Make sure that the lifting device to be used has a sufficient capacity to handle the load.***

10. Remove capscrews (1, Figure 2-3).
11. Remove four outer capscrews (4).
12. Remove four inner capscrews (4) with spacers (7).
13. Move the suspension to a clean work area for disassembly.
14. Discard the suspension mounting capscrews, washers, and nuts.

## Installation

1. Remove all paint, rust, dirt and foreign debris from the suspension mounting surfaces.



FIGURE 2-2. REMOVING DEBRIS

2. Clean and dry the mounting surfaces on both the suspension and the frame. Use a cleaning agent that does not leave a film after evaporation, such as isopropyl alcohol, acetone or lacquer thinner.
3. Inspect the suspension and frame mounting surfaces for damage or wear. Check for the following criteria:
  - ☐ Flatness of each surface must be within 0.25 mm (0.010 in.).
  - ☐ Surface finish must not exceed 0.635  $\mu\text{m}$  (250  $\mu\text{in.}$ ) roughness average ( $R_A$ ), medium tool cut.

If measurements are not within specifications, contact your Komatsu customer service manager for further instructions.
4. Clean and dry all suspension mounting hardware. Use the same cleaning agent that was used to clean the mounting surfaces.

## ⚠ WARNING

**High tightening torque is required to load the front suspension mounting cap screws. Repeated tightening will result in cap screw fatigue and damage. DO NOT reuse mounting cap screws, washers and nuts. Replace the hardware after each use.**

**Suspension mounting cap screws are specially hardened to meet or exceed grade 8 specifications. Replace only with cap screws of correct hardness. Refer to the appropriate parts book for the correct part numbers.**

## ⚠ CAUTION

**The use of dry threads in this application is not recommended. Due to the high tightening forces required to load these cap screws, dry threads or threads lubricated with anti-seize compounds may result in damage. Only use the approved lubricant specified below.**

5. Lubricate the cap screw threads, cap screw head seats, washer faces and nut seats with a multi-purpose grease with a minimum of 5% molybdenum disulphide.
6. Identify and separate the hardware according to location on the suspension. Refer to Figure 2-3.

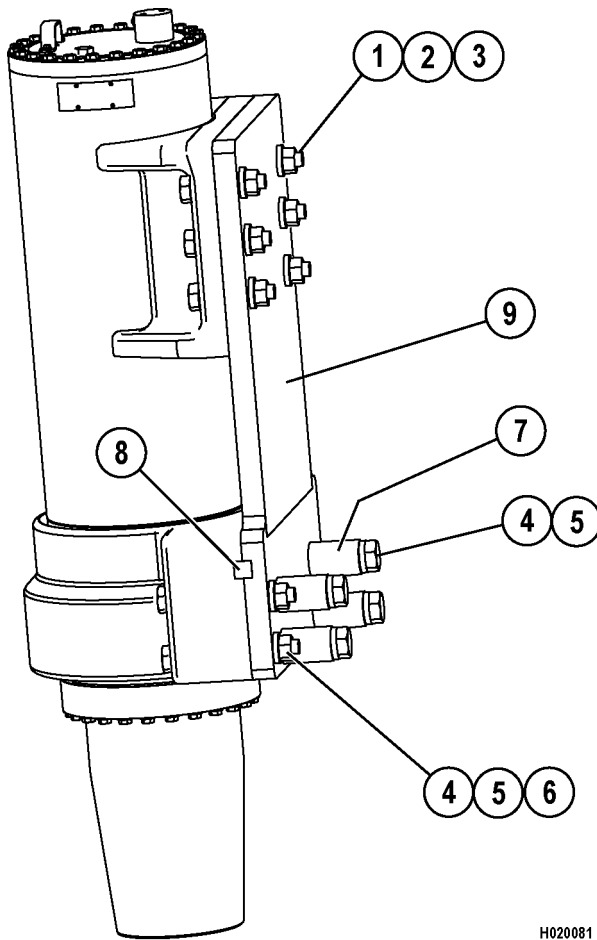


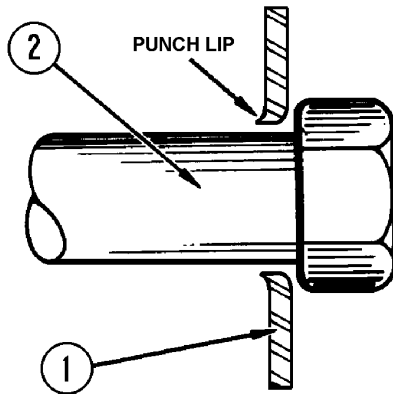
FIGURE 2-3. HARDWARE IDENTIFICATION

1. Cap Screw - 1 1/2" - 6NC x 6 1/2" (G8)
2. Flat Washer - 1 1/2" (G8)
3. Nut - 1 1/2" - 6NC (G8)
4. Cap Screw - 1 3/4" - 5NC x 14 1/2" (G8)
5. Flat Washer - 1 3/4" (G8)
6. Nut - 1 3/4" - 5NC (G8)
7. Spacer
8. Shear Key
9. Part Of Frame

H020081



7. The hardened flat washers used on the front suspensions may be punched during the manufacturing process. When assembling the cap screws and washers, position the punched lip away from the cap screw head to prevent damage. Refer to Figure 2-4.



90011

FIGURE 2-4. FLAT WASHER ORIENTATION

1. Hardened Flat Washer    2. Cap Screw

8. Lift the front suspension into position on the frame. The weight of each front suspension cylinder is approximately 2 790 kg (6,150 lb).
9. Install the mounting hardware and the shear key. Install the shear key flush with the side face of the suspension.

Install one washer under each cap screw head and one washer under each nut (where applicable).

Initially tighten the hardware to **1 356 ± 136 N·m (1000 ± 100 ft lb)** in the sequence shown in Figure 2-6. Use a properly calibrated torque wrench to ensure accuracy.

*NOTE: Do not exceed 4 rpm tightening speed. Do not hammer or jerk the wrench while tightening.*



FIGURE 2-5. SUSPENSION INSTALLATION

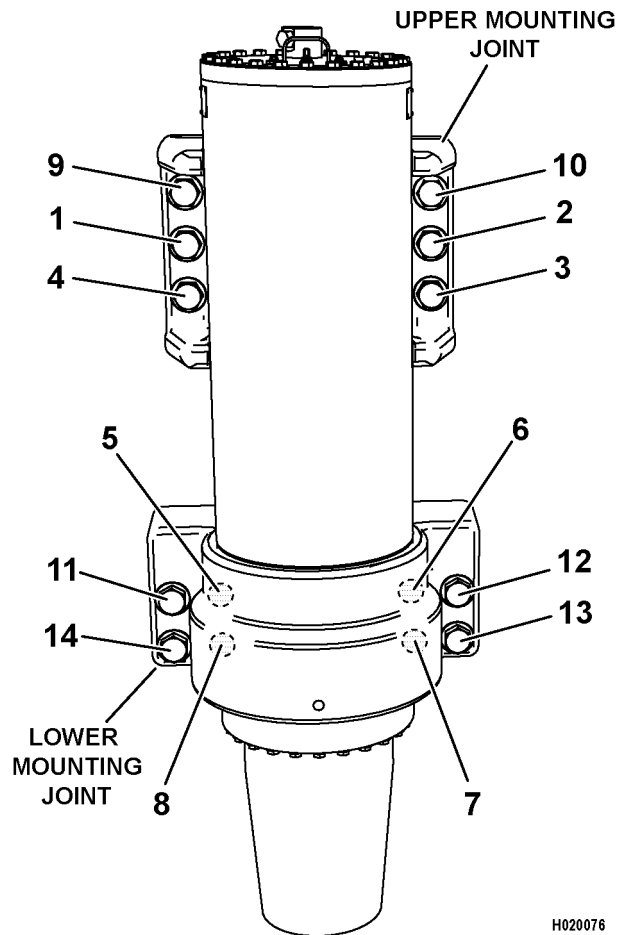


FIGURE 2-6. TIGHTENING SEQUENCE MAP

The mounting cap screws will now need to be loosened and then tightened using angle-advance. The first set of cap screws to be adjusted will be at the upper mounting joint. The cap screws must be loosened one at a time and then tightened by advancing a specified rotational degree. Use the map shown in Figure 2-6 for proper tightening sequence.

### Upper Mounting Joint - 60° Advance

The proper sequence for the upper mounting joint is 1-2-3-4-9-10. Refer to Figure 2-6.

10. Loosen the first cap screw. All other cap screws must be maintained at **1 356 ± 136 N·m (1000 ± 100 ft lb)**.
11. Tighten the cap screw to **95 N·m (70 ft lb)**.

*NOTE: Do not exceed 4 rpm tightening speed. Do not hammer or jerk the wrench while tightening.*

12. Mark a corner of the cap screw head with a paint marker as shown in Figure 2-7. Draw a reference line on the suspension surface next to the marked corner on the cap screw. Draw a reference line on the suspension 60 degrees in advance of the marked corner on the cap screw.

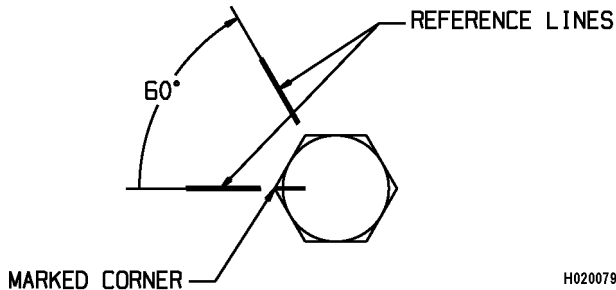


FIGURE 2-7. 60 DEGREE ADVANCE

13. Advance the cap screw to the 60° advance mark. Hold the nut at the rear of the joint stationary while tightening.

14. Make new reference lines along the cap screw, nut and frame at the rear of the joint as shown in Figure 2-8. These reference lines will be used to verify the cap screws have maintained their torque.
15. Repeat this process in sequence for the remaining cap screws in the upper mounting joint.

*NOTE: If for any reason these fasteners need to be checked for tightness after completing this procedure, replace all the hardware and repeat the entire process.*

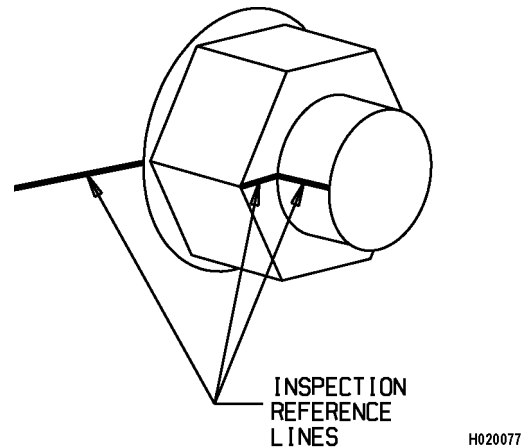


FIGURE 2-8. MAKING REFERENCE LINES

### Lower Mounting Joint - 120° Advance

The proper sequence for the lower mounting joint is 5-6-7-8-11-12-13-14. Refer to Figure 2-6.

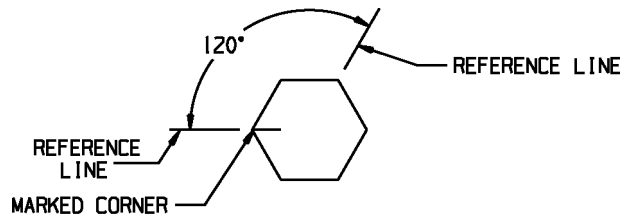
16. Loosen the first cap screw. All other cap screws must remain tightened.

17. Tighten the cap screw to the specified torque:

- cap screws 5-6-7-8.....**136 N·m (100 ft lb)**
- cap screws 11-12-13-14.....**271 N·m (200 ft lb)**

**NOTE:** Do not exceed 4 rpm tightening speed. Do not hammer or jerk the wrench while tightening.

18. Mark a corner of the cap screw head with a paint marker as shown in Figure 2-9. Draw a reference line on the suspension surface (or frame) next to the marked corner on the cap screw. Draw a reference line on the suspension surface (or frame) 120 degrees in advance of the marked corner on the cap screw.



H020082

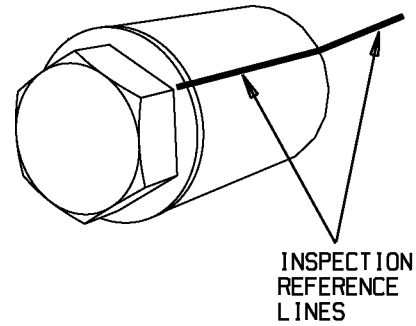
FIGURE 2-9. 120 DEGREE ADVANCE

19. Advance the cap screw to the 120° advance mark. Hold the nut at the rear of the joint (where applicable) stationary while tightening.

20. Make new reference lines along the cap screw, nut and frame at the rear of the joint as shown in Figure 2-8. For the four cap screws with spacers, refer to Figure 2-10. These reference lines will be used to verify the cap screws have maintained their torque.

21. Repeat this process in sequence for the remaining cap screws in the lower mounting joint.

**NOTE:** If for any reason these fasteners need to be checked for tightness after completing this procedure, replace all the hardware and repeat the entire process.



H020078

FIGURE 2-10. MAKING REFERENCE LINES

22. Charge the suspension with dry nitrogen to fully extend the suspension piston before installing the front wheel hub and spindle.

23. Install the front wheel hub and spindle. Refer to Section G3, Front Wheel Hub and Spindle.

24. Install the front wheel and tire. Refer to Section G2, Tires and Rims.

25. Service the suspension. For instructions refer to Section H, Oiling and Charging Procedures.

26. Install the suspension boot and secure it with the boot clamp.

### Inspection

Visual inspections of the bolted joints are necessary after the truck has been released for use. Inspect the joints at each front suspension at the following intervals: 8 hours, 50 hours, 250 hours, and 500 hours.

If the reference lines on the hardware (Figure 2-8 and Figure 2-10) have remained in alignment, the trucks may remain in use.

If at least one of the cap screws has shown signs of movement, the truck must be taken out of service. The suspension mounting cap screws must be removed, cleaned, and inspected. If any cap screws have any signs of damage, replace all cap screws. Install the cap screws again according to this installation procedure.

## LOWER BEARING & SEALS

### Removal

If only rod wiper, rod seals, bearing, O-ring and backup rings are to be replaced, refer to the following steps for lower bearing retainer removal.

1. Remove lower bearing retainer capscrews and hardened washers (1, Figure 2-11). Install pusher bolts into the tapped holes in the retainer flange.
2. Tighten the pusher bolts evenly and prepare to support the bearing retainer as it exits the suspension housing. Remove lower bearing retainer (2).
3. Remove wiper seal (3), double lip seal (4), buffer seal (5), backup rings (6), O-rings (7) and lower bearing (8).

### Installation

1. Install new buffer seal (5, Figure 2-11), double lip seal (4) and wiper seal (3).

## ! IMPORTANT !

*When installing backup rings with double lip seal (4) and buffer seal (5), make sure that the radius is positioned toward the seal and the white dot is positioned away from the seal.*

2. Install new backup rings (6) and O-rings (7) in their appropriate grooves in lower bearing retainer (2). Install new lower bearing (8).

**NOTE:** Backup rings must be positioned toward the lower bearing retainer flange.

3. Install temporary guide bolts to ensure that the bolt holes are aligned as the lower bearing retainer is seated. Carefully lift lower bearing retainer (2) into place into the suspension housing.
4. Install lower bearing retainer capscrews and hardened washers (1). Tighten the capscrews to **678 N·m (500 ft lb)**.
5. Install the front wheel hub and spindle. Refer to Section G3, Front Wheel Hub and Spindle.
6. Install the front wheel and tire. Refer to Section G2, Tires and Rims.

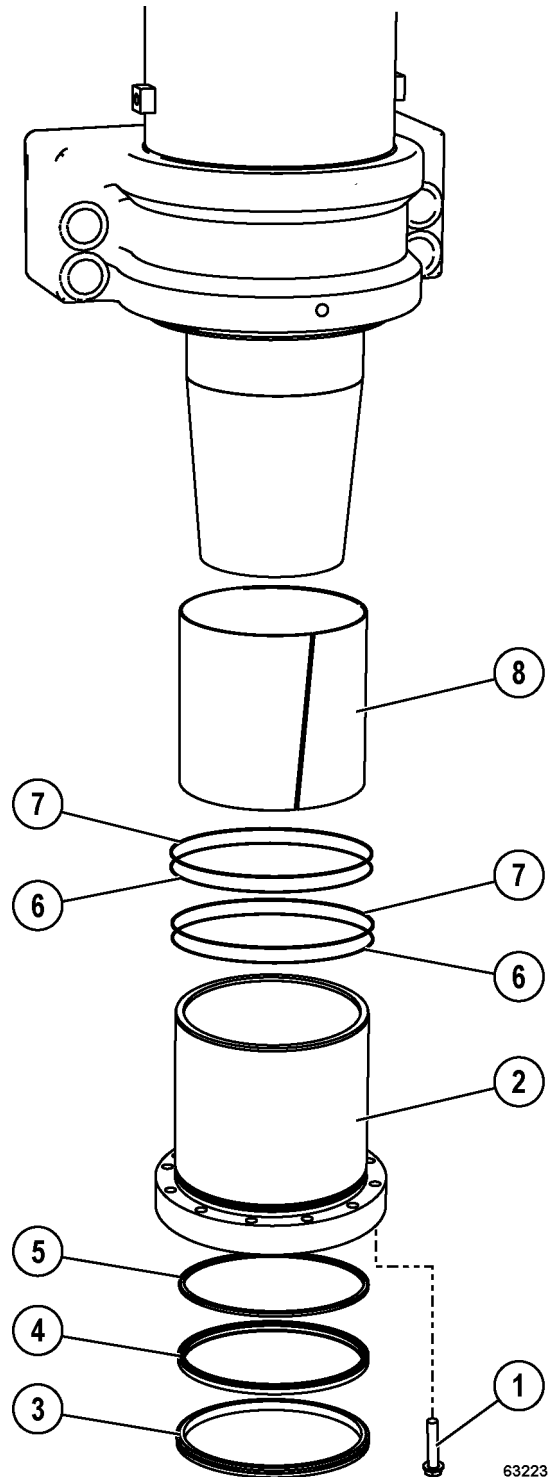


FIGURE 2-11. FRONT SUSPENSION

- |                                 |                  |
|---------------------------------|------------------|
| 1. Capscrews & Hardened Washers | 5. Buffer Seal   |
| 2. Lower Bearing Retainer       | 6. Backup Ring   |
| 3. Wiper Seal                   | 7. O-Ring        |
| 4. Double Lip Seal              | 8. Lower Bearing |

## MAJOR SUSPENSION REBUILD

### Disassembly

*NOTE: Refer to your Komatsu distributor for Hydrair®II suspension repair information and instructions not covered in this manual.*

1. With the suspension held in a vertical position (end cap up), remove capscrews (1, Figure 2-12) and hardened washers (2).

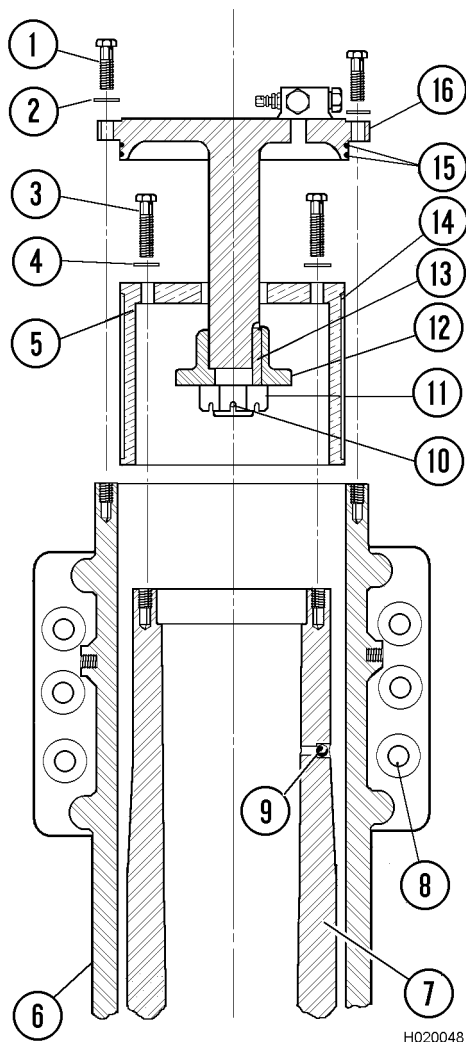


FIGURE 2-12. PISTON ROD REMOVAL

- |                           |                            |
|---------------------------|----------------------------|
| 1. Capscrew               | 9. Steel Ball (2 each)     |
| 2. Hardened Washer        | 10. Roll Pin               |
| 3. Capscrew               | 11. Nut                    |
| 4. Hardened Washer        | 12. Piston Stop            |
| 5. Upper Bearing Retainer | 13. Key                    |
| 6. Housing                | 14. Upper Bearing          |
| 7. Piston                 | 15. O-Ring and Backup Ring |
| 8. Mounting Holes         | 16. End Cap                |

2. Attach a hoist to end cap (16) and lift the end cap out of housing (6) until piston stop (12) contacts upper bearing retainer (5). Remove capscrews (3) and hardened washers (4). Lift the end cap and upper bearing (14) from the housing together.
3. Remove roll pin (10), nut (11), piston stop (12) and key (13). Separate the end cap and upper bearing.
4. Remove O-rings and backup rings (15). Remove upper bearing (14).
5. Rotate the suspension 180°.

*NOTE: Steel balls (9) will fall free when the housing is rotated.*

6. Attach a lifting device to piston (7) and carefully lift it out of the housing.
7. Remove the lower bearing and seals. Refer to Lower Bearing & Seals in this chapter.
8. If piston (7) has removable plug (1, Figure 2-13) at the bottom:
  - a. Remove capscrew (3), flat washer (4) and retaining plate (5). Push plug (1) to the inside just far enough to remove retaining ring (6).
  - b. Re-install cap screw (3) and use a slide hammer to remove the plug from the piston. Remove backup ring (7) and O-ring seal (8) from the plug.



**DO NOT use compressed air to remove the piston plug. Damage to components may occur.**

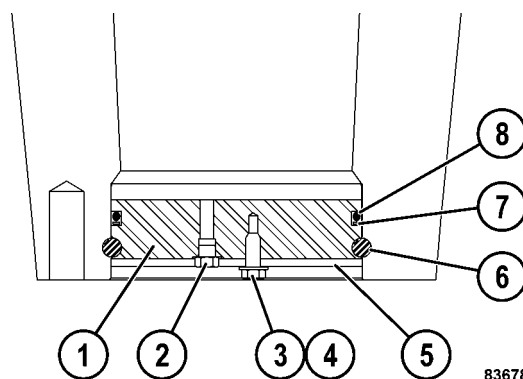


FIGURE 2-13. PISTON PLUG

- |                |                    |
|----------------|--------------------|
| 1. Piston Plug | 5. Retaining Plate |
| 2. Drain Plug  | 6. Retaining Ring  |
| 3. Capscrew    | 7. Backup Ring     |
| 4. Flat Washer | 8. O-Ring Seal     |

## Assembly

*NOTE: All parts must be completely dry and free of foreign material. Lubricate all interior parts with clean Hydrair®II suspension oil. Refer to the Oil and Nitrogen Specifications Chart in Section H4, Suspension Oiling and Charging Procedures.*

# ⚠ IMPORTANT ⚠

***Be careful not to damage the machined or plated surfaces, O-rings or seals when installing the piston assembly.***

1. If piston (7, Figure 2-12) has removable plug (1, Figure 2-13) at the bottom:
  - a. Thoroughly lubricate O-ring (8) and backup ring (7). Install new O-ring seal (8) and backup ring (7) on the plug. Use a rubber mallet to seat the piston plug.
  - b. Install plug (1) into piston just far enough to expose retaining ring groove. Install retaining ring (6). Install retaining plate (5), flat washer (4) and cap screw (3). Tighten the cap screw to **122 N·m (90 ft lb)**.
2. Install the lower bearing and seals. Refer to Lower Bearing & Seals in this chapter.
3. Install new O-rings and backup rings (15, Figure 2-12) in the end cap grooves. The backup rings must be positioned toward the flange on end cap (16).
4. Install new upper bearing (14) on upper bearing retainer (5).
5. Slide the upper bearing retainer assembly over the rod portion of end cap (16).
6. Install key (13) and piston stop (12) on the end cap rod. Make sure that the piston stop is fully seated against the rod shoulder.
7. Install nut (11) against the piston stop. Tighten the nut 1/2 turn further until the hole for roll pin (10) is in alignment. Install roll pin (10).
8. Attach a lifting device to the top side of the end cap assembly. Lower the end cap assembly onto piston (7). Insert steel balls (9) in the holes in the piston before fully seating upper bearing (14) on top of the piston. A small amount of petroleum jelly will prevent the balls from dropping out during assembly.
9. Install upper bearing retainer (5) onto the piston. Secure the upper bearing in place with new capscrews (3) and hardened washers (4). Tighten the capscrews to **678 N·m (500 ft lb)**.

*NOTE: Always use new capscrews (3) during assembly. Used capscrews will be stressed and fatigued because of loads placed on these capscrews during operation.*

10. Apply a light coating of petroleum jelly to the seals and bearings. With the suspension housing in a vertical position, carefully lower the piston and end cap assembly into the bore of housing (6) to its fully retracted position.
11. Install capscrews (1) and hardened washers and (2). Tighten the capscrews to **678 N·m (500 ft lb)**.

*NOTE: If the suspension is to be stored, put in one liter (two pints) of a rust preventive oil. This oil must be drained when the suspension is put back into service.*

12. Install charging valve (2, Figure 2-1) with a new O-ring. Lubricate the O-ring with clean Hydrair® suspension oil before threading the charging valve into the block on the end cap. Tighten charging valve (large hex) (4) to **23 N·m (17 ft lb)**.
13. Pressure test the suspension. Refer to "SUSPENSION PRESSURE TEST". Do not pressure test the suspension with oil inside.
14. After a successful pressure test, add approximately 97.6 liters (25.8 gallons) of suspension oil.
15. Apply rust preventative grease to any exposed machined surfaces.

## SUSPENSION PRESSURE TEST

The suspension assembly must be tested for leakage after rebuild procedures are completed. If leakage occurs, the cause of the leakage must be identified and repaired before the suspension is installed on the truck.



***The entire suspension assembly must be placed in a containment device that will keep the suspension piston in the retracted position and prevent it from extending during pressurization. Make sure that the containment device is capable of withstanding the applied force.***

1. Collapse the suspension until the piston is fully retracted in the housing.
2. Make sure that the charging valve and all plugs are installed. Attach the pressurization line to the charging valve.
3. Place the suspension assembly in a containment device and submerge the entire assembly in the water tank.
4. Pressurize the suspension with air or nitrogen to  $7\,585 \pm 1\,380$  kPa (1,100  $\pm$  200 psi).
5. Maintain pressure for a minimum of 20 minutes and check for bubbles at the following locations:
  - Housing bearing/housing joint
  - Piston/piston seal area
  - Charging valve and plugs
6. After the test is complete, remove the assembly from the water tank and release the air or nitrogen pressure. DO NOT remove the charging valve from suspension.
7. Remove the suspension from the containment device.
8. Coat any exposed, unpainted areas with rust preventive grease.
9. Store the suspension in a collapsed position to protect the piston chrome surface until it is installed on a truck.



# NOTES

# SECTION H3

## REAR SUSPENSIONS

### INDEX

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| Disassembly .....              | H3-6 |
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| Assembly .....                 | H3-7 |
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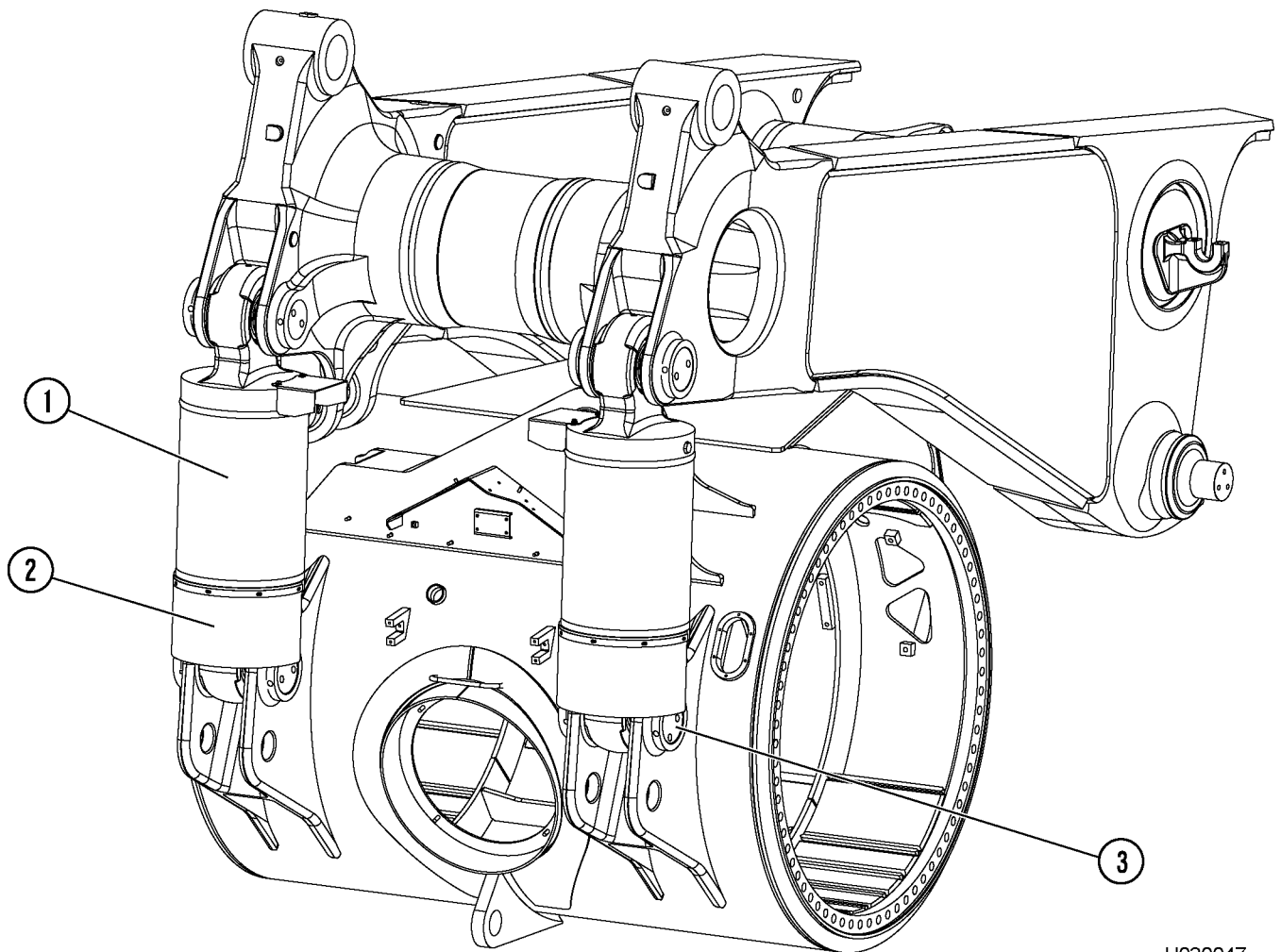
# NOTES

## REAR SUSPENSIONS

The Hydrair® II suspensions are hydro-pneumatic components containing oil and nitrogen gas. The oil and gas in the four suspensions carry the gross truck weight less wheels, spindles and final drive assembly.

The rear suspension cylinders consist of two basic components: a suspension housing attached to the frame and a suspension rod attached to the rear axle housing.

The Hydrair® II suspension cylinder requires only normal care when handling as a unit. However, after being disassembled, these parts must be handled with extreme care to prevent damage to the machined surfaces. Surfaces are machined to extremely close tolerances and are precisely fitted. All parts must be completely clean during assembly.



H030047

FIGURE 3-1. REAR SUSPENSION INSTALLATION

- 1. Suspension Cylinder
- 2. Mud Flap
- 3. Mounting Pin

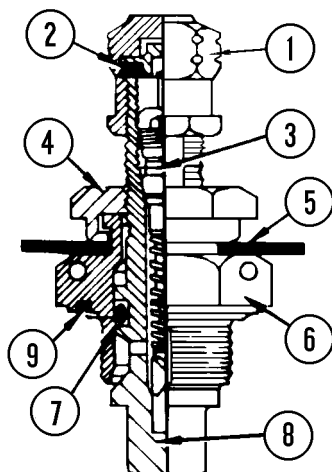
## SUSPENSION CYLINDER

### Removal

TABLE 1. TOOL LIST FOR SUSPENSION PIN REMOVAL

| Part Number | Description             | Quantity |
|-------------|-------------------------|----------|
| EJ2847      | Pin Removal Tool        | 2        |
| EJ2848      | Cylinder                | 1        |
| EJ2849      | Hand Pump               | 1        |
| EJ2850      | Shackle                 | 2        |
| MM0093      | Capscrew - M16 x 2 x 70 | 4        |

1. Remove capscrews, washers, and mud flap (2, Figure 3-1) from the suspension.



90010B

FIGURE 3-2. CHARGING VALVE

- |                  |               |
|------------------|---------------|
| 1. Valve Cap     | 6. Valve Body |
| 2. Seal          | 7. O-Ring     |
| 3. Valve Core    | 8. Valve Stem |
| 4. Swivel Nut    | 9. O-Ring     |
| 5. Rubber Washer |               |

2. Remove charging valve cap (1, Figure 3-2) loosen swivel nut (4) (small hex) on the charging valve, and turn it counterclockwise three full turns to unseat the valve seal. Connect the charging kit.

### **WARNING**

***Make sure that only swivel nut (4) turns. Turning the complete charging valve assembly may result in the valve assembly being forced out of the suspension by the gas pressure inside.***

3. If necessary, charge the suspension to be removed with dry nitrogen until the rod is exposed approximately 127 mm (5 in.).
4. Place stands or cribbing under the truck frame at each hoist cylinder mount.
5. Open the valve on the suspension charging kit to release nitrogen from the suspension. Disconnect the charging kit.
6. Disconnect the lubrication lines and the pressure sensor cable.
7. Position a fork lift under the suspension housing above the lower mounting pin. Secure the suspension to the fork lift.

### **WARNING**

***The rear Hydrair II suspension weighs approximately 883 kg (1947 lbs). Make sure that the capacity of the lifting device used is sufficient for lifting this load.***

**NOTE:** The mounting arrangement for the top and bottom pins is identical.

8. Remove locknut (4, Figure 3-4), flat washer (3) and shoulder bolt (2) from the upper and lower pins on the suspension cylinder to be removed.

**NOTE:** The pins may be retained with a capscrew and locknut as originally shipped from the factory. Upgrading to the current shoulder bolt configuration is recommended.

9. Install pin removal tool (1, Figure 3-3) to each lower pin using the capscrews listed in Table 1. Tighten the capscrews to  $240 \pm 24 \text{ N}\cdot\text{m}$  ( $177 \pm 17 \text{ ft lb}$ ).

10. Attach both shackles (2) to cylinder (3).

11. Attach each shackle to the pin removal tool, as shown.

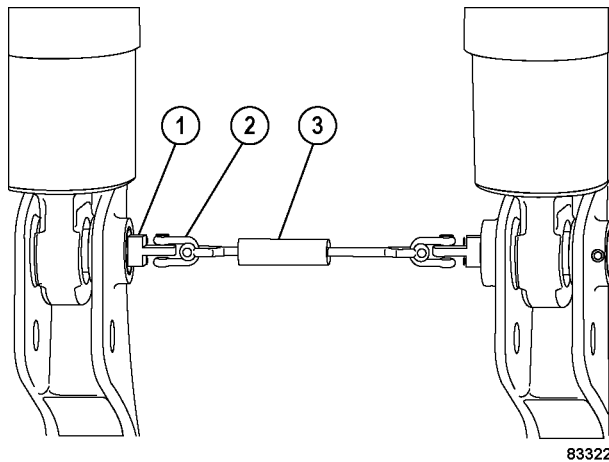


FIGURE 3-3. REAR SUSPENSION  
PIN REMOVAL TOOL

- |                     |             |
|---------------------|-------------|
| 1. Pin Removal Tool | 3. Cylinder |
| 2. Shackle          |             |

## ⚠ CAUTION

***Do not exceed 10 tons of force when applying pressure to the cylinder. Damage to the tool or suspension components may result, as well as personal injury to maintenance personnel.***

12. Apply pressure to the cylinder using the hand pump (not shown).

13. When the cylinder reaches the end of its stroke, remove one of the shackles from the cylinder and connect the cylinder shackle directly to the pin removal tool. This is necessary to pull the pin the remaining distance.

14. Remove pin (5, Figure 3-4) from the lower mounting.

**NOTE:** Spacers (7) will fall free.

15. Install the tool on the upper pin and repeat the pin removal process.

16. If it is necessary to remove the remaining rear suspension cylinder, insert the pins back into the upper and lower mountings. Secure the pins using shoulder bolts (2), flat washers (3) and locknuts (4), and repeat the removal process for the remaining cylinder.

17. Remove the cylinder from the truck.

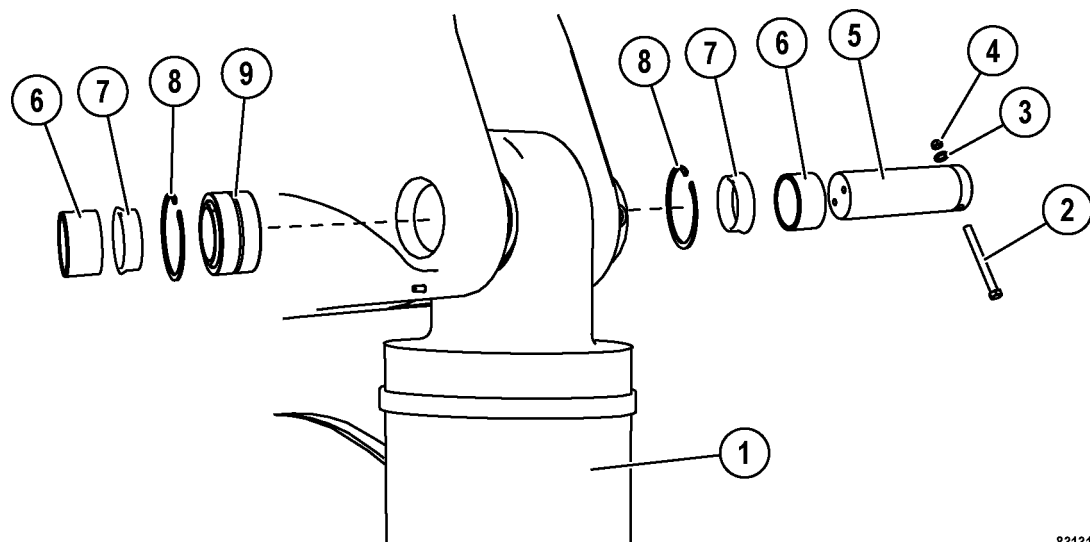


FIGURE 3-4. REAR SUSPENSION CYLINDER PIN INSTALLATION

- |                        |                   |                |
|------------------------|-------------------|----------------|
| 1. Suspension Cylinder | 5. Pin            | 9. Bearing     |
| 2. Shoulder Bolt       | 6. Sleeve         | 10. Capscrew   |
| 3. Flat Washer         | 7. Spacer         | 11. Lockwasher |
| 4. Locknut             | 8. Retaining Ring |                |

## Installation

1. Inspect mounting bore sleeves (6, Figure 3-4) and the bearing spacers for damage and wear. Check the fit of the pins in the bores before installing the suspension. Replace worn or damaged parts.
2. Secure the suspension to the fork lift and raise it into position. The suspension assembly must be retracted as far as possible before installation.
3. Position the top suspension eye and spherical bearing between the ears on the frame, as shown in Figure 3-4. Orient the cylinder so that the charging valve faces the opposite suspension cylinder, as shown in Figure 3-1.
4. Lubricate all pin-to-bearing and pin-to-sleeve contact surfaces with anti-seize compound. Lubricating the pin surfaces aids in removal and installation, as well as prevention of rust and corrosion.
5. Align the hole in pin (5, Figure 3-4) with the hole in the mounting bore. Drive in the pin far enough to hold the pin in position.
6. Insert bearing spacer (7) and continue to drive in the pin through the spherical bearing. Insert the remaining spacer and continue to drive in the pin until the hole in the axle housing is aligned with the hole in the pin.
7. Install shoulder bolt (2), flat washer (3) and locknut (4). If further alignment of the holes is necessary, install a pin removal tool onto the pin. Use the tool and a large pipe wrench or other suitable device to align the holes. Tighten the locknut to **68 N·m (50 ft lb)**.

*NOTE: If the original capscrew/locknut configuration is being used, tighten the locknut to **465 N·m (343 ft lb)**. Upgrading to the current shoulder bolt configuration is recommended.*

8. Adjust the piston rod height until the lower mount bearing aligns with the bore in the rear axle housing, and repeat the above procedure to install the bottom pin. Mounting components in the top and bottom joints are identical.
9. Install the nitrogen charging kit, and add nitrogen to raise the frame off the stands or cribbing. A lifting device may also be used.

10. Connect the lubrication lines and the pressure sensor.
11. Service the suspension. For instructions, refer to Section H, Oiling and Charging Procedures.
12. Install mud flap (2, Figure 3-1) with the capscrews, flat washers and lockwashers.

## Disassembly

*NOTE: The suspension should be placed in a fixture which will allow it to be rotated 180° vertically. Place the suspension in the fixture with the rod end down.*



**To avoid possible injury, wear a face mask or goggles while relieving nitrogen gas pressure.**

1. Depress the charging valve stem to ensure that all nitrogen gas pressure has been released before removing the charging valve.
2. Remove charging valve cover (16, Figure 3-5). Remove charging valve (17). Remove and discard the charging valve O-ring.
3. Remove pressure sensor (15).
4. With the suspension in a vertical position (piston rod down), remove drain plug (18) and drain the suspension oil. A properly charged rear suspension cylinder contains 54.5 L (14.4 gal) of oil.
5. Rotate the cylinder 180°. Remove capscrews (6) and washers (7). Lift piston rod (9) from housing (1).
6. Remove bearing retainer (5) from the piston. Remove bearing (4). Remove the seals, the O-ring and backup ring from bearing retainer (5). Discard the O-ring and backup ring.
7. Remove bearing (3) from the piston.
8. If spherical bearings (9, Figure 3-4) require replacement, remove retaining rings (8) and press the bearing from the cylinder bore.

## Cleaning and Inspection

1. Clean all parts thoroughly in fresh cleaning solvent. Use a solvent that does not leave a film after evaporation, such as trichlorethylene, acetone or laquer thinner.



***When using cleaning agents follow the solvent manufacturer's instructions.***

2. Dry all parts completely using only dry, filtered compressed air and lint free wiping materials.
3. Inspect all parts for evidence of wear or damage. Inspect plated surfaces for scratches, nicks or other defects. Replace or repair any damaged parts.

*NOTE: If other repairs are necessary, refer to your local Komatsu distributor for repair information and instructions not covered in this manual.*

## Assembly

Assembly must be performed in a clean, dust free work area. All parts must be completely clean, dry and free of rust or scale. Lubricate all interior parts and bores with fresh suspension oil. Refer to the Oil and Nitrogen Specifications Chart in Section H, Oiling and Charging Procedures.

1. If removed, install spherical bearing (9, Figure 3-4) in the eye of the piston rod and cylinder housing.
2. Install retaining rings (8) to secure the bearings.
3. Install wiper seal (12, Figure 3-5), double lip seal (11), buffer seal (10), O-ring (13) and backup ring (14) onto bearing retainer (5).

*NOTE: Refer to the seal installation illustration and details (Figure 3-5) for proper orientation.*

4. Install rod bearing (4).
5. Slide piston bearing (3) onto the lubricated piston rod.
6. With the lubricated housing (1) held in a vertical position, slide the piston assembly partially into the housing. Slide retainer (5) onto the housing and fasten with capscrews (6) and washers (7). Tighten the capscrews to standard torque. Use care during piston installation to prevent damage to machined and chrome surfaces.

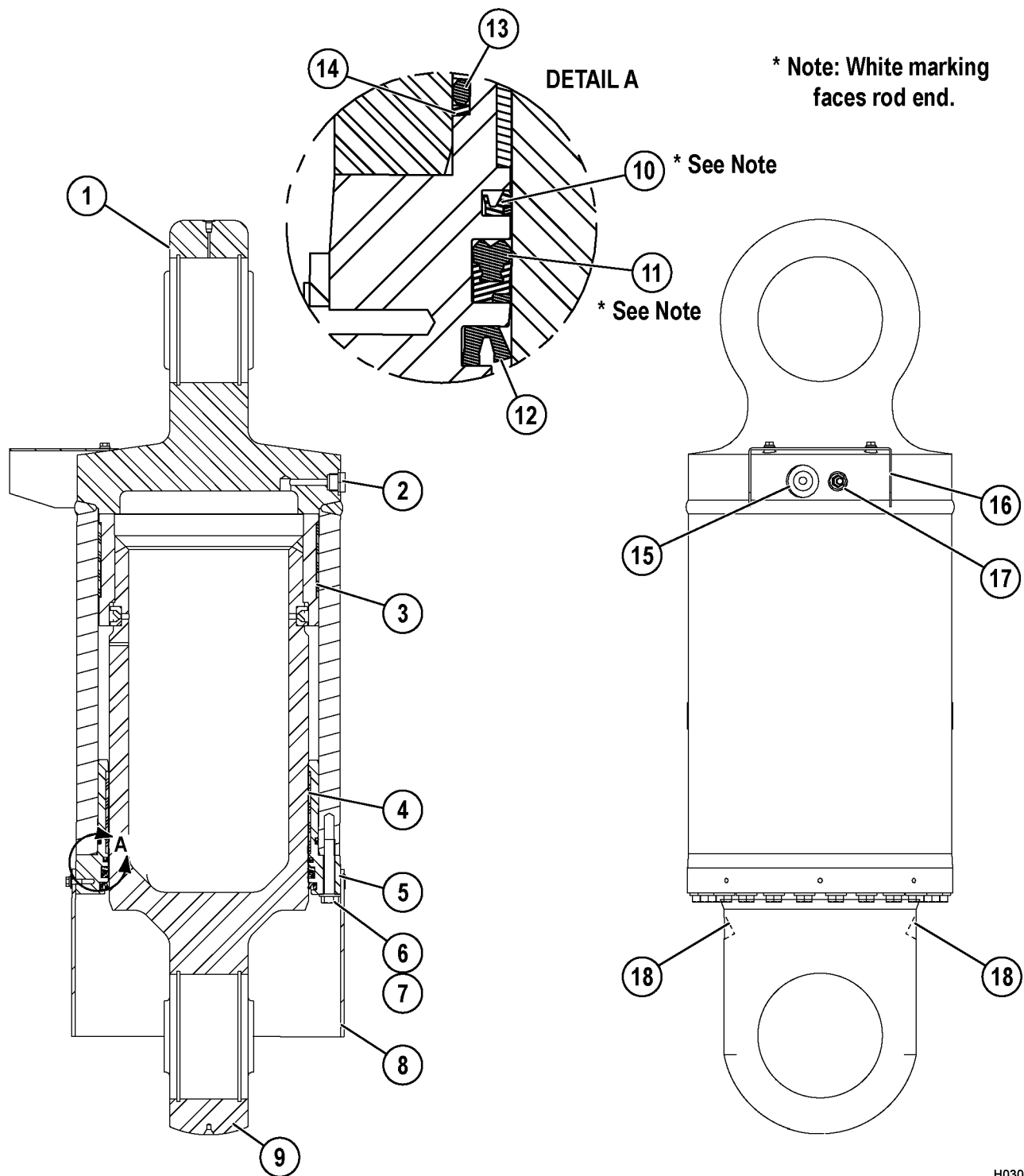
*NOTE: If the suspension is to be stored, fill with one liter (two pints) of a rust preventive oil. This oil must be drained when the suspension is put back into service.*

7. Install a new O-ring onto the charging valve, and install the charging valve onto the cylinder. Tighten the large hex of the charging valve to **23 N·m (17 ft lb)**.

If a new charging valve is being used, tighten the swivel nut to **15 N·m (11 ft lb)**, then loosen and retighten the swivel nut to **15 N·m (11 ft lb)**. Loosen the swivel nut again, and retighten it to **6 N·m (50 in lb)**. Install the valve cap finger-tight.

8. Install the charging valve, pressure sensor, and plugs.
9. Install cover (16) onto the suspension.
10. Pressure test the suspension. Refer to Suspension Pressure Test later in this section.





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FIGURE 3-5. REAR SUSPENSION ASSEMBLY

- |                     |                         |                     |
|---------------------|-------------------------|---------------------|
| 1. Housing          | 7. Hardened Washer      | 13. O-ring          |
| 2. Vent Plug        | 8. Mud Flap             | 14. Backup Ring     |
| 3. Piston Bearing   | 9. Piston Rod           | 15. Pressure Sensor |
| 4. Rod Bearing      | 10. Rod Buffer Seal     | 16. Cover           |
| 5. Bearing Retainer | 11. Rod Double Lip Seal | 17. Charging Valve  |
| 6. Capscrew         | 12. Wiper Seal          | 18. Drain Plug      |

## SUSPENSION PRESSURE TEST

The suspension assembly should be tested for leakage after rebuild procedures are completed. If leakage occurs, the cause of the leakage must be identified, and repaired before the suspension is installed on the truck.



***The entire suspension assembly must be placed in a containment device that will keep the suspension piston in the retracted position and prevent it from extending during pressurization. Make sure that the containment device is capable of withstanding the applied force.***

1. Collapse the suspension until the piston is fully retracted in the housing.
2. Make sure that the charging valve and all plugs are installed. Attach the pressurization line to the charging valve.
3. Place the suspension assembly in a containment device and submerge the entire assembly in the water tank.

4. Pressurize the suspension with air or nitrogen to  $7585 \pm 1380$  kPa ( $1100 \pm 200$  psi).
5. Maintain pressure for a minimum of 20 minutes and check for bubbles at the following locations:
  - Housing bearing/housing joint
  - Piston/piston seal area
  - Charging valve and plugs
6. After the test is complete, remove the assembly from the water tank and release the air or nitrogen pressure. DO NOT remove the charging valve from suspension.
7. Remove the suspension from the containment device.
8. Coat any exposed, unpainted areas with rust preventive grease.
9. Store the suspension in a collapsed position to protect the piston chrome surface until it is installed on a truck.

# NOTES

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**SUSPENSION OILING AND CHARGING PROCEDURES**  
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# NOTES

# SUSPENSION OILING AND CHARGING PROCEDURE

## GENERAL

These procedures cover the oiling and charging of Hydrair® II suspensions on Komatsu electric drive dump trucks.

Suspensions which have been properly charged will provide improved handling and ride characteristics while also extending the fatigue life of the truck frame and improving tire wear.

*NOTE: Inflation pressures and exposed piston lengths are calculated for a normal truck gross vehicle weight (GVW). Additions to truck weight by adding body liners, tailgates, water tanks, etc, should be considered part of the payload. Keeping the truck GVW within the specification shown on the grade/speed chart in the operator cab will extend the service life of the truck main frame and allow the suspensions to produce a comfortable ride.*



***All Hydrair® II suspensions are charged with compressed nitrogen gas with sufficient pressure to cause injury and/or damage if improperly handled. Follow all safety instructions, cautions, and warnings provided in the following procedures to prevent any accidents during oiling and charging.***

Proper charging of Hydrair® II suspensions requires that three basic conditions be established in the following order:

1. The oil level must be correct.
2. The suspension piston rod extension for nitrogen charging must be correct.
3. The nitrogen charge pressure must be correct.

For best results, Hydrair® II suspensions should be charged in pairs (fronts together and rears together). If rear suspensions are to be charged, the front suspensions should be charged first.

*NOTE: For longer life of suspension components, a friction modifier should be added to the suspension oil. See the Specifications Charts at the end of this section.*

*NOTE: Setup dimensions in the Specifications Charts must be maintained during oiling and charging procedures. However, after the truck has been operated, these dimensions may vary.*

## EQUIPMENT LIST

- Hydrair® Charging Kit
- Jacks and/or Overhead Crane
- Support Blocks for:
  - Oiling Height Dimensions (Front and Rear)
  - Charging Height Dimensions (Front Only)
- Hydrair® Oil (see Specifications Chart)
- Friction Modifier (see Specifications Chart)
- Dry Nitrogen (see Specifications Chart)

If both the front and rear suspensions are to be serviced at the same time, service the front suspensions first. Do not remove the front suspension nitrogen charging blocks until after the rear suspensions have been completely serviced

## HYDRAIR® CHARGING KIT

Assemble the charging kit as shown in Figure 4-1, and attach it to a container of pure dry nitrogen (8).

### Installation of Charging Kit

1. Remove the protective covers and charging valve caps from the suspensions.
2. Turn "T" handles (1) of adapters (2) completely counterclockwise.
3. Make sure that outlet valves (3) and inlet valve (4) are closed (turned completely clockwise).
4. Turn the swivel nut (small hex) on the charging valve three full turns counterclockwise to unseat the valve.

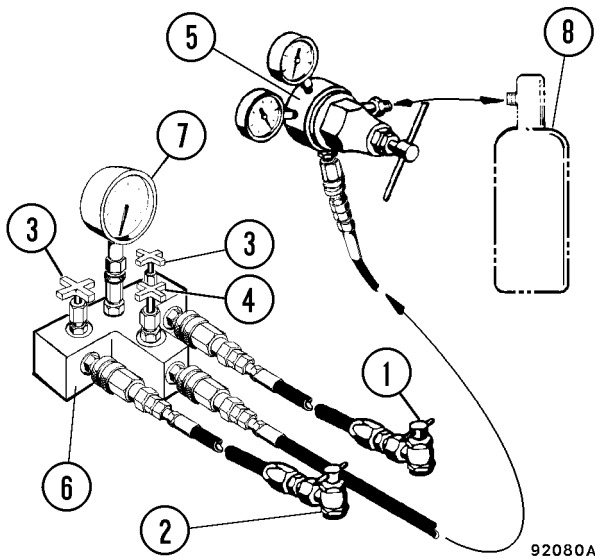


FIGURE 4-1. HYDRAIR CHARGING KIT

**NOTE:** NOTE: The arrangement of parts may vary from the illustration depending on the kit part

1. "T" Handle Valve
2. Charging Valve Adapter
3. Manifold Outlet Valves (from gauge)
4. Inlet Valve (from regulator)
5. Regulator Valve (Nitrogen Pressure)
6. Manifold
7. Charging Pressure Gauge (Suspensions)
8. Dry Nitrogen Gas Container

5. Attach charging valve adapters (2) to each suspension charging valve stem.
6. Turn "T" handles (1) clockwise. This will depress the core of the charging valve and open the gas chamber of the suspension.
7. Open both outlet valves (3).

**NOTE:** By selective opening and closing of outlet valves (3), and inlet valve (4), suspensions may be charged separately or together.

### Removal of Charging Kit

1. Close both outlet valves (3).
2. Turn "T" handles (1) counterclockwise to release the charging valve cores.
3. Remove charging valve adapters (2) from the charging valves.
4. Tighten the swivel nut (small hex) on the charging valve. If a new charging valve is being used, tighten the swivel nut to **15 N·m (11 ft lb)**, then loosen and retighten the swivel nut to **15 N·m (11 ft lb)**. Loosen the swivel nut again and retighten to **6 N·m (50 in lb)**. Install the valve cap finger-tight.
5. Install the charging valve caps and protective covers on both suspensions.

## SUPPORT BLOCKS FOR OILING AND CHARGING DIMENSIONS

Before starting the oiling and charging procedures, supports should be fabricated to maintain the correct exposed piston rod extensions.

### **IMPORTANT**

***Nitrogen charging support blocks for the rear suspension are no longer necessary. However, oiling blocks are necessary to properly set the oiling height.***

Exposed piston rod extensions are specified for both oil level and nitrogen charging for Hydrair II suspensions. These dimensions are listed in the tables below Figures 4-2 and 4-5. Measure the dimensions from the face of the cylinder gland to the machined surface on the spindle at the front suspension. At the rear suspension, measure from the face of the cylinder gland to the piston flange.

Support blocks may be made in various forms. Mild steel materials are recommended. Square stock or pipe segments at least 25 mm (1 in.) may be used. The blocks must be capable of supporting the weight of the truck during oiling and charging procedures while avoiding contact with plated surfaces and seals on the suspension. Refer to Figure 4-2 for front suspension support block placement and Figure 4-4 for rear support block placement.

## FRONT SUSPENSION

### **WARNING**

***All Hydrair® II suspensions are charged with compressed nitrogen gas with sufficient pressure to cause injury and/or damage if improperly handled. Follow all the safety notes, cautions and warnings in these procedures to prevent accidents during servicing and charging.***

1. Park the unloaded truck on a hard, level surface. Place the directional control lever in PARK.
2. Place wheel chocks in front of and behind both sets of rear tires to prevent roll away.

***NOTE: Do not place wheel chocks around the front tires. The front tires will roll forward and backward a small amount as the suspension travels up and down during the suspension charging process.***

3. Thoroughly clean the area around the charging valve on the suspensions. Remove the protective covers from the charging valves.

### Front Suspension Oiling

### **WARNING**

***Make sure that all personnel are clear and support blocks are secured with a strap or other means before relieving nitrogen pressure from the suspension. An unsecured block could fly loose as weight is applied, presenting the possibility of serious injury to nearby personnel and/or damage to the equipment.***

***Overhead clearance may be reduced rapidly and suddenly when nitrogen pressure is released.***

***NOTE: For longer life of suspension components, a friction modifier should be added to the suspension oil. See the Specifications Charts at the end of this section.***



1. Position and secure oiling height dimension blocks 180° apart to provide stability (see Figure 4-2). When nitrogen pressure is released, the suspensions will lower and rest on the blocks. Ensure that the blocks do not mar or scratch the plated surfaces of the pistons or damage wiper seals in the lower bearing retainer. Support blocks must seat on the spindle and the cylinder housing.

## ⚠ WARNING

***Wear a face mask or goggles while relieving nitrogen pressure.***

2. Remove the charging valve cap. Turn the swivel nut (small hex) counterclockwise three full turns to unseat the valve seal. DO NOT turn the large hex. The charging valve body has a bleeder groove in its mounting threads, but for safety of all personnel, the valve body must not be loosened until all nitrogen pressure has been released from the suspension.
3. Depress the charging valve core to release nitrogen pressure from the suspension. When all nitrogen pressure has been released, the suspension should have collapsed slowly and be seated solidly on the support blocks.
4. Remove the top fill plug next to the charging valve (see Figure 4-2).
5. Fill the suspension with clean Hydrair® oil (with 6% friction modifier) until the cylinder is full to the top of the fill plug bore. Use drip pans and clean all spillage from the outside of the suspension.

Allow the suspension to settle for at least 15 minutes to clear any trapped nitrogen and bubbles from the oil. Add more suspension oil if necessary. Install a new O-ring on the fill plug and install the plug.

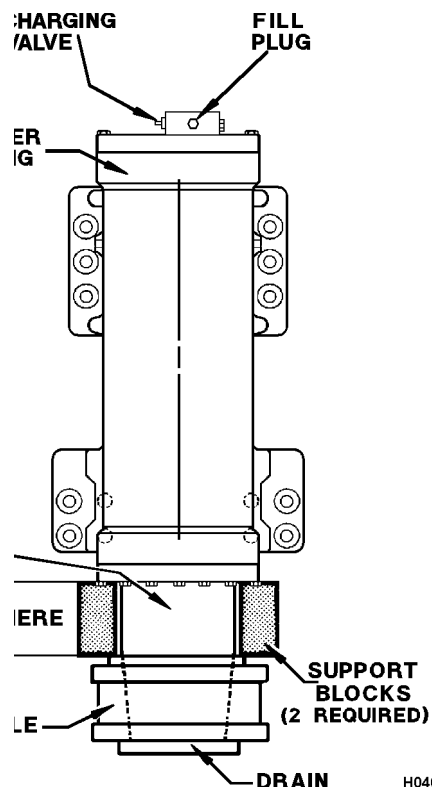


FIGURE 4-2. FRONT SUSPENSION

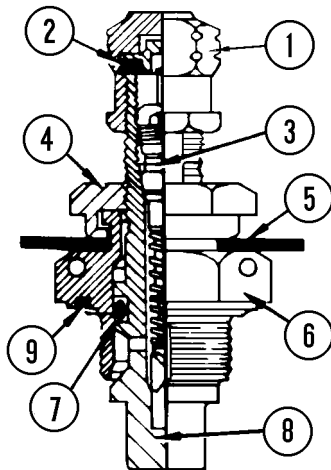
| FRONT SUSPENSION DIMENSIONS (EMPTY)                                                                                                                                                                                                                          |                       |                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|------------------------|
| OILING HEIGHT                                                                                                                                                                                                                                                | CHARGING HEIGHT       | CHARGING PRESSURE      |
| 25.4 mm<br>(1.0 in.)                                                                                                                                                                                                                                         | **229 mm<br>(9.0 in.) | *2930 kPa<br>(425 psi) |
| <p>* with standard Rock Body</p> <p>** For trucks utilizing extreme duty front suspensions, the charging height is 195 mm (7.5 in.).</p> <p><b>Note: If the truck starts to lift off the blocks before charging pressure is attained, STOP CHARGING.</b></p> |                       |                        |

## Front Suspension Nitrogen Charging

### **⚠ WARNING**

*Lifting equipment (crane or hydraulic jacks) must be of sufficient capacity to lift the truck weight. Make sure that all personnel are clear of the lift area before the lift is started. Clearances under the truck may be suddenly reduced.*

1. If removed, install the charging valve with new O-ring (9, Figure 4-3). Lubricate the O-ring with clean Hydrair® oil.
2. Tighten valve body (6) (large hex) to **23 N·m (17 ft lb)**. Swivel nut (4) (small hex) must be unseated by turning it counterclockwise three full turns.



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FIGURE 4-3. CHARGING VALVE

- |                  |               |
|------------------|---------------|
| 1. Valve Cap     | 6. Valve Body |
| 2. Seal          | 7. O-Ring     |
| 3. Valve Core    | 8. Valve Stem |
| 4. Swivel Nut    | 9. O-Ring     |
| 5. Rubber Washer |               |

### **⚠ WARNING**

*Dry nitrogen is the only gas approved for use in Hydrair® II suspensions. Charging of these components with oxygen or other gases may result in an explosion which could cause fatalities, serious injuries and/or major property damage. Use only nitrogen gas meeting the specifications shown in the Specifications Chart at the end of this section.*

3. Install the charging kit and a bottle of pure dry nitrogen. Refer to "Installation of Charging Kit" earlier in this section.
4. Charge the suspensions with nitrogen gas to a height just high enough to install the nitrogen charging blocks, but not to exceed 279 mm (11 in.). Close inlet valve (4, Figure 4-1).
5. Remove the oiling blocks from the suspensions and install the nitrogen charging blocks. Secure the blocks to prevent accidental dislodging.

*NOTE: Use caution to prevent damage to plated cylinder surfaces and oil seals when installing the blocks.*

6. Remove the center hose from manifold (6).

### **⚠ WARNING**

*Be aware when relieving nitrogen gas that the truck may drop suddenly onto the charging blocks.*

7. To lower the truck onto the charging blocks, open inlet valve (4) until the pressure drops below the pressure listed in the table below Figure 4-2, then close the valve.
8. Install the center hose to manifold (6).

9. Charge the suspensions to the pressure listed in Figure 4-2. DO NOT use an overcharge of nitrogen to lift the suspensions off the charging blocks.
10. Close inlet valve (4, Figure 4-1). Leave outlet valves (3) open for five minutes to allow the pressures in the suspensions to equalize.
11. Close outlet valves (3). Remove the charging kit components. Refer to "Removal of Charging Kit" earlier in this section.
12. If the charging valve is being reused, tighten swivel nut (4, Figure 4-3) to **6 N·m (50 in lb)**.
13. If a new charging valve is being used, tighten the swivel nut to **15 N·m (11 ft lb)**, then loosen and retighten the swivel nut to **15 N·m (11 ft lb)**. Loosen the swivel nut again and retighten to **6 N·m (50 in lb)**. Install the valve cap finger-tight.
14. Install the protective guard over the charging valve.

*NOTE: If the rear suspensions are to be serviced at this time, leave the nitrogen charging blocks in place on the front suspensions until after the rear suspensions are completely serviced.*

15. Raise the truck body in order to extend the front suspensions, allowing for removal of the nitrogen charging blocks. Make sure that sufficient overhead clearance exists before raising the body. If the suspensions do not extend after raising the body, turn the steering wheel left and right in very small movements several times. If the suspensions still do not extend enough to allow for removal of the blocks, use a crane or floor jacks to raise the truck and remove the blocks.

Any time the suspensions are recharged, the calibration of the Payload Meter system is affected. Refer to the Payload Meter III section and perform a "Clean Truck Tare". This will ensure accurate payload records.

The front Hydrair® II suspensions are now ready for operation. Visually check piston extension with the truck both empty and loaded. Record the extension dimensions. Maximum downward travel is indicated by the dirt ring at the base of the piston. Operator comments on steering response and suspension rebound should also be noted.

*NOTE: The empty ride height may be less than the charging ride height. Also, the suspension height between the left and right suspensions may not always be the same due to the fuel level in the fuel tank.*

## REAR SUSPENSION

### **WARNING**

***All Hydrair® II suspensions are charged with compressed nitrogen gas with sufficient pressure to cause injury and/or damage if improperly handled. Follow all the safety notes, cautions and warnings in these procedures to prevent accidents during servicing and charging.***

1. Park the unloaded truck on a hard, level surface. Place the directional control lever in PARK.
2. Place wheel chocks in front of and behind both sets of rear tires to prevent roll away.

*NOTE: Do not place wheel chocks around the front tires. The front tires will roll forward and backward a small amount as the suspension travels up and down during the suspension charging process.*

3. Thoroughly clean the area around the charging valve on the suspensions. Remove the protective covers from the charging valves and the rubber covers from the suspension piston.

## Rear Suspension Oiling

### **⚠ WARNING**

***Make sure that all personnel are clear and support blocks are secured with a strap or other means before relieving nitrogen pressure from the suspension. An unsecured block could fly loose as weight is applied, presenting the possibility of serious injury to nearby personnel and/or damage to the equipment.***

***Overhead clearance may be reduced rapidly and suddenly when nitrogen pressure is released.***

*NOTE: For longer life of suspension components, a friction modifier should be added to the suspension oil. See the Specifications Charts at the end of this section.*

1. Position and secure the oiling blocks in place so the blocks are seated between the frame and the rear axle housing (see Figure 4-4). A block should be used on both the left and right sides of the truck.

### **⚠ WARNING**

***Wear a face mask or goggles while relieving nitrogen pressure.***

2. Remove charging valve cap. Turn the swivel nut (small hex) counterclockwise three full turns to unseat the valve seal. DO NOT turn the large hex. The charging valve body has a bleeder groove in its mounting threads, but for safety of all personnel, the valve body must not be loosened until all nitrogen pressure has been released from the suspension.

3. Depress the charging valve core to release nitrogen pressure from the suspension. When all nitrogen pressure has been released, loosen and remove the fill plug. The suspension should have collapsed slowly as gas pressure was released. The weight of the truck is now supported by the support blocks.

*NOTE: A plastic tube can be used to help bleed off trapped air inside the piston.*

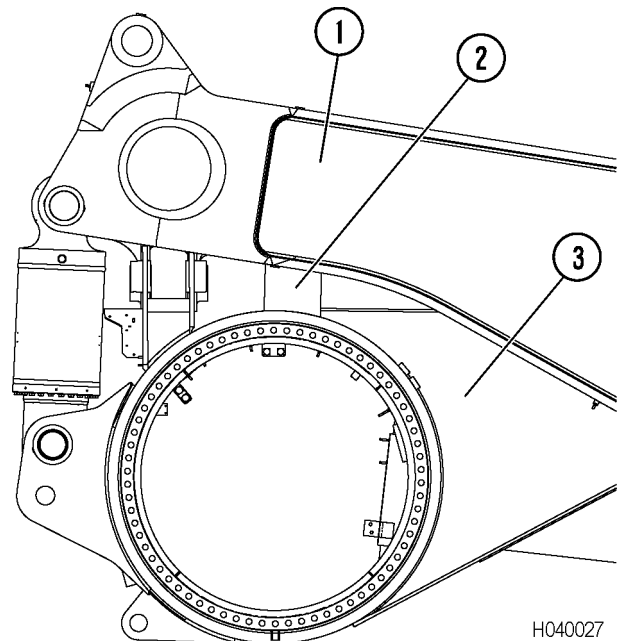


FIGURE 4-4. OILING BLOCK INSTALLATION

- |                         |                      |
|-------------------------|----------------------|
| 1. Main Frame           | 3. Rear Axle Housing |
| 2. Oiling Support Block |                      |

4. Remove the vent plug, pressure sensor, and charging valve (see Figure 4-5). Use one of the open ports to fill the suspension with clean Hydrair® oil (with 6% friction modifier). Fill until clean oil seeps from the open ports. Use drip pans and clean all spillage from the outside of the suspension.

Allow the suspension to settle for at least 15 minutes to clear any trapped nitrogen and bubbles from the oil. Add more suspension oil if necessary.

5. Install the vent plug and pressure sensor onto the suspension.
6. Install a new O-ring onto the charging valve. Lubricate the O-ring with clean Hydrair oil.
7. Install the charging valve onto the suspension. Tighten valve body (6, Figure 4-3) to **23 N·m (17 ft lb)**.

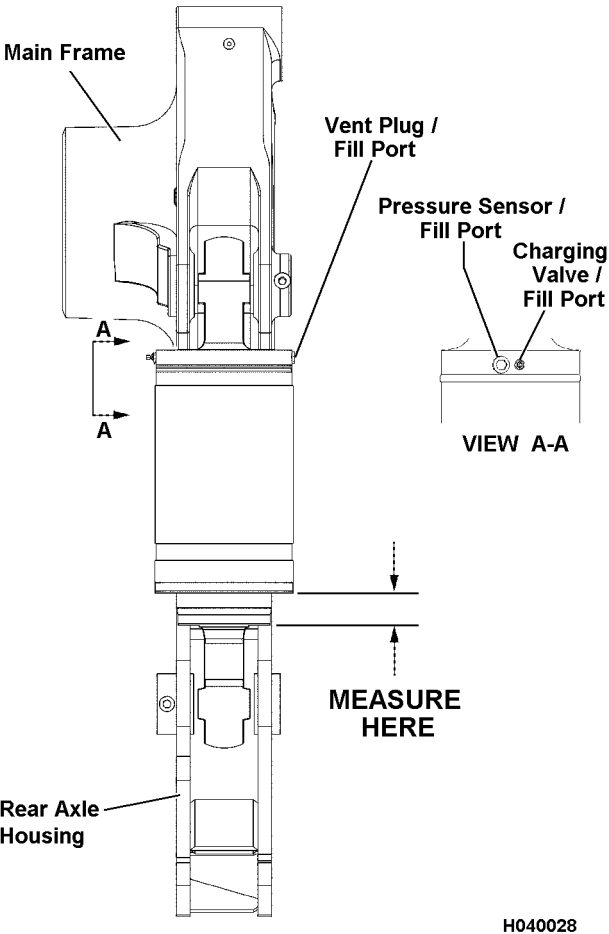


FIGURE 4-5. REAR SUSPENSION

1. Main Frame
2. Charging Valve
3. Rear Axle Housing

| REAR SUSPENSION DIMENSIONS (EMPTY)                                                  |                       |                        |
|-------------------------------------------------------------------------------------|-----------------------|------------------------|
| OILING HEIGHT                                                                       | CHARGING HEIGHT       | CHARGING PRESSURE      |
| 54.2 mm<br>(2.1 in.)                                                                | 219.3 mm<br>(8.6 in.) | *1296 kPa<br>(188 psi) |
| * Charging pressures are for reference only and may vary depending on body weights. |                       |                        |

## Rear Suspension Nitrogen Charging

### **WARNING**

*Lifting equipment must be of sufficient capacity to lift the truck weight. Ensure that all personnel are clear of lift area before lifting is started. Clearances under the truck may be suddenly reduced.*

### **IMPORTANT**

*Ensure that the automatic apply circuit has not applied the service brakes during truck maintenance. If the front brakes are applied during rear suspension charging, the axle cannot pivot for frame raising/lowering, and the rear suspension may be unable to move up or down.*

1. If removed, install charging valve with new O-ring (9, Figure 4-3). Lubricate the O-ring with clean Hydrair® oil.
2. Tighten valve body (6) (large hex) to **23 N·m (17 ft lb)**. Swivel nut (4) (small hex) must be unseated by turning it counterclockwise three full turns.

### **WARNING**

*Dry nitrogen is the only gas approved for use in Hydrair® II suspensions. Charging of these components with oxygen or other gases may result in an explosion which could cause fatalities, serious injuries and/or major property damage. Use only nitrogen gas meeting the specifications shown in the Specifications Chart at the end of this section.*

3. Install the charging kit and a bottle of pure dry nitrogen. Refer to "Installation of Charging Kit" in this section.
4. Charge the suspensions with nitrogen gas to a height just high enough to install the nitrogen blocks, but not to exceed 269.2 mm (10.6 in.).

### **WARNING**

*Be aware when relieving nitrogen gas that the truck may drop suddenly onto the charging blocks.*

5. Slowly release gas until the suspensions match the pressure listed in the table below Figure 4-5.
6. Close inlet valve (4, Figure 4-1). Leave outlet valves (3) open for five minutes to allow the pressures in the suspensions to equalize.
7. Ensure that both of the suspension cylinders are extended the same distance  $\pm 10$  mm (0.39 in.). If the difference in the extension from side to side exceeds this amount, check the front suspensions for equal extension. Adjust the front suspensions as necessary.

*NOTE: A low left front suspension will cause the right rear suspension to be high. A low right front suspension will cause the left rear suspension to be high.*

8. Close outlet valves (3). Remove the charging kit components. Refer to "Removal of Charging Kit" in this section.
9. If the charging valve is being reused, tighten swivel nut (4, Figure 4-3) to **6 N·m (50 in lb)**.
10. If a new charging valve is being used, tighten the swivel nut to **15 N·m (11 ft lb)**, then loosen and retighten the swivel nut to **15 N·m (11 ft lb)**. Loosen the swivel nut again and retighten to **6 N·m (50 in lb)**. Install the valve cap finger-tight.
11. Install the protective guards over the charging valves. Install the rubber covers over the piston rods.

Any time the suspensions are recharged, the calibration of the Payload Meter system is affected. Refer to the Payload Meter III section and perform a "Clean Truck Tare". This will ensure accurate payload records.

The rear Hydrair® II suspensions are now ready for operation. Visually check piston extension with the truck both empty and loaded. Record the extension dimensions. Maximum downward travel is indicated by the dirt ring at the base of the piston. Operator comments on steering response and suspension rebound should also be noted.

## OIL AND NITROGEN SPECIFICATIONS CHARTS

| HYDRAIR® II OIL SPECIFICATIONS                                                                                                                                      |                                      |                                                                                     |                                                                                                                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ambient Temperature Range                                                                                                                                           | Part No.                             | Approved Sources                                                                    |                                                                                                                                                             |
| -34.5°C & above<br>(-30°F & above)                                                                                                                                  | VJ3911<br>(need to add 6% of AK3761) | Mobilfluid 424<br>Mobil DTE 15M<br>Texaco TDH Oil<br>AMOCO ULTIMATE Motor Oil 5W-30 | Sunfleet TH Universal Tractor Fluid<br>Chevron Tractor Hydraulic Fluid<br>Conoco Power Tran III Fluid Petro Canada Duratran Fluid<br>Shell Canada Donax TDL |
|                                                                                                                                                                     | AK4063                               | Suspension Oil (premixed with 6% Friction Modifier)                                 | 5 gallon container                                                                                                                                          |
|                                                                                                                                                                     | AK4064                               |                                                                                     | 55 gallon container                                                                                                                                         |
| -48.5°C & above<br>(-55°F & above)                                                                                                                                  | VJ5925<br>(need to add 6% of AK3761) | Emery 2811, SG-CD, 5W-30<br>Mobil Delvac I, 5W-40                                   | Petro Canada Super Arctic Motor Oil, 0W-30<br>Conoco High Performance Synthetic Motor Oil, 5W-30                                                            |
|                                                                                                                                                                     | AK4065                               | Suspension Oil (premixed with 6% Friction Modifier)                                 | 5 gallon container                                                                                                                                          |
|                                                                                                                                                                     | AK4066                               |                                                                                     | 55 gallon container                                                                                                                                         |
| NOTE: <b>NOTE:</b> VJ3911 and VJ5925 oils are not compatible and must not be mixed in a suspension.VJ3911 and VJ5925 oils are supplied in 5 gallon (19 Liter) cans. |                                      |                                                                                     |                                                                                                                                                             |

| FRICITION MODIFIER                                       | FRICITION MODIFIER Mixing Instructions<br>(94% Suspension Oil, 6% Friction Modifier) |                                    |
|----------------------------------------------------------|--------------------------------------------------------------------------------------|------------------------------------|
| Part Number                                              | Suspension Oil                                                                       | Amount of Friction Modifier to add |
| AK3761<br>(5 Gallon container of 100% Friction Modifier) | 1 gallon of suspension oil                                                           | add 7.7 oz.                        |
|                                                          | 5 gallons of suspension oil                                                          | add 38.4 oz.                       |
|                                                          | 55 gallons of suspension oil                                                         | add 3.3 gal.                       |

| NITROGEN GAS (N <sub>2</sub> ) SPECIFICATIONS                                                                                       |           |                       |
|-------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------------------|
| Nitrogen gas used in HYDRAIR® II Suspension Cylinders must meet or exceed CGA specification G-10.1 for Type 1, Grade F Nitrogen Gas | Property  | Value                 |
|                                                                                                                                     | Nitrogen  | 99.9% Minimum         |
|                                                                                                                                     | Water     | 32 PPM Maximum        |
|                                                                                                                                     | Dew Point | -55°C (-68°F) Maximum |
|                                                                                                                                     | Oxygen    | 0.1% Maximum          |

# SECTION J BRAKE CIRCUIT INDEX

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# NOTES

# SECTION J2 BRAKE CIRCUIT INDEX

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# NOTES

## BRAKE CIRCUIT

The Komatsu truck is equipped with an all-hydraulic actuated wet disc service brake system. The brake system utilizes Type C-4 hydraulic oil provided by the brake/steering pump from the main hydraulic tank for brake application. Brake disc cooling during truck operation is provided by the hoist circuit pump through the hoist valve. A disc type parking brake, located in the rear axle housing, is attached to each wheel motor. The fundamental function of the brake system is to provide an operator with the necessary control for stopping the truck in either a slow modulating fashion or in as short a distance as reasonably possible.

Outlined below are the functions that Komatsu feels are necessary for safe truck operation:

- Warn the operator as soon as practical of a serious or potentially serious loss of brake pressure so that proper action can be taken to stop the truck before the secondary system is exhausted of its power.
- Provide secondary brake circuits so that any single failure leaves the truck with sufficient stopping power.
- Automatically apply the service brakes if low pressure warnings are ignored and pressures continue to decrease.
- Provide a wheel brake lock to relieve the operator from holding the brake pedal while at the dump or shovel.
- Provide a spring-applied parking brake for *holding* (not *stopping*) the truck during periods other than loading or dumping.
- The brake system should be easy to diagnose and service.

The following brake circuit description should be used in conjunction with the hydraulic brake system schematic located in Section R.

The brake system consists of several major valve components: the foot-operated dual circuit treadle valve, hydraulically-operated dual relay valves, brake manifold and two accumulators. The dual circuit treadle valve is the only component located in the operator's cab. The dual relay valve for the front brakes, the brake manifold, and the electrical components are located in the hydraulic brake cabinet behind the cab (See Figure 2-1).

The dual relay valve for the rear brakes is located in the rear axle housing. The two accumulators are mounted on the frame rail, behind the right front tire.

The brake manifold contains dual circuit isolation check valves, accumulator bleed down valves, and valves for the brake lock, parking brake and automatic apply functions. All of these components are screw-in cartridge type valves.

There are two independent means of brake actuation provided for the operator: the service brake/retarder pedal and the wheel brake lock switch. Additionally, the brakes will apply automatically if the brake system supply drops below a predetermined pressure.

### SERVICE BRAKE CIRCUIT

This portion of the system provides the operator with the precise control that is necessary to modulate (feather) brake pressure to slowly stop the truck or develop full brake effort to stop as quickly as possible. The heart of this circuit is the foot-operated dual circuit treadle valve. This valve enables the operator to control the relatively high pressure energy within the brake accumulators directed to the brakes. There are two valves in the dual brake valve. One supplies pressure to a dual relay valve to provide apply pressure for the brakes on the front axle. The other supplies pressure to a dual relay valve to provide apply pressure for the brakes on the rear axle.

As the brake pedal is depressed, each valve within the dual circuit brake valve simultaneously delivers fluid from its respective accumulator to the dual relay valves, which deliver fluid to the wheel brakes at a pressure proportional to both pedal position and force. The further the pedal is depressed, the higher the brake force, which gives the operator a very positive feel of control.

Brake accumulators have two functions: storing energy for reserve braking in the event of a failure and providing rapid oil flow for good brake response.

Depressing the brake pedal also actuates the stop light pressure switch, which in turn actuates the brake light on top of the cab, the stop/tail lights on the rear axle housing and the propulsion interlock.

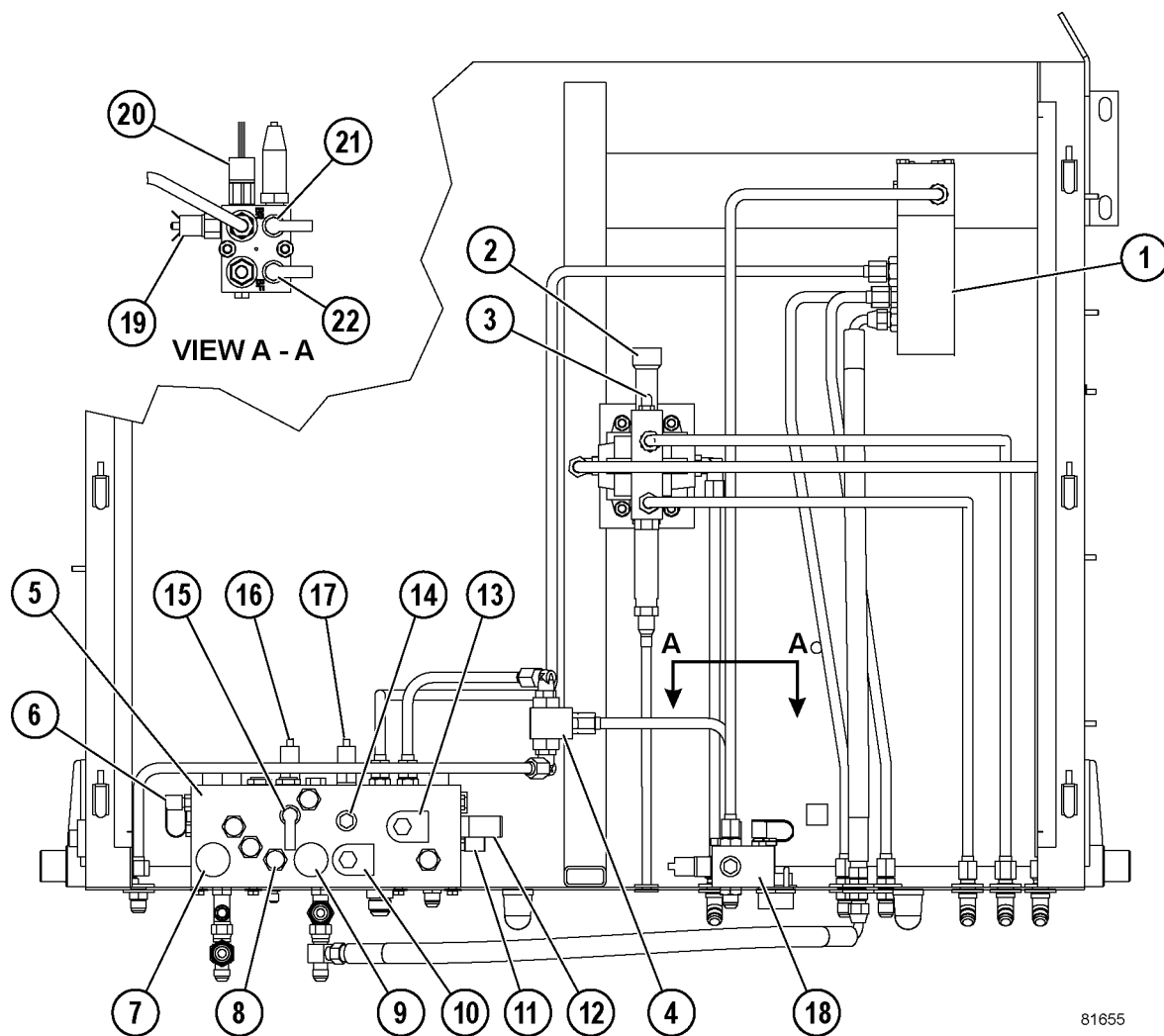


FIGURE 2-1. HYDRAULIC BRAKE CABINET

- |                                               |                                          |
|-----------------------------------------------|------------------------------------------|
| 1. Dual Relay Valve                           | 12. Pressure Reducing Valve (PR)         |
| 2. Hoist Pilot Valve                          | 13. Brake Lock Solenoid Valve (SV1)      |
| 3. Relief Valve (Hoist Power Down)            | 14. Parking Brake Release Pressure (PK2) |
| 4. Brake Lock Shuttle Valve                   | 15. Low Accumulator Test Port (LAP1)     |
| 5. Brake Manifold                             | 16. Low Brake Pressure Switch            |
| 6. Brake Oil Supply Pressure Test Port (SP3)  | 17. Parking Brake Pressure Switch        |
| 7. Rear Brake Accumulator Bleed Valve         | 18. Junction Manifold                    |
| 8. Automatic Apply Valve                      | 19. Stop Light Pressure Switch           |
| 9. Front Brake Accumulator Bleed Valve        | 20. Brake Lock Degradation Switch        |
| 10. Parking Brake Solenoid Valve (SV2)        | 21. Rear Brake Pressure Test Port (BR)   |
| 11. Brake Lock Apply Pressure Test Port (PP3) | 22. Front Brake Pressure Test Port (BF)  |

## PARKING BRAKE CIRCUIT

The parking brakes are spring applied and hydraulically released. Whenever the parking brake solenoid is de-energized, a spring in the solenoid valve will shift the spool, diverting oil pressure from the parking brakes to direct the oil back to the hydraulic tank.

### Normal Operation

(key switch ON, engine running)

- **Directional control lever in PARK**

Parking brake solenoid (10) is de-energized. The oil pressure in the parking brake lines returns to tank and the springs in the parking brake will apply the brake. Parking brake pressure switch (17) will close, completing a path to ground and illuminating the parking brake light on the overhead panel.

- **Directional control lever in F, R, or N**

Parking brake solenoid (10, Figure 2-1) is energized. The oil flow is routed from the parking brake solenoid to the parking brake pistons for release. The parking brake circuit is protected against accidental application by monitoring a wheel motor speed sensor to determine truck ground speed. The parking brake will not apply until the truck is virtually stopped. This eliminates parking brake damage and will extend brake adjustment intervals.

If the key switch is turned OFF, the parking brake will not apply until the truck speed is less than 0.5 km/h (1/3 mph).

If a loss of hydraulic supply pressure occurs with the directional control lever in F, R, or N, the parking brake solenoid will still be energized. The supply circuit that lost pressure is still open to the parking brake pistons. To prevent parking brake pressure oil from returning to the supply circuit, a check valve in the parking brake circuit traps the oil, holding the parking brake in the released position.

*NOTE: Normal internal leakage in the parking brake solenoid may allow leakage of the trapped oil to return back to tank and eventually allow parking brake application.*

If 24 volt power to the parking brake solenoid is interrupted, the parking brake will apply at any truck speed. The spring in the solenoid will cause it to shift, opening a path for the oil pressure in the parking brake line to return to tank, and the springs in the parking brake will apply the brake. Parking brake pressure switch (17) will close, completing a path to ground, illuminating the parking brake light on the overhead panel and interrupting propulsion.

## BRAKE LOCK CIRCUIT

The primary function of the brake lock is to provide a means for the operator to hold the truck while at the shovel or dump. **The brake lock only applies the rear service brakes.** It may also provide a secondary means of stopping the truck in the event of a brake valve malfunction. By turning on the dash mounted wheel brake lock switch, brake lock solenoid valve (13, Figure 2-1) and pressure reducing valve (12) will apply unmodulated pressure oil at 17 237 kPa (2,500 psi) to fully actuate the rear brakes. Shuttle valve (4) in the rear brake line provides the independence from the brake treadle valve for brake application.

## SECONDARY BRAKING AND AUTOMATIC APPLY

A fundamental function of the secondary brake system is to provide reserve braking in the event of any single failure. For this reason, the system is divided into multiple circuits, each with its own isolation check valve, accumulators, and circuit regulator. The secondary system becomes whatever circuit(s) is operable after a failure. If the failure is a jammed treadle valve, then the brake lock becomes the secondary system. Otherwise, either of the two brake circuits would be the secondary system.

The brake accumulators perform two functions: provide rapid flow for good response and store energy for secondary braking. The check valves ensure that this energy is retained if a failure should occur in the brake system supply or an accumulator circuit. An additional check valve, located between the supply line from the brake/steering pump and the brake manifold, provides additional protection against pressure loss if the oil supply is interrupted.

If a failure occurs in the pump, steering, or either brake accumulator circuit, a low brake pressure warning light located on the overhead panel in the cab will illuminate, and an audible alarm will sound, indicating that the truck should be stopped as soon as practical. When the pressure in one accumulator circuit is lower than the preset level, all the service brakes will be automatically applied. Automatic brake application is accomplished by the automatic apply valve (PS1) located in the brake manifold. This valve senses the lower brake accumulator pressure. When the pressure is less than 11 400 kPa (1650 psi), the valve shifts, operating the brake treadle valve hydraulically which, in turn, applies pressure to the dual relay valves, applying all the brakes.

Regardless of the nature or location of a failure, sensing the lowest brake accumulator circuit pressure ensures two to four full brake applications after the low brake pressure warning light and alarm activate and before automatic apply activates. This allows the operator the opportunity to safely stop the truck after the warning occurs.

## WARNING CIRCUIT

The brake warning circuit is equipped with a low brake pressure warning light on the overhead panel and an audible alarm in the cab to alert the operator to low brake pressures. Several electrical sensors, a relay and delay timer are used to detect brake system problems. See Figure 2-1 for the following component references.

- **Brake Warning Relay**

When the brake lock switch is turned on, the brake warning relay is energized and switches the electrical connection from the terminal to the brake lock degradation switch. When the brake lock switch is turned off, the relay is de-energized and switches the connection from the brake lock degradation switch to the terminal.

- **System Supply Pressure Switch**

Located on the pump pressure sensing manifold. When system supply pressure drops below 15 800 kPa (2300 psi), the low steering pressure light, low brake pressure warning light and buzzer will turn on.

- **Low Brake Pressure Switch (16)** Located on brake manifold (5). When the accumulator with the lower pressure falls below 12 700 kPa (1850 psi), the low brake pressure warning light and buzzer will turn on.

- **Brake Lock Degradation Switch (21)**

Located on junction manifold (19) in the hydraulic components cabinet. When the brake lock switch is turned on, brake lock solenoid (13) and the brake warning relay are energized. The brake warning relay switches the electrical connection from the terminal to the brake lock degradation switch. If the brake lock apply pressure is less than 6 900 kPa (1000 psi), a path to ground will be completed and the low brake pressure warning light and buzzer will turn on.

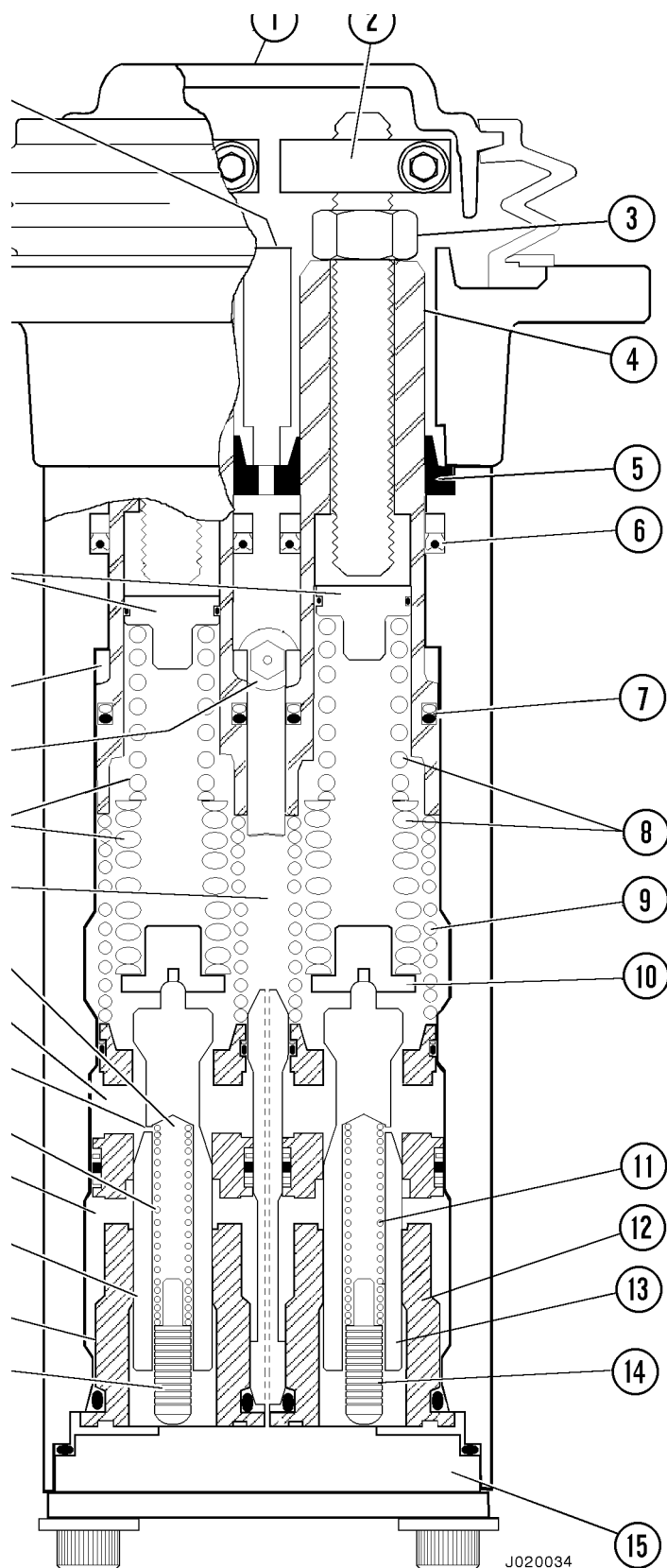


FIGURE 2-2. BRAKE VALVE

1. Actuator Cap
2. Adjustment Collar
3. Nut
4. Actuator Plunger
5. Wiper Seal
6. Poly-Pak Seal Assembly
7. Glyde Ring Assembly
8. Regulator Springs (B1)
9. Plunger Return Spring
10. Spring Seat
11. Spool Return Spring (B1)
12. Regulator Sleeve (B1)
13. Regulator Spool (B1)
14. Reaction Plunger (B1)
15. Base Plate
16. Reaction Plunger (B2)
17. Regulator Sleeve (B2)
18. Regulator Spool (B2)
19. Spool Return Spring (B2)
20. Regulator Springs (B2)
21. Staging Seat

- A. Adjustment Collar Maximum Pressure Contact Area  
 B. Automatic Apply Piston Area  
 C. PX Port  
 D. Tank Port  
 E. Reactionary Pressure Area  
 F. Brake Apply Port  
 G. Orifice  
 H. Supply Port

*Note:*  
 B1 - Rear Brakes  
 B2 - Front Brakes



Refer to Figure 2-3 for the brake lines that connect to the rear axle housing.

*NOTE: If hoses (4) and (8) are switched, the rear brakes will be slow to apply and slow to release.*

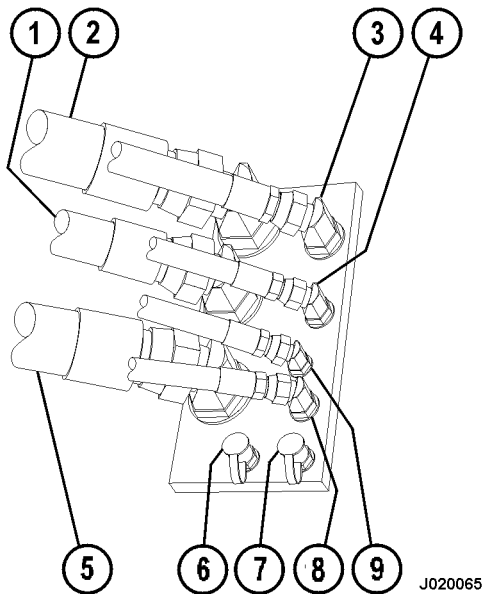


FIGURE 2-3. REAR BRAKE HOSES

1. BS - Left & Right Brake Cooling Oil Supply
2. LBR - Left Brake Cooling Return Line
3. T - Return To Tank
4. P1 - Pressure Inlet From Accumulator
5. RBR - Right Brake Cooling Return Line
6. RBP - Right Brake Pressure Test Port
7. LBP - Left Brake Pressure Test Port
8. PX - Pilot Inlet/Brake Apply Line
9. PB - Parking Brake

# SECTION J3

## BRAKE CIRCUIT COMPONENT SERVICE

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# BRAKE CIRCUIT COMPONENT SERVICE

## BRAKE VALVE

The brake pedal assembly combines the dynamic retarding control with service brake control functions.

As the pedal is depressed, the first portion of pedal travel actuates an internal potentiometer which provides an electrical signal to the propulsion system, signalling a request by the operator for retarding. The electrical signal is modulated as the operator continues to depress the pedal, increasing the level of retarding effort from minimum to maximum. Further pedal depression after maximum retarding request is achieved results in service brake application from minimum to maximum when the pedal is fully depressed. A slight increase in pedal resistance can be felt when service brake actuation occurs. Additionally, indicator lights on the overhead panel in the cab will turn on as dynamic retarding is applied. A second light will turn on when the service brakes are applied.

*NOTE: The following information discusses the operation, test and repair of the hydraulic brake valve assembly only. Refer to Section E2, Electric Propulsion System Components, for information concerning the retarding control portion of the pedal.*

The brake valve is a pressure modulating valve that is actuated mechanically through the brake pedal in the operator cab or hydraulically through automatic apply valve (2, Figure 3-1) in the hydraulic brake cabinet.

The brake valve controls the pressure delivered to the front and rear service brake dual relay valves which provide the apply pressure for the front wheel and rear wheel disc brake assemblies. Apply pressure can be modulated from zero to maximum braking effort by use of the foot pedal.

## Rebuild Criteria

If any one of the following conditions exist, the brake valve should be removed and repaired:

- Excessive cam rock in the pedal actuator.
- Any sign of external leakage.
- Internal leakage at the tank port must be less than 100 cc/minute with the valve in the released position and system pressure supplied to the "P1" and "P2" inlet ports.
- Tank port leakage must be less than 250 cc/minute with the valve pilot or manual applied at 18 960 kPa (2,750 psi) system pressure.
- Failure of the pedal to return to the full release position.
- The valve holds pressure when in the neutral position.
- Varying output pressure with the pedal fully depressed.

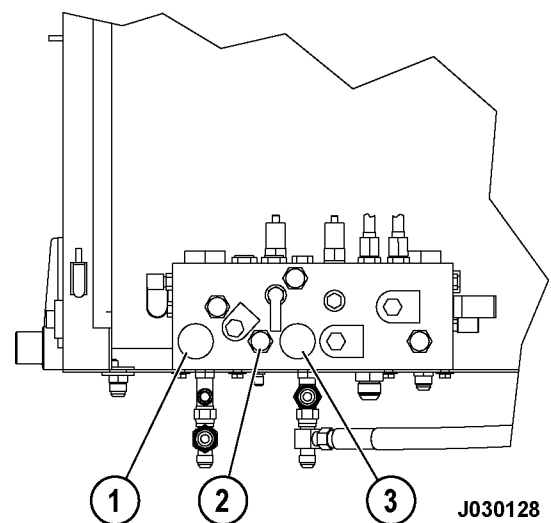


FIGURE 3-1. BRAKE MANIFOLD

1. Accumulator Bleed Down Valve (Rear Brake)
2. Automatic Apply Valve
3. Accumulator Bleed Down Valve (Front Brake)

## Removal

*NOTE: If the brake valve is to be removed from the truck for repair or adjustment, additional equipment will be required as outlined in the disassembly and assembly procedures. Minor repairs and adjustment may not require the removal of the brake valve.*

## **⚠ WARNING**

***Before disconnecting pressure lines, replacing components in the hydraulic circuits, or installing test gauges, always bleed down the steering and brake accumulators.***

1. Securely block the wheels to prevent the truck from rolling away.
2. Place the directional control lever in PARK and turn the rest switch ON. Turn the key switch OFF to shut down the engine and allow 90 seconds for the steering system accumulators to bleed down. Open bleed down valves (1) and (3, Figure 3-1) to bleed down the brake accumulators.
3. Remove the access panel at the front of the operator cab.
4. Tag and remove all hydraulic lines from the brake valve. Plug all lines and ports to prevent possible contamination. Remove all valve fittings except the fitting at port "PX".

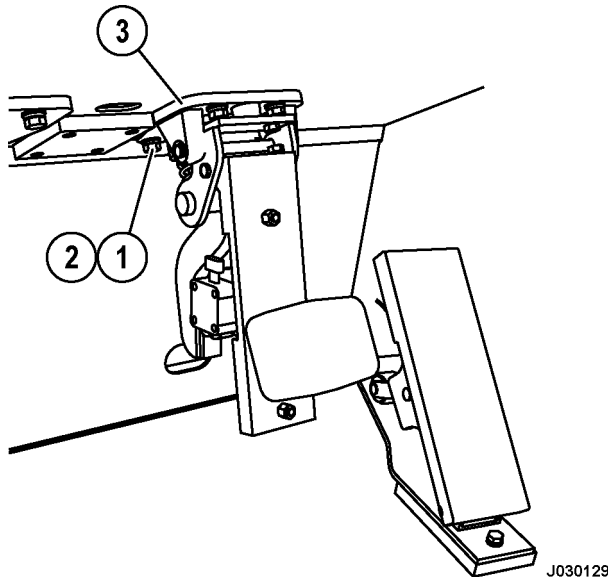


FIGURE 3-2. BRAKE VALVE REMOVAL

- |                |                |
|----------------|----------------|
| 1. Capscrews   | 3. Brake Valve |
| 2. Lockwashers |                |

5. Disconnect the retarder pedal harness.
6. Remove capscrews (1, Figure 3-2) and lockwashers (2) that secure brake valve assembly (3) to the mounting structure.
7. Slide the brake valve assembly downward and remove it from the cab. Move the brake valve assembly to a clean work area for disassembly.

## Installation

1. Place brake valve assembly (3, Figure 3-2) into position in the cab and secure it with capscrews (1) and lockwashers (2). Tighten the capscrews to the standard torque.
2. Remove the plugs from the brake valve assembly and hydraulic lines. Install all fittings and connect the lines securely to the brake valve assembly.

*NOTE: Before checking the brake valve operation, the steering system must have the proper nitrogen precharge in the steering accumulators. Refer to Section L6, Steering Circuit Component Repair, for the steering accumulator precharge procedure. Also, the air must be bled from the brake system lines and the brake accumulators must be precharged with nitrogen. Refer to Accumulator Charging Procedure later in this section.*

3. Connect the retarder pedal harness to the truck harness.
4. With the engine off and the key switch OFF, open both brake accumulator bleed down valves (1) and (3, Figure 3-1). Precharge both brake accumulators to 9 650 kPa (1400 psi). Refer to Accumulator Charging Procedure later in this section.

*NOTE: For best performance, charge the accumulators in the temperature conditions that the vehicle is expected to operate in. During the precharge, allow the temperature of the nitrogen gas to come into equilibrium with the ambient temperature.*

5. Close both accumulator bleed down valves after the accumulators have been properly charged.
6. Start the engine. Partially apply the brakes and open the supply plug at each wheel to the bleed air from brake lines and brakes. Close the supply plug after the bubbles disappear. Refer to Wet Disc Brake Bleeding Procedure later in this section for additional information.
7. Check for oil leaks at the brake valve.

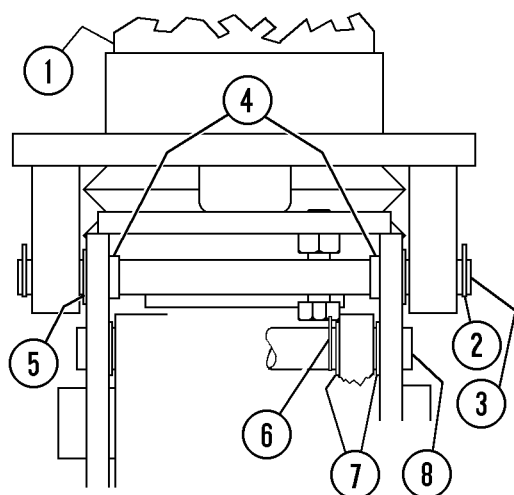
## Disassembly

1. If not already removed, remove electronic retard pedal (16, Figure 3-3) from the brake pedal by removing pivot shaft (8).

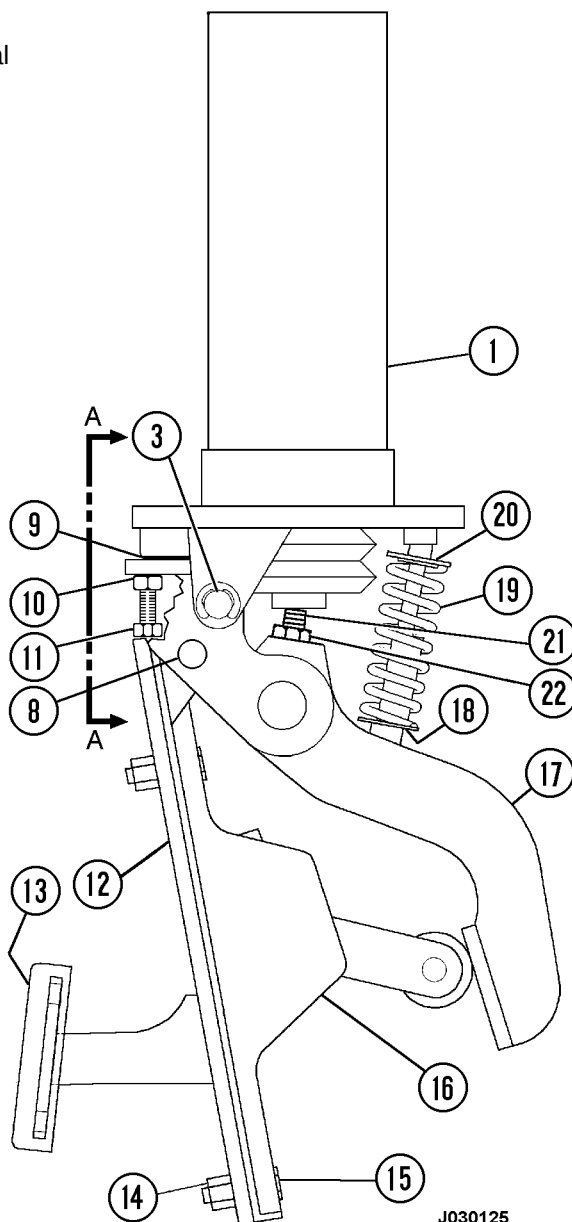
*NOTE: During disassembly, precision machined parts should be ink marked or tagged to ensure proper reassembly and minimize adjustment time. All items must be placed back into the bores from which they were removed.*

2. Match mark each section of the brake valve before disassembly.
3. Drain the oil from all ports of the brake valve by rotating the valve over a suitable container.
4. Secure the brake valve in an upright position in a vice.
5. Remove brake pedal actuator (17) by removing retainer clips (2), then removing pivot shaft (3) with a punch and hammer.

- |                     |                             |
|---------------------|-----------------------------|
| 1. Brake Valve      | 16. Electronic Retard Pedal |
| 2. Retainer Clip    | 17. Brake Pedal Actuator    |
| 3. Pivot Shaft      | 18. Spring Pivot (Lower)    |
| 4. Bushings         | 19. Spring                  |
| 5. Shims            | 20. Spring Pivot (Top)      |
| 6. Retainer Clip    | 21. Set Screw               |
| 7. Nylon Bearing    | 22. Jam Nut                 |
| 8. Pivot Shaft      |                             |
| 9. Shim             |                             |
| 10. Jam Nut         |                             |
| 11. Capscrew        |                             |
| 12. Pedal Structure |                             |
| 13. Pad             |                             |
| 14. Nut             |                             |
| 15. Capscrew        |                             |



VIEW A-A



J030125

FIGURE 3-3. BRAKE VALVE WITH RETARD PEDAL

6. Remove four button head allen screws (3, Figure 3-4) that secure boot retainer plate (4).
7. Remove boot retainer plate (4), boot (2), and actuator cap (1) as an assembly by grasping the boot and gently lifting it from the valve body.

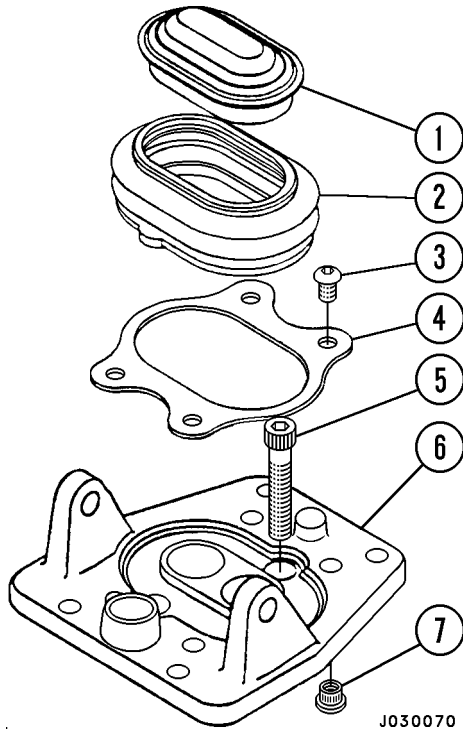


FIGURE 3-4. ACTUATOR CAP & BOOT

- |                   |                    |
|-------------------|--------------------|
| 1. Actuator Cap   | 5. Capscrew        |
| 2. Boot           | 6. Actuator Base   |
| 3. Capscrew       | 7. Threaded Insert |
| 4. Retainer Plate |                    |

8. Remove capscrews (31, Figure 3-5) and plate (30).
9. Remove and discard O-ring (22) and face seals (23).
10. Loosen the locknuts on actuator plunger (2). Loosen the socket head capscrew from adjustment collars (1).
11. Unscrew and remove the adjustment collars.
12. Remove two socket head capscrews (5, Figure 3-4) that retain actuator base (6) to the valve body.
13. Remove the actuator base from the valve body.
14. Remove the controller from the vice.
15. Remove four capscrews (29, Figure 3-5) and washers (28) from the base of the valve.

16. Remove base plate (27).

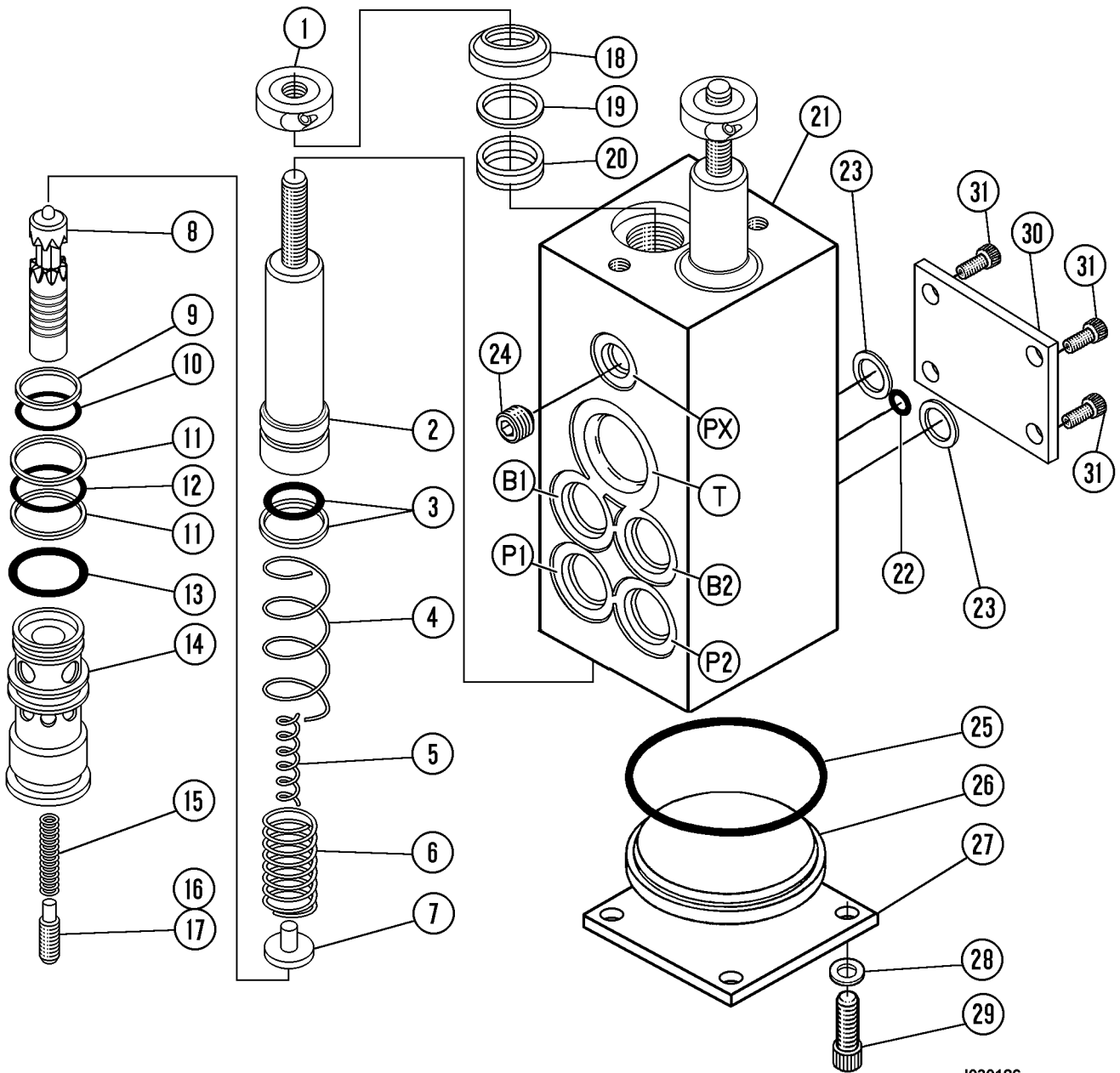
17. With the valve upright, retaining plug (26) should fall out. If the plug does not fall out, lightly tap the valve body to dislodge the plug.

*NOTE: Spools (8), reaction plungers (16) and (17), and spool return springs (15) may fall out at this time. Keep the parts separate so they can be installed in the same bores from which they were removed. "B1" reaction plunger (16) is larger than "B2" reaction plunger (17).*

18. Remove O-ring (25) from the counterbore in the base of the valve body. Discard the O-ring.
19. With the controller upright on the work bench, hold the valve with one hand and push "B1" actuator plunger (2) down with the other hand until regulator sleeve (14) becomes loose.
20. Repeat the Step 19 to loosen the "B2" regulator sleeve.
21. Turn the valve on its side on the work bench and remove both regulator sleeves (14) from the valve body.

*NOTE: Throughout the following steps, it is important to keep the circuits and circuit components identified as to which side of the unit they came from. For a given circuit, all the components have a tolerance stack which could vary. Keep the "B1" and "B2" parts separate. Springs (5) and (6) are also different in the "B1" and "B2" bores.*

22. Remove regulator spools (8), reaction plungers (16) and (17), and spool return springs (15) from regulator sleeves (14).
23. Remove plunger return springs (4), regulator springs (5) and (6), and spring seats (7) from the valve body.
24. Remove actuator plungers (2) by pushing down (toward the bottom of the valve) on the actuator plunger with your hand until the actuator plunger slides out.
25. Remove Glyde ring assembly (3) from the actuator plunger.
26. Remove O-rings (10), (12), and (13), and teflon backup rings (9) and (11) from the regulator sleeves. Discard the O-rings and backup rings.
27. Remove wiper seals (18), poly-pak seals (20), and orange backup rings (19) from the actuator section of the valve. Discard the seals and backup rings.



J030126

FIGURE 3-5. BRAKE VALVE

- |                          |                           |                            |                    |
|--------------------------|---------------------------|----------------------------|--------------------|
| 1. Adjustment Collar     | 9. Backup Ring            | 17. Reaction Plunger (B2)  | 25. O-Ring         |
| 2. Actuator Plunger      | 10. O-Ring                | 18. Wiper Seal             | 26. Retaining Plug |
| 3. Glyde Ring Assembly   | 11. Backup Ring           | 19. Backup Ring            | 27. Base Plate     |
| 4. Plunger Return Spring | 12. O-Ring                | 20. Poly-Pak Seal          | 28. Washer         |
| 5. Regulator Springs     | 13. O-Ring                | 21. Valve Body             | 29. Capscrew       |
| 6. Regulator Springs     | 14. Regulator Sleeve      | 22. O-Ring                 | 30. Plate          |
| 7. Spring Seats          | 15. Spool Return Spring   | 23. Face Seals             | 31. Capscrew       |
| 8. Regulator Spool       | 16. Reaction Plunger (B1) | 24. Set Screw Orifice Plug |                    |



## Cleaning and Inspection

1. Clean all metal parts with solvent and air dry.
2. Inspect plunger (2, Figure 3-5) for wear on the sides where it moves through the seals. If axial grooves are seen or if any wear is apparent, replace the plunger.
3. Place regulator spool (8) into its sleeve (14). Push the spool lightly through the sleeve. The spool must be able to move freely and smoothly the entire length of the sleeve. If it cannot, it must be replaced. Never replace just the spool or sleeve. They must be replaced as a matched set.
4. Inspect each spring carefully for cracks and breaks. Any spring with a crack or break must be replaced. Also, if the valve was not reaching proper regulated pressure, replace all regulator springs.
5. Inspect threaded inserts (7, Figure 3-4) in actuator base (6). If any of the threads are damaged, the inserts must be replaced.
6. Lubricate all parts with a thin coat of clean type C-4 hydraulic oil. Take care to keep the components protected from contamination.

## Assembly

### Actuator Base Threaded Inserts

1. If any threaded inserts (7, Figure 3-4) were removed from actuator base (6), position the actuator base upside down on the work bench and support it directly under each of the four mounting holes.
2. Install the threaded inserts into the actuator base by tapping lightly with a small hammer until the insert flanges become flush with the actuator base. To avoid breaking the base, make sure that the base is supported.
3. Thoroughly clean the actuator base and set it aside.

### Boot and Cap

1. Examine boot (2, Figure 3-4) for any cracks, tears, or other damage. If damage is evident, the boot must be replaced.
2. If damaged, remove the boot from actuator cap (1) and discard the old boot. Thoroughly clean the sides of the cap by using a knife or suitable scraper to scrape the lip where the cap contacts the boot. Clean the cap thoroughly to remove all adhesive or particles of the old boot.

3. Apply a thin bead of Loctite® Prism 410 onto the upper sides of the cap. Apply the bead to the two long sides only. Do not apply it to the rounded ends. The rounded ends must not be sealed to allow the boot to "breathe".
4. Carefully position the cap into the new boot groove. Wipe off any excess glue.
5. Position the boot so that it conforms to the contour of the cap, then set it aside. The adhesive requires about 30 minutes to cure.

### Valve Body Seal Installation

1. Install poly-pak seal (3, Figure 3-6) in the seal groove first. Position the seal in the groove so that the internal O-ring inside the poly-pak seal is facing down toward the bottom of the valve.
2. Make sure that the internal O-ring is still seated inside the poly-pak seal and did not get dislodged during installation. Position the poly-pak seal to the bottom of the groove.

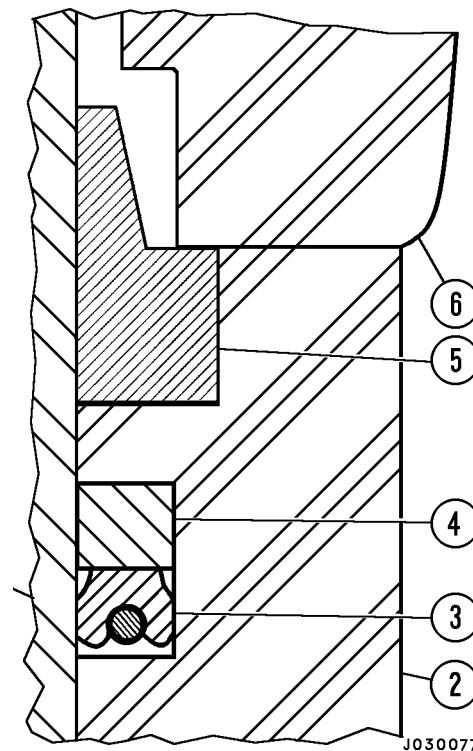


FIGURE 3-6. VALVE BODY SEAL INSTALLATION

- |                     |                  |
|---------------------|------------------|
| 1. Actuator Plunger | 4. Backup Ring   |
| 2. Valve Body       | 5. Wiper Seal    |
| 3. Poly-Pak Seal    | 6. Actuator Base |

3. Install orange backup ring (4) on top of the poly-pak seal. Start it by hand and then continue to work into the groove either by hand or by using an O-ring installation tool.
4. Install wiper seal (5) in the top counterbore. Position the seal in the groove so that the register lip is facing up toward the actuator.
5. Repeat Steps 1- 4 for the second bore.

#### Regulator Sleeve O-Ring Installation

1. Install O-ring (2, Figure 3-7) onto the smallest groove (on the top) of regulator sleeve (3). Install O-ring (5) onto the middle groove on the regulator sleeve. Install O-ring (6) onto the largest groove (on the bottom) on the regulator sleeve.
2. Install split nylon backup ring (4) onto each side of O-ring (5) located in the middle of the regulator sleeve.
3. Install one split nylon backup ring behind O-ring (2) located at the top end of the sleeve. (This O-ring is the smallest of the three O-rings.) Position the backup ring so that it is next to the top of the regulator sleeve. The top of the sleeve is the end with the smallest outside diameter.
4. Repeat Steps 1-3 for the second regulator sleeve.

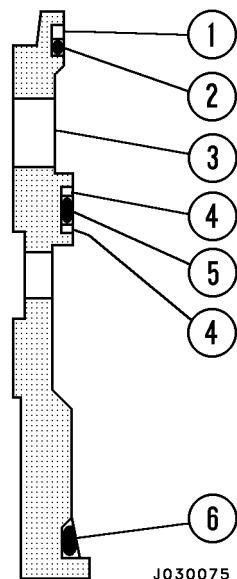


FIGURE 3-7. SLEEVE SEAL PLACEMENT

- |                     |                |
|---------------------|----------------|
| 1. Backup Ring      | 4. Backup Ring |
| 2. O-Ring           | 5. O-Ring      |
| 3. Regulator Sleeve | 6. O-Ring      |

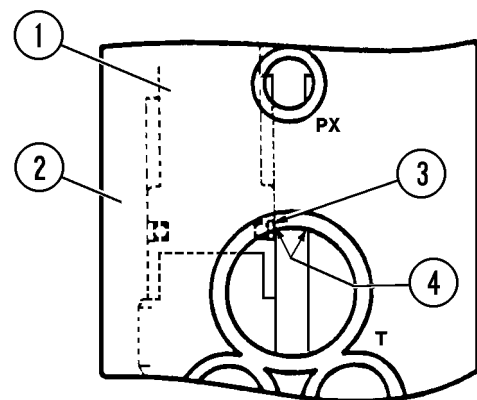
#### Actuator Plunger O-Ring Installation

1. Install an O-ring (3, Figure 3-5) into the O-ring groove located at the large diameter end of actuator plunger (2).
2. Install a split Glyde ring (3) over the O-ring. Twist and squeeze the split Glyde ring into a small circle before installing it to ensure a tight fit over the O-ring.
3. Repeat Steps 1 and 2 for the second plunger.

#### Assembly of Valve

*NOTE: Start with either side (circuit) of the valve and build that side complete through Step 4 before starting on the other side (circuit). Be careful to assemble components into the circuit from which they were removed.*

1. Lightly lubricate actuator plunger Glyde ring (3, Figure 3-5).
2. Install "B1" actuation plunger (2) into the "B1" circuit. Be careful not to damage or cut the Glyde ring during installation. Observe the Glyde ring assembly through the tank port as the plunger is being installed. (Refer to Figure 3-8) It may be necessary to work the Glyde rings past the sharp edge in the body to prevent damage to the seal. Ensure that the actuation plunger is completely seated and bottomed.



J030076

FIGURE 3-8. GLYDE RING INSTALLATION

- |                     |                |
|---------------------|----------------|
| 1. Actuator Plunger | 3. Glyde Ring  |
| 2. Valve Body       | 4. Sharp Edges |

3. Repeat Steps 1 and 2 for the "B2" actuation plunger.
4. Install the plunger return spring (4, Figure 3-5), regulator springs (5) and (6), and spring seat (7) into the appropriate circuit. If spring seat does not seat correctly on top of the control spring, lightly shake the valve to correctly position the spring seat.
5. Lightly lubricate regulator spool (8).
6. Install the regulator spool into the regulator sleeve (14). The spherical end of the spool should be at the top of the regulator sleeve. The top of the sleeve is the end with the smallest outside diameter.
7. Push the spool lightly through the sleeve. The spool must be able to move freely and smoothly the entire length of the sleeve. If it cannot, it must be replaced. Never replace just the spool or sleeve. They must be replaced as a matched set.
8. Remove the spool from sleeve before installing the sleeve into the valve body.
10. Lightly lubricate O-rings (10), (12) and (13) on the regulator sleeve.
11. Install the regulator sleeve assembly into the correct circuit in the valve. Make sure that the spring seat is correctly seated in the regulator spring before installing the regulator sleeve assembly. Push the sleeve into the bore until the sleeve retaining flange at the base of the sleeve contacts the valve body.
12. Install spool return spring (15) into regulator spool (8).
13. Insert reaction plunger (16) or (17) into the regulator spool.
14. Install regulator spool (8) into regulator sleeve (14).
15. Repeat Steps 6 - 14 for the second circuit.
16. Lightly lubricate large retainer plate O-ring (25) and install it into the counterbore in the bottom end of the valve.
17. Install retainer plug (26) into the counterbore on the bottom of the valve. Make sure that the steps on the retainer plug are facing the counterbore or toward the top of the valve.
18. Install base plate (27) on top of the retainer plug. Tighten four allen screws (29) evenly, alternating diagonally, in order to evenly seat the regulator sleeve assembly. Then tighten the screws to a final torque of **16 - 17 N·m (140 - 150 in lb)**.
19. Using a new O-ring (22, Figure 3-5) and seal (28), install plate (30) on valve body.
20. Install actuator base (6, Figure 3-4) on top of the valve. Be sure to position it properly for correct port direction. Install and tighten two socket head capscrews (5) to **20 - 21 N·m (180 - 190 in lb)**.
21. Screw adjustment collars (1, Figure 3-5) onto the top of actuator plungers (2). Screw them all the way down until they bottom on the threads.

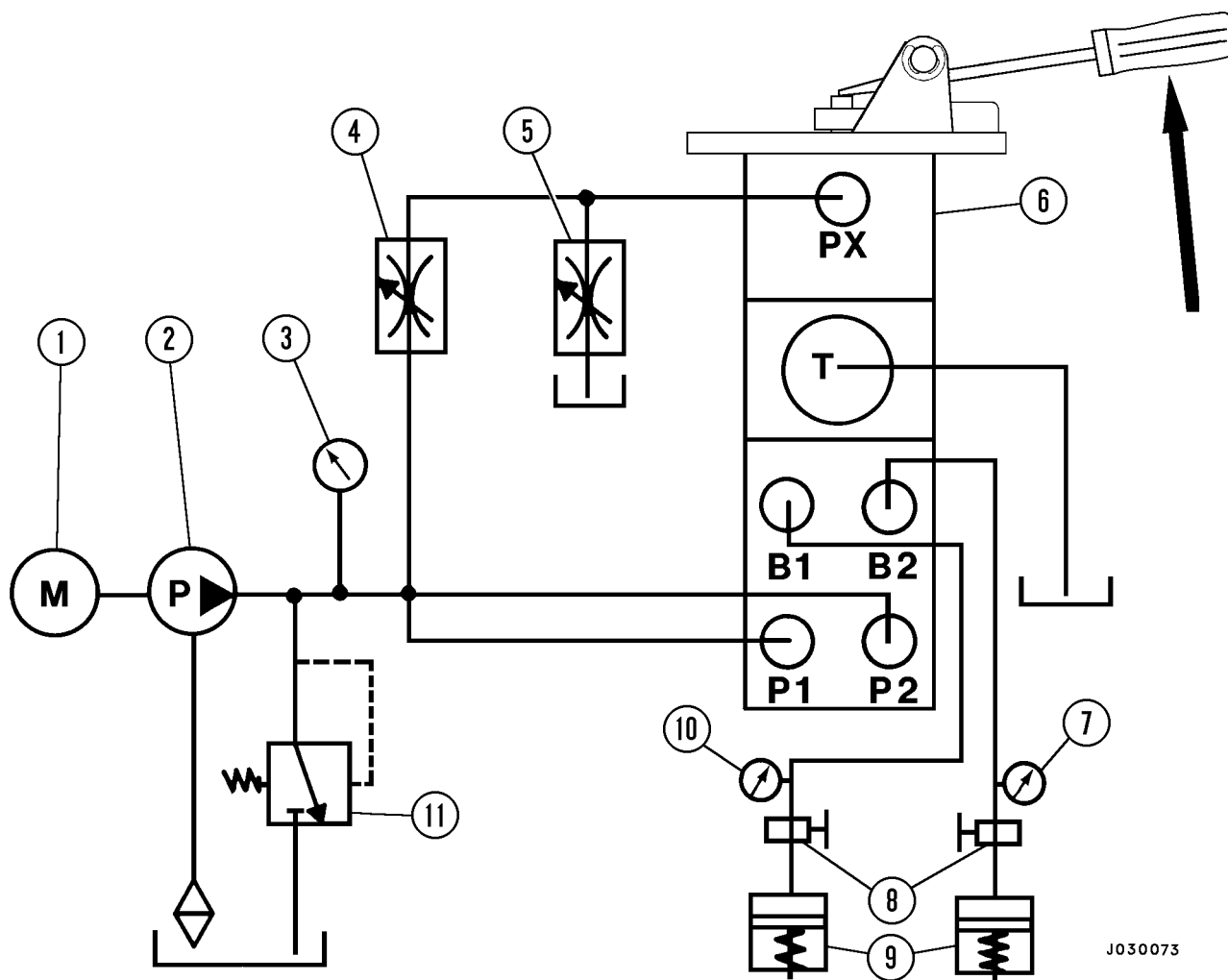
## BRAKE VALVE BENCH TEST AND ADJUSTMENT

The following parts and test equipment will be required to completely bench test and adjust the brake valve.

- Three 20 680 kPa (3,000 psi) pressure gauges
- Hydraulic pressure supply, regulated to 18 960 kPa (2,750 psi)
- Hydraulic test stand (Refer to Figure 3-9)

- Hose fittings for valve ports:  
 Port PX . . . . . 7/16 in., # 4 SAE  
 Ports P1, P2, B1 and B2. . . . . 3/4 in., #8 SAE  
 Port T . . . . . 1 1/16 in., #12 SAE
- Ohmmeter or continuity tester

*NOTE: It is possible to check the pressures with the brake valve installed and connected to the vehicle. Remove the brake pedal assembly and actuator cap and boot assembly to adjust individual brake circuit pressures.*



J030073

FIGURE 3-9. TEST BENCH SETUP

- |                          |                               |                               |
|--------------------------|-------------------------------|-------------------------------|
| 1. Motor                 | 5. Needle Valve               | 9. Simulated Brake Volume     |
| 2. Pump                  | 6. Brake Valve                | 10. Rear Brake Pressure Gauge |
| 3. System Pressure Gauge | 7. Front Brake Pressure Gauge | 11. Relief Valve              |
| 4. Needle Valve          | 8. Shutoff Valves             |                               |

## Test Setup Procedure

1. Position the valve in the fixture to allow plungers to be activated by hand using a lever. Refer to Figure 3-9.
2. Attach the pilot input supply pressure to the pilot port labeled "PX" on the rear of the valve.
3. Attach the main supply input pressure to the O-ring ports on the rear of the valve labeled "P1" and "P2".
4. Attach the tank return line to the O-ring port labeled "T" on the rear of the valve.
5. Attach the regulated output ports "B1" and "B2" to the test lines. Pressure monitoring devices in these two lines must be capable of 20 680 kPa (3,000 psi). Connect all ports according to the diagram shown in Figure 3-9. All ports must be used and connected.

## WARNING

***All ports must be used. Relieve pressure before disconnecting hydraulic and other lines. Tighten all connections before applying pressure.***

***Avoid spillage and contamination! Avoid contact with hot oil if the machine has been operating. The oil will be at very high pressure.***

***Hydraulic fluid escaping under pressure can have sufficient force to enter a person's body by penetrating the skin and cause serious injury and possibly death if proper medical treatment by a physician familiar with this injury is not received immediately.***

6. Start the hydraulic pump and regulate output pressure to 18 960 kPa (2,750 psi) at pressure gauge (3). Pressure gauges (7) and (10) should read zero.
7. Pilot supply circuit pressure must also be 18 960 kPa (2750 psi).
8. Return line pressure during this test is not to exceed 34 kPa (5 psi).
9. Test the valve with ISO 32 grade hydraulic oil at  $49^{\circ} \pm 3^{\circ} \text{ C}$  ( $120^{\circ} \pm 10^{\circ} \text{ F}$ ).

## Brake Valve Output Pressure Adjustment

1. Install the pedal pivot shaft in the actuator base by itself without installing the pedal assembly.
2. By placing a screwdriver or pry bar under the pivot pin and on top of the threaded plunger assembly, each circuit can be actuated individually. Refer to Figure 3-9.
3. Gradually apply pressure on each circuit one at a time to check for leaks around the plunger. Make sure that the adjustment collar is screwed all the way down on the threads.
4. **"B1" Adjustment:** Adjust the adjustment collar up (counterclockwise), starting with one-turn increments until the output pressure at port "B1" is  $17\,235 \pm 517 \text{ kPa}$  ( $2,500 \pm 75 \text{ psi}$ ) with the adjustment collar contacting the actuator base (fully actuated). Fine adjustment will require turning the collar only in 1/8 turn increments.
5. **"B2" Adjustment:** Adjust the adjustment collar up (counterclockwise), starting with one-turn increments until the output pressure at port "B2" is  $17\,235 \pm 517 \text{ kPa}$  ( $2,500 \pm 75 \text{ psi}$ ) with the adjustment collar contacting the actuator base (fully actuated). Fine adjustment will require turning the collar only in 1/8 turn increments.
6. Tighten the set screws in the adjustment collars to **3 - 4 N·m (25 - 30 in lb)**. The entire plunger may have to be rotated to get to the set screws.
7. Check the pressures again after tightening the set screws. If the pressures have moved out of specified range, loosen the appropriate set screw and re-adjust.
8. Cycle each circuit 50 times using pilot apply. This is done by closing needle valve (5, Figure 3-9) and opening needle valve (4). Read the pressure on gauges (7) and (10). Close valve (4) and open valve (5). The pressure gauges should read zero.
9. Recheck the pressures after cycling. If they have changed, re-adjust the pressures.

## Final Test and Adjustment

The brake pedal actuator must be installed on the brake valve body before the final test and adjustment. Refer to Installation of Brake Pedal Actuator to Brake Valve.

*NOTE: The final test and adjustment procedure can also be performed with the brake valve installed in the truck. Install the gauges at the "B1" and "B2" diagnostic test connectors in the brake cabinet.*

1. Reinstall the brake valve with the actuator pedal attached on the test stand according to the previous Test Setup Procedure.
2. With the test stand pump adjusted for 18 960 kPa (2750 psi) or with the engine running and the brake system supply pressure at or above 18 960 kPa (2750 psi), depress the pedal as quickly as possible. The pressure on the output circuits must reach at least  $17\,235 \pm 517$  kPa ( $2500 \pm 75$  psi) at port "B1" and port "B2" within one second. Measurement of time begins the moment force is applied to move the pedal.
3. With "B1" and "B2" plugged into a strip chart recorder (if available), check the modulation by slowly applying pressure until the maximum pressure is reached. Make sure that the pressure increase is smooth and no sticking of the spools is observed. Fully depress the pedal. The pressures must remain within specification at "B1" and "B2" for 20 seconds.
4. Adjust set screw (21, Figure 3-3) until the set screw is not touching the actuator cap. Apply Loctite® 242 to the adjustment screw before setting the deadband.
5. Set the deadband by placing a 0.254 mm (0.010 in.) thick shim at location (9) between the pedal structure and the return stop boss on the pivot structure.
6. Adjust the set screw until it is just touching the cap.
7. Continue turning the set screw until the pressure begins to rise on one of the brake apply pressure gauges.
8. Back off the set screw by 1/8 turn.
9. Tighten jam nut (22) and remove the shim that was inserted in Step 5.
10. Fully stroke the brake pedal actuator to verify that the output pressures at port "B1" and "B2" are within specifications.

*NOTE: If the pedal is adjusted properly, the spring and spring pivots will not interfere with pedal travel.*

11. If the pressures are not within specifications, re-adjust the set screw. If the pressure is within specifications, apply a few drops of Loctite® to the jam nut.
12. Check for internal leakage at port "T". Leakage must be less than 100 cc/minute with the valve in the released position and system pressure supplied to the "P1" and "P2" inlet ports.
13. "T" port leakage must be less than 250 cc/minute with valve pilot pressure or manually applied.

## Installing Brake Pedal Actuator Assembly to Brake Valve

1. Install jam nut (22, Figure 3-3) and set screw (21) into brake pedal actuator (17).
2. Insert nylon bushings (4) into the brake pedal actuator.
3. Install one retainer clip (2) to one end of pivot shaft (3).
4. Align pedal structure (12) with brake valve (1) and partially insert pivot shaft (3). Move the pedal structure to the "B2" side of the valve and insert shims (5) between the pedal structure and the brake valve ear to fill the gap. Fully insert the pivot shaft. Install the remaining retainer clip.
5. Assemble spring assembly (19) and install the complete assembly onto the brake pedal actuator as shown.

## IMPORTANT

***Install the spring assembly with the larger ball socket end pointing to the pedal structure and smaller end toward the valve assembly.***

*NOTE: When the pedal is adjusted properly, the spring assembly will not interfere with pedal travel.*

## Installing Retard Pedal To Brake Pedal

1. Install nylon bearings (7, Figure 3-3) in retard pedal (16).
2. Install retard pedal (16) onto brake pedal actuator (17) with pivot shaft (8). Install two retainer clips (6).
3. With jam nut (10) loose, adjust capscrew (11) until the roller on the retard pedal just contacts the brake pedal actuator. Tighten jam nut (10).

## DUAL RELAY VALVE

The dual relay valves (one for the front brake circuits and one for the rear brake circuits) supply the apply pressure for each disc brake assembly. When the operator depresses the brake valve, hydraulic pressure (proportional to the amount of brake valve application) applied to the pilot pressure circuit of each dual relay valve. Regulated pressure (proportional to the pilot pressure applied) is then delivered from the "B1" and "B2" ports of each dual relay valve to each wheel.

The dual relay valve for the front brake circuit is located in the hydraulic brake cabinet behind the cab. The dual relay valve for the rear brake circuit is located in the rear axle housing. Access to the rear brake circuit valve requires removal of the wheel motor cooling air duct components.



***Before disconnecting pressure lines, replacing components in the hydraulic circuits, or installing test gauges, always bleed down the steering and brake accumulators.***

### Removal

1. Securely block the wheels to prevent the truck from rolling away.
2. Place the directional control lever in PARK and turn the rest switch ON. Turn the key switch OFF to shut down the engine and allow 90 seconds for the steering system accumulators to bleed down. Open bleed down valves (1) and (4, Figure 3-1) to bleed down the brake accumulators.

***NOTE: Follow Steps 3 and 4 below for the front valve or Steps 5 through 9 for the rear valve.***

#### *Front Brake Circuit:*

3. Tag and remove all hydraulic lines from the dual relay valve. Plug all lines and ports to prevent possible contamination.
4. Remove two capscrews and washers that secure the valve to the rear wall of the cabinet. Remove the valve and move it to a clean work area for disassembly.

#### *Rear Brake Circuit:*

5. Open the access hatch on the rear axle housing. Remove the capscrews that secure the duct tube to the axle housing. Remove the tube.
6. Remove the wheel motor cooling air exhaust duct from between the wheel motors.
7. Tag and remove all hydraulic lines from the dual relay valve. Plug all lines and ports to prevent possible contamination.
8. Remove two capscrews and washers that secure the valve to the mounting bracket. Remove the valve and move it to a clean work area for disassembly.

### Installation

1. Install the dual relay valve in the hydraulic brake cabinet (front brake circuit) or rear axle housing (rear brake circuit). Install the two mounting capscrews and lockwashers to secure the valve. Tighten the capscrews to the standard torque.
2. Unplug all hoses and fittings and attach the hoses to the proper valve ports.
3. Start the engine and check for leaks and proper brake operation. Shut off the engine.
4. For the rear valve, reinstall the wheel motor cooling air duct between the wheel motors. Reinstall the tube in the axle access opening and close the hatch.

### Disassembly

***NOTE: The parts installed in the valve body for the "B1" and "B2" bores are identical. However, the parts must not be interchanged between the two bores.***

1. Thoroughly clean the valve to remove any dirt accumulation. Drain the oil from all valve ports by rotating the valve over a suitable container.
2. Use a felt tip pen to mark manifold body (1, Figure 3-10) and valve body (2) to ensure correct reassembly.

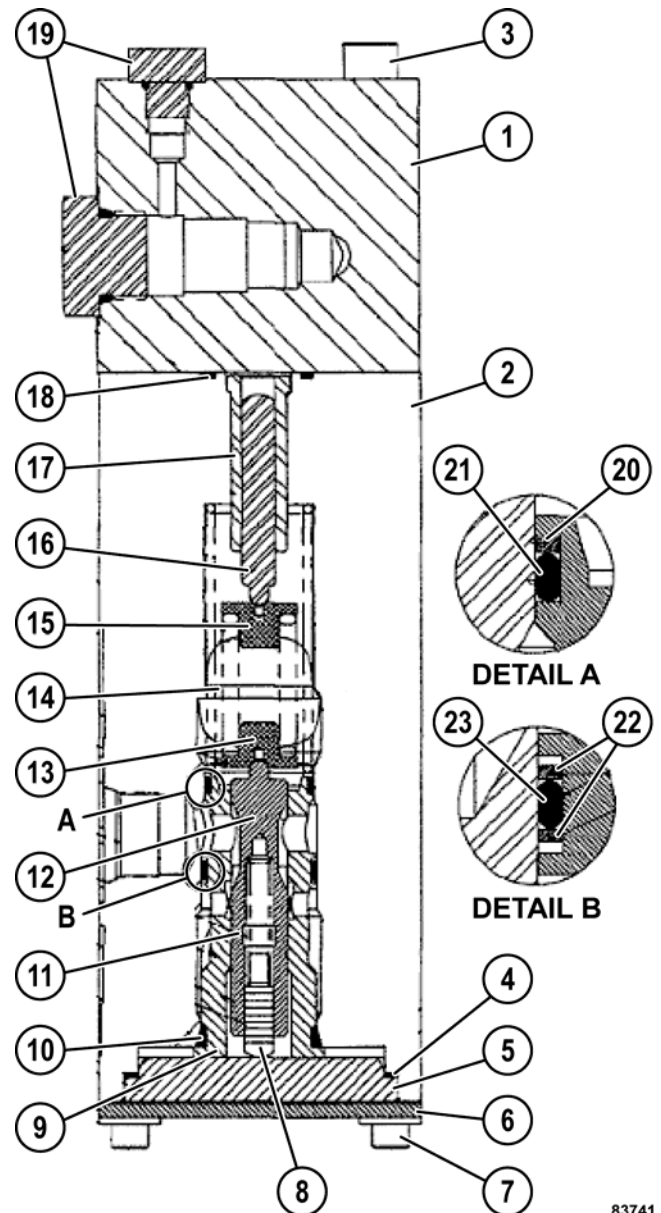
***NOTE: As the valve is disassembled, lay out the parts in the order of disassembly. Note the valve body bore from which the parts are removed. The parts must be reinstalled in the same bore from which they are removed.***

3. Secure the valve in an upright position in a vice.
4. Remove two socket head capscrews (3) that hold manifold body (1) to valve body (2). Remove the manifold body and discard O-rings (18).

5. Remove plungers (16) and sleeves (17).
6. Remove the controller from the vice.
7. Remove four capscrews and washers (7) from the base of the valve.
8. Remove sleeve retainer (6).
10. With the valve upright, plug (5) should fall out. If not, tap the valve body lightly to dislodge it.
11. Remove spools (12), reaction plungers (8) and spool return springs (11). Keep the parts separate so they can be installed in the same spool from which they were removed.
12. Remove and discard packing (4) from the counterbore in the base of the valve body.
13. Turn the valve on its side on the work bench and remove sleeves (9) from the valve body.
14. Remove and discard seal (10), O-rings (22) and (24), and backup rings (21) and (23).
15. Remove spring seats (13) and (15) and regulator springs (14).

### Cleaning and Inspection

1. Clean all metal parts with solvent and air dry.
2. Apply a light film of type C-4 hydraulic oil to plungers (16, Figure 3-10) and insert them into sleeves (17). The sleeves must slide smoothly and freely in the sleeve bores. If the parts do not slide smoothly or excessive wear is apparent, replace both the sleeve and plunger.
3. Apply a light film of oil to regulator spools (12) and slide them into sleeves (9). The spools must slide smoothly and freely in the sleeve bores. If the parts do not slide smoothly or excessive wear is apparent, replace both the sleeve and spool.
4. Inspect each spring carefully for cracks and breaks. Any spring with a crack or break must be replaced. If the valve was not reaching the proper regulated pressure, replace the regulator springs.
5. Lubricate all parts with a thin coat of clean type C-4 hydraulic oil. Take care to keep the components protected from contamination.



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FIGURE 3-10. DUAL RELAY VALVE

- |                        |                       |
|------------------------|-----------------------|
| 1. Manifold Body       | 13. Lower Spring Seat |
| 2. Valve Body          | 14. Regulator Spring  |
| 3. Capscrew            | 15. Upper Spring Seat |
| 4. Packing             | 16. Plunger           |
| 5. Plug                | 17. Sleeve            |
| 6. Sleeve Retainer     | 18. O-Ring            |
| 7. Capscrews & Washers | 19. Plug              |
| 8. Reaction Plunger    | 20. Backup Ring       |
| 9. Sleeve              | 21. O-Ring            |
| 10. Seal               | 22. Backup Ring       |
| 11. Spool Spring       | 23. O-Ring            |
| 12. Regulator Spool    |                       |



## Assembly

1. Install sleeves (17, Figure 3-10) in the bores in the top of valve body (2).
2. Install plungers (16) in the sleeves as shown in Figure 3-10.
3. Apply a film of oil to O-rings (18) and position them in the grooves on top of the valve body.
4. Position manifold body (1) on valve body (2) by aligning the marks that were made during disassembly.
5. Secure the manifold to the valve body with two socket head capscrews (20). Tighten the capscrews finger-tight only.
6. Preassemble upper spring seat (15), spring (14) and lower spring seat (13). Insert this assembly into the bore from the bottom of the valve. Make sure that the upper spring seat is positioned against plunger (16). Repeat for the other bore.
7. Install sleeve packing seal (10). Refer to Details "A" and "B" in Figure 3-10 and install O-rings (22) and (24) and backup rings (21) and (23) in the grooves of sleeve (9).
8. Apply a light film of oil to the sleeve seals. Carefully push sleeves (9) into their respective bores in the valve body until the flange at the base of each sleeve contacts the valve body.
9. Preassemble regulator spool (12) as follows:
  - a. Insert spool springs (11) into the spool bore.
  - b. Insert reaction plungers (8) into the spool bores and springs.
10. Install the regulator spool assemblies into their respective sleeve bores. The spherical end must be inserted toward the spring seat. Push them into the bore until contact is made with the lower spring seat.
11. Install sleeve retainer plug packing (4) in the valve body.
12. Verify that spring seats (13) and (15) are positioned properly in regulator springs (14) and that reaction plunger (8) slides smoothly in its bore. Install retaining plug (5) in the valve body counterbore.
13. Position sleeve retainer (6) on the valve body. Install four capscrews and washers (7). Tighten the capscrews evenly to properly seat plug (5) in the counterbore. Then tighten the capscrews to a final torque of **16 - 17 N·m (140 - 150 in lb)**.
14. Tighten two capscrews (3) that hold manifold body (1) to valve body (2) to **20 - 21 N·m (180 - 190 in lb)**.
15. Install plugs (19) in the manifold body ports. Tighten the larger plugs to **31 - 34 N·m (275 - 300 in lbs)**. Tighten the smaller plugs (installed in the "TC1" and "TC2" ports) to **10 - 11 N·m (90 - 100 in lb)**.

## DUAL RELAY VALVE BENCH TEST AND ADJUSTMENT

The following parts and test equipment will be required to completely bench test the dual relay valve.

- Four 20 680 kPa (3,000 psi) pressure gauges
- Hydraulic pressure supply, regulated to 18 960 kPa (2,750 psi)
- Hydraulic test stand (Refer to Figure 3-11)

- Hose fittings for valve ports:  
Port PX . . . . . 7/16 in., # 4 SAE  
Ports B1 and B2 . . . . . 3/4 in., #8 SAE  
Port T . . . . . 1 1/16 in., #12 SAE

- Ohmmeter or continuity tester

*NOTE: It is possible to check the pressures with the dual relay valve installed on the truck by using the brake treadle valve to modulate pilot pressure and monitoring brake apply pressure in the appropriate brake apply pressure lines.*

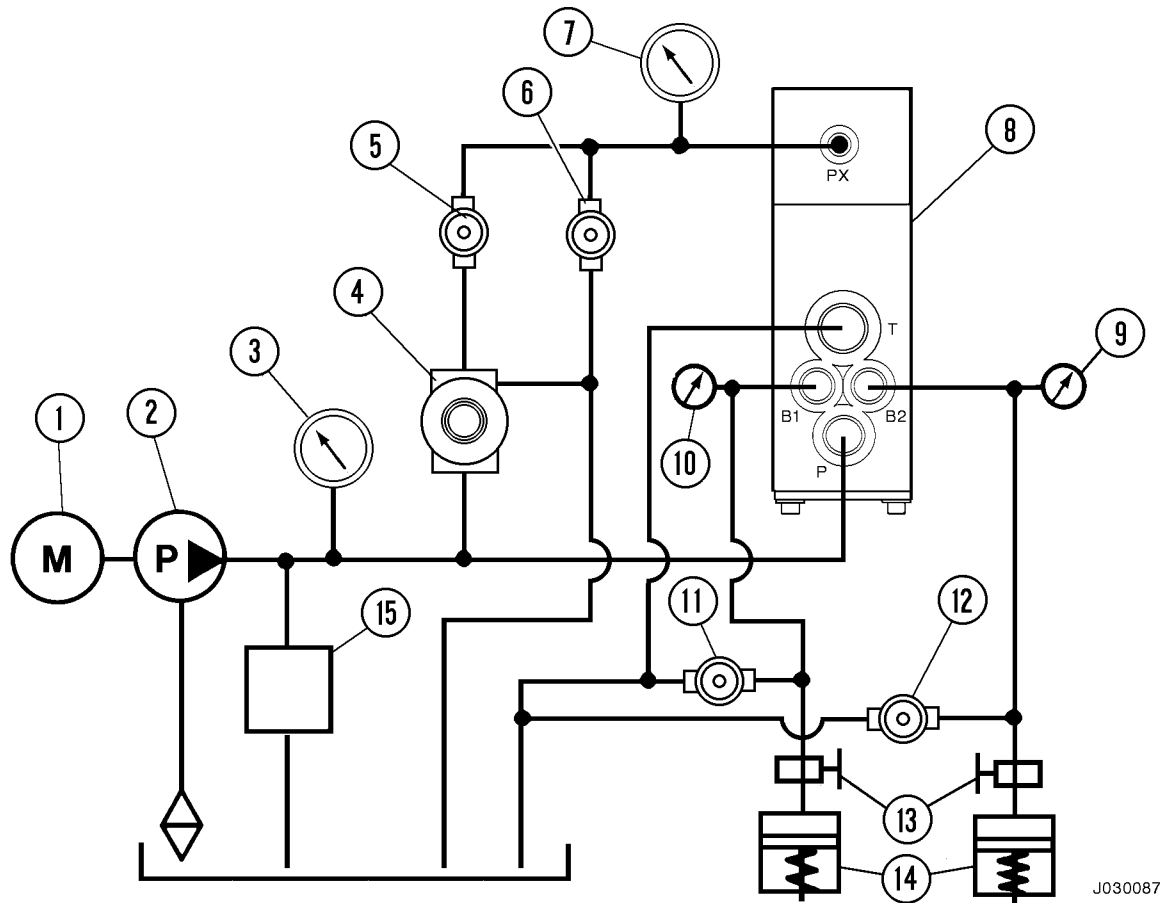


FIGURE 3-11. DUAL RELAY VALVE BENCH TEST SETUP

- |                                             |                                             |                                      |
|---------------------------------------------|---------------------------------------------|--------------------------------------|
| 1. Motor                                    | 6. Needle Valve<br>(Pressure Bleed to Tank) | 10. RH Brake Apply Pressure<br>Gauge |
| 2. Pump                                     | 7. Pilot Pressure Gauge                     | 11. Needle Valve                     |
| 3. Main Pressure Gauge                      | 8. Dual Relay Valve                         | 12. Needle Valve                     |
| 4. Pressure Regulator<br>(Pilot Pressure)   | 9. LH Brake Apply Pressure<br>Gauge         | 13. Shutoff Valves                   |
| 5. Needle Valve<br>(Pilot Pressure Release) |                                             | 14. Simulated Brake Volume           |
|                                             |                                             | 15. Relief Valve                     |

## Test Setup Procedure

1. Position the valve on the test stand as shown in Figure 3-11.
2. Attach the pilot input supply line to the port labeled "PX" on the side of the valve.
3. Attach the main supply input pressure line to the port on the front of the valve labeled "P".
4. Attach the tank return line to the port labeled "T".
5. Attach the regulated output ports "B1" and "B2" to the test lines. Pressure monitoring devices in these two lines must be capable of 20 680 kPa (3,000 psi). Connect all ports according to the diagram shown in Figure 3-11. All ports must be used and connected.

# DANGER

***All ports must be used. Relieve pressure before disconnecting hydraulic and other lines. Tighten all connections before applying pressure.***

***Avoid spillage and contamination! Avoid contact with hot oil if the machine has been operating. The oil will be at very high pressure.***

***Hydraulic fluid escaping under pressure can have sufficient force to enter a person's body by penetrating the skin and cause serious injury and possibly death if proper medical treatment by a physician familiar with this injury is not received immediately.***

6. Start the hydraulic pump and regulate output pressure at 18 960 kPa (2,750 psi) at pressure gauge (3). Pressure gauges (9) and (10) should read zero.
7. Adjust pressure regulator (4) to set pilot supply pressure to 17 235 kPa (2,500 psi) on gauge (7).
8. Return line pressure during this test is not to exceed zero.
9. Test the valve with ISO 32 grade hydraulic oil at  $49^{\circ} \pm 3^{\circ} \text{ C}$  ( $120^{\circ} \pm 10^{\circ} \text{ F}$ ).

## Relay Valve Output Pressure Adjustment

1. With the pump operating, and supply pressure and pilot pressure adjusted as described in the Test Setup Procedure, inspect the valve for leakage.
2. With 17 235 kPa (2,500 psi) of pilot pressure applied, verify that pressure gauges (9) and (10) read  $17\,235 \pm 517 \text{ kPa}$  ( $2,500 \pm 75 \text{ psi}$ ).
3. Close pilot supply needle valve (5) and open pilot pressure release needle valve (6) to bleed pressure back to the reservoir. Pilot pressure gauge (7) and regulated output pressure gauges (9) and (10) should drop to zero.
4. Repeat Steps 2 and 3 approximately 50 times to cycle the valve from minimum to maximum apply pressure.
5. Verify that the output pressure remains within specifications. If not, the valve must be rebuilt.
6. While observing pilot pressure gauge (7) and regulated output pressure gauges (9) and (10), apply pilot pressure slowly and steadily until 17 235 kPa (2,500 psi) maximum pilot pressure is obtained.

Pilot pressure and regulated output pressure must track within 345 kPa (50 psi) after the pilot pressure reaches 690 kPa (100 psi).

7. Reduce pilot pressure to zero. Apply 17 235 kPa (2,500 psi) of pilot pressure as quickly as possible. Regulated output pressure must increase to  $17\,235 \pm 517 \text{ kPa}$  ( $2,500 \pm 75 \text{ psi}$ ) within one second after pressure is applied to the pilot line.
8. Check for internal valve leakage from port "T" with full supply pressure (port "P") applied.
  - ☐ With pilot pressure released, leakage must not exceed 100 cc/minute.
  - ☐ With 17 235 kPa (2,500 psi) of pilot pressure applied, leakage must not exceed 150 cc/minute.

## BRAKE MANIFOLD

### Removal



**Accumulators maintain high pressure. DO NOT disconnect any hydraulic line or service any brake system components until all hydraulic pressure has been manually drained from the accumulators. Open the manual drain valves located on the brake manifold in the brake cabinet to relieve pressurized oil. The manual bleddown valve for the rear accumulator is identified as "NV1". The manual bleddown valve for the front accumulator is identified as "NV2".**

**NOTE:** If the brake manifold is leaking oil, a single O-ring or cartridge can be replaced while the manifold is still installed on the truck.

1. Securely block the wheels to prevent the truck from rolling away.
2. Place the directional control lever in PARK and turn the rest switch ON. Turn the key switch OFF to shut down the engine and allow 90 seconds for the steering system accumulators to bleed down. Open bleed down valves (1) and (2, Figure 3-12) to bleed down the brake accumulators.
3. Verify that the brake accumulators are bled down by pressing the brake lock switch (key switch ON, engine off) and applying the service brake pedal. The service brake light should not come on.
4. Close the bleed down valves by rotating them clockwise.
5. If a brake manifold cartridge is faulty or leaking oil externally, remove the cartridge. Inspect the O-rings and the seats in the manifold. Replace the O-rings and any defective parts.

**NOTE:** If the manifold must be removed from the truck, disconnect only the hydraulic lines and wires that are necessary to allow removal.

6. Plug all lines and ports to prevent possible contamination.
7. Remove the mounting hardware and move the brake manifold to a clean work area for disassembly.

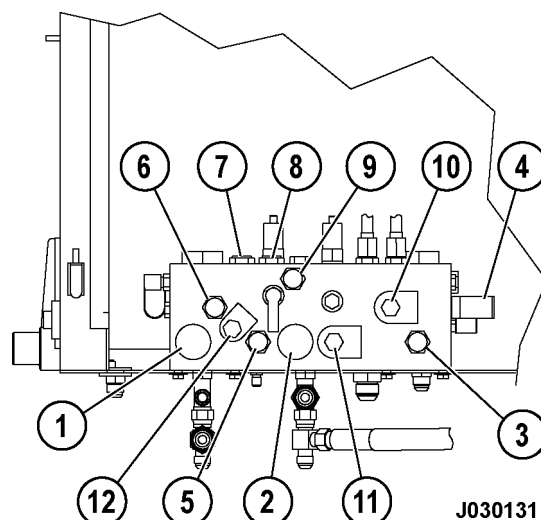


FIGURE 3-12. BRAKE MANIFOLD

1. Accumulator Bleed Down Valve (Rear Brake)
2. Accumulator Bleed Down Valve (Front Brake)
3. Check Valve
4. Pressure Reducing Valve

### Installation

1. Place the brake manifold into position in the hydraulic brake cabinet. Install and tighten the mounting hardware to the standard torque.
2. Connect all lines and electrical connections to the proper locations.
3. Start the engine. Check for leaks and for proper operation.
4. Shut off the engine and make sure that the hydraulic tank is filled to the proper level.

### Disassembly

1. Mark all plugs, valves and cartridges before removal to ensure proper assembly.
2. Remove the plugs, valves and cartridges as needed.

**NOTE:** Check valve (3, Figure 3-12) and reducing valve (4) both have an orifice disc located below them. The two orifices are different sizes. Therefore, it is very important to properly match each orifice with the correct installation location.

## Cleaning and Inspection

1. After disassembly, clean all parts with an approved cleaning solution.
2. Blow all parts dry with air and protect from dust and any foreign matter until installation.
3. Replace all O-rings and any other items deemed unsuitable for further usage.

## Assembly

1. Install new O-rings on all components that were removed from the manifold.
2. Coat all bores, cartridges and O-rings with clean C-4 hydraulic oil.
3. Before installing check valve (3, Figure 3-12) or reducing valve (4), refer to Figure 3-13 for proper orifice disc installation. The orifice discs must be installed in the direction shown for proper operation.
  - Check valve orifice - 1.58 mm (0.062 in.)
  - Reducing valve orifice - 2.54 mm (0.100 in.)
4. Install all cartridges in the bores from which they were removed. Tighten them securely.

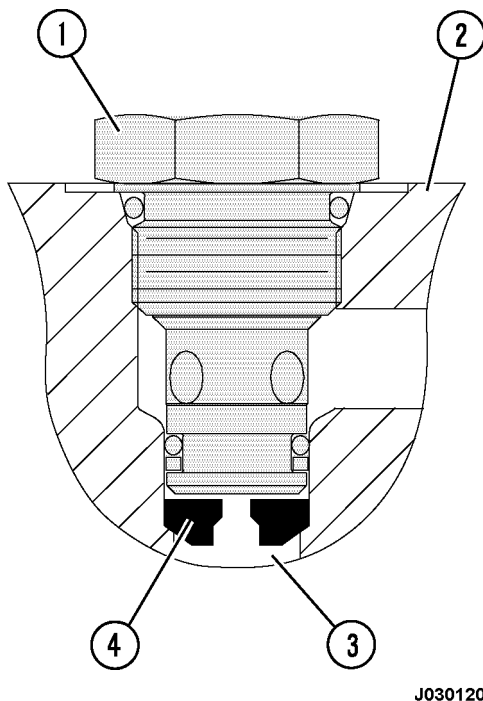


FIGURE 3-13. ORIFICE INSTALLATION

- |                   |                 |
|-------------------|-----------------|
| 1. Cartridge      | 3. Cavity       |
| 2. Brake Manifold | 4. Orifice Disc |

## BRAKE ACCUMULATORS

Two bladder type accumulators provide a volume of oil for the service brake system. The accumulators each have a capacity of 37.9 liters (10 gallons) of oil and are located on the right frame rail to the rear of the front tire. The front accumulator is for the front brake circuit and the rear accumulator is for the rear brake circuit.

## **⚠ DANGER**

***Accumulators maintain high pressure. DO NOT disconnect any hydraulic line from the accumulators or brake system until all hydraulic pressure has been manually drained from accumulators. Open manual drain valves located on the brake manifold in the brake cabinet to drain pressurized oil. The manual bleddown valve for the rear accumulators is identified as "NV1". The manual bleddown valve for the front accumulators is identified as "NV2".***

### Brake Accumulator Bleeddown Procedure

The front and rear brake circuit accumulators can be bled down by rotating the manual bleddown valves counterclockwise. The valves are located on the brake manifold in the hydraulic brake cabinet.

1. Turn accumulator bleed down valves (1) and (2, Figure 3-12) counterclockwise to open the valves. Left valve (1) will bleed the rear brake circuit and right valve (2) will bleed the front brake circuit accumulator.
2. Verify that the brake accumulators are bled down by pressing the brake lock switch (key switch ON, engine off) and applying the service brake pedal. The service brake light should not come on.
3. Close the bleed down valves by rotating them clockwise.

## Removal

1. Securely block the wheels to prevent the truck from rolling away.
2. Place the directional control lever in PARK and turn the rest switch ON. Turn the key switch OFF to shut down the engine and allow 90 seconds for the steering system accumulators to bleed down. Open bleed down valves (1) and (2, Figure 3-12) to bleed down the brake accumulators.
3. Remove protective cover (3, Figure 3-14) from the charging valve guard on top of the accumulator.
4. Install a charging valve kit onto the accumulator charging valve. Use the charging kit to release nitrogen from the accumulator that is to be removed.
5. Disconnect oil line (6) from the bottom hydraulic port. Cap the port and hose fitting to prevent contamination.
6. Attach a lifting device to accumulator.



***Each accumulator weighs approximately 100 kg (220 lb). Ensure that the lifting device has adequate capacity for handling the load.***

7. Remove accumulator mounting clamps (5) and lift the accumulator off the mounting pad. Move the accumulator to a clean work area.

## Installation

1. Position the accumulator on mounting bracket (7, Figure 3-14) with warning label visible.
2. Install mounting clamps and hardware. Tighten capscrews to standard torque.
3. Install oil line (6) at bottom (hydraulic) port.
4. Refer to Accumulator Charging Procedure in this section.
5. Install protective cover (3) over charging valve on top of accumulator.

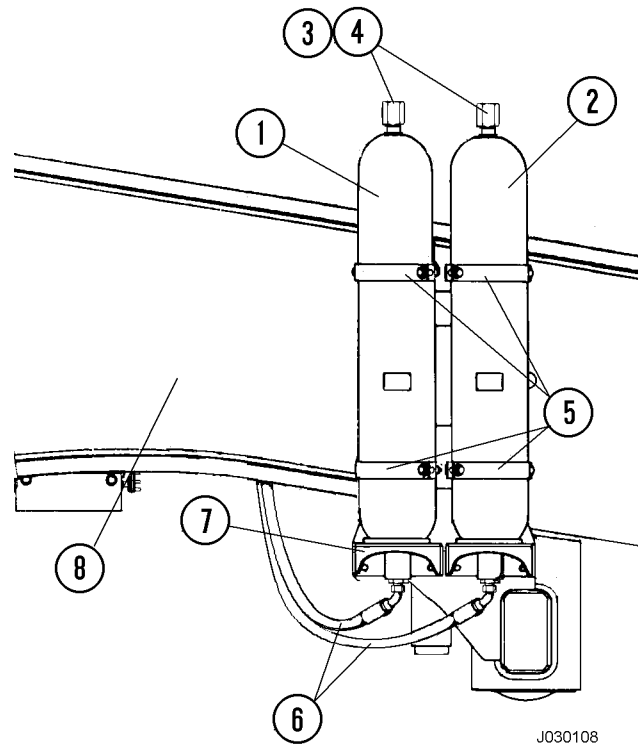
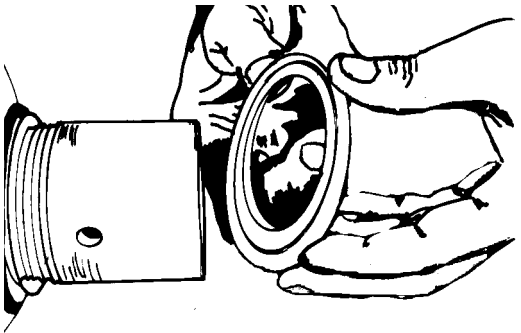


FIGURE 3-14. BRAKE SYSTEM ACCUMULATORS

- |                                    |                     |
|------------------------------------|---------------------|
| 1. Rear Brake Circuit Accumulator  | 4. Charging Valve   |
| 2. Front Brake Circuit Accumulator | 5. Mounting Clamps  |
| 3. Protective Cover                | 6. Oil Lines        |
|                                    | 7. Mounting Bracket |
|                                    | 8. R.H. Frame Rail  |

## Disassembly

1. Secure accumulator shell (10, Figure 3-17) with a chain wrench or similar device to prevent rotation during disassembly.
2. Remove cap (14). Verify that the nitrogen has been released and remove charging valve (11). Remove bleed plug (2) from hydraulic port assembly (1).
3. Use a spanner wrench to remove locking ring (3) from the hydraulic port assembly. Use an adjustable wrench on the flats of the hydraulic port assembly to prevent the port assembly from rotating.
4. Remove spacer (4) as shown in Figure 3-15. Then push the hydraulic port assembly into the shell.



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FIGURE 3-15. SPACER REMOVAL

5. Insert a hand into the accumulator shell and remove O-ring backup (5), O-ring (6) and metal backup washer (7).
6. Separate anti-extrusion ring (8) from the hydraulic port.
7. Fold the anti-extrusion ring and remove it from the shell as shown in Figure 3-16.

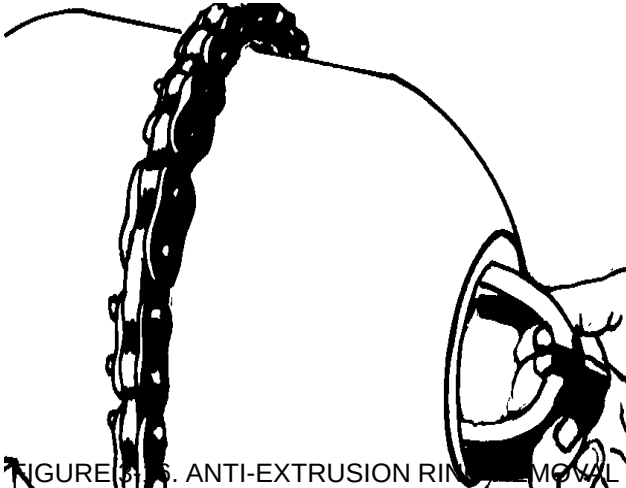
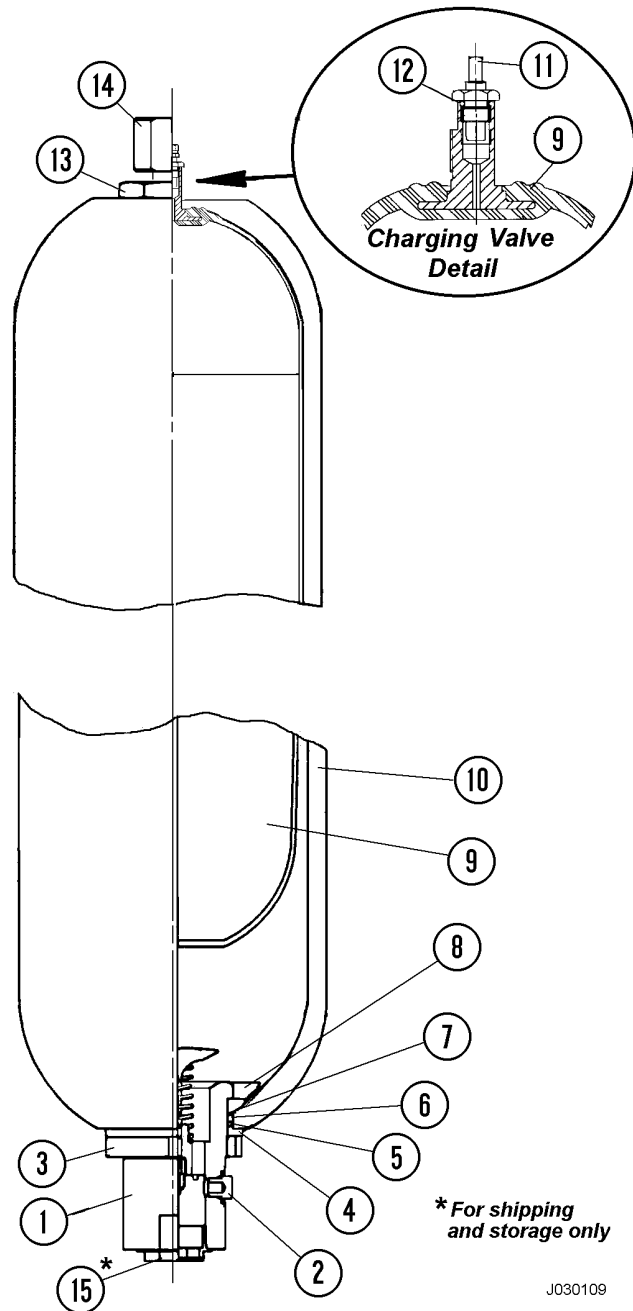


FIGURE 3-16. ANTI-EXTRUSION RING REMOVAL



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FIGURE 3-17. ACCUMULATOR ASSEMBLY

- |                            |                        |
|----------------------------|------------------------|
| 1. Hydraulic Port Assembly | 8. Anti-Extrusion Ring |
| 2. Bleed Plug              | 9. Bladder Assembly    |
| 3. Locking Ring            | 10. Shell              |
| 4. Spacer                  | 11. Charging Valve     |
| 5. O-Ring Backup           | 12. O-Ring             |
| 6. O-Ring                  | 13. Locknut            |
| 7. Metal Backup Washer     | 14. Protective Cap     |
|                            | 15. Port Protector*    |

8. Remove the hydraulic port from the shell as shown in Figure 3-18.

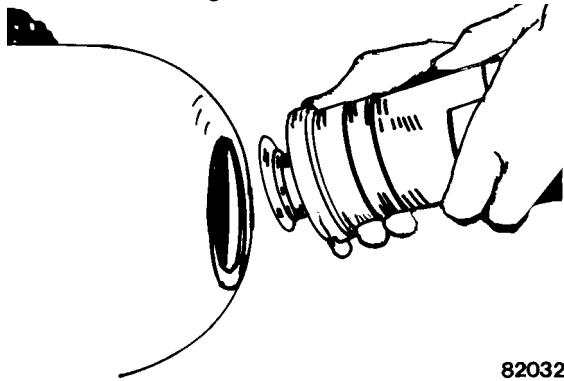


FIGURE 3-18. HYDRAULIC PORT REMOVAL

9. At the opposite end of the accumulator assembly, remove locknut (13) from the bladder valve stem.
10. Reach inside the shell at the hydraulic port end and compress the bladder to expel as much air as possible.
11. Fold the bladder and pull it out of the bottom of the accumulator shell using a twisting motion. A cloth may keep your hand from slipping due to the oil film on the bladder.

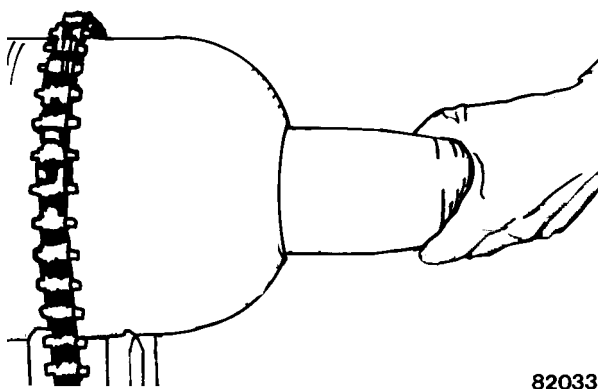


FIGURE 3-19. BLADDER REMOVAL

### Cleaning and Inspection

1. All metal parts should be cleaned with a cleaning agent.
2. Seals and soft parts should be wiped clean.
3. Inspect the hydraulic port assembly for damage. Check the poppet plunger to see that it spins freely and functions properly.
4. Check the anti-extrusion ring and soft seals for damage and wear. Replace all worn or damaged seals with original equipment seals.
5. After the shell has been cleaned with a cleansing agent, check the inside and outside of the shell. Pay special attention to the area where the gas valve and hydraulic assembly pass through the shell. Any nicks or damage in this area could destroy the accumulator bladder or damage new seals. If this area is pitted, consult your Komatsu service manager.

### Assembly

1. After shell (10, Figure 3-17) has been cleaned and inspected, secure it in place to prevent rotation during assembly.
2. Apply 2 liters (64 oz.) of clean type C-4 hydraulic oil inside the shell to lubricate and provide a cushion for the bladder.
3. With all gas completely exhausted from bladder (9), collapse the bladder and roll it longitudinally into a compact roll. To keep the bladder rolled up, insert the gas valve core to prevent air from entering the bladder.
4. Insert the bladder pull rod through the valve stem opening and through the shell hydraulic port. Attach the bladder pull rod to the bladder valve stem.
5. With one hand, pull the bladder pull rod while feeding the bladder into the shell with the other hand. A slight twisting of the bladder will ease installation.
6. Once the bladder valve stem has been pulled through the valve stem opening in the shell, install the name plate (if used) over the valve stem and install valve stem nut (13) by hand.
7. Once valve stem nut (13) is in place, remove the bladder pull rod. Tighten the nut to **76 N·m (56 ft lb)**.
8. Grasp hydraulic port assembly (1) at the threaded end and insert the poppet end into the shell. Lay the assembly inside the shell.



9. Fold anti-extrusion ring (8) to enable insertion through the shell opening, then insert the ring into the shell. Once the anti-extrusion ring has cleared the shell opening, place the ring on the poppet assembly with the steel collar facing toward the shell hydraulic oil port.
10. Pull the threaded end of the hydraulic port assembly through the shell until it seats solidly into position in the shell fluid port opening.
11. With the hydraulic port assembly firmly in place, install the charging valve into the bladder stem.
12. Slowly pressurize the bladder with dry nitrogen. Use a sufficient pressure of 275 - 345 kPa (40 - 50 psi) to hold the poppet assembly in place.
13. Install metal backup washer (7) over the poppet assembly. Push the washer into the shell fluid port until it has bottomed out on anti-extrusion ring (8).
14. Install O-ring (6) over the poppet assembly. Push it into the shell fluid port until it has bottomed out against washer (7). DO NOT twist the O-ring.
15. Install O-ring backup (5) over the poppet assembly. Push it until it bottoms against O-ring (6).
16. Insert spacer (4) with the smaller diameter of the shoulder facing the shell.
17. Install locking ring (3) on the poppet assembly and tighten it. This will squeeze the O-ring into position. Use a wrench on the flats of the port assembly to prevent it from rotating. Tighten the nut to a final torque of **373 N·m (275 ft lb)**.

18. Release all of the nitrogen from the bladder.
19. Install bleed plug (2) and tighten it to **14 N·m (10 ft lb)**.
20. Pour approximately 4 liters (1 gallon) of clean Type C-4 hydraulic oil into the accumulator through the hydraulic port.

*NOTE: The hydraulic oil added in Step 20 will act as a cushion when the accumulator is installed on the truck and precharged with nitrogen.*

21. Precharge the accumulator to 690 - 827 kPa (100 - 120 psi). Refer to Accumulator Charging Procedure for more details.
22. After precharging, install a plastic cover over the hydraulic port to prevent contamination. DO NOT use a screw-in type plug.



***Always store bladder accumulators with a 690 - 827 kPa (100 - 120 psi) nitrogen precharge pressure. Do not exceed 827 kPa (120 psi). Storing accumulators with more than 827 kPa (120 psi) pressure is not safe in case of leaks.***

*NOTE: Bladder accumulators should be stored with a 690 - 827 kPa (100 - 120 psi) precharge, which fully expands the bladder and holds oil against the inner walls for lubrication and to prevent rust formation.*

23. Tighten cap (14) to **19 N·m (14 ft lb)**.

## Accumulator Charging Procedure

### **WARNING**

**Do not loosen or disconnect any hydraulic line or component until the engine is stopped and the key switch has been OFF for at least 90 seconds.**

**Pure dry nitrogen is the only gas approved for use in the brake accumulator. The accidental charging of oxygen or any other gas in this compartment may cause an explosion. Ensure that pure dry nitrogen gas is being used to charge the accumulator.**

**Before charging or discharging nitrogen gas in the accumulator, carefully read and understand the warning labels and instructions regarding the charging valve.**

**Only precharge accumulators to 9 653 kPa (1,400 psi) while installed on the truck. Never handle an accumulator with lifting equipment if the nitrogen precharge pressure is over 827 kPa (120 psi). Always set the precharge to 690 - 827 kPa (100 - 120 psi) before removing or installing the accumulators.**

**Ensure that the nitrogen supply is shut off before attaching the charging manifold to the nitrogen container.**

**NOTE:** If one accumulator is low on nitrogen, it is recommended that both accumulators be checked and charged at the same time. Correct precharge pressure is the most important factor in prolonging accumulator life.

**NOTE:** Use only nitrogen that meets or exceeds CGA (Compressed Gas Association) specification G-10.1 for type 1, grade F. The nitrogen should be 99.9% pure. Use only nitrogen cylinders with standard CGA number 580 connections with the appropriate high pressure regulator.

1. With the engine off and key switch in the OFF position, allow at least 90 seconds for the accumulator to bleed down. Turn the steering wheel to ensure that no pressurized oil remains in the accumulators.
2. Open bleed valves bleed down valves (1) and (2, Figure 3-12) in the hydraulic brake cabinet to release hydraulic pressure from the brake accumulators. Ensure that all hydraulic pressure has been relieved from the hydraulic system. Remove the bleeder plugs in the steering accumulator stems to vent any residual pressure.

**NOTE:** If a new or rebuilt accumulator (or any bladder accumulator with all nitrogen discharged) is being precharged while installed on the truck and connected to the hydraulic system, the oil side of the accumulator must be vented to allow proper bladder expansion. Trapped air or oil on the hydraulic side of the bladder will prevent the proper precharge pressure to be obtained for safe operation.

3. Remove the protective cap from the accumulator gas valve.
4. Close all shutoff valves on brake manifold (4, Figure 3-20).
5. Connect the charging kit to nitrogen container (1), and open the valve on the tank.
6. Turn the "T" handle on charging valve adapter (6) all the way out (counterclockwise).

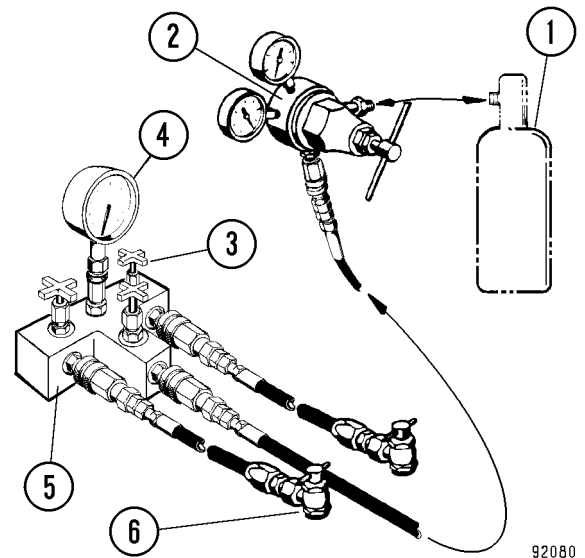


FIGURE 3-20. NITROGEN CHARGING KIT

- |                       |                           |
|-----------------------|---------------------------|
| 1. Nitrogen Container | 4. Pressure Gauge         |
| 2. Regulator          | 5. Brake Manifold         |
| 3. Shutoff Valve      | 6. Charging Valve Adaptor |

7. Close any bleed valves that are open.
8. Attach the charging valve adapter to the accumulator gas valve. Make sure that the hose does not loop or twist. Tighten the swivel nut to **14 - 20 N·m (10 - 15 in lb)**.
9. Turn the "T" handle clockwise to open the accumulator gas valve. Do not turn the "T" handle all the way down as it will damage the valve core.
10. Set the regulator for 172 kPa (25 psi), then slightly open the nitrogen container and slowly fill the accumulator. The proper fill rate for these 10 gallon accumulators is approximately three minutes.

## CAUTION

*If the precharge is not added slowly, the bladder may suffer permanent damage. A "starburst" rupture in the lower end of the bladder is a characteristic failure caused by charging too quickly.*

11. When 172 kPa (25 psi) of precharge pressure is obtained, close the nitrogen valve. Set the regulator for the operating precharge pressure based on the current ambient temperature. Refer to Table 1. Then, open the nitrogen container and fill the accumulator.
12. Let the precharge set for 15 minutes. This will allow the gas temperature to stabilize. If the desired precharge is exceeded, close the nitrogen container valve and slowly open the bleed valve until the correct precharge pressure is obtained. Refer to Table 1.

## CAUTION

*Do not reduce the precharge by depressing the valve core with a foreign object. High pressure may rupture the rubber valve seat.*

13. Hold the gas valve stationary and loosen the swivel nut to remove the assembly. Use a common leak reactant to check for nitrogen leaks.
14. Install the gas valve cap finger-tight. The gas valve cap serves as a secondary seal.

Temperature variation can affect the precharge pressure of an accumulator. As the temperature increases, the precharge pressure increases. Conversely, decreasing temperature will decrease the precharge pressure. In order to ensure the accuracy of the accumulator precharge pressure, the temperature variation must be accounted for.

A temperature variation factor is determined by the ambient temperature when charging the accumulator on a truck that has been shut down for one hour. Refer to Table 1 for charging pressures in different ambient operating conditions that the truck is currently exposed to during the charging procedure.

For example, assuming that the ambient temperature is 10°C (50°F), charge the accumulator to 9 294 kPa (1,348 psi).

**TABLE 1. Relationship Between Charging Pressure and Ambient Temperature**

| Ambient Temperature     | Charging Pressure<br>± 70 kPa (10 psi) |
|-------------------------|----------------------------------------|
| -23°C (-10°F) and below | 8 232 kPa (1194 psi)                   |
| -17°C (0°F)             | 8 412 kPa (1220 psi)                   |
| -12°C (10°F)            | 8 584 kPa (1245 psi)                   |
| -7°C (20°F)             | 8 763 kPa (1271 psi)                   |
| -1°C (30°F)             | 8 943 kPa (1297 psi)                   |
| 4°C (40°F)              | 9 122 kPa (1323 psi)                   |
| 10°C (50°F)             | 9 294 kPa (1348 psi)                   |
| 16°C (60°F)             | 9 473 kPa (1374 psi)                   |
| 21°C (70°F)             | 9 653 kPa (1400 psi)                   |
| 27°C (80°F)             | 9 832 kPa (1426 psi)                   |
| 32°C (90°F)             | 10 011 kPa (1452 psi)                  |
| 38°C (100°F)            | 10 184 kPa (1477 psi)                  |
| 43°C (110°F)            | 10 363 kPa (1503 psi)                  |
| 49°C (120°F)            | 10 542 kPa (1529 psi)                  |

*NOTE: A precharge pressure below 8232 kPa (1194 psi) is not recommended because of low precharge pressure warnings. The low accumulator precharge warning switch activates at 7584 ± 310 kPa (1100 ± 45 psi).*

## Precharge Maintenance



***If the low accumulator precharge warning light is illuminated when the key switch is turned ON, do not attempt to start the truck. Permanent bladder damage may result. Check the accumulator precharge pressure and adjust it if necessary.***

1. When starting the truck, turn key switch to the RUN position and wait to confirm that the low accumulator precharge warning light does not stay illuminated after system check is complete. If the warning light stays illuminated, do not start or drive the truck and notify maintenance personnel.
2. Check the accumulator precharge pressure every 500 hours. If the precharge pressure is too low, the bladder will be crushed into the top of the shell by hydraulic system pressure and can extrude into the gas stem and become punctured. This condition is known as "pick out". One such cycle is sufficient to destroy a bladder.

3. Check all sealing areas on the nitrogen side of the accumulator (charging valve, pressure switch, manifold, etc.) during every precharge maintenance interval to ensure that the seals do not leak. Replace all faulty or leaking seals, valves, etc. Failure to repair leaking nitrogen seals may result in a failed accumulator bladder or low performance from the accumulator.

***NOTE: If precharge pressure continues to decline frequently between precharge maintenance intervals, and if all nitrogen sealing areas are free of leaks, then the accumulator bladder most likely has a small hole in it and must be replaced.***

4. Check all heat shields and exhaust blankets, as provided, during every precharge maintenance interval to ensure that they are in place and good condition.

## Accumulator Storage Procedures

### **WARNING**

***Always store bladder accumulators with 690 - 827 kPa (100 - 120 psi) nitrogen precharge pressure. This amount of pressure fully expands the bladder and holds oil against the inner walls for lubrication and rust prevention. Do not exceed 827 kPa (120 psi). Storing accumulators with too much pressure is not safe due to possible leakage.***

***Only precharge accumulators to 9 653 kPa (1400 psi) while installed on the truck. Never handle an accumulator with lifting equipment if the nitrogen precharge pressure is over 827 kPa (120 psi). Always set the precharge to 690 - 827 kPa (100 - 120 psi) before removing or installing the accumulators.***

1. If the accumulator was just rebuilt, ensure that there is approximately 2 liters (64 oz.) of clean C-4 hydraulic oil inside the accumulator before adding 690 kPa (100 psi) of nitrogen precharge pressure.
2. Bladder accumulators should always be stored with 690 - 827 kPa (100 - 120 psi) of nitrogen precharge pressure, which fully expands the bladder and holds a film of oil against the inner walls for lubrication and rust prevention.
3. The hydraulic port should always be covered with a plastic plug to prevent contamination. **Do not use a screw-in type plug.**
4. Always store the accumulator in an upright position.

## Bladder Storage Procedures

The shelf life of bladders under normal storage conditions is one year. Normal storage conditions consist of the bladder being heat sealed in a black plastic bag and placed in a cool dry place away from the sun, ultraviolet and fluorescent lights, and electrical equipment. Direct sunlight or fluorescent light can cause the bladder to weather check and dry rot, which will appear on the bladder surface as cracks.

## Installing A Bladder Accumulator From Storage

1. Refer to the Charging Procedure to install the pressure gauges on the accumulator and to check the precharge pressure.
  - a. If the precharge pressure is 172 - 690 kPa (25 - 100 psi), set the regulator to 690 kPa (100 psi) and slowly charge the accumulator to 690 kPa (100 psi). Disconnect the pressure gauges from the accumulator and install the accumulator on the truck. Refer to Charging Procedure in this section to fully charge accumulator to the correct operating precharge pressure.
  - b. If the precharge pressure is less than 172 kPa (25 psi), slowly drain off any nitrogen precharge and proceed to Step 2.
2. Remove the gauges from the accumulator.
3. Lay the accumulator on a suitable work bench so that the hydraulic port is higher than the other end of the accumulator. Remove the plastic dust cap from the hydraulic port.
4. Pour approximately 2 liters (64 oz.) of clean C-4 hydraulic oil into the accumulator through the hydraulic port. Allow time for the oil to run down the inside of the accumulator to reach the other end.
5. Lay the accumulator flat on the work bench (or floor) and slowly rotate the accumulator two complete revolutions. This will thoroughly coat the accumulator walls with a film of oil necessary for bladder lubrication during precharging.
6. Stand the accumulator upright. Install the pressure gauges and refer to the Charging Procedure for instructions about charging the accumulator to 690 kPa (100 psi). Remove the gauges from the accumulator and install a plastic dust cap over the hydraulic port.
7. Install the accumulator on the truck.
8. Precharge the accumulator to the correct operating precharge pressure. Refer to Table 1.

## RETARDER CONTROL LEVER (STEERING COLUMN-MOUNTED)

Due to frequent use and wear, retarder control lever (5, Figure 3-22) may occasionally require adjustment or repair.

### Removal

Adjustment of the retarder control lever or replacement of the potentiometer requires removal of the assembly from the steering column.

1. Tilt the steering wheel downward and telescope the wheel toward the operator's seat. Remove the top cover from steering column assembly (4).
2. Remove capscrews (1, Figure 3-22), lockwashers (2) and bracket (3) from steering column (4).
3. Disconnect the wiring harness from harness connector (6) on the retarder control lever.
4. Remove retarder control lever (5).

### Installation

1. Connect the wiring harness to harness connector (6) on the retarder control lever.
2. Place retarder control lever (5) into position on steering column assembly (4).
3. Install capscrews (1), lockwashers (2) and bracket (3). Tighten the capscrews to **4 N·m (36 in lb)**.
4. Place the cover on the top of the steering column and return the steering wheel to its original position.

### Disassembly

1. Remove capscrews (7, Figure 3-23) and lockwashers (8) from housing (15).
2. Remove potentiometer (9).
3. Bend the tangs on washer (11) away from the slots in locknut (10).
4. If the retarder control lever is to be completely disassembled, loosen and remove locknut (10). Remove tang washer (11), spring (12), and washer (13).
5. Remove set screw (14).
6. Remove the lever and shaft assembly. If necessary, unscrew lever (17) from shaft (16) and handle (18).
7. Wash the mechanical parts in clean solvent and inspect for excessive wear, burrs, or scratches. Replace any defective parts.

### Lever Adjustments

#### Rotational Friction Adjustment

The lever assembly should be adjusted so that the frictional forces will hold the lever firmly in the position that is selected by the operator. At the same time, the adjustment should not be so tight as to cause the operator to use undue force to move the lever.

Loosen or tighten locknut (10) to attain the proper frictional force of **0.25 - 0.35 N·m (2 - 3 in lb)** at the handle. The position of the lever should remain stationary without moving from its own weight or due to machine vibrations during truck operation.

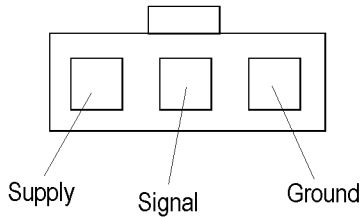
When the desired adjustment is obtained, bend the tang on washer (11) into the slots on locknut (10).

#### Detent Adjustment

The detent that holds the lever in the OFF position can be adjusted. Loosen or tighten set screw (14) to adjust the detent. The breakout force of the detent must be between **2.00 - 2.25 N·m (17 - 20 in lb)**.

## Potentiometer Check

Potentiometer (9, Figure 3-22) is spring-loaded to the OFF position. With the switch assembly removed from the retarder control lever, perform the following checks:



J030111

FIGURE 3-21. POTENTIOMETER CONNECTOR

1. Obtain a 10 volt power supply for testing the potentiometer.
2. Connect the positive lead of the 10 volt power supply to the SUPPLY terminal on the potentiometer connector as shown in Figure 3-21.
3. Connect the negative lead of the 10 volt power supply to the GROUND terminal on the potentiometer connector as shown in Figure 3-21.
4. Connect a voltmeter to the SIGNAL and GROUND connections.
5. Measure and record the signal voltage in the OFF position.
6. Rotate the potentiometer clockwise to the full ON position. Measure and record the signal voltage.
7. Determine whether the potentiometer falls within the specifications. When the potentiometer is in the OFF position, the signal voltage must be within 5 - 15% of input voltage. When the potentiometer is in the full ON position, the signal voltage must be within 75 - 95% of input voltage.

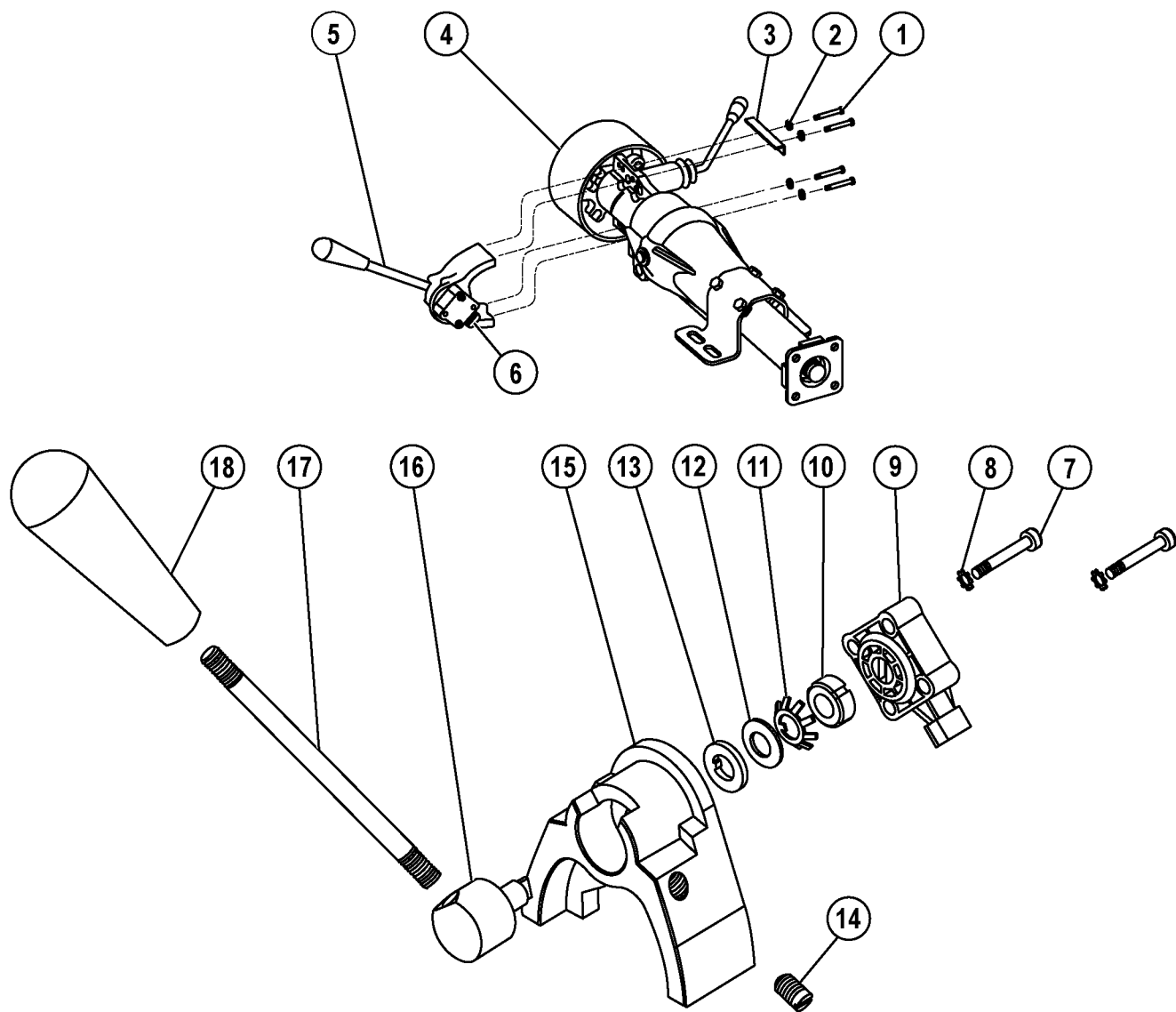
$$\% \text{ of input voltage} = \left( \frac{\text{signal voltage}}{\text{input voltage}} \right) \times 100$$

OFF position . . . . . 0.5 - 1.5 volts  
Full ON position . . . . . 7.5 - 9.5 volts

8. Replace the potentiometer if it does not meet these specifications.

## Assembly

1. Inspect the shaft bore and interior friction faces in housing (15). Remove any scratches or burrs, or replace the housing. Lightly lubricate the surfaces with a Multi-Purpose EP NLGI Consistency #2 grease.
2. If handle (18) or lever (17) has been removed from shaft (16), assemble as follows:
  - a. Apply Loctite™ #271 to the threads on each end of lever (17).
  - b. Install the lever into shaft (16). Hand tighten only.
  - c. Install handle (18) onto the lever. Hand tighten only.
3. Insert the lever, handle, and shaft assembly into housing (15).
4. Install washer (13). Install new spring (12) with the outer spring diameter against washer (13). Install tang washer (11) and locknut (10) onto shaft (16).
5. Tighten and adjust locknut (10) for proper lever resistance as described in the Rotational Friction Adjustment procedure.
6. Move the lever to the full OFF (up) position. Align the slot in potentiometer (9) with the key on shaft (16), and rotate the potentiometer until the capscrew holes line up with the housing. Install washers (8) and capscrews (7) to secure the potentiometer to the housing. Tighten the socket head capscrews to **1.35 - 1.70 N·m (12 - 15 in lb)**.
7. Install set screw (14). Refer to the Detent Adjustment procedure for proper detent adjustment.
8. Install the retarder control lever on the steering column. Refer to the Installation procedure.



J030112

FIGURE 3-22. COLUMN-MOUNTED RETARDER LEVER ASSEMBLY

- |                             |                   |                          |
|-----------------------------|-------------------|--------------------------|
| 1. Capscrew                 | 7. Capscrew       | 13. Internal Tang Washer |
| 2. Lockwasher               | 8. Lockwasher     | 14. Set Screw            |
| 3. Bracket                  | 9. Potentiometer  | 15. Housing              |
| 4. Steering Column Assembly | 10. Locknut       | 16. Shaft                |
| 5. Retarder Control Lever   | 11. Tanged Washer | 17. Lever                |
| 6. Harness Connector        | 12. Disc Spring   | 18. Handle               |



# NOTES

# **SECTION J4** **BRAKE CIRCUIT CHECK-OUT PROCEDURE** **INDEX**

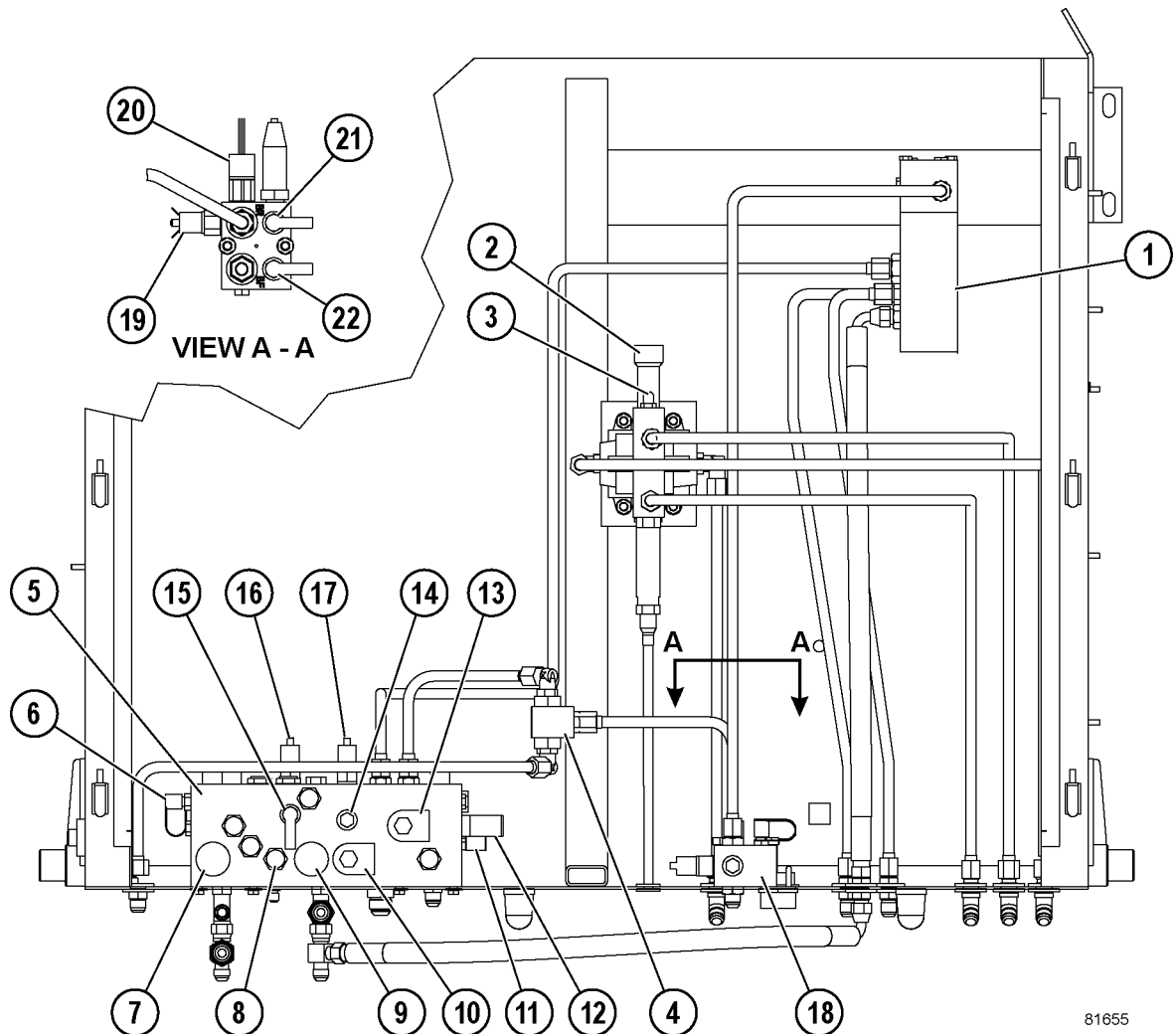
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# NOTES

## BRAKE CIRCUIT CHECK-OUT PROCEDURE

This section outlines a procedure to test the functionality of the entire brake system. Pressure specifications and other criteria that must be duplicated for the braking system to be considered ready for daily operations are provided. Refer to Brake Circuit and Brake Valve Troubleshooting for help in diagnosing a malfunctioning system.

Since the steering system supplies the brake system with operating pressure, it is important that the steering system be checked for proper operation before checking the braking system. Refer to Section L10, Hydraulic Check-Out Procedure, for information about checking the steering system.



81655

FIGURE 2-1. HYDRAULIC BRAKE CABINET

- |                                               |                                          |
|-----------------------------------------------|------------------------------------------|
| 1. Dual Relay Valve                           | 12. Pressure Reducing Valve (PR)         |
| 2. Hoist Pilot Valve                          | 13. Brake Lock Solenoid Valve (SV1)      |
| 3. Relief Valve (Hoist Power Down)            | 14. Parking Brake Release Pressure (PK2) |
| 4. Brake Lock Shuttle Valve                   | 15. Low Accumulator Test Port (LAP1)     |
| 5. Brake Manifold                             | 16. Low Brake Pressure Switch            |
| 6. Brake Oil Supply Pressure Test Port (SP3)  | 17. Parking Brake Pressure Switch        |
| 7. Rear Brake Accumulator Bleed Valve         | 18. Junction Manifold                    |
| 8. Automatic Apply Valve                      | 19. Stop Light Pressure Switch           |
| 9. Front Brake Accumulator Bleed Valve        | 20. Brake Lock Degradation Switch        |
| 10. Parking Brake Solenoid Valve (SV2)        | 21. Rear Brake Pressure Test Port (BR)   |
| 11. Brake Lock Apply Pressure Test Port (PP3) | 22. Front Brake Pressure Test Port (BF)  |

# **⚠ WARNING**

***Before disconnecting pressure lines, replacing components in the hydraulic circuits, or installing test gauges, always bleed down the steering and brake accumulators.***

The steering circuit can be isolated from the brake circuit by removing the brake supply line from the bleed down manifold. Plug the brake supply line and cap the port in the bleed down manifold.

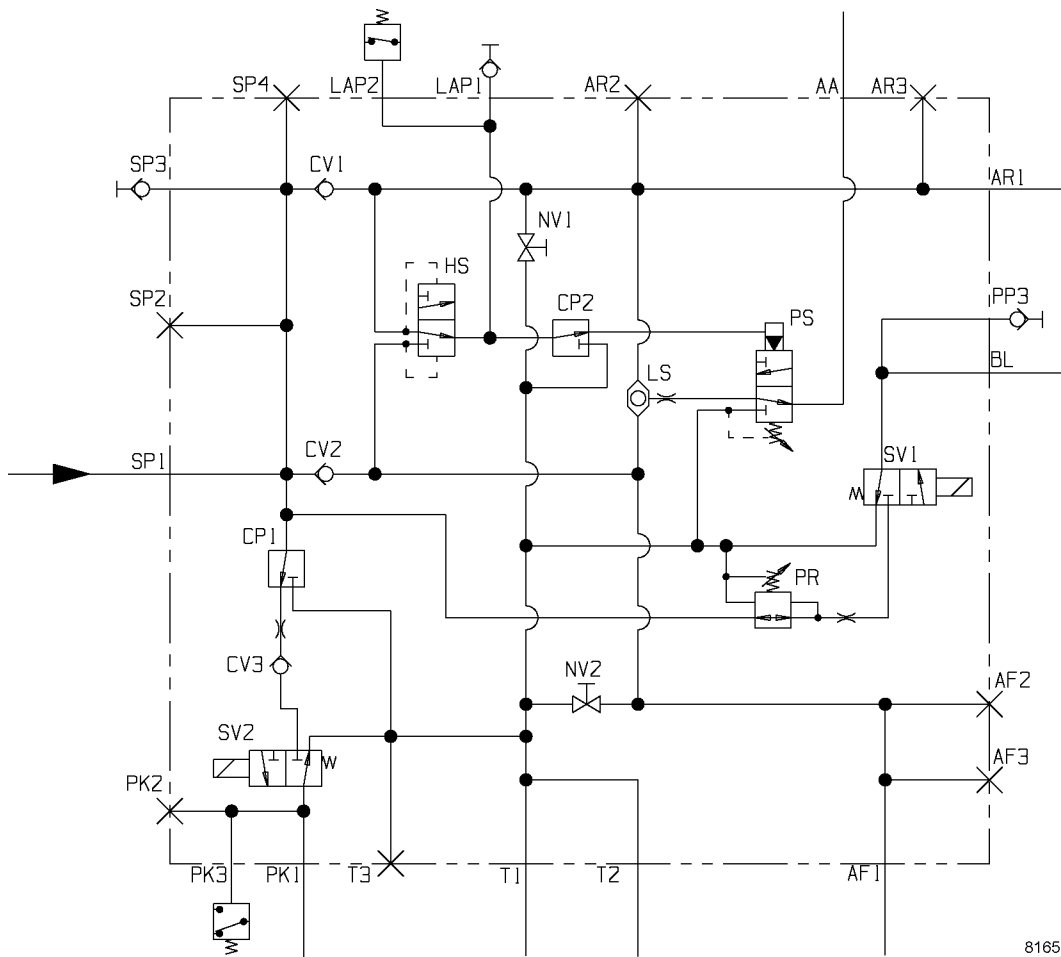
Included on the last pages of this module is a data sheet to record the information observed during the hydraulic brake system check-out procedure. The data sheet can be removed, copied, and used during the check-out procedure.

\* Steps indicated in this manner should be recorded on the data sheet for reference.

## **REQUIRED EQUIPMENT**

The following equipment will be necessary to properly check-out the hydraulic brake circuit:

- Hydraulic brake schematic (refer to Section R).
- Clear plastic hose and bucket for brake bleeding.
- Three 20 685 kPa (3,000 psi) pressure gauges.
- One PB6039 female quick disconnect and hose long enough to reach from brake cabinet to the inside of the operator's cab for each gauge.
- Accumulator charging kit (EB1759 or equivalent)



81654

FIGURE 4-2. BRAKE MANIFOLD SCHEMATIC

| BRAKE CIRCUIT ABBREVIATIONS |                                                                             |
|-----------------------------|-----------------------------------------------------------------------------|
| AA                          | Automatic Apply Pressure                                                    |
| AR1                         | Supply Pressure to Brake Valve for Rear Brakes, Rear Accumulator            |
| AF1                         | Supply Pressure to Brake Valve for Front Brakes, Front Accumulator          |
| BF                          | Pressure Tap Test Port Front Brake Pressure                                 |
| BL                          | Brake Lock Apply Pressure                                                   |
| BR                          | Pressure Tap Test Port Rear Brake Pressure                                  |
| CP1<br>CP2                  | Cavity Plug, One Direction Flow<br>No specific function in this application |
| CV1<br>CV2<br>CV3           | Check Valve                                                                 |
| HS1                         | High Pressure Shuttle Valve                                                 |
| LS1                         | Low Pressure Shuttle Valve                                                  |
| LAP1                        | Pressure Tap Test Port Low Accumulator Pressure                             |
| LAP2                        | Low Brake Pressure Switch<br>N.C., 12 750 ± 515 kPa (1,850 ± 75 psi)        |
| NV1                         | Rear Accumulator Manual Drain Valve                                         |
| NV2                         | Front Accumulator Manual Drain Valve                                        |
| PK1                         | Park Brake Release Pressure                                                 |
| PK3                         | Park Brake Pressure Switch<br>N.C., 8 620 kPa (1,250 psi)                   |
| PP3                         | Pressure Tap Test Port Brake Lock Pressure                                  |
| PR                          | Brake Lock Pressure Regulator<br>17 237 kPa (2,500 psi)                     |
| PS                          | Automatic Apply Valve<br>11 375 kPa (1,650 psi)                             |
| SP1                         | Supply Oil Inlet                                                            |
| SP3                         | Pressure Tap Test Port Supply Oil                                           |
| SV1                         | Brake Lock Solenoid                                                         |
| SV2                         | Parking Brake Solenoid                                                      |
| T1                          | Return To Tank                                                              |

## INITIAL SYSTEM SETUP

Before checking the brake system, the hydraulic steering system must be operating properly, have the correct accumulator precharge, and be at normal operating temperatures. Refer to Section L6, Steering Circuit Component Repair, for steering system operation troubleshooting procedures and specifications.

Ensure that the brakes have been properly bled to remove any trapped air. Refer to Section J5, Wet Disc Brake Assembly, for the brake bleeding procedure. Also, before checking the brake system, make sure that the parking brake is functioning properly.

1. Securely block the wheels to prevent the truck from rolling away.
2. Place the directional control lever in PARK and turn the rest switch ON. Turn the key switch OFF to shut off the engine and allow 90 seconds for the steering system accumulators to bleed down. Make sure that all hydraulic pressure is bled by turning the steering wheel.

**NOTE:** Leave rest switch in the ON position and the GF cutout switch in the CUTOUT position throughout brake tests.

3. Open bleed down valves (7) and (9, Figure 4-1) on the brake manifold to bleed down the brake accumulators.
4. Precharge both brake accumulators to 9 650 kPa (1,400 psi). Allow the gas temperature to approach ambient temperature before completing precharge process.

**NOTE:** For best performance, charge the accumulators in the ambient conditions in which the machine will be operating.

5. Close both accumulator bleeddown valves.
6. Install pressure gauges at:
  - a. Front brake test port "BF" (23, Figure 4-1) - 20 685 kPa (3,000 psi) gauge.
  - b. Rear brake test port "BR" (22, Figure 4-1) - 20 685 kPa (3,000 psi) gauge.
  - c. Low accumulator pressure test port "LAP1" (15, Figure 4-1) - 20 685 kPa (3,000 psi) gauge.
7. Release the brake lock.

8. Start the engine. Observe the rising brake pressures as the system charges. The auto apply valves should release the front and rear brakes at approximately 11 375 kPa (1,650 psi).

*\* Record on data sheet.*

9. Partially depress the brake pedal to bleed air from each brake.

10. Slowly depress the brake pedal. Force feedback of the pedal on your foot should be smooth with no abnormal noise or mechanical roughness.

- ☐ Verify that the stop lights illuminate at  $517 \pm 34$  kPa ( $75 \pm 5$  psi) rear brake pressure.

*\* Record on data sheet.*

11. Quickly and completely depress the brake pedal. Verify that within one second after the brake is applied:

- ☐ Rear brake (BR) pressure reads  $17\ 235 \pm 517$  kPa ( $2,500 \pm 75$  psi).
- ☐ Front brake (FR) pressure reads  $17\ 235 \pm 517$  kPa ( $2,500 \pm 75$  psi).
- ☐ Both pressures must remain above their minimum values for a minimum of 20 seconds.

*\* Record on data sheet.*

## **BRAKE LOCK / SECONDARY BRAKE CHECK-OUT**

12. Disconnect the lead wire from brake lock solenoid valve (13, Figure 4-1) located on the brake manifold in the hydraulic brake cabinet.

13. Apply the brake lock. The warning buzzer should sound and the service brake status light should illuminate.

14. Depress the brake pedal until the buzzer stops, then very slowly release the pedal until the buzzer sounds again.

- ☐ Verify that the rear brake (BR) pressure is  $6\ 895 \pm 172$  kPa ( $1,000 \pm 25$  psi) when the warning resumes.

*\* Record on data sheet.*

*NOTE: There is a two second delay between the time that the brake lock degradation switch senses the pressure and the status light and buzzer activate.*

15. Connect the lead wire to brake lock solenoid valve (13).

16. Cycle the brake lock switch several times to ensure crisp application and release of oil pressure and proper function of the service brake status light.

- ☐ Verify that the rear brake (BR) pressure is  $17\ 237 \pm 690$  kPa ( $2,500 \pm 100$  psi) when the brake lock is applied.

*\* Record on data sheet.*

17. To further check the functionality of the brake lock hydraulic and electrical systems (but not the rear service brakes themselves):

- a. Move the directional control lever to NEUTRAL. Cycle the brake lock switch several times to ensure proper function of the service brake status light.
- b. Move the directional control lever to PARK. Try to apply the brake lock. Verify that the parking brake status light remains illuminated but the service brake status light does not illuminate.
- c. Move the directional control lever to NEUTRAL. Apply the brake lock. Verify that the service brake status light remains illuminated but the parking brake status light does not illuminate. Move the directional control lever to PARK. Verify that the parking brake status light is illuminated but the service brake status light does not illuminate.
- d. Move the directional control lever to NEUTRAL, then pull up the engine shutdown switch. Verify that the parking brake status light is illuminated but the service brake status light does not illuminate.
- e. Start the engine and move the directional control lever to PARK. Energize the timed engine shutdown relay (RB7-K8) by grounding circuit 712M on TB28-X. (Ignore any faults that are generated.) Try to apply the brake lock. Verify that the service brake status light does not illuminate. Remove the ground connection.
- f. Pull up the engine shutdown switch to stop the engine. Connect a computer running wPTU software to the IM diagnostic port on the DID panel. Turn on the "Fwd True" output and apply the brake lock. Verify that the service brake status light remains illuminated. Turn off the brake lock. Verify that the service brake status light is off.

## PARKING BRAKE CHECK-OUT

18. Move one of the pressure gauges (BR or BF) to parking brake release (PK2) pressure port (14, Figure 4-1).

19. Apply the brake lock. Move the directional control lever to NEUTRAL. The parking brake status light should illuminate.

- ❑ Parking brake release (PK2) pressure should read  $20\ 685 \pm 690$  kPa ( $3,000 \pm 100$  psi).

*\* Record on data sheet.*

20. Cycle the parking brake several times (move the directional control lever from NEUTRAL to PARK) to ensure crisp application and release of oil pressure and proper function of the status light.

21. To further check the functionality of the brake hydraulic and electrical systems (but not the parking brake itself):

- a. Move the directional control lever to NEUTRAL, then turn off the engine. The parking brake status light should illuminate until the interface module powers down.
- b. Start the engine, move the directional control lever to PARK, then turn off the engine. The parking brake status light should illuminate until the interface module powers down.
- c. Start the engine, move the directional control lever to NEUTRAL, then pull up the engine shutdown switch. The parking brake status light should illuminate until the interface module powers down.



## SERVICE BRAKE CHECK-OUT

22. Move the directional control lever to PARK and release the brake lock.
23. Move both 20 685 kPa (3,000 psi) pressure gauges to "LBP" (7, Figure 4-3) and "RBP" (6) test ports on the manifold on the left side of the rear axle housing.
24. Quickly and completely depress the brake pedal. Verify that within one second after the brake is applied:
  - ☐ Left rear brake pressure reads  $16\,545 \pm 517$  kPa ( $2,400 \pm 75$  psi).
  - ☐ Right rear brake pressure reads  $16\,545 \pm 517$  kPa ( $2,400 \pm 75$  psi).
  - ☐ Both pressures must remain above their minimum values for a minimum of 20 seconds.

*\* Record on data sheet.*

25. Release the brake pedal. Both brake pressures should return to zero within two seconds and there should be no residual pressure trapped in the brakes.

*\* Record on data sheet.*

26. Move both 20 685 kPa (3,000 psi) pressure gauges to the test ports on the front brake backplates.

27. Quickly and completely depress the brake pedal. Verify that within one second after the brake is applied:

- ☐ Left rear brake pressure reads  $16\,545 \pm 517$  kPa ( $2,400 \pm 75$  psi).
- ☐ Right rear brake pressure reads  $16\,545 \pm 517$  kPa ( $2,400 \pm 75$  psi).
- ☐ Both pressures must remain above their minimum values for a minimum of 20 seconds.

*\* Record on data sheet.*

28. Release the brake pedal. Both brake pressures should return to zero within two seconds and there should be no residual pressure trapped in the brakes.

*\* Record on data sheet.*

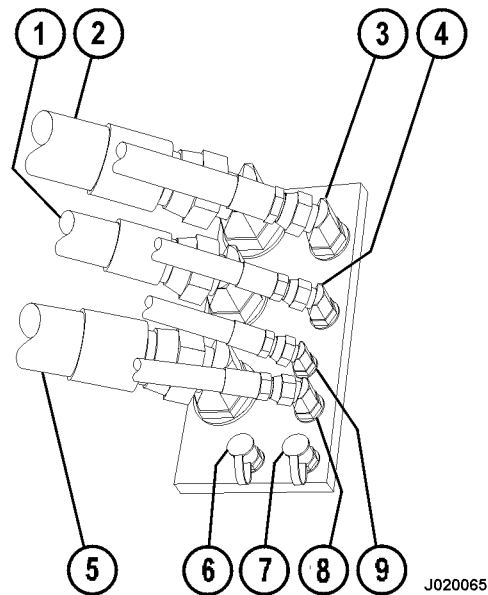


FIGURE 4-3. REAR BRAKE HOSES

1. BS - Left & Right Brake Cooling Oil Supply
2. LBR - Left Brake Cooling Return Line
3. T - Return To Tank
4. P1 - Pressure Inlet From Accumulator
5. RBR - Right Brake Cooling Return Line
6. RBP - Right Brake Pressure Test Port
7. LBP - Left Brake Pressure Test Port
8. PX - Pilot Inlet/Brake Apply Line
9. PB - Parking Brake

## LOW ACCUMULATOR PRESSURE AND AUTO APPLY CHECK-OUT

29. Move both 20 685 kPa (3,000 psi) pressure gauges back to test ports "BF" (21, Figure 4-1) and "BR" (22) in the hydraulic brake cabinet.
30. Start the engine and allow the low brake accumulator pressure to stabilize at or above 18 615 kPa (2,700 psi).
31. Turn the key switch OFF to shut off the engine and allow 90 seconds for the steering system accumulators to bleed down.
32. Disable the steering pressure switch by unplugging the diode between circuits #33 and #33F on diode board DB1 in the auxiliary control cabinet or by disconnecting the wire harness at the steering pressure switch.
33. Turn the key switch ON, but do not start the engine. Wait for two minutes, then check the low accumulator pressure reading at low accumulator pressure (LAP1) test port (15).
  - ☐ If the pressure is below 14 480 kPa (2,100 psi), there is excessive leakage in the system. The source of the leakage must be identified and repaired.

*\* Record on data sheet.*
34. Very slowly open front brake accumulator bleed valve (9) a small amount while observing the decreasing pressure at low accumulator pressure (LAP1) test port (15).
  - ☐ The low brake pressure warning light and buzzer must activate when the pressure drops to  $12\,755 \pm 517$  kPa ( $1,850 \pm 75$  psi).

*\* Record on data sheet.*

  - ☐ When the low accumulator pressure reaches 11 375 kPa (1,650 psi), brake pressures "BF" and "BR" should begin to rise (auto apply).

*\* Record on data sheet.*
35. Close front brake accumulator bleed valve (9, Figure 4-1). Record the front and rear brake apply pressures after auto apply has occurred.

*\* Record on data sheet.*
36. Start the engine and allow the low brake accumulator pressure to stabilize at or above 18 615 kPa (2,700 psi).
37. Turn the key switch OFF to shut off the engine and allow 90 seconds for the steering system accumulators to bleed down.
38. Turn key switch ON, but do not start the engine.
39. Very slowly open rear brake accumulator bleed valve (7) a small amount while observing the decreasing pressure at low accumulator pressure (LAP1) test port (15).
  - ☐ Verify that the low accumulator pressure warning activation pressure and the auto apply set point are within 690 kPa (100 psi) of the pressures that were recorded previously for Step 33.
40. Close rear brake accumulator bleed valve (7). Record the front and rear brake apply pressures after auto apply has occurred.

*\* Record on data sheet.*
41. Reinstall the diode on diode board DB1 or reconnect the wire harness at the steering pressure switch.
42. Start the engine and allow the low brake accumulator pressure to stabilize at or above 18 615 kPa (2,700 psi).
43. Use the emergency shutdown switch on the center console to shut off the engine. Do not turn the key switch OFF. The steering accumulators must not be allowed to bleed down.
44. While observing the pressure gauges, make repeated, slow brake applications until auto apply comes on.
  - ☐ Auto apply must not occur before the sixth pedal application.
  - ☐ Auto apply must not occur until after the low accumulator pressure warning activates.

*\* Record on data sheet.*
45. Turn the key switch OFF and allow the steering accumulators to bleed down.
46. Open both accumulator bleed valves and bleed down the entire brake system. Close the valves after all pressure is released.
47. Remove all test equipment from the truck.

## BRAKE CIRCUIT AND BRAKE VALVE TROUBLESHOOTING CHART

| TROUBLE                                                            | POSSIBLE CAUSE                                                                                                                                                                                                                    | SUGGESTED CORRECTIVE ACTION                                                                                                                                                      |
|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Service brakes and/or parking brake are locked.                    | Parking brake solenoid is de-energized.<br>Connections to tank and pressure ports are reversed.<br>Parking brake solenoid coil is defective.<br>Parking brake solenoid valve is defective.<br>Tank line is plugged or restricted. | Check power to solenoid.<br>Correct the plumbing.<br>Replace coil.<br>Replace solenoid valve.<br>Remove restriction.                                                             |
| Both brake circuits are dragging.                                  | Tank line has back pressure.<br>Pedal set screw is out of adjustment (residual pressure).                                                                                                                                         | Ensure that tank line has no back pressure.<br>Adjust pedal deadband with set screw.                                                                                             |
| One brake circuit is dragging.                                     | There is an obstruction in the brake valve subassembly.<br>Brake valve is out of balance.<br>Actuator piston is defective.<br>Brake valve is defective.<br>A dual relay valve is defective.                                       | Remove obstruction.<br>Adjust balance according to instructions.<br>Replace piston.<br>Rebuild or replace brake valve assembly.<br>Rebuild or replace dual relay valve assembly. |
| Brakes are not going to full pressure.                             | Supply pressure is low.<br>Modulating section of brake valve has internal malfunction.<br>Collar on brake valve is improperly adjusted.                                                                                           | Check the steering/brake pump system and accumulators.<br>Remove, disassemble, clean, and inspect brake valve.<br>Adjust collars according to instructions.                      |
| Low brake pressure warning occurs when the brakes are not applied. | There is a short in electrical system.<br>Brake accumulator is bleeding down.<br>Brake warning relay is defective.                                                                                                                | Check wiring.<br>Valve is open. Close valve.<br>Replace relay.                                                                                                                   |
| Low brake pressure warning occurs when brakes are applied.         | There is a leak or other malfunction in one brake circuit.<br>Brake valve is out of balance.<br>Dual relay valve(s) is defective.                                                                                                 | Inspect brake system and repair leaks.<br>Adjust balance according to instructions.<br>Inspect and repair dual relay valve(s).                                                   |

| <b>TROUBLE</b>                                                                | <b>POSSIBLE CAUSE</b>                                                                                                                                                                                                                                                                                                                            | <b>SUGGESTED CORRECTIVE ACTION</b>                                                                                                                                                                                                                               |
|-------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Low brake pressure warning circuit not operating properly                     | The low brake pressure warning light is burned out.<br>The electrical circuit is open.<br>Pressure switch is defective.                                                                                                                                                                                                                          | Replace bulb.<br><br>Check switch circuit wiring.<br>Replace pressure switch.                                                                                                                                                                                    |
| Low brake pressure warning is on but system pressure is correct.              | There is a short in electrical system.<br>Pressure switch is defective.                                                                                                                                                                                                                                                                          | Check wiring.<br>Replace the switch.                                                                                                                                                                                                                             |
| Low brake pressure warning comes on and pressure is low.                      | Steering circuit is malfunctioning.<br>Steering/brake pump is worn.                                                                                                                                                                                                                                                                              | Check steering circuit pressures.<br>Rebuild or replace pump.                                                                                                                                                                                                    |
| A brake accumulator bleeds off quickly when supply pressure is cut off.       | Accumulator bleiddown valve is open.<br>Accumulator precharge is low.<br>There is a leak in one circuit.<br>There is a malfunction in brake valve.                                                                                                                                                                                               | Close valve, check precharge.<br>Recharge accumulator.<br>Check plumbing.<br>Disassemble and clean, or replace brake valve.                                                                                                                                      |
| A "squeal" is heard when controller is operated.                              | Controller operates rapidly.<br>Brake valve assembly is damaged.<br>Hydraulic oil is too hot.                                                                                                                                                                                                                                                    | Normal condition.<br>Replace brake valve assembly.<br>Check entire hydraulic system for restriction or obstruction.                                                                                                                                              |
| Output pressure at controller is correct but brakes are not applying.         | Brake lines are blocked or improperly connected.                                                                                                                                                                                                                                                                                                 | Check plumbing.                                                                                                                                                                                                                                                  |
| Brake pressures drift excessively while pedal is held steady.                 | Contamination in brake valve assembly.<br>Brake valve assembly is damaged.<br>Dual relay valve is malfunctioning.                                                                                                                                                                                                                                | Disassemble and clean, or replace brake valve.<br>Repair or replace brake valve assembly.<br>Repair or replace dual relay valve.                                                                                                                                 |
| Oil is leaking around the pedal base.                                         | Seal on top of brake valve is defective.                                                                                                                                                                                                                                                                                                         | Replace seal.                                                                                                                                                                                                                                                    |
| The pump cycles too often or low pressure warning comes on at low engine rpm. | Excessive internal leakage in a component.<br><br>Steering accumulator precharge too high or too low.<br>Brake Valve plumbed incorrectly.<br>There is internal leakage in brake valve assembly.<br>There is internal leakage in dual relay valve.<br><br>Steering/brake pump is worn.<br>Pump unloader or compensator is not adjusted correctly. | Check all steering and brake system components.<br><br>Check steering accumulator precharge.<br>Correct plumbing.<br>Replace brake valve assembly.<br><br>Rebuild or replace dual relay valve.<br><br>Rebuild or replace pump.<br>Adjust pump pressure controls. |

# HYDRAULIC BRAKE SYSTEM CHECK-OUT PROCEDURE DATA SHEET

MACHINE MODEL \_\_\_\_\_ UNIT NUMBER \_\_\_\_\_ SERIAL NUMBER \_\_\_\_\_

## **INITIAL SYSTEM SETUP**

Operate hydraulic steering system to obtain proper operating temperature.

STEP 8 \_\_\_\_\_ Rear brake (BR) pressure when auto apply releases

\_\_\_\_\_ Front brake (BF) pressure when auto apply releases

STEP 10 \_\_\_\_\_ Rear brake (BR) pressure when stop lights come on

STEP 11 \_\_\_\_\_ Rear brake (BR) pressure within one second

\_\_\_\_\_ Front brake (BF) pressure within one second

## **BRAKE LOCK / SECONDARY BRAKE CHECK-OUT**

STEP 14 \_\_\_\_\_ Rear brake (BR) pressure when brake lock degradation warning occurs

STEP 16 \_\_\_\_\_ Rear brake (BR) pressure when brake lock is applied

## **PARKING BRAKE CHECK-OUT**

STEP 19 \_\_\_\_\_ Parking brake release (PK2) pressure

## **SERVICE BRAKE CHECK-OUT**

STEP 24 \_\_\_\_\_ Left rear brake (LBP) pressure when pedal is applied

\_\_\_\_\_ Right rear brake (RBP) pressure when pedal is applied

STEP 25 \_\_\_\_\_ Left rear brake (LBP) pressure when pedal is released

\_\_\_\_\_ Right rear brake (RBP) pressure when pedal is released

STEP 27 \_\_\_\_\_ Left front brake pressure when pedal is applied

\_\_\_\_\_ Right front brake pressure when pedal is applied

STEP 28 \_\_\_\_\_ Left front brake pressure when pedal is released

\_\_\_\_\_ Right front brake pressure when pedal is released

## HYDRAULIC BRAKE SYSTEM CHECK-OUT PROCEDURE DATA SHEET

### ***LOW ACCUMULATOR PRESSURE AND AUTO APPLY CHECK-OUT***

- STEP 33 \_\_\_\_\_ Low accumulator pressure after 2 minutes
- STEP 34 \_\_\_\_\_ Pressure when low brake accumulator pressure warning occurs  
\_\_\_\_\_ Pressure when low brake accumulator pressure warning occurs
- STEP 35 \_\_\_\_\_ Front brake (BF) apply pressure after auto apply occurs  
\_\_\_\_\_ Rear brake (BR) apply pressure after auto apply occurs
- STEP 40 \_\_\_\_\_ Front brake (BF) apply pressure after auto apply occurs  
\_\_\_\_\_ Rear brake (BR) apply pressure after auto apply occurs
- STEP 44 \_\_\_\_\_ Number of brake applications before low accumulator pressure  
warning occurs

Name of Technician or Inspector Performing Check-Out \_\_\_\_\_

DATE \_\_\_\_\_

# NOTES

**SECTION J5**  
**WET DISC BRAKE ASSEMBLY**  
**INDEX**

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# NOTES

## WET DISC BRAKE ASSEMBLY

The front and rear oil cooled, wet disc brake assemblies are similar in design. The rear wheel brakes differ from the front brakes in their mounting arrangement only. The rear brake assembly requires hub adapter (1, Figure 5-1) for installation on the wheel motor.

Each disc brake assembly consists of the following basic components:

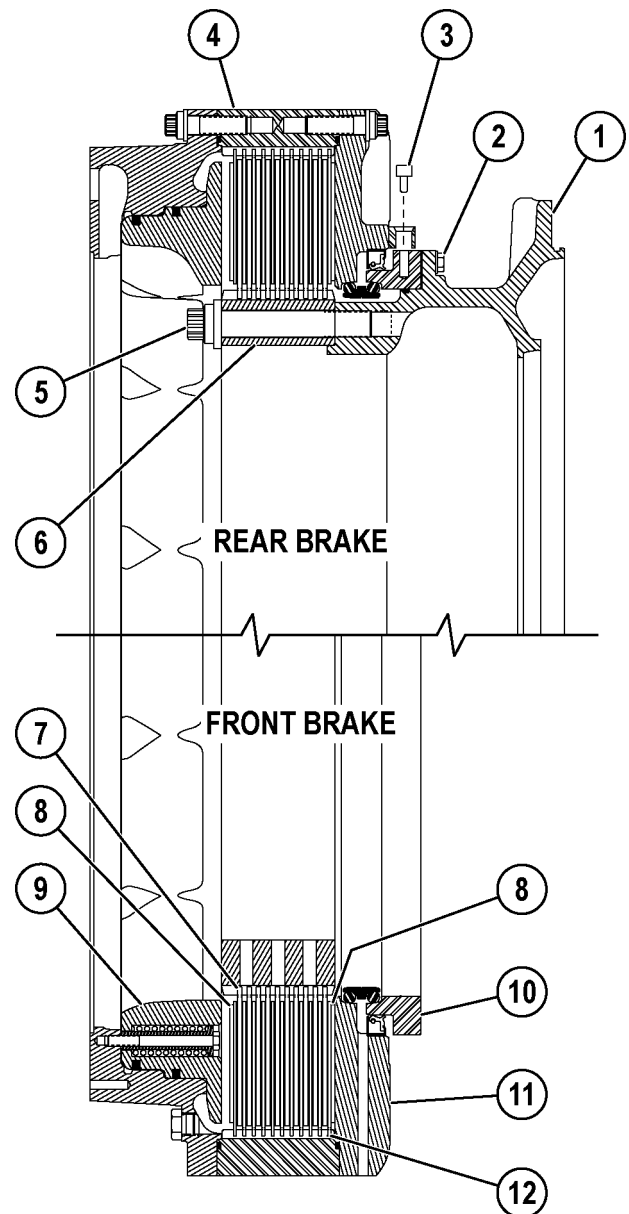
- Ten friction discs
- Nine separator plates
- Two damper discs
- Piston assembly
- Stationary ring gear
- Rotating inner gear
- Floating ring oil seal assembly

### OPERATION

Ring gear (4, Figure 5-1) is internally splined to retain dampers (8) and separator plates (12). The separator plates are alternately placed between friction discs (7), which are splined to inner gear (6). The inner gear mounts directly to the wheel hub on the front wheel brakes. On the rear brake, the inner gear requires a hub adapter which mounts on the wheel hub.

The inboard side of the assembly contains piston (9), which is activated by hydraulic pressure supplied by the dual relay valves (controlled by the brake valve). As hydraulic pressure is applied, the piston moves to compress the rotating friction discs against the stationary steel discs. The friction forces that are generated resist the rotation of the wheels. As hydraulic pressure increases, friction forces are increased and wheel rotation is slowed until the maximum force is reached and the wheel is stopped.

The entire brake disc pack is cooled by hydraulic oil. The low pressure cooling circuit is completely isolated from the high pressure piston apply circuit. Cooling oil flows from the hydraulic tank to the hoist pump, through the hoist circuit high pressure filters, through the hoist valve, and into the brake housings. A parallel circuit from the hoist valve outlet is connected to relief valves at the hydraulic tank which limit brake cooling circuit pressure to 241 kPa (35 psi). Oil that is routed to the front brakes passes through the oil cooler before entering the front brake housings. In addition, the brake cooling oil provides lubrication for the front wheel bearings. Oil that exits the brake housings returns to the hydraulic tank.



J050056

FIGURE 5-1. WET DISC BRAKE ASSEMBLIES

- |                                  |                    |
|----------------------------------|--------------------|
| 1. Hub Adapter (Rear Only)       | 6. Inner Gear      |
| 2. Capscrew & Lockwasher         | 7. Friction Disc   |
| 3. Socket Head Screw (Temporary) | 8. Damper          |
| 4. Ring Gear                     | 9. Piston Assembly |
| 5. Capscrew & Hardened Washer    | 10. Seal Carrier   |
|                                  | 11. Back Plate     |
|                                  | 12. Separator Disc |

## BRAKE DISC WEAR INSPECTION

Brake disc wear should be checked every 1000 hours by using the wear indicator tool (EF9302), that is included in the tool group that is shipped with the truck. Refer to Figure 5-2.

The brake disc wear indicator tool is inserted in a port which is open to cooling oil. Removal of the inspection hole plug to insert the brake disc wear indicator will cause the loss of some of this oil. Advance planning will help to minimize oil loss.

Consider scheduling front brake disc wear inspections along with the recommended 1000 hr. change of hydraulic oil and filters. Rear brakes should be inspected for wear whenever the rear tires are removed. Also consider obtaining additional brake disc wear indicator tools for permanent installation on truck. If all brake assemblies are equipped with disc wear indicators, future checks will not require installation and removal.

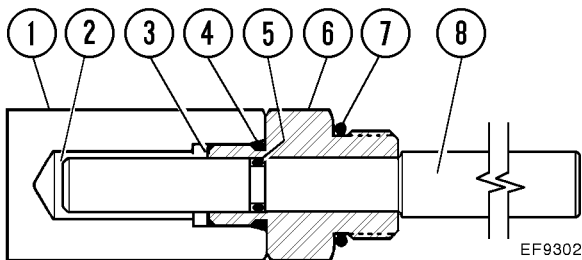


FIGURE 5-2. BRAKE WEAR INDICATOR

- |                           |                  |
|---------------------------|------------------|
| 1. Cover                  | 5. O-Ring        |
| 2. Indicator Pin End Face | 6. Tool Housing  |
| 3. Housing Face           | 7. O-Ring        |
| 4. O-Ring                 | 8. Indicator Pin |

1. Place the directional control lever in PARK and turn the rest switch ON. Turn the key switch OFF to shut off the engine and wait 90 seconds to allow the steering accumulators to bleed down completely. Block the truck wheels.
2. Open the bleeddow valves on the brake manifold in the hydraulic brake cabinet and bleed all pressure from the brake accumulators. Close the valves after the pressure is released.
3. Thoroughly clean the brake assemblies, especially the area surrounding cooling oil lines.
4. Remove the hex head O-ring plug from port (5, Figure 5-3). Quickly insert the brake wear indicator tool and tighten it securely.

**NOTE:** The front wheel brakes have a diagnostic connector installed for measuring brake apply pressure. The rear brakes do not have a diagnostic connector installed on the brake backplate, but instead have an O-ring plug installed. Do not attempt to install the brake wear indicator in this port.

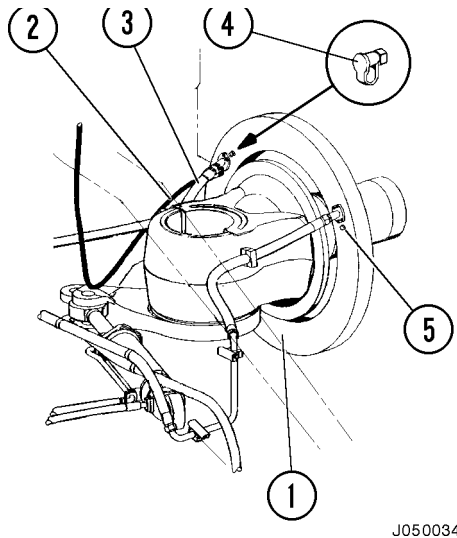
5. Start the engine. Allow the steering system pressure to stabilize and the brake accumulators to fill.
  6. While fully applying the service brake pedal, check the brake disc wear as follows:
    - a. Remove cover (1, Figure 5-2) from the wear indicator.
    - b. Push pin (8) inward until it stops against the brake piston.
    - c. Measure the distance from indicator pin end face (2) to housing face (3).
- ☐ If indicator pin end face (2) is even with housing face (3) or below, disc pack is worn to the maximum safe wear limits. The brakes should be scheduled for rebuild.
- ☐ If indicator pin end face (2) extends out beyond housing face (3), brake disc wear is still within allowable limits.
7. Pull pin (8) outward until it stops against tool housing (6). Install cover (1).

8. Release the brakes. Shut off the engine and wait 90 seconds to allow the steering accumulators to bleed down.
9. Open the bleeddow valves on the brake manifold in the hydraulic brake cabinet and bleed all pressure from the brake accumulators. Close the valves after the pressure is released.
10. Remove the brake disc wear indicator tool and reinstall the O-ring plug in the port.
11. To check the remaining brake assemblies, repeat Steps 4 through 10.

**NOTE:** Checking the disc wear in all the brake assemblies is recommended. Disc wear in one brake assembly may be different from another due to dissimilar operation of parts and/or haul profiles which require repeated braking while steering in one direction only.

12. Refill the hydraulic tank as required.
13. If brake repairs are necessary, refer to Brake Rebuild later in this section.

**NOTE:** If any leakage is observed around the brake disc wear indicator tool, replace O-rings (4, 5, and 7, Figure 5-2).



**FIGURE 5-3. BRAKE DISC WEAR INDICATOR  
INSTALLATION  
(Left Front Brake Shown)**

- |                       |                       |
|-----------------------|-----------------------|
| 1. Brake Assembly     | 4. Diagnostic Coupler |
| 2. Brake Apply Line   | 5. Wear Indicator     |
| 3. Brake Cooling Line | Installation Port     |

## BRAKE REBUILD

The brake assembly should be disassembled and reassembled on a clean, dry work surface. The surface should be wooden if possible. If it is metal, the surface should be covered with padding to prevent damaged to machined surfaces. Match mark individual parts for correct orientation before disassembly.

If a rear wheel brake is to be disassembled, start with Step 1 below. If a front wheel brake is to be disassembled, start with Step 5.

## **⚠ WARNING**

**The front brake assembly weighs approximately 1 460 kg (3210 lb). The rear brake assembly weighs approximately 1 820 kg (4000 lb). Make sure that any lifting devices are rated to handle the load.**

### Disassembly

#### Rear Wheel Brake Only:

Refer to Wheel Motor Rebuild instructions for removal of the rear wheel brake assembly from the wheel motor.

1. Verify that socket head capscrews (3, Figure 5-1) are installed at each of the six locations on back plate (11). These capscrews are required to retain seal carrier (10) to the back plate when the brake assembly is not installed on the truck. Shipping bars (4, Figure 5-4) must be installed.
2. Remove 12-point capscrews and hardened washers (5, Figure 5-1).
3. Remove capscrews and lockwashers (2) that retain the hub to the seal carrier.
4. Make sure that the hub and other parts are marked to ensure proper orientation during reassembly. Lift hub adaptor (1) from the brake assembly. Note the number of shim packs that are installed at six locations between the seal carrier and hub.

*Both Front and Rear Wheel Brakes:*

Remove the front wheel/brake assembly according to the instructions in Section G, Front Wheel Hub and Spindle.

5. Position the brake assembly on a work surface with the ring gear retainer bars on the bottom as shown in Figure 5-5.

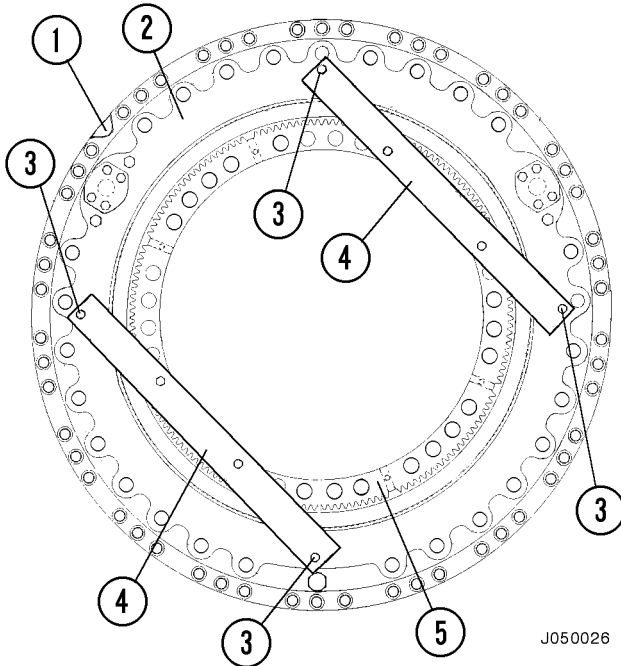


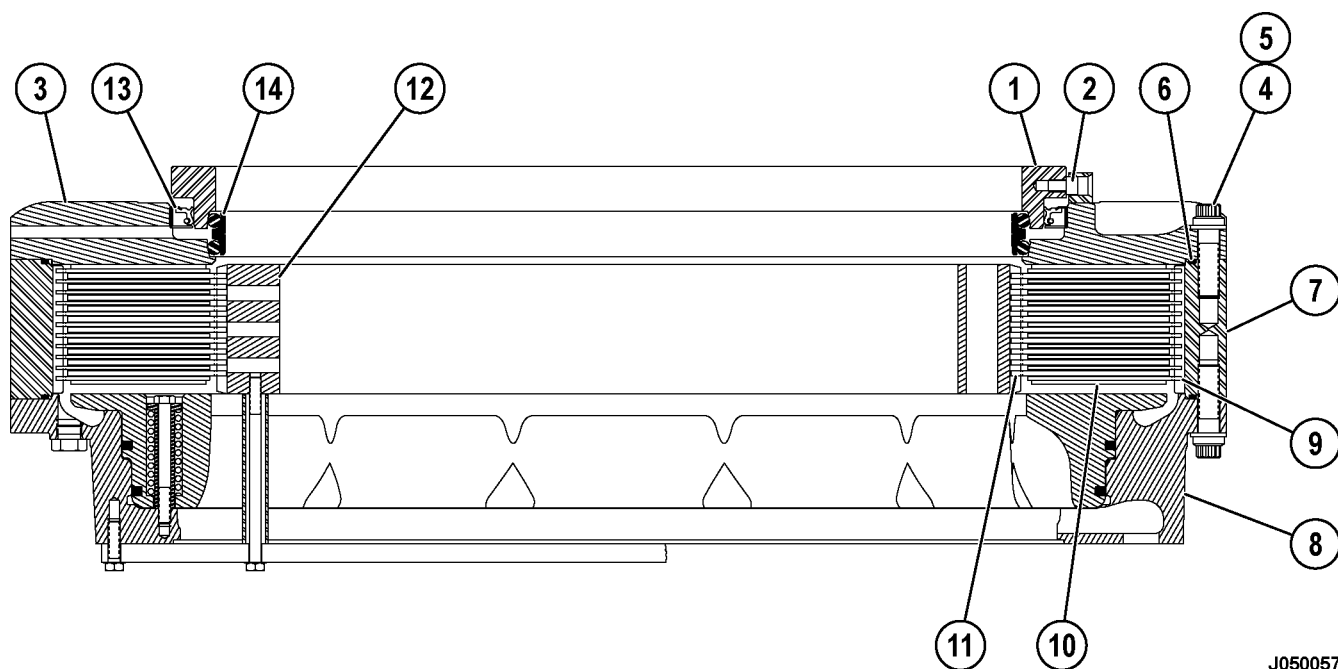
FIGURE 5-4. BRAKE ASSEMBLY

- |                            |               |
|----------------------------|---------------|
| 1. Ring Gear               | 4. Flat Bar   |
| 2. Back Plate              | 5. Inner Gear |
| 3. Capscrew and Lockwasher |               |

6. Remove socket head capscrews (2, Figure 5-5). Lift seal carrier (1) off the brake assembly with oil seal (13) and seal assembly (14).

*NOTE: Socket head capscrews (2) are required to retain and position the seal carrier on the back plate when the brake assembly is not installed on the wheel hub, or during brake assembly shipping or storage. Brake assembly tool group (EK3789) must be installed to retain the inner gear inside the brake assembly.*

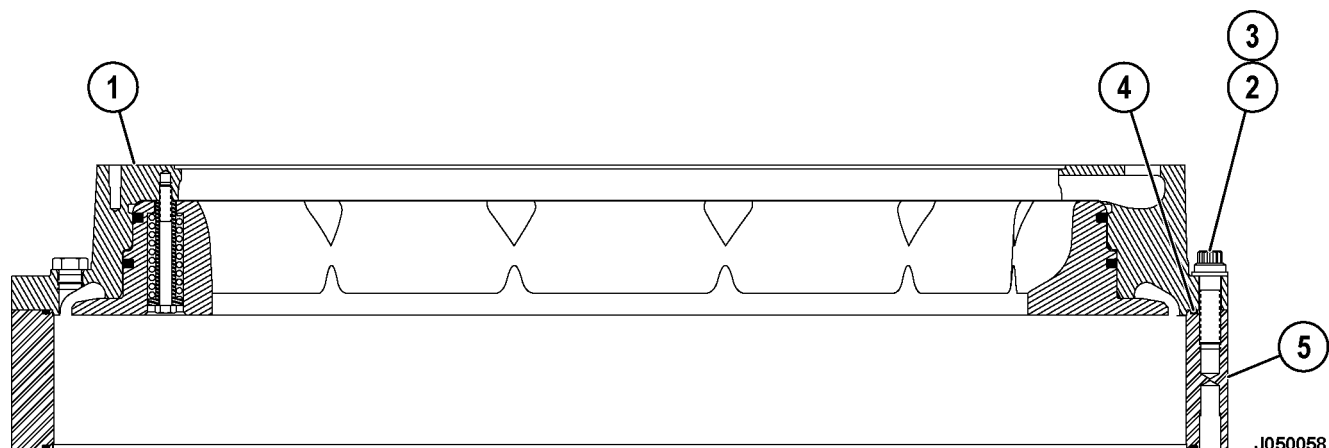
7. Remove capscrews (4, Figure 5-5) and hardened flat washers (5) from back plate (3).
8. Insert a 7/8 UNC x 2 in. pusher bolt in each of the three tapped holes in the back plate. Tighten the bolts evenly to lift the back plate from ring gear (7). Remove and discard O-ring (6).
9. Note the order of assembly of the discs, then remove damper (10) from the top of the stack. Remove friction discs (11), separator plates (9) and the remaining damper at the bottom of the stack.
10. Rotate the brake assembly to position flat bars (4, Figure 5-4) from brake assembly tool group (EK2789) on top as shown.
11. Remove short capscrews and lockwashers (3) that retain the flat bars to the housing.



J050057

FIGURE 5-5. INITIAL DISASSEMBLY

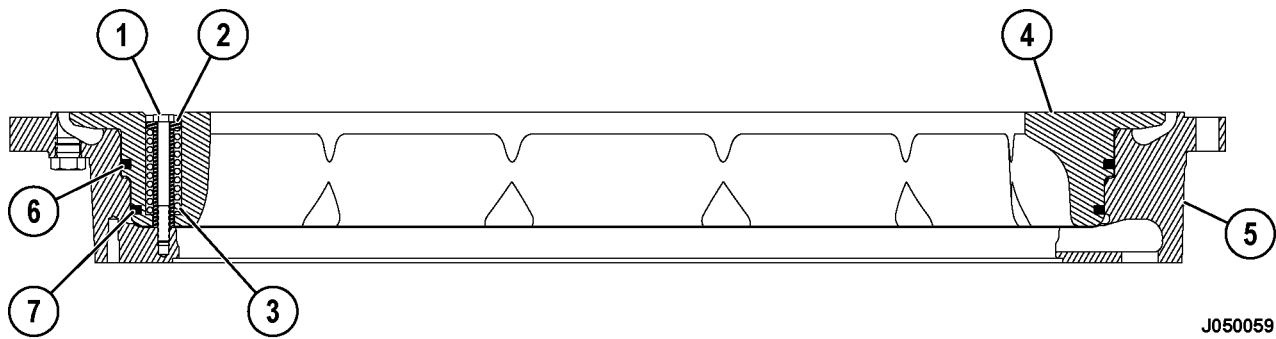
- |                    |                    |                   |
|--------------------|--------------------|-------------------|
| 1. Seal Carrier    | 6. O-Ring          | 11. Friction Disc |
| 2. Capscrew        | 7. Ring Gear       | 12. Inner Gear    |
| 3. Back Plate      | 8. Piston Housing  | 13. Oil Seal      |
| 4. Capscrew        | 9. Separator Plate | 14. Seal Assembly |
| 5. Hardened Washer | 10. Damper         |                   |



J050058

FIGURE 5-6. PISTON/HOUSING ASSEMBLY REMOVAL

- |                   |                    |              |
|-------------------|--------------------|--------------|
| 1. Piston Housing | 3. Hardened Washer | 5. Ring Gear |
| 2. Capscrew       | 4. O-Ring          |              |



J050059

FIGURE 5-7. PISTON REMOVAL

1. Capscrew  
2. Spring Guide

3. Piston Retract Spring  
4. Piston Assembly  
5. Piston Housing

6. Seal Assembly  
7. Seal Assembly

12. Attach a lifting strap through the shipping bars and lift inner gear (5) out of the brake assembly. Remove the shipping bars and spacers.
13. Remove capscrews (2, Figure 5-6) and hardened washers (3) from piston housing (1).
14. Insert a 7/8 UNC x 2 in. pusher bolt in each of the three tapped holes in the piston housing. Tighten the bolts evenly to lift the housing from ring gear (5). Remove and discard O-ring (4).

15. Position the piston assembly with piston retract springs (3) on top as shown in Figure 5-7.

16. Remove capscrews (1, Figure 5-7), spring guides (2), and piston retract springs (3).

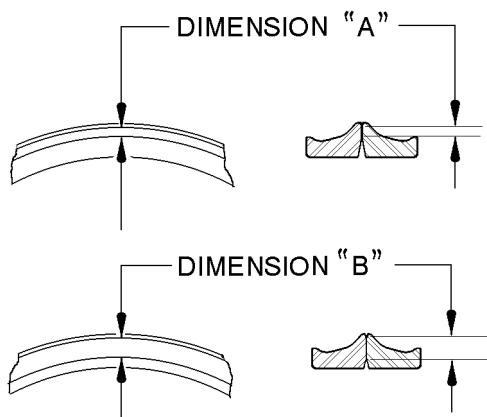
*NOTE: The threads of capscrew (1) are coated with Loctite® during assembly. A small amount of heat applied to the piston housing may be required for easier removal.*

17. Loosen or remove the plugs that are installed in the piston housing ports. Carefully lift piston (4) out of housing (5). Remove seal assemblies (6) and (7).

## ⚠ IMPORTANT ⚠

***If the brake wear indicator test indicates internal brake components are worn to the maximum allowable limit, it is recommended that all friction discs, separator plates and dampers should be replaced with new parts. Always replace seal assemblies and O-rings with new parts.***

1. Clean all parts thoroughly before inspection.
2. Remove and discard the toric rings from the floating ring oil seal assembly in seal retainer (10, Figure 5-1) and back plate (11). Inspect the polished (mating) surfaces of the seal ring for scratches and other damage. Inspect the contact band of the mating faces to determine the amount of wear.  
  
A new seal will have a contact band (dimension "A", Figure 5-8) that is approximately 1.6 mm (0.06 in.) wide. As wear occurs, the contact band will widen slightly (dimension "B") and migrate inward until the inside diameter is reached and the entire seal assembly must be replaced. The remaining seal life can be estimated by the width of the contact band.
3. Inspect the piston housing for nicks and scratches in the piston seal area. If nicks or scratches cannot be removed by polishing, replace the housing.
4. Inspect the piston seal assembly grooves for damage.



J050035

FIGURE 5-8. SEAL WEAR PROGRESSION

5. Inspect piston retract springs (3, Figure 5-7). Check for free height and test for height under load. Replace the springs if they are not within approximately 10% of specification.
  - ☐ Free Height:  
..... 101.6 mm (4.00 in.)
  - ☐ Height @ 1 000 N (225 lb) working load:  
..... 85.12 mm (3.351 in.)
  - ☐ Height @ 2 669 N (600 lb) working load:  
..... 69.9 mm (2.750 in.)
6. Inspect the friction discs for warping, tooth wear and excessive friction material wear. Replace the friction discs if wear exceeds the minimum groove depth.
  - ☐ Disc thickness including friction material:  
.....  $7.7 \pm 0.3$  mm ( $0.30 \pm 0.01$  in.)
  - ☐ Friction material thickness (new)  
..... 1.1 mm (0.04 in.)
  - ☐ Nominal friction material groove depth:  
..... 0.63 mm (0.025 in.)
  - ☐ Minimum allowable friction material groove depth:  
..... 0.25 mm (0.010 in.)
  - ☐ Flatness over friction material (new)  
..... 0.45 mm (0.018 in.)
7. Inspect the separator plates for warping and tooth wear.
  - ☐ Disc thickness (new)  
.....  $3.7 \pm 0.1$  mm ( $0.146 \pm 0.004$  in.)
  - ☐ Flatness (new)  
..... 0.5 mm (0.020 in.)
8. Inspect the damper plate for warping, tooth wear and excessive facing material wear.
  - ☐ Disc thickness, including facing material (new)  
.....  $8.2 \pm 0.5$  mm ( $0.323 \pm 0.020$  in.)
  - ☐ Disc thickness, steel plate only (new)  
.....  $3.7 \pm 0.1$  mm ( $0.146 \pm 0.004$  in.)
  - ☐ Flatness, steel plate (new)  
..... 0.5 mm (0.020 in.)
9. Inspect ring gear (4, Figure 5-1) for excessive tooth wear and nicks and scratches in O-ring seal grooves.
10. Inspect inner gear (6) for excessive tooth wear and damage at capscrow holes.



## Assembly

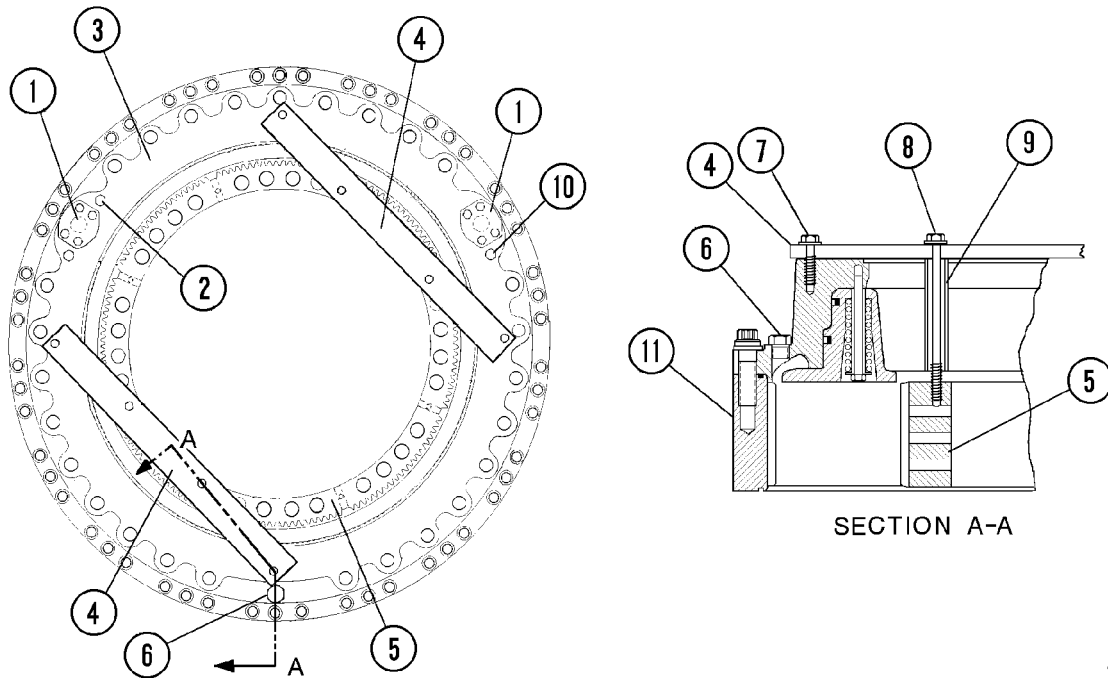


***The work area must be clean! Handle all parts carefully to avoid damage to polished sealing surfaces.***

1. Check the bore of piston housing (5, Figure 5-7) for nicks, scratches and dirt particles. Position the housing on a work surface with the bore facing up.
2. Lubricate the square section O-ring portion of piston seal assemblies (6) and (7) with type C-4 hydraulic oil and install them in the grooves of piston (4). Make sure that the O-ring is not twisted.
3. Lubricate the piston groove and outer piston seal rings. Install the seal in the grooves over the O-rings, using your fingers or a smooth rounded object to push the seal into the groove.
4. Install two equally spaced 1/2 UNC x 5 in. guide studs in the housing at the piston retract spring mount tapped holes.
5. Lubricate the housing bore. Install lifting eyes and attach an overhead hoist to the piston. Position the piston over the housing with the retract spring cavity holes aligned with the studs that were installed in Step 4. Place a spring guide over each stud to aid in alignment.
6. Carefully lower the piston straight into the piston housing bore until it is seated against the housing. If necessary, seat the piston by tapping it with a soft mallet.
7. Assemble 12 capscrews (1), spring guides (2) and piston retract springs (3). Apply Loctite® to the capscREW threads and install the assembled parts through the piston into the tapped holes in the housing. Tighten the capscrews to **122 N·m (90 ft lb)**.

## Piston Leakage Test:

8. Test the piston/housing assembly as follows:
  - a. Install a plug in brake apply pressure port (2, Figure 5-9).
  - b. Install a fitting in the remaining port. Attach a hydraulic pressure test device to the fitting.
  - c. Slowly apply pressure and loosen the apply port plug to bleed air from the piston cavity.
  - d. Cycle the piston to full stroke ten times by applying 2070 kPa (300 psi) of hydraulic pressure. Observe the piston for leakage.
  - e. Minor oil seepage (non-measurable) is permissible. If leakage is greater, disassemble the piston assembly and determine the cause.
9. After completion of the piston leakage test, release the pressure, remove the hydraulic source and drain the oil from the piston apply cavity. Plug the ports to prevent contamination.
10. Install O-ring (4, Figure 5-6) in the groove of ring gear (5).
11. Attach lifting eyes to the piston/housing assembly and lower it into position over the ring gear. Install capscrews (2) and hardened washers (3). Alternately tighten the capscrews to **780 N·m (575 ft lb)**.
12. Insert inner gear (5, Figure 5-9) into the assembly. Orient the gear as shown.
13. Install brake assembly tool group (EK3798). Place flat bars (4, Figure 5-9) over the piston housing as shown. Attach the flat bars using short capscrews and lockwashers (7) at the outer ends. Insert pipes (9) and long capscrews and lockwashers (8) as shown to retain the inner gear in position.
14. Attach lifting eyes to the assembly. Lift and rotate it to place the piston housing on the bottom.



J050029

FIGURE 5-9. INNER GEAR INSTALLATION

- |                              |                           |                                      |
|------------------------------|---------------------------|--------------------------------------|
| 1. Cooling Oil Port          | 5. Inner Gear             | 9. Pipe*                             |
| 2. Brake Apply Pressure Port | 6. Drain Plug             | 10. Wear Indicator Installation Port |
| 3. Piston Housing            | 7. Capscrew & Lockwasher* |                                      |
| 4. Flat Bar*                 | 8. Capscrew & Lockwasher* | 11. Ring Gear                        |

\* Components of Brake Assembly Tool Group (EK3798)

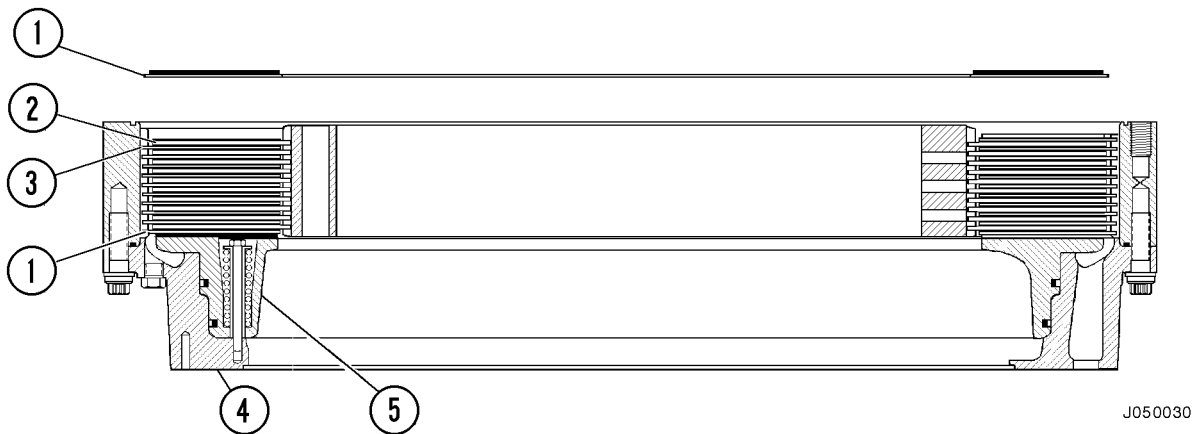


FIGURE 5-10. DISC PACK INSTALLATION

- |                  |                    |           |
|------------------|--------------------|-----------|
| 1. Damper        | 3. Separator Plate | 5. Piston |
| 2. Friction Disc | 4. Piston Housing  |           |

15. Install the brake discs as follows:

- Insert damper disc (1, Figure 5-10) into the ring gear and inner gear with the friction material (down) facing piston (5).
- Insert friction disc (2) on top of the damper.
- Install a separator plate (3).
- Continue installing the remaining friction discs and separator discs, alternating each type as installed.

**NOTE:** The disc pack contains a total of ten friction discs, nine separator plates, and two damper discs.

Install the remaining damper on top of the last friction disc with its facing material on top (unfaced side toward top friction disc).

- Using new O-ring (6, Figure 5-5), install back plate (3) over the ring gear. Make sure that the back plate is oriented properly according to the match marks that were made during disassembly.

**NOTE:** A seal carrier socket head capscrew hole on the back plate should be aligned with the drain plug on the piston housing.

- Install capscrews (4) and hardened washers (5). Alternately tighten the capscrews to **780 N·m (575 ft lb)**.

- Install seal assembly (14) in the cavities in the back plate and seal carrier. Refer to Floating Ring Seal Assembly & Installation.

- Install oil seal (13) in the back plate. Pack the area between the seal lips with grease.

- Install the seal carrier on the back plate. Install six 1/2" - 13 NC x 3/4" in. socket head capscrews to retain the seal carrier in position until the brake assembly is installed on the truck.

**NOTE:** For a front wheel brake assembly, the seal must be setup for proper seal compression after the brake assembly is installed on the truck. Refer to Seal Assembly Gap Check in Section G3, Front Wheel Hub and Spindle.

- Assembly is now complete if the brake assembly is to be installed on a front wheel. Refer to Section G3, Front Wheel Hub and Spindle, for brake installation instructions.

If the brake assembly is to be installed on a rear wheel, perform the following additional steps.

*Rear Wheel Brake Assembly Only:*

22. Check the integrity of the seals before installing the brakes on the truck.
  - a. Cap the cooling oil ports and pressurize the brake to a maximum of 103 kPa (15 psi).
  - b. Close off the air supply and monitor the interior brake pressure for 15 minutes.
  - c. If the air pressure decreases rapidly, a displaced or damaged brake seal is indicated. Disassemble the brake and reinstall or replace the seal.
23. Install new O-ring (1, Figure 5-11) in the hub groove.
24. Position the hub over the seal carrier. Orient the hub according to the marks that were made during disassembly to align the hub flange holes with the seal retainer tapped holes.
25. Insert the following shims (4) between the hub flange and seal carrier (5) at each pair of capscrews (six places):
  - ☐ Ten . . . . . 0.51 mm (0.020 in.) thick
  - ☐ One . . . . . 0.25 mm (0.010 in.) thick
26. Install capscrews and lockwashers (3) and tighten them securely.

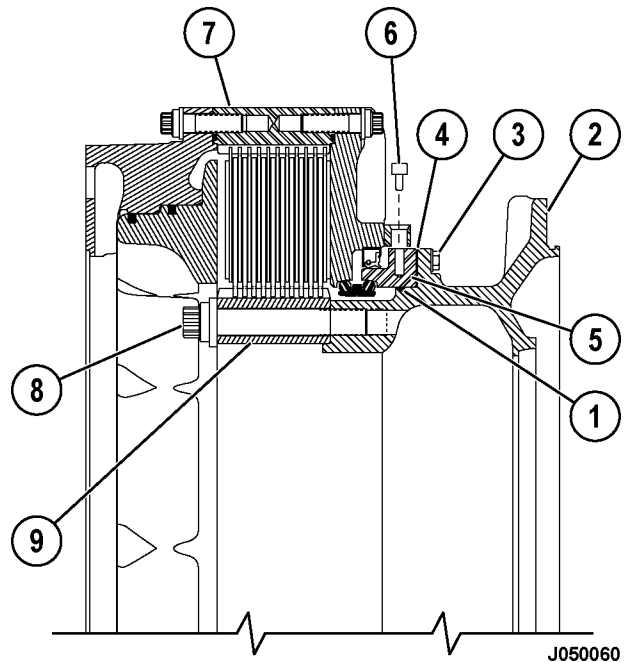


FIGURE 5-11. REAR BRAKE, HUB INSTALLATION

- |                          |                                 |
|--------------------------|---------------------------------|
| 1. O-Ring                | 6. Socket Head Capscrews        |
| 2. Hub Adapter           | 7. Ring Gear                    |
| 3. Capscrews and Washers | 8. Capscrew and Hardened Washer |
| 4. Shims                 | 9. Inner Gear                   |
| 5. Seal Carrier          |                                 |

### Seal Gap Adjustment:

27. Adjust the seal gap. Refer to Figure 5-12.

- Measure seal gap "A" at three equally spaced locations. Add the three dimensions and divide by 3 to determine an average dimension.
- Add 551.53 mm (21.714 in.) to the average dimension determined in Step a.
- Measure dimension "B" at three equally spaced locations. Add the three dimensions and divide by 3 to determine the average dimension.
- Subtract the dimension in Step c from the dimension in Step b to determine the change in the shim pack.
- Add or remove equal quantities and thicknesses of shims (as determined in Step d) at the six shim pack locations. The shim pack must provide a final dimension "B" within 0.13 mm (0.005 in.) of the dimension that was calculated in Step b.

#### Example

Step a: average gap "A" = **6 mm (0.236 in.)**

Step b: 6 mm (0.236 in.) + 551.53 mm (21.714 in.) =  
**557.53 mm (21.950 in.)**

Step c: average dimension "B" = **559 mm (22.008 in.)**

Step d: 557.53 mm (21.950 in.) - 559 mm (22.008 in.) =  
**-1.47 mm (0.058 in.)**

Step e: Removing three 0.51 mm (0.020 in.) shims would result in a dimension "B" of **557.47 mm (21.947 in.)**, which is within the tolerance limit of .13 mm (0.005 in.).

28. Tighten capscrews (3, Figure 5-11) to **122 N·m (90 ft lb)**.

29. Install capscrews and hardened washers (8). It will be necessary to remove the shipping bars from the inner gear to access some capscrews. Alternately tighten the capscrews to **2 705 N·m (1995 ft lb)**. Replace the shipping bars.

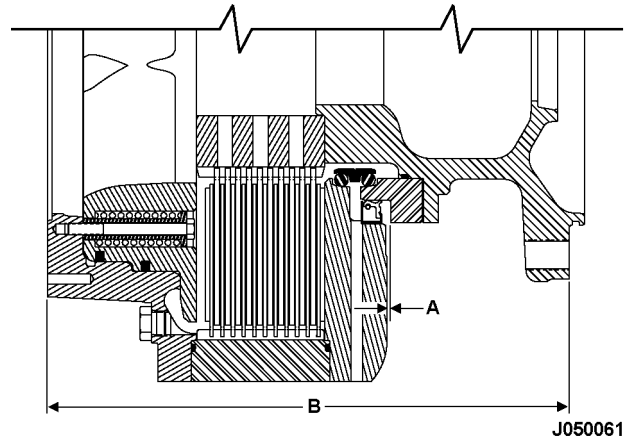


FIGURE 5-12. SEAL GAP ADJUSTMENT

## Floating Ring Seal Assembly & Installation

Failures are usually caused by combinations of factors rather than one single cause, but many failures have one common denominator: **Assembly Error!**

Floating ring seals should **always** be installed in **matched** pairs; that is, two *new* rings or two rings *that have previously run together*. Never assemble one new ring and one used ring, or two used rings that have not previously run together.

**ALWAYS USE NEW TORIC RINGS!**

1. Inspect the seal surfaces and mounting cavities for rough tool marks or nicks that may damage rubber seal rings. Hone them smooth and clean, if required.
2. Remove any oil, dust, protective coating or other foreign matter from the metal seal rings, the toric rings, and both the housing and seal ring ramps. Use tri-chloroethane #111, which is a non-petroleum based, rapid drying solvent that leaves no film. Allow the surfaces to dry completely. Use clean, lint-free material such as Micro-Wipes # 05310 for cleaning and wiping.

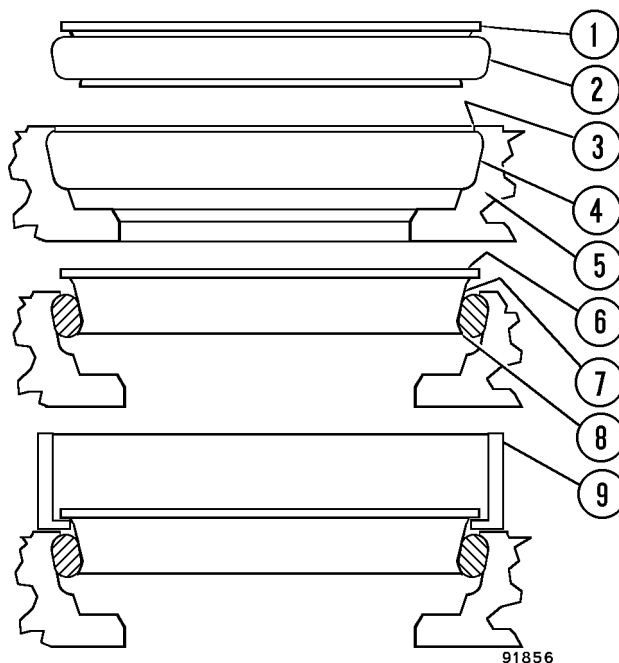


FIGURE 5-13. SEAL TERMINOLOGY

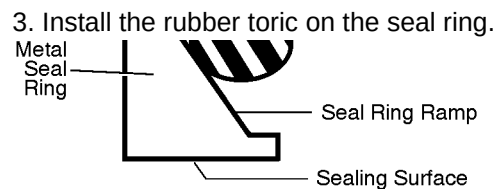
- |                         |                           |
|-------------------------|---------------------------|
| 1. Seal Ring            | 5. Seal Ring Housing      |
| 2. Rubber Toric         | 6. Seal Ring Face         |
| 3. Housing Retainer Lip | 7. Seal Ring Ramp         |
| 4. Housing Ramp         | 8. Seal Ring Retainer Lip |

## WARNING

*When using tri-chloroethane or any solvent, avoid prolonged skin contact. Use solvents only in well ventilated areas and use approved respirators to avoid breathing fumes. Do not use near open flame, welding operations or other heated surfaces exceeding 482°C (900°F). Do not smoke around solvents.*

**Both ramps must be dry.** Use clean, lint-free cloths or lint-free paper towels for wiping.

*NOTE: Oil from adjacent bearing installations or seal ring face lubrication must not get on the ramp or toric until after both seal rings are together in their final assembled position.*



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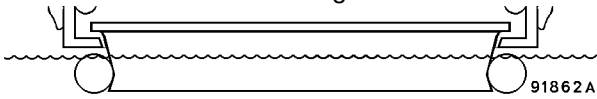
*Make sure that it is **STRAIGHT!** Make sure that the toric ring is not twisted and that it is seated against the retaining lip of the seal ring ramp. Use the flash line as a reference guide to eliminate twist. The flash line should be straight and uniform around the toric ring.*

*NOTE: Handle the seal carefully. Nicks and scratches on the seal ring face can cause leaks.*



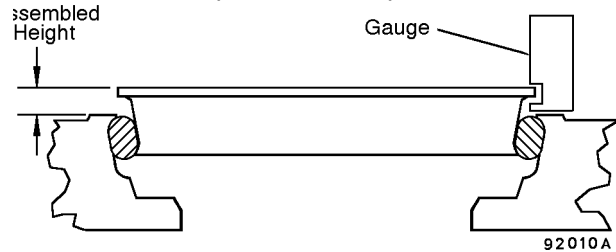
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- Place the installation tool onto the seal ring with the toric ring. Refer to Section M, Special Tools, for the installation tool part number. Lower the rings into a container of tri-chloroethane until all surfaces of the toric ring are wet.



- Use a sight gauge to check the variation in the seal ring "assembled height" in four places that are 90 degrees apart.

Height variation around the assembled ring should not exceed  $1.30 \pm 0.01$  mm ( $0.51 \pm 0.05$  in.) for the brake assembly floating seal or  $1.14 \pm 0.01$  mm ( $0.45 \pm 0.04$  in.) for the hub seal.



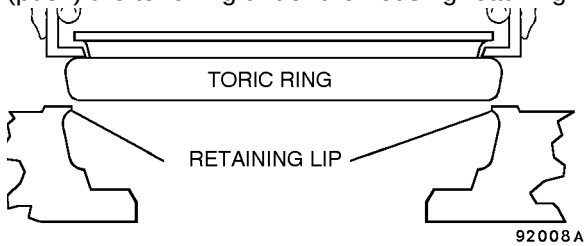
#### ALTERNATE PROCEDURE:

After positioning the seal squarely over the retaining lip, thoroughly lubricate the ring by spraying it with tri-chloroethane #111.

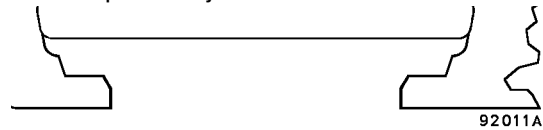
**DO NOT USE** Stanosol or any other liquid that leaves an oily film or does not evaporate quickly.

- With all surfaces of the toric ring wet, use the installation tool to position seal ring and toric ring squarely against the seal housing.

**APPLY SUDDEN AND EVEN PRESSURE** to pop (push) the toric ring under the housing retaining lip.



- If small adjustments are necessary, **do not push directly on the seal ring**. Make any required adjustments with the installation tool.



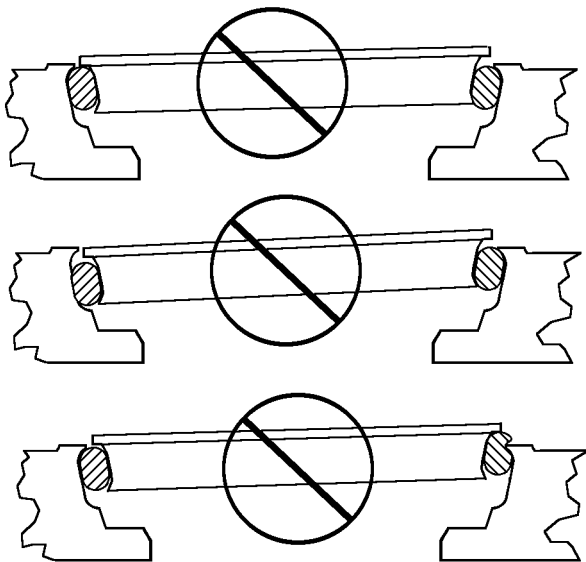
**NOTE:** The toric ring can twist if it is dry on one spot or if there are burrs or fins on the housing retaining lip. A bulging toric or cocked seal can contribute to eventual failure.

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8. The toric ring must not slip on the ramps of either the seal ring or housing. To prevent slippage, wait at least two minutes to let all the tri-chloroethane evaporate before further assembly. Once it is correctly in place, the toric ring must roll on the ramps only. If correct installation is not obvious, repeat Steps 4 through 7.

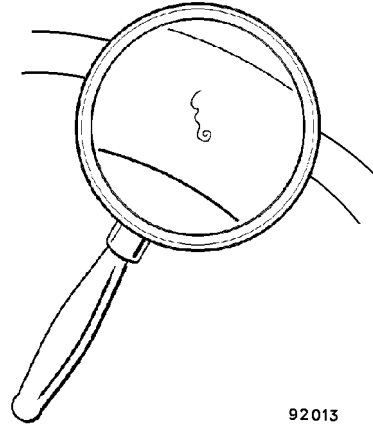


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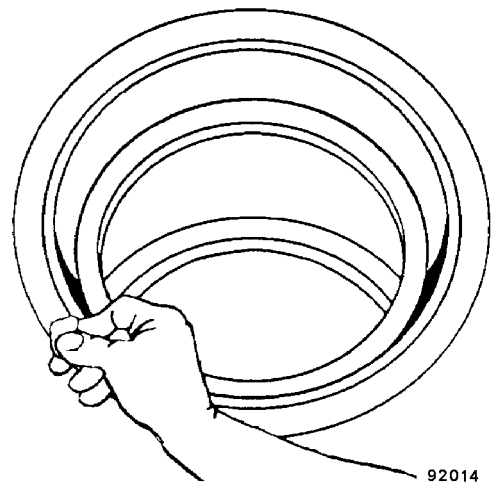
9. Wipe the polished metal seal surfaces with clean tri-chloroethane to remove any foreign material or fingerprints. No foreign particles of any kind should be on the seal ring faces. Something as small as a paper towel raveling will hold the seal faces apart and cause leakage.



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10. Apply a thin film of clean oil on the seal faces. Use a lint-free applicator or a clean finger to distribute the oil evenly. Make sure no oil comes in contact with the rubber toric rings or their mating surfaces.

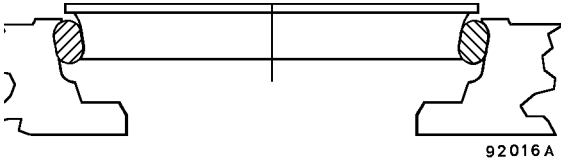
Before assembling both seals and housing together, wait at least two minutes to let all tri-chloroethane evaporate. Some may still be trapped between the toric ring and the housing ramp.



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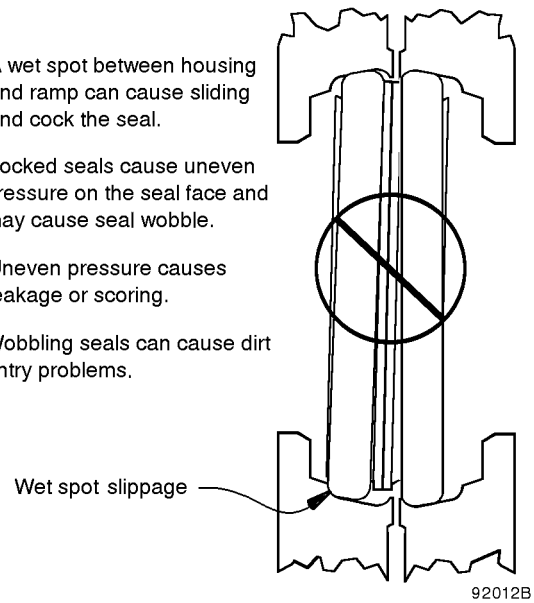
10. Ensure that both housings are in correct alignment and are square and concentric. Move the parts slowly and carefully toward each other.



**Ensure seals are square and concentric.**

**NOTE: Do not slam, bump or drop the seals together. High impact can damage the seal face and cause leakage.**

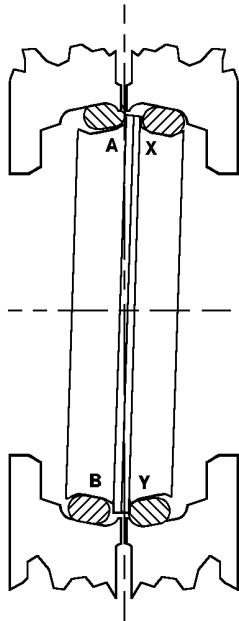
- A wet spot between housing and ramp can cause sliding and cock the seal.
- Cocked seals cause uneven pressure on the seal face and may cause seal wobble.
- Uneven pressure causes leakage or scoring.
- Wobbling seals can cause dirt entry problems.



### RESULTS OF INCORRECT ASSEMBLY

Points "A" & "B" remain stationary

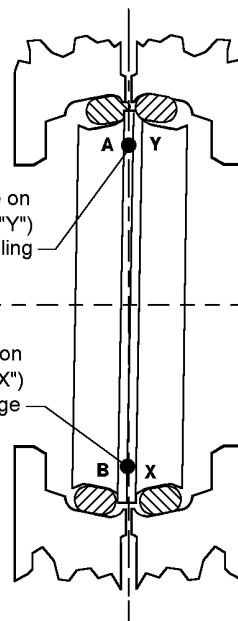
Points "X" & "Y" rotate 180°



SEALS NOT INSTALLED PERPENDICULAR TO BORE

High pressure on seal ring face ("A" & "Y") can cause galling

Inadequate pressure on seal ring face ("B" & "X") can cause oil leakage



SEALS SHOWN AFTER REVOLVING 180°

J050033

## WET DISC BRAKE BLEEDING PROCEDURE

*NOTE: Rear wheel brakes must be bled before rear tire installation.*

1. Make sure that the hydraulic brake supply (steering circuit) is operating properly.
2. If necessary, charge the brake system accumulators. Refer to Section J3, Brake Circuit Component Service.
3. Make sure that the bleed down valves on brake accumulator manifold are closed.
4. Check the hydraulic tank oil level and fill if necessary.
5. With the wheels securely blocked, start the engine and allow the accumulators to fill.
6. Slowly depress the brake pedal until the service brake is partially applied.

### *Rear Wheel Brakes:*

7. Crack open the plug in brake apply pressure port (2, Figure 5-9) on the brake back plate. Close the plug after the oil runs clear of contaminants and free of bubbles. Repeat for the other rear wheel brake.

*NOTE: The other brake cooling oil line has a single hex plug located next to it. This is the brake wear indicator port. DO NOT use this port for bleeding the brake.*

### *Front Wheel Brakes:*

*NOTE: The front wheel brakes are equipped with a diagnostic coupler in brake apply pressure port (2, Figure 5-9) on the brake back plate. A hose with a mating fitting can be attached and used to direct the oil into a container during the bleeding process.*

8. Attach a hose to the diagnostic coupler.
9. Slowly depress the brake pedal until the service brake is partially applied.
10. After the oil runs clear of contaminants and free of bubbles, remove the hose from the diagnostic coupler. Release the brake pedal.
11. Repeat Steps 8 through 10 for the other front wheel brake.
12. Shut off the engine and allow the steering accumulators to bleed down. Check the hydraulic tank oil level and fill if necessary.

# NOTES

# SECTION J7 PARKING BRAKE INDEX

|                                         |      |
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| MAINTENANCE .....                       | J7-4 |
| Inspection .....                        | J7-4 |
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| Assembly .....                          | J7-7 |
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# NOTES

# PARKING BRAKE

## OPERATION

The truck is equipped with a dry disc type parking brake assembly mounted on each wheel motor rotor shaft. The parking brake assemblies are inboard mounted and can be accessed through the rear axle housing hatch.

The parking brake is intended to prevent truck movement after the truck has stopped and the engine is shut off. The parking brake is not for use during truck loading or dumping operations. Refer to the Operation and Maintenance Manual for additional brake system operation instructions.

The parking brake is a spring applied, hydraulically released, multiple disc type brake that is actuated by the placing the directional control lever in PARK.

When the directional control lever is in PARK, the parking brake solenoid valve (SV2) located on the brake manifold in the hydraulic brake cabinet is de-energized, which removes hydraulic pressure from the parking brake assemblies. Internal belleville springs in the parking brake assemblies act on the piston to compress the disc pack, preventing rotation of gear (4, Figure 7-1), which is mounted on the wheel motor rotor shaft.

When the directional control lever is moved from PARK, pressurized oil is supplied to the brake assemblies through the now energized parking brake solenoid valve. Oil pressure applied to the parking brake piston compresses the belleville springs, releasing the discs to allow the wheel motor rotor gear to rotate.

System interlocks prevent application of the parking brake if the truck is in motion.

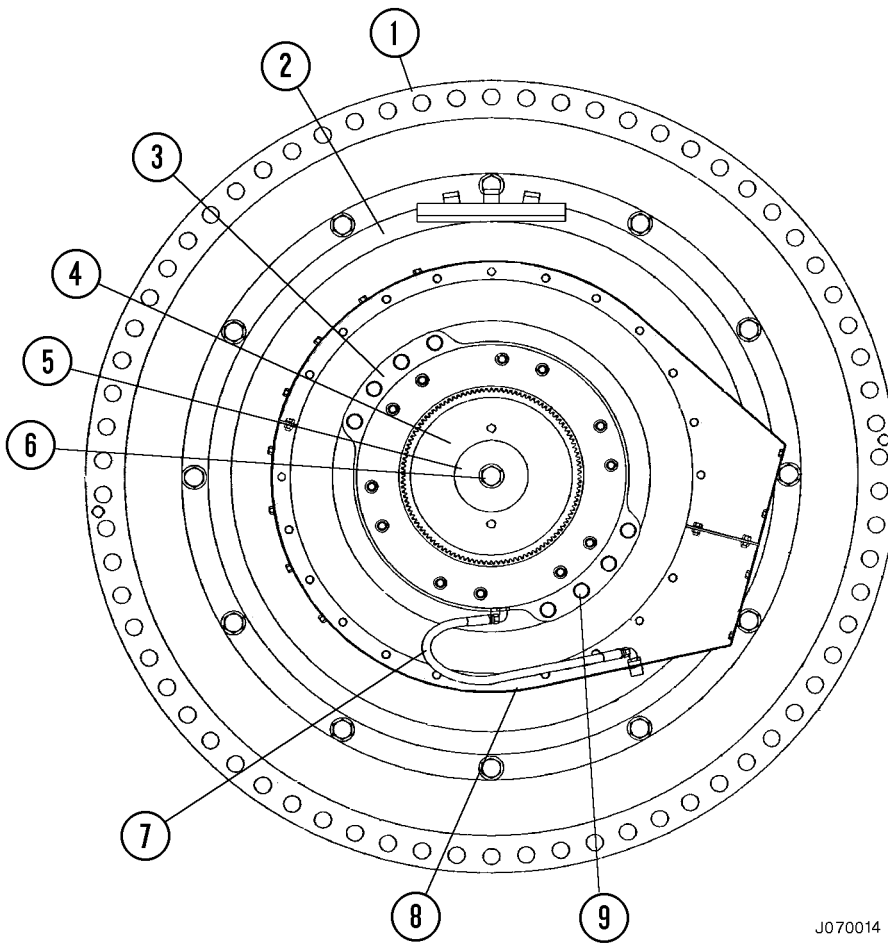


FIGURE 7-1. PARKING BRAKE INSTALLATION

1. Wheel Motor Mounting Flange
2. Wheel Motor
3. Parking Brake Assembly
4. Gear (Motor Rotor Shaft)
5. Retainer Plate
6. Capscrew & Hardened Washer
7. Park Brake Supply Hose
8. Cooling Air Duct Mount
9. Capscrew & Lockwasher

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## MAINTENANCE

The parking brakes require periodic inspection to determine the amount of wear incurred to ensure that adequate brake torque is available. The following inspection should be made at 500 hour intervals.

### Inspection

1. Securely block the wheels to prevent the truck from rolling away.
2. Place the directional control lever in PARK and turn the rest switch ON. Turn the key switch OFF to shut down the engine and allow 90 seconds for the steering system accumulators to bleed down. Open bleed down valves (1) and (3, Figure 7-2) to bleed down the brake accumulators.
3. Open the rear axle housing hatch. Check for signs of oil leakage and damage.
4. Measure the amount of disc wear on both parking brake assemblies as follows:
  - a. Insert a depth micrometer through each piston position hole (18, Figure 7-5) and record the dimensions.
  - b. Add the three measurements and divide the total by 3 to determine the average.
  - c. If the resulting average of the three measurements is greater than 24 mm (0.969 in.), the parking brake assembly must be removed and rebuilt.

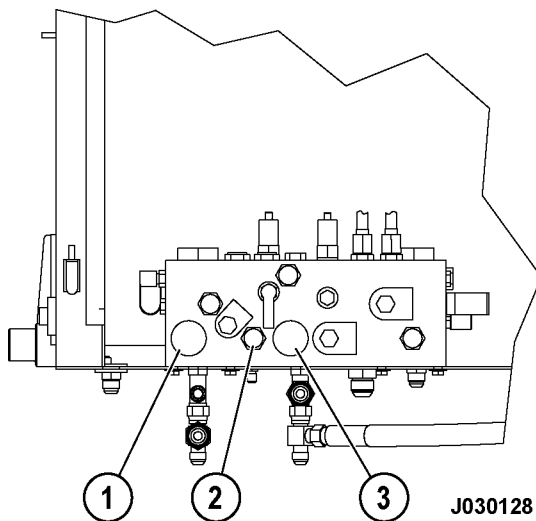


FIGURE 7-2. BRAKE MANIFOLD

1. Accumulator Bleed Down Valve (Rear Brake)
2. Automatic Apply Valve
3. Accumulator Bleed Down Valve (Front Brake)

### Removal

*NOTE: Whenever possible, parking brake repairs should be performed when the wheel motor is removed from the truck. If repairs are necessary when the wheel motor is installed, a lifting device must be set up inside the rear axle housing to support the weight of the parking brake assembly when it is removed from the wheel motor.*

## CAUTION

**The parking brake assembly weighs approximately 159 kg (350 lb). Ensure that the lifting device is capable of supporting the weight of the brake assembly when removed.**

1. Securely block the wheels to prevent the truck from rolling away.
2. Place the directional control lever in PARK and turn the rest switch ON. Turn the key switch OFF to shut down the engine and allow 90 seconds for the steering system accumulators to bleed down. Open bleed down valves (1) and (3, Figure 7-2) to bleed down the brake accumulators.
3. Open the rear axle housing access door and remove the wing nuts that retain the duct inside the door. Remove the duct tube and open the access covers inside the axle housing. Remove the cooling ducts as required to set up a lifting device for brake removal.
4. Disconnect parking brake apply supply hose (7, Figure 7-1) from the parking brake.
5. Remove capscrews and lockwashers (9). Install guide studs in two of the mounting holes to support the parking brake assembly when it is removed from the wheel motor frame.
6. Slide the parking brake assembly out of the wheel motor frame and off inner gear (4). Remove the parking brake assembly from the rear axle housing.

7. If rotor shaft gear (4, Figure 7-1) is worn, damaged or otherwise requires removal, perform the following steps:

*NOTE: The gear is shrink fit on the splined motor shaft.*

- a. Remove capscrew and hardened washer (6, Figure 7-1) from the shaft. Remove retainer plate (5).
- b. Install a gear puller using the tapped holes provided in gear (4).
- c. Apply heat around gear hub area while tightening the puller until gear is removed from shaft.

### Installation

1. If the rotor shaft gear was removed, install the gear before installing the parking brake on the wheel motor:
  - a. Thoroughly clean gear (4, Figure 7-1) and the shaft. Inspect the splines and remove any burrs that may interfere with installation.
  - b. Heat the gear to 280°C (536°F). Immediately install the gear on the shaft. The gear must be fully seated against the shoulder of the rotor shaft.
  - c. Install retainer plate (5), washer and capscrew (6). Tighten the capscrew to **595 - 670 N·m (440 - 495 ft lb)**.
2. Install two guide studs in the wheel motor frame to guide the parking brake assembly into position. Ensure that the mating surfaces are clean and free of burrs.
3. Lift the parking brake into position for installation. Note the proper orientation depending on whether the parking brake is to be installed on a left or right wheel motor. (The bleeder screw at the top of the parking brake will be tilted toward the front of the truck.)
4. Slide the parking brake over the guide studs and the gear on the wheel motor rotor shaft.
5. Install capscrews and lockwashers (9) Tighten the capscrews evenly to ensure that the parking brake housing is properly seated on the wheel motor. Tighten the capscrews to a final torque of **300 N·m (220 ft lb)**.
6. Connect parking brake supply hose (7) and remove any lifting equipment.
7. Bleed the air from the parking brake apply line and housing. Refer to Parking Brake Bleeding Procedure later in this section.
8. Install all ducts that were removed.

### Disassembly

1. Match mark parking brake housing (3, Figure 7-5), piston housing (4), and end cap (5).



***The following procedure is necessary to properly relieve the pressure exerted by the belleville springs on the end cap. Failure to follow this procedure could result in capscrew failure and personal injury during disassembly.***

2. Loosen 12-point capscrews (10) evenly in **27 N·m (20 ft lb)** increments. Alternate by selecting a successive capscrew that is located approximately 180 degrees from the previously loosened capscrew.

Continue loosening capscrews until the spring pressure is released and the capscrews and washers can be removed by hand.

3. Remove end cap (5). Lift belleville springs (7) out of the counterbore of piston (6).
4. Lift the piston/housing assembly off the dowels in housing (3).
5. Remove springs (1), separator discs (12), and friction discs (13).
6. Remove piston (6) from piston housing (4). Remove and discard piston seals (8) and (9).

### Cleaning and Inspection

1. Clean all parts thoroughly.
2. Inspect dowel pins (2). If the dowel pins are grooved from excessive wear or otherwise damaged, press them out of housing (3).
3. Check the free height of compression springs (1) and test for height under load. Replace the springs if they are not within approximately 10% of specification.

Free Height: . . . . . 11.58 mm (0.456 in.)

Height @ 162 N (36.4 lb): . . . . . 8.89 mm (0.350 in.)



4. Inspect Belleville washers (7) for cracks and damage. Replace any springs that are damaged or significantly worn.
5. Measure the unsprung total height (cone height + material thickness) of Belleville washers (7). It must be greater than or equal to 21.16 mm (0.833 in.). Discard any Belleville washers that do not meet this specification.
6. A load deflection test must now be conducted to determine whether the Belleville washers will be reused or discarded. The equipment that is needed for this test are as follows:
  - Universal Testing Machine (UTM) - capacity minimum of 18 144 kgf (40,000 lbf) in compression
  - Machined flat plate
  - Height stand and depth indicator with minimum range of 25 mm (1 in.)
  - a. Place the Belleville washer on a clean, flat and leveled testing machine bed.
  - b. Put the machined flat plate on top of the Belleville washer. Make sure that the plate will not slip from the Belleville washer when force is applied.
  - c. Adjust the height stand so that the depth indicator is at zero.
  - d. Gradually apply force. Follow all necessary safety precautions for operating the UTM. The Belleville washer will start deflecting due to compression as shown in the example in Figure 7-3.
  - e. Note the depth indicator as it moves from 10% to 100% compression of the cone height. Record the force that is required to deflect the Belleville washer in 10% increments up to 100% of the unsprung cone height as shown in Table 1.
  - f. Repeat Steps c through e three times.
  - g. Average the three recorded force values from Step f for a respective deflection percentage.
  - h. Compare the average force values with the "Minimum" values in Table 1. If all of the average force values are above or equal to the minimum specified values, the Belleville washer is acceptable for use in the parking brake assembly.
  - i. If any of the average force values are below the minimum specified values, the Belleville washer will not be of sufficient strength to cause the piston to compress and hold the disc pack. Discard the Belleville washer and replace it with a new one during assembly.
7. It is recommended that the separator discs and friction discs be replaced if wear is indicated.
 

**Min. friction disc material (each side):**  
1.52 mm (0.06 in.)
8. Check the piston and piston housing bore for scratches, nicks, pitting and other defects that may cause seal leakage. Slight defects may be repaired by polishing.

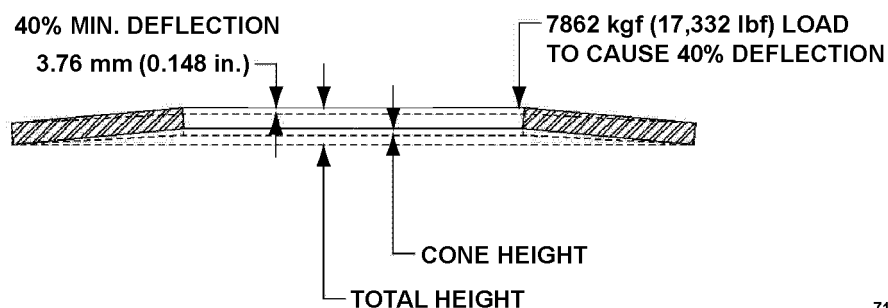


FIGURE 7-3. BELLEVILLE WASHER DEFLECTION

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**Table 1: BELLEVILLE WASHER DEFLECTION SPECIFICATIONS**

| Percentage | Deflection          | Load                    |                         |
|------------|---------------------|-------------------------|-------------------------|
|            |                     | Average                 | Minimum                 |
| 10%        | 0.94 mm (0.037 in.) | 2 285 kgf (5,038 lbf)   | 1 828 kgf (4,030 lbf)   |
| 20%        | 1.88 mm (0.074 in.) | 4 340 kgf (9,568 lbf)   | 3 472 kgf (7,654 lbf)   |
| 30%        | 2.82 mm (0.111 in.) | 6 190 kgf (13,647 lbf)  | 4 952 kgf (10,918 lbf)  |
| 40%        | 3.76 mm (0.148 in.) | 7 862 kgf (17,332 lbf)  | 6 290 kgf (13,866 lbf)  |
| 50%        | 4.70 mm (0.185 in.) | 9 380 kgf (20,679 lbf)  | 7 504 kgf (16,543 lbf)  |
| 60%        | 5.64 mm (0.222 in.) | 10 770 kgf (23,744 lbf) | 8 616 kgf (18,995 lbf)  |
| 70%        | 6.58 mm (0.259 in.) | 12 058 kgf (26,583 lbf) | 9 646 kgf (21,266 lbf)  |
| 80%        | 7.52 mm (0.296 in.) | 13 269 kgf (29,253 lbf) | 10 615 kgf (23,402 lbf) |
| 90%        | 8.46 mm (0.333 in.) | 14 429 kgf (31,811 lbf) | 11 543 kgf (25,449 lbf) |
| 100%       | 9.40 mm (0.370 in.) | 15 564 kgf (34,312 lbf) | 12 451 kgf (27,450 lbf) |

### Assembly

*NOTE: New friction discs and separator discs must be cleaned and inspected to ensure proper parking brake operation. If new discs are being installed, refer to Cleaning and Inspecting New Discs later in this section.*

1. If dowel pins (2, Figure 7-5) have been removed, press new dowel pins into the holes in housing (3) until they are fully seated against the shoulder.
2. Install one separator disc (12) over the dowel pins. Insert one compression spring (1) over the dowel pins. Insert one friction disc (13).
3. Install second compression spring (1) over the dowel pins. Install one separator disc (12) over the dowel pins. Insert second friction disc (13). Install remaining separator disc (12) over the dowel pins.

*NOTE: The internal teeth of the two friction discs must be aligned and must be concentric with the pilot diameter machined on the rear of the housing to enable the completed assembly to be installed properly. If available, a mating gear as installed on the wheel motor rotor shaft should be used to simplify alignment.*

4. Assemble seals (8) and (9) on the piston:
  - a. Install the O-ring seal in the bottom of each groove. Make sure that the O-rings are not twisted.
  - b. If available, use an expander to expand the outside diameter of each seal enough to allow it to slide onto the piston.

*NOTE: If an expander tool is not available, the seals may be heated to 204°C (400°F) MAXIMUM. This will allow ring to be manually expanded until it will fit over piston.*

- c. When each seal is over the groove, resize by compressing the seal ring. Use a full circle clamp with any sharp edges or grooves covered to prevent damage to the seal surfaces.
  - d. Install the quad ring in each seal.
5. Lubricate the piston seals and insert piston (6) into piston housing (4).
  6. Install piston/housing assembly over the dowels in housing (3).
  7. Place belleville springs (7) in the piston counterbore as shown in Figure 7-5.
  8. Place end cap (5) on the assembly and insert capscrews (10) with hardened washers (11). Do not tighten the capscrews.

## ⚠ DANGER

***The following procedure is necessary to properly compress the belleville springs. Failure to follow this procedure could result in capscrew failure and personal injury during assembly.***

9. Select three capscrews that are 120° apart and mark them.
  - a. Tighten the marked capscrews to **41 N·m (30 ft lb)**. Snug the remaining capscrews after each of the marked capscrews is tightened in the following steps.
  - b. Retighten the marked capscrews until the gap between the end cap and piston housing is equalized.
  - c. Retighten the marked capscrews in **27 N·m (20 ft lb)** increments until they are fully tightened to **122 N·m (90 ft lb)**.
  - d. Tighten the remaining capscrews to **122 N·m (90 ft lb)**.
10. Install bleeder and O-ring plug (17). Install a fitting in one of the pressure supply ports and attach a hydraulic power source. Install an O-ring plug in the remaining port.
11. Slowly apply pressure and open the bleeder valve to bleed air from the piston cavity. Close the bleeder. Apply 2 068 kPa (300 psi) of hydraulic pressure and hold it for one minute.
12. Check for oil leakage. If leakage occurs, the parking brake assembly must be disassembled and repaired.

### **Cleaning and Inspecting New Discs**

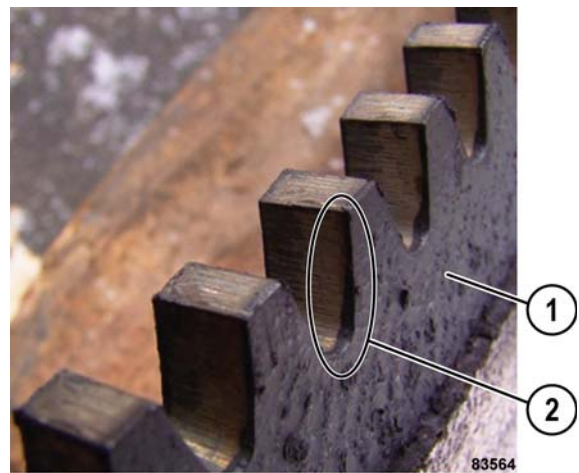
A rust preventive coating is applied to separator discs and friction discs. This coating must be removed before installing the parking brake.

**NOTE:** Use a cleaner containing organic-based solvents, such as acetone, methanol or ethanol, to clean the coating from the discs. Water-based cleaners, such as mild soaps and detergents, do not adequately remove the rust preventive coating.

1. Spray or coat both sides of the disc with the cleaner.
2. Use a cloth to quickly wipe away the coating before the cleaner evaporates.
3. Reapply the cleaner and wipe as necessary.

During the manufacture of friction discs, excess adhesive, which is used to bond the friction material to the disc, may get onto the inner teeth of the friction disc and solidify. This adhesive may prevent the friction disc from properly engaging the parking brake adapter gear.

1. Inspect the inner teeth of each friction disc. The teeth must be free of burrs and excess adhesive. Refer to Figure 7-4.
2. If adhesive is found on the teeth, use a diamond needle file (half-round, 150 grit) to slowly and carefully remove the excess adhesive from the inside of each affected tooth.
3. Thoroughly clean the teeth with an organic-based cleaning solvent as detailed previously.



**FIGURE 7-4. FRICTION DISC INNER TEETH**

1. Adhesive
2. Excess Adhesive on Inner Tooth

**NOTE:** If a new wheel motor is installed or adapter gear (14, Figure 7-5) is replaced, clean the rust preventive coating from the adapter gear teeth with an organic-based cleaning solvent as detailed previously. Do not install the parking brake until the adapter gear is completely dry.

## PARKING BRAKE BLEEDING PROCEDURE

1. Securely block the wheels to prevent the truck from rolling away.
2. Place the directional control lever in PARK and turn the rest switch ON. Turn the key switch OFF to shut down the engine and allow 90 seconds for the steering system accumulators to bleed down. Open bleed down valves (1) and (3, Figure 7-2) to bleed down the brake accumulators.
3. Open the rear axle housing access door. Close the brake accumulator bleed down valves.
4. Start the engine and allow the steering system to fully charge (pump unloads). Then move the directional control lever to NEUTRAL to disengage the parking brake.
5. Connect a clear plastic hose to bleeder screw (17, Figure 7-5). Place the other end of hose into a container.
6. Slowly open the bleeder screw and allow hydraulic oil to run until it is clear and free of bubbles. Close the bleeder screw securely.
7. Repeat Steps 5 and 6 for the other parking brake.
8. Shut off the engine. Remove the bleed equipment and close the rear axle housing hatch.

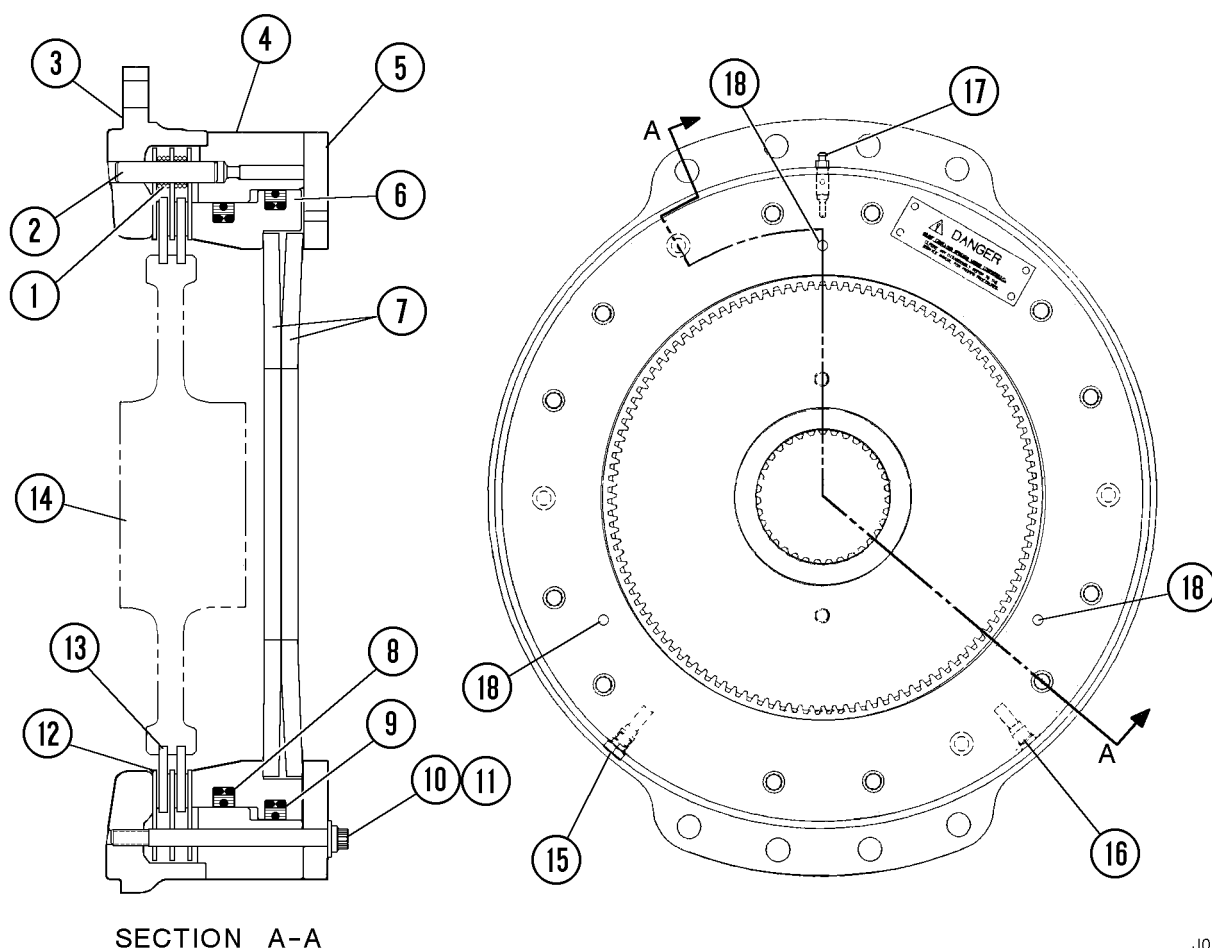


FIGURE 7-5. PARKING BRAKE ASSEMBLY

- |                          |                         |                           |
|--------------------------|-------------------------|---------------------------|
| 1. Compression Spring    | 7. Belleville Springs   | 13. Friction Disc         |
| 2. Dowel Pin             | 8. Piston Seal Assembly | 14. Adapter Gear          |
| 3. Parking Brake Housing | 9. Piston Seal Assembly | 15. Plug                  |
| 4. Piston Housing        | 10. Capscrew            | 16. Oil Supply Port       |
| 5. End Cap               | 11. Hardened Washer     | 17. Bleeder Screw         |
| 6. Piston                | 12. Separator Disc      | 18. Piston Position Holes |

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# NOTES

# SECTION L

## HYDRAULIC SYSTEM

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# NOTES

## SECTION L2

### HYDRAULIC SYSTEM

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# NOTES

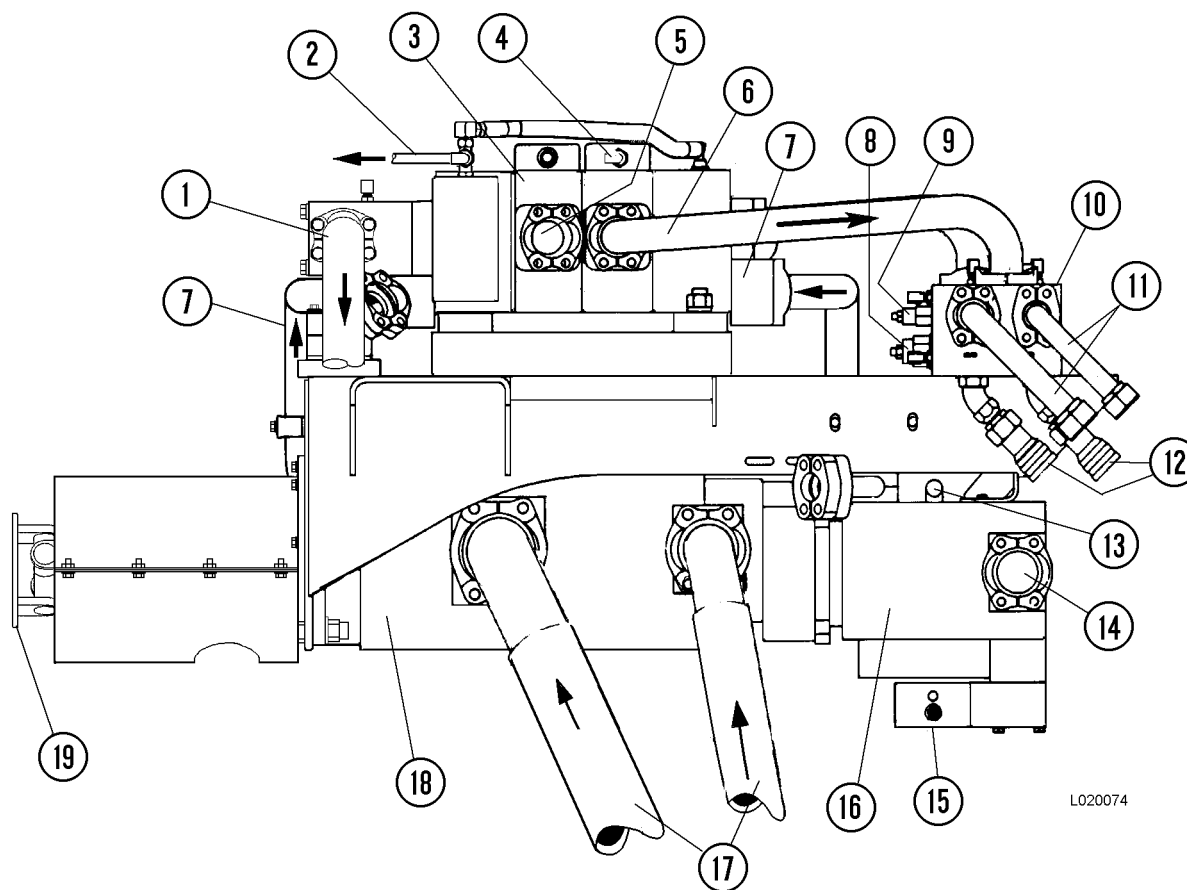
## HYDRAULIC SYSTEM

The following is a general description of the hydraulic system. Additional information concerning individual component description and operation can be found under the different system circuits such as the hoist circuit, steering circuit, and hydraulic brake circuit.

### HYDRAULIC PUMP MODULE

Hydraulic pumps (16) and (18, Figure 2-1) are mounted on a modular structure behind the traction alternator. The pump module also contains pilot operated hoist valve (3), which controls operation of the hoist cylinders during body dumping operations. Overcenter manifold (10) contains internal valves (8) and (9), which ensure smooth operation and maximum service life from the hoist cylinders.

The overcenter manifold also provides quick disconnect fittings (12) to allow operation of the hoist circuit by an external hydraulic source or to use the hoist circuit of the truck to dump the load from a disabled truck.



L020074

FIGURE 2-1. HYDRAULIC PUMP MODULE

- |                                                             |                                    |                              |
|-------------------------------------------------------------|------------------------------------|------------------------------|
| 1. Hoist Valve Tank Return (To Brake/Hoist Return Manifold) | 7. Supply From Hoist Pump & Filter | 14. Steering Pump Inlet Port |
| 2. Supply to Pilot Valve                                    | 8. Counterbalance Valve            | 15. Unloader Valve           |
| 3. Hoist Valve                                              | 9. Needle Valve                    | 16. Steering/Brake Pump      |
| 4. From Pilot Valve                                         | 10. Overcenter Manifold            | 17. Hoist Pump Inlet Hoses   |
| 5. To Brake/Hoist Return Manifold                           | 11. Hoist Cylinder Supply Tubes    | 18. Hoist Pump               |
| 6. Work Port Outlet                                         | 12. Hoist Quick Disconnects        | 19. Pump Driveshaft          |
|                                                             | 13. Pump Case Drain Line           |                              |



***Relieve pressure before disconnecting hydraulic and other lines. Tighten all connections before starting truck.***

***Hydraulic fluid escaping under pressure can have sufficient force to enter a person's body by penetrating the skin and cause serious injury and possibly death if proper medical treatment by a physician familiar with this injury is not received immediately.***

***Always maintain complete cleanliness when opening any hydraulic connection. Ensure that all system lines are capped while the component is removed from the truck.***

***NOTE: It is highly recommended that any hydraulic oil to be used for filling or adding to the hydraulic system is routed through a 3 micron filter device prior to use.***

## **HOIST CIRCUIT OPERATION**

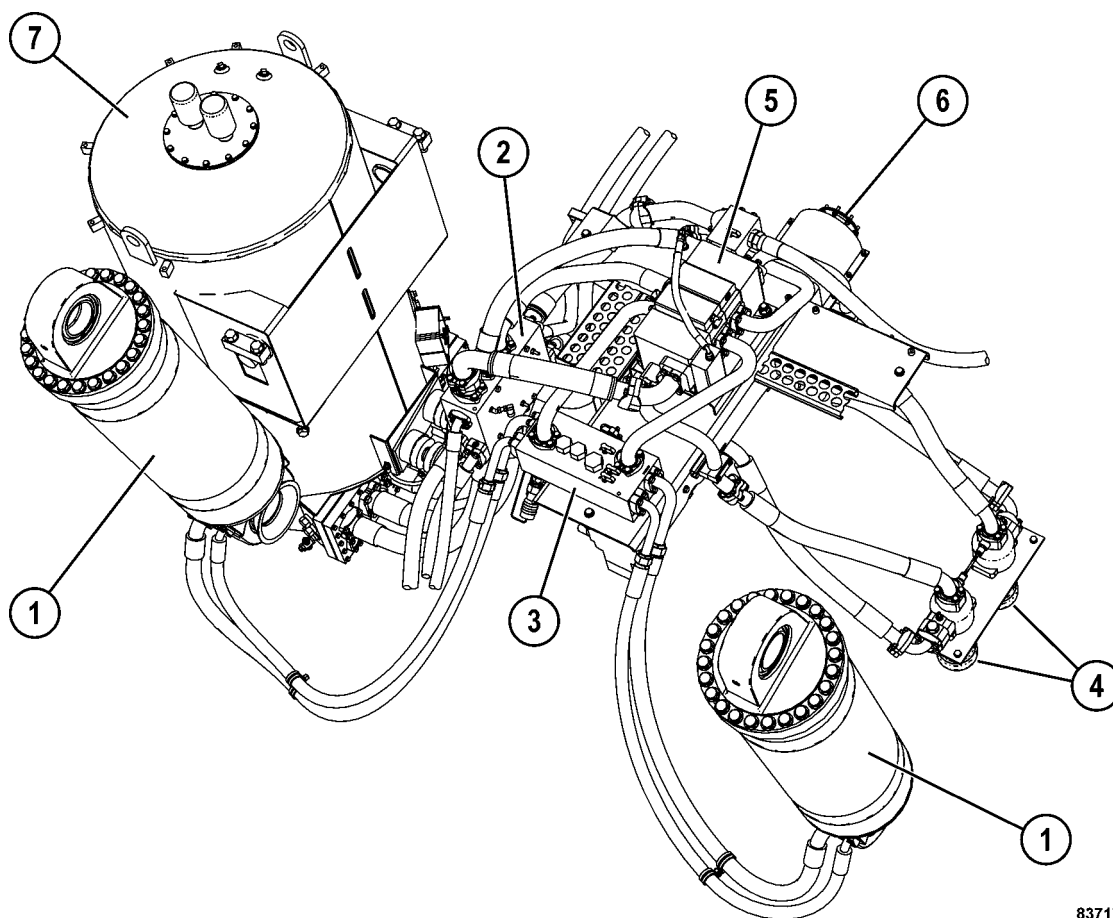
The hoist, steering and brake circuits share a common hydraulic tank (7, Figure 2-2). The tank is located on the left side of the frame in front of the rear wheels. The service capacity of the tank is 947 L (250 gal). Type C-4 hydraulic oil is recommended for use in the hydraulic system. Oil used in the hoist, steering and brake circuits flows from the bottom of the tank through 100 mesh wire suction strainers.

Hydraulic oil from the tank is supplied to a gear type hoist circuit pump (18, Figure 2-1), rated at 931 l/min (246 gpm) @ 1,900 RPM and 17 200 kPa (2,500 psi). Oil from the pump outlet port is directed to high pressure filters (4, Figure 2-2) and then enters hoist valve (5).

The hoist pump and steering/brake pump are driven by driveshaft (6) off the rear of the traction alternator.

Hoist valve (5) directs oil flow to hoist cylinders (1) when the operator moves the hoist control lever. The hoist control lever is connected to the hoist pilot valve, located in the hydraulic components cabinet, by a flexible control cable. If the control lever is in the FLOAT position, oil is returned to the hydraulic tank through brake/hoist return oil manifold (2) and also flows through the disc brake circuit to cool the wet disc brake system. Heat generated during service brake application is removed by heat exchanger (1, Figure 2-5).

Hoist system pressure is limited to a maximum of 17 200 kPa (2,500 psi).



83717

FIGURE 2-2. HOIST CIRCUIT

- |                                    |                     |
|------------------------------------|---------------------|
| 1. Hoist Cylinders                 | 5. Hoist Valve      |
| 2. Brake/Hoist Return Oil Manifold | 6. Pump Drive Shaft |
| 3. Overcenter Manifold             | 7. Hydraulic Tank   |
| 4. Hoist Circuit Filters           |                     |

## STEERING CIRCUIT OPERATION

Hydraulic oil from the tank is supplied to a piston type, pressure compensated steering and brake circuit pump (8, Figure 2-3), rated at 250 l/min (66 gpm) @ 1,900 RPM @ 17 200 kPa (2,500 psi). Unloader valve (15, Figure 2-1) maintains system pressure between 18 960 kPa (2,750 psi) and 20 860 kPa (3,025 psi).

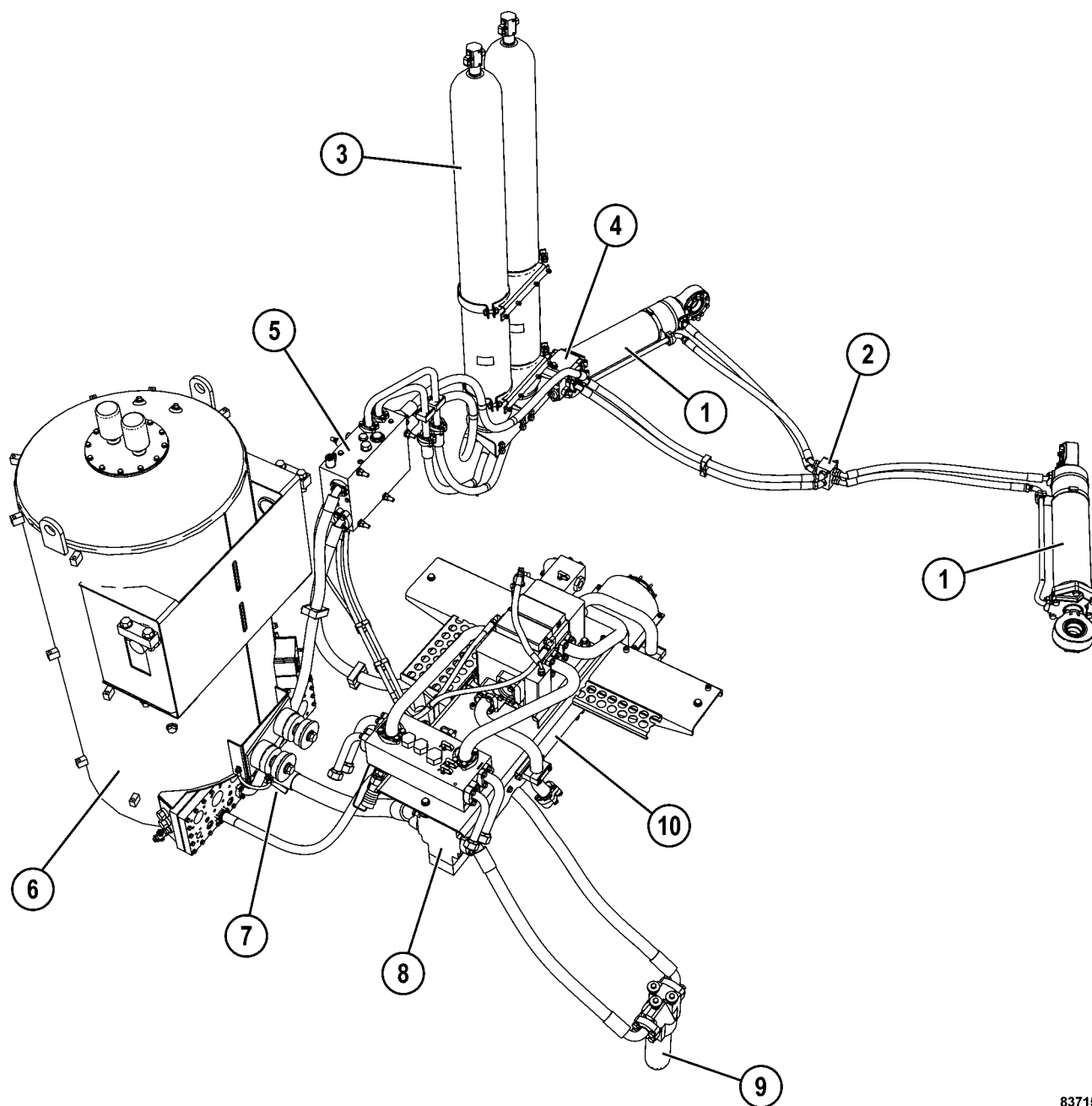
Oil from the pump is directed to high pressure filter (9, Figure 2-3) before entering bleddown manifold (5). The bleddown manifold provides several functions in the steering, brake, and hoist systems. It is used to route oil flow to the steering system and hydraulic brake system.

Refer to Section J for information regarding operation of the brake system.

Steering system oil is directed to steering accumulators (3), which store pressurized oil in the event of loss of pump pressure. Oil is also directed to flow amplifier valve (4) for use by the steering control unit and steering cylinders (1).

A relatively small volume of oil is supplied from the flow amplifier to the steering control unit, which is mounted on the steering column in the cab. When the operator moves the steering wheel, oil is routed from the steering control unit back to the flow amplifier based on the direction and rate of speed of rotation of the steering wheel.

The flow amplifier provides a large volume of oil (determined by input from the steering control unit) to the steering cylinders to turn the front wheels.



83715

FIGURE 2-3. STEERING CIRCUIT

- |                          |                         |
|--------------------------|-------------------------|
| 1. Steering Cylinders    | 6. Hydraulic Tank       |
| 2. Manifold              | 7. Shutoff Valve        |
| 3. Steering Accumulators | 8. Steering/Brake Pump  |
| 4. Flow Amplifier Valve  | 9. High Pressure Filter |
| 5. Bleeddown Manifold    | 10. Hoist Circuit Pump  |

## DISC BRAKE COOLING SYSTEM

When the hoist control valve is in the FLOAT or HOLD position, oil is supplied by the hoist valve to the disc brake assemblies of each wheel to remove heat that is generated during application of the service brakes.

Oil supplied to the rear wheel brakes flows from the hoist valve to brake/hoist return oil manifold (11, Figure 2-5), to the brake assemblies, and is then returned to the brake/hoist return oil manifold and hydraulic tank (12). Oil supplied to the front wheel brakes is routed from the hoist valve to oil/water heat exchanger (1) before entering the brake assemblies. The front brake oil is then returned to the brake/hoist return oil manifold and then to the hydraulic tank. The front brake cooling oil is also used to lubricate the front wheel hub bearings.

The brake cooling circuit pressure is limited by orificed regulator valves located in the brake/hoist return oil manifold. Figure 2-4 shows the piping and flow of oil through the brake/hoist return oil manifold. The internal check valves crack open at 241 kPa (35 psi). Their opening pressure will be slightly higher as flow through the manifold increases.

Refer to Section J for information regarding operation of the brake system.

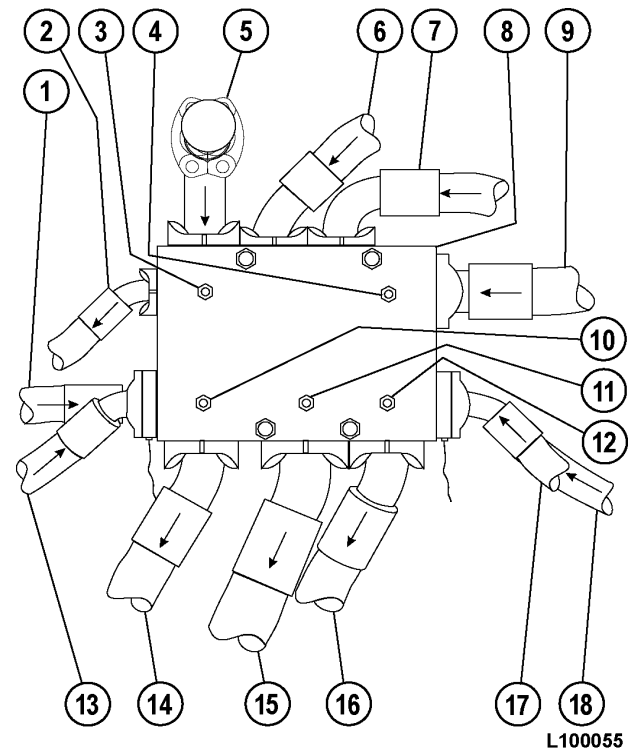
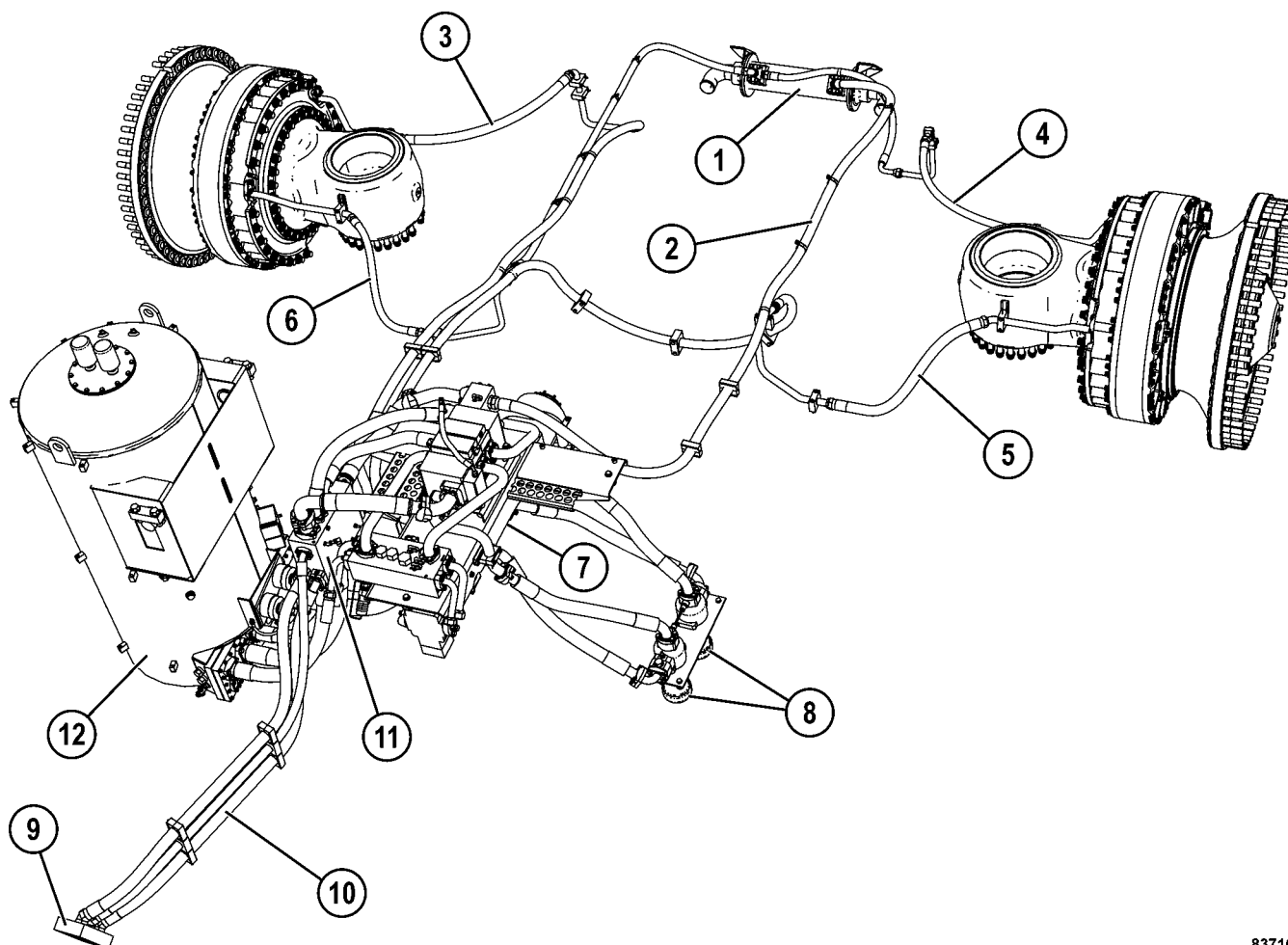


FIGURE 2-4. BRAKE/HOIST RETURN OIL MANIFOLD

1. Rear Brake Return
2. Rear Brake Supply
3. Rear Brake Supply Press.
4. Front Brake Supply Press.
5. Rear Brake Cooling Oil (From Hoist Valve)
6. Return From Hoist Valve
7. Return From Hoist Valve
8. Manifold
9. Front Brake Bypass
10. Rear Brake Return Pressure
11. Hoist Return Pressure
12. Front Brake Return Pressure
13. Rear Brake Oil Return
14. Rear Brake Oil Return to Tank
15. Hoist Return to Tank
16. Front Brake Oil Return to Tank
17. Front Brake Oil Return
18. Front Brake Oil Return



83716

FIGURE 2-5. DISC BRAKE COOLING CIRCUIT

- |                                                    |                                                |
|----------------------------------------------------|------------------------------------------------|
| 1. Heat Exchanger                                  | 7. Hoist Pump                                  |
| 2. Front Brake Cooling Oil Supply From Hoist Valve | 8. High Pressure Filters                       |
| 3. Left Front Brake Cooling Outlet Hose            | 9. Rear Axle Housing                           |
| 4. Right Front Brake Cooling Inlet Hose            | 10. Rear Brake Cooling Supply and Return Lines |
| 5. Right Front Brake Cooling Outlet Hose           | 11. Brake/Hoist Return Oil Manifold            |
| 6. Left Front Brake Cooling Inlet Hose             | 12. Hydraulic Tank                             |



# NOTES

# SECTION L3

## HYDRAULIC SYSTEM COMPONENT REPAIR

### INDEX

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# NOTES

# HYDRAULIC SYSTEM COMPONENT REPAIR

## HOIST PUMP

### Removal

*NOTE: The hoist pump can be removed without removing the steering pump from the truck if desired.*

1. Turn key switch OFF and allow approximately 90 seconds for the accumulators to bleed down.
2. If necessary, drain the hydraulic tank by using the drain valve located on the rear side of the tank.

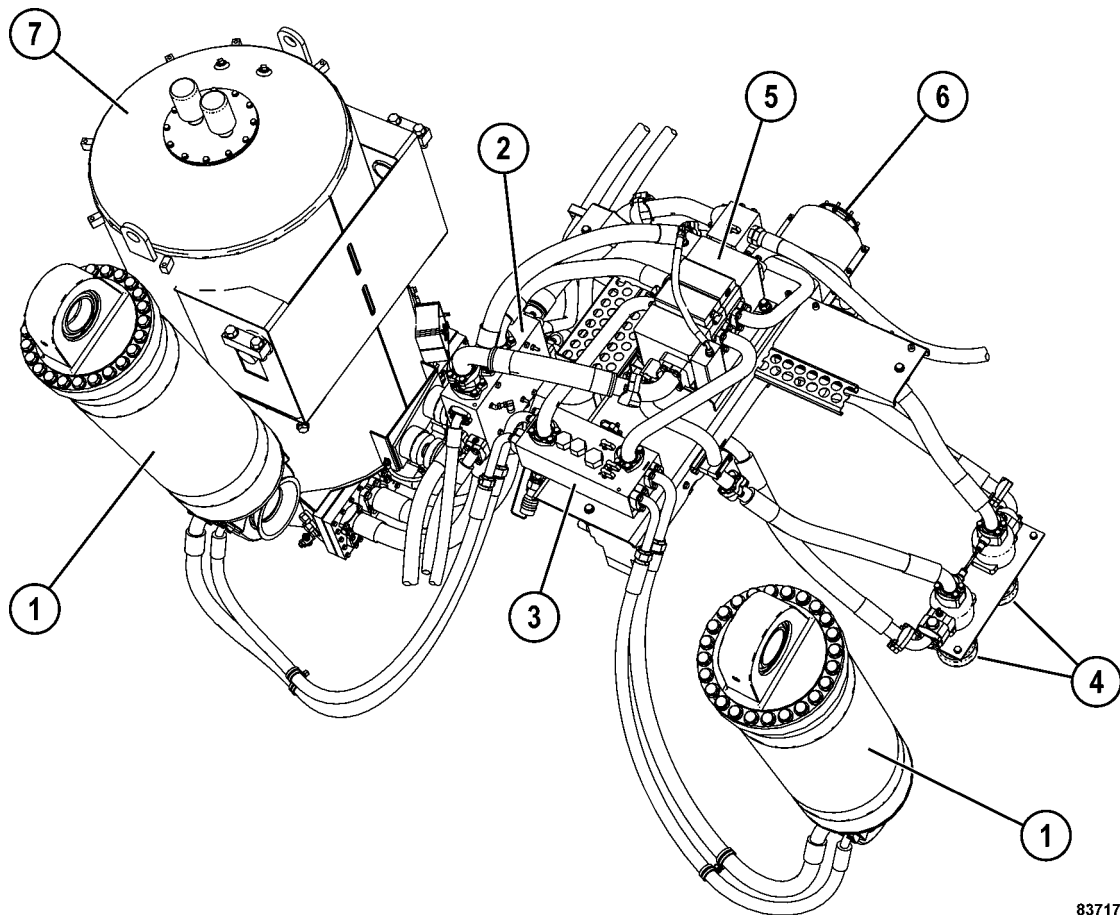
*NOTE: If oil in the hydraulic tank has not been contaminated, the shutoff valves can be closed and both pump inlet lines can be drained, eliminating the need to completely drain the tank.*

3. Remove the rear axle blower duct to allow the hoist pump to be lowered from the pump module mounting bracket for removal. Remove duct support bracket.

4. Close the pump supply shutoff valves.

## ! IMPORTANT !

*Always maintain complete cleanliness when opening any hydraulic connection. Make sure that all system lines and components are capped while the component is removed from the truck.*



83717

FIGURE 2-1. HOIST CIRCUIT

- |                                    |                     |
|------------------------------------|---------------------|
| 1. Hoist Cylinders                 | 5. Hoist Valve      |
| 2. Brake/Hoist Return Oil Manifold | 6. Pump Drive Shaft |
| 3. Overcenter Manifold             | 7. Hydraulic Tank   |
| 4. High Pressure Filters           |                     |

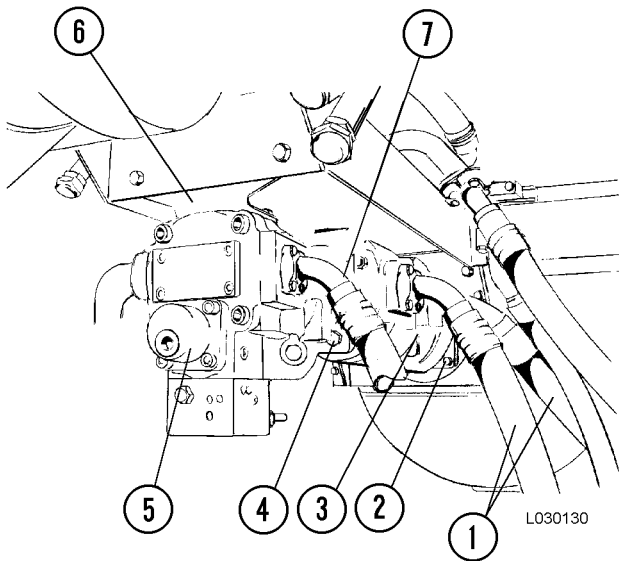


FIGURE 3-2. HYDRAULIC PUMP

- |                       |                        |
|-----------------------|------------------------|
| 1. Pump Outlet Hoses  | 5. Steering/Brake Pump |
| 2. Mounting Capscrews | 6. Pump Mount Bracket  |
| 3. Hoist Pump         | 7. Outlet Hose         |
| 4. Mounting Capscrews |                        |

5. Loosen the capscrews securing the inlet and outlet hoses (1, Figure 3-2) on the hoist pump and allow oil to drain. Remove inlet and outlet hoses. Cap or cover all lines and pump inlets and outlets to prevent contamination.

6. Remove the capscrews that secure the hoist pump drive flange to the drive shaft.

## ⚠ WARNING

*The hoist pump weighs approximately 140 kg (310 lb). The hoist and steering pump together weigh approximately 254 kg (560 lb). Use a suitable lifting or support device that can handle the load safely.*

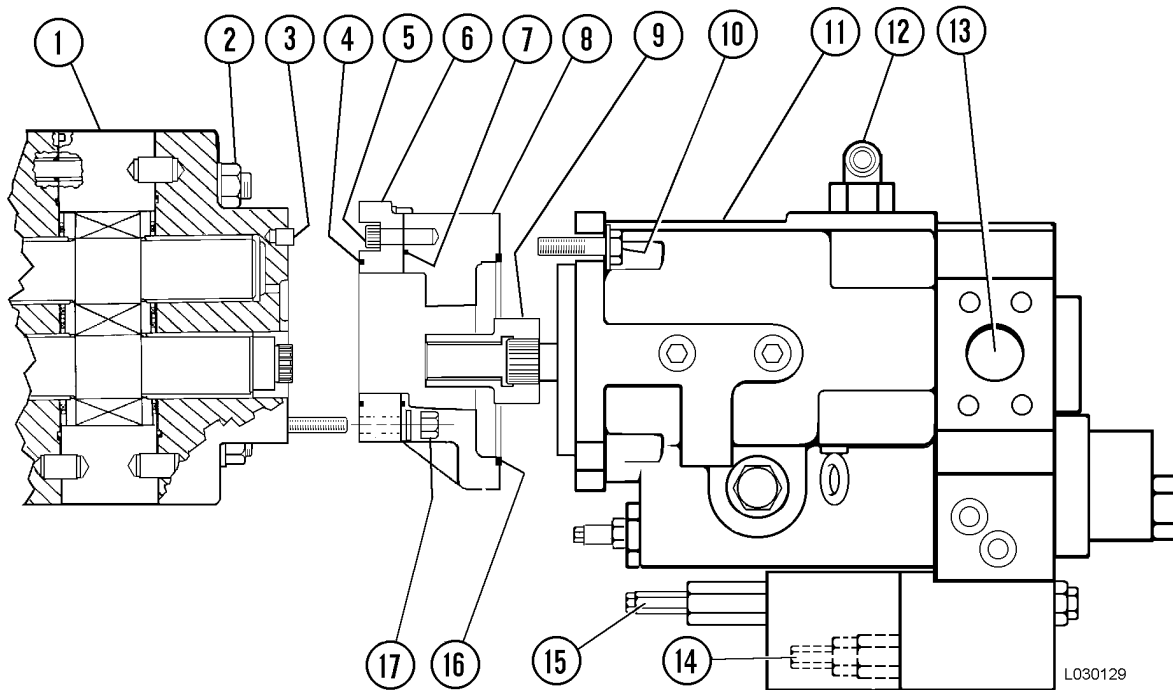


FIGURE 3-3. STEERING/BRAKE PUMP REMOVAL

- |                 |                     |                           |                  |
|-----------------|---------------------|---------------------------|------------------|
| 1. Hoist Pump   | 6. Bearing Plate    | 11. Steering & Brake Pump | 14. Compensator  |
| 2. Nut & Washer | 7. O-ring           | 12. Pump Case Return      | Adjustment Screw |
| 3. Dowel        | 8. Transition Plate | Fitting (Pump Drain)      | 15. Unloader     |
| 4. O-ring       | 9. Coupling         | 13. Inlet Port            | Adjustment Screw |
| 5. Capscrew     | 10. Capscrew        |                           | 16. O-ring       |

7. Attach a suitable lifting or support device to the hoist pump. Attach a support to the front end of the steering pump to hold it in place during removal of the hoist pump.
8. Remove four capscrews (2, Figure 3-2) that secure the hoist pump to the front support bracket. Remove the six capscrews that hold support bracket (6) to the pump module support. Remove support bracket.
9. Make sure that the lifting and support devices are in place on both pumps. Loosen (but do not remove) the rear support bracket capscrews that retain the steering pump. Lower the pumps, allowing the hoist pump to come down further than the steering pump.
10. Remove four steering/brake pump mounting capscrews (4). Slide the hoist pump forward to disengage the splines of drive coupling (9, Figure 3-3) from the steering pump.
11. Move the pump to a clean work area for disassembly.

## Installation

*NOTE: The following assumes the steering pump is already in position on the truck.*

1. Install O-ring (16, Figure 3-3) to steering pump (11). Install coupler (9) to hoist pump.



***The hoist pump weighs approximately 140 kg (310 lb). The hoist and steering pump together weigh approximately 254 kg (560 lb). Use a suitable lifting or support device that can handle the load safely.***

2. Attach a suitable lifting or support device to the hoist pump. Move the pump into position in the truck.
3. Lubricate the steering pump spline shaft and align it with coupling (9). Install the hoist pump onto the steering pump. Install capscrews (10) with hardened washers and tighten the capscrews to standard torque. Raise the pumps up into position.
4. Attach the front support bracket to the pump module support structure and to the pump with capscrews, lockwashers and nuts. Tighten the capscrews to standard torque.
5. Connect the hoist pump drive flange to the drive shaft with capscrews, lockwashers and nuts. Tighten the capscrews to standard torque.
6. Tighten the capscrews for steering pump support bracket (6, Figure 3-2) to standard torque.
7. Uncap the inlet and outlet hoses and connect them to the pumps using new O-rings. Tighten the capscrews securely.
8. Service the hydraulic tank with C-4 type hydraulic oil. Refer to the topic Hydraulic Tank later in this section for filling instructions.
9. Open the three suction line shutoff valves. Loosen the capscrews (at the pump) on the suction hoses to bleed any trapped air. Then loosen the capscrews (at the pump) on the pressure hoses to bleed any trapped air. Tighten all capscrews securely.

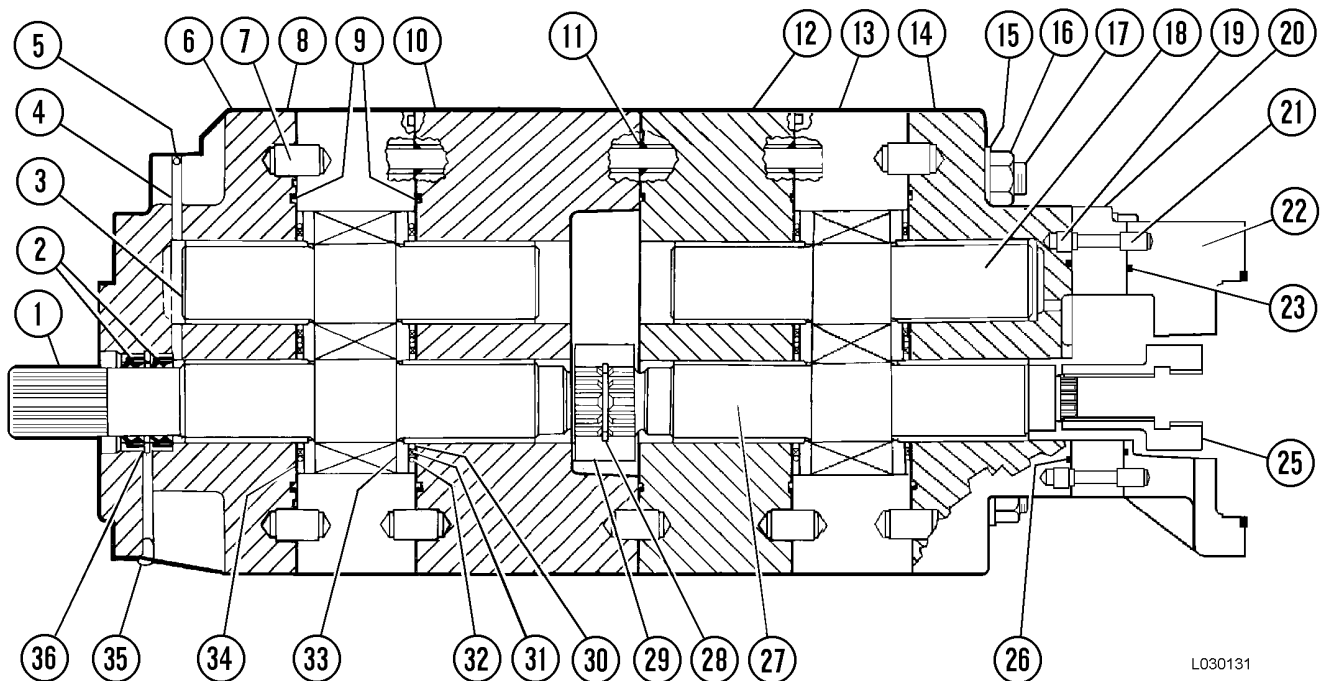
*NOTE: If trapped air is not bled from the steering pump, possible pump damage and no output may result.*

10. Reinstall the blower duct and support bracket.

## Disassembly

**NOTE:** As parts are removed, lay them out in a group in the same order in which they are removed.

1. Thoroughly clean the exterior of the pump assembly. If the steering pump is attached, remove capscrews (10, Figure 3-3) and pull the steering pump free of transition plate (8). Remove O-ring (16).
2. Remove and inspect coupling (9). Remove and discard the internal snap ring, if installed. Remove dowels (3) if damaged or if replacement of the bearing plate is necessary.
3. Support the pump by placing it on wood blocks with the input drive shaft pointing down. Mark each section nearest to the input drive gear to facilitate reassembly.
4. Remove nuts (17, Figure 3-3). Remove bearing plate (20, Figure 3-4) with transition plate (22) and O-ring (26). Remove the capscrews that secure the bearing plate to the transition plate. Remove O-ring (23). Remove the dowels if damaged or if replacement of the transition plate is necessary.



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FIGURE 3-4. HOIST PUMP

- |                       |                     |                       |                     |
|-----------------------|---------------------|-----------------------|---------------------|
| 1. Drive Gear & Shaft | 10. Connector Plate | 19. Dowel             | 28. Snap Ring       |
| 2. Seal               | 11. O-ring          | 20. Bearing Plate     | 29. Coupler         |
| 3. Idler Gear         | 12. Bearing Plate   | 21. Dowel             | 30. Steel Ring      |
| 4. Stud               | 13. Gear Plate      | 22. Transition Plate  | 31. Backup Ring     |
| 5. Steel Ball         | 14. Connector Plate | 23. O-ring            | 32. O-ring          |
| 6. Flange             | 15. Washer          | 24. (Not Used)        | 33. Pressure Plate  |
| 7. Dowel              | 16. Nut             | 25. Coupler           | 34. Isolation Plate |
| 8. Gear Plate         | 17. Stud            | 26. O-ring            | 35. Plug            |
| 9. O-ring             | 18. Idler Gear      | 27. Drive Gear (Rear) | 36. Snap Ring       |

5. Remove connector plate (9, Figure 3-5). Remove O-ring (8) and steel rings (10) and (14). Remove dowels (6) if damaged or if replacement of the connector plate is necessary.

*NOTE: If the connector plate is stuck, tap lightly with a plastic hammer to loosen.*

6. Remove backup ring (15), O-ring and retainer (16) and isolation plate (17). Grasp the drive gear (12) and idler gear (11) and pull straight up and out of the gear plate (5) bore. Remove pressure plate (18) from gears.

7. Remove gear plate (5) and pressure plate (19). Remove the steel rings, backup ring, O-ring and retainer, and isolation plate. Remove O-ring (3) and stud O-ring (4).

8. Remove bearing plate (2, Figure 3-5). Remove O-ring (23) and stud O-rings (1). Remove dowels (20) if damaged or if replacement of the bearing plate is necessary.

9. Remove coupling (21). Remove snap ring (22) if damaged or if replacement is necessary.

*NOTE: Disassembly of the rear pump section is now complete. Do not remove the thru studs at this time because the studs serve as guides for disassembly.*

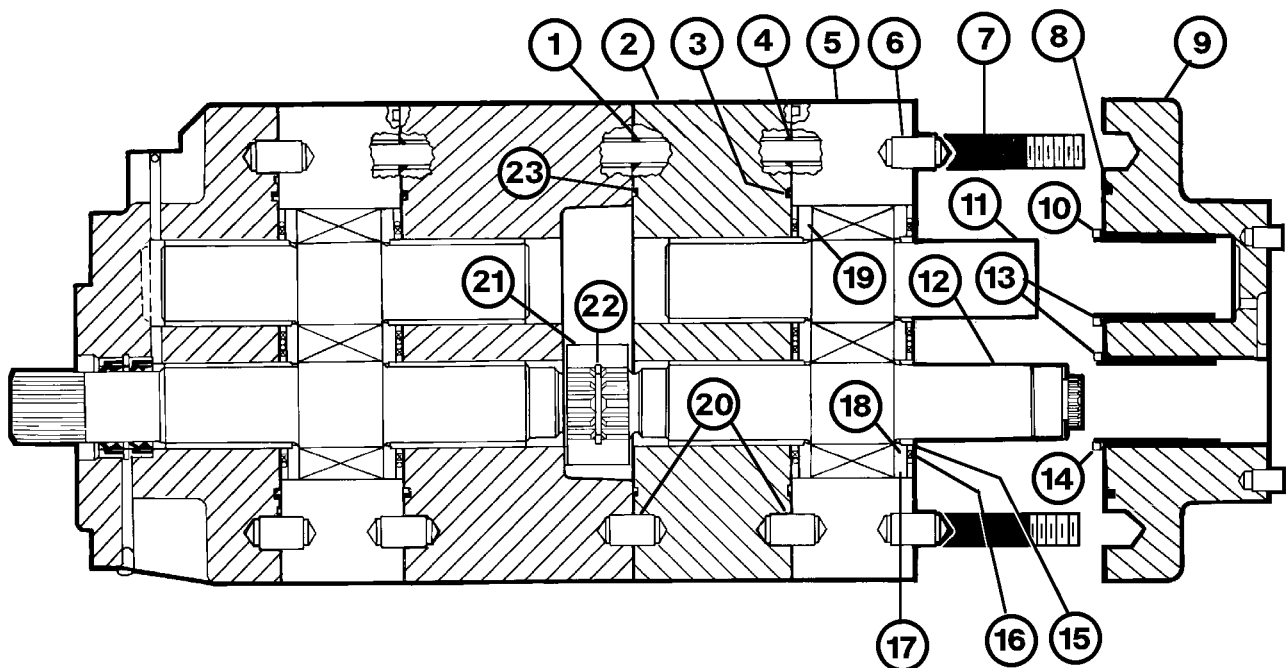


FIGURE 3-5. HOIST PUMP DISASSEMBLY (Rear Section)

- |                  |                       |                       |                    |
|------------------|-----------------------|-----------------------|--------------------|
| 1. O-ring        | 7. Stud               | 13. Bearings          | 19. Pressure Plate |
| 2. Bearing Plate | 8. O-ring             | 14. Steel Ring        | 20. Dowels         |
| 3. O-ring        | 9. Connector Plate    | 15. Backup Ring       | 21. Coupling       |
| 4. O-ring        | 10. Steel Ring        | 16. O-ring & Retainer | 22. Snap Ring      |
| 5. Gear Plate    | 11. Idler Gear        | 17. Isolation Plate   | 23. O-ring         |
| 6. Dowel         | 12. Drive Gear (Rear) | 18. Pressure Plate    |                    |



10. Remove bearing plate (10, Figure 3-6). If the bearing plate is stuck, tap lightly with a plastic hammer to loosen it. Remove O-rings (9) and (11).

11. Remove steel rings (13), backup ring (14), O-ring and retainer (15), and isolation plate (17). Remove dowels (16) if damaged or if replacement of bearing plate (10) is necessary.

12. Unscrew thru studs (12) and remove them. Remove flange (5). If the flange is stuck, tap the flange lightly with a plastic hammer to loosen. Remove O-ring (8). Remove dowels (6) if damaged or if replacement of flange (5) or gear plate (7) is necessary.

13. Remove the steel rings, backup ring, O-ring and retainer. Remove drive gear (1) and idler (3) from gear plate (7). Remove both pressure plates (18).

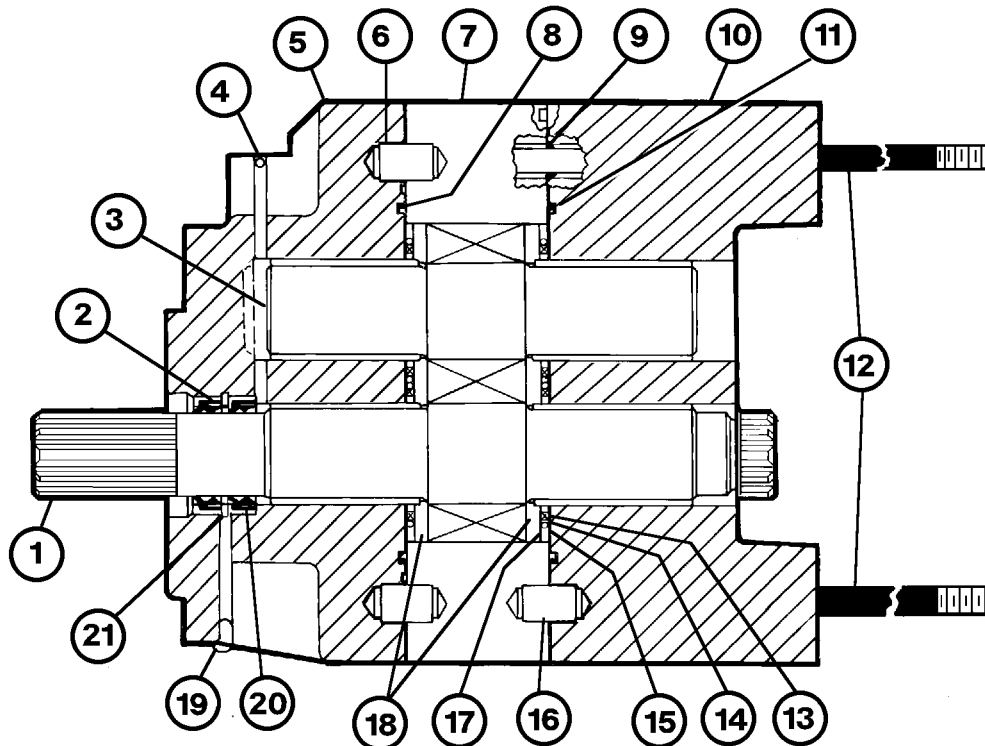


FIGURE 3-6. HOIST PUMP DISASSEMBLY (Front Section)

- |                       |                   |                       |
|-----------------------|-------------------|-----------------------|
| 1. Drive Gear & Shaft | 8. O-ring         | 15. O-ring & Retainer |
| 2. Seal               | 9. O-ring         | 16. Dowel             |
| 3. Idler Gear         | 10. Bearing Plate | 17. Isolation Plate   |
| 4. Steel Ball         | 11. O-ring        | 18. Pressure Plate    |
| 5. Flange             | 12. Thru Studs    | 19. Plug              |
| 6. Dowel              | 13. Steel Rings   | 20. Seal              |
| 7. Gear Plate         | 14. Backup Ring   | 21. Snap Ring         |

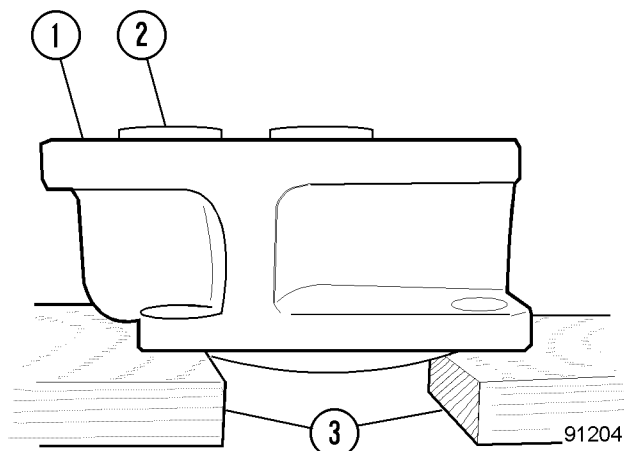


FIGURE 3-7. PREPARATION FOR SEAL REMOVAL

- 1. Flange
- 2. Bearings
- 3. Wooden Blocks

14. Remove outboard shaft seal (2), snap ring (21) and inboard shaft seal (20).

*NOTE: To aid in shaft seal removal place the flange on two small wooden blocks as shown in Figure 3-7.*

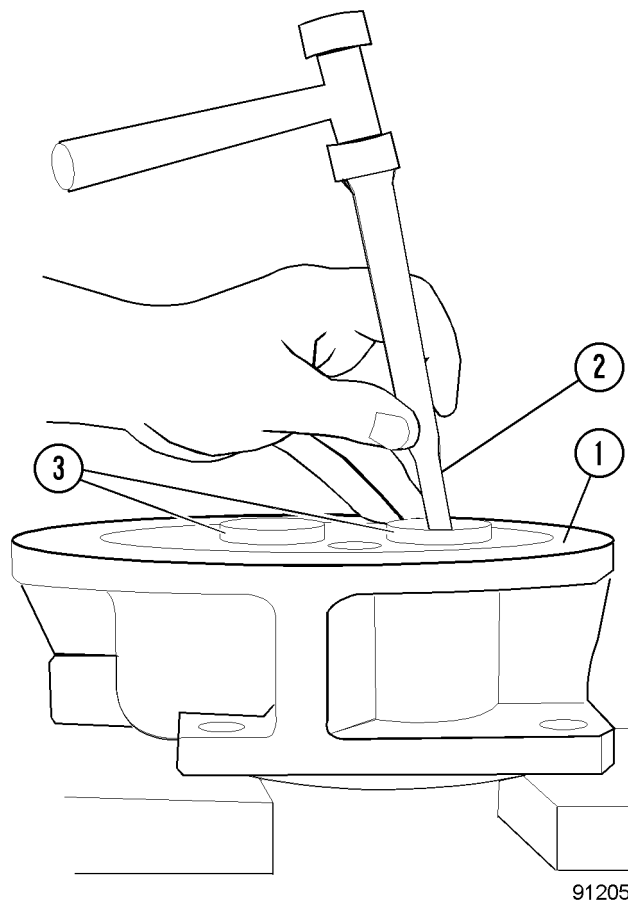


FIGURE 3-8. SHAFT SEAL REMOVAL

- 1. Flange
- 2. Punch
- 3. Bearings

15. Use a punch and hammer to tap the outboard shaft seal out of the flange bore. (Refer to Figure 3-8.) Be careful not to mar, scratch or damage the seal bore surface or bearings.
16. After the seals and snap ring have been removed, clean the bore thoroughly. If necessary, the bore may be smoothed with number 400 emery paper (only).

## Inspection

1. Examine the gear bores in both gear plates; reference Figure 3-9. During the initial break-in, the gears cut into the aluminum gear plates. The nominal depth of this cut is 0.203 mm (0.008 in.) and should not exceed 0.381 mm (0.015 in.). As the gear teeth cut into the gear plates, metal is rolled against the pressure plates. Use a knife or sharp pointed scraper to remove the metal that was rolled against the pressure plates. Remove all metal chips that were broken loose.



**When removing the rolled up metal, do not attempt to remove the gear track-in grooves.**

2. Examine the pressure plates. They should not show excessive wear on the bronzed side. If deep curved wear marks are visible, replace the pressure plates.
3. Examine the gears. If excessive wear is visible on the journals, sides, or face of the gears, or at the point where the drive gear rotates in the lip seal, replace the gears.
4. If any of the internal parts show excessive wear, replace with new parts. Replace all O-rings and seals.
5. Inspect the bearings. If they are worn beyond the gray teflon into the bronze material, the replace the flange connector plates or bearing plate.

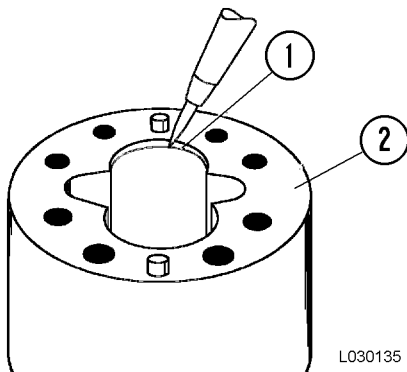


FIGURE 3-9. GEAR BORE INSPECTION

1. Gear Track-In      2. Gear Plate

**NOTE:** Replacing the bearing in the flange, connector plates or bearing plate is not recommended due to close tolerances and special tooling that is required for crimping the bearing in place to prevent bearing spin.

6. Inspect the flange seal bore for scratches or gouges which may interfere with shaft seal installation.
7. For additional pump and system inspection refer to the Troubleshooting Guide.

## Assembly

A suitable seal press ring or plug and two small wood blocks should be used. The following seal installation procedure uses a vise, but it can be adapted for use of a press if one is available.

1. Open the vise jaws wide enough to accept the combined thickness of the flange, wood blocks and press ring.
2. Place the wood blocks flat against the fixed jaw of the vise. Place the flange plate against the blocks in such a position that the bearing projections are between the blocks and clear of the vise jaw (see to Figure 3-10).
3. Lubricate the seals with hydraulic oil. Position inboard shaft seal (3, Figure 3-11) with the metal face toward the outboard end of the flange.

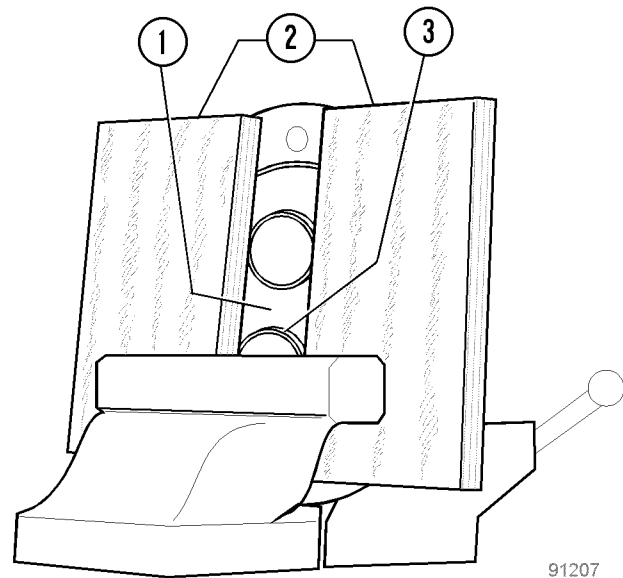


FIGURE 3-10. SHAFT SEAL INSTALLATION

1. Flange      3. Bearing Projection  
2. Wood Blocks

4. Position the press ring over the seal. Make sure that the seal stays centered and true with the bore, and start applying pressure with the vise. Continue pressing the seal until it just clears the snap ring groove in the bore.
5. Install snap ring (2, Figure 3-11) with the snap ring opening over weep hole (10).
6. Install the outboard seal (metal face out) until it just contacts the snap ring.
7. Lubricate thru stud threads (14, Figure 3-11) with hydraulic oil. Thread the studs into flange until snug. There are four long studs and four short studs. Refe to Figure 3-12 for proper stud location.
8. Lubricate and install O-ring (7). Install dowel pins (12) if removed. Install gear plate (13). Make sure that the recess in the gear plate will be toward the connector plate or facing up when the gear plate is installed.
9. Install steel rings (5, Figure 3-12). Lubricate and install backup ring (8), O-ring (7) and ring retainer (6) as shown in Figure 3-12.
10. Install isolation plate (9) on the suction side of the gear plate. The isolation plate has a relief area milled on one side. Turn that side up or toward the pressure plate.

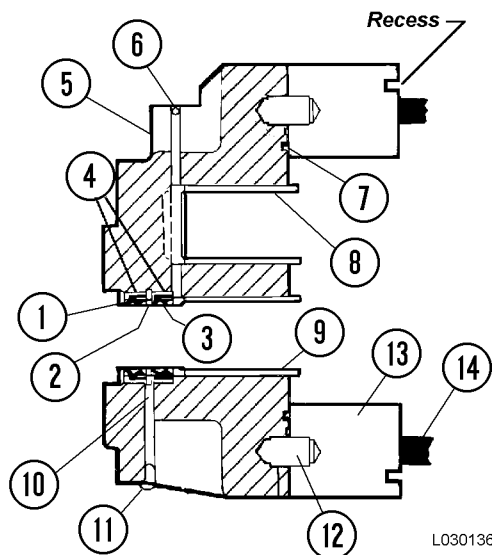


FIGURE 3-11. SHAFT SEAL INSTALLATION

- |                        |                |
|------------------------|----------------|
| 1. Outboard Shaft Seal | 8. Bearing     |
| 2. Snap Ring           | 9. Bearing     |
| 3. Inboard Shaft Seal  | 10. Weep Hole  |
| 4. Seal, Metal Face    | 11. Plug       |
| 5. Flange              | 12. Dowel      |
| 6. Steel Ball          | 13. Gear Plate |
| 7. O-ring              | 14. Thru Studs |

11. With the bronze side up and the milled slot facing toward the discharge side, slide pressure plate (2, Figure 3-13) down into the gear bores until it rests on the backup ring and O-ring. Do not force the plate down the gear bores. If it hangs up on the way down, work it back and forth until it slides freely into place.
12. Coat the inside of the gear plate and the gears with clean hydraulic oil.

**NOTE:** To ensure that the gear pump is correctly timed during reassembly, place a mark on the end of the input shaft to indicate the location of the valley between any two gear teeth. Refer to Figure 3-16 which illustrates gear timing.

13. With the extension end of the drive gear facing toward the shaft seals, install the drive gear. Do not drop the gear in the bore. Damage to the bronze face of the pressure plate could result. Be careful when pushing the drive gear extension thru the shaft seals. Install the idler gear.

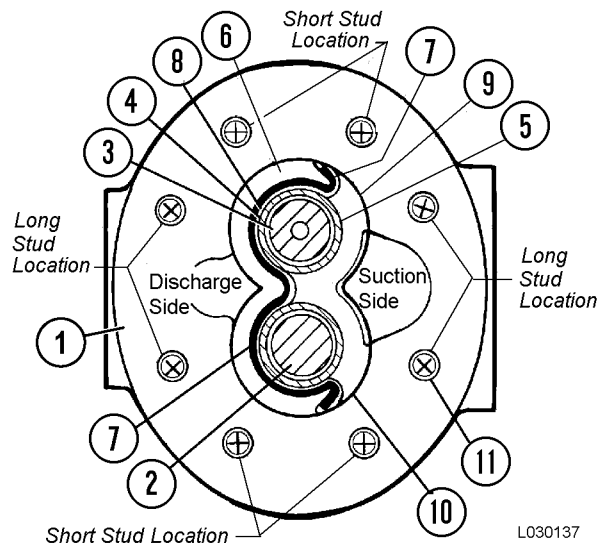


FIGURE 3-12. PUMP REASSEMBLY

- |               |                    |
|---------------|--------------------|
| 1. Gear Plate | 7. O-ring          |
| 2. Drive Gear | 8. Backup Ring     |
| 3. Idler Gear | 9. Isolation Plate |
| 4. Bearing    | 10. Relief Area    |
| 5. Steel Ring | 11. Thru Studs     |
| 6. Retainer   |                    |

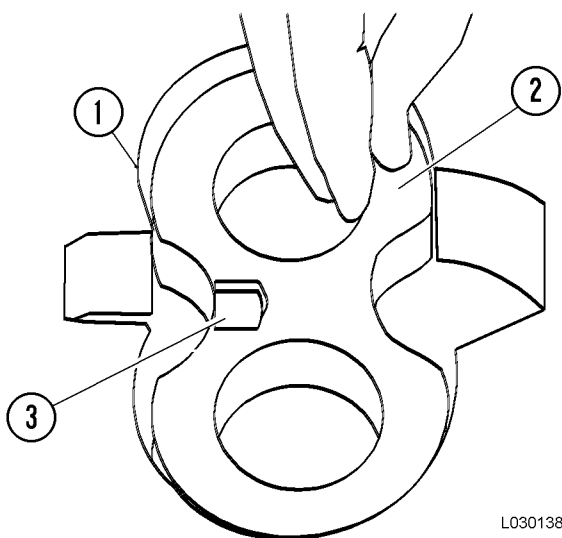


FIGURE 3-13. PRESSURE PLATE INSTALLATION

- |                   |         |
|-------------------|---------|
| 1. Gear Plate     | 3. Slot |
| 2. Pressure Plate |         |

14. Install the opposite pressure plate with the bronze side down and the milled slot facing toward the discharge side.
15. Install steel rings (11, Figure 3-14), backup ring (12), O-ring and retainer (13). Install isolation plate with its relief toward the pressure plate.
16. Lubricate and install thru stud O-rings (5) and connector plate O-ring (7). Install dowel (14) if removed. Lubricate the inside diameter of bearings (17) and install connector plate (6). Install snap ring (8) and coupling (9).

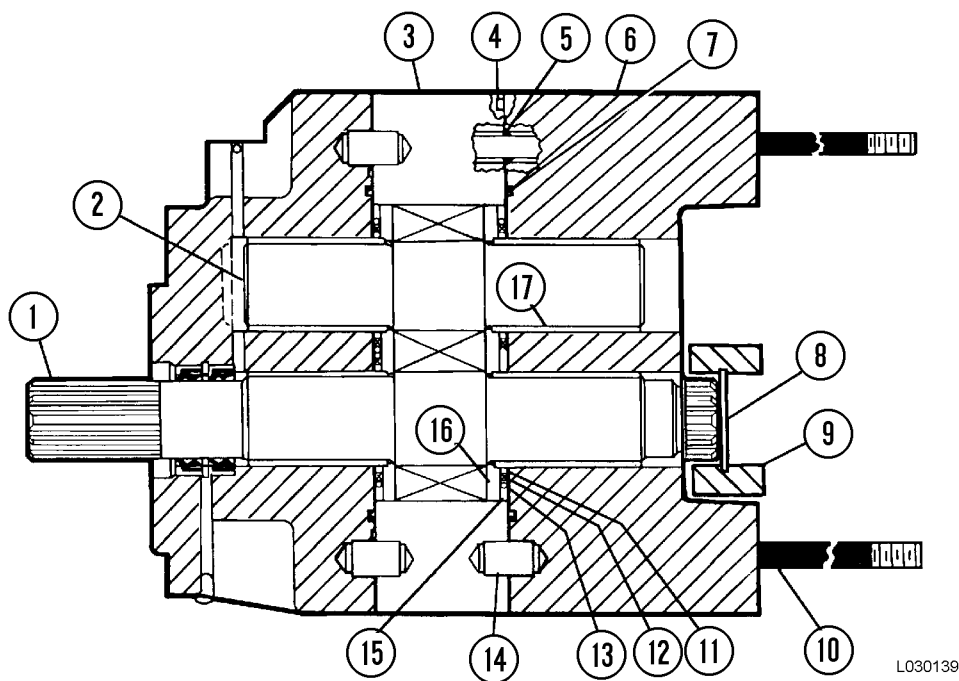


FIGURE 3-14. HOIST PUMP REASSEMBLY

- |                       |                 |                       |
|-----------------------|-----------------|-----------------------|
| 1. Drive Gear & Shaft | 7. O-ring       | 13. O-ring & Retainer |
| 2. Idler Gear         | 8. Snap Ring    | 14. Dowel             |
| 3. Gear Plate         | 9. Coupling     | 15. Isolation Plate   |
| 4. Relief             | 10. Thru Studs  | 16. Pressure Plate    |
| 5. O-ring             | 11. Steel Ring  | 17. Bearings          |
| 6. Connector Plate    | 12. Backup Ring |                       |

17. Lubricate O-ring (3, Figure 3-15) and install it in bearing plate (7). Lubricate O-rings (4) and install them over studs (12). Replace dowel (2) if removed. Install bearing plate (7).
18. Repeat Steps 9, 10 and 11 for installation of the steel rings, backup ring, O-ring, retainer, isolation plate and pressure plate.

19. Lubricate the inside diameter of bearings (26). Install O-rings (8) and (9) and dowel (25) if removed. Install gear plate (10). Make sure that the relief in the gear plate is toward bearing plate (7).

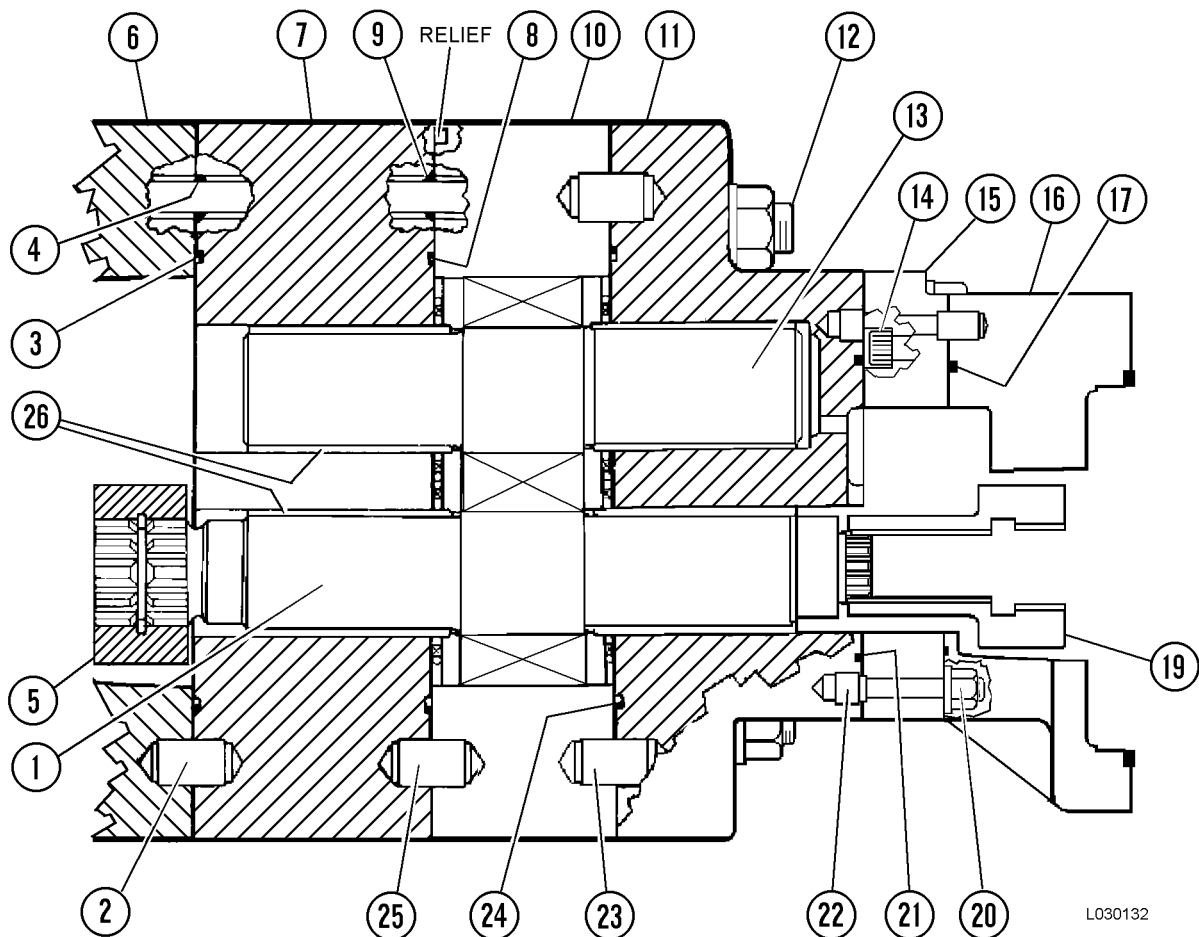


FIGURE 3-15. HOIST PUMP REASSEMBLY

- |                      |                     |                      |              |
|----------------------|---------------------|----------------------|--------------|
| 1. Drive Gear (Rear) | 8. O-ring           | 15. Bearing Plate    | 22. Dowel    |
| 2. Dowel             | 9. O-ring           | 16. Transition Plate | 23. Dowel    |
| 3. O-ring            | 10. Gear Plate      | 17. O-ring           | 24. O-ring   |
| 4. O-ring            | 11. Connector Plate | 18. (Not Used)       | 25. Dowel    |
| 5. Coupling          | 12. Stud            | 19. Coupling         | 26. Bearings |
| 6. Connector Plate   | 13. Idler Gear      | 20. Nut              |              |
| 7. Bearing Plate     | 14. Capscrew        | 21. O-ring           |              |

20. Install rear drive gear (1) and idler gear (13). The rear drive gear must be timed with the front drive gear. This is accomplished by lining up a tooth on the rear drive gear with the valley of two teeth on the front drive gear as shown in Figure 3-16.

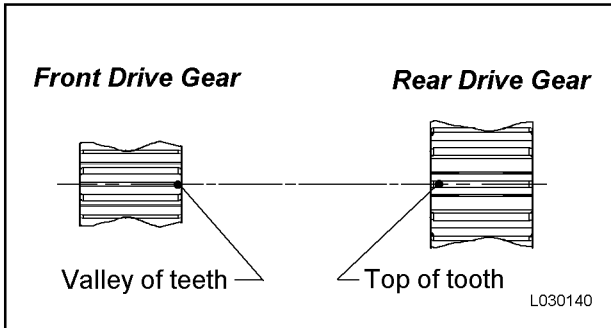


FIGURE 3-16. PUMP GEAR TIMING

21. Repeat Steps 14 and 15 for installation of the remaining pressure plate, steel rings, backup ring, O-ring and retainer, and isolation plate.
22. Lubricate and install O-ring (24, Figure 3-15) in connector plate (11). Install dowel (23) if removed. Lubricate the inside diameter of the bearing in connector plate (11). Install connector plate (11) with flat washers and nuts.
23. Install dowel (22) if removed. Lubricate and position O-ring (17) in transition plate (16). Assemble bearing plate (15) to the transition plate and install capscrews (14). Tighten the capscrews to standard torque.
24. Lubricate O-ring (21) and position it on bearing plate (15). Install the assembled bearing plate and transition plate to connector plate (11) and secure it with nuts (20). Tighten the nuts to standard torque.
25. Install coupling (19).

## CAUTION

*A snap ring may have been installed previously inside coupling (19). This snap ring is not necessary and must not be used. In some cases, depending on tolerance stack-up, the snap ring can cause excessive axial loading, leading to premature pump failure.*

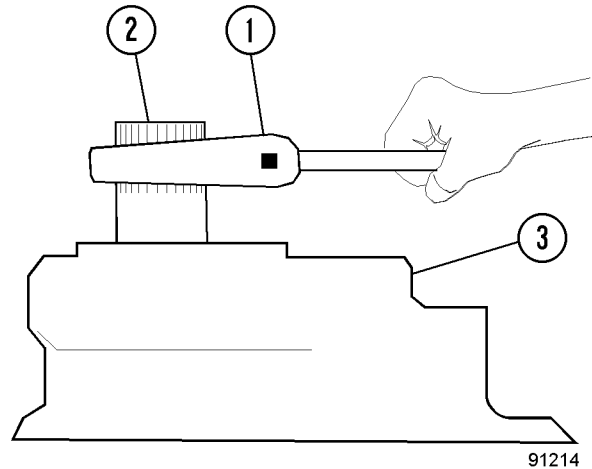


FIGURE 3-17. CHECKING PUMP DRIVE SHAFT ROTATION

1. Wrench  
2. Input Shaft  
3. Pump
26. Lubricate the thru stud threads and install two opposite stud nuts and hardened washers. Tighten the nuts to  **$332 \pm 7 \text{ N}\cdot\text{m}$  ( $245 \pm 5 \text{ ft lb}$ )**.
27. Use a 45 cm (18 inch) adjustable wrench to check the pump drive shaft rotation. The drive shaft will be tight but should turn freely with a maximum torque of  **$15 \text{ N}\cdot\text{m}$  ( $10 \text{ ft lb}$ )** after the initial surge. See Figure 3-17.
28. If the shaft will not turn properly, disassemble the pump and examine the parts for burrs or foreign material that may be causing buildup or interference between parts.
29. When the input shaft turns properly, install the remaining hardened washers and nuts. Tighten the nuts to  **$332 \pm 7 \text{ N}\cdot\text{m}$  ( $245 \pm 5 \text{ ft lb}$ )**.
30. Install a new O-ring on the steering pump flange and install the steering pump onto transition plate (16, Figure 3-15). Install the capscrews and tighten them to standard torque.

| HOIST PUMP TROUBLESHOOTING GUIDE                                                                                                                                                                                                                                      |                                                                                |                                                                                                                                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TROUBLE                                                                                                                                                                                                                                                               | POSSIBLE CAUSE                                                                 | SUGGESTED CORRECTIVE ACTION                                                                                                                                                                                                                                                 |
| 1. Sandblasted band around pressure plate bores<br>2. Angle groove on face of pressure plate<br>3. Lube groove enlarged and edges rounded<br>4. Dull area on shaft at root of tooth<br>5. Dull finish on shaft in bearing area<br>6. Sandblasted gear bore in housing | Abrasive wear caused by fine particles not visible to the eye (such as dirt)   | 1. Was clean oil used?<br>2. Was filter element change period correct?<br>3. Were correct filter elements used?<br>4. Hoist cylinder rod wiper and seals in good condition?<br>5. Cylinder rods dented or scored?<br>6. Was system flushed properly after previous failure? |
| 1. Scored pressure plates.<br>2. Scored shafts<br>3. Scored gear bore                                                                                                                                                                                                 | Abrasive wear caused by coarse metal particles visible to the eye              | 1. Was system flushed properly after previous failure?<br>2. Contaminants generated elsewhere in hydraulic system?<br>3. Contaminants generated by wearing pump components?                                                                                                 |
| 1. Any external damage to pump<br>2. Damage on rear of drive gear and rear pressure plate only                                                                                                                                                                        | Incorrect installation                                                         | 1. Did shaft bottom in mating part?<br>2. Any interference between pump and machine?                                                                                                                                                                                        |
| 1. Eroded pump housing<br>2. Eroded pressure plates                                                                                                                                                                                                                   | Aeration-cavitation caused by restricted oil flow to pump inlet or aerated Oil | 1. Tank oil level correct?<br>2. Oil viscosity as recommended?<br>3. Restriction in pump inlet line?<br>4. Air leak in pump inlet line?<br>5. Loose hose or tube connection?                                                                                                |
| 1. Heavy wear on pressure plate<br>2. Heavy wear on end of gear                                                                                                                                                                                                       | Lack of oil                                                                    | 1. Was oil level correct?<br>2. Any leaks in piping inside tank?                                                                                                                                                                                                            |
| 1. Housing scored heavily<br>2. Inlet peened and battered<br>3. Foreign object caught in gear teeth                                                                                                                                                                   | Damage caused by metal object                                                  | 1. Metal object left in system during initial assembly or previous repair?<br>2. Metal object generated by another failure in system?                                                                                                                                       |
| 1. Pressure plate black<br>2. O-rings and seals brittle<br>3. Gear and journals black                                                                                                                                                                                 | Excessive heat                                                                 | 1. Metal object left in system during initial assembly or previous repair?<br>2. Was relief valve setting too low?<br>3. Was oil viscosity correct?<br>4. Was oil level correct?                                                                                            |
| 1. Broken shaft<br>2. Broken housing or flange                                                                                                                                                                                                                        | Over-pressure                                                                  | 1. Relief valve setting correct?<br>2. Did relief valve function?                                                                                                                                                                                                           |



## HYDRAULIC TANK

### ⚠ IMPORTANT ⚠

*If a component in the hydraulic system fails, perform an oil analysis before replacing any component. If foreign particles are evident, the system must be flushed. Refer to Hydraulic System Flushing instructions.*

#### Removal

1. Turn the key switch OFF and allow at least 90 seconds for the steering accumulators to bleed down.

*NOTE: Be prepared to contain approximately 947 L (250 gal) of hydraulic oil. If the oil is to be reused, clean containers must be used with a 3-micron filtering system available for refill.*

2. Thoroughly clean the outside of the hydraulic tank and attached equipment.

### ⚠ WARNING

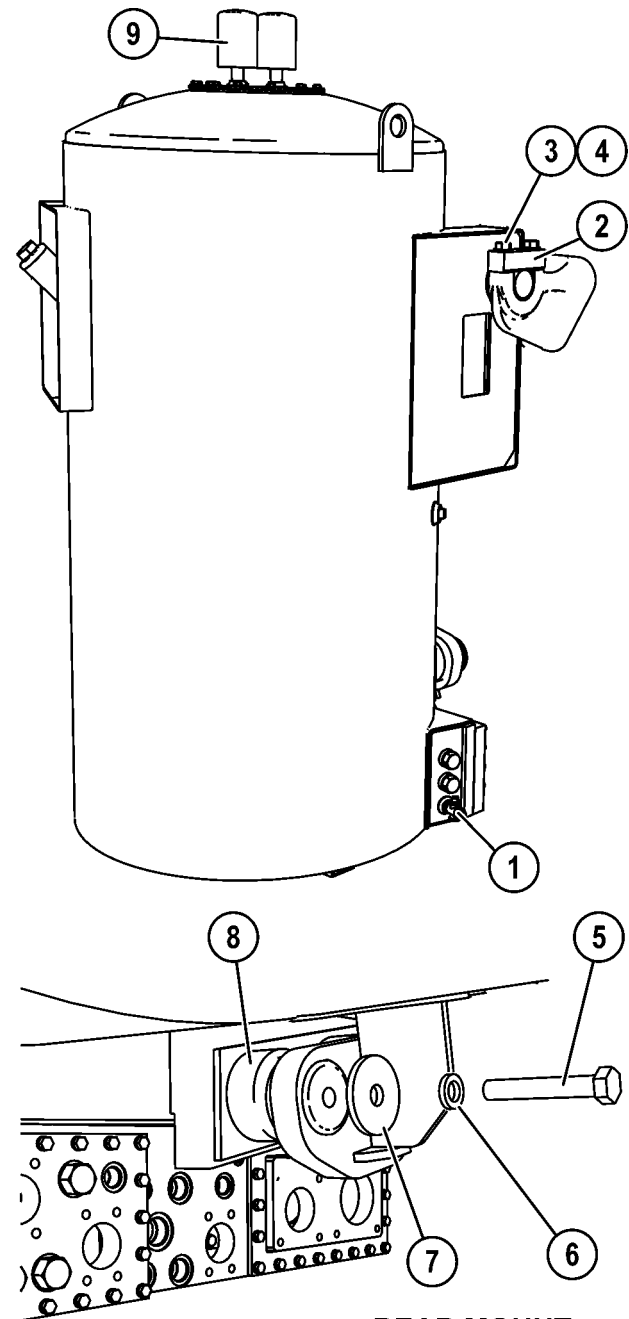
*Be careful to avoid contact with hot oil if the truck has been operating. Avoid spillage and contamination.*

3. Drain the hydraulic tank by using drain valve (1, Figure 3-18) on the rear side of the tank.
4. Disconnect the hydraulic lines. Plug all lines to prevent possible contamination to the system. Close both suction line shutoff valves. Tag each line at removal for proper identification during installation.

### ⚠ WARNING

*The weight of the hydraulic tank is approximately 590 kg (1,300 lb). Use a suitable lifting device that can handle the load safely.*

5. Attach a suitable lifting device to the hydraulic tank.
6. Remove two capscrews (5), lockwashers (6), large washers (7) and rubber mounts (8).
7. Remove capscrews (3), lockwashers (4) and both caps (2) at the top of the hydraulic tank.
8. Move the hydraulic tank to a clean work area for disassembly or repair, if necessary.



REAR MOUNT

83757

FIGURE 3-18. HYDRAULIC TANK INSTALLATION

- |                |                     |
|----------------|---------------------|
| 1. Drain Valve | 6. Lockwasher       |
| 2. Cap         | 7. Large Washer     |
| 3. Capscrew    | 8. Rubber Mount     |
| 4. Lockwasher  | 9. Breather Filters |
| 5. Capscrew    |                     |

## Installation

1. Lift the hydraulic tank into position.
2. Install both caps (2, Figure 3-18), lockwashers (4) and capscrews (3) at the top of the hydraulic tank. Tighten to **622 ± 62 N·m (459 ± 45 ft lb)**.
3. Install two rubber mounts (8), large washers (7), lockwashers (6) and capscrews (5) at the rear mount. Tighten to **800 ± 80 N·m (590 ± 59 ft lb)**.
4. Uncap the hydraulic lines and attach them to the proper connections.
5. Replace the breather filters (9), if required.
6. Fill the hydraulic tank. Refer to "Hydraulic Tank Service" in Section P2, Lubrication and Service.
7. Bleed all air from the hydraulic lines.
8. Open both suction line shutoff valves. Loosen the capscrews (at the pump) on the suction hoses to bleed any trapped air. Then loosen the capscrews (at the pump) on the pressure hoses to bleed any trapped air. Tighten all capscrews securely.



***If trapped air is not bled from the steering pump, possible pump damage and no output may result.***

## HYDRAULIC TANK BREATHERS

Two breather filters (9, Figure 3-18) located on top of the hydraulic tank allow air to flow in and out of the tank. Replace the filters at the interval that is specified on the lubrication chart.

Keep the area around the breather filters clean and free of debris. If there is any indication that the breather filters are oil-soaked, replace the filters as soon as possible and check for proper oil level in the hydraulic tank. Once the breather filters become oil-soaked, they will plug very quickly.



***Plugged breather filters can cause pressure buildup inside the hydraulic tank and can cause the service brakes to drag.***

## HYDRAULIC TANK STRAINERS

### Removal

1. Turn the key switch OFF and allow at least 90 seconds for the steering accumulators to bleed down.

*NOTE: Be prepared to contain approximately 947 L (250 gal) of hydraulic oil. If the oil is to be reused, clean containers must be used with a 3-micron filtering system available for refill.*

2. Drain the hydraulic tank by using the drain valve located on the rear side of the tank.
3. Disconnect pump suction hoses. (5, Figure 3-19).
4. Remove capscrews and lockwashers (3) that secure cover (2) to the hydraulic tank. Remove and discard the gasket.
5. Remove the capscrews and lockwashers that secure the suction strainers. Remove the suction strainers.

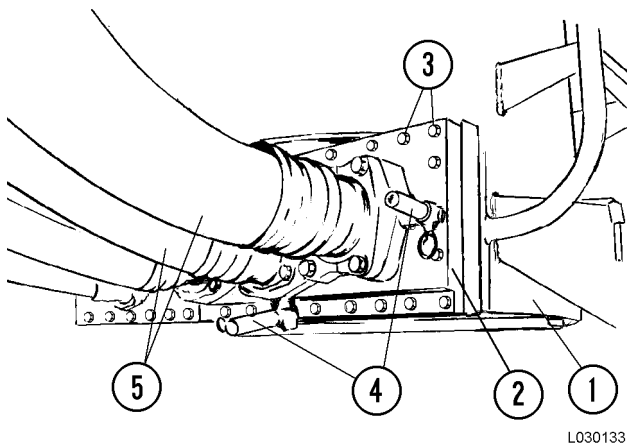


FIGURE 3-19. STRAINER REMOVAL

- |                            |                       |
|----------------------------|-----------------------|
| 1. Hydraulic Tank          | 4. Shutoff Valve      |
| 2. Cover                   | 5. Pump Suction Hoses |
| 3. Capscrews & Lockwashers |                       |

### Inspect and Clean

*NOTE: Inspect the strainers thoroughly for metallic particles and varnish build up (if oil has been overheated). The quantity and size of any particles may be an indication of excessive wear of components in the hydraulic system.*

1. Clean the strainers from the inside out with fresh cleaning solvent.
2. Inspect the strainers for cracks and wear. Replace the strainers if necessary.
3. Clean any sediment from the bottom of the hydraulic tank.

### Installation

1. Install suction strainers and secure them in place with capscrews and lockwashers. Tighten the capscrews to standard torque.
2. Using a new cover gasket, move cover (2, Figure 3-19) into position and install capscrews and lockwashers (3). Tighten the capscrews to standard torque.
3. Install the pump suction hoses.
4. Fill the hydraulic tank. Refer to Filling Instructions on the previous page.
5. Open both suction line shutoff valves (4). Loosen the capscrews (at the pump) on the suction hoses to bleed any trapped air. Then loosen the capscrews (at the pump) on the pressure hoses to bleed any trapped air. Tighten all capscrews securely.

*NOTE: If trapped air is not bled from steering pump, possible pump damage and no output may result.*

# SECTION L4

## STEERING CIRCUIT

### INDEX

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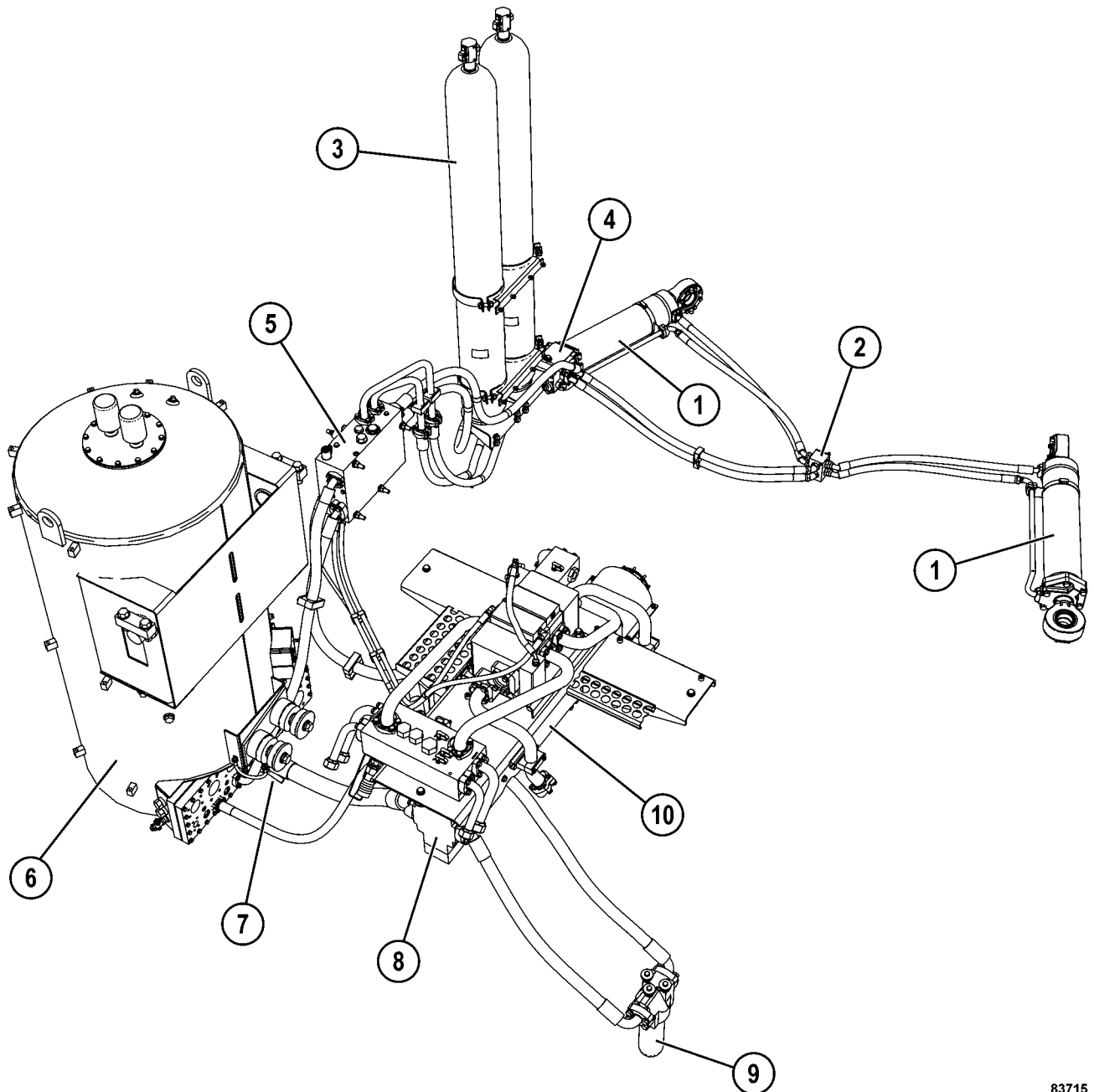
# NOTES

# STEERING CIRCUIT

## STEERING CIRCUIT OPERATION

Pump (8, Figure 4-1) supplies oil to bleeddown manifold (5) after passing through high pressure filter (9). This oil supply is then distributed to the brake system and steering system.

System pressure is regulated between 19 000 kPa (2,750 psi) and 20 900 kPa (3,025 psi) by an unloader valve located on the bottom of the steering pump case.



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FIGURE 4-1. STEERING SYSTEM COMPONENTS

- |                          |                         |                         |                        |
|--------------------------|-------------------------|-------------------------|------------------------|
| 1. Steering Cylinders    | 4. Flow Amplifier Valve | 7. Shutoff Valve        | 10. Hoist Circuit Pump |
| 2. Manifold              | 5. Bleeddown Manifold   | 8. Steering/Brake Pump  |                        |
| 3. Steering Accumulators | 6. Hydraulic Tank       | 9. High Pressure Filter |                        |

Oil pressure from the bleed down manifold is routed back to the steering/brake pump and sensed by the "ACC" port on the unloader valve located on the bottom of the steering/brake pump case. The unloader valve regulates the system pressure between 23 063 kPa (3,345 psi) and 24 132 kPa (3,500 psi). Steering system pressure switch (7, Figure 4-3) on the bleed down manifold activates a warning light on the overhead dash panel if steering system pressure drops below 15 900 kPa (2,300 psi).

There is also a 517 kPa (75 psi) pressure sensor (23) to confirm hydraulic pressure is fully relieved at bleeddown cycle completion.

Two nitrogen precharged steering accumulators (3, Figure 4-1) are mounted on the left frame rail behind the front tire. They provide an oil supply that is sufficient to allow the operator to steer the truck off the haul road and park in a safe area if a loss of steering circuit oil occurs. Steering accumulator bleed down solenoid (6, Figure 4-3) on the bleed down manifold is electrically activated to bleed the pressurized hydraulic oil from inside the accumulators back to the hydraulic tank whenever the key switch is turned to the OFF position. Approximately 90 seconds is required for the accumulator to bleed down fully.

The bleed down manifold and steering accumulators supply oil to flow amplifier (4, Figure 4-1). The flow amplifier directs oil flow to the appropriate ports on the steering cylinders as determined by inputs from steering control unit (1, Figure 4-2) as the operator steers the truck.

Oil entering the steering control unit is blocked until the steering wheel is turned in a desired direction. The steering control unit then directs oil to the flow amplifier, which in turn provides a high volume of oil to steering cylinders (1, Figure 4-1). Hydraulic oil at the opposite ends of the steering cylinders flows back through the flow amplifier and bleed down manifold to hydraulic tank (6).

Oil flow to the brake apply system flows through a check valve, preventing loss of oil in the brake circuit if a loss of oil occurs in the steering circuit. Refer to Section J for a detailed description of the brake apply system hydraulic circuit and components.

## COMPONENT DESCRIPTION

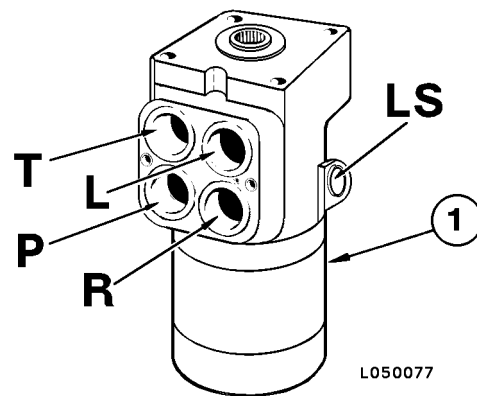


FIGURE 4-2. STEERING CONTROL UNIT

"T": Tank Port  
 "L": Left Steer Port  
 "P": Pressure (Supply) Port

"R": Right Steer Port  
 "LS": Load Sense Port  
 1. Valve Assembly

### Steering Control Unit

The steering control unit (Figure 4-2) is mounted inside the cab and is directly coupled to the lower end of the steering column.

Operation of the steering control unit is both manual and hydraulic in effect. This component incorporates a hydraulic control valve. Steering effort applied to the steering wheel by the operator actuates the valve, which in turn directs hydraulic oil through the flow amplifier valve to the steering cylinders to provide the operator with power steering.

## Bleeddown Manifold

Bleeddown manifold (5, Figure 4-1) is located on the outside of the left hand frame rail just behind the accumulators.

The bleeddown manifold receives oil from the steering/brake pump through a high pressure filter and directs oil to the accumulators, brake circuit, flow amplifier, auto lube system hydraulic pump, and the optional radiator shutters, if installed. Oil supply for the steering control unit and steering cylinders is supplied via the flow amplifier. The manifold also provides hoist circuit control when the body is raised.

The bleed down manifold receives oil from the steering/brake pump through a high pressure filter and directs oil to the steering accumulators, flow amplifier, brake circuit, and the automatic lubrication system pump. The manifold also provides hoist circuit control when the body is raised.

Relief valve (4, Figure 4-4) limits the hydraulic supply pressure to the steering and brake circuits to 27 500 kPa (4,000 psi). Relief valve (3) provides maximum pressure protection of 4 100 kPa (600 psi) for the oil returning to the hydraulic tank.

**NOTE:** The relief valves, steering accumulator bleed down solenoid, and hoist up limit solenoid are factory preset and cannot be individually rebuilt.

## Steering Accumulator Bleed Down Solenoid

Each time the key switch is turned OFF with the truck stopped, steering accumulator bleed down solenoid (6, Figure 4-3 and 2, Figure 4-4) is energized. When the solenoid is energized, all hydraulic steering pressure (including the accumulators) is bled back to the hydraulic tank. Brake pressure, however, will not bleed down due to internal check valves in the brake manifold and bleed down manifold.

After approximately 90 seconds, the solenoid will de-energize to close the return port to the tank. By this time, all the oil in the accumulators should be returned to the tank. At startup, the steering circuit and brake circuit will be charged. Steering system pressure switch (7, Figure 4-3) will activate the low steering pressure warning light and buzzer until steering pressure reaches 15 900 kPa (2,300 psi).

If steering pressure falls below 15 900 kPa (2,300 psi) during operation, the low steering pressure warning light will illuminate until pressure returns to normal.

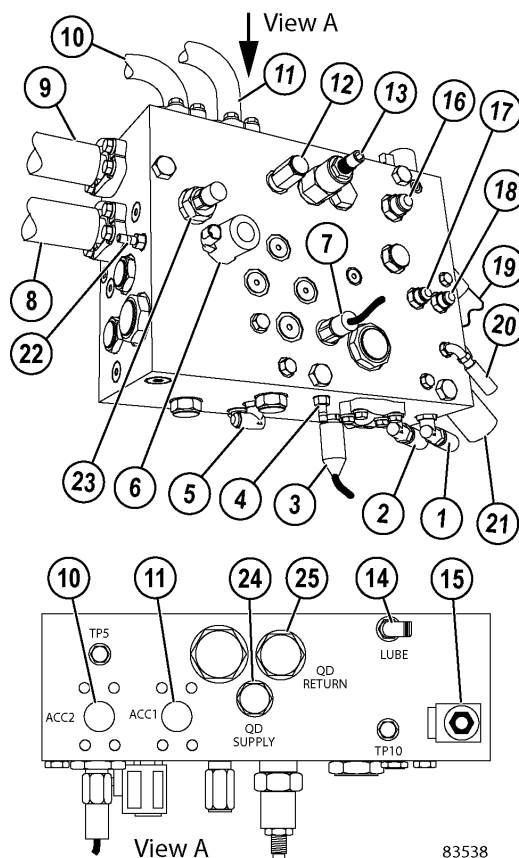


FIGURE 4-3. BLEED DOWN MANIFOLD

1. To Hoist Valve, Power Up
2. To Hoist Valve, Power Down
3. Komtrax Pressure Sensor (Port TP1)
4. Test Port (TP2)
5. To Brake System
6. Steering Accumulator Bleed Down Solenoid
7. Steering System Pressure Switch
8. Supply To Flow Amplifier
9. Return From Flow Amplifier
10. To Front Steering Accumulator
11. To Rear Steering Accumulator
12. Relief Valve, 4 100 kPa (600 psi)
13. Relief Valve, 27 500 kPa (4,000 psi)
14. Auto Lube Pump Supply Port
15. Hoist Up Limit Solenoid Valve
16. Return From Hoist Pilot Valve
17. Hoist Pilot Valve, Power Down
18. Hoist Pilot Valve, Power Up
19. Return to Tank
20. Feedback Pressure to Unloader Valve
21. Supply From Pump
22. Test Port (TP3)
23. Pressure Sensor (Bleeddown)
24. Supply Port (Quick Disconnect)
25. Return Port (Quick Disconnect)



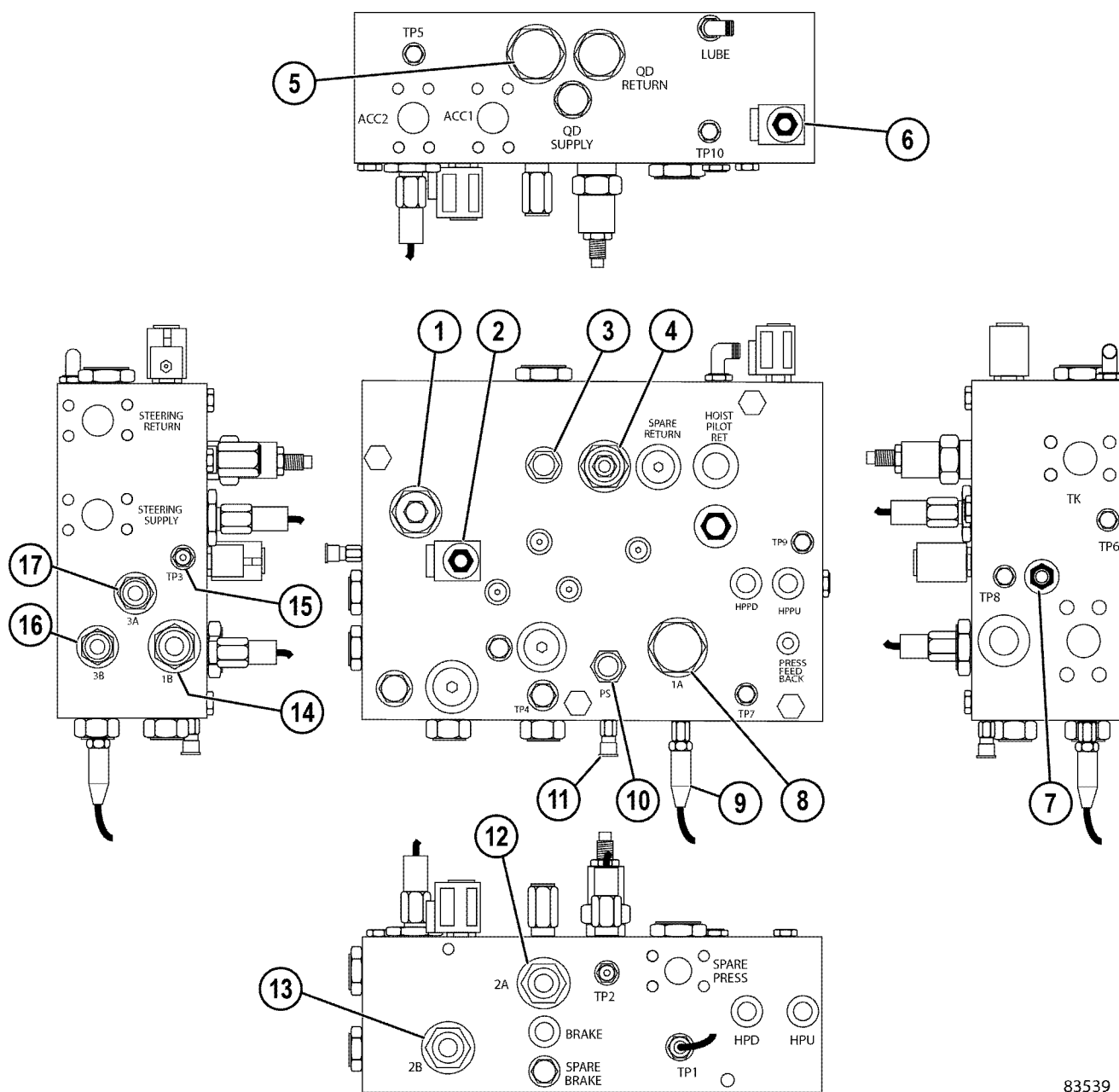


FIGURE 4-4. BLEEDDOWN MANIFOLD

- |                                              |                                     |
|----------------------------------------------|-------------------------------------|
| 1. Pressure Switch (Bleed Down)              | 10. Steering System Pressure Switch |
| 2. Steering Accumulator Bleeddown Solenoid   | 11. Test Port (TP2)                 |
| 3. Relief Valve - 4 100 kPa (600 psi)        | 12. Check Valve                     |
| 4. Relief Valve - 27 500 kPa (4,000 psi)     | 13. Check Valve                     |
| 5. Pilot Operated Check Valve                | 14. Check Valve                     |
| 6. Hoist Up Limit Solenoid                   | 15. Test Port (TP3)                 |
| 7. Pilot Operated Check Valve                | 16. Check Valve                     |
| 8. Inlet Check Valve                         | 17. Check Valve                     |
| 9. Komtrax Steering Pressure Test Port (TP1) |                                     |

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## Relief Valves

Relief valve (4, Figure 4-4), located in the bleddown manifold, limits the hydraulic supply pressure to the steering and brake circuits to 27 500 kPa (4,000 psi).

Relief valve (3) provides maximum pressure protection of 4 100 kPa (600 psi) for the oil returning to the hydraulic tank.

## Hoist Up Limit Solenoid

Hoist up limit solenoid (6) is a 3-way valve that is activated by the hoist limit switch when the hoist cylinders approach the maximum extension limit. When activated, the solenoid will close the power up pilot line to the hoist valve from the hoist pilot valve mounted in the hydraulic cabinet.

Pilot operated check valve (30, Figure 4-5) is opened by power down pilot pressure line (19) to allow oil in the raise port to bypass hoist up limit solenoid (24) for the initial power down operation while the solenoid is activated by the hoist limit switch.

Refer to Section D3, 24VDC Electrical System Components, for more information about the hoist limit switches.

## Steering Accumulators

The two steering accumulators (3, Figure 4-1) are bladder type accumulators with a capacity of 62 liters (16.5 gallons) each. The accumulators are charged to 9 700 kPa (1,400 psi) with pure dry nitrogen using the charging valve located on the top end.

Oil entering the accumulators compresses the nitrogen in the bladder. The nitrogen pressure increases directly with steering circuit pressure. When steering circuit pressure reaches 20 900 kPa (3,025 psi), the unloader valve will unload the pump. The accumulators will contain a quantity of oil, under pressure and held by check valves in the bleddown manifold, that will be available for steering the truck. When system pressure drops to 19 000 kPa (2,750 psi), the pump output will again increase to refill the accumulators and increase the steering system pressure. The accumulators also provide oil for a limited period of time to be used in case the pump becomes inoperative.

## Low Precharge Warning Switch

Pressure switches, located in the top of each accumulator, monitor the nitrogen pressure and are used to activate the accumulator precharge warning light if the nitrogen pressure drops below 7 600 kPa (1,100 psi).

The switches monitor nitrogen pressure when the key switch is turned ON before the engine is started. If nitrogen pressure is too low, the warning lamp turns on. A latching circuit prevents the warning lamp from turning off when the engine is started and steering system pressure compresses the nitrogen remaining in the accumulator.



***Do not operate the truck with less than 7 600 kPa (1,100 psi) of nitrogen precharge in the accumulator. Low nitrogen pressure may not provide an adequate supply of steering system oil in some emergency conditions. If the low precharge warning light remains on, check the accumulator precharge pressure and recharge if necessary.***

## High Pressure Filter

High pressure filter (9, Figure 4-1) filters oil for the steering and brake circuits.

If the filter element becomes restricted, a warning indicator is activated at 241 kPa (35 psi) differential, and oil will bypass the element at 345 kPa (50 psi) differential.

Refer to Section L9 for further information regarding hydraulic system filters.

## Quick Disconnect Ports

Ports on the bleddown manifold allow service personnel to connect an external hydraulic supply to allow operation of the truck steering and service brake systems if the steering/brake pump or engine is not operational.

The external supply is connected to port (24, Figure 4-3) and the return is connected to port (25). This feature should only be used for an emergency to allow operation of the truck to return to the shop for service or to move the truck out of haul road traffic.

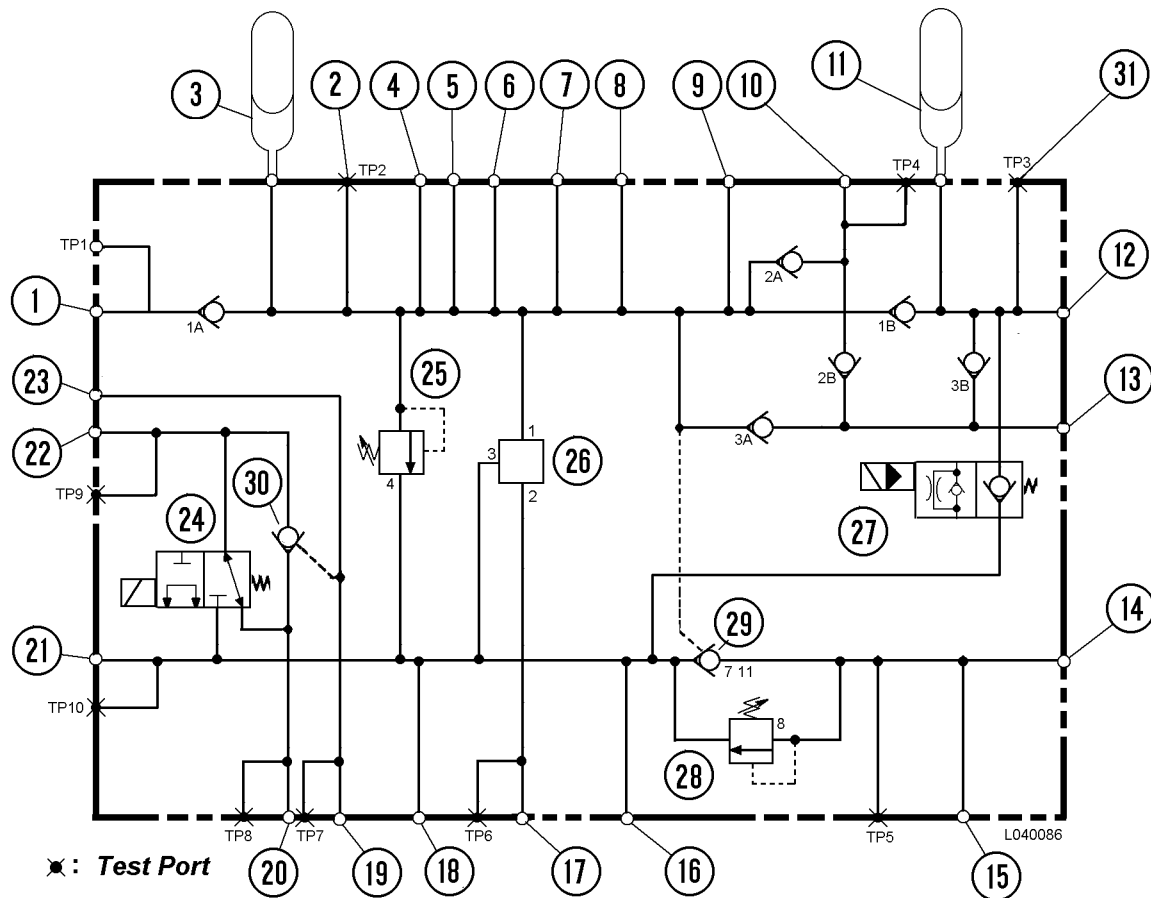


FIGURE 4-5. BLEEDDOWN MANIFOLD SCHEMATIC

- |                                                 |                                             |
|-------------------------------------------------|---------------------------------------------|
| 1. Steering/Brake Pump Inlet (Pump)             | 17. Auto Lube Pump Supply                   |
| 2. Steering Pressure Test Port (TP2)            | 18. Hoist Pilot Return                      |
| 3. Steering Accumulator                         | 19. Hoist Pilot Power Down                  |
| 4. APU Port                                     | 20. Hoist Pilot Power Up                    |
| 5. Pressure Feedback (to Steering Pump)         | 21. Tank Return Port                        |
| 6. Steering Pressure Switch                     | 22. Hoist Power Up                          |
| 7. (Not used)                                   | 23. Hoist Power Down                        |
| 8. Radiator Shutters (If installed)             | 24. Hoist Up Limit Solenoid                 |
| 9. Spare Pressure Port                          | 25. Relief Valve, 27,500 kPa (4000 psi)     |
| 10. Brake Circuit Supply                        | 26. Cavity Plug                             |
| 11. Steering Accumulator                        | 27. Steering Accumulator Bleeddown Solenoid |
| 12. Steering Circuit Supply to Flow Amplifier   | 28. Relief Valve, 4100 kPa (600 psi)        |
| 13. Quick Disconnect, Supply Port               | 29. Pilot Operated Check Valve              |
| 14. Steering Circuit Return From Flow Amplifier | 30. Pilot Operated Check Valve              |
| 15. Quick Disconnect, Return Port               | 31. Test Port (TP3)                         |
| 16. Spare Return Port                           |                                             |

## Flow Amplifier

Flow amplifier (4, Figure 4-1), located on the left frame rail in front of the accumulators, is used in the steering circuit to provide the high volume of oil required for the steering cylinders.

The flow amplifier uses the amount of flow from the steering control unit to determine a proportional amount of flow to send from the bleeddown manifold to the steering cylinders.

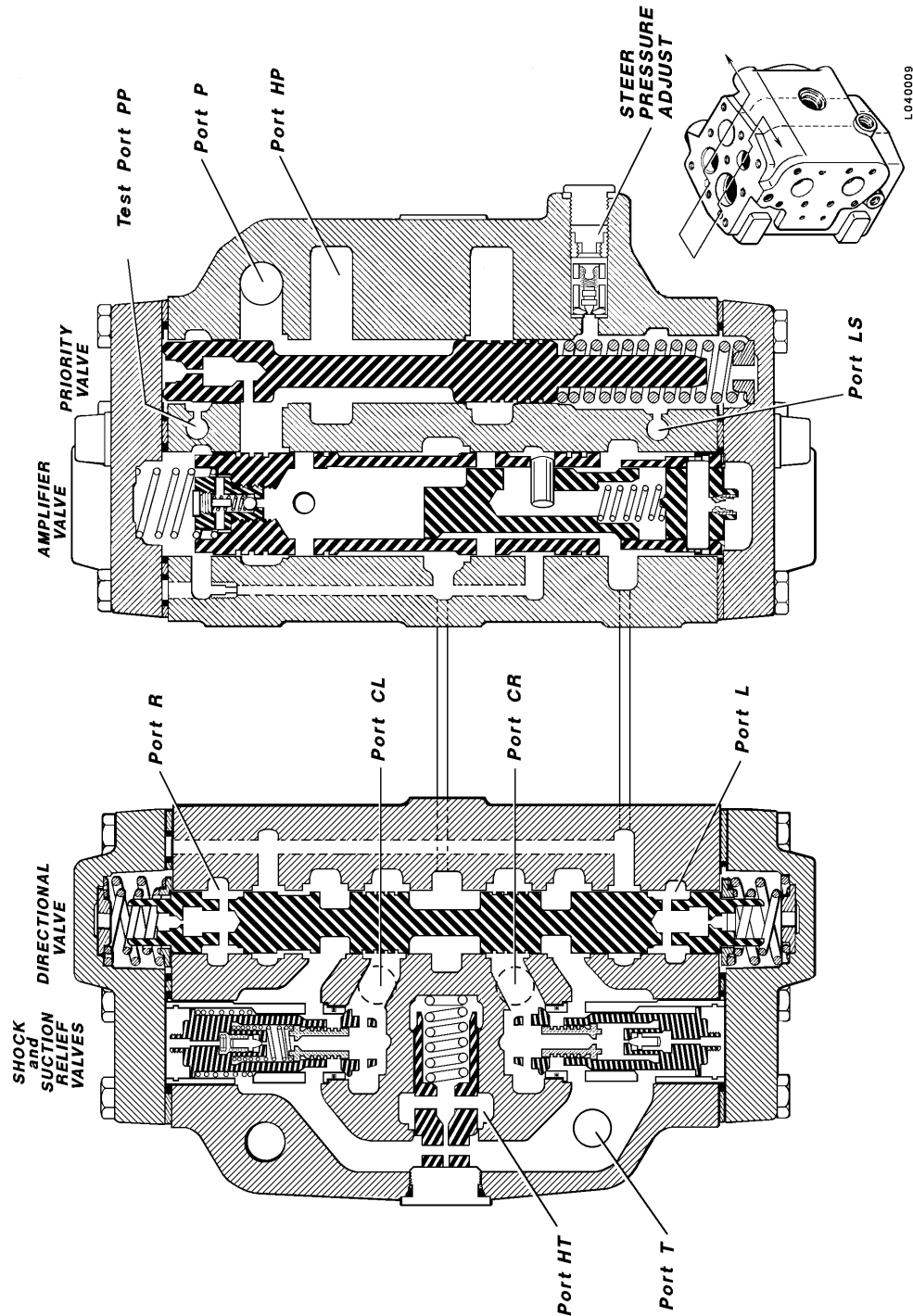


FIGURE 4-6. FLOW AMPLIFIER VALVE

## FLOW AMPLIFIER SYSTEM OPERATION

Refer to Figures 4-7 through 4-10 for oil flow paths during the following conditions:

- Neutral
- Steering Left
- Steering Right
- External shock load

### No Steer

(Refer to Figure 4-7)

High pressure oil from the steering pump and steering accumulators is available through the bleeddown manifold to the "HP" port on the flow amplifier.

Upon entering the priority valve, the oil goes past the spool to the closed amplifier valve and also out port "P" through a hose to port "P" on the steering control unit. In the steering control unit, the oil goes to a closed area in the control.

As pressure builds up in these two areas, oil passes through orifices in the end of the priority valve and builds pressure on the end of the valve and port PP. When pressure reaches approximately 3 500 kPa (500 psi), the spool moves, compressing its spring and closing off oil supply through area "A", resulting in high pressure at "PP" but only 3 500 kPa (500 psi) at the amplifier spool and steering control unit.

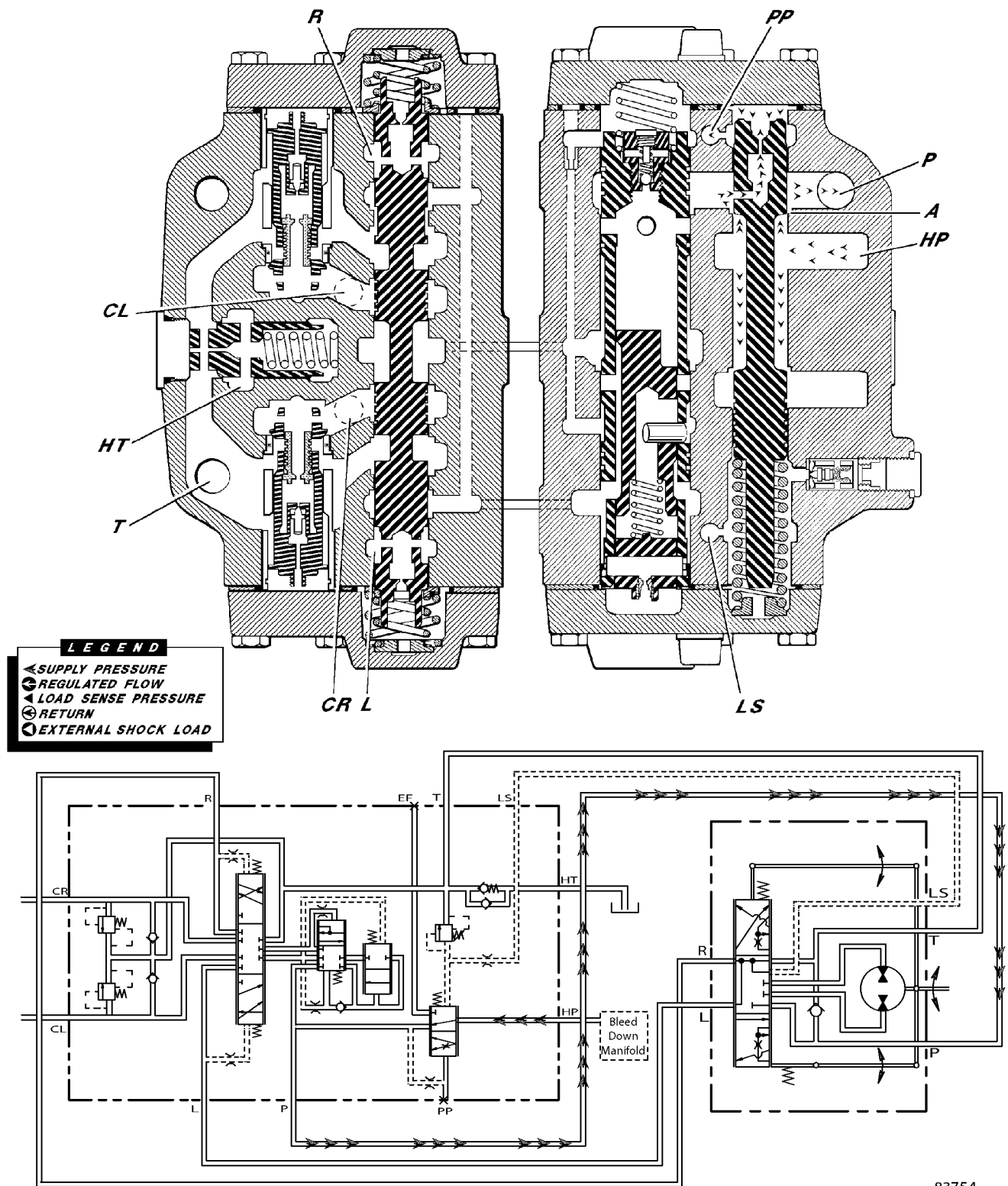


FIGURE 4-7. FLOW AMPLIFIER - NO STEER

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## Steering Left

(Refer to Figure 4-8)

When the operator turns the steering wheel to the left, the steering control unit is opened to allow oil coming in port "P" to pass to the gerotor section of the control unit to turn the rotor. Oil in the other side of the gerotor flows through other passages in the control unit valve and out steering control unit port "L". This oil enters port "L" of the flow amplifier and goes to a closed area "B" in the directional valve.

As pressure in this area builds, it also passes into the spool through orifice "C" to the spring area on the end of the directional valve. The pressure then moves the spool compressing the springs on the opposite end. This movement allows the oil entering area "B" to pass through the directional valve to area "D" of the amplifier valve through sleeve "E" holes to a passage between sleeve "E" and valve "F", through hole "G" in sleeve "E", where it initially is blocked by the valve body.

As pressure builds up in this area, oil also flows from area "D", around the outside of sleeve "E", around pin "H", through orifice "J" to build pressure on the end of the amplifier valve, and opens hole "G" just enough to allow the flow of oil coming from the steering control unit to pass to the control area of the directional valve.

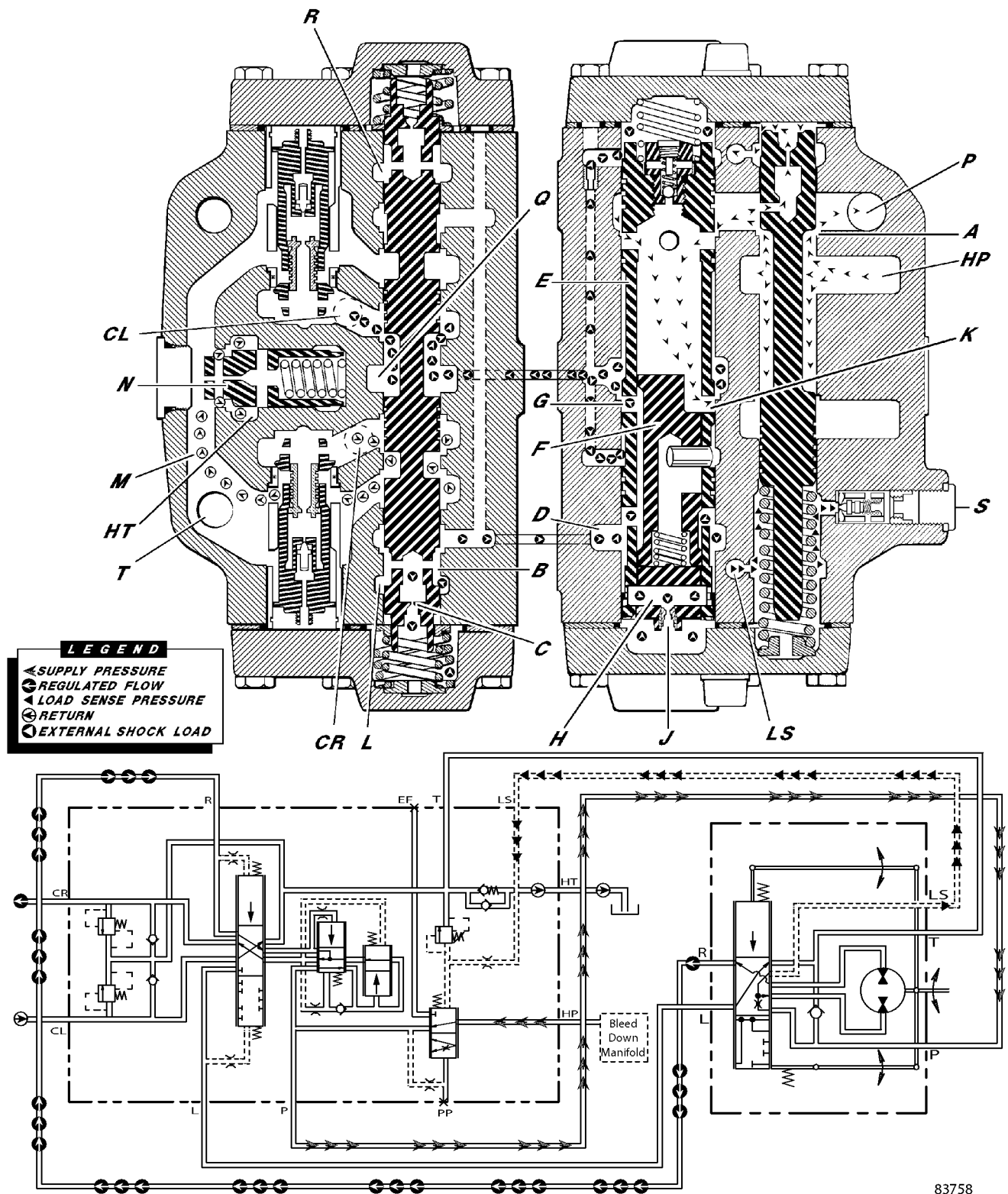
At the same time, the movement of sleeve "E" opened the holes near the spring end to allow the oil from the priority valve to flow into the center of sleeve "E".

This oil now inside sleeve "E" pushes valve "F" against its spring to give the oil access to a series of holes "K" that are in the same plane as hole "G". The passage of oil through holes "K" past the valve body is metered by holes "K" being opened the same proportion as hole "G". The number of holes "K" (7) in sleeve "E" determine the amount of additional oil that is added to the steering control unit oil passing through hole "G".

This combined oil going to the center area "Q" of the directional valve passes out port "CL" of the flow amplifier and travels to the steering cylinders to steer the front wheels to the left. As the cylinders move, oil is forced to return out the opposite ends, enter port "CR" of the flow amplifier, pass through the directional valve to area "M", pass through return check valve "N", and exit port "HT" to the hydraulic reservoir.

At the steering control unit, when the operator turned the steering wheel, supply oil from port "P" was also delivered through the control unit valve to port "LS". This oil enters the flow amplifier through its "LS" port and builds pressure in the spring area of the priority valve. This additional force on the spring end of the priority valve causes area "A" to open and allow the necessary flow and pressure to pass through the amplifier valve to operate the steering cylinders.

The flow amplifier valve includes a relief valve in the priority valve spring area that is used to control maximum steering working pressure to 19 000 kPa (2,750 psi), even though supply pressure coming in to port "HP" is higher. When 19 000 kPa (2,750 psi) is obtained, the relief valve prevents the "LS" pressure from increasing, thereby allowing the priority valve to compress the spring enough to close off area "A" when 19 000 kPa (2,750 psi) is present.



83758

FIGURE 4-8. FLOW AMPLIFIER - LEFT STEER



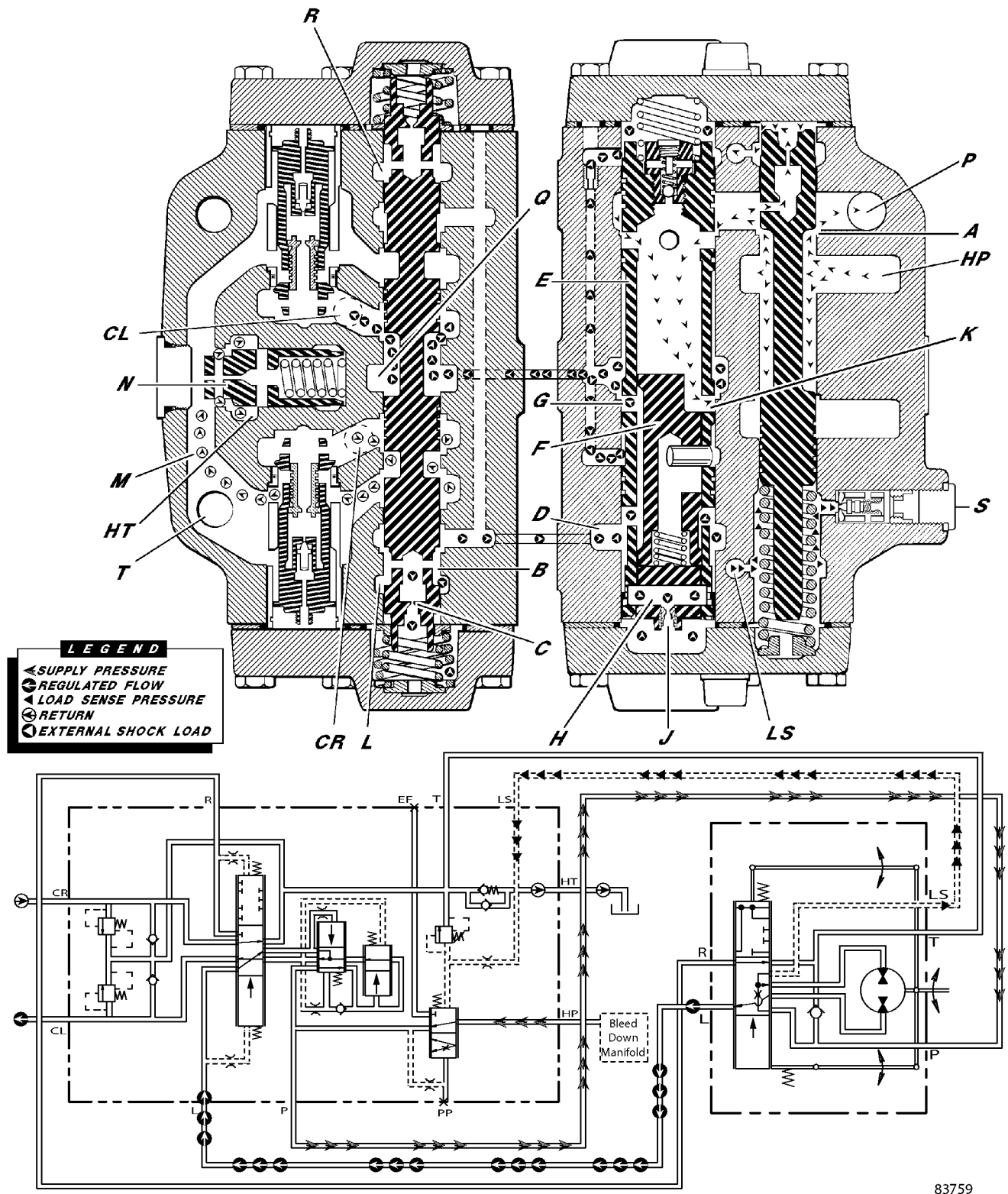
## Steering Right

(Refer to Figure 4-9)

There are only a few differences between steering left and steering right. When the operator turns the steering wheel right, oil is supplied out ports “R” and “LS” of the steering control unit.

The oil enters the flow amplifier assembly at port “R” and shifts the directional valve the opposite direction. The oil flows through the amplifier valve in exactly the same way. The combined oil from the amplifier valve passes through the center area “Q” of the directional valve to port “CR”, where it goes to the opposite ends of the steering cylinders to turn the wheels to the right.

The returning oil comes back through port “CL” to go to the tank. The “LS” oil operates exactly the same as steering left.



83759

FIGURE 4-9. FLOW AMPLIFIER - RIGHT STEER

## **No Steer, External Shock Load**

(Refer to Figure 4-10)

When the operator is not turning the steering wheel, the steering control unit valve supply is closed. The directional valve remains centered by its springs, thus closing the passages to ports "CL" and "CR". This creates a hydraulic lock on the steering cylinders to prevent their movement.

If the tires hit an obstruction that causes a large shock load, forcing the wheels to the left, increased pressure will occur in the ends of the cylinders connected to port "CR". The shock and suction relief valve inside the flow amplifier at port "CR" will open at its adjusted setting of 21 400 kPa (3,100 psi) and allow oil to escape from the pressurized ends of the cylinders, preventing a higher pressure.

As the cylinders are allowed to move, the other ends will have less than atmospheric pressure on port "CL". This low pressure permits oil that is escaping through the "CR" port relief valve to flow through the check valve portion of the shock and suction relief valve connected to port "CL". The oil then flows to the low pressure ends of the cylinders to keep the cylinders full of oil and prevent cavitation. A shock load in the opposite direction merely reverses the above procedure.

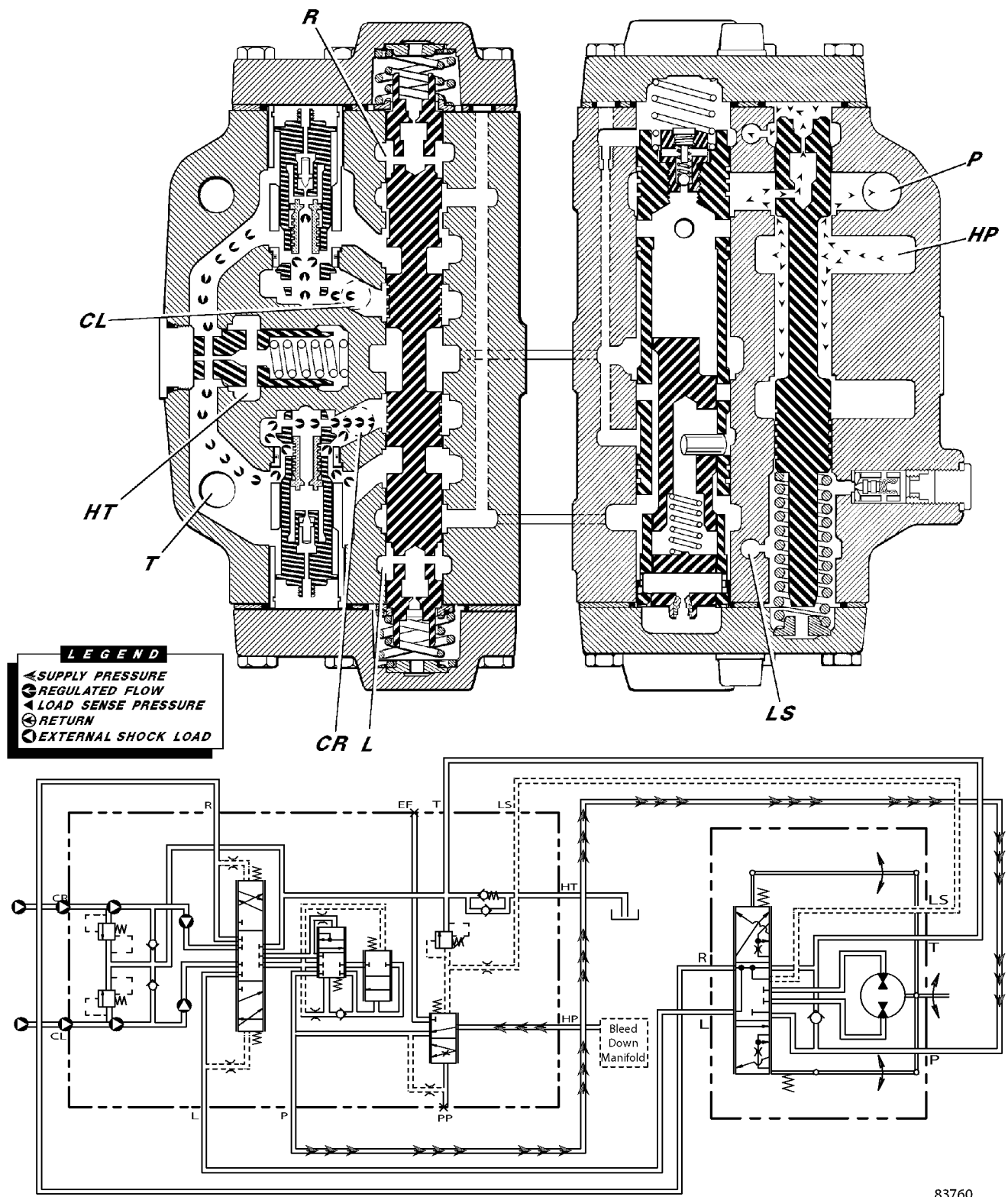


FIGURE 4-10. FLOW AMPLIFIER - NO STEER, EXTERNAL SHOCK LOAD

## STEERING PUMP

The steering/brake pump is mounted on the rear of the hoist system pump. The drive shaft mates with an internally splined coupling inside the hoist pump when the pump is bolted in place.

The steering pump is a pressure compensated piston type pump with an unloader valve. The pressure compensator is adjusted to 22 900 kPa (3,325 psi). The pump-mounted unloader valve maintains system pressure between 19 000 kPa (2,750 psi) and 20 900 kPa (3,025 psi).

Oil flow from the pump is routed through a high pressure filter, then to the bleddown manifold, where the flow is directed to the steering circuit (accumulators, flow amplifier, etc.) and to the brake apply system components located in the hydraulic brake cabinet behind the cab.

## Normal Operation

Driveshaft (35, Figure 4-11) runs through the centerline of pump housing (37) and valve plate (48). Cylinder barrel (45) is splined to the drive shaft.

Ball bearing (33) supports the outboard end of the driveshaft, and bushing type bearing (47) supports the inboard end. The pump cylinder barrel is carried in a polymerous (journal type) bearing (41).

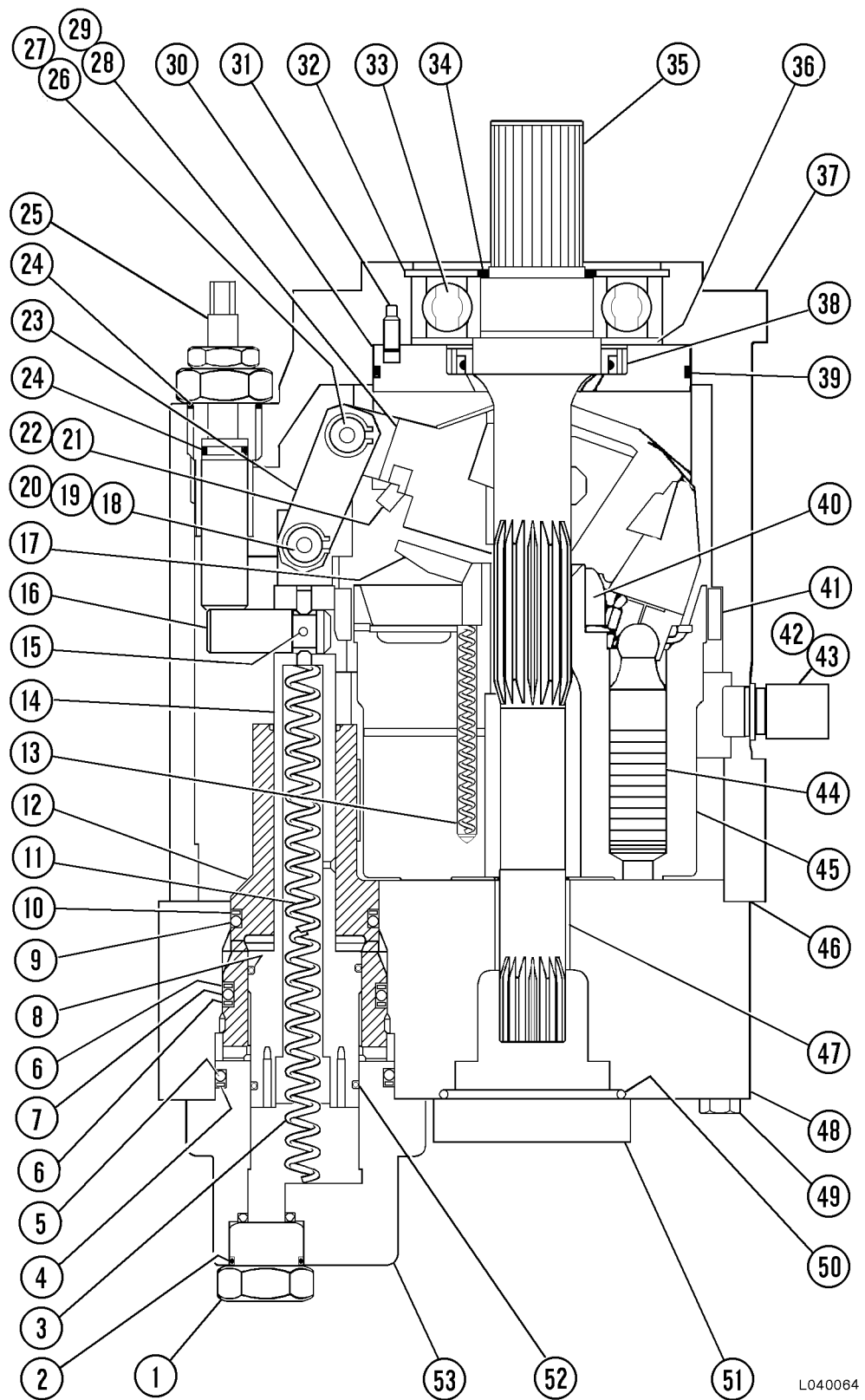
The valve plate has two crescent shaped ports (inlet and outlet). Pumping pistons (44) in the cylinder barrel are held against swashblock (28) by shoe retainer (17). The shoe retainer is held in position by fulcrum ball (40), which is forced outward by retainer springs (13). The springs act against pump cylinder barrel (45), forcing it against the valve plate while also forcing the piston shoes against the swashblock.

The semi-cylindrical shaped swashblock limits the piston stroke and can be swiveled in arc shaped saddle bearings (21). The cradle is swiveled by control piston (14).

## High Altitude Operation

The steering pump on later model trucks incorporates adjustable stroke control device (25, Figure 4-11). If the truck is operating at high altitude locations of 3 050 meters (10,000 feet) or more above sea level, it may be necessary to adjust the pump stroke to reduce oil flow to eliminate pump cavitation problems.

1. Plug
2. O-ring
3. Spring
4. Back-Up Ring
5. O-Ring
6. Back-Up Ring
7. O-Ring
8. O-Ring
9. O-Ring
10. Back-Up Ring
11. Spring
12. Control Piston Sleeve
13. Spring
14. Piston
15. Pin
16. Control Piston Stop Pin
17. Shoe Retainer
18. Pin
19. Retainer Ring
20. Washer
21. Saddle Bearing
22. Roll Pin
23. Link
24. O-Ring
25. Stroke Adjuster
26. Retainer Ring
27. Pin
28. Swashblock
29. Dowel Pin
30. Saddle
31. Roll Pin
32. Retainer Ring
33. Bearing
34. Shaft Retainer Ring
35. Shaft
36. Seal Retainer
37. Pump Housing
38. Shaft Seal
39. O-Ring
40. Fulcrum Ball
41. Cylinder Bearing
42. O-Ring
43. Elbow Fitting
44. Piston/Shoe Assembly
45. Cylinder Barrel
46. Gasket
47. Bearing
48. Valve Plate
49. Capscrew
50. O-Ring
51. Cover
52. Seal
53. Cap



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FIGURE 4-11. STEERING PUMP

## PRINCIPLE OF OPERATION

### Full Pump Volume

Control piston (1, Figure 4-12) controls the angle of swashblock (4). When the control piston moves to the full right position, the pump is at maximum pumping capacity.

The driveshaft turns splined housing (2), which contains pumping pistons (3). When the housing is rotated, the pistons move in and out of their bores and the piston shoes "ride" against the angled swashblock.

As the cylinder rotates, the individual piston bores are connected alternately to the left (port "A") and right (port "B") crescent shaped ports in the valve plate. While connected to left side (suction) port "A", each piston moves outward, drawing fluid from port "A" into the piston bore until its outermost stroke is reached. At that point, the piston bore passes from the left crescent port to the right crescent port.

While rotating across the right side crescent, each piston moves downward on the angled swashblock face. Thus, each piston is forced inward. Each piston displaces fluid through the right side crescent to port "B" until its innermost stroke is reached. At that point, the piston bore again passes from the right to the left side crescent and the operating cycle is repeated.

### Half Pump Volume

Figure 4-13 shows that the position of control piston (1) is near the center of its travel. Swashblock (4) is not angled as steep as before. Therefore, the pistons have a shorter stroke. As the piston stroke gets shorter, the pump output also decreases.

### Neutral Position

Neutral position (Figure 4-14) results when control piston (1) centers swashblock (4). The swashblock angle is now zero and the swashblock face is now parallel to cylinder face. Therefore, no inward or outward motion of pump pistons (3) exists as the piston shoes rotate around the swashblock face. The lack of inward and outward motion results in no fluid being displaced from the piston bores to the crescents in the valve plate and, subsequently, no delivery from pump ports.

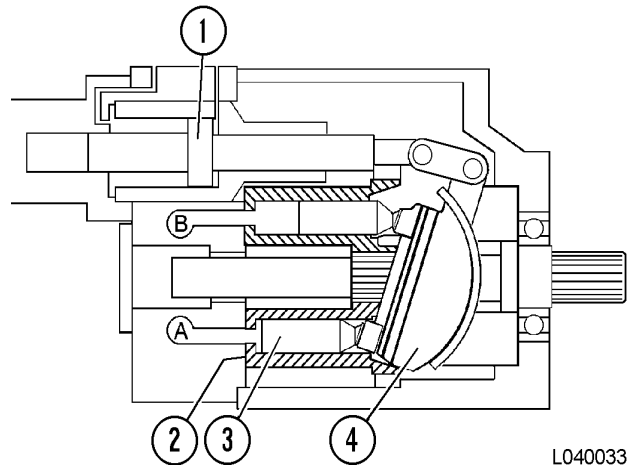


FIGURE 4-12. PUMP AT FULL VOLUME

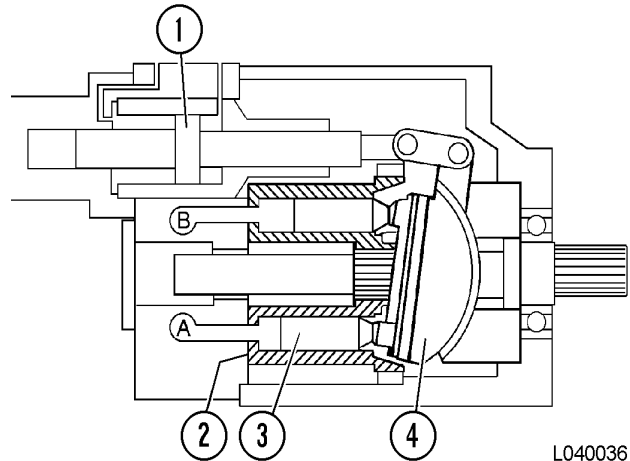


FIGURE 4-13. PUMP AT HALF VOLUME

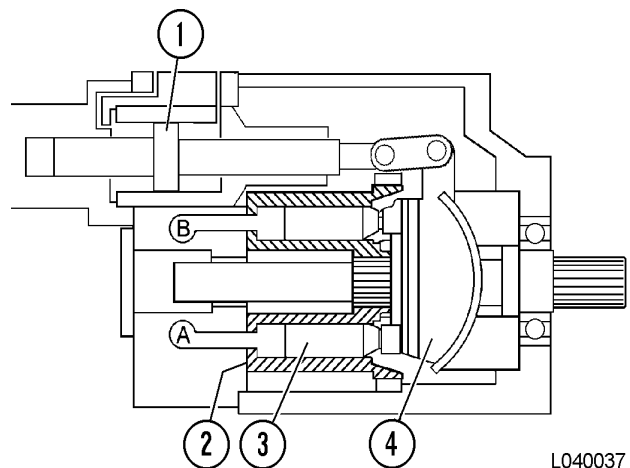


FIGURE 4-14. PUMP IN NEUTRAL POSITION

## Steering Pump

Figure 4-15 shows the steering pump and the location of the pressure control adjustments and stroke (flow) adjuster.

Stroke adjuster (3, Figure 4-15) is set at the factory to provide maximum pump flow so adjustment is not normally required. If the truck is operated at high altitude locations of 3 050 meters (10,000 feet) or more above sea level, it may be necessary to readjust the stroke control to reduce pump flow to prevent pump cavitation under certain conditions. If the truck is operating at high altitude and problems occur, consult your Komatsu service representative for adjustment procedures.

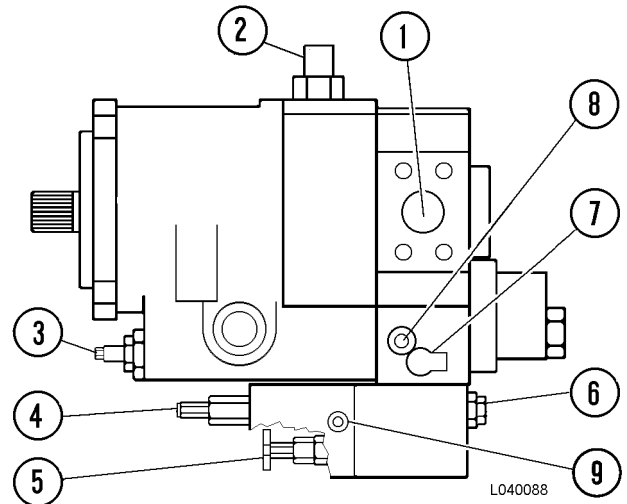


FIGURE 4-15. PUMP ASSEMBLY

- |                                     |                                  |
|-------------------------------------|----------------------------------|
| 1. Outlet Port                      | 5. Compensator Pressure Adjuster |
| 2. Case Drain                       | 6. 4-Way Valve                   |
| 3. Stroke Adjuster                  | 7. GPA Test Port                 |
| 4. Unloader Valve Pressure Adjuster | 8. GP2 Test Port                 |
|                                     | 9. "ACC" Port                    |

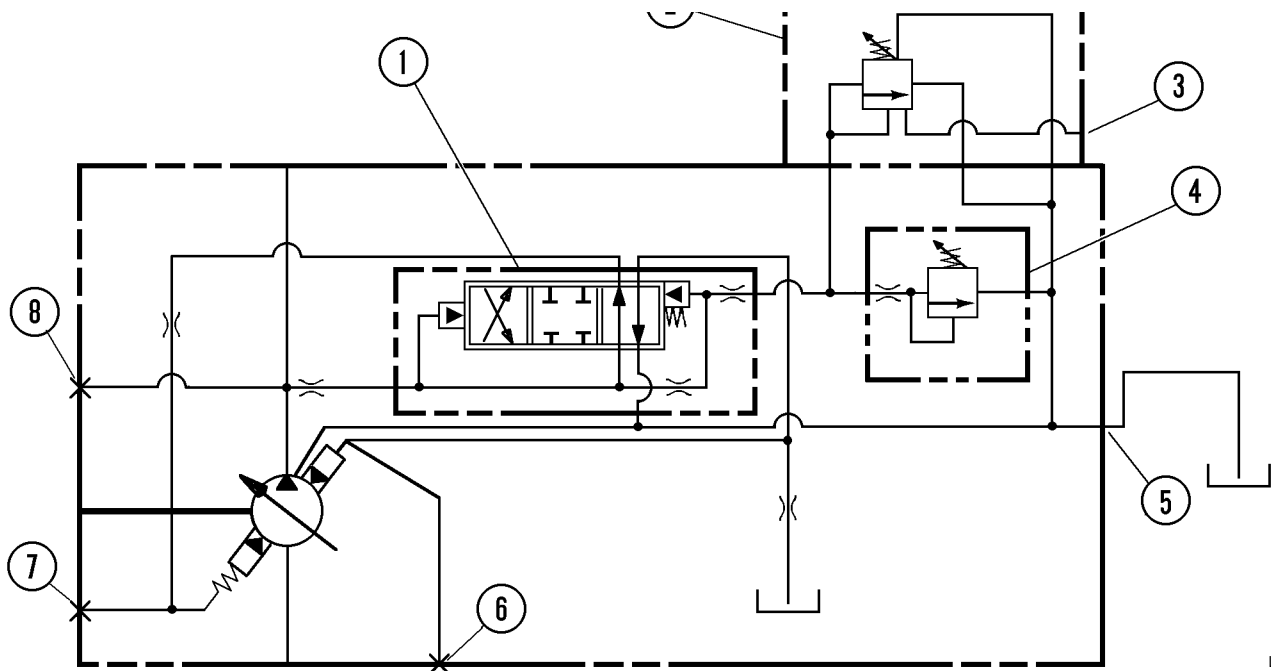


FIGURE 4-16. STEERING PUMP PRESSURE CONTROL SCHEMATIC

- |                           |                                       |                  |
|---------------------------|---------------------------------------|------------------|
| 1. 4-Way Valve            | 4. Pressure Compensator Control Block | 6. GP2 Port      |
| 2. Unloader Control Block | 5. Case Drain                         | 7. GP4 Port      |
| 3. "ACC" Port             |                                       | 8. GPA Test Port |



# NOTES

# SECTION L5

## STEERING CONTROL UNIT

### INDEX

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# NOTES

# STEERING CONTROL UNIT

## REMOVAL

### **WARNING**

*Relieve pressure before disconnecting hydraulic lines. Tighten all connections before applying pressure. Hydraulic fluid escaping under pressure can have sufficient force to enter a person's body by penetrating the skin and cause serious injury and possibly death if proper medical treatment by a physician familiar with this type of injury is not received immediately.*

1. Turn the key switch OFF and allow 90 seconds for the accumulators to bleed down. Turn the steering wheel to ensure that no oil remains under pressure.
2. Remove the capscrews and lockwashers that secure the access cover on the front side of the operator's cab. Remove the access cover.

*NOTE: Clean the steering control unit and surrounding area to help avoid contaminating the hydraulic oil when any lines are opened.*

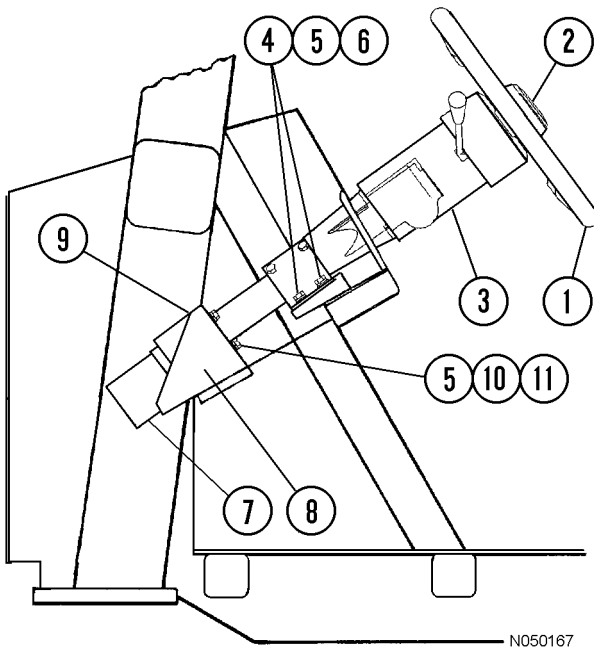


FIGURE 5-1. STEERING CONTROL UNIT

- |                    |                          |
|--------------------|--------------------------|
| 1. Steering Wheel  | 7. Steering Control Unit |
| 2. Button Horn     | 8. Bracket L.H.          |
| 3. Steering Column | 9. Bracket R.H.          |
| 4. Capscrew        | 10. Capscrew             |
| 5. Flat Washer     | 11. Lockwasher           |
| 6. Lockwasher      |                          |

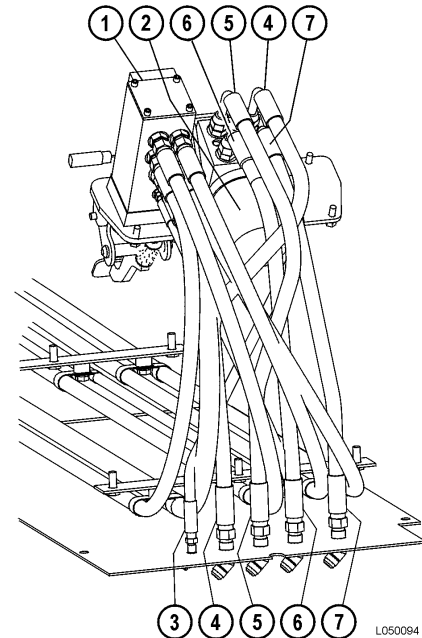


FIGURE 5-2. STEERING CONTROL UNIT

- |                          |                  |
|--------------------------|------------------|
| 1. Brake Valve           | 5. "T" Port Hose |
| 2. Steering Control Unit | 6. "P" Port Hose |
| 3. "LS" Port Hose        | 7. "R" Port Hose |
| 4. "L" Port Hose         |                  |

3. Tag all hydraulic lines for proper identification during installation. Disconnect the hydraulic lines (see Figure 5-2) at the steering control unit. Plug all hoses.
4. Remove four mounting capscrews (10, Figure 5-1), flat washers and lockwashers. Remove steering control unit (7).
5. Place the steering control unit in a clean work area for disassembly.

## SPLINE INSPECTION

Whenever the steering control unit is removed for service, the steering column shaft should be inspected for excessive wear.

1. Thoroughly clean the splines on the steering column shaft. Inspect for damage and excessive wear.
2. Use an outside micrometer or dial caliper to measure the outside diameter of the male splines on the steering column shaft.
  - **Minimum diameter:** 24.13 mm (0.950 in.)
3. If the splines are worn more than minimum diameter specification, replace the steering column. Refer to Section N.

## WARNING

***Serious personal injury to the operator or to anyone positioned near the front wheels may occur if a truck is operated with the hydraulic steering lines improperly installed. Improperly installed lines can result in uncontrolled steering and/or sudden and rapid rotation of the steering wheel as soon as the steering wheel is moved. It will turn rapidly and cannot be stopped manually.***

***After servicing the steering control unit, check the hydraulic steering lines for correct hook-up before starting the engine.***

### INSTALLATION

1. Lubricate the splines of the steering column shaft with a molybdenum disulphide or multi-purpose NLGI grease.
2. Move the steering control unit into position and align it with the steering column shaft splines. Secure the steering control unit in place using four mounting capscrews (10, Figure 5-1), flat washers and lockwashers.
3. Check for proper steering wheel rotation. Ensure that the wheel does not bind and returns to neutral after rotating 1/4 turn left and right. If necessary, adjust the steering unit and/or steering column to realign the steering column and steering control unit.

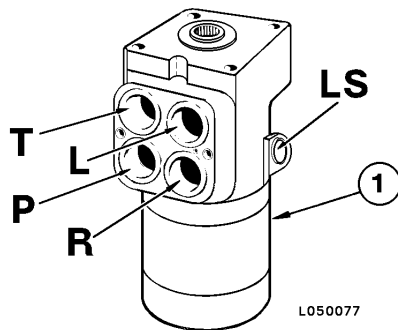


FIGURE 5-3. VALVE PORT IDENTIFICATION

- |                           |                      |
|---------------------------|----------------------|
| 1. Steering Control Valve | "L" - Left Steering  |
| "T" - Return to Tank      | "R" - Right Steering |
| "P" - Supply from Pump    | "LS" - Load Sensing  |

4. Tighten all capscrews to the standard torque. Unplug and attach the hydraulic lines to their proper ports on the steering control unit. Refer to Figure 5-2.
5. Start the engine and check for proper steering function and any leaks.
6. Install the access cover and secure it with the capscrews and lockwashers.

### DISASSEMBLY

***NOTE: The steering control unit is a precision unit manufactured to close tolerances. Therefore, complete cleanliness is essential when handling the valve assembly. Work in a clean area and use lint free wiping materials or dry compressed air. Clean type C-4 hydraulic oil should be used during reassembly to ensure initial lubrication.***

1. Allow the oil to drain from the valve ports.
2. Match mark the gear wheel set and end cover to ensure proper relocation during reassembly. Refer to Figure 5-4.

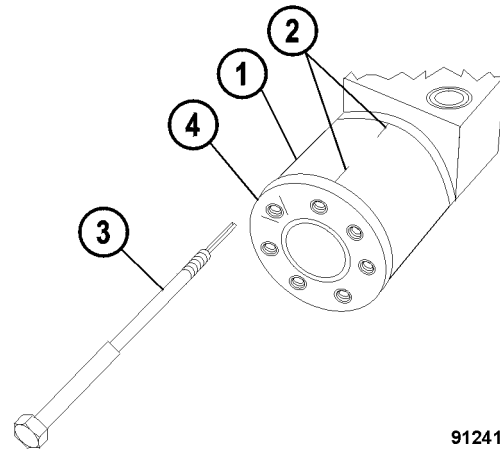


FIGURE 5-4. MARKING VALVE COMPONENTS

- |                   |                              |
|-------------------|------------------------------|
| 1. Valve Assembly | 3. Cap screw With Rolled Pin |
| 2. Match Marks    | 4. End Cover                 |
3. Remove the end cover cap screws and washers. Remove cap screw with rolled pin (3, Figure 5-4). Mark the hole location of the cap screw with rolled pin on the end cover to facilitate reassembly.

4. Remove end cover (4) and O-ring (2, Figure 5-5).

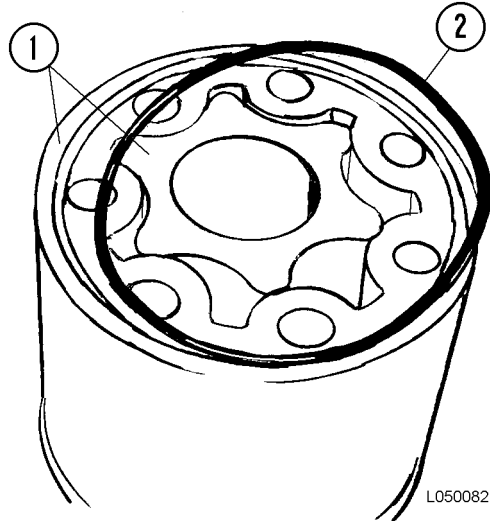


FIGURE 5-5. END COVER REMOVAL

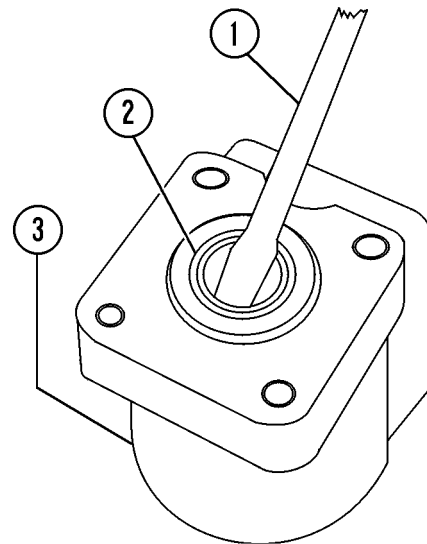
1. Gear Wheel Set
2. O-Ring

5. Remove the outer gear of gear wheel set (1) and the O-ring between the gear set and distribution plate.
6. Lift the inner gear off the cardan shaft.
7. Remove cardan shaft (11, Figure 5-7), distribution plate (15) and O-ring (14).
8. Remove threaded bushing (4) and ball (3).
9. With the valve housing positioned with the spool and sleeve vertical, carefully lift the spool assembly out of the housing bore.

## ⚠ IMPORTANT ⚠

***If housing is not vertical when spool and sleeve are removed, pin (9) may slip out of position and trap spools inside housing bore.***

10. Remove O-ring (5), kin ring (6) and bearing assembly (7).
11. Remove ring (8) and pin (9) and carefully push the inner spool out of the outer sleeve.
12. Press neutral position springs (10) out of their slot in the inner spool.
13. Remove dust seal (2, Figure 5-6) using a screwdriver. Be careful not to scratch or damage the dust seal bore.



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FIGURE 5-6. DUST SEAL REMOVAL

1. Screwdriver
2. Dust Seal
3. Housing

## CLEANING AND INSPECTION

1. Clean all parts carefully with fresh cleaning solvent.
2. Inspect all parts carefully and make any replacements necessary.

***NOTE: All O-rings, seals and neutral position springs should be replaced. Before reassembly, thoroughly lubricate all parts with clean type C-4 hydraulic oil.***

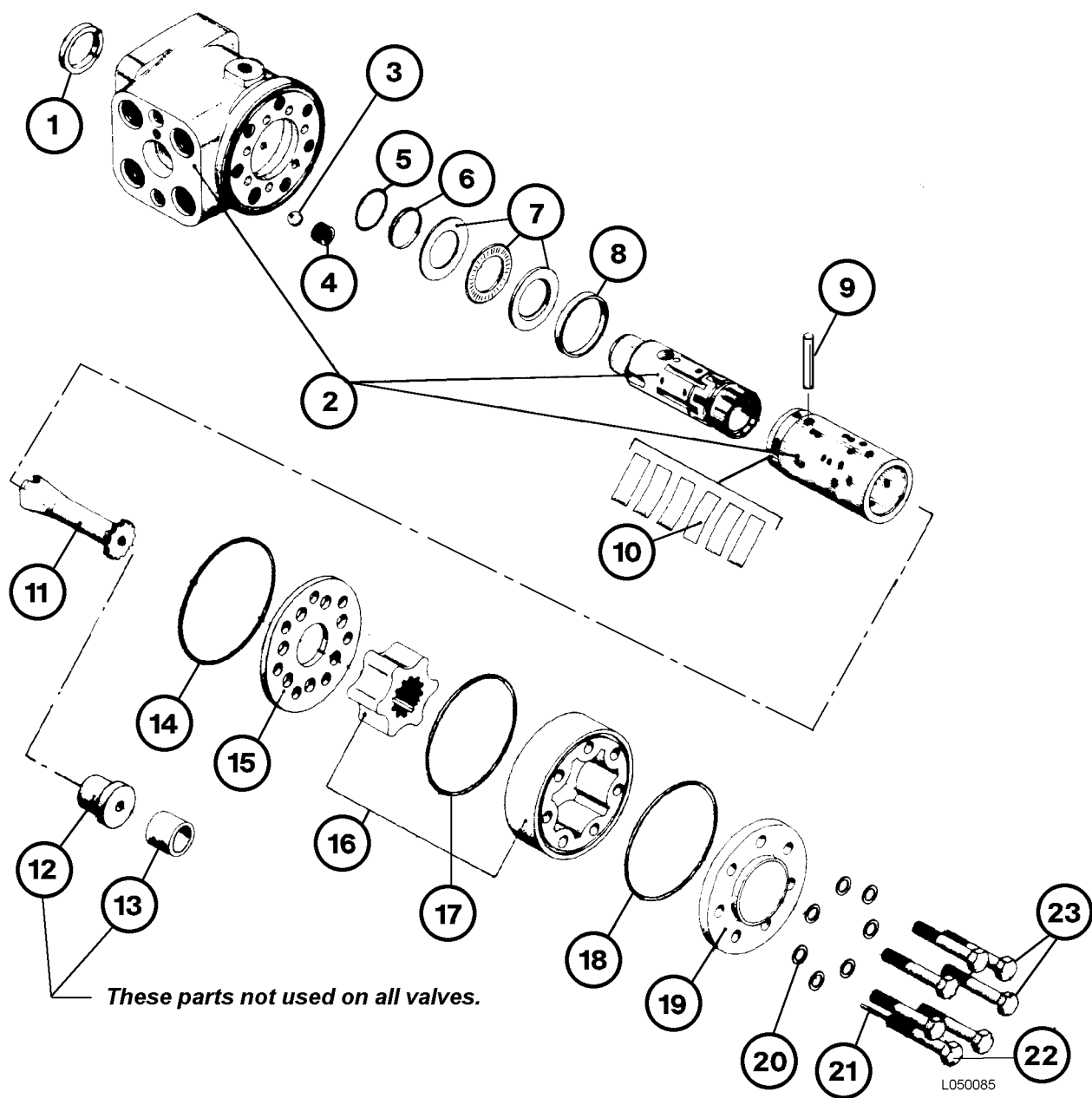


FIGURE 5-7. STEERING CONTROL VALVE

- |                     |                              |                        |                       |
|---------------------|------------------------------|------------------------|-----------------------|
| 1. Dust Seal        | 7. Bearing Assembly          | 12. Spacer             | 18. O-ring            |
| 2. Housing & Spools | 8. Ring                      | 13. Tube               | 19. End Cover         |
| 3. Ball             | 9. Cross Pin                 | 14. O- ring            | 20. Washers           |
| 4. Threaded Bushing | 10. Neutral Position Springs | 15. Distribution Plate | 21. Rolled Pin        |
| 5. O-ring           | 11. Cardan Shaft             | 16. Gear Wheel Set     | 22. Capscrew With Pin |
| 6. Kin Ring         |                              | 17. O-ring             | 23. Capscrews         |

## ASSEMBLY

**NOTE:** When assembling the spool and sleeve, only one of the two possible matching positions of the spring slots can be used because, in the other end of the sleeve and spool (opposite end of the spring slots), there are three slots in the spool and three holes in the sleeve. These must be opposite each other upon assembly so that the holes are partly visible through the slots in the spool. Refer to Figure 5-8.

1. To install the neutral position springs, place a screwdriver in the spool slot as shown in Figure 5-9.
2. Place one flat neutral position spring on each side of the screwdriver blade. Do not remove the screwdriver.
3. Push two curved neutral position springs in between one side of the screwdriver blade and a flat spring. Repeat for the opposite side. Remove the screwdriver.
4. Slide the inner spool in the sleeve. Compress the ends of the neutral position springs and push the neutral position springs in place in the sleeve.
5. Install cross pin (9, Figure 5-7).

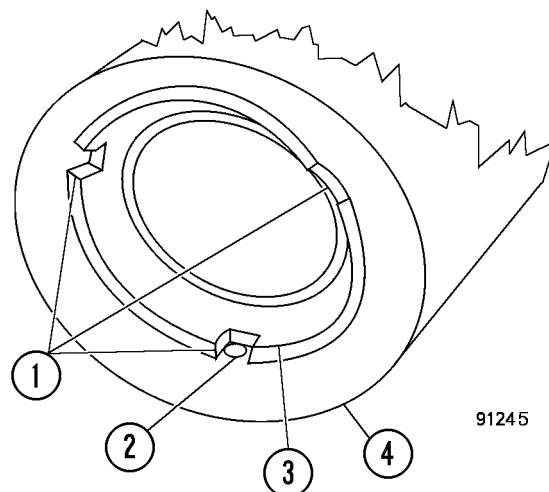


FIGURE 5-8. SPOOL AND SLEEVE ASSEMBLY

- |          |           |
|----------|-----------|
| 1. Slots | 3. Spool  |
| 2. Hole  | 4. Sleeve |

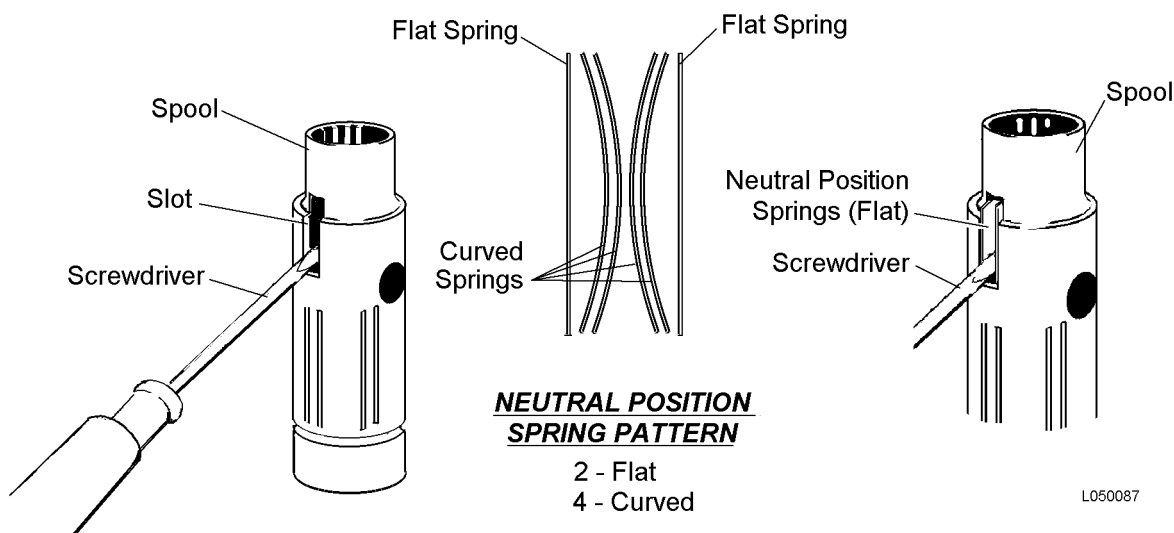
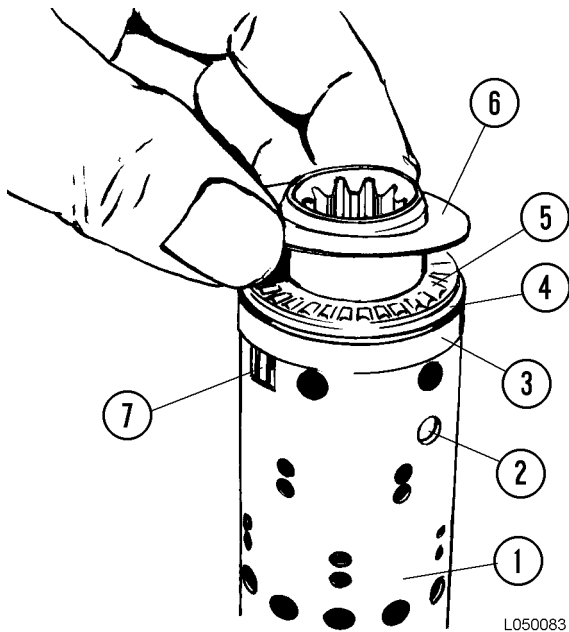


FIGURE 5-9. NEUTRAL POSITION SPRING INSTALLATION



6. With neutral position springs (7, Figure 5-10) centered in spool and sleeve, install ring (3), rear bearing race (4), thrust bearing (5) and front bearing race (6) in that order. The chamfer on the rear bearing must be facing away from the bearing.



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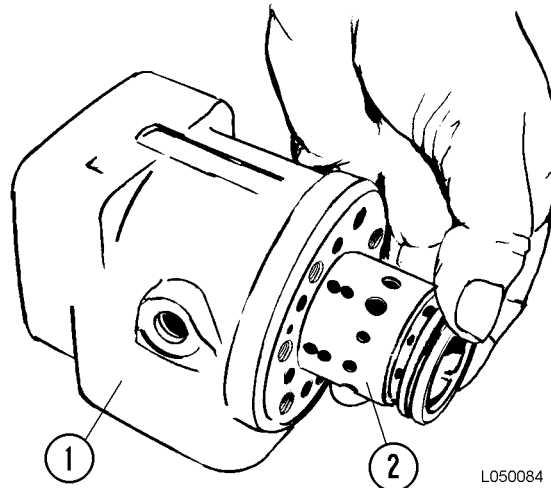
FIGURE 5-10. BEARING INSTALLATION

- |                                |                             |
|--------------------------------|-----------------------------|
| 1. Sleeve                      | 5. Thrust Bearing           |
| 2. Cross Pin                   | 6. Bearing Race             |
| 3. Ring                        | 7. Neutral Position Springs |
| 4. Bearing Race (with chamfer) |                             |

7. Place dust seal (1, Figure 5-7) in position. Using a flat iron block over the seal, tap it into the housing.

8. Position the O-ring and kin ring on the spool.

9. Position the steering control unit with the housing horizontal. Slowly guide the (lubricated) spool and sleeve with fitted parts into the bore using light turning movements. Refer to Figure 5-11.



L050084

FIGURE 5-11. SPOOL INSTALLATION

- |            |                   |
|------------|-------------------|
| 1. Housing | 2. Spool Assembly |
|------------|-------------------|

## ⚠ IMPORTANT ⚠

*The cross pin must remain horizontal when the spool and sleeve are pushed into the bore to prevent the pin from dropping out of the spool.*

10. Install the check ball in the hole shown in Figure 5-12. Install and lightly tighten the threaded bushing.
11. Lubricate O-ring (3) with Vaseline and install it in the housing groove.

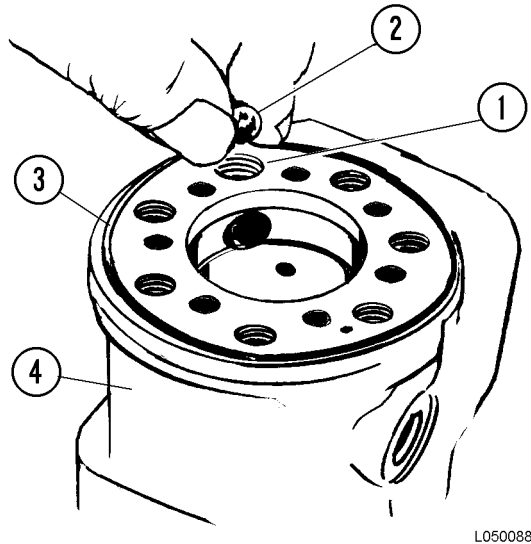


FIGURE 5-12. CHECK BALL INSTALLATION

- |                    |            |
|--------------------|------------|
| 1. Check Ball Hole | 3. O-ring  |
| 2. Check Ball      | 4. Housing |

12. Install distribution plate (15, Figure 5-7) so that the plate holes match the corresponding holes in the housing.
13. Guide cardan shaft (11) down into the bore with the slot in the cardan shaft aligned with cross pin (9).
14. Position the inner gear wheel onto the cardan shaft. It may be necessary to rotate the gear slightly to find the matching splines on the cardan shaft. Splines are machined to ensure proper alignment of the cardan shaft and inner gear wheel.
15. Lubricate O-rings (17) and (18) on both sides of the outer gear wheel with Vaseline and install them.
16. Align the outer gear wheel bolt holes with the tapped holes in the housing and the match marks.
17. Align cover (19), using the match marks as a reference. Install capscrews (23) and washers (20).
18. Install capscrew with pin (22) into the proper hole.
19. Install the end cover. Install and tighten the capscrews with washers hand-tight in a criss-cross pattern.

# NOTES

# SECTION L6

## STEERING CIRCUIT COMPONENT REPAIR

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# NOTES

# STEERING CIRCUIT COMPONENT REPAIR

## BLEEDDOWN MANIFOLD VALVE

### Removal

#### **⚠ WARNING**

***Do not loosen or disconnect any hydraulic line or component connection until engine is stopped and the key switch has been off for at least 90 seconds. Hydraulic fluid escaping under pressure can have sufficient force to enter a person's body by penetrating the skin and cause serious injury and possibly death if proper medical treatment by a physician familiar with this type of injury is not received immediately.***

***NOTE: It may not be necessary to remove the bleed down manifold from the truck for component replacement. If the problem area has been isolated, remove the inoperative component and replace it.***

1. Park the truck on a hard level surface. Turn the key switch OFF and allow at least 90 seconds for the steering accumulators to depressurize completely. Turn the steering wheel in both directions to ensure that no oil remains under pressure. Chock the wheels
2. Open brake accumulator bleed down valves (1, Figure 50-1) on brake manifold (2) in the hydraulic brake cabinet to depressurize the brake accumulators.
3. To verify that the brake accumulators are depressurized, press the brake lock switch (key switch ON, engine off) and apply the service brake pedal. The service brake status light should not come on.
4. Close the brake accumulator bleed down valves by rotating them clockwise.

5. Remove the protective cover from the bleed down manifold.
6. Disconnect the electrical leads from solenoids (6, Figure 50-2) and (15), pressure switch (7), and pressure sensors (3) and (23).
7. Disconnect, identify and plug each hydraulic line from the bleed down manifold.

#### **⚠ WARNING**

***The bleed down manifold weighs approximately 170 kg (375 lb). Use a suitable lifting device that has sufficient capacity to handle the load safely.***

8. Remove the mounting capscrews and the bleed down manifold. Clean the exterior of the manifold before removing any components from the manifold.

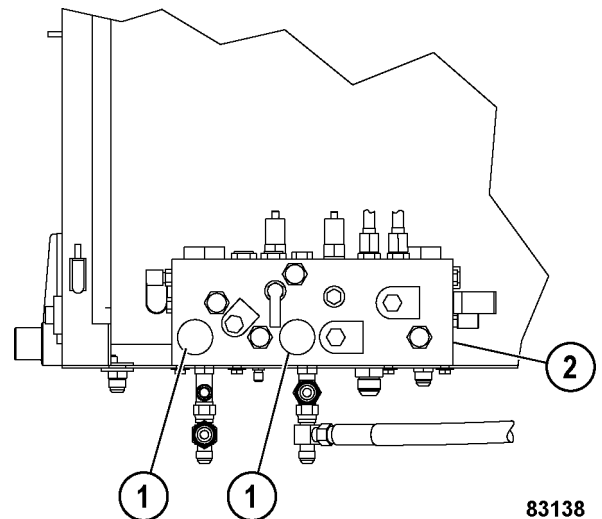


FIGURE 50-1. BRAKE MANIFOLD

1. Accumulator Bleed Down Valves
2. Brake Manifold

## Installation

1. Position the bleed down manifold on the truck. Install and tighten the mounting capscrews to the standard torque.
2. Unplug the hydraulic lines. Install new O-rings at the flange fittings and attach the hoses to the proper ports (See Figure 50-2). Tighten the connections securely.
3. Attach the electrical leads to solenoids (6, Figure 50-2) and (15), pressure switch (7), and pressure sensors (3) and (23). If the check valves or relief valves were removed, install new O-ring seals before installing them in the bleed down manifold.
4. Start the engine. Check the steering system and brake system for proper operation and leaks.
5. Install the protective cover.

**NOTE:** Relief valves are factory preset. Do not attempt to reset the relief valves or repair defective relief valves. Replace them as a unit. The check valves are also replaced only as units.

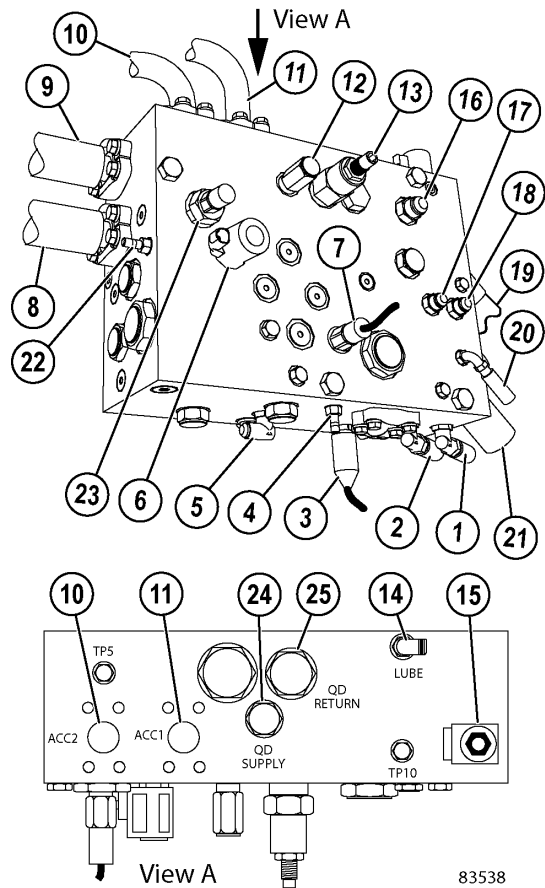


FIGURE 50-2. BLEED DOWN MANIFOLD

1. To Hoist Valve, Power Up
2. To Hoist Valve, Power Down
3. Komtrax Pressure Sensor (Port TP1)
4. Test Port (TP2)
5. To Brake System
6. Steering Accumulator Bleed Down Solenoid
7. Steering Pressure Switch
8. Supply To Flow Amplifier
9. Return From Flow Amplifier
10. To Front Steering Accumulator
11. To Rear Steering Accumulator
12. Relief Valve, 4 100 kPa (600 psi)
13. Relief Valve, 27 500 kPa (4,000 psi)
14. Auto Lube Pump Supply Port
15. Hoist Up Limit Solenoid Valve
16. Return From Hoist Pilot Valve
17. Hoist Pilot Valve, Power Down
18. Hoist Pilot Valve, Power Up
19. Return to Tank
20. Feedback Pressure to Unloader valve
21. Supply From Pump
22. Test Port (TP3)
23. Pressure Sensor (Bleeddown)
24. Quick Disconnect, Supply Port
25. Quick Disconnect, Return Port

## FLOW AMPLIFIER



***Do not loosen or disconnect any hydraulic line or component connection until engine is stopped and the key switch has been off for at least 90 seconds.***

***Hydraulic fluid escaping under pressure can have sufficient force to enter a person's body by penetrating the skin and cause serious injury and possibly death if proper medical treatment by a physician familiar with this type of injury is not received immediately.***

### Removal

1. Perform normal truck shutdown procedures. Turn key switch OFF and allow 90 seconds for the accumulators to bleed down.
2. Disconnect, plug, identify, and tag each hydraulic line.
3. Support the flow amplifier valve and remove the mounting capscrews. Remove the valve. The weight of the valve is 29 kg (64 lb).
4. Move the valve to a clean work area for disassembly.

### Installation

1. Support the flow amplifier and move it into position.
2. Install the mounting capscrews and tighten them to the standard torque.
3. Unplug the hydraulic lines and connect them at their proper locations. Tighten the fittings securely. Use new O-rings on the flange fittings.

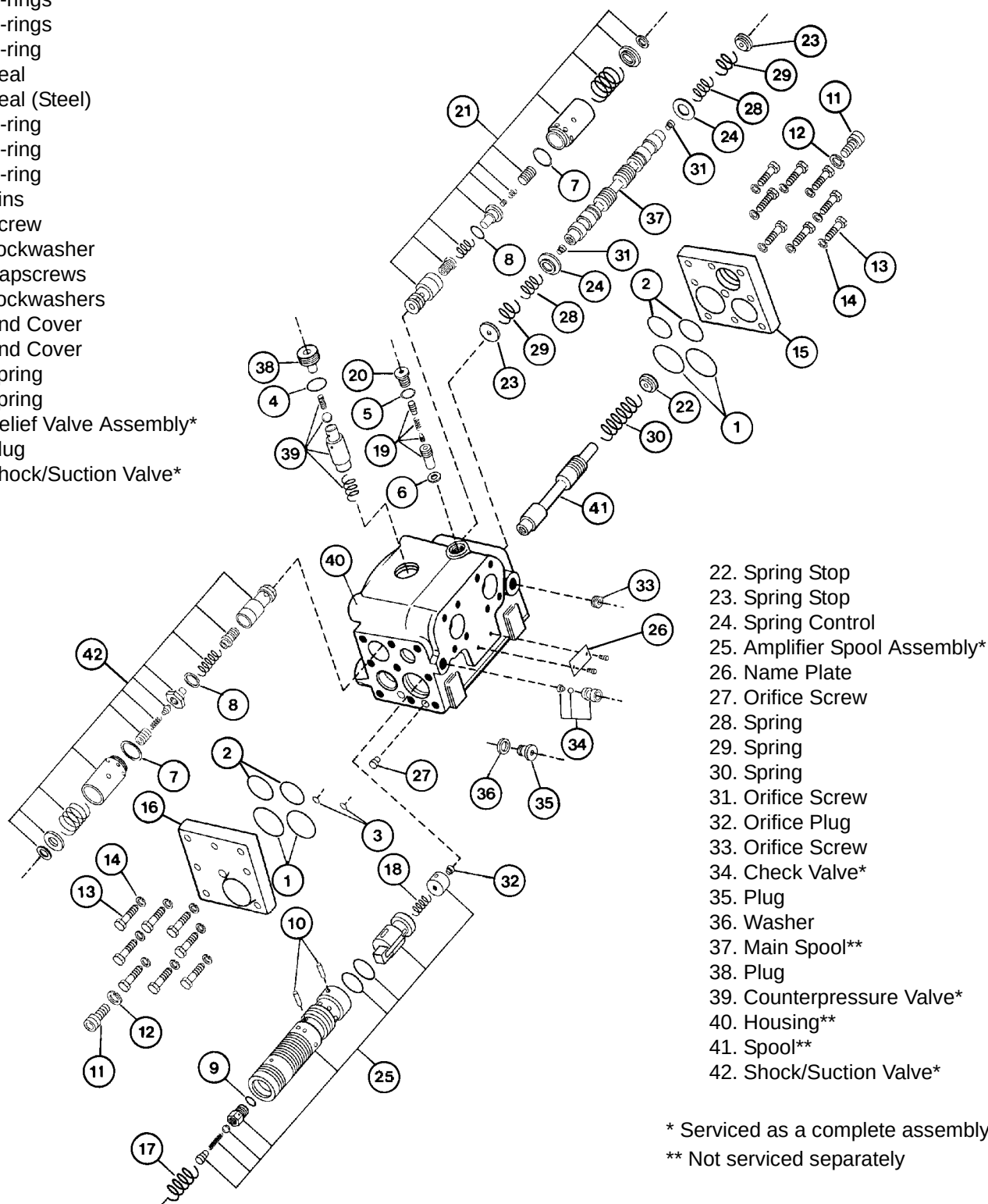
### Disassembly

***NOTE: The flow amplifier valve is a precision unit manufactured to close tolerances. Therefore, complete cleanliness is essential when handling the valve. Work in a clean area and use lint free wiping materials or dry compressed air. Use a wire brush to remove foreign material and debris from around the exterior of the valve before disassembly. Clean solvent and type C-4 hydraulic oil should be used to ensure cleanliness and initial lubrication.***

1. Remove counterpressure valve plug (38, Figure 6-3) and O-ring (4). Remove counterpressure valve (39).
2. Remove relief valve plug (20) and seal (5). Use an 8 mm hex allen head wrench to remove relief valve assembly (19). Remove steel seal (6).
3. Use a 13 mm hex head allen wrench to remove screw (11) and lockwasher (12) from end cover (15). Use a 13 mm hex head allen wrench to remove screws (13) and lockwashers (14). Remove end cover (15).
4. Remove spring stop (22) and spring (30). Remove spring stop (23) and springs (28) and (29). Remove O-rings (1) and (2) from end cover (15).
5. Remove spring control (24) and main spool (37). Remove priority valve spool (41). Remove spring control (24), springs (28) and (29), and spring stop (23) from housing (40).
6. Remove amplifier spool assembly (25) and set it aside for further disassembly, if required (see Figure 6-5).
7. Remove shock and suction valve (21) and set it aside for further disassembly, if required.
8. Use a 13 mm hex head allen wrench to remove screw (11) and lockwasher (12) from end cover (16). Use a 13 mm hex head allen wrench to remove screws (13) and lockwashers (14). Remove end cover (16).
9. Remove O-rings (1), (2) and (3). Remove spring (17).
10. Remove shock and suction valve (42) and set it aside for further disassembly, if required.
11. Remove orifice screw (27), orifice screw (33), and check valve (34).



1. O-rings
2. O-rings
3. O-rings
4. O-ring
5. Seal
6. Seal (Steel)
7. O-ring
8. O-ring
9. O-ring
10. Pins
11. Screw
12. Lockwasher
13. Capscrews
14. Lockwashers
15. End Cover
16. End Cover
17. Spring
18. Spring
19. Relief Valve Assembly\*
20. Plug
21. Shock/Suction Valve\*



\* Serviced as a complete assembly

\*\* Not serviced separately

FIGURE 6-3. FLOW AMPLIFIER VALVE

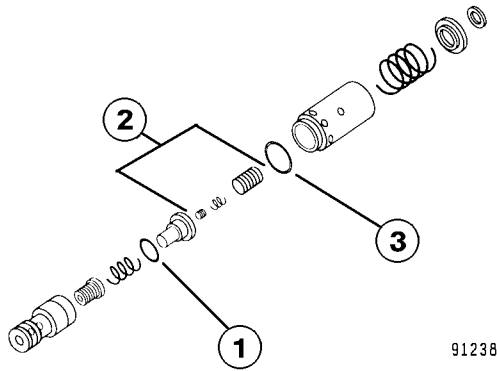


FIGURE 6-4. SHOCK & SUCTION VALVE ASSEMBLY

- |                  |           |
|------------------|-----------|
| 1. O-Ring        | 3. O-Ring |
| 2. Pilot Section |           |

**NOTE:** The flow amplifier valve is equipped with two identical shock and suction valves which are only serviced as complete valve assemblies. Only O-rings (7, Figure 6-4) and (8) are replaceable. Check valve (34) and counterpressure valve (39) are also serviced only as assemblies.

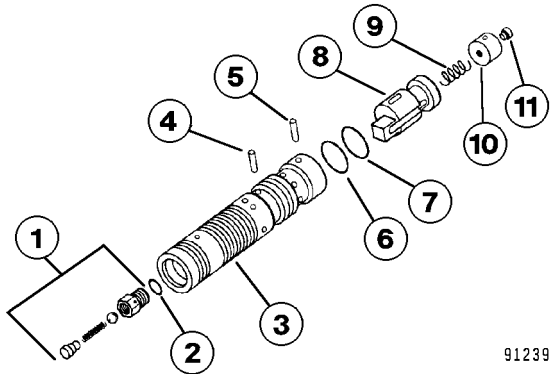


FIGURE 6-5. AMPLIFIER SPOOL ASSEMBLY

- |                   |                   |
|-------------------|-------------------|
| 1. Check Valve    | 7. Retaining Ring |
| 2. O-Ring         | 8. Inner Spool    |
| 3. Spool          | 9. Spring         |
| 4. Pin            | 10. Plug          |
| 5. Pin            | 11. Orifice Plug  |
| 6. Retaining Ring |                   |

**NOTE:** Disassembly of the amplifier spool assembly is only necessary if O-ring (2, Figure 6-5), spring (9) or orifice plug (11) require replacement. Otherwise, replace the amplifier spool assembly as a complete unit. Refer to Steps 12 and 13.

12. Remove retaining ring (7, Figure 6-5), remove pin (5). Remove plug (10) and spring (9). Remove retaining ring (6) and pin (4) and remove inner spool (8).
13. Unscrew check valve (1) and remove. Remove O-ring (2). Remove orifice screw (11) from plug (10).
14. Clean and inspect all parts carefully. Make any replacements that are necessary.

### Assembly

1. Use clean type C-4 hydraulic oil to thoroughly lubricate each internal part before installation.
2. Reassemble the amplifier spool assembly in reverse order. Refer to Steps 13 and 12 in the Disassembly procedure.
3. Install orifice screw (27, Figure 6-3) finger-tight.
4. Install check valve (34) and tighten it to **1 N·m (8 in. lb)**.
5. Install orifice screw (33) and tighten it to **1 N·m (8 in. lb)**.
6. Install steel seal (6), relief valve assembly (19), seal (5), and plug (20). Tighten the plug to **3 N·m (25 in. lb)**.
7. Install counterpressure valve assembly (39). Install plug (38) with new O-ring (4).
8. Install both shock and suction valves (21) and (42) as complete units.
9. Install spring stop (23) springs (28) and (29), and spring control (24) into housing (40). Install orifice screws (31) into main spool (37), if removed. Install main spool (37).
10. Install amplifier spool assembly (25). Install priority valve spool (41) and spring (30). Install spring (17).
11. Install spring control (24), springs (28) and (29), and spring stop (23).
12. Lubricate O-rings (1), (2) and (3) with molycote grease and install them on end covers (15) and (16). Install end covers (15) and (16), capscrews (13) and lockwashers (44). Tighten the capscrews to **3 N·m (25 in. lb)**. Install screws (11) and lockwashers (12). Tighten screws to **8 N·m (71 in. lb)**.
13. To prevent contamination, fit plastic plugs to each open valve port.

## STEERING CYLINDERS

Figure 6-6 illustrates details of the steering cylinder mounting. Refer to Section G for removal and installation instructions.

### **WARNING**

**Relieve pressure before disconnecting hydraulic and other lines. Tighten all connections before applying pressure.**

**Hydraulic fluid escaping under pressure can have sufficient force to enter a person's body by penetrating the skin and cause serious injury and possibly death if proper medical treatment by a physician familiar with this injury is not received immediately.**

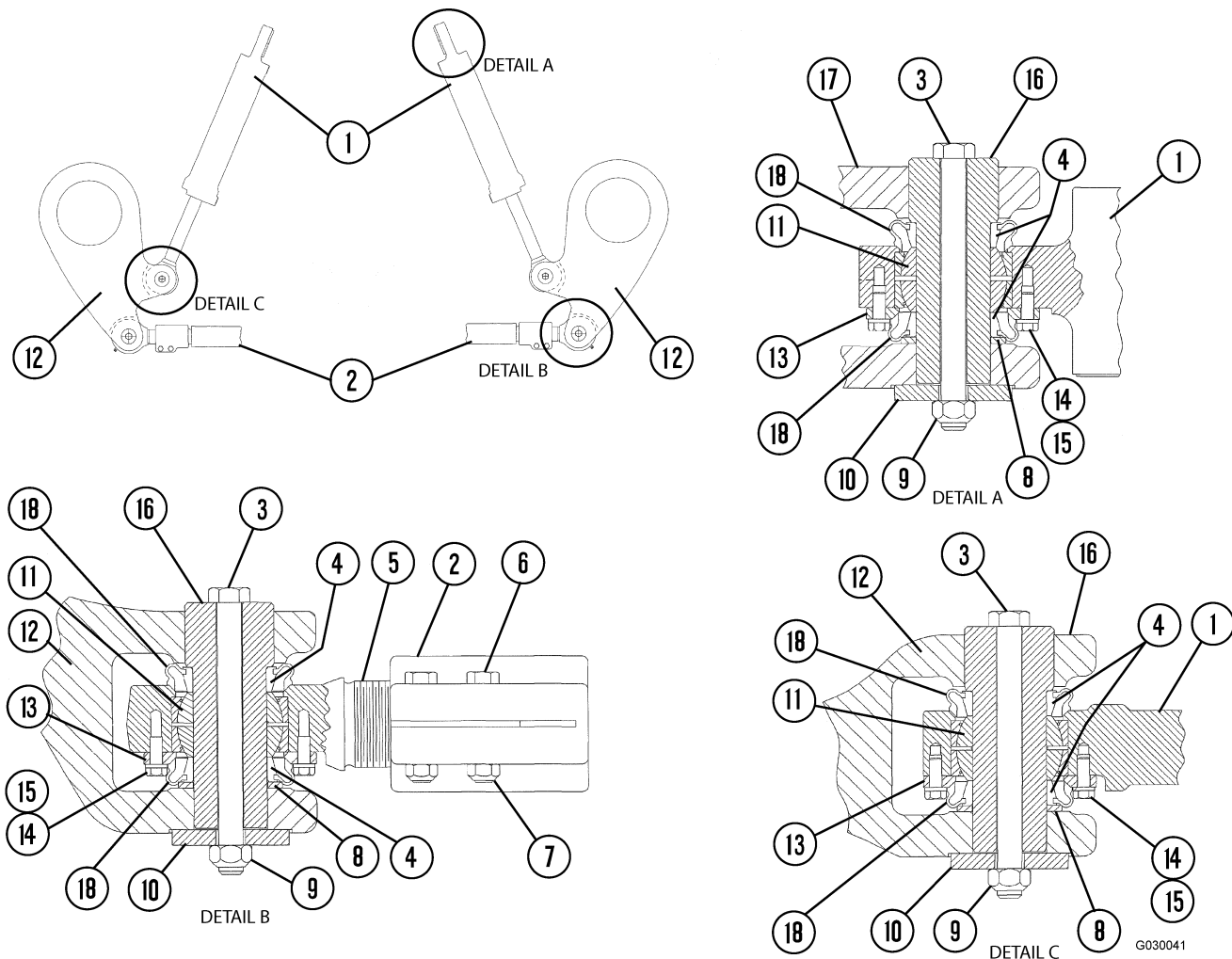


FIGURE 6-6. STEERING CYLINDER MOUNTING

- |                      |                 |                            |
|----------------------|-----------------|----------------------------|
| 1. Steering Cylinder | 7. Locknut      | 13. Bearing Retainer       |
| 2. Tie Rod           | 8. Washer       | 14. Capscrew               |
| 3. Capscrew          | 9. Locknut      | 15. Washer                 |
| 4. Spacer            | 10. Retainer    | 16. Pin                    |
| 5. Tie Rod End       | 11. Bearing     | 17. Frame                  |
| 6. Capscrew          | 12. Spindle Arm | 18. Spherical Bearing Seal |

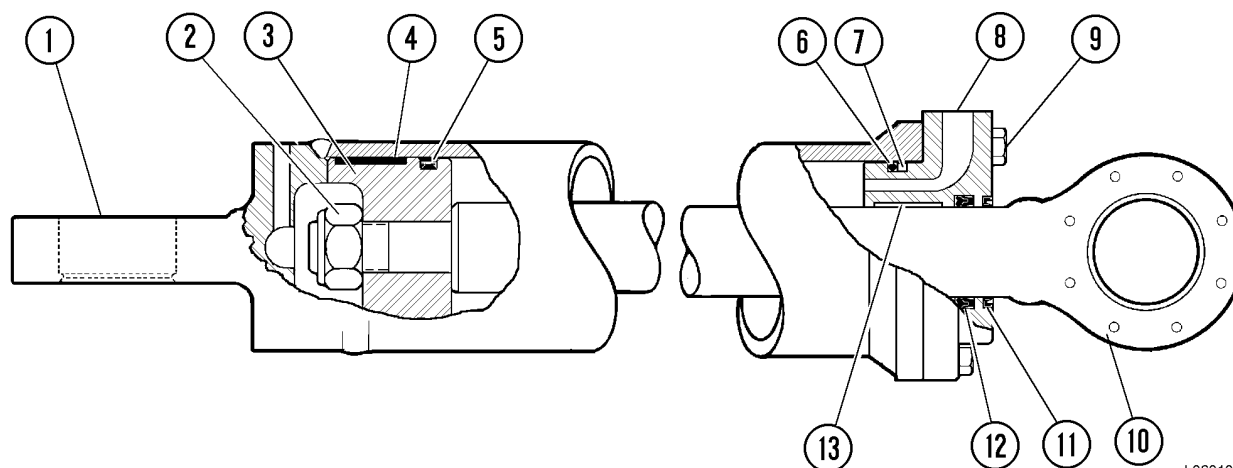


FIGURE 6-7. STEERING CYLINDER ASSEMBLY

- 1. Housing
- 2. Locknut
- 3. Piston

- 4. Piston Bearing
- 5. Piston Seal Assembly
- 6. O-Ring

- 7. Backup Ring
- 8. Gland
- 9. Capscrew

- 10. Rod Structure
- 11. Rod Wiper
- 12. Rod Seal
- 13. Bearing

### Disassembly

1. Remove capscrews (9, Figure 6-7) and pull rod (10) and gland (8) out of cylinder housing (1).
2. Remove locknut (2) and piston (3). Remove piston bearing (4) and piston seal (5) from the piston.
3. Pull rod (10) free of gland (8). Remove O-ring (6) and backup ring (7). Remove rod seal (12) and rod wiper (11). Remove bearing (13).
4. Inspect the cylinder housing, gland, piston and rod for signs of pitting, scoring and excessive wear. Clean all parts with fresh cleaning solvent and lubricate with clean Type C-4 hydraulic oil.

### Piston Seal & Bearing Installation

1. Install new piston seal assembly (5, Figure 6-7) on piston (3) as follows:
  - a. Heat the piston seal in boiling water for 3 to 4 minutes.
  - b. Remove the seal from the water and assemble it on the piston. **Do not take longer than five seconds to complete** because the seal will take a permanent set. Piston bearing (4) may be used to position the seal in the groove. Apply pressure evenly to avoid cocking the seal.
  - c. If the seal is loose on the piston, a belt type wrench or similar tool can be used to compress the outside diameter of the seal until it fits tightly on the piston.
2. Install bearing (4) in the piston groove.

### Assembly

1. Install new bearing (13, Figure 6-7), rod seal (12), rod wiper (11), backup ring (7) and O-ring (6) in gland (8).
2. Push rod (10) through the top of the gland. Slowly advance the rod over the rod seal and rod wiper.
3. Install piston assembly (3) on the rod. Secure the piston to rod with locknut (2). Tighten the locknut to **3 390 N·m (2,500 ft lb)**.
4. Carefully install the rod and gland assembly into cylinder (1). Make sure that the backup ring and O-ring are not damaged during installation of the gland.
5. Install capscrews (9). Tighten the capscrews evenly to **420 N·m (310 ft lb)**.

### Testing

After the cylinder is assembled, perform the following tests to verify that performance is within acceptable limits.

1. Piston leakage must not exceed 1.6 cm<sup>3</sup>/min. (1.0 in<sup>3</sup>/min.) at 17 250 kPa (2,500 psi), port to port.
2. Rod seal leakage must not exceed one drop of oil in eight cycles of operation.
3. Piston break-away force should not exceed 690 kPa (100 psi).

## STEERING AND BRAKE PUMP

### Removal

*NOTE: Clean the steering pump and surrounding area carefully to help avoid contamination of hydraulic oil when lines are opened.*

## **⚠ DANGER**

**Relieve pressure before disconnecting hydraulic lines. Tighten all connections before applying pressure.**

**Hydraulic fluid escaping under pressure can have sufficient force to enter a person's body by penetrating the skin and cause serious injury and possibly death if proper medical treatment by a physician familiar with this injury is not received immediately.**

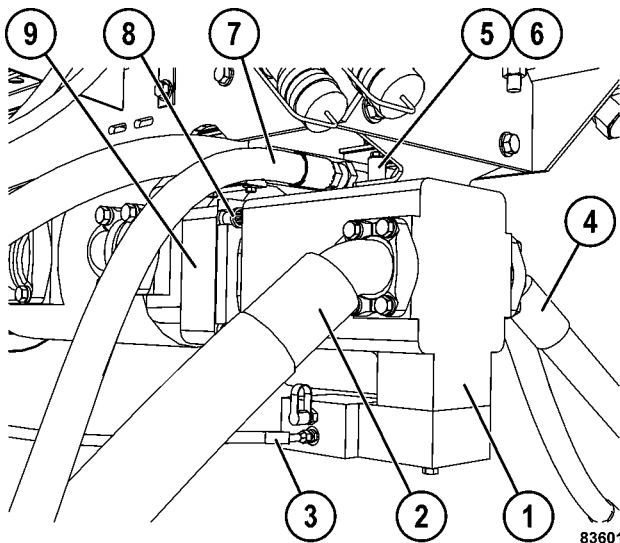


FIGURE 6-8. STEERING PUMP REMOVAL

- |                           |                            |
|---------------------------|----------------------------|
| 1. Steering/Brake Pump    | 6. Spacer                  |
| 2. Supply Hose            | 7. Case Drain Hose         |
| 3. Unloader Feedback Hose | 8. Pump Mounting Capscrews |
| 4. Steering Filter Hose   |                            |
| 5. Capscrew               | 9. Hoist Pump              |

1. Perform the normal truck shutdown procedure. Turn key switch OFF and allow 90 seconds for the accumulators to bleed down. Turn the steering wheel to ensure that no oil remains under pressure.

*NOTE: As an alternative to draining the hydraulic oil, a vacuum can be placed on the hydraulic tank to hold the oil in the tank.*

2. Drain the hydraulic tank by using the drain located on the bottom side of the tank.

*NOTE: Be prepared to contain approximately 947 liters (250 gallons) of hydraulic oil. If the oil is to be reused, clean containers must be used with a 3 micron filtering system available for refill.*

3. Disconnect supply hose (2, Figure 50-8) and steering filter hose (4) at steering/brake pump (1). Disconnect case drain hose (7) and unloader feedback hose (3) from the fittings on the pump housing. Plug all hoses and ports to prevent oil contamination.
4. If used, turn off vacuum pump kit (XB0887) after all hydraulic lines have been plugged.

## **⚠ WARNING**

**The steering pump weighs approximately 113 kg (250 lb). Use a suitable lifting device that has sufficient capacity to handle the load safely.**

5. Support steering/brake pump (1) and the rear section of hoist pump (9). Remove two capscrews (5) and spacers (6) from the pump mount bracket. Remove four pump mounting capscrews (8).
6. Move the steering/brake pump rearward to disengage the drive coupler splines from the hoist pump. Remove the steering/brake pump.
7. Clean the exterior of the pump. Move the pump to a clean work area for disassembly.

## Installation

1. Install a new O-ring on the pump mounting flange.
2. Ensure that the steering pump splined coupler is in place inside the hoist pump before steering pump installation.

### **WARNING**

***The steering pump weighs approximately 113 kg (250 lb). Use a suitable lifting device capable of handling the load safely.***

3. Move steering/brake pump (1, Figure 50-8) into position. Engage the steering/brake pump shaft with the hoist pump spline coupler.
4. Install two spacers (6) and capscrews (5). Do not tighten the capscrews at this time.
5. Align the capscrew holes and install four pump mounting capscrews (8). Tighten the capscrews to the standard torque. Now tighten capscrews (5) to the standard torque.
6. If the hydraulic tank is full of oil, install vacuum pump kit (XB0887) on the hydraulic tank to prevent the loss of oil when reconnecting hydraulic lines.
7. Remove all plugs from the hoses and ports. Install hoses (2), (3) and (4) on the steering/brake pump with new O-rings. Tighten the capscrews securely. Do not connect case drain hose (7) at this time (see Step 8).
8. Remove the case drain fitting from the top of the pump housing. Add clean C-4 hydraulic oil to the pump through the opening until the pump housing is full. This may require 2 to 3 liters (2 to 3 qt.) of oil.
9. Install the case drain fitting and tighten it to the standard torque. Connect case drain hose (7) to the fitting.
10. If used, turn off vacuum pump kit (XB0887) and remove the vacuum pump from the hydraulic tank.
11. Replace the hydraulic filter elements. Refer to Section L9, Hydraulic System Filters.

### **IMPORTANT**

***Use only Komatsu filter elements, or elements that meet the Komatsu hydraulic filtration specification of Beta 12 = 200.***

12. With the body down and the engine stopped, fill the hydraulic tank with clean C-4 hydraulic oil (as specified on the truck Lubrication Chart) to the upper sight glass level.
13. With suction line shutoff valve open, loosen the suction (inlet) hose capscrews at the pump to bleed any trapped air. Retighten the capscrews to the standard torque.

### **IMPORTANT**

***If trapped air is not bled from steering pump, possible pump damage and no output may result.***

14. If required, top off the oil in the hydraulic tank to the level of the upper sight glass.
15. In the hydraulic brake cabinet, open both brake accumulator needle valves completely to allow the steering pump to start under a reduced load.
16. Move the hoist control lever to the FLOAT position.
17. Start the truck engine and operate at low idle for two minutes.

### **WARNING**

***Do not allow the engine to run with the needle valves in the open position for longer than this recommendation. Excessive hydraulic system heating will occur.***

***DO NOT start any hydraulic pump for the first time after an oil change or pump replacement with the truck dump body raised. The oil level in the hydraulic tank may be below the level of the pump(s), causing extreme pump wear during this initial pump startup.***

18. Turn off the engine. Fully close both brake accumulator needle valves.
19. Verify that the oil level in the hydraulic tank is at the upper sight glass when the engine is off and the body is resting on the frame. If the hydraulic oil level is not at the upper sight glass, follow the instructions for filling/adding oil in Section L3, Hydraulic System Component Repair.
20. Start the engine and check for proper pump operation. If necessary, refer to Steering Circuit Checkout Procedure in Section L10, or the Troubleshooting Chart at the end of this section.

### **IMPORTANT**

*When disassembling or assembling the pump, choose a work area where no traces of dust, sand or other abrasive particles which could damage the unit are in the air. Do not work near welding, sand blasting, grinding benches, etc. Place all parts on a clean surface. To clean parts which have been disassembled, it is important that clean solvents are used. All tools and gauges should be clean before working with these units and new, clean, threadless rags should be used to handle and dry parts.*

1. Drain off excess hydraulic oil from the pump inlet and discharge ports. It may be necessary to loosen four valve plate capscrews (8, Figure 6-11) in order to pull back on the valve plate and allow fluid to seep out of the case.
2. Thoroughly clean and dry the outside surface of the pump housing.

*NOTE: Depending upon what part or parts are to be inspected, it may not be necessary to completely disassemble all components.*

#### Control Piston Group

3. Remove two large plugs (24, Figure 6-9) with O-rings (25) from both sides of the pump.
4. Control piston (11, Figure 6-10) must be in the "neutral" position. Control link pin (16) should be centered in the plug opening.
5. Use snap ring pliers to remove retaining rings (14) from both sides of the pin. Remove control link washers (15).
6. For handling purposes, insert a 1/4" - 20 UNC capscrew into the threaded end of control link pin (16).
7. Use a brass rod and hammer to tap on the end opposite the capscrew to remove control link pin (16).

### **IMPORTANT**

*Stroke adjuster assembly (items 12, 13, 14, 15 and 16, Figure 6-9) must be removed before further disassembly of the control piston.*

8. Without disturbing jam nut (16, Figure 6-9), unscrew gland (15) and remove the stroke adjuster as a complete assembly.
9. Back out capscrews (1, Figure 6-11), then remove cap (4).
10. Remove bias control spring (1, Figure 6-10).

#### Valve Plate Group

*NOTE: Valve plate (11, Figure 6-11) is a slight press fit in the pump housing.*

11. Support valve plate (11) from an overhead hoist. (Lifting lug holes are provided.) Remove four capscrews (8) from the valve plate.
12. Remove valve plate (11) from the pump housing by tapping it away from the housing with a mallet until the valve plate pilot diameter is disengaged from the case by 6 mm (0.25 in).
13. With the weight of the valve plate still suspended from the overhead hoist, slide the valve plate back until it disengages from the driveshaft. Set the valve plate aside. Be careful not to damage the wear face of the valve plate.
14. To further disassemble the control piston assembly, move control piston (11, Figure 6-10) into sleeve (4) until stop pin (8) contacts the sleeve. Use a large mallet to drive the piston and sleeve assembly outward from the valve plate.
15. When all sleeve seals (5), (6), and (7, Figure 6-10) are clear of the valve plate, re-extend the control piston. While tipping the assembly enough to clear the hole, pull the assembly from the valve plate.
16. Remove pin (18) from the control piston by pressing or tapping it out through the hole on the opposite side. Stop pin (8) can be removed and control piston (11) can be slipped out of sleeve (4).
17. Remove capscrews (12, Figure 6-12) and compensator block (10).
18. Remove capscrews (7, Figure 6-11) and cover plate (9).

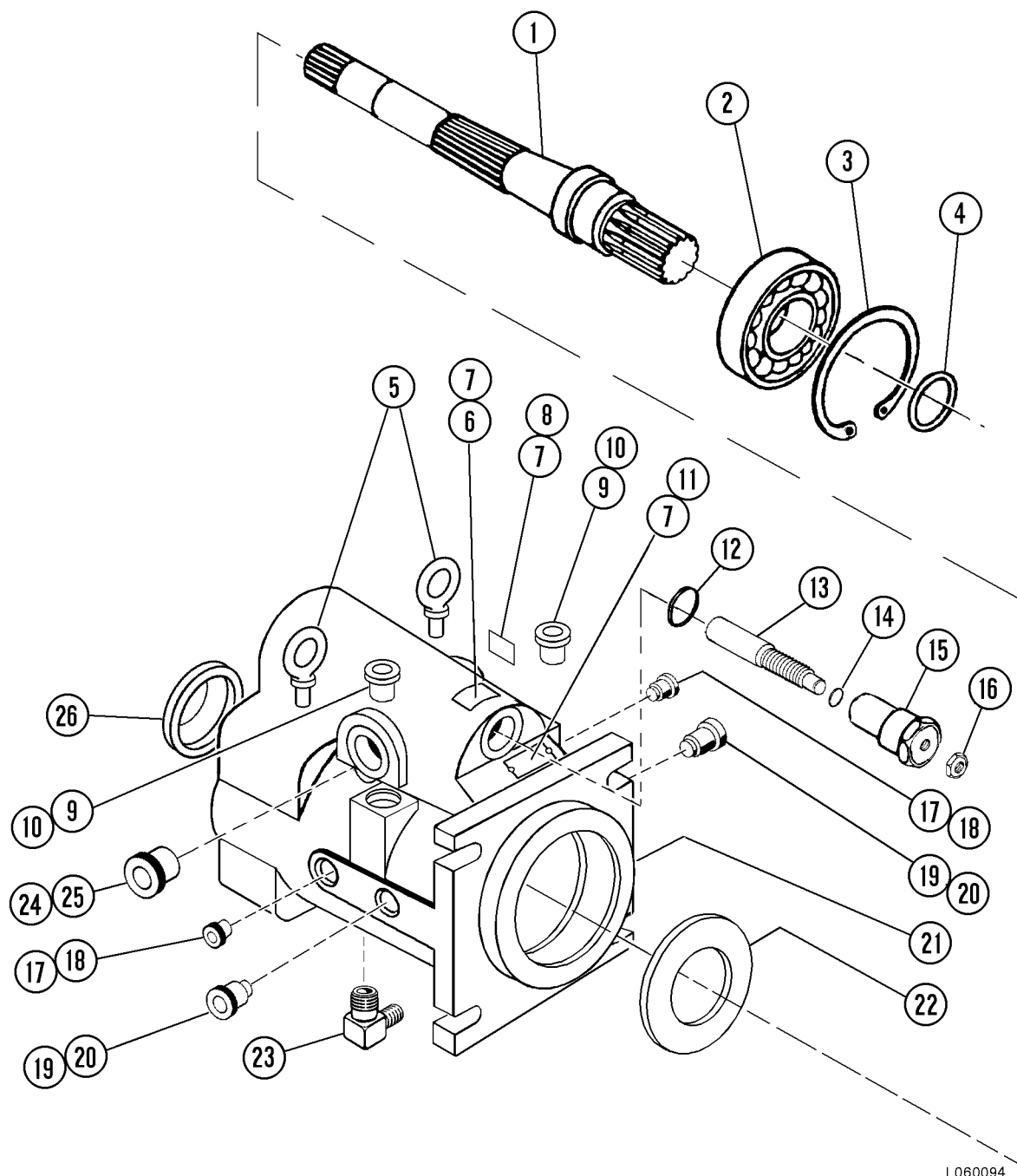
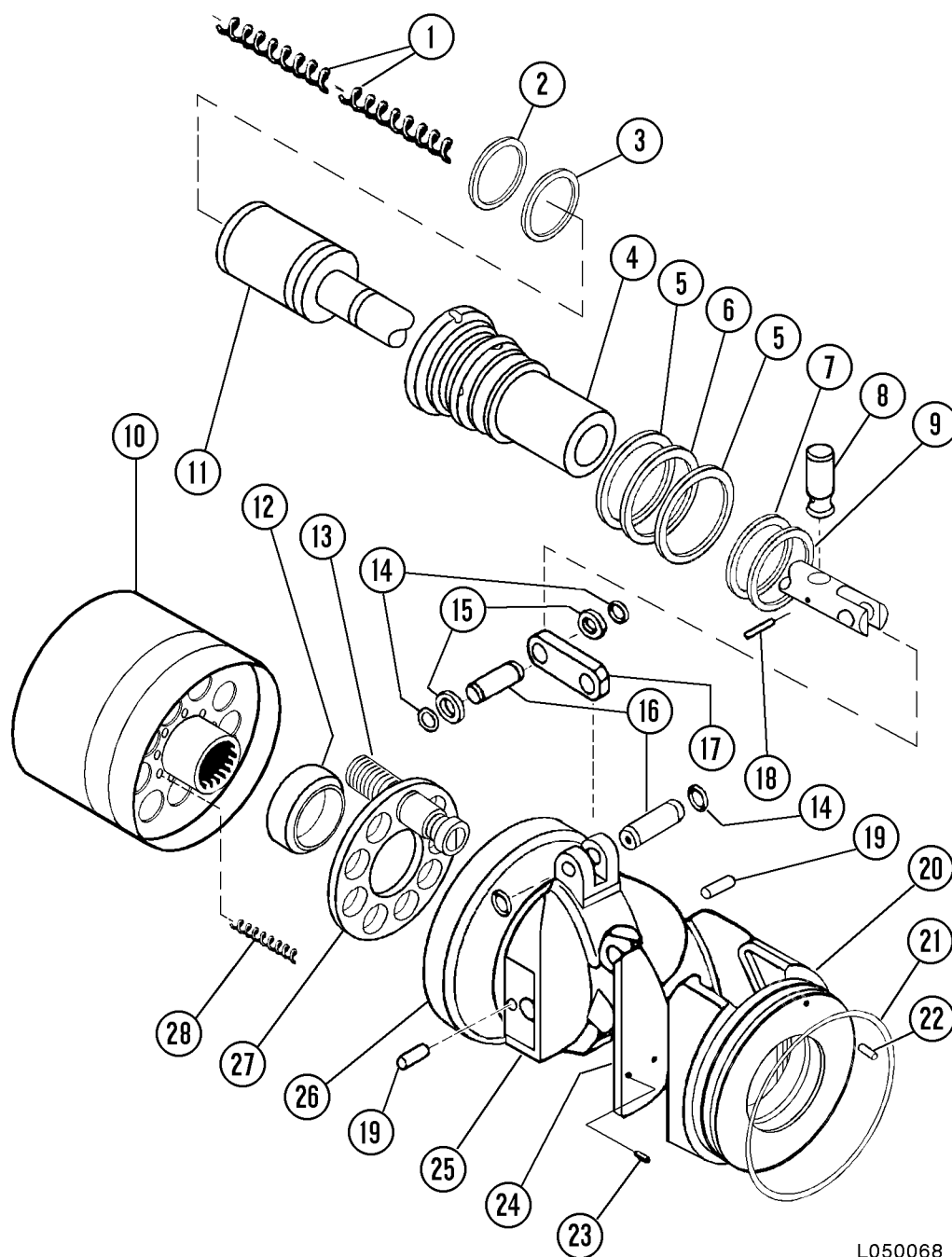


FIGURE 6-9. PUMP, FRONT HOUSING

- |                  |               |             |                   |
|------------------|---------------|-------------|-------------------|
| 1. Shaft         | 8. Name Plate | 15. Gland   | 22. Seal Retainer |
| 2. Bearing       | 9. Plug       | 16. Jam Nut | 23. Elbow Fitting |
| 3. Snap Ring     | 10. O-ring    | 17. Pin     | 24. Plug          |
| 4. Retainer Ring | 11. Plate     | 18. O-ring  | 25. O-ring        |
| 5. Lifting Eyes  | 12. O-ring    | 19. Pin     | 26. Seal          |
| 6. Name Plate    | 13. Stem      | 20. O-ring  |                   |
| 7. Drive Screw   | 14. O-ring    | 21. Housing |                   |





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FIGURE 6-10. PUMP, ROTATING GROUP

- |                |                          |               |                      |
|----------------|--------------------------|---------------|----------------------|
| 1. Spring      | 8. Stop Pin              | 15. Washer    | 22. Roll Pin         |
| 2. Seal        | 9. Backup Ring           | 16. Pin       | 23. Roll Pin         |
| 3. Piston Ring | 10. Cylinder Barrel      | 17. Link      | 24. Saddle Bearing   |
| 4. Sleeve      | 11. Control Piston       | 18. Pin       | 25. Swash Block      |
| 5. Backup Ring | 12. Ball                 | 19. Dowel Pin | 26. Cylinder Bearing |
| 6. O-ring      | 13. Piston Shoe Assembly | 20. Saddle    | 27. Retainer         |
| 7. O-ring      | 14. Retainer Ring        | 21. O-ring    | 28. Spring           |

## Rotating Group



***The rotating group weighs approximately 14 kg (30 lbs). Be careful not to damage the cylinder wear face or cylinder wear plate face, bearing diameters or piston shoes. Assistance from others and use of proper lifting techniques is strongly recommended to prevent personal injury as well.***

19. To remove the rotating group, firmly grasp cylinder barrel (10, Figure 6-10) and pull the assembly outward until the cylinder spline disengages from the driveshaft spline about 63.5 mm (2.5 in.). Then rotate the cylinder barrel one or two revolutions to break any contact between piston/shoe assemblies (13) and the wear face of swashblock (25).
20. Slide the rotating group off the driveshaft and out of the pump housing and place it on a clean, protective surface with piston shoes facing upward.
21. Mark each piston, its cylinder bore and location in shoe retainer for ease of inspection and assembly.
22. Piston/shoe assemblies can be removed individually or as a group by pulling upward on shoe retainer (27).
23. Remove fulcrum ball (12).
24. If shoe retainer springs (28) are removed, mark which spring came from which bore. Each spring must be returned to its particular bore upon assembly.
25. Remove two pins (17, Figure 6-9) and pull cylinder bearing (26, Figure 6-10) straight out of the pump housing.

## Swashblock Group

26. Remove two swashblock retaining pins (19, Figure 6-9). Tilt the bottom of swashblock (25, Figure 6-10) outward and remove the swashblock from the pump case.
27. Saddle bearings (24) can be removed by using a very short screwdriver or back hammer to pry them loose, or continue to the next step for further disassembly which will make their removal easier.

## Driveshaft Group

28. Remove bearing retaining ring (3, Figure 6-9). Use a mallet on the tail shaft to tap driveshaft (1) out from the front of the pump housing.
29. Remove seal retainer (22) from the pump housing. Use a mallet to tap saddle (20, Figure 6-10) out from the inside of the pump housing. Saddle bearings (24) can then be easily removed. Saddle O-ring (21) may also be removed at this time.

## Inspection



***Always wear safety goggles when using solvents or compressed air. Failure to wear safety goggles could result in serious personal injury.***

1. Clean all parts thoroughly.
2. Replace all seals and O-rings with new parts.
3. Check all locating pins for damage and all springs for cracking and signs of fatigue.

## Control Piston Group

4. Control piston (11, Figure 6-10) must slide smoothly in sleeve (4).
5. The linkage to the cradle should operate smoothly but not loosely (with slop). Check the piston and bore in the sleeve for signs of scratching and galling. Polish with a fine emery, if needed.

## Valve Plate Group

6. Closely examine the mating faces of valve plate (10, Figure 6-11) and cylinder barrel (10, Figure 6-10) for flatness, scratches and grooves. If the faces are not flat and smooth, the cylinder side will "lift off" from the valve plate, resulting in delivery loss and damage to the pump. Replace if necessary.

## Rotating Group

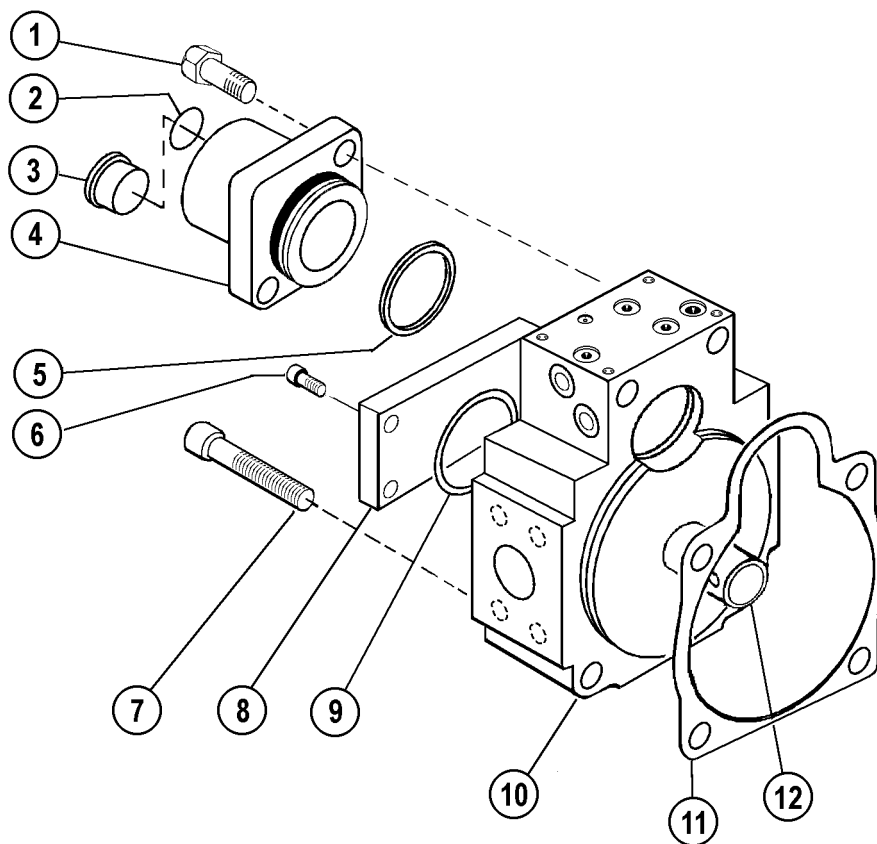
7. Check all pump piston assemblies (13, Figure 6-10) for smooth action in their bores.
8. Check the piston walls and bores for scratches or other signs of excessive wear. (The pistons should not have more than a few thousandths of an inch of clearance). Replace if necessary.
9. The piston shoes must pivot smoothly, but the end play must not exceed 0.076 mm (0.003 in.).

Check end play as follows:

- a. Place the square end of the piston on a bench and hold it down firmly. Pull on the end of the shoe with your other hand and note the end play.

A good piston/shoe fit will have no end play, but the shoe may rotate and pivot on the piston ball. Inspect each shoe face for nicks and scratches.

- b. Measure the shoe thickness between retainer (27, Figure 6-10) and the cradle. All shoes must have equal thickness within 0.003 mm (0.0001 in.). If even one piston/shoe assembly (13) is out of specification, all piston/shoe assemblies must be replaced.
- c. Inspect cylinder bearing (26) and the matching cylinder barrel bearing mating surface for galling, pitting and roughness. Replace if necessary.



L060078

FIGURE 6-11. PUMP, REAR HOUSING

1. Capscrew
2. O-ring
3. Plug
4. End Cap

5. Seal (red)
6. Capscrew
7. Capscrew
8. Cover Plate

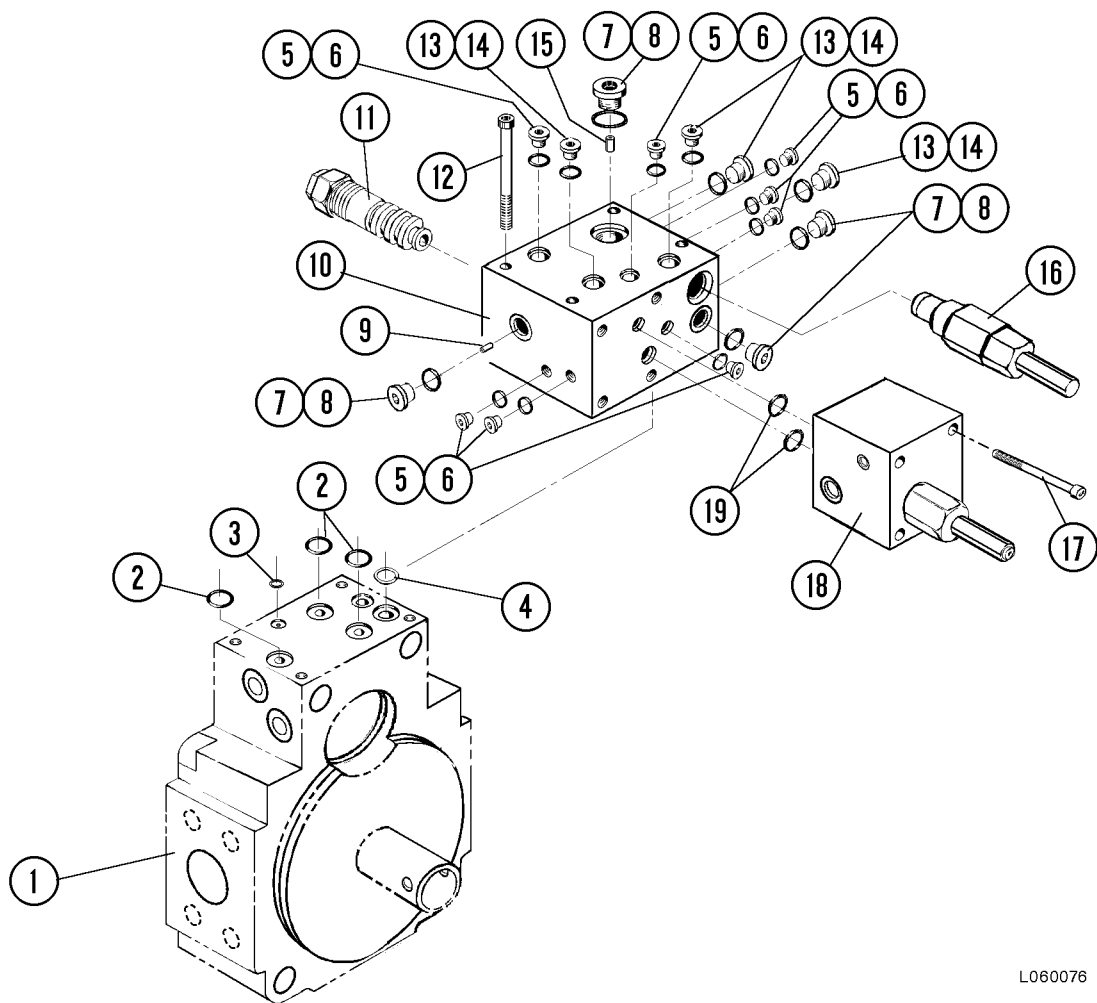
9. O-ring
10. Valve Plate
11. Gasket
12. Bearing

## Swashblock Group

10. Inspect swashblock (25, Figure 6-10) for scratches, grooves, cracks and uneven surface. Replace if necessary.

*NOTE: The wear face is coated with a gray colored, epoxy-based, dry film lubricant for break-in purposes. Scratching or wearing of this coating is not detrimental as long as the metal surface underneath the coating is not scored or "picked-up".*

11. Compare saddle bearing (24) thickness in the wear area to the thickness in a non-wear area. Replace the saddle bearings if the difference is greater than 0.102 mm (0.004 in.).
12. Check the mating surface of the swashblock for cracks and excessive wear. Replace if necessary.
13. Swashblock movement in the saddle and saddle bearing must be smooth.



L060076

FIGURE 6-12. UNLOADER & COMPENSATOR CONTROLS

- |                |                               |                         |
|----------------|-------------------------------|-------------------------|
| 1. Valve Plate | 8. O-Ring                     | 15. Orifice (0.032 in.) |
| 2. O-Ring      | 9. Orifice (0.062 in.)        | 16. Compensator         |
| 3. O-Ring      | 10. Compensator Control Block | 17. Screw               |
| 4. O-Ring      | 11. 4-Way Valve               | 18. Unloader Module     |
| 5. Plug        | 12. Screw                     | 19. O-Ring              |
| 6. O-Ring      | 13. Plug                      |                         |
| 7. Plug        | 14. O-Ring                    |                         |

### Driveshaft Group

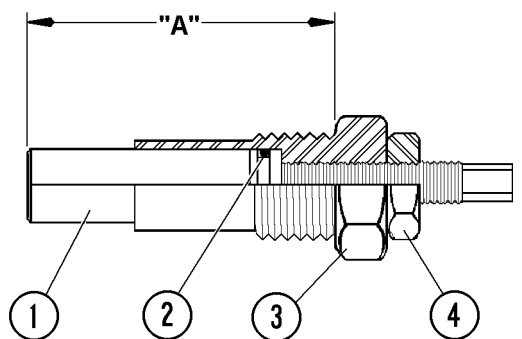
14. Remove shaft seal (26, Figure 6-9).
15. Check shaft bearing (2) for galling, pitting, binding and roughness. Replace if necessary.
16. Check the shaft and its splines for wear. Replace parts as necessary.

### Compensator Block and Unloader Module

17. Remove screws (17, Figure 6-12) and separate unloader module (18) from compensator block (10).
18. Remove 4-way valve (11) and compensator (16) from the compensator block. Remove all plugs and orifices (9) and (15). Clean the block in solvent and inspect all passages and orifices for obstructions.
19. Remove the valve from unloader module (18). Remove the plugs and clean the block passages. If the valve is inoperative, replace entire unloader module.

### Stroke Adjuster Assembly

20. Measure and record dimension "A" of the stroke adjuster assembly as shown in Figure 6-13.
21. Loosen jam nut (4). Separate stem (1) from gland (3). Remove and discard O-ring (2).
22. Inspect the parts for damage and excessive wear.
23. Install new O-ring (2) on stem (1) and reassemble it to gland (3). Adjust the stem-to-gland distance to dimension "A" recorded in Step 21. Tighten jam nut (4).



L060096

FIGURE 6-13. STROKE ADJUSTER ASSEMBLY

- |           |            |
|-----------|------------|
| 1. Stem   | 3. Gland   |
| 2. O-ring | 4. Jam Nut |

### Assembly

**NOTE:** The procedures for assembling the pump are basically the reverse order of the disassembly procedures.

1. During assembly, install new gaskets, seals, and O-rings.
2. Apply a thin film of clean grease or hydraulic oil to the sealing components to ease the assembly procedure. If a new rotating group is used, lubricate it thoroughly with clean hydraulic oil. Apply oil generously to all wear surfaces.

### Swashblock Group

3. Press or tap roll pin (22, Figure 6-10) into pump housing (18, Figure 6-9).
4. Use an arbor press to press new shaft seal (26) into saddle (20, Figure 6-10). Install O-ring (21) into the groove in the saddle.
5. Press four roll pins (23) into saddle (20) until they bottom. Then press saddle bearing (24) onto the pins to locate the bearing in the saddle.

## ! IMPORTANT !

**Be careful not to damage the saddle bearing surfaces while installing the saddle into the pump housing.**

6. Use a long brass bar and a mallet (or an arbor press) to install the saddle and bearing assembly into the pump housing. Tap or press only on the area of the saddle that is exposed between the saddle bearings. **Do not tap on the bearing surfaces.** The saddle is fully seated when a distinct metallic sound is heard when installing the saddle into the pump housing.
7. Fasten control link (17, Figure 6-10) to swashblock (25) using link pin (16) and two retaining rings (14).
8. Ensure that both dowel pins (19) are pressed into swashblock (25).
9. Insert the swashblock into the pump housing until it engages in the saddle bearing. Allow the swashblock to settle to its lowest natural position.
10. Retain the swashblock by installing two retaining pins (19, Figure 6-9) and O-rings (20). Once pinned, ensure that the swashblock strokes smoothly in the saddle by pulling firmly on the free end of the control link.

## Driveshaft Group

**NOTE:** Ensure that the punch marks on cylinder bearing (26, Figure 6-10) will face toward the shaft end of the pump.

11. Insert cylinder bearing (26) straight into the pump housing. Ensure that the bearing is positioned so that bearing retainer pins (17, Figure 6-9) can be inserted in the case and into the bearing.
12. Install O-rings (18) on pins (17). Install the pins.
13. An arbor press is required to install shaft bearing (2, Figure 6-9) onto driveshaft (1). Press only on the inner race of the bearing. Press the bearing until it contacts the shoulder on the driveshaft.
14. Use a long - 153 mm (6 in.) - sleeve with an inside diameter that is slightly larger than the retaining ring inside diameter to press retaining ring (4) toward the bearing until it seats in the groove.
15. Place seal retainer (22) over seal (26) inside pump housing (21). Lubricate the shaft seal with clean hydraulic oil.
16. Install the entire driveshaft assembly through the front of the pump housing. A mallet will be required to install the driveshaft through the shaft seal.
17. Once the driveshaft assembly is fully seated in the pump housing, install snap ring (3).

## Rotating Group

18. The mating surfaces should be greased. Place the cylinder assembly on a clean work bench with the valve plate side down.
19. Assemble the rotating group by inserting shoe retainer springs (28, Figure 6-10) into the same spring bores in cylinder barrel (10) that they came out of.
20. Slide fulcrum ball (12) over the nose of cylinder barrel (10).
21. Place shoe retainer (27) over the fulcrum ball and align the holes in the retainer with the corresponding holes (marked during disassembly) in the cylinder barrel. Once aligned, insert piston/shoe assemblies (13) into their corresponding holes.

## WARNING

**The assembled rotating group weighs approximately 14 kg (30 lb). Assistance from others and use of proper lifting techniques is strongly recommended to prevent personal injury.**

22. The rotating group can now be carefully installed over the end of the driveshaft and into the pump housing.
23. When installing the rotating group, support the weight of cylinder barrel (10, Figure 6-10) as the cylinder spline is passed over the end of the driveshaft to avoid scratching or damage.
24. Push the cylinder barrel forward until the cylinder spline reaches the driveshaft spline. Rotate the cylinder slightly to engage the shaft splines.
25. Continue to slide cylinder barrel forward until it encounters the cylinder bearing (26). Lifting the driveshaft slightly helps cylinder barrel and cylinder bearing engagement. Continue pushing cylinder forward until the piston shoes contact swashblock (25).
26. At this point, the back of the cylinder barrel should be located approximately 6 mm (0.25 in.) inside the back of the pump housing.

## Control Piston Group

27. Install seal (2, Figure 6-10) and piston ring (3) into their respective grooves on control piston (11).
28. Insert the control piston assembly into sleeve (4).
29. While supporting the control piston, press or slip in pin (8) and secure it with cotter or roll pin (18).

**NOTE:** The order of piston sleeve seal installation starts at the widest end of the sleeve.

30. Install backup ring (1, Figure 6-14), O-ring (2) and back-up ring (3) in rearmost groove on the piston sleeve. Install O-ring (4) and backup ring (5) in the remaining groove.

**NOTE:** Ensure that the grooves in sleeve (4, Figure 6-10) are at the 12 o'clock and 6 o'clock positions when inserted into the valve plate.

31. Insert the piston and sleeve assembly into valve plate (10, Figure 6-11).

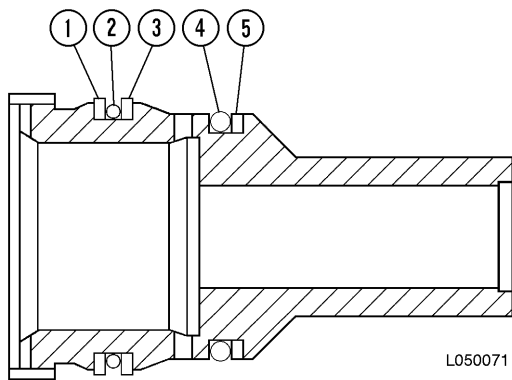


FIGURE 6-14. O-RING LOCATION ON PISTON SLEEVE

- |                |                |
|----------------|----------------|
| 1. Backup Ring | 4. O-ring      |
| 2. O-ring      | 5. Backup Ring |
| 3. Backup Ring |                |

32. Install O-ring (10) in the rear of the valve plate. Use four capscrews (7) to fasten cover plate (9) over the opening in the valve plate.
33. Pull the free end of control link (17, Figure 6-10) toward the rear of the pump housing until the open hole in the link lines up with the open ports on the sides of the pump case.
34. Install stroke adjuster assembly (Figure 6-13) to hold the swashblock in place.

#### Valve Plate Group

35. Ensure that driveshaft bearing (12, Figure 6-11) is in place. Using assembly grease to hold the desired position, place valve plate gasket (12) in position on valve plate (11).
36. Support the valve plate assembly from an overhead hoist (lifting lug holes are provided) in preparation for mating to the pump housing.
37. Assemble one control link retainer ring (14, Figure 6-10) and one control link washer (15) onto the threaded hole side of control link pin (16). Then thread a 1/4" - 20NC capscrew into the pin to ease holding.
38. Carefully maneuver the valve plate assembly (supported by overhead hoist) over the driveshaft and into the pump housing so that the slot on control piston (11) engages control link (17).
39. With the hole in the control piston lined up with the hole in the link, carefully insert control link pin (16).

## ! IMPORTANT !

*During this next step, be careful to prevent the washer and retaining ring from falling into the pump housing.*

40. Install second control link washer (15) and control link retaining ring (14) onto the pin. Remove the capscrew from the pin.

*NOTE: The valve plate is a slight press fit into the pump housing. Ensure that the pilot diameter on the valve plate is aligned with the mating diameter on the pump housing before assembly.*

41. Insert four capscrews (7, Figure 6-11) and alternately tighten them until the valve plate is drawn up to the pump housing. Tighten the capscrews evenly to **330 N·m (244 ft lb)**.
42. Clean end cap (4) and ensure the seal groove is clean and free of any flaws.
43. Lightly lubricate new seal (5) with clean hydraulic oil and install the seal onto the end cap. Refer to Figure 6-14 for the correct orientation of the seal. **DO NOT** use any tools to install the seal. Damage to the seal may occur and oil leakage may result.

*NOTE: The previous seal was an O-ring with a back-up ring. The current seal is a lip seal (red) without a back-up ring.*

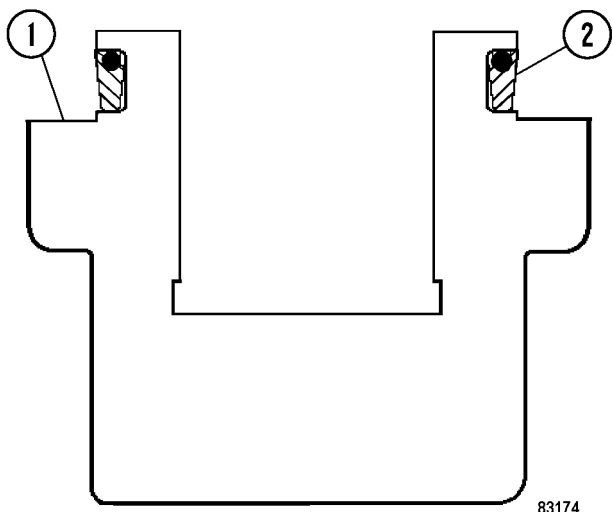


FIGURE 6-15. SEAL INSTALLATION

- |            |         |
|------------|---------|
| 1. End Cap | 2. Seal |
|------------|---------|

44. Insert bias control piston springs (1, Figure 6-10) into control piston (11).
45. Lightly lubricate the piston outer seal with clean hydraulic oil. Position the seal ring (2, Figure 5-16) gap on top so that it is visible.
46. Place end cap (1) over the end of control piston (3). Beginning at the seal ring edge opposite the gap, use end cap movement and your fingers to compress and push seal ring (2) into the end cap. Slip the end cap over the control piston until spring pressure is felt. Do not use excessive force. The piston seal ring is easily broken.

*NOTE: A small amount of vaseline may be used to hold the gap above its groove and aid this step. Apply the vaseline 180° from the gap on the hidden (far) side of the seal ring.*

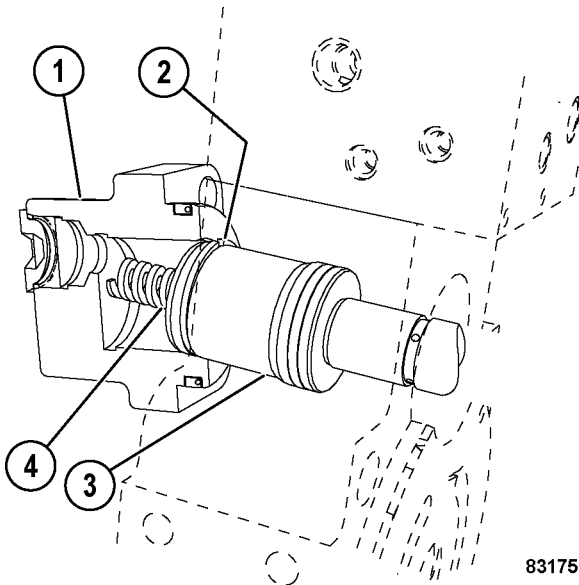


FIGURE 6-16. SEAL INSTALLATION

- |              |                   |
|--------------|-------------------|
| 1. End Cap   | 3. Control Piston |
| 2. Seal Ring | 4. Spring         |

47. Install two cap screws (1, Figure 6-11) in diagonally opposite holes of the end cap. Progressively tighten the cap screws until the end cap flange is seated against the pump housing. Spring pressure will increase as the cap screws are tightened. Install the remaining cap screws. Progressively tighten all cap screws to **187 N·m (138 ft lb)**.
48. Install cover plate (8) with new O-ring (9) and capscrews (6).
49. Install O-rings (2), (3) and (4, Figure 6-12) in their proper locations on the top of the valve plate. Install block (10) to the valve plate with socket head capscrews (12). Tighten the capscrews to **21 N·m (183 in. lb)**.
50. Install 4-way valve (11) and compensator valve (16).
51. Install orifice (9) and plug (7) with new O-ring (8) in the side of the compensator block as shown in Figure 6-12. Install orifice (15), plug (7) and O-ring (8) in the top of the block.
52. Install the remaining plugs with new O-rings.
53. Install unloader module (18) on the compensator block with new O-rings (19) and socket head capscrews (17). Tighten the capscrews to **10 N·m (88 in. lb)**.
54. Install plugs (9) and (24, Figure 6-9) and O-rings (10) and (25) in the pump housing.
55. Measure the pump rotation torque. The rotation torque should be approximately **20 N·m (15 ft lb)**.



## STEERING ACCUMULATORS

### Removal

1. Ensure that the key switch has been OFF for at least 90 seconds to allow the accumulator oil to drain back to tank. Turn the steering wheel to ensure that all pressurized oil is released.
2. Remove cover (1, Figure 6-15). Disconnect the electrical wiring from nitrogen precharge pressure switch (10).
3. Remove the charging valve guard and loosen the small hex on charging valve (9) three complete turns. Depress the valve core until all nitrogen pressure has been relieved.

### ⚠ WARNING

**Ensure that only the small swivel hex nut turns. Turning the complete charging valve assembly may result in the valve assembly being forced out of the accumulator by the nitrogen pressure inside. Wear a protective face mask when discharging nitrogen gas.**

4. Remove oil line (8) from the bottom of the accumulator that is being serviced. Plug the hoses and cover the opening in the accumulator to prevent possible contamination of the system. **Do not use a screw-in type plug.**

### ⚠ WARNING

**The accumulator weighs approximately 172 kg (380 lb). Use a suitable lifting device with adequate capacity to handle the load safely.**

5. Attach a lifting device to the accumulator.
6. Loosen the capscrews and remove mounting clamp (4).
7. Raise the accumulator until it is clear of mounting bracket (7). Move the accumulator to a clean work area for disassembly.

### Installation

1. Lift the accumulator into position it on mounting bracket (7, Figure 6-15). The accumulator should be positioned to allow access to charging valve (9).

2. Secure the accumulator to mounting bracket (7) using mounting clamps (4), capscrews, lockwashers and nuts. Tighten the capscrews to the standard torque.
3. Connect the electrical wiring to nitrogen precharge pressure switch (10). Reconnect oil line (8) to the bottom of the accumulator.
4. Precharge both accumulators with pure dry nitrogen. Refer to the Steering Accumulator Charging Procedure in this section.
5. Check charging valve (9), pre-charge switch (10) and check valve assembly (11) for leaks using a soap solution.

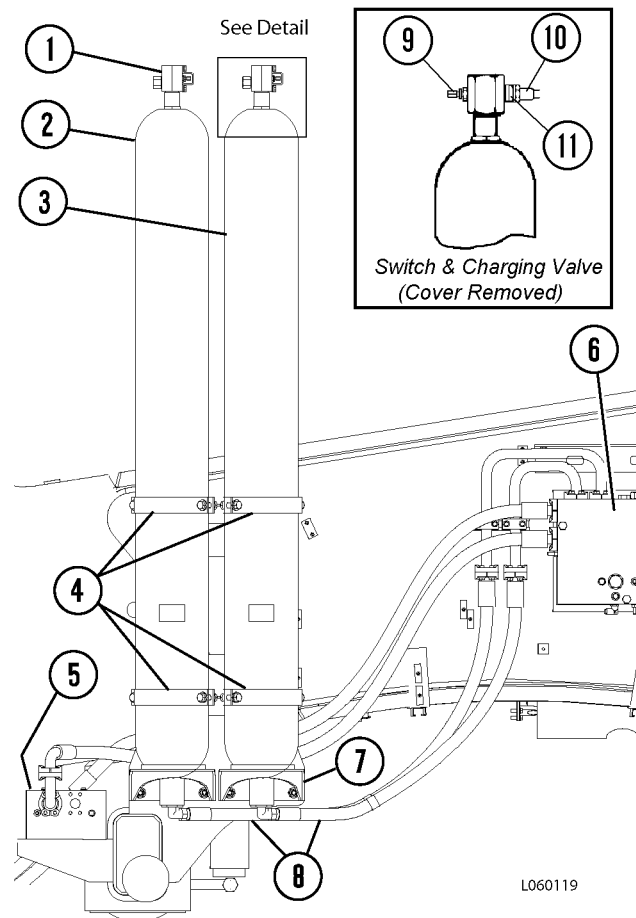


FIGURE 6-17. STEERING ACCUMULATORS

- |                               |                                        |
|-------------------------------|----------------------------------------|
| 1. Cover                      | 6. Bleeddown Manifold Valve            |
| 2. Front Steering Accumulator | 7. Mounting Bracket                    |
| 3. Rear Steering Accumulator  | 8. Oil Line                            |
| 4. Mounting Clamp             | 9. Charging Valve                      |
| 5. Flow Amplifier Valve       | 10. Nitrogen Precharge Pressure Switch |
|                               | 11. Check Valve                        |

## Disassembly

1. After the accumulator has been removed from the equipment, the accumulator body should be secured in a vise, preferably a chain vise. If a standard jaw vise is used, brass inserts should be used to protect the hydraulic port assembly from damage. Clamp on wrench flats only when using a jaw vise to prevent the accumulator from turning.
2. Remove bleed plug (12, Figure 6-17) on the hydraulic port assembly. Use a spanner wrench to remove locking ring (10) from the hydraulic port assembly. Use an adjustable wrench on the flats located on the port assembly to prevent the port assembly from rotating.
3. Remove spacer (9), then push the hydraulic port assembly into the shell.
4. Insert your hand into the accumulator shell and remove O-ring backup (8), O-ring (7), and metal backup washer (6). Separate anti-extrusion ring (3) from the hydraulic port. Fold the anti-extrusion ring and remove it from the shell (Figure 6-15).
5. Remove the hydraulic port from the accumulator shell.
6. To prevent the bladder valve stem from twisting, secure it with an appropriate wrench applied to the valve stem flats. Remove gas valve manifold (14, Figure 6-17). Then remove nut (5) while still holding the bladder valve stem.
7. Fold the bladder and pull it out of the accumulator shell. A slight twisting motion while pulling on the bladder reduces the effort required (Figure 6-16). If the bladder is slippery, hold it with a cloth.

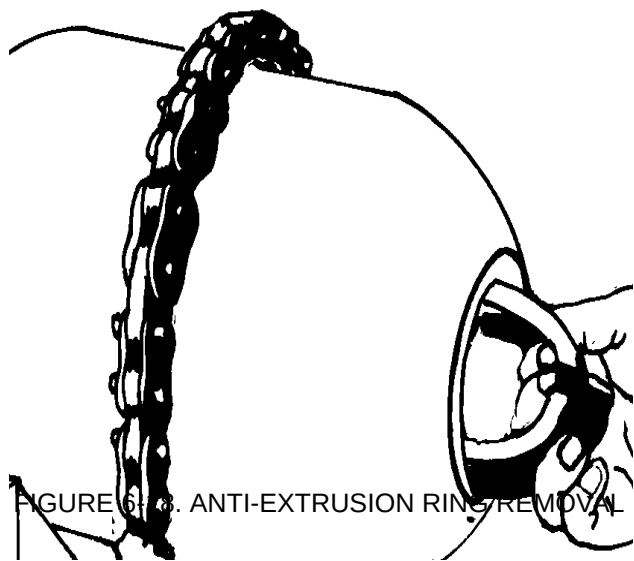


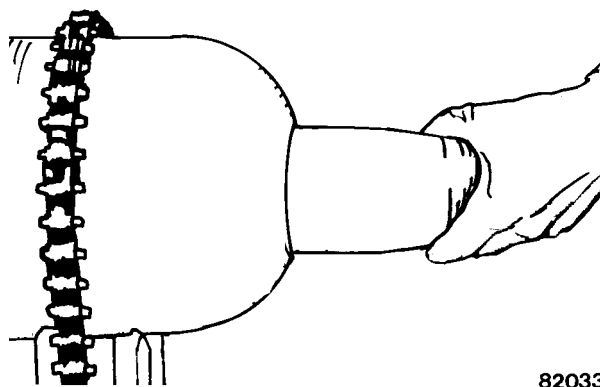
FIGURE 6-18. ANTI-EXTRUSION RING REMOVAL

## Cleaning and Inspection

1. Clean all metal parts with a cleaning agent.
2. All seals and soft parts should be wiped clean.
3. Inflate the bladder to normal size. Wash the bladder with a soap solution. If the bladder causes bubbles in the soap solution, discard the bladder. After testing, deflate the bladder immediately.
4. Inspect the hydraulic port assembly for damage. Check the poppet plunger to see that it spins freely and functions properly.
5. Check the anti-extrusion ring and soft seals for damage and wear. Replace all worn or damaged seals with original equipment seals.
6. After the shell has been cleaned with a cleaning agent, check the inside and outside of the shell. Special attention should be given to the area where the gas valve and hydraulic assembly pass through the shell. Any nicks or damage in this area could destroy the accumulator bladder or damage new seals. If this area is pitted, consult your Komatsu Service Manager.

## **⚠ DANGER**

***DO NOT repair the housing by welding, machining or plating to salvage a worn area. These procedures may weaken the housing and result in serious injury to personnel when pressurized.***



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FIGURE 6-19. BLADDER REMOVAL

## Assembly

**NOTE:** Assemble the accumulator(s) in a dust and lint free area. Maintain complete cleanliness during assembly to prevent possible contamination.

Use only nitrogen that meets or exceeds CGA (Compressed Gas Association) specification G-10.1 for type 1, grade F. The nitrogen should be 99.9% pure. Use only nitrogen cylinders with standard CGA number 580 connections with the appropriate high pressure regulator.

1. After shell (4, Figure 6-18) has been cleaned and inspected, place the accumulator shell in a vise or on a table.
2. Thoroughly spray the inside of the accumulator shell with 3.1 L (106 oz.) of clean C-4 hydraulic oil to lubricate and cushion the bladder. Ensure that the entire internal surface of the shell is lubricated.
3. With all gas completely exhausted from the bladder, collapse the bladder and fold it longitudinally in a compact roll.
4. Install the gas valve on the bladder.
5. Insert the bladder pull rod into the valve stem opening and out through the shell fluid port. Attach the bladder pull rod to the bladder valve stem.
6. With one hand, pull the bladder pull rod while feeding the bladder into the shell with the other hand. Slight twisting of the bladder will assist in this insertion.
7. Once the bladder valve stem has been pulled through the valve stem opening in the shell, install valve stem nut (5) by hand. Remove the bladder pull rod.
8. Use a wrench to hold the bladder valve stem on the flats, then tighten nut (5) securely.
9. If removed, install pressure switch (15) into valve assembly (11). Tighten the switch to **41 ± 5 N·m (30 ± 4 ft lb)**. Then install valve assembly (11) onto gas valve manifold (14) and tighten it to **18 ± 4 N·m (13 ± 3 ft lb)**.

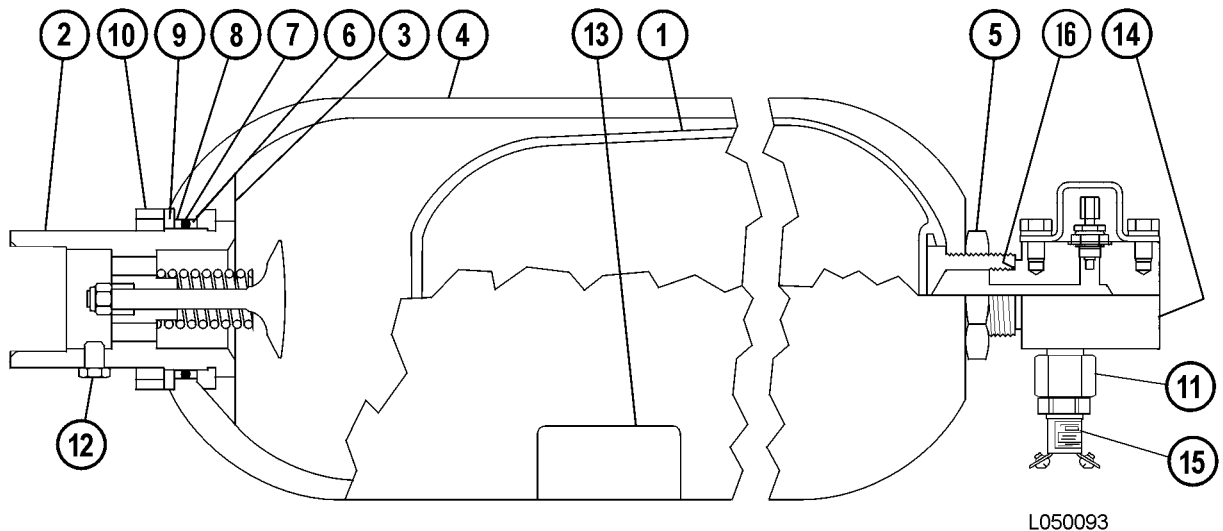


FIGURE 6-20. ACCUMULATOR ASSEMBLY

- |                            |                    |                        |
|----------------------------|--------------------|------------------------|
| 1. Bladder Assembly        | 7. O-ring          | 13. Warning Plate      |
| 2. Hydraulic Port Assembly | 8. O-ring Backup   | 14. Gas Valve Manifold |
| 3. Anti-Extrusion Ring     | 9. Spacer          | 15. Pressure Switch    |
| 4. Shell                   | 10. Locking Ring   | 16. O-ring             |
| 5. Nut                     | 11. Valve Assembly |                        |
| 6. Metal Backup Washer     | 12. Bleed Plug     |                        |

10. If removed, install the charging valve onto gas manifold (14). Then tighten small hex nut (4, Figure 6-19) to **5 N·m (45 in. lb)**.

If a new charging valve was installed, the valve stem must be seated as follows:

- a. Tighten small hex swivel nut (4, Figure 6-19) to **14 N·m (10 ft lb)**.
- b. Loosen the small hex swivel nut.
- c. Retighten the small hex swivel nut to **14 N·m (10 ft lb)**.
- d. Loosen the small hex swivel nut again.
- e. Finally, tighten small hex swivel nut to **5 N·m (45 in. lb)**.
11. Install charging valve cap (1) finger-tight. Install the charging valve cover,. Tighten the capscrews to **40 N·m (25 ft lb)**.
12. Install new O-ring (16, Figure 6-17) on gas valve manifold (14). Hold the bladder valve stem with a wrench and install gas valve manifold (14) securely.
13. While holding hydraulic port assembly (2) by the threaded end, insert the poppet end into the shell fluid port. Lay the complete assembly inside the shell.
14. Fold anti-extrusion ring (3) to enable insertion into the shell. Once the anti-extrusion ring has cleared the fluid port opening, place the anti-extrusion ring on the hydraulic port assembly with the steel collar facing toward the shell fluid port.
15. Pull the threaded end of the port assembly through the shell fluid port until it seats solidly into position on the shell fluid port opening.
16. Connect the nitrogen charging kit to the charging valve. With the hydraulic port assembly firmly in place, slowly pressurize the bladder using dry nitrogen and a sufficient pressure of approximately 275 - 345 kPa (40 - 50 psi) to hold the port assembly in place so that both of your hands are free to continue with the assembly.
17. Install metal O-ring backup washer (6) over the hydraulic port assembly. Push it into the shell fluid port to bottom it out on the anti-extrusion ring.
18. Install O-ring (7) over the hydraulic port assembly. Push it into the shell fluid port until it bottoms out against metal O-ring backup washer (6). Ensure that the O-ring does not twist.

19. Install O-ring back-up (8) over the hydraulic port assembly. Push it into the shell fluid port until it bottoms out against the O-ring.
20. Insert spacer (9) with the smaller diameter of the shoulder facing the accumulator shell.
21. Install locking ring (10) on the hydraulic port assembly and tighten it securely. This will squeeze the O-ring into position. Use an appropriate wrench on the flats on the port assembly to ensure that the unit does not turn.
22. Install bleed plug (12) into the hydraulic port assembly.
23. Verify that all warning labels are attached to the shell and that they are legible. Install new labels as required.
24. Precharge the accumulator to 690 - 827 kPa (100 - 120 psi). Refer to Steering Accumulator Charging Procedure.
25. After precharging is completed, install a plastic cover over the hydraulic port to prevent contamination. **Do not use a screw-in type plug.**

## **WARNING**

***Always store bladder accumulators with 690 - 827 kPa (100 - 120 psi) nitrogen precharge pressure. This amount of pressure fully expands the bladder and holds oil against the inner walls for lubrication and rust prevention. Do not exceed 827 kPa (120 psi). Storing accumulators with too much pressure is not safe due to possible leakage.***

***Refer to Accumulator Storage Procedures in this section for more information about properly storing bladder accumulators.***

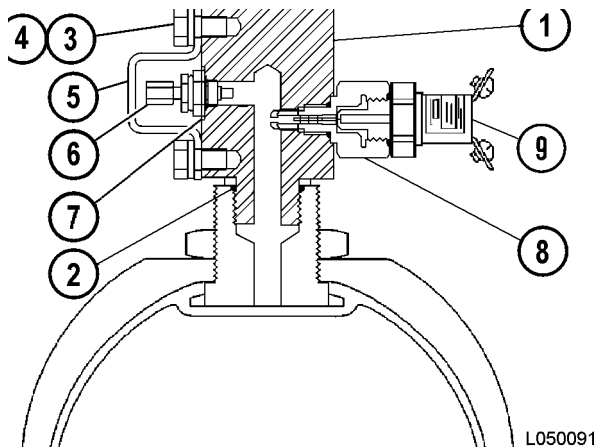


FIGURE 6-21. GAS END COMPONENTS

- |                       |                            |
|-----------------------|----------------------------|
| 1. Gas Manifold Valve | 6. Cap                     |
| 2. O-ring             | 7. Flat Gasket             |
| 3. Capscrew           | 8. Charging Valve Assembly |
| 4. Lockwasher         | 9. Pressure Switch         |
| 5. Guard              |                            |

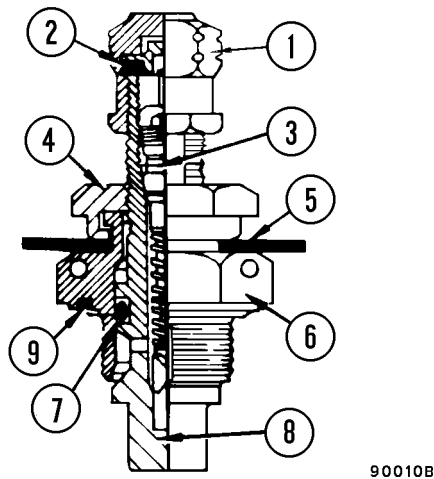


FIGURE 6-22. CHARGING VALVE

- |                  |               |
|------------------|---------------|
| 1. Valve Cap     | 6. Valve Body |
| 2. Seal          | 7. O-ring     |
| 3. Valve Core    | 8. Valve Stem |
| 4. Swivel Nut    | 9. O-ring     |
| 5. Rubber Washer |               |

## Leak Testing

Leak testing entails checking for internal and external leaks at high pressure. A source of 24 130 kPa (3,500 psi) hydraulic pressure and nitrogen pressure of 9 653 kPa (1,400 psi) will be required. A small water tank will be necessary for a portion of the test.

1. Remove charging valve guard (5, Figure 6-18) and charging valve cap (6).
2. Close the bleed valve.
3. Attach a gauging assembly to the charging valve and tighten the swivel nut finger-tight.
4. Hold valve body (6, Figure 6-19) with one wrench while unscrewing swivel nut (4) three complete turns with a second wrench. This will open the poppet inside the gas valve.
5. Slowly charge the accumulator to 690 kPa (100 psi). After 690 kPa (100 psi) is obtained, the charging rate can be increased until the accumulator is fully charged at 9 653 kPa (1,400 psi).
6. Tighten swivel nut (4) finger-tight to close the internal poppet.
7. Submerge the accumulator assembly under water and observe it for 20 minutes. No leakage (bubbles) is permitted. If leakage is present, proceed to Step 14.
8. Hold charging valve (6) with a wrench and remove the swivel nut assembly.
9. Install valve cap (1) on the charging valve finger-tight and install the gas valve guard.
10. Connect a hydraulic power supply to the oil port on the accumulator. Ensure that bleed plug (12, Figure 6-18) is installed.
11. Pressurize the accumulator with oil to 24 130 kPa (3,500 psi). This may take 23 to 30 liters (6 to 8 gallons) of oil. No external oil leakage is permitted.
12. Slowly relieve oil pressure and remove the hydraulic power supply. Install a plastic cover over the hydraulic port to prevent contamination.
13. If any gas or oil leakage was present, discharge all nitrogen gas using the charging equipment and repair as necessary. If there were no leaks of any kind, use the nitrogen charging equipment and adjust the nitrogen precharge pressure to 690 - 827 kPa (100 - 120 psi).
14. Verify that all warning and caution labels are attached and legible. Install new labels as required.

## Charging Procedure



***Do not loosen or disconnect any hydraulic line or component until the engine is stopped and the key switch has been OFF for at least 90 seconds.***

***Pure dry nitrogen is the only gas approved for use in the steering accumulator. The accidental charging of oxygen or any other gas in this compartment may cause an explosion. Ensure that pure dry nitrogen gas is being used to charge the accumulator.***

***Before charging or discharging nitrogen gas in the accumulator, carefully read and understand the warning labels and instructions regarding the charging valve.***

***Only precharge accumulators to 9653 kPa (1400 psi) while installed on the truck. Never handle an accumulator with lifting equipment if the nitrogen precharge pressure is over 827 kPa (120 psi). Always set the precharge to 690 - 827 kPa (100 - 120 psi) before removing or installing the accumulators.***

***Ensure that the nitrogen supply is shut off before attaching the charging manifold to the nitrogen container.***

***NOTE: If one accumulator is low on nitrogen, it is recommended that both accumulators be checked and charged at the same time. Correct precharge pressure is the most important factor in prolonging accumulator life.***

***NOTE: Use only nitrogen that meets or exceeds CGA (Compressed Gas Association) specification G-10.1 for type 1, grade F. The nitrogen should be 99.9% pure. Use only nitrogen cylinders with standard CGA number 580 connections with the appropriate high pressure regulator.***

1. With the engine off and key switch in the OFF position, allow at least 90 seconds for the accumulator to bleed down. Turn the steering wheel to ensure that no pressurized oil remains in the accumulators.
2. Open the needle valves in the brake cabinet to release hydraulic pressure from the brake accumulators. Ensure that all hydraulic pressure has been relieved from the hydraulic system. Remove bleeder plugs (12, Figure 6-17) in the steering accumulator stems to vent any residual pressure.

***NOTE: If a new or rebuilt accumulator (or any bladder accumulator with all nitrogen discharged) is being precharged while installed on the truck and still connected to the hydraulic system, the oil side of the accumulator must be vented to allow proper bladder expansion. Trapped air or oil on the hydraulic side of the bladder will prevent the proper precharge pressure from being obtained for safe operation.***

3. Remove charging valve guard (5, Figure 6-18) and charging valve cap (6).
4. Turn the gas valve "T" handle all the way out (counterclockwise) before attaching the charging assembly to the accumulator gas valve.
5. Close the bleed valve.
6. Attach the swivel nut to the gas valve fingertight. Do not to loop or twist the hose.
7. Hold valve body (6, Figure 6-19) with one wrench while unscrewing swivel nut (4) three complete turns with a second wrench. This will open the poppet inside the gas valve.
8. Set the regulator for 172 kPa (25 psi), then slightly open the nitrogen container. Slowly fill the accumulator. The fill rate time for this accumulator is approximately four minutes.



***If the precharge is not filled slowly, the bladder may suffer permanent damage. A "starburst" rupture in the lower end of the bladder is characteristic of precharging too rapidly.***

9. When 172 kPa (25 psi) of precharge pressure is obtained, close the nitrogen valve. Set the regulator for the operating precharge pressure based on the current ambient temperature. Refer to Table 1. Then, open the nitrogen container and fill the accumulator.
10. Let the precharge set for 15 minutes. This will allow the gas temperature to stabilize. If the desired precharge is exceeded, close the nitrogen container valve and slowly open the bleed valve until the correct precharge pressure is obtained. Refer to Table 1.

## **CAUTION**

***Do not reduce the precharge by depressing the valve core with a foreign object. High pressure may rupture the rubber valve seat.***

11. Turn the top hex to close the internal poppet. Tighten the hex nut to **7 - 11 N·m (5 - 8 ft lb)**.
12. Hold the gas valve stationary and loosen the swivel nut to remove the assembly. Use a common leak reactant to check for nitrogen leaks.
13. Install the gas valve cap finger-tight. The gas valve cap serves as a secondary seal.
14. Install the charging valve guard.

Temperature variation can affect the precharge pressure of an accumulator. As the temperature increases, the precharge pressure increases. Conversely, decreasing temperature will decrease the precharge pressure. In order to ensure the accuracy of the accumulator precharge pressure, the temperature variation must be accounted for.

A temperature variation factor is determined by the ambient temperature when charging the accumulator on a truck that has been shut down for one hour. Refer to Table 1 for charging pressures in different ambient operating conditions that the truck is currently exposed to during the charging procedure.

For example, assuming that the ambient temperature is 10°C (50°F), charge the accumulator to 9 294 kPa (1,348 psi).

**TABLE 1. Relationship Between Charging Pressure and Ambient Temperature**

| <b>Ambient Temperature</b> | <b>Charging Pressure<br/>± 70 kPa (10 psi)</b> |
|----------------------------|------------------------------------------------|
| -23°C (-10°F) and below    | 8 232 kPa (1,194 psi)                          |
| -17°C (0°F)                | 8 412 kPa (1,220 psi)                          |
| -12°C (10°F)               | 8 584 kPa (1,245 psi)                          |
| -7°C (20°F)                | 8 763 kPa (1,271 psi)                          |
| -1°C (30°F)                | 8 943 kPa (1,297 psi)                          |
| 4°C (40°F)                 | 9 122 kPa (1,323 psi)                          |
| 10°C (50°F)                | 9 294 kPa (1,348 psi)                          |
| 16°C (60°F)                | 9 473 kPa (1,374 psi)                          |
| 21°C (70°F)                | 9 653 kPa (1,400 psi)                          |
| 27°C (80°F)                | 9 832 kPa (1,426 psi)                          |
| 32°C (90°F)                | 10 011 kPa (1,452 psi)                         |
| 38°C (100°F)               | 10 184 kPa (1,477 psi)                         |
| 43°C (110°F)               | 10 363 kPa (1,503 psi)                         |
| 49°C (120°F)               | 10 542 kPa (1,529 psi)                         |

**NOTE:** A precharge pressure below 8 232 kPa (1194 psi) is not recommended because of low precharge pressure warnings. The low accumulator precharge warning switch activates at 7 584 ± 310 kPa (1,100 ± 45 psi).

## Precharge Maintenance



***If the low steering accumulator precharge warning light is illuminated when the key switch is turned ON, do not attempt to start the truck. Permanent bladder damage may result. Check the precharge pressure and adjust if necessary.***

1. When starting the truck, turn the key switch to the RUN position and wait to confirm that the low accumulator precharge warning light does not stay illuminated after the system check is complete. If the warning light stays illuminated, **do not** start the truck. Notify maintenance personnel immediately.
2. Check the accumulator precharge pressure every 500 hours. If the precharge pressure is too low, the bladder will be crushed into the top of the shell by hydraulic system pressure and can extrude into the gas stem and become punctured. This condition is known as "pick out". One such cycle is sufficient to destroy a bladder.

3. Check all sealing areas on the nitrogen side of the accumulator (charging valve, pressure switch, manifold, etc.) during every precharge maintenance interval to ensure that the seals do not leak. Replace all faulty or leaking seals, valves, etc. Failure to repair leaking nitrogen seals may result in a failed accumulator bladder or low performance from the accumulator.

***NOTE: If precharge pressure continues to decline frequently between precharge maintenance intervals, and if all nitrogen sealing areas are free of leaks, then the accumulator bladder most likely has a small hole in it and must be replaced.***

4. Check all heat shields and exhaust blankets, as provided, during every precharge maintenance interval to ensure that they are in place and good condition.

***NOTE: If the truck is equipped with cold weather bladder accumulators, the precharge pressure must be checked every 100 hours because the nitrogen permeates this bladder material at a much greater rate than the standard bladder material.***



## Accumulator Storage Procedures



***Always store bladder accumulators with 690 - 827 kPa (100 - 120 psi) nitrogen precharge pressure. This amount of pressure fully expands the bladder and holds oil against the inner walls for lubrication and rust prevention. Do not exceed 827 kPa (120 psi). Storing accumulators with too much pressure is not safe due to possible leakage.***

***Only precharge accumulators to 9 653 kPa (1,400 psi) while installed on the truck. Never handle an accumulator with lifting equipment if the nitrogen precharge pressure is over 827 kPa (120 psi). Always set the precharge to 690 - 827 kPa (100 - 120 psi) before removing or installing the accumulators.***

1. If the accumulator was just rebuilt, ensure that there is approximately 3.1 L (106 oz.) of clean C-4 hydraulic oil inside the accumulator before adding 690 kPa (100 psi) of nitrogen precharge pressure.
2. Bladder accumulators should always be stored with 690 - 827 kPa (100 - 120 psi) of nitrogen precharge pressure, which fully expands the bladder and holds a film of oil against the inner walls for lubrication and rust prevention.
3. The hydraulic port should always be covered with a plastic plug to prevent contamination. **Do not use a screw-in type plug.**
4. Always store the accumulator in an upright position.

## Bladder Storage Procedures

The shelf life of bladders under normal storage conditions is one year. Normal storage conditions consist of the bladder being heat sealed in a black plastic bag and placed in a cool dry place away from the sun, ultraviolet and fluorescent lights, and electrical equipment. Direct sunlight or fluorescent light can cause the bladder to weather check and dry rot, which will appear on the bladder surface as cracks.

## Installing A Bladder Accumulator From Storage

1. Refer to the Charging Procedure earlier in this chapter to install the pressure gauges on the accumulator and to check the precharge pressure.
  - a. If the precharge pressure is 172 - 690 kPa (25 - 100 psi), set the regulator to 690 kPa (100 psi) and slowly charge the accumulator to 690 kPa (100 psi). Disconnect the pressure gauges from the accumulator and install the accumulator on the truck. Refer to Charging Procedure in this section to fully charge accumulator to the correct operating precharge pressure.
  - b. If the precharge pressure is less than 172 kPa (25 psi), slowly drain off any nitrogen precharge and proceed to Step 2.
2. Remove the gauges from the accumulator.
3. Lay the accumulator on a suitable work bench so that the hydraulic port is higher than the other end of the accumulator. Remove the plastic dust cap from the hydraulic port.
4. Pour approximately 3.1 L (106 oz.) of clean C-4 hydraulic oil into the accumulator through the hydraulic port. Allow time for the oil to run down the inside of the accumulator to reach the other end.
5. Lay the accumulator flat on the work bench (or floor) and slowly rotate the accumulator two complete revolutions. This will thoroughly coat the accumulator walls with a film of oil necessary for bladder lubrication during precharging.
6. Stand the accumulator upright. Install the pressure gauges and refer to the Charging Procedure for instructions about charging the accumulator to 690 kPa (100 psi). Remove the gauges from the accumulator and install a plastic dust cap over the hydraulic port.
7. Install the accumulator on the truck.
8. Precharge the accumulator to the correct operating precharge pressure. Refer to Table 1.

## TROUBLESHOOTING CHART

| TROUBLE                                                                                              | POSSIBLE CAUSE                                                                                                                                                                                                                                                                                                                                                              | SUGGESTED CORRECTIVE ACTION                                                                                                                                                                                                                                                                                                                 |
|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>STEERING CIRCUIT</b>                                                                              |                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                             |
| Slow steering, hard steering or loss of power assist                                                 | <ol style="list-style-type: none"> <li>1. Overloaded steering axle</li> <li>2. Malfunctioning relief valve preventing adequate system pressure build-up</li> <li>3. Worn or malfunctioning pump</li> <li>4. Restricted high pressure filter or suction strainer</li> </ol>                                                                                                  | <ol style="list-style-type: none"> <li>1. Reduce axle loading.</li> <li>2. Check system pressure. Adjust or replace relief valve.</li> <li>3. Replace pump.</li> <li>4. Replace filter element or clean strainer.</li> </ol>                                                                                                                |
| <b>Drift</b> - truck veers slowly in one direction.                                                  | <ol style="list-style-type: none"> <li>1. Rod end of cylinder slowly extends without turning the steering wheel</li> <li>2. Worn or damaged steering linkage</li> </ol>                                                                                                                                                                                                     | <ol style="list-style-type: none"> <li>1. A small rate of extension may be normal on a closed center system.</li> <li>2. Inspect and replace linkage if necessary. Check alignment or toe-in of the front wheels.</li> </ol>                                                                                                                |
| <b>Wander</b> - truck will not stay in straight line                                                 | <ol style="list-style-type: none"> <li>1. Air in system due to low oil level, pump cavitation, leaking fitting, pinched hoses</li> <li>2. Loose cylinder piston</li> <li>3. Broken centering springs (spool valve, steering valve)</li> <li>4. Worn mechanical linkage</li> <li>5. Bent linkage or cylinder rod</li> <li>6. Severe wear in steering control unit</li> </ol> | <ol style="list-style-type: none"> <li>1. Correct oil supply problem or bleed air.</li> <li>2. Repair or replace steering cylinder.</li> <li>3. Repair or replace steering control unit.</li> <li>4. Repair or replace.</li> <li>5. Repair or replace defective components.</li> <li>6. Repair or replace steering control unit.</li> </ol> |
| <b>Slip</b> - a slow movement of steering wheel fails to steer front wheels                          | <ol style="list-style-type: none"> <li>1. Leakage of steering cylinder piston seals</li> <li>2. Worn steering control unit meter</li> </ol>                                                                                                                                                                                                                                 | <ol style="list-style-type: none"> <li>1. Repair or replace steering cylinder.</li> <li>2. Replace steering control unit.</li> </ol>                                                                                                                                                                                                        |
| Spongy or soft steering                                                                              | <ol style="list-style-type: none"> <li>1. Low oil level</li> <li>2. Air in hydraulic system. Probably air trapped in cylinders or lines.</li> </ol>                                                                                                                                                                                                                         | <ol style="list-style-type: none"> <li>1. Service hydraulic tank and check for leakage.</li> <li>2. Bleed air from system.</li> </ol>                                                                                                                                                                                                       |
| Erratic steering                                                                                     | <ol style="list-style-type: none"> <li>1. Air in system due to low oil level, cavitating pump, leaky fittings, pinched hose, etc.</li> <li>2. Loose steering cylinder piston</li> </ol>                                                                                                                                                                                     | <ol style="list-style-type: none"> <li>1. Correct condition and add oil as necessary.</li> <li>2. Repair or replace steering cylinder.</li> </ol>                                                                                                                                                                                           |
| <b>Free wheeling</b> - steering wheel turns freely with no back pressure - front wheels do not steer | <ol style="list-style-type: none"> <li>1. Splines of steering column/steering control unit coupling may be disengaged or damaged</li> <li>2. No flow to steering control unit: <ol style="list-style-type: none"> <li>a. Low oil level</li> <li>b. Ruptured hose</li> <li>c. Broken steering control unit gerotor drive pin</li> </ol> </li> </ol>                          | <ol style="list-style-type: none"> <li>1. Repair or replace steering column or coupling.</li> <li>2. Repair as required: <ol style="list-style-type: none"> <li>a. Add oil and check for leakage.</li> <li>b. Replace hose.</li> <li>c. Repair or replace steering control unit.</li> </ol> </li> </ol>                                     |

| TROUBLE                                                                      | POSSIBLE CAUSE                                                                                                                                                                                                                                                                                                                                                       | SUGGESTED CORRECTIVE ACTION                                                                                                                                                                                                                                                                                                                                             |
|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Excessive free play at steered wheels                                        | <ol style="list-style-type: none"> <li>1. Broken or worn linkage between steering cylinder and steered wheels</li> <li>2. Leaky steering cylinder seals</li> </ol>                                                                                                                                                                                                   | <ol style="list-style-type: none"> <li>1. Check for loose fitting bearings at anchor points in steering linkage between cylinder and steered wheels.</li> <li>2. Repair or replace steering cylinder.</li> </ol>                                                                                                                                                        |
| Binding or poor centering of steered wheels                                  | <ol style="list-style-type: none"> <li>1. Binding or misalignment in steering column or splined coupling and steering control unit</li> <li>2. High back pressure in tank can cause slow return to center - should not exceed 2 068 kPa (300 psi)</li> <li>3. Large particles can cause binding between the spool and sleeve in the steering control unit</li> </ol> | <ol style="list-style-type: none"> <li>1. Align column to steering control unit.</li> <li>2. Remove restriction in the lines or circuit. Check for obstruction or pinched lines.</li> <li>3. Clean steering control unit and filter the oil. If another component has malfunctioned (generating contaminating materials), flush the entire hydraulic system.</li> </ol> |
| Steering control unit locks up                                               | <ol style="list-style-type: none"> <li>1. Large particles in meter section</li> <li>2. Insufficient hydraulic power</li> <li>3. Severe wear and/or broken pin</li> </ol>                                                                                                                                                                                             | <ol style="list-style-type: none"> <li>1. Clean the steering control unit.</li> <li>2. Check hydraulic system pressure.</li> <li>3. Repair or replace steering control unit.</li> </ol>                                                                                                                                                                                 |
| Steering wheel oscillates or turns by itself                                 | <ol style="list-style-type: none"> <li>1. Lines connected to wrong ports</li> <li>2. Parts assembled wrong; steering control unit improperly timed</li> </ol>                                                                                                                                                                                                        | <ol style="list-style-type: none"> <li>1. Check line routing and connections.</li> <li>2. Reassemble correctly and re-time control valve.</li> </ol>                                                                                                                                                                                                                    |
| Steered wheels turn in opposite direction when operator turns steering wheel | <ol style="list-style-type: none"> <li>1. Lines connected to wrong cylinder ports</li> </ol>                                                                                                                                                                                                                                                                         | <ol style="list-style-type: none"> <li>1. Inspect and correct line connections.</li> </ol>                                                                                                                                                                                                                                                                              |
| <b>STEERING PUMP</b>                                                         |                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                         |
| No pump output                                                               | <ol style="list-style-type: none"> <li>1. Air trapped inside steering pump</li> <li>2. Broken pump drive shaft</li> <li>3. Excessive circuit leakage</li> <li>4. No oil to pump inlet</li> </ol>                                                                                                                                                                     | <ol style="list-style-type: none"> <li>1. Bleed air from pump and/or steering system.</li> <li>2. Inspect hoist and steering pump; replace drive shaft.</li> <li>3. Check for external leaks or internal system pressure loss.</li> <li>4. Check hydraulic tank oil level. Ensure that pump inlet shutoff valve is open.</li> </ol>                                     |

| <b>TROUBLE</b>                   | <b>POSSIBLE CAUSE</b>                                                                                                                                                                                                                                         | <b>SUGGESTED CORRECTIVE ACTION</b>                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Low pump output                  | <ol style="list-style-type: none"> <li>1. Low pump pressure</li> <li>2. Internal pump wear</li> <li>3. Restricted inlet</li> <li>4. Insufficient oil supply</li> <li>5. High pressure filter restricted</li> </ol>                                            | <ol style="list-style-type: none"> <li>1. Check unloader valve operation and adjust pressure if necessary. Check compensator adjustment pressure. Check system relief valves. Replace as necessary.</li> <li>2. Repair or replace steering pump.</li> <li>3. Ensure that shutoff valve is open. Check suction hose. Clean tank strainers.</li> <li>4. Check hydraulic tank oil level. Ensure that shutoff valve is open.</li> <li>5. Replace filter element.</li> </ol> |
| Loss of pressure                 | <ol style="list-style-type: none"> <li>1. Defective pressure control</li> <li>2. Internal steering pump wear</li> <li>3. Excessive leakage in brake system</li> </ol>                                                                                         | <ol style="list-style-type: none"> <li>1. Check unloader valve operation and adjust pressure if necessary. Check compensator adjustment pressure. Check system relief valves. Replace as necessary.</li> <li>2. Repair or replace pump.</li> <li>3. Inspect brake system and check pressures.</li> </ol>                                                                                                                                                                |
| Excessive or high peak pressure  | <ol style="list-style-type: none"> <li>1. Defective or improperly adjusted pressure control(s)</li> </ol>                                                                                                                                                     | <ol style="list-style-type: none"> <li>1. Check unloader valve operation and adjust pressure if necessary. Check compensator adjustment pressure. Check system relief valves. Replace as necessary.</li> </ol>                                                                                                                                                                                                                                                          |
| Noise or squeal                  | <ol style="list-style-type: none"> <li>1. Low compensator or unloader valve setting</li> <li>2. Fluid too cold or viscosity too high</li> <li>3. Air leak at pump inlet</li> <li>4. Insufficient inlet oil supply</li> <li>5. Internal pump damage</li> </ol> | <ol style="list-style-type: none"> <li>1. Check pressure and adjust pressure controls.</li> <li>2. Warm oil before starting or install proper viscosity oil.</li> <li>3. Inspect inlet hose, connections and shutoff valve.</li> <li>4. Check hydraulic tank level. Clean suction strainer. Ensure that shutoff valve is open.</li> <li>5. Repair or replace pump.</li> </ol>                                                                                           |
| Erratic pump (load/unload) cycle | <ol style="list-style-type: none"> <li>1. Pressure compensator adjusted incorrectly or defective</li> <li>2. Excessive internal leakage in steering circuit</li> <li>3. Unloader valve pilot seat damaged</li> </ol>                                          | <ol style="list-style-type: none"> <li>1. Adjust pressure compensator or repair if necessary. Check unloader pressure settings.</li> <li>2. Measure component leakage rates and replace defective components.</li> <li>3. Replace unloader valve module.</li> </ol>                                                                                                                                                                                                     |
| Excessive heat                   | <ol style="list-style-type: none"> <li>1. Excessive system pressure</li> <li>2. Low hydraulic fluid level</li> <li>3. Worn steering or hoist pump</li> </ol>                                                                                                  | <ol style="list-style-type: none"> <li>1. Adjust system pressures.</li> <li>2. Service hydraulic tank.</li> <li>3. Repair or replace pump(s).</li> </ol>                                                                                                                                                                                                                                                                                                                |

| TROUBLE                                          | POSSIBLE CAUSE                                                                                                         | SUGGESTED CORRECTIVE ACTION                                                                                                                                                                                         |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>FLOW AMPLIFIER</b>                            |                                                                                                                        |                                                                                                                                                                                                                     |
| Slow or hard steer                               | 1. Stuck amplifier spool<br>2. Obstruction in orifice, directional spool<br>3. Incorrect relief valve pressure setting | 1. Disassemble valve and check amplifier spool for damage or obstruction.<br>2. Disassemble valve and check directional spool and orifices for damage or obstruction.<br>3. Check and correct relief valve setting. |
| Free wheeling<br>(no end stop)                   | 1. Leaking shock and suction relief valve<br>2. Pressure setting of shock valve too low                                | 1. Disassemble valve, check and clean shock and suction valves. Replace valves if necessary.<br>2. Adjust pressure settings                                                                                         |
| Inability to steer<br>(no pressure build-up)     | 1. Defective steering control unit                                                                                     | 1. Repair or replace steering control unit.                                                                                                                                                                         |
| Resistance when initially turning steering wheel | 1. Air in "LS" line<br>2. Priority valve spring compression weak<br>3. Obstruction in orifice in "LS" or "PP" port     | 1. Bleed air from "LS" line.<br>2. Disassemble flow amplifier. Replace priority valve spring.<br>3. Disassemble flow amplifier. Inspect and clean orifice.                                                          |

# SECTION L7

## HOIST CIRCUIT

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# NOTES

## HOIST CIRCUIT

### BASIC OPERATION

The following information describes the basic hoist system circuit as shown in Figure 7-1. Detailed component operation is outlined under the individual component descriptions.

Hydraulic fluid is supplied by a tank (1) located on the left frame rail. Hydraulic oil is routed to a tandem gear type pump (2). A second pump, coupled to the rear of the hoist pump, supplies oil for the steering and brake systems. The pumps are driven by an accessory drive at the end of the traction alternator.

Hoist pump output is directed to a pair of high pressure filters (3) mounted to the inboard side of the fuel tank.

Hydraulic oil from the hoist filters is directed to the hoist valve (7), mounted above the pumps. The hoist valve directs oil to the body hoist cylinders (10) for raising and lowering of the dump body. Hoist valve functions are controlled by the operator through the lever connected to the hoist pilot valve (4) located in the hydraulic components cabinet. A hoist limit solenoid (5) located in the bleeddown manifold shifts the hoist valve out of POWER UP before the hoist cylinders extend to their maximum physical limit.

When the hoist valve is in the HOLD or FLOAT position, hoist circuit oil flows to the front (8) and rear (9) service brakes, cooling the wet disc brakes during truck operation.

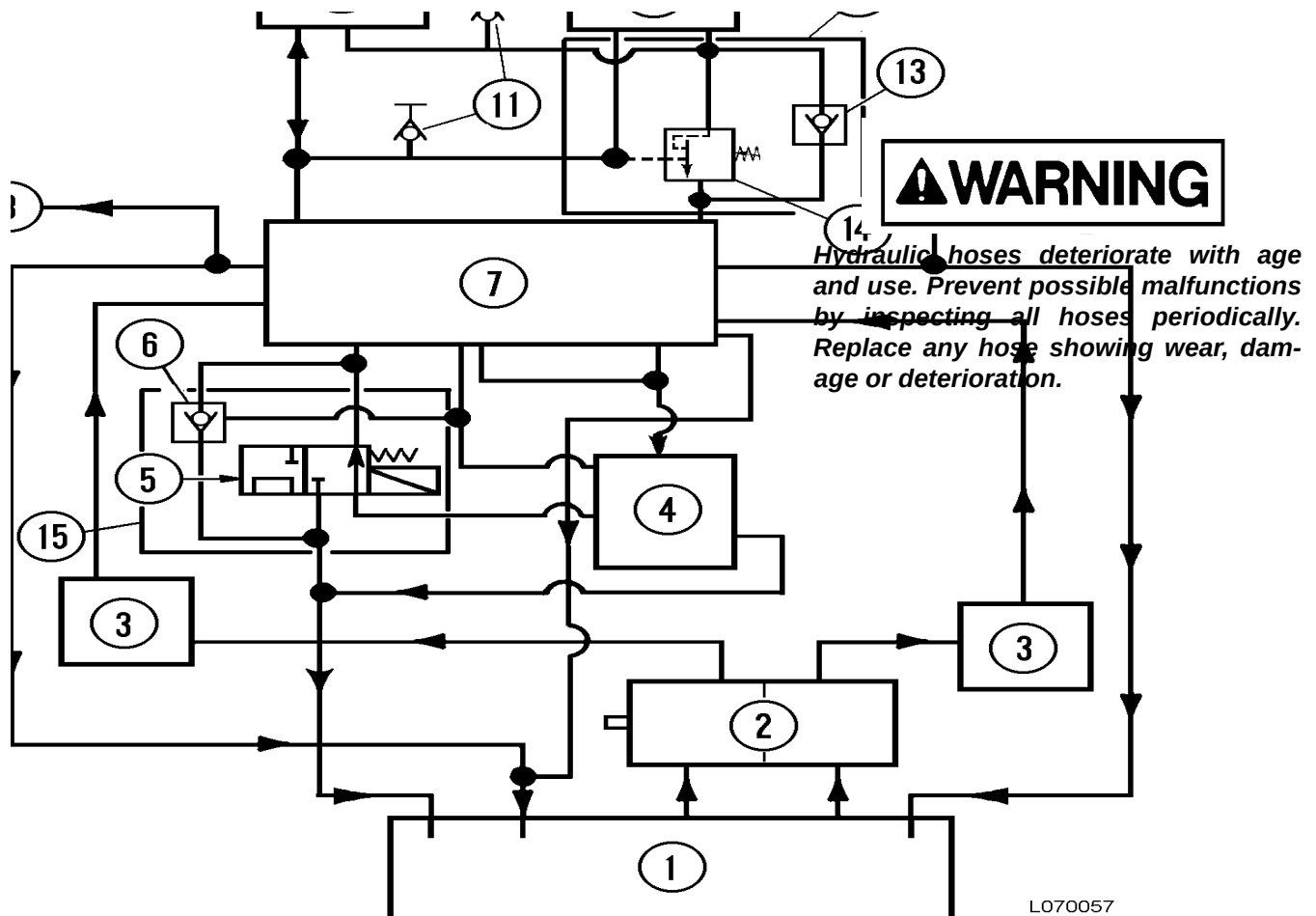


FIGURE 7-1. HOIST CIRCUIT SCHEMATIC

- |                         |                                     |                          |
|-------------------------|-------------------------------------|--------------------------|
| 1. Hydraulic Tank       | 6. Pilot Operated Check Valve       | 11. Quick Disconnect     |
| 2. Hoist Pump           | 7. Hoist Valve                      | 12. Overcenter Manifold  |
| 3. High Pressure Filter | 8. Brake Cooling Oil Supply (Front) | 13. Check Valve          |
| 4. Hoist Pilot Valve    | 9. Brake Cooling Oil Supply (Rear)  | 14. Counterbalance Valve |
| 5. Hoist Limit Solenoid | 10. Hoist Cylinder                  | 15. Bleeddown Manifold   |



## COMPONENT DESCRIPTION

### Hydraulic Tank

The hydraulic tank supplies hydraulic oil for the hoist, steering and brake circuits. The tank is located on the left hand frame rail forward of the rear wheels. The tank's service capacity is approximately 947 liters (250 gallons). Type C-4 hydraulic oil is recommended for use in the hydraulic system. Oil used in the hoist circuit flows through 100 mesh wire suction strainers to the inlet hoses of the pump. Air drawn into the tank during operation is filtered by air filters located on the top of the tank. Oil level can be checked visually at sight glasses located on the face of the tank.

### Hydraulic Pump

The hoist system hydraulic pump is a tandem, gear type pump driven by a drive shaft on the end of the traction alternator. The pump has a total output of 931 l/m (246 gpm) at 1,900 rpm and 17 235 kPa (2,500 psi).

A smaller piston type, pressure compensated pump, rated at 250 l/m (66 gpm) @ 1,900 rpm, is coupled to the hoist pump. It supplies oil to the steering system and brake apply system.

Hoist pressure is limited to 17 235 kPa (2,500 psi) by internal relief valves located in the hoist control valve inlet sections.

### High Pressure Filters

Hoist pump output oil is directed to the high pressure filters mounted on the inboard side of the fuel tank. The filter assemblies are equipped with a bypass valve which permits oil flow if the filter element becomes excessively restricted. Flow restriction through the filter element is sensed by an indicator switch. This switch will turn on a red warning light in the cab to indicate that filter service is required. The indicator light will illuminate when the restriction reaches approximately 241 kPa (35 psi). Actual filter bypass will result when the filter element restriction reaches approximately 345 kPa (50 psi).

### Hoist Valve

The hoist valve (Figure 7-2) is mounted on a modular assembly containing the hoist valve, overcenter manifold, and both hydraulic pumps. This assembly is bolted to brackets attached to the frame rails behind the main alternator. The hoist valve is a split spool design. The term "split spool" describes the spool section of the valve.

The hoist valve precisely follows differential pressure input signals generated by the hoist pilot valve as the operator moves the hoist control lever while raising and lowering the dump body.

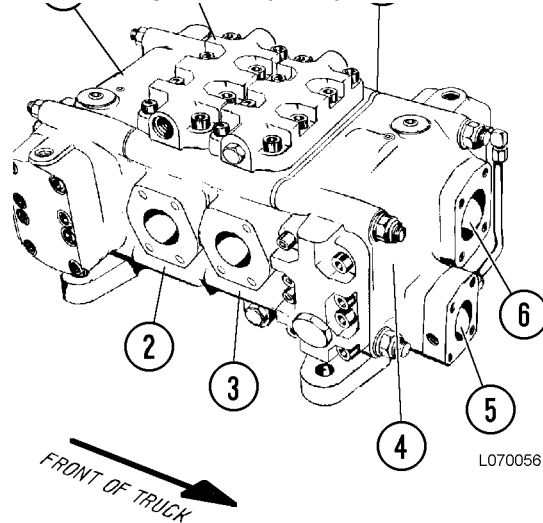


FIGURE 7-2. HOIST VALVE

- |                             |                        |
|-----------------------------|------------------------|
| 1. Inlet Section            | 4. Inlet Section       |
| 2. Work Ports Spool Section | 5. Inlet Port          |
| 3. Tank Ports Spool Section | 6. Outlet Port         |
|                             | 7. Spool Section Cover |
|                             | 8. Separator Plate     |

### Inlet Sections

Each of the front and rear inlet sections of the hoist valve contains the following components:

- Flow control valve and main relief valve
- Secondary Low pressure relief valve
- Anti-void check valve
- Load check valve

The flow control portion of the flow control and main relief valves allow pump flow to the service brake cooling circuit unless the body is being raised or lowered. The relief portion of the valve is direct-acting and has the capacity to limit the working pressure at full pump flow.

The secondary low pressure relief is located between the low pressure core and the outlet to the brake cooling circuit. It provides for pressure relief if pressure spikes occur in the low pressure passage area.

The load check allows free flow from the inlet to the high pressure core and prevents flow from the high pressure core to the inlet.

The anti-void check valve allows free flow from the low pressure core to the high pressure core and prevents flow from the high pressure core to the low pressure core.

### Work Ports (Rear) Spool Section

The rear spool section of the hoist valve consists of the following components:

- Pilot ports
- Main spools
- Work ports
- Check poppets

The pilot ports are located in the spool section cover. These ports provide connections for a pilot line to the hoist pilot valve. Each work port has a corresponding pilot port.

The work ports provide for a line connection between the spool section and the hoist cylinders. One main spool for each work port is spring biased at both end to block the work port from the high and low pressure cores when there is no flow through the spool cross holes.

When there is flow through the pilot ports to the spools, a positive differential pressure at the top of the spool will overcome the bottom spring bias and the spool will shift to connect the work port to the high pressure core.

When there is flow from the main valve work port to the pilot port through the cross-holes, a positive differential pressure at the bottom of the spool will overcome the top spring bias and the spool will shift to connect the work port to the low pressure core.

The check poppets located in the spool section permit free flow from the work port to the pilot port and restrict flow from the pilot port to the work port. These check poppets control spool response and spool movements during void conditions.

### Tank Ports (Front) Spool Section

The primary low pressure valves are located in the front spool section of the hoist valve. These valves maintain back pressure on the low pressure passage and direct the hoist cylinder return oil back to the hydraulic tank.

### Hoist Pilot Valve

The hoist pilot valve (1, Figure 7-3) is mounted in the hydraulic components cabinet behind the operator's cab. The hoist pilot valve spool is spring centered to the neutral position and is controlled directly by the operator through a lever mounted on the console between the operator and passenger seat. A cable connects the cab mounted lever to the hoist pilot valve in the hydraulic components cabinet.

When the operator moves the lever, the pilot valve spool moves and directs pilot flow to the appropriate pilot port on the hoist valve. The pilot flow causes the main spool to direct oil flow to the hoist cylinders.

The hoist pilot valve is equipped with a one way load check valve which allows free flow from the center passage to bridge core and prevents reverse flow. The valve also contains power down relief valve (2), which is used to limit the power down pressure to 10 400 kPa (1,500 psi).

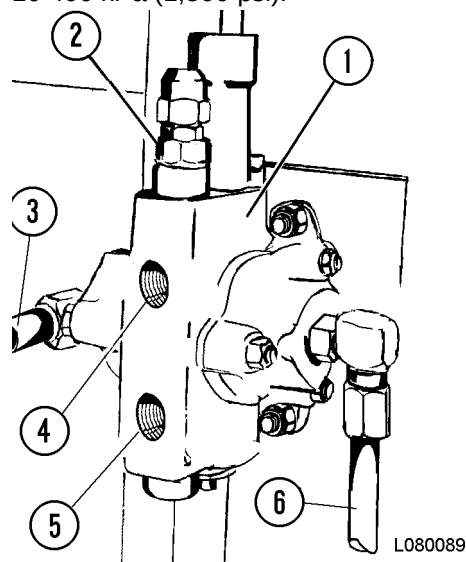


FIGURE 7-3. HOIST PILOT VALVE

- |                             |                              |
|-----------------------------|------------------------------|
| 1. Hoist Pilot Valve        | 5. To Hoist Valve (Base End) |
| 2. Relief Valve             | 6. Return to Tank            |
| 3. Supply Port              |                              |
| 4. To Hoist Valve (Rod End) |                              |

## Bleeddown Manifold

The pilot valve/hoist valve hydraulic circuit is routed through the bleeddown manifold, located on the left frame rail. The hoist up limit solenoid and pilot operated check valve described below are contained in the manifold. Refer to Section L4 for additional information about these components.

### Hoist Up Limit Solenoid

The hoist up limit solenoid (5, Figure 7-1), located on the bottom of the bleeddown manifold, is used in the hydraulic circuit to prevent maximum hoist cylinder extension.

The solenoid valve is a 3-way valve, and is controlled by a proximity switch (hoist limit switch) located near the body pivot and the right rear suspension upper mount.

When the body is nearly fully raised, the body activates the magnetic proximity switch, signalling the solenoid to open the “raise” pilot line to the tank, and close the “raise” pilot line to the hoist valve, stopping further oil flow to the hoist cylinders.

Refer to Section D3, 24VDC Electrical System Components, for the adjustment procedure of the hoist limit switches.

### Pilot Operated Check Valve

The pilot operated check valve (6, Figure 7-1), located in the bleeddown manifold, is opened by power down pilot pressure to allow oil in the raise port to bypass the hoist up limit solenoid for initial power down operation while the solenoid is activated by the hoist limit switch.

## Overcenter Manifold

The overcenter manifold (4, Figure 7-4), located on the pump module at the rear of hoist valve (1), contains counterbalance valve (2). This valve controls the pressure of the cushion of oil in the annulus area of the hoist cylinder when the body approaches the maximum dump angle. The valve limits the maximum pressure build-up by relieving pressure in excess of 20 400 kPa (3,000 psi), preventing possible seal damage.

Quick disconnect fittings (5), mounted on the bottom of the manifold, allow dumping of the load in a disabled truck by connecting jumper hoses to the hydraulic system of an operational truck.

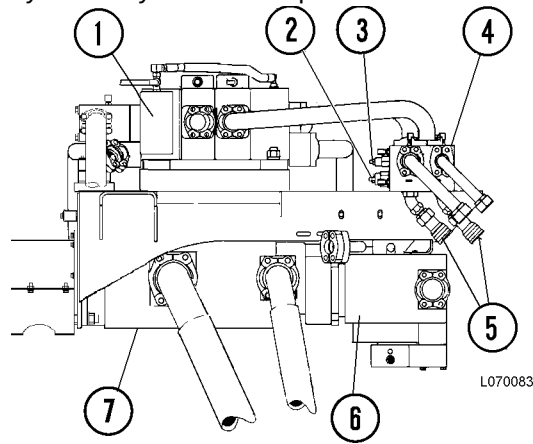


FIGURE 7-4. PUMP MODULE

- |                         |                        |
|-------------------------|------------------------|
| 1. Hoist Valve          | 5. Quick Disconnects   |
| 2. Counterbalance Valve | 6. Steering/Brake Pump |
| 3. Needle Valve         | 7. Hoist Pump          |
| 4. Overcenter Valve     |                        |

## **HOIST CIRCUIT OPERATION**

**The following pages describe the hoist circuit operation in the FLOAT, POWER UP, HOLD, and POWER DOWN positions. (Refer to Figures 7-5 through 7-9.)**

### **Float Position Of Pilot Valve With Truck Body On Frame**

Figure 7-5 shows the position of the hoist pilot valve when the truck body is resting on the frame. However, the hoist valve internal valves and spools will be in the position shown in Figure 7-7.

Oil from the hoist pump enters the inlet sections of the hoist valve in port (11), passes through check valve (18), and stops at the closed high pressure passage (19) at the two main spools (7) and (8) as shown in Figure 7-5.

Pressure builds to approximately 414 kPa (60 psi) on the pilot of flow control valve (2), causing the valve to compress the spring and open, which allows the oil to flow through hoist valve port (23) to the service brakes for brake disc cooling and eventually back to the tank. Between hoist valve port (23) and the brakes or heat exchanger, the cooling supply tee's to orificed regulator valves. If supply pressure attempts to increase above 240 kPa (35 psi), the regulator valve opens and dumps to the tank.

Oil also flows out of hoist valve port (12) to port (12) on the pilot valve, through the hoist pilot valve spool, and out through pilot valve port (10) to the tank. This oil flow is limited by orifices in the inlet sections of the hoist valve and, therefore, has no pressure buildup.

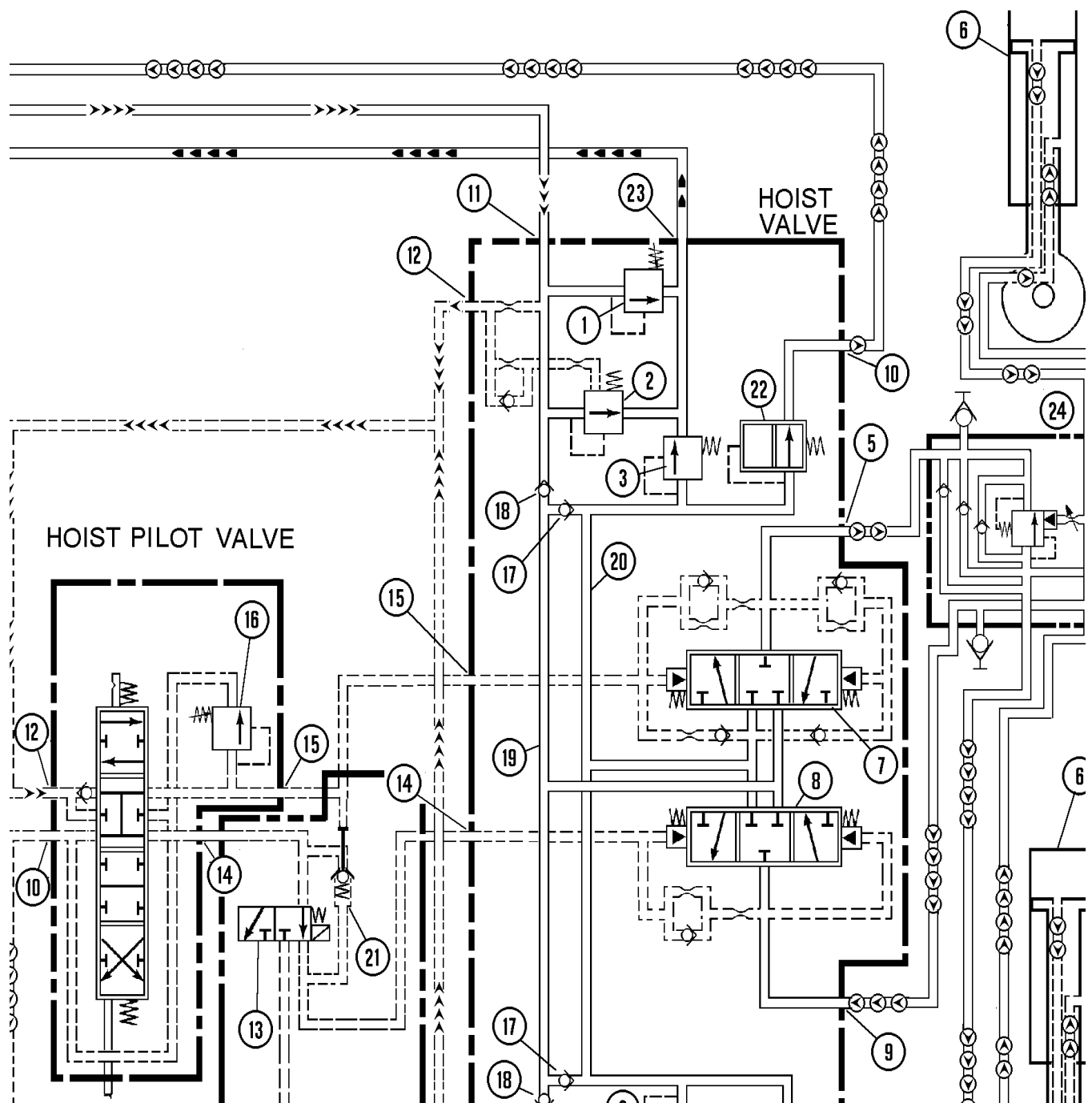


FIGURE 7-5. FLOAT POSITION

- |                                           |                                         |                                         |
|-------------------------------------------|-----------------------------------------|-----------------------------------------|
| 1. Hoist Relief Valve (2,500 psi)         | 9. Head End Work Port                   | 17. Anti-void Check Valve               |
| 2. Flow Control Valve                     | 10. Tank Return Port                    | 18. Load Check Valve                    |
| 3. Secondary Low Pressure Valve (250 psi) | 11. Supply Port                         | 19. High Pressure Passage               |
| 4. Snubber Valve                          | 12. Pilot Supply Port                   | 20. Low Pressure Passage                |
| 5. Rod End Work Port                      | 13. Hoist Limit Solenoid                | 21. Pilot Operated Check Valve          |
| 6. Hoist Cylinders                        | 14. Raise Pilot Port                    | 22. Primary Low Pressure Valve (26 psi) |
| 7. Rod End Spool                          | 15. Down Pilot Port                     | 23. Brake Cooling Circuit Port          |
| 8. Head End Spool                         | 16. Power Down Relief Valve (1,500 psi) | 24. Overcenter Manifold                 |

## Power Up Operation (Figure 7-6)

The hoist pilot valve spool is moved to the power up position when the operator moves the lever rearward in the cab. The pilot supply oil coming in port (12) is prevented from returning to the tank and, instead, is directed out port (14) through hoist limit solenoid (13) in the bleeddown manifold and into port (14) of the hoist valve.

From there it goes to the top of the head end spool (8), builds pressure on the end of the spool, causes the spool to move down compressing the bottom spring, and connects the high pressure passage (19) to head end port (9). Working oil flow in the high pressure passage is now allowed to flow through spool (8) and out port (9) to extend the hoist cylinders.

Even though a small amount of oil flows through the check poppet in the top of spool (8), raise pilot pressure at ports (14) increases to slightly higher pressure than the required hoist cylinder pressure. As a result, the pilot supply pressure in ports (12) also increases causing back pressure to occur in the spring area of the flow control valve (2). This overcomes the pilot pressure on the other end of the flow control valve causing it to close and direct the incoming pump oil through the head end spool (8) to the hoist cylinders to extend them. At this time, oil flow to the brake cooling circuit is stopped.

If at any time the resistance to the flow of the pump oil coming into the inlet section causes the pressure to increase to 17 235 kPa (2,500 psi), the pilot pressure against hoist relief valve (1) causes it to open and allow flow to exit out port (23) to the brake cooling circuit and to return to the tank.

As the hoist cylinders extend, oil in the annulus area of all three stages must exit sequentially from the cylinders. Initially, rod end spool (7) ports are closed. As the returning oil entering port (5) builds low pressure, it flows through the check-poppet in the top of the spool, through ports (15), through the pilot valve spool, and out port (10) of the pilot valve to the bleed-down manifold and to the tank.

No pressure is present on the top of spool (7). Cylinder return pressure passes through the check-poppet in the bottom of spool (7) to build pressure under the spool, which moves the spool upward, compressing the top spring. This movement allows the returning cylinder oil to flow into the low pressure passage (20) to the primary low pressure relief valve (22). Approximately 179 kPa (26 psi) causes this valve to open, allowing the oil to flow out port (10) to the tank.

If the load passing over the tail of the body during dumping attempts to cause the body to raise faster than the oil being supplied by the pump, the oil returning from the annulus area of the hoist cylinders passing through the counterbalance valve manifold controls how fast the hoist cylinders can extend due to the external force of the load. The speed is controlled by counterbalance valve.

During the normal raise process, the raise pilot pressure supplied to the counterbalance valve (24) will open the cylinder return ports fully any time raise pressure is above 4 600 kPa (666 psi). As the load shifts toward the tail of the body, the raise pressure decreases. As it drops below 4 600 kPa (666 psi), the valve starts restricting oil flow, causing a back pressure to the annulus area. If raise pressure were to drop to 0, return pressure would increase to 20 400 kPa (3,000 psi) maximum.

When the operator releases the hoist control lever, the valves change to the HOLD position. If the body raises to the position that activates the hoist limit switch, located above the right rear suspension, before the operator releases the lever, hoist limit solenoid (13) is energized. The solenoid valve closes raise pilot port (14) on the hoist and releases the hoist pilot valve raise pilot pressure at port (14) to the tank, allowing head end spool (8) to center and shut off the supply of oil to the hoist cylinders. This prevents maximum extension of the hoist cylinders.

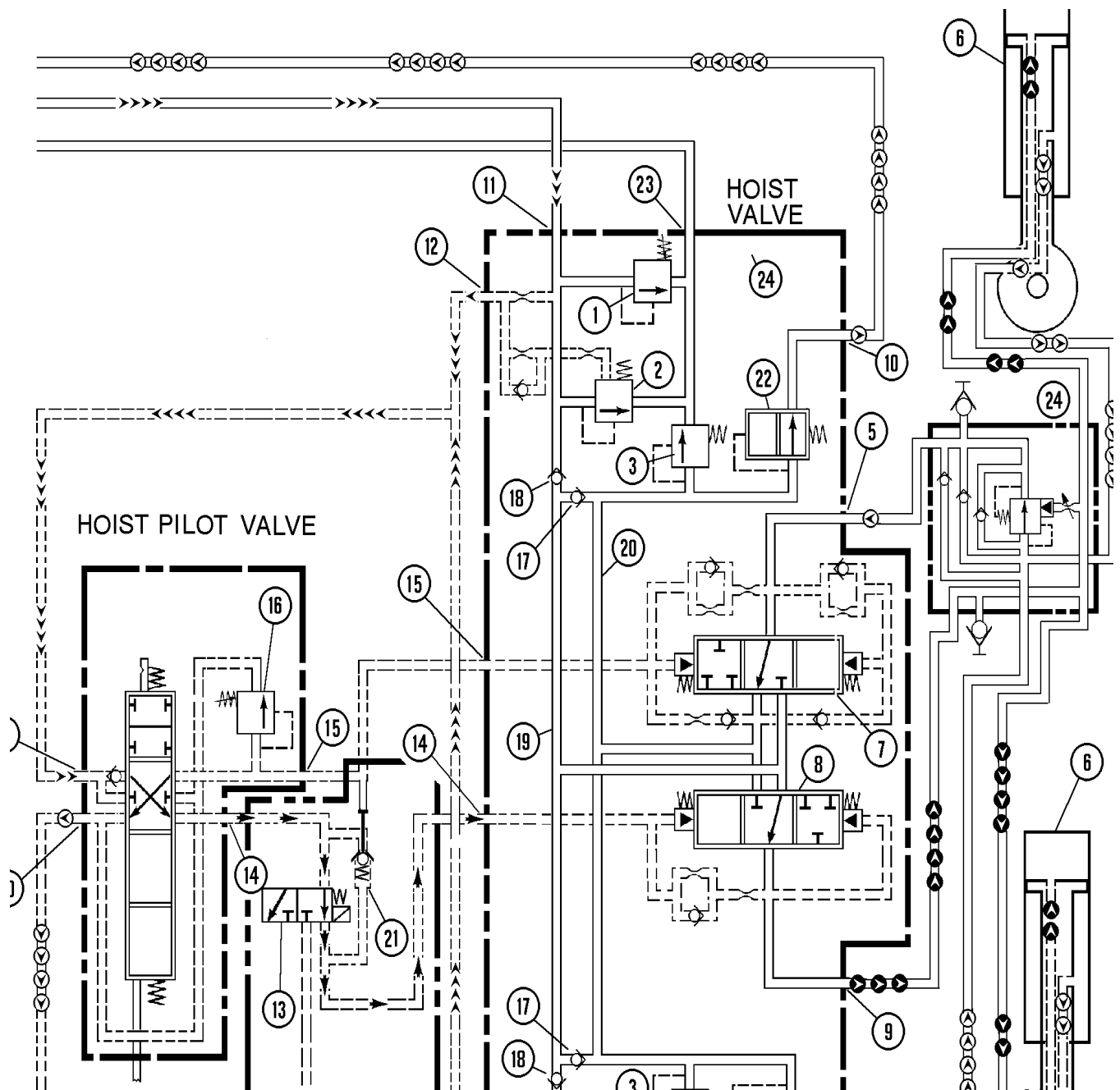


FIGURE 7-6. POWER UP POSITION

- |                                           |                                         |                                         |
|-------------------------------------------|-----------------------------------------|-----------------------------------------|
| 1. Hoist Relief Valve (2,500 psi)         | 9. Head End Work Port                   | 17. Anti-void Check Valve               |
| 2. Flow Control Valve                     | 10. Tank Return Port                    | 18. Load Check Valve                    |
| 3. Secondary Low Pressure Valve (250 psi) | 11. Supply Port                         | 19. High Pressure Passage               |
| 4. Snubber Valve                          | 12. Pilot Supply Port                   | 20. Low Pressure Passage                |
| 5. Rod End Work Port                      | 13. Hoist Limit Solenoid                | 21. Pilot Operated Check Valve          |
| 6. Hoist Cylinders                        | 14. Raise Pilot Port                    | 22. Primary Low Pressure Valve (26 psi) |
| 7. Rod End Spool                          | 15. Down Pilot Port                     | 23. Brake Cooling Circuit Port          |
| 8. Head End Spool                         | 16. Power Down Relief Valve (1,500 psi) | 24. Overcenter Manifold                 |



## **Hold Operation (Figure 7-7)**

The pilot valve spool is positioned to allow the pilot supply oil entering port (12) to return to the tank through port (10). Pilot supply pressure in ports (12) then decreases to no pressure, allowing flow control valve (2) to open and route the incoming pump oil to the brake cooling circuit through port (23) and back to the tank.

Both pilot ports (14) and (15) in the pilot valve are closed by the pilot valve spool. In this condition, pressure is equalized on each end of each main spool (7) and (8), allowing the springs to center the spools and close all ports to trap the oil in the cylinders and hold the body in its current position.

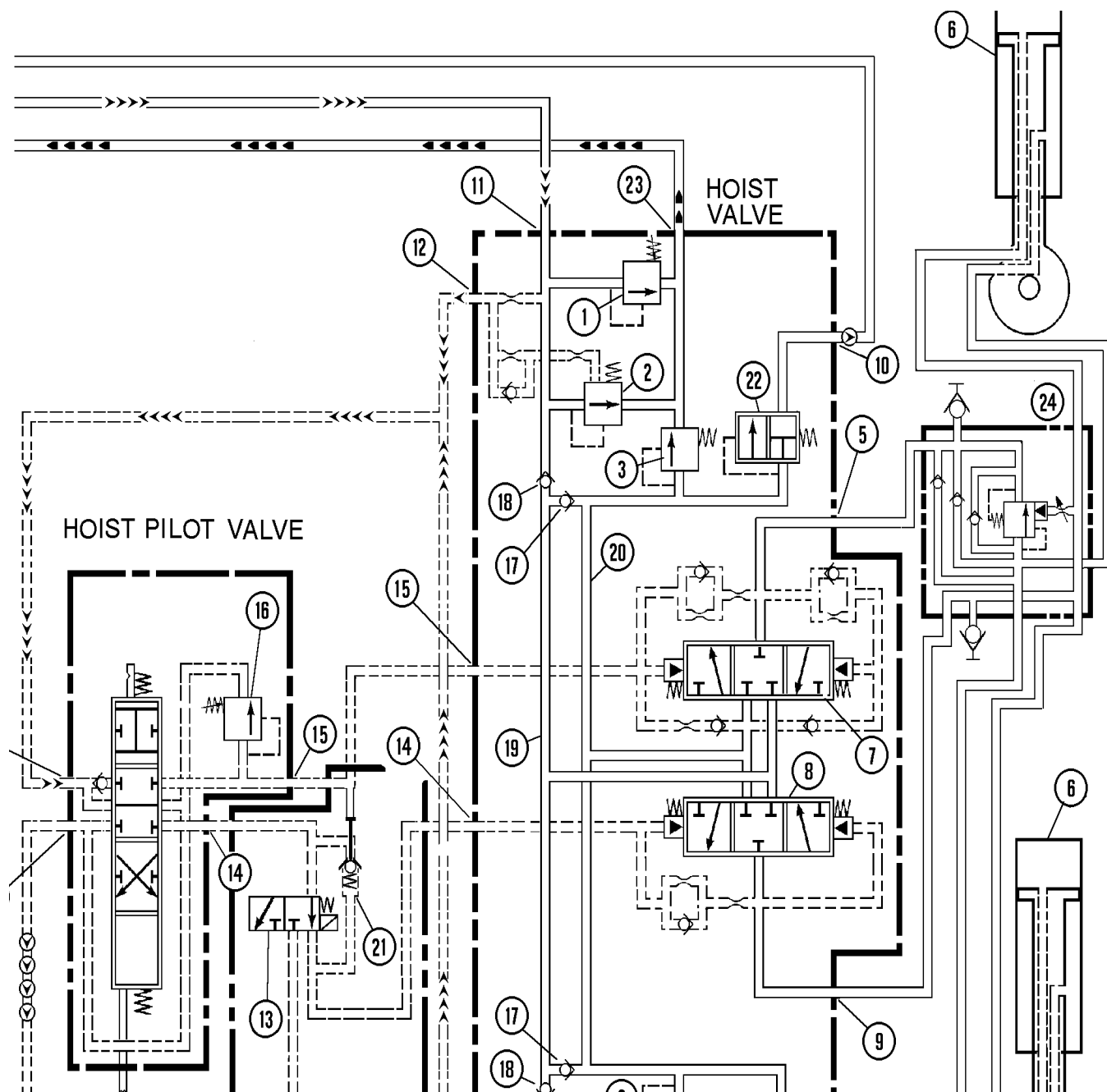


FIGURE 7-7. HOLD POSITION

- |                                           |                                         |                                         |
|-------------------------------------------|-----------------------------------------|-----------------------------------------|
| 1. Hoist Relief Valve (2,500 psi)         | 9. Head End Work Port                   | 17. Anti-void Check Valve               |
| 2. Flow Control Valve                     | 10. Tank Return Port                    | 18. Load Check Valve                    |
| 3. Secondary Low Pressure Valve (250 psi) | 11. Supply Port                         | 19. High Pressure Passage               |
| 4. Snubber Valve                          | 12. Pilot Supply Port                   | 20. Low Pressure Passage                |
| 5. Rod End Work Port                      | 13. Hoist Limit Solenoid                | 21. Pilot Operated Check Valve          |
| 6. Hoist Cylinders                        | 14. Raise Pilot Port                    | 22. Primary Low Pressure Valve (26 psi) |
| 7. Rod End Spool                          | 15. Down Pilot Port                     | 23. Brake Cooling Circuit Port          |
| 8. Head End Spool                         | 16. Power Down Relief Valve (1,500 psi) | 24. Overcenter Manifold                 |

## **Power Down Operation (Figure 7-8)**

When the operator moves the hoist control lever to lower the body, the hoist pilot valve is positioned to direct the pilot supply oil in ports (12) through ports (15) to the top of rod end spool (7).

Pilot pressure increases to move the spool down, compressing the bottom spring. Movement of the spool connects high pressure passage (19) to the rod end (annulus area) of the hoist cylinders. At the same time, flow control valve (2) is forced to close as pilot pressure increases, thus directing the incoming pump oil to the hoist cylinders through spool (7) and the check valve in the overcenter manifold rather than back to the tank. Oil flow to the brake cooling circuit ceases when the flow control valve closes.

If the body is at the maximum up position, the hoist limit switch has the hoist up limit solenoid activated, therefore closing raise port (14) on the hoist valve. Power down pilot pressure in ports (15) pushes open pilot operated check valve (21), located in the bleed-down manifold, so the pilot pressure in ports (14) is open to the tank through the pilot valve spool.

As oil attempts to return from the head end of the hoist cylinders, it initially encounters closed head end spool (8). Pressure increases on the bottom end of the spool, causing it to move upward. This allows the returning oil to go into the low pressure passage (20), build up 179 kPa (26 psi) to open primary low pressure relief (3), and exit the hoist valve through port (10) to the tank.

As the body descends and the hoist up limit solenoid is no longer activated, the pilot operated check valve is no longer necessary.

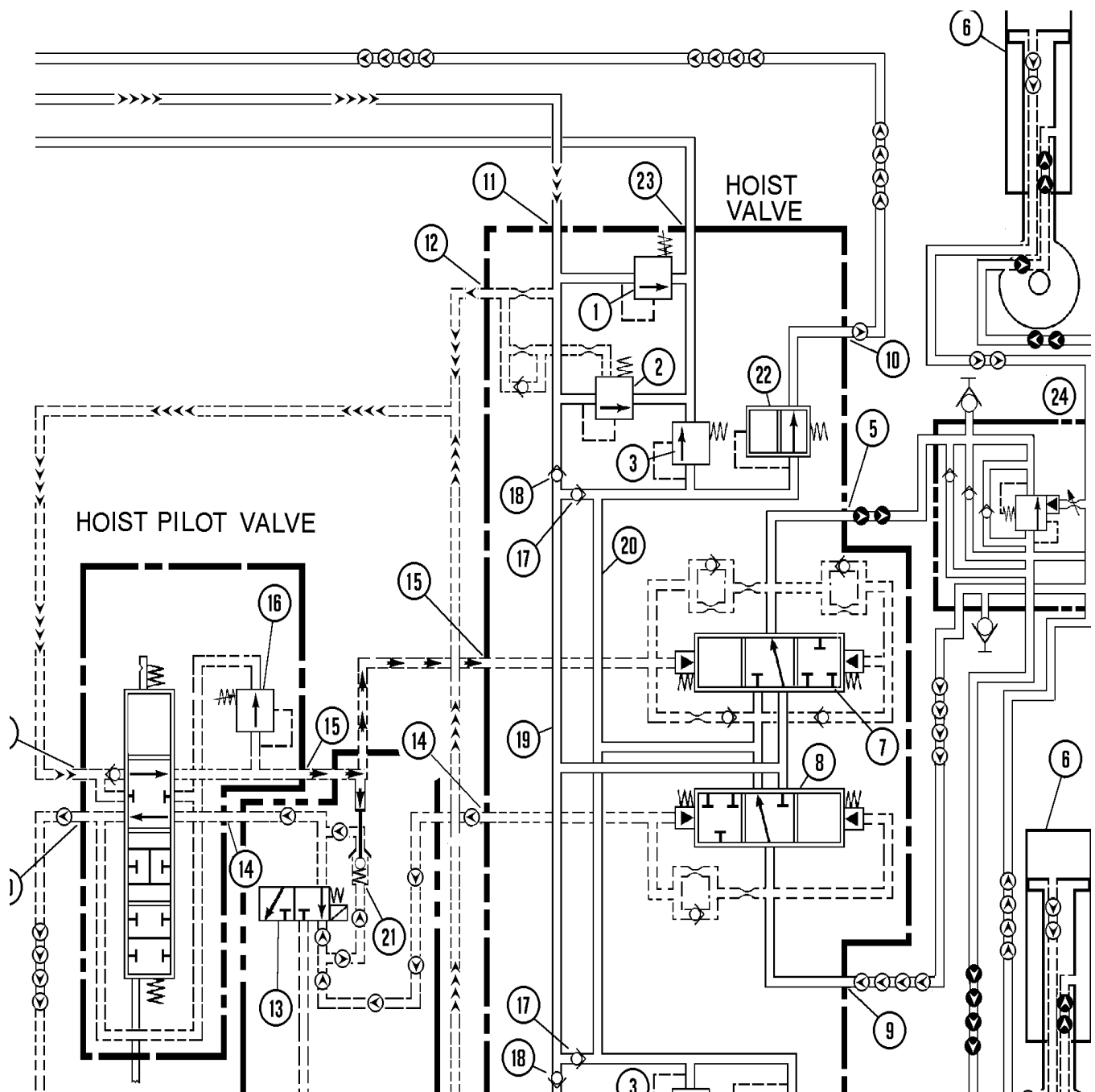


FIGURE 7-8. POWER DOWN POSITION

- |                                           |                                         |                                         |
|-------------------------------------------|-----------------------------------------|-----------------------------------------|
| 1. Hoist Relief Valve (2,500 psi)         | 9. Head End Work Port                   | 17. Anti-void Check Valve               |
| 2. Flow Control Valve                     | 10. Tank Return Port                    | 18. Load Check Valve                    |
| 3. Secondary Low Pressure Valve (250 psi) | 11. Supply Port                         | 19. High Pressure Passage               |
| 4. Snubber Valve                          | 12. Pilot Supply Port                   | 20. Low Pressure Passage                |
| 5. Rod End Work Port                      | 13. Hoist Limit Solenoid                | 21. Pilot Operated Check Valve          |
| 6. Hoist Cylinders                        | 14. Raise Pilot Port                    | 22. Primary Low Pressure Valve (26 psi) |
| 7. Rod End Spool                          | 15. Down Pilot Port                     | 23. Brake Cooling Circuit Port          |
| 8. Head End Spool                         | 16. Power Down Relief Valve (1,500 psi) | 24. Overcenter Manifold                 |

## **Float Operation (Figure 7-9)**

When the operator releases the lever as the body travels down, the hoist pilot valve spool returns to the FLOAT position. In this position, all ports (10, 12, 14, & 15) are common with each other. Therefore, the pilot supply oil is returning to tank with no pressure build-up, thus allowing flow control valve (2) to remain open to allow the pump oil to flow through hoist valve port (23) to the brake cooling circuit and eventually return to the tank.

With no blockage of either RAISE or DOWN pilot ports (14) and (15) in the pilot valve, there is no pressure on the top of either main spool. The oil returning from the head end of the hoist cylinders builds pressure on the bottom of the head end spool (8) exactly like in POWER DOWN, allowing the returning oil to transfer to low pressure passage (20). The back pressure in the low pressure passage, created by primary low pressure relief valve (22), causes pressure under rod end spool (7) to move the spool upward. This connects the low pressure passage to the rod end of the hoist cylinders.

The 179 kPa (26 psi) in the low pressure passage causes oil to flow to the rod end of the cylinders to keep them full of oil as they retract. When the body reaches the frame and there is no more oil flow from the cylinders, the main spools center themselves and close the cylinder ports and the high and low pressure passages.



# NOTES

# SECTION L8

## HOIST CIRCUIT COMPONENT REPAIR

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## HOIST CIRCUIT COMPONENT REPAIR

*NOTE: Sometimes it is necessary to dump a load from the body of a truck when the hoist system is inoperable. Refer to Section A3, General Safety and Operating Instructions - Disabled Truck Dumping Procedure.*

### HOIST VALVE

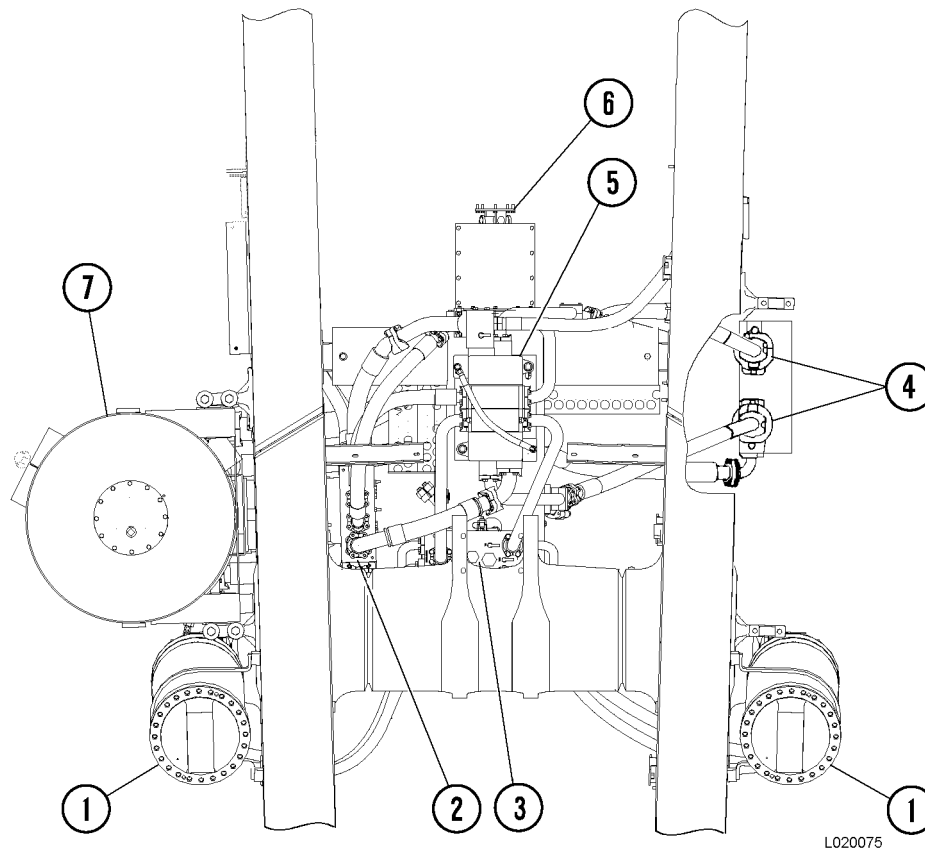
#### Removal



**Relieve all pressure before disconnecting hydraulic lines. Tighten all connections securely before applying pressure.**

**Hydraulic fluid escaping under pressure can have sufficient force to enter a person's body by penetrating the skin and cause serious injury and possibly death if proper medical treatment by a physician familiar with this type of injury is not received immediately.**

1. Make sure that there is adequate overhead clearance and raise the truck body. Secure the body in the raised position with the body-up retention cable. Refer to Section A3, General Safety and Operating Instructions - Securing The Dump Body.
2. Shut down the machine. Refer to Section A3, General Safety and Operating Instructions - Normal Engine Shutdown Procedure.
3. Slowly move the hoist lever to the LOWER position to allow the body to lower against the body-up retention cable and relieve hoist cylinder pressure.
4. Thoroughly clean the exterior of the hoist valve.
5. Mark, disconnect and plug all hydraulic lines and ports to help prevent contamination.
6. Remove the capscrews and lockwashers that secure hoist valve (5, Figure 8-1).



1. Hoist Cylinder
2. Brake/Hoist Return Oil Manifold
3. Overcenter Manifold
4. Hoist Circuit Filters
5. Hoist Valve
6. Pump Drive Shaft
7. Hydraulic Tank

FIGURE 8-1. HOIST VALVE & PIPING

## ⚠ WARNING

**The hoist valve weighs approximately 193 kg (425 lb). Use a suitable lifting device that can handle the load safely.**

7. Attach a lifting device to the hoist valve and remove the valve from the truck. Move the hoist valve to a clean work area for disassembly.

### Installation

1. Attach a suitable lifting device to the hoist valve. Move the hoist valve into position and secure it in place with capscrews, nuts and lockwashers. Tighten the capscrews to the standard torque.

**NOTE:** The hoist valve must be positioned with separator plate (8, Figure 8-2) toward the front of the truck.

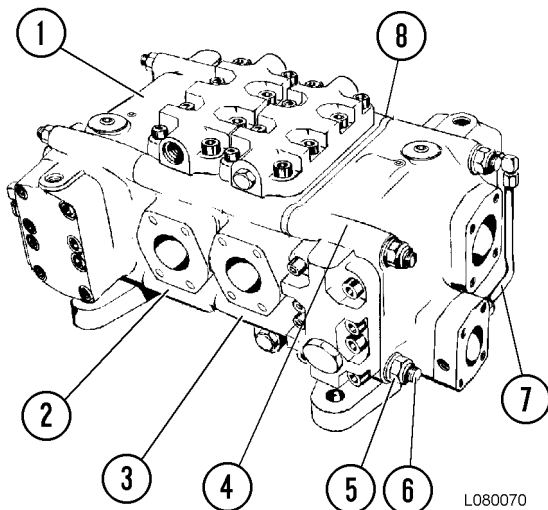


FIGURE 8-2. HOIST VALVE ASSEMBLY

- |                               |                          |
|-------------------------------|--------------------------|
| 1. Inlet Section (Rear)       | 4. Inlet Section (Front) |
| 2. Spool Section (Work Ports) | 5. Nuts and Washers      |
| 3. Spool Section (Tank Ports) | 6. Tie Rods              |
|                               | 7. Tube                  |
|                               | 8. Separator Plate       |

2. Using new O-rings at the flange fittings, connect the hydraulic lines. Tighten the flange capscrews to the standard torque. Refer to Figure 8-1 for hydraulic line location.

3. Connect the pilot supply lines. Tighten the fittings securely.
4. Start the engine. Raise the body and remove the body-up retention cable. Lower and raise the body to check for proper operation and leaks.
5. Check the hydraulic tank oil level. Refer to Section P2, Lubrication and Service - Hydraulic Tank Service.

### O-Ring Replacement

**NOTE:** It is not necessary to remove the individual valve sections to accomplish repairs unless emergency field repair is required to replace the O-rings between sections to prevent leakage. Loosening and retightening the main valve tie rod nut could cause distortion, resulting in binding or severely sticking plungers, poppet and spools.

To replace the O-rings between the valve sections:

1. Remove four tie rod nuts and washers (5, Figure 8-2) from one end of the valve. Slide the tie rods from the valve and separate the sections.
2. Inspect the machined sealing surfaces for scratches and nicks. If scratches or nicks are found, remove them by lapping on a smooth flat steel surface with fine lapping compound.
3. Lubricate the new O-rings lightly with multipurpose grease. Replace the O-rings between the sections. Stack the sections together, making sure that the O-rings between the sections are properly positioned.
4. Install the four tie rods with the dished washer between the nut and housing. See Figure 8-3.

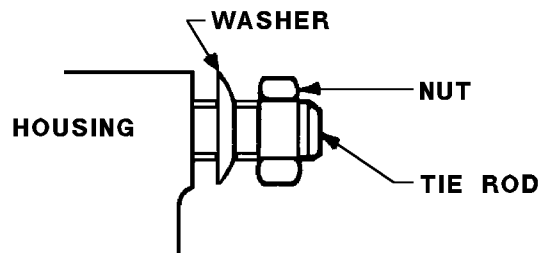


FIGURE 8-3. TIE ROD INSTALLATION

5. Use a torque wrench to tighten the nuts in the pattern shown in Figure 8-4. Tighten the tie rod nuts evenly in the following sequence:

- First, tighten the nuts to **20 N·m (15 ft lb)** in order 1, 4, 2, 3.
- Next, tighten the nuts to **43 N·m (32 ft lb)** in order 1, 4, 2, 3.
- Finally, tighten the nuts to **142 N·m (105 ft lb)** in order 1, 4, 2, 3.

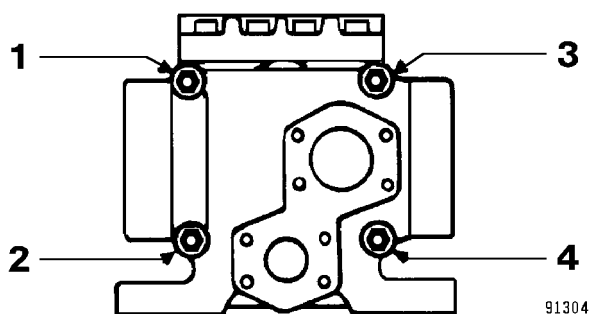


FIGURE 8-4. TIGHTENING SEQUENCE

## INLET SECTION

1. Match mark or identify each part when removed in respect to its location or mating bore to aid in reassembly.
2. Disconnect external tube (7, Figure 8-2) at the cover end. Remove capscrews (14, Figure 8-5), and cover (13). Remove springs (12), check valves (11), and O-rings (10).

*NOTE: The inlet section is shown removed from the main valve body for clarity.*

3. Remove capscrews (1, Figure 8-5) and cover (2). Remove springs (3) and (5) and flow control/main relief valve (4). Remove sleeve (6), low pressure relief valve (7), and O-rings (8).

*NOTE: If removal of the restrictor poppet in cover (2) is required, refer to Step 4 and Figure 8-6.*

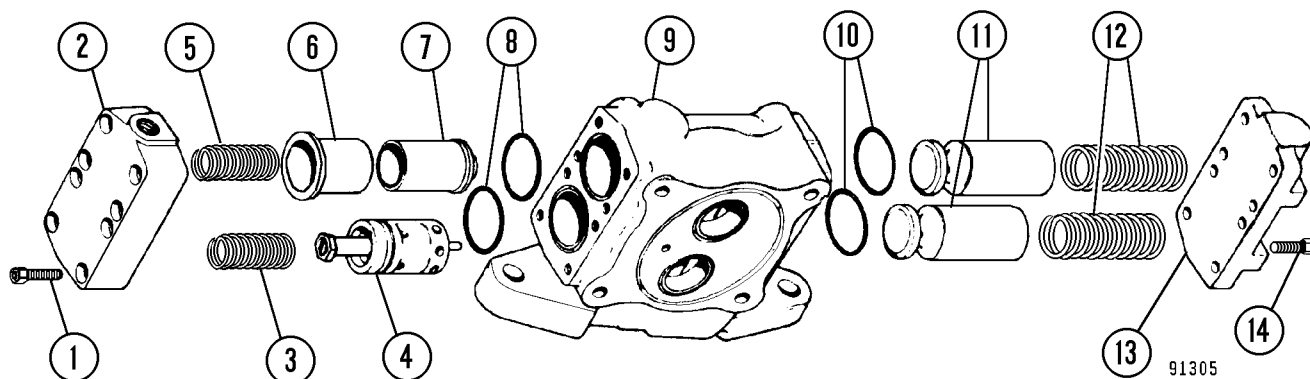


FIGURE 8-5. INLET SECTION DISASSEMBLY

- |                                     |                                        |                     |               |
|-------------------------------------|----------------------------------------|---------------------|---------------|
| 1. Capscrew                         | 5. Spring                              | 8. O-Rings          | 12. Springs   |
| 2. Inlet Cover                      | 6. Sleeve                              | 9. Inlet Valve Body | 13. Cover     |
| 3. Spring (Orange)                  | 7. Secondary Low Pressure Relief Valve | 10. O-Rings         | 14. Capscrews |
| 4. Flow Control & Main Relief Valve |                                        | 11. Check Valves    |               |

4. Remove sleeve (9, Figure 8-6), backup ring (8), O-ring (7), backup ring (6). Remove backup ring (5), O-ring (4), backup ring (3) and restrictor poppet (2).
5. Repeat Steps 1 through 4 for the opposite inlet section if disassembly is required.

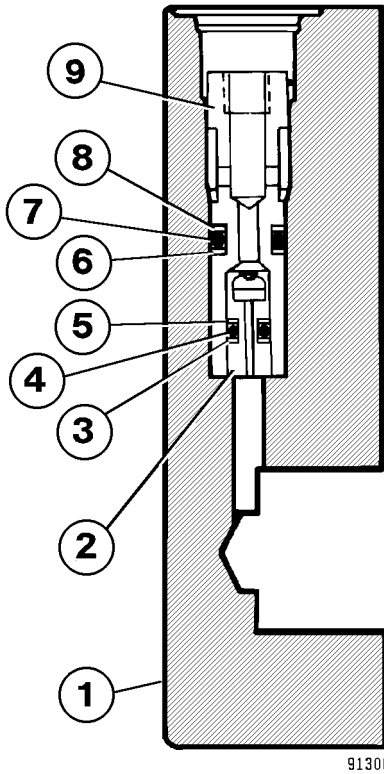


FIGURE 8-6. RESTRICTOR POPPET REMOVAL  
(Inlet Cover)

- |                      |                |
|----------------------|----------------|
| 1. Inlet Cover       | 6. Backup Ring |
| 2. Restrictor Poppet | 7. O-ring      |
| 3. Backup Ring       | 8. Backup Ring |
| 4. O-Ring            | 9. Sleeve      |
| 5. Backup Ring       |                |

#### Cleaning and Inspection

1. Discard all O-rings and backup rings. Clean all parts in solvent and dry with compressed air.
2. Inspect all springs for breaks and distortion. Inspect poppet seating surfaces for nicks or excessive wear. All seats must be sharp and free of nicks.
3. Inspect all bores and surfaces of sliding parts for nicks, scores and excessive wear.
4. Inspect the poppets in their respective bore for proper fit. Poppets should move freely through a complete revolution without binding.
5. Inspect the fit and movement between the sleeve and low pressure relief valve.

#### Assembly

1. Coat all parts, including the housing bores, with clean, type C-4 hydraulic oil. Lightly lubricate the O-rings with a multipurpose grease.
2. If restrictor poppet (2, Figure 8-6) was removed, reassemble in the order shown.
3. Install check valves (11, Figure 8-5) in their respective bores. Install springs (12).
4. Install O-rings (10) and cover (13). Install and tighten capscrews (14) to **81 N·m (60 ft lb)**.
5. Install low pressure relief valve (7) in sleeve (6), then install the assembly in housing (9). Install flow control/main relief valve (4). Install springs (3) and (5). Install cover (2). Install and tighten capscrews (1) to **81 N·m (60 ft lb)**. Connect the external tube and tighten the nuts to **34 N·m (25 ft lb)**.

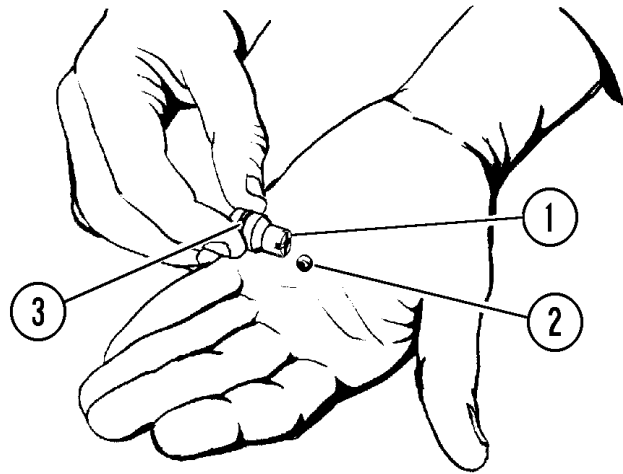


FIGURE 8-7. POPPET AND BALL

1. Poppet
2. Steel Ball
3. O-Ring

## REAR SPOOL SECTION (Work Ports)

### Disassembly

**NOTE:** It is not necessary to remove inlet sections (1) or (4, Figure 8-2) to accomplish disassembly of spool section (2) or (3).

1. Match mark or identify each part when removed in respect to its location or respect to its mating bore to aid reassembly.
2. Remove capscrews and lift spool section cover (1, Figure 8-9) from the housing.
3. Remove poppet (1, Figure 8-7) from spool cover. Remove and discard O-ring (3).

**NOTE:** The poppet (1) contains a small steel ball. Do not misplace.

4. Remove and discard O-rings (4) and (5, Figure 8-8).
5. Remove restrictor poppet (1). Remove and discard O-ring (2) and backup ring (3), if used. Note the position of the restrictor when removed to ensure correct reassembly.
6. Remove spool assembly (20, Figure 8-9). Note the color of the lower spring (blue) to ensure proper location during reassembly. Also note the "V" groove on the top end of the spool.

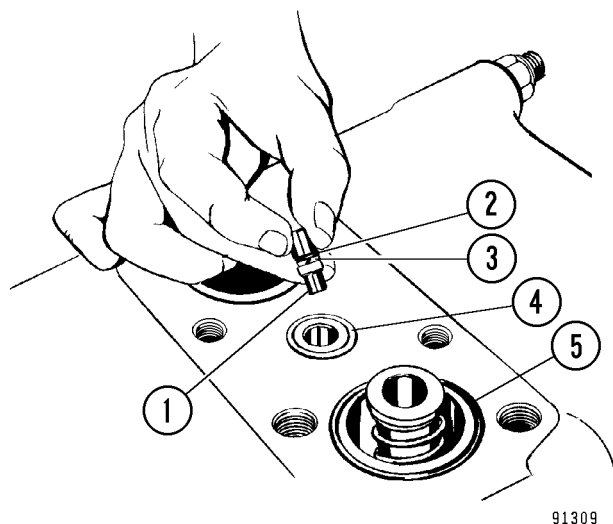


FIGURE 8-8. RESTRICTOR POPPET REMOVAL

- |                      |           |
|----------------------|-----------|
| 1. Restrictor Poppet | 4. O-Ring |
| 2. O-Ring *          | 5. O-Ring |
| 3. Backup Ring *     |           |

**\*Note:** Items 2 and 3 not used on all valves.

7. Remove plug (3) from the end of the spool. Remove spring seat (2) and spring (11). Remove poppet (21) and spool end (15).

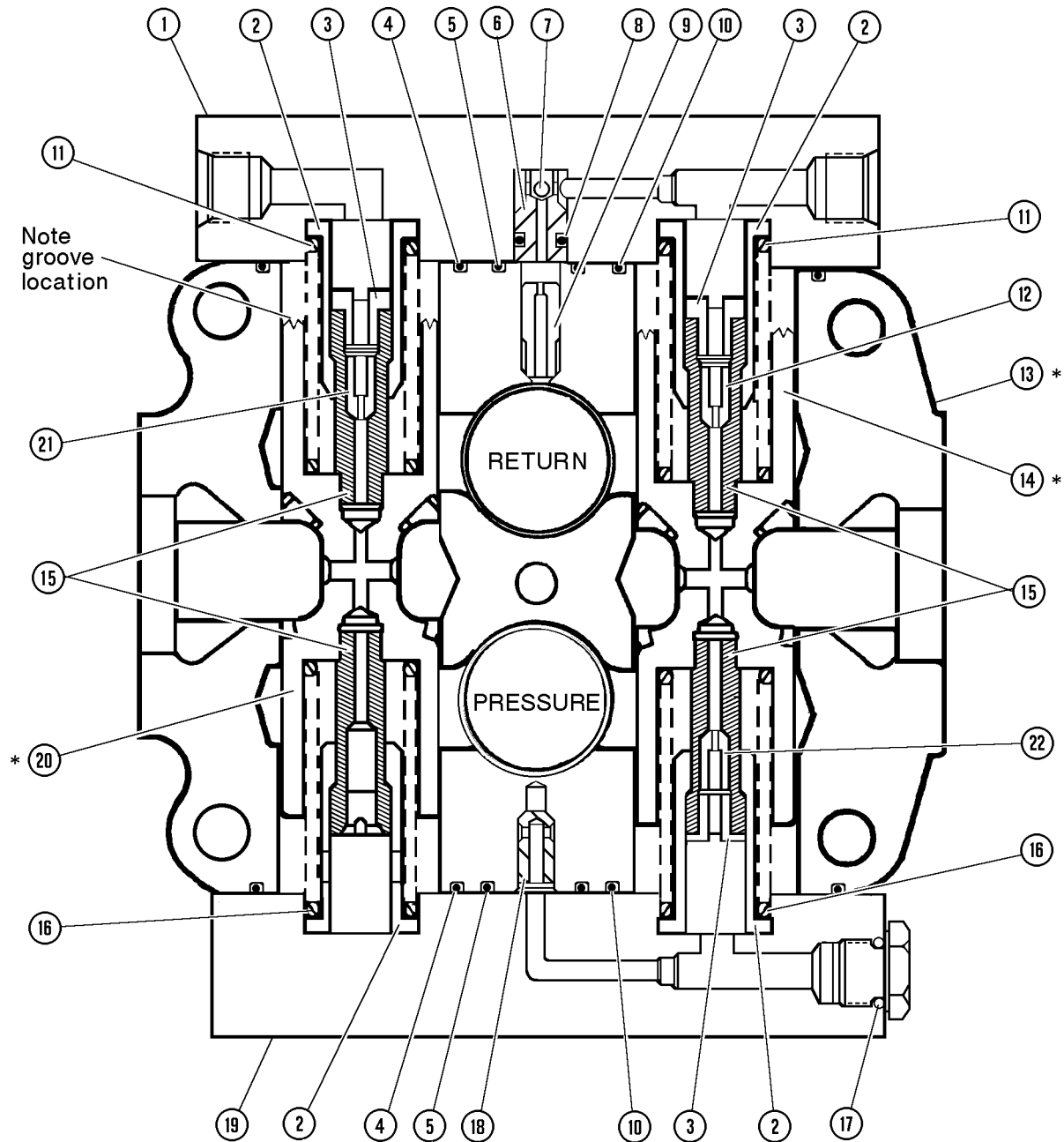
**NOTE:** Pay special attention to poppets (12), (21) and (22, Figure 8-9) during removal to ensure proper location during reassembly. Poppets may be identified with a colored dot (red, green or white). If the poppets are not color coded, use the table below and the specified drill bit to measure the orifice diameter for proper identification.

| POPPET COLOR | ORIFICE DIAMETER    | DRILL SIZE |
|--------------|---------------------|------------|
| Red          | 3.556 mm (.140 in.) | #28        |
| Green        | 2.362 mm (.093 in.) | #42        |
| White        | 1.600 mm (.063 in.) | #52        |

8. Repeat Step 7 to disassemble the opposite end of spool (20). Note that there is no plug or restrictor poppet in the opposite end and the spring is blue.
9. Remove spool assembly (14). At the top end of the spool, remove plug (3). Remove spring seat (2) and spring (11). Remove poppet (12) and spool end (15).
10. At the opposite end, remove plug (3), spring seat (2) and spring (16). Remove restrictor poppet (22) and spool end (15).
11. Remove cover (19) and O-rings (4), (5) and (10). Remove poppet (18).

### Cleaning and Inspection

1. Discard all O-rings and backup rings. Clean all parts in solvent and blow dry with compressed air.
2. Inspect all springs for breaks and distortion. Inspect the poppet seating surfaces for nicks and excessive wear. All seats must be sharp and free of nicks.
3. Inspect all bores and surfaces of sliding parts for nicks, scores and excessive wear.
4. Inspect all poppets in their respective bore for proper fit. Poppets should move freely through a complete revolution without binding.



\* Not Serviceable

L080043A

FIGURE 8-9. WORK PORTS SPOOL SECTION ASSEMBLY

- |                |                             |                               |
|----------------|-----------------------------|-------------------------------|
| 1. Spool Cover | 9. Restrictor Poppet        | 17. O-Ring                    |
| 2. Spring Seat | 10. O-Ring                  | 18. Poppet                    |
| 3. Plug        | 11. Spring                  | 19. Spool Cover               |
| 4. O-Ring      | 12. Restrictor Poppet (Red) | 20. Spool Assembly            |
| 5. O-Ring      | 13. Spool Housing           | 21. Restrictor Poppet (Green) |
| 6. Poppet      | 14. Spool Assembly          | 22. Restrictor Poppet (White) |
| 7. Ball        | 15. Spool End               |                               |
| 8. O-Ring      | 16. Spring (Blue)           |                               |

## Assembly

1. Lubricate O-rings (4), (5) and (10, Figure 8-9) with clean hydraulic oil. Install the O-rings in the spool housing. Install poppet (18). Install cover (19) and secure it in place with capscrews. Tighten the capscrews to **81 N·m (60 ft lb)**.
2. Install spring (11) in spool (20). Install spring seat (2). Apply Loctite to the threads of spool end (15). Install the spool end and tighten it to **34 N·m (25 ft lb)**. Install green poppet (21). Apply Dri-loc #204 to the threads of plug (3). Install and tighten the plug to **20 N·m (15 ft lb)**.

*NOTE: Poppets (12), (21) and (22) may be color coded and must be installed in their original location.*

3. Repeat Step 2 for the opposite end of spool (20) to install the spring, spring seat, and spool end. Ensure that spring (16) is blue. A poppet and plug are not installed in the lower end.
4. Lubricate the spool assembly and carefully install it in the spool housing as shown in Figure 8-10. Make sure that "V" groove (1) in spool assembly (2) is positioned **up** as noted.
5. Install spring (11, Figure 8-9) in the top (grooved) end of remaining spool (14). Install spring seat (2). Apply Loctite to the threads of spool end (15). Install spool end and tighten to **34 N·m (25 ft lb)**. Install red poppet (12). Apply Dri-loc #204 to the threads of plug (3). Install and tighten the plug to **20 N·m (15 ft lb)**.
6. Repeat Step 5 for the bottom end of spool (14). Install spring (16) which is blue in color, spring seat (2) and spool end (15). Apply Loctite to spool end threads. Install spool end (15) and tighten to **34 N·m (25 ft lb)**. Install white poppet (22). Apply Dri-loc #204 to the threads of plug (3). Install and tighten the plug to **20 N·m (15 ft lb)**.
7. Lubricate the assembled spool and install it in the spool housing as shown in Figure 8-10. Make sure that "V" groove (1) in spool assembly (2) is positioned **up** as noted.

*NOTE: Spools (14) and (20, Figure 8-9) are physically interchangeable. Ensure that spool (14) is installed toward the base port of the spool housing.*

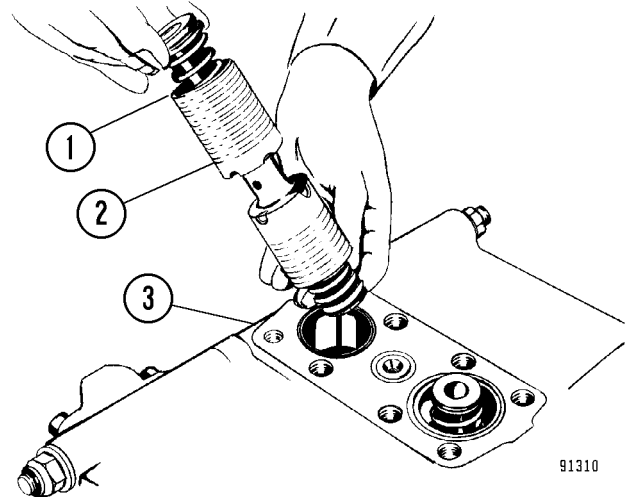


FIGURE 8-10. SPOOL INSTALLATION

- |                   |                             |
|-------------------|-----------------------------|
| 1. "V" Groove     | 3. Work Ports Spool Section |
| 2. Spool Assembly |                             |
8. If used, install O-ring (2, Figure 8-8) and backup ring (3) on restrictor poppet (1). Install the poppet in the spool housing as shown in Figure 8-9.
  9. Install new O-rings (4), (5) and (10).
  10. Install a new O-ring (and backup ring if used) on poppet (6, Figure 8-9). Make sure that small steel ball (7) is installed in the poppet. Install the poppet in cover (1).
  11. With new O-rings (4, 5 and 10) installed, position cover (1) over the spool housing. Install and tighten the capscrews to **81 N·m (60 ft lb)**.

*NOTE: The cover must be positioned according to the match marks that were made during disassembly. Do not reverse the cover position.*



## FRONT SPOOL SECTION (Tank Ports)

### Disassembly

*NOTE: It is not necessary to remove inlet sections (1) or (4, Figure 8-2) to accomplish spool section disassembly and repair.*

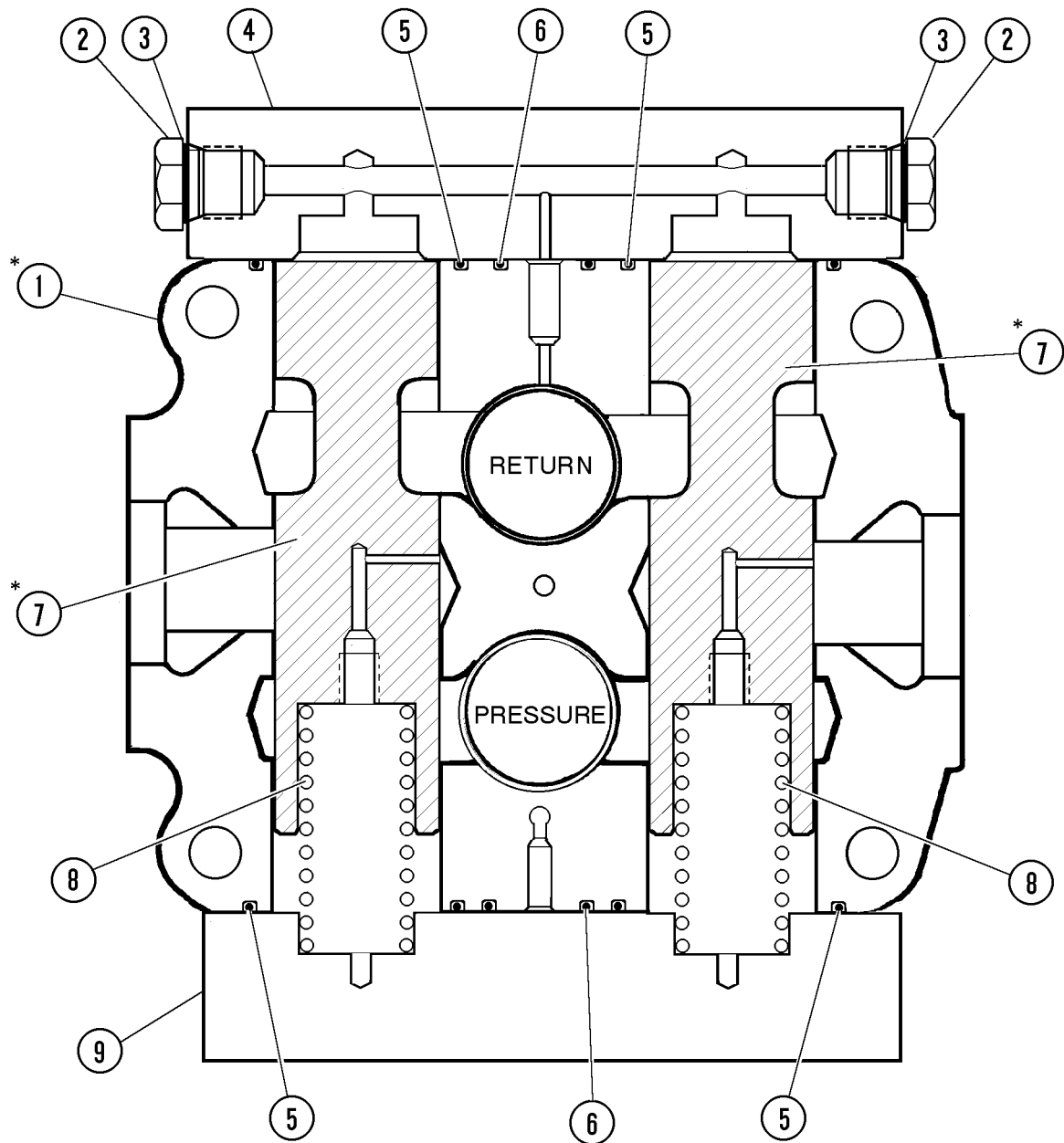
1. Match mark or identify each spool in respect to its mating bore when removed.
2. Remove the capscrews and lift spool section cover (4, Figure 8-11) from the housing.
3. Remove and discard O-rings (5) and (6).
4. Remove the capscrews and bottom cover (9) from the spool housing. Remove and discard the O-rings.
5. Remove springs (8). Push spools (7) out of the housing bores.

### Cleaning and Inspection

1. Discard all O-rings. Clean all parts in solvent and blow dry with compressed air.
2. Inspect the springs for breaks and distortion.
3. Inspect the housing bores and surfaces of the spools for nicks, scoring and excessive wear.

### Assembly

1. Lubricate spools (7, Figure 8-11) with clean hydraulic oil and reinstall them in their original bores. Install springs (8).
2. Lubricate O-rings (5) and (6) with clean hydraulic oil. Install the O-rings in the bottom of the spool housing. Install bottom cover (9) and secure it in place with capscrews. Tighten the capscrews to **81 N·m (60 ft lb)**.
3. Lubricate O-rings (5) and (6) and install them in the top of the spool housing. Install top cover (4) and secure it in place with capscrews. Tighten the capscrews to **81 N·m (60 ft lb)**.
4. If removed, install plugs (2) with new O-rings (3).



\* Not Serviceable

L080051

FIGURE 8-11. TANKS PORTS SPOOL SECTION ASSEMBLY

1. Spool Housing  
2. Plug  
3. O-Ring

4. Spool Cover  
5. O-Ring  
6. O-Ring

7. Spool  
8. Spring  
9. Spool Cover

## HOIST PILOT VALVE

### Removal

1. Place the hoist control lever in the body down position. Ensure that the body is in the full down position and resting on the frame. Release the hoist control lever to return the hoist valve spool to the FLOAT position.
2. Disconnect hydraulic lines (3, Figure 8-12) at hoist pilot valve (1) located in the hydraulic components cabinet at the rear of the cab. Remove capscrews (5).
3. Loosen and unthread jam nut (8). Unthread sleeve (9) until cotter pin (6) and pin (10) are exposed.
4. Remove cotter pin and pin.
5. Remove hoist pilot valve mounting hardware (2), and remove the valve from the cabinet. Refer to the hoist pilot valve disassembly procedure for repair instructions.

### Installation

1. Place hoist pilot valve (1, Figure 8-12) into position on the mounting bracket. Secure the valve in place with mounting hardware (2).
2. Position hydraulic lines (3) over the valve ports and assemble the fittings. Tighten the hydraulic line connections securely.
3. Place the hoist control lever in the spring-centered position. Adjust the pilot valve spool until the centerline of the cable attachment hole extends 29.5 mm (1.16 in.) from the face of the valve body.
4. Align the control cable eye with the pilot valve spool hole and insert pin (10). Secure the pin in place with cotter pin (6).
5. Thread sleeve (9) upward until contact is made with the valve body. Move flange (4) into position and secure it in place with capscrews (5).
6. Thread jam nut (8) against the sleeve. Tighten the jam nut securely.
7. Start the engine and check for proper hoist operation and leaks.

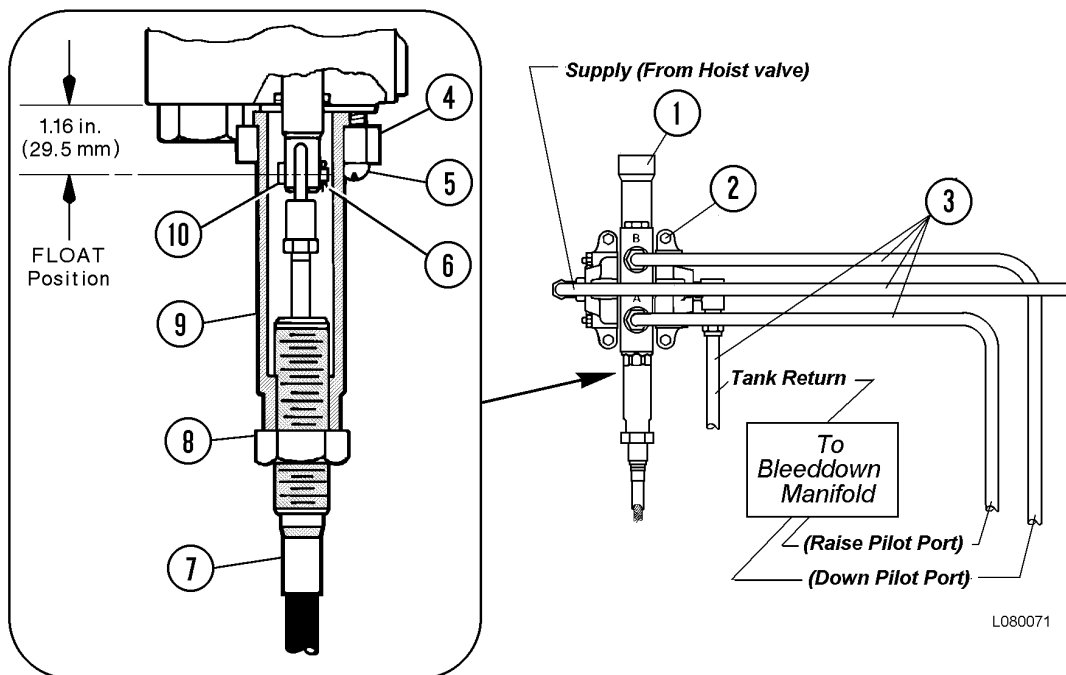


FIGURE 8-12. HOIST PILOT VALVE & PIPING  
(Located in Hydraulic Brake Cabinet)

- |                      |                  |           |
|----------------------|------------------|-----------|
| 1. Hoist Pilot Valve | 5. Capscrew      | 9. Sleeve |
| 2. Mounting Hardware | 6. Cotter Pin    | 10. Pin   |
| 3. Hydraulic Tubes   | 7. Control Cable |           |
| 4. Flange            | 8. Jam Nut       |           |

## Disassembly

1. Thoroughly clean the exterior of the valve. Match mark the components to ensure proper reassembly.

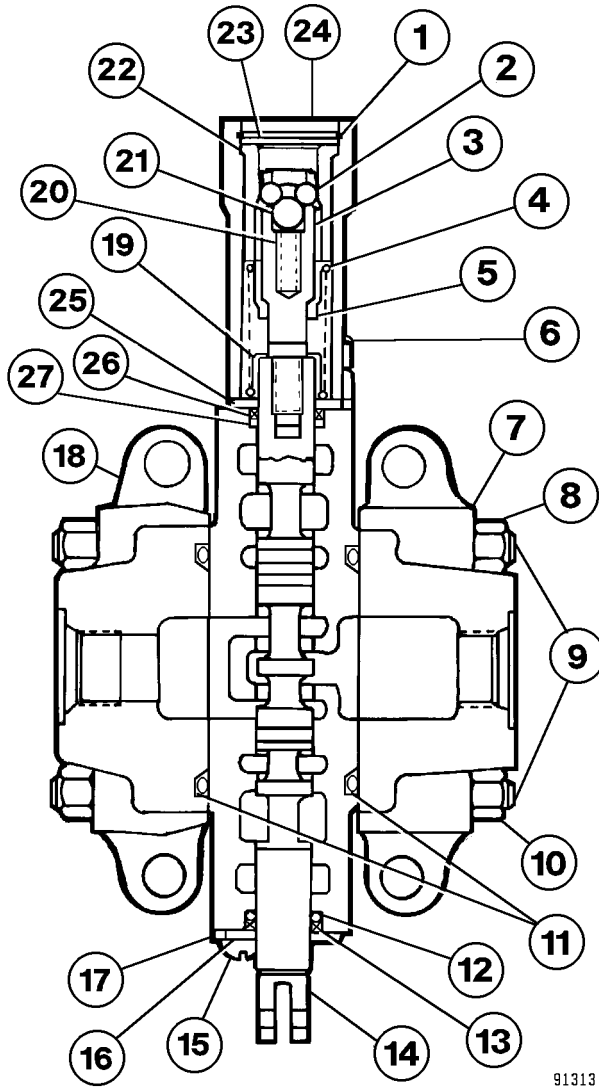


FIGURE 8-13. HOIST PILOT VALVE

- |                   |                   |
|-------------------|-------------------|
| 1. Snap Ring      | 15. Machine Screw |
| 2. Ball (4)       | 16. Seal Plate    |
| 3. Detent Pin     | 17. Spool Housing |
| 4. Spring         | 18. Inlet Housing |
| 5. Spacer         | 19. Spring Seat   |
| 6. Capscrew       | 20. Spring        |
| 7. Outlet Housing | 21. Ball (1)      |
| 8. Nut            | 22. Detent Sleeve |
| 9. Tie Rod        | 23. Spacer        |
| 10. Nut           | 24. Cap           |
| 11. O-Ring        | 25. Seal Retainer |
| 12. O-Ring        | 26. Wiper         |
| 13. Wiper         | 27. O-Ring        |
| 14. Spool         |                   |

2. Remove machine screw (15, Figure 8-13), seal plate (16), wiper (13) and O-ring (12).
3. Remove snap ring (1), capscrews (6), cap (24), spacer (23), and detent sleeve (22). Detent balls (2) and (21) will fall free when the cap and detent sleeve are removed. Separate cap (24), spacer (23) and detent sleeve (22), as this will be necessary for reassembly.
4. Carefully slide spool (14) out of spool housing (17). Remove seal retainer (25), wiper (26) and O-ring (27) from the spool.
5. Insert a rod in the cross holes of detent pin (3) and unscrew it from spool (14). Slight pressure should be exerted against the detent pin as it disengages and spring tension is released.
6. Remove spring seats (19), spring (4) and spacer (5).
7. Remove relief valve (2, Figure 8-14) from spool housing (1).
8. Match mark the inlet and outlet housings in relation to the spool housing to ensure correct location during reassembly.
9. Remove nuts (8) and (10, Figure 8-13) and remove tie rods (9). Separate the valve housings. Remove O-ring (11). Remove the poppet check and spring (located on the outlet housing side of the spool housing) from the spool housing.

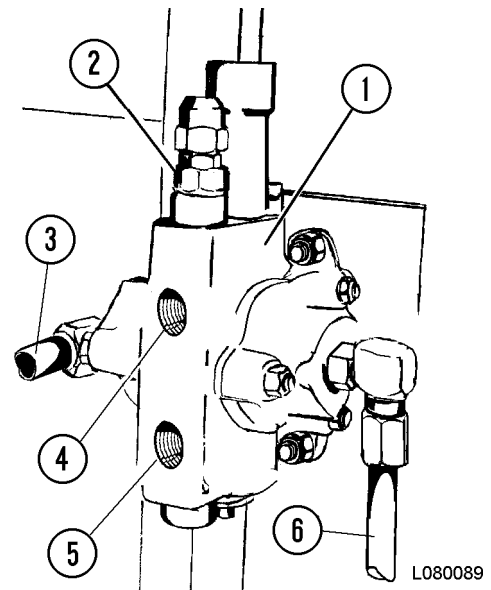


FIGURE 8-14. RELIEF VALVE

- |                             |                              |
|-----------------------------|------------------------------|
| 1. Spool Housing            | 5. To Hoist Valve (Base End) |
| 2. Relief Valve             | 6. Return to Tank            |
| 3. Supply Port              |                              |
| 4. To Hoist Valve (Rod End) |                              |

## Cleaning and Inspection

1. Clean all parts, including the housings, in solvent and blow dry with compressed air.
2. Inspect the seal counter bores. They must be free of nicks and grooves.
3. Examine springs for breaks or distortion.
4. Inspect spool (14, Figure 8-13). The spool must be free of longitudinal score marks, nicks or grooves.
5. Test spool (14) in spool housing for fit. The spool must fit and rotate freely through a complete revolution without binding.

*NOTE: Spool housing (17), spool (14), inlet housing (18) and outlet housing (7) are not serviced separately. If any of these parts require replacement, the entire control valve must be replaced.*

## Assembly

1. Thoroughly coat all parts, including the housing bores, with clean type C-4 hydraulic oil.
2. If the inlet and outlet housings were removed, perform Steps 3 through 5 for reassembly.
3. Install check poppet (2, Figure 8-15) and spring (3) in spool housing (1).
4. Install new O-ring (4) in spool housing. Place the inlet and outlet housings on the spool housing.

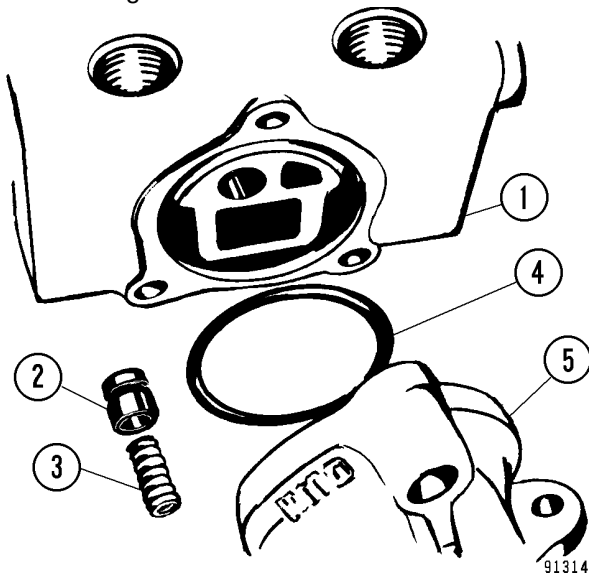


FIGURE 8-15. HOIST PILOT VALVE REASSEMBLY

- |                  |                   |
|------------------|-------------------|
| 1. Spool Housing | 4. O-Ring         |
| 2. Check Poppet  | 5. Outlet Housing |
| 3. Spring        |                   |

5. Install the tie rods. Install and tighten the tie rod nuts to the torques shown in Figure 8-16.
6. Install new O-ring (27, Figure 8-13) and wiper (26). Install seal retainer (25).
7. Install spacer (5), spring seats (19), and spring (4). Thread detent pin (3) into spool (14). Slight pressure will be required to compress the detent spring. Tighten the detent pin to **10 N·m (89 in. lb)**. Install spring (20). Carefully install the spool into the spool housing.
8. Apply grease to the cross holes of the detent pin (3) to hold balls (21) and (2).
9. Slide detent sleeve (22) into cap (24) and place it over a punch. Use the punch to depress ball (21) and insert balls (2) in detent pin cross holes.
10. While holding down ball (21), slide detent sleeve (22) and cap (24) as an assembly over detent pin (3). Continue to insert detent sleeve (22) until it contacts spring seat (19).
11. Secure cap (24) in place with capscrews (6). Tighten capscrews (6) to **7 N·m (62 in. lb)**. Install spacer (23) and snap ring (1).
12. Install new O-ring (12) and wiper (13). Install seal plate (16) and machine screws (15).
13. Using new O-rings, install relief valve (2, Figure 8-14) in the spool housing.

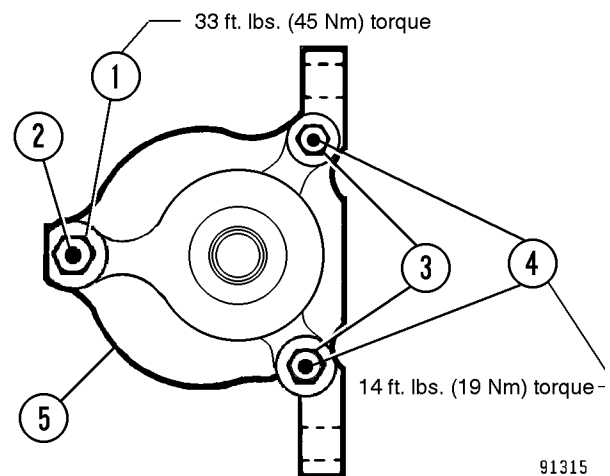


FIGURE 8-16. TIE ROD NUT TORQUE

- |            |                   |
|------------|-------------------|
| 1. Nut     | 4. Tie Rod        |
| 2. Tie Rod | 5. Outlet Housing |
| 3. Nut     |                   |

## HOIST CYLINDERS

### Removal

# **⚠ DANGER**

**Relieve pressure before disconnecting hydraulic lines. Tighten all connections securely before applying pressure.**

**Hydraulic fluid escaping under pressure can have sufficient force to enter a person's body by penetrating the skin and cause serious injury and possibly death if proper medical treatment by a physician familiar with this type of injury is not received immediately.**

1. Ensure that the engine and key switch have been OFF for at least 90 seconds to allow the accumulator to bleed down. Ensure that the park brake is applied.
2. Disconnect the lubrication lines to the upper and lower bearings of the hoist cylinder.
3. Remove the capscrew and lockwashers from the clamps that secure the hydraulic hoses to the hoist cylinder. Cap and plug all lines and ports to prevent excessive spillage and contamination. Secure the cylinder to the frame to prevent movement.

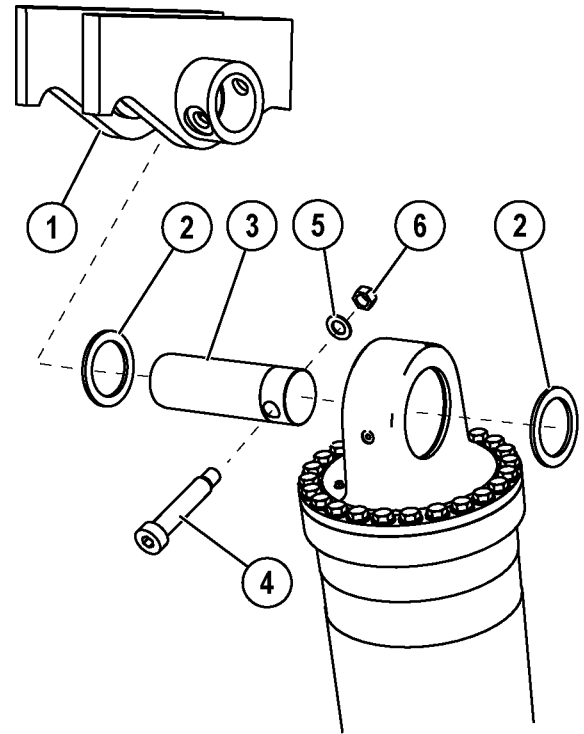
# **⚠ WARNING**

**The hoist cylinder weighs approximately 1 135 kg (2,500 lb). Some means of support is necessary to prevent it from falling or causing injury when removing it from the truck. Use a suitable lifting device that can handle the load safely.**

4. At the hoist cylinder upper mounting bracket, remove nut (6, Figure 8-17), flat washer (5) and shoulder bolt (4). Use a brass drift and hammer to drive pivot pin (3) from mounting bracket (1).

**NOTE:** The pivot pins may be retained with a capscrew and locknut as originally shipped from the factory. Upgrading to the current shoulder bolt configuration is recommended.

**NOTE:** Do not lose two spacers (2) between the cylinder bearing and mounting bracket.



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FIGURE 8-17. HOIST CYLINDER UPPER MOUNT

- |                     |                  |
|---------------------|------------------|
| 1. Mounting Bracket | 4. Shoulder Bolt |
| 2. Spacer           | 5. Flat Washer   |
| 3. Pin              | 6. Nut           |

5. Carefully lower the cylinder until it lies against the inside dual tire. Attach a suitable lifting device to the upper cylinder mounting eye.
6. Install a retaining strap or chain to prevent the cylinder from extending while handling.
7. At the lower mount, straighten the locking plate tabs to allow removal of the capscrews. Remove all capscrews (1, Figure 8-18), locking plate (2) and retaining plate (3).
8. Carefully remove the cylinder from the frame pivot by pulling it outward. Move the cylinder to a clean area for disassembly.

**NOTE:** Do not lose spacer (6, Figure 8-18) between the cylinder bearing and frame.

9. Clean the exterior of the cylinder thoroughly.

## Installation

### **⚠ WARNING**

**Install a retaining strap or chain to prevent the cylinder from extending during handling. The hoist cylinder weighs approximately 1135 kg (2,500 lb). Use a suitable lifting device that can handle the load safely.**

1. Raise the cylinder into position over the pivot point on the frame. The cylinder should be positioned with the air bleed vent plug on top, toward the front of the truck. Install spacer (6, Figure 8-18). Align the bearing eye with the pivot point and push the cylinder into place.
2. Install retaining plate (3), locking plate (2) and new capscrews (1). Tighten the capscrews to **465 N·m (343 ft lb)**. Bend the locking plate tabs over the capscrew flats.
3. Align the hoist cylinder bearing eye with the bore of hoist cylinder upper mounting bracket (1, Figure 8-17).
4. Align the bolt hole in pivot pin (3) with the bolt hole in mounting bracket (1). Install the pivot pin through the first spacer (2), then align the second spacer with the bore on the other side of hoist cylinder bearing eye. Finish installing the pin through second spacer (2).
5. Install shoulder bolt (4) through each pin.

**NOTE:** Do not use a flat washer under the head of the shoulder bolt. This will ensure that the shoulder bolt does not run out of threads when tightening.

6. Install flat washer (5) and nut (6) onto the threaded end of each shoulder bolt. Tighten the nuts to **339 N·m (250 ft lb)**.

If self-locking nut (PC3201) is being used, tighten the nut and flat washer securely against the shoulder of the bolt.

**NOTE:** If the original capscrew/locknut configuration is being used, tighten the locknut to **203 N·m (150 ft lb)**. Upgrading to the current shoulder bolt configuration is recommended.

7. Lubricate new O-rings with clean hydraulic oil and install the O-rings in the grooves on the hose flange connections. Position the flange clamps over the hoist cylinder ports and secure the flange clamps with capscrews and lockwashers. Tighten the capscrews to the standard torque.

8. Reconnect the lubrication lines for the upper and lower hoist cylinder bearings.
9. Start the engine. Raise and lower the body several times to bleed air from the cylinder. Check for proper operation and leaks.
10. Check the hydraulic tank oil level. Refer to Section P2, Lubrication and Service - Hydraulic Tank Service.

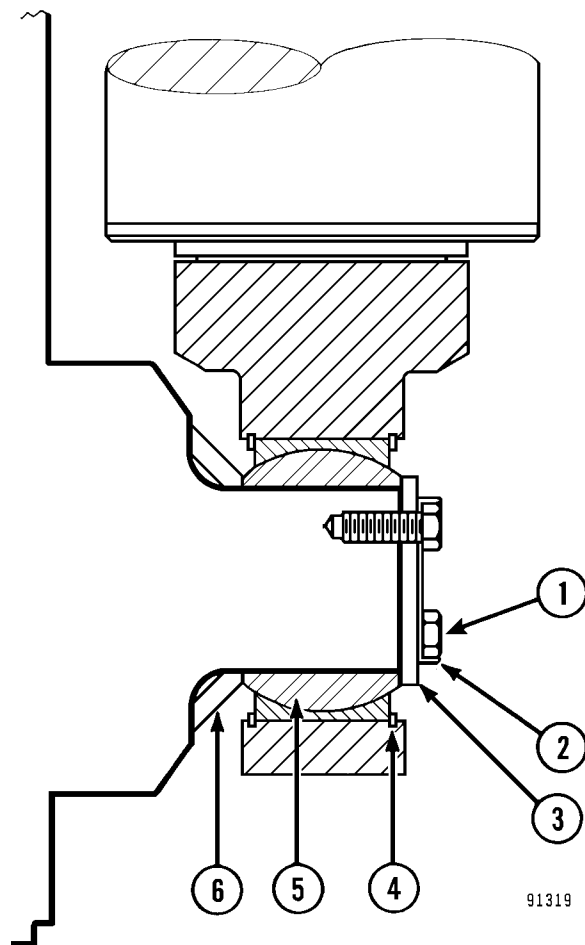
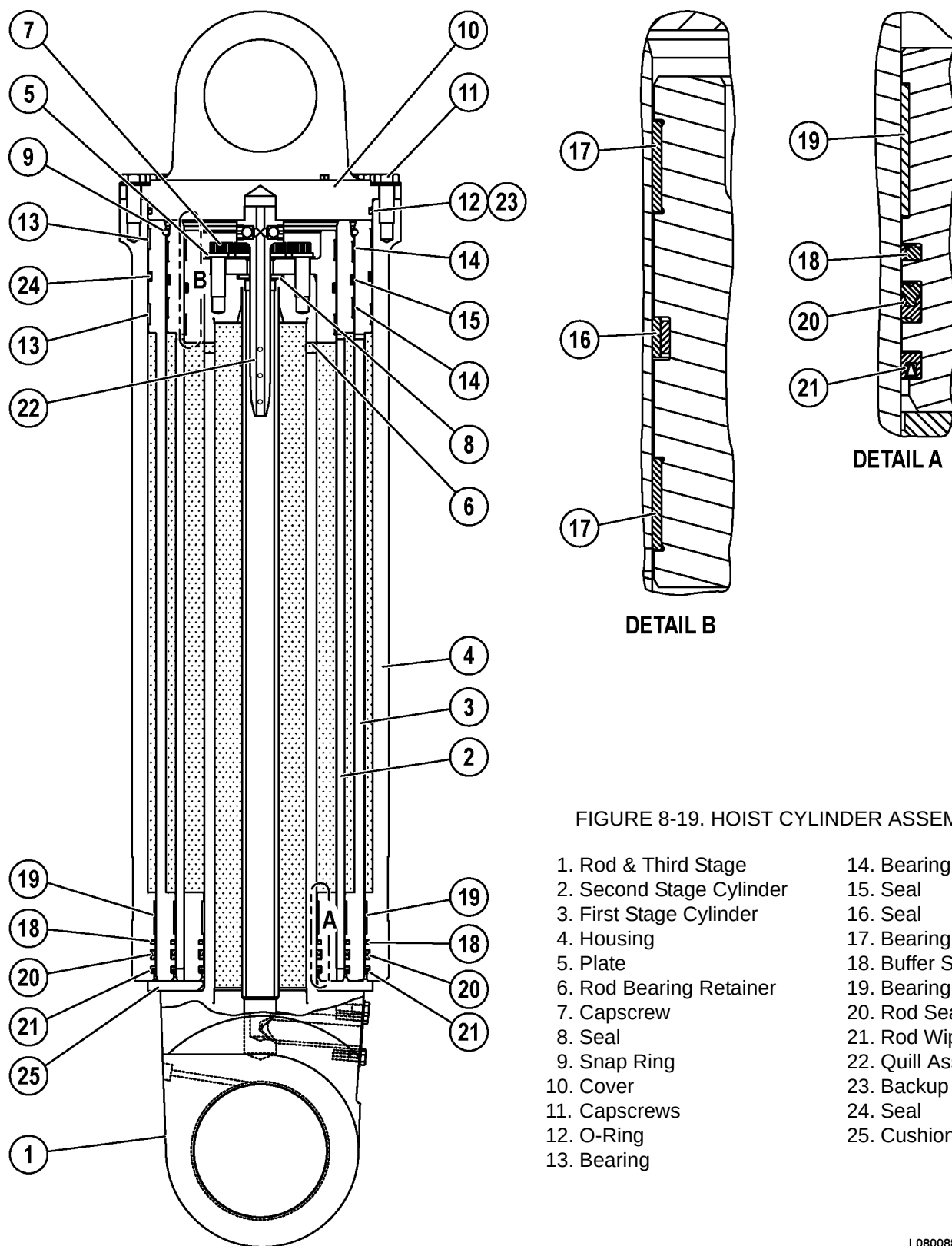


FIGURE 8-18. HOIST CYLINDER LOWER MOUNT

- |                  |                   |
|------------------|-------------------|
| 1. Capscrew      | 4. Retaining Ring |
| 2. Locking Plate | 5. Bearing        |
| 3. Retainer      | 6. Spacer         |



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## Disassembly

1. If removal of the hoist cylinder eye bearings is necessary, remove retainer ring (4, Figure 8-18) and press out bearing (5).
2. Mount the hoist cylinder in a fixture which will allow it to be rotated 180 degrees.
3. Position the cylinder with the mounting eye on cover (10, Figure 8-19) at the top. Remove capscrews (11) and lockwashers the retain the cover to housing (4).
4. Install two threaded capscrews - 22 mm (0.88 in.) diameter x 229 mm (9 in.) length - into the two threaded holes in cover (10). Screw the capscrews in evenly until the cover can be removed. Lift the cover straight up until quill assembly (22) is clear. Remove O-ring (12) and backup ring (23).
5. Remove capscrews (7) and plate (5) that secure rod bearing retainer (6) to rod (1). Remove seal (8).
6. Fabricate a retainer bar using a 6 x 25 x 460 mm (1/4" x 1" x 18") steel flat. Drill holes in the bar to align with a pair of tapped holes spaced 180 degrees apart in the housing. Attach the bar to the housing using capscrews (11).

*NOTE: A retainer bar is required to prevent the first and second stage cylinders from dropping out when the housing is inverted.*

7. Rotate the cylinder assembly 180 degrees to position the lower mounting eye at the top. Hook a lifting device to the eye on rod (1) and lift the rod and third stage cylinder assembly out of the cylinder housing. Remove cushion (25).

*NOTE: As internal parts are exposed during disassembly, protect machined surfaces from scratches or nicks.*

8. Rotate the cylinder housing 180 degrees. Remove the retainer bar that was installed in Step 6.
9. Fabricate a round disc - 318 mm (12.5 in.) diameter x 10 mm (0.38 in.) thick with a 14 mm (0.56 in.) hole in the center. Align the disc over second stage cylinder (2) and first stage cylinder (3) at the bottom of the cylinder housing.
10. Insert a threaded rod - 13 mm (0.50 in.) diameter x 1 350 mm (53 in.) - through the top and through the hole in the round disc. Thread a nut onto the bottom end of the threaded rod below the disc.
11. Screw a lifting eye into the top end of the rod. Attach it to a lifting device, then lift the second and first stage cylinders out of the housing.
12. Remove the lifting tools from the second and first stage cylinder assembly.
13. Slide second stage cylinder (2) down inside first stage cylinder (3). Remove snap ring (9) from inside the first stage cylinder.
14. Remove the second stage cylinder from the first stage cylinder by sliding it out the top.
15. Remove all old bearings, O-rings, and seals from the hoist cylinder parts.

## Cleaning and Inspection

**NOTE:** Use only fresh cleaning solvent, a lint-free wiping cloth and dry, filtered compressed air when cleaning and handling hydraulic cylinder parts. Immediately after cleaning and inspection, coat all surfaces and parts with clean Type C-4 hydraulic oil.

1. Thoroughly clean and dry all parts.
2. Inspect all parts for damage and excessive wear.
3. If the cylinder bores or plated surfaces are excessively worn or grooved, the parts must be replaced or, if possible, replated and machined to original specifications.
4. Check the tightness of quill assembly (2, Figure 8-20) if it has not previously been tack welded.
  - a. Check the tightness of the quill by using special tool SS1143 (Figure 8-20) and applying a tightening torque of **1 356 N·m (1,000 ft lb)**.
  - b. If the quill moves, remove the quill from cap assembly (1). Clean the threads in the cap and on the quill.
  - c. To install the quill, perform the Quill Installation procedure on the next page.
5. When a cylinder assembly is dismantled, check capscrews (7, Figure 8-19) carefully for distress. Replace the capscrews if in doubt.

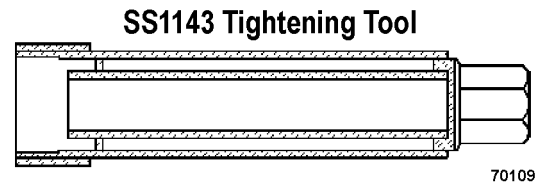
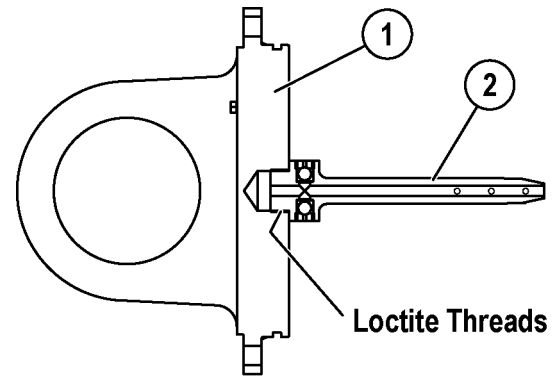


FIGURE 8-20. QUILL INSTALLATION

1. Cap Assembly      2. Quill Assembly

**NOTE:** SS1143 Tightening Tool can be made locally. Request the following drawings from your Area or Regional Service Manager:

### SS1143 Tightening Tool - Assembly Drawing

- SS1144 - Square Tube  
(3.50" x 3.50" x 0.19" wall x 2.0" long)
- SS1145 - Plate  
(2.50" x 2.50" x 0.25" thick)
- SS1146 - Square Tube  
(3.00" x 3.00" x 0.25" wall x 15.50" long)
- SS1147 - Tube, Brass  
(1.75" O.D. x 1.50" I.D. x 13.50" long)
- SS1148 - Square Cut  
(2.50" x 2.50" x 0.75" thick)
- SS1149 - Hex Drive  
(1.75" Hex stock x 2.50" long)

All materials are 1020 Steel except SS1147.

## Assembly - Quill

**NOTE:** Use only new seals, bearings and O-rings during reassembly. Thoroughly lubricate all parts and seals with hydraulic oil to aid in assembly and to provide lubrication during initial operation.

### Quill Installation

1. Check plugs (3, Figure 8-21) and the check balls (4) in the quill during any cylinder repair to ensure that the plugs are tight and the ball seats are not damaged. Refer to Installation of Check Balls and Plugs in Quill on the next page.
2. Secure cap assembly (1) in a sturdy fixture. Ensure that the threads in the cap and the threads on the quill are clean, dry and free of oil and solvent.
3. Spray the mating threads of both cap assembly (1) and quill assembly (2) with LOCQUIC® Primer T (Komatsu part number TL8753) or equivalent. Allow the primer to dry 3 to 5 minutes.
4. Apply Loctite® Sealant #277 (Komatsu part number VJ6863) or equivalent to the mating threads of both the cap assembly and quill assembly.
5. Install the quill. Use tightening tool SS1143 (Figure 8-20) to tighten the quill to **1 356 N·m (1,000 ft lb)**. Allow the parts to cure for two hours before exposing the threaded areas to oil.

**NOTE:** If LOCQUIC Primer T (TL8753) was not used, the cure time will require 24 hours instead of two hours.

6. Tack weld the quill in two places as shown in Figure 8-21.
7. Remove all slag and foreign material from the tack weld area before assembling the cylinder.

During future cylinder rebuilds, removal of the quill will not be necessary unless it has loosened or is damaged. Removal will require a break-loose force of at least **2 712 N·m (2,000 ft lb)** after the tack welds are ground off.

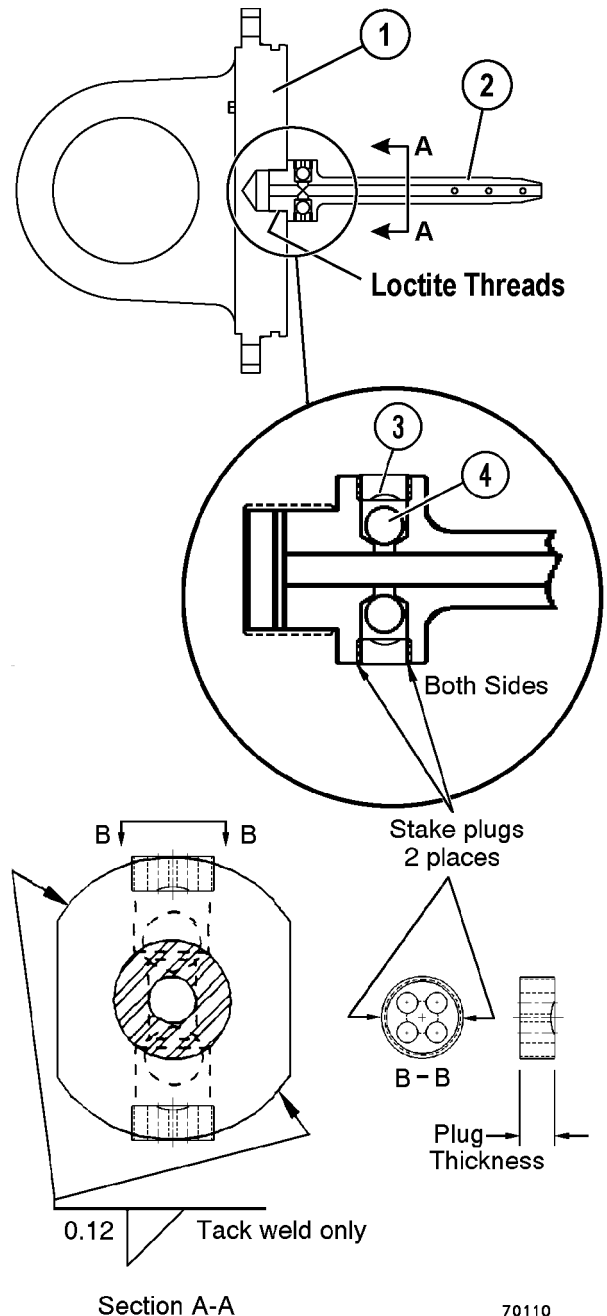


FIGURE 8-21. PLUG & CHECK BALL INSTALLATION

- |                   |               |
|-------------------|---------------|
| 1. Cap Assembly   | 3. Plug       |
| 2. Quill Assembly | 4. Check Ball |

## Installation Of Check Balls And Plugs In Quill

Check balls (4, Figure 8-21) in the side of quill assembly (2) are held in place with threaded plugs (3).

If a plug is missing and the check ball is not found in the cylinder, the opposite side hoist cylinder and the plumbing leading to the hoist valve should be examined for damage. The hoist valve itself should also be checked to see whether the ball or plug has caused internal damage to the spool. Peening of the necked down sections of the spool may result. Spool sticking may also occur.

Refer to Figure 8-22 for SS1158 tool that can be fabricated for installing and removing the check ball plugs.

Check the plugs during any cylinder repair to ensure that they are tight. If they move, remove the plugs and check whether the ball seat in the quill is deformed.

- If deformation of the ball seat has occurred, the quill should be replaced.
- If the ball seat area is not deformed, measure the plug thickness as shown in Figure 8-21:

**Older plug:** 6.35 ± 0.50 mm (0.25 ± 0.02 in.)

**Newer plug:** 9.65 ± 0.50 mm (0.38 ± 0.02 in.)

1. Use the newer plugs. Ensure that the threads in the quill tube and on the plugs are clean, dry and free of oil and solvent.
2. Spray the mating threads of both plugs (3, Figure 8-21) and quill assembly (2) with LOCQUIC® Primer T (Komatsu part number TL8753) or equivalent. Allow the primer to dry 3 to 5 minutes.
3. Apply Loctite® Sealant #277 (Komatsu part number VJ6863) or equivalent to mating threads of both plugs and quill assembly.
4. Place check balls (4) in the quill tube and install plugs (3) with the concave side facing the ball. Use the SS1158 tool to tighten the plugs to **95 N·m (70 ft lb)**. Allow the parts to cure for two hours before exposing the threaded areas to oil.

**NOTE:** If LOCQUIC® Primer T (TL8753) was not used, the cure time will require 24 hours instead of two hours.

5. Stake the plug threads in two places between the holes, as shown in Figure 8-21, to prevent loosening of the plug.

If removal of the plug is necessary in a later rebuild, it will be necessary to carefully drill out the stake marks and destroy the plug. A new plug should be installed and staked as previously detailed.

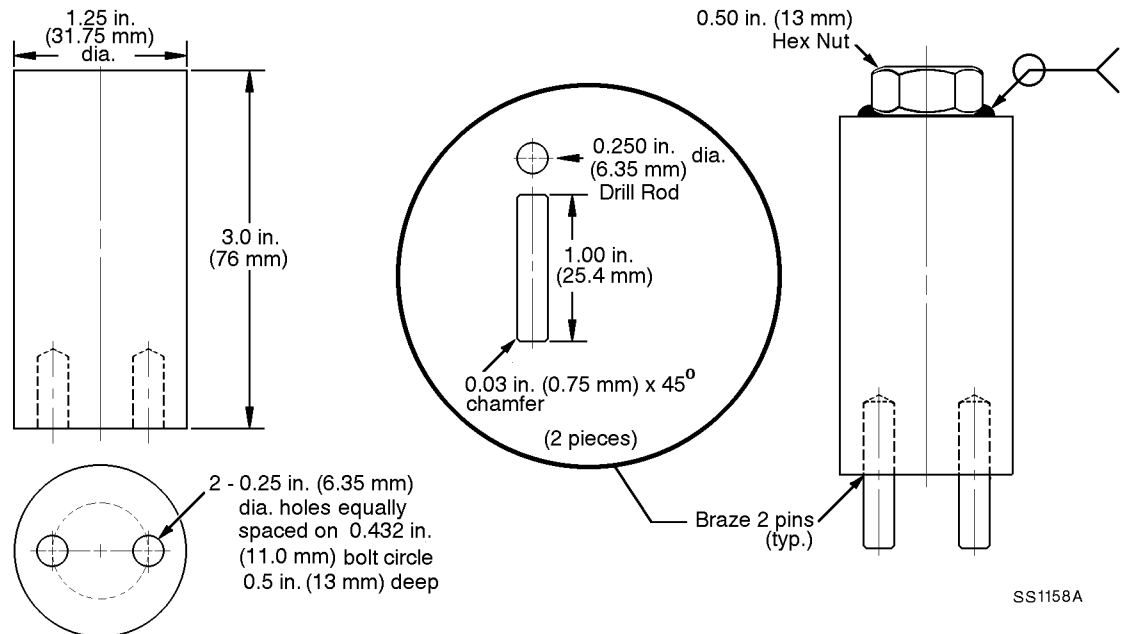


FIGURE 8-22. SS1158 PLUG INSTALLATION & REMOVAL TOOL

## Assembly - Cylinder

1. Install seals (15, Figure 8-19) and bearing (14) on the second stage cylinder. Install bearings (19) and buffer seal (18), rod seal (20) and rod wiper (21) on the first stage cylinder. Lubricate with clean Type C-4 hydraulic oil.

*NOTE: Seal installation ring (PC2709) will ease assembly of the first and second stages, but it is not required.*

- Lubricate seal installation ring (PC2709) with clean Type C-4 hydraulic oil or petroleum jelly and install it in the snap ring groove inside first stage cylinder (3). The seal installation ring allows the seals on the second stage to glide over the snap ring groove without being damaged.
2. Align and slide second stage cylinder (2) inside first stage cylinder (3). Allow the second stage to protrude far enough to install snap ring (9) on the inside of the first stage cylinder.
3. Mount housing (4) in the fixture with the cover end positioned at the top. Install bearings (19), buffer seal (18), rod seal (20) and rod wiper (21) in the housing.
4. Install the lifting tool that was used during disassembly in the second and first stage cylinder assembly.
5. Install bearings (13) and (24) on first stage cylinder (3). Lift and align this assembly over housing (4). Lower the second and first stage cylinders into the housing.
6. Install the retainer bar that was used during disassembly to hold the second and first stage cylinder in place when the housing is rotated. Rotate the housing 180 degrees to position the lower mounting eye at the top.
7. Install bearings (19), buffer seal (18), rod seal (20) and rod wiper (21) in second stage cylinder (2).
8. Attach a lifting device to the eye of rod (1) and align it over housing (4). Install cushion (25) on the rod with the chamfered corner toward the rod eye. Lower the rod into the housing. Lubricate the rod with hydraulic oil.
9. Rotate the housing 180 degrees to position the cover end at the top. Remove the retainer bar that was installed in Step 6. Install bearings (17) and seal (16) on rod bearing retainer (6).
10. Thread two guide bolts - 100 mm (4 in.) long - in the end of rod (1). Install seal (8) on the end of the rod.

11. Align piston rod bearing retainer (6) over the guide bolts and lower it over the end of rod (1). Remove the guide bolts.
12. Check capscrews (1, Figure 8-23) carefully for distress. Replace the capscrews if in doubt.
13. Lubricate capscrews (1) with a lithium-based grease. Install capscrews and plate (2). Tighten the capscrews to **780 N·m (575 ft lb)**.
14. Install O-ring (12, Figure 8-19) and backup ring (23) on cover (10). Align and lower the cover onto housing (4). Lubricate capscrews (11) and install them with the lockwashers. Tighten the capscrews to **678 N·m (500 ft lb)**.
15. Install hoist cylinder eye bearing (5, Figure 8-17) and retainer rings (4), if removed.

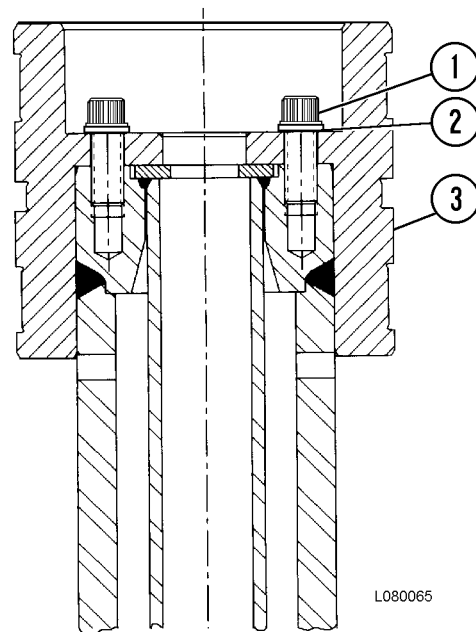


FIGURE 8-23. 3rd STAGE PISTON

- |                    |           |
|--------------------|-----------|
| 1. 12 Pt. Capscrew | 3. Piston |
| 2. Plate           |           |

## Testing

After the cylinder is assembled, perform the following tests to verify performance is within acceptable limits.

1. With the rod fully extended, piston leakage must not exceed 1.64 cm<sup>3</sup>/min. (1.0 in<sup>3</sup>/min.) at 17 250 kPa (2,500 psi), port to port.
2. With the rod fully retracted, piston leakage must not exceed 3.28 cm<sup>3</sup>/min. (2.0 in<sup>3</sup>/min.) at 17 250 kPa (2,500 psi), port to port.
3. Rod seal leakage must not exceed one drop of oil in eight cycles of operation.

## OVERCENTER VALVE MANIFOLD

The overcenter valve manifold is located at the rear of the hoist valve. The internal counterbalance valve relieves excessive pressure that can develop in the annulus area of the hoist cylinders if the load sticks to the tail of the body as the body goes overcenter while dumping.

Figure 8-24 through Figure 8-26 show the proper placement of the O-rings and backup rings on the needle valve, counterbalance valve and the cavity plug.

For information on how the overcenter valve functions, see Hoist Circuit Operation in this section. For adjustment of the counterbalance valve, refer to the Hydraulic Checkout Procedure in this section.

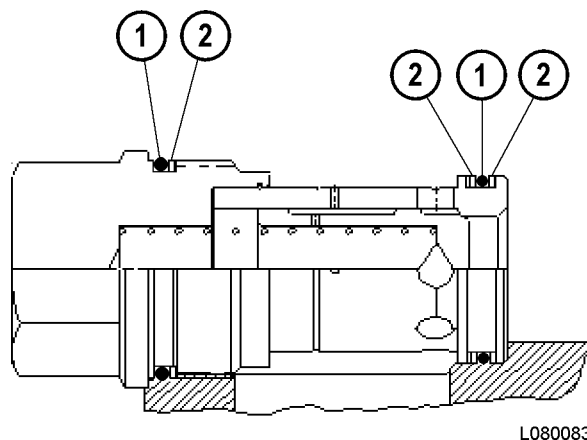


FIGURE 8-24. CAVITY PLUG

1. O-Rings                      2. Backup-Rings

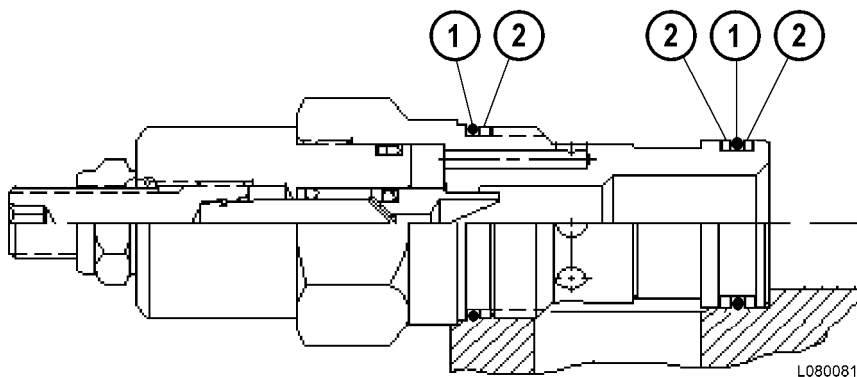


FIGURE 8-25. NEEDLE VALVE

1. O-Rings                      2. Backup-Rings

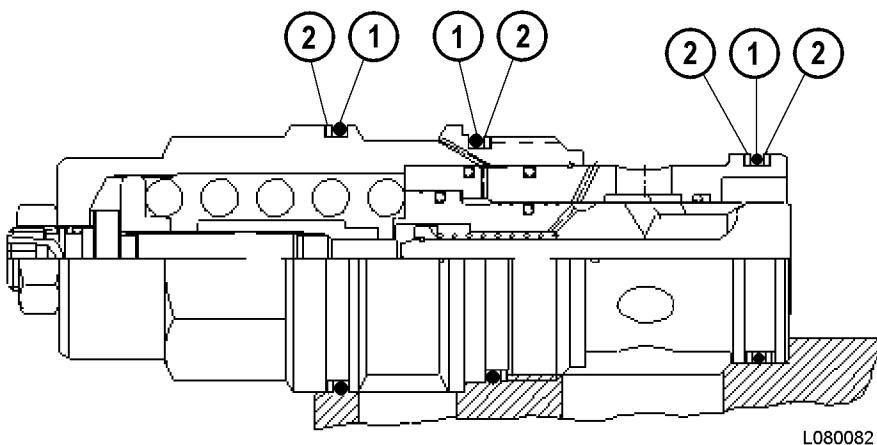


FIGURE 8-26. COUNTERBALANCE VALVE

1. O-Rings                      2. Backup-Rings

# NOTES

# SECTION L9

## HYDRAULIC SYSTEM FILTERS

### INDEX

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# NOTES

# HYDRAULIC SYSTEM FILTERS

## HOIST CIRCUIT FILTER

The hoist circuit filters (Figure 9-1) are located at the back of the fuel tank below the right frame rail. The filter provides secondary filtering protection for hydraulic oil flowing to the hoist valve and hoist circuit components.

Indicator switch (5) is designed to alert the operator to a filter restriction before actual bypass occurs. The switch contacts close at 240 kPa (35 psi) to actuate a warning lamp on the overhead display panel. Actual filter bypass occurs at 345 kPa (50 psi).

*NOTE: When the engine is initially started and the hydraulic oil is cold, the warning lamp may illuminate. Allow the hydraulic system oil to reach operating temperature before using the warning lamp as an indicator to change the element.*

Refer to Section P2, Lubrication and Service, for recommended normal filter element replacement interval. Earlier replacement may be required if the restriction indicator lamp turns on.

Premature filter restriction may indicate a system component failure and signal a service requirement before extensive secondary damage can occur.

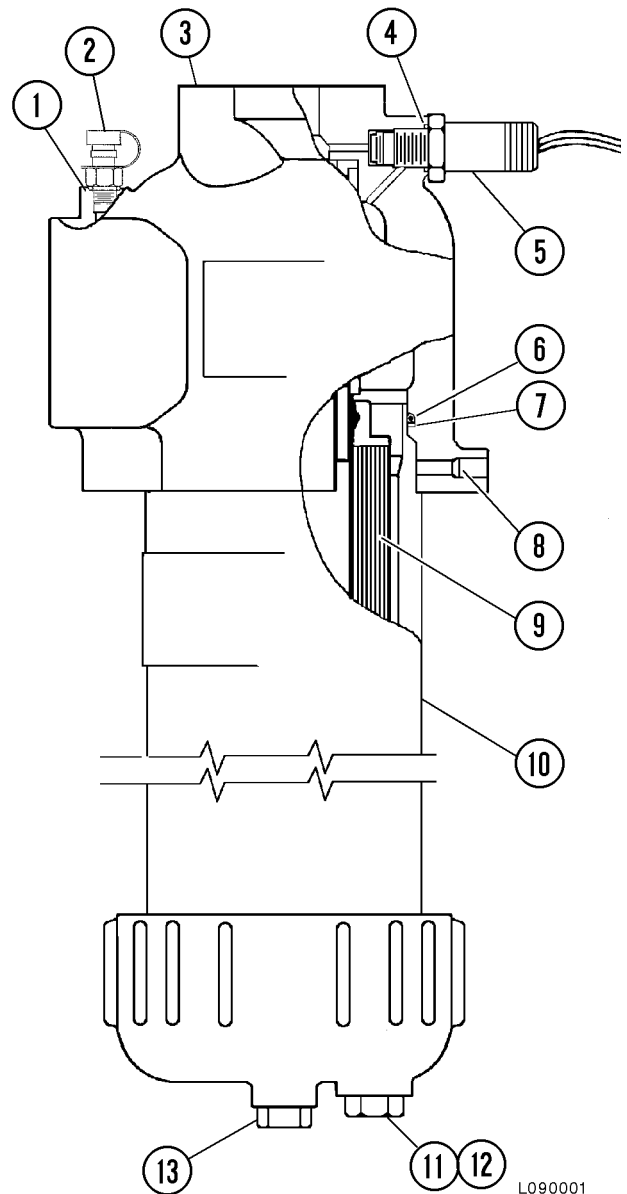
*NOTE: An early indication of the filter warning light at first installation may be due to restriction in the filter as it cleans the system. Unless the fluid appears contaminated or has a strong foul odor, do not change the oil. Replace only the filter element.*

### Filter Element Replacement

#### **⚠ WARNING**

**Relieve pressure before disconnecting hydraulic and lines. Tighten all connections before applying starting the machine.**

**Hydraulic fluid escaping under pressure can have sufficient force to enter a person's body by penetrating the skin and cause serious injury and possibly death if proper medical treatment by a physician familiar with this injury is not received immediately.**



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FIGURE 9-1. HOIST CIRCUIT FILTER

- |                     |                   |
|---------------------|-------------------|
| 1. O-Ring           | 8. Set Screw      |
| 2. Pressure Tap     | 9. Filter Element |
| 3. Filter Head      | 10. Bowl          |
| 4. O-Ring           | 11. Drain Plug    |
| 5. Indicator Switch | 12. O-Ring        |
| 6. O-Ring           | 13. Bottom Plug   |
| 7. Backup Ring      |                   |

### Removal - Hoist Circuit Filter

1. With the key switch OFF, allow at least 90 seconds for the accumulators to bleed down.
2. Remove drain plug (11, Figure 9-1). Remove bottom plug (13) and drain the oil from the housing into a suitable container.



***If the truck has been operating, avoid contact with hot oil. Avoid spillage and contamination.***

3. Loosen setscrew (8). Remove bowl (10).
4. Remove filter element (9).
5. Remove and discard backup ring (7) and O-ring (6).
6. Clean bowl in solvent and dry thoroughly.

### Installation - Hoist Circuit Filter

1. Install new element (9). Install new O-ring (6) and backup ring (7).
2. Install bowl (10) on filter head (3) and tighten. Lock the bowl in place with setscrew (8).
3. Install bottom plug (13) and drain plug (11).

### Indicator Switch - Hoist Circuit Filter

Indicator switch (5, Figure 9-1) is factory preset to actuate at 240 kPa (35 psi). The switch and warning light should be routinely tested as a part of the regular vehicle maintenance program. Refer to Indicator Switch Test Procedure in this section.

## STEERING CIRCUIT FILTER

The steering circuit filter (Figure 9-2) is located on the inboard side of the fuel tank. The filter provides secondary filtering protection for hydraulic oil flowing to the bleeddown manifold valve for the steering and brake systems.

Indicator switch (1) is designed to alert the operator of filter restriction before actual bypass occurs. The switch contacts close at  $240 \pm 35$  kPa ( $35 \pm 5$  psi) to actuate a warning lamp on the overhead display panel. Actual filter bypass occurs at 345 kPa (50 psi).

*NOTE: When the engine is initially started and the hydraulic oil is cold, the warning lamp may actuate. Allow the hydraulic system oil to reach operating temperature before using the warning lamp as an indicator to change the element.*

Refer to Section P2, Lubrication and Service for recommended normal filter element replacement interval. Earlier replacement may be required if the restriction indicator lamp turns on.

Premature filter restriction may indicate a system component failure and signal a service requirement before extensive secondary damage can occur.

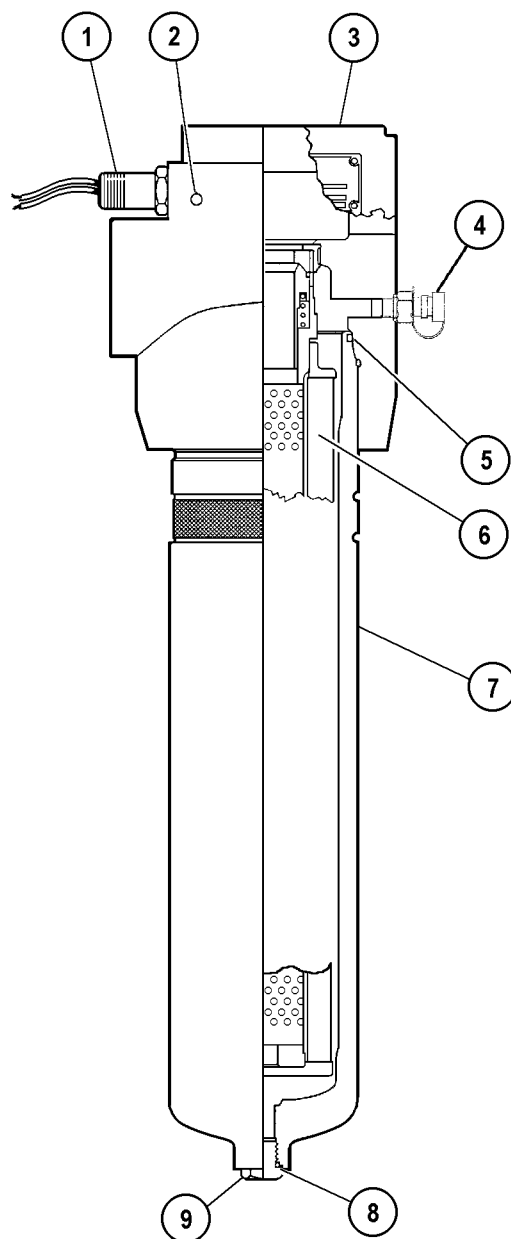
*NOTE: An early indication of the filter warning light at first installation may be due to restriction in the filter as it cleans the system. Unless the fluid appears contaminated or has a strong foul odor, do not change the oil; replace only the filter element.*

### Filter Element Replacement



**Relieve pressure before disconnecting hydraulic and other lines. Tighten all connections before applying pressure.**

**Hydraulic fluid escaping under pressure can have sufficient force to enter a person's body by penetrating the skin and cause serious injury and possibly death if proper medical treatment by a physician familiar with this injury is not received immediately.**



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FIGURE 9-2. STEERING CIRCUIT FILTER

- |                     |                   |
|---------------------|-------------------|
| 1. Indicator Switch | 6. Filter Element |
| 2. Setscrew         | 7. Bowl           |
| 3. Head             | 8. O-Ring         |
| 4. Pressure Tap     | 9. Drain Plug     |
| 5. O-Ring           |                   |

### Removal - Steering Circuit Filter

1. With the key switch OFF, allow at least 90 seconds for the accumulators to bleed down.
2. Remove drain plug (9, Figure 9-2) and drain the oil from the housing into a suitable container.



***If the truck has been operating, avoid contact with hot oil. Avoid spillage and contamination.***

3. Remove bowl (7) and element (6).
4. Remove and discard O-ring (5) on filter bowl.
5. Clean bowl in solvent and dry thoroughly.

### Installation - Steering Circuit Filter

1. Install new element (6). Install new O-ring (5) on bowl.
2. Install bowl on filter head and tighten.
3. Install drain plug (9), and O-ring (8).

### Indicator Switch - Steering Circuit Filter

Indicator switch (1, Figure 9-2) is factory preset to actuate at  $240 \pm 35$  kPa ( $35 \pm 5$  psi). The switch and warning light should be routinely tested as a part of the regular vehicle maintenance program. Refer to Indicator Switch Test Procedure in this section.

### INDICATOR SWITCH

The hydraulic filter indicator switch is factory preset to actuate at  $240 \pm 35$  kPa ( $35 \pm 5$  psi). Switch adjustment is not necessary or recommended. However, the switch and warning lamp should be tested periodically for proper operation. The indicator switch is not repairable and, if inoperative, must be replaced.

Figure 9-3 shows a pressure switch tester, and Figure 9-4 shows a test block to accept the switch during testing. The test block may be fabricated as shown.

### Test Procedure

1. With the key switch OFF, allow at least 90 seconds for the accumulators to bleed down.
2. Disconnect the wiring harness and remove the switch from the filter head.
3. Install the switch in the test block. Connect a pressure tester to the pipe nipple. Reconnect the wires to the switch.
4. Turn the key switch ON. Pump up pressure to the test block while observing the tester gauge and hydraulic oil filter warning light in the cab. The light should turn on at  $240 \pm 35$  kPa ( $35 \pm 5$  psi). If not, test the bulb and switch as follows:
  - a. If the lamp fails to light, remove the wire (circuit #39) from the switch and short to ground. If the warning lamp does not light, replace the bulb.
  - b. If the lamp lights when wire #39 is grounded in the previous step, use an ohmmeter to verify that the switch contacts close at  $240 \pm 35$  kPa ( $35 \pm 5$  psi).
5. If the switch contacts fail to close, replace the switch assembly.

***NOTE:*** The hydraulic filter warning lamp may be activated by either the hoist or steering/brake circuit filter indicator switches.

A schematic diagram of the pressure release valve assembly. It shows a pump connected to a pressure release valve, which is in turn connected to a pressure gauge. A 0.125 in. pipe adapter is shown connected to the pump. The diagram is labeled with the following components:

- PRESSURE GAUGE
- PRESSURE RELEASE VALVE
- PUMP
- 0.125 in. PIPE ADAPTER

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Phone: (313) 774-9500

## TEST BLOCK for DIFFERENTIAL PRESSURE SWITCH



# NOTES

# SECTION L10

## HYDRAULIC CHECKOUT PROCEDURE

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# NOTES

# HYDRAULIC CHECKOUT PROCEDURE

## GENERAL INFORMATION

The hydraulic checkout procedure is intended to help the technician check, adjust, and diagnose problems in the steering and hoist circuits. The technician should read the entire check-out procedure to become familiar with the procedures and all the warnings and cautions before performing any steps. The check-out procedure begins by checking the basic system before checking individual components.

A data sheet is included on the last page of this section to record the information observed during the hydraulic system check-out procedure. The data sheet is designed to be removed, copied, and used during the check-out procedure.

*\* Steps indicated in this manner should be recorded on the data sheet for reference.*

### **WARNING**

***Hydraulic fluid escaping under pressure can have sufficient force to enter a person's body by penetrating the skin and cause serious injury, and possibly death, if proper medical treatment by a physician familiar with this type of injury is not received immediately.***

***Before disconnecting pressure lines, replacing components in the hydraulic circuits, or installing test gauges, ALWAYS bleed down hydraulic steering accumulators.***

***The steering accumulators can be bled down by shutting off the engine, turning the key switch OFF, and waiting 90 seconds. Confirm that the steering pressure is released by turning the steering wheel. No front wheel movement should occur.***

***Bleed down brake system accumulators before removing any hoses supplying oil to the brake system.***

### **IMPORTANT**

***The AC drive system rest switch, located on the instrument panel, must be in the ON position and the GF cutout switch in the CUTOUT position during test procedures. See Section E3, Propulsion System, for GF switch location.***

## HYDRAULIC SYSTEM FLUSHING PROCEDURE

When a hydraulic system component is repaired or replaced, the hydraulic system must be flushed before the engine is started to remove contamination that may have been introduced during the repair process. DO NOT start the engine unless all flushing equipment is installed and ready for use. The flushing procedure will help eliminate harmful contamination that may lead to premature pump or valve failures. Starting the engine just one time without performing the flushing procedure may contaminate the hydraulic system.

Special hydraulic system flushing tools for the 930E truck have been released. Hoist system bypass blocks (PC3074) and steering system flushing hose (XB5777) must be on-site before or during field assembly, so when the truck is started for the first time, the hydraulic system can be flushed and any debris can be removed from the hydraulic system. Fitting assembly (XB7075) shown as (3, Figure 10-6) will also be required.

All flushing hoses must be rated to withstand 27 580 kPa (4,000 psi) pressure. The hoses must be 1 inch inside diameter and meet SAE 100R12 specifications.

**Table 1: XB5777 HOSE ASSEMBLY**

| Part Number | Description               | Qty. |
|-------------|---------------------------|------|
| HA2185      | Hose                      | 1    |
| WB0546      | Fitting, Elbow - 45 Deg   | 1    |
| WB0599      | Fitting, Straight Adapter | 1    |

## **WARNING**

***Hydraulic fluid escaping under pressure can have sufficient force to enter a person's body by penetrating the skin and cause serious injury, and possibly death, if proper medical treatment by a physician familiar with this type of injury is not received immediately.***

***Before disconnecting pressure lines, removing fittings or replacing components in the hydraulic circuits, or installing test gauges, ALWAYS bleed down the steering and brake accumulators.***

***NOTE: The hoist cylinder bypass blocks should be saved for possible use later in case the hydraulic system needs to be flushed after a component failure.***

1. If the tires are installed on the truck, chock the left and right rear wheels securely on both sides to prevent truck movement. If the truck is sitting on cribbing, ensure it is secure enough to allow safe operation of the engine and clearance for turning the front spindles lock to lock without risk of collapsing the cribbing.
2. Inspect all hydraulic hoses to ensure that they are properly routed and securely connected.
3. Connect a laptop PC to the interface module. Start the Interface Realtime Data Monitor program.
4. Ensure that both brake accumulators and both steering accumulators are properly precharged with nitrogen to 9 650 kPa (1,400 psi). Refer to the shop manual for the accumulator charging procedure.
5. Confirm that the steering pressure is released by turning the steering wheel. No steering movement should occur.
6. If any fault codes are displayed, troubleshoot the system and repair the problem before proceeding.

7. Install accurate pressure gauges at the following locations:

- Pressure test port "TP3" (4, Figure 10-1) on the bleed down manifold - 35 000 kPa (5,000 psi).
- Pressure test port (1, Figure 10-2) for the front brake cooling circuit - 3 450 kPa (500 psi).
- Pressure test ports (2, Figure 10-2) on both high pressure hoist filters - 24 000 kPa (3,500 psi).
- Pressure test port (1, Figure 10-3) for the rear brake cooling circuits - 3 450 kPa (500 psi).
- Pressure test port "GPA" (2, Figure 10-4) located on the steering pump - 35 000 kPa (5,000 psi).

## **WARNING**

***If any hydraulic pressure is detected on the pressure gauge installed at TP3, there is residual oil pressure in the system. Do not loosen or remove any hydraulic hoses, fittings or components. The steering accumulators have not been properly depressurized. Locate and correct the depressurization problem. Always relieve all hydraulic pressure before disconnecting any hydraulic lines or fittings. Always tighten all connections securely before applying pressure. Confirm that the steering pressure is released by turning the steering wheel. No steering movement should occur.***

8. Ensure pressure gauge connected at "TP3" reads 0 kPa (0 psi) before removing plugs (1 & 2, Figure 10-1). Remove plugs (1 & 2) from the bleed down manifold. Connect jumper hose assembly (XB5777) between "QD supply" port and the "QD return" port on the bleed down manifold. Tighten all connections securely before applying pressure.
9. Connect the four hoist cylinder hoses (3, Figure 10-2) together as shown using bypass blocks (4). The hardware for connecting the hoses to the blocks is the same hardware for connecting the hoses to the hoist cylinders. Ensure that all of the connections are secure. Cover all open ports to prevent contamination.

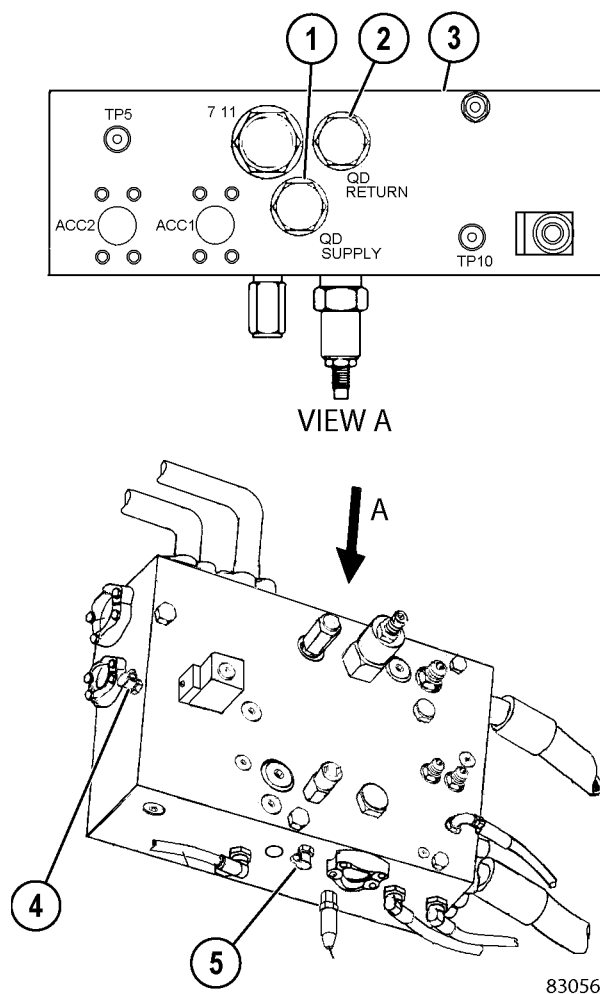


FIGURE 10-1. BLEED DOWN MANIFOLD

- |                          |                    |
|--------------------------|--------------------|
| 1. Plug - QD Supply Port | 4. Test Port (TP3) |
| 2. Plug - QD Return Port | 5. Test Port (TP2) |
| 3. Bleed Down Manifold   |                    |

10. At the rear axle housing, connect brake cooling lines (1, 2 & 3, Figure 10-5) together using fitting assembly (3, Figure 10-6). Cap the three open fittings on the axle housing with clean caps. Several hose clamps may have to be removed to connect the three hoses to the fitting assembly.

11. At the left front brake assembly, connect cooling hoses (1 & 3, Figure 10-7) together using adapter (2). Cap the open fittings on the brake assembly. At the right front brake assembly, connect the two cooling hoses together using another adapter. Cap the open fittings on the brake assembly.

**NOTE:** The adapters are installed on the truck when it is shipped from the factory.

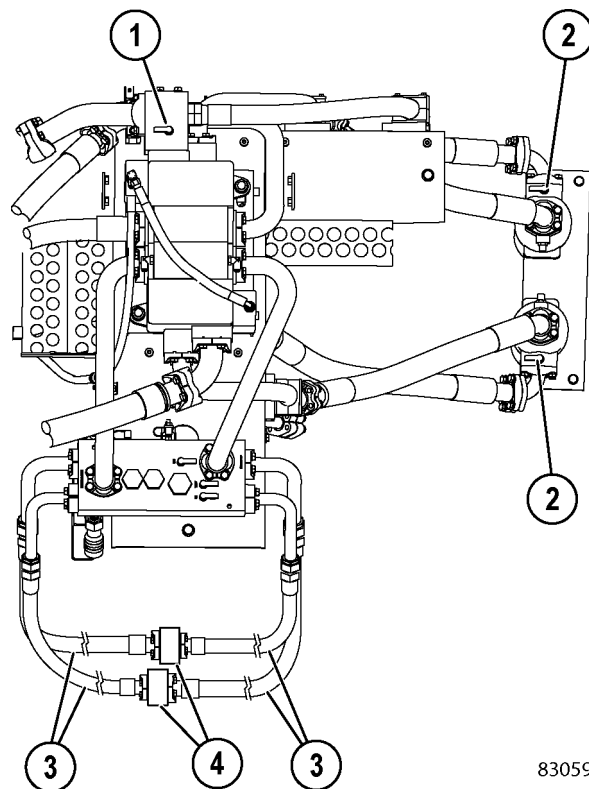


FIGURE 10-2. TEST PORT LOCATIONS AND HOIST FLUSHING HOOKUP

1. Test Port (Front Brake Cooling Circuit)
2. Test Port (Hoist Filters)
3. Hoist Cylinder Hoses
4. Bypass Blocks (PC3074)

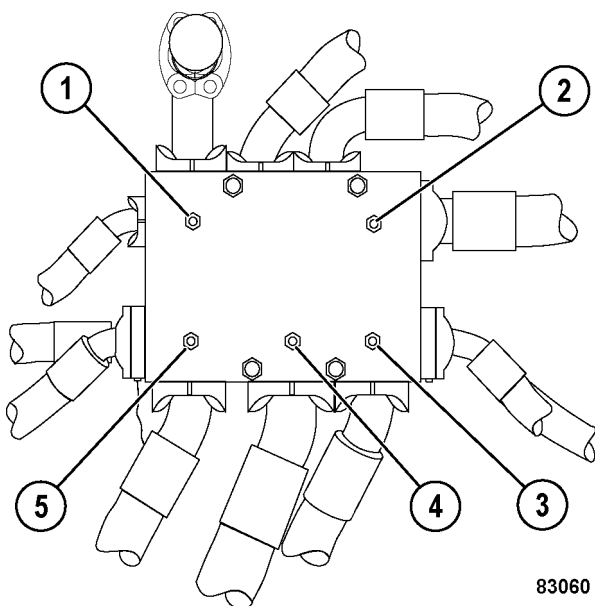


FIGURE 10-3. BRAKE COOLING MANIFOLD

1. Test Port (Rear Brake Cooling Pressure)
2. Test Port (Front Brake Supply Pressure)
3. Test Port (Front Brake Return Pressure)
4. Test Port (Hoist Return Pressure)
5. Test Port (Rear Brake Return Pressure)

12. The steering pump case must be filled with oil before startup to avoid pump damage. Disconnect the hose that is connected to fitting (1, Figure 10-4). Then remove fitting (1). Fill the steering pump case with clean type C-4 hydraulic oil through the case drain port. It may require up to 2.9 liters (3 quarts) of oil. Re-install fitting (1) and reconnect the hose.

**NOTE:** Filling the steering pump case with oil does not eliminate the need to bleed air from the pump suction line.

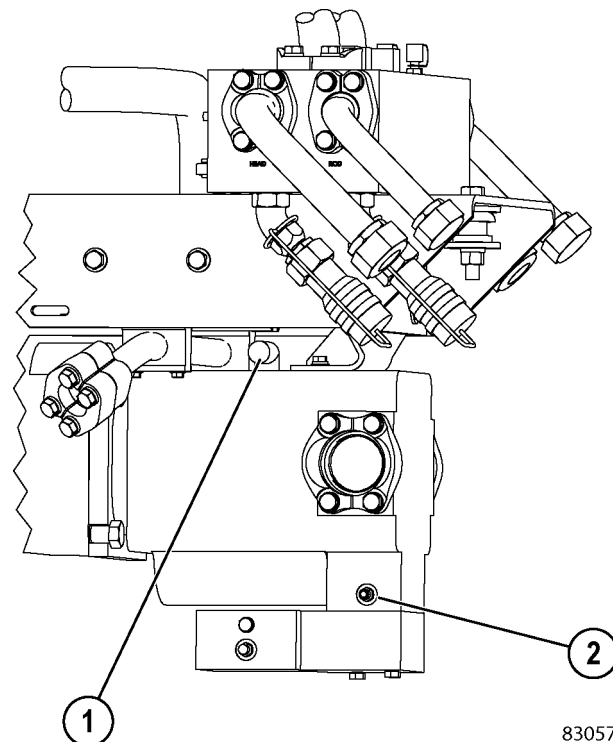
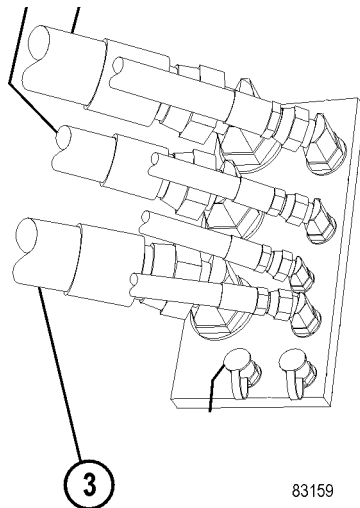


FIGURE 10-4. STEERING PUMP & HOIST SYSTEM QUICK DISCONNECTS

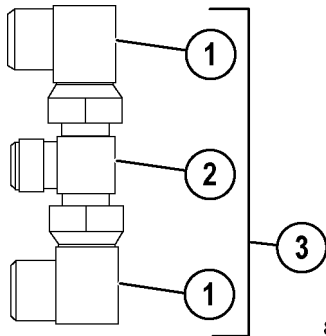
1. Fitting (Pump Case Drain)
2. Test Port (GPA)



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FIGURE 10-5. REAR AXLE BRAKE LINES

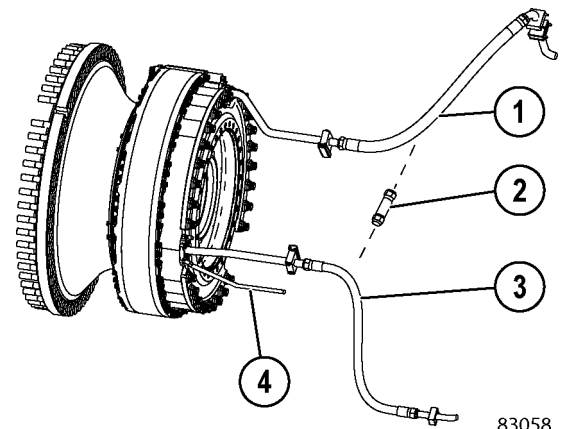
1. BS - Left & Right Brake Cooling Oil Supply
2. LBR - Left Brake Cooling Return Line
3. RBR - Right Brake Cooling Return Line



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FIGURE 10-6. FITTING ASSEMBLY

1. 90° Swivel Elbow (PC3062)
2. T-Fitting (WB0426)
3. Fitting Assembly (XB7075)



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FIGURE 10-7. FRONT WHEEL BRAKE LINES

1. Cooling Hose
2. Adapter (EG1794)
3. Cooling Hose
4. Brake Apply Hose

13. Fully open all hydraulic shutoff valves in the pump suction lines. The valves are fully open when the valve handle is in-line with the hose.



***Serious pump damage will occur if any shutoff valve is not fully open when the engine is started.***

14. Fill the hydraulic tank with clean type C-4 hydraulic oil. Ensure that the oil level is visible in the center of the upper sight glass.
15. To bleed trapped air from inside the pump suction hoses, loosen the cap screws where the three suction hoses attach to the inlet of each pump. Ensure that the O-ring does not fall out of the groove. After oil appears, retighten the cap screws to the standard torque.
16. Check the hydraulic tank oil level. If necessary, add oil.

## ⚠ CAUTION

***Do not apply the brakes or turn the steering wheel at any time during this procedure until Step 33.***

17. Ensure both brake accumulator bleed down valves (1, Figure 10-8) are open.
18. Move the hoist control lever to the FLOAT position.
19. Turn the key switch ON but do not start the engine. Verify that the low steering accumulator precharge warning is not displayed. If the warning is displayed, correct the problem before proceeding.

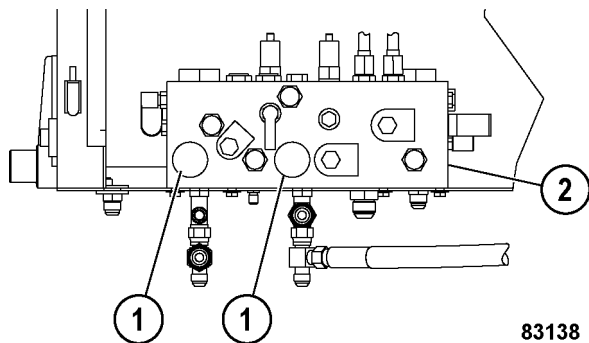


FIGURE 10-8. BRAKE MANIFOLD

1. Accumulator Bleed Down Valves
2. Brake Manifold

20. Start the engine and operate at low idle. Within 30 seconds after starting the engine, check the two oil pressures listed below. If the oil pressure readings are not correct, stop the engine. Locate and repair the problem before proceeding. If the pressure readings are correct, proceed to the next step. Do not turn the steering wheel or apply the brakes at this time.

- With the engine at low idle, and the oil temp at 21° C (70° F), oil pressure at the hoist filters should be approximately 552 kPa (80 psi) or less.
- Oil pressure at steering pump test port "GPA" should be approximately 690 kPa (100 psi).

## ⚠ CAUTION

***Do not operate the engine at low idle for more than 30 seconds. The steering pump pressure at test port "GPA" must be at or above 1 725 kPa (250 psi) during the flushing process to provide adequate internal pump lubrication. Steering pump damage will result if the pressure is below 1 725 kPa (250 psi) for more than 30 seconds.***

21. If the oil pressure readings in Step 20 are correct, increase the engine speed to obtain oil pressure at test port "GPA" of 1 725 kPa (250 psi). Move the hoist control lever to the POWER UP or to the POWER DOWN position. The pressure readings at the front brake cooling port (1, Figure 10-2) and rear brake cooling test port (1, Figure 10-3) should be 0 kPa (0 psi). If the pressure is not 0 kPa (0 psi), the hoist valve piping (or pilot lines) may be connected incorrectly, or the hoist valve could be defective. Correct the plumbing problem or replace the hoist valve and repeat this step.

22. If all pump pressures and brake cooling oil supply pressures are correct as stated in Steps 20 and 21, increase engine speed to 1500 rpm and flush the system for 20 minutes. During this time, move the hoist control lever to the POWER UP position for 30 seconds, and then move it to the POWER DOWN position for 30 seconds. Repeat this hoist control cycle five times. This process will carry any system contaminants to the hydraulic filters.

Monitor the pressure gauges at the front and rear brake cooling test ports during this step. The pressure readings should always be 0 kPa (0 psi) when the hoist valve is in the POWER UP or POWER DOWN positions, and approximately 345 kPa (50 psi) while in the HOLD or FLOAT positions.

23. When the 20 minute flushing process is complete, turn the key switch to the OFF position and allow at least two minutes after the engine has stopped for the steering accumulators to completely depressurize.
24. Remove the adapters from both front wheel brake cooling hoses and re-connect the cooling hoses to the front brake assemblies.
25. Remove the tee fitting assembly from the three rear brake cooling hoses. Re-connect the three rear brake cooling hoses to the axle housing.
26. Check the pressure gauge connected to the "TP3" port on the bleed down manifold. If any pressure is still in the system, the bleed down process did not complete. Locate and repair the problem. If the pressure is 0 kPa (0 psi), remove the jumper hose and fittings installed in the bleed down manifold. Install the two plugs (1 & 2, Figure 10-1) previously removed back into the bleed down manifold.
27. Check the hydraulic tank oil level. If necessary, add oil.
28. Close both brake accumulator bleed down valves (1, Figure 10-8).

*NOTE: Do not combine Step 29 with any other steps. Do not apply the brakes or steer the truck during Step 29.*

29. To flush the accumulators, start the engine and operate the engine until the pressure at the "TP3" port on the bleed down manifold is approximately 21 200 kPa (3,075 psi). This is the pressure where the steering pump unloader valve shifts and unloads the pump.

- If the accumulator pressure exceeds 21 546 kPa (3,125 psi), immediately turn the engine off and open the brake accumulator bleed down valves. Refer to Steering pump pressure control adjustments in the shop manual, then repeat Step 29.
- If the accumulator pressure does not reach 20 856 kPa (3,025 psi), check for faults on the Interface Realtime Data Monitor program. If there is a bleed down fault active, locate and correct the problem. If no faults are active, refer to Steering pump pressure control adjustments in the shop manual, then repeat Step 29.

30. When the pressure gauge connected to "TP3" reads approximately 21 200 kPa (3,075 psi):

a. For the first time only through this step:

- Leave the key switch in the ON position, and use the engine shut down switch on the center console to stop the engine.
- After the engine has stopped, confirm that steering accumulator bleed down does not occur, and that no bleed down faults or any other faults are displayed on the Interface Realtime Data Monitor.

b. The second and subsequent times through this step:

- Turn the key switch to the OFF position. Confirm that after the engine has stopped that the steering accumulator bleed down process begins.
- Confirm that accumulator bleed down completes within two minutes. The pressure gauge at the bleed down manifold pressure test port "TP3" will be 103 kPa (15 psi) or less after a successful bleed down.



31. If Step 30-b results are satisfactory, open both brake accumulator bleed down valves (1, Figure 10-8) on the brake manifold to completely depressurize the brake accumulators. After all hydraulic pressure has been released, close both accumulator bleed down valves. Repeat Steps 29 - 31 five times.
32. To flush the steering system, start the engine. Allow the accumulators to fully charge and the steering pump to unload.
33. Turn the steering wheel full left and then full right ten times.
34. Turn the key switch to the OFF position. Allow at least two minutes after the engine has stopped for the steering accumulators to completely depressurize. The pressure gauge at the bleed down manifold pressure test port "TP3" will be 103 kPa (15 psi) or less after a successful bleed down.
35. The hydraulic system flushing procedure has been completed. All hydraulic system high pressure filter elements must be replaced.
  - a. Close both hoist pump suction line shutoff valves and the steering pump suction line shutoff valve.
  - b. Remove the hoist and steering hydraulic filter elements. Clean the housings and install new filter elements. Detailed instructions may be found in Section L9, Hydraulic System Filters.
  - c. Open both hoist pump suction line shutoff valves and the steering pump suction line shutoff valve.
36. Remove all pressure gauges and re-install the dust caps on all quick disconnect fittings.
37. Remove hoist cylinder bypass blocks (4, Figure 6-2) and connect the hoist cylinder hoses to the hoist cylinders. Ensure the hoses and ports are clean to prevent contamination.
38. Check the hydraulic tank oil level. If necessary, add oil.

**NOTE:** Do not to extend the hoist cylinders until the body is installed.

## STEERING CIRCUIT CHECK-OUT & ADJUSTMENT PROCEDURE

The steering circuit hydraulic pressure is supplied from the piston pump and steering accumulators. Some steering system problems, such as spongy or slow steering or abnormal operation of the low steering pressure warning light, can sometimes be traced to internal leakage of steering components. If internal leakage is suspected, perform the steering component leakage tests outlined in this section.

**NOTE:** Excessive internal leakage within the brake circuit may contribute to problems within the steering circuit. Ensure that brake circuit leakage is not excessive before troubleshooting the steering circuit.

The steering circuit can be isolated from the brake circuit by first releasing all steering system pressure, then releasing all pressure from the brake accumulators and removing brake system supply line (4, Figure 10-10) from the bleeddown manifold. Plug the brake supply hose and cap the fitting at the bleeddown manifold to prevent high pressure leakage.

Before checking the steering system, the hydraulic steering and brake systems must have the correct accumulator precharge and be at normal operating temperatures. Refer to the steering accumulator charging procedure in Section L6 for accumulator charging instructions.

### **IMPORTANT**

***If the steering and brake pump has just been installed, it is essential that the steering pump case is full of oil before starting the engine. Refer to Section L6, Steering and Brake Pump, for instructions.***

## Equipment Requirements

The following equipment will be necessary to properly check-out the hydraulic steering circuit:

- Hydraulic schematic (see Section R)
- Two 35 000 kPa (5,000 psi) calibrated pressure gauges and hoses
- A graduated container marked to measure liquid volume in cubic inches or milliliters

## Pump Pressure Control Adjustments

With the brake system functioning properly and the parking brake on, proceed as follows:

1. Place the directional control lever in PARK. Place the rest switch in the ON position and the GF cutout switch in the CUTOFF position. Turn the key switch OFF and wait 90 seconds for the steering accumulators to completely bleed down before opening any hydraulic circuits.
2. Check the hydraulic oil level in the tank and add oil if required. Ensure that the suction line shutoff valves are open.

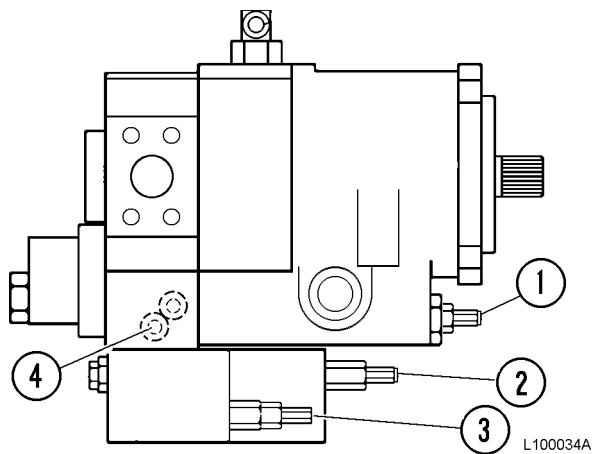
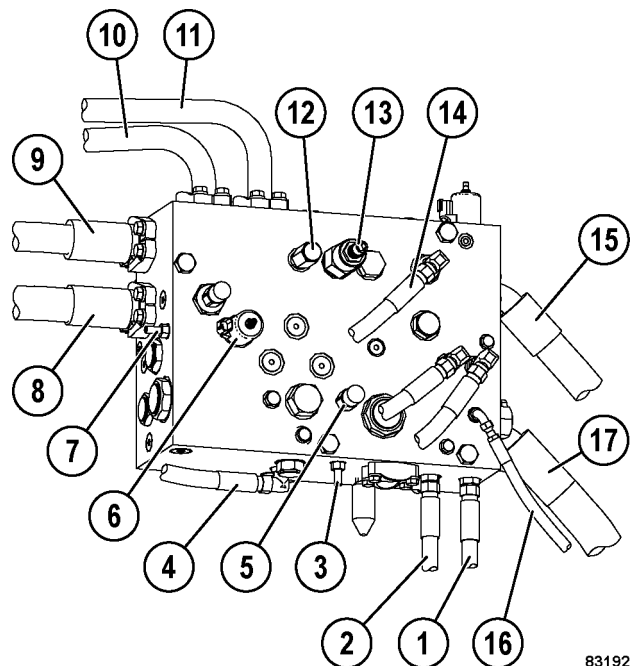


FIGURE 10-9. STEERING/BRAKE PUMP ADJUSTMENTS

1. Pump Stroke Adjustment Screw
2. Unloader Valve Adjustment Screw
3. Compensator Adjustment Screw
4. GPA Port



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FIGURE 10-10. BLEEDDOWN MANIFOLD

1. To Hoist Valve, Power Up
2. To Hoist Valve, Power Down
3. Pressure Test Port (TP2)
4. Brake System Supply Line
5. Steering System Pressure Switch
6. Steering Accumulator Bleeddown Solenoid
7. Pressure Test Port (TP3)
8. Supply to Flow Amplifier
9. Return from Flow Amplifier
10. To Front Steering Accumulator
11. To Rear Steering Accumulator
12. Relief Valve, 600 psi (4.1 MPa)
13. Relief Valve, 4000 psi (28.0 MPa)
14. Hoist Pilot Valve Return Line
15. Tank Return Line
16. Feedback Pressure to Unloader valve
17. Supply from Pump

3. Install an accurate 35 000 kPa (5,000 psi) pressure gauge at steering pressure "GPA" test port (4, Figure 10-9) located on the suction side of the pump. Install a second 35 000 kPa (5,000 psi) gauge in bleeddown manifold test port "TP2" (3, Figure 10-10).
4. Turn unloader valve adjustment screw (2, Figure 10-9) clockwise until it bottoms out.
5. Back out compensator adjustment screw (3) by turning it counterclockwise.

6. Start the engine and adjust pressure compensator (3) to obtain 22 925 kPa (3,325 psi) on the pressure gauge at the "GPA" pump port. Tighten the compensator jam nut.

*\* Record on Data Sheet*

7. Shut off the engine and allow the accumulators to bleed down.
8. Back out unloader valve adjustment screw (2, Figure 10-9) completely.
9. Start the engine and allow pump pressure to build until the pump unloads.

*NOTE: When the steering pump unloads, the pump output pressure at the "GPA" pressure test port will drop to approximately 1 380 to 2 760 kPa (200 to 400 psi).*

10. Steer the truck to cause the accumulator pressure to drop until the pump loads.
11. Adjust the unloader valve to reload the accumulators when the pressure falls to 18,960 kPa (2750 psi). The unload pressure will be approximately 20 680 to 21 025 kPa (3,000 to 3,050 psi).

- ☐ When the unload pressure increases to 20 680 to 21 025 kPa (3,000 to 3,050 psi), the adjustment has been successfully completed.

*\* Record on Data Sheet*

12. Steer the truck again slowly while observing the "TP2" steering pressure gauge.
- ☐ The pump must reload when the pressure drops to 18 960 kPa (2,750 psi) minimum.

*\* Record on Data Sheet*

13. If the minimum reloading pressure in Step 12 is not correct, repeat Steps 11 and 12.

*NOTE: The critical pressure setting is the 18 960 kPa (2,750 psi) when the pump loads. The unloading pressure follows the loading pressure adjustment and should occur at approximately 20 855 kPa (3,025 psi).*

14. Tighten the jam nut to lock the pressure setting.
15. Steer the truck several times to verify that the system pressure begins to increase (pump cycles from unload to load) once pressure has dropped to 18 960 kPa (2,750 psi).
16. Shut off the engine and allow the accumulators to bleed down completely. Turn the steering wheel to verify that no pressure remains.
17. Remove all gauges and cap the test ports.

## **Steering Control Valve and Flow Amplifier Leakage Test**

1. Disconnect flow amplifier return line (9, Figure 10-10) at the bleddown manifold. Plug the port in the bleddown manifold.
2. Disconnect the steering control unit return line at the flow amplifier. Plug the port in flow amplifier. **Do not turn steering wheel while hoses are disconnected!**
3. Start the engine and allow the steering system to reach normal operating pressure.
4. Measure the leakage from the flow amplifier return hose that was removed in Step 1.

- ☐ Maximum permissible leakage is 820 ml/min. (50 in<sup>3</sup>/min.).

*\* Record on Data Sheet*

If leakage is excessive, the flow amplifier should be repaired or replaced.

5. Measure the leakage from the steering control unit return hose removed in Step 2.

- ☐ Maximum permissible leakage is 164 ml/min. (10 in<sup>3</sup>/min.).

*\* Record on Data Sheet*

If leakage is excessive, the steering control valve should be repaired or replaced.

6. Shut off the engine and allow the accumulators to bleed down completely. If all steering system hoses are connected, steer the truck to verify that all pressure has bled off.

## Bleeddown Manifold Leakage Test

1. With the hydraulic lines still disconnected from the Steering Control Unit and Flow Amplifier Leakage Test, disconnect hoist pilot valve return line (14, Figure 10-10) located on the side of the bleeddown manifold. Plug the open port on the bleeddown manifold.
2. Before performing the next step, it will be necessary to draw a vacuum on the hydraulic tank to prevent oil loss when the bleeddown manifold tank return line is disconnected.

### **IMPORTANT**

*Hydraulic tank oil level is above the level of this return line. It is necessary to draw a vacuum on the hydraulic tank to prevent a large amount of oil draining from the tank with the return line disconnected from the bleeddown manifold.*

3. Disconnect tank return line (15, Figure 10-10) from the bleeddown manifold. Connect the hoist pilot valve return hose that was removed in Step 1 to the tank return line, or connect it to the hydraulic tank fill port. Do not plug the hoist pilot return hose.
4. Remove the vacuum on the hydraulic tank.
5. Start the engine and allow the system to build pressure until the unloader valve unloads the pump. Measure the leakage at the open "tank" return port on the bleeddown manifold.

- ❑ Maximum permissible leakage is 541 ml/min. (33 in<sup>3</sup>/min.).

#### *\* Record on Data Sheet*

If leakage is excessive, bleeddown solenoid (6, Figure 10-10), steering system relief valve (13), or the piloted check valve located on top of bleeddown manifold is defective and should be replaced.

### **WARNING**

*In the next step, DO NOT use the key switch to shut off the engine. If the key switch is turned OFF, all accumulator oil will be discharged through the open port on the bleed down manifold!*

*A suggested method to provide steering accumulator bleed down when return hoses are disconnected is to connect a hose/needle valve assembly between bleeddown manifold port TP3 and the hydraulic tank filler tube. This hose/valve assembly cannot be connected or disconnected if the accumulators are pressurized. The opposite end of the hose can be connected to a fitting which screws into the hydraulic tank filler tube in place of the filler cap. Both accumulators will be bled down when port TP3 is opened to tank, reducing the possibility of either accumulator being accidentally left pressurized. The hose can be 1/4" SAE 100R2, rated for 5000 psi, or 3/8" SAE 100R2, rated for 4000 psi. Needle valves are suggested to allow simple opening/closing, but must be rated for 4000 psi or above.*

6. After test is complete, shut off the engine by using the engine shutdown switch located on the center console in the cab.
7. Reattach the vacuum source to the hydraulic tank and draw a vacuum before disconnecting the tank return hose from the pilot valve return hose. Attach the tank return hose to the bleeddown manifold. Attach the hoist pilot return hose to the fitting on the manifold.
8. Remove the vacuum source. Remove all test equipment and connect all remaining hoses to their proper locations.
9. Turn the key switch to OFF to allow the steering accumulators to bleed down.

## STEERING/BRAKE PUMP STROKE ADJUSTMENT

Pump stroke adjustment screw (1, Figure 10-9) controls the volume of oil produced by the steering/brake pump. Use of this feature is required only if the truck is operated at high altitude locations (greater than 3 050 meters (10,000 ft) above sea level). Adjusting the pump stroke improves pump inlet suction by reducing pump displacement. The improved suction helps prevent damage from pump cavitation.

For trucks operating at lower altitudes, this adjustment should not be changed from the factory setting (adjusted fully counterclockwise). The factory setting provides full pump flow for maximum performance.

*NOTE: Changing the settings for the pressure compensator and the unloader valve is not necessary for the high altitude adjustment.*

1. Park the truck on a hard, level surface and apply the parking brake.
2. Depressurize the brake and steering system accumulators.

For the steering circuit, turn the key switch to OFF and allow 90 seconds for the accumulators to depressurize. After 90 seconds, turn the steering wheel to verify that pressure has been purged from the circuit. If the wheels do not move, the steering circuit is safe to service.

For the brake circuit, first, chock the wheels. Then, open the shut-off valves on the brake manifold. Opening the valves allows accumulator pressure to be released to the hydraulic tank.

3. Locate pump stroke adjustment screw (1, Figure 10-9). Loosen the locknut securing the adjustment screw.
4. Turn the adjustment screw counterclockwise until the screw reaches the stop.
5. Turn the screw clockwise until resistance is felt. Once resistance is felt, continue to turn the screw clockwise for 6 1/2 revolutions.
6. Hold the adjustment screw in place and tighten the locknut.
7. Drive the truck and verify that the steering and brake systems are functioning properly.

## SHOCK AND SUCTION VALVES

### Equipment Requirements

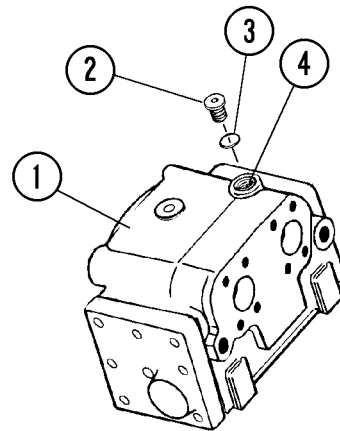
The following equipment will be necessary to properly check the steering circuit shock and suction valves:

- Hydraulic schematic (refer to Section R)
- Three 35 000 kPa (5,000 psi) range calibrated pressure gauges and hoses

### Pressure Tests

Before the shock and suction valves in the steering circuit can be tested, steering system pressure must be increased to obtain sufficient pressure for testing. In addition, the steering relief valve pressure must be raised above the pressure required to actuate the shock and suction valves.

1. Install a 35 000 kPa (5,000 psi) pressure gauge at "TP2" (3, Figure 10-10) in the steering bleeddown manifold.
2. Install a 35 000 kPa (5,000 psi) gauge in each steering cylinder manifold test port located on frame cross member under the engine.



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FIGURE 10-11. STEERING RELIEF VALVE ADJUSTMENT

- |                         |                                  |
|-------------------------|----------------------------------|
| 1. Flow Amplifier Valve | 4. Relief Valve Adjustment Screw |
| 2. Plug                 |                                  |
| 3. O-Ring               |                                  |

3. Raise steering relief valve pressure as follows:
  - a. Use an 8 mm allen wrench to remove external plug (2, Figure 10-11) on the flow amplifier valve.
  - b. Insert a 5 mm allen wrench into the opening and gently bottom out adjustment screw (4) by turning it clockwise.
4. Start the engine and allow the steering system to build pressure.
5. While observing the gauge at the "TP2" steering pressure test port, turn unloader valve adjustment screw (2, Figure 10-9) clockwise until approximately 22 750 kPa (3,300 psi) is obtained.

*NOTE: This pressure is near the pump compensator pressure setting and the pump may not unload. However, it is not necessary for the pump to unload during this test.*

6. While observing the two gauges installed on the steering manifold, steer the truck against the left stop.
  - ☐ Pressure on one of the gauges should read 21 370 -0/+2068 kPa (3,100 -0/+300 psi).

*\* Record on Data Sheet*

7. Steer the truck to the opposite stop.
  - ☐ The pressure on the other gauge should read 21 370 -0/+2068 kPa (3,100 -0/+300 psi).

*\* Record on Data Sheet*

8. If the pressure is incorrect during Step 6 or 7, the shock and suction valves must be replaced.

*NOTE: The shock and suction valves are only serviced as complete units and cannot be adjusted while installed in the flow amplifier valve.*

9. After the test is complete, lower the steering relief pressure to 18 960 kPa (2,750 psi) as follows:
  - a. Steer full left or right and maintain a slight pressure against the steering wheel.
  - b. Use a 5 mm allen wrench to adjust the steering relief valve to obtain 18 960 kPa (2,750 psi) on the gauge.

*\* Record on Data Sheet*

10. After the adjustment is complete, install plug (2, Figure 10-11) with O-ring (3) on the valve body.
11. Reset the unloader valve to the specified unload pressure. Back out the unloader valve adjustment screw completely counterclockwise.
12. Steer the truck to reduce pressure in the steering circuit and cause the pump to load.
  - a. Observe the increasing pressure readings on the gauge installed at steering pressure "TP2" test port (3, Figure 10-10).
    - ☐ When the unload pressure increases to 20 680 to 21 025 kPa (3,000 to 3,050 psi), the adjustment has been successfully completed.
  - b. Steer the truck again while observing the steering pressure feedback gauge.
    - ☐ The pump must reload when the pressure drops to 18 960 kPa (2,750 psi) minimum.

*\* Record on Data Sheet*

13. If the minimum reloading pressure in Step 12 is not correct, repeat the unloader valve adjustment.

*NOTE: The critical pressure setting is the 18 960 kPa (2,750 psi) when the pump loads. The unloading pressure follows the loading pressure adjustment and should occur at approximately 20 855 kPa (3,025 psi).*

## HOIST SYSTEM RELIEF VALVE AND BRAKE COOLING CIRCUIT PRESSURES

*NOTE: If the relief valve or hoist valve assembly has been replaced or rebuilt, the hoist valve “power up” pressure should be checked. Also, check the brake cooling circuit for correct pressures. Check the hoist system “power down” relief pressure if the hoist pilot valve has been replaced or rebuilt.*

### **WARNING**

**Relieve pressure before disconnecting any hydraulic lines. Tighten all connections securely before applying pressure.**

#### Equipment Requirements

The following equipment will be necessary to properly check the hoist relief and brake cooling circuit pressures:

- Hydraulic schematics (refer to Section R)
- Three 25 000 kPa (3,500 psi) range calibrated pressure gauges and hoses for hoist circuit pressure readings
- Two 1 000 kPa (100 psi) low pressure gauges and hoses for brake cooling circuit pressure readings

*NOTE: Hydraulic oil temperature should be approximately 21°C (70°F) during test.*

#### Pressure Gauge Locations

1. Install a 1 000 kPa (100 psi) low pressure gauge at brake cooling circuit test port (1, Figure 10-12) at the front of the hoist valve to monitor the front brake cooling oil pressure.
2. Install a 25 000 kPa (3,500 psi) gauge in power down test port (3, Figure 10-17) (marked “TPD”) located on the front face of the overcenter manifold.
3. Install a 25 000 kPa (3,500 psi) gauge in each hoist pump filter pressure test port (2, Figure 10-12).
4. Install a 1 000 kPa (100 psi) low pressure gauge at test port (3, Figure 10-15) in the brake/hoist return manifold to monitor the rear brake cooling supply pressure.

#### Brake Cooling Circuit Test

*NOTE: Hydraulic oil temperature should be approximately 21°C (70°F) during this test. Do not turn the steering wheel or apply the brakes. Do not operate the engine at low idle for more than 30 seconds.*

1. Start the engine and operate at low idle. Place the hoist control lever in the FLOAT position.
  - ☐ Hoist pump outlet pressures at the filters should be approximately 550 kPa (80 psi).

*\* Record on Data Sheet*

- ☐ Pressure at front brake cooling circuit (1, Figure 10-12) and rear brake cooling circuit (3, Figure 10-15) should be approximately 172 kPa (25 psi) or less.

*\* Record on Data Sheet*

2. With the engine at low idle, move the hoist control lever to POWER UP.

- ☐ Pressure at the front and rear brake cooling circuits should drop to 0 kPa (0 psi) while the body rises.

If the pressure is not correct, the hoist control valve may be plumbed incorrectly or defective.

*\* Record on Data Sheet*

3. Increase engine speed to 1,500 rpm. Place hoist control lever in HOLD or FLOAT.

- ☐ Pressure at the front and rear brake cooling circuits should be approximately 344 kPa (50 psi) or less.

*\* Record on Data Sheet*

4. With engine at 1,500 rpm, move the hoist control lever to POWER DOWN.

- ☐ Pressure at the front and rear brake cooling circuits should drop to 0 kPa (0 psi) while the body lowers.

*\* Record on Data Sheet*

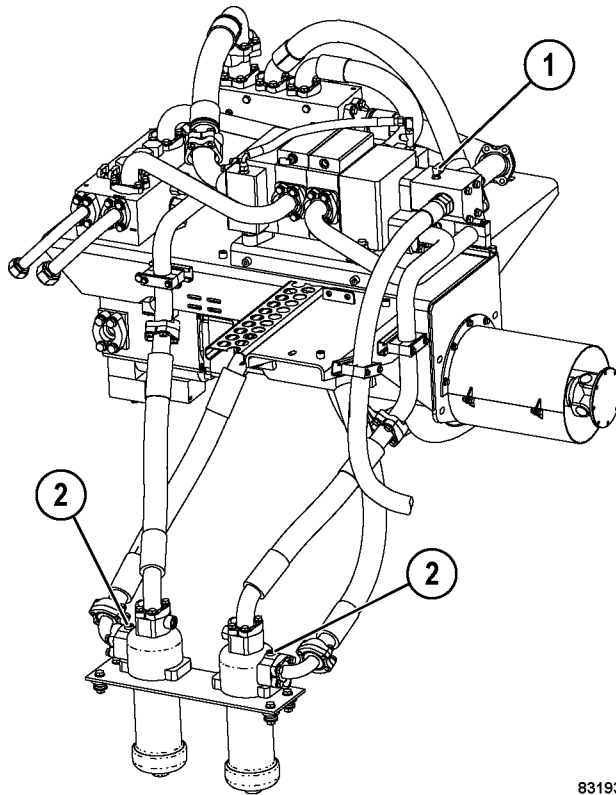


FIGURE 10-12. PRESSURE TEST PORTS  
1. Test Port - Front Brake Cooling  
2. Test Port - Hoist Pump Pressure at Filter

### Power Up Relief Pressure Test

The hoist control valve contains two relief valves. The rear inlet section contains the relief valve for the rear section of the hoist pump. The front inlet section contains the relief valve for the front section of the hoist pump.

1. To allow full extension of the hoist cylinders, disconnect the hoist up limit solenoid (located on the bottom of the bleeddown manifold) from the wiring harness.



**Ensure that there is adequate overhead clearance before raising the body.**

*NOTE: Ensure that the gauges are properly identified (front or rear section of the pump) when the pressures are read.*

2. Start the engine and operate at low idle. Move the hoist control lever to POWER UP until the body is fully raised.

- Pressure at both hoist pump filter test ports should be  $17\,235 \pm 690$  kPa ( $2,500 \pm 100$  psi).

*\* Record on Data Sheet*

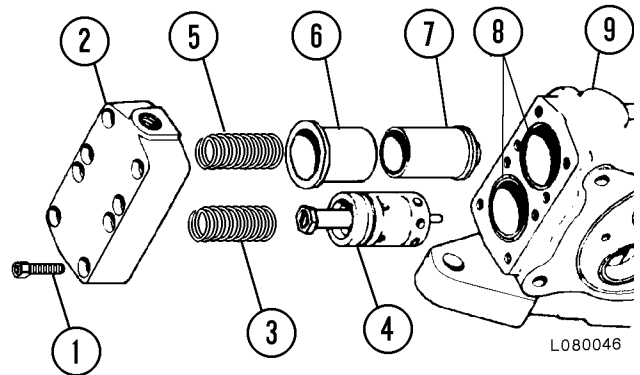


FIGURE 10-13. INLET SECTION  
(Front & Rear Inlet Sections are Identical)

- |                      |                                 |
|----------------------|---------------------------------|
| 1. Capscrew          | 6. Sleeve                       |
| 2. Inlet Cover       | 7. Secondary Low Pressure Valve |
| 3. Spring            | 8. O-Rings                      |
| 4. Main Relief Valve | 9. Inlet Valve body             |
| 5. Spring            |                                 |

3. If the hoist power up relief pressure is incorrect on either gauge, the corresponding relief valve should be replaced.

- a. Lower the body until it is resting on the frame rails. Shut off the engine and wait at least 90 seconds for the accumulators to bleed down. Turn the steering wheel to ensure that system pressure has been relieved.
- b. Relieve all hydraulic pressure from the hoist system.
- c. Disconnect the tube from inlet cover (2, Figure 10-13). Disconnect the hose(s) at the top port.
- d. Remove capscrews (1) and cover (2). Remove spring (3) and relief valve (4).
- e. Install new relief valve (4), spring (3), cover (2), and new O-rings (8). Install and tighten capscrews (1). Connect the tube to the fitting on cover (2). Install the hose(s) at the top port.

4. Check the pressure again by repeating Step 2. If necessary, repeat the adjustment procedure until the correct pressure is obtained.



## Power Down Relief Pressure Test

**NOTE:** The hoist system power down relief valve is located on the hoist pilot valve in the hydraulic brake cabinet behind the cab.

1. Start the engine and operate at low idle. Allow the accumulators to fill and the steering pump to unload.
2. With the body resting on the frame, move the hoist control lever to POWER DOWN.

- Pressure at the overcenter manifold test port "TPD" should be  $10\,340 \pm 517$  kPa ( $1,500 \pm 75$  psi).

\* Record on Data Sheet

3. If power down relief pressure is not within specifications, adjust the relief valve as follows:
  - a. Remove the cap from power down relief valve (2, Figure 10-14).
  - b. To increase the power down relief pressure, turn the adjusting screw inward (clockwise). To decrease the power down relief pressure, turn the adjusting screw outward (counterclockwise).

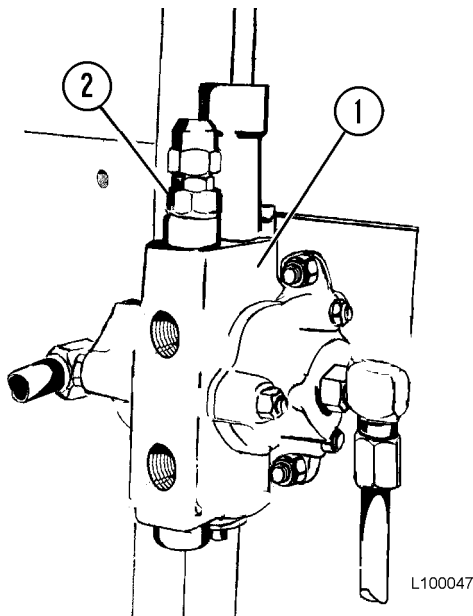


FIGURE 10-14. POWER DOWN RELIEF VALVE

1. Spool Housing
2. Power Down Relief Valve

4. Recheck the power down relief pressure by repeating Step 2 after adjusting the relief valve. Install the cap on the relief valve.
5. Shut off the engine and allow 90 seconds for the accumulators to bleed down completely. Remove the gauges and hoses.

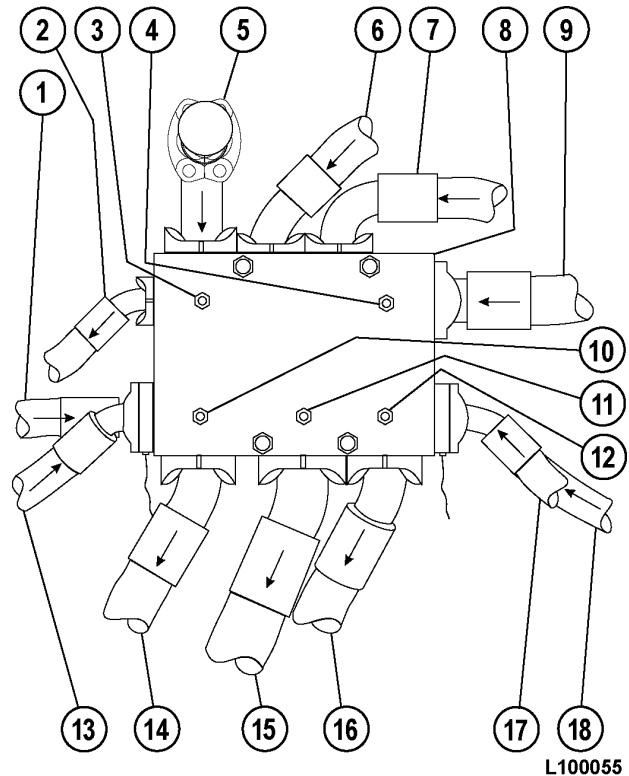


FIGURE 10-15. BRAKE/HOIST RETURN OIL MANIFOLD

- |                                              |                                 |
|----------------------------------------------|---------------------------------|
| 1. Rear Brake Return                         | 9. Front Brake Bypass           |
| 2. Rear Brake Supply                         | 10. Rear Brake Return Pressure  |
| 3. Rear Brake Supply Pressure                | 11. Hoist Return Pressure       |
| 4. Front Brake Supply Pressure               | 12. Front Brake Return Pressure |
| 5. Rear Brake Cooling Oil (From Hoist Valve) | 13. Rear Brake Return to Tank   |
| 6. Return From Hoist Valve                   | 14. Rear Brake Return to Tank   |
| 7. Return From Hoist Valve                   | 15. Hoist Return to Tank        |
| 8. Brake/Hoist Return Oil Manifold           | 16. Front Brake Return to Tank  |
|                                              | 17. Front Brake Return          |
|                                              | 18. Front Brake Return          |

### Counterbalance Valve Pressure Check

1. Ensure that the engine is off, the body is resting on the frame or properly secured in its raised position, the hoist control valve is in the FLOAT position, and all hydraulic system pressure is bled down.
2. Loosen the locknut on the adjustment stem of needle valve (9, Figure 10-17) on the overcenter manifold. Turn the adjustment stem fully clockwise.
3. Remove the plug from "PILOT VENT" port (8) on the overcenter manifold. This port will remain open to the atmosphere during the adjustment. Do not allow dirt to enter the open port.

*NOTE: It is suggested that a clean SAE #4 (1/4") hydraulic hose be installed in the open port and pointed downward.*

4. Install a 35 000 kPa (5,000 psi) gauge at test port "TR" (7) on the overcenter manifold. Use a hose that is long enough for the technician to read the pressure gauge while in the cab of the truck. Do not read the gauge near the overcenter manifold or make adjustments while the engine is running. The gauge will measure the rod end pressure (the pressure controlled by the counterbalance valve).

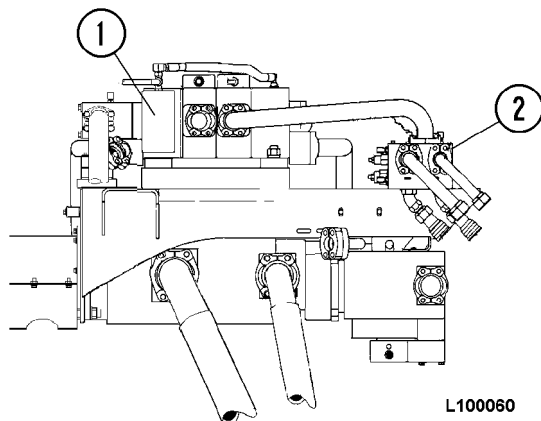


FIGURE 10-16. COUNTERBALANCE VALVE ADJUSTMENT

1. Hoist Valve

2. Overcenter Manifold

5. Start the engine and operate at low idle. If the body is in the raised position, lower the body to the frame.
6. Raise the body and, as it extends to the third stage, read the pressure on the gauge connected to the "TR" port. (All counterbalance valve pressures are read and adjusted while the hoist cylinders are in the third stage.)
  - a. If pressure is 20 700 kPa (3,000 psi) or above, stop raising the body immediately. The pressure is adjusted too high and must be lowered. Refer to "Counterbalance Valve Adjustment".
  - b. If pressure is below 20 700 kPa (3,000 psi), lower the body, increase engine speed by approximately 300 rpm and repeat Step 6 until the engine is at high idle or 20,700 kPa (3000 psi) is obtained.
  - c. If the gauge indicates 20 700 kPa (3,000 psi) with the engine at high idle, the hoist control lever in POWER UP and the hoist cylinders in the third stage, the counterbalance valve adjustment is correct.

#### \* Record on Data Sheet

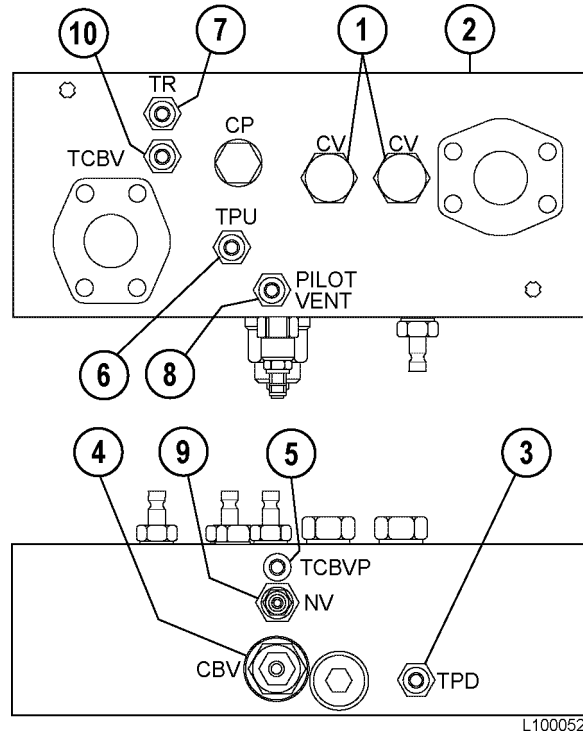
- d. If the gauge still does not indicate 20 700 kPa (3,000 psi) with the engine at high idle, the hoist control lever in POWER UP and the hoist cylinders in the third stage, or if 20 700 kPa (3,000 psi) was obtained before the engine reached high idle, adjust the counterbalance valve.

## Counterbalance Valve Adjustment

1. With the hoist control lever in the FLOAT position and the body resting on the frame, loosen the locknut on the adjustment stem of counterbalance valve (4, Figure 10-17). Turn the adjustment stem fully inward (clockwise) so the counterbalance valve pressure is as low as possible.
2. Start the engine and operate at high idle. Observe the pressure gauge while raising the body.
3. Carefully turn the adjustment stem outward (counterclockwise) to obtain 20 700 kPa (3,000 psi) with the engine at high idle, the hoist control lever in POWER UP and the hoist cylinders in the third stage.

**NOTE:** Complete valve adjustment range is three full turns.

4. Lower the body to the frame and operate the engine at low idle. Repeat Step 6 of the "Counterbalance Valve Pressure Check" procedure to verify proper adjustment.
5. With the body resting on the frame, install the plug in "PILOT VENT" port (8). Remove the pressure gauge.
6. Turn needle valve adjustment stem (9) fully outward and tighten the locknut securely. The needle valve must be fully open for normal operation.



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FIGURE 10-17. OVERCENTER MANIFOLD VALVE

1. Check Valves
2. Overcenter Valve Manifold
3. Power Down Test Port (TPD)
4. Counterbalance Valve
5. Counterbalance Valve Test Port (TCBVP)
6. Power Up Test Port (TPU)
7. Test Port (TR)
8. Pilot Vent Port
9. Needle Valve
10. Counterbalance Valve Port (TCBV)

## 930E CHECK-OUT PROCEDURE - STEERING SYSTEM DATA SHEET

MACHINE MODEL \_\_\_\_\_ UNIT NUMBER \_\_\_\_\_ SERIAL NUMBER \_\_\_\_\_

\_\_\_\_\_ Steering accumulators charged to 9650 kPa (1400 psi).

Operate the hydraulic steering system to obtain the proper operating temperature.  
Refer to the Check-out Procedures.

### ***PUMP PRESSURE CONTROL ADJUSTMENTS***

STEP 6 \_\_\_\_\_ Steering pump compensator pressure

STEP 11 \_\_\_\_\_ Steering pump unload pressure

STEP 12 \_\_\_\_\_ Steering pump load pressure

### ***STEERING CONTROL VALVE AND FLOW AMPLIFIER LEAKAGE TESTS***

STEP 4 \_\_\_\_\_ Flow amplifier return hose leakage

STEP 5 \_\_\_\_\_ Steering control unit return hose leakage

### ***BLEEDDOWN MANIFOLD LEAKAGE TEST***

STEP 5 \_\_\_\_\_ Bleeddown solenoid, steering relief, and piloted check valve leakage

### ***SHOCK AND SUCTION VALVES TEST***

STEP 6 \_\_\_\_\_ Shock and suction valve pressure, left steer

STEP 7 \_\_\_\_\_ Shock and suction valve pressure, right steer

STEP 9 \_\_\_\_\_ Steering relief valve pressure setting

STEP 12 \_\_\_\_\_ Steering pump reload pressure

Name of Technician or Inspector Performing Check-Out \_\_\_\_\_

Date \_\_\_\_\_

## 930E CHECK-OUT PROCEDURE - BRAKE COOLING & HOIST SYSTEM DATA SHEET

MACHINE MODEL \_\_\_\_\_ UNIT NUMBER \_\_\_\_\_ SERIAL NUMBER \_\_\_\_\_

Operate the brake cooling & hoist hydraulic system to obtain the proper operating temperature.  
Refer to the Check-out Procedures.

### ***BRAKE COOLING CIRCUIT TEST***

STEP 1 \_\_\_\_\_ Hoist pump outlet pressures, low idle, FLOAT position

\_\_\_\_\_ Brake cooling circuit pressure, low idle, FLOAT position

STEP 2 \_\_\_\_\_ Brake cooling circuit pressure, low idle, POWER UP/DOWN position

STEP 3 \_\_\_\_\_ Brake cooling circuit pressure, 1500 RPM, HOLD/FLOAT position

STEP 4 \_\_\_\_\_ Brake cooling circuit pressure, 1500 RPM, POWER UP/DOWN position

### ***POWER UP RELIEF PRESSURE TEST***

STEP 2 \_\_\_\_\_ Hoist relief pressure, front pump section

\_\_\_\_\_ Hoist relief pressure, rear pump section

### ***POWER DOWN RELIEF PRESSURE TEST***

STEP 2 \_\_\_\_\_ Power down relief pressure

### ***COUNTERBALANCE VALVE ADJUSTMENT***

STEP 6 \_\_\_\_\_ Counterbalance valve pressure

Name of Technician or Inspector Performing Check-Out \_\_\_\_\_

DATE \_\_\_\_\_

|                                 |       |
|---------------------------------|-------|
| HOT START SYSTEM.....           | M7-1  |
| SPECIAL TOOLS .....             | M8-1  |
| RADIATOR SHUTTERS .....         | M19-1 |
| PAYLOAD METER III.....          | M20-1 |
| RESERVE ENGINE OIL SYSTEM ..... | M31-1 |

# NOTES

**SECTION M7**  
**HOT START SYSTEM**  
**INDEX**

HOT START SYSTEM ..... M7-3

    SYSTEM OPERATION ..... M7-3

    HEATER INFORMATION ..... M7-4

    TROUBLESHOOTING GUIDELINES ..... M7-5



# NOTES

## HOT START SYSTEM

The information in this chapter applies to trucks equipped with the -40° C hot start system. Other cold weather packages may not use all of the heating elements described in this chapter.

### SYSTEM OPERATION

The hot start system is a 240V system and is equipped with heating elements in the following components:

- Hydraulic tank
- Engine oil pan
- Cooling system

An outside power source is connected to a power receptacle to operate the system. Power receptacle (1, Figure 7-1) is mounted under the LH platform.

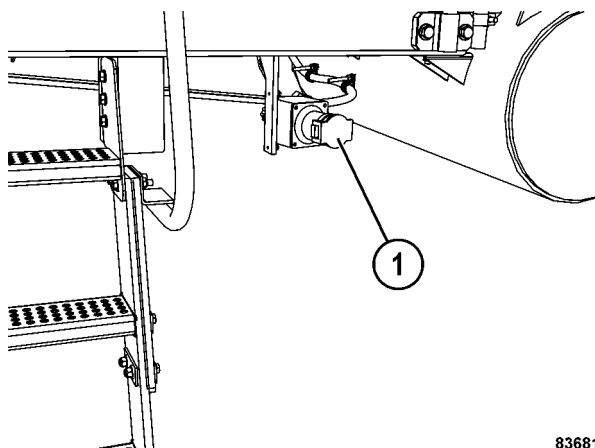


FIGURE 7-1. HEATING SYSTEM POWER RECEPTACLE

1. Power Receptacle

Power from the receptacle travels to a junction box containing a terminal block. The terminal block distributes power to the various heaters in the system. Junction box (1, Figure 7-2) is located on the rear of the engine sub-frame.

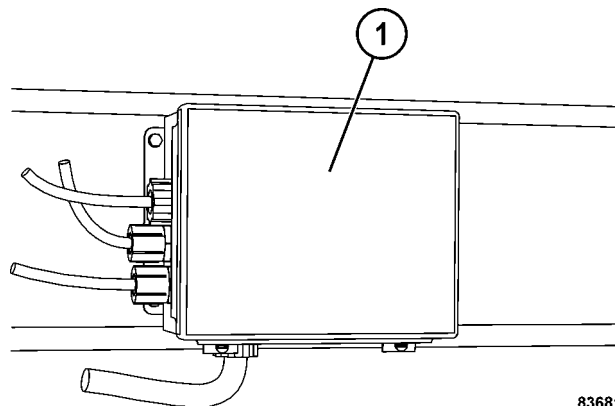


FIGURE 7-2. POWER JUNCTION BOX

1. Junction Box

The heaters in each location are controlled by thermostats. The thermostats turn on the heaters when temperature falls below a preset value. After the heating element raises the oil or coolant temperature above the preset value, the thermostats remove power to the heating elements.

### **CAUTION**

*Do not operate the engine while the cooling system heater is plugged in. The flow check valve eliminates coolant flow through the heater while the engine is on. This will cause a lack of circulation in the heater and burn out the heating elements.*

### **CAUTION**

*Close the shut off valves before replacing a coolant heater. Failure to close the valve will result in a loss of coolant from the cooling system.*

HEATER INFORMATION

Hydraulic Tank

Heater Quantity .....2  
Operating Range:..... 16° - 27° C (60° - 80° F)  
Wattage .....500W

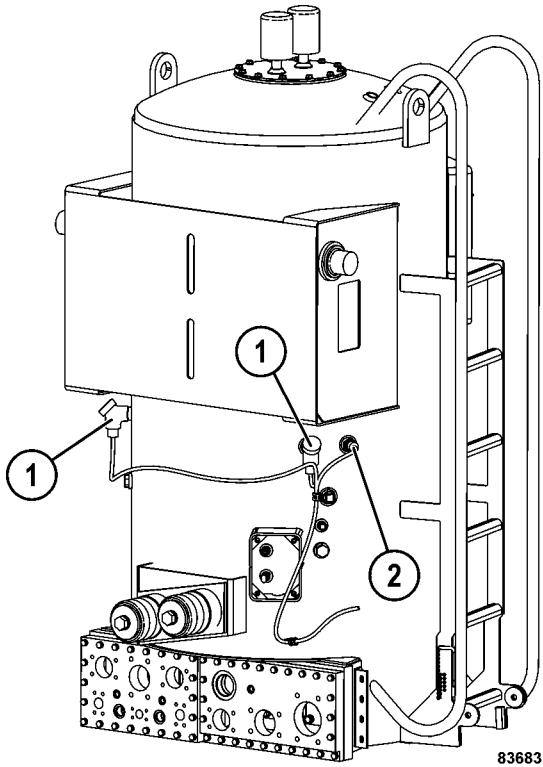


FIGURE 7-3. HYDRAULIC TANK HEATERS  
1. Oil Heater                      2. Thermostat

Engine Oil Pan

Heater Quantity ..... 2  
Operating Range: ..... 27° - 38° C (80° - 100° F)  
Wattage ..... 500W

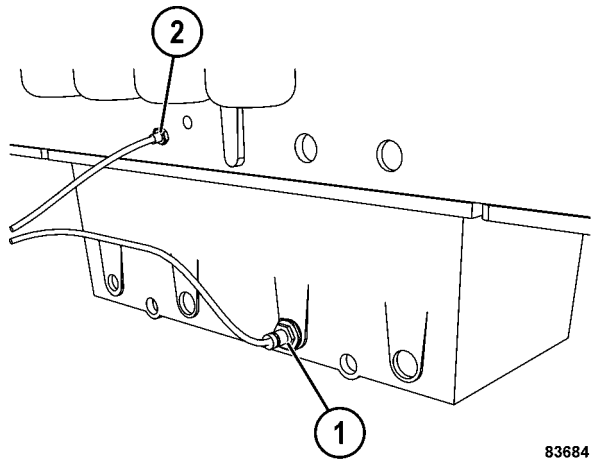


FIGURE 7-4. ENGINE OIL HEATERS  
1. Oil Heater                      2. Thermostat

Cooling System

Heater Quantity ..... 2  
Operating Range ..... 38° - 49° C (100° - 120° F)  
Wattage ..... 2500W

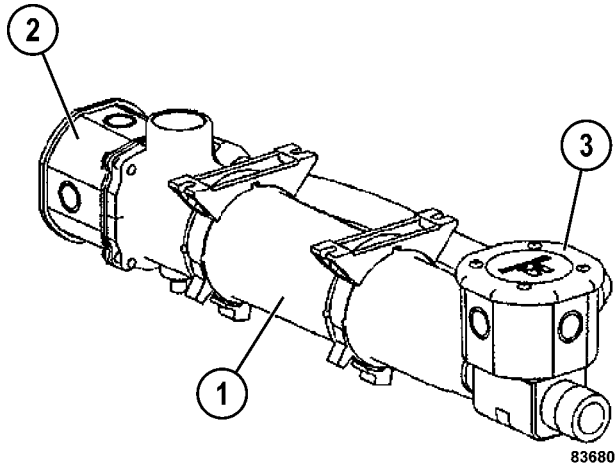


FIGURE 7-5. COOLANT HEATERS  
1. Coolant Heater                      3. Control Thermostat  
2. High Limit Thermostat

## TROUBLESHOOTING GUIDELINES

Refer to the electrical schematic located in Section R for a diagram of the circuits.

### SYMPTOM

### RECOMMENDED CHECKS

**All of the heating elements do not heat.**

Check for power going into the main power receptacle.

Check for power at the junction box.

Check for loose connections at the terminal board.

**Some of the heating elements do not heat.**

Check for loose connections at the jumper cables in between each terminal board in the junction box.

**Some of the heating elements operate outside of the proper temperature range.**

Check for loose connections.

Replace the thermostat that controls the affected heating elements.

**A heating element will not heat up.**

Connect a power source directly to the heating element and check. If the element does not heat, replace the element.

Check the wiring for an open circuit.

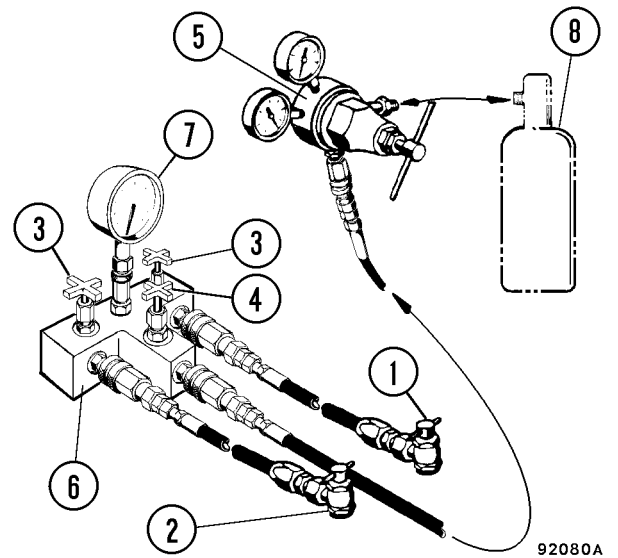
# NOTES

## SPECIAL TOOLS

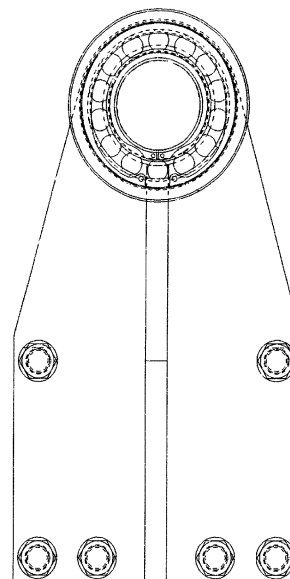
| Part Number | Description           | Use                                        |
|-------------|-----------------------|--------------------------------------------|
| EB1759      | Nitrogen Charging Kit | Suspension & Accumulator Nitrogen Charging |

1. "T" Handle Valve
2. Charging Valve Adapter
3. Manifold Outlet Valves (from gauge)
4. Inlet Valve (from regulator)
5. Regulator Valve (Nitrogen Pressure)
6. Manifold
7. Charging Pressure Gauge (Suspensions)
8. Dry Nitrogen Gas

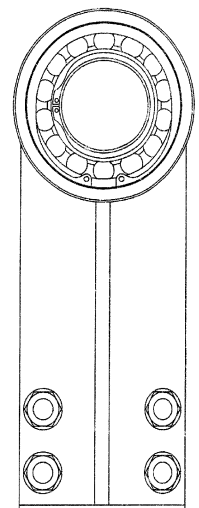
*NOTE: Arrangement of parts may vary from illustration shown, depending on Charging Kit part number.*



| Part Number                                      | Description                     | Use                                             |
|--------------------------------------------------|---------------------------------|-------------------------------------------------|
| EJ2626                                           | Roller Assembly                 | 930E-4                                          |
| EH8687<br>(No longer available as complete unit) | Roller Assembly                 | 930E-4SE<br>Power Module Removal & Installation |
| EJ2271                                           | Roller Mount                    | 930E-4                                          |
| EH8681                                           | Roller Mount                    | 930E-4SE                                        |
| PC0706                                           | Bearing                         |                                                 |
| TH9449                                           | Bearing Retainer Ring           |                                                 |
| TG1680                                           | Roller Retainer Ring            |                                                 |
| C1645                                            | Capscrew - 3/4" - 10NC x 2 1/4" |                                                 |
| C1542                                            | Lockwasher - 3/4"               |                                                 |
| EH8686                                           | Roller Ring                     |                                                 |

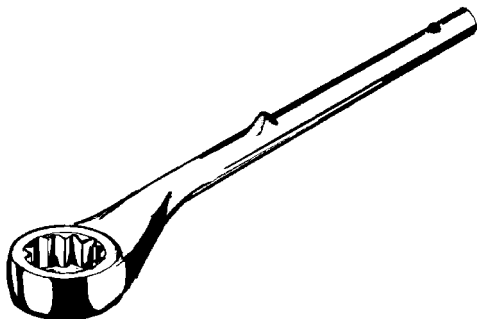


EH8687

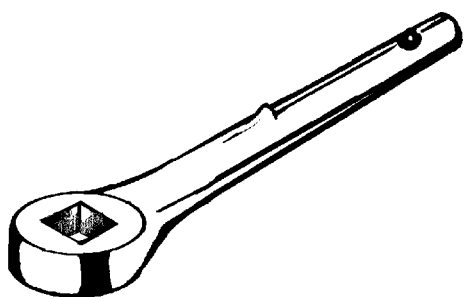


EJ2626

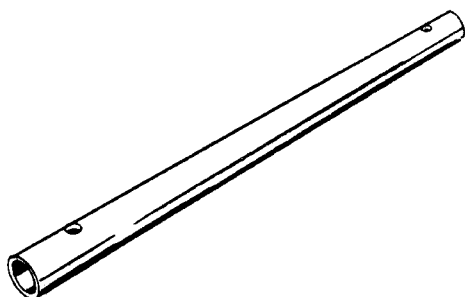
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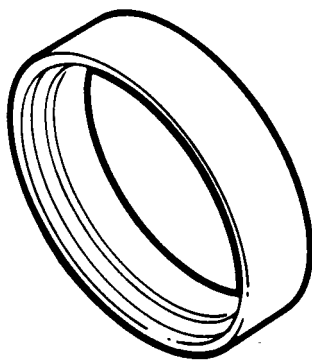
| Part Number | Description                     | Use                          |
|-------------|---------------------------------|------------------------------|
| PB8326      | Offset Box End Wrench - 1 7/16" | Cab Mounting & Miscellaneous |



| Part Number | Description         | Use           |
|-------------|---------------------|---------------|
| TZ2734      | 3/4" Torque Adapter | Miscellaneous |

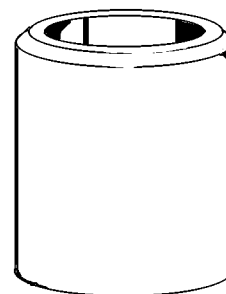


| Part Number | Description    | Use                      |
|-------------|----------------|--------------------------|
| TZ2733      | Tubular Handle | Use with PB8326 & TZ2734 |

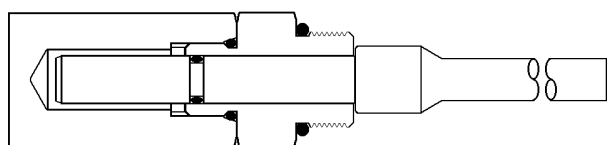


| Part Number | Description            | Use                                                     |
|-------------|------------------------|---------------------------------------------------------|
| BF4117      | Seal Installation Tool | Front & Rear Disc Brake Floating Ring Seal Installation |
| ED3347      | Seal Installation Tool | Rear Axle/Hub Adapter Floating Ring Seal Installation   |

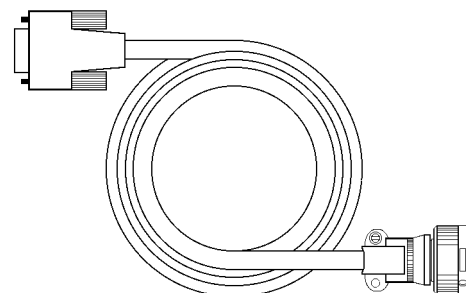
| Part Number | Description           | Use                                   |
|-------------|-----------------------|---------------------------------------|
| EH4638      | Sleeve Alignment Tool | Steering Linkage and Tie Rod Assembly |



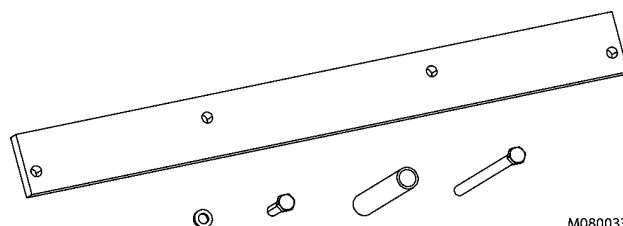
| Part Number | Description           | Use             |
|-------------|-----------------------|-----------------|
| EF9302      | Wear Indicator        | Brake Disc Wear |
| EB1723      | Cap, Indicator        |                 |
| EF9301      | Pin, Indicator        |                 |
| WA0010      | O-Ring, Indicator Pin |                 |
| TL3995      | O-Ring, Indicator Cap |                 |
| EB4813      | Housing, Indicator    |                 |
| SV9812      | O-Ring, Housing       |                 |



| Part Number | Description | Use                    |
|-------------|-------------|------------------------|
| PC2525      | Harness     | Payload Meter Download |



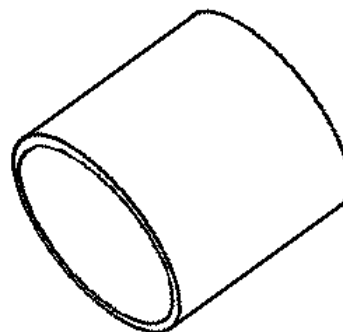
| Part Number | Description               | Use                          |
|-------------|---------------------------|------------------------------|
| EK3798      | Brake Assembly Tool Group | Brake Assembly & Disassembly |
| ED3757      | Bar, Flat                 |                              |
| ED3758      | Pipe                      |                              |
| C1540       | Lockwasher                |                              |
| C1615       | Capscrew, Short           |                              |
| D9105       | Capscrew, Long            |                              |



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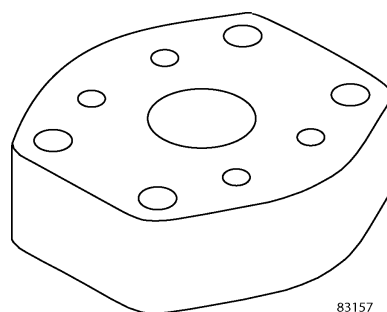
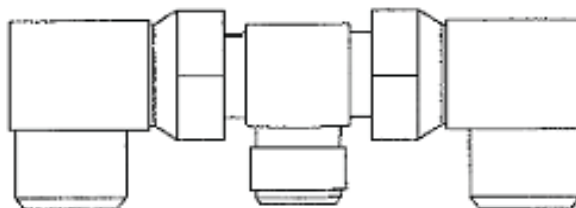


| Part Number | Description    | Use             |
|-------------|----------------|-----------------|
| EH7817      | Alignment Tool | Upper Hoist Pin |



| Part Number | Description    | Use             |
|-------------|----------------|-----------------|
| EH7817      | Alignment Tool | Upper Hoist Pin |

| Part Number | Description      | Use                    |
|-------------|------------------|------------------------|
| XB7075      | Fitting Assembly | Hoist Circuit Flushing |
| PC3074      | Flushing Block   | Hoist Circuit Flushing |

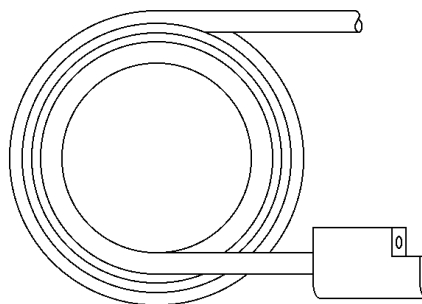


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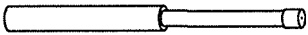
| Part Number | Description   | Use                       |
|-------------|---------------|---------------------------|
| XB5777      | Hose Assembly | Steering Circuit Flushing |



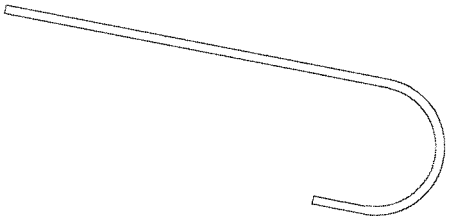
| Part Number | Description        | Use            |
|-------------|--------------------|----------------|
| PB9067      | Bulkhead Connector | Battery Jumper |



| Part Number | Description         | Use                    |
|-------------|---------------------|------------------------|
| PC2061      | Belt Tension Tester | A/C Belt Tension Check |



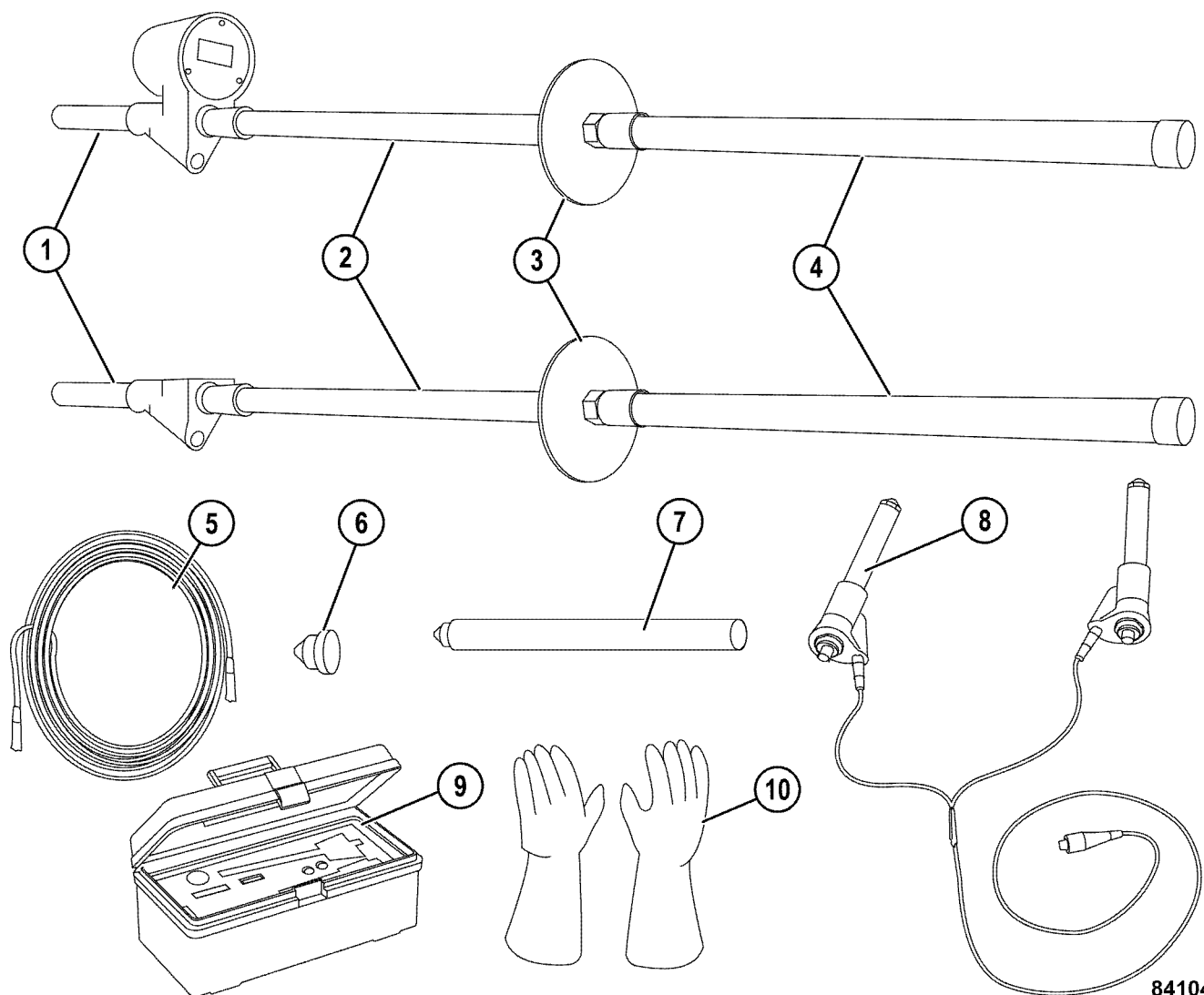
| Part Number | Description           | Use                |
|-------------|-----------------------|--------------------|
| EL8868      | V-Belt Alignment Tool | A/C Belt Alignment |



## HIGH VOLTAGE TOOLS

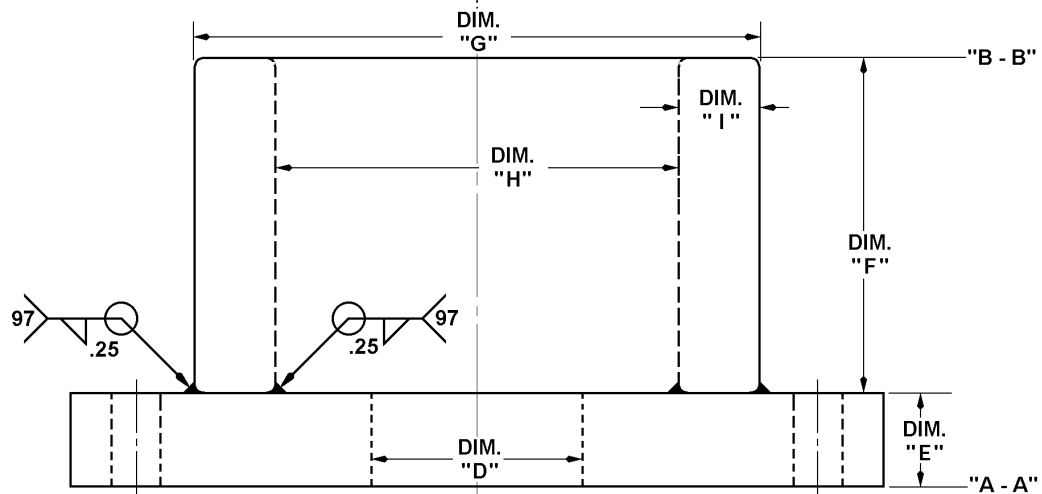
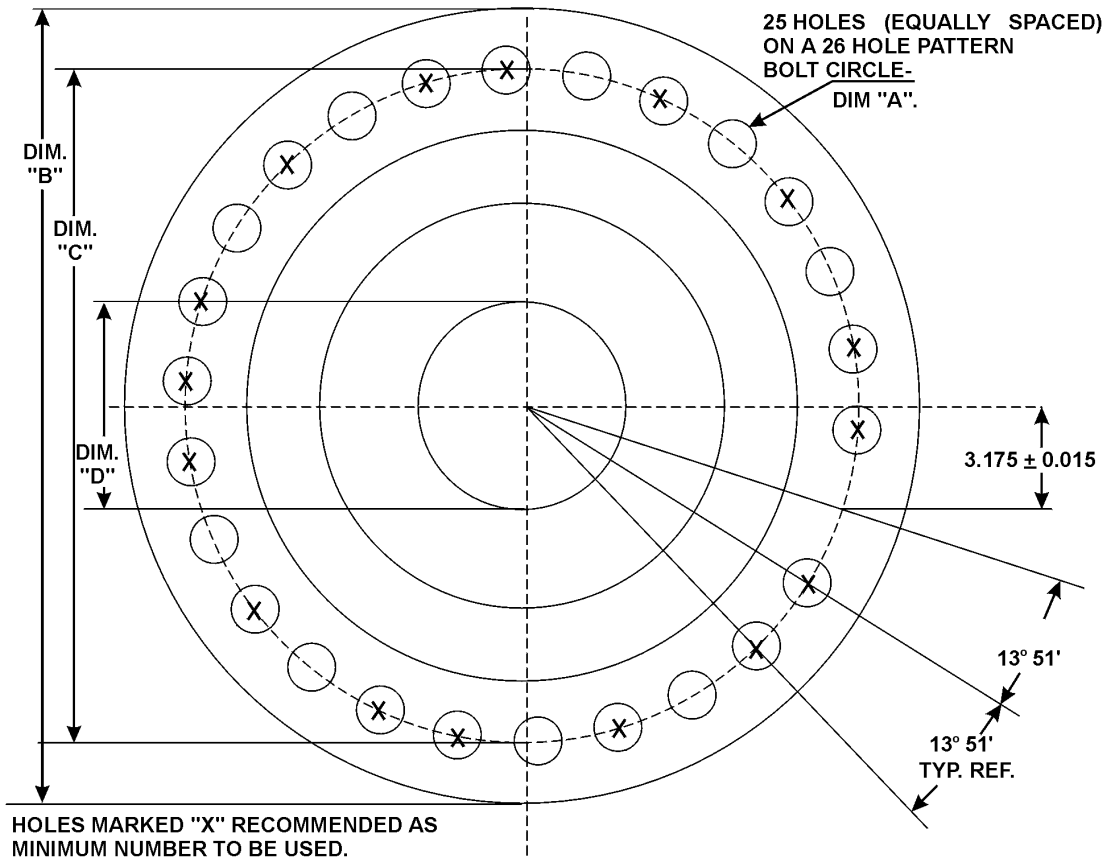
### 0-2000 VDC CUSTOM METER KIT (58B-06-00800)

| Item Number | Part Number  | Description                | Use                       |
|-------------|--------------|----------------------------|---------------------------|
| 1           | 58B-06-00700 | 2000 V Meter & Probe       | Servicing control cabinet |
| 2           | 58B-06-00730 | Handle Extension           | Servicing control cabinet |
| 3           | 58B-06-00740 | Hand Guard                 | Servicing control cabinet |
| 4           | 58B-06-00720 | Handle                     | Servicing control cabinet |
| 5           | 58B-06-00710 | RCDC Lead                  | Servicing control cabinet |
| 6           | 58B-06-00750 | Probe Tip                  | Servicing control cabinet |
| 7           | 58B-06-00770 | Power Supply (Tester)      | Servicing control cabinet |
| 8           | 58B-06-00780 | Capicitor Discharge Sticks | Servicing control cabinet |
| 9           | 58B-06-00760 | Box w/Foam                 | Servicing control cabinet |
| 10          | 58B-06-00790 | Gloves                     | Servicing control cabinet |



84104

0-2000 VDC CUSTOM METER KIT (58B-06-00800)



**MATERIAL - T-1 STEEL OR EQUIVALENT**

M080022

### SPINDLE REMOVAL TOOL

Dimension "A" - Bolt Hole Diameter  
Dimension "B" - Plate Outside Diameter  
Dimension "C" - Bolt Circle Diameter  
Dimension "D" - Plate Inside Diameter  
Dimension "E" - Plate Thickness

Dimension "F" - Cylinder Height  
Dimension "G" - Cylinder Outer Diameter  
Dimension "H" - Cylinder Inner Diameter  
Dimension "I" - Cylinder Wall Thickness

*NOTE: Surface A-A must be parallel to surface B-B  
within 0.062 in. after welding.*

| <b>TABLE 1. SPINDLE REMOVAL TOOL DIMENSIONS</b> |                          |                      |
|-------------------------------------------------|--------------------------|----------------------|
| <b>Dimension</b>                                | <b>Description</b>       | <b>Measurement</b>   |
| <b>"A"</b>                                      | Plate Bolt Hole Diameter | 34.8 mm (1.37 in.)   |
| <b>"B"</b>                                      | Plate O.D.               | 577.9 mm (22.75 in.) |
| <b>"C"</b>                                      | Bolt Circle Diameter     | 517.7 mm (20.38 in.) |
| <b>"D"</b>                                      | Plate I.D.               | 50.8 mm (2.00 in.)   |
| <b>"E"</b>                                      | Plate Thickness          | 34.8 mm (1.37 in.)   |
| <b>"F"</b>                                      | Pusher Cylinder Height   | 133.4 mm (5.25 in.)  |
| <b>"G"</b>                                      | Pusher Cylinder O.D.     | 320.6 mm (12.62 in.) |
| <b>"H"</b>                                      | Pusher Cylinder I.D.     | 228.6 mm (9.00 in.)  |
| <b>"I"</b>                                      | Cylinder Wall Thickness  | 46.0 mm (1.81 in.)   |

Refer to Section G for information about using the spindle removal tool.

The following table lists more special tools that will be necessary for various service procedures:

| <b>Part No.</b> | <b>Description</b>         | <b>Use</b>                  |
|-----------------|----------------------------|-----------------------------|
| EK6089          | Front Brake Tool Group     | Front Brake Removal         |
| EK6090          | Rear Brake Tool Group      | Rear Brake Removal          |
| EJ2847          | Pin Removal Tool           | Rear Suspension Pin Removal |
| EJ2848          | Cylinder                   | Rear Suspension Pin Removal |
| EJ2849          | Hand Pump                  | Rear Suspension Pin Removal |
| EJ2850          | Shackle                    | Rear Suspension Pin Removal |
| MM0093          | Capscrew - M16 x 2.00 x 70 | Rear Suspension Pin Removal |
| XB0887          | Vacuum Pump                | Hydraulic System Service    |

| <b>Part No.</b> | <b>Description</b>                     | <b>Use</b>                                         |
|-----------------|----------------------------------------|----------------------------------------------------|
| EC6027          | Oil Charge Kit                         | Suspension Oil Charging                            |
| SS1143          | Hoist Cylinder Quill Installation Tool | Hoist Cylinder Quill Installation                  |
| SS1153          | Hoist Cylinder Quill Plug Tool         | Hoist Cylinder Quill Plug Removal and Installation |
| VJ6567          | Radiator Tube Installation Tool        | Radiator Tube Installation                         |
| XA2307          | Breaker Tool                           | Radiator Tube Removal                              |
| BF4810          | Bearing Tool                           | 24V Alternator                                     |
| BF4818          | Front Bearing Removal Tool             | 24V Alternator                                     |
| BF4820          | Bobbin Removal Tool                    | 24V Alternator                                     |
| BF4821          | Bearing Insertion Tool                 | 24V Alternator                                     |
| BF4822          | Stator Service Tool                    | 24V Alternator                                     |

The tools in the following table can be ordered from Kent-Moore:

| <b>Part No.</b> | <b>Description</b>                       | <b>Use</b>  |
|-----------------|------------------------------------------|-------------|
| J-24092         | Puller Legs                              | A/C Service |
| J-25030         | Clutch Hub Holding Tool                  | A/C Service |
| J-33884-4       | Indicator Switch Tester                  | A/C Service |
| J-8092          | Universal Handle                         | A/C Service |
| J-8433          | Pulley Puller                            | A/C Service |
| J-9395          | Puller Pivot                             | A/C Service |
| J-9399          | Thin Wall Socket                         | A/C Service |
| J-9401          | Clutch Plate & Hub Assembly Removal Tool | A/C Service |
| J-9403          | Spanner Wrench                           | A/C Service |
| J-9480-01       | Drive Plate Installer                    | A/C Service |
| J-9480-02       | Spacer                                   | A/C Service |
| J-9481          | Puller and Bearing Installer             | A/C Service |

**SECTION M19**  
**RADIATOR SHUTTERS**  
**INDEX**

RADIATOR SHUTTERS ..... M19-3

    OPERATION. .... M19-3

        Hydraulic Circuit ..... M19-3

        Electrical Circuit ..... M19-4

    MAINTENANCE AND REPAIR..... M19-4

    TROUBLESHOOTING ..... M19-5

# NOTES

## RADIATOR SHUTTERS

The optional radiator shutters aid in maintaining proper engine coolant temperature, primarily in cold climate operations.

The shutter system contains the following components:

- Shutter assembly with hydraulic control actuator cylinder
- Hydraulic oil supply pressure reducing valves and cylinder control solenoid
- Solenoid control relays

### OPERATION

#### Hydraulic Circuit

Hydraulic oil pressure to operate the shutter assembly actuator cylinder is supplied from the truck steering circuit. Hose (2, Figure 19-1), which is attached to a "T" fitting at the automatic lubrication system pump supply port, routes oil to pressure reducing valve (3). This valve reduces the steering circuit pressure from 24 132 kPa (3,500 psi) to 10 342 kPa (1,500 psi). The pressure is further reduced to 517 kPa (75 psi) by pressure reducing valve (4). This valve supplies the oil to shutter solenoid valve (5) to pressurize the head end of the shutter actuator cylinder (10), which closes the shutters during engine warm-up. When the shutters are signalled to open, the solenoid valve will be de-energized and the spool will shift to direct oil to the rod end of the actuator cylinder. Oil returns to the hydraulic tank through hose (9).

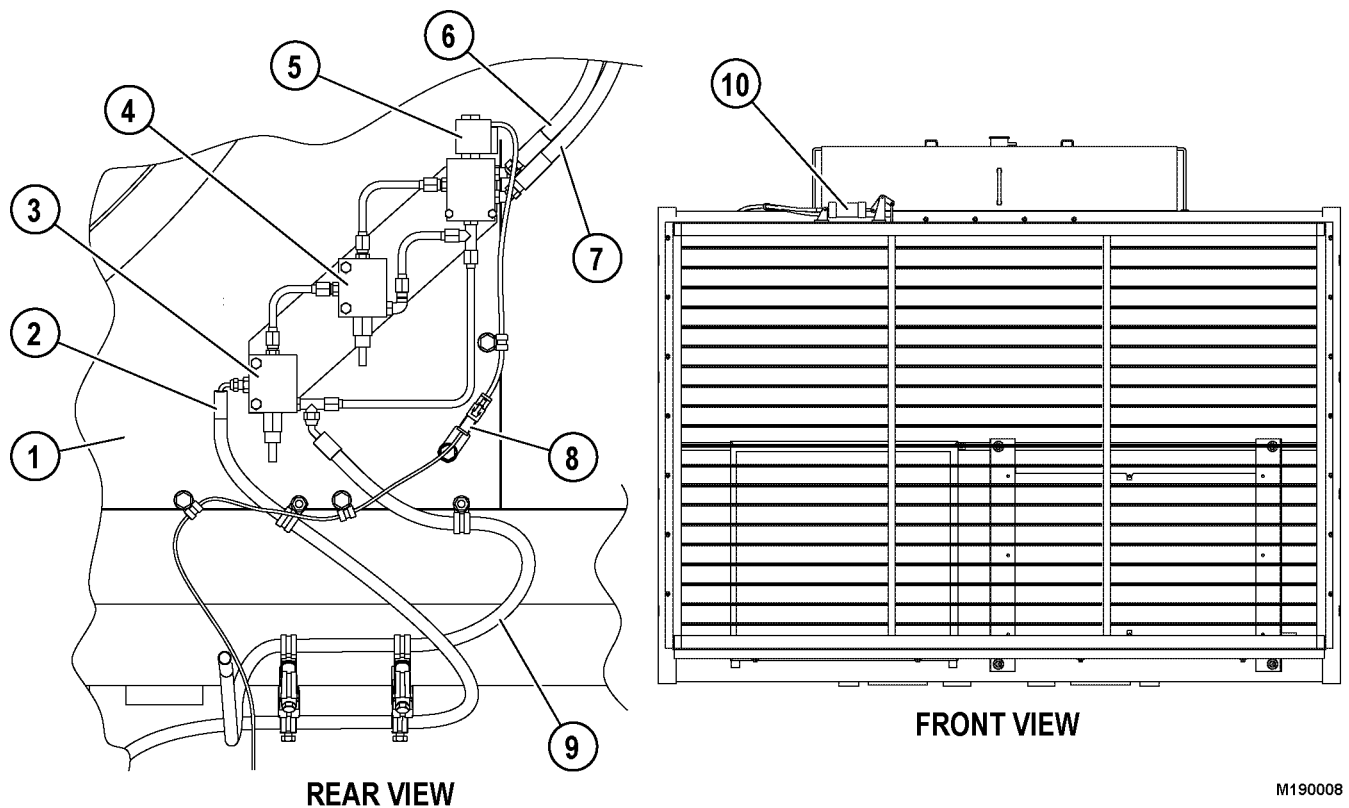


FIGURE 19-1. SHUTTER ACTUATOR CYLINDER OIL SUPPLY VALVES

- |                            |                           |                               |
|----------------------------|---------------------------|-------------------------------|
| 1. Radiator Shroud         | 5. Shutter Solenoid Valve | 9. Oil Return Hose            |
| 2. Oil Supply Hose         | 6. To Cylinder Head End   | 10. Shutter Actuator Cylinder |
| 3. Pressure Reducing Valve | 7. To Cylinder Rod End    |                               |
| 4. Pressure Reducing Valve | 8. Electrical Harness     |                               |



## Electrical Circuit

The shutters are normally activated by the engine ECM. The engine controller monitors coolant temperature and provides a signal to close the shutters when the temperature is too low. See Table 1 for specific temperatures at which the ECM will send a signal to close the shutters.

The engine ECM will provide a +24VDC signal to energize the relay K3 located on relay board RB8. This will close the N.O. relay contacts fed by relay K3, which provides +24VDC when the key switch is ON. Current through relay K5 will energize the shutter solenoid valve to direct oil to the head end of the shutter actuator cylinder to close the shutters.

If the cab air conditioner is switched on, relay K3 will energize, opening the +24VDC circuit that supplies the shutter control solenoid and preventing the shutters from closing.

Refer to the electrical schematic in Section R for additional wiring information.

## MAINTENANCE AND REPAIR

Solenoid valve (5, Figure 19-1) and pressure reducing valves (3) and (4) in the hydraulic circuit are factory set and are not adjustable. If a valve is inoperative, replace the complete valve and body assembly.

The relays are plug-in devices that may easily be replaced if defective. The relays are located on relay board RB8 in the auxiliary control cabinet.

The shutter assembly should be inspected for physical damage and to ensure that it opens and closes completely without binding. If necessary, adjust actuator cylinder linkage to ensure proper operation.

## TROUBLESHOOTING

All hydraulic testing is to be performed with hydraulic oil at normal operating temperature and engine speed at low idle.

Refer to Figure 19-2 for hydraulic troubleshooting test points shown with arrows.

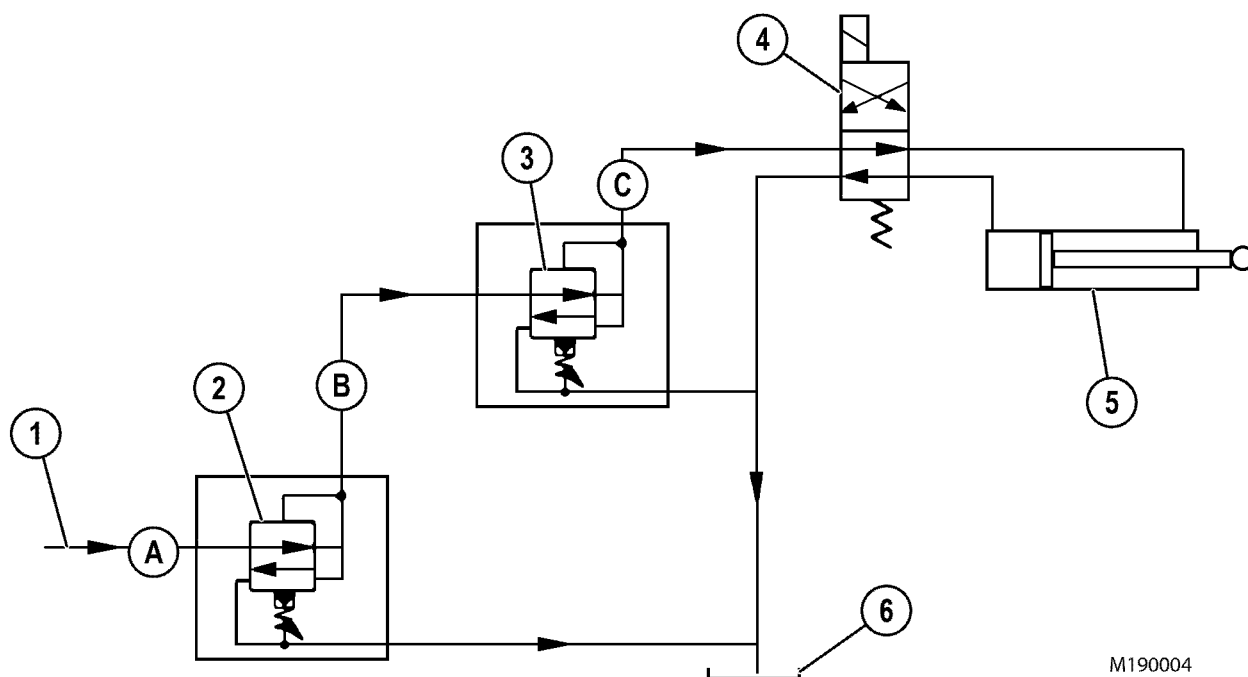
Refer to Table 1 for shutter open and close specifications for Komatsu engines.

**⚠ DANGER**

*Ensure that the engine and key switch have been OFF for at least 90 seconds to allow accumulator pressure to bleed down.*

*Relieve pressure before disconnecting hydraulic lines. Tighten all connections securely before applying pressure.*

*Hydraulic fluid escaping under pressure can have sufficient force to enter a person's body by penetrating the skin, causing serious injury and possibly death if proper medical treatment by a physician familiar with this type of injury is not received immediately.*



M190004

FIGURE 19-2. HYDRAULIC SCHEMATIC

| TABLE 1. Komatsu 16V160/18V170 Engines      |                                             |                                             |
|---------------------------------------------|---------------------------------------------|---------------------------------------------|
| Coolant Temperature                         | Intake Manifold Temperature                 | Fuel Temperature                            |
| Open 88° C (190° F)<br>Close 85° C (185° F) | Open 60° C (140° F)<br>Close 57° C (135° F) | Open 68° C (155° F)<br>Close 66° C (150° F) |

**TABLE 2. Troubleshooting Procedure**

|                                                                                                         |                                                                                                                                                                                                                                                           |                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Test operation of shutters.</b>                                                                   | With engine at slow idle, connect a 24 VDC power source to solenoid valve (5, Figure 19-1).                                                                                                                                                               | Shutters close - <b>GO TO STEP 11</b><br>Shutters stay open - <b>GO TO STEP 2</b>                                                     |
| <b>2. Check hydraulic pressure to cylinder rod end.</b>                                                 | Install tee fittings in lines (6) and (7). Connect 0-3 500 kPa (0-500 psi) pressure gauges to tee fittings. Disconnect electrical harness (8) from solenoid valve (5). There should be 517 kPa (75 psi) pressure in hose (7) and NO pressure in hose (6). | In Spec - <b>GO TO STEP 3</b><br>Out of Spec - <b>GO TO STEP 4</b>                                                                    |
| <b>3. Check hydraulic pressure to cylinder head end.</b>                                                | Connect 24 VDC power source to solenoid valve (5). There should be 517 kPa (75 psi) pressure in hose (6) and NO pressure in hose (7).                                                                                                                     | In Spec - <b>GO TO STEP 11</b><br>Out of Spec - <b>GO TO STEP 4</b><br>If nothing changed as compared to Step 2 - <b>GO TO STEP 7</b> |
| <b>4. Check hydraulic pressure between pressure reducing valve (4) and solenoid valve (5).</b>          | Install hose with tee fitting between pressure reducing valve (4) and solenoid valve (5). Connect 0-3 500 kPa (0-500 psi) pressure gauge to tee fitting. Pressure should be 517 kPa (75 psi).                                                             | In Spec - <b>GO TO STEP 7</b><br>Out of Spec - <b>GO TO STEP 5</b>                                                                    |
| <b>5. Check hydraulic pressure between pressure reducing valve (3) and pressure reducing valve (4).</b> | Install hose with tee fitting between pressure reducing valve (3) and pressure reducing valve (4). Connect pressure gauge with a 0-35 000 kPa (0-5,000 psi) range to tee fitting. Pressure should be 10 342 kPa (1,500 psi).                              | In Spec - <b>GO TO STEP 6</b><br>Out of Spec - <b>GO TO STEP 8</b>                                                                    |
| <b>6. Pressure reducing valve (4) is defective.</b>                                                     | Replace pressure reducing valve (4).                                                                                                                                                                                                                      | <b>GO TO STEP 5</b>                                                                                                                   |
| <b>7. Solenoid valve (5) is defective.</b>                                                              | Replace solenoid valve (5).                                                                                                                                                                                                                               |                                                                                                                                       |
| <b>8. Check oil supply pressure to pressure reducing valve (2).</b>                                     | Install tee fitting between oil supply hose (2) and pressure reducing valve (3). Connect 0-35000 kPa (0-5000 psi) pressure gauge to tee fitting. Pressure should be 20 684 - 24 131 kPa (3,000-3,500 psi) the same as steering system standby pressure.   | In Spec - <b>GO TO STEP 9</b><br>Out of Spec - <b>GO TO STEP 10</b>                                                                   |
| <b>9. Pressure reducing valve (2) is defective.</b>                                                     | Replace pressure reducing valve (2).                                                                                                                                                                                                                      | <b>GO TO STEP 8</b>                                                                                                                   |
| <b>10. Hydraulic supply to shutters is incorrect.</b>                                                   | Perform troubleshooting procedures in appropriate shop manual to correct truck hydraulic system.                                                                                                                                                          | <b>GO TO STEP 8</b>                                                                                                                   |
| <b>11. Check for proper voltage on relay board RB8.</b>                                                 | With the key switch ON, use a V.O.M. to check for 24 VDC in circuit 712R at relay K3.                                                                                                                                                                     | In Spec - <b>GO TO STEP 13</b><br>Out of Spec - <b>GO TO STEP 12</b>                                                                  |
| <b>12. Improper voltage on relay board RB8.</b>                                                         | Perform troubleshooting procedures in appropriate shop manual to obtain proper voltage in circuit 712R with key switch ON.                                                                                                                                | After repair, <b>GO TO STEP 11</b>                                                                                                    |

| <b>TABLE 2. Troubleshooting Procedure</b>          |                                                                                                                                                                                          |                                                                      |
|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| <b>13. Check relay K3 for proper voltage.</b>      | Make sure that there is 0 voltage in circuit 65T. If there is voltage, turn off air conditioner switch or disconnect wire 65T. Then check for 24VDC in circuit 712SP on relay board RB8. | In Spec - <b>GO TO STEP 15</b><br>Out of Spec - <b>GO TO STEP 14</b> |
| <b>14. Relay K3 is defective.</b>                  | Replace K3 relay.                                                                                                                                                                        | After repair, <b>GO TO STEP 13</b>                                   |
| <b>15. Check relay K5 for proper voltage.</b>      | Jumper 24VDC power source to circuit 65S. Then check for 24VDC in circuit 65SS on relay board RB8.                                                                                       | In Spec - <b>GO TO STEP 17</b><br>Out of Spec - <b>GO TO STEP 16</b> |
| <b>16. Relay K5 is defective.</b>                  | Replace K5 relay.                                                                                                                                                                        | <b>GO TO STEP 15</b>                                                 |
| <b>17. Check for 24VDC at solenoid valve (5).</b>  | Check for 24VDC at solenoid valve in circuit 65SS.                                                                                                                                       | In Spec - <b>GO TO STEP 19</b><br>Out of Spec - <b>GO TO STEP 18</b> |
| <b>18. Check for open circuit in circuit 65SS.</b> | Locate and repair open connection in circuit 65SS.                                                                                                                                       | <b>GO TO STEP 17</b>                                                 |
| <b>19. Check for ground at solenoid valve (5).</b> | Check for ground at solenoid valve terminal. Repair as necessary.                                                                                                                        |                                                                      |

# NOTES

# **SECTION M20**

## **PAYLOAD METER III™**

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# OPERATION SECTION

## INTRODUCTION

Payload Meter III (PLMIII) measures, displays and records the weight of material being carried by an off-highway truck. The system generally consists of a payload meter, a gauge display, deck-mounted lights, and sensors. The primary sensors are four suspension pressures and an inclinometer. Other inputs include a body up signal, brake lock signal, and speed.

### Data Summary

5208 haul cycles can be stored in memory. The following information is recorded for each haul cycle:

- Payload
- Operator ID number (0000-9999)
- Distance traveled loaded and empty
- The amount of time spent empty run/stop, loading, loaded run/stop, and dumping
- Maximum speed loaded and empty with time of day
- Average speed loaded and empty
- Empty carry-back load
- Haul-cycle, loading, dumping start time of day.
- Peak positive and peak negative frame torque with time of day
- Peak sprung load with time of day
- Tire ton-mph for each front and average per rear tires

The payload meter stores lifetime data that cannot be erased. This data includes:

- Top 5 maximum payloads and time stamps.
- Top 5 positive and negative frame torque and time stamps
- Top 5 maximum speeds and time stamps

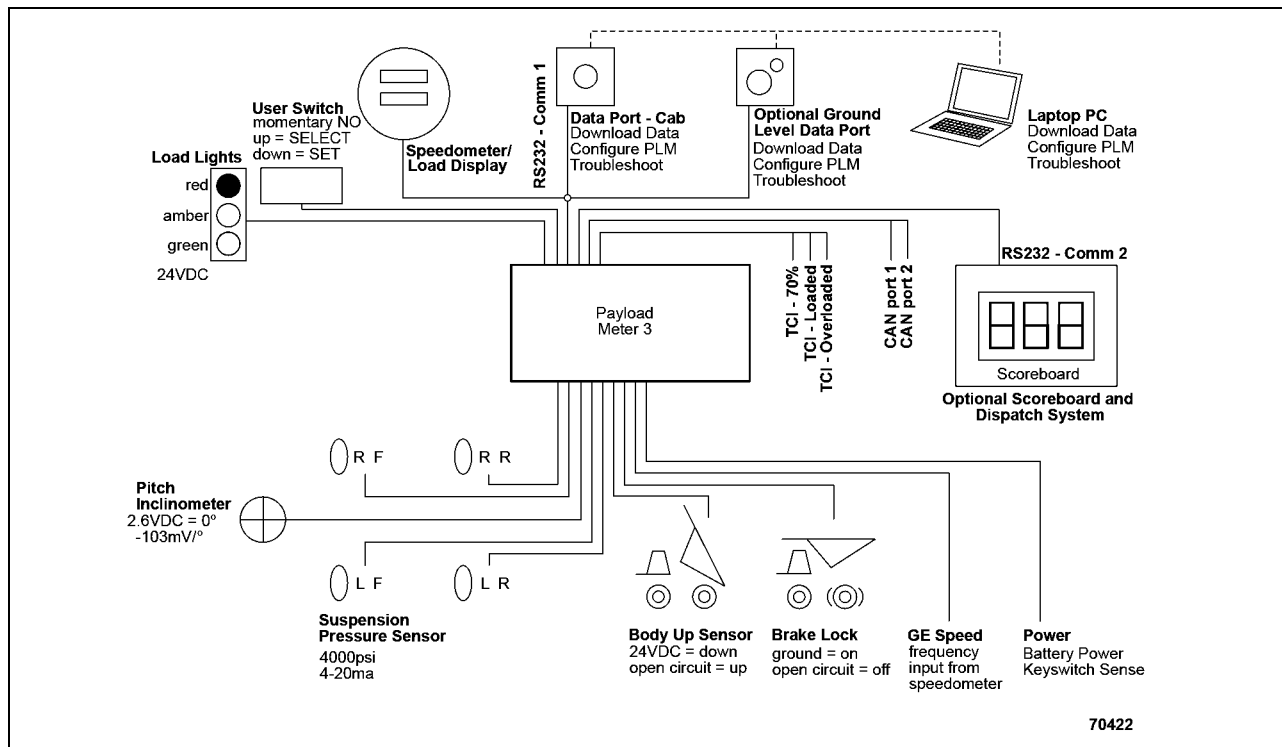
### Data Gathering

Windows 95/98/NT software is available to download, store and view payload and fault information. The PC software will download an entire truck fleet into one Paradox database file. Users can query the database by date, time, truck type and truck number to produce reports, graphs and export the data. The software can export the data in '.CSV' format that can be easily imported into most spreadsheet applications. The Windows software is not compatible with the Payload Meter II system.

It is important that each payload meter be configured for each truck using the PC software. The information for frame serial number and truck number is used by the database program to organize the payload data. In addition, the payload meter must be configured to make calculations for the proper truck model. Improper configuration can lead to data loss and inaccurate payload calculations.

## COMPONENT DESCRIPTION

### System Diagram



### Suspension Pressure Sensors

PLMIII uses a two-wire pressure sensor. The range for the pressure sensor is 281 kg/cm<sup>2</sup> (4000 psi) and the overload limit is 700 kg/cm<sup>2</sup> (10,000 psi). One wire to the sensor is the supply voltage and the other is the signal. The 0-4000 psi range is converted into an electrical current between 4-20 ma. The supply voltage for the sensor is nominally +18VDC. Each pressure sensor has an 3000 mm (118 in.) length of cable. The cable is specially shielded and reinforced to provide mechanical strength and electronic noise immunity.

### Inclinometer

The inclinometer is used to increase the accuracy of load calculations on an incline. The inclinometer uses three wires. For the sensor, red is the +18VDC supply voltage, black is ground and the white is the signal. The incline signal is a voltage between 1 and 4 volts. Zero degrees of incline is represented by 2.6VDC on the signal line. The voltage signal will be decreased by 0.103VDC for every degree of nose up incline.

### Operator Display

The speedometer/display gauge is used as a speedometer and payload display. The top display is used for speed and can display metric (km/h) or English (mph) units. Grounding terminal #4 on the back of the speedometer will switch the meter to display metric units. Leaving terminal #4 unconnected will cause the gauge to display English units. The speedometer can be adjusted using a calibration potentiometer in the back just like existing speedometers.

The payload meter uses the lower display for payload information. The normal display mode shows the current payload. The display can be changed to show the load and total tons counter or the Operator ID. Using the operator switch on the dash panel, the current suspension pressures and incline can be displayed. The units for display are set using the PC software. Payloads can be displayed in short tons, long tons or metric tons.

## Operator Switch

The payload operator switch is used to set, view and clear the total load counter and total ton counter. It is also used to enter the operator ID number (0-9999). This switch can also be used to view the suspension pressures and inclinometer. The payload meter operator switch is located on the dashboard. It is a two-way momentary switch. The top position is the SELECT position. The SELECT position is used step through the different displays. The lower position is the SET position. The SET position is used to set the operator ID or clear the load and total ton counters. Normally the inputs from the switch to the payload meter are open circuit. The switch momentarily connects the circuit to ground.

## Speed Input

PLMIII uses a speed signal to calculate speed, distance, and other performance data. This input is critical to the proper operation of the system. PLMIII receives this signal from the speedometer/operator display on the dashboard. The same signal displayed to the operator is used by the system. Distance calculations are made based on the rolling radius of the tires for a particular truck.

## Body-Up Switch

The body-up input signal is received from a magnetic switch located on the inside of the truck frame, forward the pivot pin of the truck body. This is the same switch typically used for input to the drive system. When the body is down, the switch closes and completes the circuit to 71-control power. 24VDC indicates the body is down. Open circuit indicates that the body is up.

## Brake Lock Switch

The brake lock is used to lock the rear brakes on the truck. It is necessary for the accurate calculation of swingloads during the loading process. Without the brake lock applied, the payload meter will not calculate swingloads during the loading process. Without the brake lock, the payload meter will assume that the truck was loaded using a continuous loader and flag the haul cycle record. All other functions will be normal regardless of brake lock usage. The brake lock input comes from the switch located on the dash panel. The brake lock switch connects the circuit to ground. Open circuit indicates brake lock off. Ground indicates brake lock on.

## Payload Meter

The payload meter is housed in a black aluminum housing. There is a small window on the face of the unit. Status and active alarm codes can be viewed through the window. During normal operation, a two-digit display flashes 0 back and forth. Active fault codes will be displayed for two seconds. These codes are typically viewed using the laptop computer connected to the serial communications port.

There is one 40-pin connector on the payload meter. A jack-screw is used to hold the payload meter and wire harness connector housings together. This screw requires a 4mm or 5/32 hex wrench. The correct tightening torque for this screw is **3 N·m (25 in. lb)**. Four bolts hold the payload meter housing to its mounting bracket in the cab.

The circuit board inside the payload meter housing is made from multi-layer, dual-sided surface-mount electronics. There are no field serviceable components inside. The electronics are designed to withstand the harsh operating environment of the mining industry. Opening the payload meter housing will result in voiding the warranty.

## Communications Ports

The payload meter has two RS232 serial communications ports and two CAN ports. Connections for the two serial ports are available inside the payload meter junction box. The two CAN ports are available for future electronics systems.

Serial port #1 is used to communicate with the dashboard display. It is also used to connect to the laptop computer. The display gauge will remain blank when the PC is using the serial port. This port initially operates with serial settings at 9600,8,N,1. These settings change automatically to increase the communications rate when the PC is using the port. This serial port uses a 3-wire hardware connection.

Serial port #2 is used to communicate to other on-board electronics like Modular Mining's Dispatch® system or the scoreboard from Komatsu. This port uses a 3-wire hardware connection. Connections to this serial port need to be approved by Komatsu. Several protocol options are available and detailed technical information is available depending on licensing.

## Key Switch Input

PLMIII monitors the status of the key switch. 24VDC indicates that the key switch is on, open indicates the key switch is off. The payload meter does not receive its electrical power from the key switch circuit. The payload meter will remain on for several seconds after key switch is removed. When the key switch power is removed, payload meter performs a series of internal memory operations before turning itself off. To allow for these operations, the key switch should be turned off for at least 15 seconds before turning the key switch back on. The payload meter will automatically reset itself without error if not enough time is given for these operations. The display may blink briefly.

## Payload Meter Power

The payload meter receives its power from the battery circuit on the truck. Removing battery power from the payload meter before removing key switch and waiting 15 seconds may result in lost haul cycle data. The payload meter turns itself off approximately 15 seconds after the key switch power is removed. Some haul cycle data will be lost if battery power is removed before waiting 15 seconds. The payload meter system operates at a nominal voltage of 24VDC at 1 to 2 amps depending on options. The payload meter is designed to turn itself off if the supply voltage rises above 36VDC. The payload meter is also protected by a 5 amp circuit breaker located in the junction box.

Power to the load lights comes from the same battery circuit. The load lights are powered through a relay. The key switch circuit controls the relay. The load lights are also protected by a 15 amp circuit breaker in the junction box.

## Load Lights

PLMIII uses load lights to indicate to the shovel operator the approximate weight of the material in the truck. The load lights are illuminated only when the brake lock is applied. The lights are controlled by the payload meter through a series of relays in the junction box. The payload meter controls the relays with 24VDC outputs. A 24VDC signal from the payload meter powers the relay coil and connects battery power to the load light. When the relay is not powered by the payload meter, a pre-warm resistor connects the load light to a reduced voltage. This circuit pre-warms the load light filaments and reduces the inrush current when the light is fully illuminated. This lengthens the operating life of the load lights.

The load lights progressively indicate to the shovel operator the approximate weight of the material in the truck.

A flashing green light indicates the next swingload will make the measured load greater than 50% of rated load. A solid green light indicates that the current load is greater than 50% of rated capacity.

A flashing amber light indicates the next swingload will make the measured load greater than 90% of rated load. A solid amber light indicates that the current load is greater than 90% of rated capacity.

A flashing red light indicates the next swingload will make the measured load greater than 105% of rated load. A solid red light indicates that the current load is greater than 105% of rated capacity.

The optimal loading target is a solid green and amber lights with a flashing red light. This indicates that the load is between 90% and 105% of rated load for the truck and the next swingload will load the truck over 105%.

## **Wiring and Termination**

Most of the PLMIII truck connections use a heavy-duty cable. This yellow multi-conductor cable uses a 16AWG, finely stranded wire designed for continuous motion operations. The conductors are protected by a foil and braided shield for electronic noise immunity and physical strength. This wire is typically terminated with a #10 ring terminal. Most connections for the PLMIII system are made in the payload meter junction box.

## **TCI Outputs**

The GE drive system on the 930E/960E requires information from the payload meter regarding the loaded condition of the truck. There are three outputs from the payload meter to GE to indicate the relative load in the truck. 24VDC on the 73MSL circuit indicates that the load is 70% of rated load. 24VDC on the 73FSL circuit indicates the truck is 100% loaded. The 73OSL circuit is not currently used.

## OPERATOR'S DISPLAY AND SWITCH

### Reading the Speedometer

The top window of the speedometer/display gauge is the speedometer section. The display shows the speed indicated by the frequency being received by the gauge. This can be adjusted using the potentiometer on the back of the gauge. In addition, the units for the display can be changed. Terminal #4 controls the displayed units. If #4 is grounded, the display will be metric. If terminal #4 is left open, the display will be in English units.

### Reading the Load Display

The lower display on the speedometer/display gauge is used for payload information. The SELECT position on the operator switch allows the user to scroll through a number of useful displays. The order for the displays is as follows:

- $PL$  = Payload
- $ID$  = Operator ID
- $TL$  = Total Shift Tons
- $LC$  = Shift Load Counter
- $LF$  = Left Front Suspension Pressure
- $RF$  = Right Front Suspension Pressure
- $LR$  = Left Rear Suspension Pressure
- $RR$  = Right Rear Suspension Pressure
- $IN$  = Inclinator

The display holds the displayed information until the SELECT switch is pressed again. The suspension pressures, inclinometer, and payload displays are based on current sensor inputs.

Communications to the display use the same serial link as the download connection. Whenever another computer is connected to serial port #1 to download or configure the system, the lower display will blank. This is not the same connection used by mine dispatch systems.

### Using the Operator ID

The current operator ID number is recorded with each haul cycle. The number can be between 0 and 9999.

To set the Operator ID:

1. Press the "SELECT" switch until  $ID$  is displayed.
2. Hold the "SET" button until  $□□□□$  is displayed. The first digit should be flashing.
3. Press the "SET" button again to change the digit.
4. Press the "SELECT" button once to adjust the second digit.
5. Use the "SET" button again to change the digit.
6. Press the "SELECT" button once to adjust the third digit.
7. Use the "SET" button again to change the digit.
8. Press the "SELECT" button once to adjust the fourth digit.
9. Use the "SET" button again to change the digit.
10. Press the "SELECT" button one more time to enter the ID.

If no buttons are pressed for 30 seconds, the display will return to normal operation. The number being entered will be lost and the ID number returns to the previous ID number.

### Using the Load and Ton Counter

PLMIII allows the truck operator to monitor and track the total tons hauled and the number of haul cycles during the shift. This display can be cleared at the beginning of each shift to allow the operator to record how many loads and tons have been hauled during the shift.

#### Total Ton Counter

The total ton counter records the number of tons hauled since the last time it was cleared. This display is in 100's of tons. For example, if the display shows  $432$  the total tons is 43,200. This display can be cleared at the beginning of each shift to allow the operator to record how many tons have been hauled during the shift. The units are selected using the PC software.

- To view the total ton counter press and release the "SELECT" switch until  $TL$  is displayed on the gauge.

## Total Load Counter

The total load counter records the number of loads hauled since the last time it was cleared. This display can be cleared at the beginning of each shift to allow the operator to record how many loads have been hauled during the shift.

- To view the total load counter press and release the "SELECT" switch until  $\overline{L}\overline{C}=$  is displayed on the gauge.

## Clearing the Counters

Clearing the total ton counter or total load counter clears both records.

To clear the total ton and total load counter:

1. Press the "SELECT" switch until  $\overline{L}\overline{L}=$  or  $\overline{L}\overline{C}=$  is displayed.
2. Hold the "SET" button until the display clears.

## Viewing Live Sensor Data

The display can also be used to quickly show the current readings from the four suspension pressure sensors and the inclinometer. This can be used during regularly scheduled service periods to check the state of the suspensions. These displays are live and will update as the values change.

The live displays cannot be cleared and the SET button will have no effect.

The units for the display are controlled by the configuration of the payload meter. If the payload meter is set to display metric units, the pressures will be displayed in tenths of  $\text{kg}/\text{cm}^2$ . For example, if the display shows  $\overline{2}\overline{0}\overline{2}$  the actual value is  $20.2 \text{ kg}/\text{cm}^2$ . If the payload meter is set to display short tons, the pressures will be displayed in psi ( $\text{lbs}/\text{in}^2$ ). Multiply by 14.2 to convert  $\text{kg}/\text{cm}^2$  to psi. (example --  $1 \text{ kg}/\text{cm}^2 \times 14.2 = 14.2 \text{ psi}$ ). There is no way to detect the units setting for the gauge without the PC software.

The inclinometer displays whole degrees of incline. Positive incline is truck nose up. The gauge will quickly display the type of information shown every 10 seconds. For example, if the left-front pressure is being displayed,  $\overline{L}\overline{F}=$  will flash on the display every minute. Only the payload display,  $\overline{P}\overline{L}=$  does not display this information.

- Left Front Pressure - To display the pressure in the left-front suspension, press and release the "SELECT" switch until  $\overline{L}\overline{F}=$  is displayed.
- Right Front Pressure - To display the pressure in the right-front suspension, press and release the "SELECT" switch until  $\overline{R}\overline{F}=$  is displayed.
- Left Rear Pressure - To display the pressure in the left-rear suspension, press and release the "SELECT" switch until  $\overline{L}\overline{R}=$  is displayed.
- Right Rear Pressure - To display the pressure in the right-rear suspension, press and release the "SELECT" switch until  $\overline{R}\overline{R}=$  is displayed.
- Inclinometer - To display the truck incline, press and release the "SELECT" switch until  $\overline{I}\overline{N}=$  is displayed.

## Other Display Messages

On startup of the payload meter system, the gauge display will scroll the truck type that the PLMIII is configured for. For example, on a 930E, the gauge will scroll  $\text{---}\overline{9}\overline{3}\overline{0}\overline{E}\text{---}$ .

If the PLMIII encounters memory problems, it will display  $\overline{E}\overline{R}\overline{8}\overline{8}$ , where 88 is the specific memory error. In this very rare circumstance, the system should be turned off for 30 seconds and restarted.



## PAYLOAD OPERATION & CALCULATION

### Description of Haul Cycle States

The typical haul cycle can be broken down into eight distinct stages or states. Each state requires the payload meter to make different calculations and store different data.

"States" or stages of a typical haul cycle

1. Tare Zone
2. Empty
3. Loading
4. Maneuvering
5. Final Zone
6. Hauling
7. Dumping
8. After Dump

### Haul Cycle Description

A new haul cycle is started after the load has been dumped from the previous cycle. The payload meter will stay in the after\_dump state for 10 seconds to confirm that the load has actually been dumped. If the current payload is less than 20% of rated load, the payload meter will switch to the tare\_zone and begin calculating a new empty tare. If, after dumping, the payload has not dropped below 20% of rated load the meter will return to the maneuvering or hauling states. In this case, the false\_body\_up flag will be recorded in the haul cycle record.

While in the tare\_zone state, and moving faster than 5 km/h (3 mph), the payload meter calculates the empty sprung weight of the truck. This tare value will be subtracted from the loaded sprung weight to calculate the final payload. The payload meter will switch from the tare\_zone or empty to the loading state if swingloads are detected. By raising the dump body while in the empty state the payload meter can be manually switched back to the tare\_zone to calculate a new tare.

From the empty state, the payload meter will switch to the loading state through one of two means. If the brake lock is applied, the payload meter will be analyzing the suspension pressures to detect a swingload. If a swingload is detected, the meter will switch to the loading state. The minimum size for swingload detection is 10% of rated load. Swingload detection usually takes 4-6 seconds. The second method to switch from empty to loading is through continuous loading. This can happen if the brake lock is not used during loading. If the load increases above 50% of

rated load for 10 seconds without the brake lock applied, the meter will switch to loading and record the continuous\_loading flag in the haul cycle.

The payload meter switches from loading to maneuvering as soon as the truck begins moving. The maneuvering zone is 160 m (0.1 miles) and is designed to allow the operator to reposition the truck under the shovel. More payload can be added anytime within the maneuvering zone. Once the truck travels 160 m (0.1 miles) the payload meter switches to the final\_zone and begins calculating payload. If the body is raised while the payload meter is in the maneuvering state, the no\_final\_load flag will be recorded in the haul cycle record, no payload will be calculated, and the meter will switch to the dumping state.

While in the final\_zone moving faster than 5 km/h (3 mph), the payload meter calculates the loaded sprung weight of the truck. The same advanced algorithm is used to calculate the empty and loaded sprung weights. The payload meter will switch from the final\_zone to the dumping state if the Body-Up signal is received. If the truck has moved for less than 1 minute in the final\_zone, the payload meter will calculate the final payload using an averaging technique which may be less accurate. If this happens, the average\_load flag will be recorded in the haul cycle.

The payload meter switches to the dumping state when the dump body rises. The payload meter will switch from dumping to after\_dump when the dump body comes back down.

From the after\_dump, the payload meter will switch to one of three states:

1. If the average payload is greater than 20% of rated load and no final payload has been calculated, the payload meter will return to the maneuvering state. After the truck travels 160m (0.1 mile) the meter will switch to the final\_zone and attempt to calculate the payload again. The false\_body\_up flag will be recorded in the haul cycle record.
2. If the average payload is greater than 20% of rated load and the final payload has been calculated, the payload meter will switch back to the hauling state. The false\_body\_up flag will be recorded in the haul cycle record.
3. If the average payload is less than 20% of rated load, the payload meter will switch to the tare\_zone and begin to calculate a new empty tare.

## Load Calculation

The final load calculation is different from the last swingload calculation. The accuracy of the swingload calculation depends on loading conditions and the position of the truck during loading. The last swingload calculation is not the value recorded in memory as the final load. The final load is determined by a series of calculations made while the truck is traveling to the dump site.

## Carry Back

Carry back is calculated as the difference between the current truck tare and the clean truck tare. The clean truck tare is calculated using the PC software. When the suspensions are serviced or changes are made that may affect the sprung weight of the truck, a new clean truck tare should be calculated.

## Measurement Accuracy

Payload measurements are typically repeatable within 1%. Accuracy for a particular scale test depends on specific combinations of pressure sensors and payload meters as well as the specifics of each scale test. Comparisons from different scale tests are often made without considering the differences introduced by the specific installation and operation of the scales for each test. In addition, each pressure sensor and payload meter introduces its own non-linearity. Each truck becomes an individual combination of sensors and payload meter. Errors from these sources can introduce up to a  $\pm 7\%$  bias in the payload meter calculations for a specific scale test, for an individual truck.

Because the PLMIII calculates a new empty tare for each payload, a detailed scale test must weigh the trucks empty and loaded for each haul cycle. Using a simple average of 2 or 3 empty truck weights as an empty tare for the entire scale test will introduce significant error when comparing scale weights to PLMIII weights.

## SOURCES FOR PAYLOAD ERROR

### Payload Error

The number one source of error in payload calculation is improperly serviced suspensions. The payload meter calculates payload by measuring differences in the sprung weight of the truck when it is empty and when it is loaded. The sprung weight is the weight of the truck supported by the suspensions. The only method for determining sprung weight is by measuring the pressure of the nitrogen gas in the suspensions. If the suspensions are not properly maintained, the payload meter cannot determine an accurate value for payload. The two critical factors are proper oil height and proper nitrogen charge.

If the suspensions are overcharged, the payload meter will not be able to determine the empty sprung weight of the truck. The suspension cylinder must be able to travel up and down as the truck drives empty. The pressure in an overcharged suspension can push the suspension rod to full extension. In this case, the pressure inside the cylinder does not accurately represent the force necessary to support that portion of the truck.

If the suspensions are undercharged, the payload meter will not be able to determine the loaded sprung weight of the truck. The suspension cylinder must be able to travel up and down as the truck drives loaded. If the pressure in an undercharged suspension cannot support the load, the suspension will collapse and make metal-to-metal contact. In this case, the pressure inside the cylinder does not accurately represent the force necessary to support that portion of the truck.

Low oil height can also introduce errors by not correctly supporting a loaded truck. This is why the correct oil height and nitrogen charge are the most critical factors in the measurement of payload. If the suspensions are not properly maintained, accurate payload measurement is not possible. In addition, suspension maintenance is very important to the life of the truck.

## Loading Conditions

The final load calculation of the PLMIII system is not sensitive to loading conditions. The final load is calculated as the truck travels away from the shovel. Variations in road conditions and slope are compensated for in the complex calculations performed by the payload meter.

## Pressure Sensors

Small variations in sensors can also contribute to payload calculation error. Every pressure sensor is slightly different. The accuracy differences of individual sensors along the range from 0 to 4000 psi can add or subtract from payload measurements. This is also true of the sensor input circuitry within individual payload meters. These differences can stack up 7% in extreme cases. These errors will be consistent and repeatable for specific combinations of payload meters and sensors on a particular truck.

## Swingloads

Swingload calculations can be affected by conditions at the loading site. Parking the truck against the berm or large debris can cause the payload meter to inaccurately calculate individual swingloads. While the PLMIII system uses an advanced calculation algorithms to determine swingloads, loading site conditions can affect the accuracy.

## Speed and Distance

The payload meter receives the same speed signal as the speedometer. This signal is a frequency that represents the speed of the truck. The payload meter uses this frequency to calculate speeds and distances. The meter assumes a single value for the rolling radius of the tire. The rolling radius may change at difference speeds by growing larger at higher speeds. The actual rolling radius of the tire will also change between a loaded and empty truck. The payload meter does not compensate for these changes.

*NOTE: Earlier 730E and 830E models are subject to incorrect speed data due to electrical interference. The incorrect speeds are generated while the truck is stopped. An attenuator was added to newer production models to prevent this error from occurring. A kit was released to update older PLMIII systems with the attenuator. Consult your area service representative for details.*

## HAUL CYCLE DATA

PLMIII records and stores data in its on-board flash memory. This memory does not require a separate battery. The data is available through the download software.

PLMIII can store 5208 payload records. When the memory is full, the payload meter will erase the oldest 745 payload records and continue recording.

PLMIII can store 512 alarm records in memory. When the memory is full, the payload meter will erase the oldest 312 alarm records and continue recording.

All data is calculated and stored in metric units within the payload meter. The data is downloaded and stored in metric units within the Paradox database on the PC. The analysis program converts units for displays, graphs and reports.

The units noted in the Table 1 are the actual units stored in the data file. The value for the haul cycle start time is the number of seconds since January 1, 1970 to the start of the haul cycle. All other event times are referenced in seconds since the haul cycle start time. The PC download and analysis program converts these numbers into dates and times for graphs and reports.

The following information is recorded for each haul cycle:

| Table 1: HAUL CYCLE DATA   |               |                                                                                                                                                                                                                                        |
|----------------------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data                       | Unit          | Remark                                                                                                                                                                                                                                 |
| Truck #                    | alpha-numeric | Up to 22 characters can be stored in this field to identify the truck. Typically this field will be just the truck number.                                                                                                             |
| Haul Cycle Start Date/Time | seconds       | Number of seconds from 1/1/70 to the start of the haul cycle, haul cycle starts when the meter transitions from dumping to empty state after the previous haul cycle, download program converts seconds into date and time for display |
| Payload                    | tons          | Stored as metric, download program allows for conversion to short or long tons.                                                                                                                                                        |
| Number of Swingloads       | number        | The number of swingloads detected by the payload meter                                                                                                                                                                                 |
| Operator ID                | number        | This is a 4 digit number that can be entered by the operator at the start of the shift.                                                                                                                                                |
| Warning Flags              | alpha         | Each letter represents a particular warning message about the haul cycle, details are located on page 19.                                                                                                                              |
| Carry-back load            | tons          | The difference between the latest empty tare and the clean truck tare                                                                                                                                                                  |
| Empty haul time            | seconds       | Number of seconds in the tare_zone and empty states with the truck moving                                                                                                                                                              |
| Empty stop time            | seconds       | Number of seconds in the tare_zone and empty states with the truck stopped                                                                                                                                                             |
| Loading time               | seconds       | Number of seconds in the loading state                                                                                                                                                                                                 |
| Loaded haul time           | seconds       | Number of seconds in the maneuvering, final_zone and loaded states with the truck moving                                                                                                                                               |
| Loaded stop time           | seconds       | Number of seconds in the maneuvering, final_zone and loaded states with the truck stopped                                                                                                                                              |
| Dumping time               | seconds       | Number of seconds in the dumping state                                                                                                                                                                                                 |
| Loading start time         | seconds       | Number of seconds from the start of the haul cycle to when the meter transitions from empty to loading state                                                                                                                           |
| Dump start time            | seconds       | Number of seconds from the start of the haul cycle to the time when the meter switches from loaded to dumping state                                                                                                                    |
| Loaded haul distance       | m             | Distance traveled while loaded                                                                                                                                                                                                         |
| Empty haul distance        | m             | Distance traveled while empty                                                                                                                                                                                                          |
| Loaded max speed           | km/h          | Maximum speed recorded while the truck is loaded                                                                                                                                                                                       |
| Loaded max speed time      | seconds       | Number of seconds from the start of the haul cycle to the time when the max speed occurred                                                                                                                                             |
| Empty max speed            | km/h          | Maximum speed recorded while the truck is empty                                                                                                                                                                                        |
| Empty max speed time       | seconds       | Number of seconds from the start of the haul cycle to the time when the max speed occurred                                                                                                                                             |
| Peak positive frame torque | ton-meter     | Positive frame torque is measured as the frame twists in the clockwise direction as viewed from the operator's seat.                                                                                                                   |
| Peak frame torque time     | seconds       | Number of seconds from the start of the haul cycle to the peak torque, download program converts to time for display                                                                                                                   |
| Peak negative frame torque | ton-meter     | Negative frame torque is measured as the frame twists in the counter-clockwise direction as viewed from the operator's seat.                                                                                                           |
| Peak frame torque time     | seconds       | Number of seconds from the start of the haul cycle to the peak torque, download program converts to time for display                                                                                                                   |
| Peak sprung load           | tons          | Peak dynamic load calculation                                                                                                                                                                                                          |
| Peak sprung load time      | seconds       | Number of seconds from the start of the haul cycle to the peak instantaneous load calculation                                                                                                                                          |
| Front-left tire-ton-km/h   | t-km/h        | Tire ton-km/h for haul cycle                                                                                                                                                                                                           |
| Front-right tire-ton-km/h  | t-km/h        | Tire ton-km/h for haul cycle                                                                                                                                                                                                           |
| Average rear tire-ton-km/h | t-km/h        | Tire ton-km/h for haul cycle                                                                                                                                                                                                           |
| Truck Frame Serial Number  | alpha         | The truck serial number from the nameplate on the truck frame                                                                                                                                                                          |
| Reserved 1-10              | number        | These values are internal calculations used in the continued development of the PLMIII system and should be ignored                                                                                                                    |

## **Haul Cycle Warning Flags**

The payload meter expects haul cycles to progress in a particular way. When something unexpected takes place, the system records a warning flag. Several events within the haul cycle can cause a warning flag to be generated. Each one indicates an unusual occurrence during the haul cycle. They do not necessarily indicate a problem with the payload meter or payload calculation.

### **A: Continuous Loading**

This message is generated when the truck is loaded over 50% full without the payload meter sensing swingloads. This indicates that a continuous loading operation was used to load the truck. It may also indicate that the payload meter did not receive the brake lock input while the truck was being loaded. There may be a problem with the wiring or the brake lock was not used. The payload meter will not measure swingloads unless the brake lock is used during the loading process.

### **B: Loading to Dumping Transition**

This message is generated when the payload meter senses a body up input during the loading process. This message is usually accompanied by a `no_final_load` flag.

### **C: No Final Load**

This message is generated when the payload meter is unable to determine the final payload in the truck. Typically, this means that the payload meter switched from a loaded state to the dumping state before the load could be accurately measured.

### **D: Maneuvering to Dumping Transition**

This message is generated when the payload meter senses a body-up input during the maneuvering or repositioning process indicating that the operator has dumped the load. It may also be generated if the body-up signal is not properly reaching the payload meter and the weight in the truck falls dramatically while the truck is maneuvering or repositioning.

### **E: Average Load or Tare Used**

This message indicates that the recorded payload may not be as accurate as a typical final load calculation. Typically, this is recorded when loading begins before an accurate tare is calculated or the load is dumped before the load can be accurately measured.

## **F: Final Zone to Dumping Transition**

This message is generated when the payload meter senses a body-up while it is calculating the final payload indicating that the operator has dumped the load. It may also be generated if the body-up signal is not properly reaching the payload meter and the weight in the truck falls dramatically while the truck is calculating the final payload.

### **H: False Body Up**

This message indicates that the body was raised during the haul cycle without the load being dumped. The body-up signal indicated that the truck was dumping, but the weight of the truck did not fall below 20% of the rated load.

### **I: Body Up Signal Failed**

This message indicates that the load was dumped without a body-up signal being received by the payload meter. The weight of the truck fell below 20%, but the payload meter did not receive a body-up signal from the sensor.

### **J: Speed Sensor Failed**

This message indicates that the payload meter sensed the truck loading and dumping without receiving a speed signal.

### **K: New Tare Not Calculated**

The payload meter was not able to accurately calculate a new empty sprung weight for the truck to use as the tare value for the haul cycle. The tare value from the last haul cycle was used to calculate payload.

### **L: Incomplete Haul Cycle**

The payload meter did not have proper data to start the haul cycle with after powering up. When the PLMIII powers off, it records the data from the haul cycle in progress into memory. This flag indicates that this data was not recorded the last time the payload meter was shut down. This can happen when the main battery disconnect is used to shut the truck down instead of the key switch. A haul cycle with this warning flag should not be considered accurate. Haul cycles with this warning are displayed in red on the Payload Summary window and are not included in the summary statistics for reports or display.

### **M: Haul Cycle Too Long**

The haul\_cycle\_too\_long flag indicates that the haul cycle took longer than 18.2 hours to complete. The times stored for particular events may not be accurate. This does not affect the payload calculation.

### **N: Sensor Input Error**

An alarm was set for one of the 5 critical sensor inputs during the haul cycle. The five critical sensors are the four pressure sensors and the inclinometer. Without these inputs, the payload meter cannot calculate payload. A haul cycle with this warning flag should not be considered accurate. Haul cycles with this warning are displayed in red on the Payload Summary window and are not included in the summary statistics for reports or display.

### **Frame Torque Data**

Payload meter records the top 5 peak positive and negative frame torque values and the time they occurred. The frame torque is a measure of the twisting action along the centerline of the truck. Positive frame torque is measured when the suspension forces on the front of the truck act to twist the frame in the clockwise direction as viewed from the operator's seat. Negative frame torque is measured when the forces from the suspensions act in the opposite direction.

For example, if the left front and right rear pressure rises as the right front and left rear pressure drops, the truck frame experiences a twisting motion along the longitudinal centerline. In this case, the payload meter will record a positive frame torque.

The 5 highest values in the positive and negative direction are stored in permanent memory within the payload meter.

### **Sprung Weight Data**

The payload meter is constantly monitoring the live payload calculation. This value naturally rises and falls for a loaded truck depending on road and driving conditions. The payload meter records the top 5 highest payload calculations and the time they occurred. This information is stored in permanent memory inside the meter.

### **Maximum Speed Data**

The payload meter records the top 5 highest speeds and the time they occurred. This information is stored in permanent memory inside the meter.

### **Alarm Records**

The payload meter stores alarm records to give service personnel a working history of the system. All codes are viewed using the PC connected to the payload meter. Active codes are also displayed on the two-digit display on the meter itself. Each code has a specific cause and should lead to an investigation for correction. Some failures can be overcome by the payload meter. Haul cycle data will indicate if an alarm condition was present during the cycle. Failures with the suspension or inclinometer sensors cannot be overcome.

## Fault Code Data

| Table 2:   |                            |                                                                                                                                                                                              |
|------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fault Code | Name                       | Description                                                                                                                                                                                  |
| 1          | Left front pressure high   | Input current > 22 ma                                                                                                                                                                        |
| 2          | Left front pressure low    | Input current < 2 ma                                                                                                                                                                         |
| 3          | Right front pressure high  | Input current > 22 ma                                                                                                                                                                        |
| 4          | Right front pressure low   | Input current < 2 ma                                                                                                                                                                         |
| 5          | Left rear pressure high    | Input current > 22 ma                                                                                                                                                                        |
| 6          | Left rear pressure low     | Input current < 2 ma                                                                                                                                                                         |
| 7          | Right rear pressure high   | Input current > 22 ma                                                                                                                                                                        |
| 8          | Right rear pressure low    | Input current < 2 ma                                                                                                                                                                         |
| 9          | Inclinometer high          | Input voltage < 0.565 VDC                                                                                                                                                                    |
| 10         | Inclinometer low           | Input voltage > 5.08 VDC                                                                                                                                                                     |
| 11         | Speed input failure        | Not Used                                                                                                                                                                                     |
| 12         | Brake lock input failure   | Not Used                                                                                                                                                                                     |
| 13         | Body-up input failure      | Payload meter detected dumping activity without receiving a body up signal                                                                                                                   |
| 16         | Memory write failure       | Indicates possible memory problem at power start up. Cycle power and recheck.                                                                                                                |
| 17         | Memory read failure        | Indicates possible memory problem at power start up. Cycle power and recheck.                                                                                                                |
| 18         | Rear right suspension flat | Payload meter detected an undercharged suspension condition on the rear right suspension.                                                                                                    |
| 19         | Rear left suspension flat  | Payload meter detected an undercharged suspension condition on the rear left suspension.                                                                                                     |
| 20         | Time change                | Payload meter time was changed by more than 10 minutes. The Alarm Set time indicates original time. The Alarm Clear time indicates the new time.                                             |
| 21         | Tare value reset           | The user manually forced the payload meter to reset the haul cycle empty (tare) sprung weight. This forced the meter into the tare_zone state and lost all data for the previous haul cycle. |
| 22         | Excessive carryback        | The payload meter detected an empty carryback load in excess of the user-defined carryback threshold on two consecutive haul cycles.                                                         |
| 26         | User switch fault - SELECT | Select switch on for more than 2 minutes, may indicate short to ground                                                                                                                       |
| 27         | User switch fault - SET    | Set switch on for more than 2 minutes, may indicate short to ground                                                                                                                          |

## PC SOFTWARE OVERVIEW

The PC software has several basic functions:

- Configure the PLMIII system on the truck.
- Troubleshoot and check the PLMIII system.
- Download data from the PLMIII system.
- Analyze data from the payload systems.

Configuration, troubleshooting and downloading require a serial connection to the payload meter on the truck. Analysis can be done at any time without a connection to the payload meter.

Payload data is downloaded from several trucks into one database on the PC. The database can be queried to look at the entire fleet, one truck or truck model. The data can be graphed, reported, imported or exported. The export feature can take payload data and save it in a format that spreadsheet programs like Excel or word processing programs can easily import.

### System Configuration

PLMIII needs to be configured for operation when it is first installed on the truck. This process requires several steps and uses the laptop computer to make the necessary settings. The setup procedure can be broken down into several steps:

- Connecting the laptop to the PLMIII system.
- Starting communications
- Setting the time & date
- Setting the truck type
- Setting the truck ID
- Setting the speedometer/display gauge units

### Installing the PLMIII Software

The CD ROM containing the Payload Data Management (PDM) Software will automatically begin installation when it is inserted into the drive on the PC. If this does not happen, the software can be installed by running the Setup.exe program on the CD ROM.

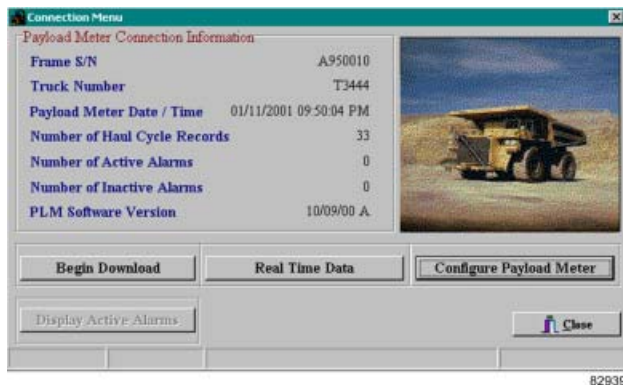
The minimum PC requirements for running the software is a Pentium 133Mhz with 64 MB of ram and at least 300 MB of free hard drive space available. For improved performance, the recommended PC would be a Celeron, AMD K6-2 or better processor with 128 MB of ram running at 400 Mhz. The PDM Software uses a powerful database to manipulate the large amounts of data gathered from the PLMIII system. Using a more powerful computer and added memory to run the software can result in a significant improvement in performance. The software is written to use a minimum 800 x 600 screen resolution.



## DOWNLOADING DATA

PLMIII records many types of data. The PLMIII PC software is designed to download the data from a whole truck fleet. Instead of creating one data file for each truck, the PC software combines all the data from many trucks into one database on the hard drive of the computer. The software then allows users to query the database to create custom reports and graphs. Data for individual trucks or groups of trucks can be easily analyzed. This same data can be exported for use in other software applications like word processors and spreadsheet applications.

As the database grows, performance of the PC software for analysis will slow down. It may be helpful to periodically export data. For example, query the database to show the oldest quarter, month, or half year and print out a summary report. Then export the data to a compressed format and save the file in a secure location. Once the data is exported, delete the entire query results from the database. If necessary, the data can easily be imported back into the main database for analysis at a future date. Removing this older data will improve performance.



To download the payload meter:

1. Connect to the payload meter and start the PC software.
2. From the main menu, select "Connect to Payload Meter". The PC will request the latest status information from the payload meter. The number of haul cycles and alarms will be displayed.
3. Select the "Begin Download" button. The PC will request the payload and alarm data from the payload meter and save it into the database. This may take several minutes. A progress bar at the bottom will show the approximate time left.

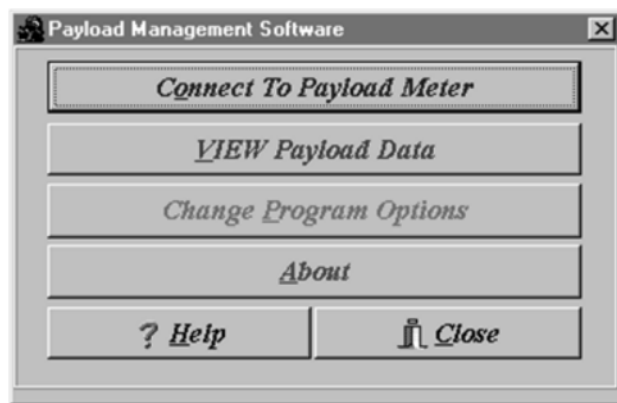
The PC software downloads the data from the payload meter into a single Paradox database. The data from all the trucks is added to the same database. Downloading the payload meter can take several minutes. The data is added to the database on the laptop used to download.

To move the data to another computer, a query must be run to isolate the particular data for export. Do not press the operator switch on the dashboard while downloading

## PLM III SYSTEM CONFIGURATION

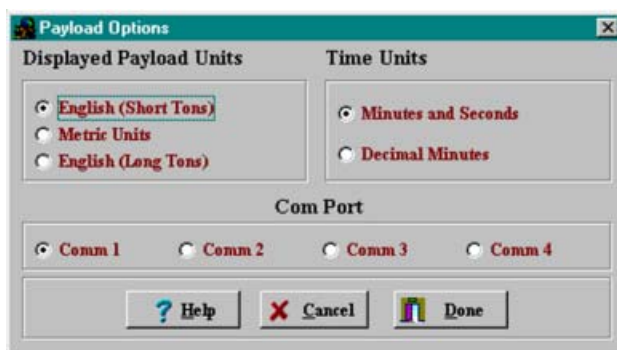
### Starting Communications

The PDM software allows users to download and configure the system.



82941

Before connecting to the payload meter, select "Change Program Options" and confirm that the program has selected the correct laptop serial port. Most laptop computers use Comm 1 for serial communications. The units displayed for reports and graphs by the PC software can be set on this form. Click "Done" to return to the main menu.



82940

From the main menu, click the "Connect to Payload Meter" button. The PC will try to connect to the payload meter and request basic information from the system. In the event of communications trouble, the PC will try 3 times to connect before "timing-out". This may take several seconds.

### Displayed Payload Units

Three options are available for the display of units in the PC software, reports, and graphs:



82949

**Short Tons:** Payload is displayed in short tons, distances and speeds will be displayed in Miles

**Metric Tons:** Payload is displayed in metric tons, distances and speeds are displayed in Kilometers

**Long Tons:** Payload is displayed in long tons, distances and speeds are displayed in Miles

### Time Units

Minutes and Seconds Example:

Five minutes and thirty-two seconds = 5:32

Decimal Minutes Example:

Five minutes and thirty-two seconds = 5.53

### Connection Menu



82939

The connection screen displays basic system information to the user.

- Frame S/N should agree with the truck serial number from the serial plate located on the truck frame.
- Truck Number is an ID number assigned to the truck by the mine.
- The Payload Meter Date / Time values come from the payload meter at the moment of connection.
- Number of Haul Cycle Records is the number of haul cycles records stored in memory and available for download.
- Number of Active Alarms shows how many alarms are currently active in the system at the time of connection. If there are active alarms, the "Display Active Alarms" button is available.
- Number of Inactive Alarms shows how many alarms have been recorded in memory and are available for download.
- PLM Software Version displays the current version of software in the payload meter.

The information on the connection menu comes from the configuration of the payload system on the truck.

There are also many configuration and download options available from this screen. The Connection Menu is updated only when the connection is first made. It does not update automatically. To view changes made while connected, the user must close the window and reconnect to the payload meter.

The connection menu is displayed after a serial connection has been established and the PC software has connected to the payload meter.

## Connecting to the Payload Meter

Communications to the PLMIII requires a laptop computer running the PDM software. The software connects to the payload meter through the meter's serial port #1. This is the same port used by the speedometer/display gauge. When the laptop is using the serial port, the lower display on the operator gauge on the dashboard will be blank. This does not affect the operation of the speedometer.

- Connect the laptop to the system using the EF9160 or PC2525 communications harness. The download connector is typically located on the housing mounted in the cab to the back wall. The PLMIII system uses the same connection as the Payload Meter II system.

## Configure the Payload Meter

Configuration of the payload meter requires a serial connection to the PLMIII system. Clicking the "Configure Payload Meter" button will bring up the Truck Configuration screen and menu. This screen displays the latest configuration information stored on the payload meter.

When changes are made to the configuration, the "Save Changes" button must be pressed to save the changes into the payload meter. To confirm the changes, exit to the main menu and re-connect to the payload meter.

The screenshot shows the 'Truck Configuration' window. It has a 'PLM Date and Time' section with 'Edit Time' set to 10:59:58 AM and 'Edit Date' set to 4/11/01. There is a 'Gauge Units' section with radio buttons for 'Metric Tons' (selected), 'Short Tons', and 'Long Tons'. Below this is a 'Truck Type' dropdown menu showing '930E'. Further down are text input fields for 'Frame Serial Number' (A950010), 'Truck Number' (T611), 'KMS Distributor' (RMS), and 'KMS Customer' (Big Mine). At the bottom are four buttons: 'Save Changes', 'Clean Truck Tare', 'Inclinometer', and 'Close'.

## Setting the Date and Time

The screenshot shows a dialog box titled 'PLM Date and Time'. It has a 'Change To' label. Below it are two input fields: 'Edit Time' showing '9:51:36 PM' and 'Edit Date' showing '1/11/2001'.

The time shown on the form is the time transmitted from the payload when the connection was first established.



The date and time are maintained by a special chip on the PLMIII circuit board. The memory for this chip is maintained by a large capacitor when the power is removed from the payload meter. This will maintain the date and time settings for approximately 30 days. After this time, it is possible for the payload meter to lose the date and time setting. It is recommended that the system be powered every 20 days to maintain the date and time. If the date and time is lost, reset the information using this procedure. It takes approximately 90 minutes to recharge the capacitor.

Changing the date and time will affect the haul cycle in progress and may produce unexpected results in the statistical information for that one haul cycle.

**NOTE:** If the truck is equipped with KOMTRAX Plus, do not set the time or date in the PLM III controller. The PLM III clock is synchronized by the KOMTRAX Plus clock.

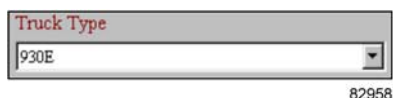
To change the time:

1. Click on the digit that needs to be changed.
2. Use the up/down arrows to change or type in the correct value.
3. Press the "Save Changes" button to save the new time in the payload meter.

To change the date:

1. Click on the digit that needs to be changed.
2. Type in the correct value or use the pull-down calendar to select a date.
3. Press the "Save Changes" button to save the new time in the payload meter.

## Setting the Truck Type



1. From the Truck Configuration screen, use the pull-down menu to select the truck type that the payload meter is installed on.
2. Press the "Save Changes" button to program the change into the meter.

## Setting the Gauge Display Units

The payload meter speedometer / display gauge displays the speed on the upper display. The units for the speed display are selected using a jumper on the rear of the case.

The payload units on the lower display can be changed from metric to short tons or long tons using the Truck Configuration screen. This selection also switches between metric (kg/cm<sup>2</sup>) and psi (lbs/in<sup>2</sup>) for the live display of pressure on the gauge.

1. From the Truck Configuration screen, select the payload units to be used on the lower display of the speedometer/display gauge.
2. Press the "Save Changes" button to program the change into the payload meter.

## Setting the Frame Serial Number

|                     |          |
|---------------------|----------|
| Frame Serial Number | A950010  |
| Truck Number        | T3444    |
| KMS Distributor     | RMS      |
| KMS Customer        | Pit Mine |

The frame serial number is located on the plate mounted to the truck frame. The plate is outboard on the lower right rail facing the right front tire. It is very important to enter the correct frame serial number. This number is one of the key fields used within the haul cycle database. The field will hold 20 alpha-numeric characters.

1. On the Truck Configuration screen, enter the truck frame serial number in the appropriate field.
2. Press the "Save Changes" button to program the change into the payload meter.

## Setting the Truck Number

Most mining operations assign a number to each piece of equipment for quick identification. This number or name can be entered in the Truck Number field. It is very important to enter a unique truck number for each truck using the PLMIII system. This number is one of the key fields used within the haul cycle database. The field will hold 20 alpha-numeric characters.

1. On the Truck Configuration screen, enter the truck number in the appropriate field.
2. Press the "Save Changes" button to program the change into the payload meter.

## Setting the Komatsu Distributor

This field in the haul cycle record can hold the name of the Komatsu distributor that helped install the system. Komatsu also assigns a distributor number to each distributor. This number is used on all warranty claims. This Komatsu distributor number can also be put into this field. The field will hold 20 alpha-numeric characters.

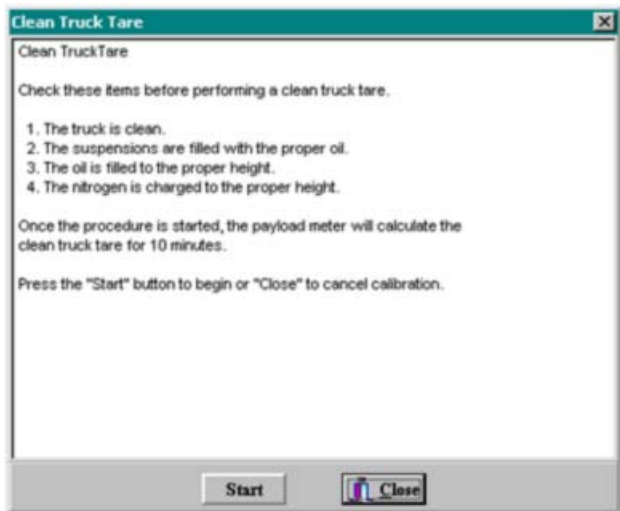
1. On the Truck Configuration screen, enter the distributor name or number in the appropriate field.
2. Press the "Save Changes" button to program the change into the payload meter.

## Setting the Komatsu Customer

This field in the haul cycle record can hold the name of the mine or operation where the truck is in service. Komatsu also assigns a customer number to each customer. This number is used on all warranty claims. This Komatsu customer number can also be put into this field. The field will hold 20 alpha-numeric characters.

1. On the Truck Configuration screen, enter the customer name or number in the appropriate field.
2. Press the "Save Changes" button to program the change into the payload meter.

## Clean Truck Tare Calibration



82944

The payload meter uses the clean truck tare value to calculate carry-back load for each haul cycle. The carry-back stored in the haul cycle record is the new empty tare minus the clean truck tare.

This procedure should be performed after service to the suspensions or when significant changes are made to the sprung weight of the truck. Before performing this procedure, make sure that the suspensions are properly filled with oil and charged. It is critical to payload measurement that the proper oil height and gas pressure be used.

Once the clean tare process is started, the payload meter will begin to calculate the clean empty sprung weight of the truck. This calculation continues while the truck drives to the next loading site. Once the procedure is started, there is no reason to continue to monitor the process with the PC. The truck does not need to be moving to start this procedure.

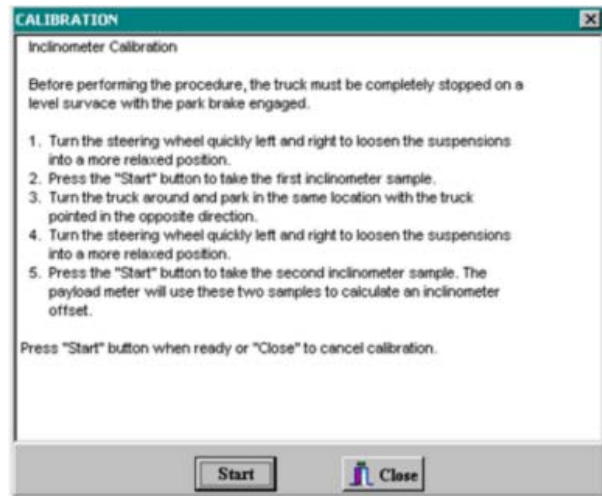
1. Clean all debris from the truck and dump body.
2. Ensure that there are no faults recorded in the system and all sensors are reading correctly.
3. Use the PDM software to connect to the payload meter. From the Truck Configuration screen, select "Clean Truck Tare".
4. Follow the on-screen instructions.

**NOTE:** The truck must be driven continuously for ten minutes above 5 kph (3 mph). The PLM module does not calibrate while the machine is stopped.

**NOTE:** Drive on actual haul roads and do not use the wheel brake lock during calibration. Otherwise, inaccurate payload readings may result.

5. Once the calibration is complete, confirm that the PLM module states "Empty" in the Haul Cycle State parameter. If the calibration is not complete or has failed, the Haul Cycle State parameter will state "Tare Zone".

## Inclinometer Calibration



82945

The inclinometer calibration procedure is designed to compensate for variations in the mounting attitude of the inclinometer. The inclinometer input is critical to the payload calculation.

This procedure should be performed on relatively flat ground. The maintenance area is often an ideal location for this procedure.

1. Clean all debris from the truck and dump body.
2. Ensure that there are no faults recorded in the system and all sensors are reading correctly.
3. Use the PDM software to connect to the payload meter. From the Truck Configuration screen, select "Inclinometer".
4. With the truck stopped and the brake lock on, click on "Start". This instructs the payload meter to sample the inclinometer once.
5. Turn the truck around. Drive the truck around and park in the exact same spot as before, facing the other direction.
6. With the truck stopped and the brake lock on, click on "Start". This instructs the payload meter to sample the inclinometer again. The payload meter will average the two samples to determine the average offset.
7. Follow the on-screen instructions.



## DATA ANALYSIS

### PAYLOAD SUMMARY FORM

**Date:** Sorts the data within a date range. eg. "Dec 1, 2000 through Dec 31, 2000"

**Truck Number:** Sorts the data by the truck unit number, eg. "374"

**Payload Data Summary:** Summary statistical analysis of the payloads from the selected query.

**Output Options:** Use to create reports, graphs and export data from the selected query.

**Time:** Sorts the data within a time for each day within the data range. "8:00 AM to 5:00PM"

**Truck Type:** Sorts the data by the truck type, eg. "930E" or "830E"

**Query Database & Display:** Sorts the data by the selected query options (unit, type, date, time) and displays the results.

**Cycle Summary:** Cycle time summary from the selected query.

**Haul Cycle Records:** Summary view of the haul cycle records from the selected query. Double Click to view the details for individual haul cycle records. Haul cycles in red area are not included in the summary statistics..

| Truck Number | Date       | Time       | Payload | # Swings | Total Time |
|--------------|------------|------------|---------|----------|------------|
| 351          | 07/10/2000 | 8:22:40 AM | 323.8   | 5        | 7:09       |
| 351          | 07/10/2000 | 8:36:12 AM | 323.8   | 5        | 5:23       |
| 351          | 07/10/2000 | 8:41:35 AM | 323.8   | 5        | 5:24       |
| 351          | 07/10/2000 | 8:46:58 AM | 321.5   | 1        | 30:11      |
| 351          | 07/10/2000 | 9:17:06 AM | 321.9   | 0        | 0:32       |
| 351          | 07/10/2000 | 9:17:37 AM | 318.0   | 0        | 0:32       |

The data analysis tools allow the user to monitor the performance of the payload systems across the fleet. Analysis begins when the "View Payload Data" button is pressed. This starts an "all trucks, all dates, all times" query of the database and displays the results in the Payload Summary Form.

The user can change the query by changing the dates, times, or trucks to include in the query for display.

Haul cycles in the data grid box at the bottom can be double-clicked to display the detailed results of that haul.

#### Creating a Query

The program defaults to show all trucks, all types, all dates and all times for the initial query. The display can be narrowed by selecting which trucks or types to view and for what dates and times.

The query items are added in the "AND" condition. If the user selects a truck # and date range, the query will sort the data for that truck number AND the date range.

#### Sorting on Truck Unit Number

The truck unit number is the truck unit number entered into the payload meter when it was configured at installation. The query can be set to look for all trucks or one particular truck number. When the program begins, it searches through the database for all the unique truck numbers and creates a list to select from.

Choosing one particular truck number will limit the data in the displays, summaries and reports to the one selected truck. To create reports for truck number 374, select 374 from the pull-down menu and hit the "Query Database and Display" button.

#### Sorting on Truck Type

The truck type is the size of the truck from the family of Komatsu trucks. This allows the user to quickly view results from different types of trucks on the property. For example, a separate report can be generated for 830E and 930E trucks.

## Sorting on Date Range

The default query starts in 1995 and runs through the current date on the PC. To narrow the range to a specific date, change the "From" and "To" dates. For example, to view the haul cycle reports from truck 374 for the month of July, 2000:

1. Select truck 374 from the Truck Unit pull-down menu.



2. Change the "From" date to July 1, 2000.



3. Change the "To" date to July 31, 2000.
4. Press the "Query Database and Display" to view the results.

## Sorting on Time Range

The time range sorts the times of the day for valid dates. Changing the time range to 6:00AM to 6:00PM will limit the payloads displayed to the loads that occurred between those times for each day of the date range. Times are entered in 24:00 format. To view the haul cycle reports from the first shift for truck 374 from January 5, 2000 to January 8, 2000:

1. Select truck 374 from the Truck Unit pull-down menu.



2. Change the "From" date to January 5, 2000.

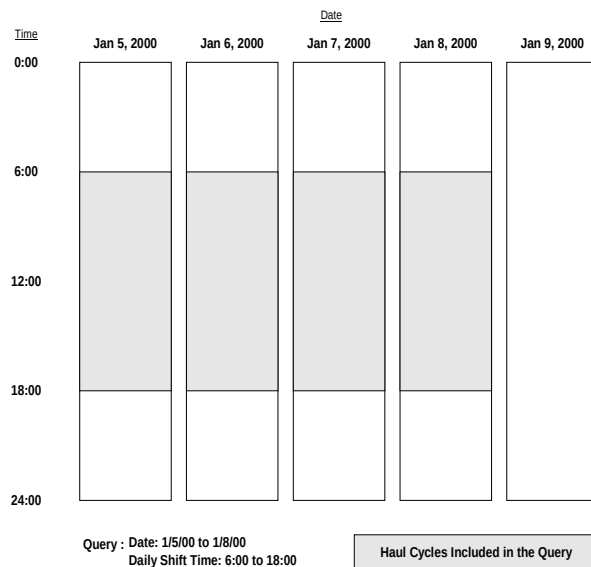


3. Change the "To" date to January 8, 2000.
4. Change the "From" time to 06:00.
5. Change the "To" time to 18:00.

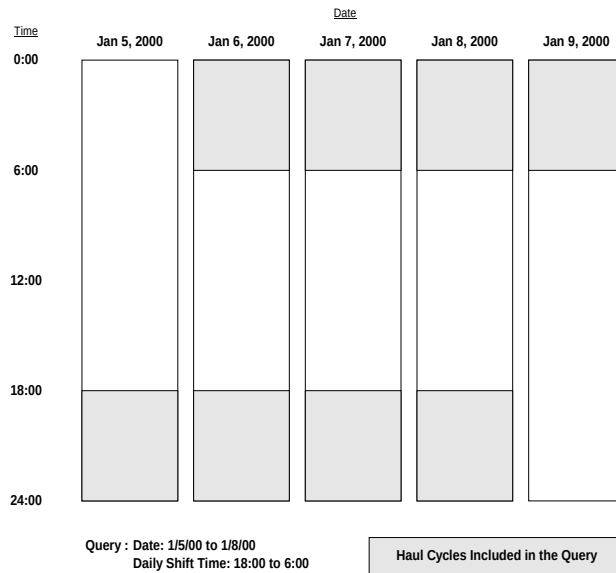
Daily Shift Time  To

6. Press the "Query Database and Display" to view the results.

This query will display haul cycles from January 5 to January 8, from 6:00 AM to 6:00 PM.



The shift times selected can extend the query past the original date. If the dates set for the query are January 5 to January 8 and the times were changed to query the 6:00 PM (18:00) to 6:00 AM (06:00) shift, the results would extend into the morning of the 9th. This can be seen in the following example:



## Creating Reports

Reports can be generated and viewed on the screen or printed. These reports are generated from the query displayed on the Payload Summary Screen. From the example in "Sorting on Time Range", the report printed would only contain data from truck 374 during the month of July 2000, from 8:00 AM to 5:00 PM.

It is important to carefully select the query data and press the "Query Database & Display" button before printing a report.

## Payload Detail Screen

The Payload Detail screen gives the details for any individual haul cycle. From the "Payload Summary" screen, double-click on any haul cycle to display the detail.

**PAYLOAD** (Short tons)

|                      |            |
|----------------------|------------|
| Truck ID             | 374        |
| Operator ID          | 4200       |
| Payload Start Date   | 11/06/2000 |
| Dump Date            | 11/06/2000 |
| Carry Back           | -0.77      |
| Swingloads           | 5          |
| Final Payload        | 317        |
| Payload Est @ Shovel | 322        |

**CYCLE TIMES** (Min:Sec)

|                      |       |
|----------------------|-------|
| Load Time            | 2:26  |
| Haul Time            | 12:03 |
| Haul Stop Time       | 0:31  |
| Dump Time            | 0:49  |
| Return Time          | 5:42  |
| Return Stop Time     | 1:23  |
| Total Operating Time | 22:54 |

**START TIME** (hh:mm:ss)

|                    |          |
|--------------------|----------|
| Payload Start Time | 12:26:19 |
| Load Start Time    | 12:33:12 |
| Dump Start Time    | 12:48:10 |

**CYCLE DISTANCE** (mi)

|                 |      |
|-----------------|------|
| Haul Distance   | 1.76 |
| Return Distance | 1.68 |
| Total Distance  | 3.44 |

**SPRUNG LOAD and TIRE TON MPH** (hh:mm:ss)

|                  |                             |          |
|------------------|-----------------------------|----------|
| Peak Sprung Load | 562.40 (Short Tons)         | 12:35:59 |
| Left Front Tire  | 460.97 (Tire Short Ton MPH) |          |
| Right Front Tire | 507.54 (Tire Short Ton MPH) |          |
| Rear Tire        | 374.66 (Tire Short Ton MPH) |          |

**SPEEDS** (mph)

|               |                |
|---------------|----------------|
| <b>LOADED</b> | <b>TIME</b>    |
| Max Speed     | 25.02 12:44:50 |
| Avg Speed     | 8.43           |
| <b>EMPTY</b>  |                |
| Max Speed     | 28.03 12:27:20 |
| Avg Speed     | 14.21          |

? Help    ✓ Warning Flags    Close



**NOTE:** Some haul cycles may contain the Sensor Input warning flag. This indicates that one of the four pressure sensors or inclinometer was not functioning properly during the haul cycle. Haul cycles with this warning are displayed in red on the Payload Summary window and are not included in the summary statistics for reports or display.

## Summary - one page report

A summary of the queried data can be printed onto 1 page. The cycle data is summarized onto one sheet. Displayed is the speeds, cycle times, load statistics, frame and tire data.

## Detailed - multi-page report

The detail report starts with the summary report and follows with pages of data for each haul cycle. The detailed report prints date, time, payload, cycle times, and cycle distances, speeds and the number of swing loads.

|                                  |             |             |                                        |
|----------------------------------|-------------|-------------|----------------------------------------|
| <b>Payload Summary Report</b>    |             |             |                                        |
| Report Date: October 17, 2000    |             |             |                                        |
| Query used for summary:          |             |             |                                        |
| Truck Number: ALL                |             |             |                                        |
| Truck Type: ALL                  |             |             |                                        |
| Dates: 8/28/00 to 10/3/00        |             |             |                                        |
| Daily Shift Time: 00:00 to 24:00 |             |             |                                        |
|                                  |             |             | <b>Cycle Distance Averages (miles)</b> |
|                                  |             |             | Loaded 2.2                             |
|                                  |             |             | Empty 2.4                              |
|                                  |             |             | Avg Total Distance 4.6                 |
| <b>Time Period</b>               | <b>Date</b> | <b>Time</b> |                                        |
| First Load                       | 08/28/2000  | 03:54:13 PM |                                        |
| Last Load                        | 10/03/2000  | 12:08:36 PM |                                        |
|                                  |             |             | <b>Payload Data</b>                    |
|                                  |             |             | Average 326                            |
|                                  |             |             | Standard Deviation 27.1                |
|                                  |             |             | Maximum 417                            |
|                                  |             |             | Minimum 10                             |
|                                  |             |             | Total Tons 373,959                     |
|                                  |             |             | Number of Payloads 1,148               |
|                                  |             |             | Avg Carry Back 2.3                     |
|                                  |             |             | <b>Frame Data</b>                      |
|                                  |             |             | Max Peak Pos 12,958                    |
|                                  |             |             | Max Peak Neg 13,352                    |
|                                  |             |             | Avg Peak Pos 5,767                     |
|                                  |             |             | Avg Peak Neg 6,588                     |
|                                  |             |             | <b>Cycle Time Averages (Min:Sec)</b>   |
|                                  |             |             | Load 3:00                              |
|                                  |             |             | Haul Run 11:36                         |
|                                  |             |             | Haul Stop 2:48                         |
|                                  |             |             | Empty Run 7:20                         |
|                                  |             |             | Empty Stop 4:05                        |
|                                  |             |             | Dump 1:17                              |
|                                  |             |             | Avg Total Cycle Time 30:09             |
|                                  |             |             | <b>Avg Tire (Ton-MPH)</b>              |
|                                  |             |             | Left Front 528                         |
|                                  |             |             | Right Front 527                        |
|                                  |             |             | Rear 377                               |
|                                  |             |             | <b>SPEED DATA (mph)</b>                |
|                                  |             |             | Loaded Max 47.24                       |
|                                  |             |             | Loaded Avg 9.30                        |
|                                  |             |             | Empty Max 47.24                        |
|                                  |             |             | Empty Avg 12.52                        |

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## Payload Detail Report

Report Date: October 17, 2000

Query used for summary:

Truck Number: ALL

Truck Type: ALL

Dates: 8/28/00 to 10/3/00

Daily Shift Time: 00:00 to 24:00

Page 1

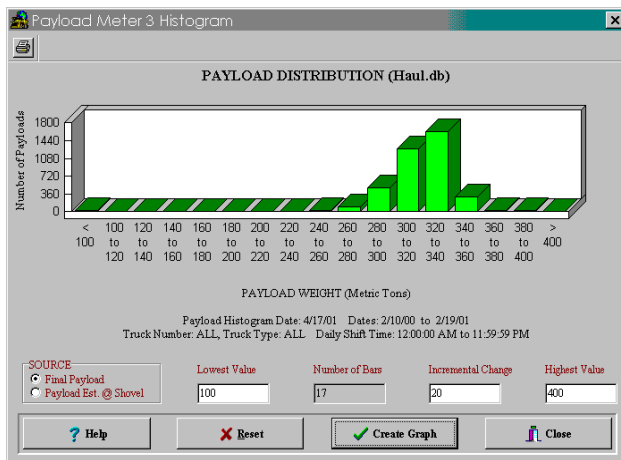
|     |         |        |                |         | CYCLE Times(min:sec) |       |           |      |        | CYCLE DISTANCE(mi.) |        |      |        | SPEEDS(mph) |       |           |       |             |
|-----|---------|--------|----------------|---------|----------------------|-------|-----------|------|--------|---------------------|--------|------|--------|-------------|-------|-----------|-------|-------------|
| No. | Date    | Time   | Payload S.Tons | #Swings | Load                 | Haul  | Haul Stop | Dump | Return | Return Stop         | Total  | Haul | Return | Total       | Max.  | Haul Avg. | Max.  | Return Avg. |
| 1   | 8/28/00 | 3:54P  | 347            | 7       | 4.40                 | 11.11 | 0.06      | 0.52 | 7.54   | 14.28               | 39.11  | 1.9  | 2.0    | 4.0         | 21.99 | 10.14     | 31.03 | 5.48        |
| 2   | 8/28/00 | 5:18P  | 301            | 6       | 3.39                 | 11.10 | 0.25      | 0.43 | 8.12   | 3.43                | 27.52  | 1.9  | 1.9    | 3.8         | 23.02 | 9.91      | 26.05 | 9.42        |
| 3   | 8/28/00 | 5:46P  | 300            | 6       | 3.32                 | 11.32 | 2.32      | 0.56 | 7.07   | 0.09                | 25.48  | 1.9  | 1.9    | 3.8         | 23.02 | 8.30      | 25.02 | 15.49       |
| 4   | 8/28/00 | 6:11P  | 330            | 4       | 1.58                 | 17.51 | 2.49      | 1.58 | 11.34  | 1.41                | 37.51  | 3.9  | 2.6    | 6.5         | 36.21 | 11.37     | 24.07 | 11.73       |
| 5   | 8/28/00 | 6:49P  | 344            | 4       | 1.54                 | 18.26 | 13.04     | 1.50 | 11.20  | 0.24                | 46.58  | 4.0  | 3.9    | 7.9         | 34.97 | 7.56      | 37.06 | 20.15       |
| 6   | 8/28/00 | 7:36P  | 380            | 5       | 2.12                 | 19.52 | 1.07      | 1.39 | 11.32  | 0.48                | 37.10  | 4.0  | 4.0    | 8.0         | 30.00 | 11.28     | 34.03 | 19.59       |
| 7   | 8/28/00 | 8:13P  | 352            | 4       | 2.26                 | 18.17 | 0.23      | 1.54 | 11.54  | 1.56                | 36.50  | 3.9  | 4.0    | 8.0         | 31.97 | 12.62     | 34.07 | 17.49       |
| 8   | 8/28/00 | 8:50P  | 356            | 4       | 1.56                 | 18.52 | 0.21      | 1.55 | 11.59  | 0.38                | 35.41  | 4.0  | 4.0    | 8.0         | 36.17 | 12.34     | 36.21 | 19.09       |
| 9   | 8/28/00 | 9:25P  | 370            | 5       | 2.18                 | 16.17 | 2.18      | 0.53 | 12.25  | 0.25                | 34.36  | 2.7  | 4.1    | 6.7         | 34.97 | 8.55      | 36.17 | 19.14       |
| 10  | 8/28/00 | 10:00P | 363            | 5       | 3.07                 | 16.56 | 5.48      | 0.52 | 9.02   | 0.33                | 36.18  | 2.7  | 2.7    | 5.3         | 32.97 | 6.99      | 32.97 | 16.61       |
| 11  | 8/28/00 | 10:36P | 341            | 7       | 3.58                 | 12.23 | 0.23      | 0.53 | 7.26   | 0.57                | 26.00  | 2.0  | 2.0    | 3.9         | 21.03 | 9.20      | 32.00 | 14.10       |
| 12  | 8/28/00 | 11:02P | 321            | 4       | 2.24                 | 16.19 | 5.06      | 0.50 | 10.17  | 1.05                | 36.01  | 2.7  | 2.6    | 5.3         | 34.03 | 7.49      | 31.97 | 13.71       |
| 13  | 8/28/00 | 11:38P | 316            | 5       | 2.49                 | 15.21 | 7.06      | 0.58 | 13.01  | 1.31                | 52.26  | 3.9  | 4.0    | 7.9         | 40.12 | 10.30     | 40.12 | 9.16        |
| 14  | 8/29/00 | 12:56A | 320            | 5       | 2.23                 | 15.18 | 0.07      | 0.55 | 13.16  | 1.43                | 43.42  | 3.9  | 3.8    | 7.7         | 36.21 | 14.99     | 41.10 | 9.18        |
| 15  | 8/29/00 | 1:40A  | 302            | 4       | 1.55                 | 14.12 | 0.08      | 0.54 | 10.34  | 0.11                | 27.54  | 3.9  | 3.9    | 7.7         | 40.12 | 16.15     | 41.10 | 21.64       |
| 16  | 8/29/00 | 2:08A  | 310            | 5       | 2.21                 | 15.14 | 4.55      | 1.01 | 9.46   | 4.59                | 38.16  | 3.9  | 3.9    | 7.7         | 40.12 | 11.51     | 41.16 | 15.67       |
| 17  | 8/29/00 | 2:46A  | 318            | 5       | 2.25                 | 14.44 | 0.15      | 0.54 | 10.36  | 0.19                | 29.13  | 3.9  | 3.9    | 7.7         | 41.10 | 15.50     | 41.10 | 21.28       |
| 18  | 8/29/00 | 3:15A  | 278            | 4       | 2.01                 | 15.53 | 1.12      | 0.46 | 10.38  | 2.08                | 32.38  | 4.3  | 3.8    | 8.1         | 38.13 | 15.06     | 41.10 | 18.05       |
| 19  | 8/29/00 | 3:48A  | 328            | 5       | 2.25                 | 17.38 | 3.08      | 0.57 | 10.07  | 0.20                | 34.35  | 4.3  | 3.9    | 8.2         | 34.03 | 12.41     | 41.10 | 22.19       |
| 20  | 8/29/00 | 4:22A  | 287            | 4       | 2.49                 | 17.00 | 15.44     | 1.02 | 11.59  | 4.49                | 53.23  | 4.3  | 4.0    | 8.2         | 40.12 | 7.87      | 41.10 | 14.09       |
| 21  | 8/29/00 | 1:41P  | 285            | 4       | 2.15                 | 16.08 | 6.14      | 5.21 | 9.02   | 14.43               | 53.43  | 4.0  | 3.0    | 6.9         | 34.97 | 10.58     | 34.03 | 7.55        |
| 22  | 8/29/00 | 2:50P  | 340            | 7       | 3.38                 | 10.14 | 0.06      | 0.54 | 8.41   | 7.10                | 30.43  | 1.8  | 3.3    | 5.2         | 27.07 | 10.54     | 37.07 | 12.65       |
| 23  | 8/29/00 | 3:20P  | 326            | 5       | 3.32                 | 9.47  | 0.08      | 1.09 | 5.21   | 0.16                | 20.13  | 1.8  | 1.8    | 3.6         | 27.05 | 10.98     | 30.02 | 18.98       |
| 24  | 8/29/00 | 3:41P  | 342            | 6       | 3.59                 | 11.24 | 0.08      | 0.47 | 19.49  | 15.18               | 51.25  | 1.9  | 6.2    | 8.1         | 23.02 | 10.02     | 32.00 | 10.56       |
| 25  | 8/29/00 | 4:43P  | 349            | 7       | 4.46                 | 11.12 | 0.18      | 0.57 | 7.32   | 3.12                | 27.57  | 1.8  | 1.9    | 3.8         | 23.02 | 9.50      | 27.07 | 10.87       |
| 26  | 8/29/00 | 5:11P  | 322            | 4       | 2.11                 | 5.29  | 19.58     | 1.29 | 15.38  | 59.19               | 104.04 | 1.1  | 5.1    | 6.1         | 42.20 | 2.52      | 42.20 | 4.05        |
| 27  | 8/30/00 | 4:27A  | 330            | 7       | 4.27                 | 10.33 | 0.09      | 1.02 | 21.08  | 15.27               | 196.46 | 1.7  | 4.3    | 6.0         | 23.02 | 9.69      | 28.01 | 1.42        |
| 28  | 8/30/00 | 4:55P  | 320            | 6       | 3.52                 | 10.42 | 0.14      | 1.00 | 5.43   | 1.42                | 23.13  | 1.7  | 1.7    | 3.4         | 24.07 | 9.55      | 29.06 | 13.47       |
| 29  | 8/30/00 | 5:18P  | 338            | 6       | 3.16                 | 10.45 | 0.51      | 0.58 | 6.00   | 2.34                | 24.24  | 1.7  | 1.6    | 3.3         | 23.02 | 8.71      | 29.04 | 11.36       |
| 30  | 8/30/00 | 5:42P  | 343            | 6       | 3.52                 | 10.28 | 0.08      | 1.01 | 6.06   | 0.25                | 22.00  | 1.7  | 1.6    | 3.3         | 24.08 | 9.57      | 28.03 | 14.44       |

## Creating Graphs

The PLMIII software can generate graphs that quickly summarize payload data. These graphs can be customized for printing. Just like the reports, the graphs are generated from the query displayed on the "Payload Summary" screen. From the "Sorting on Time Range" example, the graph that is printed would only contain data from truck 374 during the month of July 2000, from 8:00 AM to 5:00 PM.

It is important to carefully select the query data and press the "Query Database & Display" button before creating a graph.

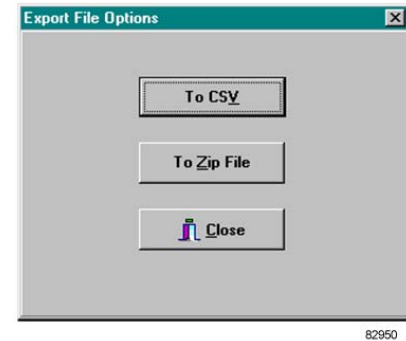
1. From the Payload Summary Screen select the "Graph" button at the bottom. The Histogram Setup screen will display



2. Enter the "Lowest Value". This will be the lowest payload on the graph. Any payloads less than this value will be summed in the first bar.
3. Enter the "Highest Value". This will be the highest value on the graph. Payloads over this value will be summed in the last bar.
4. Enter the "Incremental Change". This will determine the number of bars and the distance between them. The program limits the number of bars to 20. This allows graphs to fit on the screen and print onto 1 page.
5. Press the "Create Graph" button.

The graph will be displayed based on the query settings from the Payload Summary screen. The graph can be customized and printed.

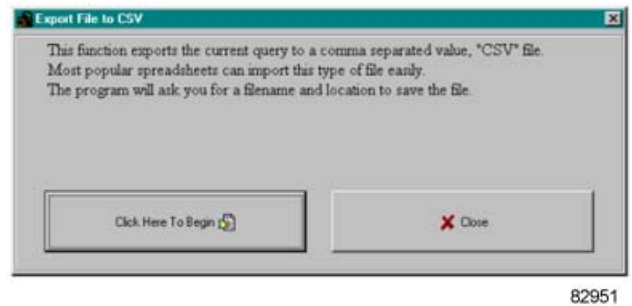
## Exporting Data



The data from the database can be exported for use with other software applications. The data is selected from the currently displayed query. The exported data can be put into a ".csv" file or a compressed ".zip" file.

- The ".csv" format allows data to be easily imported into spreadsheet applications and word processing applications.
- The ".zip" format allows data to be transferred from one computer to the PDM Software database on another computer. This offers a compact way to transfer data from one computer to another.

## CSV Export



CSV stands for Comma Separated Value. This is an ASCII text file format that allows spreadsheet applications like Excel and Lotus 123 to import data easily. To export the data into a ".csv" file, press the "Export" button at the bottom of the payload summary screen and select "To CSV". The program will request a filename and location for the file.

| Truck # | Date    | Time       | Payload | Swingload | Operator | Status | Fla | Carry | Bac  | Total Cyc | E-Run | Tin | E-Stop | Tin | Loading |
|---------|---------|------------|---------|-----------|----------|--------|-----|-------|------|-----------|-------|-----|--------|-----|---------|
| 374     | 2/11/00 | 4:06:48 PM | 294.4   | 2         | 0        | E      |     | -1.2  | 2.7  | 1         | 0.2   | 0   |        |     |         |
| 374     | 2/11/00 | 4:10:09 PM | 299.6   | 2         | 0        | I      |     | -1.4  | 7.2  | 0.6       | 0.6   | 0   |        |     |         |
| 374     | 2/11/00 | 4:41:01 PM | 265.5   | 3         | 0        |        |     | 15.6  | 20.1 | 4.6       | 3.9   | 2   |        |     |         |
| 374     | 2/11/00 | 5:24:54 PM | 277.5   | 0         | 0        | AK     |     | 0     | 27   | 4.6       | 6.4   | 1   |        |     |         |
| 374     | 2/11/00 | 6:21:27 PM | 231     | 3         | 0        |        |     | 15.5  | 27.7 | 4.5       | 5     | 1   |        |     |         |
| 374     | 2/11/00 | 7:09:08 PM | 265.5   | 3         | 0        | H      |     | 17.1  | 23.9 | 4.2       | 3.2   | 1   |        |     |         |
| 374     | 2/11/00 | 7:59:53 PM | 262.4   | 4         | 0        |        |     | 14.1  | 29.7 | 4.3       | 3.7   | 2   |        |     |         |
| 374     | 2/11/00 | 8:29:32 PM | 225.1   | 3         | 0        |        |     | 15.6  | 27.7 | 4.5       | 5     | 1   |        |     |         |
| 374     | 2/11/00 | 8:57:11 PM | 265.5   | 3         | 0        |        |     | 15.6  | 20.1 | 4.6       | 3.9   | 2   |        |     |         |
| 374     | 2/11/00 | 9:17:13 PM | 265.4   | 3         | 0        | H      |     | 17    | 23.9 | 4.2       | 3.1   |     |        |     |         |
| 374     | 2/11/00 | 9:41:05 PM | 277.6   | 0         | 0        | AK     |     | 0     | 27   | 4.6       | 6.4   | 1   |        |     |         |

Two sets of data are exported. At the top of the file will be the haul cycle data. The columns, left to right are:

- Truck number
- Haul cycle start date
- Haul cycle start time
- Payload
- Swingloads
- Operator ID
- Warning Flags
- Carry Back
- Total Haul Cycle time
- Empty Running Time
- Empty stop time
- Loading time
- Loaded running time
- Loaded stopped time
- Dumping time
- Loading start time
- Dumping start time
- Loaded haul distance
- Empty haul distance
- Loaded maximum speed
- Time when loaded maximum speed occurred
- Empty maximum speed
- Time when loaded maximum speed occurred
- Maximum + frame torque
- Time when the maximum + frame torque occurred
- Maximum - frame torque
- Time when the maximum - frame torque occurred
- Maximum sprung weight calculation
- Time when the maximum sprung weight calculation occurred
- Left Front Tire-kilometer-hour
- Right Front Tire-kilometer-hour
- Average Rear Tire-kilometer-hour
- Frame serial number

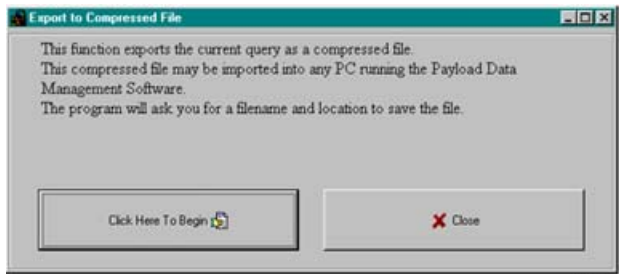
- Reserved 1-5, 7-10: These values are internal calculations used in the continued development of PLMIII and should be ignored.
- Reserved 6: This value is the payload estimate at the shovel just before the truck begins to move.

| Alarm Type | Date Set | Time Set    | Description    | Date Cleared | Time Cleared |
|------------|----------|-------------|----------------|--------------|--------------|
| 10         | 12/5/99  | 8:51:25 AM  | INCL SENSOR LO | 12/5/99      | 8:51:29 AM   |
| 10         | 12/5/99  | 8:56:07 AM  | INCL SENSOR LO | 12/5/99      | 8:56:12 AM   |
| 10         | 12/5/99  | 9:00:49 AM  | INCL SENSOR LO | 12/5/99      | 9:00:54 AM   |
| 10         | 12/5/99  | 9:05:31 AM  | INCL SENSOR LO | 12/5/99      | 9:05:36 AM   |
| 10         | 12/5/99  | 9:10:13 AM  | INCL SENSOR LO | 12/5/99      | 9:10:17 AM   |
| 10         | 12/5/99  | 9:14:54 AM  | INCL SENSOR LO | 12/5/99      | 9:14:59 AM   |
| 10         | 12/5/99  | 9:19:37 AM  | INCL SENSOR LO | 12/5/99      | 9:19:41 AM   |
| 10         | 12/5/99  | 9:24:19 AM  | INCL SENSOR LO | 12/5/99      | 9:24:24 AM   |
| 10         | 12/5/99  | 9:29:01 AM  | INCL SENSOR LO | 12/5/99      | 9:29:06 AM   |
| 10         | 12/5/99  | 9:33:44 AM  | INCL SENSOR LO | 12/5/99      | 9:33:48 AM   |
| 10         | 12/5/99  | 9:38:26 AM  | INCL SENSOR LO | 12/5/99      | 9:38:30 AM   |
| 10         | 12/5/99  | 9:43:08 AM  | INCL SENSOR LO | 12/5/99      | 9:43:13 AM   |
| 10         | 12/5/99  | 9:47:50 AM  | INCL SENSOR LO | 12/5/99      | 9:47:55 AM   |
| 10         | 12/5/99  | 9:52:32 AM  | INCL SENSOR LO | 12/5/99      | 9:52:37 AM   |
| 10         | 12/5/99  | 9:57:13 AM  | INCL SENSOR LO | 12/5/99      | 9:57:17 AM   |
| 10         | 12/5/99  | 10:01:55 AM | INCL SENSOR LO | 12/5/99      | 10:01:59 AM  |

The second series of data below the haul cycle data is the alarms. The alarm columns, left to right are:

- The alarm type
- The date the alarm was set
- The time the alarm was set
- Alarm description
- The date the alarm was cleared
- The time the alarm was cleared

### Compressed



82954

This export function allows the data from one laptop to be transferred to another computer. This can be useful when a service laptop is used to download multiple machines and transfer the data to a central computer for analysis. This can also be used to copy haul data from a particular truck onto a diskette for analysis.

The file format is a compressed binary form of the displayed query. The file can only be imported by another computer running the PDM Software.

To export data in ZIP format:

1. Confirm that the data displayed is the query data that needs to be exported.
2. From the payload summary screen, press the "EXPORT" button and select "To ZIP".
3. The program will ask for a filename and location.

### Importing Data

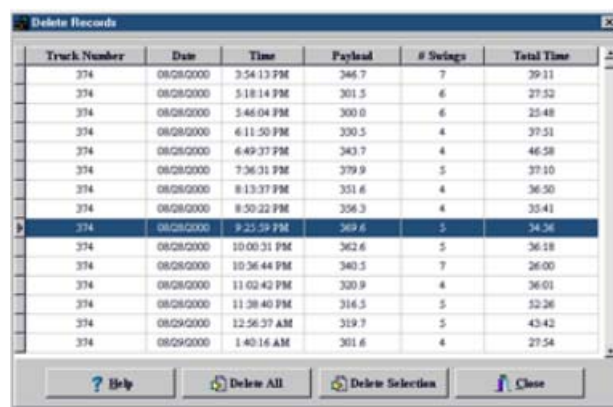
This import function allows the data from one laptop to be transferred to another computer. This can be useful when a service laptop is used to download multiple machines and transfer the data to a central computer for analysis. This can also be used to copy haul data from a particular truck from a diskette into a database for analysis.

To import data, press the "IMPORT" button at the bottom of the "Payload Summary" screen. The program will ask for a ".zip" file to import, locate the file and press "Open". The program will only import ".zip" files created by another computer running the PDM Software.

### Deleting Haul Cycle Records

To delete haul cycle records from the main database, press the "Delete" button at the bottom of the "Payload Summary" screen. The program will display a summary of the records from the displayed query. To delete a record, select one at a time and press the "Delete" button. It is recommended that records be exported to a .zip file for archival purposes before deletion. Multiple records may be selected by holding down the Shift key. Pressing the "Delete All" button will select all the records from the current query and delete them.

*NOTE: There is no recovery for records that have been deleted from the main database. It is highly recommended that all records be exported and archived in a compressed file format for future reference before being deleted.*

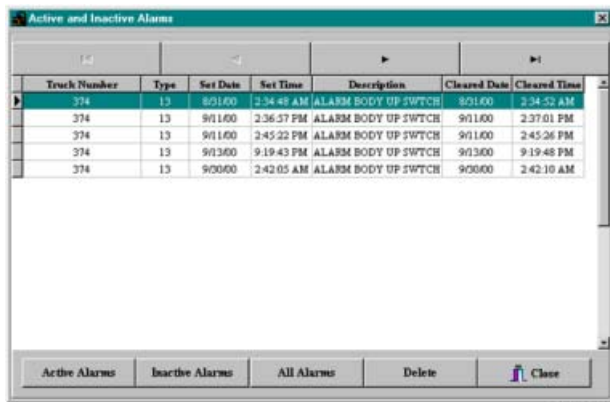


| Truck Number | Date       | Time        | Payload | # Swings | Total Time |
|--------------|------------|-------------|---------|----------|------------|
| 374          | 08/08/2000 | 3:54:13 PM  | 346.7   | 7        | 39:11      |
| 374          | 08/08/2000 | 5:18:14 PM  | 301.5   | 6        | 27:52      |
| 374          | 08/08/2000 | 5:46:04 PM  | 300.0   | 6        | 25:48      |
| 374          | 08/08/2000 | 6:11:50 PM  | 330.5   | 4        | 37:51      |
| 374          | 08/08/2000 | 6:49:37 PM  | 343.7   | 4        | 46:58      |
| 374          | 08/08/2000 | 7:36:31 PM  | 379.9   | 5        | 37:10      |
| 374          | 08/08/2000 | 8:13:37 PM  | 351.6   | 4        | 36:50      |
| 374          | 08/08/2000 | 8:50:22 PM  | 356.3   | 4        | 35:41      |
| 374          | 08/08/2000 | 9:25:59 PM  | 349.6   | 5        | 34:56      |
| 374          | 08/08/2000 | 10:00:31 PM | 362.6   | 5        | 36:18      |
| 374          | 08/08/2000 | 10:36:44 PM | 340.5   | 7        | 26:00      |
| 374          | 08/08/2000 | 11:02:42 PM | 320.9   | 4        | 36:01      |
| 374          | 08/08/2000 | 11:38:40 PM | 316.5   | 5        | 52:26      |
| 374          | 08/09/2000 | 12:56:37 AM | 319.7   | 5        | 43:42      |
| 374          | 08/09/2000 | 1:40:16 AM  | 301.6   | 4        | 27:54      |

82955

## Viewing Alarms

From the Payload Summary screen, click the "Alarms" button to display the alarm screen. The alarms are sorted by the query settings from the Payload Summary screen. Alarms can be displayed as Active or Inactive.



The screenshot shows a window titled "Active and Inactive Alarms". It contains a table with the following columns: Truck Number, Type, Set Date, Set Time, Description, Cleared Date, and Cleared Time. The table lists five alarm records for truck number 374. The first record is highlighted in blue.

| Truck Number | Type | Set Date | Set Time   | Description          | Cleared Date | Cleared Time |
|--------------|------|----------|------------|----------------------|--------------|--------------|
| 374          | 13   | 8/1/00   | 2:34:48 AM | ALARM BODY UP SWITCH | 8/1/00       | 2:34:52 AM   |
| 374          | 13   | 9/1/00   | 2:36:37 PM | ALARM BODY UP SWITCH | 9/1/00       | 2:37:01 PM   |
| 374          | 13   | 9/1/00   | 2:45:22 PM | ALARM BODY UP SWITCH | 9/1/00       | 2:45:26 PM   |
| 374          | 13   | 9/13/00  | 9:19:43 PM | ALARM BODY UP SWITCH | 9/13/00      | 9:19:48 PM   |
| 374          | 13   | 9/30/00  | 2:42:05 AM | ALARM BODY UP SWITCH | 9/30/00      | 2:42:10 AM   |

At the bottom of the window, there are buttons for "Active Alarms", "Inactive Alarms", "All Alarms", "Delete", and "Close".

82946

## Deleting Alarm Records

To delete alarm records from the main database, press the "Delete" button at the bottom of the "Alarm Display" screen. The program will display a summary of the alarms from the query. To delete an alarm, select one at a time and press the "Delete" button. It is recommended that the query data be exported to a ".zip" file for archival purposes before deletion. Multiple records may be selected by holding down the Shift key. Pressing the "Delete All" button will select all the alarms from the current query and delete them.

*NOTE: There is no recovery for alarms that have been deleted from the main database. It is highly recommended that all records be exported and archived in a compressed file format for future reference before being deleted.*



The screenshot shows a window titled "Delete Alarm Records". It contains a table with the following columns: Truck Number, Alarm Type, Set Date, Set Time, Description, Cleared Date, and Cleared Time. The table lists five alarm records for truck number 374. The first record is highlighted in blue.

| Truck Number | Alarm Type | Set Date | Set Time   | Description         | Cleared Date | Cleared Time |
|--------------|------------|----------|------------|---------------------|--------------|--------------|
| 374          | 13         | 8/1/00   | 2:34:48 AM | ALARM BODY UP SWITC | 8/1/00       | 2:34:52 AM   |
| 374          | 13         | 9/1/00   | 2:36:37 PM | ALARM BODY UP SWITC | 9/1/00       | 2:37:01 PM   |
| 374          | 13         | 9/1/00   | 2:45:22 PM | ALARM BODY UP SWITC | 9/1/00       | 2:45:26 PM   |
| 374          | 13         | 9/13/00  | 9:19:43 PM | ALARM BODY UP SWITC | 9/13/00      | 9:19:48 PM   |
| 374          | 13         | 9/30/00  | 2:42:05 AM | ALARM BODY UP SWITC | 9/30/00      | 2:42:10 AM   |

At the bottom of the window, there are buttons for "? Help", "Delete All", "Delete Selection", and "Close".

82957

# TROUBLESHOOTING SECTION

## TROUBLESHOOTING

Troubleshooting the PLMIII system is done through the PC software you can:

- View active alarms.
- View the sensor inputs using the Real-Time Data Display.
- Test the payload lights.
- Create log files of sensor inputs for further analysis.
- These activities require a connection to the PLMIII system.

### Viewing Active Alarms

Active alarms are alarms that have been set, but not yet cleared. Each alarm is set when the conditions for activation are held for 5 seconds. Each alarm is cleared when the condition has been returned to normal range for 5 seconds. For example, 5 seconds after the left-rear pressure sensor is disconnected, the LR-Pressure Lo alarm will be activated. This can be viewed using the "Connect to Payload Meter" screen. 5 seconds after the pressure sensor is re-connected, the alarm will clear and be recorded in memory.

Active alarms are recorded in memory as "cleared" when the key switch is turned off. When power is restored to the payload meter, the alarms will be re-activated if the conditions still exist for 5 seconds.

To view active alarms:

1. Connect to the payload meter and start the PC software.
2. From the main menu, select "Connect to Payload Meter". The PC will request the latest status information from the payload meter.
3. If there are active alarms, the "Display Active Alarms" button in the lower left corner will be available. If the button is not available, there are no active alarms at the time of connection. The screen does not automatically refresh. If a condition changes to cause an alarm, the user must exit and re-enter the "Connect to Payload Meter" screen.

### Real-Time Data Display

| Real Time Data                                                                                               |      |                          |           |
|--------------------------------------------------------------------------------------------------------------|------|--------------------------|-----------|
| <b>Pressure PSI</b>                                                                                          |      | <b>Truck Status</b>      |           |
| Left Front                                                                                                   | 304  | Inclinometer             | 0.5       |
| Right Front                                                                                                  | 287  | Body Up                  | No        |
| Left Rear                                                                                                    | 299  | Brake Lock               | Off       |
| Right Rear                                                                                                   | 293  | Haul Cycle State         | Tare Zone |
| <b>Speed MPH</b>                                                                                             |      | <b>Weight Short Tons</b> |           |
| Truck                                                                                                        | 20.0 | Sprung Weight            | 107       |
| <b>Lights</b>                                                                                                |      |                          |           |
| <input type="checkbox"/> Green Light <input type="checkbox"/> Amber Light <input type="checkbox"/> Red Light |      |                          |           |
| Set FileName                                                                                                 |      | Start Log                | Stop Log  |
| Set Lights                                                                                                   |      | Close                    |           |

The PC software can be used to view the 'live' input readings from the payload meter. The numbers displayed are 1-second averages.

1. Connect to the payload meter and start the PC software.
2. From the main menu, select "Connect to Payload Meter". The PC will request the latest status information from the payload meter.
3. Select the "Real Time Data" button. The data screen will pop up. The PC will request the payload meter to begin transmitting data.
4. To exit, press the "Close" button.

The units for each measurement are determined by the setting in the Program Options for the PC software. The four suspension pressures and inclinometer are shown. The status of the Body-Up and Brake-Lock inputs is also shown. The haul cycle state and speed is displayed. The current sprung weight is displayed. This differs from pure payload. The sprung weight is a measurement of the weight of the truck above the suspension. It does not include the tires, spindles, wheel motors, drive case, or anything below the suspensions.

### Testing the Payload Lights

The real time data display also allows the user to individually power the payload lights. This can be useful for testing the lights. To turn on a particular color payload light:

1. Click the check box beside the color light to power.
2. Press the "Set Lights" button to turn on the light.
3. Uncheck the box and press "Set Lights" to turn off the light.

The lights will return to their normal state when the real time data display is closed.

## Creating Log Files of Inputs

The PC software can create a text file of the live data stream from the payload meter. This can be very useful for diagnostic purposes. The data is written into a text data file in comma separated value format. The data is recorded in metric units at 50 samples per second. The data file can grow large very quickly. Each sample writes one line into the ASCII file in comma separated format.

The order for each line of data is:

- Date
- Time
- Sprung Weight
- LF Pressure
- RF Pressure
- LR Pressure
- RR Pressure
- Incline
- Speed
- Body Up State (1=up)
- Brake Lock State (1=on)
- Payload State
- Status Flags
- Spare

To create a log file:

1. Connect to the payload meter and start the PC software.
2. From the main menu, select "Connect to Payload Meter". The PC will request the latest status information from the payload meter.
3. Select the "Real Time Data" button. The real time data screen will pop up. The PC will request the payload meter to begin transmitting data.
4. Click the "Set File Name" button and enter a name and location for the text file. The default extension is ".txt". This data can be easily imported into spreadsheets as a comma separated value (.csv) format.
5. Once the filename has been entered, the Start Log and Stop Log buttons will be active.
6. Press the "Start Log" button to start taking data and recording into the file. Once a file is started, it cannot be stopped and started again.
7. Press the "Stop Log" button to stop recording data. Attempting to start the log file again will overwrite the previous file and erase the previous data. To gather more data, close the real time data window, start it again and create a new log file.

## Daily Inspections

An important part of maintaining the Payload Meter III (PLMIII) system is monitoring the basic inputs to the system. It is recommended that the truck operator walk around the truck and visually inspect the following:

- Charging condition of the suspensions - not flat, not overcharged.
- Pressures in the suspensions - check suspensions by using the operator gauge and the operator switch.

## Periodic Maintenance

It is recommended that the following items be checked every 500 hours:

- Confirm the suspension pressures using external gauges.
- Confirm proper suspension height.
- Confirm suspensions do not collapse and make metal-to-metal contact when the truck is loaded.
- Confirm that inclinometer indicates positive (+) values for truck nose up, and negative (-) values for truck nose down.

In addition, it may be useful to confirm proper operation of the suspensions by riding the truck during a complete haul cycle. Record the suspension pressures using the CSV log file tool in the Payload Data Manager software for the PC. The suspension pressures in this log file can be graphed to inspect for flat or overcharged suspensions.

## Abnormal Displays at Power-Up

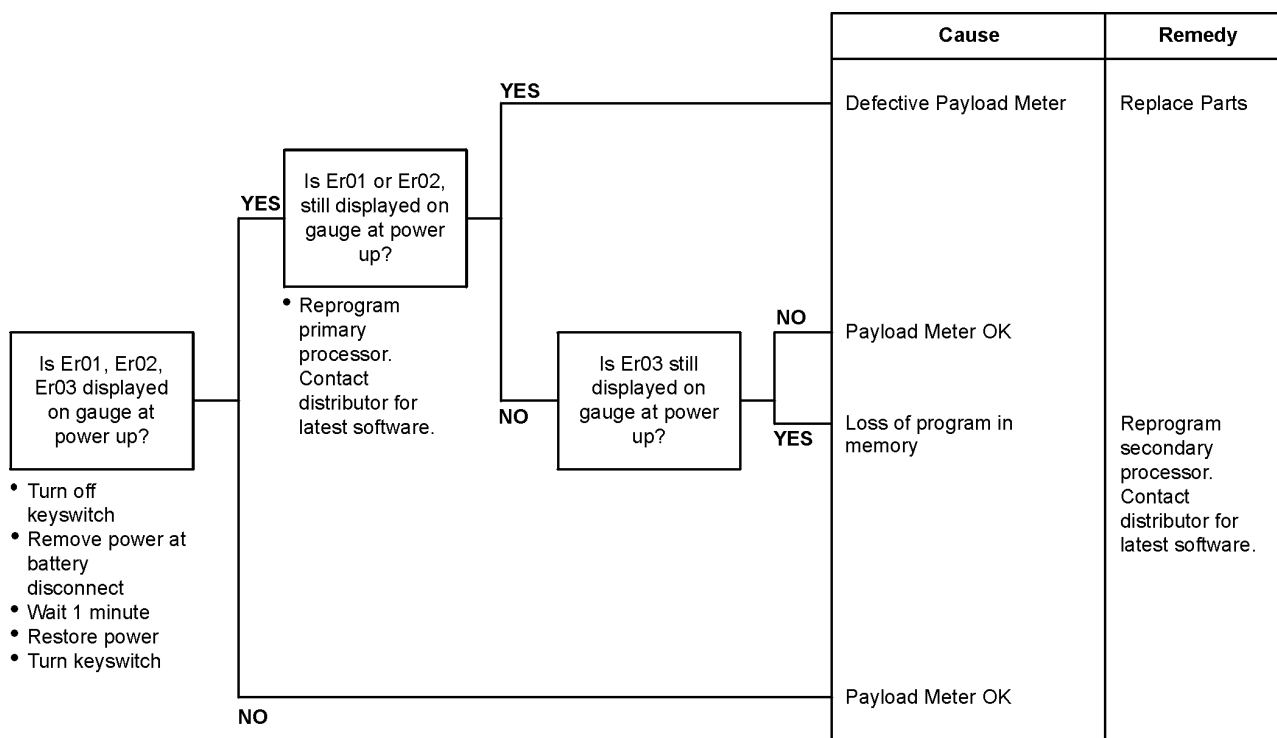
The payload meter performs several internal memory system checks every time it powers-up. In case of error, the operator gauge may display an error code when power is applied to the PLMIII system.

**Er:01** - Bad Truck Configuration error indicates that the meter encountered an error while reading the current truck configuration record from memory.

**Er:02** - Bad Calibration Record error indicates that the meter encountered an error while passing messages between the microprocessors on the circuit board.

**Er:03** - Interprocessor Communications error indicates that the meter encountered an error while passing messages between the microprocessors on the circuit board.

To resolve these errors:

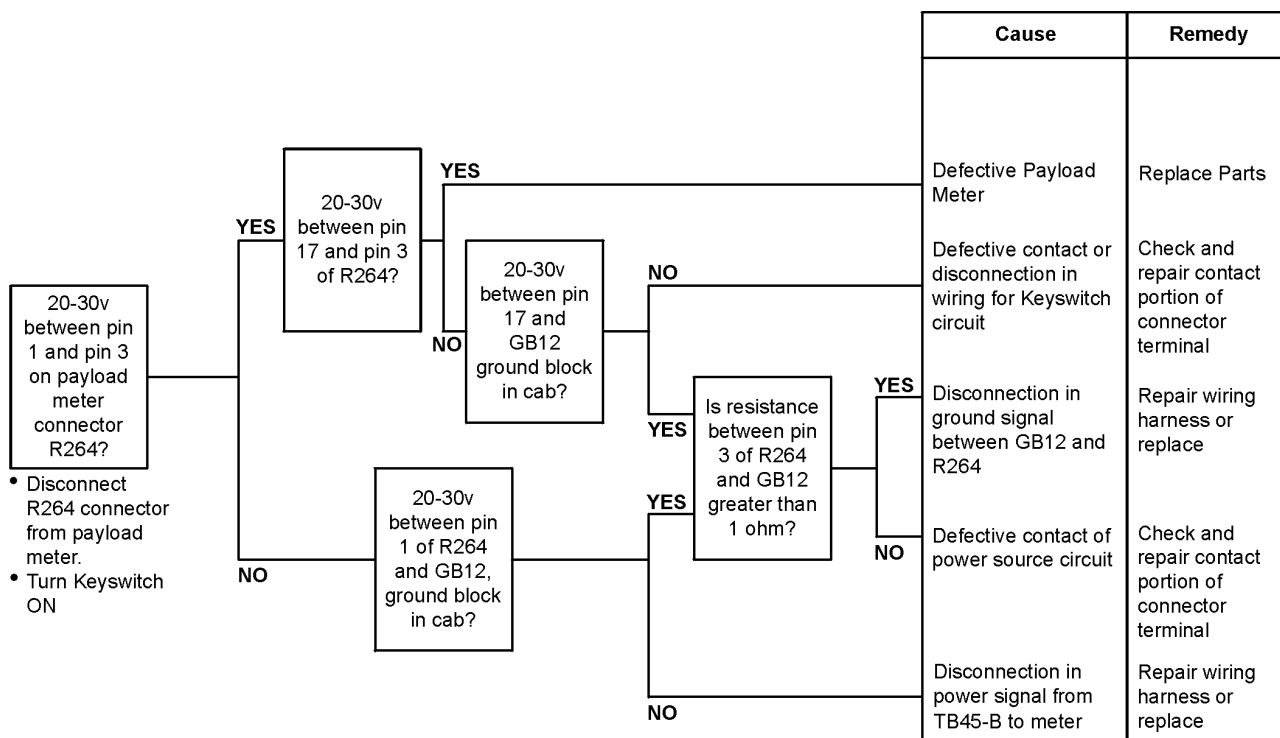


- If these errors persist after reprogramming, the primary and secondary processors then the payload meter must be replaced.



## No Payload Display When Key Switch is Turned ON

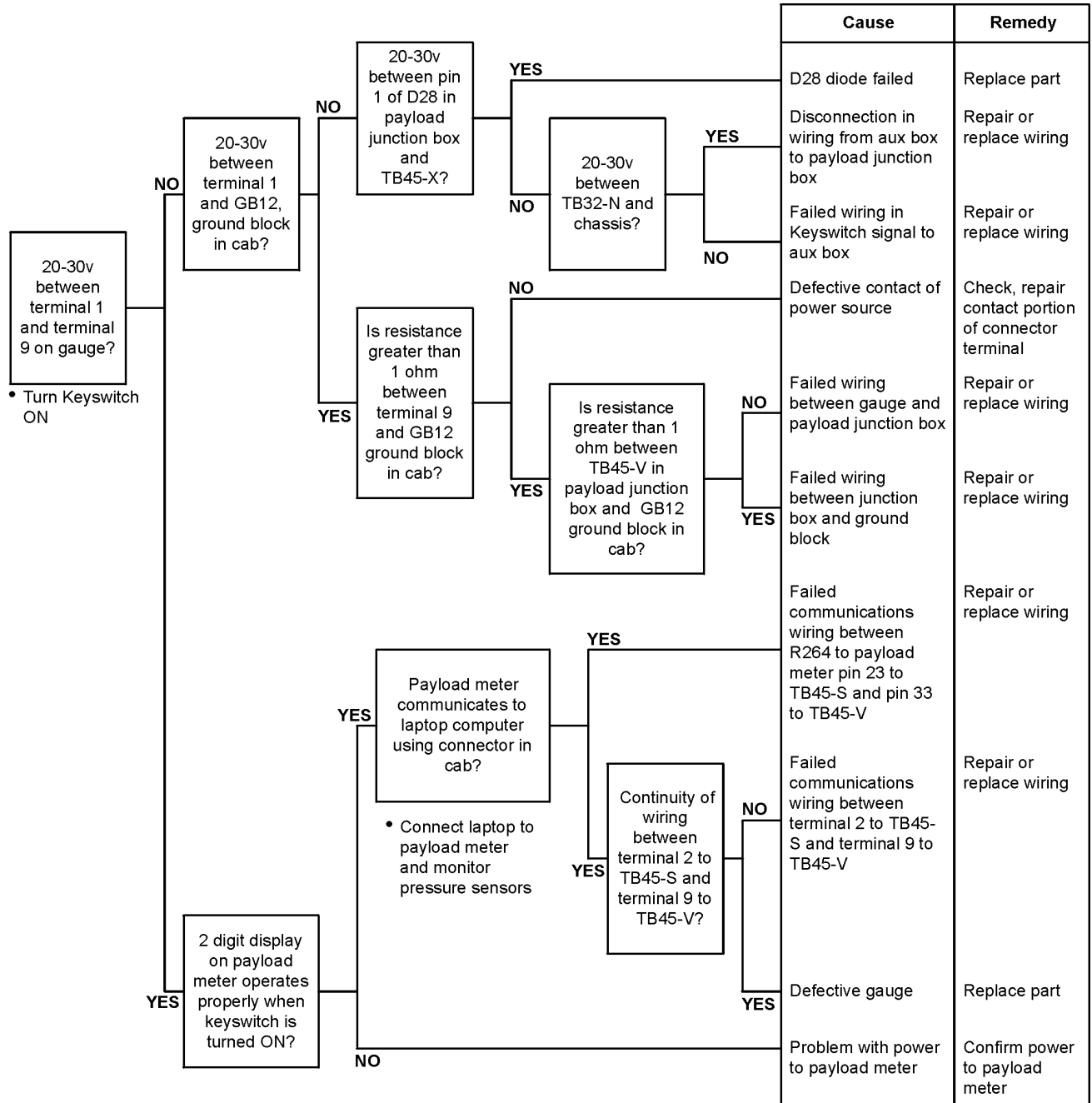
- Confirm battery voltage in PLMIII junction box between TB45-A (positive) and TB45-X (ground).
- Check the 5A circuit breaker (CB A) in PLMIII junction box.
- Check all connectors and terminal connectors in the power circuits to the payload meter.
- If two digit display on payload meter displays 00 then 88 on power up, continue to “No Display on Operator Display”. This two digit display normally alternates 0 on each display. In the case of active alarms, this display will show the code for each active alarm. The alarm codes are in the operation section.



## No Display on Speedometer

## No Display on Operator Display

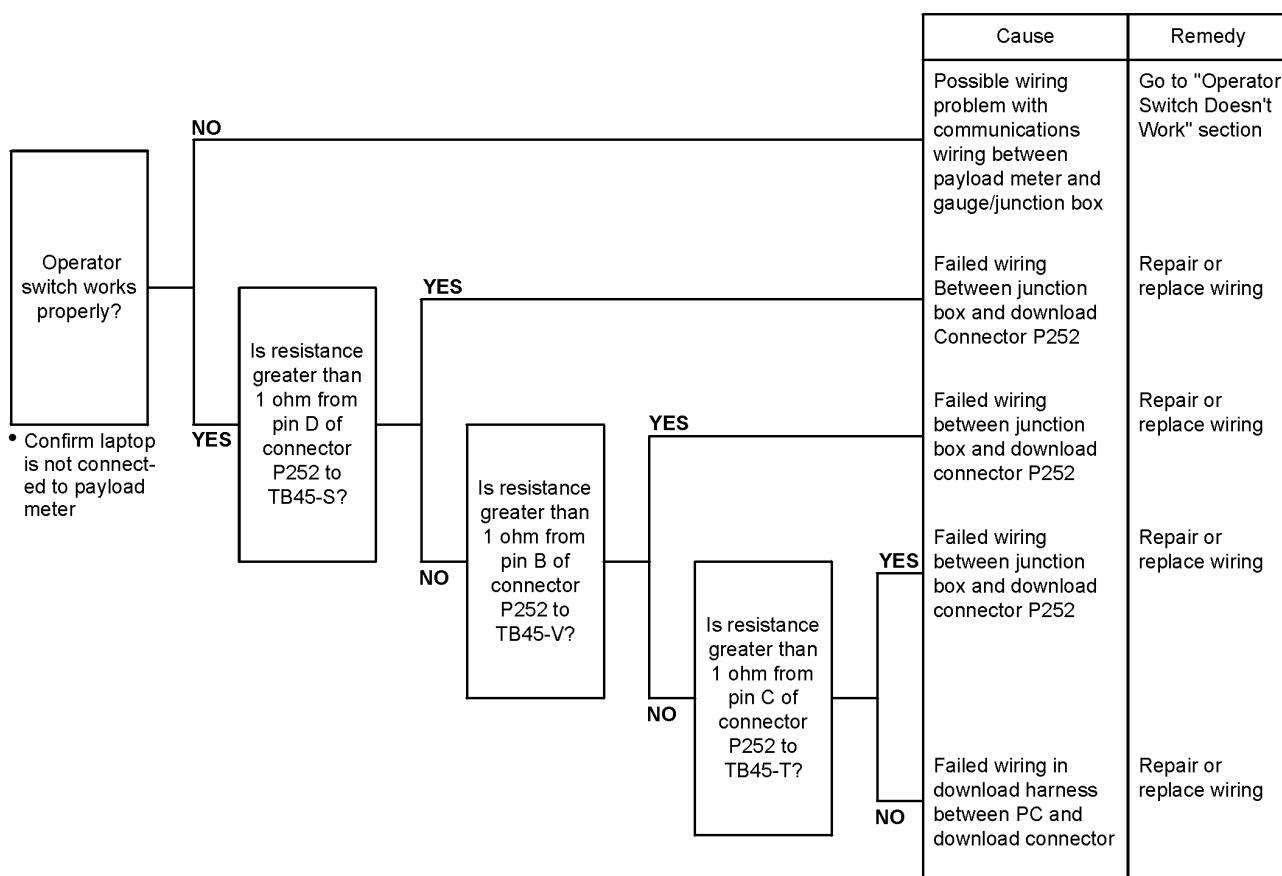
- If the speedometer works but the operator displays remain blank, confirm payload connections at “No Payload Display When Key Switch is Turned ON”.



## No Communications With PC

In a case where the laptop PC will not properly connect to the PLMIII system:

- Confirm power to the payload meter.
- Confirm laptop serial port setting using the PC software. From the main menu select "Change Program Options". Confirm that the selected serial port is correct and that it is available.
- Confirm that a Personal Digital Assistants (PDA) synchronization software is not using the serial port. Often, PDA software like Palm Pilot's HotSync software will take control of the serial port and not let other applications use the serial connection. Close the synchronization software and retry the Payload Data Manager software.
- Confirm Operator Switch use. If this switch works properly, it confirms that the communications wiring between the payload meter and the junction box is functional.
- Before the beginning of troubleshooting, turn key switch OFF. Wait 1 minute and turn key switch ON.



### **Load Lights Don't Light During Loading**

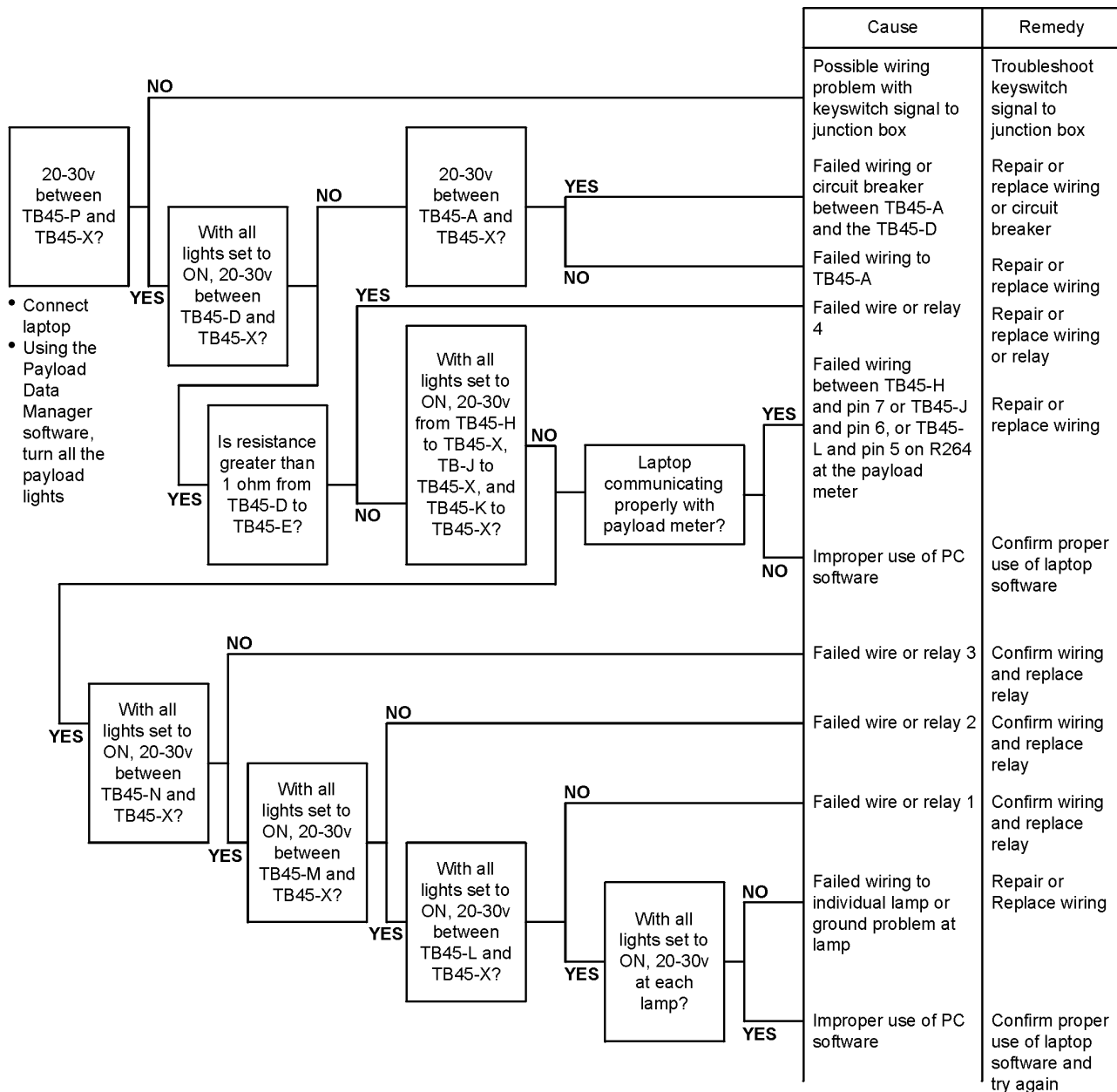
- Confirm that the truck operator uses the brake lock switch (NEUTRAL) during loading. Without this input, the payload meter will not properly recognize swingloads.
- Confirm bulbs in payload lights by using lamp check mode.
- Confirm 15A breaker CB-B in payload junction box.
- To continue troubleshooting, turn on all the payload lights using the lamp check mode of the Payload Data Manager software.

## Load Lights Remain ON

## Load Lights Remain ON During Dumping

## Display Doesn't Clear When The Load Is Dumped

- Confirm the body up switch signal. When the body up signal is not properly received during dumping, the payload meter may maintain the lights after the body is lowered.
- Confirm the payload light wiring using the procedures in "Load Lights Don't Light During Loading".



## Calibration Problems

- Confirm that the truck is empty and clean.
- Confirm that the payload meter is in the proper haul state. The payload meter must be in the empty, or tare zone states to begin calibration. This can be checked by using the real-time monitor mode of the Payload Data Manager software.
- The payload meter can be reset to acknowledge the beginning of a new haul cycle by raising the body when the truck is empty. This may be necessary after servicing the suspensions.

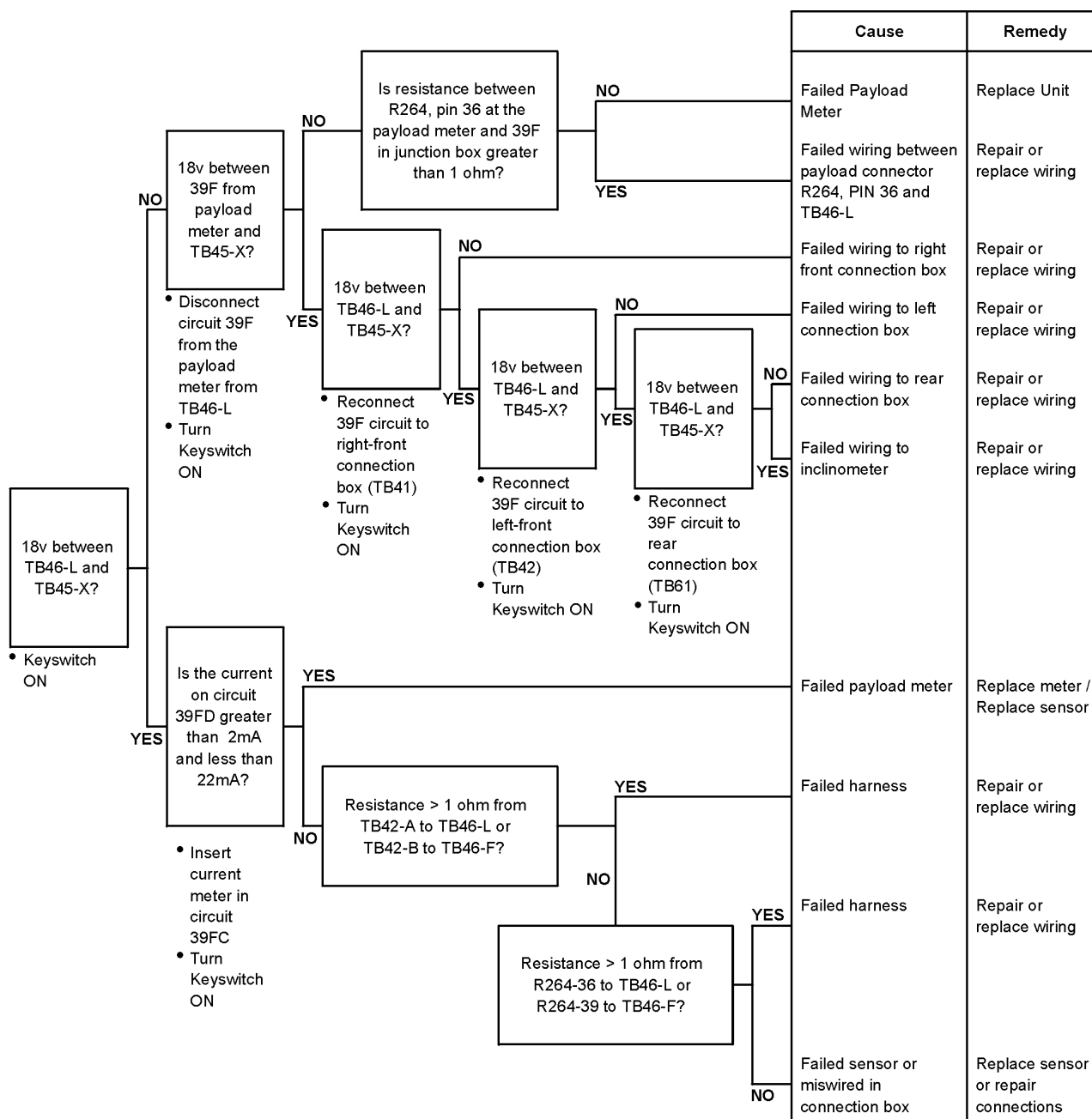
## Alarm 1 - Left Front Pressure High

## Alarm 2 - Left Front Pressure Low

### Troubleshoot Wiring to Left Front Suspension

These alarms indicate that the current being read by the payload meter is higher than 22ma or lower than 2ma. The pressure sensor is designed to output 4-20ma over a pressure range of 27 579 kPa (4000 psi).

- Confirm 18V sensor supply at TB46-L in payload junction box.
- Confirm proper connection of signal circuit 39FD from left suspension connection box, TB42-B to payload junction box TB46-F to payload meter connector R264, pin 39.



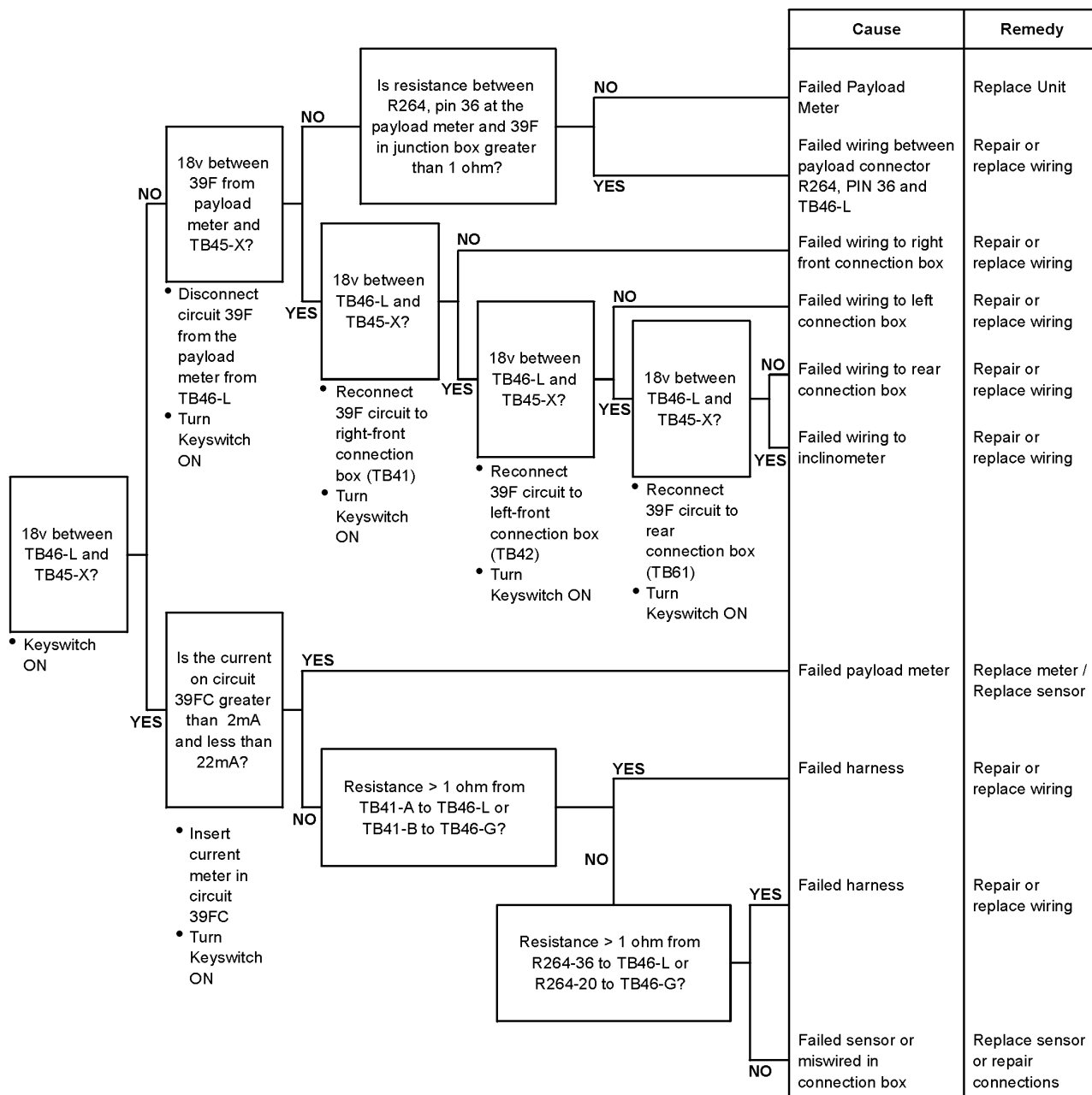
### Alarm 3 - Right Front Pressure High

### Alarm 4 - Right Front Pressure Low

### Troubleshoot Wiring to Right Front Suspension

These alarms indicate that the current being read by the payload meter is higher than 22ma or lower than 2ma. The pressure sensor is designed to output 4-20ma over a pressure range of 27 579 kPa (4000 psi).

- Confirm 18V sensor supply at TB46-L in payload junction box.
- Confirm proper connection of signal circuit 39FC from right suspension connection box, TB41-B to payload junction box TB46-G to payload meter connector R264, pin 20.





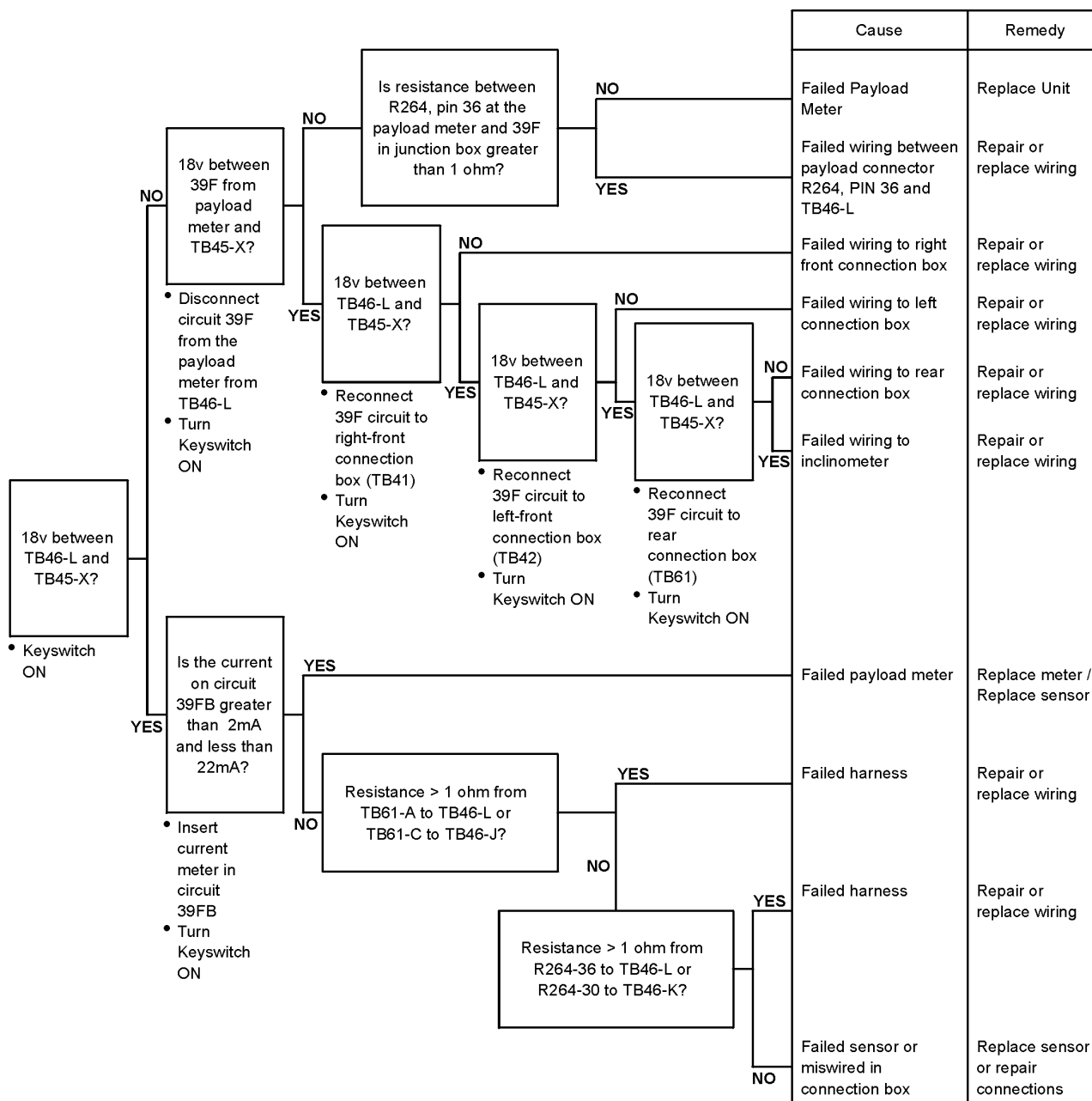
## Alarm 5 -Left Rear Pressure High

## Alarm 6 - Left Rear Pressure Low

### Troubleshoot Wiring to Left Rear Suspension

These alarms indicate that the current being read by the payload meter is higher than 22ma or lower than 2ma. The pressure sensor is designed to output 4-20ma over a pressure range of 27 579 kPa (4000 psi).

- Confirm 18V sensor supply at TB46-L in payload junction box.
- Confirm proper connection of signal circuit 39FC from right suspension connection box, TB41-B to payload junction box TB46-G to payload meter connector R264, pin 20.



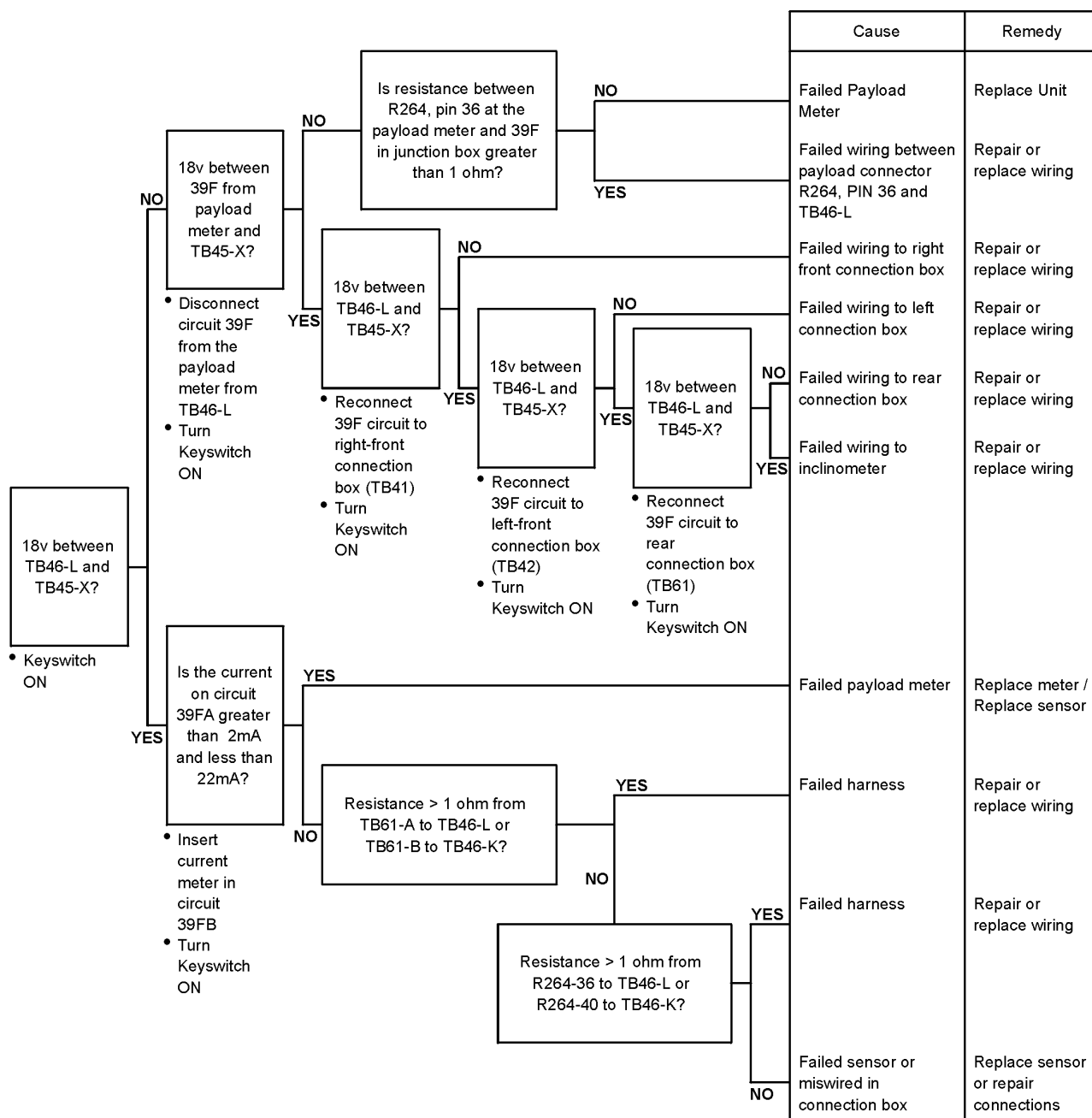
## Alarm 7 - Right Rear Pressure High

## Alarm 8 - Right Rear Pressure Low

### Troubleshoot Wiring to Right Rear Suspension

These alarms indicate that the current being read by the payload meter is higher than 22ma or lower than 2ma. The pressure sensor is designed to output 4-20ma over a pressure range of 27 579 kPa (4000 psi).

- Confirm 18V sensor supply at TB46-L in payload junction box.
- Confirm proper connection of signal circuit 39FC from right suspension connection box, TB41-B to payload junction box TB46-G to payload meter connector R264, pin 20.

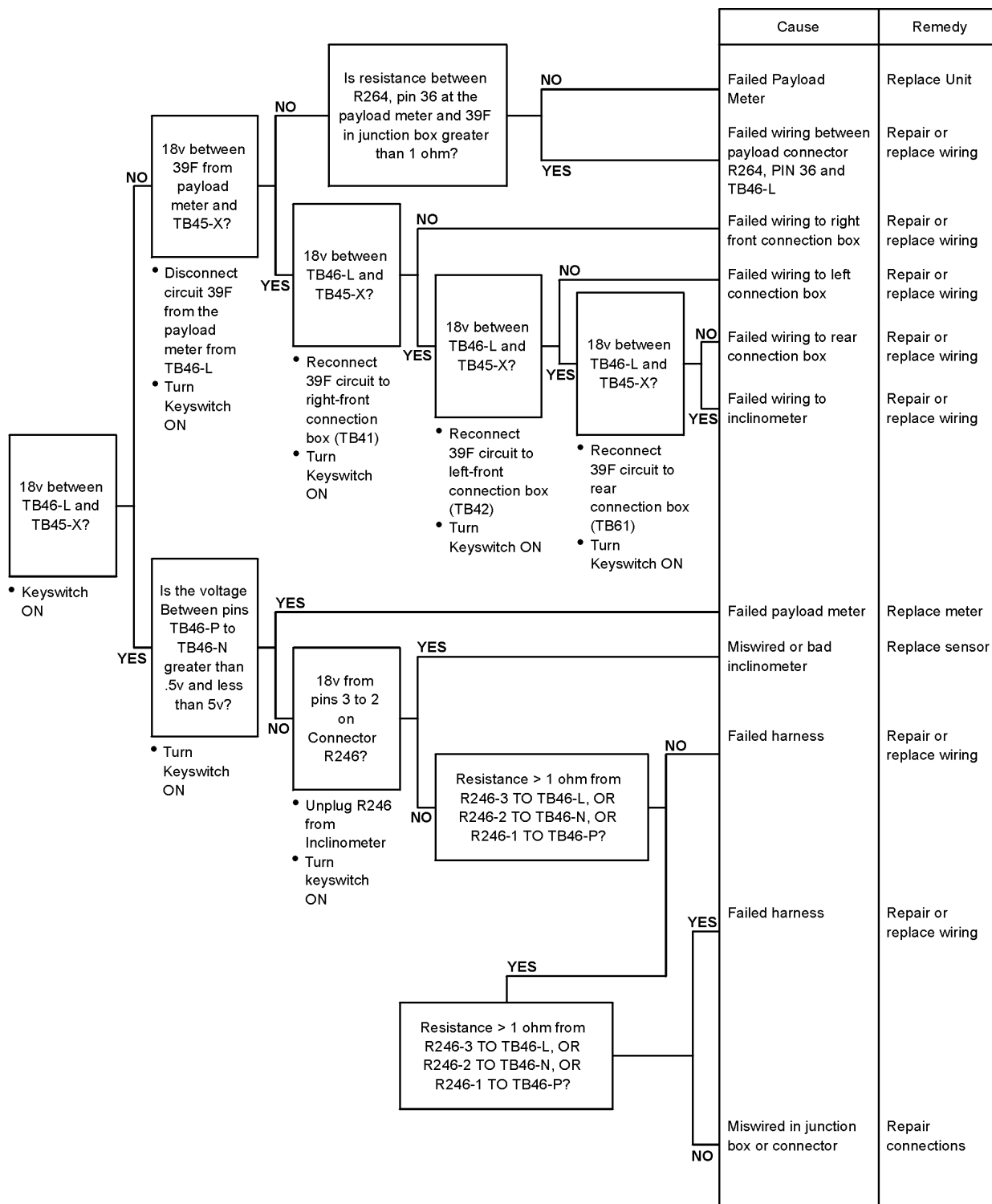


## Alarm 9 - Inclinator High

## Alarm 10 - Inclinator Low

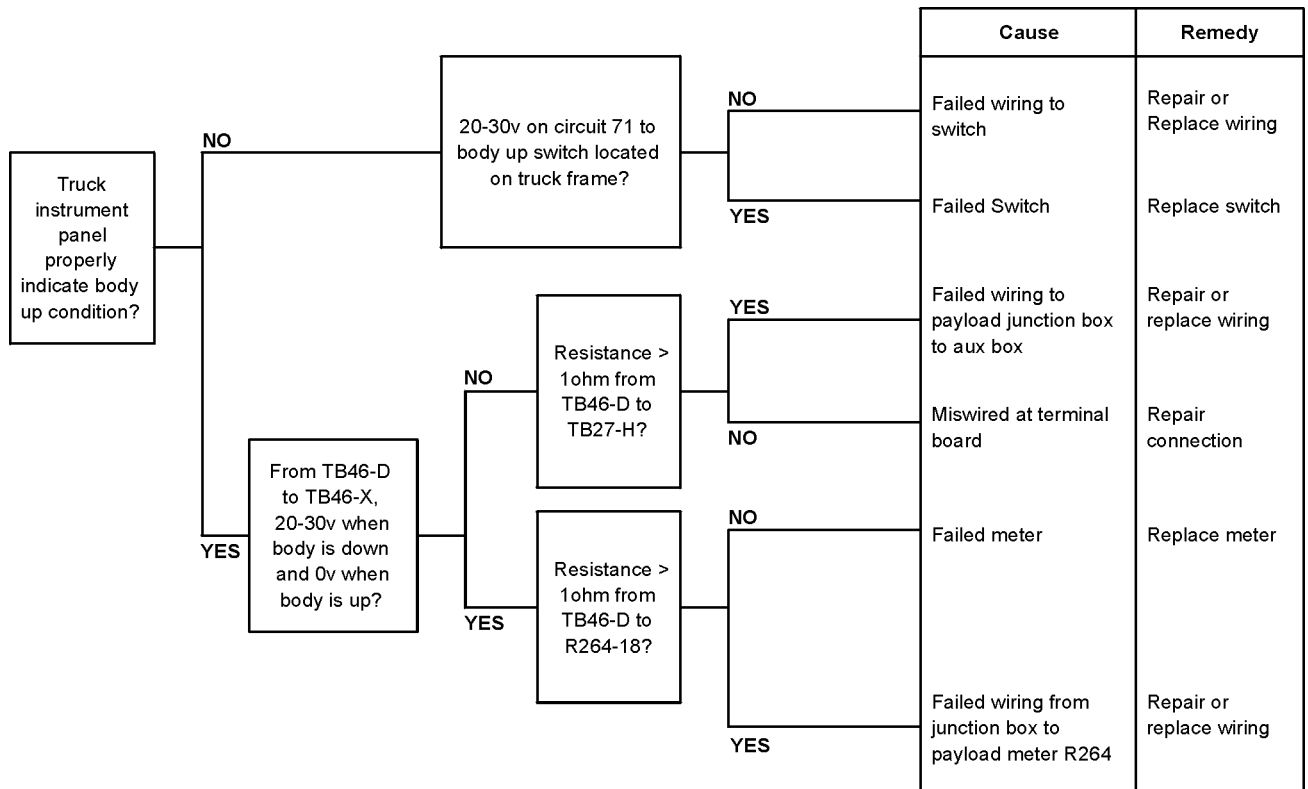
### Troubleshoot Inclinator Wiring

These alarms indicate that the voltage to the payload meter from the inclinometer is out of range. The voltage on signal 39FE should be greater than 0.5V and less than 5.0V as measured in the junction box between TB46-.



## Alarm 13 - Body Up Input Failure

The payload meter senses when the load is dumped without receiving a body-up signal. When the load quickly drops below 50% without the body up signal, Alarm 13 is set. The alarm will be cleared when a normal dump cycle is detected. A normal dump cycle will be detected when the body up signal is received, the load drops quickly and the body down signal is received.



### **Alarm 16 - Memory Write Failure**

### **Alarm 17 - Memory Read Failure**

These alarms indicate that the payload meter has encountered a problem internally with its memory. It is recommended that power to the payload meter be removed for 1 minute. First turn the key switch OFF. Wait 30 seconds, then turn the battery disconnect OFF. Wait 1 minute before restoring power.

In cases where re-powering the payload meter does not restore normal operation, it may necessary to reprogram the payload meter. All current data in memory will be lost. This will effectively restart the payload meter.

- See “Troubleshooting Abnormal Displays at Power-Up” for more information.

**Alarm 18**

Payload meter detected an undercharged suspension condition on the rear right suspension.

The suspension may be in need of servicing. Refer to Section H in the shop manual for information on charging the suspensions.

**Alarm 19**

Payload meter detected an undercharged suspension condition on the rear left suspension.

The suspension may be in need of servicing. Refer to Section H in the shop manual for information on charging the suspensions.

**Alarm 22**

The payload meter detected an empty carryback load in excess of the user-defined carryback threshold on two consecutive haul cycles.

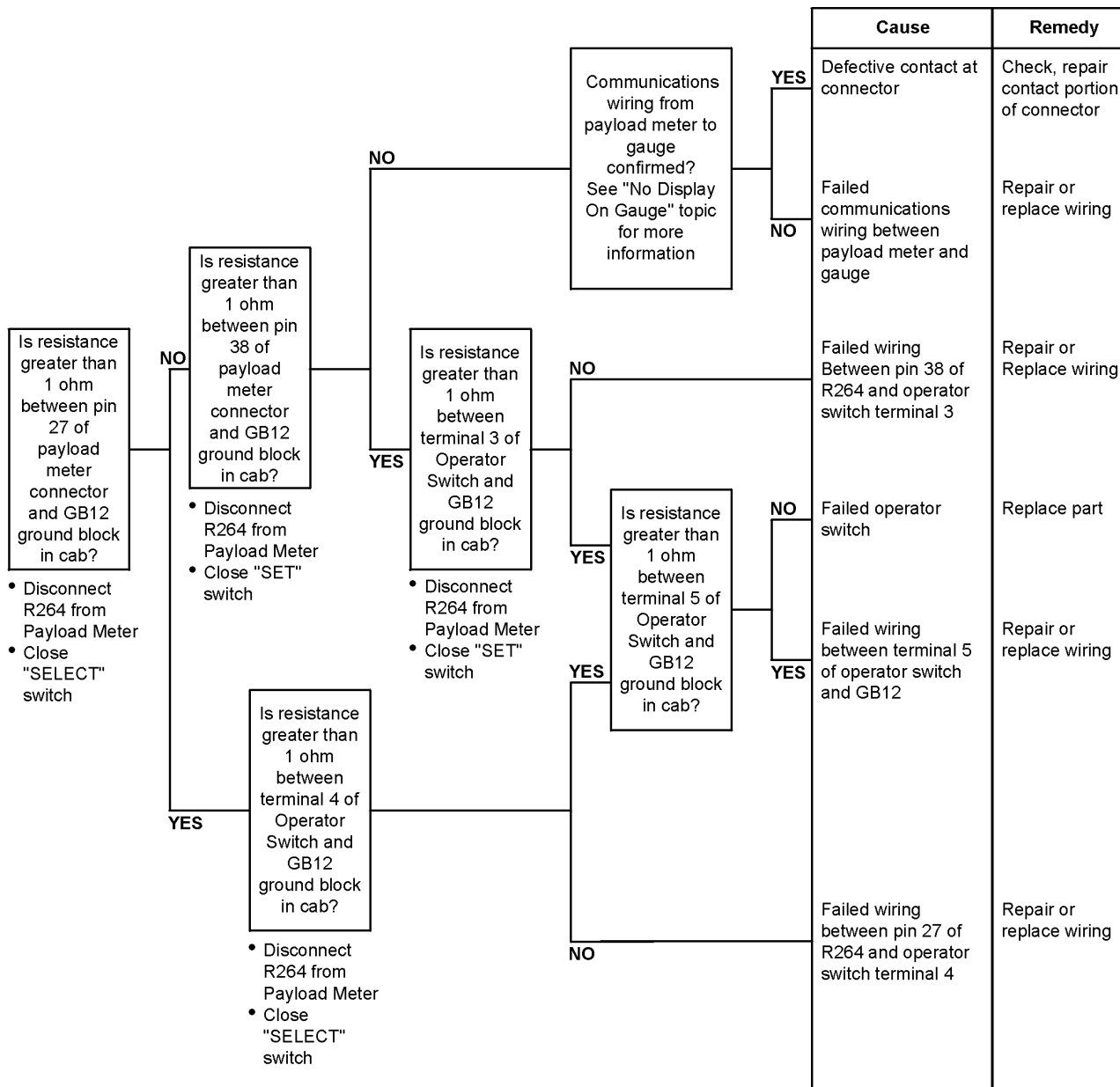
Stop the truck and clean any stuck material from the truck body.

## Operator Switch Doesn't Work

### Alarm 26 - User Switch Fault - SELECT

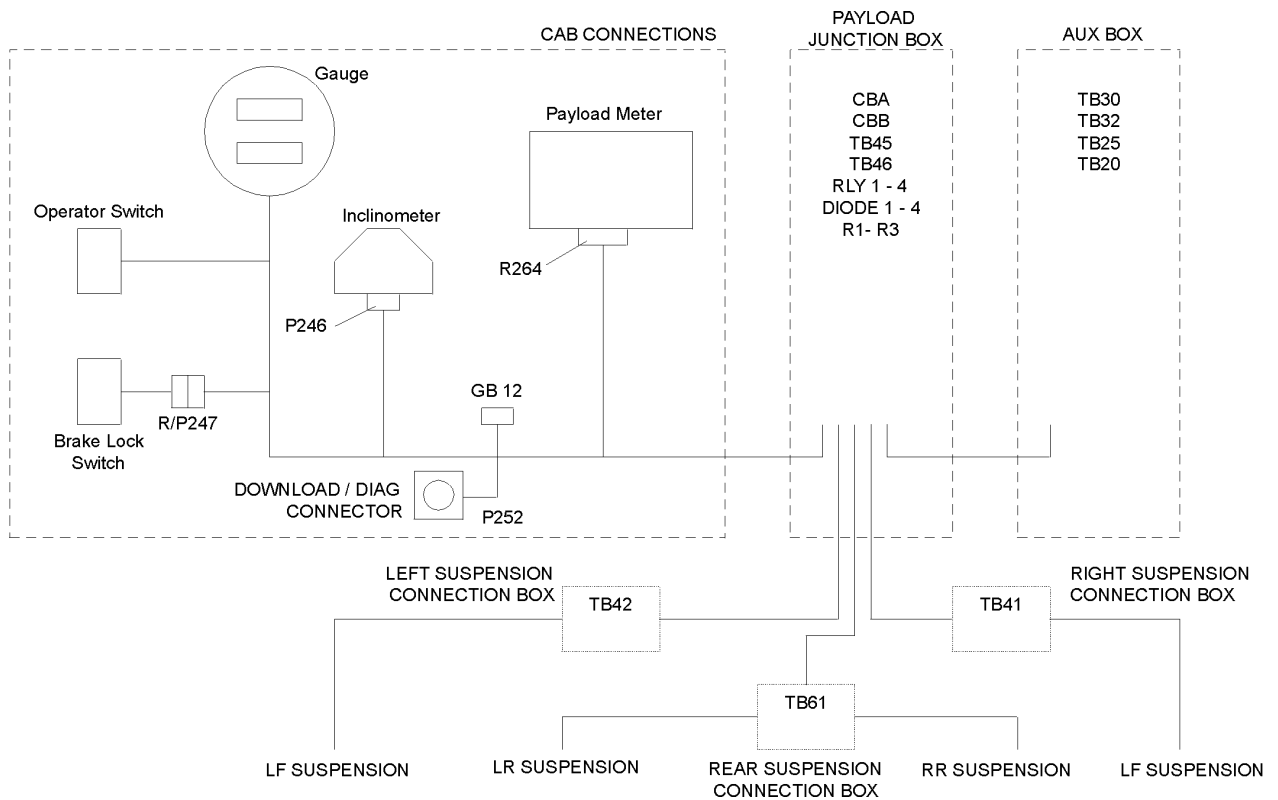
### Alarm 27 - User Switch Fault - SET

- Confirm power to the payload meter speedometer and display gauge.
- Confirm that a laptop is not connected to the PLMIII system.
- Turn key switch OFF. Wait 1 minute and turn key switch ON. Confirm problem still exists.



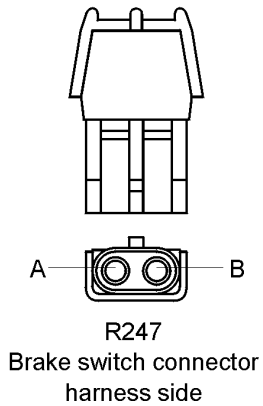
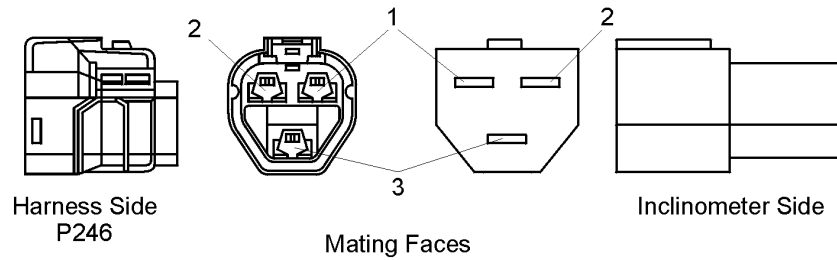
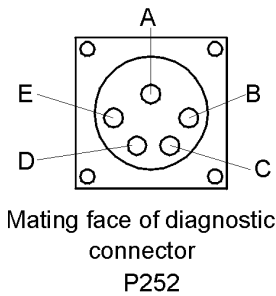
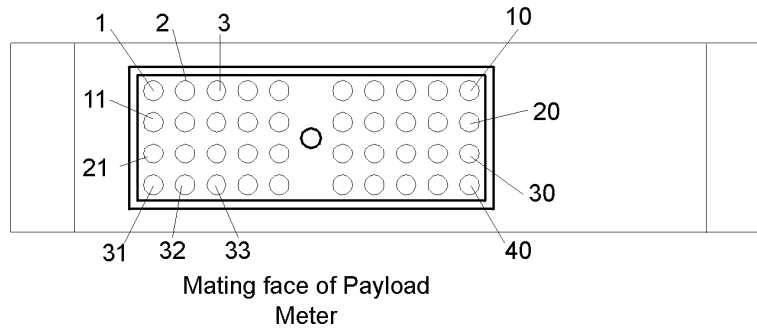
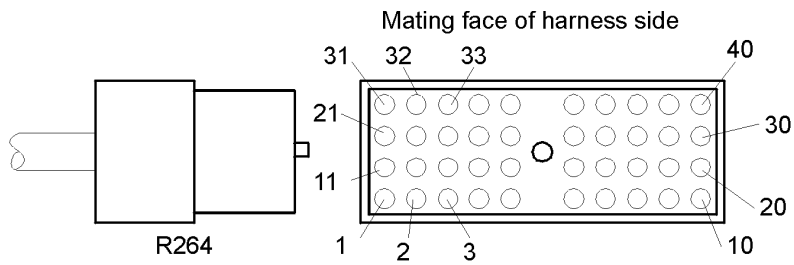
## Connector Map

This diagram shows the general location of connectors, terminal boards and miscellaneous connections.





## Connectors



# PLMIII CHECK OUT PROCEDURE

## General Description

The process consists of attaching dummy loads in place of the suspension pressure sensors and checking the pressures indicated by the payload meter. In addition, connecting to the payload meter using a laptop PC in order to confirm the latest software version and the rest of the inputs and outputs of the system.

## Tools Required

- Payload Data Manager software
- EF9160 or PC2525 - Download Harness
- EJ3057 - Harness Str, PLMIII test (4 needed).

## Checkout Procedure

1. Attach one EJ3057 harness to the left-front suspension connection box. The red alligator clip attaches to the 39F circuit at TB42-A. The white alligator clip attaches to the 39FD circuit at TB42-B. The EJ3057 acts as a dummy load to simulate a suspension pressure sensor for the payload system.
2. Attach one EJ3057 harness to the right-front suspension connection box. The red alligator clip attaches to the 39F circuit at TB41-A. The white alligator clip attaches to the 39FC circuit at TB41-B.
3. Attach one EJ3057 harness to the left-rear suspension connection in the rear suspension connection box. The red alligator clip attaches to the 39F circuit at TB61-A. The white alligator clip attaches to the 39FB circuit at TB61-C.
4. Attach one EJ3057 harness to the right-rear suspension connection in the rear suspension connection box. The red alligator clip attaches to the 39F circuit at TB61-A. The white alligator clip attaches to the 39FA circuit at TB61-B.
5. In the PLMIII junction box, check the input voltage on circuit 39G between TB45-B and TB45-X. This voltage should be 24VDC from the batteries.
6. Turn the key switch ON. The speedometer/display gauge on the dashboard will scroll the truck type across the lower display. The payload meter defaults to 930E.
7. In the PLMIII junction box, check the sensor supply voltage on circuit 39F between TB46-L and TB45-X. This voltage should be 18VDC  $\pm$ 1VDC.

8. Return to the cab and check the speedometer/display gauge. The gauge will display the current payload. With the EJ3057 harnesses attached at the sensor locations, the payload should be 0.

*NOTE: The display can be used to quickly show the current readings from the four suspension pressure sensors and the inclinometer. This can be used during regularly scheduled service periods to check the state of the suspensions. These displays are live and will update as the values change. The display is changed by pressing the "SELECT" button on the dashboard. The sequence of displays is:*

- $P L$  = Payload
- $I d$  = Operator ID
- $T L$  = Total Shift Tons
- $L C$  = Shift Load Counter
- $L F$  = Left Front Suspension Pressure
- $r F$  = Right Front Suspension Pressure
- $L r$  = Left Rear Suspension Pressure
- $r r$  = Right Rear Suspension Pressure
- $I n$  = Inclinometer

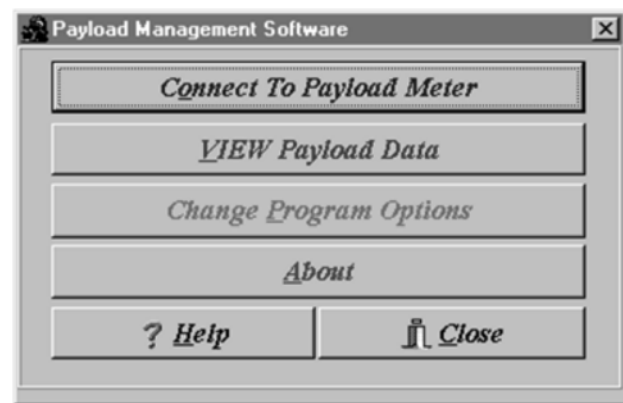
*NOTE: The live displays cannot be cleared and the SET button will have no effect.*

*NOTE: The units for the display are controlled by the configuration of the payload meter. The payload meter defaults to display metric units, the pressures will be displayed in tenths of  $kg/cm^2$ . For example, if the display shows  $20.2$  the actual value is  $20.2 kg/cm^2$ . If the payload meter is set to display short tons, the pressures will be displayed in psi ( $lbs/in^2$ ). To convert from  $kg/cm^2$  to psi, multiply by 14.2233.  $14.2233 psi (lbs/in^2) = 1 kg/cm^2$ .*

*NOTE: The inclinometer displays whole degrees of incline. Positive incline is when front of truck is pointing up.*

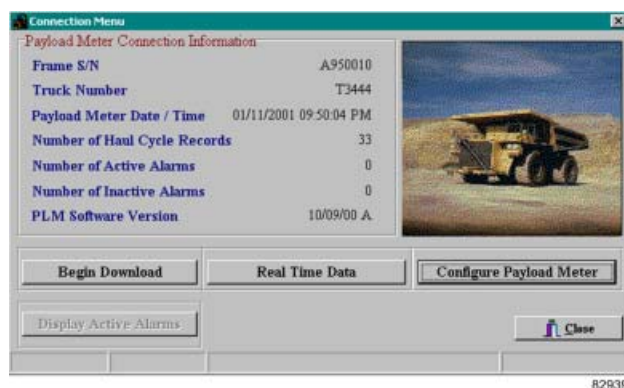
*NOTE: The gauge will quickly display the type of information being displayed every 1 minute. For example, if the left-front pressure is being displayed,  $L F$  will quickly display every minute. Only the payload display,  $P L$  does not display this information.*

9. Press and hold the "SELECT" button on the dashboard.  $Id=$  will be displayed. Release the button and the Operator ID will be displayed. This value should be 0.
10. Press and hold the "SELECT" button on the dashboard.  $L=$  will be displayed. Release the button and the total tons will be displayed. This value should be 0.
11. Press and hold the "SELECT" button on the dashboard.  $Lc=$  will be displayed. Release the button and the number of loads will be displayed. This value should be 0.
12. Press and hold the "SELECT" button on the dashboard.  $LF=$  will be displayed. Release the button and the left-front pressure will be displayed. This value should be displayed in metric units. Refer to Section H4, Suspension Oiling and Charging Procedures, for the nominal charging pressure. The acceptable pressure range varies according to truck model.
13. Press and hold the "SELECT" button on the dashboard.  $rF=$  will be displayed. Release the button and the right-front pressure will be displayed. This value should be displayed in metric units. Refer to Section H4, Suspension Oiling and Charging Procedures, for the nominal charging pressure. The acceptable pressure range varies according to truck model.
14. Press and hold the "SELECT" button on the dashboard.  $Lr=$  will be displayed. Release the button and the left-rear pressure will be displayed. This value should be displayed in metric units. Refer to Section H4, Suspension Oiling and Charging Procedures, for the nominal charging pressure. The acceptable pressure range varies according to truck model.
15. Press and hold the "SELECT" button on the dashboard.  $rr=$  will be displayed. Release the button and the right-rear pressure will be displayed. This value should be displayed in metric units. Refer to Section H4, Suspension Oiling and Charging Procedures, for the nominal charging pressure. The acceptable pressure range varies according to truck model.
16. Press and hold the "SELECT" button on the dashboard.  $In=$  will be displayed. Release the button and the inclinometer value will be displayed. This value is in degrees. The incline will depend on how the truck is set during assembly. Values between  $\pm 3^\circ$  are acceptable. It is not necessary to zero this reading by adjusting the attitude of the inclinometer in the buddy seat.
17. Press and hold the "SELECT" button on the dashboard.  $PL=$  will be displayed. Release the button and the current payload will be displayed.
18. Connect a laptop to the PLMIII system. Typically an EF9160 or PC2525 download cable is used. Refer to Section D11 for the location of the payload meter connector. The laptop must have the Payload Data Manager software installed.
19. Run the PC software.
20. From the main menu, select "Connect to Payload Meter".



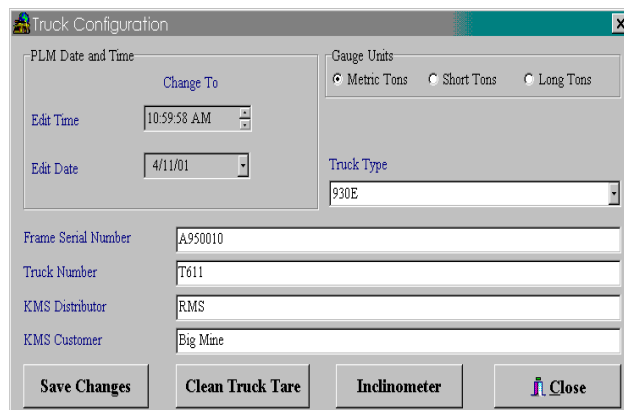
82941

21. The Connection Menu will be displayed. Select "Configure Payload Meter".



22. Confirm that the PLMIII software version matches the latest available version. As of 26-June-06 the EJ0575-5 software version will display as "EJ0575-5". The latest version can be found at the Komatsu Extranet web site. If the version does not match the latest indicated on the internet, download the latest and update the PLMIII software using the Flashburn software. See Checkout Procedure Confirmation for more information.

23. Using the Truck Configuration menu, set the following:



- Set the time.
- Set the Date to today's date.
- Set the Gauge display units to Metric, Short Tons or Long Tons according to the final destination of the vehicle. If nothing has been specified, set to Metric Tons.
- Set the truck type to the proper truck model.
- Press the "Save Changes" button to program the change into the payload meter.

24. Setting the Frame Serial Number.

|                     |          |
|---------------------|----------|
| Frame Serial Number | A950010  |
| Truck Number        | T3444    |
| KMS Distributor     | RMS      |
| KMS Customer        | Pit Mine |

**NOTE:** The frame serial number is located on a plate mounted to the truck frame. The plate is outboard on the lower right rail facing the right front tire. It is very important to enter the correct frame serial number. This number is one of the key fields used within the haul cycle database. The field will hold 20 alpha-numeric characters.

- On the Truck Configuration screen, enter the frame serial number in the appropriate field.
- Press the "Save Changes" button to program the change into the payload meter.

25. Setting the Customer Unit Number.

**NOTE:** Most mining operations assign a number to each piece of equipment for quick identification. This number or name can be entered in the Customer Unit Number field. It is very important to enter customer unit number. This number is one of the key fields used within the haul cycle database. The field will hold 20 alpha-numeric characters. If no truck number has been specified, enter the frame serial number.

- On the Truck Configuration screen, enter the truck number in the appropriate field.
- Press the "Save Changes" button to program the change into the payload meter.

26. Setting the Komatsu Distributor.

**NOTE:** This field in the haul cycle record can hold the name of the Komatsu distributor that helped install the system. Komatsu also assigns a distributor number to each distributor. This number is used on all warranty claims. This Komatsu distributor number can also be put into this field. This number is one of the key fields used within the haul cycle database. The field will hold 20 alpha-numeric characters. If the distributor is not known, enter "UNKNOWN".

- On the Truck Configuration screen, enter the distributor name or number in the appropriate field.
- Press the "Save Changes" button to program the change into the payload meter.

## 27. Setting the Komatsu Customer.

*NOTE: This field in the haul cycle record can hold the name of the mine or operation where the truck is in service. Komatsu also assigns a customer number to each customer. This number is used on all warranty claims. This Komatsu customer number can also be put into this field. This number is one of the key fields used within the haul cycle database. The field will hold 20 alpha-numeric characters. If the customer is not known, enter "UNKNOWN"*

- On the Truck Configuration screen, enter the customer name or number in the appropriate field.
- Press the "Save Changes" button to program the change into the payload meter.

28. Press "Save Changes" and close the Truck Configuration screen and the Connection Menu.

29. From the main menu select "Connect to Payload Meter".

30. From the Connection Menu select "Configure Payload Meter". Confirm that all previous changes have been saved and close the Truck Configuration form.

31. From the Connection Menu select "Real Time Data".

| Real Time Data                                                                                               |      |                          |           |
|--------------------------------------------------------------------------------------------------------------|------|--------------------------|-----------|
| <b>Pressure PSI</b>                                                                                          |      | <b>Truck Status</b>      |           |
| Left Front                                                                                                   | 304  | Inclinometer             | 0.5       |
| Right Front                                                                                                  | 287  | Body Up                  | No        |
| Left Rear                                                                                                    | 299  | Brake Lock               | Off       |
| Right Rear                                                                                                   | 293  | Haul Cycle State         | Tare Zone |
| <b>Speed MPH</b>                                                                                             |      | <b>Weight Short Tons</b> |           |
| Truck                                                                                                        | 20.0 | Sprung Weight            | 107       |
| <b>Lights</b>                                                                                                |      |                          |           |
| <input type="checkbox"/> Green Light <input type="checkbox"/> Amber Light <input type="checkbox"/> Red Light |      |                          |           |
| Set FileName                                                                                                 |      | Start Log                | Stop Log  |
| Set Lights                                                                                                   |      | Close                    |           |

*NOTE: The weight shown on the real time data screen is the sprung weight and includes the weight of the truck. Given the suspension pressure dummy loads, the nominal value shown should be 101 metric tons (112 short tons).*

32. Confirm that the suspension pressures are within range. The nominal value should be 23.4 kg/cm<sup>2</sup> (332 psi). Values between 17.6 and 29.2 kg/cm<sup>2</sup> (250 and 416 psi) are acceptable. Record the values displayed.

33. Confirm that the inclinometer is within range and record the value.

34. Confirm that the body up input is working correctly. Place a steel washer on the body up switch. The real time data screen should indicate "No". Remove the washer and the real time data screen should indicate "Yes". The Haul Cycle State should change to "Dumping".

35. Confirm that the brake lock input is working correctly. Turn the brake lock on using the switch on the dashboard. The real time data screen should indicate ON. Turn the brake lock off. The real time data screen should indicate OFF.

36. Turn on the green payload lights by checking "Green Light" and pressing the "Set Lights" button. Check to be sure that only the green payload lights on the truck are illuminated.

37. Uncheck the green light and turn on the amber payload lights by checking "Amber Light" and pressing the "Set Lights" button. Check to be sure that only the amber payload lights are illuminated.

38. Uncheck the amber light and turn on the red payload lights by checking "Red Light" and pressing the "Set Lights" button. Check to be sure that only the red payload lights are illuminated.

39. Uncheck all the payload lights and press the "Set Lights" button. Confirm that all the lights are off.

40. Use the procedure for speedometer calibration for the particular truck type to simulate a 40.2 km/h (25 mph) speed signal. Confirm that this value is displayed by the speedometer on the dashboard and the real time data screen. The value can be  $\pm 2$  km/h ( $\pm 1$  mph). The brake lock must be off for the PLMIII to recognize speed input.

41. On the PC, close the Real Time Screen and the Connection Menu and return to the Main Menu.

42. Remove the EJ3057 harness from the left front suspension junction box, TB42-A and TB42-B.
43. Wait at least 1 minute, then remove the EJ3057 harness from the left rear connections in the rear junction box, TB61-B and TB61-C.
44. Wait at least 1 minute, then remove the EJ3057 harness from the right-rear connections in the rear junction box, TB61-A and TB61-C.
45. Wait at least 1 minute, then remove the EJ3057 harness from the right front connections in the right-front junction box, TB61-B and TB61-C.
46. Wait at least 1 minute.
47. From the main menu of the PC software, press the "Connect to Payload Meter" button.
48. From the Connection Menu select "Display Active Alarms". Confirm that the four alarms displayed occurred in the proper order:
  - Left-front suspension low
  - Left-rear suspension low
  - Right-rear suspension low
  - Right-front suspension low
49. Close all screens and disconnect the laptop from the PLMIII system.

## PLMIII CHECKOUT PROCEDURE CONFIRMATION

### Flashburn Programming

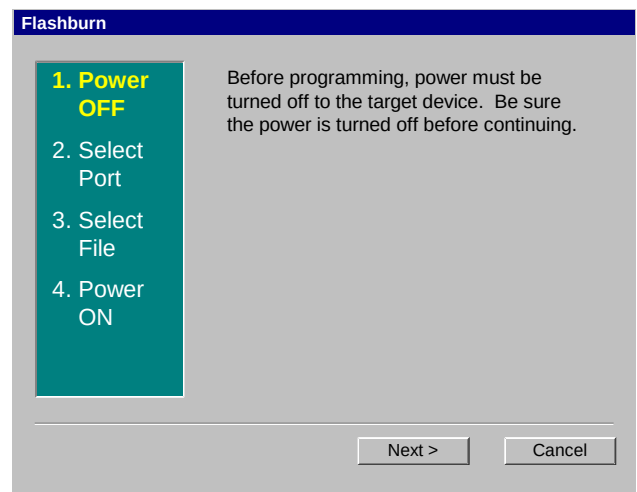
Before beginning, make sure that you have the .kms file required to program the product and you know where to find it on your computer.

Programming will reset all the truck configuration information.

*NOTE: Before starting this procedure, record the Payload Meter configuration information.*

This information can be found using the Payload Data Manager software. After programming, it will be necessary to restore this information in the payload meter configuration.

1. Turn off power to the payload meter by turning the key switch OFF.
2. Start the Flashburn software installed on the laptop.



3. Confirm that the payload meter power is OFF and press "NEXT".

4. Confirm the proper communications port for the programming laptop. This is usually COM 1. Press "NEXT".

**Flashburn**

1. Power OFF  
2. **Select Port**  
3. Select File  
4. Power ON

Select the serial communications port to use between the computer and the target device. For most computers this will be COM 1.

COM 1

< Back   Next >   Cancel

5. Press "BROWSE" and select the ".kms" file to program into the payload meter. Press "NEXT".

**Flashburn**

1. Power OFF  
2. Select Port  
3. **Select File**  
4. Power ON

Select the file that will be used to program the target device. This file will end with the ".KMS" extension.

Comm Port: COM 1

Filename: \*.KMS   Browse

< Back   Next >   Cancel

6. When instructed, turn the key switch ON in order to power-up the payload meter. The PC will begin to reprogram the payload meter. This process takes approximately 5 minutes.

**Flashburn**

1. Power OFF  
2. Select Port  
3. Select File  
4. **Power ON**

Turn on power to the target device. This will start the programming process.

Comm Port: COM 1

Filename: 071000A.KMS

| Steps         | Status   |
|---------------|----------|
| Connection:   | Complete |
| Preparation:  | Complete |
| Programming:  | Complete |
| Verification: | Complete |

61 %

< Back   Cancel

7. After successful programming, turn the key witch OFF.
8. Wait 20 seconds and turn the key switch ON.
9. The payload meter will need to be configured as instructed in the manual using the Payload Data Manager software on the laptop computer.

**Confirmation Checklist**

Use the Real Time Data Screen in order to verify the checklist items in the table below.

| Checklist Item                         | Value | Initials |
|----------------------------------------|-------|----------|
| PLMIII Software Version                |       |          |
| User switch and display works properly |       |          |
| Left - Front Pressure                  |       |          |
| Right - Front Pressure                 |       |          |
| Left - Rear Pressure                   |       |          |
| Right - Rear Pressure                  |       |          |
| Inclinometer                           |       |          |
| Green light works properly             |       |          |
| Amber light works properly             |       |          |
| Red light works properly               |       |          |
| Brake Lock input works properly        |       |          |
| Body Up input works properly           |       |          |
| Speed input works properly             |       |          |
|                                        |       |          |
| Date                                   |       |          |
| Truck                                  |       |          |
| Signature                              |       |          |



# NOTES

PORTIONS OF THIS PRODUCT RELATING TO PAYLOAD MEASURING  
SYSTEMS ARE MANUFACTURED UNDER LICENSE FROM  
L.G. HAGENBUCH, holder of U.S. Patent Numbers  
5,416,706; 5,528,499; 5,631,832; 5,631,835; 5,644,489; 5,650,928; 5,650,930; 5,742,914

# **SECTION M31**

## **RESERVE ENGINE OIL SYSTEM**

### **INDEX**

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# NOTES

## RESERVE ENGINE OIL SYSTEM

The reserve engine oil system is designed to add more oil capacity to the engine sump and to reduce frequent servicing of the engine oil. The constant circulation of oil between the engine sump and reserve tank (9, Figure 31-1) increases the total volume of working oil. This dilutes the effects of contamination and loss of additives and maintains the oil quality over longer periods. The system adds or removes oil from the engine as required to maintain a constant level which prevents over fills or under fills.

The normal reserve system capacity of oil carried in the tank can be one half or nearly the same (depending on tank size) volume inside the engine. In the process of continuous adjustment of the engine oil level, there is a constant circulation of oil between the engine and the reserve tank. The volume of oil in the tank becomes part of the working oil for the engine. Oil change intervals may usually be extended in proportion to the increased working oil volume. Extension beyond a proportional increase is often possible, but should be undertaken only as determined by oil sampling and analysis. Local conditions such as engine application, climate, and fuel quality should be taken into consideration before determining permissible oil life.

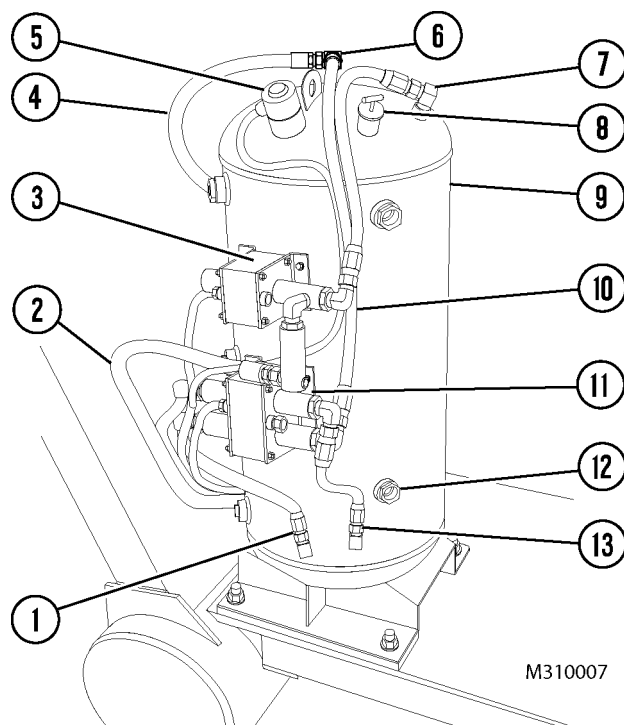


FIGURE 31-1. OIL RESERVE TANK

- |                     |                     |
|---------------------|---------------------|
| 1. Suction Line     | 8. Tank Fill Cap    |
| 2. Remote Fill Line | 9. Reserve Tank     |
| 3. Fill Valve       | 10. Engine Add Line |
| 4. Engine Add Line  | 11. Pumping Unit    |
| 5. Fill Switch      | 12. Sight Gauge     |
| 6. Air Relief Valve | 13. Tank Add Line   |
| 7. Tank Fill Line   |                     |

## Operation

Engine oil is circulated between engine oil pan (1, Figure 31-2) and reserve tank (2) by two electrically driven pumps (pump 1 and pump 2) within a single pumping unit (4). The pumping unit is mounted on the side of the reserve tank. The pump unit is equipped with an LED monitor light on one side.

Pump 1 draws oil from the engine oil pan (1) at a pre-set control point determined by the height of the suction tube (3). Oil above this point is withdrawn and transferred to the reserve tank. This lowers the level in the engine oil pan until air is drawn.

Air reaching the pumping unit activates pump 2, which returns oil from reserve tank (2) and raises the engine oil level until air is no longer drawn by pump 1. Pump 2 then turns off. The running level is continuously adjusted at the control point by alternating between withdrawal and return of oil at the engine oil pan. The oil returning to the engine oil pan is below the normal operating level to prevent aeration of the oil.

## LED Monitor Light

- Steady - Pump 1 is withdrawing oil from the engine sump and bringing down the oil level.
- Regular pulsing - Pump 2 is returning oil to the engine sump and raising the oil level.

Irregular pulsing - Oil is at the correct running level.

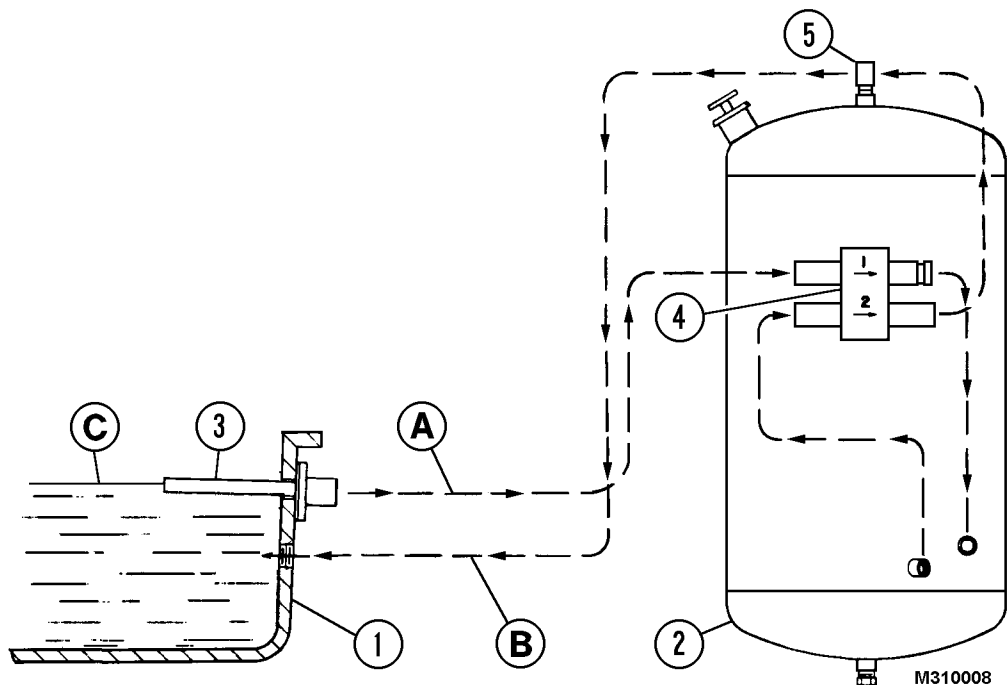


FIGURE 31-2. RESERVE SYSTEM SCHEMATIC

1. Engine Oil Pan (Sump)
2. Oil Tank
3. Suction Tube

4. Pumping Unit (1 & 2)
5. Air Relief Valve

- A. Oil Suction Line
- B. Oil Return Line
- C. Engine Oil Level

## Tank Fill Control (Optional)

With the reserve oil system the engine oil level is held constant, with only the reserve tank needing routine filling. The fill system automatically controls the filling of remote tank from a convenient ground level position. Filling of the tank to the proper "full" level is fast and accurate and accomplished in 2 to 3 minutes.

The reserve oil tank for the engine is designed to add more oil capacity to the engine to reduce the frequent servicing of the engine oil. The engine oil level must still be checked every shift using the dipstick. The oil level in the reserve tank should also be checked using the filler cap dipstick.

If the engine oil has been drained from the engine oil pan, the new oil must be added through the engine fill tube. After an oil change, both engine and reserve tank must be at the full level of oil before starting the engine.

**NOTE:** DO NOT use the oil in the reserve tank to fill the engine oil pan.

System switch (2, Figure 31-3) is an illuminated push-pull POWER-ON switch that powers the fill system. Start switch (3) is a momentary push button switch that opens the fill valve mounted on the reserve tank to begin the automatic filling of the reserve tank. Supply oil under pressure flows through the fill valve and into the tank.

## Filling Procedure (Remote Fill Feature)

**NOTE:** This procedure adds oil to the reserve tank.

1. Connect the pressure supply hose from the new oil supply to the quick coupler on the truck. Open valve on supply hose to apply pressure.
2. Pull out on system switch (2, Figure 31-3) to turn the fill system on.
3. Push start switch (3). VALVE OPEN light (5) should illuminate and the filling process will begin.
4. When the tank is full, the VALVE OPEN light will turn off and FULL light (4) will illuminate.
5. Close the oil supply valve in the fill hose.
6. Press and hold start switch (3) for a couple of seconds to relieve oil pressure in the line.
7. Disconnect the new oil supply hose.
8. Push switch (2) in to turn fill system power off.

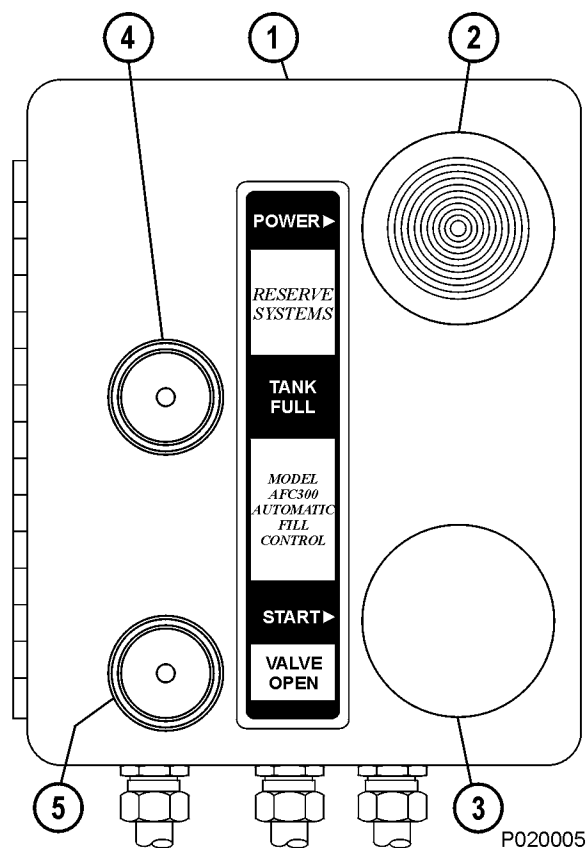


FIGURE 31-3. RESERVE OIL TANK  
REMOTE FILL

- |                       |                     |
|-----------------------|---------------------|
| 1. Remote Control Box | 4. FULL Light       |
| 2. System Switch      | 5. VALVE OPEN Light |
| 3. Start Switch       |                     |

## SERVICE

Between oil drains, the only normal servicing required is routine replenishment of oil at the reserve supply tank. Maintenance of running levels should be checked routinely; manually before starting the engine and with the LED system monitor on the reserve tank pumping unit (11, Figure 31-1) when the engine is running.

There is also an in-line filter (screen) installed at the inlet of the fill valve (3, Figure 31-1). This filter requires no periodic maintenance, but it can be cleaned by removing it from the system and back flushing through the filter.



***Always check the engine oil level before starting engine. Use the engine dipstick.***

### **Every 10 Hours, or once each shift:**

1. Before starting engine, check oil level using the engine dipstick. Oil level should be in normal operating range. If not, check the reserve system for proper operation.
2. The engine oil quality will be best if the reserve tank is kept reasonably full. Check the oil level in the reserve tank. As a minimum guideline, if the oil is below the half-full level, fill the tank manually according to the filler cap dipstick or by using the automatic fill control method.
3. After starting and warm-up, check engine oil level signal (LED) to verify that the engine is being maintained at the preset running oil level. The signal should alternate between periods of 'steady on' and 'flashing'.

***NOTE: Oil should always be visible in the lower sight gauge. If the tank is equipped with three sight gauges, oil should always be visible in the middle sight gauge.***

### **Every 500 Hours**

1. Change all engine filters, if applicable.
2. More system failures result from bad electrical connections than all other causes combined. Check electrical system connections for tightness, corrosion and physical damage. Check battery, alternator, oil pressure switch, junction boxes, remote control fill box and circuit breakers.
3. Examine electrical cables over their length for possible damage.
4. Small hose leaks can cause system malfunction. Examine all hoses, including those on the reserve tank and the ones leading to and from the engine for leaks, cracks or damage. Check all fittings for tightness, leakage or damage.

### **Changing Oil**

1. Drain both the engine sump and the reserve tank. Refill both engine and reserve tank with new oil to proper levels.
2. Change engine oil filters as required.
3. Start engine and check for proper operation.

***NOTE: Do not use the oil in the reserve tank to fill the engine sump. Both must be at proper level before starting engine.***

The engine oil level should be checked with the engine dipstick at every shift change. The oil level in the reserve tank must also be checked at every shift change using the fill cap dipstick.

## TROUBLESHOOTING

It is important to understand the LED signal for the pumping unit. It is used primarily to verify that the system is maintaining the oil level at the level of the open end of the withdrawal tube in the engine oil pan. The signal is also a valuable tool in troubleshooting the system.

When the signal is STEADY (not flashing), pump 1 is running and oil is being withdrawn from the engine and being transferred to the reserve tank.

When the signal is FLASHING, pump 1 is drawing air from the suction tube which triggers operation of the pump 2 to operate and transfer oil back to the engine from the tank (the flashing is actually the pulses of pump 2). When the oil is at the correct level in the engine, air and oil are alternatively entering the suction tube, with pump 1 commanding operation of pump 2 with each portion of air that comes through the line.

This is a complete test for proper operation of the pumping unit. This operation can be accomplished without running the engine by jumping the oil pressure switch that activates the system.

1. If the signal light is STEADY, pump 1 should be pumping oil. Verify by loosening the hose at pump 1 outlet to verify that oil is coming through (pump 1 is marked by a groove on its outlet).
2. Loosen the hose at the inlet of pump 1 to admit air. Pump 2 should then run and the signal should be flashing. Verify proper pumping of pump 2 by loosening the hose at its outlet to see that oil is coming through.
3. Re-tighten the inlet hose on pump 1. The pump should again receive oil and the flashing should stop.

*NOTE: There is a condition that would show a level higher than the controlled point. If both the engine and reserve tank are overfilled, there is no room in the tank to draw the oil level down in the engine. In this case, the LED signal would never start FLASHING because pump 1 is never receiving air. It will continue to pump oil from the engine to the tank, but because the tank is full, the oil will be routed back to the engine via the air relief valve on top of the tank.*

There are two explanations for an overfilled tank and engine:

- When the tank is filled to "FULL" and the engine is overfilled.
- When oil is added directly to the engine between oil changes. The system transfers the oil to the reserve tank until it can not receive any more and the engine remains overfilled. It is, therefore, important that oil should be added only to the reserve tank between oil changes (except if the engine oil level is extremely low).

### Circuit Fuse

The reserve system fill control unit is protected by a 15 amp fuse (Fuse Block 2, position 10) located in the auxiliary control cabinet.

The pump unit is protected by a 15 amp fuse (Fuse Block 2, position 9) located in the auxiliary control cabinet.

For circuit information, refer to the system schematic in the back of the shop manual.



## SYSTEM ELECTRICAL SCHEMATICS

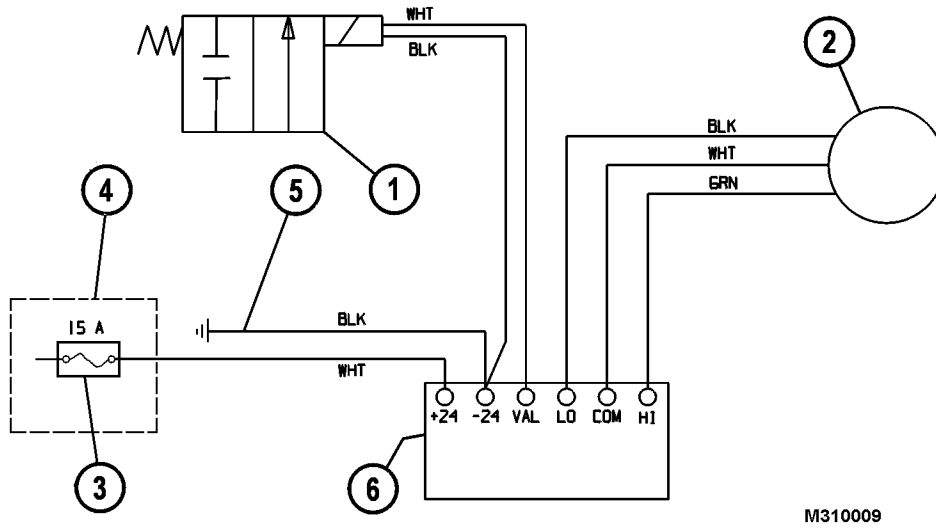


FIGURE 31-4. FILL SYSTEM SCHEMATIC

- |                                               |                            |
|-----------------------------------------------|----------------------------|
| 1. Fill Valve                                 | 4. Auxiliary Control Box   |
| 2. Oil Level Sensor (top of reserve oil tank) | 5. Ground Wire             |
| 3. 15 Amp Fuse                                | 6. Remote Fill Control Box |

# SECTION N OPERATOR CAB INDEX

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# NOTES

## SECTION N2

### TRUCK CAB AND COMPONENTS

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# NOTES

## TRUCK CAB AND COMPONENTS

### TRUCK CAB

#### Description

The truck cab is a fully insulated design incorporating an integral ROPS structure for maximum operator comfort and safety. All gauges, switches, and controls have been designed to simplify operation and are placed within easy reach of the operator. Servicing of cab and associated electrical systems is simplified by use of heavy duty connectors on the various wiring harnesses. Hydraulic components are located outside the cab interior and are accessed through cover (1, Figure 2-1) on the front of the cab.

#### **⚠ WARNING**

***DO NOT attempt to modify or repair damage to the ROPS structure without written approval from the manufacturer. Unauthorized repairs to the ROPS structure will void certification. If modification or repairs are required, contact the servicing Komatsu Distributor.***

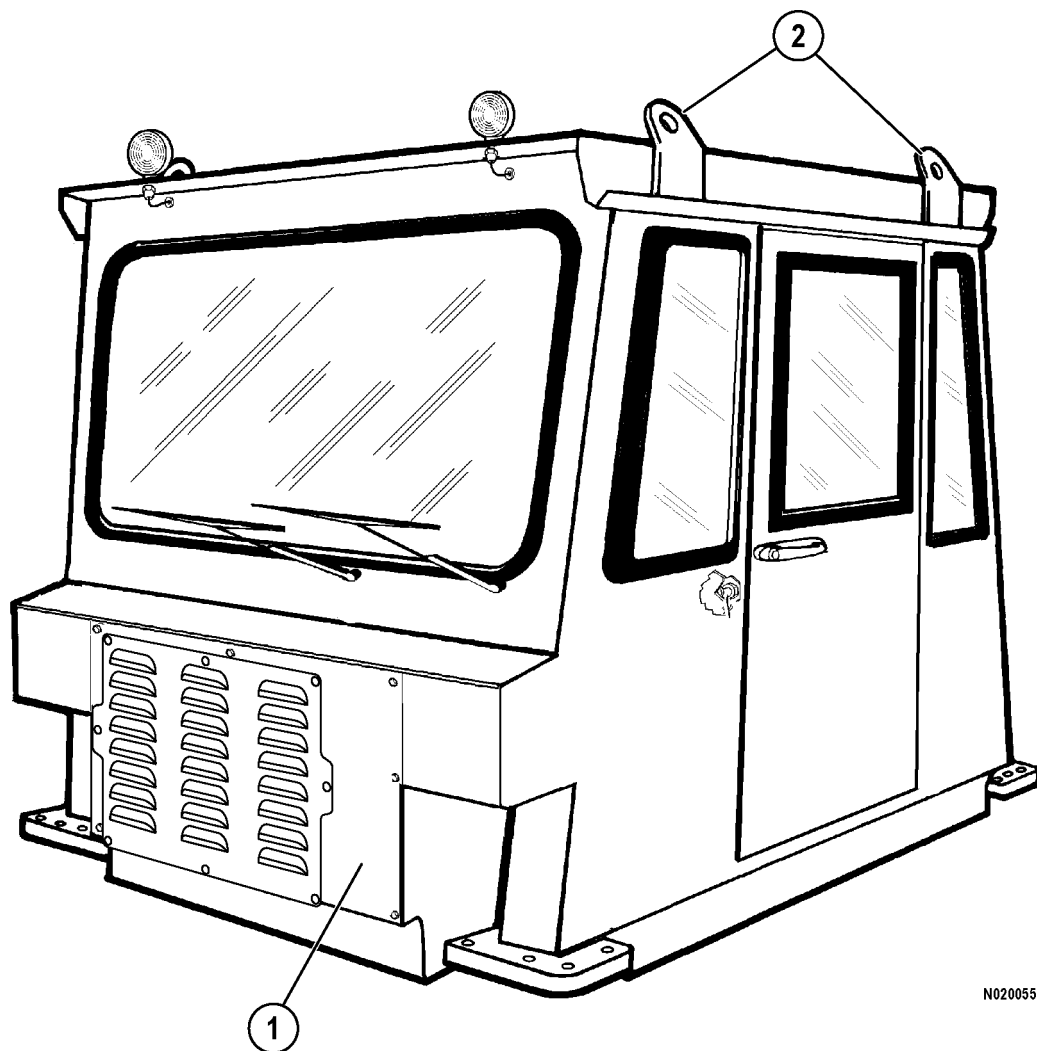


FIGURE 2-1. CAB ASSEMBLY (FRONT VIEW)

1. Access Cover

2. Lifting Eyes

*NOTE: Prior to cab removal or repair procedures, it may be necessary to remove the body to provide clearance for lifting equipment to be used. If body removal is not required, the body should be raised and the safety cables installed at the rear of the truck.*

## **WARNING**

**Do not attempt to work in the deck area until the body safety cables have been installed.**

### **Removal**

*NOTE: The following procedure describes removal of the cab as a complete module with the hydraulic brake cabinet attached. All hoses and wire harnesses should be marked prior to removal for identification to ensure correct reinstallation.*

1. Turn the key switch to the OFF position and allow at least 90 seconds to bleed the steering accumulator. Turn the steering wheel to ensure that no pressure remains.
2. Block the truck securely, and open the brake accumulator bleed down valves on the accumulators located in the cabinet mounted on the rear of the cab. Allow sufficient time for the accumulators to bleed down completely.
3. Activate the battery disconnect switches located at one end on the battery box.
4. Disconnect all hydraulic hoses that are routed to the frame from the fittings on the bottom of the hydraulic brake cabinet. (It is not necessary to disconnect hoses that are routed under the cab.) Cap all fittings and plug the hoses to prevent contamination.
5. Disconnect the wire harnesses from cab connector plate (1, Figure 2-2) at the rear of the cab.
6. Remove cable clamps and hose clamps as needed for cab removal.
7. Close the heater shutoff valves located at the water pump inlet housing on the right side of the engine and at the water manifold. Disconnect the heater hoses at each valve and drain the coolant into a container.
8. Remove clamps and heater hoses from the fittings on the underside of the deck, below the heater.

## **IMPORTANT**

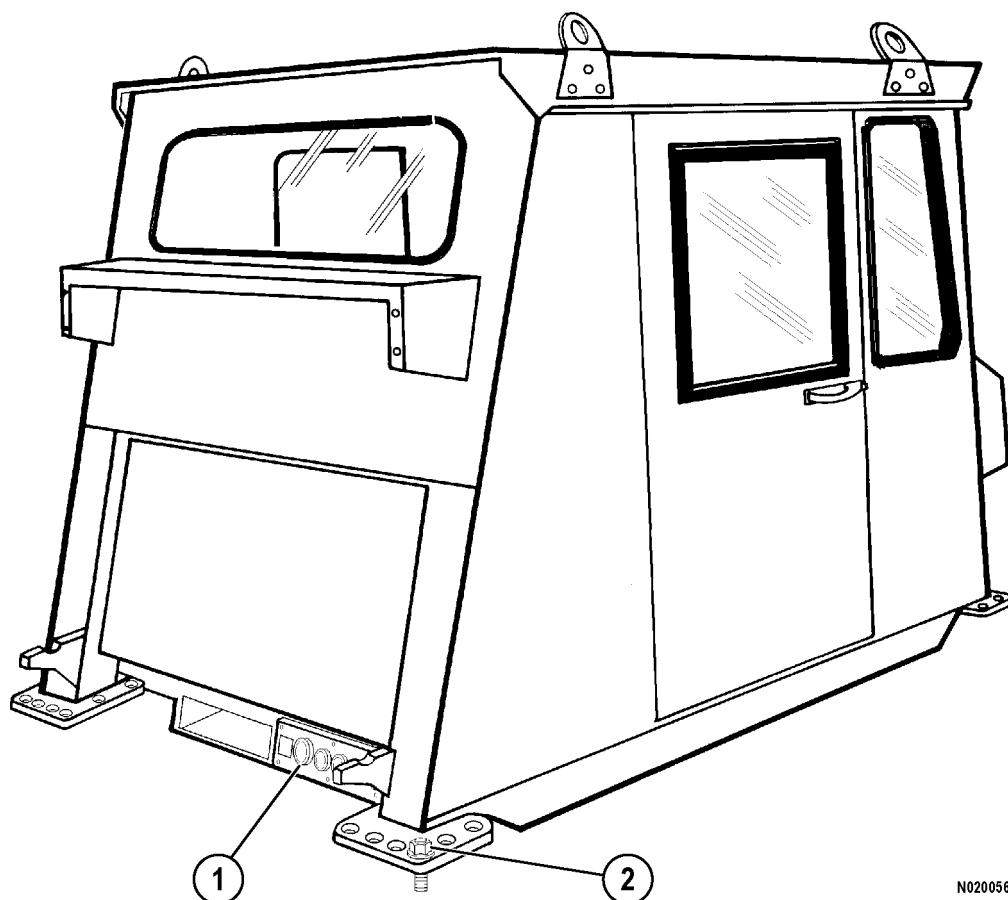
***Federal regulations prohibit venting air conditioning system refrigerants into the atmosphere. An approved recovery/recycle station must be used to remove the refrigerant from the air conditioning system.***

9. Evacuate the air conditioning system as follows:
  - a. Attach a recycle/recovery station at the air conditioning compressor service valves. Refer to "Heater/Air Conditioning System" in this section for detailed instructions.
  - b. Evacuate the air conditioning system refrigerant.
  - c. Remove the air conditioner system hoses that are routed to the cab from the receiver/drier and compressor. Cap all hoses and fittings to prevent contamination.
10. Attach a lifting device to lifting eyes (2, Figure 2-1) provided on top of the cab.

## **WARNING**

***The cab assembly weighs approximately 2 270 kg (5,000 lb). Ensure the lifting device has adequate capacity for lifting the load.***

11. Remove cab mounting hardware (2, Figure 2-2) from the mounting pads at each corner of the cab.
12. Check for any other hoses or wiring which may interfere with cab removal.
13. Lift the cab assembly off the truck and move it to a work area for further service.
14. Place blocking under each corner of the cab to prevent damage to the floor pan and hoses before the cab is lowered to the floor.



N020056

FIGURE 2-2. CAB ASSEMBLY (REAR VIEW)

1. Cab Connector Plate

2. Cab Mounting Hardware

## Installation

1. Lift the cab assembly and align the mounting pad holes with tapped pads. Insert at least one cap screw and hardened washer at each of the four pads prior to lowering cab onto the truck.

*NOTE: The tool group shipped with the truck contains the following tools which may be used to install the inner cap screws, as clearance is limited:*

- PB8326 - 1-1/2" offset wrench
- TZ2733 - Tubular Handle
- TZ2734 - 3/4" torque wrench Adaptor

2. After cab is positioned, measure the gap between each cab mounting pad and the frame. Insert 1 mm (0.040 in.) shims as required so gap at each pad is less than 1 mm (0.040 in.).
3. After the cab is positioned, install the remaining capscrews and hardened washers (32 total). Tighten the capscrews to **800 N·m (590 ft lb)**.
4. Route the wire harnesses to cab connector plate (1, Figure 2-2) on the rear corner of the cab. Align the cable connector plug key with the receptacle key and push the plug onto the receptacle. Carefully thread the retainer onto the receptacle and tighten it securely. Install any clamps that were removed during cab removal.



5. Remove the caps from the hydraulic hoses and tubes and reinstall the hoses and tubes. Reinstall any hose clamps as required.
6. Install the heater hoses and clamps on the fittings on the underside of the cab. Connect the other end of each hose to the fittings at the shutoff valves on the engine. Open the heater shutoff valves. Connect the air cleaner restriction indicator hoses.
7. Remove the caps and reinstall the air conditioning system hoses to the compressor and receiver/drier. Refer to "Heater/Air Conditioning System" for detailed instructions about evacuation and recharging with refrigerant.
8. Close the brake accumulator bleed down valves.
9. Deactivate the battery disconnect switches.
10. Service the hydraulic tank and engine coolant as required.
10. Start the engine and verify proper operation of all controls. Ensure that the air conditioning system is properly recharged.

## **CAB DOOR**

The cab door assemblies are similar except for the hinge side. Each is hinged on the rear edge with a heavy duty hinge. For repairs on the door latches or window controls, it is usually better (but not necessary) to remove the door from the cab and lower it to the floor for service.

### **Removal**

1. If overhead space is available, raise the body to allow access to the door with an overhead hoist. Secure the body in the raised position with safety cables.
2. Lower the door glass far enough to allow insertion of a lifting sling when the door is removed.
3. Disconnect the door harness from the floor.
4. Remove the retainer clip and bolt clip from the door travel limiting strap.
5. Insert a lifting sling through the door and attach it to a hoist. Remove the capscrews (a swivel socket works best) that secure the door hinge to the cab. Lift the door from the cab.
6. Place the door on blocks or on a work bench to protect the window glass and allow access to internal components for repair.

### **Installation**

1. Attach a lifting sling and hoist to the door assembly. Lift the door up to the deck and position the door hinges on the cab.
2. Align the door hinges with the cab and install the capscrews that secure the door to the cab.
3. Attach the door travel limiting strap with the bolt and clip that were removed previously. The strap has three holes sewn into it at each end to allow adjustment of the strap.
4. Connect the door harness to the receptacle mounted in the cab floor. Ensure that the harness runs through the loop sewn into the bottom of the door travel limiting strap.
5. Verify proper operation of the power window and door latch adjustment.
6. If an adjustment is necessary to ensure tight closure of the door, refer to "Door Jamb Bolt Adjustment".
7. A rubber sealer strip is mounted with adhesive around the perimeter of the door assembly to keep out dirt and drafts. This sealer strip should be replaced if it becomes damaged.

## Door Jamb Bolt Adjustment

Over time, the door latch mechanism and door seals may wear and allow dirt and moisture to enter the cab. To ensure proper sealing of the door seals, the door jamb bolt may need to be adjusted periodically.

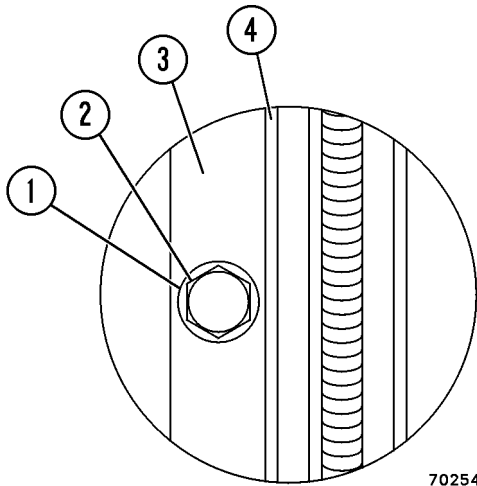


FIGURE 2-3. DOOR JAMB BOLT ADJUSTMENT

- |                 |          |
|-----------------|----------|
| 1. Washer       | 3. Frame |
| 2. Striker Bolt | 4. Seal  |

- If the door closes, but not tightly enough to give a good seal between the seal on the door and the cab skin:

1. Mark washer location (1, Figure 2-3) portion of the door jamb bolt with a marker, pen, or pencil by circumscribing the outside edge of the washer onto the jamb.
2. Loosen door jamb bolt (2) and move it straight inwards 1.5 mm (1/16 inch) and retighten the bolt.
3. Hold a piece of paper between where door seal (4) will contact the skin of the cab and firmly close the door.

**NOTE:** The door latch mechanism has a double catch mechanism, so ensure that it latches on the second catch.

4. The door seal should firmly grip the paper all along the top, front, and bottom edge of the door. If the paper is loose all around, repeat Step 2. If the paper is firmly gripped, but can be removed without tearing it, open the door and tighten the jamb bolt completely without affecting the adjustment.

5. If the paper slips out from the door seal easily along the top but the bottom (or vice versa), the door itself will have to be adjusted.

- a. If the seals are tight at the bottom of the door but not at top, place a 4 x 4 block of wood at the bottom edge of the door, below the handle. Close the door on the wood block and press firmly inward on the top corner of the door. Press in one or two times, then remove the wood block and check the seal tension again using the paper method. Seal compression should be equal all the way around the door. If the seal is still loose at the top, repeat the procedure again until seal compression is the same all the way around.

- b. If the seals are tight at the top of the door but not at the bottom, place a 4 x 4 block of wood at the top corner of the door and press firmly inward on the lower corner of the door. Press in one or two times, then remove the wood block and check seal compression again using the paper method. Seal compression should be equal all the way around the door. If seal is still loose at the bottom, repeat the procedure again until seal compression is the same all the way around.

- If the door springs back when trying to close it, striker bolt (2, Figure 2-4) has probably loosened and slipped down from where the catch can engage with the bolt.

1. Open the door and close both latches (3) and (5) on the catch until they are both fully closed.
2. Transfer the center of this opening onto the skin of the cab nearest where the door jamb bolt is located. Use a T-square or other measuring equipment and mark on the cab with a pencil.

**NOTE:** Release the door catch before trying to close the door.

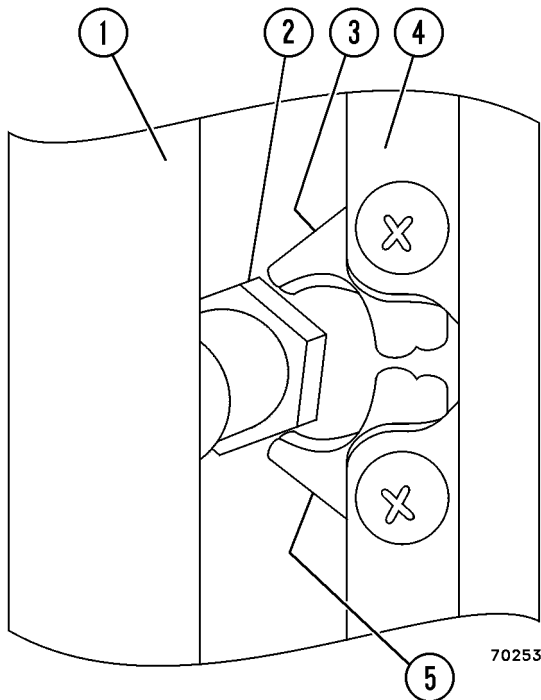


FIGURE 2-4. DOOR JAMB BOLT ADJUSTMENT

- |                 |                |
|-----------------|----------------|
| 1. Cab          | 4. Door        |
| 2. Striker Bolt | 5. Lower Latch |
| 3. Upper Latch  |                |

3. Loosen and vertically align (center) the door jamb bolt with the mark and tighten it firmly enough to hold it in place but still allow some slippage.
4. Carefully try to close door (4) and determine whether this adjustment has helped the "springing" problem. If the door latches but not firmly enough, follow the procedure in Step 1. If the door latch does not catch, move the bolt outward and try again. When corrected, follow the adjustment procedures in Step 1 to ensure a good seal.

By design, if both seals are in good condition, proper adjustment of the outside seal will ensure good contact on the inside seal to prevent dust and moisture from entering the cab.

## Door Handle Plunger Adjustment

If the door handle does not function, it can either be adjusted or replaced. The following is a procedure for adjusting the exterior door handle plunger.

1. Determine the amount of free play in the door release plunger by pushing in on the plunger until it just contacts the door release mechanism. Measure the distance that the plunger travels from this position to where the plunger is fully released (See Figure 2-5).

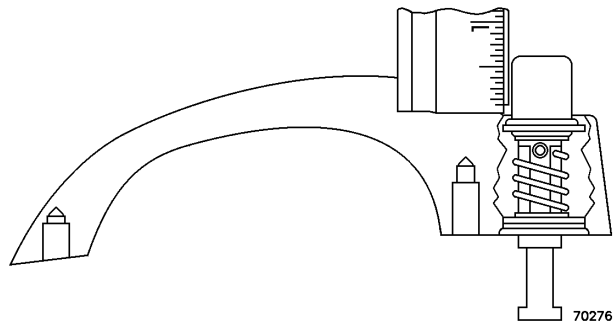


FIGURE 2-5. MEASURING TRAVEL DISTANCE OF PLUNGER

2. Remove the door panel as follows:
  - a. Remove hair pin clip (1, Figure 2-6) and bolt (2) from the door check strap closest to the door.
  - b. Remove two capscrews (3) which secure the door strap bracket to the door.
  - c. Disconnect wiring harness (4) from the window regulator.
  - d. Open the door as far as possible in order to remove the internal door panel.
  - e. Before removing all door panel mounting screws, support the panel to prevent the assembly from dropping. Remove 15 mounting screws (5).

*NOTE: Remove the panel screws across the top last.*



***The door glass and internal door panel will drop when the door panel screws are removed.***

- f. Carefully lower the door panel a few inches. Hold the glass at the top to prevent it from dropping. Slide the door panel toward the cab to disengage the window regulator roller (Figure 2-7) from the track on the bottom of the glass. Slide the panel away from the cab to disengage the other top roller and lower roller from its tracks. Place the panel out of the way after removal.
- g. Lift the door glass and support it at the top of the frame.
- h. Remove two screws (Figure 2-8) holding the roller track to the bottom of the door glass.
3. Remove the capscrew and nut from inside release lever (1, Figure 2-14).
4. Remove four mounting screws (2) that hold the latch mechanism in the door.
5. Remove door latch mechanism (4). Check whether the door latch mechanism works properly by performing the following test:
  - a. Close the latch mechanism pawls.
  - b. Operate the inside door release lever to see whether the pawls open. If the pawls do not open, replace the door latch assembly.
  - c. Close the pawls again.
  - d. Press the outside door button to see whether the pawls open.
  - e. If the latch mechanism operates properly, proceed to Step 6. If the latch mechanism does not operate properly, install a new door latch assembly, then continue with Step 6.
6. Remove mounting screws (3) from the outside door handle. With the door handle removed, adjust the plunger counterclockwise to increase the height of the door handle release button. Lock the plunger capscrew with the locknut. Apply Loctite® to prevent the screw from loosening.
7. Reassemble the door assembly by reversing the previous steps.

## Replacing the Door Glass

1. Remove hair pin clip (1, Figure 2-6) and strap pin (2) from door check strap (3).

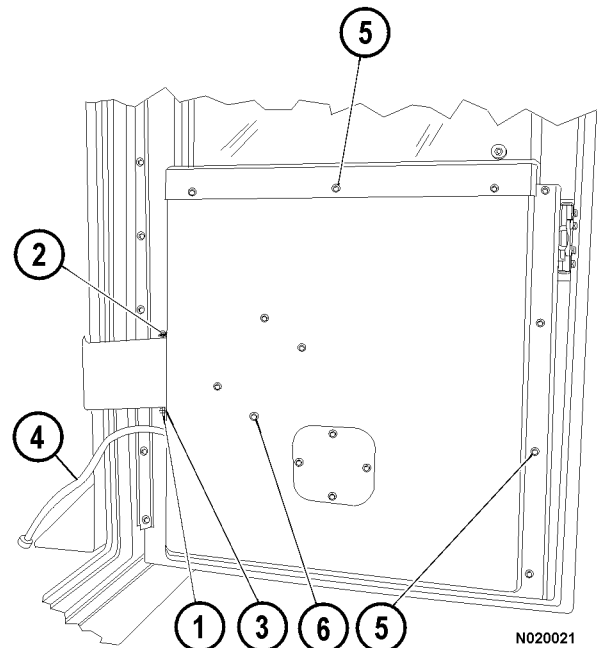


FIGURE 2-6.

- |                      |                      |
|----------------------|----------------------|
| 1. Hair Pin Clip     | 5. Door Panel Screws |
| 2. Strap Pin         | 6. Window Regulator  |
| 3. Door Check Strap  | Mounting Screws      |
| 4. Regulator Harness |                      |

2. Disconnect regulator harness (4) from the floor of the cab.
3. Open the door as far as possible in order to remove the inner door panel.
4. Before removing all of the door panel screws, support the panel to prevent the assembly from dropping. Remove 15 door panel screws (5).

**NOTE:** Remove panel screws across the top last.



**The door glass and inner door panel will drop when door panel screws are removed. Secure the glass at the top to prevent it from dropping.**

5. Carefully lower the door panel a few inches. Secure the glass with duct tape at the top to prevent it from dropping. Slide the door panel toward the cab to disengage the window regulator roller (Figure 2-7) from the track on the bottom of the glass. Slide the panel away from the cab to disengage the other top roller and lower roller from its tracks.

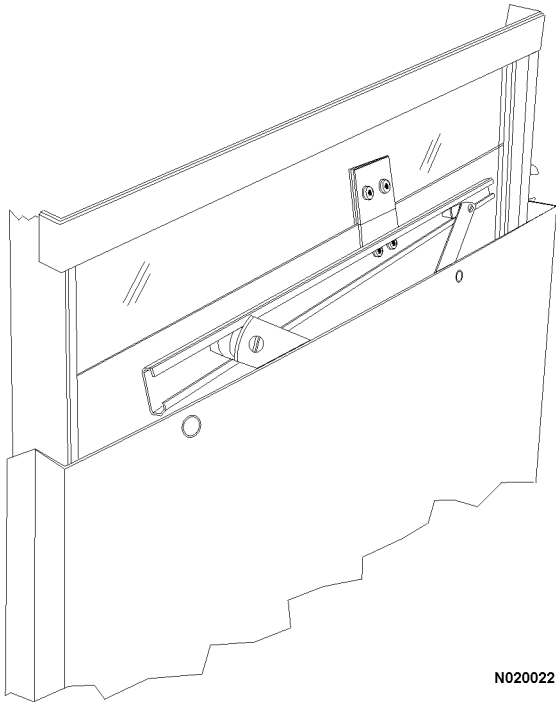


FIGURE 2-7.

6. Remove two screws (Figure 2-8) holding the roller track to the bottom of the door glass.

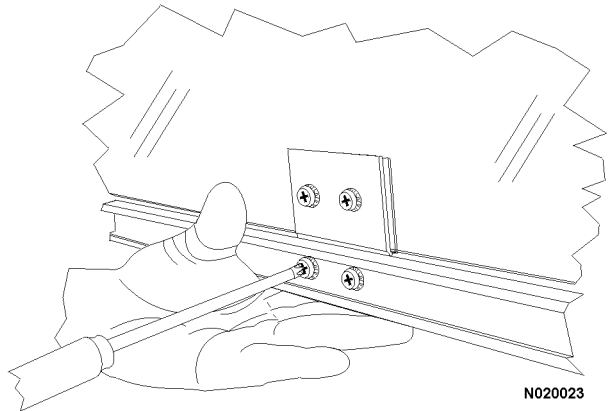


FIGURE 2-8.

7. Support the glass in the door frame with support block (1, Figure 2-9) as shown. Remove screws (2) that hold the adapter for the window regulator track.

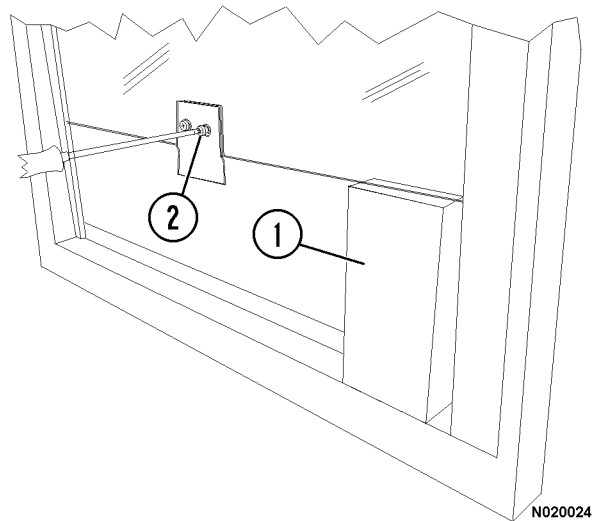


FIGURE 2-9.

1. Support Block      2. Screws

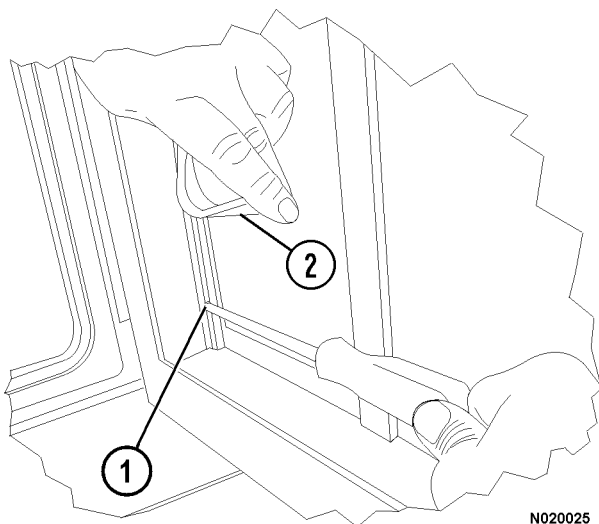


FIGURE 2-10.

1. Screws                      2. Rubber Felt Insert

8. Remove the screw at the lower end of the window channels. It is necessary to pull rubber felt insert (2, Figure 2-10) out of the channel to be able to remove the screws.

9. Remove the trim material covering screws (1, Figure 2-11) that are holding the window frame to the door. Remove the screws.

**NOTE:** The screws along the bottom of the window frame may be shorter than the screws along the top and sides.

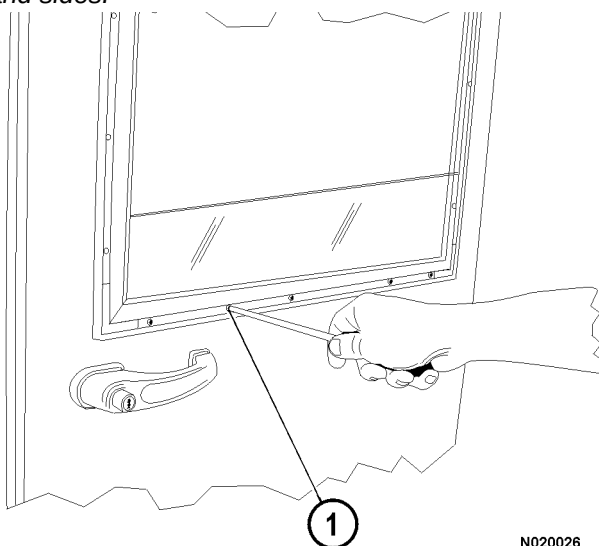


FIGURE 2-11.

1. Screws

10. Lift the door glass up in frame (1, Figure 2-12) so that it is near the top. While holding the glass in place, tilt the frame out at the top. Lift the frame and glass straight up and out of the door.



**Bracket (2, Figure 2-12) at bottom of glass must clear the door frame if still on the glass.**

11. Move the window glass and frame to a work area where the glass can be removed. Slide the glass down and out of the window channels.

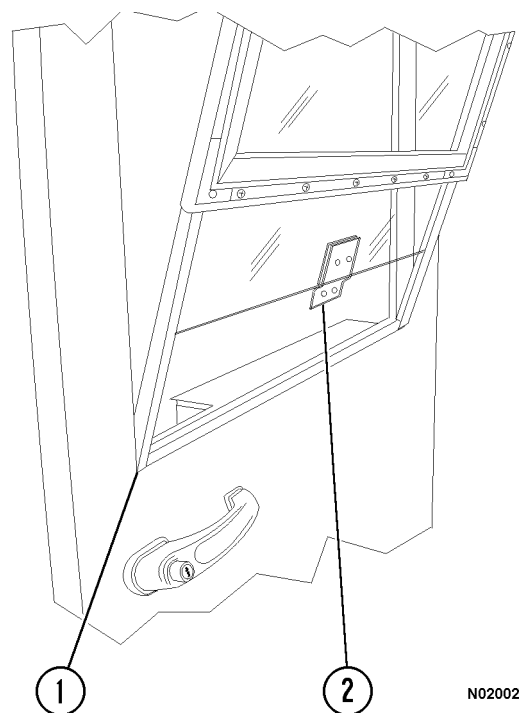


FIGURE 2-12.

1. Window Frame                      2. Window Bracket

12. Before installing new window glass, inspect the window frame. In each corner there is an "L" shaped bracket (1, Figure 2-13) with two screws in it to hold the corners of the frame together. Check the screws to ensure that they are tight. Also ensure that rubber felt insert (2, Figure 2-10) in the window channels is in good condition. Replace it if necessary.
13. Slide the new window glass into the window frame glass channels. Move the glass to the top of the frame.
14. While holding the glass at the top of the frame, lift the window frame and lower the assembly into the door.

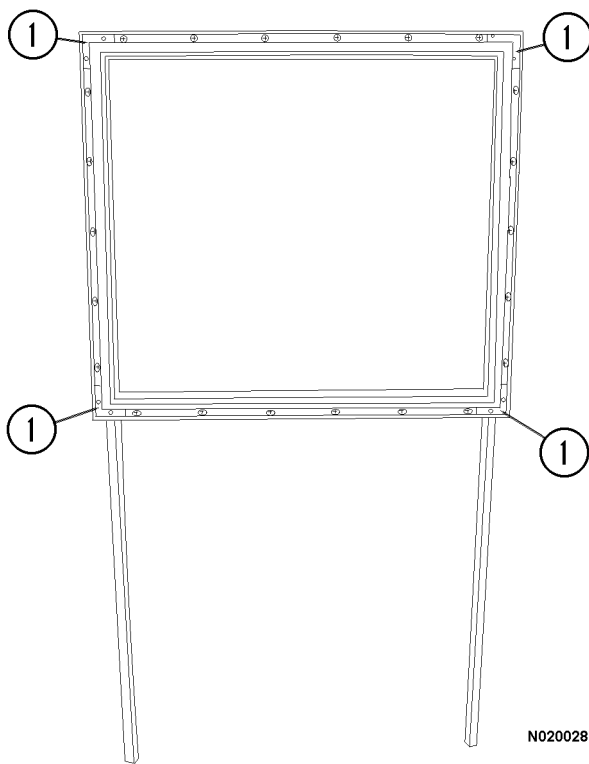


FIGURE 2-13.  
1. "L" Shaped Brackets

## ⚠ IMPORTANT ⚠

**Ensure that the one channel in window frame (5, Figure 2-14) that is next to the door latch passes to the inside of latch assembly (4).**

15. Lower the glass in the frame and support it with a support block as shown in Figure 2-9.

16. Install screws (1, Figure 2-10) that secure the window frame to the door frame.

## ⚠ CAUTION

**The screws along the outer bottom of the window frame may be shorter than the ones along the sides and top. These screws must be used in this area to prevent the window glass from being scratched or cracked. See Figure 2-11.**

17. Install the trim material over the top of screws that hold the window frame to the door. Use a flat blade screwdriver to assist with installing the trim material. See Figure 2-15.

**NOTE: Be careful not to cut the retainer lip on the trim material.**

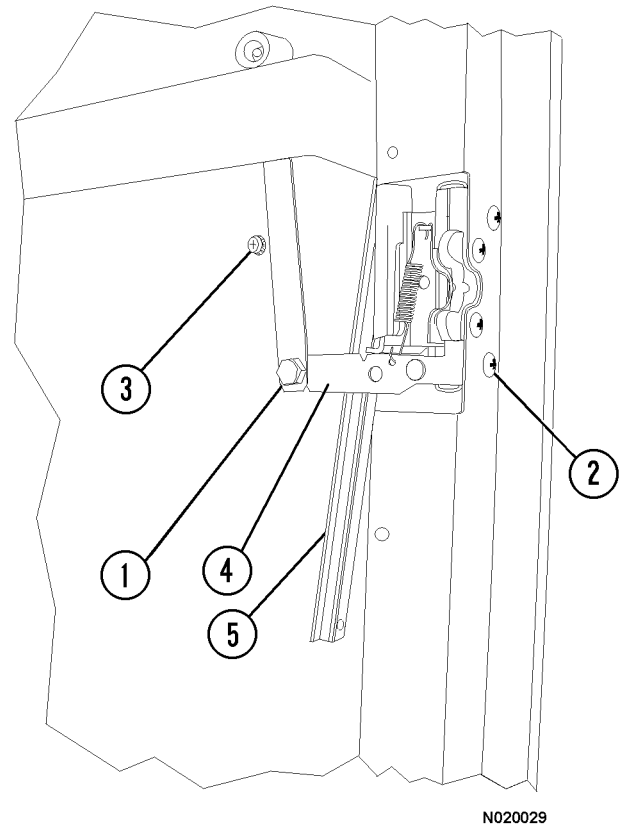


FIGURE 2-14.

- |                      |                     |
|----------------------|---------------------|
| 1. Capscrew & Nut    | 3. Mounting Screw - |
| 2. Mounting Screws - | Outside Door Handle |
| Latch                | 4. Latch Assembly   |
|                      | 5. Window Frame     |

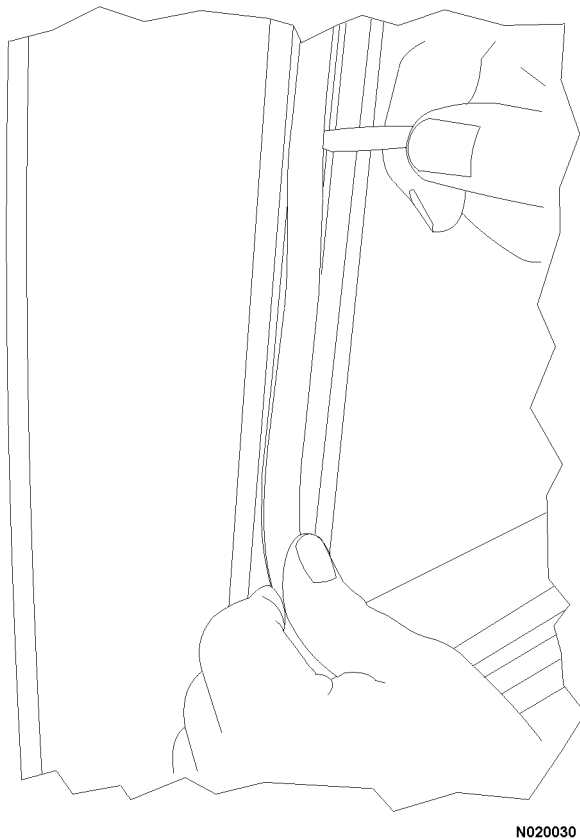


FIGURE 2-15.

18. Install the two screws that were removed in Step 8. Ensure that the rubber felt insert is back in place after the screws are installed.
19. Install the window regulator track bracket as shown in Figure 2-7. Ensure that the nylon bushings and gaskets are installed properly to prevent damage to the glass.
20. Lift the window glass in the frame and install the window regulator roller track onto the bracket installed in Step 20. See Figure 2-8.
21. Secure the window glass with duct tape a few inches from the top (Figure 2-9). Install the lower and upper regulator rollers in their tracks. Start by moving the door panel away from the cab just far enough to allow the rollers to enter their tracks. Then, with the rollers in the tracks, slide the panel toward the cab. Move the panel just far enough to allow the upper regulator roller to go into the track on the bottom of the glass.

22. Lift up the inner door panel, window regulator and glass to align the screw holes in the panel with the holes in the door frame. Install door panel screws (5, Figure 2-6) that secure the panel to the door frame.
23. Connect regulator harness (4) to the floor of the cab.
24. Align the opening of door check strap (3) with the holes in the bracket and install strap pin (2). Install hair pin clip (1).

### Replacing the Door Handle or Latch Assembly

The cab doors are equipped with serviceable latch handle assemblies (inner and outer). If they become inoperative, they should be replaced. The outer latch handle assembly on each door has a key-operated lock.

1. Perform Steps 1 through 5 in the procedure "Replacing the Door Glass".
2. Remove capscrew and nut (1, Figure 2-14) from the inside door handle.
3. Remove four mounting screws (2) for the latch. Remove the latch assembly.
4. If replacing the latch assembly, proceed to Step 5.

If replacing the outside door handle, remove three screws (3) that secure the handle to the door panel.

*Note: Only one screw is shown. The other two are behind the latch assembly.*

5. Install a new latch assembly and align the mounting holes. Install four mounting screws (2, Figure 2-14). Ensure that they are tight.
6. Align the inside door handle and install capscrew and nut (3).
7. Perform Steps 21 through 24 in the procedure "Replacing the Door Glass".



## Replacing the Door Window Regulator

### **CAUTION**

*Do not replace a 24VDC window regulator with a 12VDC window regulator. A 12VDC motor will burn out quickly and stop working if supplied with 24VDC power.*

*Do not replace a 12VDC window regulator with a 24VDC window regulator. A 24VDC motor will not raise the window enough if supplied with 12VDC power.*

1. Perform Steps 1 through 5 in the procedure "Replacing the Door Glass".
2. Move the inner door panel to a work area.
3. Disconnect regulator harness (2, Figure 2-16) at window regulator motor (1).
4. Remove four window regulator mounting screws (6, Figure 2-6).
5. If replacing only the window regulator motor, ensure that the worm gear on the motor is engaged properly into the regulator gear. Also, the regulator must be in the UP position before replacing the motor assembly. Ensure that the motor mounting screws are tight.

If replacing the entire window regulator assembly, the new regulator must be in the UP position before being mounted.

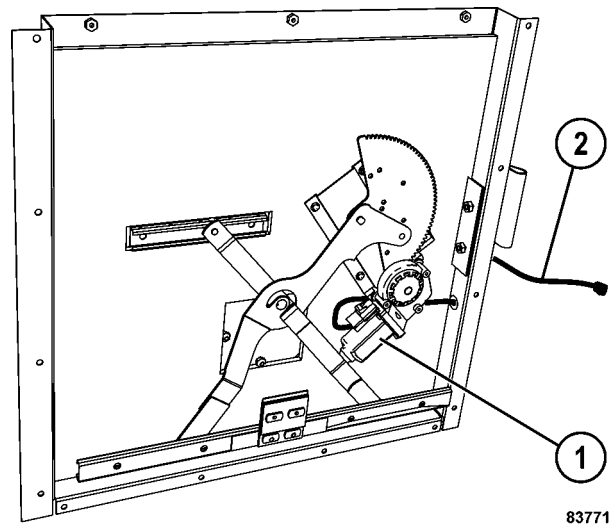


FIGURE 2-16. WINDOW REGULATOR

1. Window Regulator Motor
  2. Regulator Harness
6. Mount the window regulator to the inner panel using the four mounting screws that were removed in Step 4. Tighten the screws to **12 N·m (9 ft lb)**.
  7. Connect regulator harness (2, Figure 2-16) to window regulator motor (1).
  8. Perform Steps 21 through 24 in the procedure "Replacing the Door Glass".

### Replacing the Door and Door Hinge Seal

1. The door assembly seal has only three members to it (sides and top) and is glued on the door. This seal can be replaced by peeling the seal away from the door frame. Then use a suitable cleaner to remove the remaining seal and glue material.
2. The area where the door seal mounts should be free of dirt and oil. Spread or spray a glue which is quick drying and waterproof onto the area where the seal is to be installed.
3. Install the seal so that the corners of the seal fit up into the corners of door frame (3, Figure 2-17).
4. Door hinge seal (2, Figure 2-18) is glued to the hinge. Use the same procedure as above for this seal.

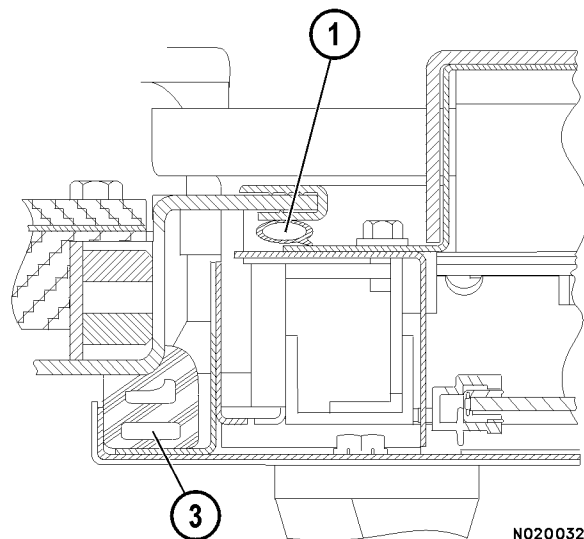


FIGURE 2-17.

### Replacing the Door Opening Seal

1. Starting at the lower center of the door opening, pull up on one end of the seal. The seal should pull loose from the cab opening lip. Pull the seal loose all the way around opening (1, Figures 2-17 and 2-18).
2. Inspect the cab opening lip for damage, dirt, and oil. Repair or clean the cab opening as necessary. Remove all dirt and old sealant. Ensure that the perimeter of the opening is clean and free of burrs.
3. Install the seal material around the door opening in the cab. Start at the bottom center of the cab opening and work the seal lip over the edge of the opening. Go all the way around the opening. Ensure that the seal fits tightly in the corners. A soft face tool may be used to work the seal up into the corners.
4. Continue going all the way around the opening. When the ends of the seal meet at the bottom center of the cab opening, it may be necessary to trim off some of the seal.

**NOTE:** The ends of the seal material need to be square-cut to assure a proper fit.

5. Fit both ends so that they meet squarely. Then while holding the ends together, push them firmly into the center of the opening.

1. Door Opening Seal      3. Door Assembly Seal

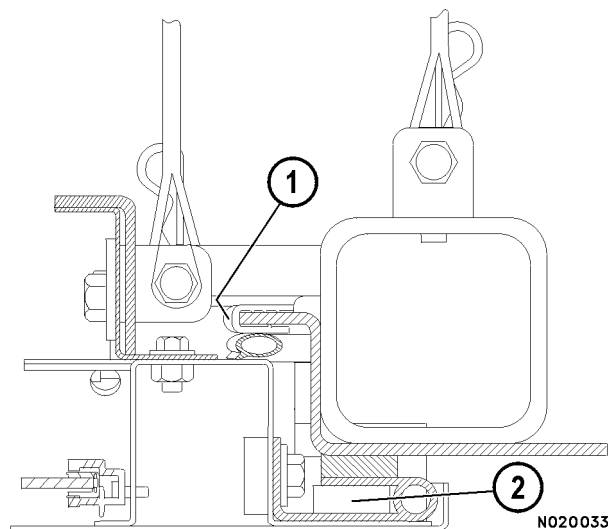


FIGURE 2-18.

1. Door Opening Seal      2. Door Hinge Seal

## GLASS REPLACEMENT (Adhesive-Bonded Windows)

### Recommended Tools and Supplies

- Cold knife, pneumatic knife, or a piano wire cutting device, long knife. Cutout tools are available at an auto glass supply store.
- Heavy protective gloves
- Safety eyeglass goggles
- Windshield adhesives, proper cleaners, primers and application gun
- SM2897 glass installation bumpers (6 or 7 per window)
- Window glass (Refer to the Parts Catalog.)

### Recommended adhesives:

- SikaTack Ultrafast or Ultrafast II (both heated). Vehicle can be put into service in 4 hours under optimum conditions. Heated adhesives require a Sika approved oven to heat adhesive to 80° C (176° F).
- Sikaflex 255FC or Drive (unheated). Vehicle can be put into service in 8 hours under optimum conditions.

Sika Corporation  
30800 Stephenson Hwy.  
Madison Heights, MI 48071  
Toll Free Number: 1-800-688-7452  
Fax number: 248-616-7452  
<http://www.sika.com> or  
<http://www.sikasolutions.com>

## WARNING

***Due to the severe duty application of off-highway vehicles, the cure times listed by the adhesive manufacturer should be doubled before a truck is moved. If the cure time is not doubled, vibration or movement from a moving truck will weaken the adhesive bond before it cures, and the glass may fall off the cab.***

***If another adhesive manufacturer is used, follow that manufacturer's instructions for use, including the use of any primers, and double the allowances for proper curing time.***

## Replacement Procedure

## IMPORTANT

***The first concern with all glass replacement is SAFETY! Wear heavy protective gloves and safety eyeglass goggles when working with glass.***

1. Use a cut-out tool to slice into the existing urethane adhesive and remove the window.
2. Carefully clean and remove all broken glass chips from any remaining window adhesive. The surface should be smooth and even. Use only clean water.

***NOTE: Removal of all old adhesive is not required. Remove just enough to provide an even bedding base.***

3. Use a long knife to cut the remaining urethane from the opening, leaving a bed 2 - 4 mm (0.08 - 0.15 in.) thick. If the existing urethane is loose, completely remove it. Leave the installation bumpers in place, if possible.
4. Clean the metal with Sika Aktivator. Allow it to dry for ten minutes. Then paint on a thin coat of Sika Primer 206G+P and allow it to dry for ten minutes.
5. Using only the new side window(s) which are to be bonded in place, center the new glass over the opening in the cab. Use a permanent marker to mark on the cab skin along all the edges of the new glass that is to be installed. All edges must be marked on the cab in order to apply the adhesive in the proper location.
6. Use Sika Primer 206G+P to touch up any bright metal scratches on the metal frame of the truck. Do not prime the existing urethane bed. Allow it to dry for ten minutes.
7. Use a clean, lint free cloth to apply Sika Aktivator to the black ceramic Frit surrounding the new window. Use a clean, dry cloth and wipe off the Sika Aktivator. Allow it to dry for ten minutes.

8. For the **side windows**, use six or seven glass installation bumpers (SM2897). Space them equally around the previously marked glass perimeter, approximately 19 mm (0.75 in.) inboard from where the edge of the glass will be when it is installed.

*NOTE: Be careful not to place the adhesive too far inboard, as it will make any future glass replacement more difficult.*

9. Apply a continuous, even bead of adhesive, approximately 10 mm (0.38 in.) in diameter, to the cab skin at a distance of 13 - 16 mm (0.50 - 0.63 in.) inboard from the previously marked final location of the glass edges from Step 4.
10. Immediately install the glass. Carefully locate the glass in place with the black masking side toward the adhesive. Press firmly, but not abruptly, into place to ensure that the glass is properly seated. **Do not pound the glass into place.**
11. Use a wooden prop and duct tape to hold the glass in place for at least two hours or double the adhesive manufacturer's curing time, whichever time is longer.
12. Remove the tape or prop from the glass only after the cure time has expired.

## WINDSHIELD AND REAR GLASS

*NOTE: Two people are required to remove and install the windshield or rear glass. One person inside the cab, and the other person on the outside.*

Special tools that are helpful in removing and installing automotive glass are available from local tool suppliers.

### Removal

1. If the windshield is to be replaced, lift the windshield wiper arms out of the way.
2. Starting at the lower center of the glass, pull out weatherstrip locking lip (2, Figure 2-19 or 2-20). Use a non-oily rubber lubricant and a screwdriver to release the locking lip.
3. Remove glass (1) from weatherstrip (3) by pushing it out from inside the cab.
4. Clean all dirt and old sealant from the weatherstrip grooves. Ensure that the perimeter of the window opening is clean and free of burrs.

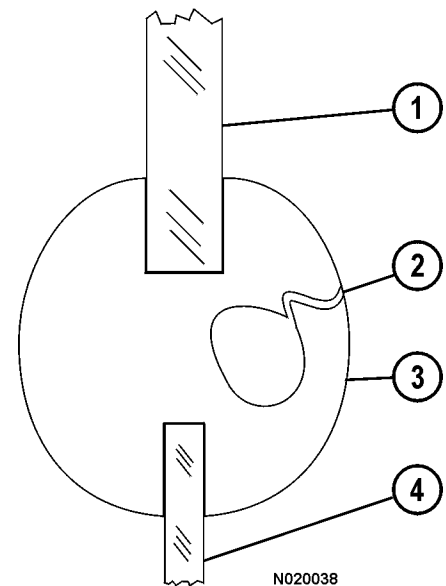


FIGURE 2-19. FRONT WINDSHIELD

- |                |                 |
|----------------|-----------------|
| 1. Glass       | 3. Weatherstrip |
| 2. Locking Lip | 4. Sheet Metal  |

## Installation

1. If the weatherstrip that was previously removed is broken, weathered, or damaged in any way, install a new rubber weatherstrip.

*NOTE: Using a non-oily rubber lubricant on the weatherstrip material and cab opening will make the following installation easier:*

- a. Install weatherstrip (3, Figure 2-19 or 2-20) around the window opening. Start with one end of the weatherstrip at the center, lower part of the window opening and press the weatherstrip over the edge of the opening.
- b. Continue installing the weatherstrip all around the opening. When the ends of the weatherstrip meet at the lower, center part of the window opening, there must be 12.7 mm (0.5 in.) of overlapping material.

*NOTE: The ends of the weatherstrip material need to be square-cut to assure a proper fit.*

- c. Lift both ends so that they meet squarely. Then, while holding the ends together, force them back over the lip of the opening.
2. Lubricate the groove of the weatherstrip where the glass is to be seated.
  3. Lower the glass into the groove along the bottom of the window opening.
  4. Two people should be used for glass installation. Have one person on the outside of the cab pushing in on the glass against the opening, while one person on the inside uses a soft flat tool (such as a plastic knife) to work the weatherstrip over the edge of the glass all the way around.

5. After the glass is in place, go around the weatherstrip and push in on locking lip (2) to secure the glass in the weatherstrip.
6. If the windshield was replaced, lower the windshield wiper arms back onto the glass.

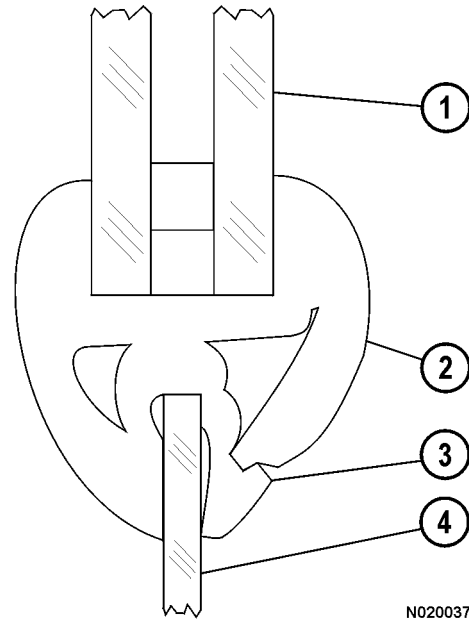


FIGURE 2-20. REAR WINDOW

- |                |                 |
|----------------|-----------------|
| 1. Glass       | 3. Weatherstrip |
| 2. Locking Lip | 4. Sheet Metal  |

# SECTION N3 CAB COMPONENTS INDEX

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# NOTES

# CAB COMPONENTS

## WINDSHIELD WIPERS

The windshield wipers are operated by a 24 volt electric motor. The wipers can be adjusted for a variable intermittent delay or a constant low or high speed by the switch mounted on the turn signal lever.

### WIPER MOTOR

#### Removal

1. Remove the large access panel from the front of the cab.
2. Disconnect the wiper motor harness connector.

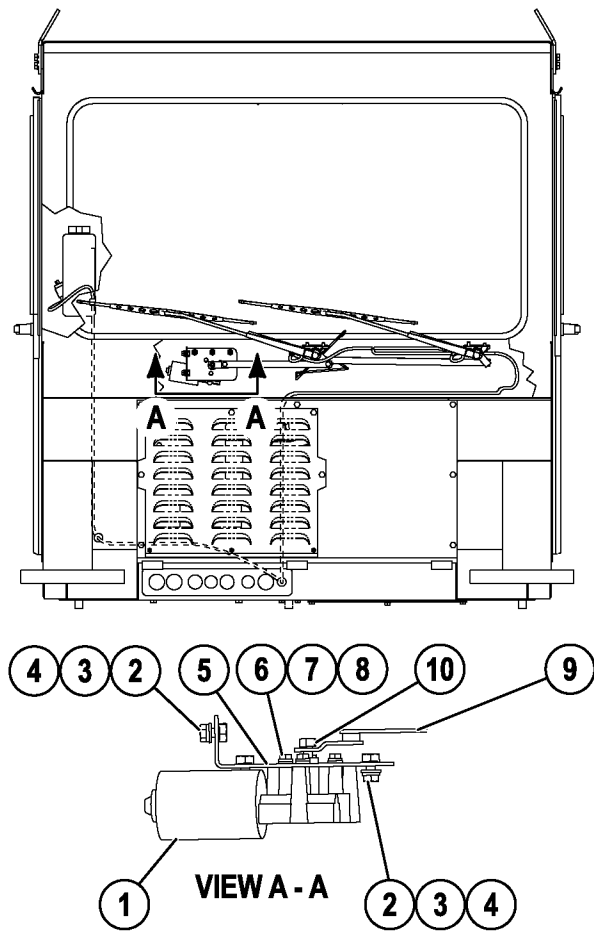


FIGURE 3-1. WINDSHIELD WIPER

- |                |                |
|----------------|----------------|
| 1. Wiper Motor | 6. Cap Screw   |
| 2. Cap Screw   | 7. Flat Washer |
| 3. Flat Washer | 8. Lock Washer |
| 4. Lock Washer | 9. Linkage     |
| 5. Plate       | 10. Nut        |

3. While holding the linkage stationary, remove nut (10, Figure 3-1) and disconnect the linkage from the motor.
4. Remove three cap screws (6) with washers attaching the wiper motor to plate (5). Remove the motor assembly.

#### Installation

1. Place wiper motor (1, Figure 3-1) into position on plate (5).
2. Install three cap screws (6) with flat washers (7) and lock washers (8). Tighten cap screws to **8-9 N·m (71-79 in. lbs)** torque.
3. Align the motor output shaft with the wiper linkage. Install nut (10) and while holding the linkage stationary, tighten nut to **22-24 N·m (16-18 ft. lbs)** torque.
4. Reconnect the wiper motor harness connector.
5. Verify the wipers operate properly and park in the proper position. Refer to Figure 3-3.

### WIPER ARM

#### Removal

1. Note the parked position of wiper arm (1, Figure 3-2).
2. Lift the wiper arm cover and remove nut (2) and washer (3).
3. Disconnect the washer hose, and remove the wiper arm.

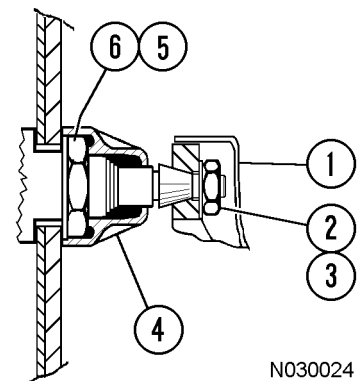


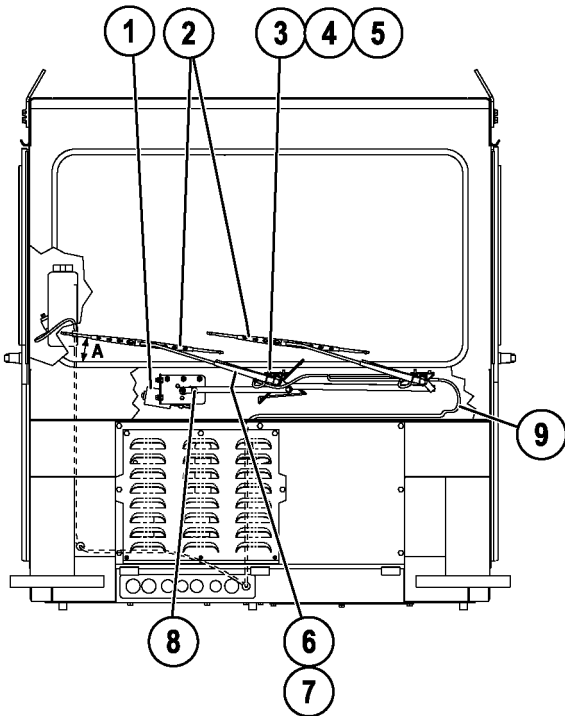
FIGURE 3-2. WIPER ARM DETAIL

- |                  |           |
|------------------|-----------|
| 1. Wiper Arm     | 4. Cap    |
| 2. Nut           | 5. Washer |
| 3. Spring Washer | 6. Nut    |



## Installation

1. Place wiper arm (1, Figure 3-2) into the position noted during removal. Install washer (3) and nut (2). Tighten the nut to **16-20 N·m (142-177 in. lbs)** torque. Close the cover.
2. Connect the washer hose to the wiper arm.
3. Ensure the wipers arms operate properly and park in the proper position after installation is complete. Refer to Figure 3-3.



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FIGURE 3-3. PARK POSITION

- |                       |                |
|-----------------------|----------------|
| A. Park Position (7°) | 5. Flat Washer |
| 1. Wiper Motor        | 6. Wiper Arm   |
| 2. Wiper Blade        | 7. Nozzle      |
| 3. Cap Screw          | 8. Retainer    |
| 4. Lock Washer        | 9. Hose        |

## WIPER LINKAGE

### Removal

1. Remove the wiper arms. Refer to Wiper Arm Removal in this section.
2. Remove wiper retainer (8, Figure 3-3) and disconnect the wiper linkage from the wiper motor drive arm.
3. Remove nut (6, Figure 3-2) and the washer (5) from each wiper shaft.
4. Remove cap screws (3, Figure 3-3) with washers.
5. Remove wiper assembly from cab.

### Installation

1. Place the wiper assembly into position in the wiper compartment.
2. Install cap screws (3, Figure 3-3) with lock washers (4) and flat washers (5) and tighten cap screws.
3. Install washer (5, Figure 3-2) and nut (6) on each wiper shaft and tighten finger-tight. Tighten the nuts to **18-20 N·m (160-177 in. lbs) torque. Do not overtighten. The threads on the shafts are easily stripped when improperly tightened.** Install cap (4) over nut (6).
4. Align the linkage and attach to the wiper motor drive arm using retainer (8, Figure 3-3).

*NOTE: When the motor is parked, the drive arm will be in the 3 o'clock position as shown in Figure 3-3.*

5. Install the wiper arms. Refer to Wiper Arm Installation. Ensure the wipers arms operate properly and park in the proper position after installation is complete.

## WINDSHIELD WASHER

### Operation

The windshield washer, mounted on the right side of the hydraulic components cabinet behind the cab, has a 3.8 liter (1 gal) plastic reservoir (1, Figure 3-4) with a 24 volt electric pump (2).

The washer is controlled by the windshield wiper switch mounted on the turn signal lever. When the switch is activated, washing solution is pumped through the outlet hose (3) and fed to a jet located in each of the windshield wiper arms.

### Service

If windshield washer maintenance is required, check the strainer opening for obstructions and inspect the hoses for damage. Check the voltage to the pump from the control switch. If the pump is inoperable, replace it with a new pump assembly.

*Note: The pump is only available as an assembly and cannot be repaired.*

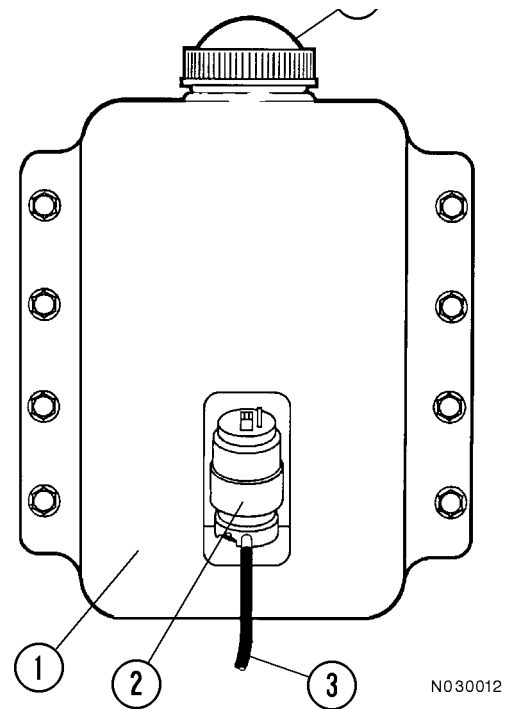


FIGURE 3-4. WINDSHIELD WASHER FLUID RESERVOIR AND PUMP

- |              |                |
|--------------|----------------|
| 1. Reservoir | 3. Outlet Hose |
| 2. Pump      | 4. Filler Cap  |

## OPERATOR SEAT

The operator's seat provides a fully adjustable cushioned ride for driver comfort and easy operation.

### Seat Belts

- On both driver and passenger seats, check the seat belt fabric, buckle, all belt retractors and hardware for damage or wear. Replace any worn or damaged parts immediately.
- Even if there are no signs of damage, replace both driver and passenger seat belts 5 years after seat belt manufacture, or every 3 years after start of use, whichever comes first. The passenger seat belt date of manufacture label is sewn into the seat belt near the buckle. The driver seat belt date of manufacture label is sewn into the shoulder harness belt, near the retractor end.

### Adjustment

The following adjustments must be made while sitting in the seat.

1. **Headrest:** (1, Figure 3-5) will move up, down, fore, or aft by moving headrest to desired position.
2. **Armrests:** Rotate adjusting knob until the armrest is in the desired position.
3. **Backrest:** Lift backrest adjustment handle (3) to release and select the backrest angle. Release the control handle to set.
4. **Seat Belt:** The operator must have seat belt (4) buckled in place and properly adjusted whenever vehicle is in motion.
5. **Seat Slope:** Lift seat slope lever (5) and hold to adjust the slope of the seat. Release the lever to lock the adjustment.
6. & 7. **Air Lumbar Support:** Each rocker switch (6 or 7) controls an air pillow. Switch (7) controls the lower air pillow and switch (6) controls the upper air pillow. To inflate, press on top of the rocker switch and hold for desired support, then release. To deflate, press on the bottom of the rocker switch and hold for desired support, then release. Adjust each pillow for desired support.
8. **Seat Suspension:** Press rocker switch (8) on top to increase ride firmness. Press on lower part of rocker switch to decrease ride firmness.
9. **Fore/Aft Location of Seat:** Lift fore and aft lever (9) and hold. Bend knees to move seat to a comfortable position. Release the control lever to lock the adjustment.
12. **Seat Height:** Lift lever (12) and hold to adjust the height of the seat.

Release the lever to lock the adjustment. Optimum ride height is obtained when the suspension is at the middle of its stroke. To properly adjust, raise the seat to its highest level. Then, lower the seat approximately 51 mm (2 in.) to the middle of the suspension stroke.

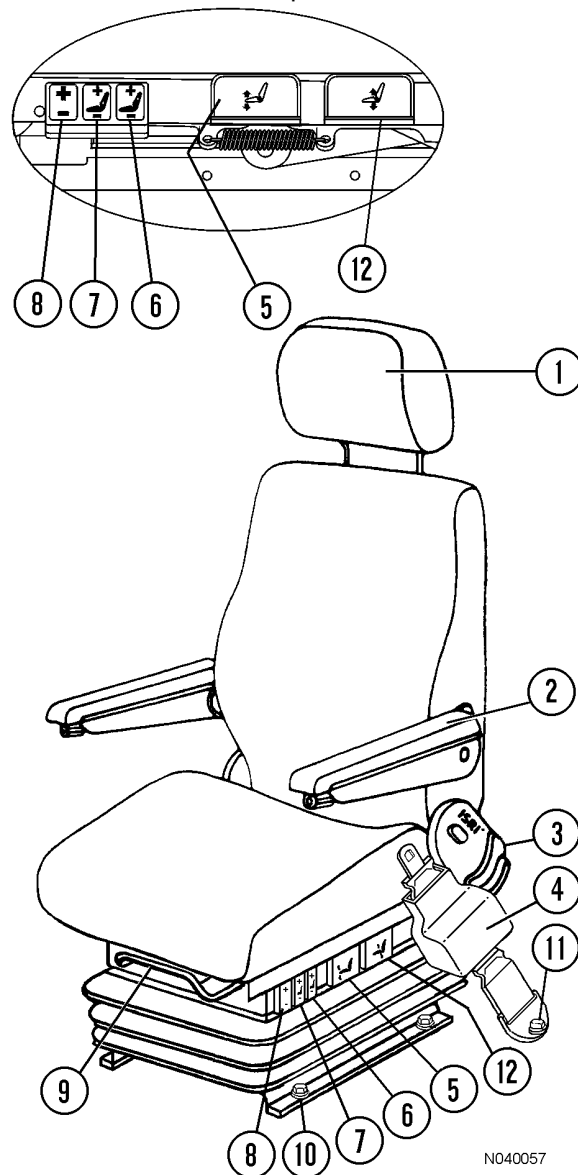


FIGURE 3-5. OPERATOR'S SEAT ADJUSTMENT CONTROLS

- |                                    |                            |
|------------------------------------|----------------------------|
| 1. Headrest                        | 8. Height Adjustment       |
| 2. Armrest Adjustment              | 9. Fore and Aft Adjustment |
| 3. Backrest Adjustment             | 10. Mounting Hardware      |
| 4. Seat Belt                       | 11. Seat Tether Cap screw  |
| 5. Seat Slope Adjustment           | 12. Seat Height Adjustment |
| 6. Upper Air Pillow Lumbar Support |                            |
| 7. Lower Air Pillow Lumbar Support |                            |

## Inspection

1. Inspect the seat for wear or damage. Replace worn or damaged parts on the seat.
2. Inspect the seat belts and buckles for proper operation. Also check for worn or frayed fabric. Replace all worn or damaged parts.
3. Inspect the date of manufacture on the seat belt. If the seat belt is over five years old, replace the seat belt. Or, if the seat belt has been in service for more than three years, replace the seat belt.
4. Inspect the seat tethers. If worn or damaged, replace the tethers.



***Do not remove the handrail from the deck to allow room to remove the operator seat from the cab. Remove the cab door to gain clearance for seat removal. There is a risk to personnel of falling if the handrail is removed.***

## Removal

1. Disconnect the door stop strap so the door can swing open wider. If necessary, remove the door from the cab to allow room to remove the seat. Refer to the index in this chapter for door removal instructions.
2. Remove cap screws (11) so that seat belt tethers can be removed from the cab floor.
3. Disconnect seat harness at the cab floor.
4. Remove mounting hardware (10) that secures the riser to the floor.
5. The seat weighs approximately 58 kg (128 lb). Remove the seat from the cab. Then use a suitable lifting device to lift the seat from the deck down to the ground. Do not lift the seat by using the arm rests, as they could be damaged.

## Installation

1. The seat weighs approximately 58 kg (128 lb). Use a suitable lifting device to lift the seat from the ground up to the deck. Place seat assembly in position in the cab.
2. Install mounting hardware (10) consisting of cap screws, flat washers and lock washers to secure the riser to the floor. Tighten the cap screws to the standard torque.
3. Connect seat harness.
4. Fasten seat belt tethers to the floor using cap screws (11) with lock washers and flat washers. Tighten the cap screws to **84 Nm (62 ft lb)**.
5. If the door was removed, install the door. Refer to the index in this chapter for door installation instructions.

## PASSENGER SEAT

### Inspection

1. Inspect the seat (4, Figure 3-6) for wear or damage. Replace worn or damaged parts on the seat.
2. Inspect the seat belts and buckles for proper operation. Also check for worn or frayed fabric. Replace all worn or damaged parts.
3. Inspect the date of manufacture on the seat belt. If the seat belt is over five years old, replace the seat belt. Or, if the seat belt has been in service for more than three years, replace the seat belt.
4. Inspect the seat tethers. If worn or damaged, replace the tethers.

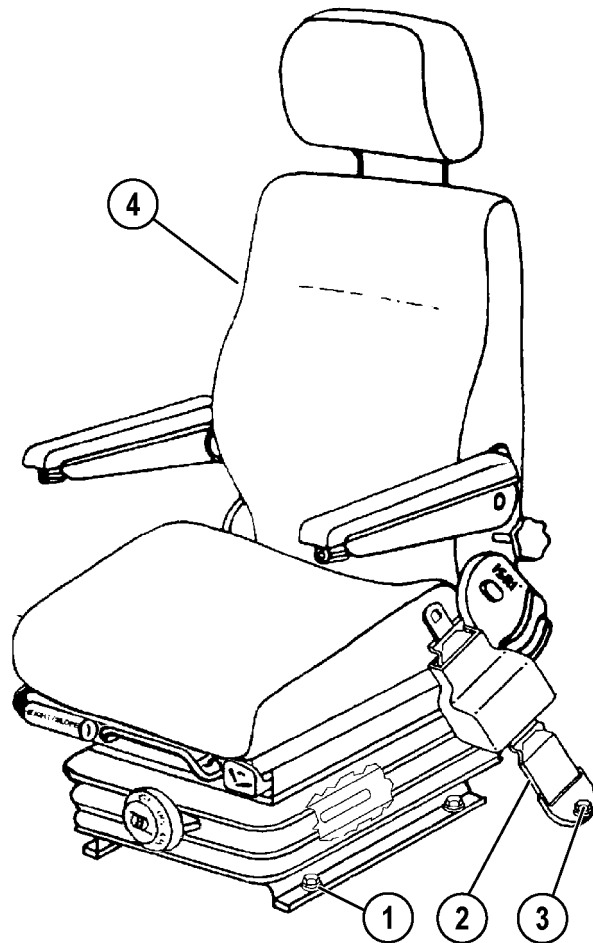
### Removal

1. Disconnect the door stop strap so the door can swing open wider.
2. Remove cap screws (3) so that seat belt tethers (2) can be disconnected from the cab floor.
3. Remove cap screws (1), flat washers and lock washers that secure the riser to the floor.
4. The seat weighs approximately 55 kg (121 lb). Remove the seat from the cab. Then use a suitable lifting device to lift the seat from the deck down to the ground.

### Installation

1. The seat weighs approximately 55 kg (121 lb). Use a suitable lifting device to lift the seat from the ground up to the deck. Place seat assembly in position in the cab.
2. Install cap screws (1), with flat washers and lock washers that secure the riser to the floor. Tighten the cap screws to the standard torque.

3. Fasten seat belt tethers (2) to the cab floor, using cap screws (3) with flat washers and lock washers. Tighten cap screws to **84 Nm (62 ft lb)**.
4. Install the door stop strap. The strap has three holes on each end for adjustment of how wide the door can swing open. Install the pin in the door stop strap where desired.
5. If disconnected, route power window wires through the loop in the bottom of the door stop strap.



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FIGURE 3-6. PASSENGER SEAT

- |              |              |
|--------------|--------------|
| 1. Cap Screw | 3. Cap Screw |
| 2. Tether    | 4. Seat      |

## SECTION N4

### HEATER / AIR CONDITIONER

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# HEATER/AIR CONDITIONER

## OPERATION

The heater/air conditioner compartment contains heater/air conditioner controls and some of the heater/air conditioner components, such as the blower motor assembly and the heater coil. Optimum cab air climate can be selected by using the following controls in various combinations.

### Fan Speed Control Knob

Fan speed control knob (1, Figure 4-1) is provided to control the cab air fan motor. The fan motor is a 3-speed motor (low, medium and high). Speeds are selected by rotating the control knob clockwise to the desired position. OFF is in the full counterclockwise position. The control knob and the air conditioner compressor switch must be switched ON for the air conditioner to function.

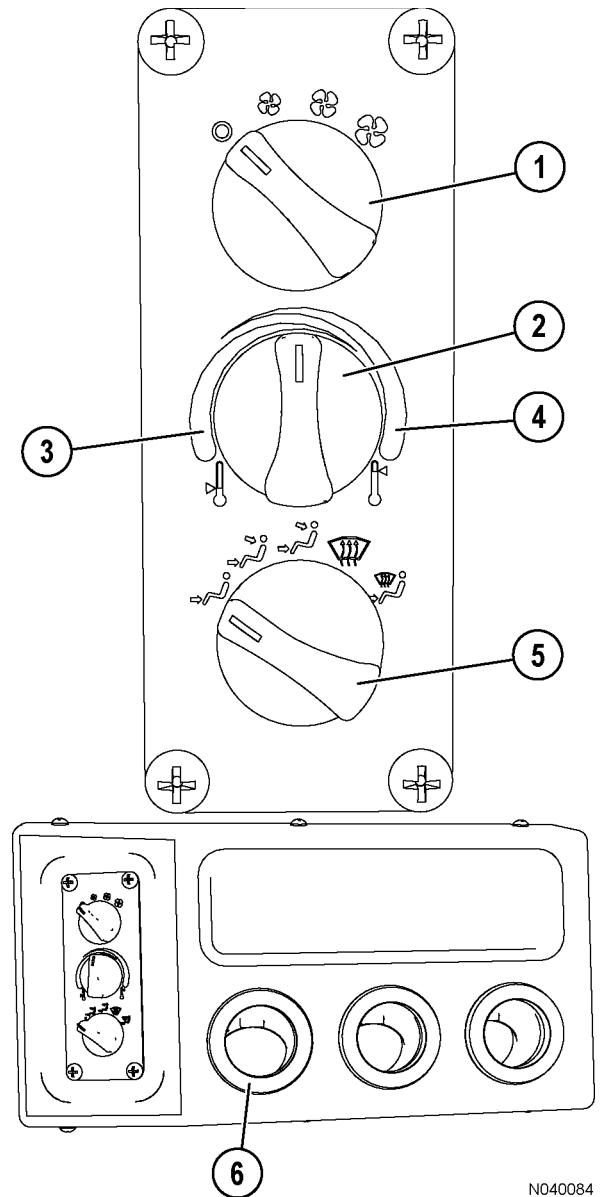
### Temperature Control Knob

Temperature control knob (2, Figure 4-1) allows the operator to select a comfortable air temperature.

The control knob determines the operation of the air conditioning and heater modes.

Rotating the control knob counterclockwise into blue zone (3) will cause the A/C compressor to operate and result in cooler air temperatures. Full counterclockwise position is the coldest air setting.

Rotating the control knob clockwise into red zone (4) will affect coolant flow through the heater core and result in warmer air temperatures. The full clockwise position is the warmest heater setting.



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FIGURE 4-1. A/C & HEATER CONTROLS

- |                             |                              |
|-----------------------------|------------------------------|
| 1. Fan Speed Control Knob   | 4. Red Zone                  |
| 2. Temperature Control Knob | 5. Air Flow Directional Knob |
| 3. Blue Zone                | 6. Vents                     |



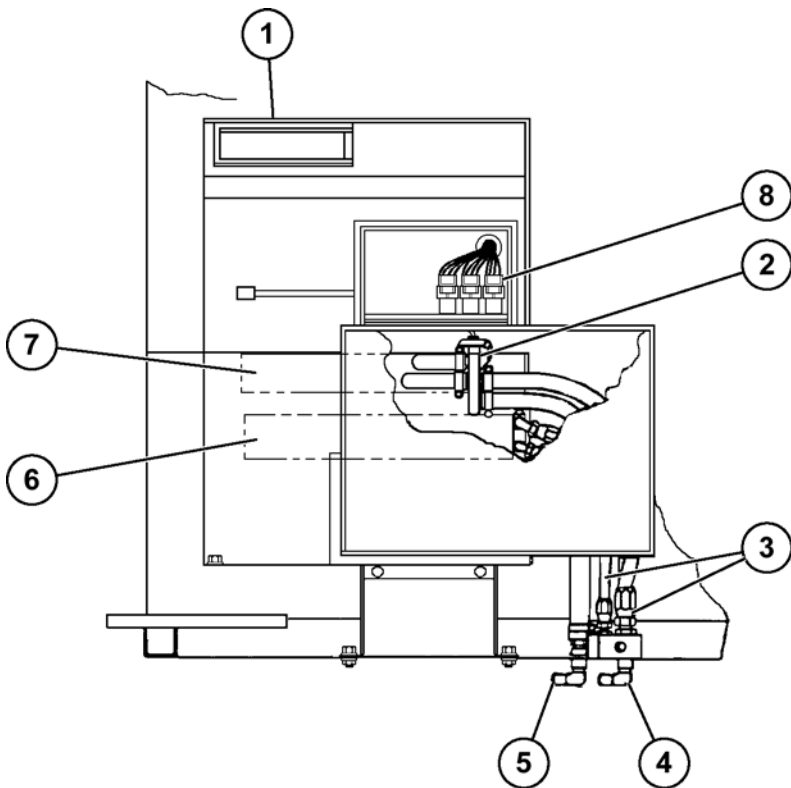
**Air Flow Directional Knob**

Air flow directional knob (5, Figure 4-1) controls the direction of airflow as follows:

|                                                                                   |                                                                               |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
|  | Provides airflow to the floor vents only.                                     |
|  | Provides airflow to the upper vents and the floor vents.                      |
|  | (Blue Icon) Provides dehumidified air to the upper vents and the floor vents. |
|  | Defrost - Provides dehumidified air to the upper vents only.                  |
|  | Defrost - Provides dehumidified air to the upper vents and the floor vents.   |

**Heater/Air Conditioner Vents**

Heater/air conditioner vents (6, Figure 4-1) may be rotated 360°. There are three vents in the heater/air conditioner compartment, four vents across the top of the instrument panel, and one vent each in the RH and LH instrument panels. There are also an additional four vents under the instrument panel. Air flow through the vents is controlled by manually opening, closing or turning the louvers.



**COMPONENTS**

Figure 4-2 and Figure 4-4 illustrate both the heater system and air conditioning system parts contained in the cab mounted enclosure. Refer to the air conditioning system topics later in this section for additional information regarding air conditioning system components, maintenance and repair.

**Fuse and Circuit Breaker**

Before attempting to troubleshoot the electrical circuit in the heater enclosure, turn the key switch ON and verify that the fuse at location FB1-1 (located in the auxiliary control cabinet) is not burned out. Refer to the electrical schematic in Section R for more detailed information.

**Relays**

Three relays (8, Figure 4-2) control the damper doors, A/C compressor clutch, temperature control switch, water (heater) valve and the actuator motors.

FIGURE 4-2. CAB HEATER/AIR CONDITIONER COMPONENTS

- 1. Enclosure
- 2. Water (Heater) Valve
- 3. A/C Refrigerant Hoses
- 4. Water Outlet (To Engine)
- 5. Water Inlet (From Engine Water Pump)
- 6. Evaporator Core
- 7. Heater Core
- 8. Relays

## Heater Core

Heater core (2, Figure 4-4) receives engine coolant through water valve (33) when heat is selected. If temperature control knob (2, Figure 4-1) is placed in between the red and blue area, or turned counterclockwise to the blue area, coolant flow should be blocked.

If temperature control knob (2) and water valve (33, Figure 4-4) appear to be working properly, yet no heat is apparent in heater core (2), the core may be restricted. Remove and clean or replace the core.

## Fan Motor And Speed Control

Fan speed is controlled by the position of the fan speed control knob (1, Figure 4-1).

If blower assembly (3, Figure 4-4) does not operate at any of the speed selections, verify that battery voltage is available at the switches and relay. Refer to the electrical schematic in Section R. If voltage is present, the blower assembly is probably defective and should be removed and replaced.

## Actuators

Two rotary actuator motors (9, Figure 4-4) are installed inside the heater housing and are used to actuate the damper doors for the following:

- Defroster outlet
- Bi-level or floor outlets

A failure to switch one of the above modes of operation may be caused by a faulty actuator.

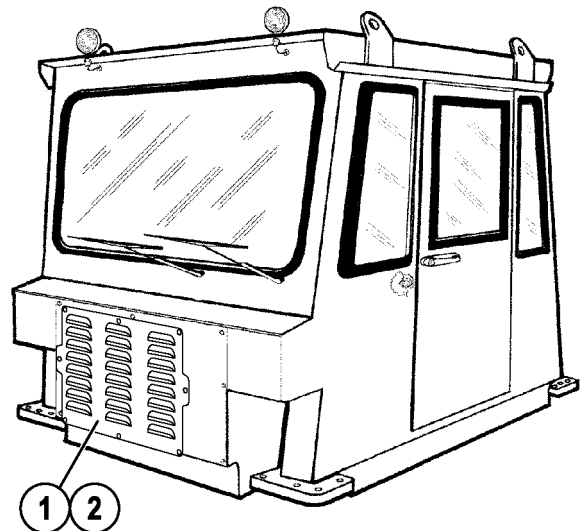
Visually inspect damper doors (35, 36) and the linkage for the function being diagnosed. Ensure that the damper door is not binding or obstructed, preventing movement from one mode to the other.

Verify that 24 VDC is present at the actuator when air flow directional knob (5, Figure 4-1) is moved through its various positions of operation.

If the correct voltages are present during operation of the air flow knob, disconnect the actuator from the damper door and verify that actuator force is comparable to a known (new) actuator. If it is not, install a new actuator.

## Cab Air Filter

Recirculation air filter (5, Figure 4-4) and fresh air filter (2, Figure 4-3) in the front access panel of the cab need periodic cleaning to prevent restrictions in air circulation. Restricted filters will decrease the performance of the heater and air conditioner. The recommended interval for cleaning and inspection is 250 hours, but in extremely dusty conditions the filters may need daily service and inspection, especially the outer panel filter on the cab shell. The filter elements should be cleaned with water and dried in a dust free environment before reinstallation. Replace the filter element every 2000 hours, or sooner if inspection indicates a clogged or damaged filter.



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FIGURE 4-3. CAB FILTER LOCATION

1. Access Cover

2. Cab Filter

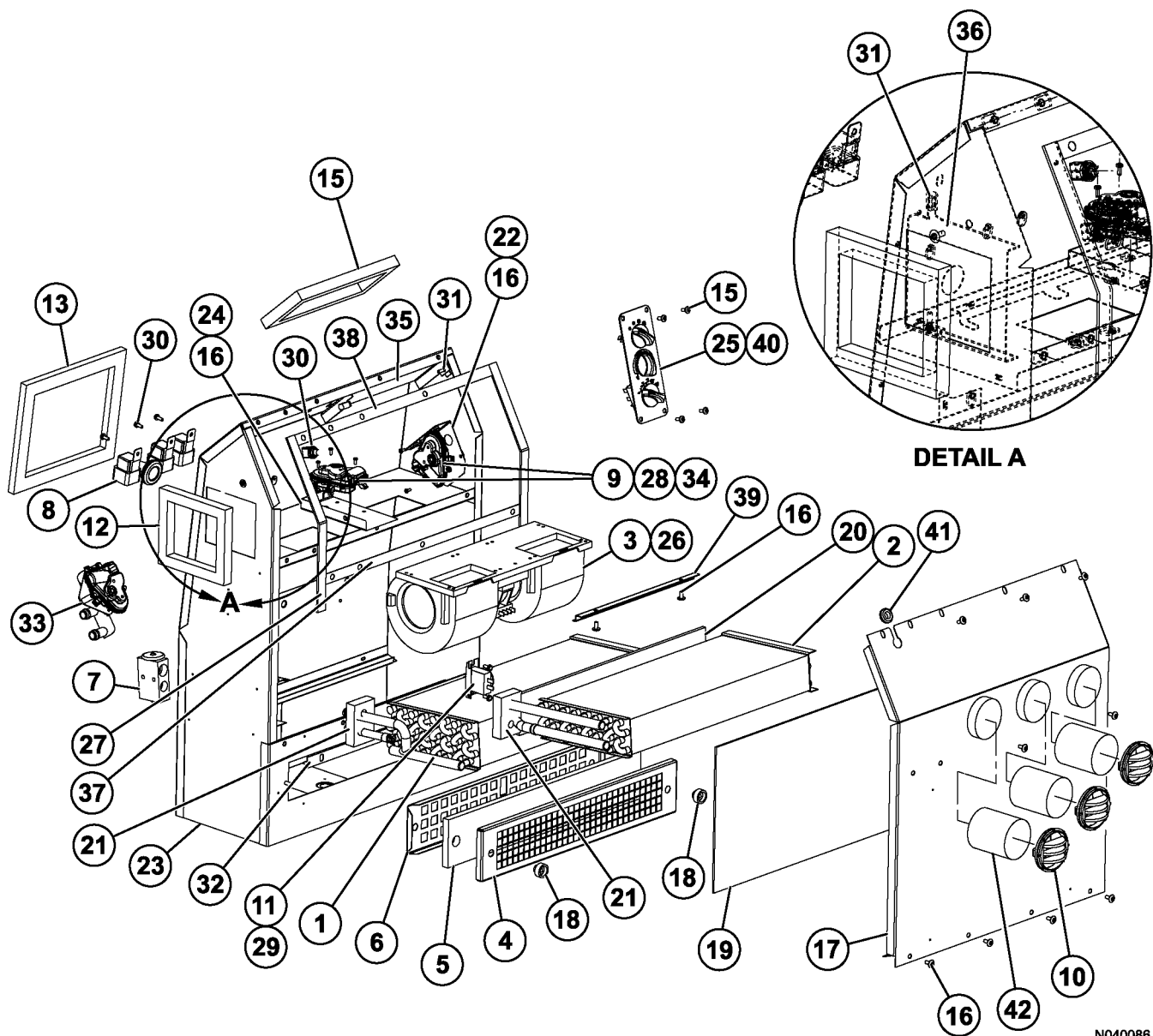
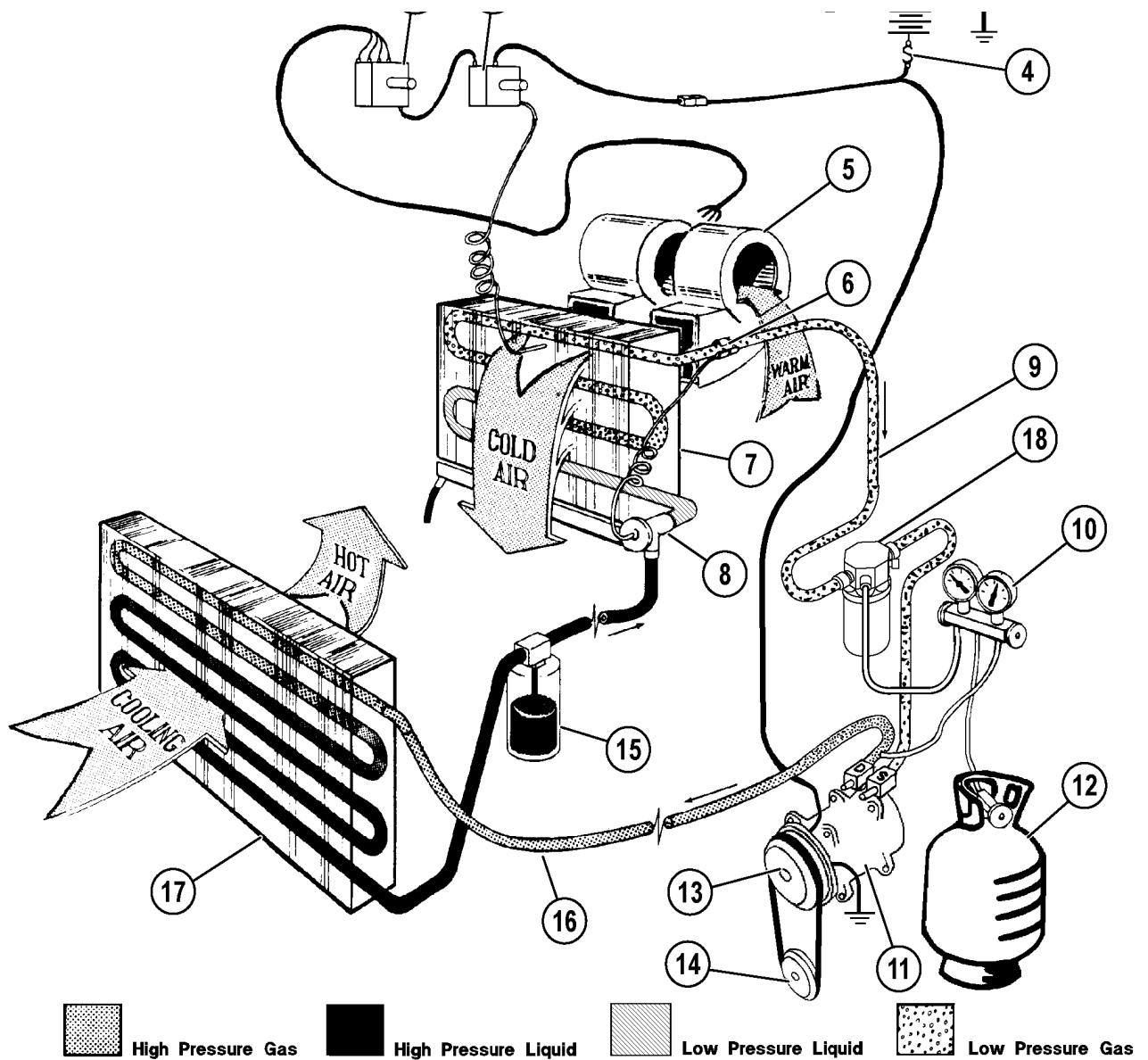


FIGURE 4-4. HEATER/AIR CONDITIONER ASSEMBLY

- |                               |                |                           |                        |
|-------------------------------|----------------|---------------------------|------------------------|
| 1. Evaporator Core            | 10. Louver     | 22. Bracket               | 33. Water Valve        |
| 2. Heater Core                | 11. Thermostat | 23. Case                  | 34. Drive Adapter      |
| 3. Blower Assembly            | 12. LH Seal    | 24. Bracket               | 35. Damper Door (Up)   |
| 4. Outer Recirculation Grille | 13. Front Seal | 25. Control Panel Harness | 36. Damper Door (Side) |
| 5. Recirculation Filter       | 14. Top Seal   | 26. Main Unit Harness     | 37. Punched Strip      |
| 6. Inner Recirculation Grille | 15. Screw      | 27. Seal                  | 38. Punched Strip      |
| 7. Expansion Block Valve      | 16. Screw      | 28. Screw                 | 39. Blower Retainer    |
| 8. Relay                      | 17. Cover      | 29. Screw                 | 40. Control Panel      |
| 9. Actuator                   | 18. Knob       | 30. Bulb                  | 41. Grommet            |
|                               | 19. Seal       | 31. Pivot Bushing         | 42. Duct Adapter       |
|                               | 20. Foam       | 32. Punched Setup         |                        |



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FIGURE 4-5. BASIC AIR CONDITIONING SYSTEM

- |                        |                            |                             |
|------------------------|----------------------------|-----------------------------|
| 1. Blower Switch       | 7. Evaporator              | 13. Magnetic Clutch         |
| 2. Thermostatic Switch | 8. Expansion Valve         | 14. Compressor Drive Pulley |
| 3. Battery Supply      | 9. Suction Line            | 15. Receiver-Drier          |
| 4. Circuit Breaker     | 10. Test Gauges & Manifold | 16. Discharge Line          |
| 5. Blower              | 11. Compressor             | 17. Condenser               |
| 6. Temperature Sensor  | 12. Refrigerant Container  | 18. Accumulator             |

## **ENVIRONMENTAL IMPACT OF AIR CONDITIONING**

Environmental studies have indicated a weakening of the earth's protective Ozone (O<sub>3</sub>) layer in the outer stratosphere. Chloro-flouro-carbon compounds (CFC's), such as R-12 refrigerant (Freon), commonly used in mobile equipment air conditioning systems, have been identified as a possible contributing factor of the Ozone depletion.

Consequently, legislative bodies in more than 130 countries have mandated that the production and distribution of R-12 refrigerant be discontinued after 1995. Therefore, a more "environmentally-friendly" hydro-flouro-carbon.

(HFC) refrigerant, commonly identified as HFC-134a or R-134a, is being used in most current mobile air conditioning systems. Additionally, the practice of releasing either refrigerant to the atmosphere during the charging/recharging procedure is prohibited.

These restrictions require the use of equipment and procedures which are significantly different from those traditionally used in air conditioning service techniques. The use of new equipment and techniques allows for complete recovery of refrigerants, which will not only help to protect the environment, but through the "recycling" of the refrigerant will preserve the physical supply, and help to reduce the cost of the refrigerant.

## **AIR CONDITIONING FOR OFF-HIGHWAY VEHICLES**

Mining and construction vehicles have unique characteristics of vibration, shock-loading, operator changes, and climate conditions that present different design and installation challenges for air conditioning systems. Off-highway equipment, in general, is unique enough that normal automotive or highway truck engineering is not sufficient to provide the reliability to endure the various work cycles encountered.

The cab tightness, insulation, and isolation from heat sources is very important to the efficiency of the system. It is advisable to close all vents, even the intakes of pressurization systems, when there are high humidity conditions.

The general cleanliness of the system and components is important. Dust and dirt collected in the condenser, evaporator, and air filters decreases the system's cooling capacity.

The compressor, condenser, evaporator units, hoses and fittings must be installed clean and tight and be capable of withstanding the strain and abuse they are subjected to on off-highway vehicles.

Equipment downtime costs are high enough to encourage service areas to perform preventive maintenance at regular intervals on vehicle air conditioning systems (cleaning, checking belt tightness, and checking operation of electrical components).

## **PRINCIPLES OF REFRIGERATION**

A brief review of the principles of air conditioning is necessary to relate the function of the components, the technique of troubleshooting, and the corrective action necessary to put the air conditioning unit into top operating efficiency.

Too frequently, the operator and the service technician overlook the primary fact that no air conditioning system will function properly unless it is operated within a completely controlled cab environment. The circulation of air must be a directed flow. The cab must be sealed against seepage of ambient air. The cab interior must be kept free of dust and dirt which, if picked up in the air system, will clog the intake side of the evaporator coil.

### **Air Conditioning**

Air conditioning is a form of environmental control. As applied to the cab, it refers to the control of temperature, humidity, cleanliness, and circulation of air. In the broad sense, a heating unit is as much an air conditioner as is a cooling unit. The term "air conditioner" is commonly used to identify an air cooling unit. To be consistent with common usage, the term "air conditioner" will refer to the cooling unit utilizing the principles of refrigeration, sometimes referred to as the evaporator unit.

## Refrigeration - The Act Of Cooling

- There is no process for producing cold; there is only heat removal.
- Heat always travels toward cooler temperatures. This principle is the basis for the operation of a cooling unit. As long as one object has a temperature lower than another, this heat transfer will occur.
- Temperature is the measurement of the intensity of heat in degrees. The most common measuring device is the thermometer.
- All liquids have a point at which they will turn to vapor. Water boiling is the most common example of heating until vapor is formed. Boiling is a rapid form of evaporation. Steam is a great deal hotter than boiling water. The water will not increase in temperature once brought to a boil. The heat energy is used in the vaporization process. The boiling point of a liquid is directly affected by pressure. By changing pressure, we can control the boiling point and temperature at which a vapor will condense. When a liquid is heated and vaporizes, the gas will absorb heat without changing pressure.
- Reversing the process, when heat is removed from water vapor, it will return to the liquid state. Heat from air moves to a cooler object. Usually the moisture in the cooled air will condense on the cooler object.
- Refrigerant - Only R-134a should be used in the new mobile systems which are designed for this refrigerant.

## The Refrigeration Cycle

In an air conditioning system, the refrigerant is circulated under pressure through the five major components in a closed circuit. At these points in the system, the refrigerant undergoes predetermined pressure and temperature changes.

The compressor (refrigerant pump) takes in low pressure heat laden refrigerant gas through the suction valve (low side), and as its name indicates, pressurizes the heat laden refrigerant and forces it through the discharge valve (high side) on to the condenser.

Ambient air, passing through the condenser removes heat from the circulating refrigerant resulting in the conversion of the refrigerant from gas to liquid.

The liquid refrigerant moves on to the receiver drier where impurities are filtered out, and moisture removed. This component also serves as the temporary storage unit for some liquid refrigerant.

The liquid refrigerant, still under high pressure, then flows to the expansion valve. This valve meters the amount of refrigerant entering the evaporator. As the refrigerant passes through the valve, it becomes a low temperature, low pressure liquid and saturated vapor. This causes the refrigerant to become cold.

The remaining low pressure liquid immediately starts to boil and vaporize as it approaches the evaporator, adding to the cooling. The hot, humid air of the cab is pulled through the evaporator by the evaporator blower. Since the refrigerant is colder than the air, it absorbs the heat from the air producing cool air which is pushed back into the cab. The moisture in the air condenses upon movement into the evaporator and drops into the drain pan from which it drains out of the cab.

Refrigerant leaving the evaporator enters the accumulator. The accumulator functions as a sump for liquid refrigerant in the system. Because of its design, the accumulator only allows vaporized refrigerant to return to the compressor, preventing compressor slugging from occurring. Desiccant is located at the bottom of the accumulators to remove moisture that is trapped in the system.

The cycle is completed when the heated low pressure gas is again drawn into the compressor through the suction side.

This simplified explanation of the principles of refrigeration does not call attention to the fine points of refrigeration technology. Some of these will be covered in the following discussions of the components, controls, and techniques involved in preparing the unit for efficient operation.

## **AIR CONDITIONER SYSTEM COMPONENTS**

### **Compressor (Refrigerant Pump)**

The compressor is where the low pressure side of the system changes to high pressure. It concentrates the refrigerant returning from the evaporator (low side), creating high pressure and a temperature much higher than the outside air temperature. The high temperature differential between the refrigerant and the outside air is necessary to aid rapid heat flow in the condenser from the hot refrigerant gas to much cooler outside air.

To create high pressure concentration, the compressor draws in refrigerant from the evaporator through the suction valve and, during compression strokes, forces it out through the discharge valve to the condenser. The pressure from the compressor action moves the refrigerant through the condenser, receiver-drier and connecting hoses to the expansion valve.

The compressor is driven by the engine through a V-belt driving an electrically operated clutch mounted on the compressor drive shaft.

### **Service Valves**

Quick-connect hose end fittings with integral service valves attach to system service ports for servicing the unit. A manifold gauge set is connected into the system at the service valve ports and all procedures, such as discharging, evacuating and charging the system, are performed through the service valves.

### **Condenser**

The condenser receives the high pressure, high-temperature refrigerant vapor from the compressor and condenses it to high pressure, hot liquid.

It is designed to allow heat movement from the hot refrigerant vapor to the cooler outside air. The cooling of the refrigerant changes the vapor to liquid. Heat exchange is accomplished using cooler air flowing through the condenser. Condenser cooling is achieved by air flowing from the radiator fan along with ram air provided by vehicle movement. The radiator fan moves more than 50% of condenser air flow unless travel speed is at least 40 kph (25 mph).

Ram air condensers depend upon the vehicle movement to force a large volume of air past the fins and tubes of the condenser. The condenser is usually located in front of the radiator or on the roof of the truck.

Condensing of the refrigerant is the change of state of the refrigerant from a vapor to a liquid. The action is affected by the pressure of the refrigerant in the coil and air flow through the condenser. Condensing pressure in an air conditioning system is the controlled pressure of the refrigerant which affects the temperature at which it condenses to liquid, giving off large quantities of heat in the process. The condensing point is sufficiently high to create a wide temperature differential between the hot refrigerant vapor and the air passing over the condenser fins and tubes. This difference permits rapid heat transfer from the refrigerant to ambient air.

### **Receiver-Drier**

The receiver-drier is an important part of the air conditioning system. The drier receives the liquid refrigerant from the condenser and removes any moisture and foreign matter present which may have entered the system. The receiver section of the tank is designed to store extra refrigerant until it is needed by the evaporator. The storage of this refrigerant is temporary and is dependent on the demand of the expansion valve.

A desiccant is a solid substance capable of removing moisture from gas, liquid or solid. It is held in place within the receiver between two screens, which also act as strainers.

The receiver-drier is also equipped with a sight glass and a moisture indicator. The sight glass can give a good indication of the charge of the system. If the sight glass is not clear, the system is low on refrigerant.

The moisture indicator is a device to notify service personnel that the drier is full of moisture and must be replaced. The indicator is blue when the component is free from moisture. When the indicator turns beige or tan, the drier must be replaced.

## Expansion Block Valve

The expansion block valve controls the amount of refrigerant entering the evaporator coil. Both internally and externally equalized valves are used.

The expansion valve is located near the inlet of the evaporator and provides the functions of throttling, modulating, and controlling the liquid refrigerant to the evaporator coil.

The refrigerant flows through a restriction creating a pressure drop across the valve. Since the expansion valve also separates the high side of the system from the low side, the state of the refrigerant entering the valve is warm to hot high pressure liquid; exiting it is low pressure liquid and gas. The change to low pressure allows the flowing refrigerant to immediately begin changing to gas as it moves toward the evaporator. This produces the desired cooling effect.

The amount of refrigerant metered into the evaporator varies with different heat loads. The valve modulates from wide open to the nearly closed position, seeking a point between for proper metering of the refrigerant.

As the load increases, the valve responds by opening wider to allow more refrigerant to pass into the evaporator. As the load decreases, the valve reacts and allows less refrigerant into the evaporator. It is this controlling action that provides the proper pressure and temperature control in the evaporator.

This system uses an internally equalized, block type expansion valve. With this type valve, the refrigerant leaving the evaporator coil is also directed back through the valve so the temperature of the refrigerant is monitored internally rather than by a remote sensing bulb. The expansion valve is controlled by both the temperature of the power element bulb and the pressure of the liquid in the evaporator.

*NOTE: It is important that the sensing bulb, if present, is tight against the output line and protected from ambient temperatures with insulation tape.*

## Accumulator

As the accumulator receives vaporized refrigerant from the evaporator, moisture and/or any residual liquid refrigerant is collected at the bottom of the component. The moisture is absorbed by the desiccant where it is safely isolated from the rest of the system.

The storage of the liquid refrigerant is temporary. When the liquid vaporizes into a gas it will be pulled from the bottom of the accumulator into the compressor. This process not only allows the accumulator to act as a storage device, but also protects the compressor from liquid slugging.

The low side service port is also located on the accumulator.

## Evaporator

The evaporator cools and dehumidifies the air before it enters the cab. Cooling a large area requires that large volumes of air be passed through the evaporator coil for heat exchange. Therefore, a blower becomes a vital part of the evaporator assembly. It not only draws heat laden air into the evaporator, but also forces this air over the evaporator fins and coils where the heat is surrendered to the refrigerant. The blower forces the cooled air out of the evaporator into the cab.

Heat exchange, as explained under condenser operation, depends upon a temperature differential of the air and the refrigerant. The greater the temperature differential, the greater will be the amount of heat exchanged between the air and the refrigerant. A high heat load condition, as is generally encountered when the air conditioning system is turned on, will allow rapid heat transfer between the air and the cooler refrigerant.

The change of state of the refrigerant in and going through the evaporator coil is as important as that of the air flow over the coil.

All or most of the liquid that did not change to vapor in the expansion valve or connecting tubes boils (expands) and vaporizes immediately in the evaporator, becoming very cold. As the process of heat loss from the air to the evaporator coil surface is taking place, any moisture (humidity) in the air condenses on the cool outside surface of the evaporator coil and is drained off as water.

At atmospheric pressure, refrigerant boils at a point lower than water freezes. Therefore, the temperature in the evaporator must be controlled so that the water collecting on the coil surface does not freeze on and between the fins and restrict air flow. The evaporator temperature is controlled through pressure inside the evaporator, and temperature and pressure at the outlet of the evaporator.



## **ELECTRICAL CIRCUIT**

The air conditioner's electrical circuit is fed from an accessory circuit and is fused with a 30-ampere circuit breaker.

The blower control is a switch which provides a range of blower speeds from fast to slow. When the blower switch is turned on, current is available at the compressor clutch. Once the blower is turned on, fan speeds may be changed without affecting the thermostat sensing level.

The thermostat reacts to changing temperatures which cause electrical contacts to open and close. The thermostat has a capillary tube extended into the evaporator coil to sense temperature.

When the contacts are closed, current flows to the clutch field and energizes the clutch, causing the swash plate inside the compressor to turn which starts the refrigeration cycle. When the temperature of the evaporator coil drops to a predetermined point, the contacts open and the clutch disengages.

When the clutch is disengaged, the blower remains at the set speed. After the evaporator temperature rises about twelve degrees above the cutout point, the contacts in the thermostat close and the refrigeration cycle resumes.

### **Thermostat**

An electromagnetic clutch is used on the compressor to provide a means of constant temperature control of the cab. The clutch is controlled by a thermostat in the evaporator which is set initially by the driver to a predetermined point. Evaporator temperature is then maintained by the cycling action of the clutch.

The thermostat is simply a thermal device which controls an electrical switch. When warm, the switch is closed; when cold, it is open. Most thermostats have a positive OFF position as a means to turn the clutch off regardless of temperature.

The bellows type thermostat has a capillary tube connected to it which is filled with refrigerant. The capillary tube is attached to the bellows inside of the thermostat. Expansion of the gases inside the capillary tube exerts pressure on the bellows, which in turn closes the contacts at a predetermined temperature.

## **Compressor Clutch**

An electromagnetic clutch is used in conjunction with the thermostat to disengage the compressor when it is not needed, such as when a defrost cycle is indicated in the evaporator, or when the system or blower is turned off.

The stationary field clutch is the most desirable type since it has fewer parts to wear. The field is mounted to the compressor by mechanical means depending on the type field and compressor. The rotor is held on the armature by a bearing and snap rings. The armature is mounted on the compressor body.

When no current is fed to the field, there is no magnetic force applied to the clutch and the rotor is free to rotate on the armature, which remains stationary on the crankshaft.

When the thermostat or switch is closed, current is fed to the field. This sets up a magnetic force between the field and armature, pulling it into the rotor. When the armature becomes engaged with the rotor, the complete unit rotates while the field remains stationary. This causes the compressor crankshaft to turn, starting the refrigeration cycle.

When the switch or thermostat is opened, current is cut off. The armature snaps back out and stops while the rotor continues to turn. Pumping action of the compressor is stopped until current is again applied to the field. In addition, safety switches in the compressor clutch electrical circuit control clutch operation, disengaging the clutch if system pressures are abnormal.

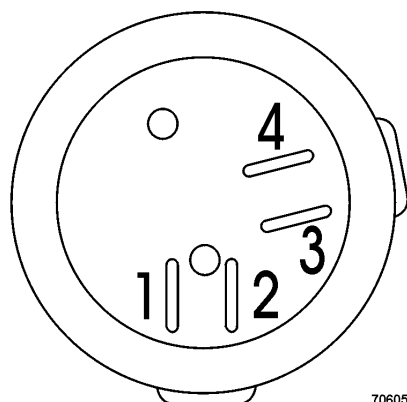
## Trinary™ Switch

This switch is mounted on the receiver-drier and has three functions, as implied by the name:

1. Disengage the compressor clutch when system pressure is too high.
2. Disengage the compressor clutch when system pressure is too low.
3. Engage and disengage the radiator fan drive clutch during normal variation of system pressure.

The Trinary™ switch performs three distinct functions to monitor and control refrigerant pressure in the system. This switch is installed on the receiver-drier. The switch functions are:

Terminals 1 and 2 are connected internally through two, normally closed pressure switches in series, the low pressure switch and the high pressure switch.



Terminals 3 and 4 are connected internally through a normally open switch that is used to control the clutch that drives the radiator fan. This switch closes and causes the cooling fan clutch to engage when system pressure rises to 1 379 - 1 586 kPa (200 - 230 psi). When pressure falls to 965 - 1 344 kPa (140 - 195 psi), the switch contacts open and the cooling fan clutch disengages.

- Low Pressure - This switch opens and disengages the compressor clutch if system pressure drops into the 103 - 207 kPa (15 - 30 psi) range. When pressure rises above 276 kPa (40 psi), the switch contacts close, and the clutch engages the compressor. Since temperature has a direct effect on pressure, if the ambient temperature is too cold, system pressure will drop below the low range and the pressure switch will disengage the clutch.

- Fan Clutch - The mid-range function actuates the engine fan clutch, if installed.
- High Pressure - This switch opens and disengages the compressor clutch if system pressure rises above the 2 068 - 2 413 kPa (300 - 350 psi) range. After system pressure drops to 1 448 - 1 724 kPa (210 - 250 psi), the switch contacts will close and the clutch will engage.

The switch functions will automatically reset when system pressure returns to normal.

|               | OPENS                                                    | CLOSES                                                |
|---------------|----------------------------------------------------------|-------------------------------------------------------|
| Low Pressure  | 103 - 207 kPa<br>(15 - 30 psi)<br>descending pressure    | 276 kPa<br>(40 psi)<br>rising pressure                |
| High Pressure | 2068 - 2413 kPa<br>(300 - 350 psi)                       | 1448 - 1724 kPa<br>(210 - 250 psi)                    |
| Fan Clutch    | 241 - 414 kPa<br>(35 - 60 psi)<br>below closing pressure | 1379 - 1586 kPa<br>(200 - 230 psi)<br>rising pressure |

## ⚠ IMPORTANT ⚠

*The pressures listed above are typical of pressures at the receiver-drier. Due to normal system flow losses and the distance between the service port and the receiver-drier, it is expected that actual system pressure displayed on the gauge will normally be approximately 138 kPa (20 psi) higher. This factor should be observed when checking for proper operation of the switch.*

*NOTE: One other pressure controlling device is installed within the compressor. A mechanical relief valve is located on the back of the compressor. The relief valve will open at 3 447 - 3 792 kPa (500 - 550 psi). The purpose of this valve is to protect the compressor in the event that pressure should be allowed to rise to that level. Damage to the compressor will occur if pressure exceeds 550 psi.*

## AIR CONDITIONING SYSTEM SERVICING WARNINGS

Servicing an air conditioning system really means closely monitoring refrigerant flow. For this reason, the following procedures deal extensively with the proper use, handling, care and safety factors involved in the R-134a refrigerant quality and quantity in an air conditioning system.

Because the refrigerant in an air conditioning system must remain pressurized and sealed within the unit to function properly, safety is a major consideration when anything causes this pressurized, sealed condition to change. The following warnings are provided here to alert service personnel to their importance before learning the correct procedures. Read, remember, and observe each warning before beginning actual system servicing.

*NOTE: If the mine operates a fleet with some trucks using R-12 and others using R-134a refrigerant, it is essential that servicing tools that come into contact with the refrigerant (gauge sets, charging equipment, recycle/recovery equipment etc.) be dedicated to one type of refrigerant only in order to prevent cross contamination.*

### CAUTION

*Trucks operating in cold weather climates must continue to keep the A/C system charged during cold weather months. Keeping the system charged helps prevent moisture intrusion into system oil and desiccants.*

### WARNING

*Federal regulations prohibit venting R-12 and R-134a refrigerant into the atmosphere. An SAE and UL approved recovery/recycle station must be used to recover refrigerant from the A/C system. Refrigerant is stored in a container on the unit for recycling, reclaiming, or transporting. In addition, technicians servicing A/C systems must be certified they have been properly trained to service the system.*

*Although accidental release of refrigerant is a remote possibility when proper procedures are followed, the following warnings must be observed when servicing A/C systems:*

- *Provide appropriate protection for your eyes (goggles or face shield) when working around refrigerant.*
- *A drop of the liquid refrigerant on your skin will produce frostbite. Wear gloves and use extreme caution when handling refrigerant.*
- *If even the slightest trace of refrigerant enters your eye, flood the eye immediately with cool water and seek medical attention as soon as possible.*
- *Ensure that there is sufficient ventilation whenever refrigerant is being discharged from a system. Keep in mind that refrigerant is heavier than air and will fall to low-lying areas.*
- *When exposed to flames or sparks, the components of refrigerant change and become deadly phosgene gas. This poison gas will damage the respiratory system if inhaled. NEVER smoke in an area where refrigerant is used or stored.*
- *Never direct a steam cleaning hose or torch in direct contact with components in the air conditioning system. Localized heat can raise the pressure to a dangerous level.*
- *Do not heat or store refrigerant containers above 49° C (120° F).*
- *Do not flush or pressure test the A/C system using shop air or another compressed air source. Certain mixtures of air and R-134a refrigerant are combustible when slightly pressurized. Shop air supplies also contain moisture and other contaminants that could damage system components.*

## SERVICE TOOLS AND EQUIPMENT

### Recovery/Recycle Station

Whenever refrigerant must be removed from the system, a dual purpose station, as shown in Figure 4-6, performs both recovery and recycle procedures which follows the new guidelines for handling used refrigerant. The recovered refrigerant is recycled to reduce contaminants, and can then be reused in the same machine or fleet.

To accomplish this, the recovery/recycle station separates the oil from the refrigerant and filters the refrigerant multiple times to reduce moisture, acidity, and particulate matter found in a used refrigerant.

*NOTE: To be re-sold, the gas must be "reclaimed" which leaves it as pure as new, but requires equipment normally too expensive for all but the largest refrigeration shops.*

Equipment is also available to just remove or extract the refrigerant. Extraction equipment does not clean the refrigerant; it is used to recover the refrigerant from an A/C system prior to servicing.

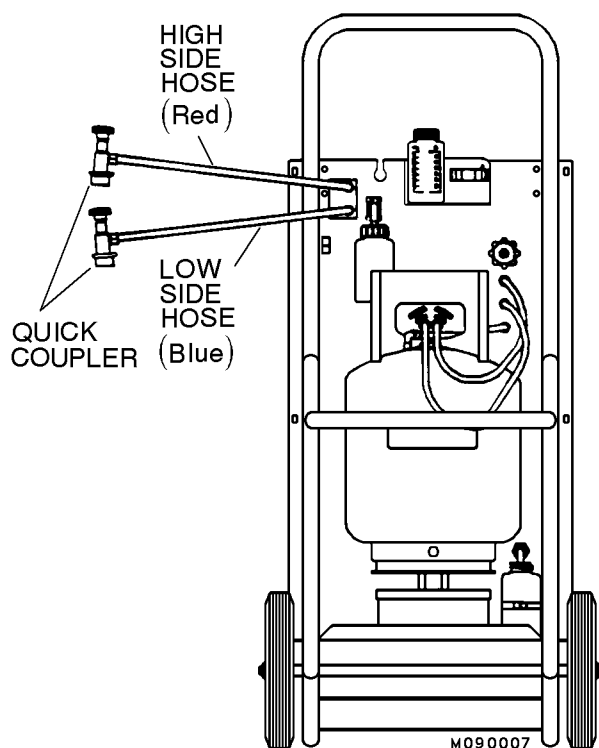


FIGURE 4-6. RECOVERY/RECYCLE STATION

## CAUTION

**Mixing different types of refrigerant will damage equipment. Dedicate one recovery/recycle station to each type of refrigerant processing to avoid equipment damage. DISPOSAL of the gas removed requires laboratory or manufacturing facilities.**

Test equipment is available to confirm that the refrigerant in the system is actually the type intended for the system and has not been contaminated by a mixture of refrigerant types.

Recycling equipment must meet certain standards as published by the Society of Automotive Engineers (SAE) and carry a UL approved label. The basic principals of operation remain the same for all machines, even if the details of operation differ somewhat.

### Leak Detector

The electronic leak detector (Figure 4-7) is very accurate and safe. It is a small hand-held device with a flexible probe used to seek refrigerant leaks. A buzzer, alarm or light will announce the presence of even the smallest leak.

Some leak detectors are only applicable to one type of refrigerant. Ensure that the leak detector being used applies to the refrigerant in the system.

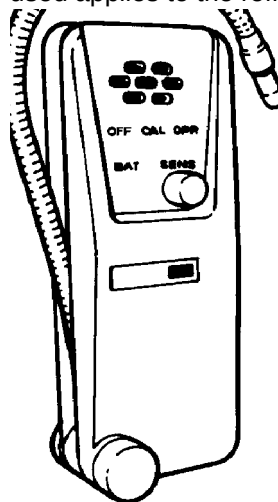


FIGURE 4-7. TYPICAL ELECTRONIC LEAK DETECTOR

## Manifold Gauge Set

A typical manifold gauge set (Figure 4-8) has two screw type hand valves to control access to the system, two gauges and three hoses. The gauges are used to read system pressure or vacuum. The manifold and hoses are for access to the inside of an air conditioner, to remove air and moisture, and to put in, or remove, refrigerant from the system. Shutoff valves are required within 305 mm (12 in.) of the hose end(s) to minimize refrigerant loss.

A gauge set for R-134a will have a blue hose with a black stripe for the low side, a red hose with a black stripe for the high side, and a yellow hose with a black stripe for the utility (center) hose. The hoses use a 1/2 in. ACME female nut on the gauge end. Special quick disconnect couplings are normally combined with a shutoff valve on the high and low side hoses. The free end of the center hose contains a 1/2 in. ACME female nut and a shutoff device within 305 mm (12 in.) of the hose end. These special hoses and fittings are designed to minimize refrigerant loss and to preclude putting the wrong refrigerant in a system.

**NOTE:** When hose replacement becomes necessary, the new hoses must be marked "SAE J2916 R-134a".

Functions of the manifold gauge set are included in many of the commercially available recovery or recovery/recycle stations.

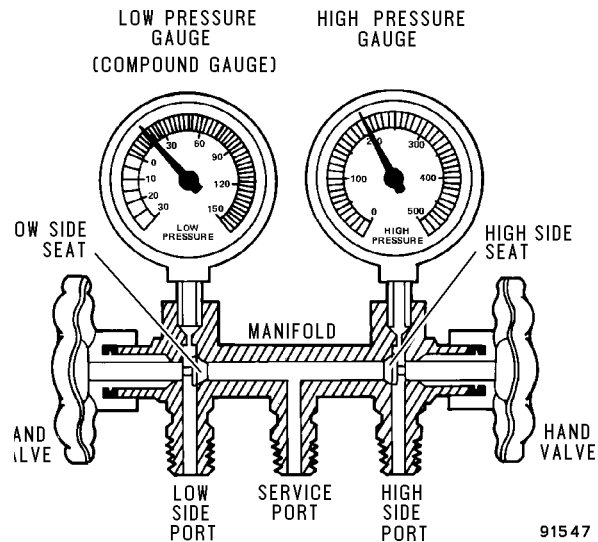


FIGURE 4-8. MANIFOLD GAUGE SET

### Low Side Gauge

The low side gauge, registers both vacuum and pressure. The vacuum side of the scale is calibrated from 0 to 30 inches of mercury (in. Hg). The pressure side of the scale is calibrated to 150 psi.



**Never open the hand valve to the high side when the air conditioning system is operating. High side pressure, if allowed, may rupture charging containers and potentially cause personal injury.**

### High Side Gauge

The high side gauge is used to measure pressure only on the discharge side of the compressor. The scale is calibrated to 500 psi.

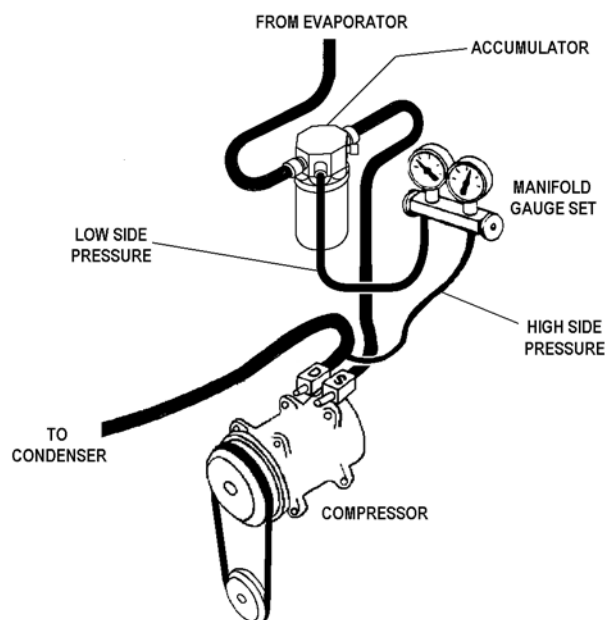
## Installing Manifold Gauge Set

Before attempting to service the air conditioning system, a visual inspection of both the engine and air conditioning system components is recommended. Particular attention should be given to the belts, hoses, tubing and all attaching hardware as well as the radiator cap, fan clutch, and thermostat. Inspect both the condenser and the radiator for any obstructions or potential contamination. Minimize all the possibilities for error or malfunction of components in the air conditioning system.



***Shut off engine. DO NOT attempt to connect service equipment when the engine is running.***

1. Ensure all valves on the manifold are closed all the way (turn them clockwise).
2. Check the hose connections on the manifold for tightness.
3. Locate the low and high side system service fittings and remove their protective caps.



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FIGURE 4-9. SERVICE HOSE HOOK-UP

4. Connect the two service hoses from the manifold to the correct service valves on the compressor and accumulator, as shown in Figure 4-9. (High side to compressor discharge valve and low side to accumulator.) Do not open the service valves at this time.

This gauge hook-up process will be the same, regardless of the gauge set being installed. Whether it is a recovery station or individual gauges, the connections are the same. The procedures performed next will vary depending what type of equipment is being used. If a recovery/recycling station is being used, complete servicing can be accomplished. Using only a set of gauges will limit the servicing to only adding refrigerant or observing pressures.

## Purging Air From Service Hoses

The purpose of this procedure is to remove all the air trapped in the hoses prior to actual system testing. Environmental regulations require that all service hoses have a shutoff valve within 12 inches of the service end. These valves are required to ensure only a minimal amount of refrigerant is lost to the atmosphere. R-134a gauge sets have a combination quick disconnect and shutoff valve on the high and low sides. The center hose also requires a valve.

The initial purging is best accomplished when connected to recovery or recycle equipment. With the center hose connected to the recovery station, service hoses connected to the high and low sides of the system, we can begin the purging. The manifold valves and service valves should be closed. Activating the vacuum pump will now pull any air or moisture out of the center hose. This will require only a few minutes of time. The hose is the only area that is being placed in a vacuum and this will not require a lengthy process. Closing the valve will then insure the hose is purged. It is now safe to open the other manifold valves.

## Service Valves

Because an air conditioning system is a sealed system, two service valves are provided on the compressor to enable diagnostic tests, system charging or evacuation. Connecting the applicable hoses from the manifold gauge set to the compressor service valves enables each of these to be readily performed.

New and unique service hose fittings (Figure 4-10) have been specified for R-134a systems. Their purpose is to avoid accidental cross-mixing of refrigerants and lubricants with R-12 based systems. The service ports on the system are quick disconnect type with no external threads. They do contain a Schrader type valve. The low side fitting has a smaller diameter than the high side attachment.

Protective caps are provided for each service valve. When not being used these caps should be in place to prevent contamination or damage to the service valves.

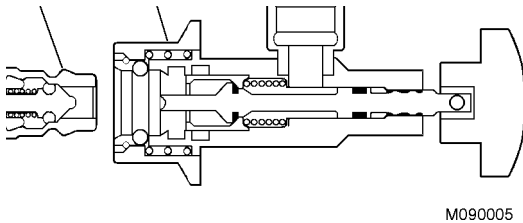


FIGURE 4-10. R-134a SERVICE VALVE

- |                                |                            |
|--------------------------------|----------------------------|
| 1. System Service Port Fitting | 3. Service Hose Connection |
| 2. Quick Connect               |                            |

## Vacuum Pump

The vacuum pump (Figure 4-11) is used to completely evacuate all of the refrigerant, air, and moisture from the system by deliberately lowering the pressure within the system to the point where water turns to a vapor (boils) and together with all air and refrigerant is withdrawn (pumped) from the system.

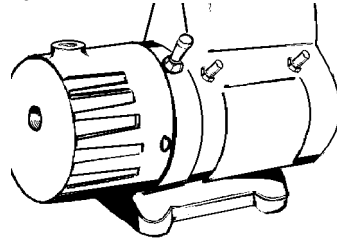


FIGURE 4-11. VACUUM PUMP

## SYSTEM PERFORMANCE TEST

This test is performed to establish the condition of all components in the system. Observe these conditions during testing:

1. Place a fan in front of the condenser to simulate normal ram air flow and allow the system to stabilize.
2. Install a thermometer into the air conditioning vent closest to the evaporator.
3. Start the engine and operate at 1,000 rpm.
4. Evaluate the readings obtained from the gauges to see if they match the readings for the ambient temperature.
5. Set the air conditioning system at maximum cooling and maximum blower speed operation.
6. Close all windows and doors to the cab.
7. Carefully feel the hoses and components on the high side. All should be warm or hot to the touch. Check the inlet and outlet of receiver-drier for even temperatures. If outlet is cooler than inlet, a restriction is indicated.

### **WARNING**

***Use extreme caution when placing hands on high side components and hoses. Under most normal conditions, these items can be extremely hot.***

8. Feel the hoses and components on the low side. They should be cool to the touch. Check the connections near the expansion valve. The inlet side should be warm and the outlet side should be cold.
9. After a minimum of 10 minutes has elapsed and the system has stabilized, observe the gauge readings. Compare the readings to the specifications in Table 1.

***NOTE: Pressures may be slightly higher in very humid conditions and lower in very dry conditions. Pressures listed in the table are during compressor clutch engagement.***

10. Check the cab vents for cool air. Outlet air temperature should be approximately 16 - 22° C (30 - 40° F) below ambient air temperature.
11. If pressures and temperatures are not within the specified ranges, the system is not operating properly. Refer to "Preliminary Checks" on page N4-35 for tips on diagnosing poor system performance.

**Table 1: NOMINAL R-134a PRESSURE RANGES**

| Ambient Air Temperature | High Side Pressure              | Low Side Pressure           |
|-------------------------|---------------------------------|-----------------------------|
| 21° C (70° F)           | 820 - 1300 kPa (120 - 190 psi)  | 70 - 138 kPa (10 - 20 psi)  |
| 27° C (80° F)           | 950 - 1450 kPa (140 - 210 psi)  | 70 - 173 kPa (10 - 25 psi)  |
| 32° C (90° F)           | 1175 - 1650 kPa (170 - 240 psi) | 105 - 210 kPa (15 - 30 psi) |
| 38° C (100° F)          | 1300 - 1850 kPa (190 - 270 psi) | 105 - 210 kPa (15 - 30 psi) |
| 43° C (110° F)          | 1450 - 2075 kPa (210 - 300 psi) | 105 - 210 kPa (15 - 30 psi) |

***NOTE: All pressures in this chart are for reference only. Weight is the only absolute means of determining proper refrigerant charge.***



## SYSTEM OIL

R-134a air conditioning systems require the use of Polyalkylene Glycol (PAG) lubricating oil. This is the only oil recommended for use in this system. The Komatsu PAG oil (PC2212) is the oil that is furnished in the system on Komatsu trucks.

### Handling and Reusing PAG Oil

- Avoid skin contact and inhalation of PAG oil, as these are normal precautions with any chemical.
- PAG oil removed from new or old components mustn't be retained for re-use. It must be stored in a marked container and properly sealed. PAG oil is an environmental pollutant and must be properly disposed of after use.
- PAG oil in containers or in an air conditioning system must not be left exposed to the atmosphere any longer than necessary. PAG oil absorbs moisture very rapidly, and therefore, any absorbed moisture could cause damage to an air conditioning system.

### Oil Quantity

It is critical to keep the correct amount of lubricant in the air conditioning system at all times. Failure to do so could result in damage to the compressor.

Damage to the compressor can be a result from not only a lack of oil, but from too much oil, also. A lack of oil will cause excess friction and wear on moving parts. Excessive oil can result in "slugging" the compressor. This condition occurs when the compressor attempts to compress liquid oil as opposed to vaporized refrigerant. Since liquid cannot be compressed, damage to internal parts results.

## Checking System Oil

### **IMPORTANT**

***The receiver-drier and accumulator must be replaced each time the system is opened.***

1. Remove the compressor from the truck. With the compressor positioned horizontally, remove the drain plug, and capture the oil in a clear graduated container. Rock the compressor back and forth, and rotate the shaft to facilitate oil removal.

### **CAUTION**

***Under no circumstances should the A/C compressor be stood upright onto the clutch assembly. Damage to the compressor clutch will result, leading to premature compressor failures.***

2. Inspect the oil for any foreign particles. If particles are found, further investigation and service are necessary to determine the source. After repair, the system will need to be flushed. Refer to "System Flushing" on page N4-24. If no particles are found, proceed to the next step.
3. Add 207 ml (7oz.) of PAG oil to the compressor sump. Add the oil through the drain port, and install the drain plug. It is important to only add the specified amount to ensure optimal system performance. Too much oil will result in a reduction in cooling. Too little oil will result in compressor failure.
4. Determine the correct amount of additional oil to add to the system by using the Replacing Oil table. Add this extra oil to the inlet side of the receiver drier or accumulator.

***NOTE: If truck is being assembled for the first time, add 207 ml (7oz.) of PAG oil to the inlet side of the receiver-drier or to the accumulator.***

**EXAMPLE** - If only the accumulator and receiver drier were replaced, then add 120 ml (4 oz.) of PAG oil to the inlet side of the receiver-drier or to the accumulator. If the evaporator was also replaced at this time, then add 150 ml (5 oz.) of PAG oil to the inlet side of the receiver-drier or to the accumulator.

***NOTE: The proper quantity of oil may be injected into the system during charging as an alternate method of adding oil.***



***When installing a new compressor, the compressor must be completely drained of its oil before installation. Add 207 ml (7 oz.) of new PAG oil to the compressor to ensure proper system oil level. Failure to adjust the amount of oil in the compressor will lead to excessive system oil and poor A/C performance. Additionally, a new receiver-drier and accumulator must be installed and oil added to either one of these components.***

| REPLACING OIL           |                                   |
|-------------------------|-----------------------------------|
| Component               | Oil to add                        |
| Condenser               | 60 - 90 ml (2 - 3 oz.)            |
| Evaporator              | 30 ml (1 oz.)                     |
| Receiver-Drier          | 60 ml (2 oz.)                     |
| Accumulator             | 60 ml (2 oz.)                     |
| Compressor              | 207 ml (7 oz.)                    |
| Block Valve (Expansion) | Adding oil is not necessary.      |
| Hoses                   | Drain and measure amount removed. |

5. Connect all hoses and components in the system. Lubricate O-rings with clean mineral oil before assembly.

***NOTE: Do not use PAG oil to lubricate O-rings or fittings. PAG oil could corrode fittings when used externally.***

6. Evacuate the system. Refer to "EVACUATING THE SYSTEM" on page N4-33.

## REFRIGERANT

### Recycled Refrigerant

Recycled refrigerant has been extracted from a mobile air conditioning system using a recovery unit. The refrigerant is cleaned by the recovery unit as it passes through filters located on the unit that meet specifications stipulated by Society of Automotive Engineers, SAE J2099. The refrigerant that has passed through the filtering process has only been cleaned of contaminants that are associated with mobile systems. Therefore, recycled refrigerant from mobile systems is only acceptable for reuse in mobile systems.

### Reclaimed Refrigerant

Reclaimed refrigerant has been filtered through a more thorough filtering process and has been processed to the same standards of purity as virgin refrigerant. Because of this, reclaimed refrigerant is acceptable for use in all systems, not just mobile. The reclaiming equipment used for this process is expensive, and therefore, not common among normal maintenance shops. Equipment such as this is more commonly found in air conditioning specialty shops.



***Always use new, recycled, or reclaimed refrigerant when charging a system. Failure to adhere to this recommendation may result in premature wear or damage to air conditioning system components and poor cooling performance.***

## Refrigerant Quantity

If not enough refrigerant is charged into the system, cooling ability will be diminished. If too much refrigerant is charged into the system, the system will operate at higher pressures and, in some cases, may damage system components. Exceeding the specified refrigerant charge will not provide better cooling.

If an incorrect charge is suspected, recover the refrigerant from the system, and charge the system with the correct operating weight of 3.4 kg (7.4 lb). This is not only the recommended procedure, but it is also the best way to ensure that the system is operating with the proper charge and providing optimum cooling. Using the sight glass to determine the charge is not an accurate method.

### **! IMPORTANT !**

***An unclear sight glass on R-134a systems can indicate that the system may be low on refrigerant. However, the sight glass should not be used as a gauge for charging the system. Charging the system must be done with a scale to ensure the proper amount of refrigerant has been added.***

## R-134a Refrigerant Containers

Two basic, readily available containers are used to store R-134a: the 14 or 28 kg (30 or 60 lb) bulk canisters (Figure 4-12). Always read the container label to verify the contents are correct for the system being serviced. Note the containers for R-134a are painted light blue.

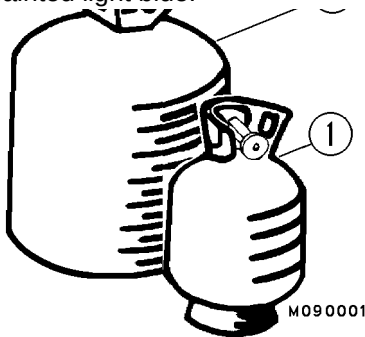


FIGURE 4-12. R-134a CONTAINERS

- |                              |                              |
|------------------------------|------------------------------|
| 1. 14 kg (30 lb)<br>Cylinder | 2. 28 kg (60 lb)<br>Cylinder |
|------------------------------|------------------------------|

## SYSTEM LEAK TESTING

Refrigerant leaks are probably the most common cause of air conditioning problems, resulting from improper or no cooling, to major internal component damage. Leaks most commonly develop in two or three places. The first is around the compressor shaft seal, often accompanied by an indication of fresh refrigerant oil. If a system is not operated for a while (winter months), the shaft seal may dry out and leak slightly. The centrifugal force of the clutch pulley spinning can also cause the problem. When the system is operated and lubricant wets the seal, the leak may stop. Such leaks can often be located visually or by feeling with your fingers around the shaft for traces of oil. The R-134a itself is invisible, odorless, and leaves no trace when it leaks, but has a great affinity for refrigerant oil.

A second common place for leaks is the nylon and rubber hoses where they are crimped or clamped to the fittings, or where routing allows abrasion. Other threaded joints or areas where gaskets are used should be visually and physically examined. Moving your fingers along the bottom of the condenser and evaporator, particularly near the drain hole for the condensate will quickly indicate the condition of the evaporator. Any trace of fresh oil here is a clear indication of a leak.

Usually, a 50% charged system is enough to find most leaks. If the system is empty, connect the manifold gauge set to the system and charge at least 1.6 kg (3.5 lb) of refrigerant into the system.

### **! WARNING !**

***Use extreme caution when leak testing a system while the engine is running. In its natural state, refrigerant is a harmless, colorless gas. But when combined with an open flame, it will generate toxic fumes (phosgene gas) which can cause serious injuries or death.***

***NOTE: The refrigerant is heavier than air and will move downward when it leaks. Apply pickup hose or test probe on the under-surface of all components to locate leaks.***

## Electronic Leak Detector

Refer to Figure 4-7. As the test probe is moved into an area where traces of refrigerant are present, a visual or audible announcement indicates a leak. Audible units usually change tone or speed as intensity changes.

## Tracer Dyes

Tracer dyes are available that can be added to the system as refrigerant is added. The system is then operated to thoroughly circulate the dye. As refrigerant escapes, it leaves a trace of the dye at the point of leakage, which is then detected using an ultraviolet light ("black light"), revealing a bright fluorescent glow.

## Soap and Water

Soap and water can be mixed together and applied to system components. Bubbles will appear to pinpoint the specific location of leaks.

After determining the location or source of leak(s), repair or replace leaking component(s).

*NOTE: The length of the hose will affect the refrigerant capacity. When replacing hoses, always use the same hose length, if possible.*



**Before system assembly, check the compressor oil level and fill to specifications.**

## RECOVERING AND RECYCLING THE REFRIGERANT

### Draining Oil from the Previous Recovery Cycle

1. Place the power switch and the controller on the recovery unit in the OFF position.
2. Plug in the recovery station to the correct power source.
3. Drain the recovered oil through the valve marked "oil drain" on the front of the machine.
4. Place the controller knob in the ON position. The low pressure gauge will show a rise.
5. Immediately switch to the OFF position and allow the pressure to stabilize. If the pressure does not rise to 34 - 69 kPa (5 - 10 psi), switch the controller ON and OFF again.

6. When the pressure reaches 34 - 69 kPa (5 - 10 psi), open the "oil drain" valve, collect the oil in an appropriate container, and dispose of container as indicated by local, state or federal regulation. The oil is not reusable due to contaminants that were absorbed during use.

### Performing the Recovery Cycle

1. Ensure that the equipment being used is designed for the refrigerant you intend to recover.
2. Observe the sight glass oil level. Having been drained, it should be at zero.
3. Check the cylinder refrigerant level before beginning recovery to make sure you have enough capacity.
4. Confirm that all shut-off valves are closed before connecting to the A/C system.
5. Attach the appropriate hoses to the system being recovered.
6. Start the recovery process by operating the equipment as per the manufacturer's instructions.
7. Continue extraction until a vacuum exists in the A/C system.
8. If an abnormal amount of time elapses after the system reaches 0 psi and does not drop steadily into the vacuum range, close the manifold valves and check the system pressure. If it rises to 0 psi and stops, there is a major leak.
9. Check the system pressure after the recovery equipment stops. After five minutes, system pressure should not rise above "0" gauge pressure. If the pressure continues to rise, restart and begin the recovery sequence again. This cycle should continue until the system is void of refrigerant.
10. Check the sight glass oil level to determine the amount of oil that needs to be replaced. (The amount of oil that was lost during the recovery cycle must be replaced back into the system).
11. Mark the cylinder with a "RECOVERED" (red) magnetic label to reduce the chance of charging a system with contaminated refrigerant. Record the amount of refrigerant recovered.

## Performing the Recycling Procedure

The recovered refrigerant contained in the cylinder must undergo the recycle procedure before it can be reused. The recycle or clean mode is a continuous loop design and cleans the refrigerant rapidly. Follow the equipment manufacturer's instructions for this procedure.

## Evacuating and Charging the A/C System

Evacuate the system once the air conditioner components are repaired or replacement parts are secured, and the A/C system is reassembled. Evacuation removes air and moisture from the system. Then, the A/C system is ready for the charging process, which adds new refrigerant to the system.

## SYSTEM REPAIR

The following service and repair procedures are not any different than typical vehicle service work. However, A/C system components are made of soft metals (copper, aluminum, brass, etc.). Comments and tips that follow will make the job easier and reduce unnecessary component replacement.

### WARNING

***All of the service procedures described are only performed after the system has been discharged. Never use any lubricant or joint compound to lubricate or seal any A/C connections.***

***NOTE: To help prevent air, moisture or debris from entering an open system, cap or plug open lines, fittings, components and lubricant containers. Keep all connections, caps, and plugs clean.***

### CAUTION

***Never leave A/C components, hoses, oil, etc. exposed to the atmosphere. Always keep sealed or plugged until the components are to be installed and the system is ready for evacuation and charging. PAG oil and receiver-drier desiccants attract moisture. Leaving system components open to the atmosphere will allow moisture to invade the system, resulting in component and system failures.***

## System Flushing

If any contaminants are found in system hoses, components or oil, the entire system must be flushed. Major components such as the compressor are extremely susceptible to foreign particles and must be replaced. If contaminated, the evaporator and condenser must also be replaced. The evaporator and condenser are multi-pass units, and they can not be properly cleaned by flushing.

### CAUTION

***Only SAE and/or Mobile Air Conditioning Society (MACS) approved flushing methods with the appropriate refrigerants are to be performed when removing debris from the system. Other methods may be harmful to the environment, as well as air conditioning components.***

1. Remove the compressor, receiver-drier, expansion valve, and accumulator.
2. Inspect all other components such as the condenser, evaporator, hoses and fittings. If any of these items are damaged or highly contaminated, replace the components.
3. Flush the remaining hoses with a flushing unit. Use only R134a as a flushing agent.
4. After flushing, blow out the system with dry shop air for 5 to 10 minutes.
5. If the expansion valve has been removed of all foreign contamination, it may be reinstalled back into the system. If contamination is still present, replace the valve.
6. Install a new compressor, receiver-drier, and accumulator.
7. Add oil to the system as outlined in "Checking System Oil" on page N4-20.

## A/C DRIVE BELT CHECKOUT PROCEDURE

This procedure must be performed each time any component in the accessory drive is serviced, such as replacing a belt or removing the compressor. In addition, a 250 hour inspection of the AC drive belt is mandatory. The belts must be inspected for indications of wear and damage that may hinder performance. Replace as necessary and perform the following procedure.

### Pulley Alignment

1. Install alignment tool (EL8868) onto the pulleys to check the alignment. Refer to Figure 4-15. If misalignment of the pulleys exceeds 3 mm (0.13 in.), the position of the compressor must be adjusted.

### Belt Tension Check

*NOTE: This procedure has been written for use with belt tension tool (XA3379), shown in Figure 4-13. Other tension tools may differ in functionality.*



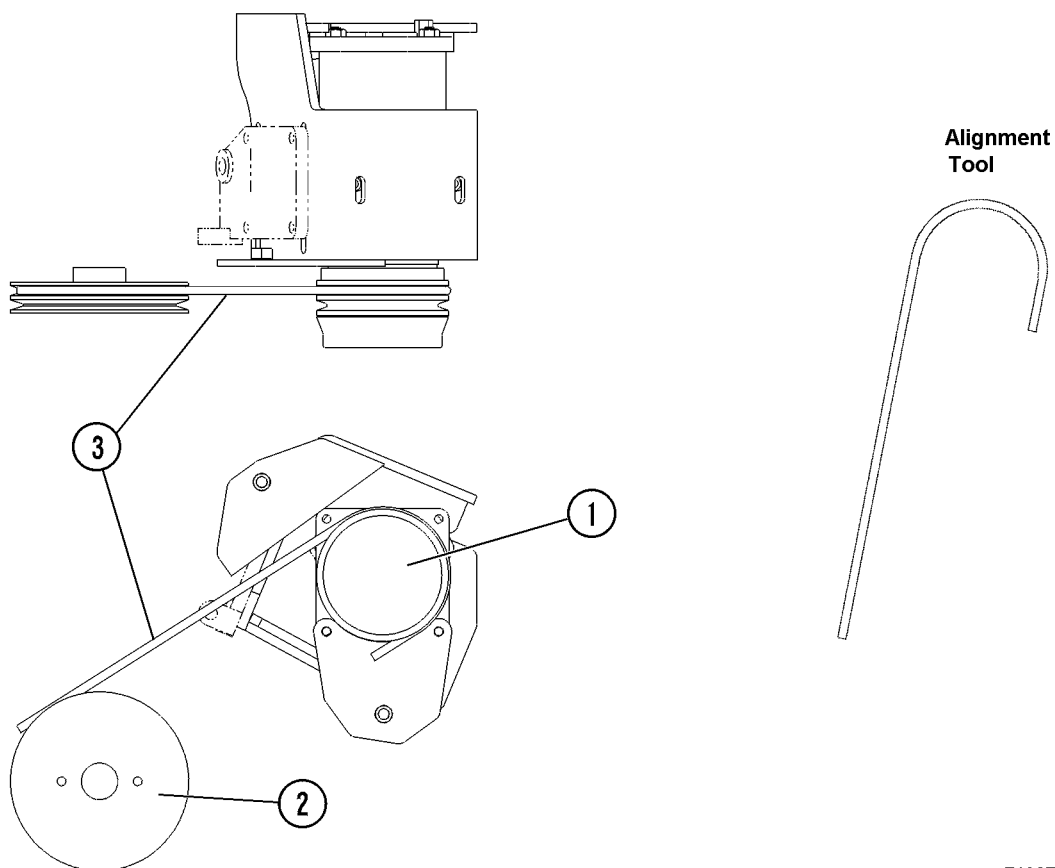
FIGURE 4-13. BELT TENSION TOOL - XA3379

2. Refer to Figure 4-16 for the proper distance from the centerline of the drive pulley to the centerline of the compressor pulley. Set the tension tool accordingly on the "deflection" scale by moving the deflection O-ring to the corresponding distance on the scale.
3. Slide the O-ring for the "force" scale to zero.



FIGURE 4-14. DEFLECTION MEASUREMENT

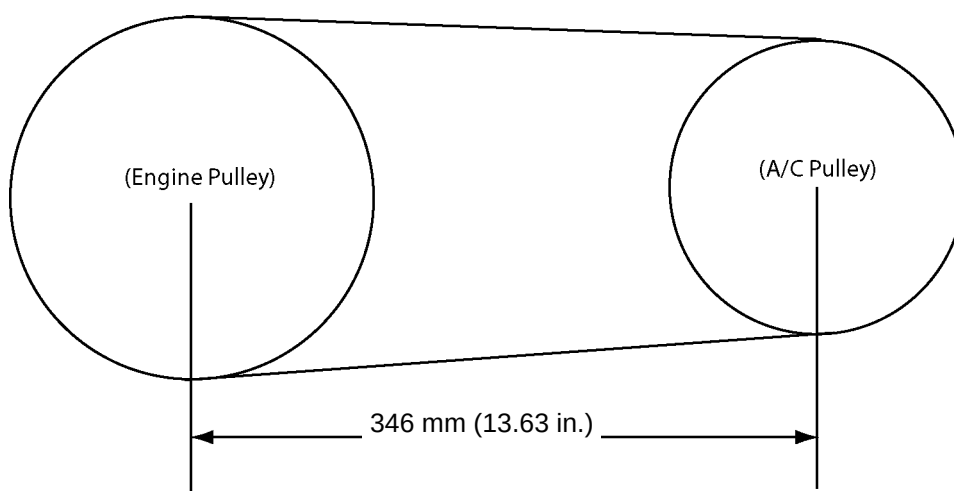
4. Find the approximate center of the belt between the two pulleys. Place the tip of the tool onto the outer face of the belt and apply pressure, as shown in Figure 4-14. The tool must be perpendicular to the belt. Push on the tool until the bottom edge of the deflection scale O-ring is even with the outer face of the adjacent drive belt. If only one belt is used, rest a straight edge across both pulleys to serve as the indicating plane.
5. The O-ring on the force scale indicates the force used to deflect the belt. The belt must deflect 5.3 mm (0.21 in.) under a force of  $1.6 \pm 0.1$  kgf ( $3.44 \pm 0.11$  lbf). If not, adjust the belt accordingly and recheck the tension.



71327

FIGURE 4-15. BELT ALIGNMENT TOOL

1. AC Compressor Pulley      2. Drive Pulley      3. Alignment Tool



N040074

FIGURE 4-16. BELT TENSION DIMENSIONS

## COMPONENT REPLACEMENT

### Hoses and Fittings

When replacing hoses, use the same type and ID hose you removed. When hoses or fittings are shielded or clamped to prevent vibration damage, ensure that these are in position or secured.

### Lines

Always use two wrenches when disconnecting or connecting A/C fittings attached to metal lines. Copper and aluminum tubing can kink or break very easily. When grommets or clamps are used to prevent line vibration, ensure that these are in place and secure.

## ! IMPORTANT !

*It is important to always torque fittings to the proper torque. Failure to do this may result in improper contact between mating parts and leakage may occur. Refer to the following torque chart for tightening specifications.*

| Fitting Size | Foot Pounds     | Newton Meters |
|--------------|-----------------|---------------|
| 6            | 10 - 15 ft.lbs. | 14 - 20 Nm    |
| 8            | 24 - 29 ft.lbs. | 33 - 39 Nm    |
| 10           | 26 - 31 ft.lbs. | 36 - 42 Nm    |
| 12           | 30 - 35 ft.lbs. | 41 - 47 Nm    |

Installation torque for the single M10 or 3/8 in. capscrews securing the inlet and outlet fittings onto the compressor ports is **15 - 34 N·m (11 - 25 ft lb)**.

### Expansion Valve

When removing the expansion valve from the system, remove the insulation, clean the area and disconnect the line from the receiver-drier. Detach the capillary (bulb) and external equalizer tube (if present) from their mounting locations. Remove the expansion valve from the evaporator inlet. Expansion valve service is limited to cleaning or replacing the filter screen. If this is not the problem, replace the valve. Secure the capillary and equalizer, if used, to clean surfaces and replace or attach any insulating material.

### Receiver-Drier

The receiver-drier can not be serviced or repaired. It must be replaced whenever the system is opened for any service. The receiver-drier has a pressure switch to control the clutch, and should be removed and installed onto the new unit.

### Thermostat

A thermostat can be stuck open or closed due to contact point wear or fusion. The thermostat temperature sensing element (capillary tube) may be broken or kinked closed and therefore unable to sense evaporator temperature.

When thermostat contact points are stuck open or the sensing element can not sense temperature in the evaporator, the clutch will not engage (no A/C system operation). Causes are a loss of charge in the capillary tube or a kink, burned thermostat contact or just no contact. When troubleshooting, bypass the thermostat by hot wiring the clutch coil with a fused lead. If the clutch engages, replace the thermostat.

Thermostat contact points may be fused (burned) closed and the clutch will not disengage. Causes are a faulty switch that could be due to fatigue. The thermostat must be replaced. When the clutch will not disengage you may also note that condensate has frozen on the evaporator fins and blocked air flow. There will also be below normal pressure on the low side of the system. Side effects can be compressor damage caused by oil accumulation (refrigeration oil tends to accumulate at the coldest spot inside the system) and lower than normal suction pressure that can starve the compressor of oil.



## Compressor

The compressor can fail due to shaft seal leaks (no refrigerant in the system), defective valve plates, bearings, or other internal parts or problems associated with high or low pressure, heat, or lack of lubrication. Ensure that the compressor is securely mounted and the clutch pulley is properly aligned with the drive pulley.

Use a mechanic's stethoscope to listen for noises inside the compressor.

### CAUTION

*When installing a new compressor, the compressor must be completely drained of its oil before installation. Add 207 ml (7 oz.) of new PAG oil to the compressor to ensure proper system oil level. Failure to adjust the amount of oil in the compressor will lead to excessive system oil and poor A/C performance.*

### CAUTION

*Under no circumstances should the A/C compressor be stood upright onto the clutch assembly. Damage to the compressor clutch will result, leading to premature compressor failures.*

## Accumulator

The accumulator can not be serviced or repaired. It must be replaced whenever the system is opened for any service.

## Clutch

Clutch problems include electrical failure in the clutch coil or lead wire, clutch pulley bearing failure, worn or warped clutch plate or loss of clutch plate spring temper. Defective clutch assembly parts may be replaced or the whole assembly replaced. If the clutch shows obvious signs of excessive heat damage, replace the whole assembly.

The fast way to check electrical failure in the lead wire or clutch coil is to hot wire the coil with a fused lead. This procedure enables you to bypass clutch circuit control devices.

Clutch pulley bearing failure is indicated by bearing noise when the A/C system is off or the clutch is not engaged. Premature bearing failure may be caused by poor alignment of the clutch and clutch drive pulley.

Sometimes it may be necessary to use shims or enlarge the slots in the compressor mounting bracket to achieve proper alignment.

Excessive clutch plate wear is caused by the plate rubbing on the clutch pulley when the clutch is not engaged or the clutch plate slipping when the clutch coil is energized. A gap that is too small or too large between the plate and clutch pulley or a loss of clutch plate spring temper are possible causes. The ideal air gap between the clutch pulley and the clutch plate is  $1.02 \pm 0.043$  mm (0.023 to 0.057 in.). If the gap is too wide, the magnetic field created when the clutch coil is energized will not be strong enough to pull and lock the clutch plate to the clutch pulley.

### IMPORTANT

*Some compressors may be discarded because it is suspected that internal components within the compressor have seized. Ensure that the compressor clutch is working properly before discarding a compressor for internal seizure. The normal compressor life span should be about twice as long as the normal life span of the compressor clutch.*

It is important to note that often times a weak clutch coil may be mistaken for a seized compressor. When a coil's resistance has increased over time and the magnetic field weakens, the coil may not be able to pull the load of the compressor. Failure of the coil to allow the compressor shaft to be turned, may appear as though the compressor is locked up.

Before a compressor is dismissed as being seized, a check for proper voltage to the coil should be performed. In addition, the coil should be ohm checked for proper electrical resistance. The coil should fall within the following range:

**12.0  $\pm$  0.37 Ohms @ 20° C (68° F)**

**16.1  $\pm$  0.62 Ohms @ 116° C (240° F)**

The temperatures specified above are roughly typical of a summer morning before first start-up and the heat beside an engine on a hot day. At temperatures in between those listed above, the correct resistance is proportionate to the difference in temperature.

## Servicing the Compressor Clutch

| <b>* RECOMMENDED TOOLS FOR COMPRESSOR CLUTCH REMOVAL AND INSTALLATION</b> |                                       |
|---------------------------------------------------------------------------|---------------------------------------|
| J-9399                                                                    | Thin Wall Socket                      |
| **J-9403                                                                  | Spanner Wrench                        |
| **J-25030                                                                 | Clutch Hub Holding Tool               |
| J-9401                                                                    | Clutch Plate and Hub Assembly Remover |
| J-8433                                                                    | Pulley Puller                         |
| J-9395                                                                    | Puller Pilot                          |
| ***J-24092                                                                | Puller Legs                           |
| J-8092                                                                    | Universal Handle                      |
| J-9481                                                                    | Pulley and Bearing Installer          |
| J-9480-01                                                                 | Drive Plate Installer                 |
| J-9480-02                                                                 | Spacer, Drive Plate Installer         |

\* Tools are available through your local Kent-Moore dealer.

\*\* These tools are interchangeable.

\*\*\*For use on multiple groove pulleys.



**Use the proper tools to remove and replace clutch components. Using the recommended tooling helps prevent damage to compressor components during maintenance.**

**Do not drive or pound on the clutch plate, hub assembly, or shaft. Internal damage to the compressor may result.**

1. Remove the belt guard from the front of the air conditioning compressor.

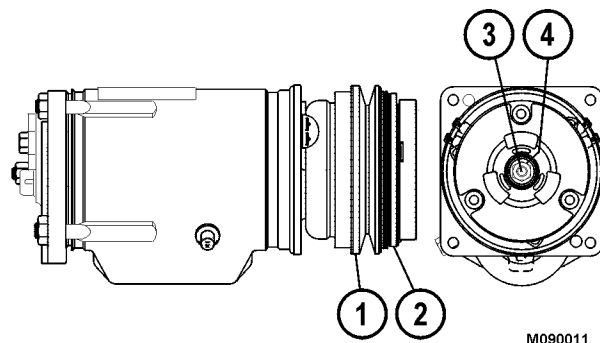


FIGURE 4-17.

1. Belt Pulley
2. Clutch Hub/Drive Plate
3. Shaft
4. Locknut

2. Remove the drive belt from compressor belt pulley (1, Figure 4-17).

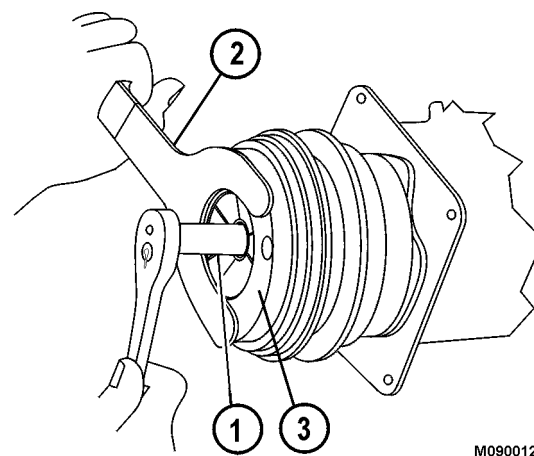


FIGURE 4-18.

1. Thin Wall Socket
2. Clutch Hub Holding Tool
3. Clutch Hub

3. Remove locknut (4) using thin wall socket (1, Figure 4-18) or the equivalent. Use clutch hub holding tool (2), spanner wrench (J-9403), or the equivalent to hold clutch hub (3) while removing the locknut. It is recommended that the locknut be replaced after it has been removed.

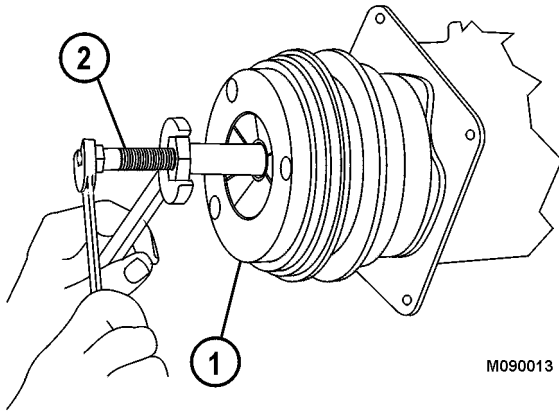


FIGURE 4-19.

1. Clutch Assembly      2. Clutch Plate & Hub Assembly Remover

4. Thread clutch plate and hub assembly remover (2, Figure 4-19) into the hub of clutch assembly (1). Hold the body of the remover with a wrench and tighten the center screw to pull the clutch plate and hub assembly from the compressor.

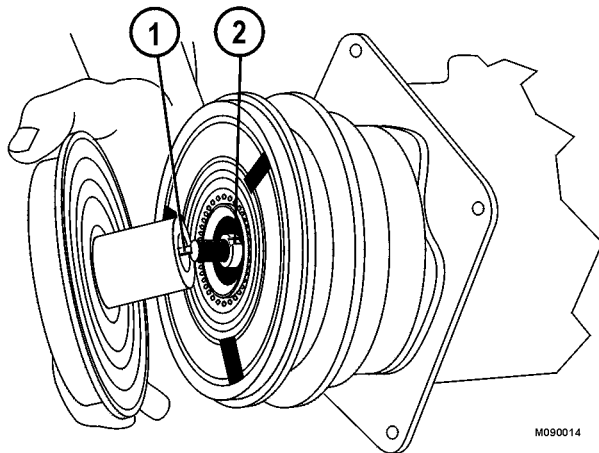


FIGURE 4-20.

1. Square Key      2. Keyway in Shaft

5. Remove square key (1, Figure 4-20) from the keyways.

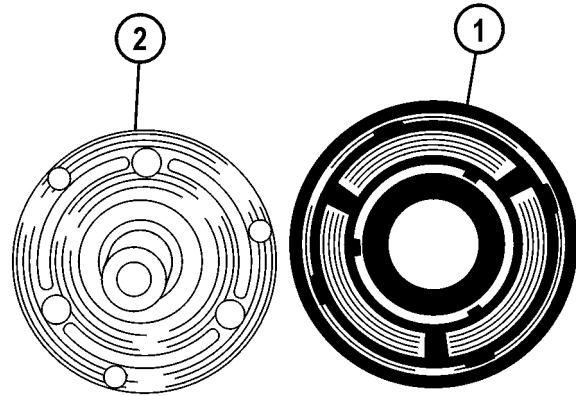


FIGURE 4-21.

1. Clutch Hub      2. Pulley

6. Inspect the friction surfaces on clutch hub (1, Figure Figure 4-21) and pulley (2). Scoring on the friction surfaces is normal. DO NOT replace these components for this condition only.

### ⚠ IMPORTANT ⚠

*Inspect the steel friction surface on the clutch and ensure that it is not damaged by excessive heat. Inspect the other components near the clutch for damage due to heat. If signs of excessive heat are evident, it may be necessary to replace the compressor. Excessive heat may cause leakage in the seals and damage to internal components as well as external components.*

## Pulley Removal

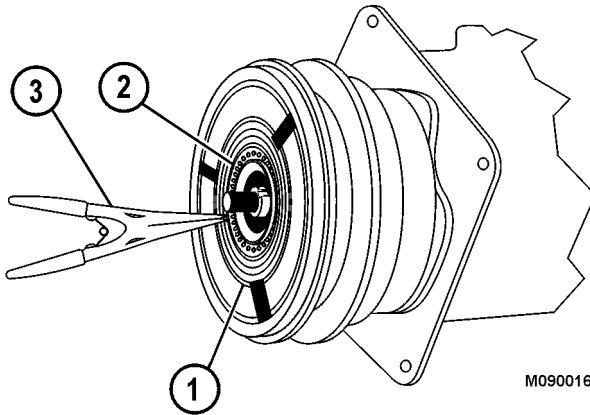


FIGURE 4-22.

- 1. Pulley Assembly
- 2. Pulley Retainer Ring
- 3. Retaining Ring Pliers

7. Use retaining ring pliers (3, Figure 4-22) to remove pulley retainer ring (2) from pulley (1).
8. Pry the absorbent sleeve retainer from the neck of the compressor, and remove the sleeve.

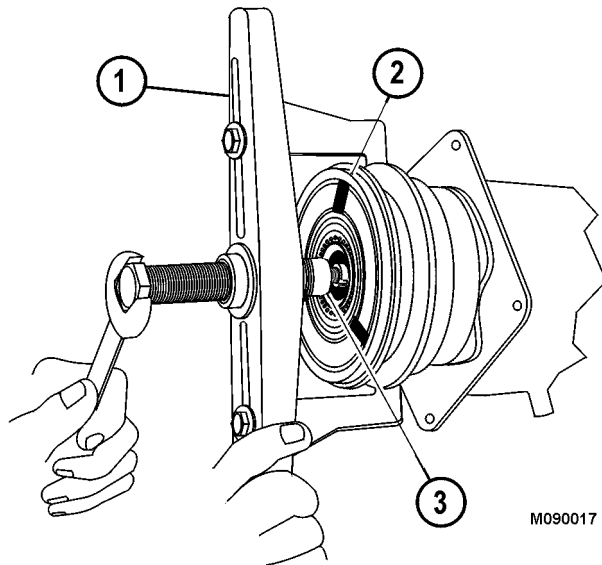


FIGURE 4-23.

- 1. Pulley Puller
- 2. Pulley Assembly
- 3. Puller Pilot

9. Install pulley puller (1, Figure 4-23) and puller pilot (3) onto the compressor, as shown. If a multiple groove pulley is used, install puller legs (J-24092) onto the puller in place of the standard legs. Extend the puller legs to the back side of the pulley. DO NOT use the belt grooves to pull the pulley from the compressor.

10. Tighten the center screw on the puller against the shaft of the compressor to remove the pulley.
11. Clean the pulley and pulley bearing with solvent. Inspect the assembly for damage. Check the bearing for brinneling, excessive looseness, noise, and lubricant leakage. Replace the assembly if any of these warning signs are evident.

## Clutch Coil Check

12. Use a multi-meter to ohm check the clutch coil. The resistance should be as follows:
  - $12 \pm 0.37$  ohms @  $20^{\circ}\text{C}$  ( $68^{\circ}\text{F}$ )
  - $16.1 \pm 0.62$  ohms @  $115^{\circ}\text{C}$  ( $239^{\circ}\text{F}$ )

If the resistance of the coil is not within the specifications, the clutch will not operate properly. Remove the retaining ring and replace the coil.

## Pulley Installation

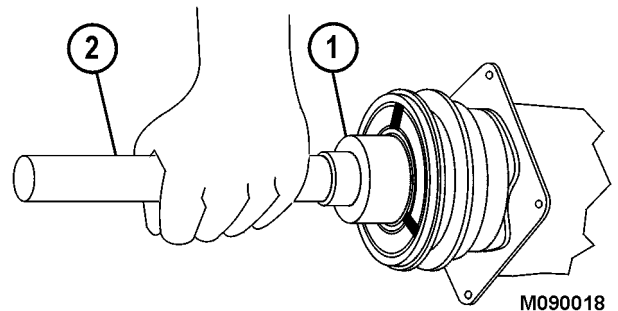


FIGURE 4-24.

- 1. Bearing Installer
- 2. Universal Handle

1. Place the pulley assembly into position on the compressor. Use bearing installer (1, Figure 4-24), universal handle (2), and a hammer to lightly tap the pulley assembly onto the compressor until it seats. Use of the installer or the equivalent ensures that the force driving the bearing into position acts on the inner race of the bearing. Applying force to the outer race of the bearing will result in bearing damage.
2. Ensure the pulley rotates freely. If the pulley does not rotate freely, remove the pulley and check for damaged components. Replace any damaged components and reinstall the pulley.
3. Install the pulley retainer ring and ensure that the ring is properly seated.
4. Install the absorbent sleeve into the neck of the compressor. Install the sleeve retainer.

## Clutch Assembly Installation

1. Insert square key (1, Figure 4-20) into the keyway in the clutch hub. Allow the key to protrude about 4.5 mm (0.18 in.) from the outer edge of the hub. Use petroleum jelly to hold the key in place.

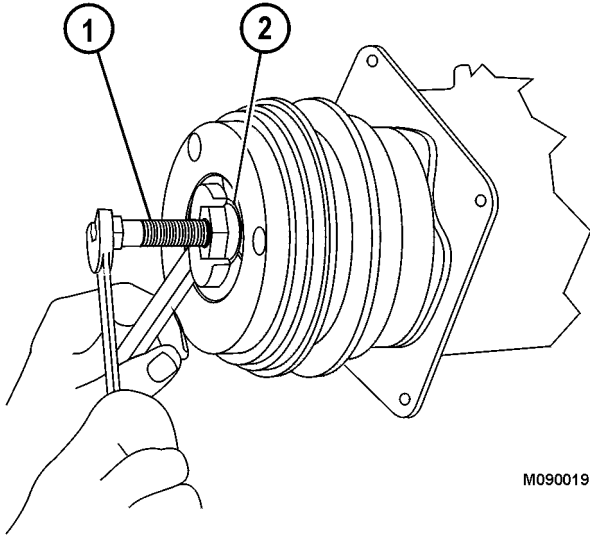


FIGURE 4-25.

1. Drive Plate Installer      2. Spacer

2. Place the clutch assembly into position on the compressor. Align the square key with the keyway on the shaft.
3. Thread drive plate installer (1, Figure 4-25) onto the shaft of the compressor. Spacer (2) should be in place under the hex nut on the tool.
4. Press the clutch onto the compressor using drive plate installer (1). Continue to press the clutch plate until a 2 mm (0.08 in.) gap remains between the clutch friction surface and the pulley friction surface. Refer to Figure 4-26.

*NOTE: The outer threads of installer (J-9480-01) are left handed threads.*

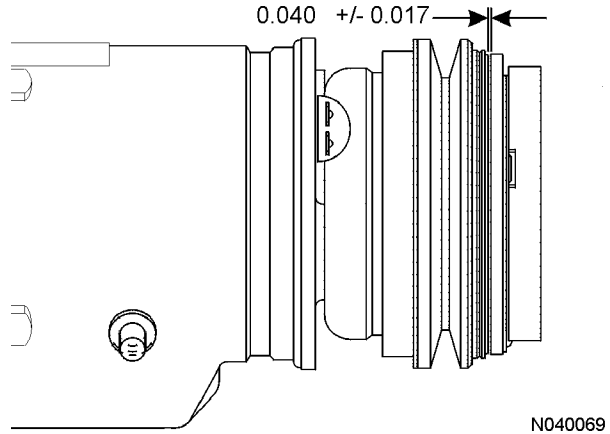


FIGURE 4-26. CLUTCH GAP

5. Install locknut (4, Figure 4-17) and tighten the nut until it seats. The gap should now measure  $1.02 \pm 0.043$  mm ( $0.040 \pm 0.017$  in.). If the gap is not within the specification, check for proper installation of the square key.
6. Install the drive belt onto the compressor. Ensure that the proper tension on the belt is attained. Refer to the belt tension chart in the appropriate engine manual for the proper specifications.
7. After assembly is complete, burnish the mating parts of the clutch by operating the air conditioning system at maximum load conditions with the engine at high idle. Turn the air conditioning control ON and OFF at least 15 times for one second intervals.
8. Install the belt guard if no further servicing is required.

## EVACUATING THE SYSTEM

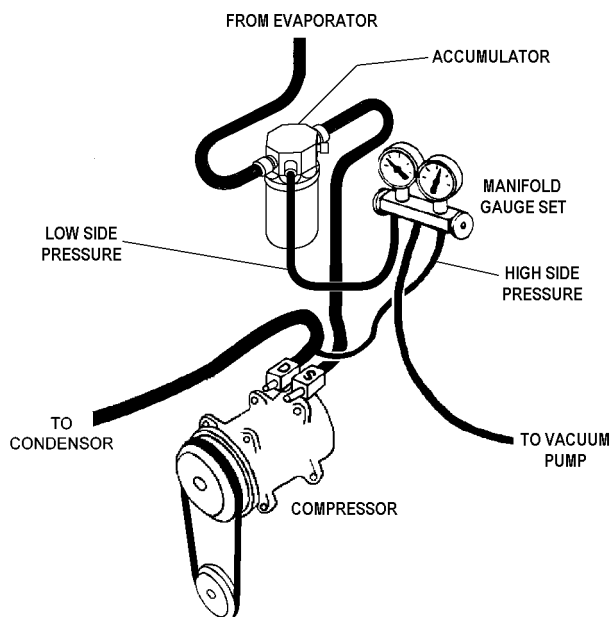
Evacuating the complete air conditioning system is required for all new system installations, when repairs are made on systems requiring a component replacement (system opened), or when a major loss of refrigerant has occurred. All these conditions will require that a vacuum be pulled using a vacuum pump that completely removes any moisture from the system. Once properly evacuated, the system can be recharged again.

Using a pump to create a vacuum in the air conditioning system effectively vaporizes any moisture, allowing the water vapor to be easily drawn out by the pump. The pump does this by reducing the point at which water boils (100°C, 212°F at sea level with 14.7 psi). In a vacuum, water will boil at a lower temperature depending upon how much of a vacuum is created.

As an example, if the ambient air outside the truck is 24°C (75°F) at sea level, by creating a vacuum in the system so that the pressure is below that of the outside air (in this case, at least 749.3 mm (29.5 in.) of vacuum is needed), the boiling point of water will be lowered to 22°C (72°F). Thus any moisture in the system will vaporize and be drawn out by the pump if the pump is run for approximately an hour. The following steps indicate the proper procedure for evacuating all moisture from the heavy duty air conditioning systems.



***Do not attempt to use the air conditioning compressor as a vacuum pump or the compressor will be damaged.***



N040071

FIGURE 4-27. VACUUM PUMP HOOKUP

**NOTE:** Refer to Table 2 for optimal vacuum specifications at various altitudes.

1. With the manifold gauge set still connected (after discharging the system), connect the center hose to the inlet fitting of the vacuum pump as shown in Figure 4-27. Then open both hand valves to maximum.
2. Open the discharge valve on the vacuum pump or remove the dust cap from the discharge outlet. Turn on the pump and watch the low side gauge. The pump should pull the system into a vacuum. If not, the system has a leak. Find the source of the leak, repair, and attempt to evacuate the system again.

3. Allow the vacuum pump to run for at least 45 minutes.
4. Shut off the vacuum pump and observe the gauges. The system should hold the vacuum within 5 cm Hg (2 in. Hg) of the optimal vacuum for five minutes. If the vacuum does not hold, moisture may still be present in the system. Repeat the previous step. If the vacuum still does not hold, a leak may be present in the system. Find the source of the leak, repair, and evacuate the system again.

*NOTE: In some cases, 45 minutes of evacuation may not be sufficient to vaporize all of the moisture and draw it out of the system. If it has been verified that no system leaks exist and gauge readings increase after 45 minutes, extend the evacuation time to ensure total moisture removal.*

**Table 2: ALTITUDE VACUUM VARIATIONS**

| Altitude Above Sea Level | Optimal Vacuum              |
|--------------------------|-----------------------------|
| 0 m (0 ft)               | 76.0 cm Hg. (29.92 in. Hg.) |
| 305 m (1000 ft)          | 73.5 cm Hg. (28.92 in. Hg.) |
| 610 m (2000 ft)          | 70.7 cm Hg. (27.82 in. Hg.) |
| 914 m (3000 ft)          | 68.1 cm Hg. (26.82 in. Hg.) |
| 1219 m (4000 ft)         | 65.6 cm Hg. (25.82 in. Hg.) |
| 1524 m (5000 ft)         | 63.3 cm Hg. (24.92 in. Hg.) |
| 1829 m (6000 ft)         | 60.8 cm Hg. (23.92 in. Hg.) |
| 2134 m (7000 ft)         | 58.5 cm Hg. (23.02 in. Hg.) |
| 2438 m (8000 ft)         | 56.4 cm Hg. (22.22 in. Hg.) |
| 2743 m (9000 ft)         | 54.2 cm Hg. (21.32 in. Hg.) |

*NOTE: The chart indicates the expected gauge readings at altitude to obtain the optimal vacuum.*

## CHARGING THE A/C SYSTEM

The proper method for charging refrigerant into a R-134a system is to first, recover all of the refrigerant from the system. The charging refrigerant should then be weighed on a scale to ensure the proper amount is charged into the system. Most recovery units include a scale within the apparatus, thus making it very easy to charge the correct amount every time. If equipment such as this is not available, a common scale can be used to determine the weight of charge. Simply weigh the charging tank, subtract the weight of the proper charge, and charge the system until the difference is shown on the scale. On certain types of equipment, it is also possible to add any necessary lubricant when charging the system.

If a scale is not used when charging R-134a into a system, it is difficult to tell if the correct charge has been achieved. The sight glass can provide some indication, but it is not a reliable tool for determining proper charge.

*NOTE: Charging is to be performed with the engine and compressor operating. Charge the A/C system through the low side service port. Trucks equipped with accumulators may charge the refrigerant as a liquid or as a vapor.*

1. Charge the A/C system with 3.4 kg (7.4 lbs) of R-134a refrigerant.
2. Check the system for leaks. Refer to System Leak Testing.
3. If no leaks are found, verify that the system's cooling capacity meets requirements. Refer to "SYSTEM PERFORMANCE TEST" on page N4-19.

## TROUBLESHOOTING

### Preliminary Checks

If the system indicates insufficient cooling, or no cooling, the following points should be checked before proceeding with the system diagnosis procedures.

*NOTE: If equipped, ensure that the rest switch in the cab is ON. Place the GF cutout switch in the CUTOUT position.*

Some simple, but effective checks can be performed to help determine the cause of poor system performance. Check the following to ensure proper system operation.

- Compressor belt - Must be tight, and aligned.
- Compressor clutch - The clutch must engage. If it does not, check fuses, wiring, and switches.
- Oil leaks - Inspect all connection or components for refrigeration oil leaks (especially in the area of the compressor shaft). A leak indicates a refrigerant leak.
- Electrical check - Check all wires and connections for possible open circuits or shorts. Check all system fuses.
- Cooling system - Check for correct cooling system operation. Inspect the radiator hoses, heater hoses, clamps, belts, water pump, thermostat and radiator for condition or proper operation.
- Radiator shutters - Inspect for correct operation and controls, if equipped.
- Fan and shroud - Check for proper operation of fan clutch. Check installation of fan and shroud.
- Heater/water valve - Check for malfunction or leaking. With the heat switch set to COLD, the heater hoses should be cool.

- System ducts and doors - Check the ducts and doors for proper function.
- Refrigerant charge - Make sure system is properly charged with the correct amount of refrigerant.
- Cab filters - Ensure the outside air filter and inside recirculation filter are clean and free of restriction.
- Condenser - Check the condenser for debris and clogging. Air must be able to flow freely through the condenser.
- Evaporator - Check the evaporator for debris and clogging. Air must be able to flow freely through the condenser.

### Diagnosis Of Gauge Readings And System Performance

Successfully servicing an air conditioning system, beyond the basic procedures outlined in the previous section, requires additional knowledge of system testing and diagnosis.

A good working knowledge of the manifold gauge set is required to correctly test and diagnose an air conditioning system. An accurate testing sequence is usually the quickest way to diagnose an internal problem. When correctly done, diagnosis becomes an accurate procedure rather than guesswork.

The following Troubleshooting Chart lists typical malfunctions encountered in air conditioning systems. Indications and or problems may differ from one system to the next. Read all applicable situations, service procedures, and explanations to gain a full understanding of the system malfunction. Refer to information listed under "Suggested Corrective Action" for service procedures.



# TROUBLESHOOTING BY MANIFOLD GAUGE SET READINGS

## **PROBLEM: Insufficient Cooling**

---

### **Indications:**

- Low side pressure - LOW.
- High side pressure - LOW.
- Discharge air is only slightly cool.

### **Possible Causes**

---

- Low refrigerant charge, causing pressures to be slightly lower than normal.

### **Suggested Corrective Actions**

---

Check for leaks by performing leak test.

#### ***If No Leaks Are Found:***

Recover the refrigerant and use a scale to charge the proper amount into the system. Check system performance.

#### ***If Leaks Are Found:***

After locating the source of the leak, recover the refrigerant, and repair the leak. Evacuate the system and recharge using a scale. Add oil as necessary. Check A/C operation and performance test the system.

## **PROBLEM: Little or No Cooling**

---

### **Indications:**

- Low side pressure - VERY LOW
- High side pressure - VERY LOW
- Discharge air is warm.
- No bubbles observed in sight glass, may show oil streaks.

### **Possible Causes**

---

- Pressure sensing switch may have compressor clutch disengaged.
- Refrigerant excessively low; leak in system.

### **Suggested Corrective Actions**

---

Add refrigerant (make sure system has at least 50% of its normal amount) and leak test system. It may be necessary to use a jumper wire to enable the compressor to operate, if the compressor has shut down due to faulty pressure sensing switch. Repair any leaks and evacuate the system if necessary. Replace the receiver-drier if the system was opened. Recharge the system using a scale and add oil as necessary. Check A/C operation and do system performance test.

## **PROBLEM: Extremely Low Refrigerant Charge in the System**

---

### **Indications:**

Low side pressure - LOW.  
High side pressure - LOW.  
Discharge air is warm.  
The low pressure switch may have shut off the compressor clutch.

---

### **Possible Causes**

- Extremely low or no refrigerant in the system.  
Possible leak in the system.

---

### **Suggested Corrective Actions**

Check for leaks by performing leak test.

#### **No Leaks Found:**

Recover refrigerant from the system. Recharge using a scale to ensure correct charge. Check A/C operation and performance.

#### **Leaks Found:**

Add refrigerant (make sure system has at least 50% of its normal amount) and leak test system. It may be necessary to use a jumper wire to enable the compressor to operate, if the compressor has shut down due to faulty pressure sensing switch. Repair any leaks and evacuate the system if necessary. Replace the receiver-drier if the system was opened. Recharge the system using a scale and add oil as necessary. Check A/C operation and do system performance test.

## **PROBLEM: Air and/or Moisture in the System**

---

### **Indications:**

Low side pressure - Normal  
High side pressure - Normal  
Discharge air is only slightly cool.  
(In a cycling type system with a thermostatic switch, the switch may not cycle the clutch on and off, so the low pressure gauge will not fluctuate.)

---

### **Possible Causes**

Leaks in the system.

---

### **Suggested Corrective Actions**

Test for leaks, especially around the compressor shaft seal area. When the leak is found, recover refrigerant from the system and repair the leak. Replace the receiver-drier or accumulator because the desiccant may be saturated with moisture. Check the compressor and replace any refrigerant oil lost due to leakage. Evacuate and recharge the system with refrigerant using a scale. Check A/C operation and performance.

## **PROBLEM: Air and/or Moisture in the System**

---

### **Indications:**

Low side pressure - HIGH  
High side pressure - HIGH  
Discharge air is only slightly cool.

### **Possible Causes**

---

- Leaks in system.

### **Suggested Corrective Actions**

---

Test for leaks, especially around the compressor shaft seal area. After leaks are found, recover refrigerant from the system and repair leaks. Replace the receiver-drier. Check the compressor and replace any oil lost due to leakage. Evacuate and recharge the system using a scale to ensure proper quantity. Check A/C operation and performance.

## **PROBLEM: Expansion Valve Stuck or Plugged**

---

### **Indications:**

Low side pressure - VERY LOW or in a Vacuum  
High side pressure - HIGH  
Discharge air only slightly cool.  
Expansion valve body is frosted or sweaty.

### **Possible Causes**

---

An expansion valve malfunction could mean the valve is stuck in the closed position, the filter screen is clogged (block expansion valves do not have filter screens), moisture in the system has frozen at the expansion valve orifice, or the sensing bulb is not operating. If the sensing bulb is accessible, perform the following test. If not, proceed to the Repair Procedure.

### **Suggested Corrective Actions**

---

**Test:** Warm diaphragm and valve body with your hand, or very carefully with a heat gun. Activate the system and watch to see if the low pressure gauge rises. Next, carefully spray a little nitrogen, or any substance below 0° C (32° F), on the capillary coil (bulb) or valve diaphragm. The low side gauge needle should drop and read at a lower (suction) pressure on the gauge. This indicates the valve was partially open and that your action closed it. Repeat the test, but first warm the valve diaphragm or capillary with your hand. If the low side gauge drops again, the valve is not stuck.

**Repair Procedure:** Inspect the expansion valve screen (except block type valves). To do this, remove all refrigerant from the system. Disconnect the inlet hose fitting from the expansion valve. Remove, clean, and replace the screen. Reconnect the hose and replace the receiver-drier. Evacuate and recharge the system with refrigerant using a scale. Check AC operation and performance. If the expansion valve tests did not cause the low pressure gauge needle to rise and drop, and if the other procedure described did not correct the problem, the expansion valve is defective. Replace the valve.

## **PROBLEM: Expansion Valve Stuck Open**

---

### **Indications:**

Low side pressure - HIGH  
High side pressure - Normal  
Air from vents in the cab seems warm  
or only slightly cool.

### **Possible Causes**

---

The expansion valve is stuck open and/or the capillary tube (bulb) is not making proper contact with the evaporator outlet tube. Liquid refrigerant may be flooding the evaporator making it impossible for the refrigerant to vaporize and absorb heat normally. In vehicles where the expansion valve sensing bulb is accessible, check the capillary tube for proper mounting and contact with the evaporator outlet tube. Then perform the following test if the valve is accessible. If it is not, proceed to the Repair Procedure.

### **Suggested Corrective Actions**

---

**Test:** Operate the A/C system on it's coldest setting for a few minutes. Carefully spray nitrogen or another cold substance on the capillary tube coil (bulb) or head of the valve. The low pressure (suction) side gauge needle should now drop on the gauge. This indicates the valve has closed and is not stuck open. Repeat the test, but first warm the valve diaphragm by warming with hands. If the low side gauge shows a drop again, the valve is not stuck. Clean the surfaces of the evaporator outlet and the capillary coil or bulb. Make sure the coil or bulb is securely fastened to the evaporator outlet and covered with insulation material. Operate the system and check performance.

**Repair Procedure:** If the test did not result in proper operation of the expansion valve, the valve is defective and must be replaced. Recover all refrigerant from the system and replace the expansion valve and the receiver-drier. Evacuate and recharge the system with refrigerant using a scale. Check A/C operation and performance.

## **PROBLEM: High Pressure Side Restriction**

---

### **Indications:**

Low side pressure - LOW  
High side pressure - Normal to HIGH  
Discharge air is only slightly cool.  
Look for sweat or frost on high side hoses and tubing. The line will be cool to the touch near the restriction.

### **Possible Causes**

---

Kink in a line, collapsed hose liners, plugged receiver-drier or condenser, etc.

### **Suggested Corrective Actions**

---

**Repair Procedure:** After you locate the defective component containing the restriction, recover all of the refrigerant. Replace the defective component and the receiver-drier. Evacuate and recharge the system with refrigerant, then check A/C operation and performance.

## **PROBLEM: Compressor Malfunction**

---

### **Indications:**

- Low side pressure - HIGH
- High side pressure - LOW
- Compressor operates noisily.

### **Possible Causes**

---

- Defective reed valves or other internal components.

### **Suggested Corrective Actions**

---

**Repair Procedure:** If the belt is worn or loose, replace or tighten it and recheck system performance and gauge readings. If inspection of the compressor is required, all of the refrigerant must be recovered and the compressor disassembled to the point that inspection can be performed. Replace defective components or replace the compressor. If particles of desiccant are found in the compressor, flushing of the system will be required. It will also be necessary to replace the receiver-drier. Always check the oil level in the compressor, even if a new unit has been installed. Rotary compressors have a limited oil reservoir. Extra oil must be added for all truck installations. Tighten all connections and evacuate the system. Recharge the system with refrigerant using a scale. Check system operation and performance.

## **PROBLEM: Thermostatic Switch Malfunction**

---

### **Indications:**

- Low side pressure - Normal
- High side pressure - Normal
- Low side pressure may cycle within a smaller range as the compressor clutch cycles more frequently than normal. This may indicate the thermostat is set too high.

### **Possible Causes**

---

- Thermostat malfunctioning possibly due to incorrect installation.

### **Suggested Corrective Actions**

---

Replace the thermostatic switch. When removing the old thermostat, replace it with one of the same type. Take care in removing and handling the thermostat and the capillary tube that is attached to it. Use care not to kink or break the tube.

Position the new thermostat capillary tube at or close to the same location and seating depth between the evaporator coil fins as the old one. Connect the electrical leads

## **PROBLEM: Condenser Malfunction or System Overcharge**

---

### ***Indications:***

Low Side High  
High Side High  
Discharge air may be warm.  
High pressure hoses and lines are very hot.

---

### **Possible Causes**

- Lack of air flow through the condenser fins

---

### **Suggested Corrective Actions**

**Repair Procedure:** Check the engine cooling system components, fan and drive belt, fan clutch operation, and the radiator shutter. Inspect condenser for dirt, bugs, or other debris, and clean if necessary. Be sure the condenser is securely mounted and there is adequate clearance (about 38 mm) between it and the radiator. Check the radiator pressure cap and cooling system, including the fan, fan clutch, drive belts and radiator shutter assembly. Replace any defective parts and then recheck A/C system operation, gauge readings, and performance. If the problem continues, the system may be overcharged. Recover the system refrigerant. Use a scale to recharge the system using the correct amount. Recheck A/C system operation, gauge readings and performance. If the gauge readings do not change, all of the refrigerant should be recovered and the system flushed. The condenser may be partially blocked -replace condenser. The receiver-drier must also be replaced. Evacuate the system, recharge, and check operation and performance.

# PREVENTIVE MAINTENANCE SCHEDULE FOR A/C SYSTEM

Truck Serial Number \_\_\_\_\_

Last Maintenance Check: \_\_\_\_\_

Site Unit Number \_\_\_\_\_

Name of Service Technician \_\_\_\_\_

Date: \_\_\_\_\_ Hour Meter: \_\_\_\_\_

*NOTE: The compressor should be operated for at least 5 minutes at 4°C (40°F) minimum ambient temperature every month in order to circulate oil and lubricate components.*

| COMPONENT                                                                                                                                                                                                                                                                                                                                                                                       | Maintenance Interval<br>(months) |   |    |      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|---|----|------|
|                                                                                                                                                                                                                                                                                                                                                                                                 | 3                                | 6 | 12 | Done |
| <b>1. <u>Compressor</u></b><br>Check noise level<br>Check clutch pulley<br>Check oil level<br>Run system 5 minutes<br>Check belt tension<br>(80-100) lbs; V-belt<br>Inspect shaft seal for leakage<br>Check mounting bracket<br>(tighten bolts)<br>Check clutch alignment w/<br>crankshaft pulley -<br>within 1.5 mm (0.06 in.)<br>Perform manifold gauge<br>check<br>Verify clutch is engaging |                                  |   |    |      |
| <b>2. <u>Condenser</u></b><br>Clean dirt, bugs, leaves, etc.<br>from coils (w/compressed air)<br>Verify engine fan clutch is<br>engaging (if installed)<br>Check inlet/outlet for<br>obstructions or damage                                                                                                                                                                                     |                                  |   |    |      |
| <b>3. <u>Receiver-Drier</u></b><br>Check inlet line from<br>condenser (should be hot to<br>touch)<br>Replace, if system is opened                                                                                                                                                                                                                                                               |                                  |   |    |      |
| <b>4. <u>Accumulator</u></b><br>Check the inlet line from the<br>evaporator. It should be cool<br>to cold.<br>Replace the accumulator<br>each time the system is<br>opened.                                                                                                                                                                                                                     |                                  |   |    |      |

| COMPONENT                                                                                                                                                                                                                                                                                                                                              | Maintenance Interval<br>(months) |   |    |      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|---|----|------|
|                                                                                                                                                                                                                                                                                                                                                        | 3                                | 6 | 12 | Done |
| <b>5. <u>Expansion Valve</u></b><br>Inspect capillary tube (if<br>used) for leakage, damage,<br>looseness                                                                                                                                                                                                                                              |                                  |   | X  |      |
| <b>6. <u>Evaporator</u></b><br>Clean dirt, bugs, leaves, etc.<br>from fins (w/ compressed air)<br>Check solder joints on inlet/<br>outlet tubes (leakage)<br>Inspect condensation drain                                                                                                                                                                |                                  |   |    |      |
| <b>7. <u>Other Components</u></b><br>Check discharge lines<br>(hot to touch)<br>Check suction lines<br>(cold to touch)<br>Inspect fittings/clamps/hoses<br>Check thermostatic switch for<br>proper operation<br>Outlets in cab: 4 to 10° C<br>(40 to 50° F)<br>Inspect all wiring connections<br>Operate all manual controls<br>through full functions |                                  |   |    |      |

# SECTION N5

## OPERATOR CAB CONTROLS

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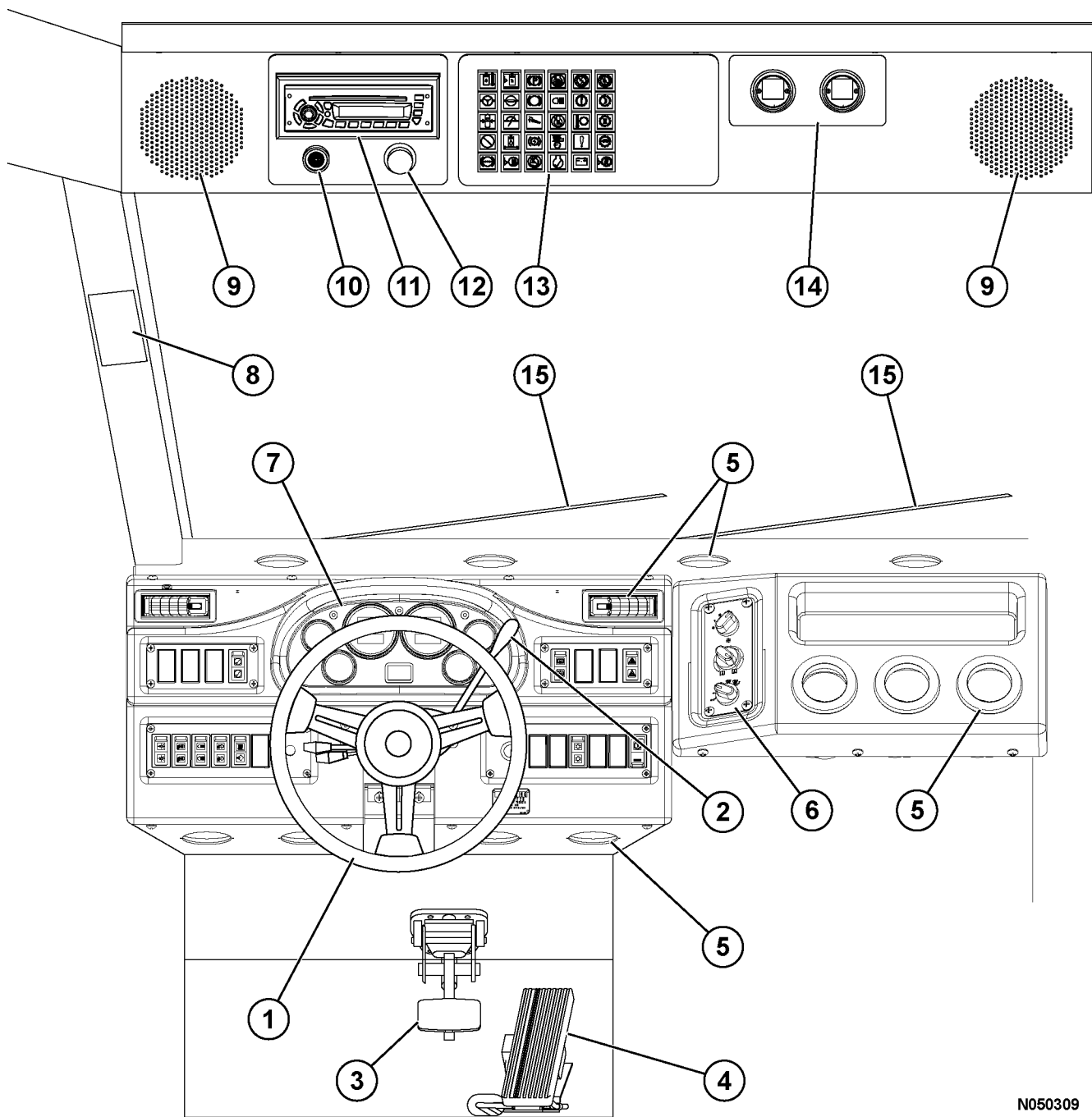
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## OPERATOR CAB CONTROLS



N050309

FIGURE 5-1. CAB INTERIOR - OPERATOR VIEW

- |                                 |                                    |                                     |
|---------------------------------|------------------------------------|-------------------------------------|
| 1. Steering Wheel               | 6. Heater/Air Conditioner Controls | 11. Radio/CD Player                 |
| 2. Retarder Lever               | 7. Instrument Panel                | 12. Warning Lights Dimmer Control   |
| 3. Brake/Retarder Pedal         | 8. Grade/Speed Chart               | 13. Warning/Status Indicator Lights |
| 4. Throttle/Accelerator Pedal   | 9. Radio Speakers                  | 14. Air Cleaner Restriction Gauges  |
| 5. Heater/Air Conditioner Vents | 10. Warning Alarm Buzzer           | 15. Windshield Wipers               |

## STEERING COLUMN

### Removal

1. Turn the key switch OFF. Allow at least 90 seconds for the steering accumulators to bleed down. Turn the steering wheel to ensure that no pressure remains.
2. Activate the battery disconnect switch.
3. Remove access cover (15, Figure 5-2) from the front of the cab.

*NOTE: Do not remove hydraulic lines from the steering control unit unless necessary.*

4. Loosen capscrews (10) on steering control unit (7) and move it out of the way.
5. Disconnect the steering column wire harness.
6. Remove the screws that secure trim cover (14) where the steering column enters the instrument panel. Remove the cover.
7. Remove capscrews (12) and brackets (8) and (9).
8. Remove four capscrews (4) with flat washers (5) and lockwashers (6). Access these capscrews from the front of the cab through the access opening.
9. Lift the steering column from the instrument panel.

### Inspection

Whenever the steering column or steering control unit is removed for service, the steering column shaft splines should be inspected for excessive wear.

1. With the column assembly removed from the truck, thoroughly clean the splines on the steering column shaft. Inspect for damage and excessive wear.
2. Use an outside micrometer or dial caliper to measure the outside diameter of the male splines on the steering column shaft.

**Minimum diameter:** 24.13 mm (0.95 in.)

3. If the splines are smaller than the minimum diameter specification, replace the steering column.

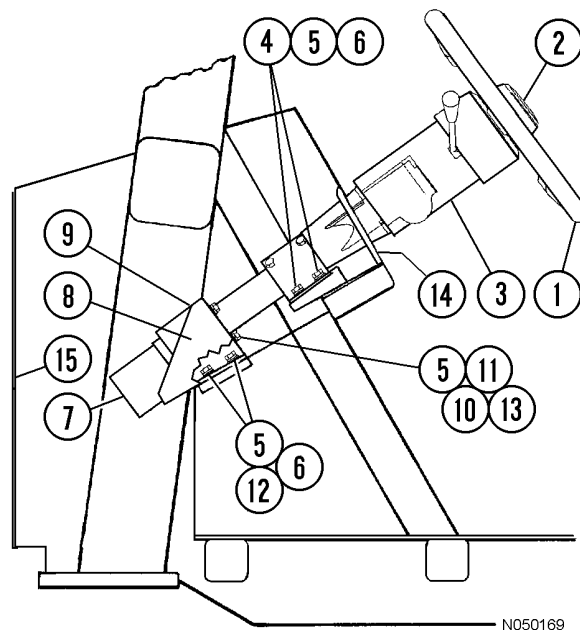


FIGURE 5-2. STEERING COLUMN INSTALLATION

- |                          |                   |
|--------------------------|-------------------|
| 1. Steering Wheel        | 9. Bracket - R.H. |
| 2. Button Horn           | 10. Capscrew      |
| 3. Steering Column       | 11. Lockwasher    |
| 4. Capscrew              | 12. Capscrew      |
| 5. Flat Washer           | 13. Nut           |
| 6. Lockwasher            | 14. Trim Cover    |
| 7. Steering Control Unit | 15. Access Cover  |
| 8. Bracket - L.H.        |                   |

## Installation

1. Insert capscrew (10, Figure 5-2) with lockwashers (11) and flat washers (5) through brackets (8) and (9), then through the steering column flange. Add second flat washer (5) and nut (13) to each capscrew to hold the parts together. Tighten the nuts securely.
2. Slide the entire assembly down the tapered blocks until brackets (8) and (9) contact the mounting surface in the cab. Install capscrews (4) and (12) with flat washers (5) and lockwashers (6). Tighten capscrews (4) only.
3. Inspect brackets (8) and (9) to see whether they contact the mounting surface evenly and are flat and inline with the surface. If they are, tighten capscrews (12). If brackets are not quite parallel, install flat washers as needed between the brackets and mounting surface to eliminate any gaps. Tighten capscrews (12) to the standard torque.
4. After capscrews (4) and (12) are tightened to the standard torque, remove nuts (13) and flat washers (5) that were holding the steering column to the two brackets. Do not remove capscrews (10) from the brackets.
5. Lubricate the male splines on the end of the steering column shaft.

*Note: There is no lower end bearing in this steering column assembly. Therefore, the male end of the shaft will have to be guided into the mating female part of the steering control unit.*

6. Without removing capscrews (10) from the holes, move steering control unit (7) into place and start each of the capscrews.
7. Tighten four capscrews (10) to the standard torque.
8. Ensure that the steering wheel turns properly without binding and that the steering wheel returns to its centered position after rotating 1/4 turn to the left and to the right.
9. If disconnected, reconnect the hoses to the steering control unit.
10. Connect the steering column wire harness to the harness in the cab.
11. Install access cover (15) and trim cover (14).

## STEERING WHEEL

### Removal

1. Turn off the battery disconnect switch to remove battery power from the horn circuit.
2. Use a pocket screwdriver to pry horn button (4, Figure 5-3) from steering wheel (1).
3. Disconnect horn wire (3) and set the horn button aside.
4. Remove nut (2).
5. Pull the steering wheel from the column. If the steering wheel will not slide off the shaft it may be necessary to install a puller into the tapped holes (5/16" - 24NF) in the steering wheel.

### Installation

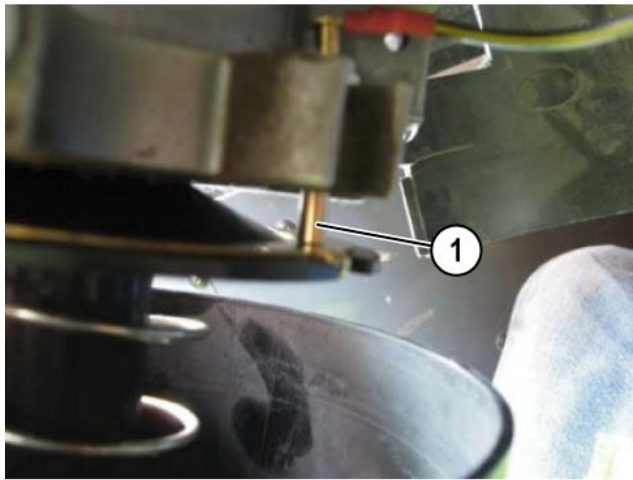
*NOTE: A new, more reliable horn contact is available. Some trucks may have the previous horn contact, which is a stick design (1, Figure 5-4). The new horn contact uses a roller (2). Install the new horn contact as shown in Figure 5-5.*

1. Place steering wheel (1, Figure 5-3) into position on the steering column shaft while guiding horn wire (3) through the access hole in the steering wheel. Align the serrations and push the steering wheel onto the shaft.
2. Install nut (2). Tighten the nut to **81 ± 7 N·m (60 ± 5 ft lb)**.
3. Connect horn wire (3) to horn button (4).
4. Install the horn button onto the steering wheel. Turn on the battery disconnect switch and verify that the horn functions properly.

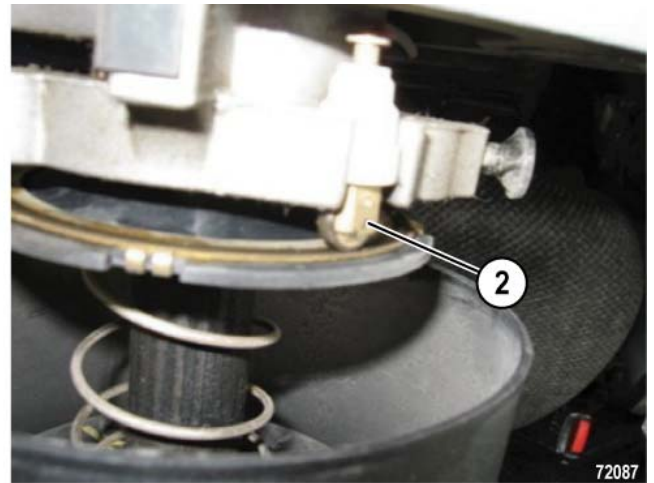


FIGURE 5-3. STEERING WHEEL RETAINER NUT

- |                   |                |
|-------------------|----------------|
| 1. Steering Wheel | 3. Horn Wire   |
| 2. Nut            | 4. Horn Button |

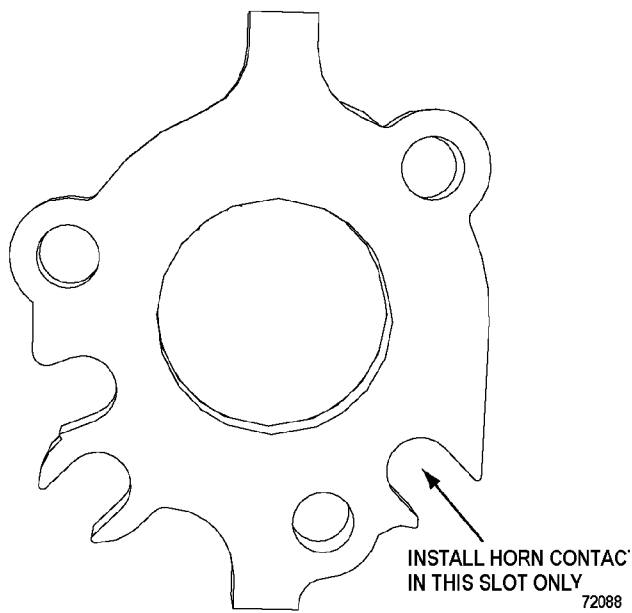


**PREVIOUS DESIGN**



**NEW DESIGN**

**FIGURE 5-4. HORN CONTACTS**



**FIGURE 5-5. INSTALLATION LOCATION**

# STEERING WHEEL AND CONTROLS

Steering wheel (1, Figure 5-6) can be telescoped in and out, and the tilt angle can be adjusted to provide a comfortable steering wheel position for most operators.

## Horn Button

Actuate the horn by pushing horn button (2, Figure 5-6) in the center of the steering wheel. Ensure that the horn operates before moving the truck. Observe all local safety rules regarding the use of the horn as a warning signal device before starting the engine and moving the vehicle.

## Tilt / Telescope Lever

The steering column can be telescoped and the steering wheel can be tilted with lever (3, Figure 5-6) in front of the turn signal lever.

Adjust the tilt of the steering wheel by pulling the lever toward the steering wheel and moving the wheel to the desired angle. Releasing the lever will lock the wheel in the desired location.

Adjust the steering column by pushing the lever forward to unlock the telescoping function. After positioning as desired, release the lever to lock the steering column in position.

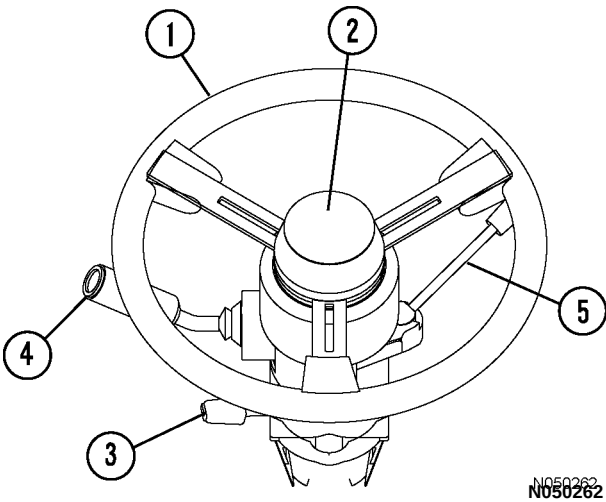


FIGURE 5-6. STEERING WHEEL & CONTROLS

1. Steering Wheel
2. Horn Button
3. Tilt/Telescope Lever
4. Multi-Function Turn Signal Switch
5. Retarder Lever

# Multi-Function Turn Signal Switch



Multi-function turn signal switch (4, Figure 5-6) is used to activate the turn signal lights, the windshield wipers, and to select high or low beam headlights.

## Turn Signal Operation

- Move the lever upward to signal a right turn. ➡
- Move the lever downward to signal a left turn. ⬅

An indicator in the top center of the instrument panel will illuminate to indicate the selected turn direction. Refer to Instrument Panel later in this section.

*NOTE: The turn signal does not automatically cancel after the turn has been completed. The turn signal lever must be manually returned to the neutral position.*

## High Beam Headlight Operation

Pulling the lever inward (toward the rear of the cab) changes the headlights to high beam. When the high beams are selected, the indicator in the top center of the instrument panel will illuminate. Moving the switch back to the original position will return the headlights to low beam.

## Windshield Wiper Operation

- Windshield Wipers OFF
- Intermittent - Long Delay
- Intermittent - Medium Delay
- Intermittent - Short Delay
- Low Speed
- High Speed
- Depressing the button at the end of the lever will activate the windshield washer.

*NOTE: The wipers will not come on automatically when activating the washer. This must be done manually.*

## DYNAMIC RETARDING

Dynamic retarding is a braking torque (not a brake) produced through electrical generation by the wheel motors when the truck motion (momentum) is the propelling force.

For normal truck operation, dynamic retarding should be used to slow and control truck speed.

Dynamic retarding is available in FORWARD/REVERSE at all truck speeds above 0 kph/mph. However, as the truck speed slows below 5 kph (3 mph), the available retarding force may not be effective. Use the service brakes to bring the truck to a complete stop.

Dynamic retarding will not hold a stationary truck on an incline. Use the parking brake or wheel brake lock for this purpose.

When in NEUTRAL, dynamic retarding is available only when truck speed is above 5 kph (3 mph).

When dynamic retarding is in operation, engine rpm will automatically go to an advance retard speed setting. This rpm will vary depending on temperature of several electrical system components.

Dynamic retarding will be applied automatically if the speed of the truck obtains the maximum speed setting programmed in the control system software.

When dynamic retarding is activated, an indicator light in the overhead display will illuminate. The grade/speed retard chart should always be used to determine safe downhill speeds. Refer to Grade/Speed Chart in this chapter.

### Retarder Lever

Retarder lever (5, Figure 5-6) mounted on the right side of the steering column can be used to modulate retarding effort. The lever will command the full range of retarding and will remain at a fixed position when released.

- When the lever is rotated to full “up” (counterclockwise) position, it is in the OFF/no retard position. An adjustable detent holds the lever in the OFF position. Refer to Section J in the service manual for adjustment procedures.
- When the lever is rotated to full “down” (clockwise) position, it is in the full ON/retard position.
- For long downhill hauls, the lever may be positioned to provide desired retarding effort, and it will remain where it is positioned.

*NOTE: The retarder control lever must be rotated back to the OFF position before the truck will resume the propel mode of operation.*

The lever and foot-operated retarder/service brake pedal can be used simultaneously or independently. The Propulsion System Controller (PSC) will determine which device is requesting the most retarding effort and apply that amount.

### Brake/Retarder Pedal

Brake/retarder pedal (3, Figure 5-1) is a single, foot-operated pedal that controls both retarding and service brake functions. The first portion of pedal travel commands retarding effort through a rotary potentiometer. The second portion of pedal travel modulates service brake pressure directly through a hydraulic valve. Thus, the operator must first apply, and maintain, full dynamic retarding in order to apply the service brakes. Releasing the pedal returns brake and retarder to the OFF position.

When the pedal is partially depressed, the dynamic retarding is actuated. As the pedal is further depressed, to where dynamic retarding is fully applied; the service brakes (while maintaining full retarding) are actuated through a hydraulic valve which modulates pressure to the service brakes. **Completely depressing the pedal causes full application of both dynamic retarding and the service brakes.** Indicator light (B3, Figure 5-10) in the overhead panel will illuminate, and an increase in pedal resistance will be felt when the service brakes are applied.

For normal truck operation, **dynamic retarding** (lever or foot-operated pedal) should be used to slow and control the speed of the truck.

**Service brakes** should be applied only when dynamic retarding requires additional braking force to slow the truck speed quickly or when bringing the truck to a complete stop.

### Throttle/Accelerator Pedal

Throttle/accelerator pedal (4, Figure 5-1) is a foot-operated pedal which allows the operator to control engine rpm depending on pedal depression.

It is used by the operator to request torque from the motors when in forward or reverse. In this mode, the propulsion system controller commands the correct engine speed for the power required. In NEUTRAL, this pedal controls engine speed directly.

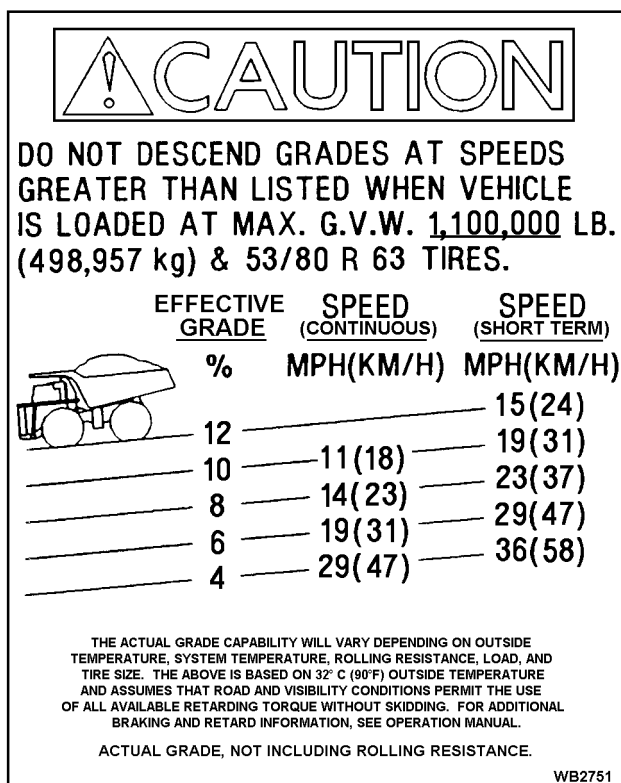


## GRADE/SPEED CHART

Grade/speed chart (8, Figure 5-1 and shown below) provides the recommended MAXIMUM retarding limits at various truck speeds and grades with a fully loaded truck.

This decal in the truck may differ from the decal below due to optional truck equipment such as: wheel motor drive train ratios, retarder grids, tire sizes, etc. Always refer to this decal in the operator's cab and follow these recommendations for truck operation.

The operator should reference this chart before descending any grade with a loaded truck. Proper use of dynamic retarding will maintain a safe speed.



Two speed lists are provided, one for *continuous* retarding, and the second for *short term* retarding. Both lists are matched to the truck at maximum Gross Vehicle Weight (GVW). The two ratings are guidelines for proper usage of the retard function on downhill grades.

The “short term” numbers listed on the chart indicate the combination of speeds and grades which the vehicle can safely negotiate for a short duration before system components reach the maximum allowable temperature during retarding. These speeds are faster than the “continuous” values, reflecting the short term thermal capacity of various system components.

**NOTE:** The “short term” rating will successfully accommodate most downhill loaded hauls.

If the vehicle is operated at “short term” grade and speed limits for a period of time, it is possible to reach the thermal capacity of the drive system components. The Propulsion System Controller (PSC) will then gradually reduce the retarding effort from “short term” to “continuous”. The “retard @ continuous” indicator light will flash for 15 seconds, then illuminate steadily alerting the operator of the reduction in retarding performance and the need for a reduction in speed. The operator must control the speed of the truck in accordance to the “continuous speeds” on the grade/speed chart. The operator must use the service brakes to **quickly** slow the truck to the maximum “continuous” retarding limits (or less) based on the grade the truck is currently on.



**Do not LIGHTLY apply the service brakes when attempting to slow the truck on a downhill grade. Overheating of the brakes will result. Apply the brakes (within safe limits for road conditions) in order to quickly slow the truck to maximum “continuous” retarding limits or less.**

The “continuous” numbers on the chart indicate the combination of speeds and grades which the vehicle can safely negotiate for unlimited time or distance during retarding.

DO NOT exceed these recommended MAXIMUM speeds when descending grades with a loaded truck.

## OVERHEAD PANEL AND DISPLAYS

The components described below are located on the overhead panel. Refer to Figure 5-1.

### Speakers

Speakers (9, Figure 5-1) for the radio/CD player are located at the far left and right of the overhead panel.

### Warning Alarm Buzzer

A warning alarm buzzer (10) will sound when activated by any one of several truck functions. Refer to Instrument Panel and Indicator Lights in this section for a detailed description of functions and indicators that will activate this alarm.

### Radio/CD Player

This panel will normally contain radio/CD player (11). Refer to Section 70 in the Operation and Maintenance Manual for a complete description of the radio/CD player and its functions. Individual customers may use this area for other purposes, such as a two-way communications radio.

### Warning Indicator Light Dimmer Control

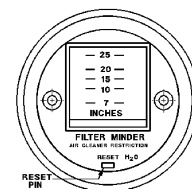
Dimmer control (12) below the radio/CD player permits the operator to adjust the brightness of the warning indicator lights.

## Status/Warning Indicator Light Panel

Status/ warning indicator panel (13) contains an array of indicator lights to provide the operator with important status messages concerning selected truck functions. Refer to Instrument Panel and Indicator Lights in this section for a detailed description of these indicators.

### Air Cleaner Restriction Gauges

Air cleaner restriction gauges (14) provide a continuous reading of the maximum air cleaner restriction reached during operation. The air cleaner(s) should be serviced when the gauge(s) shows the maximum recommended restriction of 25 inches of H<sub>2</sub>O vacuum.



**NOTE:** After service, push the reset button on face of gauge to allow the gauge to return to zero.

## CENTER CONSOLE

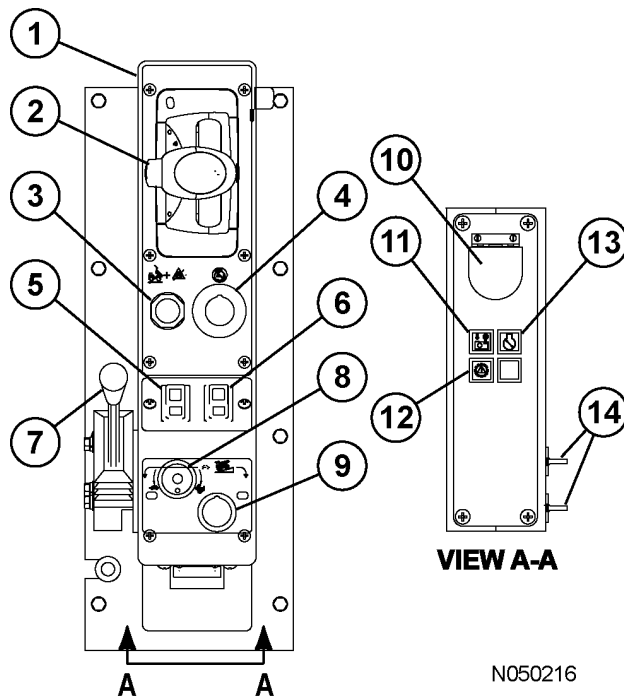


FIGURE 5-7. CENTER CONSOLE

1. Center Console
2. Directional Control Lever
3. Override/Fault Reset Switch
4. Engine Shutdown Switch
5. L.H. Window Control Switch
6. R.H. Window Control Switch
7. Hoist Control Lever
8. Retarder Speed Control Dial
9. Retarder Speed Control Switch
10. Data Store Button
11. KOMTRAX Plus Snapshot In Progress Light
12. Link Energized Light (red)
13. Service Engine Light (blue)
14. 12V Auxiliary Power Outlets

## Directional Control Lever

Directional control lever (2, Figure 5-7) is mounted on a console to the right of the operator's seat. It is a four position lever that controls the park, reverse, neutral and forward motion of the truck.

Before moving the directional control lever, apply the service brakes to completely stop the truck. Depress the button on the side to release the detent lock, then move the control lever to the desired position. When the control lever is in the center N position, it is in NEUTRAL. When the control lever is in the P position, it is in PARK, and the parking brake will be applied. The parking brake is spring applied and hydraulically released. It is designed to hold the truck stationary when the engine is off and the key switch is turned OFF. The truck must be completely stopped before moving the control lever to PARK, or damage may occur to the parking brake.

When the key switch is ON, and the control lever is in PARK, parking brake indicator light (A3, Figure 5-10) in the overhead panel will be illuminated.

## ⚠ IMPORTANT ⚠

***The directional control lever must be in PARK to start the engine.***

***NOTE:*** Do not move the directional control lever to the PARK position at the shovel or dump site.

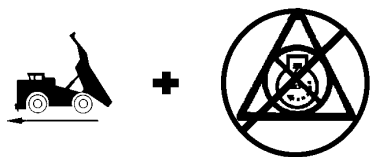
The operator can select FORWARD drive by moving the lever to the F position.

The operator can select REVERSE drive by moving the lever to the R position. Do not allow the control lever to travel too far and go into the PARK position when REVERSE is desired.

***NOTE:*** The truck must be completely stopped before the control lever is moved to a drive position or into PARK. A fault will be recorded if the control lever is placed into the PARK position while the truck is still moving.

### Override/Fault Reset Switch

This push-button switch (3, Figure 5-7) is spring-loaded to the OFF position. When pushed in and held, this switch may be used for several functions.



- The switch permits the operator to override the body-up limit switch and move the truck forward when the directional control lever is in FORWARD, the dump body is raised, and the brakes are released. **Use of the override switch for this purpose is intended for emergency situations only!**
- The push-button deactivates the retard pedal function when the truck speed is below 5 kph (3 mph).
- The switch is also used to reset an electric system fault that is indicated by a red warning light. Refer to Overhead Status/Warning Indicators later in this section.

### Engine Shutdown Switch

Engine shutdown switch (4, Figure 5-7) is used for engine shutdown. Pull the switch up to stop the engine. Push the switch back down to enable engine operation.



SWITCH UP  
ENGINE OFF



SWITCH DOWN  
ENGINE ON

Use this switch to shutdown the engine if the key switch fails or to stop the engine without turning off the 24 VDC electrical circuits.

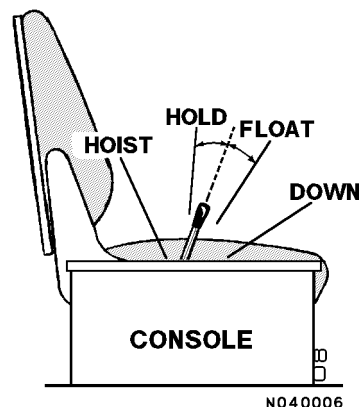
A ground level engine shutdown switch is also located at the left front corner of the truck.

### Window Control Switches

Control switches (5 and 6, Figure 5-7) are spring-loaded to the OFF position. Pushing the front of the switch raises the left hand or right hand cab window. Pushing the rear of the switch lowers the window.

### Hoist Control Lever

Hoist control (7, Figure 5-7) is a four position hand-operated lever located between the operator seat and the center console (see illustration below).



#### Raising The Dump Body

1. Pull the lever to the rear to actuate the hoist circuit. Releasing the lever anywhere during "hoist up" will place the body in HOLD at that position.
2. Raise engine rpm to increase hoist speed.
3. Reduce the engine speed as the last stage of the hoist cylinders begin to extend, then let the engine go to low idle as the last stage reaches half-extension.
4. Release the hoist lever as the last stage reaches full extension.
5. After the material being dumped clears the body, lower the body to the frame.

Refer to Section A3, General Safety & Operating Instructions - Dumping, for more complete details about this control.

#### Lowering The Dump Body

Move the hoist lever forward to the DOWN position and release. Releasing the lever places hoist control valve in the FLOAT position, allowing the body to return to the frame.

## Retarder Speed Control (RSC) Adjustment Dial

RSC adjustment dial (8, Figure 5-7) allows the operator to vary the downhill truck speed that the retard speed control system will maintain when descending a grade. This function can be overridden by the throttle pedal, retard lever, or brake/retard pedal.

When the dial is rotated counterclockwise toward this symbol, the truck will descend a grade at lower speeds.



When the dial is rotated clockwise toward this symbol, the truck speed will increase.



**NOTE:** Always refer to the grade/speed chart in the operator's cab and follow the recommendations for truck operation. DO NOT exceed these recommended MAXIMUM speeds when descending grades with a loaded truck.

Throttle pedal position will override the RSC setting. If the operator depresses the throttle pedal to increase truck speed, dynamic retarding will not come on unless the truck overspeed setting is reached or the brake/retarder pedal is used. When the throttle pedal is released and the RSC switch is ON, dynamic retarding will come on at, or above, the RSC dialed speed and will adjust the truck speed to, and maintain, the dialed speed.

To adjust the RSC control, pull up switch (9) and start with dial (8) rotated toward fastest speed while driving truck at the desired maximum speed. Release the throttle pedal to let the truck coast and turn the RSC adjusting dial slowly counterclockwise until dynamic retarding is activated. Dynamic retarding will now be activated automatically anytime the "set" speed is reached, the RSC switch is on, and the throttle pedal is released.

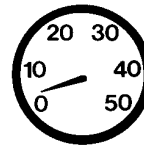
With the RSC switch on and dial adjusted, the system will function as follows:

As the truck speed increases to the "set" speed and the throttle pedal is released, dynamic retarding will apply. As the truck speed tries to increase, the amount of retarding effort will automatically adjust to keep the selected speed. When the truck speed decreases, the retarding effort is reduced to maintain the selected speed. If the truck speed continues to decrease to approximately 5 kph (3 mph) below the "set" speed, dynamic retarding will turn off automatically. If the truck speed must be reduced further, the operator can turn the dial to a new setting or depress the brake/retarder pedal.

If the operator depresses the foot operated retard pedal and the retard effort called for is greater than that from the automatic system, the foot pedal retard will override RSC.

## Retarder Speed Control (RSC) Switch

Retarder speed control switch (9, Figure 5-7) turns the system on and off. Push the knob in for OFF and pull the knob out to turn the system ON.



## Data Store Button

Button (10, Figure 5-7) is for use by qualified maintenance personnel to record in memory a "snapshot" of the AC drive system. It will also trigger KOMTRAX Plus to store a snapshot of the truck operating system. Light (13, Figure 5-7) will stay illuminated while KOMTRAX Plus is recording the snapshot.

## KOMTRAX Plus Snapshot In Progress Light

Light (11, Figure 5-7) is an indicator that will illuminate while KOMTRAX Plus is in the process of taking a snapshot of machine data.

## Link Energized Light

Light (12, Figure 5-7) is a red indicator that, when illuminated, indicates that the AC drive system is energized. No one is permitted to work on the AC drive system while this light is illuminated.

## Service Engine Light

Light (13, Figure 5-7) is a blue indicator that will illuminate if a problem is detected by the electronic engine monitoring system.

Electric propulsion and dynamic retarding will still be available.

If this light is ON, notify maintenance personnel so they can diagnose and repair the problem the next time the truck is in the shop for repairs or at the next PM (Preventive Maintenance) interval.

## 12V Auxiliary Power Outlets

Outlets (14, Figure 5-7) can be used to provide 12VDC power for tools and accessories.

## DIAGNOSTIC PORTS

### KOMTRAX Plus Diagnostic Port

Diagnostic port (1, Figure 5-8) is used to download truck operation data from the KOMTRAX Plus controller.

### Interface Module (IM) Diagnostic Port

Diagnostic port (2) is used to connect the interface module to a computer for installing software.

### Payload Meter Diagnostic Port

Diagnostic port (3) is used to download data from the payload meter system. Refer to Section M20, Payload Meter III, for a more complete description of the payload meter and its functions.

### Truck Control Interface (TCI) Diagnostic Port

Diagnostic port (4) is used to access diagnostic information for the Truck Control Interface (TCI).

### Propulsion System Controller (PSC) Diagnostic Port

Diagnostic port (5) is used to access diagnostic information for the Propulsion System Controller.

### Engine Diagnostic Port (CENSE)

3-pin diagnostic port (6) is used to access diagnostic information for the engine monitoring system.

### Engine Diagnostic Port (QUANTUM)

9-pin diagnostic port (7) is used to access diagnostic information for the engine monitoring system.

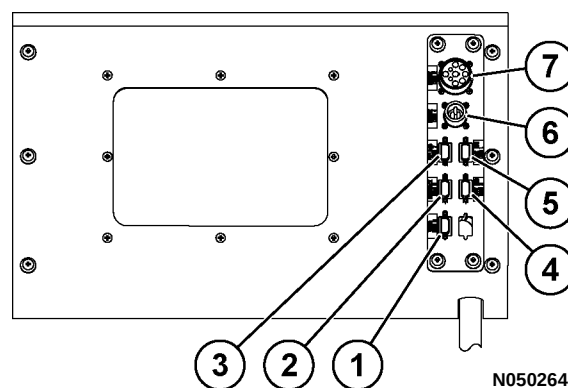


FIGURE 5-8. DID PANEL DIAGNOSTIC PORTS

- |                                  |                                     |
|----------------------------------|-------------------------------------|
| 1. KOMTRAX Plus Diagnostic Port  | 5. PSC Diagnostic Port              |
| 2. IM Diagnostic Port            | 6. Engine Diagnostic Port (CENSE)   |
| 3. Payload Meter Diagnostic Port | 7. Engine Diagnostic Port (QUANTUM) |
| 4. TCI Diagnostic Port           |                                     |

## INSTRUMENT PANEL

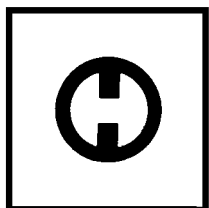
The operator must understand the function and operation of each instrument and control. Control functions are identified with “international” symbols that the operator should learn to recognize immediately. This knowledge is essential for proper and safe operation.

Items that are marked “optional” do not apply to every truck.

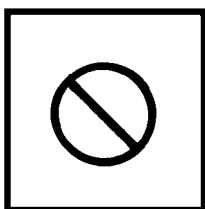
### Control Symbols

Many control functions are identified with international symbols that the operator should learn to recognize immediately. The operator must understand the function and operation of each instrument and control. This knowledge is essential for proper and safe operation of the machine.

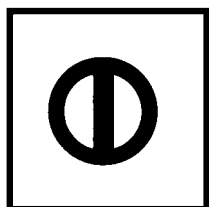
The following symbols are general indicators and may appear in multiple locations and combinations on the instrument panel.



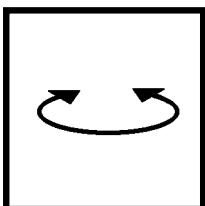
This symbol may be used alone or with another symbol. This symbol identifies the OFF position of a switch or control.



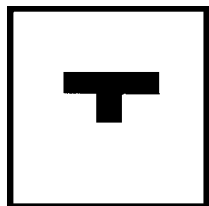
This symbol when it appears on an indicator or control identifies that this indicator or control is NOT used.



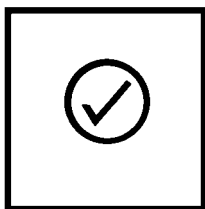
This symbol may be used alone or with another symbol. This symbol identifies the ON position of a switch or control.



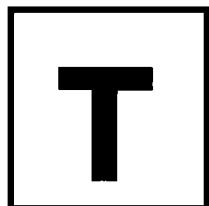
This symbol identifies a rotary control or switch. Rotate the knob clockwise or counterclockwise for functions.



This symbol identifies the “Pushed-In” position of a push-pull switch or control.



This symbol identifies a switch used to test or check a function. Press the switch on the side near the symbol to perform the test.



This symbol identifies the “Pulled-Out” position of a push-pull switch or control.

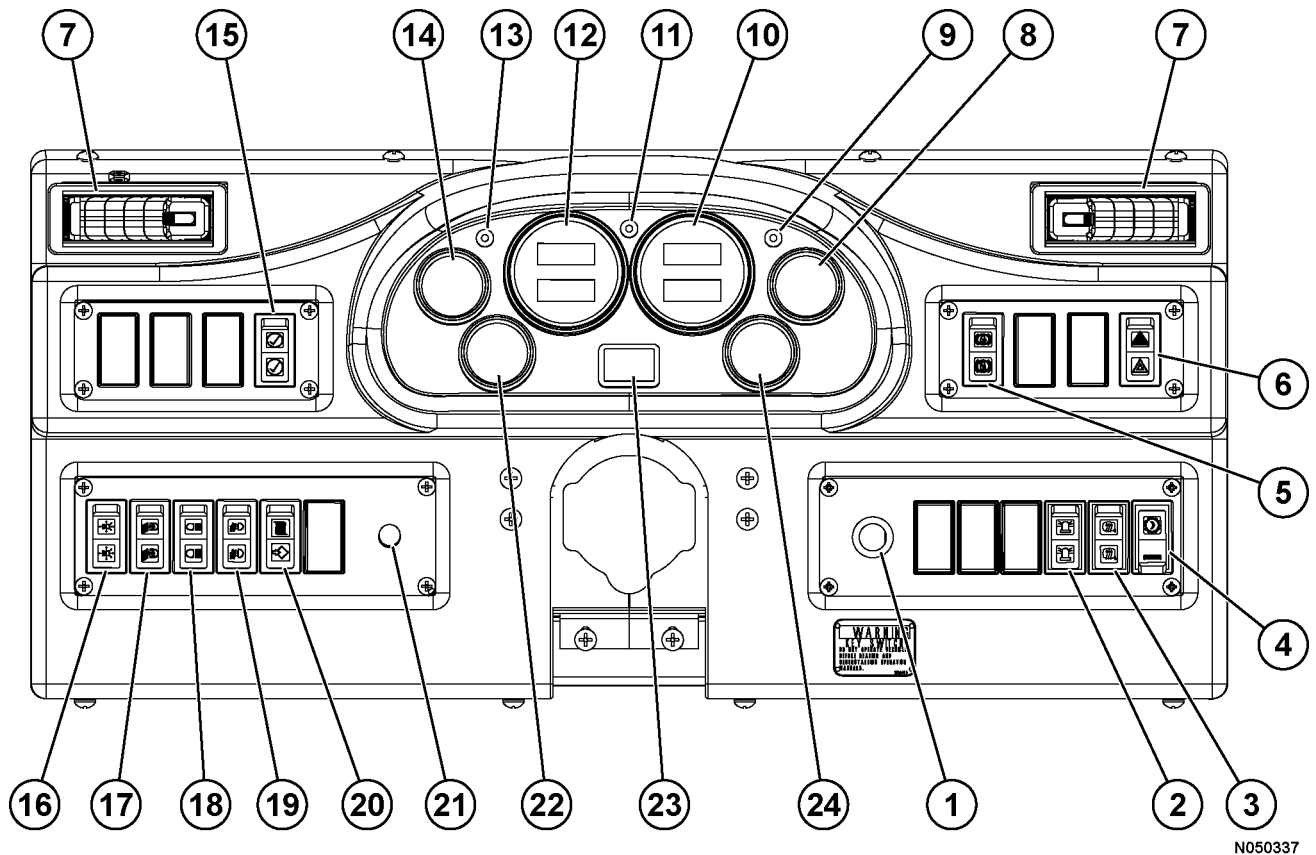
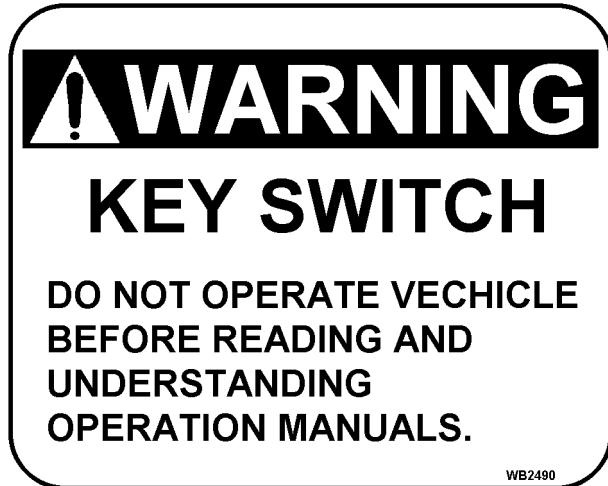


FIGURE 5-9. PANEL GAUGES, INDICATORS, AND CONTROLS

- |                                       |                                             |
|---------------------------------------|---------------------------------------------|
| 1. Key Switch                         | 13. Left Turn Signal Indicator              |
| 2. Rotating Beacon Switch (optional)  | 14. Water Temperature Gauge                 |
| 3. Heated Mirrors Switch (optional)   | 15. Lamp Check Switch                       |
| 4. AC Drive System Rest Switch        | 16. Light Switch (3-Way)                    |
| 5. Wheel Brake Lock Switch            | 17. Ladder Light Switch                     |
| 6. Hazard Lights Switch               | 18. Backup Light Switch                     |
| 7. Heater/Air Conditioner Vents       | 19. Fog Light Switch                        |
| 8. Engine Oil Pressure Gauge          | 20. Payload Meter Switch                    |
| 9. Right Turn Signal Indicator Light  | 21. Panel Illumination Lights Dimmer Switch |
| 10. Digital Tachometer                | 22. Hydraulic Oil Temperature Gauge         |
| 11. High Beam Headlight Indicator     | 23. Engine Hourmeter                        |
| 12. Speedometer/Payload Meter Display | 24. Fuel Level Gauge                        |



## Key Switch



Key switch (1, Figure 5-9) is a four-position switch (ACC, OFF, RUN, START). The ACC position is not currently used.

When the switch is rotated one position clockwise, it is in the RUN position and all electrical circuits (except START) are activated.

1. With the directional control lever in PARK, rotate key switch fully clockwise to START position, and hold this position until engine starts. START position is spring-loaded to return to RUN when key is released.
2. After the engine has started, place rest switch (4, Figure 5-9) in the OFF position to deactivate the rest mode of operation.

**NOTE:** The electric cranking motors have a 30 second time limit. If the 30 second limit is reached, cranking will be prohibited for two minutes. After two minutes, cranking will be allowed. If the 30 second limit is reached seven consecutive times, the key switch must be turned to the OFF position. This will allow the interface module to power down and reset, which requires seven minutes to complete.

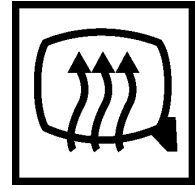
## Rotating Beacon Switch (optional)

Rotating beacon switch (2, Figure 5-9) is used to activate the rotating beacon (if equipped) on top of the operator cab.



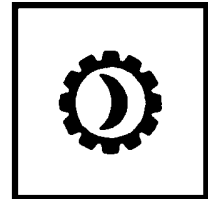
## Heated Mirrors Switch (optional)

Heated mirrors switch (3, Figure 5-9) is used to activate the heated mirrors (if equipped) on both sides of the truck.



## AC Drive System Rest Switch

Rest switch (4, Figure 5-9) is a rocker switch with a locking device for the OFF position. A small tab must be pushed to unlock the switch before it can be depressed to the ON position. When in the ON position, an amber lamp in the switch will illuminate.



The switch should be in the ON position to de-energize the AC drive system whenever the engine is to be shutdown or the truck will be parked for a length of time with the engine running. The directional control lever must be in PARK or NEUTRAL and the vehicle not moving to enable this function. This will allow the engine to continue running while the AC drive system is de-energized.

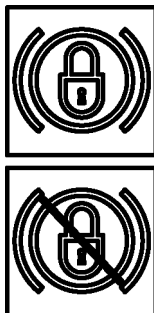
Amber indicator light (B6, Figure 5-10) in the overhead panel will illuminate when the rest mode is activated.

## WARNING

**Activation of the rest switch alone DOES NOT completely ensure that the drive system is safe to work on. Refer to Section A3, Safety, for more information on servicing an AC drive truck. Check all "link-on" or "link energized" indicator lights to verify that the AC drive system is de-energized before performing any maintenance on the drive system. DO NOT activate the rest switch while the truck is moving! The truck may unintentionally enter the rest mode after stopping.**

### Wheel Brake Lock Switch

Wheel brake lock switch (5, Figure 5-9) should be used with engine running for dumping and loading operations only. The brake lock switch actuates the hydraulic brake system which locks the **rear wheel service brakes only**. When pulling into shovel or dump area, stop the truck using the foot-operated service brake pedal. When truck is completely stopped and in loading position, apply the brake lock by pressing the top of the rocker switch. To release the brake, press the bottom of the rocker switch.



*NOTE: Use at the shovel and dump only to hold the truck in position.*



**Do not use the wheel brake lock switch to stop the truck unless foot-operated treadle valve is inoperative. Use of this switch applies rear service brakes at a reduced, unmodulated pressure. Do not use brake lock for parking. With engine stopped, hydraulic pressure will bleed down, allowing brakes to release.**

### Hazard Warning Lights

Hazard warning light switch (6, Figure 5-9) flashes all the turn signal lights. Pressing the bottom of the rocker switch activates these lights. Pressing the top of the rocker switch turns these lights off.

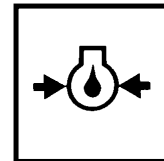


### Heater/Air Conditioner Vents

Heater/air conditioner vents (7, Figure 5-9) may be directed by the operator to provide the most comfortable cabin air flow.

### Engine Oil Pressure Gauge

Engine oil pressure gauge (8, Figure 5-9) indicates pressure in the engine lubrication system.



Normal operating pressure after engine warm-up should be:

Idle: 138 kPa (20 psi) minimum

Rated Speed: 310 - 483 kPa (45 - 70 psi)

### Right Turn Signal Indicator

Indicator (9, Figure 5-9) illuminates to indicate that the right turn signals are operating when the turn signal lever on the steering column is moved upward. Moving the lever to its center position will turn the indicator off.

### Digital Tachometer

Digital tachometer (10, Figure 5-9) registers engine crankshaft speed in hundreds of revolutions per minute (rpm).

| Governed Speed |          |
|----------------|----------|
| Low Idle       | 750 rpm  |
| High Idle      | 1910 rpm |
| Full Load      | 1900 rpm |

### High Beam Indicator

When lit, high beam indicator (11, Figure 5-9) indicates that the headlights are on high beam. To switch the headlights to high beam, push the turn indicator lever away from the steering wheel. For low beam, pull the lever toward the steering wheel.

### Speedometer/Payload Meter Display

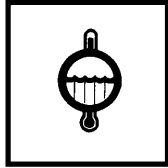
Speedometer/payload meter display (12, Figure 5-9) indicates the truck speed in kilometers per hour (kph) or in miles per hour (mph). The display also shows payload meter information. For more information, see Section M20, Payload Meter III.

### Left Turn Signal Indicator

Indicator (13, Figure 5-9) illuminates to indicate that the left turn signals are operating when the turn signal lever is moved downward. Moving the lever to its center position will turn the indicator off.

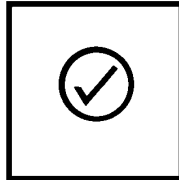
### Coolant Temperature Gauge

Coolant temperature gauge (14, Figure 5-9) indicates the temperature of the coolant in the engine cooling system. The temperature range after engine warm-up and with the truck operating under normal conditions should be 85° - 97°C (185° - 207°F).



### Lamp Check Switch

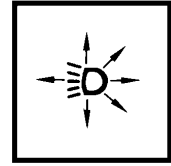
Lamp check switch (15, Figure 5-9) is provided to allow the operator to test the indicator lamps prior to starting the engine. To test the lamps and the warning horn, turn the key switch to the RUN position and press the bottom side of the rocker switch for the CHECK position. All lamps should illuminate except those which are for optional equipment that may not be installed. The warning alarm should also sound. Any lamp bulbs which do not illuminate should be replaced before operating the truck. Releasing the spring-loaded switch will allow the switch to return to the OFF position.



*NOTE: Do not use the lamp check switch while the engine is on. Pressing this switch while the engine is on will cause false electrical system faults. Warning light (D5, Figure 5-10) will illuminate and engine shutdown is required to turn it off.*

### Light Switch (3-Way)

The instrument panel lights, clearance lights, and the headlights are controlled by this three-position rocker type switch (16, Figure 5-9). OFF is selected by pressing the bottom of the switch. Press the top of the switch until it reaches the first detent to select the panel lights, clearance lights and tail lights only. Press the top of the switch again until it reaches the second detent to select headlights, panel lights, clearance lights and tail lights.



### Ladder Light Switch

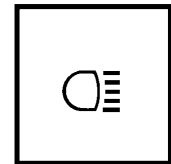
Ladder light switch (17, Figure 5-9) turns the ladder lights on or off after or before using ladder. Pressing the top of the rocker switch turns the lights on. Pressing the bottom of the switch turns the lights off. Another switch is mounted at the front left of truck near the base of ladder.



*NOTE: This switch can be used to activate the lights whenever the key switch is in the ON position.*

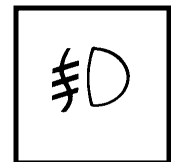
### Backup Light Switch

Backup light switch (18, Figure 5-9) allows the backup lights to be turned on for added visibility and safety when the directional control lever is not in the REVERSE position. When the switch is in the ON position, backup light indicator (B4, Figure 5-10) in the overhead panel will be illuminated.



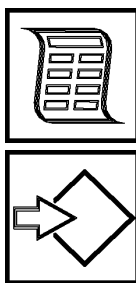
### Fog Light Switch

Pressing the top of fog light switch (19, Figure 5-9) turns the lights on. Pressing the bottom of the switch turns the lights off.



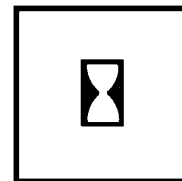
### Payload Meter Switch

Payload Meter switch (20, Figure 5-9) is a two-way momentary rocker switch. The top position is the SELECT position. The SELECT position is used to step through the different displays. The lower position is the SET position. The SET position is used to set the operator ID, or clear the load and total ton counters. Refer to Section M20, Payload Meter III, for a more complete description of the payload meter system and its functions.



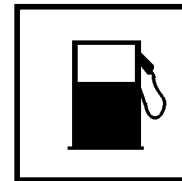
### Engine Hourmeter

Engine hourmeter (23, Figure 5-9) registers the total number of hours the engine has been in operation.



### Fuel Gauge

Fuel gauge (24, Figure 5-9) indicates how much diesel fuel is in the fuel tank. See Section P2, Lubrication & Service, for the fuel tank capacity.



### Panel Illumination Light Dimmer Switch

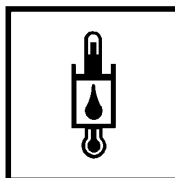
Panel illumination light dimmer switch (21, Figure 5-9) is a rheostat which allows the operator to vary the brightness of the instruments and panel illumination lights.



Rotating knob to the full counterclockwise position turns panel lights on to brightest condition. Rotating knob clockwise continually dims lights until the OFF position is reached at full clockwise rotation.

### Hydraulic Oil Temperature Gauge

Hydraulic oil temperature gauge (22, Figure 5-9) indicates the hydraulic oil temperature. There are two colored bands: green and red.



Green indicates normal operation.

As the needle approaches the red zone, minimum engine idle speed will increase to help cool the oil.

Red indicates high oil temperature. Continued operation could damage components in the hydraulic system. Warning lights (A1, Figure 5-10) and (E1) in the overhead panel will illuminate when the temperature exceeds a certain level (when the needle enters the red zone). If this occurs, the operator should safely stop the truck, move directional control lever to PARK and operate the engine at 1200 - 1500 rpm to reduce system temperature.

*NOTE: If the temperature gauge does not move into the green range after a few minutes, and the warning lights do not go out, shut down the truck and notify maintenance personnel immediately.*

## OVERHEAD STATUS / WARNING INDICATORS

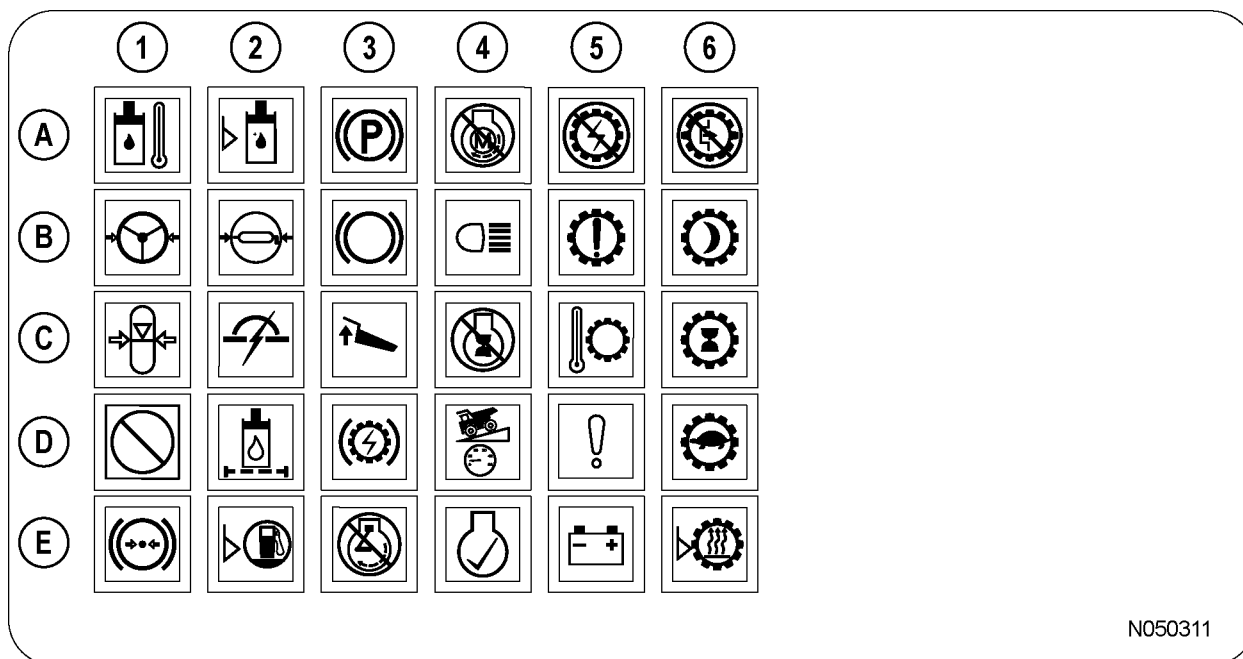


FIGURE 5-10. OVERHEAD STATUS / WARNING INDICATOR

| Row / Column | Indicator Description              | Indicator Color Wire Index |
|--------------|------------------------------------|----------------------------|
| A1*          | High Brake Oil Temp.               | Red-24VIM/34TL             |
| B1*          | Low Steering Pressure              | Red-12F/33A                |
| C1           | Low Accumulator Press.             | Red-12F/33K                |
| D1           | Not Used                           |                            |
| E1           | Low Brake Pressure                 | Red-12F/33L                |
| A2*          | Low Hydraulic Oil Level            | Red-12F4/34LL              |
| B2*          | Low Automatic Lubrication Pressure | Amber-12MD8/68LLP          |
| C2*          | Circuit Breaker Tripped            | Amber-12MD3/31CB           |
| D2*          | Hydraulic Oil Filter               | Amber-12MD/39              |
| E2*          | Low Fuel                           | Amber-24VIM/38             |
| A3*          | Park Brake Applied                 | Amber-24VIM/52AL           |
| B3*          | Service Brake Applied              | Amber-12MD/44L             |
| C3*          | Body Up                            | Amber-12MD6/63L            |
| D3*          | Dynamic Retarding                  | Amber-12MD/44DL            |
| E3           | Stop Engine                        | Red-12M/31MT               |
| A4*          | Starter Failure                    | Amber-21SL/24VIM           |
| B4*          | Manual Backup Lights               | Amber-12MD/47L             |
| C4*          | Timed Engine Shutdown              | Amber-12MD/23L1            |
| D4*          | Retard Speed Control               | Amber-12MD/31R             |
| E4*          | Check Engine                       | Amber-12MD7/419            |

| Row / Column | Indicator Description         | Indicator Color Wire Index |
|--------------|-------------------------------|----------------------------|
| A5           | No Propel/ No Retard          | Red-12M/75-6P1             |
| B5           | Propulsion System Warning     | Amber-12F/79WI             |
| C5           | Propulsion System Temperature | Amber -12F/34TW1           |
| D5           | System/Component Failure      | Red-24VIM/311MLI           |
| E5           | Battery Charge System Failure | Red-24VIM/11BCF1           |
| A6           | NO PROPEL                     | Red-12M/75NPI              |
| B6           | Propulsion System @ Rest      | Amber-12M/72PR1            |
| C6*          | Propulsion System Not Ready   | Amber-12MD/72NR1           |
| D6*          | Reduced Propulsion            | Amber-12MD/72LP1           |
| E6*          | Retard @ Continuous Level     | Amber-12MD/76LR1           |

\* Brightness for these indicator lamps can be adjusted by using dimmer control (12, Figure 5-1).

Indicator lights that are amber (yellow) in color alert the operator that the indicated truck function requires some precaution when lighted.

Indicator lights that are red in color alert the operator that the indicated truck function requires **immediate action** by the operator. Safely stop the truck and turn off the engine.

### **DO NOT OPERATE THE TRUCK WITH A RED WARNING LIGHT ILLUMINATED!**

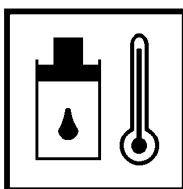
Refer to Figure 5-10 and the descriptions below for explanations of the symbols. Location of the symbols is described by rows (A - E) and columns (1 - 6).

#### **A1. High Hydraulic Oil Temperature**

This red warning light indicates that the brake oil temperature is too high. The low brake pressure warning light will also illuminate.

If these lights come on, the operator should safely stop the truck, move the directional control lever to PARK and operate the engine at 1200 - 1500 rpm to reduce system temperature.

If temperature gauge (22, Figure 5-9) does not move into the green range after a few minutes, and the warning lights do not go out, turn off the engine and notify maintenance personnel immediately.

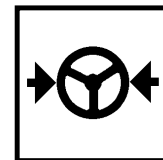


### **⚠ WARNING**

***If temperature gauge does not move into the green range after a few minutes, and the red overhead indicator light does not go out, stop the engine and notify maintenance personnel immediately.***

#### **B1. Low Steering Pressure**

When the key switch is turned ON, this red warning light will illuminate and the warning buzzer will sound until the steering system hydraulic pressure reaches 15 860 kPa (2,300 psi). Both will remain on until the accumulator has been charged.



During truck operation, the low steering pressure warning light, low brake pressure warning light, and warning alarm will turn on if steering system hydraulic pressure drops below 15 860 kPa (2,300 psi).

- If the light illuminates momentarily (flickers) while turning the steering wheel at low truck speed and low engine rpm, this may be considered normal and truck operation may continue.
- If the indicator lights illuminate at a higher truck speed and high engine rpm, **do not operate the truck.**

### **⚠ WARNING**

***If the low steering warning light continues to illuminate and the alarm continues to sound, low steering pressure is indicated. The remaining pressure in the accumulators allows the operator to control the truck to a stop. If the oil pressure continues to decrease, the brake auto-apply feature will activate and the service brakes will apply automatically to stop the truck. Do not attempt further operation until the malfunction is located and corrected.***

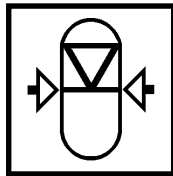
1. Stop the truck as quickly as possible by using the foot pedal to apply the service brakes. If possible, steer the truck to the side of the road while braking.
2. As soon as the truck has stopped moving, shift the directional control lever to PARK. This will apply the parking brake.
3. Slowly release the service brakes to check the capacity of the parking brake. If the parking brake can not hold the truck stationary, apply the service brakes and hold them ON. DO NOT turn the key switch OFF, and do not release the service brakes.
4. Notify maintenance personnel immediately.
5. If the truck is on level ground, or if the parking brake can hold the truck stationary and the truck is in a stable condition, it is then OK to turn the key switch OFF.

6. If safe to do so, have maintenance personnel place wheel chocks or other mechanisms in front or behind the wheels to reduce the risk of the truck rolling.
7. If traffic is excessive near the disabled machine, mark the truck with warning flags during day-light hours or use flares at night. Adhere to local regulations.

**NOTE:** If the engine is stopped, and the key switch is OFF and the light is flashing, a malfunction has been detected in the accumulator bleeddown system, indicating there is still hydraulic pressure in the system. The system/component system failure light will also be illuminated, and the warning horn will be on. Do not operate the truck, and notify maintenance personnel.

### C1. Low Accumulator Precharge Pressure

This red warning light indicates low nitrogen precharge for the steering accumulator(s). To check for proper accumulator precharge, the engine must be stopped and the hydraulic system must be completely depressurized. Turn the key switch to RUN. The warning light will not come on if the system is properly charged. The warning light will flash if the nitrogen precharge in the accumulator(s) is below  $7\ 585 \pm 310\ \text{kPa}$  ( $1,100 \pm 45\ \text{psi}$ ).

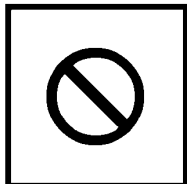


## ⚠ WARNING

**If the low accumulator precharge warning light flashes, notify maintenance personnel. Do not attempt further operation until the accumulators have been properly charged with nitrogen. Sufficient pressure for emergency steering may not be available if the system is not properly charged.**

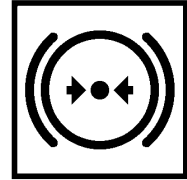
### D1. Not used

This light is reserved for future use by the AC drive system.



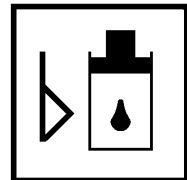
### E1. Low Brake Pressure

This red warning light indicates a malfunction within the hydraulic brake circuit. If this light comes on and warning buzzer sounds, shut down the truck and notify maintenance personnel. Adequate hydraulic oil is stored to allow the operator to safely stop the truck.



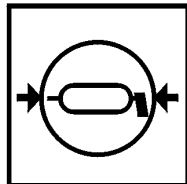
### A2. Low Hydraulic Tank Level

This red warning light indicates that the oil level in the hydraulic tank is below the recommended level. Damage to the hydraulic pumps may occur if operation continues. Shut down the truck and notify maintenance personnel immediately.



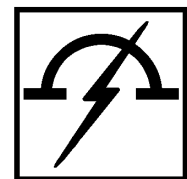
### B2. Low Automatic Lubrication System Pressure

This amber indicator light will illuminate if the automatic lubrication system fails to reach  $13\ 790\ \text{kPa}$  ( $2,000\ \text{psi}$ ) at the junction block located on the rear axle housing within one minute after the lube timer initiates a cycle of grease. To turn the light off, turn key switch OFF, then back to ON again. Notify maintenance personnel at shift change.



### C2. Circuit Breaker Tripped

This amber indicator light will illuminate if any of the circuit breakers in the relay circuit control boards are tripped. The relay circuit boards are located in the electrical control cabinet.



**NOTE:** Additional circuit breakers are in the operator cab behind the center console. However, tripping of these circuit breakers should not activate this light.

## D2. Hydraulic Oil Filter Monitor

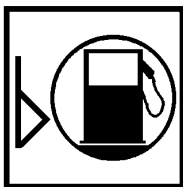
This amber indicator light indicates a restriction in the high pressure filter assembly for either the steering or hoist circuit. This light will come on before the filters start to bypass. Notify maintenance personnel as soon as possible after the light comes on.



*NOTE: The filter monitor warning light may also illuminate after the engine is initially started if the oil is cold. If the light turns off after the oil is warmed, filter maintenance is not required.*

## E2. Low Fuel

When the usable fuel remaining in the tank is approximately 950 liters (250 gallons), this amber indicator light will flash twice and the warning buzzer will also sound twice. Then, the warning buzzer will sound once more and the indicator light will remain illuminated. This will repeat every 15 minutes while the fuel level is still low.



## A3. Parking Brake

This amber parking brake indicator will illuminate when the parking brake is applied. The parking brake is applied by placing the directional control lever in the PARK position.

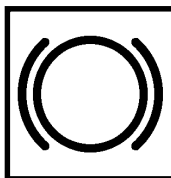


If the parking brake light is flashing, that is an indication that the parking brake is requested, but has not applied.

If the parking brake light is flashing, the operator must not leave the cab. Notify maintenance personnel immediately.

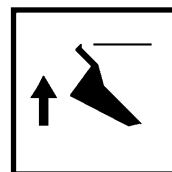
## B3. Service Brake

This amber indicator light will illuminate when the service brake pedal is applied or when wheel brake lock or secondary brake is applied. Do not attempt to drive the truck from a stopped position with the service brakes applied, except as noted in Section 30, Operating Instructions - Starting On A Grade With A Loaded Truck.



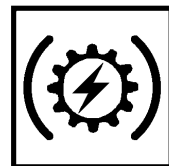
## C3. Body Up

This amber indicator light will illuminate if the body is not completely down on the frame. The truck should not be driven until body is down and the light is off.



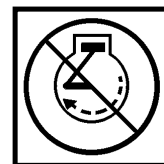
## D3. Dynamic Retarding

This amber indicator light illuminates whenever the brake/retarder pedal (or retarder lever) is operated, retarder speed control (RSC) switch is activated, or the automatic overspeed retarding circuit is energized, indicating that the dynamic retarding function of the truck is operating.



## E3. Stop Engine

This red warning light will illuminate if a serious engine malfunction is detected in the electronic engine control system. Electric propulsion to the wheel motors will be discontinued. Dynamic retarding will still be available if needed to slow or stop the truck.



## CAUTION

***Stop the truck as quickly as possible in a safe area and move the directional control lever to PARK. TURN OFF THE ENGINE IMMEDIATELY. Additional engine damage is likely to occur if operation is continued.***

Listed below are a few conditions that could cause the stop engine light to illuminate:

- Low Oil Pressure - red warning light will illuminate, but engine does not shutdown.
- Low Coolant Level - red warning light will illuminate, but engine does not shutdown.
- Low Coolant Pressure - red warning light will illuminate, but engine does not shutdown.
- High Coolant Temperature - red warning light will illuminate, but engine does not shutdown.



#### A4. Cranking Motor Failure

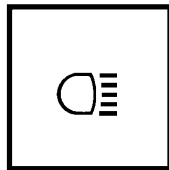
This amber indicator will illuminate when either starter motor (of two) fails to crank the engine, leaving just one cranking motor to start the engine. With only one cranking motor doing the work of two, the motor life will be shortened. If this indicator illuminates, truck operation may continue, but maintenance personnel should be alerted as soon as possible.



This light will also illuminate if the cranking motors have been operational for more than 30 seconds at a time, or if seven attempts at starting for the full 30 seconds is reached.

#### B4. Backup Lights

This amber indicator light will illuminate when backup light switch (18, Figure 5-9) is turned ON.



#### C4. Engine Shutdown Timer

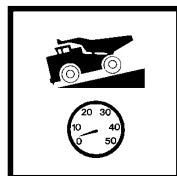
When the key switch has been turned OFF, and certain conditions have been met, this indicator light will illuminate to indicate that the shutdown timing sequence has started. The engine could operate for up to three minutes.



- If the directional control lever is moved out of PARK, the engine will stop immediately.
- If the key switch is turned back ON, the engine stop sequence will be terminated, and the engine will remain running.

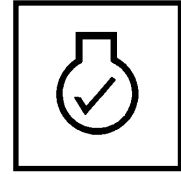
#### D4. Retard Speed Control (RSC) Indicator

This amber indicator light is illuminated when RSC switch (9, Figure 5-7) is pulled out to the ON position. The light indicates that the retarder is active. It is for feedback only and does not signal a problem.



#### E4. Check Engine

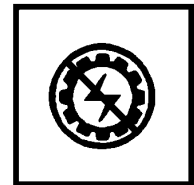
This amber indicator light will illuminate if a malfunction is detected by the engine electronic control system.



If this indicator illuminates, truck operation may continue, but maintenance personnel should be alerted as soon as possible.

#### A5. No Power

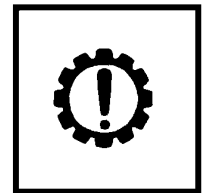
This red indicator light indicates that a fault has occurred which has eliminated the retarding and propulsion capability. A warning buzzer will also sound.



If this condition occurs, the operator should safely stop the truck, move the directional control lever to PARK, turn off the engine, and notify maintenance personnel immediately.

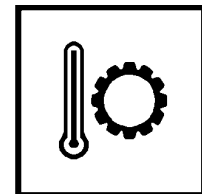
#### B5. Propulsion System Warning

This amber indicator light indicates a drive system fault has occurred. Many faults will result in a speed limit restriction, which could be as low as 16 kph (10 mph). The operator must notify maintenance personnel immediately.



#### C5. Propulsion System Temperature

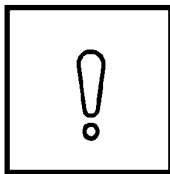
This amber indicator light indicates that the drive system temperature is above a certain level. When this condition occurs, the operator should consider modifying truck operation in order to reduce system temperature.



The operator is not required to stop the truck at this time.

### D5. System/Component Failure

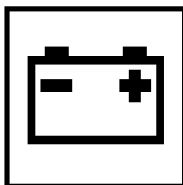
This red warning light indicates that the interface module system detected a failure somewhere on the truck. There are many conditions that could trigger the light to illuminate.



If this warning light illuminates, the operator should safely stop the truck, move selector switch to PARK, shut the engine off, and notify maintenance personnel immediately.

### E5. Battery Charging System Failure

This red warning light indicates that a problem has been detected in the charging system, and system voltage is outside the normal operating range.



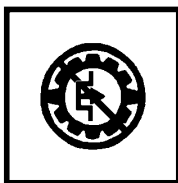
If this warning light illuminates, the safely stop the truck, move directional control lever to PARK, shut off the engine, and notify maintenance personnel immediately. If truck operation continues, and the battery voltage drops below 20 volts, the propulsion system will not operate, but retarding will still be available.

The following conditions will also illuminate this light:

- Battery voltage below 24.5V, engine rpm above 1450. Resets at 26V.
- Battery voltage below 23V with engine off. Resets at 25.5V.
- Battery voltage above 32V with engine operating. Resets at 27.5V.

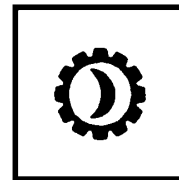
### A6. No Propel

This red warning light indicates that a fault has occurred which has eliminated the propulsion capability. If this condition occurs, the operator should safely stop the truck, move the directional control lever to PARK, turn off the engine, and notify maintenance personnel immediately.



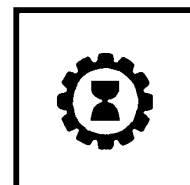
### B6. Propel System At Rest

This amber indicator light illuminates if the AC drive system is de-energized and propulsion is not available. This light is activated when rest switch (4, Figure 5-9) is turned ON and the AC drive system is de-energized. Link energized light (12, Figure 5-7) on the rear of the center console should NOT be illuminated at this time.



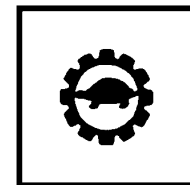
### C6. Propel System Not Ready

This amber indicator light functions during start-up much like the hourglass icon on a computer screen. This light indicates the computer is in the process of performing the self-diagnostics and set-up functions at start-up. Propulsion will not be available at this time.



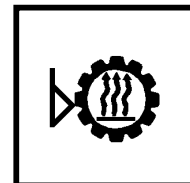
### D6. Reduced Propulsion

This amber indicator light illuminates if the full AC drive system performance in propulsion is not available. At this time, the only event that should activate this light is the use of "limp home" mode. This mode of operation requires a technician to enable.



### E6. Retard At Continuous Level

When the drive system detects that any of its components has reached maximum temperature, this amber indicator light will flash for 15 seconds, then illuminate steadily when the retarding performance is limited to the continuous level. The operator should control the speed of the truck in accordance to the "continuous speeds" on the grade/speed chart.



## REAR AXLE LIGHT BAR

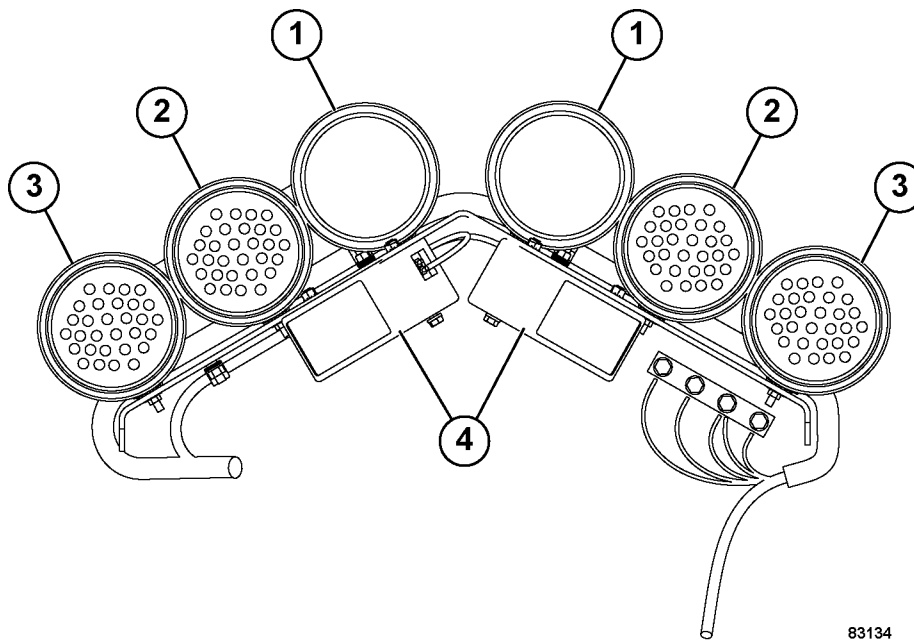


FIGURE 5-11. REAR AXLE LIGHT BAR

- |                 |                 |
|-----------------|-----------------|
| 1. Backup Light | 3. Brake Light  |
| 2. Retard Light | 4. Backup Alarm |

*NOTE: The brake light is functional at all times regardless of key switch position. All other lights (and the backup alarm) are only functional with the key switch in the RUN position.*

### Backup Lights

Backup lights (1, Figure 5-11) illuminate when the operator moves the directional control lever to REVERSE. The lights can also be activated by using the backup light switch in the cab. Using the switch turns the lights on regardless of directional control lever position.

### Retard Lights

Retard lights (2) will illuminate when the operator uses the retard function to slow the truck. The amber light signals to traffic that the truck is slowing.

### Brake Light

Brake light (3) will illuminate when the operator depresses the brake pedal. The red light signals to traffic that the truck is slowing.

### Backup Alarm

Backup alarm (4) will sound when the operator moves the directional control lever to REVERSE. The alarm can also be turned on by using the backup light switch. Using the switch turns the alarm on regardless of directional control lever position.

## KOMTRAX PLUS

### Operation

This system uses KOMTRAX Plus controller (formerly known as VHMS) (6, Figure 5-13) to gather data about the operation of the truck from sensors and other controllers installed on the truck. The data stored in KOMTRAX Plus is collected by a laptop personal computer (PC) or transmitted directly by communications satellite (utilizing the ORBCOMM controller). This data is then compiled at the Komatsu computer server. Based on this information, the servicing Komatsu distributor will suggest improvements and provide information aimed at reducing machine repair costs and downtime.

*NOTE: A contract is necessary before KOMTRAX Plus can be used. Contact your Komatsu distributor for more information.*

### CAUTION

***DO NOT disassemble, repair, or modify the KOMTRAX Plus system without proper authorization. Changes to the system may cause machine failures and fire.***

***DO NOT touch the KOMTRAX Plus system components during machine operation.***

***DO NOT pull on KOMTRAX Plus system wiring harnesses, connectors, or sensors. This may cause short circuits or open circuits and lead to machine failure or fire.***

***DO NOT allow water, dirt, or oil onto system components.***

***The Orbcomm satellite requires the installation of a pole and antenna, adding to the overall height of the machine. The height increase is 410 mm (16.2 in.). With the guard installed, the overall height increases another 260 mm (10.2 in.). Use caution when operating in areas with height restrictions.***

### DANGER

***Anyone with a pacemaker must remain a minimum of 22 cm (9 in.) from the communications antenna. Radio waves from the antenna can interfere with pacemaker operation.***

### IMPORTANT

***The KOMTRAX Plus system uses wireless components that transmit via radio waves. It is necessary to conform to local laws when using this system.***

***Proper operation of the system is dependent on good reception. Operating in tunnels, mountain ranges and covered areas may prevent communication of the system.***

***Contact your Komatsu distributor before selling or exporting a truck equipped with KOMTRAX Plus. It may be necessary to remove the system before transfer of ownership.***

***Contact your Komatsu distributor before installing equipment that may interfere with the KOMTRAX Plus system.***

***Komatsu is not responsible for any failures that result from neglecting KOMTRAX Plus system precautions and instructions.***

### DANGER

***Orbcomm is a two-way radio communication device. Wireless signals from the system can interfere with other wireless signals in the area. This interference can cause a malfunction in a blast zone resulting in an unintended detonation. Know the locations of blast zones in the area and keep a safe distance to avoid unintentional blasts. If the machine is operating within a distance of 12m (40 ft) of a blast zone, disconnect the Orbcomm harness. Failure to do so could result in serious injury or death.***

***This warning does not supersede requirements or regulations of the area or country where this machine is in operation. The following specifications are provided to ensure compliance with all of the applicable requirements or regulations:***

***Transmit power: 5-10 Watts***

***Operating Frequency Range: 148 - 150 MHz***

**NOTE:** The Orbcomm modem installed from the factory on all new trucks may not be approved for use in certain countries of the world. Local regulation may prohibit the use of the Orbcomm modem/satellite communicator. Depending on local regulation, you may need to either remove the modem or disconnect it and remove the antenna.

Check with KAC service for assistance and preferred action in your area.

The following is a list of "at risk" countries: China, Russia, Serbia, Tanzania, Senegal, Zambia, Botswana and Namibia.

When data store button (1, Figure 5-12) is pressed on the back side of the center console, it will store a "snapshot" of the electric drive system. It will also trigger the KOMTRAX Plus system to store a "snapshot" of the truck operating system. Indicator light (2, Figure 5-12) will stay illuminated while the KOMTRAX Plus system is recording the "snapshot", which lasts for 7.5 minutes.

The KOMTRAX Plus system is turned on by the truck key switch. Immediately after receiving 24V power from the key switch, the KOMTRAX Plus controller begins the power-up initialization sequence. This sequence takes about three seconds, during which time red LED digits (7, Figure 5-13) display a circular sequence of flashing LED segments.

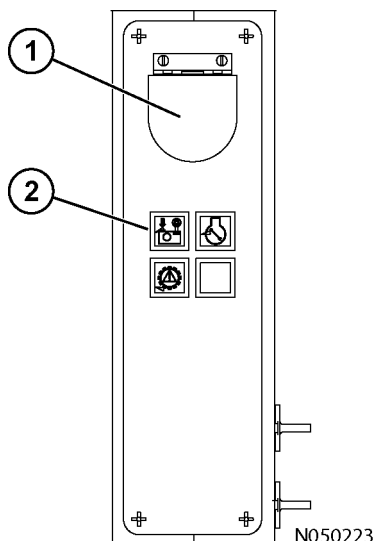


FIGURE 5-12. CENTER CONSOLE, REAR VIEW

1. Data Store Button
2. KOMTRAX Plus Snapshot In Progress Light

During normal truck operation, the red LED digits on the KOMTRAX Plus controller will count from 00-99 continuously.

When the key switch is turned OFF, the KOMTRAX Plus controller will remain on while it finishes processing internal data and saves the recent data into permanent memory. When the data has been safely stored, the two digit LED display will turn OFF. This process could take up to three minutes to complete.

## ⚠ WARNING

**If 24V power is disconnected (using the battery disconnect switches) from the KOMTRAX Plus controller before it has completed its shut down procedure, the KOMTRAX Plus controller will lose all data gathered since the key switch was last turned ON. Do not disconnect battery power until the KOMTRAX Plus controller has completed the shut down procedure and has turned the LED digits off.**

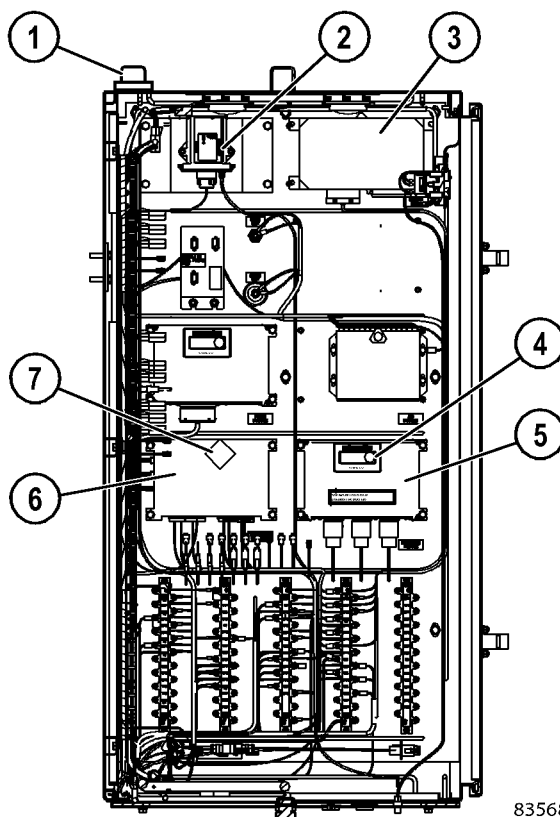


FIGURE 5-13. KOMTRAX PLUS COMPONENTS

- |                       |                     |
|-----------------------|---------------------|
| 1. Antenna            | 5. Interface Module |
| 2. Wireless Bridge    | 6. KOMTRAX Plus     |
| 3. Orbcomm Controller | Controller          |
| 4. Green LED Light    | 7. Red LED Lights   |

Orbcomm controller (3, Figure 5-13) transmits data through antenna (1, Figure 5-14) mounted on top of the cab. The antenna coaxial cable is routed through the inside of the cab to protect it from damage. If the antenna or coaxial cable is damaged, replace the parts.

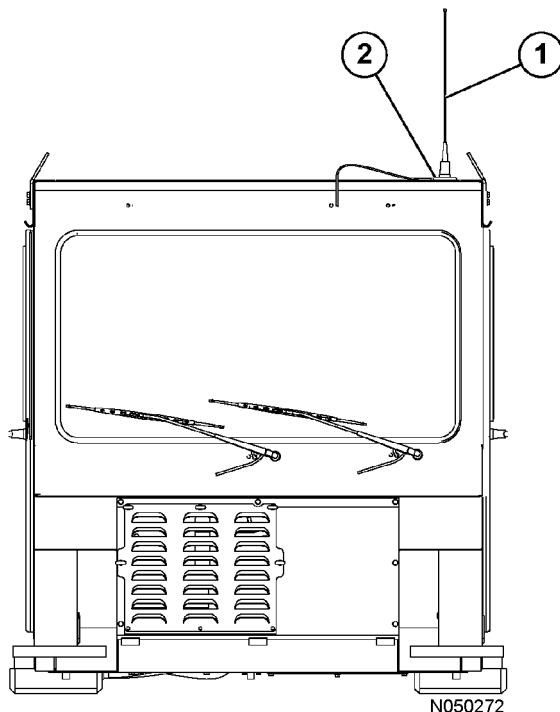


FIGURE 5-14. ORBCOMM ANTENNA

1. Orbcomm Antenna      2. Magnetic Base

### Interface Module

Interface module (3, Figure 5-13) receives data from the sensors installed on the truck and sends this information to the KOMTRAX Plus controller. There is a small green LED light on the face of the controller. With the key switch ON, the light must be blinking. If the light is continuously illuminated, there is a problem in the controller.

When a new interface module controller is installed on the truck, new software has to be installed inside the controller. IM-Diag connector (1, Figure 5-15) is used to connect the interface module to a laptop PC for installing software.

### Basic Precautions

- When using this truck, there is no particular need to operate the KOMTRAX Plus system.
- Never disassemble, repair, or modify the KOMTRAX Plus system. This may cause failure or fire on the machine or this system.
- Do not touch the system when operating the machine.
- Do not pull on the wiring harnesses, connectors, or sensors of this system. This may cause short circuits or disconnections that lead to failure or fire on the machine or this system.
- Do not get water, dirt or oil on the system controllers.
- If there is any abnormality with the KOMTRAX Plus system, please consult the servicing Komatsu distributor.

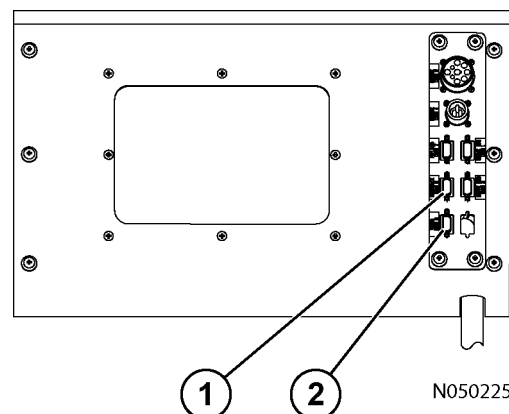


FIGURE 5-15. DIAGNOSTIC PORTS (D.I.D. PANEL AT REAR OF CAB)

1. IM Diagnostic Port  
2. KOMTRAX Plus Diagnostic Port

## KOMATSU WIRELESS BRIDGE (Optional)

### General Information

The Komatsu wireless bridge (2, Figure 5-13) allows for the transfer of information from the KOMTRAX Plus controller to a laptop computer with wireless network capabilities. The information that is downloaded through the KWB (Komatsu wireless bridge) is the same information that is gathered during a manual download from the KOMTRAX Plus system. Downloads can be performed with the laptop computer located on or near a stationary truck, or the computer can be located close to a path of moving trucks. Total download time is approximately 10 seconds. If only a portion of the data is downloaded while the truck is passing by a laptop computer, the remaining information will be downloaded the next time a data download occurs.

The wireless bridge transmits data using antenna (1).

### Communication Setup

A computer running a Microsoft® operating system (with a wireless network adaptor) is required to download data from the KWB. The individual steps required are different depending upon the computer operating system and each computer's specific wireless adaptor.

Please refer to the computer and/or adaptor manuals for further details.

#### 1. Ensure that the KWB:

- shows power and function (LEDs illuminated)
- has not been previously setup
- is within range of the computer

#### 2. Use the computer to scan for any available wireless networks. The computer should find a network with **KWB\_SETUP** as its SSID (SSID = Service Set Identifier). If this SSID is present the KWB is transmitting correctly. Switch to the KWB\_SETUP network.

If the KWB\_SETUP SSID is not detected, verify all connections to the KWB. If all connections are correct, the KWB will need to be reset. Reset the KWB by pushing the reset button (2, Figure 5-16) next to the connector plug.

*NOTE: A KWB\_SETUP SSID could be transmitted by another laptop computer and not by the desired KWB. Power down any other wireless sources with an SSID of KWB\_SETUP.*

### Switching to the KWB\_SETUP Network

The computer should be on the same network as the KWB. It will be necessary to "create a profile" for the new network. Check the wireless adaptor's manual for details.

#### 3. Configure the new profile as follows:

- Set the profile name to KWB\_SETUP.
- Set the SSID to KWB\_SETUP.
- Select the wireless network type as "ad hoc."
- In the security settings, select "no encryption" or a similar setting.
- Select channel 11 (most adaptors will automatically select a channel).

Some wireless adaptor software allows the setting of a static IP address and subnet mask for each profile. If this option is available, complete the following steps:

- a. Set the IP address to 192.168.0.xxx, where xxx is any number between 1 and 255. 123 is recommended if only one computer will be used to interact with the KWB. If multiple computers are used, each must be set to a different IP address.

Set the subnet mask to 255.255.0.0. If asked for a default gate-way, this may be assigned the same IP address as the computer.

## Setting the Computer Subnet Mask and IP Address

If the adaptor software does not allow for setting an IP address, then this must be done manually.

The Windows TCP/IP settings must be modified for the correct subnet mask and an adequate IP address. The steps to reach the point where the TCP/IP settings can be modified vary according to the operating system and are described below. The actual steps for making the changes to the settings are described for the following operating systems.

Windows 2000:

1. Click on the start menu, select settings and then "Network and Dial-up Connections."
2. Right click on the wireless network adaptor and select "properties." Skip instructions for other Operating Systems and continue to the topic of "Properties".

Windows XP:

1. Click on the start menu, right click on "My Networks" and select "Properties." A window with all available networks will be shown.
2. Right click on the wireless network and select "Properties." Skip instructions for Windows Vista and continue to the topic of "Properties".

Windows Vista:

1. Click on the Windows menu, right click on "Network" then select "Properties." A window will open with the option "Manage network connections" as one of the options on the left side. Select that option.
2. A new window showing all available networks will open. Right click on the wireless network to be used to connect to the KWB and select "Properties." Vista will inquire if this was an intentional selection; select "Continue." Continue to the topic of "Properties".

## Properties

Once the window with the list of available protocols is shown, the TCP/IP should then be selected. Click on "Properties" and bring up a window for setting the IP address and the subnet mask (alternatively, double click on TCP/IP). Before making any changes, write down the current settings. These settings will be needed to restore the original computer settings if needed.

3. Select "Use the following IP address," then set the IP address to 192.168.0.xxx, where xxx represents any number between 1 and 255. 123 is generally used. However, no two computers should use the same IP address). The subnet mask should now be set to 255.255.0.0. If a gateway address is requested, this is set to the same as the IP address.

This prepares the computer to communicate with the KWB and to interact with the automatic download software.

*NOTE: The previous steps are not necessary if the software that comes with the computer/adaptor allows an IP address and mask to be assigned to each profile. This is done automatically by selecting the profile.*

*It may be necessary to repeat these steps to restore the computer settings when connecting to another network.*

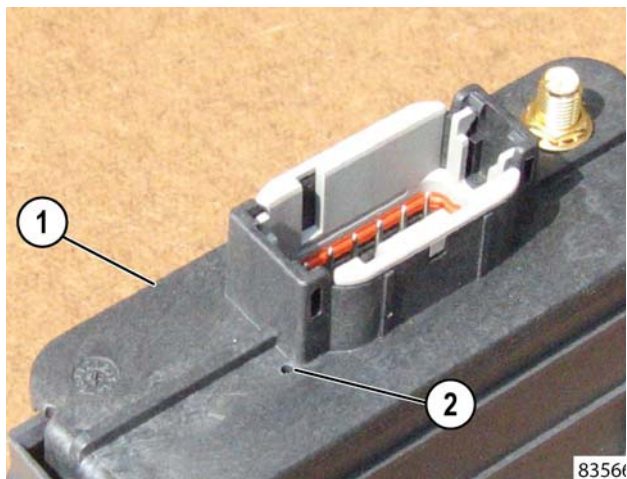


FIGURE 5-16. RESET LOCATION

- |                            |                      |
|----------------------------|----------------------|
| 1. Komatsu Wireless Bridge | 2. Reset Button Hole |
|----------------------------|----------------------|



## Setting Up the KWB

The KWB is preset from the factory. It is recommended that the SSID be changed at setup to establish a unique identity from other KWBs. No previously installed KWBs should show up as KWB\_SETUP.

Changing the network channel may be necessary. For example, if another network in the vicinity uses channel 11 for transmission, the KWB may interfere with this network.

If the information downloaded from the KOMTRAX Plus controller is considered sensitive and private, then the KWB data should be encrypted.

Once the computer has been setup correctly the KWB may be setup using a web browser as follows:

1. On the KWB, label locate the IP address.
2. Type the KWB IP address into the address bar of a web browser (just the numbers and the periods).
3. When prompted to enter a User Name and Password, enter "dpac" for both.
4. The initial screen that is displayed will show the current status of the KWB. Select "Network" from the top right corner.
5. Change the SSID to any name. (for example, "KOMTRAX controller"). The SSID is used to separate networks. If one computer will be used to download data from excavators, one from haul trucks, and one from dozers, then the KWB that will be installed on these machines could have their SSID set to "KOMTRAX\_dig," "KOMTRAX\_haul," and "KOMTRAX\_push" respectively. This would make excavators and haul trucks invisible to the computer setup to download data from dozers. All KWBs that are to be in the same network should be set to the same SSID.

If the default channel (11) is used by another network in the vicinity, change the channel. It is preferable to assign channels that are at least three channels apart.

## Adding Encryption

It is recommend that encryption be added.

1. On "Wireless Security Mode," select WEP128. In "WEP key 1", using only numbers and the letters from "A" to "F," input a 26 character long passkey, for example "0123456789ABCDEF0123456789" (a more complex combination should be used). All KWBs that are to be in the same network should have the same WEP key, SSID, and be on the same channel.
2. Keep a record of the key.

*NOTE: If a more secure encryption method is desired, consult a network professional to setup WPA encryption.*

3. If the "DHCP Client Name" is blank, input the KWB IP address in this field before pressing the "Save" button at the bottom of the page. No further network settings are required.
4. Once the KWB is reset, all the settings will be applied. Reset the KWB when prompted or, if you prefer to review the settings, click on "Networks" and, when reviewing is complete, click on "Reset." Both of these options are in the blue bar under "KWB-1".

## Final Computer Settings

Once the KWB SSID is changed, the KWB is no longer in the KWB\_SETUP network. In order to establish communications with the KWB, the computer must be setup to have the same settings as the KWB. Follow steps 1 through 3 beginning in "Communication Setup", but use the new settings to create a different network profile. Once all setting are complete, typing the KWB IP address in the browser's address bar should open the KWB settings page. This confirms that the KWB and the computer are on the same network.

Once the steps described above have been performed, the computer will have two network profiles in the program that manages the network adaptor. The KWB\_SETUP profile should remain in the list in order to setup other KWBs in the future. The other profile will be the one used to communicate with the existing KWB. For both networks, the computer IP address can remain the same. Therefore, Step 3 as described in "Switching to the KWB\_SETUP Network" is not required. If the computer is also used to connect to the internet, select "obtain IP address automatically" when connecting to the internet.

## Data Downloading

To obtain the software for data downloading from the KOMTRAX Plus controller, go to the Komatsu Extranet (<https://www.komatsuamerica.net>).

1. Select KOMTRAX / VHMS / PLM.
2. Select VHMS / PLM, then select SOFTWARE.
3. Select KWB - Komatsu Wireless Bridge (Wireless Download Software). Load and install the software. The icon "VHMS Controller Auto Download Tool" will be created during the download process.

Running the software for the first time will offer the opportunity to choose a login name and password; "user" should be used for "User Name" and "1" for "Password." This may change this the next time the software is run. Clicking "OK" will start the download tool. If any KWBs are within range and are connected to the VHMS controller, the tool will automatically download all the data from each VHMS controller.

The auto-download software will not download from KOMTRAX Plus controllers that have not had the vehicle data setup (this is a consideration only when replacing a KOMTRAX Plus Controller or installing a KOMTRAX Plus Controller for the first time nor will any KWB that is not connected to a KOMTRAX Plus controller show up in the list of available download sources. Once the data from one vehicle has been downloaded, the tool will not perform another download for that vehicle until 20 hours have passed.

It is possible to force a download by right clicking on a vehicle in "View." Vehicles in "View" are shown in the top list; vehicles that have been previously in "View" are shown in the lower list.

## Resetting the KWB

It may be necessary to reset the KWB. For example, if the WEP key was incorrectly typed when setting up the KWB, it will not be possible to connect to that KWB without the correct key.

1. Remove the power by turning OFF the key or by disconnecting the harness from the KWB.
2. Lightly insert, press and hold a paperclip into the hole on the connector face. To prevent damage to the internal switch do not use sharp objects.
3. Restore power to the KWB while keeping the reset switch depressed. The LEDs should begin to flash. The KWB is now reset.

*NOTE: After a reset, the KWB will revert to the KWB\_SETUP SSID.*

4. Switch to the KWB\_SETUP SSID network. The IP address after a reset will be 192.168.1.0.
5. When setting the KWB, the IP address must be changed from the reset IP address. Use the address on the KWB label.

## KWB Lights

**Table 1: KWB LIGHTS**

| Name | Description                                                                                                                                                                                                                       |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CONN | Reserved for future use.                                                                                                                                                                                                          |
| LINK | When ON, the KWB is operational. When OFF, the KWB may not be set for Ad-Hoc, or it may be damaged. An ON condition is not an indication that the antenna or antenna cable is in good condition or that the antenna is connected. |
| CFG  | When ON, the KWB has an IP address. When OFF, the KWB may be configured for DHCP (DHCP should not be selected for ad-hoc mode).                                                                                                   |
| POST | When ON, the KWB is receiving power and is operational. When OFF, the KWB may not be powered or it may be defective.                                                                                                              |

# NOTES

# SECTION P

## LUBRICATION AND SERVICE

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# NOTES

## SECTION P2

# LUBRICATION AND SERVICE

## INDEX

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# NOTES

# LUBRICATION AND SERVICE

## GENERAL

Recommended preventive maintenance will contribute to the long life and dependability of the truck and its components. The use of proper lubricants and the performance of checks and adjustments at the recommended intervals is most important.

Lubrication requirements are referenced to the lube key found in the Lubrication Chart (page 2-5). For detailed service requirements for specific components, refer to the shop manual section for that component (for example, Section H for suspensions, Section L for hydraulic system, etc).

Refer to the manufacturer's service manual when servicing any components of the General Electric propulsion system.

Refer to engine manufacturer's service manual when servicing the engine or any of its components.

The service intervals presented here are in hours of operation. **These intervals are recommended in place of an oil analysis program which may determine different intervals.** However, if the truck is being operated under extreme conditions, some or all of the intervals may need to be shortened and the service should be performed more frequently.

The 930E truck is equipped with a Lincoln Automatic Lubrication System. The initial setup for this system provides for nominal amounts of lubricant to be delivered to each serviced point. The lubrication injectors can be adjusted to vary the amount of lubricant delivered. In addition, the timer for lubrication intervals is normally adjustable. For adjustments to these devices, refer to Section 42, Automatic Lubrication System.

## SERVICE CAPACITIES

| Component/System                                                        | Liters | U.S. Gallons |
|-------------------------------------------------------------------------|--------|--------------|
| <b>Crankcase</b> (including 4 oil filters)<br>Komatsu SSDA16V160 Engine | 280    | 74           |
| <b>Cooling System</b><br>Komatsu SSDA16V160 Engine                      | 594    | 157          |
| <b>Hydraulic System</b><br>Refer to "Hydraulic Tank Service"            | 1325   | 350          |
| <b>Wheel Motors</b> (each side)                                         | 95     | 25           |
| <b>Fuel Tank</b> (Diesel fuel only)                                     | 4542   | 1200         |

## HYDRAULIC TANK SERVICE

There are two sight gauges (1, Figure 40-1) on the side of the hydraulic tank. With the engine stopped, key switch OFF, hydraulic system depressurized, and body down, the oil level should be visible in the center of the top sight gauge. If the oil level is not visible in the center of the top sight gauge, follow the instructions below for adding oil.

### Adding Oil

*NOTE: Keep the system open to the atmosphere only as long as absolutely necessary to lessen chances of system contamination.*

Service the tank with clean Type C-4 hydraulic oil only. All oil being put into the hydraulic tank should be filtered through 3 micron filters.

1. Make sure that the engine is stopped, key switch is OFF, hydraulic system is depressurized, and body is down
2. Remove fill cap (2) and add hydraulic oil until the oil level is visible in the center of the top sight gauge.
3. Install the fill cap.
4. Start the engine. Raise and lower the dump body three times.
5. Check the hydraulic oil level again with the engine stopped, key switch OFF, hydraulic system depressurized, and body down.
6. Repeat Steps 1 - 5 until the oil level is maintained in the center of the top sight gauge.

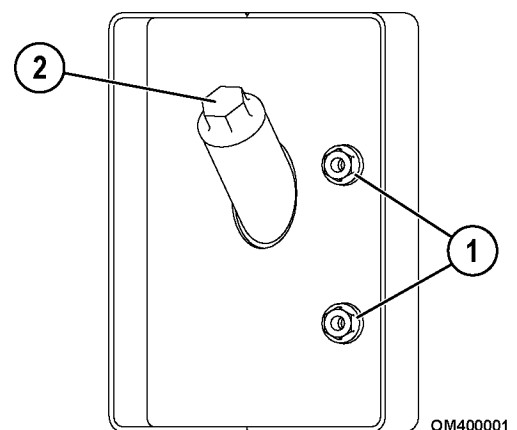


FIGURE 2-1. HYDRAULIC TANK SERVICE

1. Sight Gauges

2. Fill Cap



## COOLING SYSTEM SERVICE

Check the coolant level sight gauge. If coolant cannot be seen in the sight gauge, add coolant to the system before truck operation. Refer to the procedure below for the proper filling procedure.

### Radiator Filling Procedure



**The cooling system is pressurized due to thermal expansion of coolant. Do not remove the radiator cap while the engine and coolant are hot. Severe burns may result.**

*NOTE: If coolant is added using the Wiggins service center, the radiator cap must still be removed before adding coolant.*

Engine coolant must always be visible in the sight gauge before truck operation.

1. With the engine and coolant at ambient temperature, remove the radiator cap.
2. Fill the radiator with the proper coolant mixture until coolant is visible in the sight gauge.
3. Install the radiator cap.
4. Run the engine for five minutes, then check the coolant level.
5. If coolant is not visible in the sight gauge, repeat Steps 1 - 4. Any excess coolant will be discharged through the vent hose after the engine reaches normal operating temperature.

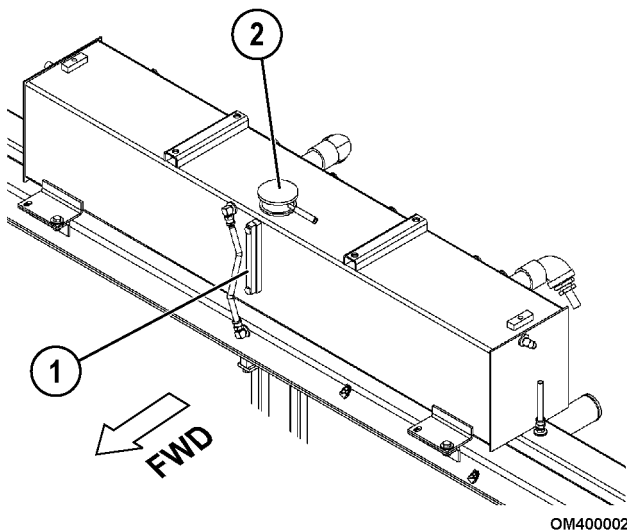


FIGURE 2-2. COOLING SYSTEM SERVICE

1. Sight Gauge

2. Radiator Cap

## Coolant Specifications

- For ambient temperatures of -25° F (-32° C) and above, use a standard 50/50 anti-freeze-to-water mixture.
- For arctic climates with ambient temperatures between -25° F (-32° C) and -65° F (-54° C), use a 60/40 anti-freeze-to-water mixture.

*NOTE: Do not use propylene glycol coolant in arctic climates. Only use ethylene glycol coolants.*

## Unacceptable Practices

- Use of high-silicate anti-freeze.
- Under concentration or over concentration of Extended Service Additive (SCA).
- Use of anti-freezes/coolants that are **not** fully formulated for extended service intervals.
- Use of sealing additives (stop-leak) in the cooling system.
- Use of soluble oils in the cooling system.
- Use of poor-quality water. See the engine manufacturer's specifications for water quality requirements.
- Use of anti-freeze, Extended Service Additive (SCA) or coolant filter(s) that do **not** meet the engine manufacturer's specifications.
- Use of Treated Water coolant.
- Use of coolants with less than 40 percent antifreeze.

## WHEEL MOTOR SERVICE

Due to differences in gear ratio and component evolution/design, wheel motor service intervals may be unit number and/or mine specific. Because of the wide variety of factors involved, it is necessary to consult your area Komatsu representative for all wheel motor service intervals and instructions. General intervals for oil service and sampling are listed in the interval charts.

## RESERVE OIL TANK SERVICE

The reserve engine oil system is designed to add more oil capacity to the engine to reduce the frequent servicing of the engine oil. The engine oil level must still be checked every shift by using the dipstick. If the engine oil level is not correct, check for proper operation of the reserve engine oil system. Never add oil to the engine unless it has been drained.

If the engine oil has been drained from the oil pan, new oil must be added to the engine oil pan before starting. Do not use the oil in the reserve tank to fill an empty engine with oil. After an oil change, both the engine and reserve tank must be full of oil before starting the engine.

With the engine running, check the operation of the red LED light on the pump.

- **Continuously on** - Pump 1 is withdrawing oil from the engine sump and bringing down the oil level.
- **Regular pulsing** - Pump 2 is returning oil to the engine sump and raising the oil level.
- **Irregular pulsing** - Oil is at the correct running level.

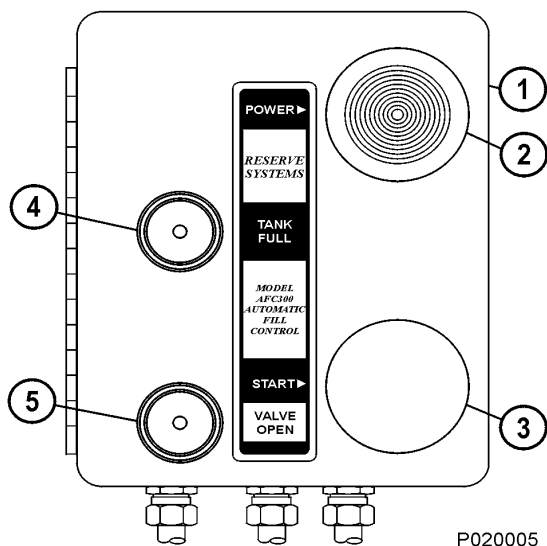


FIGURE 2-3. REMOTE FILL CONTROL BOX

- |                       |                     |
|-----------------------|---------------------|
| 1. Remote Control Box | 4. FULL Light       |
| 2. Power Switch       | 5. VALVE OPEN Light |
| 3. Start Switch       |                     |

Oil should always be visible in the bottom sight gauge. If not, add oil to the reserve oil tank until oil is visible in the top sight gauge.

- Never add oil to the engine unless it has been completely drained.
- If the engine oil has been drained from the oil pan, new oil must be added to the oil pan. **Do not use the oil in the reserve oil tank to fill an empty engine oil pan.**
- After an oil change, both the engine and reserve oil tank must be full of oil before starting the engine.

### Filling the Reserve Oil Tank (Remote Fill)

1. Connect the supply hose from the new oil supply to the quick disconnect coupling on the truck. Open the valve on the supply hose to apply pressure.
2. Pull power switch (2, Figure 40-3) to turn the system ON.
3. Push start switch (3). VALVE OPEN light (5) should illuminate and the filling process will begin.
4. When tank is full, VALVE OPEN light (5) will turn off and FULL light (4) will illuminate.
5. Close the valve in the supply hose.
6. Press and hold start switch (3) for a couple of seconds.
7. Disconnect the supply hose from the quick disconnect coupling on the truck.
8. Push power switch (2) to turn the system OFF.

### Inline Screen

There is an inline screen located at the inlet of the fill valve. This screen does not require periodic maintenance, but it can be cleaned by removing the screen and back flushing.

## QUICK FILL SERVICE CENTER

The service center (Figure 2-4) can be located on either side of the truck and is used to fill system fluids. Table 1 shows the maximum recommended fill pressures for the service center.

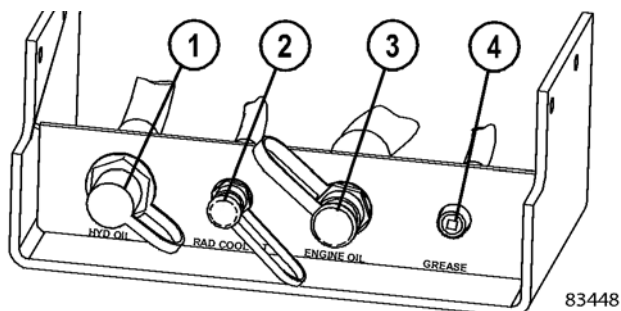


FIGURE 2-4. SERVICE CENTER

- 1. Hydraulic Oil
- 2. Radiator Coolant
- 3. Engine Oil
- 4. Grease

| TABLE 1: MAXIMUM FILL PRESSURES |        |       |
|---------------------------------|--------|-------|
| Fill System                     | kPa    | psi   |
| Engine Crankcase                | 345    | 50    |
| Reserve Tank                    | 862    | 125   |
| Hydraulic Tank                  | 345    | 50    |
| Coolant                         | 345    | 50    |
| Grease                          | 10 342 | 1,500 |

# LUBRICATION CHART

## LUBRICATION SPECIFICATIONS

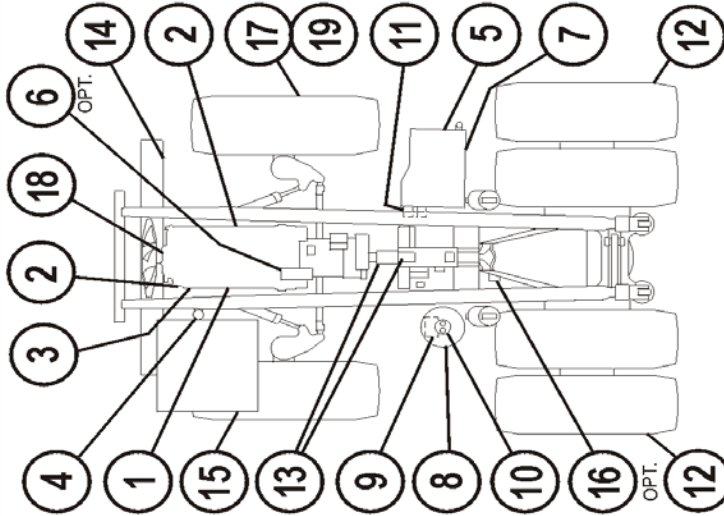
| LUBE KEY | TYPE LUBRICANT                                                                           | PTS | LUBE KEY | 10 HR | 50 HR | 100 HR | 250 HR | 500 HR | 1000 HR                 | 2000 HR | 2500 HR |
|----------|------------------------------------------------------------------------------------------|-----|----------|-------|-------|--------|--------|--------|-------------------------|---------|---------|
| A        | ENGINE OIL - ..... SEE ENGINE MANUAL                                                     |     |          |       |       |        |        |        |                         |         |         |
| B        | HYDRAULIC OIL - ..... SAE 10W C-4 - ..... AUXILIARY HEATERS REQUIRED BELOW -10°F (-23°C) |     |          |       |       |        |        |        |                         |         |         |
| C        | HEAVY-DUTY SYNTHETIC GEAR OIL - ..... SEE DRIVE SYSTEM MANUAL                            |     |          |       |       |        |        |        |                         |         |         |
| D        | MULTI - PURPOSE EXTREME PRESSURE GREASE - ..... NLGI NO.2 (5% MIN. MOLY-DISULFIDE)       |     |          |       |       |        |        |        |                         |         |         |
| E        | MULTI - PURPOSE GEAR OIL - ..... SAE 80W - 90                                            |     |          |       |       |        |        |        |                         |         |         |
| SYM      | DESCRIPTION                                                                              | PTS | LUBE KEY | 10 HR | 50 HR | 100 HR | 250 HR | 500 HR | 1000 HR                 | 2000 HR | 2500 HR |
| 1        | CRANKCASE OIL LEVEL                                                                      | 1   | A        | CHECK |       |        |        |        |                         |         |         |
| 2        | ENGINE LUBE OIL FILTERS                                                                  |     |          |       |       |        |        |        |                         |         |         |
| 3        | FUEL FILTER                                                                              |     |          |       |       |        |        |        |                         |         |         |
| 4        | FUEL SEPARATOR (DAVCO)                                                                   |     |          |       |       |        |        |        |                         |         |         |
| 5        | FUEL TANK                                                                                | 1   |          |       |       |        |        |        |                         |         |         |
| 6        | GE PREFILTER BLOWER                                                                      | 1   | D        |       |       |        |        |        |                         |         |         |
| 7        | FUEL TANK BREATHER                                                                       | 1   |          |       |       |        |        |        |                         |         |         |
| 8        | HYDRAULIC OIL LEVEL                                                                      | 1   | B        | CHECK |       |        |        |        | CLEAN                   |         |         |
| 9        | HYDRAULIC STRAINER                                                                       | 2   |          |       |       |        |        |        | *CHANGE                 |         |         |
| 10       | HYDRAULIC TANK BREATHER                                                                  | 2   |          |       |       |        |        | CHANGE | CLEAN                   |         |         |
| 11       | HYDRAULIC FILTERS                                                                        | 3   |          | **    | **    | **     | **     | CHANGE |                         |         |         |
| 12       | MOTORIZED WHEEL OIL LEVEL                                                                | 2   | C        |       |       |        |        |        | SEE DRIVE SYSTEM MANUAL |         |         |
| 13       | HYD. PUMP DRIVE SHAFT                                                                    | 2   | D        |       |       |        |        | GREASE |                         |         |         |
| 14       | CHASSIS LUBE LEVEL                                                                       | 1   | D        |       |       |        |        | GREASE |                         |         |         |
| 15       | SEAT SLIDES & STEER SHAFT                                                                | 4   | D        |       |       |        |        |        | GREASE                  |         |         |
| 16       | WHEEL MOTOR BLOWER                                                                       | 2   | D        |       |       |        |        | GREASE |                         |         |         |
| 17       | FRONT WHEEL BEARINGS **                                                                  | 2   | E        |       |       |        |        | CHECK  |                         |         | CHANGE  |
| 18       | FRONT TRUNION                                                                            | 1   | D        |       |       |        |        |        | GREASE                  |         |         |
| 19       | MAGNETIC PLUG                                                                            | 2   |          |       |       |        |        | CHECK  |                         |         |         |
|          | FRONT WHL COVER ***                                                                      |     |          |       |       |        |        |        |                         |         |         |

\* 1000 HR INTERVAL CAN BE EXTENDED TO 2500 HR PROVIDED OIL SAMPLING AND ANALYSIS IS CONDUCTED EVERY 250 HR.

\*\* ONE-TIME CHANGE AT 50, 100 AND 250 HR.

\*\*\* NOT APPLICABLE FOR 930E

\*\*\*\* APPLICABLE FOR 930E ONLY



630E, 730E, 830E, & 930E

WB2790

## LUBRICATION CHART

## 10 HOUR (DAILY) INSPECTIONS

*NOTE: "Lube Key" references are to the **Lubrication Chart** on page P2-7.*

| <b>Truck Serial Number</b> _____ <b>Site Unit Number</b> _____ <b>Date</b> _____ |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |         |          |
|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------|----------|
| <b>Hourmeter</b> _____ <b>Name of Service Technician</b> _____                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |         |          |
|                                                                                  | TASK                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | COMMENTS | CHECKED | INITIALS |
| 1.                                                                               | MACHINE - Inspect the entire machine for leaks, worn parts and damage. Repair as necessary.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |         |          |
| 2.                                                                               | FAN DRIVE AND TURBOCHARGERS - Check for leaks, vibration and unusual noise. Check the condition and alignment of the alternator and fan belt.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |          |         |          |
| 3.                                                                               | RADIATOR - Check the coolant level and fill the radiator with the proper coolant mixture according to Coolant Specifications on page P2-4. Refer to the engine service manual for additional coolant recommendations.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |         |          |
| 4.                                                                               | ENGINE -<br><br>1. Check the oil level. To obtain an accurate measurement, remove the dipstick and wipe it off. Then reinsert the dipstick and remove it again to check the oil level. Refer to the engine service manual for oil recommendations. (Lube Key "A")<br><br>2. Check the oil level in the reserve tank. The oil level should be visible in the center of the top sight gauge. If it is not, add oil to the reserve tank until the oil level is visible in the center of the top sight gauge. Also, with the engine running, check the operation of the LED monitor light. Refer to Section M31, Reserve Engine Oil System.<br><br>3. Make sure that the exhaust piping is secure.<br><br>4. Check for abnormal noises and fluid leaks.<br><br>5. Check the operating indicator of the eliminator filter. |          |         |          |
| 5.                                                                               | HYDRAULIC TANK - Check the oil level in the tank. Add oil if necessary. Refer to Hydraulic Tank Service on page P2-3. (Lube Key "B")                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |         |          |
| 6.                                                                               | WHEELS AND TIRES -<br><br>1. Inspect the tires for wear and proper inflation.<br>2. Check for embedded debris in the treads.<br>3. Check for damaged, loose or missing wheel mounting nuts and studs. Replace damaged or missing hardware and tighten to standard torque before truck operation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |          |         |          |

### 10 HOUR (DAILY) INSPECTIONS (Cont.)

|     | TASK                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | COMMENTS | CHECKED | INITIALS |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------|----------|
| 7.  | COOLING AIR DUCTWORK - Inspect the ductwork from the blower to the rear axle. Make sure that the ductwork is secure, undamaged and unrestricted.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |         |          |
| 8.  | AIR INTAKE PIPING - Check all mounting hardware, joints and connections. Make sure that there are no air leaks and all hardware is properly tightened.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |         |          |
| 9.  | AIR CLEANERS - Check the air cleaner restriction gauges in the operator cab. The air cleaners should be serviced if the gauges show a maximum restriction of 25 inches of H <sub>2</sub> O vacuum.<br><br>Refer to Section C5 for servicing instructions. Empty the air cleaner dust caps. After service, push the reset button on face of gauges to return the gauges to zero.                                                                                                                                                                                                                                                                                                                               |          |         |          |
| 10. | CAB AIR FILTER - In extremely dusty conditions, check this filter daily and service as frequently as required. See the 250 Hour Lubrication and Maintenance Checks for the cleaning procedure.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          |         |          |
| 11. | WASHER FLUID - Check the level of washer fluid in the bottle behind the cab. Fill as needed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |         |          |
| 12. | FUEL TANK -<br><br>1. Check the quick fill fuel receivers for excessive wear, leaks and mud buildup.<br><br>2. Check the vent on top of the fuel tank for mud buildup or any other obstructions. Clean if necessary. See the 1000 Hour Lubrication and Maintenance Checks for cleaning instructions.                                                                                                                                                                                                                                                                                                                                                                                                          |          |         |          |
| 13. | FUEL FILTERS - Drain any water from the bottom drain valve on each fuel separator.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |         |          |
| 14. | AUTO LUBE SYSTEM -<br><br>1. Check the reservoir grease level. Fill as required. (Lube Key "D")<br><br>2. When filling the grease reservoir, check the grease filter indicator. Clean or replace the grease filter if a problem is indicated.<br><br>3. Inspect the system for leaks and check for proper operation. Make sure that the following important areas are receiving adequate amounts of grease. (Lube Key "D")<br><ul style="list-style-type: none"> <li>• Steering Linkage</li> <li>• Final Drive Pivot Pin</li> <li>• Rear Suspension Pin Joints (Upper and Lower)</li> <li>• Body Hinge Pins</li> <li>• Hoist Cylinders Pins (Upper and Lower)</li> <li>• Anti-Sway Bar (Both Ends)</li> </ul> |          |         |          |

## 50 HOUR LUBRICATION AND MAINTENANCE CHECKS

| Truck Serial Number _____ Site Unit Number _____ Date _____<br>Hourmeter _____ Name of Service Technician _____                        |          |         |          |
|----------------------------------------------------------------------------------------------------------------------------------------|----------|---------|----------|
| TASK                                                                                                                                   | COMMENTS | CHECKED | INITIALS |
| *1. FUEL FILTERS - Replace the fuel filters. Refer to the engine operation and maintenance manual for filter replacement instructions. |          |         |          |
| *2. HYDRAULIC SYSTEM FILTERS - Replace the filter elements. Check the oil level. Add oil as necessary. (Lube Key "B")                  |          |         |          |

\* These checks are required **only after the initial 50 hours of operation**, such as the commissioning of a new truck or after installing a new or rebuilt component.

## 100 HOUR LUBRICATION AND MAINTENANCE CHECKS

| Truck Serial Number _____ Site Unit Number _____ Date _____<br>Hourmeter _____ Name of Service Technician _____       |          |         |          |
|-----------------------------------------------------------------------------------------------------------------------|----------|---------|----------|
| TASK                                                                                                                  | COMMENTS | CHECKED | INITIALS |
| *1. HYDRAULIC SYSTEM FILTERS - Replace the filter elements. Check the oil level. Add oil as necessary. (Lube Key "B") |          |         |          |

\* This check is required **only after the initial 100 hours of operation**, such as the commissioning of a new truck or after installing a new or rebuilt component.

## 250 HOUR LUBRICATION AND MAINTENANCE CHECKS

| Truck Serial Number _____ Site Unit Number _____ Date _____<br>Hourmeter _____ Name of Service Technician _____       |          |         |          |
|-----------------------------------------------------------------------------------------------------------------------|----------|---------|----------|
| TASK                                                                                                                  | COMMENTS | CHECKED | INITIALS |
| *1. HYDRAULIC SYSTEM FILTERS - Replace the filter elements. Check the oil level. Add oil as necessary. (Lube Key "B") |          |         |          |

\* This check is required **only after the initial 250 hours of operation**, such as the commissioning of a new truck or after installing a new or rebuilt component.

## 250 HOUR LUBRICATION AND MAINTENANCE CHECKS (Cont.)

All 10 Hour (Daily) Inspections should also be performed at this time.

NOTE: "Lube Key" references are to the **Lubrication Chart** on page P2-7.

| Truck Serial Number _____ Site Unit Number _____ Date _____ |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |          |         |          |
|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------|----------|
| Hourmeter _____ Name of Service Technician _____            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |          |         |          |
|                                                             | TASK                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | COMMENTS | CHECKED | INITIALS |
| 2.                                                          | <b>ENGINE -</b><br><br>1. Change the engine oil. Refer to the engine operation and maintenance manual for engine oil specifications. (Lube Key "A")<br><br><i>NOTE: If the engine is equipped with the Centinel™ oil system and/or the Eliminator™ filter system, engine oil and filter change intervals are extended beyond 250 hours. Take an engine oil sample for analysis. Refer to engine operation and maintenance manual for specific oil and filter change intervals.</i><br><br>2. Replace the engine oil filters. Follow the instructions of the filter manufacturer, which are normally printed on the outside of the filter. <b>Do not use a wrench or strap to tighten filter elements.</b><br><br>3. Change the reserve tank oil filter.<br><br>4. Check the tension and condition of each accessory drive belt. Refer to the engine operation and maintenance manual for specific adjustment instructions.<br><br>5. Check the torque on the cooling fan mounting capscrews. Tighten to <b>237 N•m (175 ft lb)</b> . |          |         |          |
| 3.                                                          | <b>COOLING SYSTEM -</b><br><br>1. Check the coolant level and fill the radiator with the proper coolant mixture according to Coolant Specifications on page P-4. Refer to the engine operation and maintenance manual for additional coolant recommendations.<br><br>2. Change the coolant filters. Refer to the engine operation and maintenance manual for coolant filter replacement instructions.<br><br>3. Check the cooling system hoses for damage and deterioration. Replace as necessary.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |          |         |          |
| 4.                                                          | <b>FUEL FILTERS -</b> Change the fuel filters. Refer to the engine operation and maintenance manual for filter replacement instructions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |         |          |
| 5.                                                          | <b>FUEL TANK -</b> Drain water and sediment from the fuel tank. Refer to Section B4, Fuel Tank - Cleaning.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |         |          |



## 250 HOUR LUBRICATION AND MAINTENANCE (Cont.)

|     | TASK                                                                                                                                                                                                                                                                                                                                                                               | COMMENTS | CHECKED | INITIALS |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------|----------|
| 6.  | STEERING LINKAGE - Check the torque on the pin retaining nuts for the steering cylinders and tie rods. Tighten to <b>1017 N•m (750 ft lb)</b> .                                                                                                                                                                                                                                    |          |         |          |
| 7.  | HYDRAULIC PUMP DRIVESHAFT & U-JOINTS - Add one or two applications of grease to each grease fitting. Make sure that each bearing of the cross and bearing assembly is receiving grease. Replace bearings if any wear is detected. (Lube Key "D")                                                                                                                                   |          |         |          |
| 8.  | <p>REAR WHEEL MOUNTING - Use a flashlight and a mirror on a long rod to inspect all inner and outer wheel mounting nuts and studs. Check for loose, damaged or missing hardware.</p> <p>If wheel mounting nuts and studs must be tightened or replaced, the outer wheel must be removed for access. Refer to Section G for the procedures.</p>                                     |          |         |          |
| 9.  | REAR AXLE HOUSING - Check the rear axle housing for oil leaks by removing the two drain plugs on the bottom of the axle housing. If oil is present, the cause of the leak must be found and corrected before operating the truck.                                                                                                                                                  |          |         |          |
| 10. | FRONT WHEEL HUB - Remove the magnetic plugs from the front wheel hub covers and check for debris. Clean the plugs and perform any necessary repairs.                                                                                                                                                                                                                               |          |         |          |
| 11. | <p>MOTORIZED WHEELS -</p> <ol style="list-style-type: none"> <li>1. Check for the correct oil level. Rotate the wheel so that the magnetic plug is at the 6 o'clock position and remove the plug. The oil level should be even with the bottom of the plug opening.</li> <li>2. Inspect the magnetic plugs for ferrous materials. Service the wheel motor as necessary.</li> </ol> |          |         |          |
| 12. | WHEEL MOTOR OIL SAMPLING - Refer to Section G5, Wheel Motor, for oil sampling information.                                                                                                                                                                                                                                                                                         |          |         |          |
| 13. | AUTOMATIC LUBRICATION SYSTEM - Perform the 250 hour checks as outlined in Automatic Lubrication System, Section P, in this manual.                                                                                                                                                                                                                                                 |          |         |          |
| 14. | DUMP BODY PADS - Check the pads, shims and mounting hardware for wear and proper tightening.                                                                                                                                                                                                                                                                                       |          |         |          |
| 15. | BATTERIES - Check the electrolyte level and add water if necessary.                                                                                                                                                                                                                                                                                                                |          |         |          |
| 16. | BODY-UP AND HOIST LIMIT SWITCHES - Check the operation of the switches. Clean any dirt from the sensing areas and inspect the wiring for damage.                                                                                                                                                                                                                                   |          |         |          |

## 250 HOUR LUBRICATION AND MAINTENANCE CHECKS (Cont.)

|     | TASK                                                                                                                                                              | COMMENTS | CHECKED | INITIALS |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------|----------|
| 17. | CAB RECIRCULATION AIR FILTER - Clean the filter element with mild soap and water. Rinse completely and allow to air dry before reinstalling the filter.           |          |         |          |
| 18. | CAB AIR FILTER - Clean the filter element with mild soap and water. Rinse completely and dry with pressure air of 275 kPa (40 psi) maximum. Reinstall the filter. |          |         |          |

## 500 HOUR LUBRICATION AND MAINTENANCE CHECKS

All 10 Hour (Daily) Inspections and 250 Hour Lubrication and Maintenance Checks should also be performed at this time.

NOTE: "Lube Key" references are to the **Lubrication Chart** on page P2-7.

| Truck Serial Number _____ Site Unit Number _____ Date _____<br>Hourmeter _____ Name of Service Technician _____ |                                                                                                                                                                                                                         |          |         |          |
|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------|----------|
|                                                                                                                 | TASK                                                                                                                                                                                                                    | COMMENTS | CHECKED | INITIALS |
| 1.                                                                                                              | FINAL DRIVE CASE BREATHERS - Replace the breather elements for the motorized wheels.                                                                                                                                    |          |         |          |
| 2.                                                                                                              | HYDRAULIC SYSTEM FILTERS - Replace the hydraulic tank breathers and high pressure filter elements. Check the oil level. Add oil as necessary. (Lube Key "B")                                                            |          |         |          |
| 3.                                                                                                              | HYDRAIR® SUSPENSION - Check for proper piston extension (front and rear).                                                                                                                                               |          |         |          |
| 4.                                                                                                              | THROTTLE AND BRAKE PEDAL - Lubricate the treadle roller and hinge pins with lubricating oil. Lift the boot from the mounting plate and apply a few drops of lubricating oil between the mounting plate and the plunger. |          |         |          |
| 5.                                                                                                              | HOIST LEVER LINKAGE - Check the lever operation. Clean, lubricate and adjust the linkage as necessary.                                                                                                                  |          |         |          |
| 6.                                                                                                              | PARKING BRAKE - Refer to Section J7, Parking Brake, for the recommended inspections.                                                                                                                                    |          |         |          |

## 500 HOUR LUBRICATION AND MAINTENANCE CHECKS (Cont.)

|     | TASK                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | COMMENTS | CHECKED | INITIALS |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------|----------|
| 7.  | <b>RESERVE ENGINE OIL SYSTEM -</b><br>1. Check the electrical system connections for tightness, corrosion and damage. Check the battery, oil pressure switch, junction boxes, remote control box and the circuit breakers.<br>2. Examine all electrical cables over their entire length for possible damage.<br>3. Inspect all reserve engine oil system hoses for leaks, cracks or other damage. Check all fittings for tightness, leakage or damage.                                                         |          |         |          |
| 8.  | <b>BLADDER ACCUMULATORS -</b> Check the precharge pressure. Failure to maintain the correct precharge pressure may result in bladder failures.                                                                                                                                                                                                                                                                                                                                                                 |          |         |          |
| 9.  | <b>TIE ROD -</b> Check the torque on the locknuts for the tie rod threaded rod ends. Refer to Tie Rod Inspection and Torque Procedure in Section G3, Front Wheel Hub and Spindle                                                                                                                                                                                                                                                                                                                               |          |         |          |
| 9.  | <b>WHEEL MOTOR OIL -</b> Change or filter the wheel motor oil. The oil must be filtered/changed earlier if oil analysis indicates contaminated oil.                                                                                                                                                                                                                                                                                                                                                            |          |         |          |
| 10. | <b>KOMTRAX PLUS DATA DOWNLOAD -</b> Perform a data download from the KOMTRAX Plus controller and send the data to WebCARE. Refer to Section D in the shop manual for more detailed instructions.                                                                                                                                                                                                                                                                                                               |          |         |          |
| 11. | <b>CONTROL CABINET -</b> Vacuum the dust and debris from inside the control cabinet.<br><br>Do not use compressed air. Compressed air causes dust particles and contaminants to be blown across the surface of the electrical cards, eroding the varnish and driving the particles into the electrical and mechanical components of the system.<br><br>Wear the appropriate Personal Protective Equipment (PPE). A clean, soft-bristle brush can be used in areas such as the ST transformer coils, if needed. |          |         |          |
| 12. | <b>FRAME AND AXLE BOX INSPECTION -</b> Wash the truck, clean all of the weld joints, and visually inspect the entire frame and axle box for cracking or damage.                                                                                                                                                                                                                                                                                                                                                |          |         |          |

## 1000 HOUR LUBRICATION AND MAINTENANCE CHECKS

All 10 Hour (Daily) Inspections, 250 Hour Lubrication and Maintenance Checks, and 500 Hour Lubrication and Maintenance Checks should also be performed at this time.

NOTE: "Lube Key" references are to the **Lubrication Chart** on page P2-7.

| Truck Serial Number _____ Site Unit Number _____ Date _____                                                                                                                                                                                                                          |          |         |          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------|----------|
| Hourmeter _____ Name of Service Technician _____                                                                                                                                                                                                                                     |          |         |          |
| TASK                                                                                                                                                                                                                                                                                 | COMMENTS | CHECKED | INITIALS |
| 1. HYDRAULIC TANK - Drain the hydraulic tank and clean the inlet strainers. Refer to Section L3. Refill the hydraulic tank. (Lube Key "B")                                                                                                                                           |          |         |          |
| 2. RADIATOR - Clean the cooling system with a good quality cleaning compound. Flush with water.<br><br>Fill the radiator with the proper coolant mixture according to Coolant Specifications on page P-4. Refer to the engine service manual for additional coolant recommendations. |          |         |          |
| 3. FUEL TANK - Remove the vent on top of the fuel tank and clean it in solvent. Dry the breather with pressurized air and install it on the tank.                                                                                                                                    |          |         |          |
| 4. OPERATOR'S SEAT - Apply grease to the slide rails. (Lube Key "D")                                                                                                                                                                                                                 |          |         |          |
| 5. AUTOMATIC BRAKE APPLICATION - Make sure that the brakes are automatically applied when the brake pressure decreases below the specified limit. Refer to Section J4, Brake Circuit Check-Out Procedure.                                                                            |          |         |          |

## 5000 HOUR MAINTENANCE CHECKS

All 10 Hour (Daily) Inspections, 250 Hour Lubrication and Maintenance Checks, 500 Hour Lubrication and Maintenance Checks, and 1000 Hour Lubrication and Maintenance Checks should also be performed at this time.

NOTE: "Lube Key" references are to the **Lubrication Chart** on page P2-7.

| Truck Serial Number _____ Site Unit Number _____ Date _____                                                                                                                                      |          |         |          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------|----------|
| Hourmeter _____ Name of Service Technician _____                                                                                                                                                 |          |         |          |
| TASK                                                                                                                                                                                             | COMMENTS | CHECKED | INITIALS |
| 1. AIR CLEANERS - Clean the Donacelone tubes in the precleaner section of the air filter. Use low pressure cold water or low pressure air to clean the tubes. Refer to Section C5, Air Cleaners. |          |         |          |
| 2. STEERING WHEEL RETAINER NUT - Check that the torque on the steering wheel retainer nut is <b>81 ± 7 N•m (60 ± 5 ft lb)</b> . Tighten if necessary.                                            |          |         |          |

# NOTES

# SECTION P3

## AUTOMATIC LUBRICATION SYSTEM

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# AUTOMATIC LUBRICATION SYSTEM

## GENERAL DESCRIPTION

The automatic lubrication system is a pressurized lubricant delivery system which delivers a controlled amount of lubricant to designated lube points. The system is controlled by an electronic timer which signals a solenoid valve to operate a hydraulic motor powered grease pump. Hydraulic oil for pump operation is supplied by the truck steering circuit.

Grease output is proportional to the hydraulic motor input flow. A pump control manifold, mounted on top of the hydraulic motor, controls input flow and pressure. A 24VDC solenoid mounted on the manifold turns the pump on and off.

The pump is driven by the rotary motion of the hydraulic motor, which is then converted to reciprocating motion through an eccentric crank mechanism. The reciprocating action causes the pump cylinder to move up and down. The pump is a positive displacement, double-acting type as grease output occurs on both the up and the down stroke.

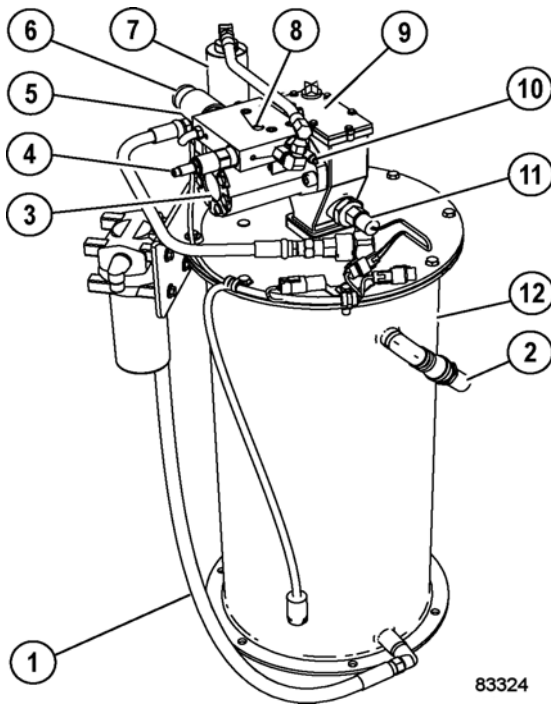


FIGURE 3-1. PUMP & RESERVOIR COMPONENTS

- |                            |                        |
|----------------------------|------------------------|
| 1. Hose From Filter        | 6. Override Switch     |
| 2. Vent Hose               | 7. Vent Valve          |
| 3. Hydraulic Motor         | 8. Pipe Plug           |
| 4. Pressure Reducing Valve | 9. Pump Assembly       |
| 5. Solenoid Valve          | 10. Flow Control Valve |
|                            | 11. Pressure Switch    |
|                            | 12. Grease Reservoir   |

During the down stroke, the pump cylinder is extended into the grease. Through the combination of shovel action and vacuum generated in the pump cylinder chamber, the grease is forced into the pump cylinder. Simultaneously, grease is discharged through the outlet of the pump. The volume of grease during intake is twice the amount of grease output during one cycle. During the upstroke, the inlet check valve closes. One half of the grease taken in during the previous stroke is transferred through the outlet check and discharged to the outlet port.

## **⚠ WARNING**

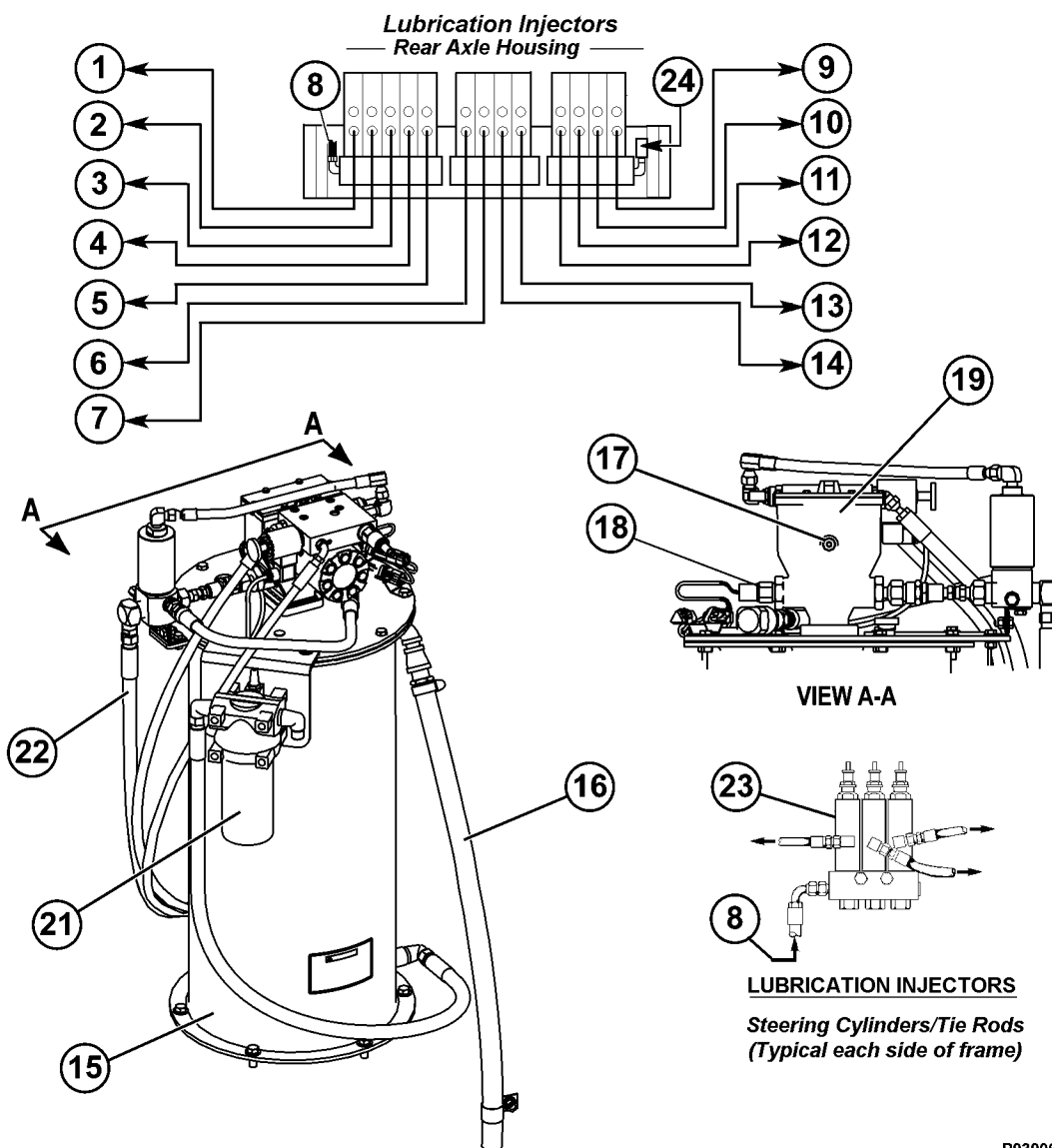
***Over-pressurizing the system, modifying parts, using incompatible chemicals and fluids, or using worn or damaged parts may result in equipment damage and/or serious personal injury.***

- ***DO NOT exceed the stated maximum working pressure of the pump or the lowest rated component in the system.***
- ***Do not alter or modify any part of this system unless approved by the factory.***
- ***Do not attempt to repair or disassemble the equipment while the system is pressurized.***
- ***Make sure that all fluid connections are securely tightened before using this equipment.***
- ***Always read and follow the fluid manufacturer's recommendations regarding fluid compatibility and the use of protective clothing and equipment.***
- ***Check all equipment regularly. Repair or replace worn or damaged parts immediately.***

***This equipment generates very high grease pressure. Extreme caution must be used when operating this equipment as material leaks from loose or ruptured components can inject fluid through the skin and into the body causing serious bodily injury including possible need for amputation. Adequate protection is recommended to prevent splashing of material onto skin or into the eyes.***

***If any fluid appears to penetrate the skin, get emergency medical care immediately! Do not treat as a simple cut. Tell attending physician exactly what fluid was injected.***





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FIGURE 3-2. AUTOMATIC LUBRICATION SYSTEM INSTALLATION

- |                                         |                                                    |
|-----------------------------------------|----------------------------------------------------|
| 1. L.H. Suspension, Bottom Bearing      | 13. R.H. Suspension, Top Bearing                   |
| 2. L.H. Hoist Cylinder, Bottom Bearing  | 14. R.H. Body Pivot Pin                            |
| 3. L.H. Anti-Sway Bar Bearing           | 15. Grease Reservoir                               |
| 4. L.H. Suspension, Top Bearing         | 16. Vent Hose                                      |
| 5. L.H. Body Pivot Pin                  | 17. Pipe Plug (Oil Level)                          |
| 6. L.H. Hoist Cylinder, Top Bearing     | 18. Pressure Switch, N.O. - 20 685 kPa (3,000 psi) |
| 7. R.H. Hoist Cylinder, Top Bearing     | 19. Grease Pump                                    |
| 8. Grease Supply From Pump              | 20. Vent Valve                                     |
| 9. R.H. Suspension, Bottom Bearing      | 21. Filter                                         |
| 10. R.H. Hoist Cylinder, Bottom Bearing | 22. Grease Supply to Injectors                     |
| 11. Rear Axle Pivot Pin                 | 23. Injectors                                      |
| 12. R.H. Anti-Sway Bar Bearing          | 24. Pressure Switch, N.O. - 13 790 kPa (2,000 psi) |

## SYSTEM COMPONENTS

### Filter

Filter assembly (21, Figure 3-2), mounted on the grease reservoir, filters the grease prior to refilling the reservoir from the shop supply. A bypass indicator alerts service personnel when the filter requires replacement.

### Hydraulic Motor and Pump

Rotary hydraulic pump (3 & 9, Figure 3-1) is a fully hydraulically operated grease pump. An integrated pump control manifold is incorporated with the motor to control input flow and pressure.

*NOTE: The pump crankcase oil level must be maintained to the level of the pipe plug (17, Figure 3-2). If necessary, refill with 10W-30 motor oil.*



***Hydraulic oil supply inlet pressure must not exceed 24 132 kPa (3500 psi). Exceeding the rated pressure may result in damage to the system components and personal injury.***

### Grease Reservoir

Reservoir (12, Figure 3-1) has an approximate capacity of 41 kg (90 lb) of grease. When the grease supply is replenished by filling the system at the service center, the grease is passed through the filter to remove contaminants before it flows into the reservoir.

### Pressure Reducing Valve

Pressure reducing valve (4, Figure 3-1), located on the manifold, reduces the hydraulic supply pressure (from the truck steering circuit) to a suitable operating pressure for the hydraulic motor used to drive the lubricant pump. **The pressure control valve has been factory adjusted and the setting should not be disturbed unless grease output pressure is outside the recommendations.**

A pressure gauge can be installed where pipe plug (8, Figure 3-1) is located. The pressure gauge will indicate hydraulic oil pressure to the inlet of the hydraulic motor. Normal pressure is 2 241 - 2 413 kPa (325 - 350 psi).

### Flow Control Valve

Flow control valve (10, Figure 3-1), mounted on the manifold, controls the amount of oil flow to the hydraulic motor. **The flow control valve has been factory adjusted and the setting should not be disturbed.**

### Solenoid Valve

Solenoid valve (5, Figure 3-1), when energized, allows oil to flow to the hydraulic motor.

### Vent Valve

With vent valve (7, Figure 3-1) closed, the pump continues to operate until maximum grease pressure is achieved. As this occurs, the vent valve opens and allows the grease pressure to drop to zero, so the injectors can recharge for their next output cycle.

### Lubrication Cycle Timer

The solid state lubrication cycle timer provides a 24 VDC timed-interval signal to energize solenoid valve (3, Figure 3-3) providing oil flow to operate the grease pump motor. This timer is mounted in the Electrical Interface Cabinet.

### Over Pressure Cut Off Switch

Pressure switch (18, Figure 3-2) is a normally open switch set at 20 685 kPa (3,000 psi). This switch de-energizes the pump solenoid relay if the grease line pressure reaches the switch pressure setting, turning off the motor and pump.

### Grease Pressure Failure Switch

Pressure switch (24, Figure 3-2) is a normally open switch set at 13 790 kPa (2,000 psi). If the appropriate grease pressure is not achieved during the normal pump cycle, the warning system will be activated, illuminating the warning lamp in the overhead display to notify the operator a problem exists in the lube system.

### Injectors

Each injector (23, Figure 3-2) delivers a controlled amount of pressurized lubricant to a designated lube point. Refer to Figure 3-2 for locations.

### Relief Valve (unloader valve)

Relief valve (14, Figure 3-3) protects the pump from high pressures. This relief valve is set at 27 580 kPa (4,000 psi).

## SYSTEM OPERATION

### Normal Operation

1. During truck operation, the lubrication cycle timer will energize the system at a preset time interval.
2. The timer provides 24VDC to energize the pump solenoid valve (3, Figure 3-3), allowing hydraulic oil provided by the truck steering pump circuit to flow to the pump motor and initiate a pumping cycle.
3. The hydraulic oil from the steering circuit is directed through the pressure reducing valve (4) and flow control valve (6) before entering the motor. Pump pressure can be read on optional pressure gauge (5) mounted on the manifold.
4. With oil flowing into the hydraulic motor, the grease pump will operate, pumping grease from the reservoir to the injectors (13), through a check valve (10) and to the vent valve (11).
5. During this period, the injectors will meter the appropriate amount of grease to each lubrication point.
6. When grease pressure reaches pressure switch (18, Figure 3-2) setting, the switch contacts will close and energize the relay RB7-K5, removing power from the hydraulic motor/pump solenoid and the pump will stop. The relay will remain energized until grease pressure drops (pressure switch opens again) and the timer turns off.
7. After the pump solenoid valve is de-energized, hydraulic pressure in the manifold drops and vent valve (11, Figure 3-3) will open, releasing grease pressure in the lines to the injector banks. When this occurs, the injectors are then able to recharge for the next lubrication cycle.
8. The system will remain at rest until the lubrication cycle timer turns on and initiates a new grease cycle.
9. During the normal lubrication cycle, if grease pressure fails to reach 13 790 kPa (2,000 psi) within 120 seconds at the pressure switch located on the rear axle housing, an amber indicator light will illuminate on the overhead panel.

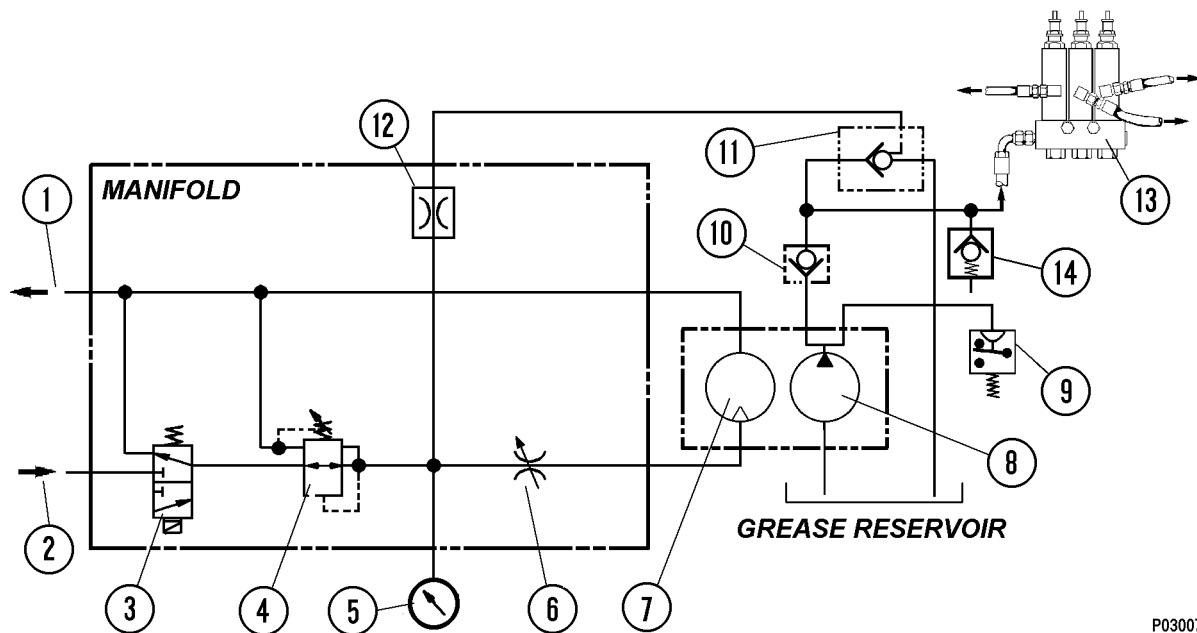


FIGURE 3-3. HYDRAULIC SCHEMATIC

- |                            |                           |                   |
|----------------------------|---------------------------|-------------------|
| 1. Hydraulic Oil Return    | 6. Flow Control Valve     | 11. Vent Valve    |
| 2. Hydraulic Oil Supply    | 7. Hydraulic Motor        | 12. Orifice       |
| 3. Pump Solenoid Valve     | 8. Grease Pump            | 13. Injector Bank |
| 4. Pressure Reducing Valve | 9. Pressure Switch (N.O.) | 14. Relief Valve  |
| 5. Motor Pressure Gauge    | 10. Check Valve           |                   |

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## GENERAL INSTRUCTIONS

### Lubricant Required For System

Grease requirements will depend on ambient temperatures encountered during truck operation:

- Above 32°C (90°F) - Use NLGI No. 2 multipurpose grease (MPG).
- -32° to 32°C (-25° to 90°F) - Use NLGI No. 1 multipurpose grease (MPG).
- Below -32°C (-25°F) - Refer to local supplier for extreme cold weather lubricant requirements.

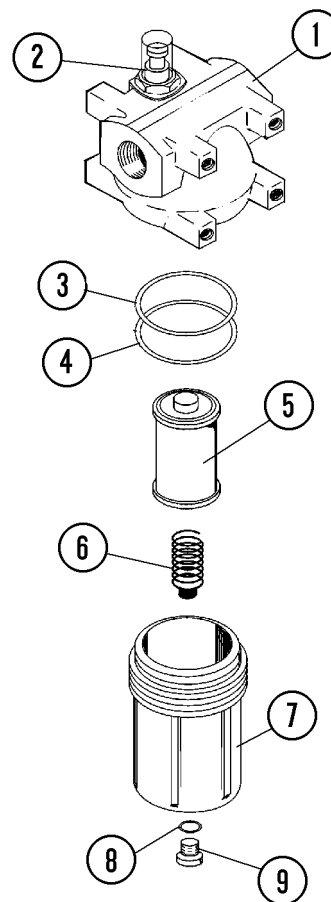
### System Priming

The system must be full of grease and free of air pockets to function properly. After maintenance, if the primary or secondary lubrication lines were replaced, it will be necessary to prime the system to eject all entrapped air.

1. Fill lube reservoir with lubricant, if necessary.
2. To purge air from the main supply line, remove the main supply line at the grease canister. Connect an external grease supply to the line.
3. Remove plugs from each injector group in sequence (right front, left front, and rear axle).
4. Using the external grease source, pump grease until grease appears at the group of injectors and re-install the pipe plug. Repeat for remaining injector groups.
5. Remove the caps from each injector and connect an external grease supply to the zerk on the injector and pump until grease appears at the far end of the individual grease hose or the joint being greased.

### Filter Assembly

Filter element (5, Figure 3-4) must be replaced if bypass indicator (2) shows excessive element restriction.



P030023

FIGURE 3-4. FILTER ASSEMBLY

- |                     |           |
|---------------------|-----------|
| 1. Housing          | 6. Spring |
| 2. Bypass Indicator | 7. Bowl   |
| 3. O-Ring           | 8. O-Ring |
| 4. Backup Ring      | 9. Plug   |
| 5. Element          |           |

## LUBRICANT PUMP

### Pump Housing Oil Level

The pump housing must be filled to the proper level with SAE 10W-30 motor oil. Check the oil level at at the 1000 hour maintenance interval. To add oil, remove dipstick (4, Figure 3-5) on top of the pump housing and fill the housing to the proper level marked on the dipstick.

*NOTE: If the pump is not equipped with dipstick (4), remove oil level plug (5) and fill the housing to the bottom of the plug hole.*

### Pump Pressure Control

High pressure hydraulic fluid from the truck steering system is reduced by the pressure reducing valve located on the manifold on top of the pump motor. This pressure can be read if pipe plug (3) is removed, and a gauge is installed on the manifold.

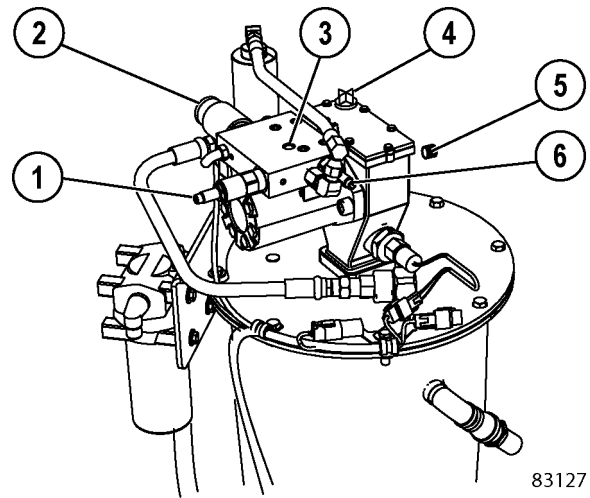


FIGURE 3-5. AUTO LUBE PUMP CONTROLS

- |                           |                       |
|---------------------------|-----------------------|
| 1. Pump Pressure Control  | 4. Dipstick           |
| 2. Override Switch        | 5. Oil Level Plug     |
| 3. Pipe Plug (Gauge Port) | 6. Flow Control Valve |

## INJECTORS (SL-1 Series “H”)

### Injector Specifications

- Each lube injector services only one grease point. In case of pump malfunction, each injector is equipped with a covered grease fitting to allow the use of external lubricating equipment.
- Injector output volume:  
Maximum output = . . . . . 1.31 cc (0.08 in<sup>3</sup>).  
Minimum output = . . . . . 0.13 cc (0.008 in<sup>3</sup>).
- Operating Pressure:  
Minimum - . . . . . 12 755 kPa (1,850 psi)  
Maximum - . . . . . 24 133 kPa (3,500 psi)  
Recommended - . . .17 238 kPa (2,500 psi)
- Maximum Vent Pressure - (Recharge)  
. . . . . 4 137 kPa (600 psi)

### Injector Adjustment

The injectors may be adjusted to supply from 0.13 - 1.31 cc (0.008 - 0.08 in<sup>3</sup>) of lubricant per injection cycle. The injector piston travel distance determines the amount of lubricant supplied. This travel is in turn controlled by an adjusting screw in the top of the injector housing.

Turn adjusting screw (1, Figure 3-6) counterclockwise to increase lubricant amount delivered and clockwise to decrease the lubricant amount.

When the injector is not pressurized, maximum injector delivery volume is attained by turning the adjusting screw (1) fully counterclockwise until the indicating pin just touches the adjusting screw. At the maximum delivery point, about 9.7 mm (0.38 in.) adjusting screw threads should be showing. Decrease the delivered lubricant amount by turning the adjusting screw clockwise to limit injector piston travel. If only half the lubricant is needed, turn the adjusting screw to the point where about 4.8 mm (0.19 inch) threads are showing. The injector will be set at minimum delivery point with about 0.22 mm (0.009 in.) thread showing.

*NOTE: The above information concerns adjustment of injector delivery volume. The timer adjustment should also be changed, if overall lubricant delivery is too little or too much. Injector output should not be adjusted to less than 1/4 capacity.*

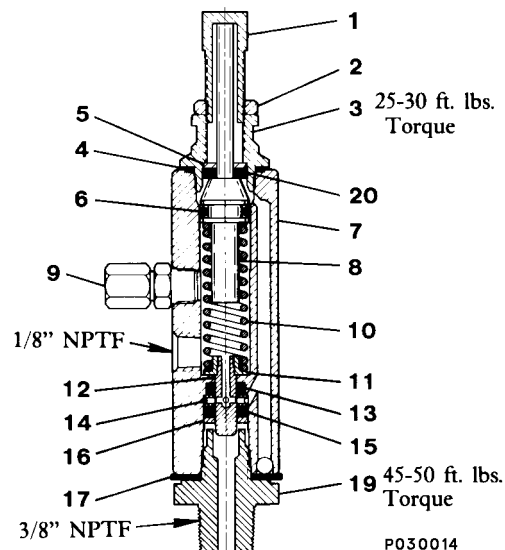


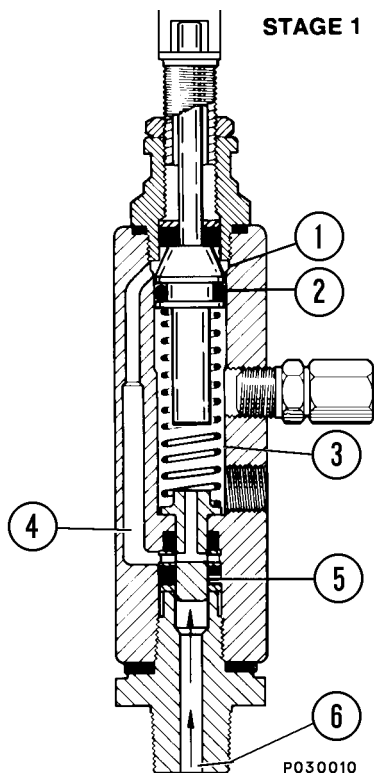
FIGURE 3-6. TYPE SL-1 INJECTOR

- |                        |                   |
|------------------------|-------------------|
| 1. Adjusting Screw     | 11. Spring Seat   |
| 2. Locknut             | 12. Plunger       |
| 3. Piston Stop Plug    | 13. Viton Packing |
| 4. Gasket              | 14. Inlet Disc    |
| 5. Washer              | 15. Viton Packing |
| 6. Viton O-Ring        | 16. Washer        |
| 7. Injector Body Assy. | 17. Gasket        |
| 8. Piston Assembly     | 18. Adapter Bolt  |
| 9. Fitting Assembly    | 19. Adapter       |
| 10. Plunger Spring     | 20. Viton Packing |

*NOTE: The piston assembly (8) has a visible indicator pin at the top of the assembly to verify injector operation.*

## INJECTOR OPERATION

**STAGE 1**



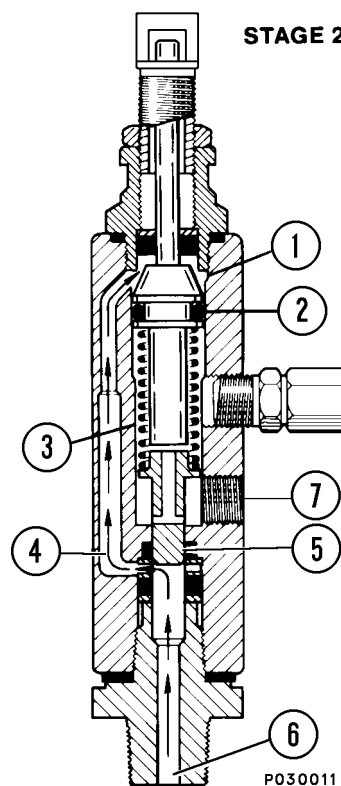
**STAGE 1.**

The injector piston (2) is in its normal or "rest" position. The discharge chamber (3) is filled with lubricant from the previous cycle. Under the pressure of incoming lubricant (6), the slide valve (5) is about to open the passage (4) leading to the measuring chamber (1) above the injector piston (2).

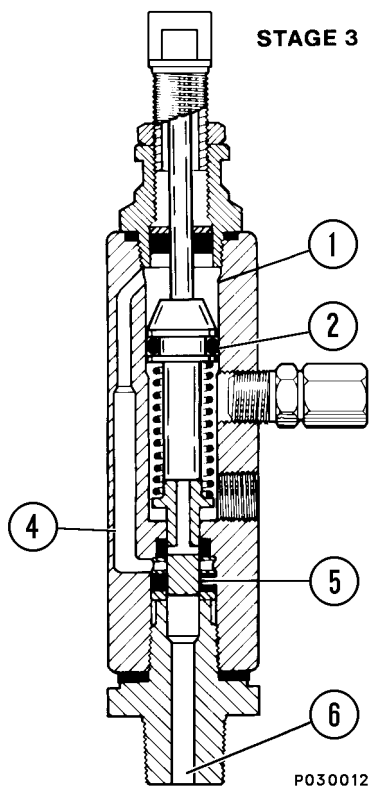
**STAGE 2.**

When the slide valve (5) uncovers the passage (4), lubricant (6) is admitted to the measuring chamber (1) above the injector piston (2) which forces lubricant from the discharge chamber (3) through the outlet port (7) to the bearing.

**STAGE 2**



**STAGE 3**



**STAGE 3.**

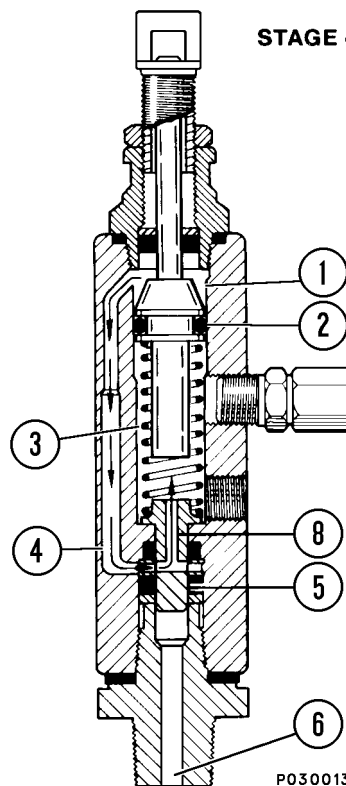
As the injector piston (2) completes its stroke, it pushes the slide valve (5) past the passage (4), cutting off further admission of lubricant (6) to the passage (4) and measuring chamber (1). The injector piston (2) and slide valve (5) remain in this position until lubricant pressure in the supply line (6) is vented.

**STAGE 4.**

After venting, the injector spring expands, causing the slide valve (5) to move, so that the passage (4) and discharge chamber (3) are connected by a valve port (8). Further expansion of the spring causes the piston to move upward, forcing the lubricant in the measuring chamber (1) through the passage (4) and valve port (8) to refill the discharge chamber (3).

Injector is now ready for the next cycle.

**STAGE 4**



## PREVENTIVE MAINTENANCE PROCEDURES

Use the following maintenance procedures to ensure proper system operation.

### Daily Lubrication System Inspection

1. Check the grease reservoir level after each shift of operation. Grease usage should be consistent from day-to-day operations. Lack of lubricant usage would indicate an inoperative system. Excessive usage would indicate a broken supply line.
2. Check the filter bypass indicator when filling the reservoir. Replace the element if bypassing.
3. Check all grease hoses from the SL-1 Injectors to the lubrication points.
  - a. Repair or replace all damaged feed line hoses.
  - b. Ensure that all air is purged and all new feed line hoses are filled with grease before returning the truck to service.
4. Inspect the key lubrication points for a bead of lubricant around the seal. If a lubrication point appears dry, troubleshoot and repair the problem.

### 250 Hour Inspection

1. Check all grease hoses from the SL-1 Injectors to the lubrication points (see Figure 3-2).
  - a. Repair or replace all worn or broken hoses.
  - b. Ensure that all air is purged and all new feed line hoses are filled with grease before returning the truck to service.
2. Check all grease supply line hoses from the pump to the SL-1 injectors.
  - a. Repair or replace all worn or broken supply lines.
  - b. Ensure that all air is purged and all new supply line hoses are filled with grease before returning the truck to service.
3. Check the grease reservoir level.
  - a. Fill the reservoir if the grease level is low. Check the filter bypass indicator when filling the reservoir. Replace the element if bypassing.
  - b. Check the reservoir for contaminants. Clean the reservoir, if required.
  - c. Ensure that all filler plugs, covers and breather vents on the reservoir are intact and free of contaminants.

4. Inspect all bearing points for a bead of lubricant around the bearing seal.

*NOTE: It is good practice to manually lube each bearing point at the grease fitting provided on each Injector. This will indicate whether there are any frozen or plugged bearings, and it will help flush the bearings of contaminants.*

### 5. System Checkout

- a. Remove all SL-1 injector cover caps to allow visual inspection of the injector cycle indicator pins during system operation.
- b. Start the engine.
- c. Actuate lube system override switch (6, Figure 3-1). The hydraulic motor and grease pump should operate.
- d. With the grease under pressure, check each SL-1 injector assembly. The cycle indicator pin should be retracted inside the injector body.
- e. When the system is at 16 203 - 17 237 kPa (2,350 - 2,500 psi), the pump should shut off and the pressure in the system should drop to zero, venting back to the grease reservoir.
- f. After the system has vented, check the SL-1 injector indicator pins. All of the pins should be visible. Replace or repair any defective injectors.
- g. Install all injector cover caps.
- h. Check the lubrication timer operation.

*NOTE: With the engine on, the lube system should activate within five minutes. The system should build to 13 790 kPa (2,000 psi) at the rear axle during normal pump cycle.*

- If the system is working properly, the machine is ready for operation.
- If the system is malfunctioning, refer to the troubleshooting chart.

### 1000 Hour Inspection

1. Check the pump housing oil level using the dipstick on top of the pump unit. If necessary, add SAE 10W-30 motor oil.



## SYSTEM CHECKOUT

To check system operation (not including timer), proceed as follows:

1. Remove the dust covers from the injectors.
2. Start the engine.
3. Actuate the manual override button at the pump assembly and observe pin movement at each injector.
4. Check for pump, hose or injector damage or leakage with the system under pressure.
5. After checking system, stop the engine. Observing normal precautions regarding high voltage present in the propulsion system before attempting to repair lube system.
6. Re-install the injector dust covers.

### Lubrication Controller Check

Pressing the manual lube button on the enclosure cover will initiate a lube event (see Figure 3-7).

### Lubrication Controller Components

Mode switch (2, Figure 3-7) consists of four dip switches. The first, second and third dip switches are not functional. These settings have been pre-set and are not changeable.

The fourth dip switch is used to select "memory off" or "memory on". When the switch is set to "memory off", a lube cycle will occur each time power is turned on. The lube cycle will start at the beginning of the on time setting.

When the switch is set to "memory on", the controller will function as follows:

1. When power is turned off during 'off time' (between cycles), the lube cycle will resume at the point of interruption after power is restored. In other words, the controller will remember its position in the cycle.
2. When power is turned off during 'on time' (during a cycle), the controller will reset to the beginning of the lube cycle after power is restored.

The time between lube events is determined by the setting of the Off Time rotary switch (1, Figure 3-7). Off time switch is used to select units of time. Possible time intervals are: 0.5, 1, 2, 4, 8, 15, 24 or 30 minutes.

Cover (3) contains three LED windows and a manual lube switch. The LEDs indicate system operation and status. When power is on, a green LED will illuminate. When the pump is on, another green LED will illuminate. A red LED will illuminate when an alarm condition occurs.

### Lubrication Controller Adjustment

The lubrication controller is factory set to the following switch settings:

Dip Switch 1 - 120 SEC (not adjustable)

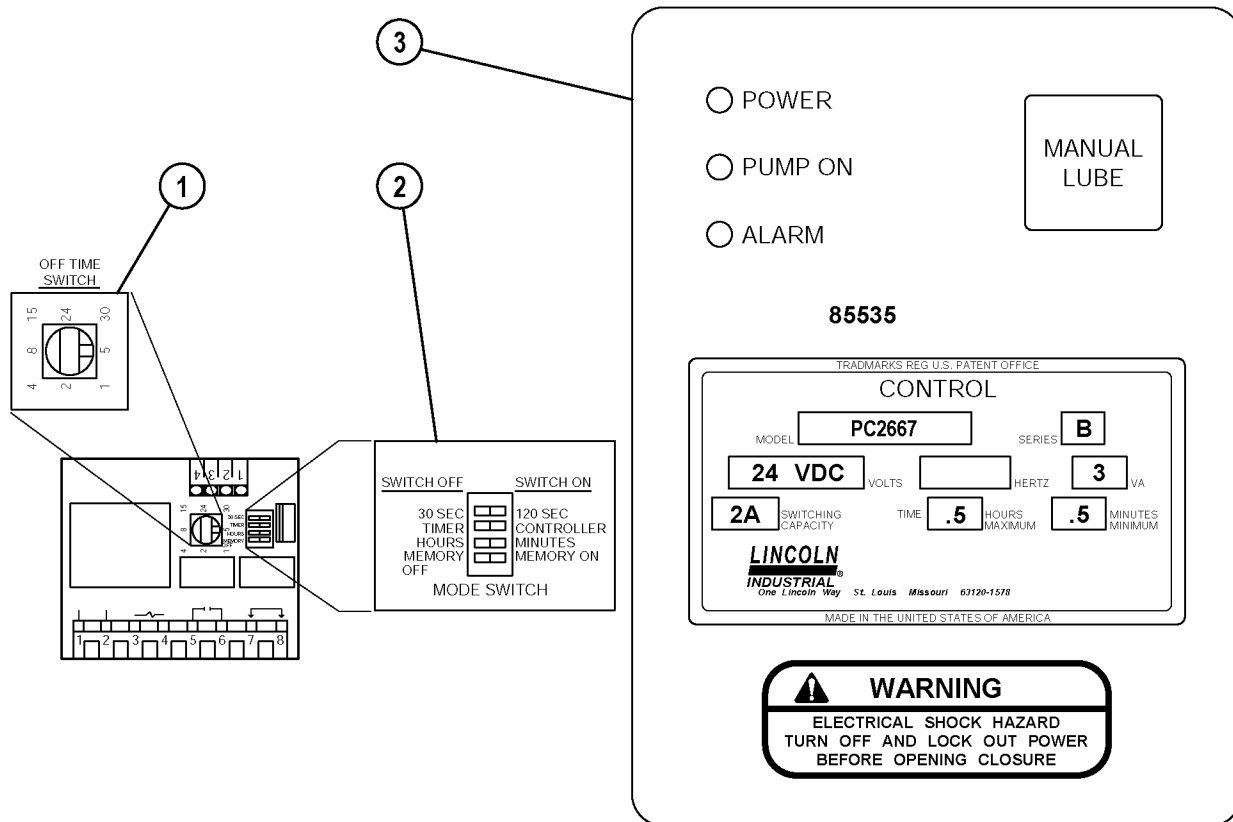
Dip Switch 2 - CONTROLLER (not adjustable)

Dip Switch 3 - MINUTES (not adjustable)

Dip Switch 4 - MEMORY OFF

Rotary Switch - 15 minutes

*NOTE: Only dip switch 4 and the rotary switch are adjustable.*



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FIGURE 3-7. LUBRICATION CYCLE TIMER

1. Off Time Switch
2. Mode Switch

3. Cover

## SYSTEM TROUBLESHOOTING CHART

*If the following procedures do not correct the problem, contact a factory authorized service center.*

| TROUBLE                                                                                  | POSSIBLE CAUSES                                                                                                                                                                                                                                      | SUGGESTED CORRECTIVE ACTION                                                                                                                                                                                                                                                                                                                                |
|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Pump Does Not Operate</b>                                                             | Lube system is not grounded.<br><br>Electrical power loss.<br><br>Timer malfunction.<br>Solenoid valve malfunction.<br>Relay malfunction.<br>Motor or pump malfunction.                                                                              | Correct grounding connections to pump assembly and truck chassis.<br><br>Locate cause of power loss and repair. 24 VDC power required. Ensure key switch is ON.<br><br>Replace timer assembly.<br>Replace the solenoid valve assembly.<br>Replace relay.<br>Repair or replace motor and/or pump assembly. (Refer to shop manual for rebuild instructions.) |
|                                                                                          | <i>NOTE: On initial startup of the lube system, the timing capacitor will not contain a charge, therefore the first timing cycle will be about double in length compared to the normal interval. Subsequent timer cycles should be as specified.</i> |                                                                                                                                                                                                                                                                                                                                                            |
| <b>Pump Will Not Prime</b>                                                               | Low lubricant supply                                                                                                                                                                                                                                 | Dirt in reservoir, pump inlet clogged, filter clogged.                                                                                                                                                                                                                                                                                                     |
| <b>Pump Will Not Build Pressure</b>                                                      | Trapped air in lubricant supply line.<br>Lubricant supply line is leaking.                                                                                                                                                                           | Prime system to remove trapped air.<br>Check lines and connections to repair leakage.                                                                                                                                                                                                                                                                      |
|                                                                                          | Vent valve is leaking.<br>Pump is worn or scored.                                                                                                                                                                                                    | Clean or replace vent valve.<br>Repair or replace pump assembly. (Refer to shop manual for rebuild instructions.)                                                                                                                                                                                                                                          |
| <b>Injector Indicator Stem Does Not Operate</b>                                          | <i>NOTE: Normally, during operation, the injector indicator stem will move into the body of the injector when pressure builds properly. When the system vents (pressure release) the indicator stem will again move out into the adjusting yoke.</i> |                                                                                                                                                                                                                                                                                                                                                            |
|                                                                                          | Malfunctioning injector - Usually indicated by the pump building pressure and then venting.<br><br>All injectors inoperative - Pump build up not sufficient to cycle injectors.                                                                      | Replace individual injector assembly.<br><br>Service and/or replace pump assembly. (Refer to shop manual for rebuild instructions.)                                                                                                                                                                                                                        |
| <b>Pressure Gauge Does Not Register Pressure (pressure gauge not included with pump)</b> | No system pressure to pump motor.<br>No 24 VDC signal at pump solenoid.<br><br>Pressure reducing valve setting is too low.<br><br>24V Relay (RB7K8 or RB7K5) may be defective.                                                                       | Check hydraulic hose from steering system.<br><br>Determine problem in 24 VDC electric system.<br><br>Refer to "Pressure Control Valve Adjustment".<br>Replace relay.                                                                                                                                                                                      |

| TROUBLE                                                                                  | POSSIBLE CAUSES                                                              | SUGGESTED CORRECTIVE ACTION                                                                                         |
|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Pump Pressure Builds Very Slowly Or Not At All</b>                                    | No signal at solenoid.                                                       | Check timer.                                                                                                        |
| <b>Controller Does Not Operate</b>                                                       | No electric power to controller.                                             | Turn on electric power to pump. "POWER" LED should light, "PUMP ON" LED should light when "MANUAL LUBE" is pressed. |
| <b>"PUMP ON" LED Lights, But Load Connected To Terminals 3 &amp; 4 Will Not Energize</b> | Printed circuit board failure.                                               | Remove and replace.                                                                                                 |
| <b>Load Connected To Terminals 3 &amp; 4 Energized, But "PUMP ON" LED Does Not Light</b> | Failure of the printed circuit board or keypad.                              | Remove and replace.                                                                                                 |
| <b>Bearing Points Excessively Lubricated</b>                                             | Controller memory mode is OFF.                                               | Switch controller memory mode to ON.                                                                                |
|                                                                                          | Injector output adjustment setting is too high.                              | Readjust to lower setting.                                                                                          |
| <b>Bearing Points Are Not Sufficiently Lubricated</b>                                    | Timer/controller cycle time setting is too low.                              | Set to longer cycle time or reevaluate lube requirements.                                                           |
|                                                                                          | Injector output adjustment setting is too low.                               | Readjust injector output setting.                                                                                   |
|                                                                                          | Timer/controller cycle time setting does not deliver lubricant often enough. | Set to shorter cycle time or reevaluate lube requirements.                                                          |
|                                                                                          | System is too large for pump output.                                         | Calculate system requirements per planning manual.                                                                  |

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ELECTRICAL SCHEMATICS . . . . .XS4800

# NOTES

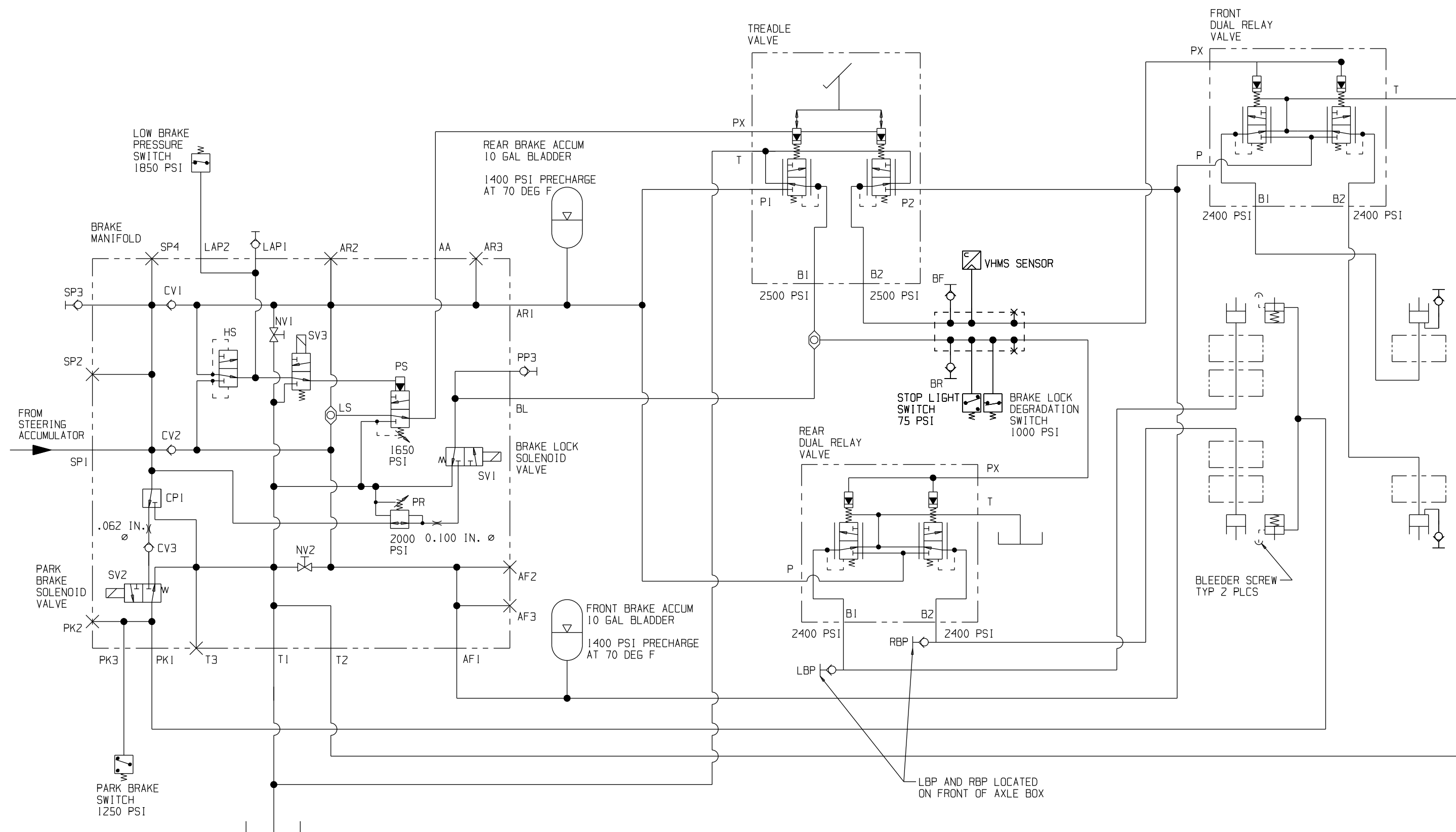


A

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**EM1358-0 OCT 06**  
**HYDRAULIC BRAKE SCHEMATIC**  
**930E-4**  
**A30462 & UP**

A

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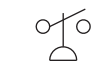
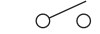
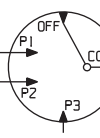
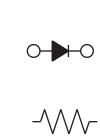
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1 | 2 | 3 | 4 | 5 | 6 | 7 |

SYMBOLS USED IN SCHEMATICS

PART NO. INDEX GENERAL TOPIC

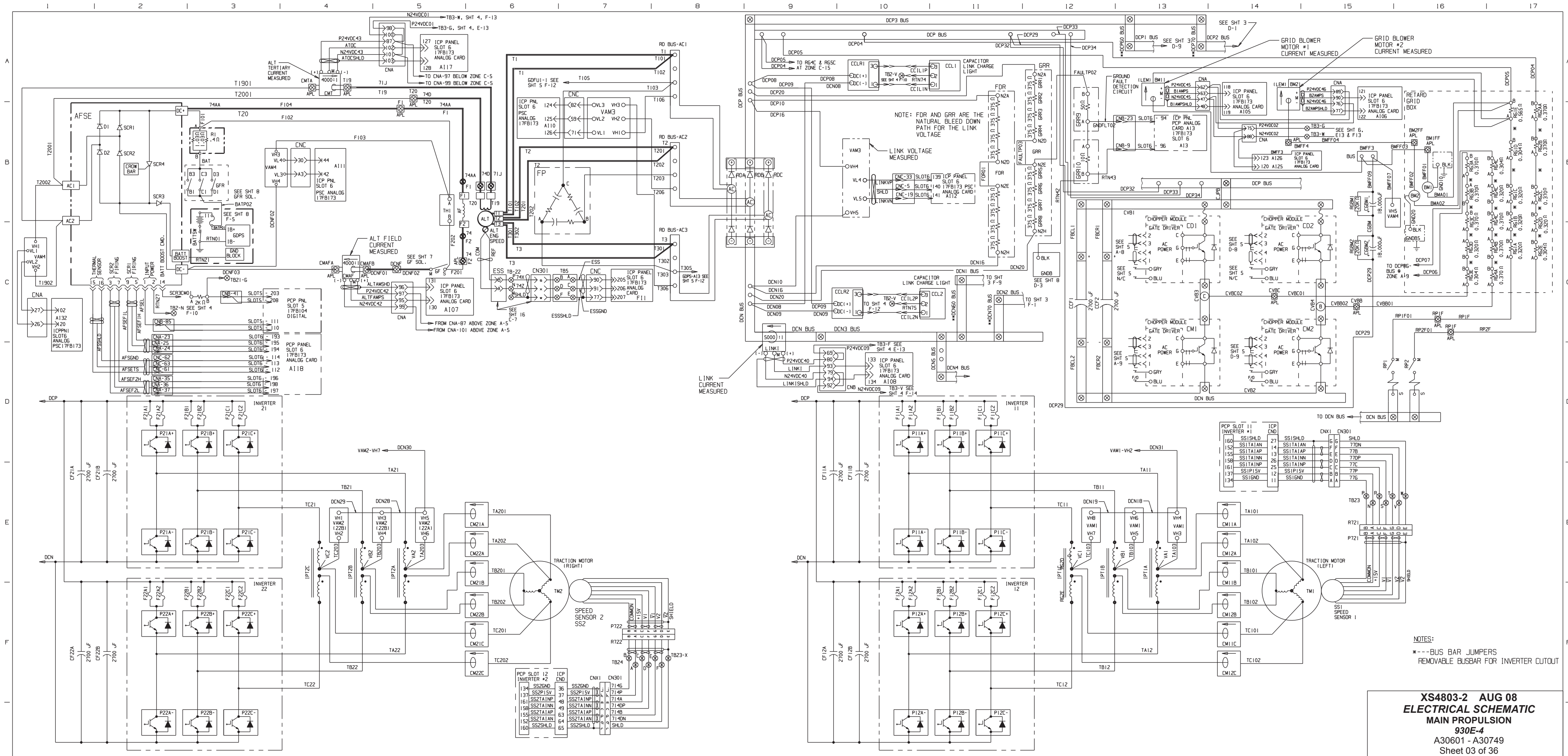
-  TERMINAL BOARD POINT
-  TERMINAL IN LOWER INSTRUMENT PANEL
-  TERMINAL IN REAR AXLE JUNCTION BOX
-  TERMINAL POINT IN FRAME AREA
-  CONNECTION POINT
-  KNIFE DISCONNECT/BULLET TERMINAL
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-  CONNECTOR, BULKHEAD (X IS CAVITY IDENTIFIER)
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-  PROXIMITY SWITCH
-  SINGLE ELEMENT LAMP
-  DUAL ELEMENT LAMP
-  HORN, BUZZER, OR SPEAKER
-  TEMPERATURE SENSOR
-  LED (LIGHT EMITTING DIODE)
-  BATTERY
-  FUSE

-  PRESSURE SWITCH - PRESSURE TO OPEN
-  PRESSURE SWITCH - PRESSURE TO CLOSE
-  TOGGLE SWITCH
-  ROTARY SWITCH
-  PUSHBUTTON SWITCH - NORMALLY CLOSED
-  CIRCUIT BREAKER - MANUAL RESET
-  PRESSURE OPERATED VARIABLE RESISTOR
-  TEMPERATURE SWITCH - INCREASE TEMP. TO OPEN
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-  LIMIT SWITCH - CAM TO OPEN
-  DIODE
-  FIXED RESISTOR
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-  TIMER MODULE
-  CNX(1-5)/P30(1-5) GE IDENTIFIER/KAC IDENTIFIER
-  X (X IS CAVITY IDENTIFIER)

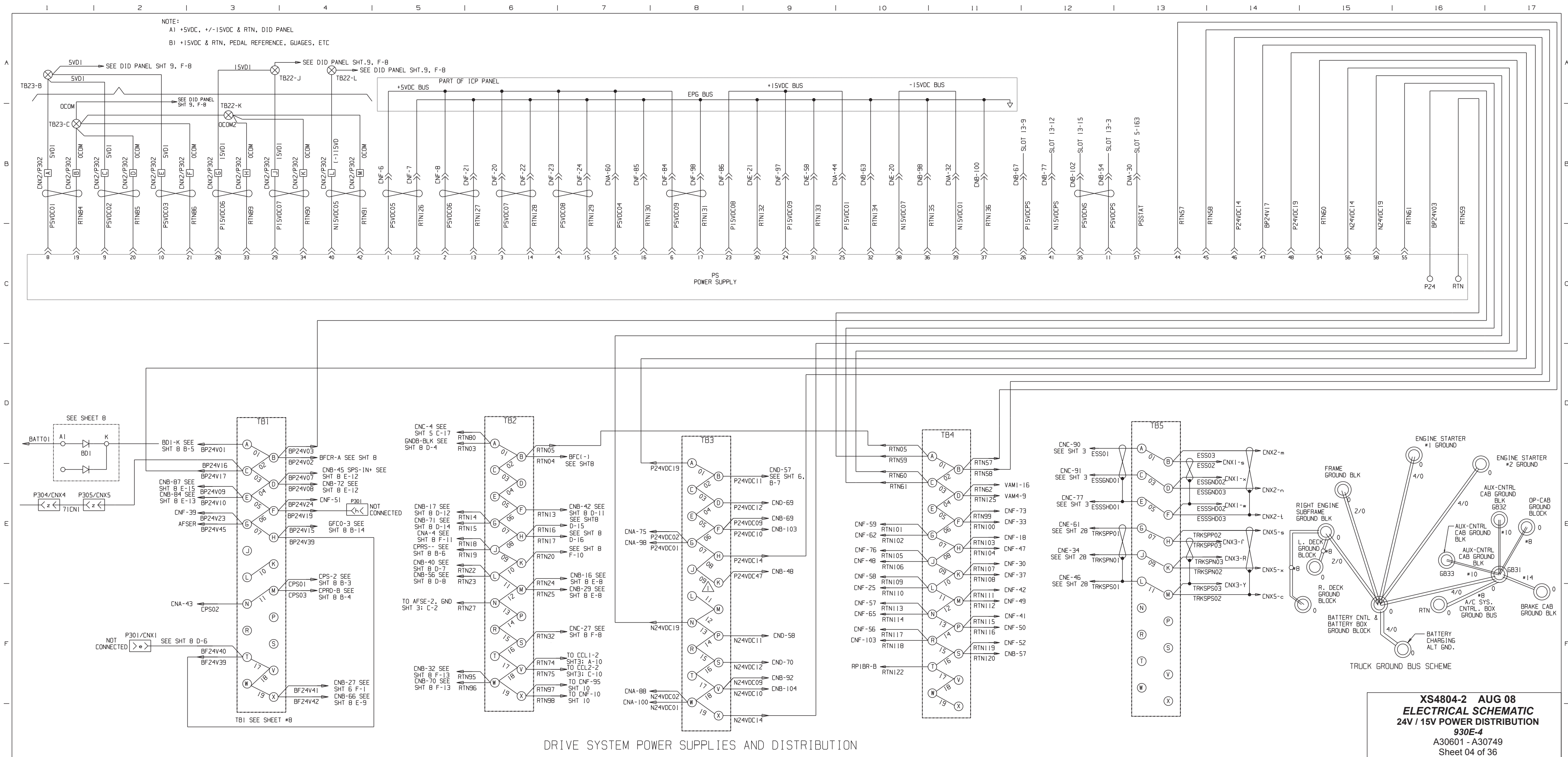
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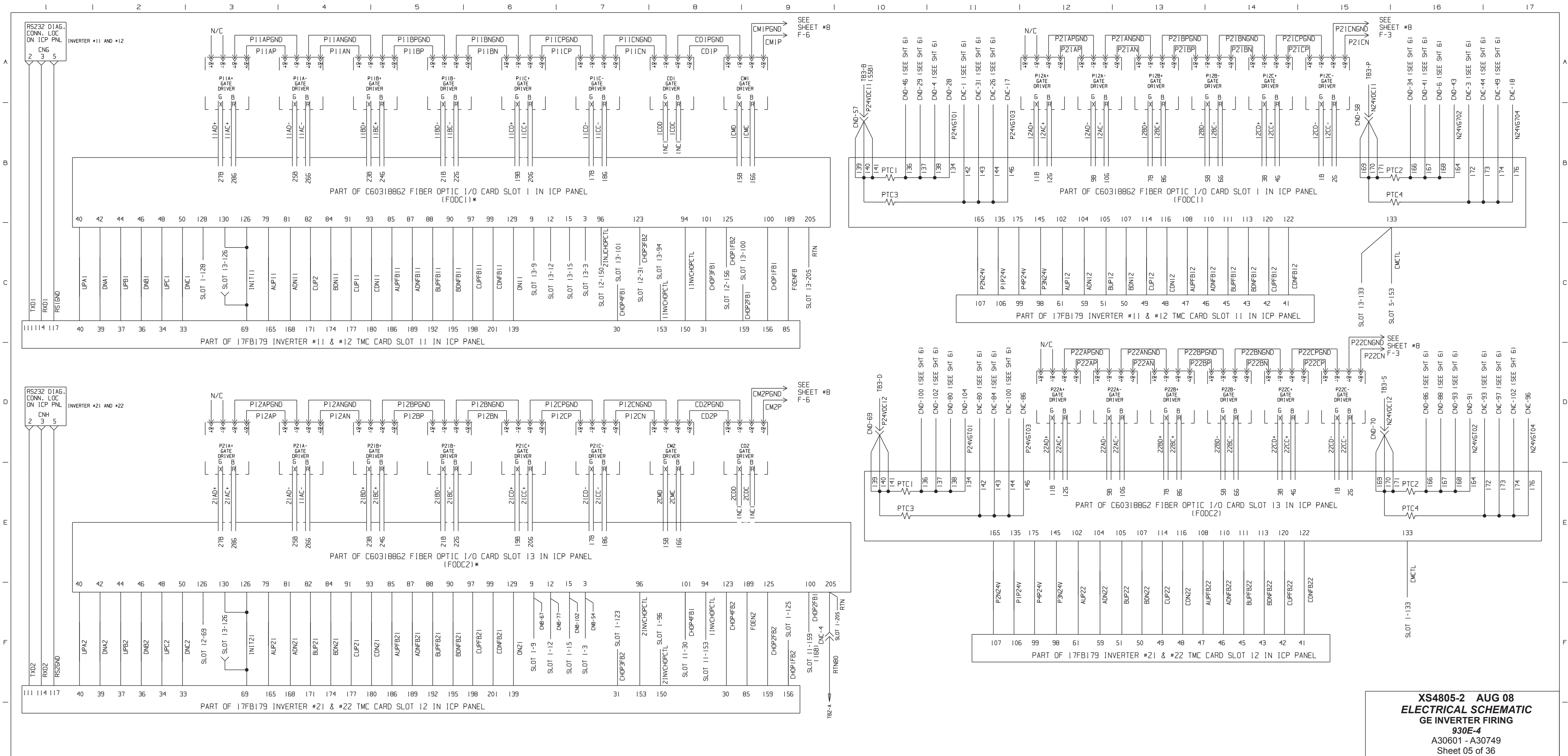


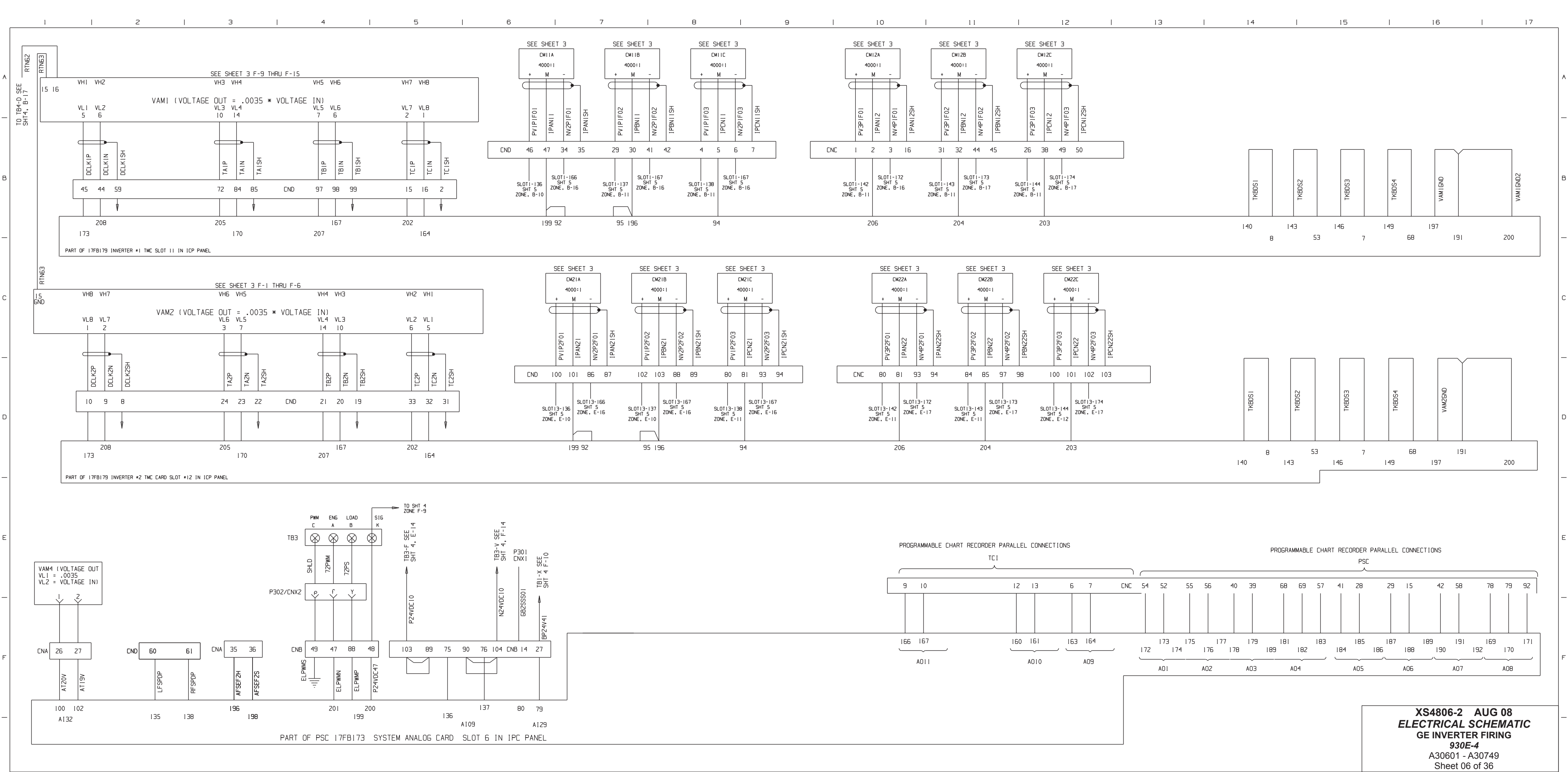










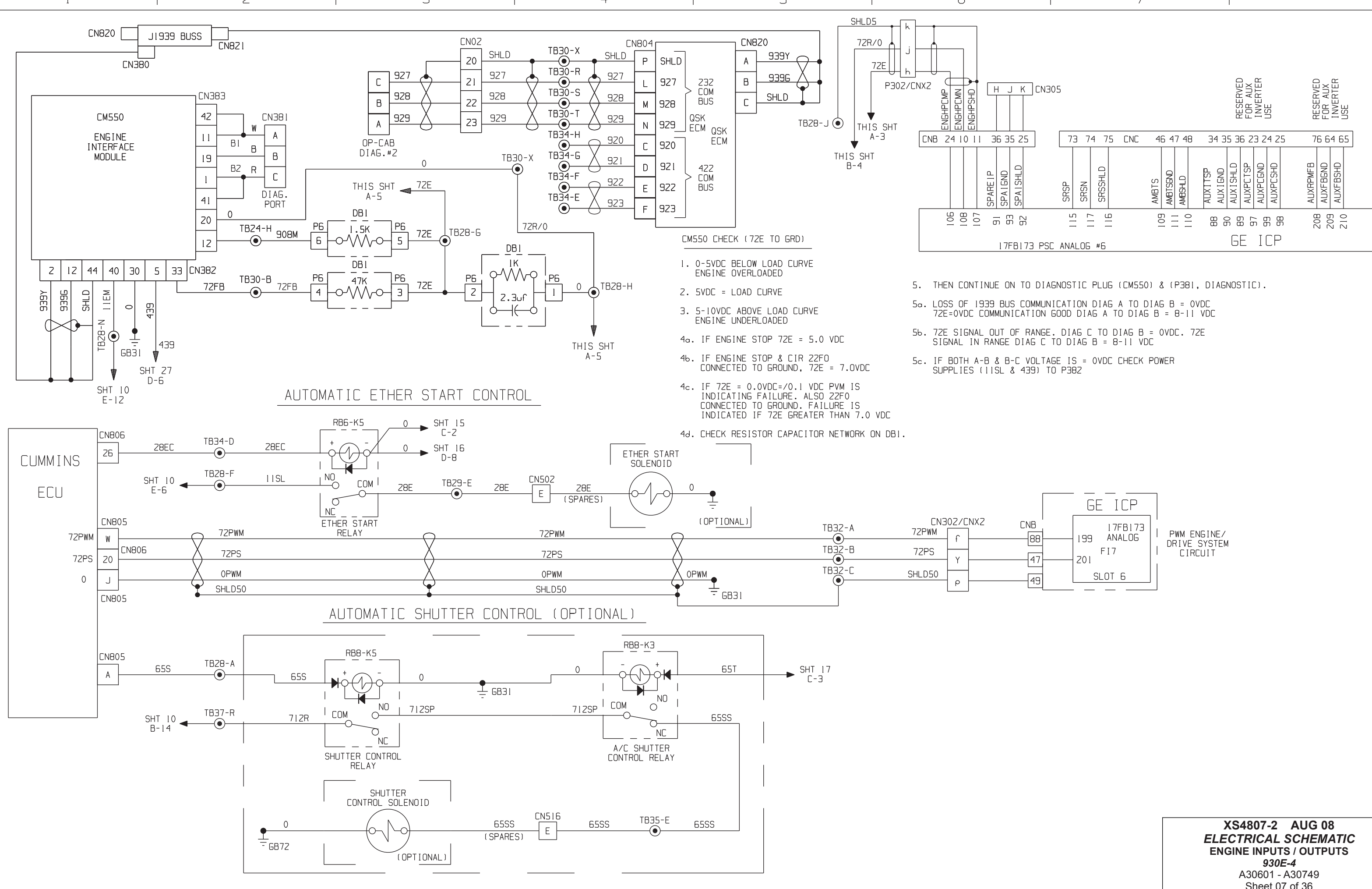


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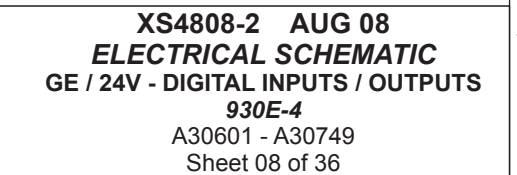
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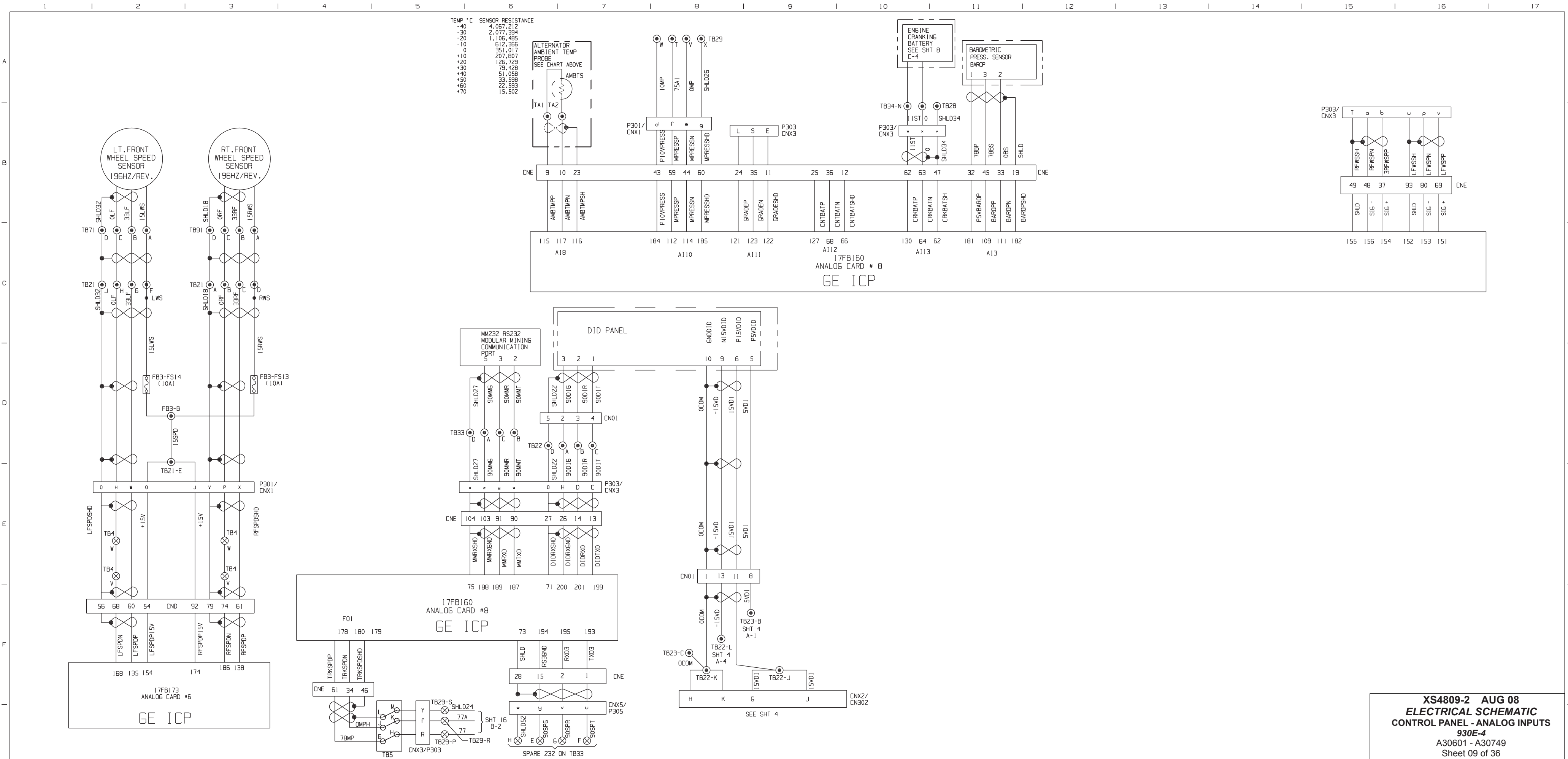
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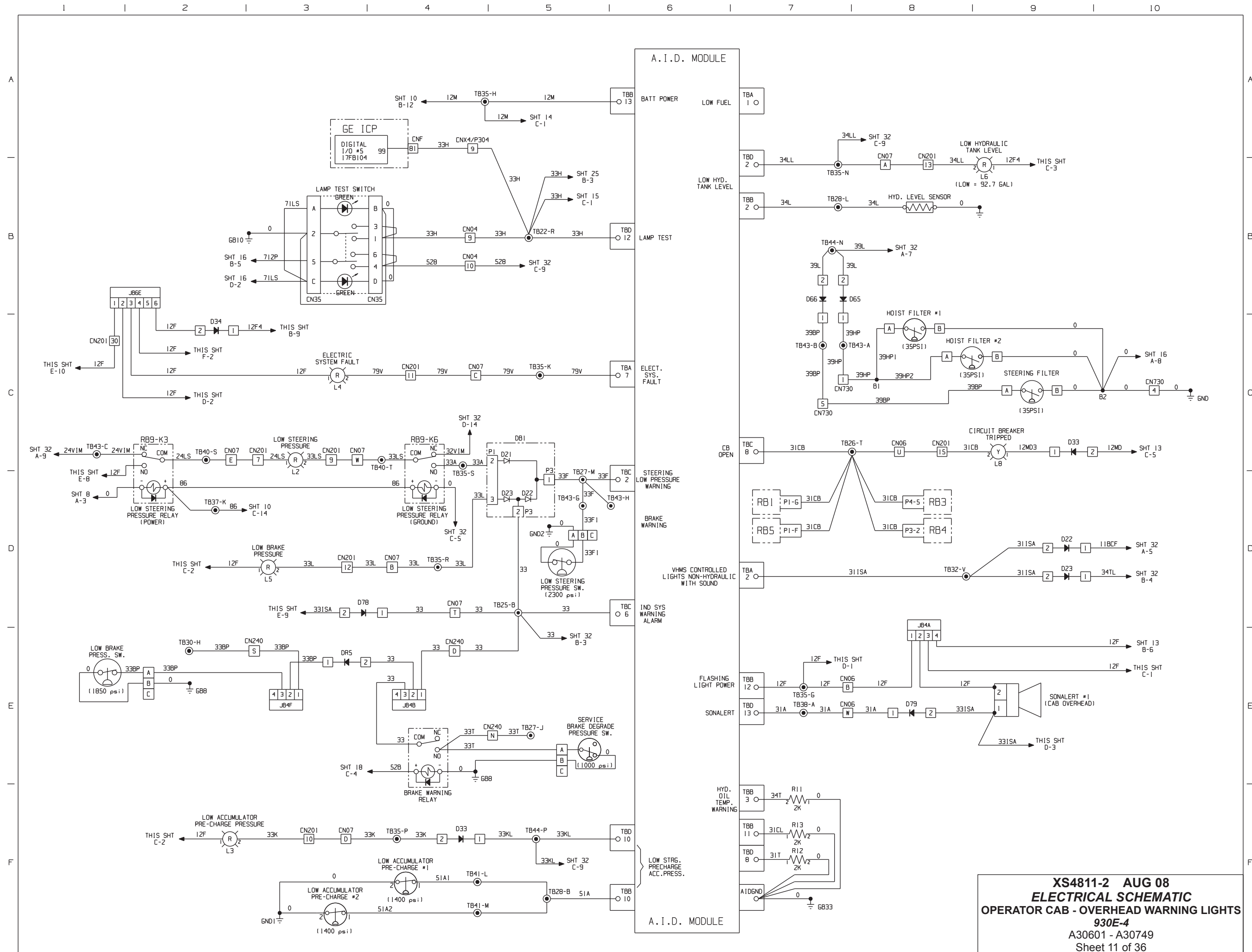








The schematic illustrates the 24V power distribution and circuit protection for a vehicle system. It features a central power source (24VDC) connected to various components through a network of wires and terminals. Key components include a 24VDC to 12VDC converter, a GE power supply, a key switch, a timed engine shutdown, a ground level power relay, a start circuit relay, and various sensors and actuators. The schematic is organized into several sections, with components labeled with alphanumeric codes and wire numbers. A legend at the bottom right identifies the document as 'XS4810-1 AUG 08 ELECTRICAL SCHEMATIC 24V POWER DISTRIBUTION & CIRCUIT PROTECTION 930E-4 A30601 - A30749 Sheet 10 of 36'.



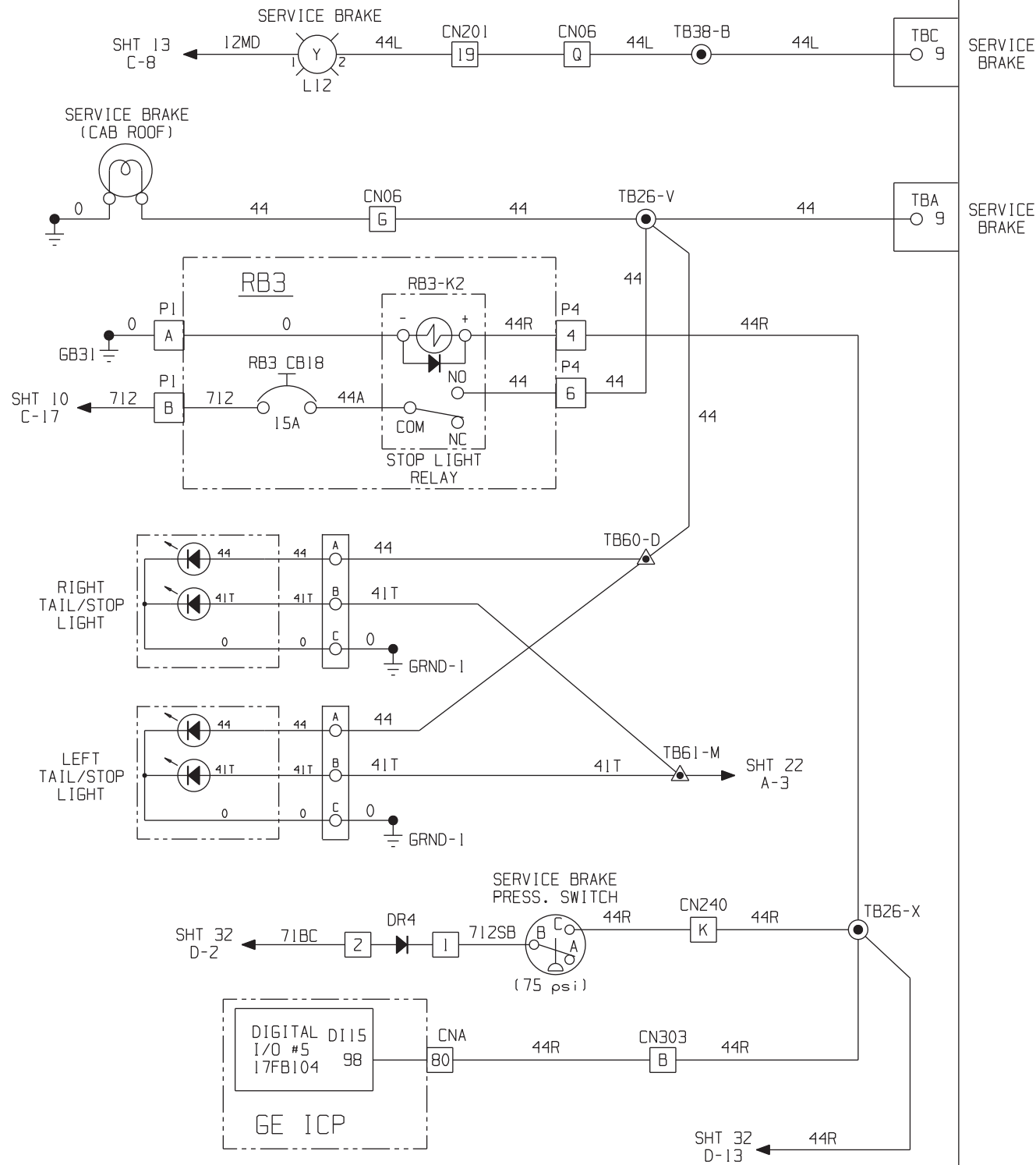


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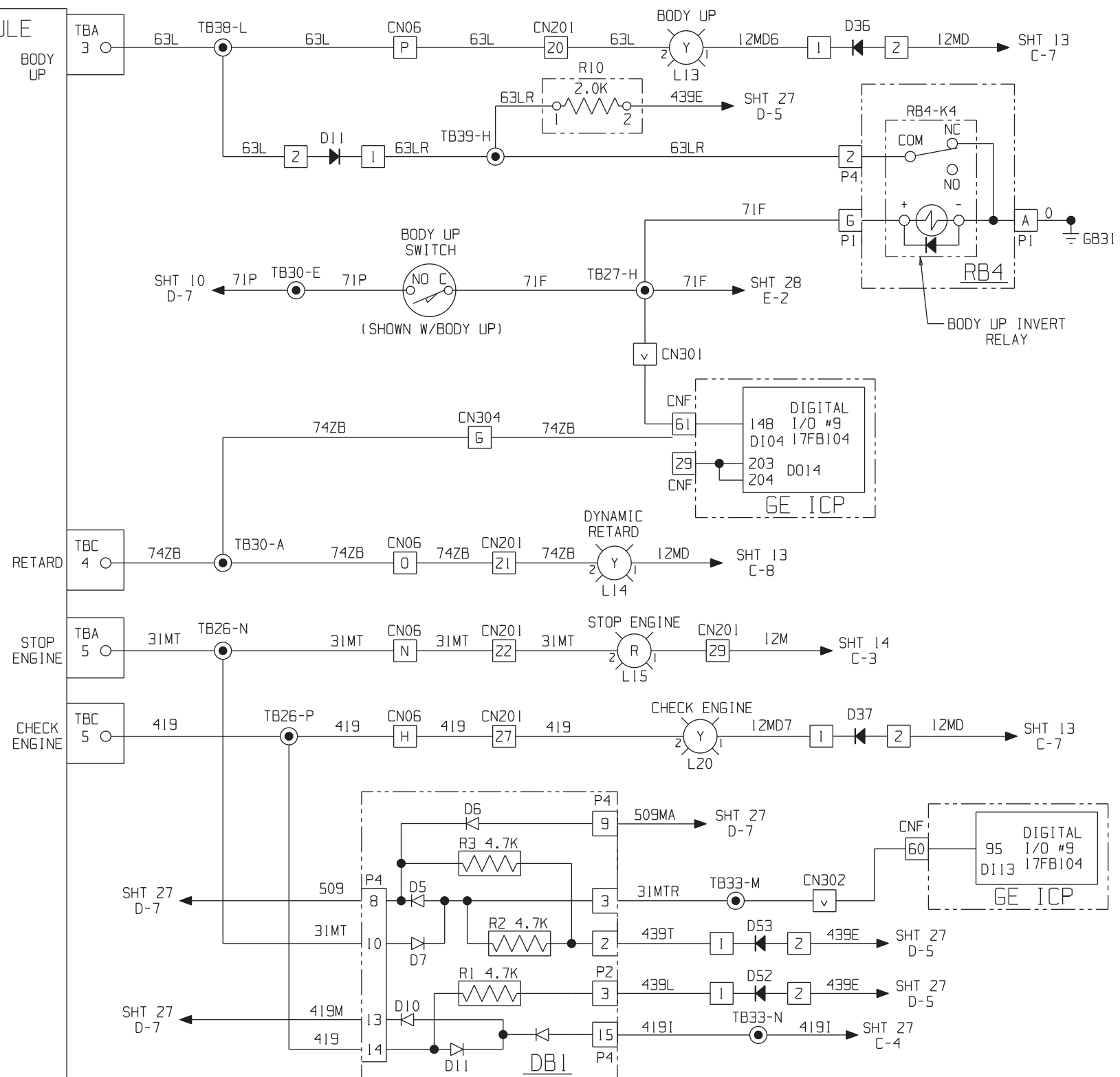
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A.I.D. MODULE

A.I.D. MODULE

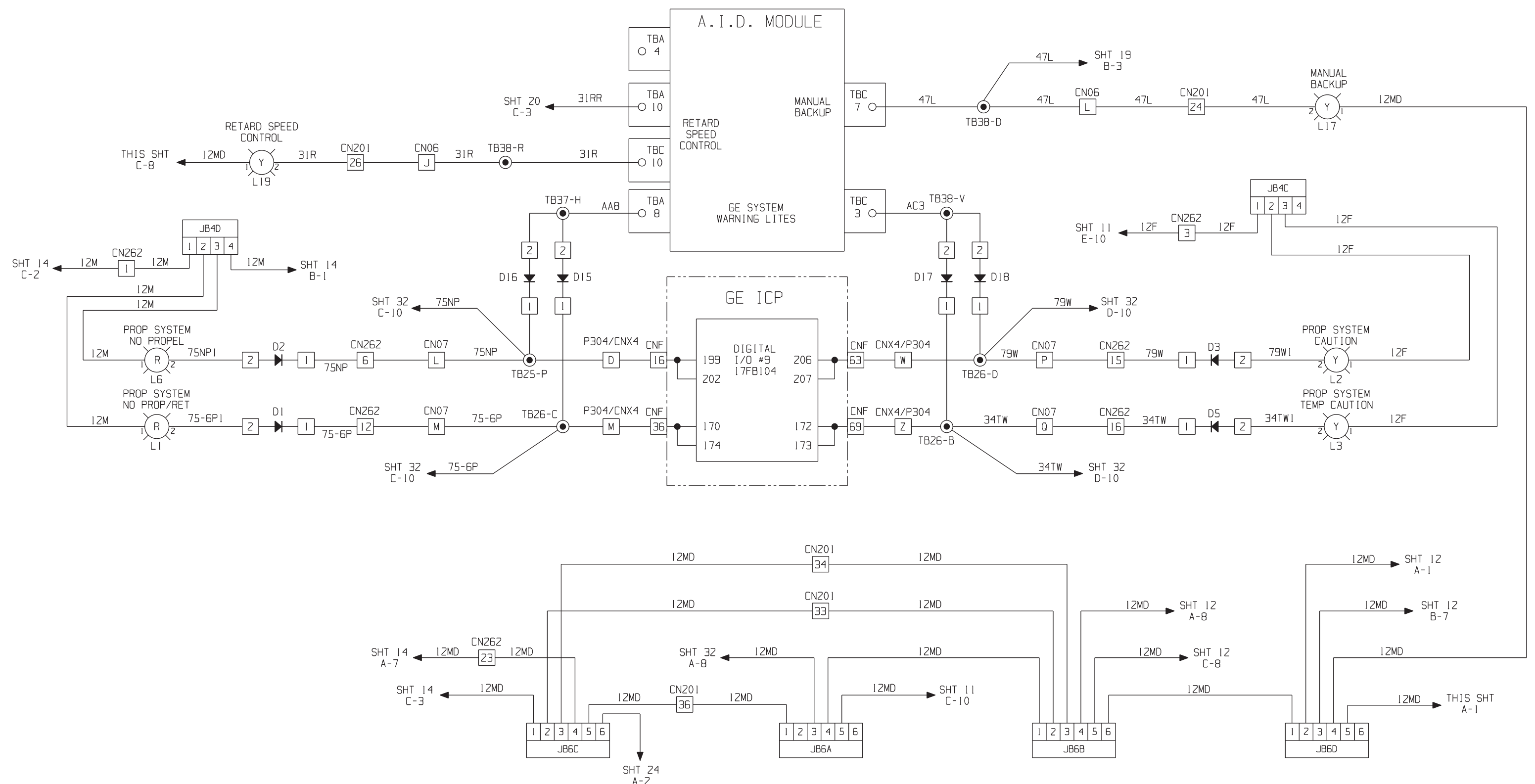


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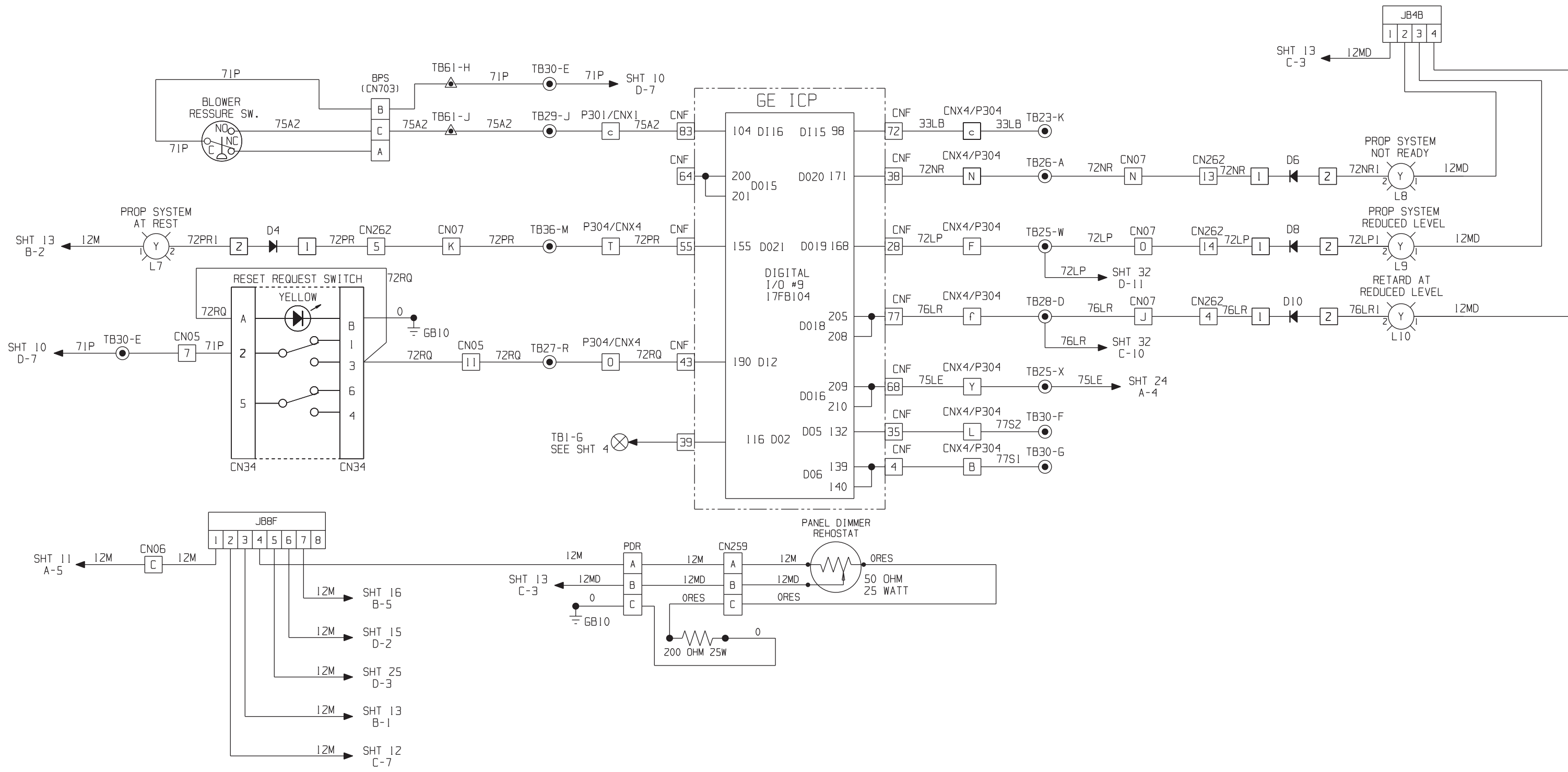


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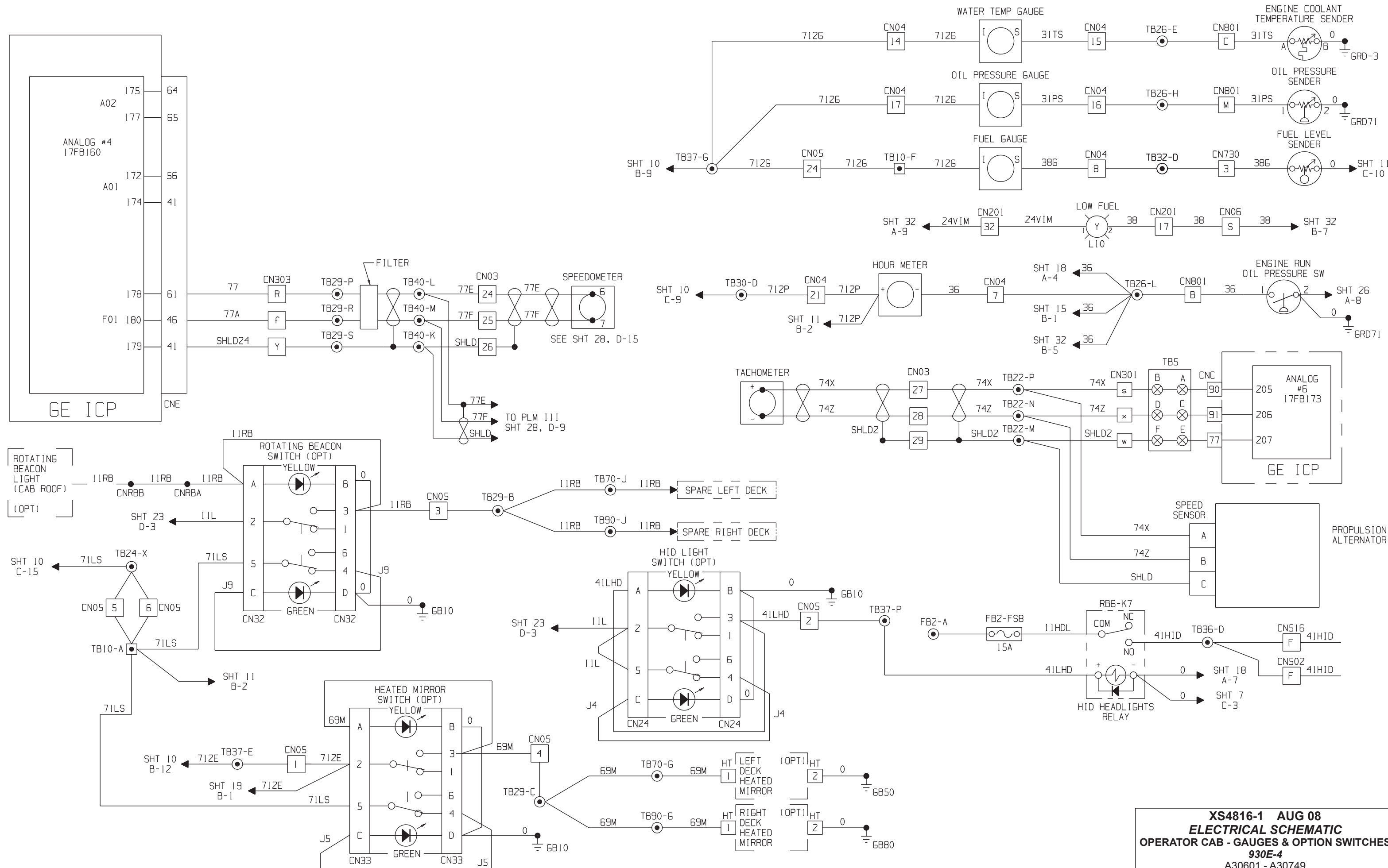
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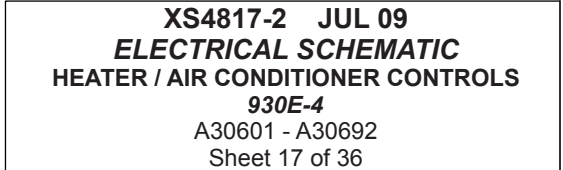
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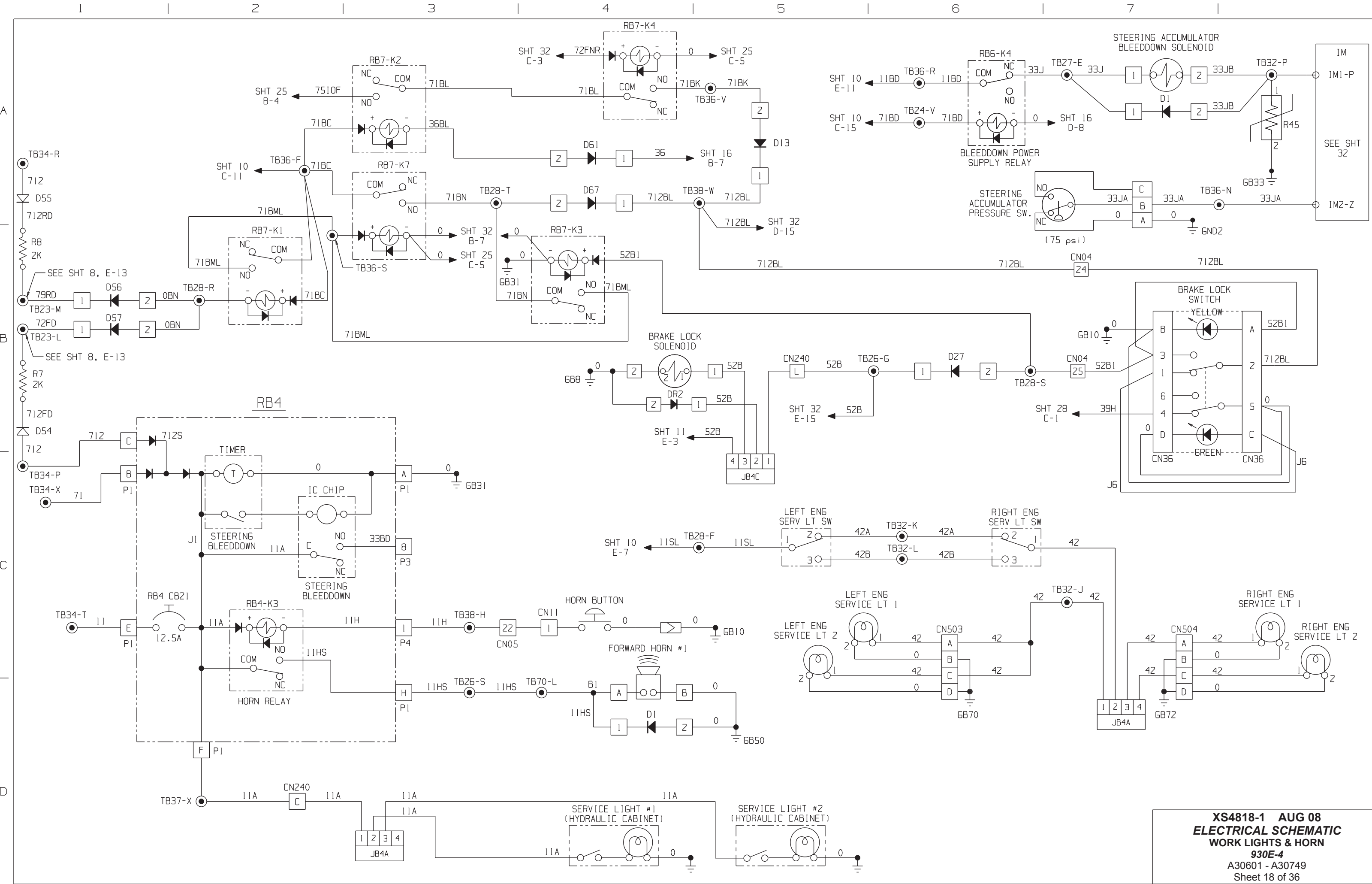










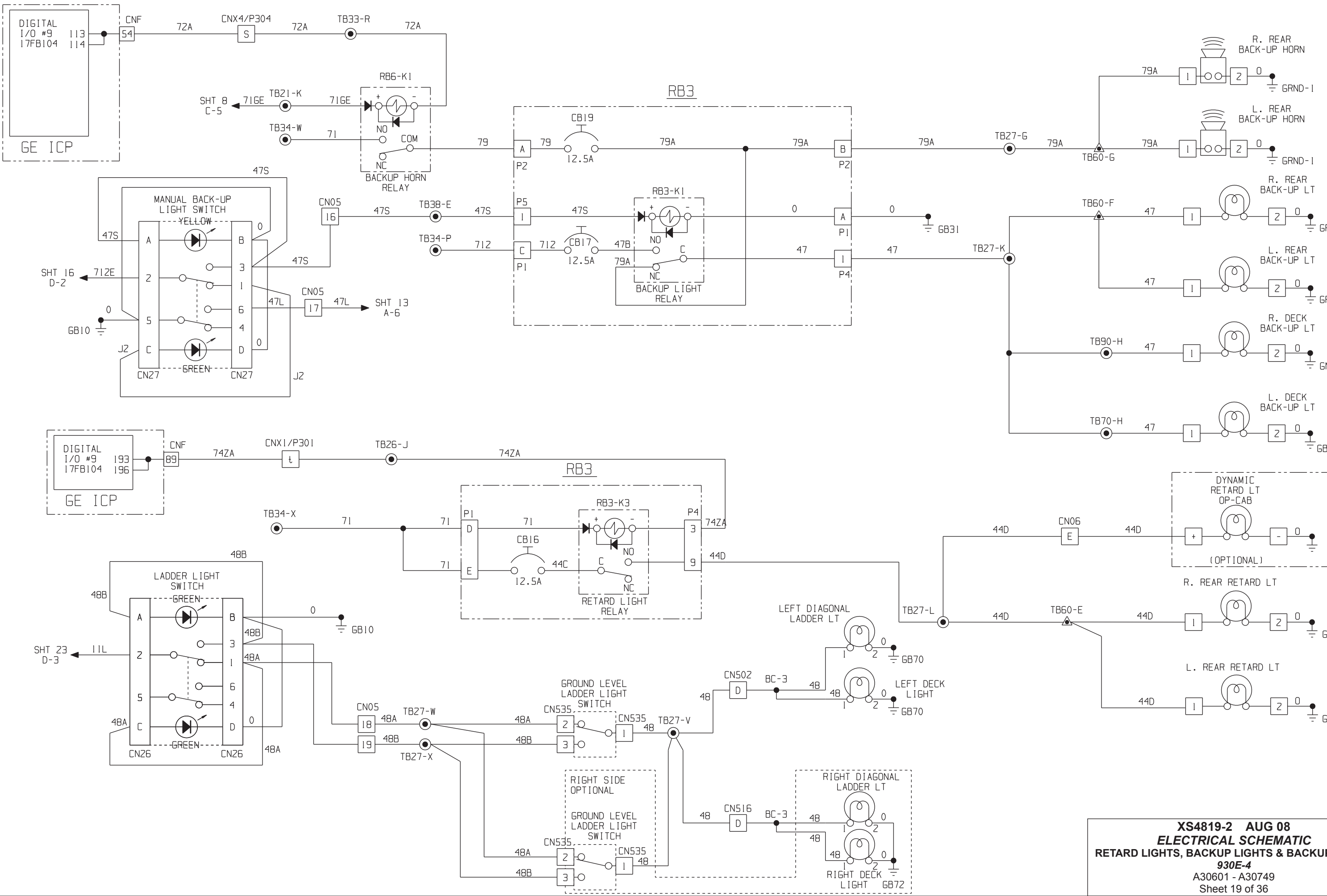


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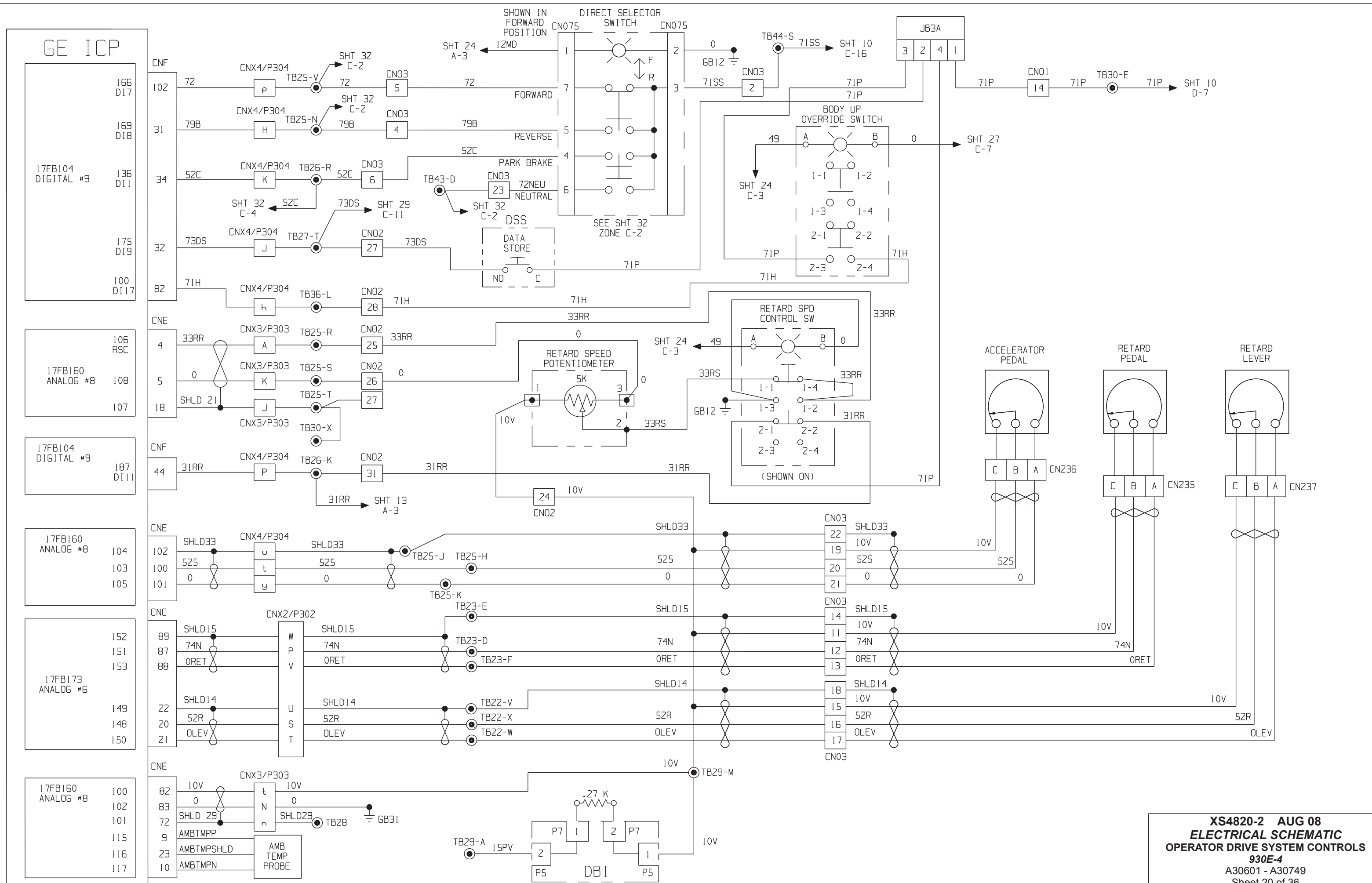
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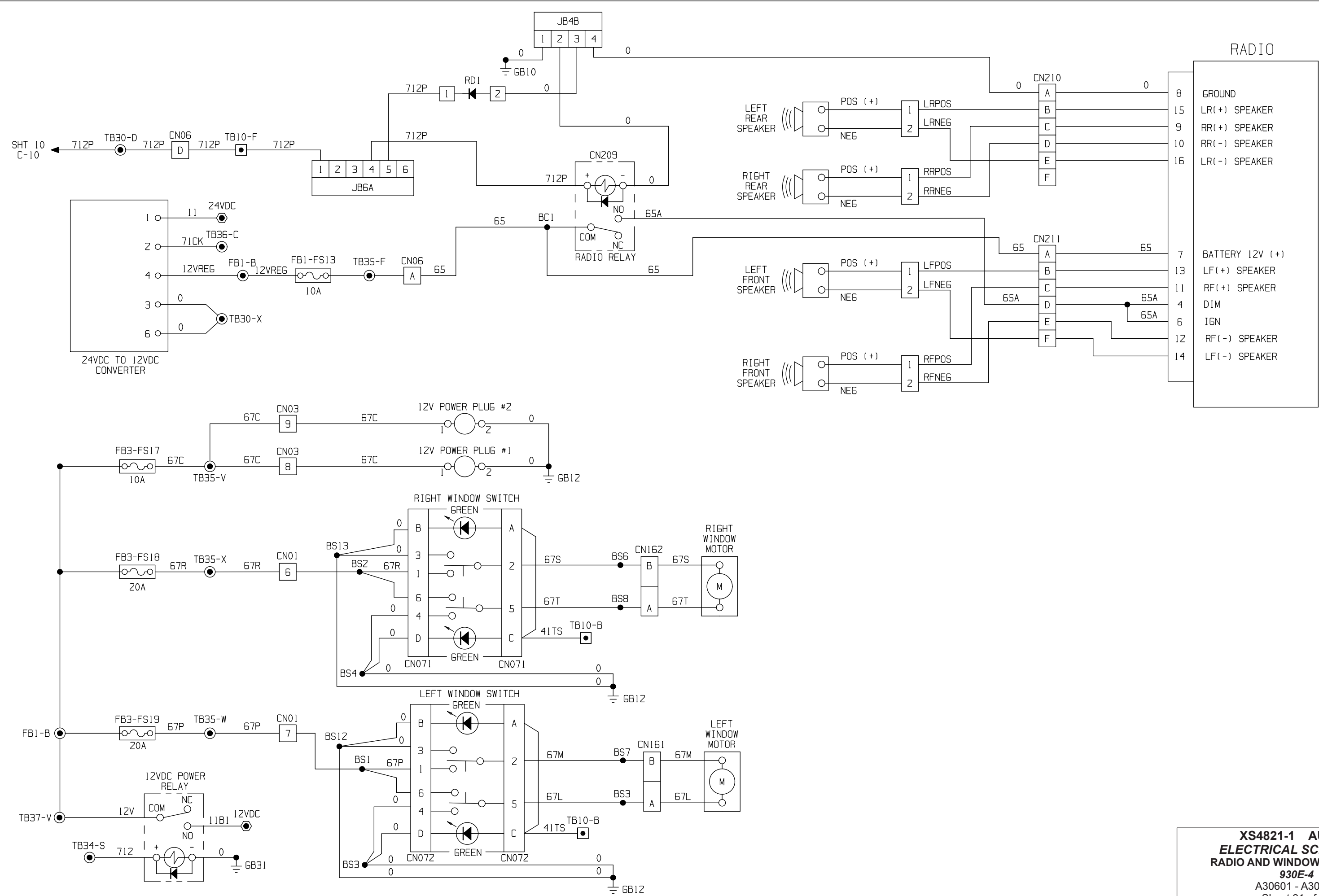


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RADIO

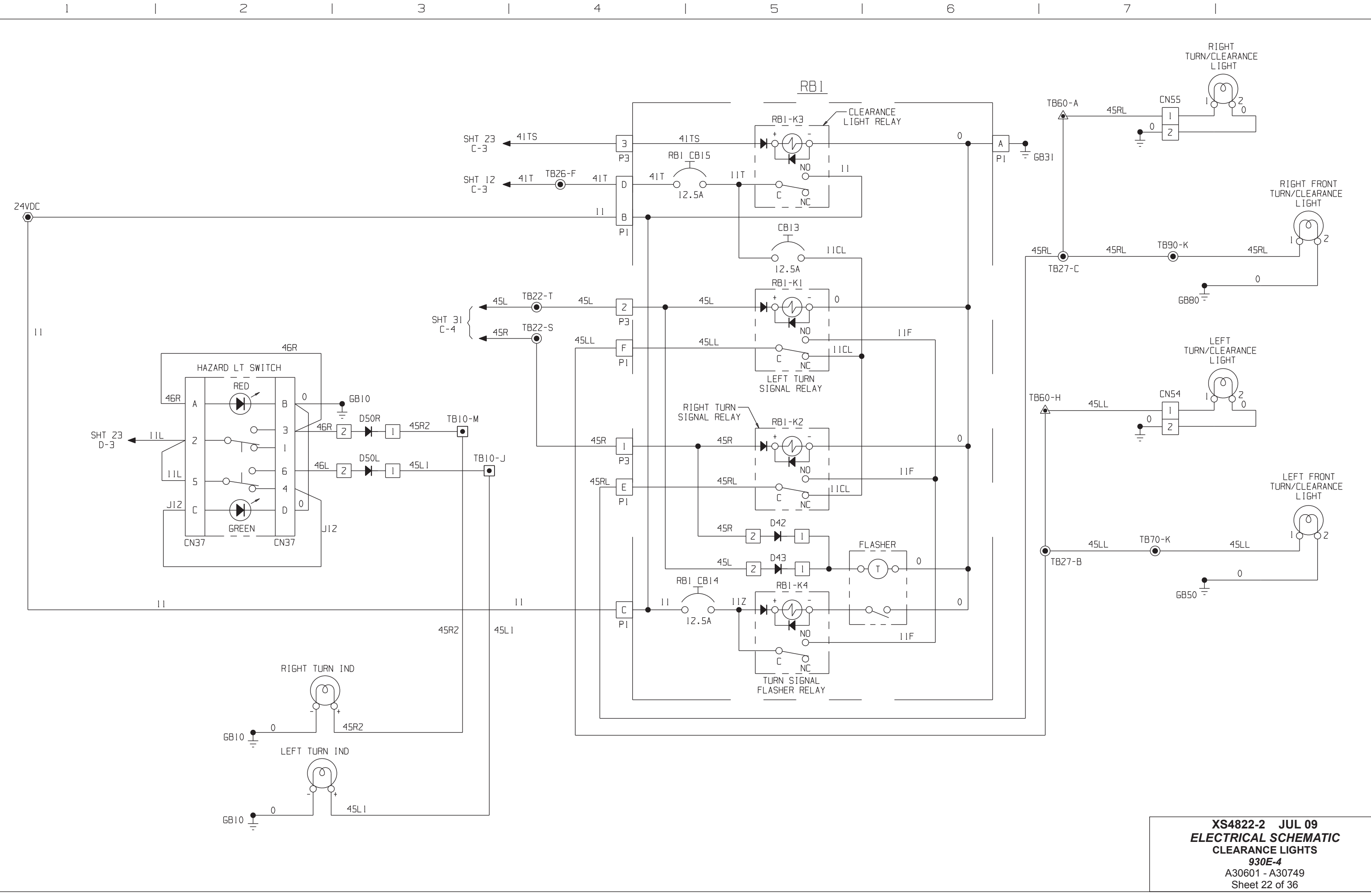
|    |                 |
|----|-----------------|
| 8  | GROUND          |
| 15 | LR(+) SPEAKER   |
| 9  | RR(+) SPEAKER   |
| 10 | RR(-) SPEAKER   |
| 16 | LR(-) SPEAKER   |
| 7  | BATTERY 12V (+) |
| 13 | LF(+) SPEAKER   |
| 11 | RF(+) SPEAKER   |
| 4  | DIM             |
| 6  | IGN             |
| 12 | RF(-) SPEAKER   |
| 14 | LF(-) SPEAKER   |

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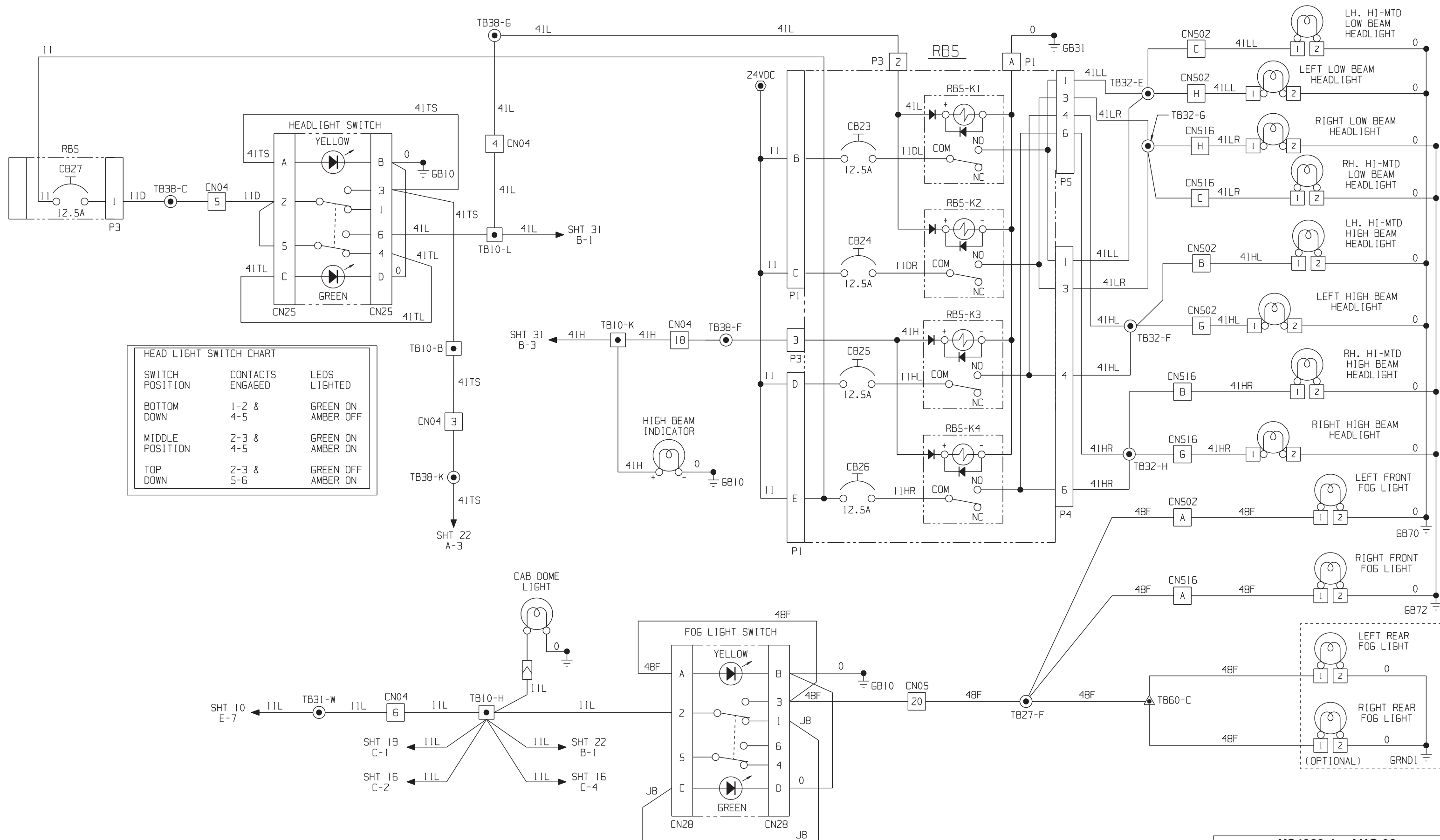


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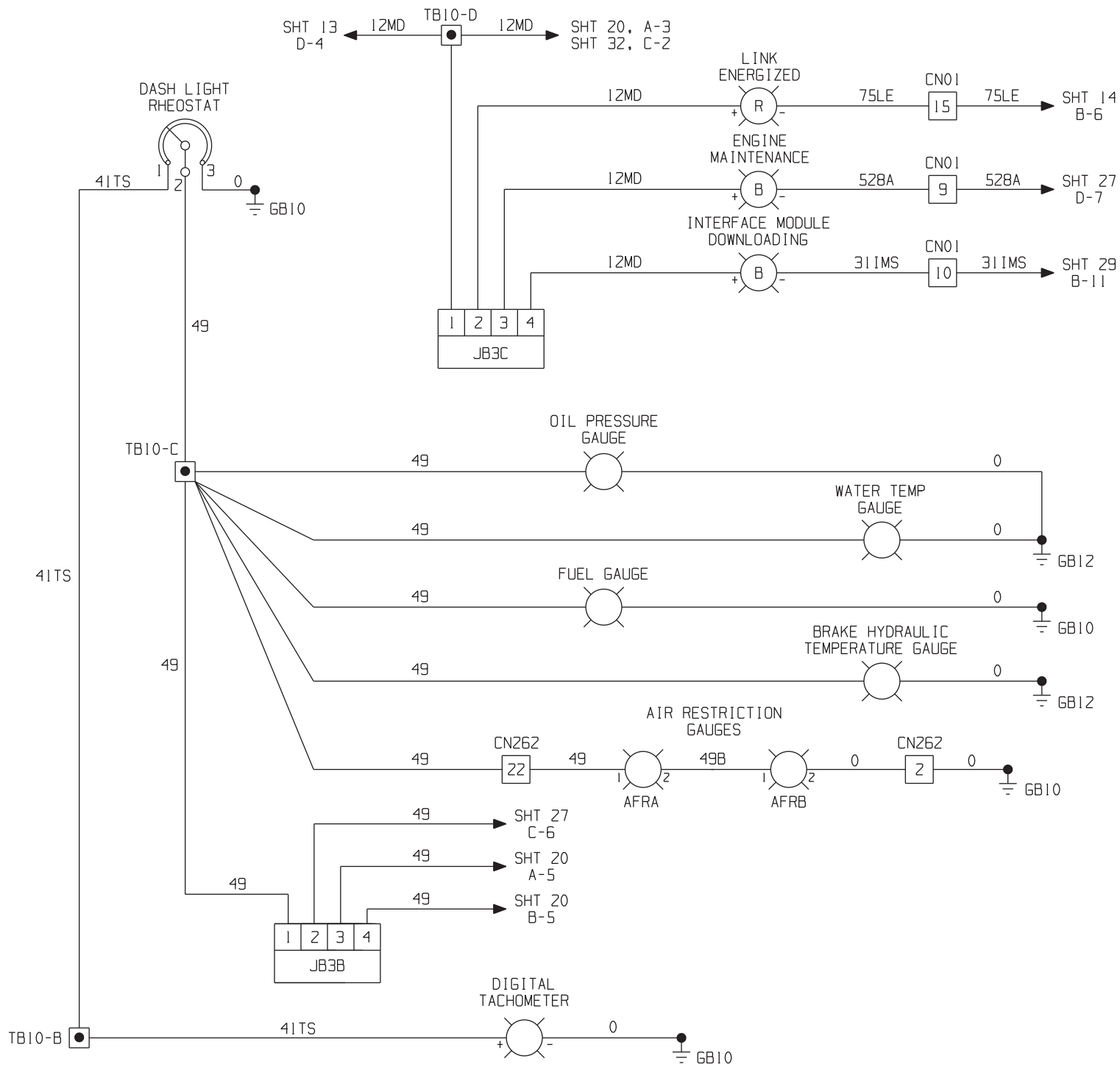


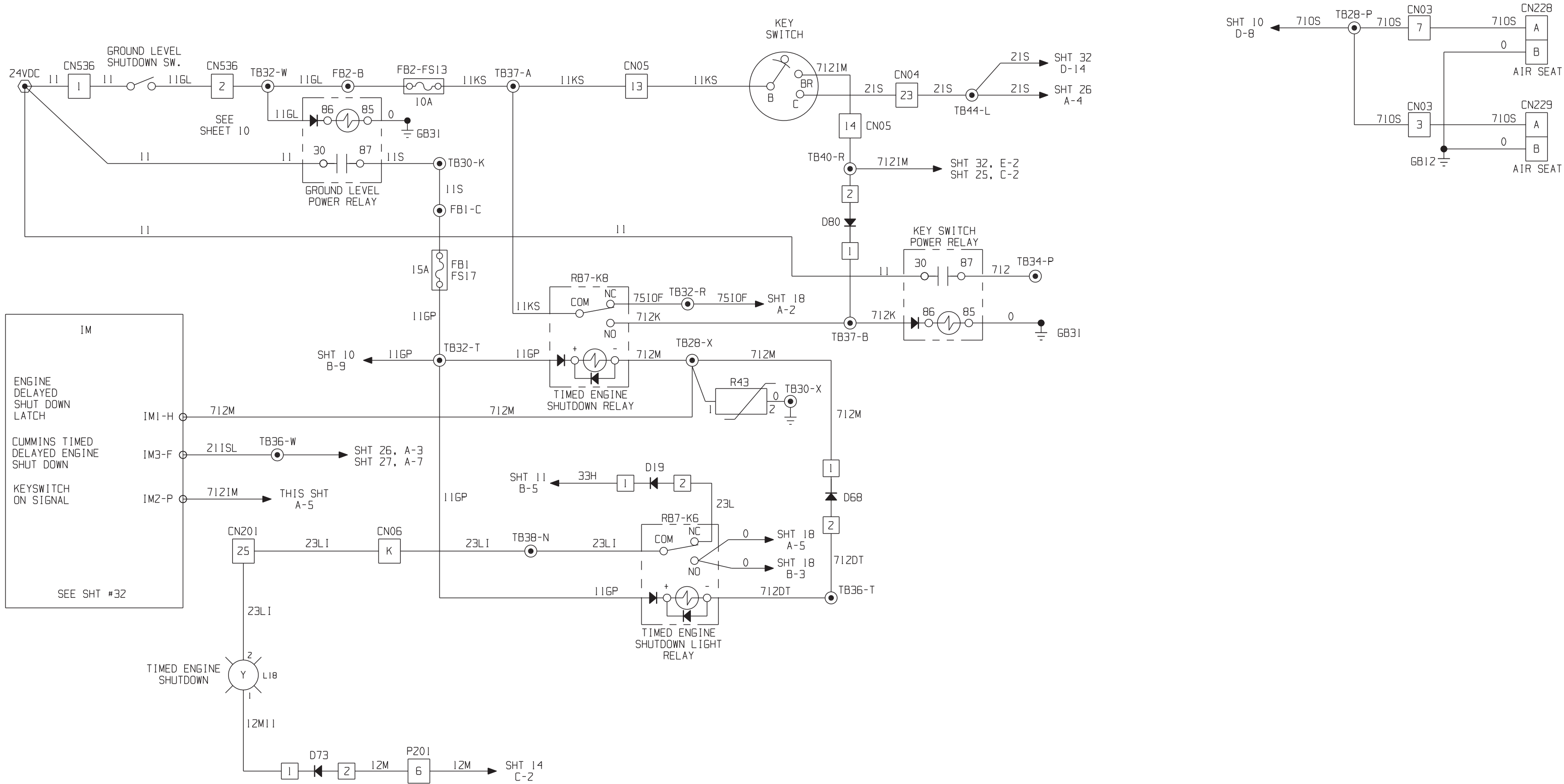
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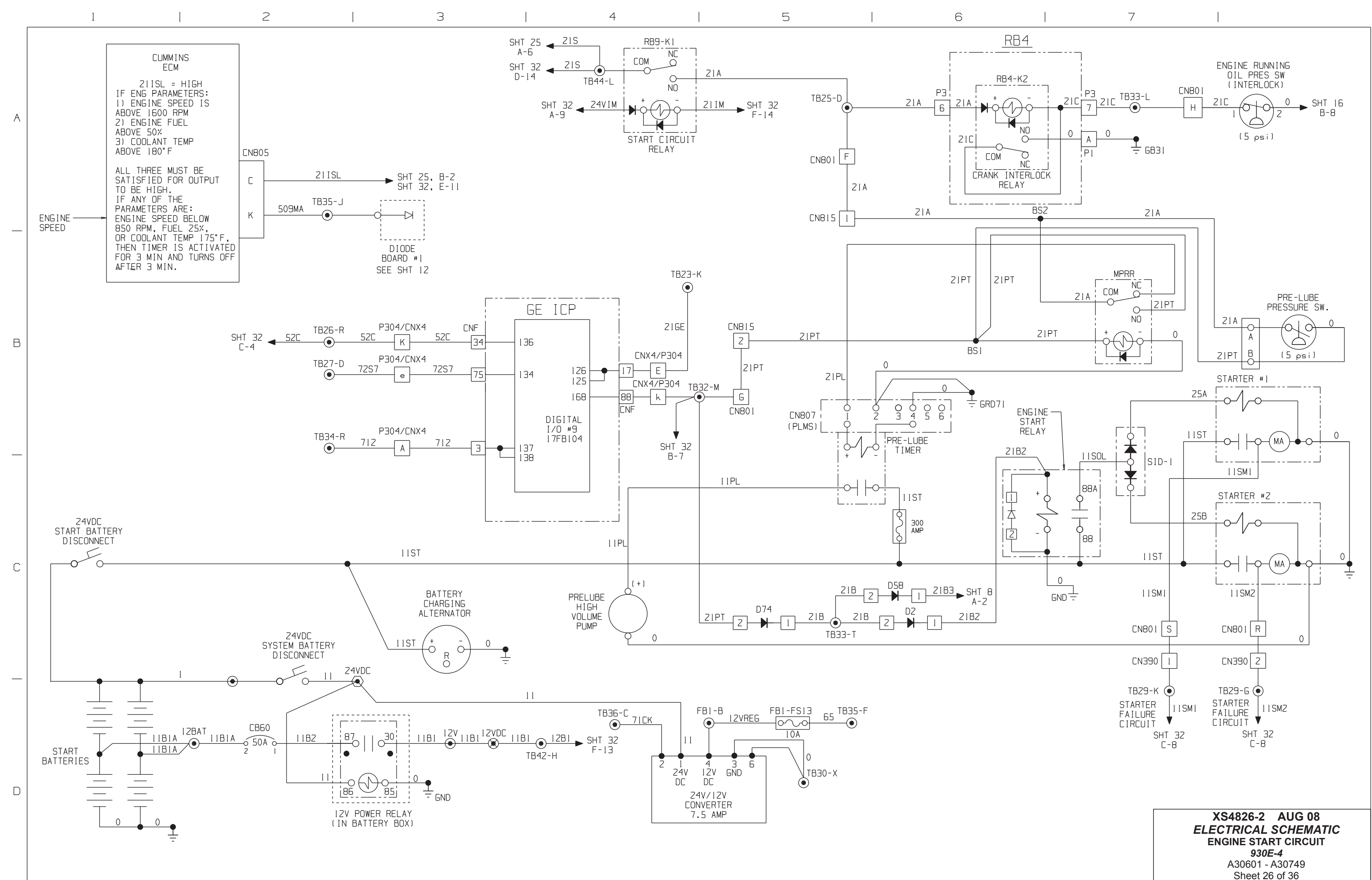
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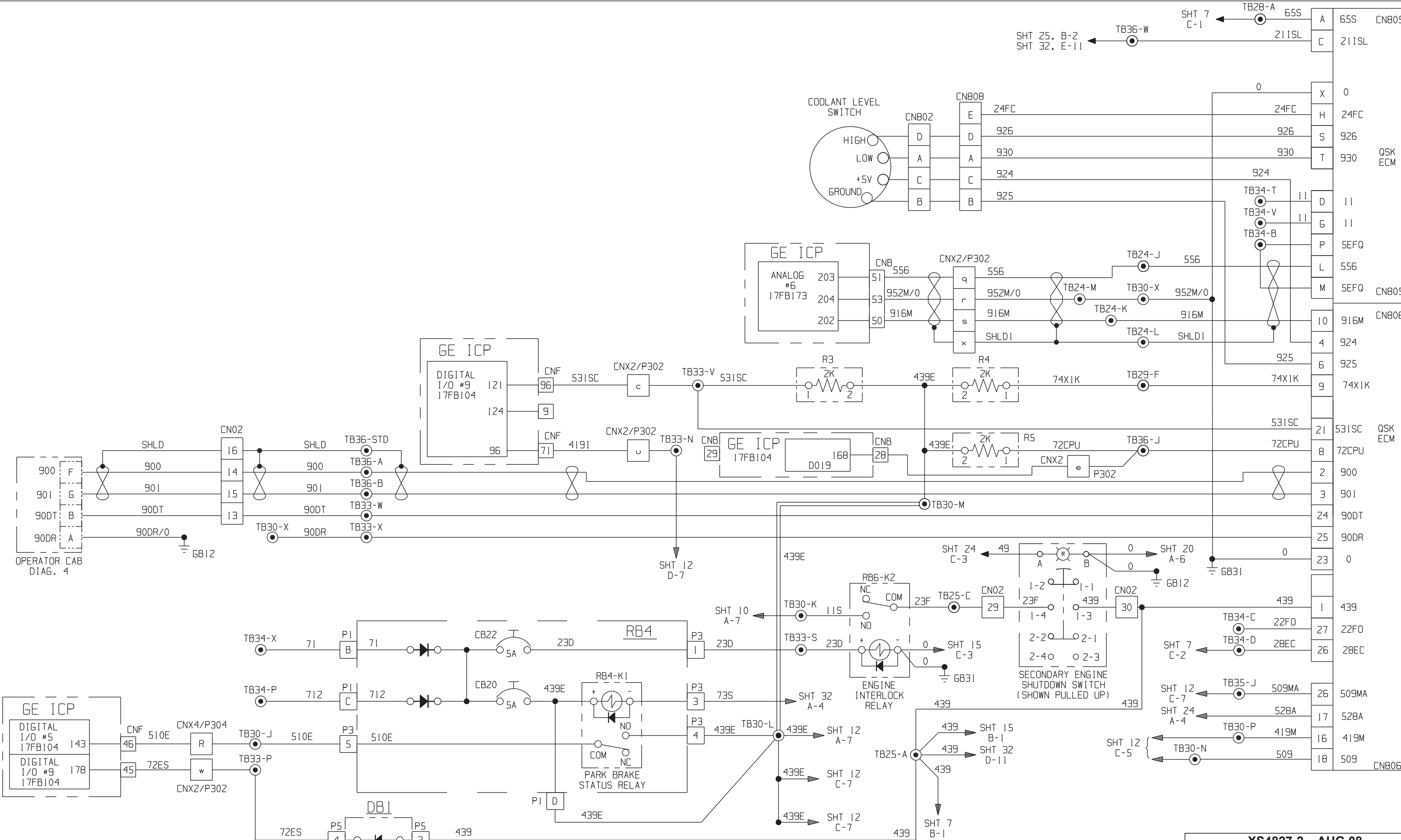


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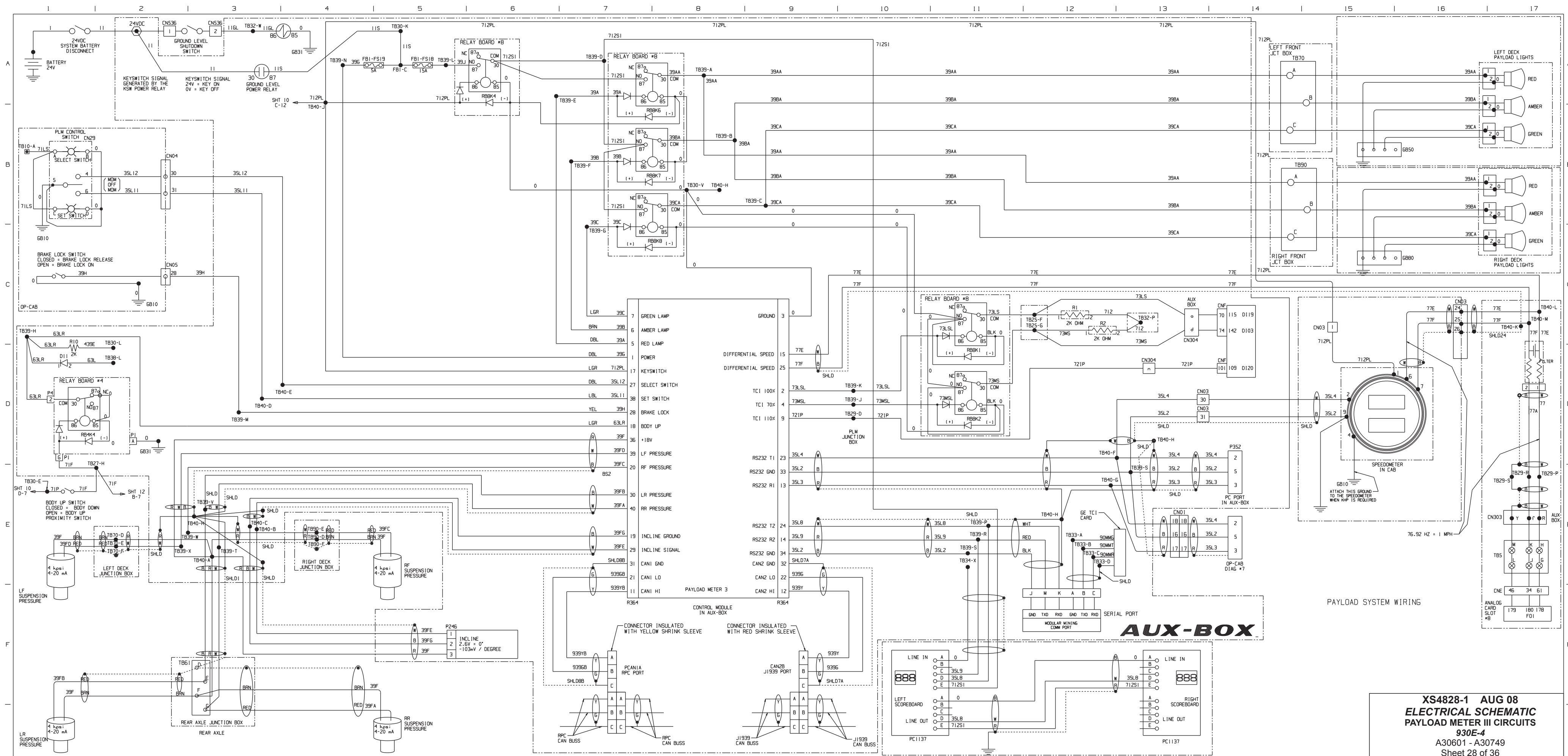
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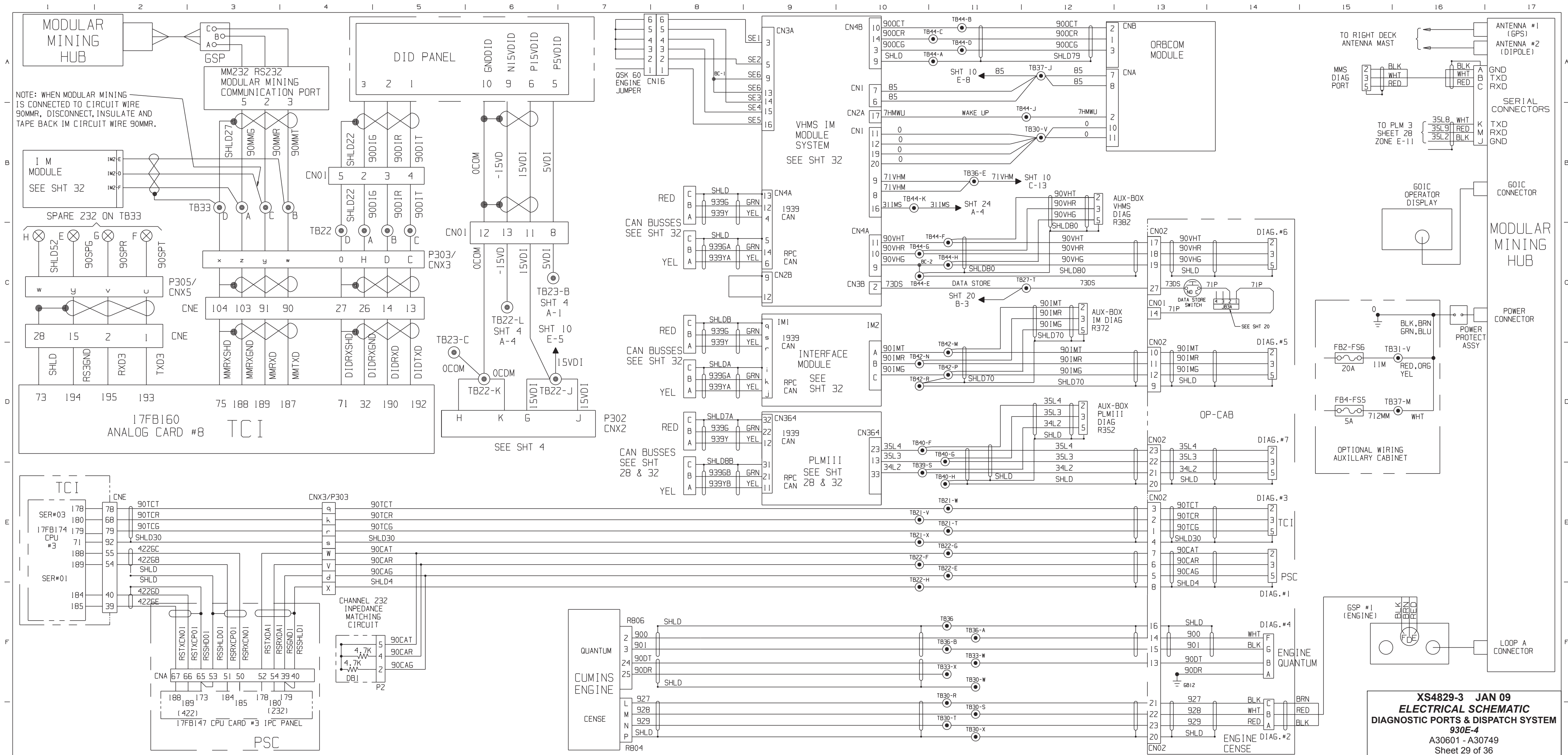
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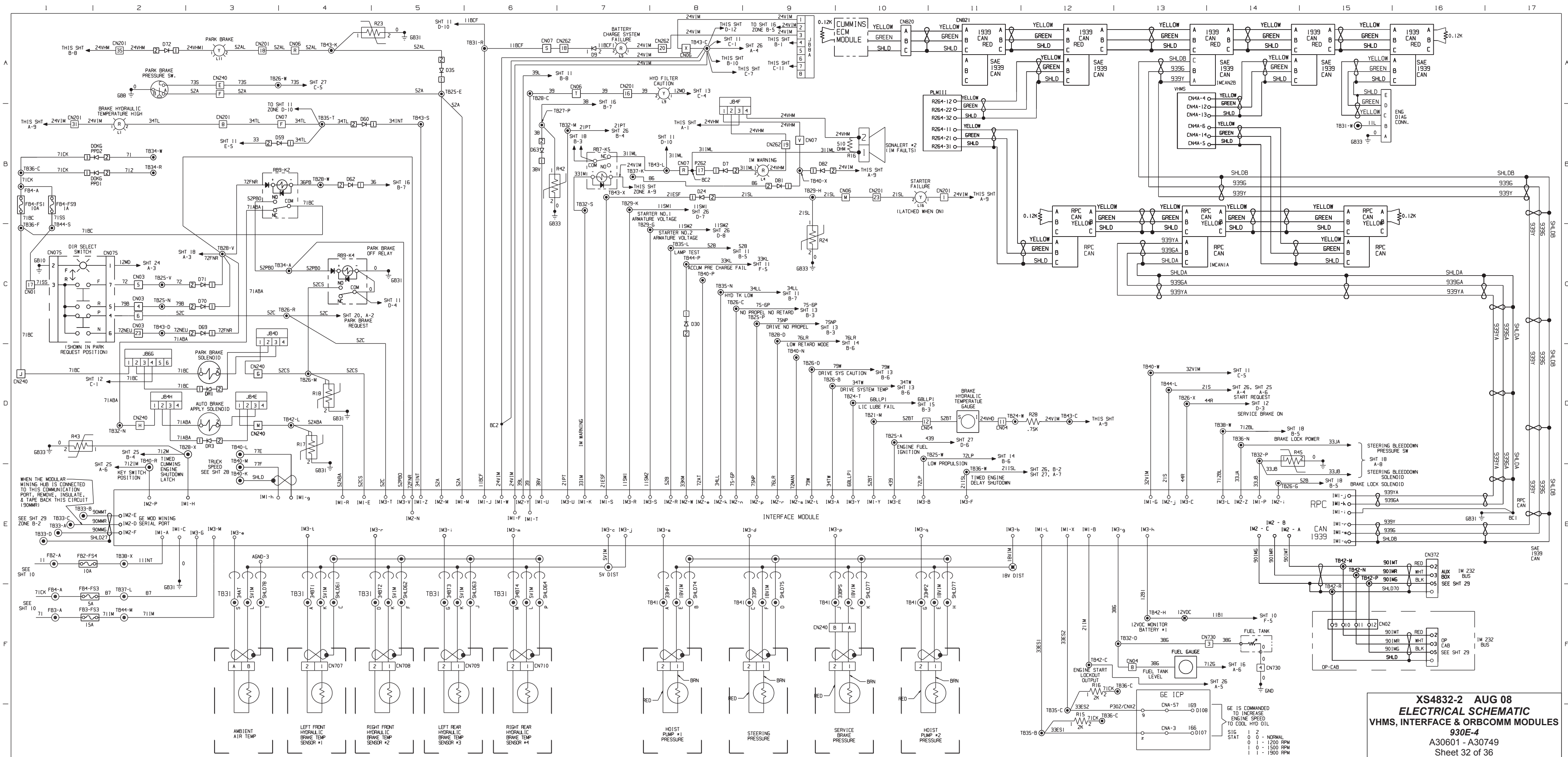




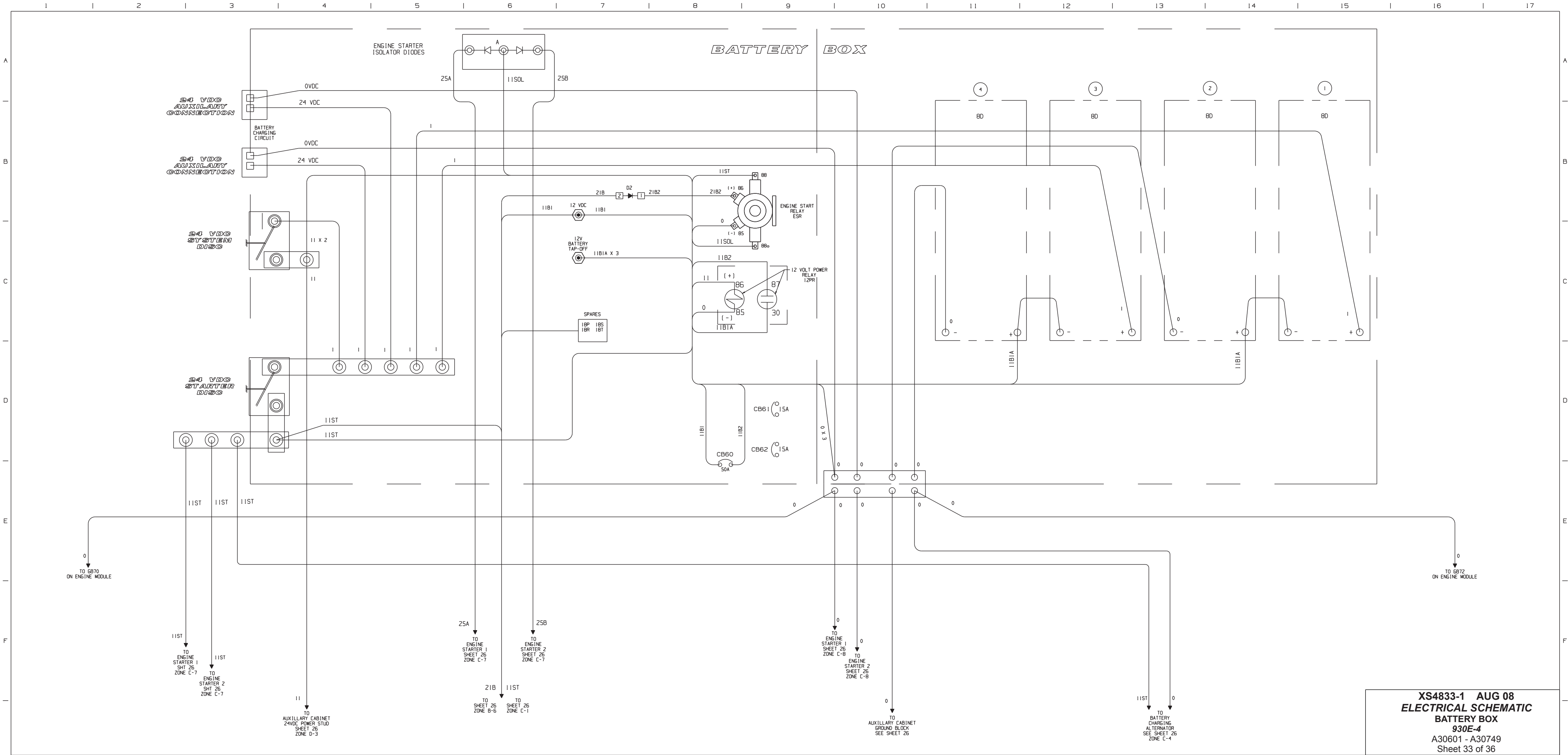












|   | 1                 | 2                            | 3                 | 4                 | 5                 | 6                 | 7                 |                   |                   |                                 |                   |                   |                   |                   |                   |                   |
|---|-------------------|------------------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|---------------------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
|   | CIRCUIT<br>NUMBER | SHEET<br>LOCATION            | CIRCUIT<br>NUMBER | SHEET<br>LOCATION | CIRCUIT<br>NUMBER | SHEET<br>LOCATION | CIRCUIT<br>NUMBER | SHEET<br>LOCATION | CIRCUIT<br>NUMBER | SHEET<br>LOCATION               | CIRCUIT<br>NUMBER | SHEET<br>LOCATION | CIRCUIT<br>NUMBER | SHEET<br>LOCATION | CIRCUIT<br>NUMBER | SHEET<br>LOCATION |
| A | 0-                | 7-33                         | 15SPD-            | 9,10              | 34BT3-            | 32                | 45R2-             | 22                | 71CN1-            | 4,8                             | 74XA-             | 8                 | 939GB-            | 28,29             | AA8-              | 13                |
|   | 0BN-              | 18                           | 15V-              | 10                | 34BT4-            | 32                | 45RL-             | 22                | 71CN2-            | 8                               | 74XS-             | 8                 | 939Y-             | 7,28,29,32        | AC3-              | 13                |
|   | 0BS-              | 9                            | -15VD-            | 4,9,29            | 34INT-            | 32                | 46-               | 10                | 71F-              | 12,28                           | 74X1K-            | 27                | 939YA-            | 29,32             | AMBTMPN-          | 9,20              |
|   | 0COM-             | 4,9,10,29                    | +15VDC-           | 10                | 34L-              | 11                | 46L-              | 22                | 71GE-             | 2,8,10,19                       | 74Z-              | 3,16              | 939YB-            | 28,29             | AMBTMPP-          | 9,20              |
|   | 0LEV-             | 20                           | 15VD1-            | 4,9,10,29         | 34LL-             | 11,32             | 46R-              | 22                | 71H-              | 20                              | 74ZA-             | 19                | 952M/0-           | 27                | AMBTMPSHLD-       | 9,20              |
|   | 0LF-              | 9                            | 15VL-             | 10                | 34T-              | 11                | 47-               | 19                | 71IM-             | 10,32                           | 74ZB-             | 12                |                   |                   | LFNEG-            | 21                |
|   | 0MP-              | 9                            | 18VIM-            | 32                | 34TL-             | 32                | 47B-              | 10,19             | 71J-              | 2,3                             | 75A1-             | 9,14              |                   |                   | LFPOS-            | 21                |
|   | 0MPH-             | 9                            | 21A-              | 10,26             | 34TW-             | 13,32             | 47L-              | 13,19             | 71LS-             | 10,11,16,28                     | 75A2-             | 14                |                   |                   | LRNEG-            | 21                |
|   | 0PWM-             | 7                            | 21B-              | 26,33             | 34TW1-            | 13                | 47S-              | 19                | 71OS-             | 10,25                           | 7510F-            | 18,25             |                   |                   | LRPOS-            | 21                |
|   | 0RES-             | 14                           | 21BSR-            | 32                | 35L2-             | 28,29             | 48-               | 19                | 71P-              | 10,12,14,20,28,29               | 75LE-             | 14,24             |                   |                   | RFNEG-            | 21                |
| B | 0RET-             | 20                           | 21B2-             | 26,33             | 35L3-             | 28,29             | 48A-              | 19                | 71SS-             | 10,20,32                        | 75NP-             | 13,32             |                   |                   | RFPOS-            | 21                |
|   | 0RF-              | 9                            | 21B3-             | 8,26              | 35L4-             | 28,29             | 48B-              | 19                | 71TC1-            | 2                               | 75NP1-            | 13                |                   |                   | RRNEG-            | 21                |
|   | 0VDC-             | 33                           | 21C-              | 26                | 35L8-             | 28,29             | 48F-              | 23                | 71VHM-            | 10,29                           | 75-6P-            | 13,32             |                   |                   | RRPOS             | 21                |
|   | 1-                | 2,26,28,33                   | 21ESF-            | 32                | 35L9-             | 28,29             | 49-               | 20,24,27          | 71Z-              | 2,8,10,12,18,19,<br>21,25-28,32 | 75-6P1-           | 13                |                   |                   | SE1-              | 29                |
|   | 5EFQ-             | 27                           | 21GE-             | 26                | 35L11-            | 28                | 49B-              | 24                |                   |                                 | 76LR-             | 14,32             |                   |                   | SE2-              | 29                |
|   | 5VDC-             | 10                           | 21IM-             | 10,26,32          | 35L12-            | 28                | 509-              | 12,27             | 71ZA-             | 10,15                           | 76LR1-            | 14                |                   |                   | SE3-              | 29                |
|   | 5VD1-             | 4,9,29                       | 21ISL-            | 25-27,32          | 36-               | 15,16,18,32       | 509MA-            | 12,26,27          | 71ZBL-            | 18,32                           | 77-               | 9,16,28           |                   |                   | SE4-              | 29                |
|   | 5VIM-             | 32                           | 21PT-             | 10,26,32          | 36BL-             | 18                | 51A-              | 11                | 71ZDT-            | 25                              | 77A-              | 9,16,28           |                   |                   | SE5-              | 29                |
|   | 7HMMU-            | 29                           | 21S-              | 10,25,26,32       | 36LL-             | 15                | 51A1-             | 11                | 71ZE-             | 10,16,19                        | 77B-              | 3                 |                   |                   | SE6-              | 29                |
|   | 10MP-             | 9                            | 21SFL-            | 32                | 36PB-             | 32                | 51A2-             | 11                | 71ZFD-            | 18                              | 77C-              | 3                 |                   |                   |                   |                   |
| C | 10V-              | 20                           | 21SL-             | 32                | 38-               | 11,16,32          | 510E-             | 27                | 71ZG-             | 10,16                           | 77DN-             | 3                 |                   |                   |                   |                   |
|   | 11-               | 2,8,10,21-23,<br>25-28,32,33 | 22C-              | 17                | 38G-              | 16,32             | 510E2-            | 8                 | 71ZGE-            | 8                               | 77DP-             | 3                 |                   |                   |                   |                   |
|   | 11A-              | 10,18                        | 22FO-             | 27                | 38L-              | 11                | 52A-              | 32                | 71ZH-             | 10,15                           | 77E-              | 16,28,32          |                   |                   |                   |                   |
|   | 11BCF-            | 11,32                        | 23D-              | 10,27             | 38V-              | 32                | 52ABA-            | 32                | 71ZIM-            | 8,10,25,32                      | 77F-              | 16,28,32          |                   |                   |                   |                   |
|   | 11BCF1-           | 32                           | 23F-              | 27                | 39-               | 32                | 52AL-             | 32                | 71ZK-             | 8,10,25                         | 77G-              | 3                 |                   |                   |                   |                   |
|   | 11BCG-            | 32                           | 23L-              | 25                | 39A-              | 28                | 52B-              | 11,18,32          | 71ZL-             | 10,15                           | 77P-              | 3                 |                   |                   |                   |                   |
|   | 11BD-             | 10,18                        | 23LI-             | 25                | 39AA-             | 28                | 52B1-             | 18                | 71ZM-             | 25,32                           | 77S1-             | 14                |                   |                   |                   |                   |
|   | 11BD-             | 10,18                        | 24FC-             | 27                | 39B-              | 28                | 52BT-             | 32                | 71ZMM-            | 10,29                           | 77S2-             | 14                |                   |                   |                   |                   |
|   | 11B1-             | 10,21,26,32,33               | 24LS-             | 11                | 39BA-             | 28                | 52C-              | 20,26,32          | 71ZP-             | 10,11,16,21                     | 78BP-             | 9                 |                   |                   |                   |                   |
|   | 11B1A-            | 26,33                        | 24VDC-            | 33                | 39BP-             | 11                | 52CS-             | 32                | 71ZPL-            | 10,28,32                        | 78BS-             | 9                 |                   |                   |                   |                   |
| D | 11B2-             | 26,33                        | 24VHD-            | 32                | 39C-              | 28                | 52PB0-            | 32                | 71ZR-             | 7,10                            | 78MP-             | 9                 |                   |                   |                   |                   |
|   | 11CL-             | 10,22                        | 24VHM-            | 32                | 39CA-             | 28                | 52R-              | 20                | 71ZRD-            | 18                              | 79-               | 10,19             |                   |                   |                   |                   |
|   | 11D-              | 10,23                        | 24VHM1-           | 32                | 39F-              | 28                | 52S-              | 20                | 71ZS-             | 10,18                           | 79A-              | 10,19             |                   |                   |                   |                   |
|   | 11DISP-           | 10                           | 24VIM-            | 10,11,26,32       | 39FA-             | 28                | 52B-              | 11,32             | 71ZSB-            | 12                              | 79B-              | 20,32             |                   |                   |                   |                   |
|   | 11DL-             | 10,23                        | 25A-              | 26,33             | 39FB-             | 28                | 52BA-             | 24,27             | 71ZSF-            | 10                              | 79RD-             | 8,18              |                   |                   |                   |                   |
|   | 11DR-             | 10,23                        | 25B-              | 26,33             | 39FC-             | 28                | 53H-              | 15                | 71ZSP-            | 7                               | 79V-              | 11                |                   |                   |                   |                   |
|   | 11EM-             | 7,10                         | 28E-              | 7                 | 39FD-             | 28                | 53ISC-            | 27                | 71ZSI-            | 28                              | 79W-              | 13,32             |                   |                   |                   |                   |
|   | 11E1-             | 10                           | 28EC-             | 7,27              | 39FE-             | 28                | 556-              | 27                | 71ZT-             | 10,31                           | 79W1-             | 13                |                   |                   |                   |                   |
|   | 11E2-             | 10                           | 31A-              | 11                | 39FG-             | 28                | 63-               | 10,31             | 71ZTI-            | 31                              | 85-               | 10,29             |                   |                   |                   |                   |
|   | 11E3-             | 10                           | 31CB-             | 11                | 39G-              | 10,28             | 63L-              | 12,28             | 714A-             | 3                               | 86-               | 10,11,32          |                   |                   |                   |                   |
| E | 11E4-             | 10                           | 31CL-             | 11                | 39H-              | 18,28             | 63LR-             | 12,28             | 714B-             | 3                               | 87-               | 10,32             |                   |                   |                   |                   |
|   | 11F-              | 10,22                        | 31IML-            | 11,32             | 39HP-             | 11                | 65-               | 10,21             | 714DN-            | 3                               | 90CAG-            | 29                |                   |                   |                   |                   |
|   | 11GL-             | 2,8,10,25,28                 | 31IML1-           | 32                | 39HP1-            | 11                | 65A-              | 21                | 714DP-            | 3                               | 90CAR-            | 29                |                   |                   |                   |                   |
|   | 11GP-             | 10,25                        | 31IMS-            | 24,29             | 39HP2-            | 11                | 65C-              | 17                | 714G-             | 3                               | 90CAT-            | 29                |                   |                   |                   |                   |
|   | 11H-              | 18                           | 31MT-             | 12                | 39J-              | 10,28             | 65S-              | 7,27              | 714P-             | 3                               | 90DR-             | 27,29             |                   |                   |                   |                   |
|   | 11HDL-            | 10,16                        | 31MTR-            | 12                | 39L-              | 11,32             | 65SS-             | 7                 | 72-               | 20,32                           | 90DT-             | 27,29             |                   |                   |                   |                   |
|   | 11HL-             | 10,23                        | 31ISA-            | 11,32             | 41H-              | 23,31             | 65T-              | 7,17              | 72A-              | 19                              | 90DIG-            | 9,29              |                   |                   |                   |                   |
|   | 11HR-             | 10,23                        | 31PS-             | 16                | 41H1D-            | 16                | 66-               | 31                | 72AT-             | 32                              | 90DIR-            | 9,29              |                   |                   |                   |                   |
|   | 11HS-             | 18                           | 31R-              | 13                | 41HL-             | 23                | 66H-              | 31                | 72AUT-            | 8                               | 90DIT-            | 9,29              |                   |                   |                   |                   |
|   | 11INT-            | 10,32                        | 31RR-             | 13,20             | 41HR-             | 23                | 66L-              | 31                | 72CPU-            | 8,27                            | 90IMG-            | 29,32             |                   |                   |                   |                   |
| F | 11KS-             | 8,10,25                      | 31T-              | 11                | 41L-              | 10,23,31          | 66L1-             | 31                | 72E-              | 7                               | 90IMR-            | 29,32             |                   |                   |                   |                   |
|   | 11L-              | 10,16,19,<br>22,23,32        | 31TS-             | 16                | 41LHD-            | 16                | 66L2-             | 31                | 72ES-             | 27                              | 90IMT-            | 29,32             |                   |                   |                   |                   |
|   | 11M-              | 10,29                        | 32VIM-            | 11,32             | 41LL-             | 23                | 66P-              | 31                | 72FB-             | 7                               | 90MMG-            | 9,28,29,32        |                   |                   |                   |                   |
|   | 11ORS-            | 10,30                        | 33-               | 11,32             | 41LR-             | 23                | 66S-              | 31                | 72FD-             | 8,18                            | 90MMR-            | 9,28,29,32        |                   |                   |                   |                   |
|   | 11PL-             | 26                           | 33A-              | 11                | 41T-              | 10,12,22          | 66T1-             | 31                | 72FNR-            | 18,32                           | 90MMT-            | 9,28,29,32        |                   |                   |                   |                   |
|   | 11POR-            | 30                           | 33BD-             | 18                | 41TL-             | 23                | 67C-              | 10,21             | 72IP-             | 28                              | 90OCG-            | 29                |                   |                   |                   |                   |
|   | 11RB-             | 16                           | 33BP-             | 11                | 41TS-             | 10,21-24          | 67L-              | 21                | 72LP-             | 14,32                           | 90OCR-            | 29                |                   |                   |                   |                   |
|   | 11RCNT-           | 10,30                        | 33BPS-            | 32                | 419-              | 12                | 67M-              | 21                | 72LPI-            | 14                              | 90OCT             |                   |                   |                   |                   |                   |

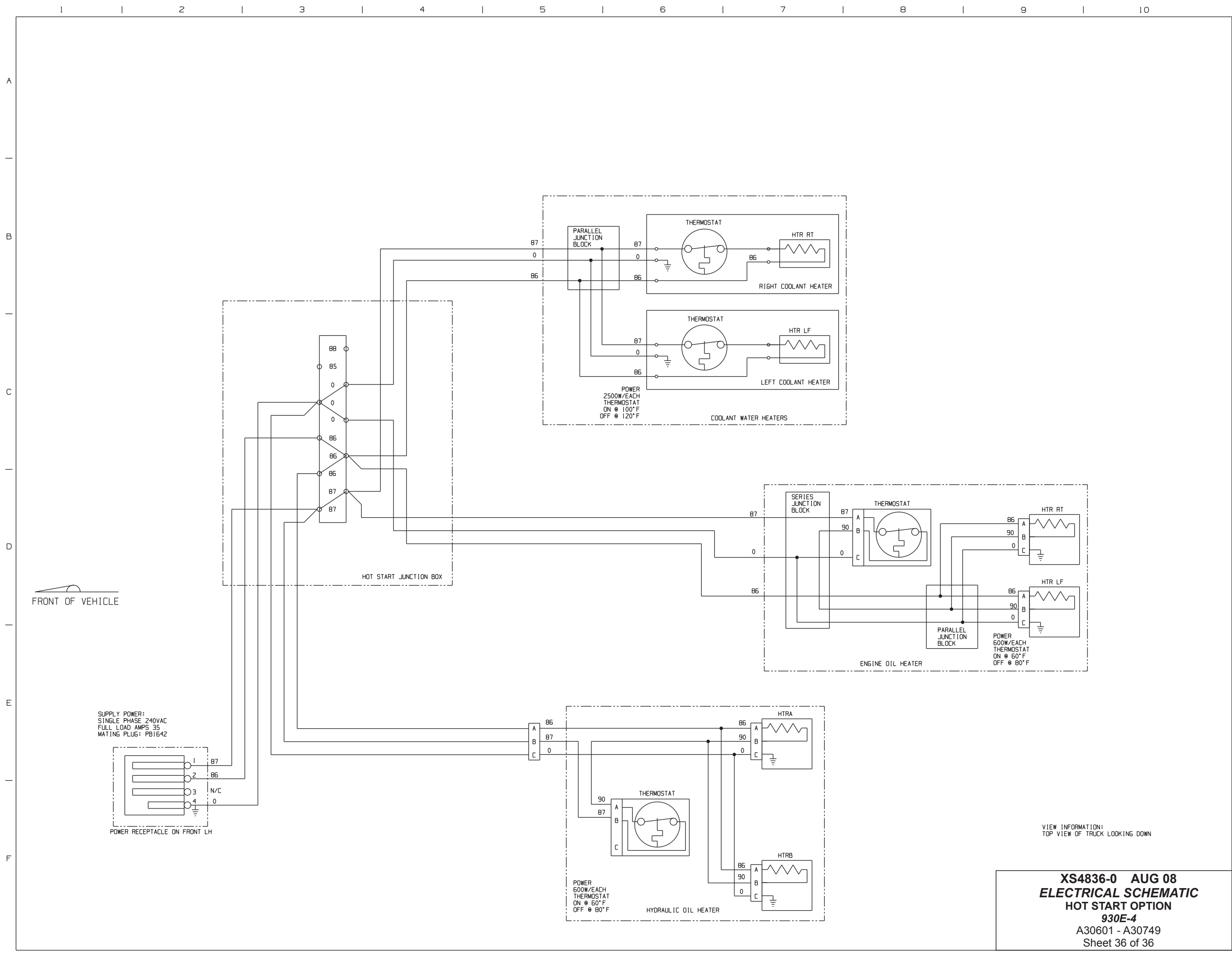
NOTE:  
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CIRCUIT NUMBERS MAY APPEAR IN  
SEVERAL PLACES ON INDIVIDUAL SHEETS

A

| COMPONENT                             | SHEET            | COMPONENT                            | SHEET    | COMPONENT                               | SHEET      | COMPONENT                             | SHEET       |
|---------------------------------------|------------------|--------------------------------------|----------|-----------------------------------------|------------|---------------------------------------|-------------|
| AID MODULE                            | 11,12,13         | GAUGE, BRAKE HYD. TEMPERATURE        | 32       | LIGHT, RIGHT FOG REAR (OPT)             | 23         | RELAY, 12V POWER (BATTERY BOX)        | 26,33       |
| ALTERNATOR, BATTERY CHARGING          | 26               | GAUGE, FUEL                          | 16,32    | LIGHT, RIGHT REAR RETARD                | 19         | RELAY BOARD 1                         | 10,11,22    |
| ALTERNATOR, PROPULSION                | 16               | GAUGE, HOURMETER                     | 16       | LIGHT, RIGHT REAR TURN/CLEARANCE        | 22         | RELAY BOARD 3                         | 10,11,12,19 |
| BATTERY, ENGINE CRANKING              | 9                | GAUGE, OIL PRESSURE                  | 16       | LIGHT, RIGHT TAIL/STOP                  | 12         | RELAY BOARD 4                         | 10,11,12,18 |
| BATTERIES, START                      | 26,33            | GAUGE, WATER TEMPERATURE             | 16       | LIGHT, RIGHT TURN INDICATING            | 22         |                                       | 26,27,28    |
| CB 13 - CLEARANCE LIGHTS (RB1)        | 10,22            | GOIC- OPERATOR DISPLAY               | 29       | LIGHT, ROTATING BEACON (OPT)            | 16         | RELAY BOARD 5                         | 10,11,23    |
| CB 14 - TURN SIGNAL FLASHER           | 10,22            | HEATER A/C SYSTEM                    | 17       | LIGHT, SERVICE BRAKE (CAB ROOF)         | 12         | RELAY BOARD 8                         | 28          |
| CB 15 - TAIL LIGHTS (RB1)             | 10,22            | HORN BUTTON                          | 18       | LIGHT, WATER TEMPERATURE GAUGE          | 24         | RETARD LEVER                          | 20          |
| CB 16 - RETARD LIGHTS (RB3)           | 10,19            | HORN, FORWARD #1                     | 18       | LIGHTS, LEFT DECK PAYLOAD               | 28         | RHEOSTAT, PANEL DIMMER                | 14          |
| CB 17 - MANUAL BACK-UP LTS (RB3)      | 10,19            | HORN, LEFT REAR BACKUP               | 19       | LIGHTS, RIGHT DECK PAYLOAD              | 28         | RHEOSTAT, DASH LIGHT                  | 24          |
| CB 18 - STOP LIGHTS (RB3)             | 10,12            | HORN, RIGHT REAR BACKUP              | 19       | MIRROR, HEATED LEFT FLAT (OPT)          | 16         | SCOREBOARD, LEFT PLM3                 | 28          |
| CB 19 - BACKUP LTS & HORN (RB3)       | 10,19            | INCLINOMETER                         | 28       | MIRROR, HEATED RIGHT CONCAVE (OPT)      | 16         | SCOREBOARD, RIGHT PLM3                | 28          |
| CB 20 - PARK BRAKE FAILURE (RB4)      | 10,27            | INDICATOR, BATT. CHG. SYS. FAILURE   | 32       | MODULAR MINING COMMUNICATION PORT       | 9,28,29    | SENDER, ENGINE TEMPERATURE            | 16          |
| CB 21 - SERV. LTS, HORN BTN. (RB4)    | 10,18            | INDICATOR, BODY UP                   | 12       | MODULAR MINING HUB                      | 29         | SENDER, FUEL LEVEL                    | 16          |
| CB 22 - ENGINE CONTROL POWER (RB4)    | 10,27            | INDICATOR, BRAKE HYD. TEMP. HIGH     | 32       | MOTOR, LEFT WINDOW                      | 21         | SENDER, OIL PRESSURE                  | 16          |
| CB 23 - HDLIGHTS LFT LO BEAM (RB5)    | 10,23            | INDICATOR, CHECK ENGINE              | 12       | MOTOR, RIGHT WINDOW                     | 21         | SENSOR, AMBIENT AIR TEMP              | 32          |
| CB 24 - HDLIGHTS RT LO BEAM (RB5)     | 10,23            | INDICATOR, CIRCUIT BREAKER TRIPPED   | 11       | MOTOR, STARTER #1                       | 26         | SENSOR, BAROMETRIC PRESSURE           | 9           |
| CB 25 - HDLIGHTS LFT HI BEAM (RB5)    | 10,23            | INDICATOR, DYNAMIC RETARD            | 12       | MOTOR, STARTER #2                       | 26         | SENSOR, HOIST PUMP #1 PRESSURE        | 32          |
| CB 26 - HDLIGHTS RT HI BEAM (RB5)     | 10,23            | INDICATOR, ELECTRIC SYS FAULT        | 11       | MOTOR, WINDSHIELD WIPER                 | 31         | SENSOR, HOIST PUMP #2 PRESSURE        | 32          |
| CB 27 - HDLIGHTS & DASH LTS (RB5)     | 10,23            | INDICATOR, ENGINE STOP               | 12       | ORBCOM MODULE                           | 29         | SENSOR, HYD. LEVEL                    | 11          |
| CB 60 - 12V POWER                     | 26,33            | INDICATOR, HYDRAULIC OIL FILTER      | 32       | PAYLOAD METER 3 CONTROL MODULE          | 10,28,29   | SENSOR, LEFT FRONT HYD. BRAKE TEMP    | 32          |
| CB 61 -                               | 33               | INDICATOR, IM WARNING                | 32       | PEDAL, ACCELERATOR                      | 20         | SENSOR, LEFT FRONT SUSPENSION PRESS.  | 28          |
| CB 62 -                               | 33               | INDICATOR, LINCOLN LUBE LOW PRESSURE | 15       | PEDAL, RETARD                           | 20         | SENSOR, LEFT FRONT WHEEL SPEED        | 9           |
| CONVERTER, 24VDC TO 12VDC             | 10,21,26         | INDICATOR, LOW ACCUM PRECHARGE       | 11       | POTENTIOMETER, RETARD SPEED             | 20         | SENSOR, LEFT REAR HYD. BRAKE TEMP     | 32          |
| DIODE BOARD 1 (DB1)                   | 7,11,12,15,20,27 | INDICATOR, LOW BRAKE PRESSURE        | 11       | POWER PLUGS, 12V                        | 21         | SENSOR, LEFT REAR SUSPENSION PRESS.   | 28          |
|                                       | 26,33            | INDICATOR, LOW FUEL                  | 16,32    | PROBE, ALTERNATOR AMB. TEMP.            | 9,20       | SENSOR, RESERVE OIL LEVEL (OPT)       | 30          |
| DIODES, STARTER ISOLATOR (SID-1)      | 9,29             | INDICATOR, LOW HYDRAULIC TK LEVEL    | 11       | PUMP, RESERVE OIL CIRCULATING (OPT)     | 30         | SENSOR, RIGHT FRONT HYD. BRAKE TEMP   | 32          |
| D.I.D. PANEL                          | 17               | INDICATOR, LOW STEERING PRESSURE     | 11       | PUMP, PRE-LUBE HI FLOW                  | 26         | SENSOR, RIGHT FRONT SUSPENSION PRESS. | 28          |
| ENGINE ECU - FAN CLUTCH SIGNAL        | 7                | INDICATOR, MANUAL BACK UP LIGHTS     | 13       | PUMP, WINDSHIELD WASHER                 | 31         | SENSOR, RIGHT FRONT WHEEL SPEED       | 9           |
| ENGINE ECU - CUMMINS                  | 7                | INDICATOR, PARK BRAKE                | 32       | RADIO                                   | 21         | SNESOR, RIGHT REAR HYD. BRAKE TEMP    | 32          |
| ENGINE INTERFACE MODULE               | 7                | INDICATOR, PROPUL. SYS AT REST       | 14       | RB1 K1 RELAY (LEFT TURN SIGNAL)         | 22         | SENSOR, RIGHT REAR SUSPENSION PRESS.  | 28          |
| FUEL TANK                             | 32               | INDICATOR, PROPUL. SYS CAUTION       | 13       | RB1 K2 RELAY (RIGHT TURN SIGNAL)        | 22         | SENSOR, SERVICE BRAKE PRESSURE        | 32          |
| FUSE                                  | 26               | INDICATOR, PROPUL. SYS NO PROPEL     | 13       | RB1 K3 RELAY (CLEARANCE LIGHT)          | 22         | SENSOR, STEERING PRESSURE             | 32          |
| FUSE-FB1-1 15A, A/C HTR BLOW MOTOR    | 10               | INDICATOR, PROPUL. SYS NO PROP/RET   | 13       | RB1 K4 RELAY (FLASHER)                  | 22         | SOLENOID, A/C COMPRESSOR              | 17          |
| FUSE-FB1-2 10A, WINDSHLD WASH/WIPE    | 10               | INDICATOR, PROPUL. SYS NOT READY     | 14       | RB3 K1 RELAY (MANUAL BACKUP LTS)        | 19         | SOLENOID, AUTO BRAKE APPLY (OPT)      | 32          |
| FUSE-FB1-3 5A, FUEL, TEMP GA. VOLTM   | 10               | INDICATOR, PROPUL. SYS REDUCED LVL   | 14       | RB3 K2 RELAY (STOP LIGHT)               | 12         | SOLENOID, BRAKE LOCK                  | 18          |
| FUSE-FB1-4 10A, CAB KEY POWER         | 10               | INDICATOR, PROPUL. SYS TEMP CAUTION  | 13       | RB3 K3 RELAY (RETARD LIGHT)             | 19         | SOLENOID, ETHER START                 | 7           |
| FUSE-FB1-5 10A, HOIST SOLENOID        | 10               | INDICATOR, RETARD AT REDUCED LEVEL   | 14       | RB4 K1 RELAY (PARK BRAKE STATUS)        | 27         | SOLENOID, HOIST LIMIT                 | 15          |
| FUSE-FB1-6 15A, TURN/CLEAR LIGHTS     | 10               | INDICATOR, RETARD SPEED CONTROL      | 13       | RB4 K2 RELAY (CRANK INTERLOCK)          | 26         | SOLENOID, LUBE PUMP                   | 15          |
| FUSE-FB1-7 10A, ENGINE OPTIONS        | 10               | INDICATOR, SERVICE BRAKE             | 12       | RB4 K3 RELAY (HORN)                     | 18         | SOLENOID, PARK BRAKE                  | 32          |
| FUSE-FB1-8 10A, AID & INDIC. LTS      | 10               | INDICATOR, STARTER FAILURE           | 32       | RB4 K4 RELAY (BODY UP INVERT)           | 12,28      | SOLENOID, SHUTTER CONTROL             | 7           |
| FUSE-FB1-9 10A, ENG START FAIL.       | 10               | INDICATOR, TIMED ENGINE SHUTDOWN     | 25       | RB5 K1 RELAY (LEFT LOW BEAM HDLT)       | 23         | SOLENOID, STEERING ACCUM. BLEEDDOWN   | 18          |
| FUSE-FB1-10 10A, ENG. SHUTTERS        | 10               | INTERFACE MODULE                     | 25,29,32 | RB5 K2 RELAY (RIGHT LOW BEAM HDLT)      | 23         | SONALERT, #1 (CAB OVERHEAD)           | 11          |
| FUSE-FB1-11 10A, DOME LIGHT SW.       | 10               | LIGHT, AIR RESTRICTION GAUGE         | 24       | RB5 K3 RELAY (LFT HI BEAM HDLT)         | 23         | SONALERT, #2 (IM FAULTS)              | 32          |
| FUSE-FB1-13 10A, RADIO RELAY          | 10,21,26         | LIGHT, BRAKE HYD. TEMP. GAUGE        | 24       | RB5 K4 RELAY (RIGHT HI BEAM HDLT)       | 23         | SPEAKER, RADIO LEFT FRONT             | 21          |
| FUSE-FB1-17 15A, GROUND LEVEL PWR     | 10,25            | LIGHT, CAB DOME                      | 23       | RB6 K1 RELAY (BACK UP HORN)             | 19         | SPEAKER, RADIO LEFT REAR              | 21          |
| FUSE-FB1-18 5A, PLM III POWER         | 10,28            | LIGHT, DIGITAL TACHOMETER            | 24       | RB6 K2 RELAY (ENGINE INTERLOCK)         | 27         | SPEAKER, RADIO RIGHT FRONT            | 21          |
| FUSE-FB1-19 5A, PLM III POWER         | 10,28            | LIGHT, DOME 1-5                      | 10,15    | RB6 K3 RELAY (LINCOLN LUBE SOLENOID)    | 15         | SPEAKER, RADIO RIGHT REAR             | 21          |
| FUSE-FB2-1 15A, SERVICE LIGHTS        | 10               | LIGHT, DYNAMIC RETARD (OPT)          | 19       | RB6 K4 RELAY (BLEEDDOWN POWER SUPPLY)   | 18         | SPEEDOMETER                           | 16,28       |
| FUSE-FB2-2 15A, DOME,FOG,LADDER LTS   | 10               | LIGHT, ENGINE MAINTENANCE            | 24       | RB6 K5 RELAY (ETHER START)              | 7          | STEERING BLEED DOWN IC CHIP           | 18          |
| FUSE-FB2-3 15A, HAZARD LIGHTS         | 10               | LIGHT, FUEL GAUGE                    | 24       | RB6 K7 RELAY (HID HEADLIGHT)            | 16         | SWITCH, A/C RECEIVER/DRIER PRESSURE   | 17          |
| FUSE-FB2-4 10A, INTERFACE MODULE      | 10,32            | LIGHT, HIGH BEAM INDICATING          | 23       | RB6 K8 RELAY (LINCLON LUBE TIMER)       | 15         | SWITCH, AIR PRESSURE DIFFERENTIAL     | 14          |
| FUSE-FB2-5 10A, ORBCOM/VHMS POWER     | 10               | LIGHT, INTERFACE MODULE DOWNLOADING  | 24       | RB7 K1 RELAY (BRAKE LOCK LIMITATION)    | 18         | SWITCH, BODY UP                       | 12,28       |
| FUSE-FB2-6 20A, M M HUB POWER         | 10,29            | LIGHT, LEFT DECK                     | 19       | RB7 K2 RELAY (BRAKE LOCK LIMITATION)    | 18         | SWITCH, BODY UP OVERRRIDE             | 20          |
| FUSE-FB2-7 10A, DISPLAY MODULE        | 10               | LIGHT, LEFT DECK BACK UP             | 19       | RB7 K3 RELAY (BRAKE LOCK LIMITATION)    | 18         | SWITCH, BRAKE LOCK                    | 18,28       |
| FUSE-FB2-8 15A, HID HEADLIGHTS        | 10,16            | LIGHT, LEFT DIAGONAL LADDER          | 19       | RB7 K4 RELAY (BRAKE LOCK LIMITATION)    | 18         | SWITCH, COOLANT LEVEL                 | 27          |
| FUSE-FB2-9 15A, OIL RESERVE PUMP      | 10               | LIGHT, LEFT ENGINE SERVICE #1        | 18       | RB7 K5 RELAY (IM WARNING)               | 32         | SWITCH, DATA STORE                    | 20,29       |
| FUSE-FB2-10 15A, RESERVE CTRL MODULE  | 10               | LIGHT, LEFT ENGINE SERVICE #2        | 18       | RB7 K6 RELAY (TIMED ENGINE SHUTDOWN LT) | 25         | SWITCH, DIMMER COMBO                  | 31          |
| FUSE-FB2-11 20A, BLEEDDOWN PWR SUPPLY | 10               | LIGHT, LEFT FRONT FOG                | 23       | RB7 K7 RELAY (BRAKE LOCK LIMITATION)    | 18         | SWITCH, DIRECT SELECTOR               | 20,32       |
| FUSE-FB2-12 10A, ENG LOAD CNTRL PWR   | 10               | LIGHT, LEFT FRONT TURN/CLEARANCE     | 22       | RB7 K8 RELAY (TIMED ENGINE SHUTDOWN)    | 10,25      | SWITCH, DOME LIGHT (AUX-BOX)          | 10,15       |
| FUSE-FB2-13 10A, KEYSWITCH POWER      | 8,10,25          | LIGHT, LEFT HIGH BEAM                | 23       | RB8 K1 RELAY (TCI-70%)                  | 28         | SWITCH, ENGINE LEFT SERVICE LIGHT     | 18          |
| FUSE-FB2-17 20A, ENGINE ECM POWER     | 10               | LIGHT, LEFT HI-MTD HIGH BEAM         | 23       | RB8 K2 RELAY (TCI-100%)                 | 28         | SWITCH, ENGINE OIL PRESSURE           | 26          |
| FUSE-FB2-18 20A, ENGINE ECM POWER     | 10               | LIGHT, LEFT HI-MTD LOW BEAM          | 23       | RB8 K3 RELAY (A/C SHUTTER CONTROL)      | 7          | SWITCH, ENGINE RIGHT SERVICE LIGHT    | 18          |
| FUSE-FB2-19 20A, ENGINE ECM POWER     | 10               | LIGHT, LEFT HYDRAULIC CABINET        | 18       | RB8 K4 RELAY (PLM3 LIGHTS)              | 28         | SWITCH, ENGINE RUN OIL PRESSURE       | 16          |
| FUSE-FB2-20 20A, ENGINE ECM POWER     | 10               | LIGHT, LEFT LOW BEAM                 | 23       | RB8 K5 RELAY (SCHUTTER CONTROL)         | 7          | SWITCH, FOG LIGHT                     | 23          |
| FUSE-FB3-1 15A, CAB DRIVE SYS. PWR    | 10               | LIGHT, LEFT REAR BACK UP             | 19       | RB8 K6 RELAY (PLM3 RED LIGHTS)          | 28         | SWITCH, GROUND LEVEL SHUTDOWN         | 2,8,10      |
| FUSE-FB3-2 10A, AUTO LUBE TIMER       | 10,15            | LIGHT, LEFT REAR FOG                 | 23       | RB8 K7 RELAY (PLM3 AMBER LIGHTS)        | 28         |                                       | 25,28       |
| FUSE-FB3-3 15A, GE PWR INTERFACE MOD  | 10,32            | LIGHT, LEFT REAR RETARD              | 19       | RB8 K8 RELAY (PLM3 GREEN LIGHTS)        | 28         | SWITCH, HAZARD LIGHT                  | 22          |
| FUSE-FB3-4 20A, OP-CAB SEAT POWER     | 10               | LIGHT, LEFT REAR TURN/CLEARANCE      | 22       | RB9 K1 RELAY (START CIRCUIT)            | 10,26      | SWITCH, HEADLIGHT                     | 10,23       |
| FUSE-FB3-13 10A, L. FRT WHEEL SPS     | 9,10             | LIGHT, LEFT TAIL/STOP                | 12       | RB9 K2 RELAY (AUTO BRAKE APPLY POWER)   | 32         | SWITCH, HEATED MIRROR                 | 16          |
| FUSE-FB3-14 10A, R. FRT WHEEL SPS     | 9,10             | LIGHT, LEFT TURN INDICATING          | 22       | RB9 K3 RELAY (LOW STEERING PRESS. PWR)  | 11         |                                       |             |
| FUSE-FB3-17 10A, 12V POWER PLUGS      | 10,21            | LIGHT, LINK ENERGIZED                | 24       | RB9 K4 RELAY (PARK BRAKE OFF)           | 32         |                                       |             |
| FUSE-FB3-18 20A, R.H. WINDOW          | 10,21            | LIGHT, OIL PRESSURE GAUGE            | 24       | RB9 K5 RELAY (KEYSW START ACTIVATION)   | 8          |                                       |             |
| FUSE-FB3-19 20A, L.H. WINDOW          | 10,21            | LIGHT, RIGHT DECK (OPT)              | 19       | RB9 K6 RELAY (LOW STEERING PRESS. GND)  | 11         |                                       |             |
| FUSE-FB4-1 10A, BRAKE CIRCUITS        | 10,32            | LIGHT, RIGHT DECK BACK UP            | 19       | RELAY, BRAKE WARNING                    | 11         |                                       |             |
| FUSE-FB4-2 5A, PLM III                | 10               | LIGHT, RIGHT DIAGONAL LADDER (OPT)   | 19       | RELAY, CONTROL POWER (CPR)              | 10         |                                       |             |
| FUSE-FB4-3 5A, INTERFACE MODULE       | 10,32            | LIGHT, RIGHT ENGINE SERVICE #1       | 18       | RELAY, ENGINE START                     | 26,33      |                                       |             |
| FUSE-FB4-4 10A, VHMS MODULE           | 10               | LIGHT, RIGHT ENGINE SERVICE #2       | 18       | RELAY, GENERATOR FIELD (GFR)            | 10         |                                       |             |
| FUSE-FB4-5 5A, M M HUB POWER          | 10,29            | LIGHT, RIGHT FRONT FOG               | 23       | RELAY, GROUND LEVEL POWER               | 8,10,25,28 |                                       |             |
| FUSE-FB4-6 5A, DISP MODULE            | 10               | LIGHT, RIGHT FRONT TURN/CLEAR        | 22       | RELAY, KEYSWITCH POWER                  | 8,10,25    |                                       |             |
| FUSE-FB4-7 15A, BLEEDDOWN POWER       | 10               | LIGHT, RIGHT HIGH BEAM               | 23       | RELAY, KONTROL POWER (KPR)              | 2,8,10     |                                       |             |
| FUSE-FB4-8 10A, LIGHTED SW POWER      | 10               | LIGHT, RIGHT HI-MTD HIGH BEAM        | 23       | RELAY, MAINTAIN PRE-LUBE RUN (MPRR)     | 26         |                                       |             |
| FUSE-FB4-9 10A, DIRECT SEL. SW.       | 10,32            | LIGHT, RIGHT HI-MTD LOW BEAM         | 23       | RELAY, RADIO                            | 21         |                                       |             |
| FUSE-FB4-17 5A, GAUGE VOLTAGE         | 10               | LIGHT, RIGHT HYDRAULIC CABINET       | 18       | RELAY, WIPER (HI)                       | 31         |                                       |             |
| FUSE-FB4-18 5A, PEDAL VOLTAGE         | 10               | LIGHT, RIGHT LOW BEAM                | 23       | RELAY, WIPER (LO)                       | 31         |                                       |             |
| FUSE-FB4-19 5A, ENGINE INTERFACE      | 10               | LIGHT, RIGHT REAR BACK UP            | 19       | RELAY, 12V POWER (AUX-BOX)              | 10,21      |                                       |             |

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